YOUR PERSONAL COACH

PHYSICS

for JEE Main and Advanced

Mechanics Vol II

Mc Graw Hill Education

Shashi Bhushan Tiwari

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Shashi Bhushan Tiwari

McGraw Hill Education (India) Private Limited

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Preface

Physics is extremely hierarchical. It is essential to have a thorough understanding of the concepts of the previous chapters in order to master a particular one. Also, the principles involved are often simple to state but very difficult to apply.

For the reasons cited above a Physics text book must have a great deal of continuity in what it discusses and must have the relevant examples to illustrate every little principle. This book has been written keeping these two basic principles in mind. I have tried my level best to unfold the concepts gradually, one by one; illustrating each of them with examples.

During my years of experience as a Physics teacher, I have realised that nothing contributes more to understand this subject than a good example. This book is a translation of this philosophy of mine.

This is a student centric book. Its basic aim is to make a student learn the basic principles of Physics by himself/herself. By saying this, I am not discouraging discussions with friends and teachers. Discussions are essential part of learning this subject. All I intend is to provide a book which will help a student understand the basics on his own while working through the book, requiring minimum external help. For this reason, I have provided solutions to almost all problems in the book.

To make problem solving enjoyable, I have tried to pick up examples from real life wherever possible.

I shall be grateful to all of you who point out any error or help me with useful suggestions.

S.B. Tiwari

How to use this Book

To make full use of this book one must go through the topics sequentially while working through the examples and in-chapter problems given under heading "***Your Turn***". By doing this you will have a fair amount of grasp over all the essentials in a chapter.

Miscellaneous examples given at the end of each chapter have problems which involve multiple concepts or have some mathematical complexity or are tricky. *If you are studying the subject for the first time or are hard pressed for time*, you *may* skip the section on miscellaneous examples.

Almost every solved example starts with explanation of physical situation and basic principles involved. This feature comes under heading ***"Concepts"*** at the beginning of each example.

I have highlighted the important points of learning under the heading ***"In short".*** Here, I have also taken important learning points from the examples. While going through the chapter it is essential to go through these points.

Physics cannot be mastered without practice. Keeping this in view I have given three Worksheets (exercises) after every chapter. ***Worksheet 1*** has multiple choice objective type questions with single correct answer. ***Worksheet 2*** has multiple choice questions having *one or more than one correct answers.* ***Worksheet 3*** has subjective problems. A good number of problems has been given in the Worksheets to give you a good practice on concepts learnt.

After few chapters, at regular intervals, you will find separate assignments on miscellaneous type problems. These are problems based on latest trend of competitive examinations and contain ***Match the Column type questions and problems based on a given paragraph***. Attempt these questions only after you gain enough confidence in the related chapters. I have kept these problems in separate chapters so that you have no bias or hint about the equation/s to use.

In the last chapter, you will find a collection of questions asked in competitive examinations since 2005.This is an ideal collection of problems for revision.

In the end of the book, solutions to all questions has been given. Solutions are quite descriptive and easy to understand.

Those who desire to practice at even higher level, I recommend my book – "*Problems in Physics for JEE Advanced*".

I hope you will enjoy this book.

S.B. Tiwari

Acknowledgements

My sincere thanks to all who supported, encouraged and helped me in preparation of the book. My special thanks to –

- My family members.
- My students, who have taught me a lot.
- Entire management team at McGraw Hill. In particular, I am grateful to Mr. Biswajit Das who worked tirelessly in bringing out the book.

S.B. Tiwari

Contents

CHAPTER 1

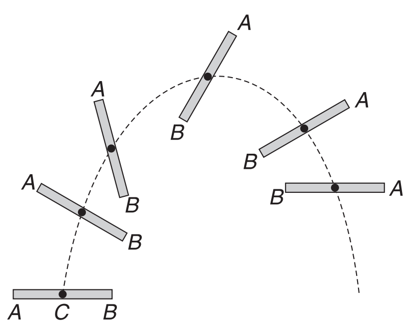
Centre of Mass

"Your centre of mass is a place you cannot visit but you always carry with you. Like memories, it is part of life's baggage."

–Neil deGrasse Tyson

1. INTRODUCTION

Think of a stick tossed in air. It spins while going up and falling down. Motion of a point, like end A, is not along a parabolic path. Moreover, paths taken by different points, like the paths of ends A and B, have different geometry. Because every part of the stick moves differently and have different speeds, we cannot represent the stick as a particle. In fact, the stick is a collection of interacting particles all of which are moving differently. A particle at the end A does not move with an acceleration equal to g because it experiences force due to neighbouring particles, apart from the gravity. However, the stick has one special point – its centre of mass – that does move on a parabolic path with acceleration g, just like a particle projectile.

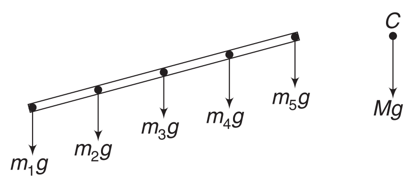


We will learn by the end of chapter on 'Rotational motion' that the motion of the stick can be described as combination of two motions – translation and rotation. We think that the entire mass of the stick is concentrated in the form of a particle kept at its centre of mass. This special point - Centre of mass - of the stick moves on a parabolic path like a projectile. All other points rotate about the Centre of mass. Knowing the motion of centre of mass and superimposing a rotational motion about the centre of mass, helps us in describing the most complex kind of motions of extended objects.

In this chapter, we will learn the procedure of locating the centre of mass of a system of particles and also learn how to predict the motion of the centre of mass of the system. It will be of immense use in the chapters to follow.

2. IMPORTANCE OF CENTRE OF MASS

The centre of mass (COM) of a system of particles moves as if the entire mass were concentrated there and all external forces were applied there. When a stick is tossed in air, each particle experiences its weight (m_1g, m_2g, m_3g...) as an external force. But, acceleration of particle of mass m_1 is not g, because it experiences forces applied by neighbouring particles also.



The special point in the stick whose acceleration is decided solely by the external forces is called its COM.

If mass of the stick is $M = m_1 + m_2 + m_3 + \ldots$, then net external force on it is

$$F_{ext} = m_1g + m_2g + m_3g + \ldots$$
$$= Mg$$

Acceleration of COM is

$$a_{cm} = \frac{F_{ext}}{M} = g$$

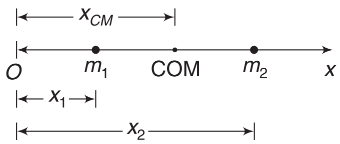
For describing translational motion of the stick, we consider its entire mass at its COM and the sum of all external forces ($m_1g + m_2g + m_3g$ ...) acting at that point.

Without going into the details of internal interactions amongst the particles, we can predict the acceleration of COM by knowing the external forces only.

3. LOCATION OF CENTRE OF MASS

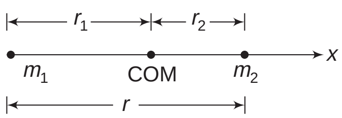
3.1 Two Particles

The given figure shows two particles of masses m_1 and m_2, placed on the x-axis, at co-ordinates x_1 and x_2 respectively. The position of COM for the system of two particles is given by



$$x_{cm} = \frac{m_1x_1 + m_2x_2}{m_1 + m_2} = \frac{m_1x_1 + m_2x_2}{M} \quad ...(1)$$

Where, $M = m_1 + m_2$ is the total mass of the system. The COM of a two-particle system lies somewhere between them on the line joining the two.



If the origin is chosen at the position of mass m_1, then x co-ordinate of COM is the distance of the COM from the particle of mass m_1. Let us assume this distance as r_1 and let the distance between the two particles be r.

$$x_{CM} = \frac{m_1x_1 + m_2x_2}{m_1 + m_2}$$

or
$$r_1 = \frac{0 + m_2r}{m_1 + m_2} = \frac{m_2r}{m_1 + m_2}$$

Distance of the COM from m_2 is

$$r_2 = r - r_1 = \frac{m_1r}{m_1 + m_2}$$

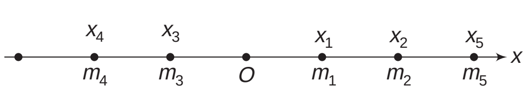
It is easy to see that $\frac{r_1}{r_2} = \frac{m_2}{m_1}$

$$\Rightarrow \quad m_1r_1 = m_2r_2 \quad ...(2)$$

The last equation leads us to conclude that the COM of the system will be closer to the heavier mass.

3.2 Many Particles on a Straight Line

Particles of masses $m_1, m_2, m_3... m_n$ are located on the x-axis at co-ordinates $x_1, x_2, x_3, ... x_n$ respectively. The COM of this collection of particles lies on x-axis with x co-ordinate of COM given by



$$x_{CM} = \frac{m_1x_1 + m_2x_2 + ... + m_nx_n}{m_1 + m_2 + ... + m_n}$$

$$= \frac{1}{M}\sum_{i=1}^{n} m_ix_i \quad \text{where, } M \text{ is total mass of the system.}$$

If the position of COM is chosen as the origin,

then $\sum_{i=1}^{n} m_ix_i = 0$

3.3 Three-Dimensional Distribution of particles

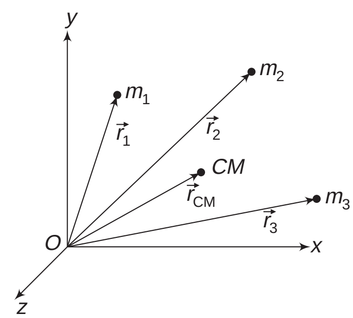
When particles in a system are scattered in a three-dimensional space, then the position of COM is defined by three co-ordinates, given as:

$$x_{CM} = \frac{1}{M}\sum_{i=1}^{n} m_ix_i;\ y_{CM} = \frac{1}{M}\sum_{i=1}^{n} m_iy_i;\ z_{CM} = \frac{1}{M}\sum_{i=1}^{n} m_iz_i \quad ...(3)$$

We can also write the above relations in terms of a single vector relation. The position vector of the COM of a system of particles is given by

$$\vec{r}_{CM} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2 + ... + m_n\vec{r}_n}{M}$$

$$= \frac{1}{M}\sum_{i=1}^{n} m_i\vec{r}_i \quad ...(4)$$



Where, $\vec{r}_1, \vec{r}_2 ... \vec{r}_n$ are position vectors of particles of masses $m_1, m_2 ... m_n$, respectively.

$$\vec{r}_i = x_i\hat{i} + y_i\hat{j} + z_i\hat{k}$$

and
$$\vec{r}_{CM} = x_{CM}\hat{i} + y_{CM}\hat{j} + z_{CM}\hat{k}$$

If we choose the COM of the system as the origin, then

$$x_{CM} = y_{CM} = z_{CM} = 0$$

$$\Rightarrow \quad \sum m_ix_i = \sum m_iy_i = \sum m_iz_i = 0$$

Example 1 A particle of mass $m_1 = 1\,\text{kg}$ is located at $(3, 0)\,\text{m}$ and another particles of mass $m_2 = 2\,\text{kg}$ is located at $(0, 4)\,\text{m}$ in xy-plane. Find the distance of the COM of the two particle-system from the particle of mass m_1.

Solution

Concepts

(i) The COM will lie on the line joining the two particles.

(ii) Distance of COM from m_1 is given by

$$r_1 = \frac{m_2 r}{m_1 + m_2},$$

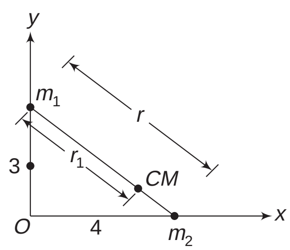
where r = distance between the particles

Distance between the particles is

$$r = \sqrt{3^2 + 4^2} = 5\,\text{m}$$

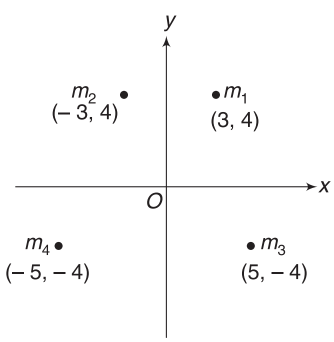
Distance of COM from m_1 is

$$r_1 = \frac{m_2 r}{m_1 + m_2} = \frac{2 \times 5}{1 + 2} = \frac{10}{3}\,\text{m}$$



Example 2 Four particles are located in xy plane at co-ordinates indicated in the figure. The co-ordinates are in metres and the masses are given as follows.

$$m_1 = 1\,\text{kg}, \quad m_2 = 4\,\text{kg}$$
$$m_3 = 2\,\text{kg}, \quad m_4 = 3\,\text{kg}$$



(i) Find the co-ordinates of the COM of the system.

(ii) A fifth particle of mass $m_5 = 1\,\text{kg}$ is added to the system and is placed at (0, 0, 4) m.

Find the co-ordinates of COM of the collection of five particles.

Solution

Concepts

(i) The COM of the four-particle system will lie in xy plane and equation (3) will be used to find its co-ordinates – x_{CM} and y_{CM}.

(ii) After addition of the fifth particle, the position of the COM can be defined by finding x_{CM}, y_{CM} and z_{CM}.

(i) $$x_{CM} = \frac{m_1x_1 + m_2x_2 + m_3x_3 + m_4x_4}{m_1 + m_2 + m_3 + m_4}$$

$$= \frac{1 \times 3 + 4(-3) + 2 \times 5 + 3(-5)}{1 + 4 + 2 + 3}$$

$$= -\frac{14}{10}m = -1.4\,\text{m}$$

$$y_{CM} = \frac{m_1y_1 + m_2y_2 + m_3y_3 + m_4y_4}{m_1 + m_2 + m_3 + m_4}$$

$$= \frac{1 \times 4 + 4 \times 4 + 2(-4) + 3(-4)}{10}$$

$$= 0$$

(ii) $$x_{CM} = \frac{1 \times 3 + 4(-3) + 2 \times 5 + 3(-5) + 1(0)}{1 + 4 + 2 + 3 + 1}$$

$$= -\frac{14}{11}\text{m}$$

$$y_{CM} = \frac{1 \times 4 + 4 \times 4 + 2(-4) + 3(-4) + 1(0)}{11} = 0$$

$$z_{CM} = \frac{1 \times 0 + 4 \times 0 + 2 \times 0 + 3 \times 0 + 1 \times 4}{11}$$

$$= \frac{4}{11}\text{m}$$

Note: For solving second part of the problem, one can replace the four masses, m_1, m_2, m_3 and m_4, with a point mass, equal to $m_1 + m_2 + m_3 + m_4 = 10\,\text{kg}$, placed at the COM of the system, at (–1.4, 0) m.

Now, the problem has only two point masses:

M = 10 kg, placed at (–1.4, 0, 0) m

and m_5 = 1 kg, placed at (0, 0, 4) m

$$\therefore \quad x_{CM} = \frac{10 \times (-1.4) + 1 \times 0}{11} = \frac{-14}{11}\,\text{m}$$

$$y_{CM} = \frac{10 \times 0 + 1 \times 0}{11} = 0$$

$$z_{CM} = \frac{10 \times 0 + 1 \times 4}{11} = \frac{4}{11}\,\text{m}$$

Co-ordinates of COM are $\left(\frac{-14}{11}, 0, \frac{4}{11}\right)$m.

Example 3 The centre of mass of a system of four particles is at the origin of the co-ordinate system. Masses of the particles are 1g, 2g, 3g and 4g. The particles of mass 1g and 3g are displaced by 2 cm and 1 cm respectively in positive x-direction. The particle of mass 2g is moved in negative x-direction by 3 cm. Find the co-ordinates of COM of the system in the new arrangement.

Solution

Concepts

$$x_{CM} = \frac{m_1x_1 + m_2x_2 + m_3x_3 + m_4x_4}{M}$$

$$\Delta x_{CM} = \frac{m_1\Delta x_1 + m_2\Delta x_2 + m_3\Delta x_3 + m_4\Delta x_4}{M}$$

$$\Delta x_1 = 2\,\text{cm};\ \Delta x_2 = 1\,\text{cm};\ \Delta x_3 = -3\,\text{cm};\ \Delta x_4 = 0$$

Change in x co-ordinate of COM is

$$\Delta x_{CM} = \frac{1 \times 2 + 2 \times 1 + 3 \times (-3) + 4 \times 0}{1 + 2 + 3 + 4}$$

$$\therefore \quad \Delta x_{CM} = \frac{-5}{10} = -0.5 \text{ cm}$$

x co-ordinate of COM was zero before movement of the particles. It changed by $\Delta x_{CM} = -0.5$ cm.

$\therefore$ New x co-ordinate is -0.5 cm.

There is no change in y and z co-ordinates.

$\therefore$ New co-ordinates of COM are (-0.5 cm, 0, 0).

Example 4 A particle of mass m has position vector $\vec{r}_1 = \hat{i} + \hat{j} + \hat{k}$ and another particle of mass $2m$ has position vector $\vec{r}_2 = \hat{i} - \hat{j} - \hat{k}$. Write the vector obtained by joining mass m to the COM of the system.

Solution

Concepts

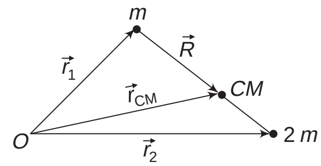
(i) $\vec{r}_{CM} = \frac{1}{M}(m_1\vec{r}_1 + m_2\vec{r}_2)$

(ii) The required vector is $\vec{r}_{CM} - \vec{r}_1$

$$\vec{r}_{CM} = \frac{m(\hat{i} + \hat{j} + \hat{k}) + 2m(\hat{i} - \hat{j} - \hat{k})}{3m}$$

$$= \hat{i} - \frac{1}{3}\hat{j} - \frac{1}{3}\hat{k}$$

Vector connecting m to COM is



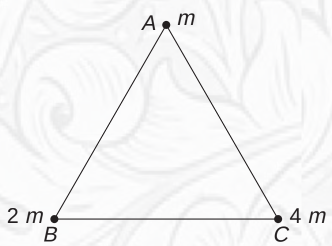
$$\vec{R} = \vec{r}_{CM} - \vec{r}_1$$

$$= -\frac{4}{3}\hat{j} - \frac{4}{3}\hat{k}$$

Your Turn

Q.1 Two stars, having mass m and $4m$, are separated by a distance d. Both of them are moving in circles about the COM of the system with equal angular speed. Find the radius of the circular path of the star of mass m. Assume the stars to be point masses.

Q.2 Three masses have been placed on the vertices of an equilateral triangle of side length $a = 10\,m$ (see figure). You are standing at vertex B. How far must you travel to reach the COM of the three-particle system?



Q.3 Find the displacement of the COM of a system of two particles, having masses m and $3m$, if the particles are displaced in opposite directions, along the line joining them, by 6 cm and 2 cm respectively.

Q.4 Particles of masses $m_1 = 1$ kg, $m_2 = 2$ kg and $m_3 = 3$ kg are placed on x-axis, at $x_1 = 4$ cm, $x_2 = 6$ cm and $x_3 = -1$ cm respectively. Where shall we place a fourth particle of mass $m_4 = 2$ kg so that the COM of the system of four particles is at

(i) the origin

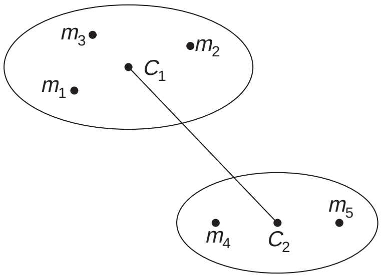
(ii) the location of fourth particle itself.

Q.5 Particles of mass $m_1 = 1$ kg, $m_2 = 2$ kg and $m_3 = 3$ kg are placed at points having position vectors $\vec{r}_1 = (\hat{i} + 4\hat{j} + \hat{k})$m, $\vec{r}_2 = (\hat{i} + \hat{j} + \hat{k})$m and $\vec{r}_3 = (2\hat{i} - \hat{j} - 2\hat{k})$m respectively. Find the position vector of the COM of the system.

3.4 Centre of Mass of Groups of Particles

Consider a group of particles having masses m_1, m_2 and m_3. The COM of this group is at C_1. Another group, comprising particles of masses m_4 and m_5 has its COM at C_2.

If you are asked to find the COM of the collection of five particles, you can replace the first group by a particle of mass $M_1 = m_1 + m_2 + m_3$, placed at C_1 and the second group can be replaced by a particle of mass $M_2 = m_4 + m_5$, placed at C_2. Now find the COM of the two-particle system, comprising masses M_1 and M_2.



One can prove this as follows.

$$\vec{r}_{CM} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2 + m_3\vec{r}_3 + m_4\vec{r}_4 + m_5\vec{r}_5}{m_1 + m_2 + m_3 + m_4 + m_5}$$

$$= \frac{\left(\dfrac{m_1\vec{r}_1 + m_2\vec{r}_2 + m_3\vec{r}_3}{m_1 + m_2 + m_3}\right)(m_1 + m_2 + m_3) + \left(\dfrac{m_4\vec{r}_4 + m_5\vec{r}_5}{m_4 + m_5}\right)(m_4 + m_5)}{m_1 + m_2 + m_3 + m_4 + m_5}$$

$$= \frac{M_1\vec{r}_{c_1} + M_2\vec{r}_{c_2}}{M_1 + M_2} \qquad \text{...(5)}$$

3.5 Centre of Mass of Continuous Bodies

An ordinary object, such as this book, contains countless number of particles. The summations in equation (3) are performed with the help of integration.

A very small element of mass dm can be regarded as a point mass and the formulae for co-ordinates of COM can be given as follows:

$$x_{CM} = \frac{1}{M}\int x\,dm,\quad y_{CM} = \frac{1}{M}\int y\,dm,\quad z_{CM} = \frac{1}{M}\int z\,dm \quad ...(6)$$

where (x, y, z) are co-ordinates of elemental mass dm. The examples to follow will clarify the procedure.

3.5.1 Use of Symmetry

We need not perform all the three integrals if a body has a point, a line or a plane of symmetry. The COM of such a body always falls on that point, on that line or on that plane. A uniform ball or a uniform cube has its geometrical centre as the point of symmetry. The COM of a uniform ball or a uniform cube is at its geometrical centre. Similarly, the COM of a uniform thin rod is at its geometrical centre. The axis of a cone is a line of symmetry and its COM lies on its axis.

It is also important to note that the COM of an object may lie outside the material. The COM of a ring is at its geometrical centre, where there is no material.

Example 5 *COM of a uniform semicircular ring*

A uniform semicircular ring has radius R. Find the position of COM of the ring.

Solution

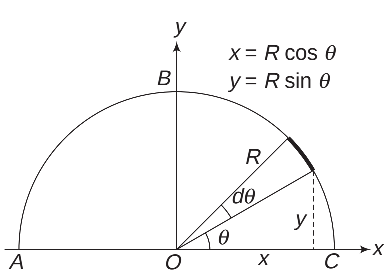
Concepts

(i) The mass is distributed in a plane. In general, we will need two co-ordinates – x_{CM} and y_{CM} – to fix the position of COM.

(ii) In the figure shown, y-axis is a line of symmetry. Distribution of mass to the left and right of y-axis is identical. The COM lies on y-axis. Hence, $x_{CM} = 0$. We need to find y_{CM} only.

(iii) $y_{CM} = \frac{1}{M}\int y\,dm$

ABC is a half-ring of radius R. We choose the co-ordinate axes as shown. Such choice of co-ordinate axes is intelligent, as y-axis happens to be the symmetry-axis and $x_{CM} = 0$.



Consider a small element subtending an angle $d\theta$ at the centre O.

Length of the element = $R\,d\theta$

If λ (kg m^{-1}) is the linear mass density of the ring then mass of the element is dm = $\lambda \cdot Rd\theta$

Co-ordinates of the element are ($R\cos\theta$, $R\sin\theta$). Remember that element is of infinitesionally small size and can be treated like a point mass. Mass of the half ring, $M = \lambda \cdot \pi R$.

Now $$y_{cm} = \frac{1}{M}\int y\,dm$$

$$= \frac{1}{\lambda\pi R}\int_{\theta=0}^{\theta=\pi}(R\sin\theta)(\lambda R d\theta)$$

[As θ changes from 0 to π, all elements in the ring gets covered]

$$\therefore \quad y_{CM} = \frac{\lambda R^2}{\lambda\pi R}\int_0^{\pi}\sin\theta\,d\theta = -\frac{R}{\pi}[\cos\ \theta]_0^{\pi}$$

$$= -\frac{R}{\pi}[\cos\pi - \cos 0] = -\frac{R}{\pi}[-1-1] = \frac{2R}{\pi}$$

Hence, COM is at a distance of $\frac{2R}{\pi}$ from O on y-axis.

Example 6 *Uniform semicircular disc*

A semicircular disc has radius R. The disc is uniform. Locate its COM.

Solution

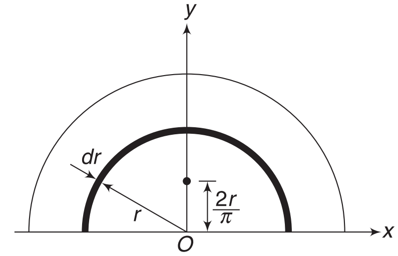
Concepts

(i) In the figure shown, y-axis is the line of symmetry. COM lies on this line. It means $x_{CM} = 0$.

(ii) We will divide the disc into countless number of ring elements. Each of these ring elements can be replaced with a point mass, placed at the respective COMs (refer to article 3.4). The system now reduces to a collection of large number of particles, distributed along the y-axis.

Consider the co-ordinate axes as shown. We need to find y_{CM}.

Consider a ring element of radius r having infinitesimally small thickness dr. Many such rings, laid one after another, will make the disc.



Let σ (kg m^{-2}) be the surface density of mass for the disc (i.e., mass per unit area of the disc). Circumference of the ring element = πr. Area of the ring element (the strip shown in dark colour)

$$dA = \pi r \cdot dr$$

Mass of the ring element, $dm = \sigma dA = \sigma\pi r dr$

y co-ordinate of COM of the ring element is

$$y = \frac{2r}{\pi}$$

The ring element can be replaced by a point mass dm placed on the y-axis at $y = \frac{2r}{\pi}$.

Total mass of disc, $M = \frac{\sigma \pi R^2}{2}$

$$\therefore\ y_{CM} = \frac{\int y\,dm}{M}$$

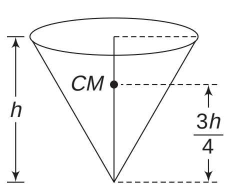
$$= \frac{1}{\frac{\sigma\pi R^2}{2}} \int_{r=0}^{R} \left(\frac{2r}{\pi}\right)(\sigma\pi r\,dr)$$

$$= \frac{4}{\pi R^2}\int_0^R r^2 dr = \frac{4}{\pi R^2}\cdot\frac{R^3}{3} = \frac{4R}{3\pi}$$

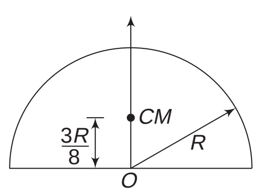
Note: It is recommended to remember the position of the COM for the following objects.

1. COM of a uniform triangular plate is at its centroid.
2. COM of a uniform solid cone of height h is on its symmetry axis, at a distance $\frac{3h}{4}$ from the apex.
3. COM of a uniform solid hemisphere is on its symmetry axis, at a distance $\frac{3R}{8}$ from the centre of its base circle.
4. COM of a uniform hemispherical shell is on its symmetry axis, at a distance $\frac{R}{2}$ from the centre of its base circle.

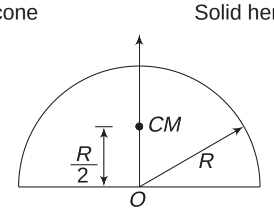
One who is interested may try to prove these results by use of integration.



Solid cone



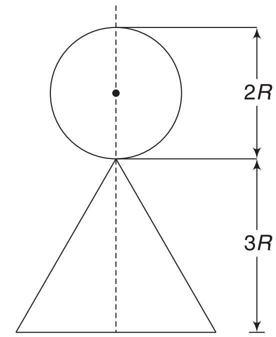
Solid hemisphere



Hemispherical shell

Example 7 *COM of combination of known geometrical shapes*

A craftsman makes a doll by joining a solid sphere to a cone, as shown [Later he painted various features like eyes, nose, etc., on the object]. The axis of the cone coincides with the diameter of the ball. Mass of the uniform ball is $2M$ and that of the cone is $3M$. Radius and height of the cone are R and $3R$ respectively. Find the position of the COM of the doll.



Solution

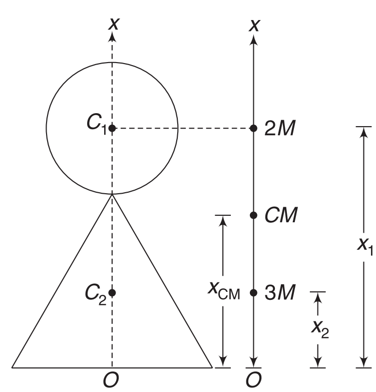
Concepts

(i) Refer to Article 3.4. We can replace the ball with a point mass $2M$ placed at its COM and the cone can be replaced with a point mass $3M$ kept at its COM.

(ii) To find COM of the resulting two-particle system. we will use

$$x_{CM} = \frac{m_1x_1 + m_2x_2}{m_1 + m_2}$$

Consider O as the origin and the axis of the cone as x-axis. x-axis happens to be the line of symmetry and the COM of the doll will lie on this line. We consider a point mass $2M$ at the centre (C_1) of the sphere. x co-ordinate of this mass is $x_1 = 4R$.



Point mass $3M$ is assumed at the COM of the cone (at C_2). x co-ordinate of this mass is

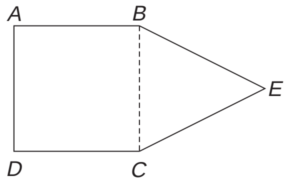
$$x_2 = \frac{1}{4}(3R) = \frac{3R}{4}$$

x co-ordinate of the COM of the doll is

$$x_{CM} = \frac{2Mx_1 + 3Mx_2}{2M + 3M}$$

$$= \frac{2M(4R) + 3M\left(\frac{3R}{4}\right)}{5M} = \frac{41R}{20} = 2.05R$$

Example 8 A piece of paper has been cut in the shape shown in figure. ABCD is a square and BEC is an equilateral triangle. Find the position of COM of the paper piece.



Solution

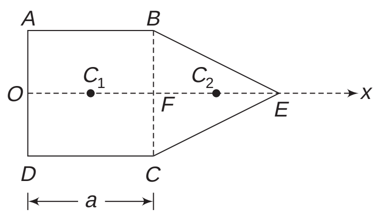
Concepts

(i) The COM of the square piece will be at its geometrical centre. The COM of the triangle will be at its centroid.

(ii) We can replace the square and triangle with point masses at their respective COMs.

(iii) A line through E and centre of the square is symmetry axis.

Let side length of the square be a and σ (kg m^{-2}) be the mass of paper per unit area. Mass of the square



$$m_1 = \sigma \cdot a^2$$

Mass of the triangle

$$m_2 = \sigma \frac{1}{2}(a)\left(\frac{\sqrt{3}}{2}a\right) = \sigma \frac{\sqrt{3}}{4}a^2$$

The x-axis shown in the figure is the line of symmetry and the COM lies on this line. With O as the origin, the x co-ordinate of COM of the square (at C_1) is

$$x_1 = \frac{a}{2}$$

COM of the triangle (at C_2) has co-ordinate

$$x_2 = a + \frac{a}{2\sqrt{3}} \qquad \left[\because \; FC_2 = \frac{1}{3}FE = \frac{1}{3}\cdot\frac{\sqrt{3}}{2}a\right]$$

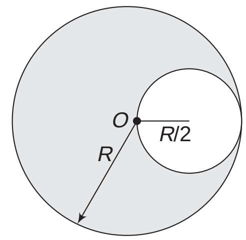
Now, we think as if point masses m_1 and m_2 are at C_1 and C_2 respectively.

$$\therefore \quad x_{CM} = \frac{m_1x_1 + m_2x_2}{m_1 + m_2}$$

$$= \frac{(\sigma a^2)\left(\frac{a}{2}\right) + \left(\sigma\frac{\sqrt{3}}{4}a^2\right)\left(1 + \frac{1}{2\sqrt{3}}\right)a}{\sigma a^2 + \sigma\frac{\sqrt{3}}{4}a^2} \simeq 0.74a$$

Example 9 *COM of an object with a cavity*

A spherical cavity of radius $\frac{R}{2}$ is carved out from a uniform sphere of radius R. The cavity has its boundary passing through the centre of the sphere, as shown in figure. Locate the COM of the sphere with cavity.



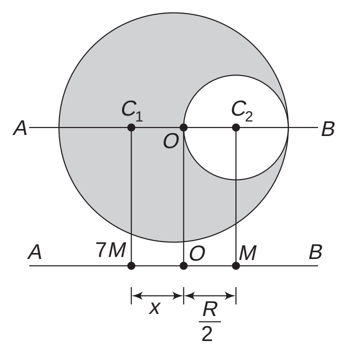
Solution

Concepts

(i) Mass of the sphere will be proportional to its volume. If volume of the removed sphere of radius $\frac{R}{2}$ is v, then volume of the original sphere of radius R will be $8v$ ($\because v \propto r^3 \Rightarrow$ If radius is doubled, volume becomes 8 times). Volume of material in the given object will be $7v$. If the mass of removed part is M, then mass of remaining part is $7M$.

(ii) Line AB shown in figure is the line of symmetry for the given object. Its COM will lie on line AB. Because mass to the left of O is more than the mass to the right of O, the COM will be to the right of O, say at C_1.

(iii) For the purpose of calculation of COM, an extended body can be replaced with a point mass placed at its COM.



Mass of removed part is M. Mass of the given object with cavity is $7M$. Imagine that the cavity is filled back to complete the sphere. We can consider the sphere to be made up of two objects – (i) a sphere of radius $\frac{R}{2}$, having mass M and (ii) a sphere of radius R, with a cavity, having mass $7M$.

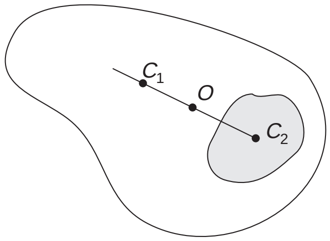
The two masses – M and $7M$ – can be replaced with point masses placed at their respective COMs, at C_2 and C_1. The system of these two masses has its COM at O.

From equation (2) in Article 3.1 we can write

$$7M \cdot x = M \cdot \frac{R}{2}$$

$$\Rightarrow \qquad x = \frac{R}{14}$$

Note: A body has its COM at O. The shaded part of the body has its COM at C_2. Now, the shaded part is removed. The COM of the remaining object is at C_1.



(i) C_1, O and C_2 lie on a straight line.

(ii) If mass of the removed part is m and mass of the remaining object is M, then

$$m(OC_2) = M(OC_1).$$

In Short:

(1) COM of a body (or collection of particles) is a point, whose motion can be predicted by knowing the external forces, only without worrying about the interaction of the particles amongst themselves.

(2) For a discrete particle system, the COM is located by finding its co-ordinates as

$$x_{CM} = \frac{1}{M}\Sigma m_i x_i;\ y_{CM} = \frac{1}{M}\Sigma m_i y_i;$$

$$z_{CM} = \frac{1}{M}\Sigma m_i z_i$$

where M is the total mass of the system.

(3) For a body having continuous mass distribution, the process of summation is done by integration.

$$x_{CM} = \frac{1}{M}\int x\,dm;\ y_{CM} = \frac{1}{M}\int y\,dm;$$

$$z_{CM} = \frac{1}{M}\int z\,dm$$

(4) For a two-particle system, $m_1 r_1 = m_2 r_2$.

Here, r_1 and r_2 are distances of the two particles (having masses m_1 and m_2 respectively) from the COM of the system.

(5) If the x co-ordinate of individual particles in a system are changed by $\Delta x_1, \Delta x_2, \Delta x_3 \ldots \Delta x_n$, then the change in x co-ordinates of the COM of the system is

$$\Delta x_{CM} = \frac{1}{M}\sum_{i=1}^{n} m_i \Delta x_i$$

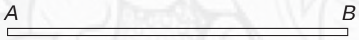
Similar equation can be written for change in y and z co-ordinates.

(6) For the purpose of calculation of position of COM, a part of a body (or a group of particles) can be replaced with a point mass, kept at its COM.

(7) It is useful to identify the line of symmetry (or a plane of symmetry) while finding the position of the COM of a body. The COM lies on the line (or plane) of symmetry.

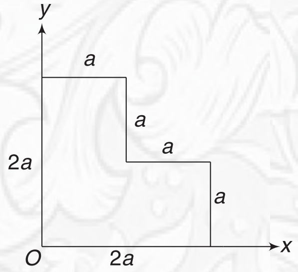
YOUR TURN

Q.6 A rod of length L has its linear mass density changing with distance (x) from one end A as $\lambda = \lambda_0\left[1 + \frac{x}{L}\right]$. Find the distance of the COM of the rod from end A.

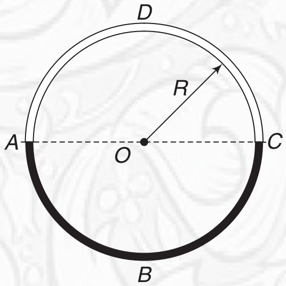


Q.7 Two identical rods are kept perpendicular to each other, with one of their ends coinciding. Length of each rod is L. Find the distance of the COM of the system from the common end of the two rods.

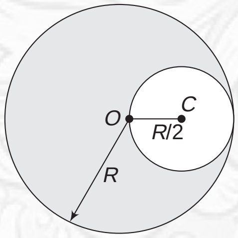
Q.8 A uniform plate is in L shape, with dimensions as shown. Find the co-ordinates of the COM of the plate.



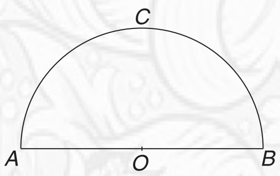
Q.9 A ring is made by joining two semicircular pieces of radius R. Semicircle ABC has mass twice that of the other. Find the distance of the COM of the ring from centre O.



Q.10 A uniform metal plate is in the shape of a disc of radius R. A circular part of radius $\frac{R}{2}$ is stamped out (i.e., removed) from the plate, such that the circumference of the hole passes through the centre O of the disc. Find the distance of the COM of the 'plate with hole' from O.

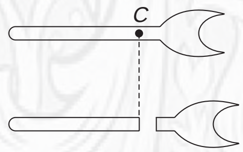


Q.11 A uniform wire of length L is used to make a semicircular closed frame ABC. Find the position of the COM of the object.



Q.12 Where could the COM of the Earth's atmosphere be?

Q.13 A metal wrench of uniform density is cut at the location of its COM (at C), as shown in figure. Can we say that the two pieces have equal mass?



4. MOTION OF CENTRE OF MASS

4.1 Velocity of COM

Equation (4) gives the position of the COM for a collection of particles. If one or more particles move and their position changes, the position vector of the COM may also change. It means that the COM of the system has a velocity.

$$\vec{r}_{CM} = \frac{1}{M}(m_1\vec{r}_1 + m_2\vec{r}_2 + \ldots + m_n\vec{r}_n)$$

Differentiating this equation with respect to time,

$$\frac{d\vec{r}_{CM}}{dt} = \frac{1}{M}\left(m_1\frac{d\vec{r}_1}{dt} + m_2\frac{d\vec{r}_2}{dt} + \ldots + m_n\frac{d\vec{r}_n}{dt}\right)$$

$$\Rightarrow \quad \vec{v}_{CM} = \frac{1}{M}(m_1\vec{v}_1 + m_2\vec{v}_2 + \ldots + m_n\vec{v}_n)$$

$$\Rightarrow \quad \vec{v}_{CM} = \frac{1}{M}\sum_{i=1}^{n} m_i\vec{v}_i \qquad \ldots(7)$$

The above equation helps us to find the velocity of the COM of a system when velocities of its constituent particles are given. If you have solid objects in your system, you can replace them with point masses moving with velocities of their respective COMs.

Let us rearrange the terms in the above equation to write it as

$$M\vec{v}_{CM} = m_1\vec{v}_1 + m_2\vec{v}_2 + \ldots + m_n\vec{v}_n$$

The right side of the equation is the sum of the linear momentum of the constituent particles of the system. It is the total momentum of the system. The above equation tells you that if you know the velocity of the COM of a system, multiply it with the mass of the system (M) to get the total momentum of the system.

$$\vec{P}_{system} = M\vec{v}_{CM} \qquad \ldots(8)$$

Switch on the ceiling fan in your room. What is momentum of the fan when it is running at its full speed? The fan has many particles. Do we need to add momentum of all the particles? Not really. The COM of the fan (it is at the geometrical centre of the fan) is at rest. $v_{CM} = 0$. Hence, momentum of the fan is zero.

4.2 Acceleration of COM

Differentiating equation (7) with respect to time

$$\frac{d\vec{v}_{CM}}{dt} = \frac{1}{M}\left(m_1\frac{d\vec{v}_1}{dt} + m_2\frac{d\vec{v}_2}{dt} + \ldots + m_n\frac{d\vec{v}_n}{dt}\right)$$

$$\Rightarrow \quad \vec{a}_{CM} = \frac{1}{M}(m_1\vec{a}_1 + m_2\vec{a}_2 + \ldots + m_n\vec{a}_n) \qquad \ldots(9)$$

This equation gives the acceleration of a system of particles, when the accelerations of individual particles are known. Let us write the equation as

$$M\vec{a}_{CM} = m_1\vec{a}_1 + m_2\vec{a}_2 + \ldots + m_n\vec{a}_n$$

$m_1\vec{a}_1(=\vec{F}_1)$ is the sum of all forces acting on particle of mass m_1. $\vec{F}_1$ is the resultant of external forces acting on m_1 and the forces applied by other particles present in the system. Similarly, $m_2\vec{a}_2$ $(=\vec{F}_2)$ is resultant force on m_2 due to external agencies and the other particles in the system (viz., m_1, m_3, m_4 ...).

$$\therefore \quad M\vec{a}_{CM} = \vec{F}_1 + \vec{F}_2 + \ldots + \vec{F}_n$$

The right side of the above equation includes all the internal interactions amongst the particles. However, from Newton's third law, these forces occur in pairs (equal and opposite) and their vector sum is zero. So the sum on the right side represents the sum of external forces only.

$$\therefore \quad M\vec{a}_{CM} = \vec{F}_{ext} \qquad \ldots(10)$$

Where, $\vec{F}_{ext}$ is the sum of all external forces on the system. This is *Newton's Second Law* for a collection of particles. It states that the acceleration of the COM of a body (or collection of particles) is equal to the net external force acting on the body divided by its total mass.

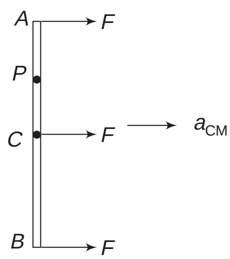
Again, think of the rotating ceiling fan in your room. What is the total external force (i.e., sum of forces by the support, force due to air, weight) acting on the fan? The answer is zero. Acceleration of the COM of the fan is zero (as it never moves even a centimeter) and therefore, we conclude that the net external force on it is zero. Even when the fan is speeding up (i.e., its angular speed is increasing), its COM has no acceleration and the net external force on the fan is zero. Can we say that every particle in the fan has zero acceleration? No, a typical particle is going in a circle and must be experiencing unbalanced force towards the centre.

Equation (10) is equivalent to three component equations along the three co-ordinate axes.

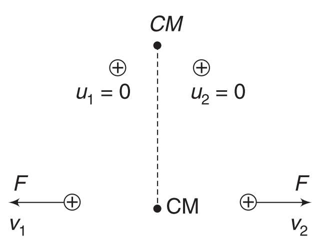
$$(F_{\text{ext}})_x = M(a_{\text{CM}})_x;\ (F_{\text{ext}})_y = M(a_{\text{CM}})_y;\ (F_{\text{ext}})_z = M(a_{\text{CM}})_z \quad ...(11)$$

Following are some important situations that need your attention.

(1) An external force $\vec{F}$ is applied to a body. *Irrespective of its point of application,* the acceleration of the COM of the body will be $\vec{a}_{\text{CM}} = \frac{\vec{F}}{M}$. A stick of mass M is lying on a smooth floor. A force $\vec{F}$ is applied to it. No matter whether the force is applied at A, B or C, the acceleration of COM of the stick will be $\frac{\vec{F}}{M}$. We will learn later that applying force at A will produce a motion of point P, which will be very different from the motion of point P that we see when the same force is applied at C.



(2) Suppose you find that the COM of a system is at rest (i.e., the velocity of the COM, $u_{\text{CM}} = 0$) and also observe that the net external force on the system is zero. It implies that a_{CM} is zero. Therefore, the COM of the system will not get displaced. It is possible that the components of the system are accelerated due to internal interactions. Consider two charges – both positive – held certain distance apart. They are released. They repel one another and move away from one another with increasing speed. But the COM of the system, comprising the two charges, does not move.



Charge move away with increasing speed.
The COM remains at rest

When charges were released, each had zero initial velocity.

$\therefore$ Initial velocity of the COM was zero ($u_{\text{CM}} = 0$)

$$F_{\text{ext}} = 0 \quad \Rightarrow \quad a_{\text{CM}} = 0$$

$\Rightarrow$ COM does not move.

How are the displacements of two particles related if their COM has no motion?

Let $\Delta\vec{r}_1$ and $\Delta\vec{r}_2$ be the displacements of the two particles, having masses m_1 and m_2 respectively. If the COM has not moved, then

$$\Delta\vec{r}_{\text{CM}} = \frac{m_1\Delta\vec{r}_1 + m_2\Delta\vec{r}_2}{m_1 + m_2} = 0$$

$$\Rightarrow \quad m_1\Delta\vec{r}_1 + m_2\Delta\vec{r}_2 = 0$$

$$\Rightarrow \quad m_1\Delta\vec{r}_1 = -m_2\Delta\vec{r}_2 \quad ...(12)$$

The negative sign indicates that the displacement $\Delta\vec{r}_1$ and $\Delta\vec{r}_2$ must be in opposite directions. If you knowingly take the two displacement in opposite directions then you need not write the negative sign and use the above equation as:

$$m_1\Delta r_1 = m_2\Delta r_2 \quad ...(13)$$

Where Δr_1 and Δr_2 are magnitudes of displacements of the two particles *in opposite directions*.

The above result can be applied in a specific direction as well. If $(u_{\text{CM}})_x = 0$ and $(F_{\text{ext}})_x = 0$ [$(u_{\text{CM}})_y$ and $(F_{\text{ext}})_y$ may or may not be zero] then

$$\Delta x_{\text{CM}} = 0$$

$$m_1\Delta x_1 + m_2\Delta x_2 = 0 \quad ...(14)$$

where Δx_1 and Δx_2 are changes in x co-ordinates of the two masses.

If we consider the displacement Δx_1 and Δx_2 in opposite directions, then

$$m_1\Delta x_1 = m_2\Delta x_2 \quad ...(15)$$

(3) The result obtained above can be extended to a system having more than two particles.

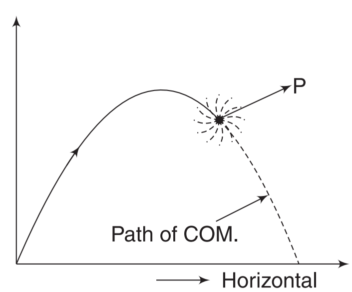
If $\quad \vec{u}_{\text{CM}} = 0$ and $\vec{a}_{\text{CM}} = 0$ ($\Rightarrow \ \vec{F}_{\text{ext}} = 0$) then

$$m_1\Delta\vec{r} + m_2\Delta\vec{r}_2 + m_3\Delta\vec{r}_3 + ... = 0$$

If $(F_{\text{ext}})_x = 0$ and $(u_{\text{CM}})_x = 0$, then

$$m_1\Delta x_1 + m_2\Delta x_2 + m_3\Delta x_3 + ... = 0.$$

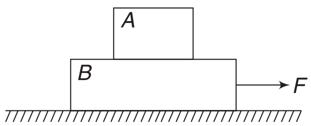
(4) Consider a diwali bomb, moving like a projectile. It explodes at point P when moving with a velocity v (this is the velocity of the COM of the bomb at the instant of explosion). The fragments move along different directions due to the force of explosion. But for the system as a whole, the forces of explosions are internal and should not affect the motion of the COM. The COM still has acceleration equal to $g(\downarrow)$, as the net external force is Mg. The COM of the fragments continues on the original parabolic path until the fragments begin to fall on the ground.



Bomb thrown as a projectile

Events like explosion, collision, etc., do not change the motion of the COM of the system, as they are phenomena of internal interactions amongst the particles.

Example 10 In the arrangement shown in the figure, the horizontal surface is smooth but there is friction between the two blocks. Coefficient of friction is μ. Mass of B is twice that of A. When a horizontal force F is applied to B, the two blocks slip on one another. Find the acceleration of the COM of the system of two blocks, A and B. Mass of A is m.



Solution

Concepts

Acceleration of the COM depends on external force only.

Friction force between the blocks is an internal interaction when we are considering ($A + B$) as our system. F is the only unbalanced external force on the system.

$$\therefore \qquad a_{CM} = \frac{F}{m + 2m} = \frac{F}{3m}$$

Example 11 A particle of mass m is projected vertically up from a point with initial velocity $20\,\text{ms}^{-1}$. Another particle of mass $2m$ is projected simultaneously from the same point, with velocity $20\,\text{ms}^{-1}$ at an angle of 30° to the horizontal.

(i) Find the velocity of the COM of the system of two particles, one second after their projection.

(ii) Final acceleration of the COM of the system, one second after their projection.

Solution

Concepts

$$\vec{v}_{CM} = \frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1 + m_2} \quad \text{and} \quad \vec{a}_{CM} = \frac{m_1\vec{a}_1 + m_2\vec{a}_2}{m_1 + m_2}$$

(i) Velocity of the first particle after 1 s is

$$v = 20 - gt = 20 - 10 \times 1 = 10\,\text{ms}^{-1}\ (\uparrow)$$

Horizontal and vertical components of initial velocity of the second particle are (Horizontal–x and vertical–y-direction)

$$u_x = 20\cos 30° = 10\sqrt{3}\,\text{ms}^{-1} \quad \text{and}$$

$$u_y = 20\sin 30° = 10\,\text{ms}^{-1}$$

Velocity components after 1 s

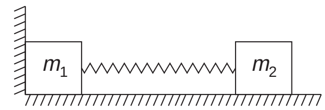
$$v_x = 10\sqrt{3}\ \text{ms}^{-1};\ v_y = 10 - gt = 10 - 10 \times 1 = 0$$

$$\therefore \quad \vec{v}_{CM} = \frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1 + m_2} = \frac{m(10\,\hat{j}) + 2m(10\sqrt{3}\ \hat{i})}{m + 2m}$$

$$= \left(\frac{20}{\sqrt{3}}\,\hat{i} + \frac{10}{3}\,\hat{j}\right)\text{ms}^{-1}$$

(ii) $$\vec{a}_{CM} = \frac{m_1 g(\downarrow) + m_2 g(\downarrow)}{m_1 + m_2} = g(\downarrow)$$

Example 12 Two blocks of masses m_1 and m_2 are connected by an ideal spring, having force constant k. The system is placed on a smooth floor, and pushed against a wall with compression in the spring equal to x. System is released from this position.



(i) Find the acceleration of the COM of the system, immediately after its release.

(ii) Find the acceleration of the COM of the system, after m_1 breaks-off the wall.

(iii) Find the velocity of the COM of the system, after a long time.

Solution

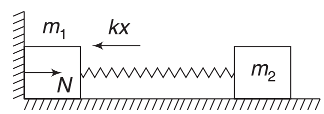
Concepts

(i) The compressed spring pushes the block of mass m_1 against the wall. The wall applies a normal force on m_1. This is the external force that causes the COM to accelerate.

(ii) Once m_1 leaves contact with the wall, there is no external force on the system. Now, $a_{CM} = 0$.

(iii) Velocity of the COM at the instant m_1 leaves the wall will be constant thereafter, as no external force acts on the system.

(i) Spring force on the block of mass m_1, at the instant the system is released, is kx. Normal force exerted by the wall on the block of mass m_1 is $N = kx$



$\therefore$ For the system, $F_{ext} = N = kx$

$$\therefore \qquad a_{CM} = \frac{F_{ext}}{m_1 + m_2} = \frac{kx}{m_1 + m_2}$$

(ii) As long as the spring is compressed, it will keep the block pressed against the wall. When the block of mass m_2 moves to the right by a distance x, the spring attains its natural length. At this instant, the spring does not exert any force on the blocks. The normal force by wall on m_1 becomes zero. The system leaves the wall. After leaving the wall, $F_{ext} = 0$

$$\Rightarrow \qquad a_{CM} = 0$$

(iii) Let us find velocity of m_2 at the instant the spring regains its natural length. (At this instant, m_1 is about to move, its velocity is zero).

The entire spring energy has got converted into KE of m_2.

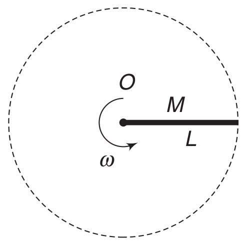
$$\therefore \quad \frac{1}{2}m_2v_2^2 = \frac{1}{2}kx^2 \quad \Rightarrow \quad v_2 = \sqrt{\frac{k}{m_2}}\cdot x$$

$$v_{CM} = \frac{m_1(0) + m_2v_2}{m_1 + m_2} = \frac{\sqrt{km_2}\cdot x}{m_1 + m_2}$$

Velocity of the COM will remain constant at the above value, as there is no external force on the system after m_1 leaves contact with the wall.

Example 13 *Force on a rotating rod*

A uniform rod of mass M and length L is rotating in a horizontal plane, about a vertical axis passing through its end O. Find the horizontal force that the axis must apply on the rod. Angular speed of the rod is ω.



Solution

Concepts

Force applied by the axis is the only external horizontal force on the rod. This must be equal to $F = Ma_{CM}$, where a_{CM} is the horizontal acceleration of the COM.

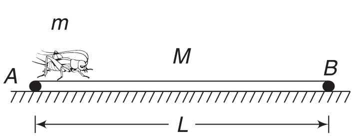
COM is moving in a circle of radius $\frac{L}{2}$ with angular speed ω.

$$a_{CM} = \omega^2\frac{L}{2}$$

$$\therefore \quad F = Ma_{CM} = \frac{M\omega^2L}{2}$$

Example 14 *Insect on a straw*

A straw of mass M and length L is lying on a smooth horizontal surface. A small insect of mass m is sitting at the end A of the straw. The insect crawls to the other end B of the straw. Will the straw move? If yes, by what distance.



Solution

Concepts

(i) Frictional force on the insect (applied by the straw) causes it to move to the right. Friction applied by the insect on the straw is to the left and makes the straw move to the left.

(ii) Friction is an internal force for the insect–straw system. There is no external force on the system. Hence, the COM of the system does not move. This can happen if $m\Delta x_1 = M\Delta x_2$, where x_1 is the displacement of the insect to the right and x_2 is displacement of the straw to the left.

Initial velocity of COM, $u_{CM} = 0$, as both the insect and the straw are at rest.

$$a_{CM} = 0, \quad \text{as} \quad F_{ext} = 0$$

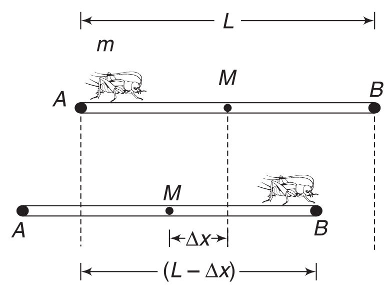
This means that the COM will not get displaced. The straw can be replaced with a point mass, placed at its centre.

Let the straw move to the left by Δx, when the insect moves from A to B. Displacement of the insect is $(L - \Delta x)$ to the right.

Using equation 15.

$$m(L - \Delta x) = M\cdot\Delta x$$

$$\Rightarrow \qquad \Delta x = \frac{mL}{M + m}$$



As the insect moves from A to B, the staw moves to the left by Δx. Displacement of insect is $(L - \Delta x)$ to the right.

Example 15 A projectile is fired at a speed of $u = 100\,\text{ms}^{-1}$ at an angle of 37° above the horizontal. At the highest point, the projectile breaks into two parts of mass ratio 1:4 and the smaller piece comes to rest. Find the distance from the launching point to the point where the heavier piece lands.

Solution

Concepts

(i) The COM continues to move on the usual parabolic path even after the projectile breaks. Total external force on the pieces is still the weight and hence, acceleration of COM is still $g(\downarrow)$.

(ii) At the highest point, the two pieces have no vertical velocity. They will both land on the ground simultaneously; at the same instant, the COM will also land on the ground.

Had there been no breakage, the range of projectile would have been

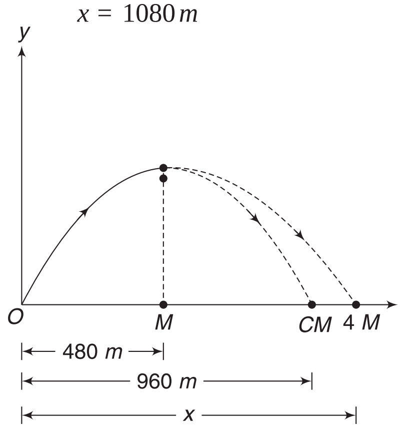
$$R = \frac{2u^2\sin\theta\,\cos\theta}{g} = \frac{2\times 100^2\times\frac{3}{5}\times\frac{4}{5}}{10} = 960\,m$$

The COM will land at 960 m distance from the point of projection. The lighter piece (having mass M) lands at a distance $\frac{960}{2} = 480\,m$ from the point of projection O. The heavier piece (having mass $4M$) lands at a distance x from O.

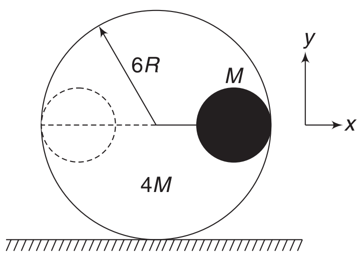
Using $\quad x_{CM} = \dfrac{m_1x_1 + m_2x_2}{m_1 + m_3}$

$$\Rightarrow \quad 960 = \frac{M \times 480 + 4M \cdot x}{5M}$$

$$\Rightarrow \quad x = 1080\,m$$



Example 16 A sphere of Mass M and radius R is held on the inner wall of a hollow sphere of mass $4M$ and radius $6R$. The bigger sphere is kept on a smooth horizontal table and the centres of the two spheres lie on a horizontal line. The system of two spheres is released from this position. The smaller sphere slides on the inner smooth wall to reach the other extreme position (shown in dotted line). What will be displacement of the centre of the larger sphere, as the smaller sphere moves from one extreme to another?



Solution

Concepts

(i) Initial velocity of both spheres = 0. Hence, initial velocity of the COM is zero. There is no horizontal force on the system. Hence,

$$(a_{CM})_x = 0. \text{ It means } (v_{CM})_x = 0, \text{ always.}$$

This implies that the x co-ordinates of the COM of the system will not change.

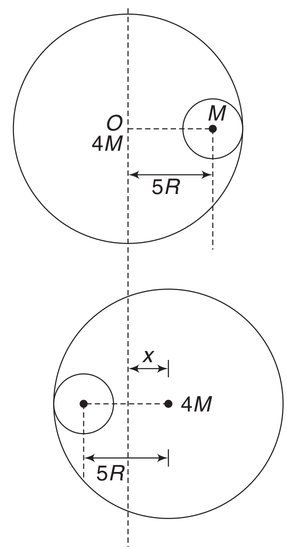
(ii) When one sphere stops the other must also stop, since $(v_{CM})_x = 0$. When KE of the system is zero, its PE must be equal to its initial value. This ensures that the smaller sphere will be at its initial height when it stops moving.

Let the centre of the bigger sphere move a distance x to the right, as the smaller sphere moves to the extreme left. From the figure, it is easy to see that the centre of the smaller sphere moves a distance $(10R - x)$ to the left. For no motion of the COM of the system,

$$m_1\Delta x_1 = m_2\Delta x_2$$

$$\Rightarrow \quad 4M \cdot x = M(10R - x)$$

$$\Rightarrow \quad x = 2R$$



The spheres can be treated as point masses, placed at their centres.

Example 17 *Rod, falling on a smooth table*

A uniform thin rod of length L is standing vertically on a smooth table. Consider the lower end of the rod as the origin and vertically upward as the positive y-direction. A slight disturbance causes the lower end to slip on the smooth table along positive x-axis and the rod begins to fall.

(i) What is path of the COM of the rod during its fall?

(ii) Write the equation of the path followed by a point P on the rod, which is at a distance $r\left(\neq \frac{L}{2}\right)$ from lower end of the rod.

Solution

Concepts

There is no friction. It means there is no horizontal force on the rod. The COM does not have any horizontal motion. It moves only in vertical direction.

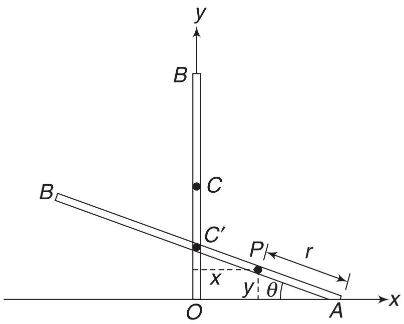
(i) The COM moves along vertical straight line.

(ii) When the rod slides to position AB, its centre is on y-axis, at C'.

$$AC' = \frac{L}{2}$$

P is a point at a distance r from end A. Co-ordinates of point P are:

$$x = C'P \cos\theta = \left(\frac{L}{2} - r\right)\cos\theta \quad ...(i)$$



And $\qquad y = r\sin\theta \quad ...(ii)$

From (i) and (ii) $\dfrac{x^2}{\left(\frac{L}{2} - r\right)^2} + \dfrac{y^2}{r^2} = 1$

This is the trajectory equation of point P and it is an equation for an ellipse.

In Short:

(i) Velocity of the COM of a system does not change if there is no external force on the system. Internal forces can change the velocities of individual particles but they cannot alter $\vec{v}_{CM}$.

(ii) Momemtum of a system is its total mass multiplied with $\vec{v}_{CM}$. When $\vec{F}_{ext} = 0$, $\vec{v}_{CM}$ does not change. This implies that the momentum of the system does not change (provided it has fixed mass) when $\vec{F}_{ext} = 0$. This is law of conservation of momentum.

(iii) Acceleration of COM is decided by the resultant of all external forces.

$$\vec{a}_{cm} = \frac{\vec{F}_{ext}}{M}$$

Point of application of external forces and shape and size of the system are not important while using the above equation.

(iv) If $(u_{CM})_x = 0$ and $(F_{ext})_x = 0$, then there is no displacement of the COM along x-direction.

$$m_1\Delta x_1 + m_2\Delta x_2 + m_3\Delta x_3 + ... + m_n\Delta x_n = 0.$$

(v) Events like explosion, collision, etc., do not change the motion of COM.

YOUR TURN

Q.14 A long boat is moving at constant speed through the water. People on the boat gathered at the front and ran together towards the back end of the boat.

(a) When the people are running, is the speed of the boat greater than it was before?

(b) People stop running when they reach the other end. What can be said about the speed of the boat after they stopped?

Q.15 Position vector of a 2.0 g particle changes with time as $\vec{r}_1 = 3t\hat{i} + 2t^2\hat{j}$. Another particle of mass 3.0 g has its position vector changing as $\vec{r}_2 = -6t\hat{i} + 3\hat{j}$, where t is in seconds and r is in centimeters. At $t = 1.0$ s, find:

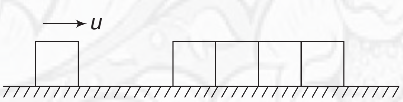
(i) Velocity of the COM of the system of two particles.

(ii) Acceleration of the COM of the system.

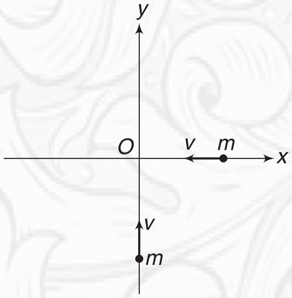
(iii) External force on the system.

Q.16 A uniform rod of length L is held vertically, with its lower end on a smooth horizontal table. The rod begins to fall from this position. Find the distance travelled by the lower end of the rod, by the time it falls completely on the table.

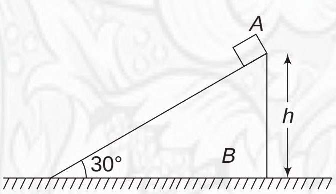
Q.17 Four identical blocks are placed on a smooth horizontal surface, stuck to each other. Another identical block, moving with velocity u, hits the blocks and sticks. Find the velocity of the COM of the system of five blocks before and after the collision.



Q.18 Two identical balls are moving with constant speed (v) along x and y axes, as shown in the figure. They collide at the origin and the ball travelling along x-direction gets deviated by 45°. Find the trajectory of the COM of the system of two balls, before and after collision.

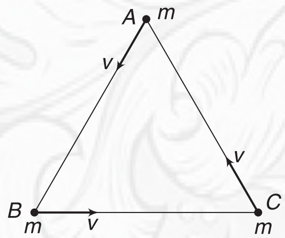


Q.19 Block A of mass m is placed at the top of a smooth inclined surface of block B, which has mass $5m$ and is placed on a smooth horizontal surface. The system is released from rest. Find the distance moved by B



by the time A reaches the bottom. Neglect size of A and take $h = 10\sqrt{3}\ m$.

Q.20 Three particles, each of mass m, are located at the vertices of an equilateral triangle ABC.

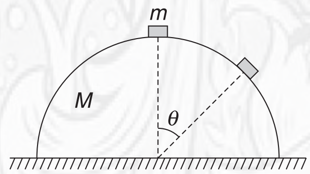


(i) Where is the COM of the system?

(ii) The three particles begin to move with constant velocities, along $\overrightarrow{AB}$, $\overrightarrow{BC}$ and $\overrightarrow{CA}$. All of them have the same speed v. How far will the COM move from its original position in time Δt?

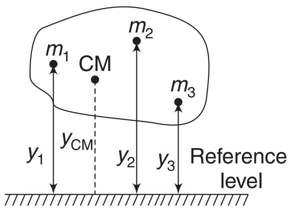
Q.21 A projectile of Mass M moving horizontally with velocity $\vec{v}_0 = (120\,\text{ms}^{-1})\ \hat{i}$ explodes into two fragments A and B – of masses $\frac{2M}{5}$ and $\frac{3M}{5}$ respectively. Taking the point of explosion as the origin and knowing that $\Delta t = 3\,\text{s}$ later, the position of fragment A is $(300, 24, -48)\,m$. Find the co-ordinates of B at the instant. Take y-direction to be vertically up.

Q.22 A small block is released from the top of a hemispherical metal object. Mass of the block is m and that of the hemisphere is M. Find the displacement of the hemisphere when the block reaches at angular position θ. Radius of hemisphere is R and there is no friction.



5. GRAVITATIONAL POTENTIAL ENERGY OF AN EXTENDED BODY

The potential energy of a body is simply the sum of potential energies of its constituent particles. In the figure shown, the potential energy of the body can be written as



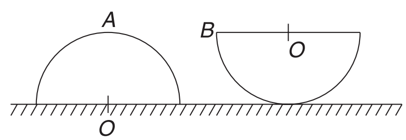
$$u = m_1 g y_1 + m_2 g y_2 + \ldots$$

$$= \frac{(m_1 y_1 + m_2 y_2 + \ldots)g}{M} \cdot M \quad [M = m_1 + m_2 + \ldots]$$

$$\therefore \quad u = Mg\,y_{CM}$$

It means, we can assume the entire body to be a point mass placed at its COM, for the purpose of writing potential energy.

Example 18 A uniform solid hemisphere is kept on a horizontal table in two ways – shown as A and B in figure. In which case is the potential energy of the hemisphere higher?



Solution

Concepts

For writing PE, we can assume that the entire mass is concentrated at its COM.

Distance of COM from centre O is $\frac{3R}{8}$

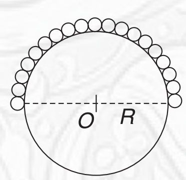
$\therefore$ PE in position A is $u_A = Mg\left(\frac{3R}{8}\right)$

PE in position B is $u_B = Mg\left(R - \frac{3R}{8}\right) = Mg\left(\frac{5R}{8}\right)$

$$\therefore \quad u_B > u_A$$

YOUR TURN

Q.23 A uniform chain is placed over a fixed sphere of radius R, as shown. The chain has a length equal to πR and its mass per unit length is λ. Write the potential energy of the chain, taking the centre of the sphere as reference level.



6. THE CENTRE OF MASS FRAME

A reference frame, which is moving with the COM of a system is often advantageous to use in solution of problems. We will call such a reference frame as the COM frame. Velocity of the COM itself is always zero in this frame and therefore, the linear momentum of the system in this frame $[=M\vec{v}_{CM}]$ is always equal to zero. For this reason, the COM frame is also known as ''zero momentum frame''.

Consider a two-particle system, having particles of masses m_1 and m_2, moving with velocities $\vec{v}_1$ and $\vec{v}_2$ respectively. The velocity of the COM of the system is

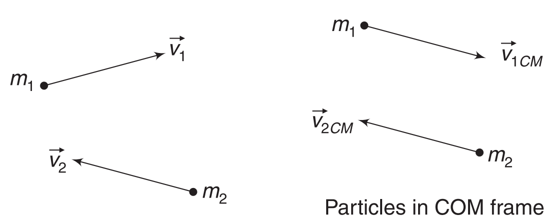
$$\vec{v}_{CM} = \frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1 + m_2}$$

Consider an observer moving with velocity $\vec{v}_{CM}$. He will find that the particle of mass m_1 is moving at velocity

$$\vec{v}_{1CM} = \vec{v}_1 - \vec{v}_{CM}$$

$$= \vec{v}_1 - \frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1 + m_2}$$

$$= \frac{m_2(\vec{v}_1 - \vec{v}_2)}{m_1 + m_2} = \frac{m_2\vec{v}_{12}}{m_1 + m_2} \qquad ...(16)$$



Two Particles In ground frame

Particles in COM frame move with equal and opposite momenta.

where, $\vec{v}_{12}$ = velocity of particle 1 relative to particle 2. Similarly, velocity of particle of mass m_2 in reference frame attached to the COM is

$$\vec{v}_{2CM} = \vec{v}_2 - \vec{v}_{CM} = \vec{v}_2 - \frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1 + m_2}$$

$$= \frac{m_1(\vec{v}_2 - \vec{v}_1)}{m_1 + m_2} = \frac{m_1\vec{v}_{21}}{m_1 + m_2} \qquad ...(17)$$

Momentum of the particles in COM frame will be

$$\vec{P}_{1CM} = m_1\vec{v}_{1CM} = \left(\frac{m_1 m_2}{m_1 + m_2}\right)\vec{v}_{12} = \mu\vec{v}_{12}$$

and $$\vec{P}_{2CM} = m_2\vec{v}_{2CM} = \left(\frac{m_1 m_2}{m_1 + m_2}\right)\vec{v}_{21} = \mu\vec{v}_{21} \qquad ...(18)$$

where $\mu = \dfrac{m_1 m_2}{m_1 + m_2}$, is known as *reduced mass* of the system.

Obviously, $\vec{P}_{1CM} + \vec{P}_{2CM} = \mu(\vec{v}_{12} + \vec{v}_{21}) = 0$.

Momentum of the system is zero in COM frame.

Note: Reduced mass is an useful concept in dealing with bonded system – a system in which particles are bound to each other. For example, two blocks connected with a spring, two stars moving under mutual gravitational pull, H_2O molecule, etc.

In general, the reduced mass of a *n*-particle system is defined as

$$\frac{1}{\mu} = \frac{1}{m_1} + \frac{1}{m_2} + \frac{1}{m_3} + ... + \frac{1}{m_n}$$

Example 19 Two identical billiard balls collide. Just before collision, they were moving with speeds u and $2u$ in opposite directions. Each ball has mass m.

(i) Find the momentum of each ball in COM frame before collision.

(ii) Find the momentum of the system of two balls in COM frame after collision.

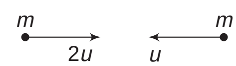
Solution

Concepts

(i) $|\vec{P}_{1CM}| = |\vec{P}_{2CM}| = \mu|\vec{v}_{12}|$

(ii) COM frame is zero-momentum frame.

(i) $$\mu = \frac{m \cdot m}{m + m} = \frac{m}{2}$$



$$v_{12} = 3u \; (=v_{21})$$

$$\therefore \quad |\vec{P}_{1CM}| = \mu v_{12} = \frac{3}{2}\,mu$$

Also $$|\vec{P}_{2CM}| = \mu v_{21} = \frac{3}{2}\,mu$$

(ii) Momentum of a system is always zero in COM frame.

6.1 Kinetic Energy

Consider a system consisting of particles of masses m_1, m_2, m_3 ... m_n and velocities $\vec{v}_1$, $\vec{v}_2$, $\vec{v}_3$ ... $\vec{v}_n$ respectively. Let the velocity of the COM of the system be $\vec{v}_{CM}$.

In reference frame attached to the COM, the velocity of i^{th} particle will be

$$\vec{v}_{iCM} = \vec{v}_i - \vec{v}_{CM} \;\Rightarrow\; \vec{v}_i = \vec{v}_{iCM} + \vec{v}_{CM}$$

kinetic energy of the system of particles in ground frame is

$$k = \frac{1}{2}\sum m_i(\vec{v}_i \cdot \vec{v}_i)$$

$$= \frac{1}{2}\sum m_i(\vec{v}_{iCM} + \vec{v}_{CM})\cdot(\vec{v}_{iCM} + \vec{v}_{CM})$$

$$= \frac{1}{2}\sum m_i v_{iCM}^2 + \frac{1}{2}\sum m_i v_{CM}^2 + \left(\sum m_i\vec{v}_{iCM}\right)\vec{v}_{CM}$$

[Since $\vec{a}\,.\,\vec{a} = a^2$]

The third term is zero, since $\sum m_i\vec{v}_{iCM}$ is the momentum of the system in COM frame and it must be zero.

$$\therefore \quad k = \frac{1}{2}\sum m_i v_{i\text{CM}}^2 + \frac{1}{2}\left(\sum m_i\right)v_{\text{CM}}^2$$

$$= \frac{1}{2}\sum m_i v_{i\text{CM}}^2 + \frac{1}{2}Mv_{\text{CM}}^2$$

$$k = k_{\text{wrt CM}} + \frac{1}{2}Mv_{\text{CM}}^2 \qquad ...(19)$$

$k_{\text{wrt CM}}$ is the kinetic energy of the system in COM frame and $\frac{1}{2}Mv_{\text{CM}}^2$ is the kinetic energy associated with the translational motion of the system as a whole.

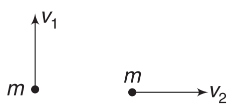
If no external force is acting on a system, v_{CM} will not change. Only $k_{\text{wrt CM}}$ can change due to internal interactions. This has a very important implication. When no external force is acting on a system, change in *KE* in ground frame will be same as the change in COM frame.

Using equations (16) and (17), one can easily prove that for a two-particle system

$$k_{\text{wrt CM}} = \frac{1}{2}\mu v_{\text{rel}}^2 \qquad ...(20)$$

where $\mu = \dfrac{m_1 m_2}{m_1 + m_2}$ and $|v_{\text{rel}}| = |\vec{v}_1 - \vec{v}_2|$

Example 20 Two identical particles, each of mass m are moving in perpendicular directions with speed v. Find the kinetic energy of the system in a reference frame attached to the COM of the system.



Solution

Concepts

$$k_{\text{wrt CM}} = \frac{1}{2}\mu v_{\text{rel}}^2$$

$$\mu = \frac{m\,m}{m+m} = \frac{m}{2}$$

$$\vec{v}_{\text{rel}} = \vec{v}_1 - \vec{v}_2 \quad \Rightarrow \quad |\vec{v}_{\text{rel}}| = \sqrt{v_1^2 + v_2^2}$$

$$\therefore \quad k_{\text{wrt CM}} = \frac{1}{2}\mu v_{\text{rel}}^2 = \frac{1}{2}\frac{m}{2}(v_1^2 + v_2^2) = \frac{1}{4}m(v_1^2 + v_2^2)$$

Example 21 Two balls of masses, m and $2m$, approach each other with a relative velocity u and collide. After collision, they separate from one another at a relative speed $\frac{u}{2}$. Find loss in kinetic energy of the system of the two balls due to collision.

Solution

Concepts

When no external force acts, change in *KE* of a system in COM frame is same as the change in *KE* in ground frame.

KE of system in COM frame before collision is

$$k_{\text{wrt CM}} = \frac{1}{2}\mu u^2 = \frac{1}{2}\left(\frac{m\cdot 2m}{3m}\right)u^2 = \frac{1}{3}mu^2$$

KE in COM frame after collision is

$$k'_{\text{wrt CM}} = \frac{1}{2}\mu\left(\frac{u}{2}\right)^2 = \frac{1}{2}\left(\frac{2}{3}m\right)\frac{u^2}{4} = \frac{1}{12}mu^2$$

Loss in *KE* in COM frame = Loss in *KE* in ground frame

$$= \frac{1}{3}mu^2 - \frac{1}{12}mu^2 = \frac{1}{4}mu^2$$

In Short:

(i) COM frame is 'zero-momentum frame'. Linear momentum of a system in its COM frame is always zero.

(ii) In a system having two particles, the particles move with equal and opposite momentum in COM frame. Magnitude of momentum of each particle in COM frame is

$$P = \mu v_{\text{rel}}$$

where $\mu = \dfrac{m_1 m_2}{m_1 + m_2}$, know as reduced mass and v_{rel} is the relative speed of the two particles.

(iii) *KE* of a system in COM frame, for a two-particle system, is $k_{\text{wrt CM}} = \frac{1}{2}\mu v_{\text{rel}}^2$

(iv) *KE* of a system in ground frame is

$$k = k_{\text{wrtCM}} + \frac{1}{2}Mv_{\text{CM}}^2$$

where M is the total mass of the system.

(v) If $F_{\text{ext}} = 0$, then change in *KE* in ground frame is same as the change in *KE* in COM frame.

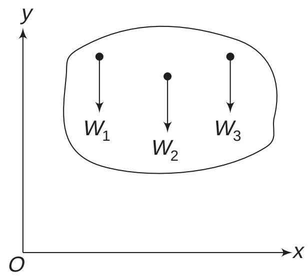
YOUR TURN

Q.24 A particle of mass $m_1 = 4$ kg moves at $5\hat{i}$ ms^{-1}, while a particle of mass $m_2 = 2$ kg moves at $2\hat{i}$ ms^{-1}. Find *KE* of the system in a reference frame attached to its COM.

Q.25 Two projectiles of equal mass move with velocities $\vec{v}_1 = (2\hat{i} + 4\hat{j})$ms^{-1} and $\vec{v}_2 = (6\hat{i} - 3\hat{j})$ms^{-1}. Find the momentum of the system in a reference frame attached to its COM.

7. CENTRE OF GRAVITY

Consider a body having many particles. Weights of the particles are w_1, w_2, w_3 ..., etc., and their y co-ordinates are y_1, y_2, y_3 ..., etc. The centre of gravity is a point where we can assume the weight of the body to be effectively acting.



Its y co-ordinate is given by

$$y_{cg} = \frac{w_1y_1 + w_2y_2 + w_3y_3 + ...}{w_1 + w_2 + w_3 + ...} \quad ...(21)$$

If acceleration due to gravity is uniform, then

$$w_1 = m_1g,\ w_2 = m_2g,\ w_3 = m_3g\ ...$$

$$\therefore \quad y_{cg} = \frac{m_1y_1 + m_2y_2 + m_3y_3}{m_1 + m_2 + m_3} = y_{CM}$$

It means there is no difference in COM and centre of gravity (CG), as long as acceleration due to gravity is uniform. *The two terms are used interchangeably for daily-life objects*.

However, when you consider very tall buildings, the acceleration due to gravity will decrease as one moves up. In such a case, the COM and the CG will be different points. We must use equation (21) to find the position of CG in such situations.

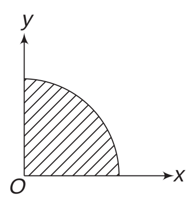
YOUR TURN

Q.26 Is it right to assume that CG of a very tall building will be slightly below the COM? Give reason.

Miscellaneous Examples

Example 22 *A quarter disc*

Find the x and y co-ordinates of the COM of a uniform plate in shape of a quarter circle, of radius R. [see fig.]



Solution

Concepts

(i) The COM will lie on the symmetry line, passing through O and making 45° with both x an y axes.

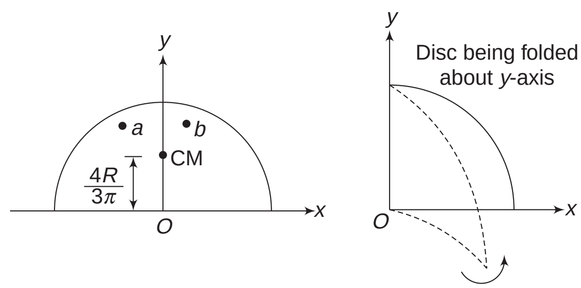
(ii) It means that x and y co-ordinate of the COM will be same.

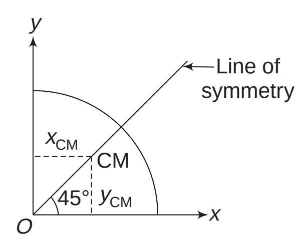
(iii) Refer to Example 6. The y co-ordinate of the COM of a semicircular disc is $\dfrac{4R}{3\pi}$.

Imagine a semicircular disc of radius R. Its COM is on y-axis, at a distance $\dfrac{4R}{3\pi}$ from O. If the disc is folded about y-axis to a make a quarter disc, the y co-ordinate of none of the particles (in the disc) change. A particle at 'a' (see figure) moves to 'b'. Therefore, y_{CM} does not change.

$$\therefore \quad \text{For quarter disc, } y_{CM} = \frac{4R}{3\pi}$$

$$\text{And} \qquad x_{CM} = y_{CM} = \frac{4R}{3\pi}$$





Example 23 *COM of a liquid-filled container*

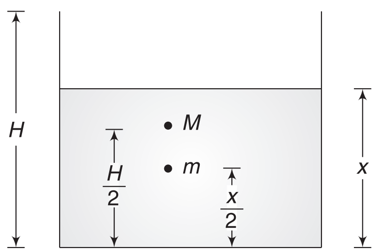
A liquid has been poured into a cylindrical vessel of mass M and height H. Mass of liquid column of unit height is α. Find the height x of the liquid column for which the COM of the liquid plus the container system is at the lowest position. Assume the COM of the empty vessel to be at height $\dfrac{H}{2}$ from the base.

Solution

Concepts

(i) COM of the empty container is at height $\frac{H}{2}$. When liquid is poured, the bottom gets heavier and the COM of the system moves lower. After a lot of liquid has been poured, the COM begins to shift up. Think of a situation where the liquid has been poured to a great height. Mass of the liquid will be much larger than the container and the COM will be close to the COM of the liquid (i.e., nearly at height $\frac{H}{2}$). We can say that the COM goes down and then rises as the liquid is poured.

(ii) We will write the height of COM (x_{CM}) when liquid column has height x. Then we will use the concept of maxima and minima $\left(\frac{dx_{CM}}{dx} = 0\right)$ to find when x_{CM} will be minimum.



The container can be replaced with a point mass (M) at height $\frac{H}{2}$. When liquid is poured to a height x, mass of liquid is $m = \alpha x$. It can be assumed to be a point mass at height $\frac{x}{2}$ from the base.

$$x_{CM} = \frac{M\left(\frac{H}{2}\right) + (\alpha x)\left(\frac{x}{2}\right)}{M + \alpha x}$$

For x_{CM} to be minimum, $\frac{dx_{CM}}{dx} = 0$

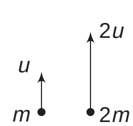
$$\Rightarrow \quad (M + \alpha x)(\alpha x) - \left(M\frac{H}{2} + \frac{\alpha x^2}{2}\right)\alpha = 0$$

$$\Rightarrow \quad \alpha x^2 + 2Mx - MH = 0$$

$$\Rightarrow \quad x = \frac{-2M \pm \sqrt{4M^2 + 4MH\alpha}}{2\alpha}$$

$$\Rightarrow \quad x = \frac{\sqrt{M(M + H\alpha)} - M}{\alpha}$$

Example 24 Two particles of masses m and $2m$ are projected vertically upward from a point, with velocities u and $2u$ respectively. Find the maximum height attained by the COM of the system of two particles, measured from the point of projection.



Solution

Concepts

(i) We can find initial velocity of COM using

$$u_{CM} = \frac{m_1u_1 + m_2u_2}{m_1 + m_2}$$

(ii) We can find acceleration of COM using

$$a_{CM} = \frac{m_1a_1 + m_2a_2}{m_1 + m_2}$$

(iii) We can use equation of kinaematics directly for motion of the COM.

Initial velocity of COM is

$$u_{CM} = \frac{mu + (2m)(2u)}{m + 2m} = \frac{5}{3}u$$

Acceleration of COM is

$$a_{CM} = \frac{m(g) + (2m)(g)}{m + 2m} = g(\downarrow)$$

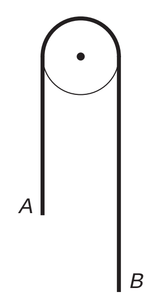
COM is an imaginary point having initial velocity $\frac{5}{3}u\,(\uparrow)$ and a constant acceleration $g(\downarrow)$. When COM is at its maximum height, its velocity is zero.

$$v_{CM}^2 = u_{CM}^2 + 2a_{CM}\cdot x_{CM}$$

$$\Rightarrow \quad 0 = \left(\frac{5u}{3}\right)^2 - 2g(x_{CM})$$

$$\Rightarrow \quad x_{cm} = \frac{25u^2}{18g}$$

Example 25 A uniform rope of length L is placed over a smooth pulley and held in a position, where 40% of its length lies on one side and the remaining 60% hangs on the other side. The rope is released from this position. [Neglect dimensions of the pulley]



(i) Find the vertical acceleration of the COM of the rope, immediately after it is released.

(ii) Find the vertical velocity of the COM, at the instant end A is $\frac{L}{5}$ above its initial position.

Solution

Concepts

(i) Pulley is of negligible dimension. The rope has two segments – one to the left of the pulley and other to its right. Each segment of the rope has a mass proportional to its length.

(ii) Acceleration of the rope can be calculated using Atwood machine formula and after that, a_{CM} can be calculated using $a_{CM} = \dfrac{m_1 a_1 + m_2 a_2}{m_1 + m_2}$

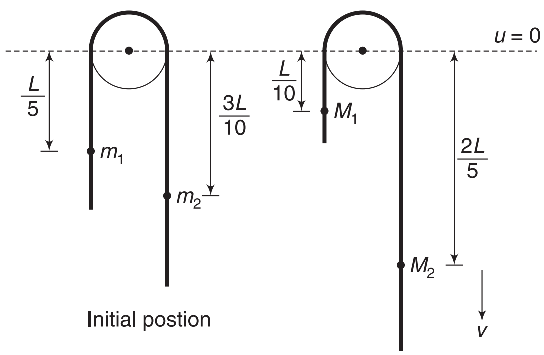
(iii) Speed of the rope after certain displacement can be calculated from conservation of energy. Then we can use $v_{CM} = \dfrac{m_1 v_1 + m_2 v_2}{m_1 + m_2}$

(i) Length of rope on left $= \dfrac{2}{5}L$

Length of rope on right $= \dfrac{3}{5}L$

Mass of rope on left and right side of the pulley respectively is $m_1 = \dfrac{2}{5}L\lambda$ and $m_2 = \dfrac{3}{5}L\lambda$. Where λ is the mass per unit length.

$$a = \frac{(m_2 - m_1)g}{m_2 + m_1} = \frac{\left(\frac{3}{5} - \frac{2}{5}\right)L\lambda g}{\lambda L} = \frac{g}{5}$$



(ii) Initially, COM of the left side of the rope is $\dfrac{L}{5}$ below the pulley and that of the right side is $\dfrac{3L}{10}$ below the pulley.

Assuming gravitational *PE* to be zero at the reference level of the pulley, we can write the *PE* of rope in its initial position by assuming point masses at respective COMs of the segments.

$$u_i = -m_1 g\frac{L}{5} - m_2 g\frac{3L}{10}$$

$$= -\left(\frac{2}{5}L\lambda\right)g\,\frac{L}{5} - \left(\frac{3}{5}L\lambda\right)g\frac{3L}{10}$$

$$= -\frac{13}{50}\lambda g L^2$$

After the rope has moved by $\dfrac{L}{5}$, the masses on two sides are

$$M_1 = \lambda\frac{L}{5} \text{ and } M_2 = \lambda\frac{4L}{5}$$

These can be assumed as point masses kept at respective COMs of the two segments. *PE* in this position is

$$U_f = -M_1 g\frac{L}{10} - M_2 g\,\frac{2L}{5}$$

$$= -\left(\frac{\lambda L}{5}\right)\left(g\frac{L}{10}\right) - \left(\lambda\frac{4L}{5}\right)\left(g\frac{2L}{5}\right)$$

$$= -\frac{17}{50}\,\lambda g L^2$$

Gain in *KE* of rope = Loss in its *PE*

$$\frac{1}{2}(\lambda L)v^2 = U_i - U_f$$

$$\frac{1}{2}\,\lambda L v^2 = \frac{2}{25}\,\lambda g L^2$$

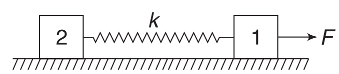
$$\Rightarrow \quad v = \frac{2}{5}\sqrt{gL}$$

Velocity of the COM of the rope is

$$v_{CM} = \frac{M_2 v - M_1 v}{M_2 + M_1} = \frac{\left(\lambda\frac{4L}{5} - \lambda\,\frac{L}{5}\right)\frac{2}{5}\sqrt{gL}}{\lambda L}$$

$$= \frac{6}{25}\sqrt{gL}$$

Example 26 Two blocks of equal mass m are connected by an unstretched spring and the system is kept at rest on a smooth horizontal surface. A constant horizontal force (F) begins to act on block 1, at time $t = 0$, so as to pull it away from the other block.



(i) Find displacement of the COM of the system of two blocks at time t.

(ii) Extension in the spring is observed to be x_0 at time t. Find the displacement of the two blocks at this time.

Solution

Concepts

(i) Acceleration of COM is decided by external forces only.

(ii) If x_1 and x_2 are displacement of the two blocks, displacement of COM will be

$$x_{CM} = \frac{mx_1 + mx_2}{m + m} = \frac{x_1 + x_2}{2}$$

(iii) Extension in spring is $x_0 = x_1 - x_2$

(i) $a_{CM} = \dfrac{F_{ext}}{M} = \dfrac{F}{2m}$

Displacement of COM in time t is

$$x_{CM} = u_{CM}\cdot t + \frac{1}{2}\,a_{CM}\cdot t^2 = \frac{1}{2}\left(\frac{F}{2m}\right)t^2 = \frac{Ft^2}{4m}$$

(ii) Since $x_{CM} = \dfrac{x_1 + x_2}{2}$

$\therefore\ x_1 + x_2 = \dfrac{1}{2}\dfrac{Ft^2}{m}$ ...(a)

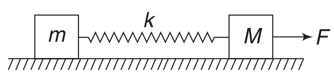
Extension in spring is $x_1 - x_2 = x_0$ ...(b)

Solving (a) and (b), we get

$$x_1 = \frac{1}{2}\left(\frac{Ft^2}{2m} + x_0\right) \text{ and } x_2 = \frac{1}{2}\left(\frac{Ft^2}{2m} - x_0\right)$$

Example 27 *Maximum extension*

A block of mass m is connected to another block of mass M by a massless spring of force constant k. The blocks are kept on a smooth horizontal plane. Initially, the blocks are at rest and the spring is relaxed. A constant horizontal force F begins to act on the block of mass M (see figure). Find the maximum extension in the spring during the subsequent motion.



Solution

Concepts

(i) Initially, acceleration of M will be higher and it will move away from the other block. The spring expands. The stretched spring will exert more and more force on m and its acceleration will increase. In fact, when the spring stretches a lot, M will begin to retard. There will be a point when both the blocks will have the same velocity (i.e., their relative velocity will be zero). At this point, the extension will be maximum.

(ii) Motion of the COM can be predicted as external force is known.

(iii) Displacement of the COM when the two blocks move by x_1 and x_2 is

$$x_{CM} = \frac{Mx_1 + mx_2}{M + m}$$

(iv) Work–energy theorem can be applied.

Initially, $u_{CM} = 0$

When the extension is maximum, velocity of both blocks $= v$ (say).

At this instant,

$$v_{CM} = \frac{Mv + mv}{M + m} = v$$

The COM moves with a constant acceleration

$$a_{CM} = \frac{F}{M + m}$$

$$\therefore \quad v_{CM}^2 = u_{CM}^2 + 2a_{CM}\, x_{CM}$$

$$\Rightarrow \quad v^2 = 2\left(\frac{F}{M + m}\right)x_{CM}$$

x_{CM} = displacement of COM when extension is maximum

$$= \frac{Mx_1 + mx_2}{M + m} \quad \left[\begin{array}{c} x_1 \text{ and } x_2 \text{ are displacement of} \\ M \text{ and } m \text{ respectively} \end{array}\right]$$

$$\therefore \quad v^2 = \frac{2F}{(M + m)^2}(Mx_1 + mx_2) \quad \text{...(a)}$$

Initially, $k_i = 0$

At maximum extension, $k_f = \dfrac{1}{2}(m + M)v^2$

Using work–energy theorem:

Work done by F = Gain in KE + Gain in PE

$$Fx_1 = \frac{1}{2}(m + M)v^2 + \frac{1}{2}kx^2$$

$$\Rightarrow \quad Fx_1 - F\left(\frac{Mx_1 + mx_2}{M + m}\right) = \frac{1}{2}kx^2 \qquad \text{[using (a)]}$$

$$\Rightarrow \quad \frac{Fm(x_1 - x_2)}{M + m} = \frac{1}{2}kx^2$$

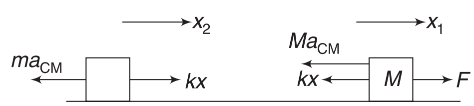
But $x_1 - x_2 = x$ (extension in spring)

$$\therefore \quad kx = \frac{2Fm}{M + m}$$

$$\Rightarrow \quad x = \frac{2Fm}{k(M + m)}$$

Alternate: Here, we will use work–energy theorem in the RF attached to the COM. This RF has an acceleration

$$a_{CM} = \frac{F}{M + m}$$



Forces on the blocks in COM frame

We apply a pseudo-force on both the blocks.

Net external force (leaving apart the spring force) on M is

$$F_1 = F - Ma_{CM} = \frac{mF}{M + m} \text{ (towards right)}$$

Net external force on m is $F_2 = ma_{CM} = \dfrac{mF}{M + m}$ (towards left)

In COM frame, initial KE is zero and final KE, when spring has maximum extension (i.e., when $v_1 = v_2 = v_{CM}$), is also zero.

$$\left[k_{\text{wrt CM}} = \frac{1}{2}\mu\, v_{\text{rel}}^2 = 0\right]$$

Using work–energy theorem in COM frame:

$$W_{F1} + W_{F2} = \Delta U + \Delta k.$$

$$\Rightarrow \quad \frac{mF}{M+m}x_1 - \frac{mF}{M+m}x_2 = \frac{1}{2}kx^2$$

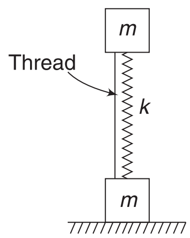
$$\Rightarrow \quad \frac{mF}{M+m}(x_1 - x_2) = \frac{1}{2}kx^2$$

$$\Rightarrow \quad \frac{mF}{M+m}x = \frac{1}{2}kx^2$$

$$\Rightarrow \quad x = \frac{2mF}{k(M+m)}$$

Example 28 *Jumping blocks*

Two blocks, each of mass m, are connected by a compressed light spring of spring constant k. The blocks are held in position by a light thread with spring compressed by x_0. The thread is burnt. Find



(i) The smallest value of x_0 for which the lower block will bounce up.

(ii) The height to which COM of the system will rise, if $x_0 = \frac{7mg}{k}$.

Solution

Concepts

(i) when the spring stretches, it pulls the lower block up. When this force becomes greater than mg, the lower block jumps.

(ii) Energy conservation can be used to find the extension in the spring after the system is released (thread is burnt) and lower block remaining fixed.

(iii) After the lower block leaves the surface, the net external force on the system is $2mg$ and acceleration of COM is $g(\downarrow)$.

(i) Let the upper block come to rest when the spring is stretched by x. Assume that the lower block has not moved till now.

Using conservation of energy,

$$\frac{1}{2}kx_0^2 = \frac{1}{2}kx^2 + mg(x_0 + x)$$

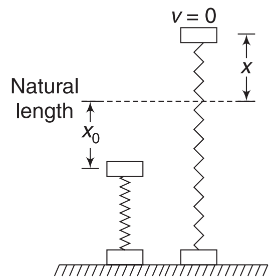
$$\Rightarrow \quad kx^2 + 2mg\,x + 2mg\,x_0 - kx_0^2 = 0$$

$$\Rightarrow \quad x = \frac{-2mg \pm \sqrt{4m^2g^2 - 4k(2mgx_0 - kx_0^2)}}{2k}$$

$$= \frac{-2mg \pm (2kx_0 - 2mg)}{2k}$$

$$= \frac{(kx_0 - 2mg)}{k} \text{ and } x_0 \text{(Not acceptable).}$$

$$\therefore \quad kx = kx_0 - 2mg$$



The lower block will just leave the ground if $kx = mg$

$$\Rightarrow \quad mg = kx_0 - 2mg$$

$$\Rightarrow \quad x_0 = \frac{3mg}{k}$$

(ii) If $x_0 = \frac{7mg}{k}\left(>\frac{3mg}{k}\right)$, the lower block will bounce off when the upper block is still moving up. Let v = speed of the upper block at the instant the lower block just begins to move (i.e., at $x = \frac{mg}{k}$).

$$\frac{1}{2}k\left(\frac{7mg}{k}\right)^2 = \frac{1}{2}mv^2 + \frac{1}{2}k\left(\frac{mg}{k}\right)^2 + mg\left(\frac{7mg}{k} + \frac{mg}{k}\right)$$

$$\Rightarrow \quad v^2 = 32\frac{mg^2}{k} \qquad \text{...(a)}$$

Velocity of COM at the instant the lower block begins to move is

$$v_{CM} = \frac{mv + 0}{2m} = \frac{v}{2}$$

Now the COM moves with a retardation g. If the COM further rises by x', then for motion of COM

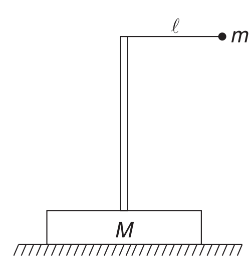
$$0^2 = \left(\frac{v}{2}\right)^2 - 2gx' \quad \Rightarrow \quad x' = \frac{v^2}{8g} = \frac{4mg}{k}$$

By the time the lower block begins to move, the upper block has gone up by $x_1 = \frac{7mg}{k} + \frac{mg}{k} = \frac{8mg}{k}$.

It means the COM has moved through a distance $\frac{mx_1 + 0}{2m} = \frac{4mg}{k}$ before the lower block starts to move.

$$\therefore \quad \text{Total rise of COM} = \frac{4mg}{k} + \frac{4mg}{k} = \frac{8mg}{k}.$$

Example 29 A point-like ball of mass m is tied to the end of a light string, which is attached to the top of a vertical massless rod. The rod is fixed to a block of mass M lying at rest on a horizontal surface. The pendulum is displaced to a horizontal position with string just taut



and released. Friction between the block and the horizontal surface is large enough to prevent the block from sliding.

(i) Find the magnitude of the horizontal component of acceleration of the COM of the entire system when the string makes an angle of $\theta = 60°$ with the vertical.

(ii) Find the friction force on the block when the string makes 60° with vertical.

(iii) Find the friction force on the block when the string becomes vertical.

Solution

Concepts

(i) Since the block does not move, friction does not perform work on it. Mechanical energy of the swinging pendulum is conserved. We will use energy conservation to find speed of the ball when $\theta = 60°$.

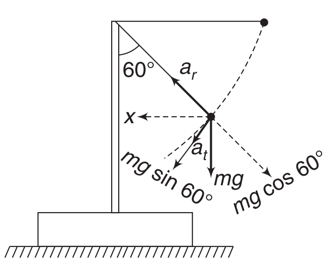
(ii) Acceleration of the bob has a radial and a tangential component. We can find these components separately. Then, we find the horizontal acceleration (a_x) of the ball by resolving a_r and a_t in horizontal direction.

(iii) $a_{CM} = \dfrac{ma_x}{m + M}$, since M is not moving.

(iv) Friction is the only horizontal force on the system.

$\therefore \quad f = (M + m)\,a_{CM}$.

(v) When the string is vertical, there is no horizontal acceleration.



(i) speed at $\theta = 60°$ can be calculated as

$$\frac{1}{2}mv^2 = mg\,l\cos 60°$$

$$\Rightarrow \quad v = \sqrt{gl}$$

$\therefore$ Radial acceleration is

$$a_r = \frac{v^2}{l} = g$$

Tangential acceleration is

$$a_t = g\sin 60° = \frac{\sqrt{3}}{2}g$$

Horizontal acceleration of the ball is:

$$a_x = a_r\cos 30° + a_t\cos 60°$$

$$= \frac{\sqrt{3}}{2}g + \frac{\sqrt{3}}{2}g\cdot\frac{1}{2} = \frac{3\sqrt{3}\,g}{4}$$

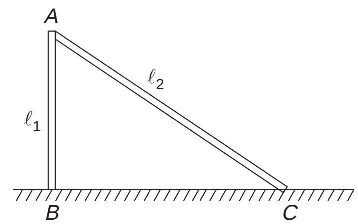
$$\therefore \quad (a_{CM})_x = \frac{M(O) + ma_x}{M + m} = \frac{3\sqrt{3}mg}{4(M + m)}$$

(ii) $f = (M + m)(a_{CM})_x = 3\sqrt{3}\,mg$

(iii) When the string is vertical, there is no tangential force on the ball ($\Rightarrow \; a_t = 0$) and the radial acceleration is vertical.

$$\therefore \quad (a_{CM})_x = 0 \quad \Rightarrow \quad f = 0$$

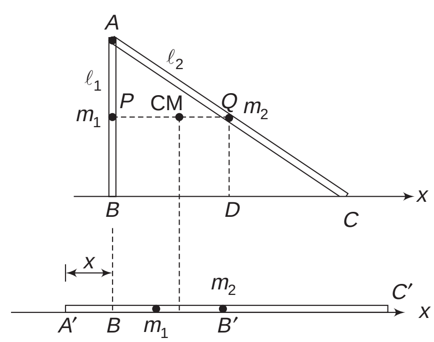
Example 30 Two thin rods of identical material and equal cross-section are connected (at A) by a frictionless joint. Lengths of the two rods are $AB = l_1$ and $AC = l_2$. The structure is kept vertical on a smooth table, as shown. It is released from this position and the ends B and C slide so that the angle between the two rods decreases. Where will the joint A hit the table? Assume that the structure always remains vertical during the fall.



Solution

Concepts

There is no horizontal force on the structure. Its COM will not have any horizontal displacement. End B moves to the right (and C also moves to the right) so that the angle between the rods decreases.



Mass of rod AB, $\quad m_1 = \lambda l_1$

Mass of rod AC, $\quad m_2 = \lambda l_2$

AB can be replaced with a point mass at its centre P and AC can be replaced with a point mass at its centre Q.

$$BD = \frac{1}{2}BC = \frac{1}{2}\sqrt{l_2^2 - l_1^2}$$

Taking B as the origin, the x co-ordinate of the COM of the structure in initial position is

$$x_{CM} = \frac{m_1(0) + m_2(BD)}{m_1 + m_2} = \frac{\lambda l_2\frac{1}{2}\sqrt{l_2^2 - l_1^2}}{\lambda l_1 + \lambda l_2}$$

$$= \frac{l_2}{2}\sqrt{\frac{l_2 - l_1}{l_2 + l_1}} \qquad \text{...(i)}$$

When the structure collapses, let end A be at a distance x from origin B (see figure).

Now, m_1 is a point mass at a distance $\frac{l_1}{2}$ from A'.

$\therefore$ Point mass m_1 is at a distance $\left(\frac{l_1}{2} - x\right)$ from B.

Similarly, point mass m_2 is at a distance $\left(\frac{l_2}{2} - x\right)$ from B.

$$\therefore \qquad x_{CM} = \frac{m_1\left(\frac{l_1}{2} - x\right) + m_2\left(\frac{l_2}{2} - x\right)}{m_1 + m_2}$$

$$\Rightarrow \qquad \frac{l_2}{2}\sqrt{\frac{l_2 - l_1}{l_2 + l_1}} = \frac{l_1\left(\frac{l_1}{2} - x\right) + l_2\left(\frac{l_2}{2} - x\right)}{l_1 + l_2}$$

$$\Rightarrow \qquad \frac{l_2}{2}\sqrt{l_2^2 - l_1^2} = \frac{1}{2}(l_1^2 + l_2^2) - (l_1 + l_2)x$$

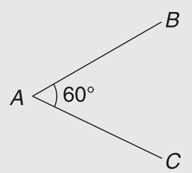
$$\Rightarrow \qquad x = \frac{l_2}{2(l_1 + l_2)}\left[l_1^2 + l_2^2 - \sqrt{l_2^2 - l_1^2}\right]$$

Worksheet 1

1. All the particles of a system are situated at a distance r from the origin. The distance of the centre of mass of the system from the origin is

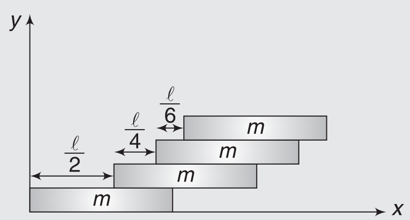
(a) $= r$ (b) $\leq r$
(c) $> r$ (d) $\geq r$

2. A uniform wire of length ℓ is bent into a 'V' shaped object as shown. The distance of its centre of mass from the vertex A is



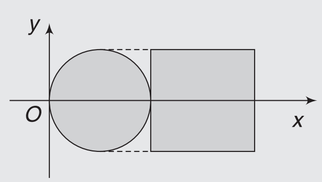
(a) $\ell/2$ (b) $\dfrac{\ell\sqrt{3}}{4}$
(c) $\dfrac{\ell\sqrt{3}}{8}$ (d) None of these

3. Find the x coordinate of the centre of mass of the arrangement of bricks shown in figure:



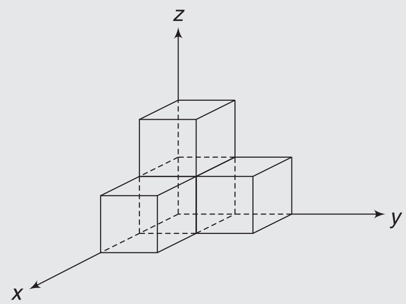
(a) $\dfrac{24}{25}\ell$ (b) $\dfrac{25}{24}\ell$
(c) $\dfrac{15}{16}\ell$ (d) $\dfrac{16}{15}\ell$

4. A composite body is made by joining a uniform disk of radius r and a uniform square plate of side length $2r$, as shown in figure. Both plates have same mass per unit area. The co ordinates of COM of the composite plate are



(a) $\left(\dfrac{r(\pi+3)}{(\pi+1)},\ 0\right)$ (b) $\left(\dfrac{r(\pi+2)}{(\pi+1)},\ 0\right)$
(c) $\left(\dfrac{r(\pi+1)}{(\pi+1)},\ 0\right)$ (d) none

5. Find the coordinates of centre of mass of the structure shown in figure made of four identical cubes. Length of each side of a cube is 1 unit.

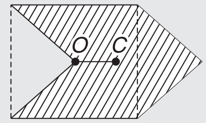


(a) (1/2, 1/2, 1/2) (b) (1/3, 1/3, 1/3)
(c) (3/4, 3/4, 3/4) (d) (1/2, 3/4, 1/2)

6. In a HCl molecule, the separation between the nuclei of the two atoms is about 1.27 Å (1 Å = 10^{-10} m). Given that a chlorine atom is about 35.5 times as massive as a hydrogen atom and nearly all the mass of an atom is concentrated in its nucleus, the position of COM of the molecule is

(a) On the line joining H and Cl and at a distance of slightly less than 1.27Å from H end
(b) On the line joining H and Cl and at a distance of nearly 1.27Å from Cl end
(c) On the line joining H and Cl and at a distance of 0.27Å from H end
(d) On the line joining H and Cl and at a distance of 0.57Å from H end

7. An arrow sign is made by cutting and rejoining a quarter part of a square plate of side 'L' as shown. The distance OC, where 'C' is the centre of mass of the arrow, is

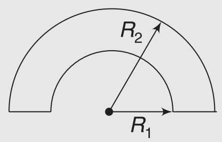


(a) $\dfrac{L}{3}$ (b) $\dfrac{L}{4}$
(c) $\dfrac{3L}{8}$ (d) None of these

8. The centre of mass of a system of particles is at the origin. From this, we conclude that

(a) The number of particles on positive x-axis is equal to the number of particles on negative x-axis
(b) The total mass of the particles on positive x-axis is same as the total mass on negative x-axis
(c) The number of particles on x-axis may be equal to the number of particles on y-axis.
(d) If there is a particle on the positive x-axis, there must be at least one particle on the negative x-axis.

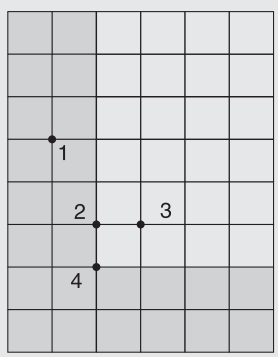
9. Distance of the centre of mass, from centre, of an annular half-disc shown in figure is



(a) $\dfrac{4(R_2^3 - R_1^3)}{3\pi(R_2^2 - R_1^2)}$ (b) $\dfrac{(R_2^3 - R_1^3)}{3\pi(R_2^2 - R_1^2)}$

(c) $\dfrac{3\pi(R_2^2 - R_1^2)}{3\pi(R_2^2 - R_1^2)}$ (d) None

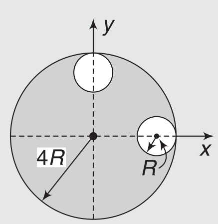
10. An L-shaped (shaded piece) is cut from a rectangular metal plate of uniform thickness. The point that corresponds to the centre of mass of the L-shaped piece is:



(a) 1 (b) 2

(c) 3 (d) 4

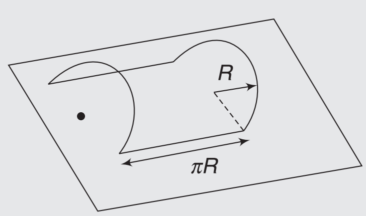
11. From the circular disc of radius $4R$, two small disc of radius R are cut off. The centre of mass of the new structure will be at:



(a) $i\frac{R}{5} + i\frac{R}{5}$ (b) $-i\frac{R}{5} + j\frac{R}{5}$

(c) $\frac{-3R}{14}(\hat{i} + \hat{j})$ (d) None of these

12. A wire is bent into a structure as shown in the figure, and placed on a table. It consists of two half-rings of radius R and two straight parts of length πR. The height of COM from the table is



(a) $\frac{2R}{\pi}$ (b) $\frac{R}{\pi}$

(c) $\frac{R}{2}$ (d) zero

13. A piece of paper (shown in figure 1) is in the form of a square. Two corners of this square are folded to make it appear like figure 2. Both corners are put together at the centre 'O' of the square. If O is taken to be (0, 0), the centre of mass of the new system will be at

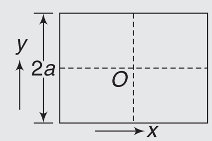


Fig. 1

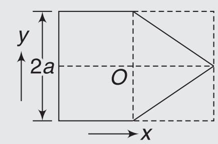
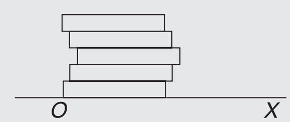


Fig. 2

(a) $\left(-\frac{a}{8}, 0\right)$ (b) $\left(-\frac{a}{6}, 0\right)$

(c) $\left(\frac{a}{12}, 0\right)$ (d) $\left(-\frac{a}{12}, 0\right)$

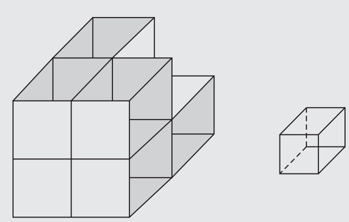
14. Five homogeneous bricks, each of length L, are arranged as shown in figure. Each brick is displaced with respect to the one in contact by $L/5$. Find the x-coordinate of the centre of mass relative to the origin O shown.



(a) $\frac{23L}{50}$ (b) $\frac{33L}{50}$

(c) $\frac{13L}{50}$ (d) $\frac{29L}{50}$

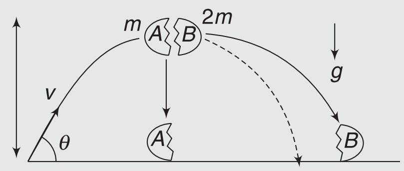
15. Eight solid uniform cubes of edge l are stacked together to form a single cube with centre O. One cube is removed from this system. Distance of the centre of mass of the remaining 7 cubes from O is



(a) $\frac{7\sqrt{3}l}{16}$ (b) $\frac{\sqrt{3}l}{16}$

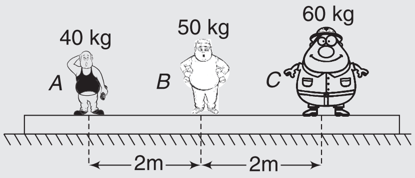
(c) $\frac{\sqrt{3}l}{14}$ (d) zero

16. A projectile is launched from the origin with speed v at an angle θ from the horizontal. At the highest point in the trajectory, the projectile breaks into two pieces, A and B, of masses m and $2m$, respectively. Immediately after the breakup, piece A is at rest relative to the ground. Neglect air resistance. Which of the following sentences most accurately describes what happens next?



(a) Piece B will hit the ground first, since it is more massive.
(b) Both pieces have zero vertical velocity immediately after the breakup, and therefore they hit the ground at the same time.
(c) Piece A will hit the ground first, because it will have a downward velocity immediately after the breakup.
(d) There is no way of knowing which piece will hit the ground first, because not enough information is given about the breakup.

17. Three men A, B & C of masses 40 kg, 50 kg & 60 kg are standing on a plank of mass 90 kg, which is kept on a smooth horizontal plane. If A & C exchange their positions, then mass B will shift



(a) 1/3 m towards left
(b) 1/3 m towards right
(c) will not move w.r.t. ground
(d) 5/3 m towards left

18. An isolated particle of mass m is moving in horizontal plane (x–y), along the x–axis, at a certain height above the ground. It suddenly explodes into two fragment of masses $\frac{m}{4}$ and $\frac{3m}{4}$. An instant later, the smaller fragment is at $y = +15$ cm. The larger fragment at this instant is at:

(a) $y = -5$ cm (b) $y = +20$ cm
(c) $y = +5$ cm (d) $y = -20$ cm

19. Two particles A and B, initially at rest, move towards each other under the mutual force of attraction. At the instant when the speed of A is v and the speed of B is $2v$, the speed of the centre of mass of the system is:

(a) $3v$ (b) v
(c) $1.5v$ (d) zero

20. A boy of mass 60 kg is standing over a platform of mass 40 kg placed over a smooth horizontal surface. He throws a stone of mass 1 kg with velocity $v = 10$ ms^{-1} at an angle of 45° with respect to the ground. Find the displacement of the platform (with boy) on the horizontal surface when the stone lands on the ground. Neglect the height of projection ($g = 10$ ms^{-2})

(a) 10 cm (b) 1 m
(c) 1.2 m (d) 50 cm

21. Two particles of equal mass have initial velocities $2\hat{i}$ ms^{-1} and $2\hat{j}$ ms^{-1}. First particle has a constant acceleration $(\hat{i} + \hat{j})$ ms^{-2} while the acceleration of the second particle is always zero. The centre of mass of the two particles moves in

(a) Circle (b) Parabola
(c) Ellipse (d) Straight line

22. A uniform thin rod of mass M and Length L is standing vertically along the y-axis on a smooth horizontal surface, with its lower end at the origin (0, 0). A slight disturbance at $t = 0$ causes the lower end to slip on the smooth surface along the positive x-axis, and the rod starts falling. The acceleration vector of the centre of mass of the rod during its fall is: [$\vec{R}$ is reaction from surface]

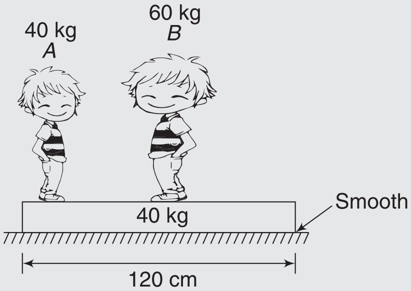
(a) $\vec{a}_{CM} = \dfrac{M\vec{g} + \vec{R}}{M}$ (b) $\vec{a}_{CM} = \dfrac{M\vec{g} - \vec{R}}{M}$

(c) $\vec{a}_{CM} = M\vec{g} - R$ (d) None of these

23. A uniform sphere is placed on a smooth horizontal surface and a horizontal force F is applied on it at a distance h above the centre. The acceleration of the centre of mass of the sphere

(a) is maximum when $h = 0$
(b) is maximum when $h = R$
(c) is maximum when $h = R/2$
(d) is independent of h

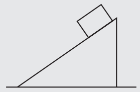
24. Two men 'A' and 'B' are standing on a plank which is placed on a smooth surface. 'B' is at the middle of the plank and 'A' is at the left end of the plank. Surface of the plank is smooth. System is initially at rest and



masses are as shown in figure. A and B start moving such that the position of 'B' remains fixed with respect to ground. The point where A meets B is located at

(a) the middle of the plank
(b) 30 cm from the left end of the plank
(c) the right end of the plank
(d) None of these

25. A large wedge rests on a horizontal frictionless surface, as shown. A block starts from rest and slides down the inclined surface of the wedge, which is rough. During the motion of the block, the centre of mass of the block and wedge system

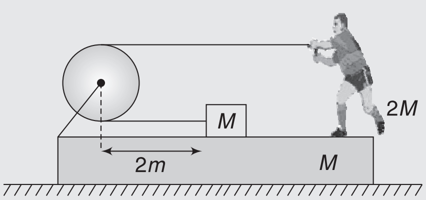


(a) does not move
(b) moves vertically with increasing speed
(c) moves horizontally with constant speed
(d) moves both horizontally and vertically

26. Two uniform non-conducting balls A & B have identical size having radius R but made of different density materials; density of A being twice the density of B. The ball A is positively charged & ball B is negatively charged. The balls are released on the horizontal smooth surface when their centres are at separation $10R$. Because of mutual attraction, the balls start moving towards each other. They will collide when the lighter ball has travelled a distance

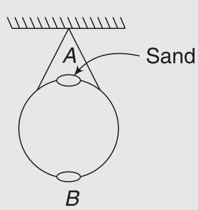
(a) $x = \dfrac{10R}{3}$ (b) $x = \dfrac{16R}{3}$
(c) $x = 5R$ (d) $x = \dfrac{7R}{5}$

27. A block of mass M is tied to one end of a massless rope. The other end of the rope is in the hands of a man of mass $2M$. Both the man and the block lie on a rough wedge of mass M, as shown in the figure. The whole system is resting on a smooth horizontal surface. The man pulls the rope. Pulley is massless and frictionless. The displacement of the wedge when the block meets the pulley is (Man does not leave his position during the pull)



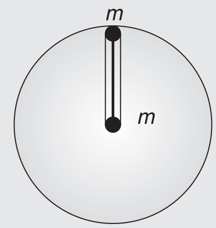
(a) 0.5 m (b) 1 m
(c) Zero (d) 2/3 m

28. A uniform metallic spherical shell is suspended from ceiling. It has two holes A and B at top and bottom respectively. Which of the following is/are true:



(a) If B is closed and sand is poured from A, centre of mass first rises and then falls
(b) If shell is completely filled with sand and B is opened then centre of mass falls initially
(c) If shell is slightly filled with sand and B is opened, then centre of mass falls.
(d) None of these

29. Imagine a massless rod of length R carrying two particles of mass m each at its end. The rod is placed inside a tunnel dug along a radius of the Earth. Find the distance between the centre of gravity and the centre of mass of a two-particle system attached to the ends of a light rod. R is the radius of earth. It is given that acceleration due to gravity at the centre of the Earth is zero.



(a) R (b) $R/2$
(c) zero (d) $R/4$

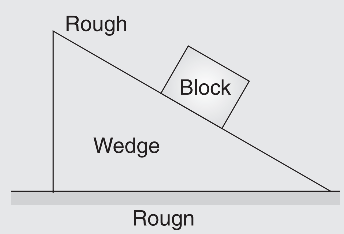
30. Two particles approach each other with different velocities. After collision, one of the particles has a momentum $\vec{p}$ in their centre of mass frame. In the same frame, the momentum of the other particle is

(a) 0 (b) $-\vec{p}$
(c) $-\vec{p}/2$ (d) $-2\vec{p}$

31. Two identical cars start at the same point, but travel in opposite directions on a circular path of radius R, each at speed v. While each car travels a distance less than $(\pi/2)R$, (one quarter circle) the centre of mass of the two cars

(a) remains at the initial point
(b) travels along a diameter of the circle at speed $< v$
(c) travels along a diameter of the circle at speed $= v$
(d) travels along a diameter of the circle at speed $> v$

32. When a block is placed on a wedge as shown in the figure, the block starts sliding down and the wedge also starts sliding on the ground. All surfaces are rough. The centre of mass of (wedge + block) the system will move



(a) leftward and downwards
(b) rightward and downwards
(c) leftward and upwards
(d) only downwards

Worksheet 2

1. In which of the following cases, the centre of mass of a rod may be at its centre?

(a) The linear mass density continuously decreases from left to right.

(b) The linear mass density continuously increases from left to right.

(c) The linear mass density decreases from left to right upto centre and then increases.

(d) The linear mass density increases from left to right upto centre and then decreases.

2. An external force $\vec{F}(\vec{F} \neq 0)$ acts on a system of particles. The velocity and the acceleration of the centre of mass are found to be v_{cm} and a_{cm} at an instant, then it is possible that

(a) $v_{cm} = 0$, $a_{cm} = 0$ (b) $v_{cm} = 0$, $a_{cm} \neq 0$

(c) $v_{cm} \neq 0$, $a_{cm} = 0$ (d) $v_{cm} \neq 0$, $a_{cm} \neq 0$

3. Consider a particle at rest, which may decay into two (daughter) particles or into three (daughter) particles. Which of the following is **true in the two-body case but false in the three-body case?**

(There are no external forces and the masses of daughter particles are known.)

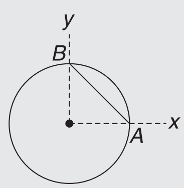
(a) Velocity vectors of the daughter particles must lie in a single plane.

(b) Given the total kinetic energy of the system, it is possible to determine the speed of each daughter particle.

(c) Given the speed (s) of all but one daughter particle, it is possible to determine the speed of the remaining particle.

(d) The total momentum of the daughter particles is zero.

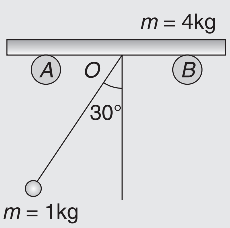
4. An object comprises a uniform ring of radius R and its uniform chord AB (not necessarily made of the same material) as shown. Which of the following cannot be the centre of mass of the object?



(a) $(R/3, R/3)$ (b) $(R/3, R/2)$

(c) $(R/4, R/4)$ (d) $(R/\sqrt{2}, R/\sqrt{2})$

5. A ball of mass 1 kg is suspended by an inextensible string, 1 m long, attached to a point O of a smooth horizontal bar resting on a fixed smooth supports A and B. The ball is released from rest from the position where the string makes an angle of 30° with the vertical. The mass of the bar is 4 kg.



(a) The bar first moves to left and then to the right, as the ball oscillates.

(b) The maximum angle that the string makes with vertical on other side is less than 30°.

(c) The displacement of the bar when the string makes the maximum angle on the other side of the vertical is 0.2 m

(d) The COM of the ball plus bar system experiences no acceleration as the ball oscillates.

6. A mercury thermometer is placed in a gravity-free hall without touching anything. As temperature rises mercury expands and ascends in thermometer. If height ascended by mercury in thermometer is h then

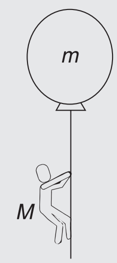
(a) The COM of the "mercury and thermometer" system may descend if mass of mercury is very small relative to the mass of glass.

(b) The COM of the "mercury and thermometer" system may ascend if mass of mercury is very large relative to the glass.

(c) The COM of the "mercury and thermometer" system may ascend even if mass of mercury is very small relative to the glass.

(d) All the above are false.

7. A man of mass M is hanging with a light rope, which is connected to a balloon of mass m. The system is at rest in air. The man climbs a distance h with respect to the rope.



(a) Due to gravity, the COM of the entire system (Balloon + man) will move down.

(b) The actual displacement of the man is $\dfrac{mh}{m+M}$

(c) The actual displacement of the balloon is $\dfrac{Mh}{m+M}$

(d) If the man just releases the rope and sets himself into a free fall, the balloon will begin to rise with an acceleration.

8. A ring of mass m and a particle of same mass are fixed on a disc of same mass such that the centre of mass of the system lies at the centre of the disc. The system rotates such that the centre of mass of the disc moves in a circle of radius R with a constant angular velocity ω. From this, we conclude that

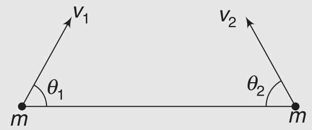
(a) An external force $m\omega^2 R$ must be applied to central particle

(b) An external force $m\omega^2 R$ must be applied to the ring

(c) An external force $3m\omega^2 R$ must be applied to central particle

(d) An external force $3m\omega^2 R$ must be applied on the system

9. Two particles of equal mass m are projected form the ground with speeds v_1 and v_2 at angles θ_1 and θ_2 (θ_1, $\theta_2 \neq 0$, 180°), as shown in figure. The centre of mass of the two particles



(a) will move in a parabolic path for any values of v_1, v_2, θ_1 and θ_2

(b) can move in a vertical line

(c) can move in a horizontal line

(d) will move in a straight line for any value of v_1, v_2, θ_1 and θ_2

10. For a two-body system, in the absence of external forces, the kinetic energy as measured from ground frame is k_o and from the centre of mass frame is k_{cm}. Pick up the wrong statement/s

(a) The kinetic energy as measured from the centre of mass frame is least.

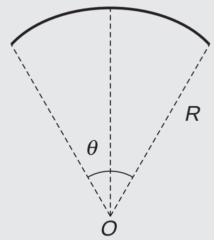
(b) Only the portion of energy k_{cm} can be transformed from one form to another due to internal changes in the system.

(c) The system always retains at least $k_o - k_{cm}$ amount of kinetic energy, as measured from ground frame, irrespective of any kind of internal changes in the system.

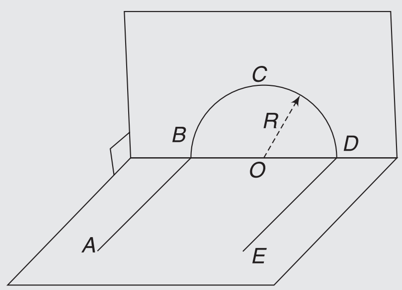
(d) The system always retains at least k_{cm} amount of kinetic energy, as measured from ground frame, irrespective of any kind of internal changes in the system.

Worksheet 3

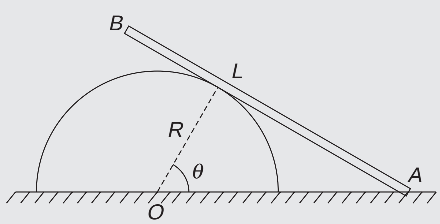
1. A uniform wire is in the shape of a circular arc, subtending an angle θ at the centre. Radius of the arc is R. Find the distance of the COM of the arc from centre of curvature O.



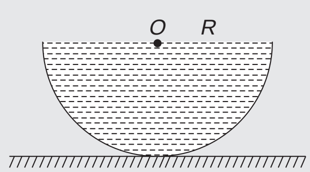
2. A uniform wire is bent into a shape shown in figure. Segments AB and ED are straight, having length $2R$ each, lying in a horizontal plane and the segment BCD is a semi-circle of radius R. Find the distance of the COM of the wire from centre O of the circle.



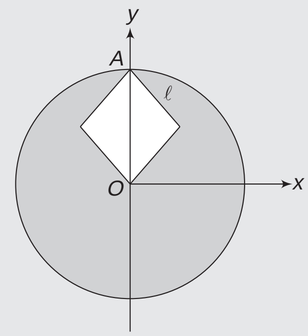
3. A smooth rod of length $L = 2R$ is leaning against a semi cylinder of radius R, which is placed on a smooth horizontal surface. In the position shown, $\theta = 45°$. The system is released from this position. Find the displacement of the semi cylinder by the time end B of the rod touches it. Both cylinder and rod have the same mass.



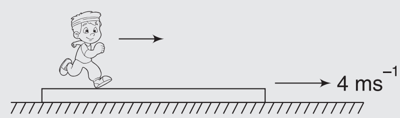
4. A hemispherical shell has mass M. A liquid is filled in it upto the brim. Mass of the liquid is $2M$. Find the height of the COM of the liquid-filled hemisphere, measured from the ground.



5. A disc of radius r is uniform. A square-hole of side length $l = \dfrac{r}{\sqrt{2}}$ is punched in the disc, as shown. One corner of the hole is at centre O of the disc and the other corner is at A on the circumference. The diagonal OA of the square is y-axis and O is the origin. Find the co-ordinates of the COM of the disc with hole.



6. A 75-kg man stands at the rear end of a platform of mass 25 kg and length $4m$, which is moving on a smooth horizontal surface, with velocity $4\,\text{ms}^{-1}$. The man begins to walk at $2\,\text{ms}^{-1}$, relative to the platform, towards the front end. He stops after reaching the front end. Find the following for the interval the man walks on the platform–

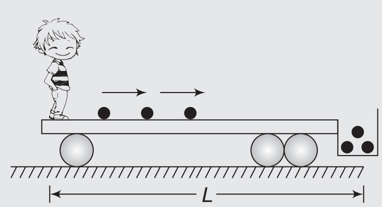


(i) Displacement of the COM of the man + platform system.

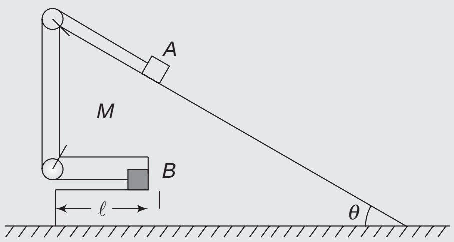
(ii) Displacement of the man, relative to the ground.

(iii) Displacement of the platform, relative to the ground.

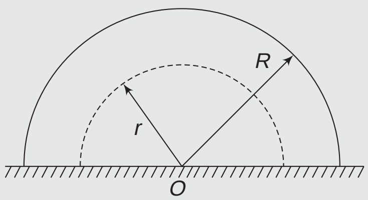
7. A flat railroad car has a boy standing at its end. The boy rolls billard balls along the surface of the car, which get collected at the other end into a small bucket (See figure) attached to the car. Length of the car is L and it stands on a frictionless track. Initially, the entire system was at rest when the boy rolled the first ball. Prove that, irrespective of the number of balls the boy has or the speed of the balls, the car cannot travel more than L.



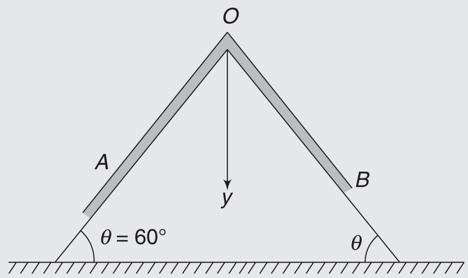
8. A wedge of mass M carries two blocks A and B, as shown in figure. Block A lies on a smooth-incline surface of the wedge inclined at angle θ to the horizontal. B lies inside a horizontal groove made in the wedge. The two blocks have mass m each and are connected by a light string. Length of the groove is l, as shown. The entire system is released from rest. Find the distance travelled by the wedge on the smooth horizontal surface by the time block B comes out of the groove.



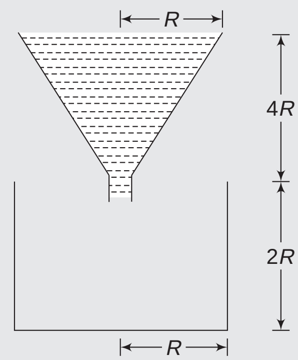
9. A solid hemisphere has radius R. Density of the solid changes with distance r from the centre O, as $\rho = \rho_O \dfrac{r}{R}$. Find the position of the COM of the hemisphere.



10. A rope has length $3L$ and is made of three uniform segments of length L each. Mass of the two end-segments are m and $2m$. The mid segment has mass $1.5\ m$. The rope is placed over a smooth equilateral wedge fixed on a horizontal table. The rope stays in equilibrium. Find y co-ordinate of the COM of the rope, measured from the apex point O.



11. A conical container has a circular base of radius R and height $4R$. It is filled with a liquid of density d. Below the cone, there is a cylindrical container of radius R. Base of the cylinder is at a depth $2R$ below the apex of the cone. Cap at the apex of the cone is opened and the liquid slowly drains into the cylindrical containers. Calculate work done by gravity in the process.



Answers Sheet

Your Turn

1. $\frac{4d}{5}$
2. 6.5 m
3. Zero
4. (i) –6.5 cm (ii) $\frac{13}{6}$ cm
5. $\left(\frac{3}{2}\hat{i} + \frac{1}{2}\hat{j} - \frac{1}{2}\hat{k}\right)$m
6. $\frac{5L}{9}$
7. $\frac{L}{2\sqrt{2}}$
8. $\left(\frac{5}{6}a, \frac{5}{6}a\right)$
9. $\frac{2R}{3\pi}$
10. $\frac{R}{6}$
11. $\frac{2R}{2+\pi}$ from O
12. At the centre of the Earth
13. No
14. (a) yes (b) Speed becomes the same as original speed
15. (i) $(-2.4\hat{i} + 1.6\hat{j})$cms^{-1} (ii) $4\hat{j}$ cms^{-2} (iii) $8\hat{j}$ dyne
16. $\frac{L}{2}$
17. $v_{CM} = \frac{u}{5}$
18. COM continues to move along a straight line, with velocity $\vec{v}_{CM} = \frac{v}{2}(-\hat{i} + \hat{j})$
19. 5 m
20. (i) At the centroid of the triangle (ii) Will not move
21. (400, – 91, 32) m
22. $\frac{mR\sin\theta}{m+M}$
23. $2\lambda g R^2$
24. 6 J.
25. 0
26. yes

Worksheet 1

1. (b)	2. (c)	3. (b)	4. (d)	5. (c)	6. (a)	7. (b)	8. (c)	9. (a)
10. (b)	11. (c)	12. (b)	13. (d)	14. (a)	15. (c)	16. (b)	17. (b)	18. (a)
19. (d)	20. (a)	21. (d)	22. (a)	23. (d)	24. (c)	25. (b)	26. (b)	27. (a)
28. (b)	29. (b)	30. (b)	31. (b)	32. (b)				

Worksheet 2

1. (c, b)	2. (b, d)	3. (b, c)	4. (b, d)	5. (a, c)	6. (d)	7. (b, c, d)	8. (d)	9. (b)
10. (d)								

Worksheet 3

1. $\frac{R\sin\left(\frac{\theta}{2}\right)}{\left(\frac{\theta}{2}\right)}$
2. $\frac{2\sqrt{5}R}{\pi + 4}$
3. $\frac{(3\sqrt{2} - \sqrt{5})R}{2\sqrt{10}}$
4. $\frac{7R}{12}$
5. $\left(O, \frac{-r}{2(2\pi - 1)}\right)$
6. (i) 8 m, (ii) 9 m, (iii) 5 cm
8. $\frac{ml(1-\cos\theta)}{2m+M}$
9. $\frac{2R}{5}$ from O on symmetry axis
10. 0.63 L
11. $W_g = \frac{52\pi}{9} R^4 \cdot d \cdot g$

CHAPTER 2

Momentum and Its Conservation

"The world is wide, and I will not waste my life in friction when it could be turned into momentum."

–Frances E. Willard

1. INTRODUCTION

A heavy truck and a car are moving with same velocity. Which has more "motion" in it? Certainly, the truck. It is very difficult to stop it. If it hits a wall, it will cause more damage then the car. We say that the truck has more "quantity of motion" than the car. The physical quantity measuring this "quantity of motion" is called momentum. Similarly, a fast-moving bullet can cause more harm than a slow-moving ball, though mass of ball is more.

Therefore, "quantity of motion" (i.e, momentum) possessed by a body depends on two things – its mass and its velocity. The product, mass × velocity, is known as momentum of a body. Momentum of a body is closely related to force acting on it.

In this chapter, we will understand that momentum of a system remains conserved in absence of any external force. This is a very strong conservation principle in physics and helps us in solving diverse kinds of problems. We will be able to understand collision problems and rocket propulsion. Later, you will learn that conservation of momentum helps us in understanding physics of sub-atomic particles also.

2. MOMENTUM

Momentum of a particle of mass m moving with a velocity $\vec{v}$ is defined as

$$\vec{P} = m\vec{v} \qquad \text{...(1)}$$

Its unit is kg m s^{-1} which is same as Ns. It is a vector and any change in direction of motion implies a change in momentum.

Momentum of an extended body or a collection of particles is the sum of momenta of individual particles. As studied in last chapter, momentum for a collection of particles is given by product of total mass and velocity of COM.

$$\vec{P} = \sum m_i \vec{v}_i = M\vec{v}_{CM} \qquad \text{...(2)}$$

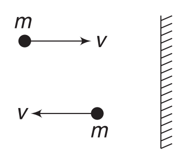
Example 1 A ball of mass 1 kg hits a wall normally at a speed of 10 ms^{-1} and rebounds at a speed of 10 ms^{-1}. Find change in momentum of the ball.

Solution

Concepts

Momentum is a vector. Change in direction of a vector implies that the vector has changed.

$$\begin{aligned}\Delta P &= P_f - P_i \\ &= mv(\leftarrow) - mv(\rightarrow) \\ &= mv(\leftarrow) + mv(\leftarrow) \\ &= 2mv(\leftarrow) = 2 \times 1 \times 10(\leftarrow) \\ &= 20 \text{ kg ms}^{-1}(\leftarrow)\end{aligned}$$



YOUR TURN

Q.1 Which has larger momentum — a 50 g bullet travelling at 500 ms^{-1} or a 80 kg bicycle (with rider) travelling at 9 kmh^{-1}?

Q.2 A car and a truck have same kinetic energy. Which has more momentum?

Q.3 A projectile of mass m is projected at an angle θ to the horizontal at speed u. Find change in its momentum by the time

(i) it reaches the top point of its trajectory.

(ii) it is about to hit the ground.

3. FORCE AND MOMENTUM (NEWTON'S SECOND LAW)

Newton, originally, expressed his second law in terms of momentum. According to this law:

The rate of change of momentum of a particle is equal to the net external force acting on it and the change in momentum always takes place in the direction of the force.

$$\frac{d\vec{P}}{dt} = \vec{F}_{\text{ext}} \qquad \text{...(3)}$$

This, means that a force is necessary to produce a change in momentum of a particle. If there is no force, momentum cannot change.

If we put $\vec{P} = m\vec{v}$ in the above equation and assume that the mass (m) of the particle is constant, we get

$$\frac{m\,d\vec{v}}{dt} = \vec{F}_{\text{ext}}$$

$$\Rightarrow \quad m\vec{a} = \vec{F}_{\text{ext}} \qquad \text{...(4)}$$

Thus, the two relations (3) and (4) are equivalent when mass of a particle stays constant.

If we have a system of particles, its momentum is given by equation (2)

$$\vec{P} = M\vec{v}_{\text{CM}}$$

$$\Rightarrow \quad \frac{d\vec{P}}{dt} = M\frac{d\vec{v}_{\text{CM}}}{dt}, \text{ assuming } M \text{ to be a constant}$$

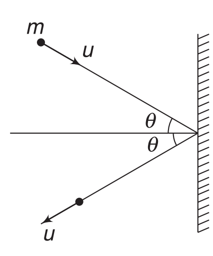
$$= M\,\vec{a}_{\text{CM}}$$

We already know that $M\,\vec{a}_{\text{CM}}$ is the net external force on the system. Therefore,

$$\frac{d\vec{P}}{dt} = \vec{F}_{\text{ext}}$$

This relation is true for a single particle or a system of particles.

Example 2 A ball of mass m impinges on a wall at a speed u, at an angle of incidence θ. The ball rebounds at the same speed at an angle of reflection θ. The ball interacted with the wall for a time interval $\Delta t = 0.01$ s. Find the average force applied by the ball on the wall if $m = 0.5$ kg and $u = 10\,\text{ms}^{-1}$, and $\theta = 60°$.



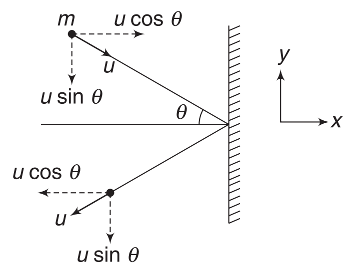
Solution

Concepts

(i) The ball experiences a change in momentum. This happens because it experienced a force (from the wall). Average value of this force is the average rate of change of momentum of the ball $\left(\frac{\Delta P}{\Delta t}\right)$

(ii) The force experienced by the ball will be in the direction of change in its momentum.

(iii) From Newton's third law, the force applied by the ball or the wall is same as the force applied by the wall on the ball.



Momentum of the ball before collision is

$$\vec{P}_1 = mu\cos\theta\,\hat{i} - mu\sin\theta\,\hat{j}$$

Momentum after collision is

$$\vec{P}_2 = -mu\cos\theta\,\hat{i} - mu\sin\theta\,\hat{j}$$

∴ Change in momentum of the ball

$$\Delta\vec{P} = \vec{P}_2 - \vec{P}_1 = -(2mu\cos\theta)\hat{i}$$

$$= 2mu\cos\theta\ (\leftarrow)$$

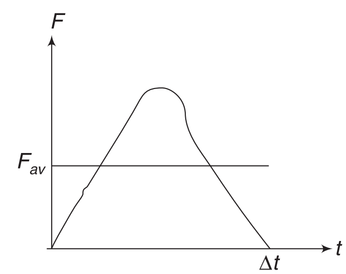
This change in momentum of the ball happened during its interaction with the wall for an interval Δt. Hence, the average force experienced by the ball is

$$F = \frac{\Delta P}{\Delta t} = \frac{2mu\cos\theta}{\Delta t}(\leftarrow) = \frac{2 \times 0.5 \times 10 \times \frac{1}{2}}{0.01}$$

$$= 500\,\text{N}\ (\leftarrow)$$

The wall experiences 500 N force towards right.

Note: Actually, the force between the ball and the wall will change in some complex way during the interaction period of Δt (see figure). What we have calculated is the average force during interval Δt. The instantaneous value of the force must have a peak value larger than 500 N. *Colliding bodies usually exert large force on one another.*



Example 3 *Bullets hitting a wall*

A gun is firing bullets on a vertical mud wall. Bullets get embedded into the wall. Each bullet has mass m and speed u. Bullets hit the wall while flying horizontally. Find force experienced by the wall if the gun fires n bullets per second.

Solution

Concepts

(i) The rate at which momentum of impinging bullets change is equal to the force applied by the wall on the group of bullets hitting it. Same amount of force is applied by the bullets on the wall.

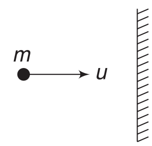
(ii) If ΔP = change in momentum of all bullets that hit the wall in a time interval Δt, then force is

$$F = \frac{\Delta P}{\Delta t}$$

Change in momentum of one bullet when it hits the wall is

$$\Delta P_{\text{one}} = P_f - P_i = 0 - mu(\rightarrow) = mu(\leftarrow)$$

Note that the bullet comes to rest inside the wall and hence its final momentum is zero.



Also, the bullet experiences a force towards left and change in its momentum is towards left.

Number of bullets hitting the wall in an interval Δt is $n\Delta t$.

$\therefore$ Change of momentum of all bullets hitting the wall in time Δt is

$$\Delta P = (n\Delta t)(\Delta P_{\text{one}}) = (n\Delta t)(mu)(\leftarrow)$$

$$\therefore \quad \text{Force} = \text{rate of change of momentum} = \frac{\Delta P}{\Delta t}$$

$$= nmu\ (\leftarrow)$$

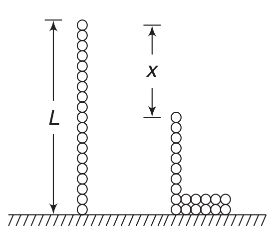
This is the force experienced by the bullets (as a group) due to the wall. The wall experiences same force towards right.

$$\therefore \quad F = nmu\ (\rightarrow)$$

Note: Tracking the direction (using $\rightarrow$ or $\leftarrow$ sign) was not essential in this problem, as it is obvious that the wall will experience a force towards right. Just focusing on magnitude of ΔP would have sufficed.

Example 4 *A falling chain*

A uniform chain of mass M and length L is held straight above a horizontal table with its lower end just touching the table. The chain is released from this position. Find the force exerted by the chain on the table when its length x has piled up on the table. Assume that each link of the chain comes to rest on hitting the table and neglect the height of pile formed on the table.



What is the maximum force that the falling chain will apply on the table at any instant?

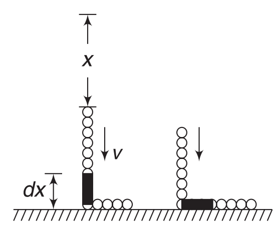
Solution

Concepts

(i) In principle, this problem is not much different from the last one. When one small piece of the falling chain hits the table, it comes to rest, just like a bullet hitting the wall and coming to rest. Momentum of a falling piece changes due to the force it experiences from the table.

In the last problem, we could count the number of particles (bullets) hitting the wall. In this problem, infinite number of tiny particles of the chain hit the table one after another. Also, each successive particle hits the table with a slightly greater speed, as the speed of falling chain is increasing.

(ii) The part of the chain that is vertical (in air) is falling down with acceleration g. There will be no tension force between two successive links. (If you place two books one over another and release them, both fall with acceleration g and there is no normal force between them.)



Just before the element lands on the table, it is moving with velocity *v*.

Just after the element lands on the table, its speed is zero.

Consider the time when x length of the chain has already landed on the table. The chain that is in air has fallen through a distance x. Its velocity is

$$v = \sqrt{2gx}$$

A very small element of length dx at the bottom of the chain will land on the table in next small interval of time dt. Momentum of this element just before it hits the table is

$$dP = \left(\frac{M}{L}dx\right)v \qquad \left[\frac{M}{L}dx \text{ is mass of the element}\right]$$

In next 'dt' time, its momentum becomes zero due to force by the table. Hence, change in momentum has magnitude

$$dP = \left(\frac{M}{L}dx\right)v$$

$\therefore$ Force $\quad F = \dfrac{dP}{dt} = \dfrac{M}{L}v.\dfrac{dx}{dt} = \dfrac{M}{L}v^2$

$$= \frac{M}{L}(2gx) = 2Mg\frac{x}{L}$$

This is thrust force applied by chain links hitting the table. Obviously, the part of chain (length x) already lying on the table is exerting a force equal to its weight on the table. This force is $\dfrac{M}{L}x.g = Mg\left(\dfrac{x}{L}\right)$.

Thus, total force due to chain (hitting + weight) on the table is

$$2Mg\,\frac{x}{L} + Mg\,\frac{x}{L} = 3Mg\,\frac{x}{L}$$

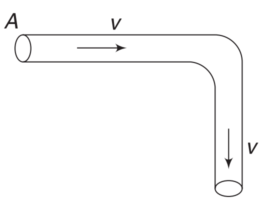
This force is maximum when x is maximum (= L), i.e., when the last links hits the table.

Maximum force is

$$F_{\max} = 3Mg\,\frac{L}{L} = 3Mg$$

Example 5 *Force on a bend in a water pipe*

Water (density ρ) is flowing at speed v through a pipe of cross-sectional area A. The pipe has a right angled bend as shown. Find the necessary force to keep the bend fixed.

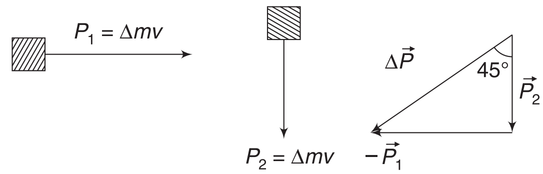


Solution

Concepts

(i) Direction of momentum of liquid particles change at the bend. This means that momentum of liquid particles change as they cross the bend. This change in momentum happens due to force applied by the pipe on the liquid. Liquid also applies equal force in opposite direction on the pipe. To keep the pipe in place, we need to apply an external force to balance this force.

(ii) Volume of liquid flowing through a cross-section in unit time is Av.



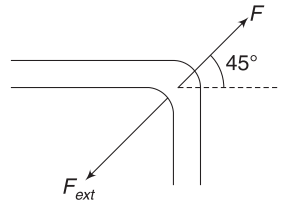
Volume flow in unit time = Av. Mass of water that flows through the pipe in unit time is ρAv. In a time interval Δt, the mass of liquid that flows is

$$\Delta m = \rho Av\,\Delta t.$$

Its momentum changes from $P_1 = \Delta mv(\rightarrow)$ to

$$P_2 = \Delta mv\ (\downarrow)$$

$\therefore$ Change in momentum has magnitude



F is force by water on bend.
F_{ext} is force by external agent.

$\Delta P = \sqrt{2}\ mv$ and is in a direction making 45° with either sides of the bend.

$\therefore$ Force by pipe on water is

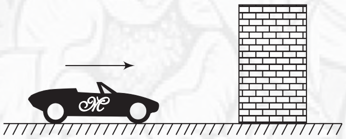
$$F = \frac{\Delta P}{\Delta t} = \frac{\sqrt{2}\ \Delta mv}{\Delta t} = \frac{\sqrt{2}\ \rho Av\Delta t\cdot v}{\Delta t}$$

$$\Rightarrow \quad F = \sqrt{2}\ \rho Av^2$$

This forces is in direction of $\overrightarrow{\Delta P}$ shown in figure. Water exerts equal force on pipe in opposite direction (shown as F in above figure). To keep the pipe in place, the external agent applies $F_{\text{ext}} = \sqrt{2}\ \rho Av^2$ in direction opposite to F.

Your Turn

Q.4 Car manufacturers undertake crash tests of cars to gauge the extent of damage caused when an accident occurs. In one such test, a car of mass M = 1500 kg travelling at 54 kmph is made to hit a temporary wall. The video shoot of the event suggested that the car interacted with the wall for a time interval of Δt = 0.15 s and maintained its straight line trajectory. Just after the collision it was travelling at 18 kmh^{-1} in same direction.

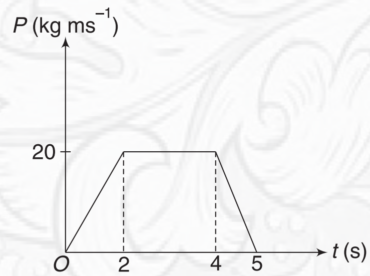


(i) Find the average force experienced by the car during collision?

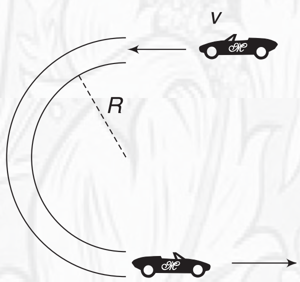
(ii) Find loss in KE of the car.

Q.5 A ball of mass m = 50 g falls from a height of 5 m and rebounds to a heights of 1.25 m. The ball was in contact with the floor for an estimated time of Δt = (0.11 ± 0.01) s. Estimate the average force applied by the floor on the ball.

Q.6 Momentum of a particle, travelling on a straight line, changes with time, as shown in the figure. Find the maximum force acting on the particle at any instant.

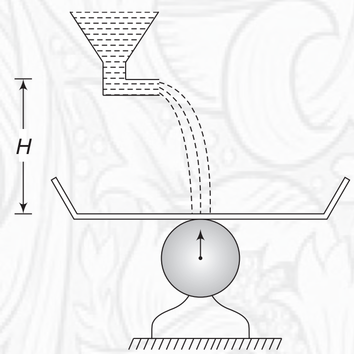


Q.7 A car of mass M enters a semi-circular track at speed v and exits out of it at speed v. The track has radius R. Find the average force experienced by the car on the track.



Q.8 Wind is blowing horizontally at a speed v. Density of air is ρ. Calculate force experienced by a vertical wall of area A. Assume that the air spreads parallel to the surface of the wall after striking it. How will the force change if speed increases to $2v$?

Q.9 A wide container of negligible mass is placed on a weighing scale. A tank at height H pours water on it. Assume that water comes out of the tank at negligible speed and at a rate of $\mu\,\text{kg s}^{-1}$. The falling water does not splash and spreads uniformly on the container. Find the reading of the weighing scale at a time t after the water starts pouring into the container.



Q.10 An army person fires 50 g bullets horizontally from a machine gun. Bullets leave the gun at a speed of $1\,\text{km s}^{-1}$ and he fires 20 bullets in 4 s. Find the average horizontal force that he exters on the gun to hold it.

4. IMPULSE

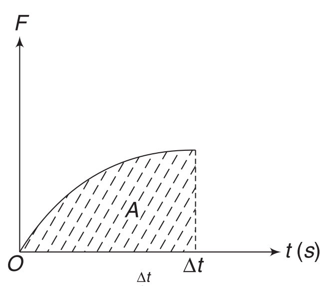
We known that a force produces acceleration in a body (of fixed mass) and thereby, changes its momentum. The greater the force acting on a body, the greater its change in velocity, and hence the greater its change in momentum. But one more thing is important in changing momentum–time. By applying a force over an extended period of time, one can produce a large change in momentum of a body. Both force and time interval for which the force is applied are important in changing momentum.

The quantity–force × time is called impulse.

Impulse $(\vec{J}) = \vec{F}\cdot\Delta t$ ...(5)

Unit of impulse is Ns. It is a vector in the direction of force $(\vec{F})$.

When force is changing with time, the impulse of the force can be written as



Area (A)= $\int_0^{\Delta t} F\,dt$

$$\vec{J} = \int_0^{\Delta t} \vec{F}.dt \quad ...(6)$$

When a graph of F versus t is given, the area enclosed between the graph and time axis gives the impulse.

From Newton's second law,

$$\frac{d\vec{P}}{dt} = \vec{F}$$

for a particle or a system of particles.

$$\Rightarrow \quad d\vec{P} = \vec{F}\,dt \quad ...(7)$$

This equation says that a force $\vec{F}$ acting on a body for an infinitesimally small time interval dt produces a change in momentum of the body, given by equation (7).

When the force acts for a finite time, change in momentum can be obtained by integrating the above equation on both sides.

$$\int_{\vec{P}_i}^{\vec{P}_f} d\vec{P} = \int_0^{\Delta t} \vec{F}\,dt$$

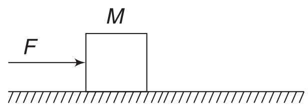
$$\Rightarrow \quad \vec{P}_f - \vec{P}_i = \int_0^{\Delta t} \vec{F}\,dt$$

$$\Rightarrow \quad \overrightarrow{\Delta P} = \vec{J} \quad ...(8)$$

This equation says that change in momentum of a body is equal to impulse of external force applied on it. Impulse and momentum both are vectors having same unit – kg ms^{-1} or Ns.

We will refer to equation (8) as *Impulse–momentum theorem.*

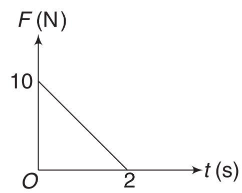
Example 6 A block of mass $M = 4$ kg is kept on a smooth horizontal surface. A horizontal force F starts acting on it. Find velocity of the block after 2 s if



(i) the force F remains constant at 10 N.

(ii) the force F varies with time as $F = (2t)$ N.

(iii) the force F varies with time, as shown in the graph.



Solution

Concepts

(i) Change in momentum = Impulse

(ii) Impulse = $F\Delta t$ for constant force

$$= \int_0^{\Delta t} F\,dt \text{ for variable force}$$

(iii) The problem can also be solved by writing acceleration $a = \dfrac{F}{M}$ and then using $v = u + at$ for constant acceleration and $\int_u^v dv = \int_0^{\Delta t} a\,dt$ for variable acceleration.

(i) Impulse $= F\Delta t = 10 \times 2 = 20$ Ns

$\therefore \quad \Delta P = 20$

$P_f - P_i = 20 \quad \Rightarrow \quad Mv - 0 = 20$

$\Rightarrow \quad v = \dfrac{20}{4} = 5 \text{ ms}^{-1}$

(ii) Impulse $= \int_0^2 F\,dt = 2\int_0^2 t\,dt = 4$ Ns

$\therefore \quad P_f - P_i = 4 \quad \Rightarrow \quad Mv - 0 = 4$

$\Rightarrow \quad v = 1 \text{ ms}^{-1}$

(iii) Impulse = Area under F–t graph

$= \dfrac{1}{2} \times 2 \times 10 = 10$ Ns

$\therefore \quad P_f - P_i = 10 \quad \Rightarrow \quad Mv - 0 = 10$

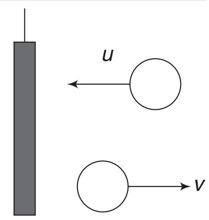
$\Rightarrow \quad v = \dfrac{10}{4} = 2.5 \text{ ms}^{-1}$

Example 7 A cricket ball of mass 150 g hits a bat normally while travelling at a speed 144 kmh^{-1}. It bounces back at a speed of 108 kmh^{-1} along the original line of motion. Find the impulse received by the ball.

Solution

Concepts

Though we do not known the exact nature of the force between the ball and the bat and also we do not know the duration for which they interacted, we can still find the impulse by knowing the change in momentum of ball. Actually, the force must have varied in a complex manner during the interval of interaction and the interaction time will be very small. But without getting into these details we can find the impulse.



For ball:

Initial momentum is

$P_i = mu = 0.15 \times 40 (\leftarrow)$

$= 6 \text{ kgms}^{-1} (\leftarrow)$

$[\because \quad 144 \text{ kmh}^{-1} = 40 \text{ ms}^{-1}]$

After the hit, final momentum of the ball is

$P_f = mv = 0.15 \times 30 (\rightarrow) = 4.5 \text{ kgms}^{-1} (\rightarrow)$

$[108 \text{ kmh}^{-1} = 30 \text{ ms}^{-1}]$

Change in momentum of the ball is

$\Delta P = P_f - P_i = 4.5(\rightarrow) - 6(\leftarrow) = 10.5 \text{ kgms}^{-1} (\rightarrow)$

$\therefore$ Impulse applied by the bat on the ball is

$J = \Delta P = 10.5 \text{ Ns } (\rightarrow)$

Note that the ball applies same impulse in opposite direction on the bat.

4.1 Impulsive Force

When you kick a football, your feet interacts for a very small time (Δt) with the ball. But the force experienced by the ball is relatively large (compared to other usual forces on the ball – like its weight). The impulse received by the ball ($F\Delta t$) is significant and the momentum of the ball changes almost abruptly (in very small time Δt) by a magnitude $\Delta P = F\Delta t$.

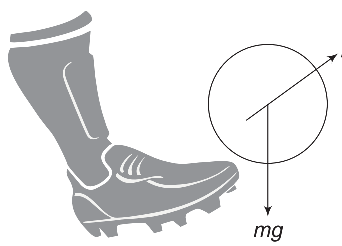
The ball certainly gets displaced during the intervals (Δt) when force acted. But this displacement is too small (to catch our attention) compared to the displacement that the ball is going to have after the kick. For us, the process of kick was nearly instantaneous. The ball suddenly acquired a momentum, in no time, while being at the same location (almost).

Such a force of relatively large magnitude acting on a body for a very short interval of time is called an impulsive force.

The duration for which the foot applies force on the ball being small, we can neglect the impulse of a finite force like weight of the ball. For the interval Δt, the applied force F was way too large compared to weight (mg). Change in momentum of the ball during interval Δt is

$$\overrightarrow{\Delta P} = \vec{F}\Delta t + m\vec{g}\ \Delta t$$
$$\simeq \vec{F}\cdot\Delta t$$

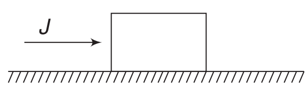
A kick, a jerk, a punch or a collision involve forces of large magnitude. They are impulsive forces. They cause large change in momentum during a very small interval of time without causing appreciable displacement of the body.



Impulse of mg is negligible compared to impulse of F.

Example 8 *Neglect impulse due to friction*

A block of mass M = 2 kg is placed on a horizontal table. Coefficient of friction is μ = 0.2. A sharp horizontal blow is applied to the block. Impulse imparted is J = 10 Ns. Find the velocity acquired by the block due to the blow.



Solution

Concepts

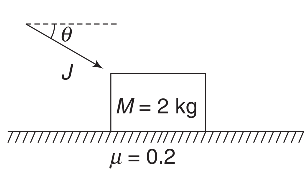
Friction is a finite force and we will neglect its impulse during the small interval for which the blow lasted. Impulse due to the applied force is J = 10 Ns. Impulse due to friction (in the interval for which the force was applied) must be << 10 Ns.

$$\Delta P = J$$

$$\Rightarrow \quad Mv - 0 = 10 \quad \Rightarrow \quad v = \frac{10}{2} = 5\,\text{ms}^{-1}$$

Example 9 *Significant impulse due to friction*

In the last example, consider the sharp impulse (J = 10 Ns) to be applied at θ = 37° to the horizontal. Find the velocity of the block immediately after the impact.

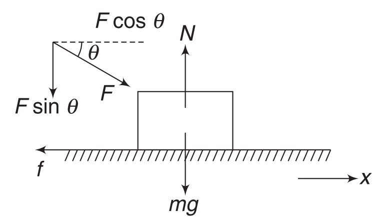


Solution

Concepts

(i) Applied force is large. Normal reaction, which depends on the applied force is also large. Therefore, friction is large and has significant impulse. This time, we cannot neglect the impulse of friction during the small time interval (Δt) for which the force was applied.

(ii) You must carefully note why the impulse of friction was insignificant in last example and why it has become significant in the present problem.



If the applied force is F, the normal reaction will be

$$N = F\sin\theta + mg$$
$$\simeq F\sin\theta = \frac{3}{5}F$$

We have neglected mg as it must be a small force compared to the large impulsive force F.

$$\therefore \quad \text{Friction} \quad F = \mu N = \frac{3}{5}\mu F$$

Net impulse in x-direction is

$$J_x = (F\cos\theta)\Delta t - f\Delta t$$
$$= \frac{4}{5}(F\Delta t) - \frac{3}{5}\mu(F\Delta t)$$

The question directly gives us the value of impulse

$$F\Delta t = J = 10 \text{ Ns}$$

$$\therefore \quad J_x = \frac{4}{5}(10) - \frac{3}{5}\times 0.2(10) = 8 - 1.2$$
$$= 6.8 \text{ Ns}$$

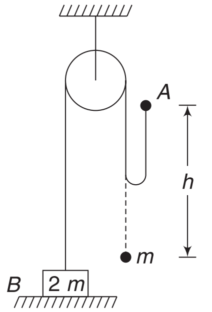
$$\Delta P_x = 6.8$$

$$\Rightarrow \quad Mv - 0 = 6.8 \quad \Rightarrow \quad v = \frac{6.8}{2} = 3.2\text{ ms}^{-1}$$

Example 10 *Impulsive tension*

In an Atwood machine, a block (B) of mass $2m$ rests on a table. It is connected to a ball of mass m using a light string passing over a smooth pulley. The ball of mass m is raised to a height h from its initial position and released.

(i) Find the speed of block B immediately after the string becomes taut.



(ii) Write the kinetic energy of the two-block system just before and just after the string gets taut. What happened to the kinetic energy? Why?

Solution

Concepts

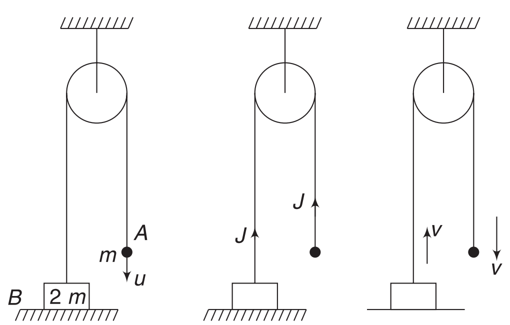
(i) The string is loose and ball *A* experiences free fall till its displacement becomes *h*.

(ii) During the period (a small time) in which the string gains tension, it applies an impulse on both the blocks. This impulse sets *B* in motion and decreases the downward momentum of *A*. Just after the string is taut, the two objects will have same speed – *B* going up and *A* going down.

(iii) We will write the impulse-momentum equation for both the bodies for interval during which the string gains tension.

(i) Speed of the ball after falling through *h* is

$$u = \sqrt{2gh} \text{ (first figure)}$$



Just before the string is taut

String is gaining tension

Just after the string is taut.

In the next interval Δt, the string will acquire full tension. After this, both objects will have same speed *v* (last figure).

During the interval Δt, the tension might have changed in a complicated way. Forget about the way the tension changes. Let us assume that its impulse $\int_0^{\Delta t} T\,dt$ is equal to *J* during this interval.

Impulse on both the bodies is upward and has same magnitude *J* as tension on both is same at any point of time.

It is important to note that impulse of weight on the two objects can be neglected for the small interval of time that we are considering

For B: $J(\uparrow) = P_f - P_i$

$\Rightarrow \quad J(\uparrow) = 2m\, v(\uparrow) - 0$

$\Rightarrow \quad J = 2mv$ ...(i)

For A: $J(\uparrow) = P_f - P_i$

$\Rightarrow \quad J(\uparrow) = mv(\downarrow) - mu(\downarrow)$

$\Rightarrow \quad J = mu - mv$ ...(ii)

Eliminating *J* between (i) and (ii)

$$3mv = mu \Rightarrow v = \frac{u}{3} = \sqrt{\frac{2gh}{3}}$$

We can now write the impulse applied by the string on the two object using (i)

(ii) *KE* just before the string has tension is

$$K_i = \frac{1}{2} mu^2$$

KE just after the string has tension is

$$K_f = \frac{1}{2}(2m)v^2 + \frac{1}{2}(m)v^2 = \frac{3}{2}mv^2$$

$$= \frac{1}{6}mu^2$$

Loss in *KE* is $\Delta K = K_i - K_f = \frac{1}{3}mu^2 = \frac{1}{3}m(2gh)$

$$= \frac{2}{3}mgh$$

Where goes the kinetic energy?

Actually, the string stretches a little when it is taut. Energy is spent in deforming the string – just like the energy spent to stretch a spring. No string can be perfectly inextensible. Though the stretch may be small enough to make any significant change in its length, it may require a lot of energy to stretch a string, even by a small amount.

In short:

(i) Momentum of a particle is $\vec{P} = m\vec{v}$ and momentum of a collection of particles is $\vec{P} = M\vec{v}_{CM}$.

(ii) Momentum of a system can change only when an external force acts on it.

(iii) Rate of change of momentum is given by $\frac{d\vec{P}}{dt} = \vec{F}_{ext}$.

This equation is same as $\vec{F}_{ext} = M\vec{a}$ (= $M\vec{a}_{CM}$ for a system is particles) if mass is constant.

(iv) If n particles strike a surface per second and each particle suffers a change in momentum equal to $\overrightarrow{\Delta P}$, then force experienced by the surface is $\vec{F} = n\overrightarrow{\Delta P}$.

(v) $\vec{J} = \int_0^{\Delta t} \vec{F}\,dt$ is known as impulse of a force. Impulse experienced by a body is equal to change in its momentum. This result is called Impulse – momentum theorem.

$$\Delta P = \vec{J}$$

(vi) A large force acting for a short interval of time is often known as an impulsive force. Usually, finite forces like weight have negligible impulse during a quick process like collision.

(vii) When a string gets taut, it consumes energy. In fact, deformation of all objects consume energy. When a car meets an accident, its body gets deformed. This consumes a lot of kinetic energy and, in a way, protects the passengers.

Your Turn

Q.11 Give reason for following:

(i) When boxing, if you move into a punch instead of away, you are in trouble.

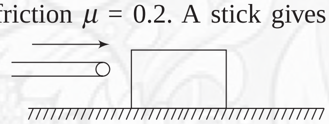
(ii) You find it convenient to catch a high-speed cricket ball while your hand moves away from the ball.

(iii) When your car is out of control, if you drive it into a concrete wall instead of a haystack, you are really in trouble.

Q.12 A block of mass $m = 2$ kg begins to slide on a smooth incline having inclination angle $\theta = 30°$. Find the impulse of all the forces acting on the block in a time of $t = 2$ s after the motion starts.

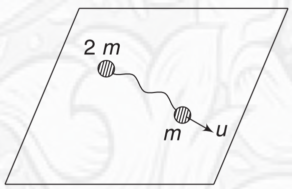
Q.13 How much impulse is needed to slow down a 1,000 kg car from a speed of $u = 20\,\text{ms}^{-1}$ to a final speed $v = 10\,\text{ms}^{-1}$?

Q.14 A block of mass $M = 10$ kg is kept on a horizontal surface having coefficient of friction $\mu = 0.2$. A stick gives a sharp horizontal blow to the block. The block begins to move and stops after travelling a distance $s = 1.0$ m. Find the impulse that the stick imparted to the block.



Q.15 Consider the situation described in Example 4. What is the impulse applied by the chain on the table during the interval the complete chain falls on the table?

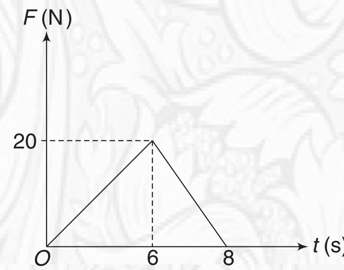
Q.16 Two balls of masses m and $2m$ are placed on a smooth surface connected with a loose massless string. The ball of mass m is imparted a sudden velocity u in a direction that is parallel to the line joining the initial positions of the balls. Find the impulse imparted by the string to set the ball of mass $2m$ into motion.



Q.17 A particle of mass 1kg is moving in free space. Its velocity is $\vec{u} = (2\hat{i} + 3\hat{j})\,\text{ms}^{-1}$ when a force $\vec{F} = 2t\,\hat{j}$ newton starts acting on it. Find the velocity of the particle at $t = 2$ s.

Q.18 A projectile of mass m is fixed from ground with a velocity $\vec{u}$. Find its momentum after time t (< time of flight).

Q.19 A particle of mass $m = 1$ kg is moving along a straight line with speed $u = 4\,\text{ms}^{-1}$. A force (F) starts acting on it in the direction of its motion. The force acts on the particle for 8 s and changes with time as shown in the figures. Find work done by the force in 8 s.



5. CONSERVATION OF LINEAR MOMENTUM

We studied, in the last chapter, that momentum of a system having n number of particles can be written as

$$\vec{P} = m_1\vec{v}_1 + m_2\vec{v}_2 + ... + m_n\vec{v}_n = M\vec{v}_{CM}$$

Where $M = m_1 + m_2 + ... + m_n$ = total mass of the system and $\vec{v}_{CM}$ = velocity of COM of the system.

We also know that velocity of COM remains constant (even when $\vec{v}_1$, $\vec{v}_2$ ... etc. are changing) when net external force on the system is zero.

This implies that *the total momentum of a system remains constant (i.e., conserved) when sum of all external forces acting on the system is zero. This is known as principle of conservation of momentum*.

It is important to note that momentum of individual particles (or parts of the system) may change due to internal interactions. But the total momentum of the system cannot change if there is no external force.

Consider two positively charged particles released from rest. Initial momentum of the system is zero. If the system is isolated, i.e., free of any external force, momentum will remain zero forever. If we look at individual particles, they acquire momentum due to force applied by the other particle. But they move such that $P_{\text{system}} = 0$

Two charged particles released from rest. Momentum of system is always zero if system is isolated.

$$\Rightarrow \quad m\vec{v}_1 + m\vec{v}_2 = 0$$

$$\Rightarrow \quad m\vec{v}_1 = -m\vec{v}_2$$

Here, negative sign indicates that $\vec{v}_1$ and $\vec{v}_2$ have opposite directions.

There may be a situation when there is no external force acting on a system in a particular direction (say x-direction) though there may be a force in other directions. One can always conserve momentum in x-direction, whenever such situation occurs. If means:

$$P_x = \text{a consant if } F_x = 0 \quad [F_y \text{ and } F_z \text{ may not be zero}] \qquad ...(9)$$

When a projectile moves through its path, horizontal components of its momentum remains constant, as there is no horizontal force on the projectile.

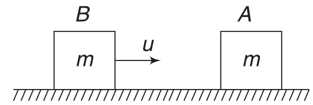
The conservation of momentum, as sated above, will be very difficult to apply in practice, as it is difficult to find systems which are completely free of any external force. Let us put the principle in a slightly different from.

$$\frac{d\vec{P}}{dt} = \vec{F} \quad \Rightarrow \quad \overrightarrow{\Delta P} = \int_0^{\Delta t} \vec{F}\, dt = \vec{J}$$

The above equation is impulse-momentum relation, which says that change in momentum of a system is equal to the impulse of external force. Consider a system, which is being acted upon by some finite-sized (non-impulsive) external force. If an event (like collision) takes place, which lasts for a very small interval of time (Δt), then the impulse ($\vec{J}$) of the force can be neglected and we can still say that $\Delta P \simeq 0$. This means momentum is (nearly) conserved.

If two projectiles collide in mid-air, we can say that the momentum of system of two objects just before collision will be equal to momentum of the system just after collision. Remember, there is an external force – weight – acting on the system. Still, we are saying that we will use momentum conservation! The impulse of weight in the short duration for which the projectiles interacts is negligible.

Example 11 Block A, having mass m, is placed on a smooth horizontal surface. Another identical block (B) is sliding towards A at a velocity u. They collide. Find their speeds if



(i) They stick (ii) B comes to rest

What happens to kinetic energy of the system?

Solution

Concepts

(i) External force on the system ($A + B$) is zero. The momentum of system is conserved. Both A and B apply large forces on one another and change each other's momentum. But there is no change in overall system's momentum.

(ii) Internal forces can perform work and change KE of the system.

(i) Let final velocity of two blocks after they stick be v.

$$P_f = P_i$$

$$\Rightarrow \quad 2m \cdot v = mu \quad \Rightarrow \quad v = \frac{u}{2}$$

Kinetic energy of system has initial and final values as:

$$K_i = \frac{1}{2}mu^2 \quad \text{and} \quad K_f = \frac{1}{2}(2m)v^2 = \frac{1}{4}mu^2$$

$$\therefore \quad KE \text{ decrease by } \frac{1}{4}mu^2.$$

(iii) Let final velocity of A be v.

Velocity of B after collision = 0

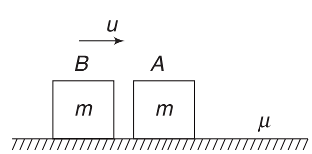
$$\therefore \quad P_f = P_i \quad \text{gives}$$

$$mv = mu$$

$$\Rightarrow \quad v = u$$

There is no change in kinetic energy.

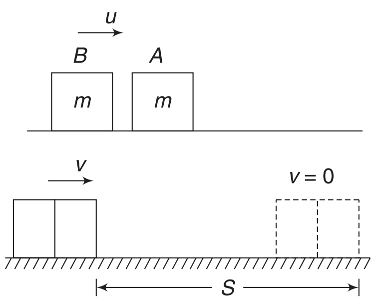
Example 12 Block A, of mass m, rests on a rough horizontal surface. Another identical block B moving along the surface hits block A and sticks to it. Speed of B, just before collision is u. The combined block travelled a distance s along the surface and stopped. Find s if the coefficient of friction between the blocks and the surface is μ.



Solution

Concepts

There is friction between the blocks and the surface. Shall we apply conservation of momentum? Yes, we can write $P_{bc} = P_{ac}$, where P_{bc} and P_{ac} are momentum of the system 'just before collision' and just after collision'. During the short time for which the blocks applied force on one-another, the impulse of friction ($f\Delta t$) is negligible.



Just before collision

$$P_{bc} = m_B u_B + m_A u_A$$

$$= mu + 0 = mu$$

Just after collision, let us assume that the blocks move with a common velocity v.

$$P_{ac} = 2mv$$

Momentum conservation: $P_{ac} = P_{bc}$

$$\Rightarrow \quad 2mv = mu \quad \Rightarrow \quad v = \frac{u}{2}$$

The friction retards the two-block system to rest as they travel through distance s. Using Work-Energy theorem:

$$W_f = K_f - K_i$$

$$-\mu(2mg)s = 0 - \frac{1}{2}(2m)\left(\frac{u}{2}\right)^2$$

$$\Rightarrow \quad s = \frac{u^2}{8\mu g}$$

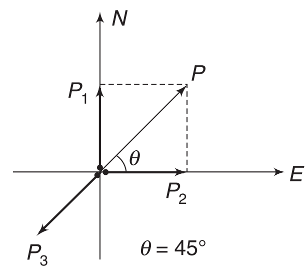
Example 13 *A bomb blast*

A bomb lies at rest. A remote is pressed and it blasts. However, the blast turned out to be a failure and the bomb could just split into three mass chunks. A piece of mass 100 g was found moving due north at a speed of 20 ms^{-1} and another piece of mass 200 g was moving towards east at a speed of 10 ms^{-1}. Find the velocity of the third piece of mass 50 g, immediately after the blast.

Solution

Concepts

Bomb is an isolated system – free of external force. It blasts due to internal reasons (some chemical reactions). The momentum of system will remain conserved equal to zero.



Momentum of first piece is

$$P_1 = (100\,\text{g})\left(20\,\frac{\text{m}}{\text{s}}\right) = 2\,\text{kg ms}^{-1} \text{ (North)}$$

Momentum of second piece is

$$P_2 = (200\,\text{g})\left(10\,\frac{\text{m}}{\text{s}}\right) = 2\,\text{kg ms}^{-1} \text{ (East)}$$

Sum of momentum of first two pieces is

$$P = \sqrt{P_1^2 + P_2^2} = 2\sqrt{2}\,\text{kg ms}^{-1} \text{ (NE)}$$

Sum of momentum of all three pieces must be zero. Hence, momentum of third piece (P_3) is equal to P in direction opposite to P.

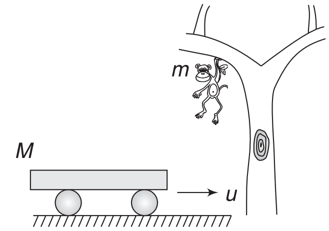
$$\therefore \quad P_3 = 2\sqrt{2} \text{ (SW)}$$

$$\left(\frac{50}{1000}\right)v_3 = 2\sqrt{2}$$

$$\Rightarrow \quad v_3 = 40\sqrt{2}\ \text{ms}^{-1}$$

Example 14 *Monkey lands on a car*

A car of mass M is moving uniformly on a smooth road at a velocity u. A monkey of mass m is hanging from a tree branch. On seeing the car, the monkey releases the branch and falls vertically. It lands on the car and quickly steadies itself on the car. Find the speed of the car after the event. How does height of the tree affect you answer?



Solution

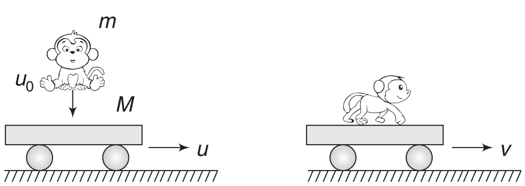
Concepts

(i) Just before the monkey lands on the car, it has a vertically downward velocity and the car has a horizontal velocity. There is no horizontal force on the (monkey + car) system and the horizontal momentum remains conserved.

(ii) Initially, the monkey has no horizontal motion but as it lands on the car, friction applied by the car on its body provides it a horizontal motion. Friction (by monkey) on car slows down the car.

But we need not think about all these details. These are internal force in our system. Conservation of momentum helps us avoid digging too much into the internal forces!

(iii) Where has the vertical momentum (possessed by the monkey) gone? The road applied a big vertical impulse on the car when the monkey lands on it. This impulse has caused the vertical momentum of the system to change and become zero.



Consider monkey + car as our system. Momentum is conserved in horizontal direction.

Momentum before landing = momentum after landing

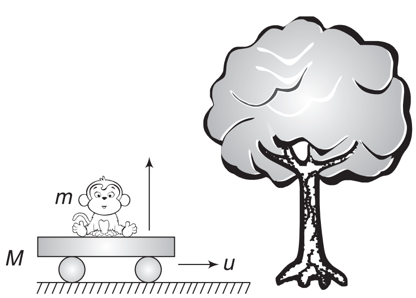
$$mu = (M + m)v$$

$$\Rightarrow \qquad v = \frac{mu}{M + m}$$

Height of the tree has no relevance in deciding v.

Example 15 *Monkey jumps out of a car*

A car of mass M is travelling on a road. A monkey of mass m is sitting on it. The monkey sees a mango tree and jumps vertically relative to the car. It catches a tree branch and clings to it. How does the jump by the monkey affect the speed of the car?

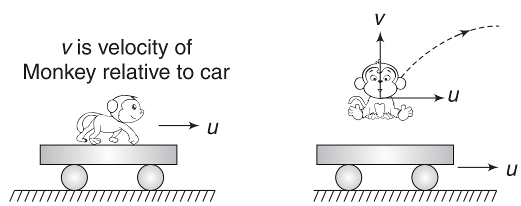


Solution

Concepts

(i) The monkey and the car, both are moving at velocity u. The interaction between the two cannot change the overall momentum of the system.

(ii) The monkey jumps *vertically relative to the car*. The monkey retains its horizontal motion during the jump (recall inertia). In addition to its horizontal motion, it acquired a vertical motion also.



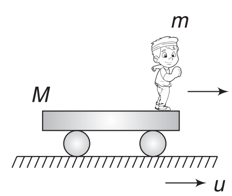
After jump, the monkey still has a horizontal velocity u.

For no change in horizontal momentum, the car must continue to move with velocity (u), which is same as velocity before jump.

An observer on the ground will find that the monkey has a horizontal as well as vertical velocity. For him, the path of the monkey will be parabolic.

Example 16 *Man jumps out of a moving car*

A flat car of mass M is moving on a smooth horizontal surface with velocity u. A man of mass m stands at the front edge of the car. The man jumps horizontally at a velocity u_r relative to the car in forward direction. Find velocity of the car after the jump.

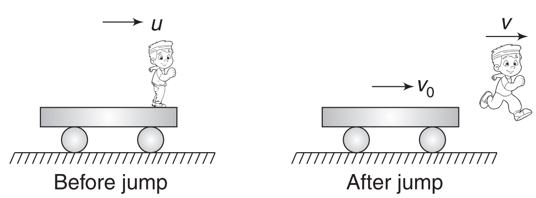


Solution

Concepts

(i) Momentum of man + car system is conserved in horizontal direction.

(ii) After the man jumps, velocity of the car changes. Now, the velocity of man as seen from car is u_r.



Let velocity of the man and the car be v and v_0 respectively after the jump. Given:

$$u_r = v - v_0$$

$$\Rightarrow \qquad v = (u_r + v_0)$$

Momentum conservation in horizontal direction gives

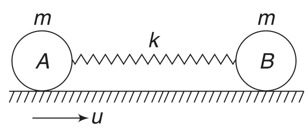
$$P_{\text{after jump}} = P_{\text{before jump}}$$

$$\Rightarrow \quad mv + mv_0 = (M + m)u$$

$$\Rightarrow \quad m(u_r + v_0) + Mv_0 = (M + m)u$$

$$\Rightarrow \quad v_0 = \frac{(M + m)u - mu_r}{(M + m)} = u - \frac{mu_r}{M + m}$$

Example 17 Two identical balls, A and B, have mass m each. They are placed on a smooth horizontal surface connected by a spring of force constant k. The spring is relaxed. Ball A is kicked toward B so as to impart it a velocity u.



(i) Find velocity of two balls when the spring is compressed the most.

(ii) Find potential energy stored in the spring when velocity of B becomes $\frac{u}{4}$ towards right.

Solution

Concepts

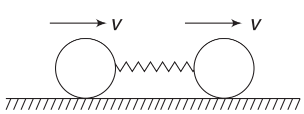
(i) Momentum of the entire system is conserved, though momentum of the individual balls change.

(ii) The spring will have maximum compression when both the balls have same velocity.

Initially, the spring compresses and the spring force accelerates B and retards A. As long as speed of A is greater than that of B, the spring compresses. When velocity of the two balls becomes equal, further compression of the spring stops.

(iii) Energy of the system is conserved, as there is no non-conservative force performing work.

(i) Maximum compression takes place when both A and B have same velocity (say v).



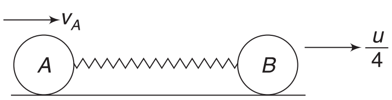
Maximum Compression.

Momentum conservation gives

$$mv + mv = mu \quad \Rightarrow \quad v = \frac{u}{2}$$

(ii) Let velocity of A be v_A when

$$v_B = \frac{u}{4}$$



Momentum conservation gives:

$$mv_A + m\frac{u}{4} = mu \quad \Rightarrow \quad v_A = \frac{3u}{4}$$

Kinetic energy of the system at this instant is

$$KE = \frac{1}{2}m\left(\frac{u}{4}\right)^2 + \frac{1}{2}m\left(\frac{3u}{4}\right)^2 = \frac{5}{16}mu^2$$

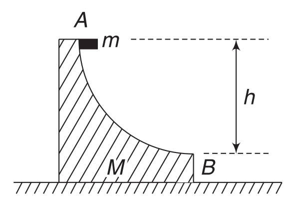
Conservation of mechanical energy gives:

$$U_i + K_i = U_f + K_f$$

$$0 + \frac{1}{2}mu^2 = U_f + \frac{5}{16}mu^2$$

$$\Rightarrow \quad U_f = \frac{3}{16}mu^2$$

Example 18 A wedge of mass M is placed on a smooth horizontal table. It has a curved smooth track from A to B. The track is horizontal at end B. A small block of mass m is released from the top of the track at A. It slides down from A to B and leaves the wedge while travelling horizontally with velocity v_1. Find v_1 if height of A above B is h.



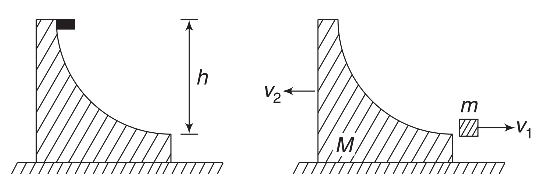
Solution

Concepts

(i) As the block slides, the wedge will begin to move towards left. [Think of the normal force between the two objects]

(ii) There is no external horizontal force on the system (block + wedge). Hence, horizontal momentum of the system will remain conserved.

(iii) Mechanical energy is conserved.



Initial momentum is zero | Final momentum is also zero.

Let the velocity of the block (when it leaves the wedge) be $v_1(\rightarrow)$ and the velocity of the wedge be $v_2(\leftarrow)$.

Initial momentum is zero. Momentum is conserved in horizontal direction.

$\therefore$ Final horizontal momentum is also zero.

$$\Rightarrow \quad mv_1 = Mv_2 \qquad ...(1)$$

Conservation of energy gives

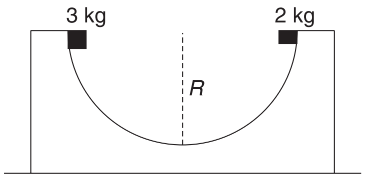
$$\frac{1}{2}mv_1^2 + \frac{1}{2}Mv_2^2 = mgh$$

$$\Rightarrow \quad mv_1^2 + M\left(\frac{mv_1}{M}\right)^2 = 2mgh$$

$$\therefore \quad v_1^2 = \frac{2mMgh}{mM + m^2} = \frac{2Mgh}{M + m}$$

$$\therefore \quad v_1 = \sqrt{\frac{2Mgh}{M + m}}$$

Example 19 A fixed wedge has a hemispherical trough in it. Radius of the hemisphere is $R = 2m$. Two small objects of masses $m_1 = 3\,\text{kg}$ and $m_2 = 2\,\text{kg}$ are released at the same moment at the rim of the hemispherical trough. The two objects are initially at diametrically opposite ends. The two objects collide and stick. To what height the combined mass will rise after collision. Neglect friction.



Solution

Concepts

(i) The two blocks slide and collide at the bottom of the trough with same speed.

(ii) Energy conservation can give us speed of either objects, just before they collide.

(iii) Momentum conservation can be used to find speed of the combined object just after collision.

(iv) After collision, energy conservation can once again be used to find the maximum height attained.

Remember, kinetic energy just before collision is not same as kinetic energy just after collision. Energy is lost in deformation of the blocks.

Speed of each block just before collision is given by energy conservation as:

$$\frac{1}{2}mu^2 = mgR$$

$$\Rightarrow \quad u = \sqrt{2gR} = \sqrt{2 \times 10 \times 2} = \sqrt{40}\ \text{ms}^{-1}$$

Let v = speed of the blocks immediately after they stick to each other. Momentum conservation gives

$$P_{\text{after collision}} = P_{\text{before collision}}$$

$$5v = 3u - 2u \quad \Rightarrow \quad v = \frac{u}{5} = \frac{\sqrt{40}}{5}\ \text{ms}^{-1}$$

Let the combined object rise to a height h after collision. Energy conservation gives

$$Mgh = \frac{1}{2}Mv^2 \quad [\text{Where,} \quad M = 3 + 2 = 5\,\text{kg}]$$

$$\Rightarrow \quad h = \frac{v^2}{2g} = \frac{40}{25 \times 2 \times 10} = 0.08\,\text{m}$$

In short:

(i) If external force on a system is zero, its momentum remains conserved.

(ii) If there is no external force on the system in a particular direction, its momentum will remain conserved along that direction.

$F_x = 0$ implies P_x = a constant (F_y, F_z may not be zero)

(iii) If an event lasts for a very short duration (like collision), the usual external forces (like weight etc), which are not impulsive, will not produce any significant change in momentum of the system.

(iv) Conservation of momentum and conservation of mechanical energy together help us in solving a wide variety of problems in mechanics.

YOUR TURN

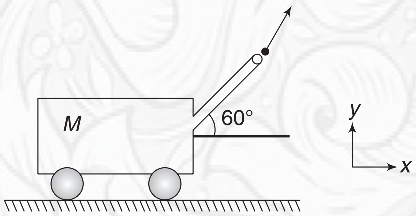
Q.20 An empty gun of mass M is loaded with a bullet of mass m. The gun is kept on a smooth surface and it fires. Velocity of the bullet relative to ground is u.

(i) Find the recoil speed of the gun.

(ii) The gun fires due to a small amount of the gun powder getting ignited. What can you say about the energy released due to ignition of the powder?

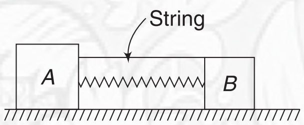
Q.21 An empty gun of mass M is loaded with a bullet of mass m. The gun is kept on a smooth surface and it fires. Velocity of the bullet relative to the gun is u. Find the recoil speed of the gun.

Q.22 An empty canon of mass M is kept on a smooth horizontal surface. It is loaded with a shell of mass m. It fires with its barrel inclined at 60° to the horizontal. Speed



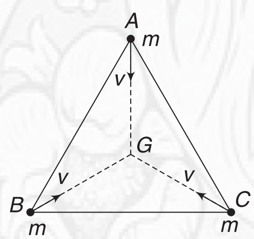
of the shell relative to the cannon is u. Find the recoil speed of the gun.

Q.23 Two blocks A and B have masses $2m$ and m respectively. A spring of force constant k is tied to block A and block B is moved towards A so as to compresss the spring by x. In this position, the blocks are held fixed by a string connected to both of them. Surface is smooth and B is not tied to the spring. The string snaps suddenly. Find the final velocities of the two blocks.

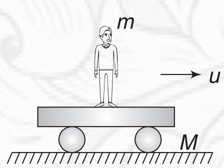


Q.24 A toy ball is made of three pieces of masses in the ratio 1 : 2 : 1. The ball at rest explodes on its own. Reason is not known. The two equal pieces move with velocities $(2\hat{i} + 5\hat{j} - 6\hat{k})$ and $(-4\hat{i} + 3\hat{j} + 2\hat{k})\,\mathrm{ms}^{-1}$. Find the speed of the third mass.

Q.25 Three particles of mass m each, are located at vertices (A, B and C) of an equilateral triangle. All of them are travelling uniformly with same speed v towards the centroid (G) of the triangle. The three particles collide at G. After collision, one of the particle was found to be at rest and the other was travelling along GB. In which direction was the third particle travelling after collision?



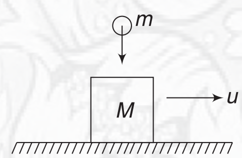
Q.26 A man of mass m is standing on a flat rail road car of mass M. The car is travelling on a smooth straight track with speed u. The man begins to run with a velocity v relative to the car in a direction opposite to the direction of motion of the car. Find the change in speed of the car.



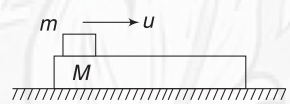
Q.27 In the situation given in the last problem, how should the man run to bring the car to rest?

Q.28 A man is standing on a railroad car travelling on a smooth track with velocity u. (see the figure in problem 26). The man has lots of balls with him. He begins to throw the balls horizontally, in a direction perpendicular to the motion of the car, relative to himself. He also throws some balls in vertical direction relative to himself. How will the velocity of car change if it is constrained to move along a straight line.

Q.29 A block of mass M is travelling horizontally at a speed u. A liquid drop of mass m falls on it and sticks. Find the change in velocity of the block if the liquid drop was travelling vertically.



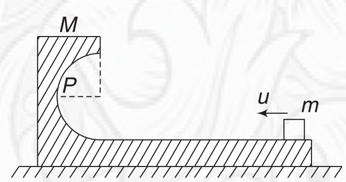
Q.30 A long plank of mass M is kept on a smooth horizontal surface. A small block of mass m is kept over it and imparted a velocity u. Due to friction between the block and the plank, the sliding stops after some time. Calculate:



(i) The final speed of the block.

(ii) Work done by the friction on the system.

Q.31 A block of mass M has a long straight part and a vertical part having a semi-circular cut in it. A small block of mass m is placed on the straight part and given a horizontal velocity u. Find the speed of bigger block when the smaller block reaches point P (Where radius is horizontal) on the curved part. Neglect friction.



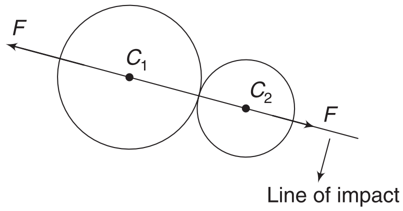
6. COLLISION

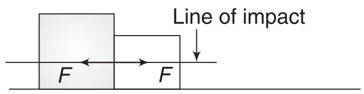
Collision is a short-term interaction between two bodies during which they exert relatively large force on one another. This sudden impulse causes a big change in momentum of the individual bodies. However, the total momentum of two bodies remains conserved if the system is isolated from external impulses. Momentum conservation plays a key role in analysing collision problems.

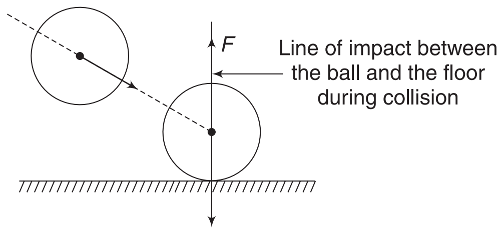
6.1 Line of Impact

The line of common normal to the surfaces in contact during the collision is called the line of impact. The two bodies apply impulsive force on one another along this line. Figures given below show line of impacts in few cases. In case of collision of two balls, the line of impact (LOI) is a line joining their centres.

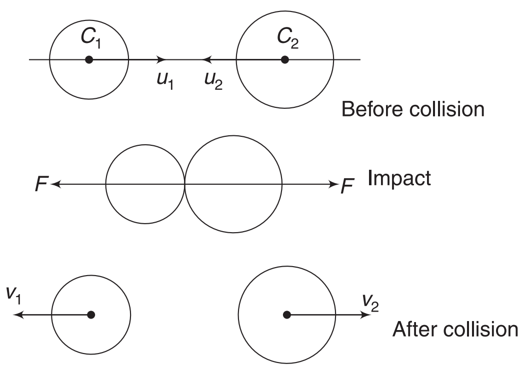
The two objects experience change in momentum along the line of impact. Change in momentum of two bodies will



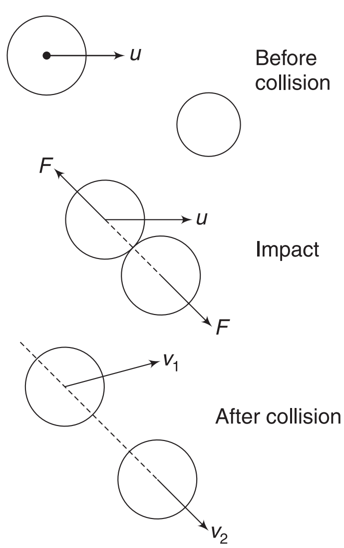




be equal and opposite (due to equal and opposite impulse applied on one another), but there is no change in momentum of the system.



One-D Collision (Fig. (a))



Oblique collision (Fig. (b))

When the colliding objects move along the line of impact before and after collision, the collision is said to be *One-Dimensional or, Head on or, Direct Collision*.

When velocities of the colliding objects, before and after collision, are not all along the same line, collision is said to be *oblique collision or two-dimensional collision*.

Consider two balls moving with velocities u_1 and u_2 along the line joining their centres (i.e., LOI). During collision, they exert force on one another along this line only. Obviously, they will continue to move along the same line after impact. This is a direct collision (see fig. (a)).

Now consider a billiard ball hitting another ball at rest, as shown in Fig (b). Initial velocity u is not along the line of impact. The two balls move in different directions after collision. In this example, u, v_1 and v_2 are not along same line. This is an example of oblique collision.

6.2 Elastic and Inelastic Collision

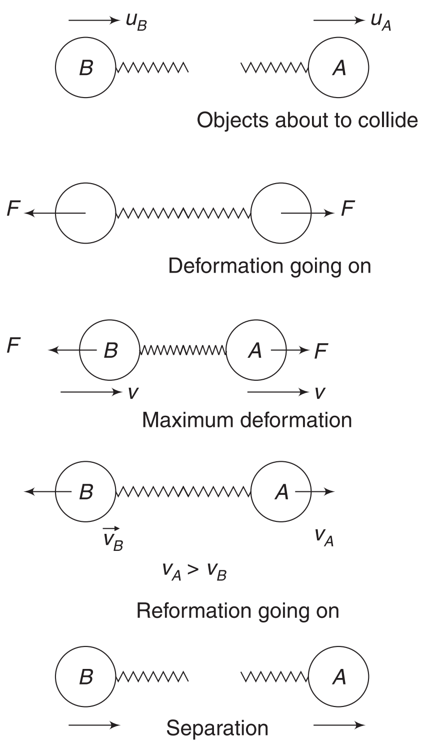
External forces can deform a body. When deforming forces are removed, bodies have a tendency to regain their shape. A body is said to be *perfectly elastic* if it can regain its original shape completely. A body is said to be *perfectly inelastic* if it has no tendency to go back to its original shape. However, real-life objects lie between these two extremes and could regain their shape to some extent. Such objects are called *inelastic*. Steel is very close to being a perfectly elastic material and a clay ball is very close to a perfectly inelastic material.

A collision between perfectly elastic bodies is said to be elastic. A collision between the two perfectly inelastic bodies is said to be perfectly inelastic. All other collisions are inelastic.

In an elastic collision between two bodies, the kinetic energy remains conserved. In inelastic collision, there is loss in kinetic energy and in perfectly inelastic collision, there is maximum loss in kinetic energy. Let us make a simple *model* and understand what happens during a collision.

Consider a ball A, travelling at speed u_A being hit head on by another ball B travelling at speed u_B. Assume two imaginary springs tied to the balls as shown. We are assuming the springs to model the interaction between the balls. The interaction between the balls can be understood in terms of spring force. Deformation in the spring is actually representing deformation in the balls. Potential energy stored in the spring is actually energy stored in deformed balls.

Initially, $u_B > u_A$ (in figure shown). This means that the two balls are getting closer. Springs (actually the balls) get compressed (deformed). They push the balls as shown. Ball A accelerates and B retards. At some point in time, velocities of the two balls become equal. This is when deformation stops. Now the springs (balls) try to regain their shape. Once again they push each other. Ball A further speeds up and B slows down. They get separated.



Model explaining what happens during a deformation and reformation phase of a collision

Now imagine that the springs (balls), completely regain their natural shape as they separate. The energy that was stored as deformation energy for a moment, gets back as kinetic energy. [Everything happens quite fast.] If so is the case, collision is elastic.

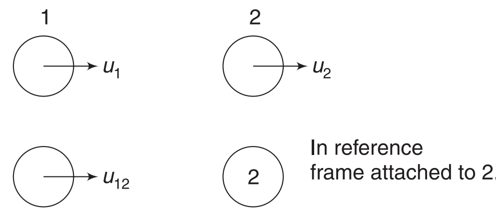
If the springs (the balls), after experiencing maximum deformation (when both balls have same velocity), have no tendency to go back to their original shape, the two balls will keep moving with same velocity after deforming each other. In this case, a large chunk of original kinetic energy is lost as deformation potential energy. Such collision is perfectly inelastic.

If the springs (the balls), after experiencing maximum deformation try to regain their shape but fail to go back to original state, then the two balls will move separately after collision but some kinetic energy will be lost in deformation. Such collision is inelastic.

When two steel balls collide, they deform each other and quickly get back to their original shape. Collision is elastic. When two aluminium balls collide they retain some deformation. Collision is inelastic. Collision of two mud balls is almost perfectly inelastic. They deform each other but have no tendency to regain their shape.

6.3 Coefficient of Restitution

Consider two balls having velocities u_1 and u_2 just before collision. To an observer attached to ball 2, the ball 1 appears to approach at a velocity



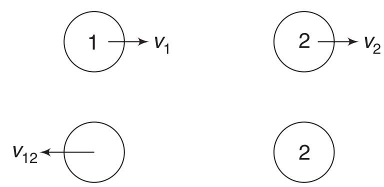
Before collision.

$$u_{12} = u_1 - u_2$$

After collision, the velocities of the two balls are v_1 and v_2. Now, an observer in 2 sees ball 1 receding away. The relative velocity is

$$v_{12} = v_1 - v_2$$

It is obvious that u_{12} and v_{12} are oppositely directed.



In *RF* attached to 2.

After collision 1 is moving away from 2.

The ratio of magnitude of v_{12} and u_{12} is known as coefficient of restitution (e).

$$e = \frac{|v_{12}|}{|u_{12}|} = \frac{|v_1 - v_2|}{|u_1 - u_2|} = -\left(\frac{v_1 - v_2}{u_1 - u_2}\right) \qquad ...(10)$$

In perfectly inelastic collision, the two balls move with a common velocity after collision. It means, $v_1 = v_2$.

Hence, $e = 0$ for perfectly inelastic collision.

We known that kinetic energy for the system of balls can be written as

$$K = \frac{1}{2}\mu v_{\text{rel}}^2 + \frac{1}{2}Mv_{\text{CM}}^2$$

Due to collision, v_{CM} cannot change as there is no external force on the system. Kinetic energy of the system will remain same before and after collision if

$$\left|(u_{\text{rel}})_{\text{before collision}}\right| = \left|(v_{\text{rel}})_{\text{after collision}}\right|$$

$$\Rightarrow \qquad |u_1 - u_2| = |v_1 - v_2|$$

This implies that *coefficient of restitution is $e = 1$ for an elastic collision.*

All other collisions, which are called inelastic, result in loss in kinetic energy. It means

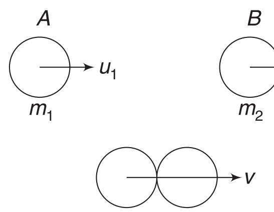
$$\left|(u_{\text{rel}})_{\text{before collision}}\right| > \left|(v_{\text{rel}})_{\text{after collision}}\right|$$

$$\therefore \qquad |u_1 - u_2| > |v_1 - v_2|$$

This implies that the *coefficient of restitution is < 1 for inelastic collisions.*

6.4 Head on, Perfectly Inelastic Collision

Consider two balls, A and B, made of perfectly inelastic materials and hitting each other head on. u_1 and u_2 are velocities before collision.



Velocities of the two balls are same ($v_1 = v_2 = v$) after collision. (It means $e = 0$).

Now, we need only conservation of momentum to find v.

$$(m_1 + m_2)v = m_1 u_1 + m_2 u_2$$

$$\Rightarrow \qquad v = \frac{m_1 u_1 + m_2 u_2}{m_1 + m_2}$$

Loss in KE is large. We can find the loss as:

Loss in $KE = K_i - K_f$

$$= \frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 - \frac{1}{2}(m_1 + m_2)\left(\frac{m_1 u_1 + m_2 u_2}{m_1 + m_2}\right)^2$$

$$= \frac{1}{2}\left[\frac{(m_1 + m_2)(m_1 u_1^2 + m_2 u_2^2) - (m_1 u_1 + m_2 u_2)^2}{m_1 + m_2}\right]$$

$$= \frac{1}{2}\left(\frac{m_1 m_2}{m_1 + m_2}\right)(u_1 - u_2)^2 = \frac{1}{2}\mu u_{\text{rel}}^2$$

This result is easy to guess using $K = \frac{1}{2}\mu v_{\text{rel}}^2 + \frac{1}{2} M v_{\text{CM}}^2$ (Try yourself).

6.5 Head on Perfectly Elastic Collision

Velocities of two balls before and after collision are as shown in figure.

Before collision

After collision.

Conservation of momentum gives:

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2 \qquad \text{...(i)}$$

Use of equation (10) gives:

$$-\left(\frac{v_1 - v_2}{u_1 - u_2}\right) = 1 \qquad [\because \ e = 1 \text{ for elastic collision}]$$

$$\Rightarrow \qquad v_1 - v_2 = u_2 - u_1 \qquad \text{...(ii)}$$

Solving (i) and (ii) for v_1 and v_2 gives

$$\left.\begin{aligned} v_1 &= \left(\frac{m_1 - m_2}{m_1 + m_2}\right)u_1 + \left(\frac{2m_2}{m_1 + m_2}\right)u_2 \\ v_2 &= \left(\frac{m_2 - m_1}{m_1 + m_2}\right)u_2 + \left(\frac{2m_1}{m_1 + m_2}\right)u_1 \end{aligned}\right\} \qquad \text{...(11)}$$

In elastic collision, the KE of the system of balls is conserved.

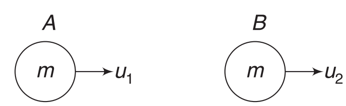
6.5.1 *Special Cases*

It is worth remembering the following cases in head on elastic collision of two objects:

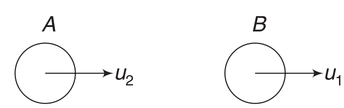
(i) When both objects have same mass (i.e, $m_1 = m_2$), final velocities are (using 11)

$$v_1 = u_2 \quad \text{and} \quad v_2 = u_1$$

It means the velocities of the two objects get exchanged.



Before collision



After collision.
A acquires Velocity of *B* and
B acquires Velocity of *A*.

(ii) When the body of mass m_2 is originally at rest (i.e., $u_2 = 0$), equation (11) gives:

$$v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)u_1 \quad \text{and}$$

$$v_2 = \left(\frac{2m_1}{m_1 + m_2}\right)u_1 \qquad \text{...(12)}$$

(iii) A light body hits a very heavy object at rest and collision is head on – elastic. It means $m_2 >> m_1$. From (12), we get

$$v_1 \simeq -u_1 \quad \text{and} \quad v_2 \simeq 0$$

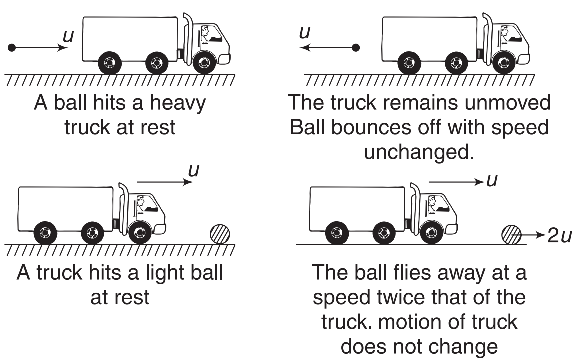
It means that light object bounces back with same speed. [Negative value of v_1 indicates that direction of motion has flipped]. There is no effect on the very heavy object, which was at rest ($v_2 = 0$).

Imagine a ball hitting a heavy truck. The ball bounces off with unchanged speed but the truck does not move.

(iv) A heavy body (a truck) hits a light object (a ball) at rest and collision is head on-elastic. On putting $m_1 >> m_2$, equation (12) gives:

$$v_1 \simeq u_1 \quad \text{and} \quad v_2 \simeq 2u_1$$

The heavy object (the truck) experiences no change in its velocity and the lighter object (ball) flies away with twice the velocity of the truck.



6.6 Head on, Inelastic Collision

If u_1 and u_2 refer to velocities before collision and v_1 and v_2 refer to velocities after collision (see fig given in article 6.5), then

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2 \qquad \text{...(i)}$$

And
$$-\left(\frac{v_1 - v_2}{u_1 - u_2}\right) = e \qquad \text{...(ii)}$$

Solving these two equation gives

$$\left.\begin{aligned} v_1 &= \left(\frac{m_1 - em_2}{m_1 + m_2}\right)u_1 + \left(\frac{(1+e)m_2}{m_1 + m_2}\right)u_2. \\ v_2 &= \left(\frac{(1+e)m_1}{m_1 + m_2}\right)u_1 + \left(\frac{m_2 - em_1}{m_1 + m_2}\right)u_2. \end{aligned}\right\} \qquad \text{...(13)}$$

Example 20 A ball is flying horizontally at speed v. It hits a heavy wall travelling with velocity u in opposite direction. Find the speed at which the ball rebounds if coefficient of restitution is e. What will be your answer if collision is elastic?

Solution

Concepts

(i) $-\left(\dfrac{v_1 - v_2}{u_1 - u_2}\right) = e$

In words: (speed of approach) × e = speed of separation.

(ii) For elastic collision, $e = 1$

Let velocity of the ball on rebounding be $v'(\leftarrow)$. Velocity of the wall will remain practically unchanged because it is heavy.

Taking left as positive direction and using

$$-\left(\frac{v_1 - v_2}{u_1 - u_2}\right) = e$$

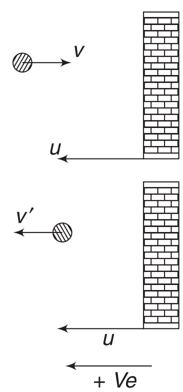
$$\Rightarrow \quad -\left(\frac{v' - u}{-v - u}\right) = e$$

$$\Rightarrow \quad v' - u = ev + eu$$

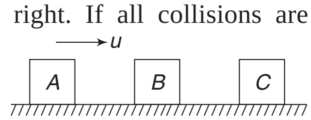
$$\Rightarrow \quad v' = ev + (e+1)u$$

For elastic collision $e = 1$

$$\therefore \quad v' = v + 2u.$$



Example 21 Three blocks A, B and C have equal masses and are placed on a horizontal smooth surface. Block A is imparted a velocity u towards right. If all collisions are elastic, find the final velocities of all three blocks.

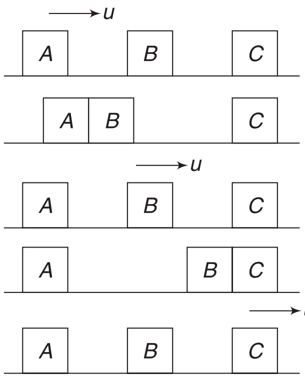


Solution

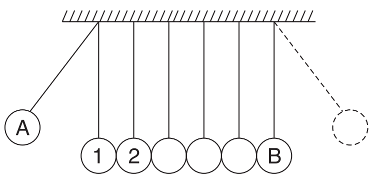
Concepts

In head-on elastic collision of equal mass, velocities get exchanged.

When A hits B, A stops and B acquires velocity u. Now, B hits C. Block C acquires velocity u and B stops. Finally, only C is moving and A and B are at rest.



Note: Consider number of simple pendulums made of steel bobs and placed side by side as shown. All pendulums are identical and touching one another. If one of the extreme pendulum (A) is taken aside and released, we find that none of the pendulums appear to move after A hits 1 and soon, the pendulum (B) at other extreme swings. Then B comes back to hit the system. Only A moves. Therefore, only A and B appear to move and all other bobs remain stationary. Actually, A transfers all its momentum to 1 and 1 transfers it to 2 and so on. (elastic collision of equal masses). Finally, it is B which acquires all the speed and moves. This puzzling device is known as Newton's Cradle.

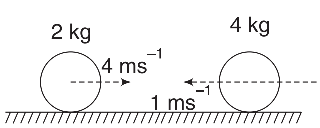


Example 22 Two balls are moving as shown. They collide and coefficient of restitution is $e = 0.5$.

(i) Find the velocity of the two balls after collision.

(ii) Find loss in kinetic energy

(iii) Find the smallest kinetic energy of the system when collision was going on.



Solution

Concepts

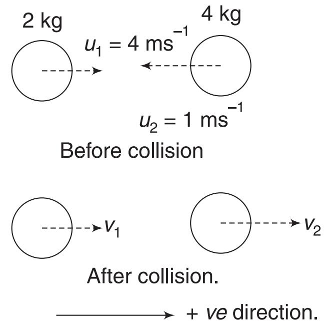
(i) Conservation of momentum and definition of coefficient of restitution give us two equations, which can be solved for getting final velocities or else, you can directly use equation (13).

When deformation is maximum, the two balls have least kinetic energy. This happens when both have same velocity at a point during the ongoing collision process.

(i) Let v_1 and v_2 be velocities of the two balls after collision. Momentum conservation gives:

$$2v_1 + 4v_2 = 2u_1 + 4u_2$$

$$\Rightarrow \quad 2v_1 + 4v_2 = 2 \times 4 + 4 \times (-1)$$



$$\Rightarrow \quad v_1 + 2v_2 = 2 \qquad \text{...(i)}$$

$$e = 0.5$$

$$-\left(\frac{v_1 - v_2}{u_1 - u_2}\right) = 0.5 \quad \Rightarrow \quad -\left(\frac{v_1 - v_2}{4 - (-1)}\right) = 0.5$$

$$\Rightarrow \quad v_1 - v_2 = -2.5 \qquad \text{...(ii)}$$

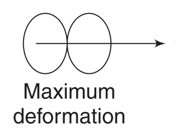
Solving (i) and (ii) gives: $v_2 = 1.5\ \text{ms}^{-1}$ and $v_1 = -1.0\ \text{ms}^{-1}$ Negative sign of v_1 indicates that the 2 kg ball will travel to left with a velocity of 1.0 ms^{-1}.

(ii) $KE_i = \frac{1}{2} \times 2 \times 4^2 + \frac{1}{2} \times 4 \times 1^2 = 18\,\text{J}$

$$KE_f = \frac{1}{2} \times 2 \times 1^2 + \frac{1}{2} \times 4 \times 1.5^2 = 5.5\,\text{J}$$

$$\therefore \quad \text{Loss in } KE = 18 - 5.5 = 12.5\,\text{J}$$

(iii) Maximum deformation (minimum KE) occurs when $v_1 = v_2 = v$ (say)



Conservation of momentum gives:

$$(2 + 4)v = 2 \times 4 - 4 \times 1$$

$$\Rightarrow \quad v = \frac{2}{3}\ \text{ms}^{-1}$$

$$\therefore \quad KE_{\text{minimum}} = \frac{1}{2}(2 + 4)\left(\frac{2}{3}\right)^2 = \frac{4}{3}\ \text{J}.$$

Example 23 *A ball dropped on a hard floor*

A ball of mass m is dropped from a height h on a hard floor. The ball hits the floor and rebounds. It again falls back and hits the floor. It again rebounds. The process continues till the ball comes to rest. Coefficient of restitution is $e(<1)$ and the ball keeps moving along a vertical straight line. Find

(i) the time for which the ball keeps moving after it is dropped.

(ii) the total distance travelled by the ball before it stops.

(iii) the total impulse imparted by the ball to the floor before it stops.

Solution

Concepts

(i) Coefficient of restitution is $e(<1)$. If speed before a collision is u, it is simply the speed of approach as the floor is not moving. Speed after collision (v) is nothing but speed of separation.

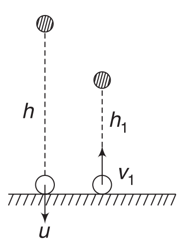
$$\therefore \quad \frac{v}{u} = e \quad \Rightarrow \quad v = eu$$

(ii) The balls will bounce repeatedly with its speed getting reduced. If its speed is $0.001\,\mathrm{ms^{-1}}$ before collision, it will become $e(0.001)$ after collision. It will again bounce and come back to hit the floor. When its speed becomes $0.000001\,\mathrm{ms^{-1}}$, then also it bounces and comes back! Theoretically, there will be infinite many collisions.

But the time that the ball takes between two collisions goes on decreasing. When its speed is too small, it will rise to a very small height and will take a very small time to return back. As we shall see, this makes the total time and total distance a finite quantity.

(iii) Initially, the ball is released from rest and ultimately it will comes to rest. Impulse of all the forces acting on it must sum up to zero.

Velocity of ball just before it hits the floor is



$$u = \sqrt{2gh}$$

Time of fall $t_1 = \sqrt{\dfrac{2h}{g}}$

Now the ball rebounds with speed

$v_1 = eu$ after collision.

Time of flight before it returns for next collision is $t_2 = \dfrac{2v_1}{g}$

$$\Rightarrow t_2 = \frac{2eu}{g} = \frac{2e\sqrt{2gh}}{g} = 2e\sqrt{\frac{2h}{g}}$$

Height attained after first collision is $h_1 = \dfrac{v_1^2}{2g} = \dfrac{e^2u^2}{2g}$

$$= e^2h$$

The balls comes back to the floor with speed v_1 for the next collision. Speed after this collision is

$$v_2 = ev_1 = e^2u$$

Time of flight before it returns back for next collision is

$$t_3 = \frac{2v_2}{g} = \frac{2e^2u}{g} = 2e^2\sqrt{\frac{2h}{g}}$$

And height attained after second collision is

$$h_2 = \frac{v_2^2}{2g} = e^4h$$

(i) Total time is

$$T = t_1 + t_2 + t_3 + \ldots$$

$$= \sqrt{\frac{2h}{g}}\,[1 + 2e + 2e^2 + 2e^3 + \ldots \infty \text{ terms}]$$

$$= \sqrt{\frac{2h}{g}}\,[1 + 2(e + e^2 + e^3 + \ldots)]$$

$$= \sqrt{\frac{2h}{g}}\,[1 + 2e/(1 - e)] \text{ [Sum of infinite G.P]}$$

$$= \sqrt{\frac{2h}{g}}\left[\frac{1 + e}{1 - e}\right]$$

(ii) Total distance travelled is

$$s = h + 2h_1 + 2h_2 + \ldots$$

$$= h + 2e^2h\,[1 + e^2 + e^4 + \ldots \infty \text{ terms}]$$

$$= h + \frac{2e^2\,h}{1 - e^2} = h\left(\frac{1 + e^2}{1 - e^2}\right)$$

(iii) Total impulse on the ball = change in momentum of the ball

$$J_{\text{floor}} + J_{\text{gravity}} = 0$$

$$\therefore \quad J_{\text{floor}} = -J_{\text{gravity}} = -mgT = -m\sqrt{2gh}\left[\frac{1 + e}{1 - e}\right]$$

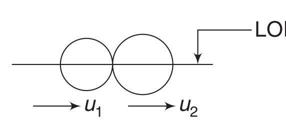
Negative sign implies that impulse on the ball due to the floor is upward.

The ball applies same impulse on the floor in downward direction.

$$\therefore \quad J = m\sqrt{2gh}\left[\frac{1 + e}{1 - e}\right] (\downarrow)$$

Example 24 *Another way to define coefficient of restitution*

Two balls, having masses m_1 and m_2 collide head on while travelling with velocities u_1 and u_2 respectively. During collision,



both deform each other. By the time deformation becomes maximum, they apply an impulse J_D (Impulse of deformation) on one another. Now they begin to regain back their shape and push one another. During this phase of reformation, they apply an impulse J_R (Impulse of reformation) on one another before separating. Prove that coefficient of restitution is $e = \frac{J_R}{J_D}$

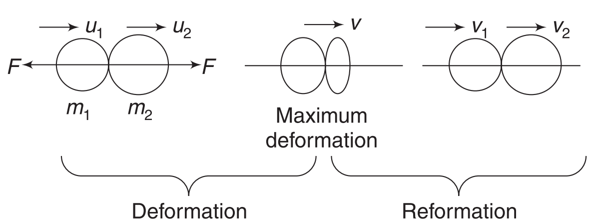
Solution

Concepts

(i) Momentum of the system before collision = Momentum during collision = Momentum after collision.

(ii) When deformation is maximum, both balls have same velocity.

(iii) Impulse = ΔP



Let velocity of both balls at the instant of maximum deformation be v.

Momentum conservation gives

$$m_1u_1 + m_2u_2 = m_1v + m_2v = m_1v_1 + m_2v_2$$

$$\Rightarrow \quad v = \frac{m_1u_1 + m_2u_2}{m_1 + m_2} = \frac{m_1v_1 + m_2v_2}{m_1 + m_2} \quad ...(i)$$

During deformation phase, impulse on m_2 is

$$J_D = m_2v - m_2u_2 = m_2(v - u_2)$$

During reformation phase, impulse on m_2 is

$$J_R = m_2v_2 - m_2v = m_2(v_2 - v)$$

$$\therefore \quad \frac{J_R}{J_D} = \frac{v_2 - v}{v - u_2}$$

using (i)

$$\frac{J_R}{J_D} = \frac{v_2 - \frac{m_1v_1 + m_2v_2}{m_1 + m_2}}{\frac{m_1u_1 + m_2u_2}{m_1 + m_2} - u_2} = \frac{m_1(v_2 - v_1)}{m_1(u_1 - u_2)}$$

$$= -\left(\frac{v_1 - v_2}{u_1 - u_2}\right) = e$$

In short:

(i) Line along which two colliding objects apply force on one another is called line of impact (LOI). For two balls, LOI is the line joining their centres. For two blocks, LOI is a line normal to the common contact wall.

When an object hits a wall, LOI is normal to the wall.

(ii) If initial velocities of colliding objects (u_1 and u_2) are along LOI, the collision is head on.

(iii) Collision in which u_1 and u_2 are not along LOI, are called oblique collisions.

(iv) Colliding objects apply huge imulse on one another during collision and deform each other. Deformation is maximum when both have same velocity along the LOI.

If the objects quickly regain their shape and KE of the system remains conserved, the collision is called elastic. In such a collision

$$\frac{\text{Relative speed of separation after collision}}{\text{Relative speed of approach before collision}} = 1$$

This ratio is called coefficient of restitution (e).

(v) If colliding objects have no tendency to regain their shape after deforming each other, they travel with equal velocity after head-on collision ($v_1 = v_2$). Such collision is called perfectly inelastic ($e = 0$). Loss in KE is maximum.

(vi) If colliding objects are able to regain their shape partially, collision is called inelastic. In such a collision, $0 < e < 1$. There is some loss in KE.

(vii) Usually, conservation of momentum and defining equation of coefficient of restitution are sufficient to solve a problem on head-on collision.

(viii) If a ball hits a fixed wall (or floor) with velocity u while travelling normally to the wall, velocity of ball after collision is $v = eu$ with direction of motion reversed.

(ix) When two equal masses undergo head-on elastic collision, there is exchange of velocity.

(x) In a perfectly inelastic head-on collision, loss in KE is

$$\Delta K = \frac{1}{2}\mu\, u_{rel}^2 \quad \text{where}$$

$$\mu = \frac{m_1m_2}{m_1 + m_2} \quad \text{and} \quad u_{rel} = u_1 - u_2.$$

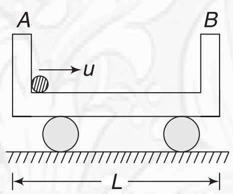
YOUR TURN

Q.32 Block A, travelling on a smooth horizontal surface with velocity u, hits another identical block, B which is at rest. Each block has mass m.

(i) Find the kinetic energy of each block, after collision, if the collision is elastic.

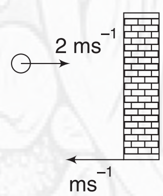
(ii) Find loss in kinetic energy of the system if the collision is perfectly inelastic.

Q.33 A cart of mass M rests on a smooth horizontal surface. At one end of the cart there is a ball of mass M. The ball is imparted a sharp hit so as to give it a velocity u towards the other end of the cart. The ball travels along the smooth cart surface and hits the front wall (B) elastically. After how much time (from the start) will the ball hit the wall A?



Q.34 A ball of mass m moving with velocity v makes a head-on collision with an identical ball at rest. The system loses 25% of its kinetic energy in collision. Find the coefficient of restitution.

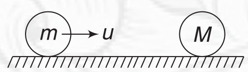
Q.35 A ball is moving with velocity $2\,\text{ms}^{-1}$ towards a heavy wall which is moving towards the ball with speed $1\,\text{ms}^{-1}$. They collide head on. Find velocity of ball immediately after collision, if



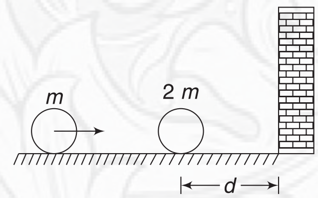
(i) Collision is elastic

(ii) Coefficient of restitution is $e = 0.5$

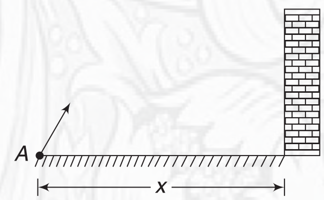
Q.36 A ball of mass m is travelling at speed u along a smooth horizontal surface. It collides head on with another ball at rest. Find the mass (M) of the other ball so that the direction of motion of the first balls gets reversed due to collision. Coefficient of restitution is 0.8.



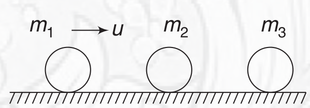
Q.37 A particle A of mass m moving on a smooth horizontal surface collides head on with another stationary particle B of mass $2m$, which is located at a distance d from a wall. Coefficient of restitution for collision between the balls and that between a ball and the wall is $e = 0.25$. Find distance from the wall where the balls collide for the second time. All motion takes place on a straight line, perpendicular to the wall.



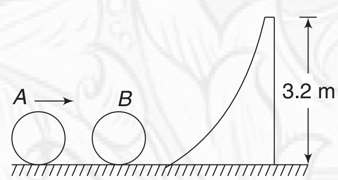
Q.38 A ball is fixed from point A on ground. There is a vertical wall at a distance x from point A. The projectile hits the wall while its velocity is horizontal. The coefficient of restitution is $e = 0.5$. At what distance from the wall will the ball land on the ground?



Q.39 Three balls of masses m_1, m_2 and m_3 are lying on a smooth horizontal surface, as shown. Ball of mass m_1 is given a velocity towards m_2. It hits the ball of mass m_2 and itself comes to rest. The middle balls hits the third ball of mass m_3 and comes to rest. Coefficient of restitution for each collision is e. Find m_2 in terms of m_1 and m_3



Q.40 Two identical balls A and B lie on a smooth horizontal surface. Ahead of B, there is a fixed smooth track of height 3.2 m. Ball A is given a velocity $u = 10\,\text{ms}^{-1}$ towards B. It collides head on with B, which then begins to climb the curved track. Find minimum coefficient of restitution (e) between the balls so that B can reach the top of the track.

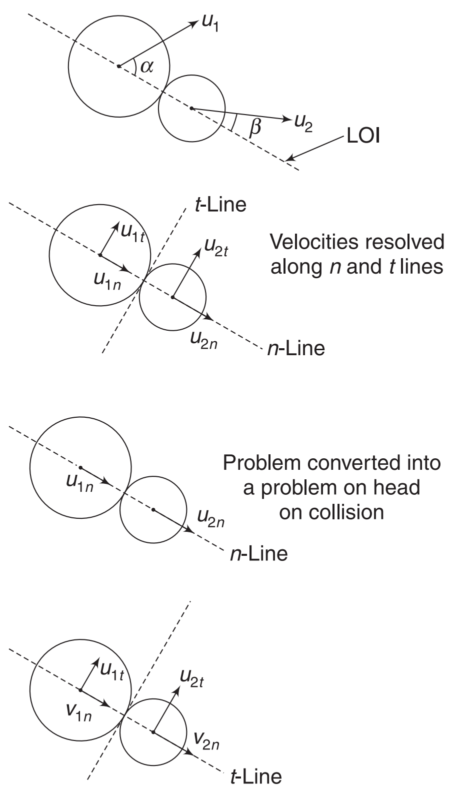


6.7 Oblique Collision

Consider two balls moving with velocities u_1 and u_2, as shown. When they collide, they apply impulse along the line joining their centre (Line of Impact), assuming that the balls are smooth. The effect of collision is to impart an impulse to each ball along the LOI. There will be no effect on the motion of two balls in a direction that is perpendicular to LOI.

For ease of discussion, we will call the LOI as n-line ('n' for normal) and the line perpendicular to LOI as t-line ('t' for tangent).

Before collision the velocity component of two balls along n-line are u_{1n} (= $u_1 \cos\alpha$ in shown figure) and u_{2n} (= $u_2 \cos\beta$ in shown figure).

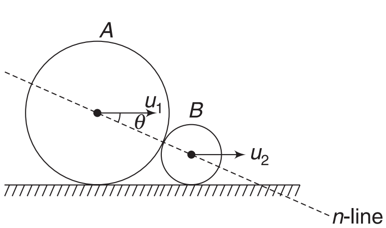


Add tangential velocity to normal velocity to get the resultant.

Velocity components along t-line are $u_{1t}(= u_1 \sin \alpha)$ and $u_{2t}(= u_2 \sin \beta)$

The effect of collision is to change velocities along n-line only. For the time being, forget about velocity components along t-line. Take this as a problem of one-dimensional collision with balls moving with velocities u_{1n} and u_{2n}. Use what you have studied in last section about head-on collisions and find velocities (v_{1n} and v_{2n}) after collision. Now, you need to recall that u_{1t} and u_{2t} are unchanged. Add u_{1t} and v_{1n} vectorially to get final velocity of the first ball. Similarly, add u_{2t} and v_{2n} to get final velocity of the second ball.

The above procedure will work if the system is free of external impulses. At times, you may face problems when there are external impulses on the system. If there is an impulse along n-line then momentum cannot be conserved along this line. However, *for writing coefficient of restitution, we consider velocity components along n-line in all cases*.



Consider two balls A and B moving horizontally on a floor with velocities u_1 and u_2, as shown. When A hit B, the floor applies a large impulsive normal reaction on B. This impulsive force has a component along n-line. Momentum of the system of two balls is not conserved along the n line. [However, it is conserved along horizontal direction].

If v_1 and v_2 are velocities after collision and coefficient of restitution is e then

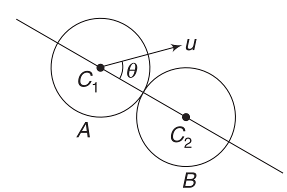
$$e = -\left(\frac{v_{1n} - v_{2n}}{u_{1n} - u_{2n}}\right) \qquad ...(14)$$

Note: In present example, $v_{1n} = v_1 \cos \theta$, $v_{2n} = v_2 \cos \theta$, $u_{1n} = u_1 \cos \theta$, $u_{2n} = u_2 \cos \theta$

$$\therefore \qquad e = -\left(\frac{v_1 - v_2}{u_1 - u_2}\right)$$

Example 25 *Oblique elastic collision between balls of equal mass*

A billiard ball B is at rest. Another identical and smooth ball A hit it while travelling with a velocity u, making an angle θ with the line joining the centre of the two balls. Collision is elastic. Find velocity of the two balls after the impact.



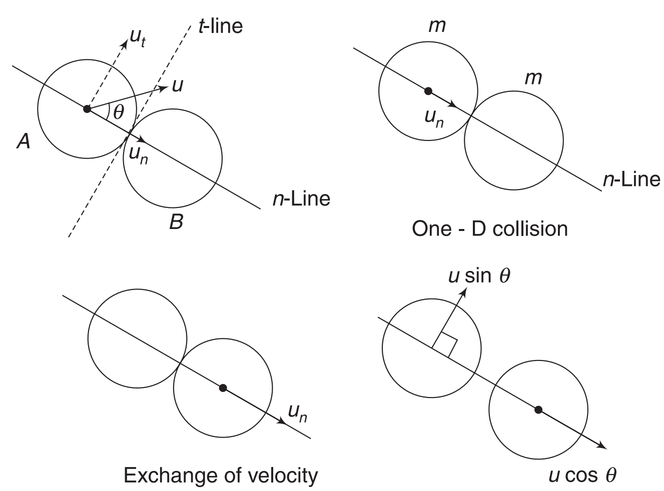
Solution

Concepts

(i) Line joining the centre (C_1C_2) is the n-line.

(ii) Velocity of ball will remain unchanged along t-line (a line ⊥ to the n-line).

(iii) In a head-on elastic collision of two balls, there will be exchange of velocities.



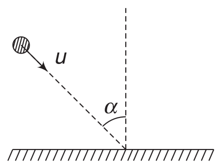
Velocity component of A along the n-line and t-line are $u_n = u \cos \theta$, $u_t = u \sin \theta$. Forget about u_t for a moment. Assume that B is at rest and A hits it head on with a velocity u_n. What will happen? Velocities will get exchanged. A comes to rest and B moves along n-line with velocity $u_n = u\cos\theta$.

Now recall that A has a velocity $u_t = u\sin\theta$, which remained unaffected during collision.

Thus, A will have a final velocity $u\sin\theta$ along t–line and B will move with velocity $u\cos\theta$ along n-line. Both balls will move in directions perpendicular to each other.

Example 26 *A ball colliding obliquely with a smooth floor*

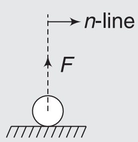
A ball hits a smooth floor at an angle of incidence α while travelling with velocity u (see figure). Coefficient of restitution is e. Find the velocity of the ball after the impact. Discuss the extreme cases of elastic and perfectly inelastic collision.



Solution

Concepts

(i) The LOI (the n-line) is normal to the floor. It does not make any difference whether the ball came from right or left, whether it was incident at angle 0° or 30°; the line along which the floor and the ball apply force on one another will be normal to the floor passing through the center of the ball.



(ii) There is no horizontal impulse on the ball. Its horizontal momentum will not change.

(iii) $$e = \left|\frac{\text{relative velocity of separation along } n\text{-line}}{\text{relative velocity of approach along } n\text{-line}}\right|$$

$$= \left|\frac{\text{velocity of ball in vertically upward direction after impact}}{\text{velocity of ball in vertically downward direction before impact}}\right|$$

Note that the floor does not move.

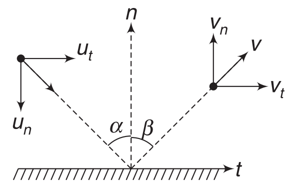
The n-line and t-line for the given collision are as shown.

Component of velocity before collision are:

$$u_t = u\sin\alpha,\ u_n = u\cos\alpha$$

Velocity component after collision are:

$$v_t \text{ and } v_n.$$



Since there is no change in momentum of the ball along the t-line

$$\therefore \quad v_t = u_t = u\sin\alpha$$

$$\text{And} \quad \frac{v_n}{u_n} = e \ \Rightarrow\ v_n = e\,u_n = eu\cos\alpha$$

$$\therefore \quad v = \sqrt{v_t^2 + v_n^2} = u\sqrt{\sin^2\alpha + e^2\cos^2\alpha} \qquad \text{...(i)}$$

Angle β can be calculated as

$$\tan\beta = \frac{v_t}{v_n} = \frac{u\sin\alpha}{e\,u\cos\alpha} = \frac{1}{e}\tan\alpha \qquad \text{...(ii)}$$

When collision is elastic, $e = 1$

$$\therefore \quad v = u$$

$$\text{and} \quad \beta = \alpha \qquad \text{[from (i) and (ii)]}$$

It means there is no change in speed and angle of reflection = angle of incidence.

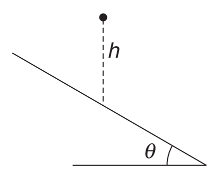
When collision is perfectly inelastic ($e = 0$):

$$v_n = 0 \text{ and } v_t = u\sin\alpha.$$

The ball will move along the floor (will not bounce back) with velocity $u\sin\alpha$.

Example 27 *A ball hitting an inclined plane*

A ball is dropped from a point. After falling through a distance $h = 5\,\text{m}$, it hits an inclined plane. The inclination angle of the plane is $\theta = 60°$. Coefficient of restitution is $e = 0.5$. Find the velocity of the ball immediately after the impact. What angle does the velocity of ball (after impact) makes with incline surface?



Solution

Concepts

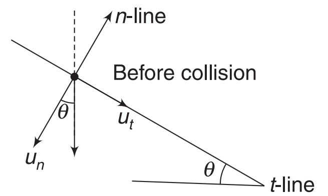
The concepts required are essentially same as in the last example.

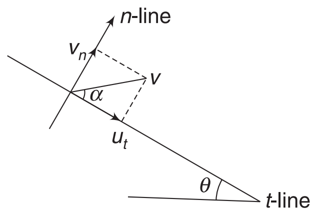
Velocity of the ball just before impact is

$$u = \sqrt{2gh} = 10\,\text{ms}^{-1}(\downarrow)$$

n-line and t-line are shown in figure.

$$u_t = u\sin\theta = 10 \times \frac{\sqrt{3}}{2} = 5\sqrt{3}\ \text{ms}^{-1}$$





After collision

$$u_n = u\cos\theta = 10 \times \frac{1}{2} = 5\,\text{ms}^{-1}$$

v is velocity after collision as there is no impulse on the ball along the t-line

$$v_t = u_t = 5\sqrt{3}\ \text{ms}^{-1}.$$

And $$v_n = e\,u_n = \frac{5}{2}\ \text{ms}^{-1}$$

$\therefore$ $$\tan\alpha = \frac{v_n}{v_t} = \frac{1}{2\sqrt{3}}$$

$\Rightarrow$ $$\alpha = \tan^{-1}\left(\frac{1}{2\sqrt{3}}\right)$$

Example 28 *Impact parameter*

A ball of mass m and radius R travelling with velocity u, hits another ball of mass $2m$ and radius $2R$. The impact parameter of the collision is $1.8R$. Coefficient of restitution is 0.5. Find velocity of two balls after the collision.

Solution

Concepts

(i) Draw a diameter of the second ball that is parallel to $\vec{u}$. The distance between line of $\vec{u}$ (through the centre of the first ball) and the diameter drawn is known as impact parameter (b).

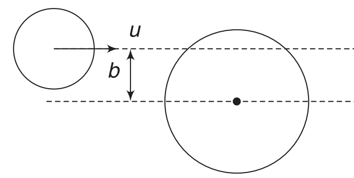
(ii) Knowing b, we can find the angle that u makes with the LOI (i.e., n-line).

(iii) We can treat the collision problem as a head-on collision problem, considering velocities along n-line only.

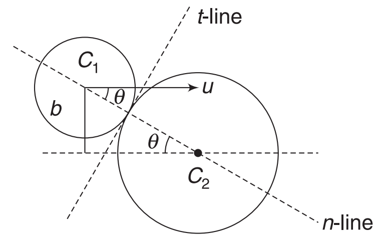
(iv) Velocity along t-line does not change.

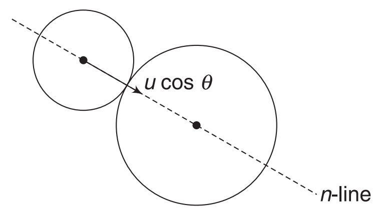
At the time of impact, the velocity u makes an angle θ with the n-line where

$$\sin\theta = \frac{b}{3R} = \frac{1.8R}{3R} = 0.6$$

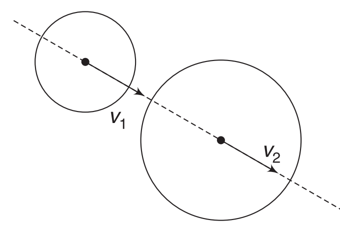


b = impact parameter





One-D collision problem



$\Rightarrow$ $$\theta = 37°$$

Velocity component of smaller ball along n-line and t-line are

$$u_n = u\cos\theta = 0.8u$$
$$u_t = u\sin\theta = 0.6u$$

u_t does not change due to collision.

Consider a head-on collision problem along n-line.

$$u_1 = u\cos\theta,\ u_2 = 0$$

Let v_1 and v_2 be velocities after collision.

Momentum conservation gives:

$$mv_1 + 2mv_2 = mu_1$$

$\Rightarrow$ $$v_1 + 2v_2 = u_1 \qquad \text{...(i)}$$

And $$-\left(\frac{v_1 - v_2}{u_1 - u_2}\right) = e$$

$\Rightarrow$ $$-\left(\frac{v_1 - v_2}{u_1 - 0}\right) = \frac{1}{2}$$

$$\Rightarrow \qquad v_2 - v_1 = \frac{u_1}{2} \qquad \text{...(ii)}$$

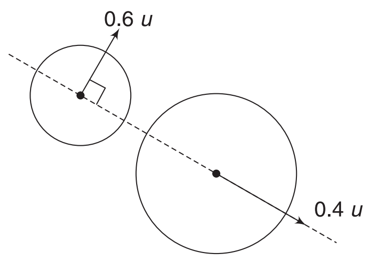
Solving (i) and (ii) gives

$$v_2 = \frac{u_1}{2} \quad \text{and} \quad v_1 = 0$$

$$\therefore \qquad v_1 = 0 \quad \text{and} \quad v_2 = \frac{0.8u}{2} = 0.4u$$

$$[\because u_1 = u\cos\theta = 0.8\ u]$$

First ball has its velocity along t-line unchanged. Thus final velocities of the two balls are as shown. The two balls will be moving perpendicular to one another.



Example 29 *Line of impact not known*

A ball of mass m travelling with velocity u hits another ball of mass $2m$, which was at rest. After impact, the first ball was found to be travelling at velocity $\frac{u}{2}$ in a direction perpendicular to the original line of motion. Find the velocity and direction of motion of the second ball after the impact.

Solution

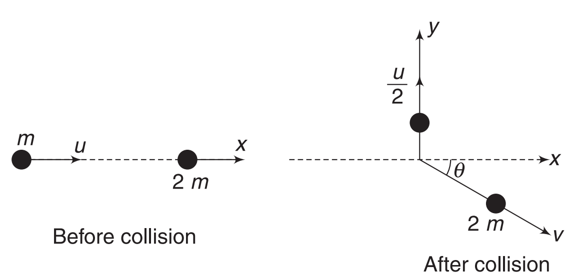
Concepts

(i) In the last problem, we knew the angle between initial velocity (u) and the LOI. This allowed us to resolve the velocities along n-line and t-line.

In this question, geometry of collision is not known. We do not know the angle between initial velocity of the ball and LOI.

(ii) If we have no knowledge regarding the motion of either balls after collision, the problem cannot be solved. However, the present question tells us the velocity of first ball after collision. It can be solved.

(iii) Apply conservation of momentum along two perpendicular directions.



Let the initial direction of motion of the first ball be x-direction. After collision, this ball moves along y-direction.

Let the other ball move with velocity v making an angle θ with x-direction, as shown in the figure.

Using conservation of momentum along x-direction:

$$P_{ix} = P_{fx}$$

$$mu = 2mv\cos\theta \quad \Rightarrow \quad u = 2v\cos\theta \qquad \text{...(i)}$$

Using conservation of momentum along y-direction

$$P_{iy} = P_{fy}$$

$$0 = m\frac{u}{2} - 2m\ v\sin\theta \quad \Rightarrow \frac{u}{2} = 2v\sin\theta \qquad \text{...(ii)}$$

Squaring and adding (i) and (ii) gives

$$u^2 + \left(\frac{u}{2}\right)^2 = 4v^2(\cos^2\theta + \sin^2\theta)$$

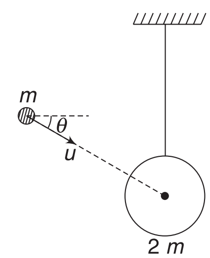
$$\Rightarrow \qquad v = \frac{\sqrt{5}}{4}u$$

And dividing (ii) with (i) gives

$$\tan\theta = \frac{1}{2} \quad \Rightarrow \quad \theta = \tan^{-1}\left(\frac{1}{2}\right)$$

Example 30 *External impulsive force present on the system*

A ball of mass $2m$ is suspended using an inextensible string. Another ball of mass m hits it while travelling with velocity u in a direction making an angle θ with the horizontal. The collision in elastic. Find the velocity of the smaller ball after the impact.



Solution

Concepts

(i) During collision, the string applies a large impulsive tension on the ball. This tension force has a component along n-line. Therefore, we cannot use momentum conservation along n-line.

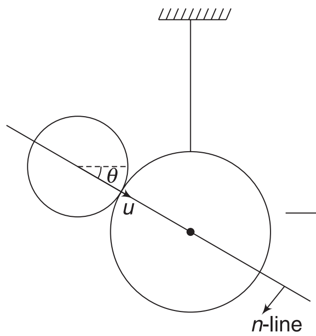
(ii) However, we can use equation (14) along n-line.

(iii) Momentum is conserved along horizontal direction, as there is no external force on the system of balls in horizontal direction.

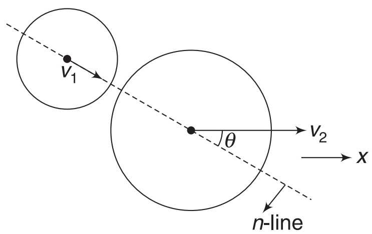
Let velocity of the bigger ball be v_2 along horizontal direction. This ball cannot have vertical motion. Let velocity of the smaller ball be v_1 along its original direction of motion. This ball has experienced an impact along this line (the n-line) only. Thus, it will move along this line only. [There is no t-component of velocity]

Momentum of conservation along x-direction gives

$$mu\cos\theta = mv_1\cos\theta + 2mv_2$$



Before collision



After collision

$$\Rightarrow \qquad u\cos\theta = v_1\cos\theta + 2v_2 \qquad \text{...(i)}$$

$$\text{And} \qquad -\left(\frac{v_{1n} - v_{2n}}{u_{1n} - u_{2n}}\right) = e$$

$$\Rightarrow \qquad -\left(\frac{v_1 - v_2\cos\theta}{u - 0}\right) = 1$$

$$\Rightarrow \qquad v_2\cos\theta - v_1 = u \qquad \text{...(ii)}$$

Solving (i) and (ii) gives

$$v_1 = -\frac{(2 - \cos^2\theta)u}{(2 + \cos^2\theta)} \quad \text{and} \quad v_2 = \frac{2u\cos\theta}{2 + \cos^2\theta}$$

Negative value of v_1 indicates that its direction is opposite to that indicated in figure.

In short:

(i) The line that is normal to contact surface of the two colliding bodies may be termed as *n*-line and a line perpendicular to that is termed as *t*-line.

(ii) When angle between initial velocities (u_1, u_2) and the *n*-line is known, we can resolve u_1 and u_2 along *n* and *t*-lines.

Velocity along *t*-line does not change if the system is free of any external impulsive force along this line.

In this case, we can consider the problem as a problem of head-on collision along *n*-line. After finding velocities along *n*-line we add the unchanged velocity along *t*-line to get the resultant final velocity of the body.

(iii) In all cases, $e = -\left(\frac{v_{1n} - v_{2n}}{u_{1n} - u_{2n}}\right)$ ...(14)

v_{1n}, v_{2n} are velocity components along *n*-line after collision and u_{1n}, u_{2n} are velocity components along *n*-line before collision.

(iv) When a ball hits a smooth fixed wall (vertical, horizontal or inclined), velocity parallel to the wall remains unchanged. The velocity normal to the wall reverses in direction and is equal to $v_n = e \cdot u_n$.

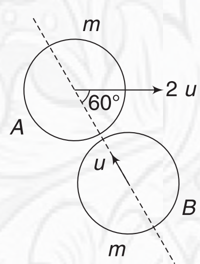
(v) When angle between velocities and LOI is not known we cannot use equation (14). However, conservation of momentum can be used.

(vi) When external impulsive force acts on the system of colliding bodies, apply conservation of momentum in a direction perpendicular to the impulse and use equation (14).

YOUR TURN

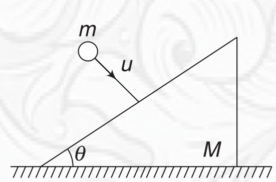
Q.41 A ball travelling with velocity $\vec{u} = (2\hat{i} + 3\hat{k})\,\text{ms}^{-1}$ hits a smooth wall. The plane of the wall is *xy* plane. Find velocity of the ball immediately after impact if collision is (i) Elastic (ii) Perfectly inelastic (iii) Inelastic with coefficient of restitution $e = 0.5$.

Q.42 Two balls collide as shown in figure. Find their velocities after collision if the collision is elastic.



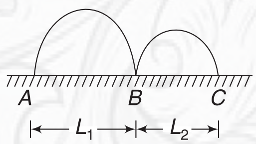
Q.43 A ball is flying horizontally when it hits an inclined surface. After impact, the ball was found to be moving vertically up. If collision is elastic, find the inclination of the incline surface with horizontal.

Q.44 A wedge of mass $M = 4$ kg is placed on a smooth horizontal surface. Its inclined face is inclined at $\theta = 37°$ to the horizontal. A ball of mass $m = 1$ kg hits the smooth inclined surface normally, with a velocity of $u = 10\,\text{ms}^{-1}$. The wedge begins to move horizontally at a velocity $2\,\text{ms}^{-1}$. Find the coefficient of restitution for the collision.



Q.45 A smooth ball hits a vertical wall with its direction of motion making 30° with the normal to the wall. The ball rebounds in a direction that is perpendicular to its original direction of motion. Find the coefficient of restitution.

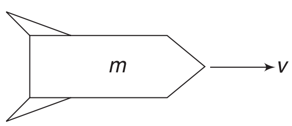
Q.46 A projectile is launched from point A and it lands back on the horizontal floor at B. It bounces at B and again lands back on the floor at C. Floor is smooth and coefficient of restitution is e. Find the ratio $\frac{L_1}{L_2}$.

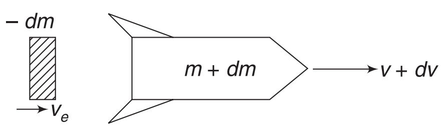


Q.47 A ball of mass m is moving along x-axis with speed u. It hits another ball of mass $2m$ at rest. As a result of collision, the ball of mass m loses half its speed but maintains its line of motion. The other ball breaks into two equal pieces with one piece travelling in y-direction with velocity $\frac{u}{2}$. Find the velocity of the other piece.

7. SYSTEM WITH VARYING MASS

Application of Newton's second law of motion is little tricky when mass of a system is changing. A rocket is a very good example of varying mass system. Most of its mass is fuel which is eventually ejected as exhaust gases. Consider a rocket of mass m travelling at a velocity v in space. In an infinitesimally small time dt its mass changes by dm. Note that dm is negative as its mass is decreasing due to burning of fuel and throwing it out. [In other words, during time dt, a negative mass dm gets added to the rocket and a positive mass $(-dm)$ is thrown out].





Speed of rocket change to $v + dv$.

Let the exhaust gases come out of the rocket at a velocity u relative to the rocket. What we are saying is that the engine shoots out the gases at a fixed speed relative to itself whatever be its own speed. Actual velocity of exhaust $v_e = (v + dv - u)$, which may be positive or negative depending on values of v and u. [Note that we have taken u to be directed towards left in our diagram].

Change in momentum of ejected mass $(-dm)$ (in time dt) is

$$dp = (-dm)v_e - (-dm)v$$

$$= -dm[v_e - v] = -dm[dv - u] = u\,dm$$

Where we have neglected $dm \cdot dv$

$\therefore$ Force on ejected mass is

$$F = \frac{dp}{dt} = u\frac{dm}{dt}$$

Same force is applied by the outgoing mass on the rocket in opposite direction.

$\therefore$ Thrust force acting on the rocket is

$$F_{\text{th}} = -u\,\frac{dm}{dt}$$

In the above expression, F_{th} is positive, i.e., towards right, since dm/dt is negative.

We can also have a variable mass system in which mass is getting added. The general formula for thrust force acting on a system having variable mass is:

$$\vec{F}_{\text{th}} = \vec{u}\,\frac{dm}{dt} \qquad \text{...(15)}$$

Here, $\vec{u}$ = velocity of incoming/outgoing mass relative to the system and $\frac{dm}{dt}$ = rate of change of mass of the system.

We must be very careful about the sign of $\vec{u}$ and $\frac{dm}{dt}$.

If the variable mass system is experiencing an external force ($\vec{F}_{\text{ext}}$) then equation for its motion becomes:

$$m\,\frac{d\vec{v}}{dt} = \vec{F}_{\text{th}} + \vec{F}_{\text{ext}}$$

$$m\,\frac{d\vec{v}}{dt} = \vec{u}\,\frac{dm}{dt} + \vec{F}_{\text{ext}} \qquad \text{...(16)}$$

Example 31 A rocket in space is having mass M_0 and is travelling at a velocity v_0. It is ejecting gases at a velocity u relative to itself. Find its velocity when its mass reduces to m.

Solution

Concepts

(i) Rocket is in space. Hence, external force on it is zero.

(ii) Thrust force is $F_{\text{th}} = u\frac{dM}{dt}$.

Here, u (velocity of exhaust, relative to the rocket) is negative; i.e., opposite to the velocity of the rocket.

$$\therefore \qquad M\frac{dv}{dt} = -u\frac{dM}{dt}$$

$$\Rightarrow \quad dv = -u\frac{dM}{M}$$

$$\Rightarrow \quad \int_{v_0}^{v} dv = -u\int_{M_0}^{m}\frac{dM}{M}$$

$$\Rightarrow \quad v - v_0 = -u\,ln\left(\frac{m}{M_0}\right) \Rightarrow v = v_0 + u\,ln\left(\frac{M_0}{m}\right)$$

Example 32 A rocket of mass M_0 is fired vertically from the ground. It ejects gases at a velocity u relative to itself and throws out mass at a rate b kgs^{-1}. Find its velocity at time t. Assume that the acceleration due to gravity remains constant at g.

Solution

Concepts

(i) Apart from the thrust force, the rocket experiences an external force Mg.

(ii) At time t, its instantaneous mass is $M = M_0 - bt$

$$M\frac{dv}{dt} = -u\frac{dM}{dt} - Mg$$

Here $\dfrac{dM}{dt} = -b$, thus the thrust force is positive.

$$\frac{dv}{dt} = -\frac{u}{M}\frac{dM}{dt} - g$$

$$\Rightarrow \quad dv = -u\frac{dM}{M} - g\,dt$$

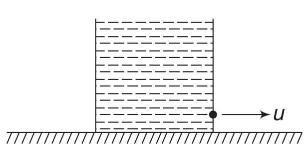
$$\Rightarrow \quad \int_{0}^{v} dv = -u\int_{M_0}^{m}\frac{dM}{M} - g\int_{0}^{t} dt$$

Where m = mass at time $t = M_0 - bt$

$$\Rightarrow \quad v = -u\,ln\frac{m}{M_0} - gt$$

$$\Rightarrow \quad v = u\,ln\left(\frac{M_0}{M_0 - bt}\right) - gt$$

Example 33 A container filled with a liquid of density ρ has a mass M (including liquid). It is at rest on a smooth horizontal surface. A small hole of area A is punched in its side wall through which the liquid gushes out at speed u. Find the initial acceleration of the container.



Solution

Concepts

(i) This is a variable mass system as the mass of tank decreases continuously.

(ii) Initial velocity of tank = 0

$\therefore$ Relative velocity of outgoing mass = u

(iii) Volume of liquid flowing out per unit time through the hole = Au.

Mass flowing out per unit time = ρAu

$$F_{\text{th}} = u\frac{dM}{dt} = u(\rightarrow)\cdot(-\rho Au)$$

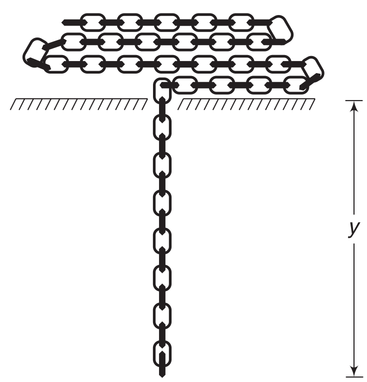
[$\because$ Mass is decreasing]

$$= \rho Au^2\ (\leftarrow)$$

$$\therefore \quad M\frac{dv}{dt} = \rho Au^2(\leftarrow) \Rightarrow \frac{dv}{dt} = \frac{\rho Au^2}{M}(\leftarrow)$$

Example 34 *Chain falling through a hole*

A chain has mass per unit length equal to λ. It is piled up on a table. There is a small hole in the table through which the chain begins to fall. Find its acceleration when its free end has fallen though a distance y.



Solution

Concepts

(i) We can always solve this problem using Newton's second law as $\vec{F} = \dfrac{d\vec{p}}{dt}$; but here, we will do it by treating the hanging part of the chain as a system to which mass is continuously getting added (From the heap).

(ii) If speed of chain is v at an instant, then $v\,dt$ length of chain is added form the heap to the falling chain (which is our system) in time dt.

$\therefore$ Mass added in time dt is $dm = \lambda v\,dt$

$$\therefore \quad \frac{dm}{dt} = \lambda v$$

$\frac{dm}{dt}$ is positive since mass is getting added.

(iii) Velocity of incoming mass (a small segment on table) = 0. Velocity of our system(hanging chain) = v ($\downarrow$)

$\therefore$ Relative velocity, $u = 0 - v(\downarrow) = v(\uparrow)$

When y length is hanging, mass of our system is

$$m = \lambda y$$

Thrust force due to incoming mass is

$$F_{th} = u\frac{dm}{dt} = v(\lambda v)(\uparrow) = \lambda v^2(\uparrow)$$

$$\therefore \quad m\frac{dv}{dt} = mg - \lambda v^2$$

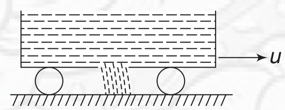
$$\Rightarrow \quad \frac{dv}{dt} = g - \frac{\lambda}{m}v^2 = g - \frac{\lambda}{\lambda y}v^2 = g - \frac{v^2}{y}$$

Your Turn

Q.48 A rocket has a mass of 500 kg (including its fuel). It can have a maximum exhaust speed of 2000 ms^{-1} (relative to itself). Find the minimum rate of fuel consumption for it to just lift off the launch pad.

Q.49 In the last problem, find the rate of fuel burning if we want the initial acceleration of the rocket to be 5 ms^{-2}.

Q.50 A cart is moving on a smooth horizontal road at speed u. It is carrying sand. Sand begins to leak at a constant rate b kgs^{-1} through a small hole at the bottom of the cart. Find the acceleration of the cart.



Q.51 A balloon of negligible mass has a gas of mass m filled in it. It suddenly gets released and the gas leaks from it at a constant velocity u = 2 ms^{-1} relative to it. Find the velocity of the balloon when its mass is reduced to half. Neglect gravity and buoyancy.

Miscellaneous Examples

Example 35 A cannon of mass m starts sliding freely down a smooth inclined plane at an angle α to the horizontal. After the cannon covered a distance l, a shot was fired, the shell leaving the canon in horizontal direction with momentum P. As a consequence, the canon stops. Assume the mass of the shell to be negligible compared to that of the canon and find the duration of the shot.

Solution

Concepts

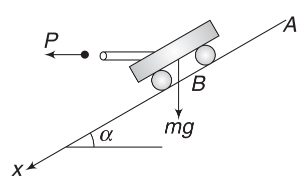
(i) Duration of the shot implies the time duration for which the cannon exerted force on the shell. This duration can be deliberately prolonged (by making the barrel long) so as to impart huge impulse to the shell.

(ii) Change in momentum of the cannon + shell system along the incline = impulse of external force along the incline. We will use this relation for the duration of the shot.

The cannon started from A and moved a distance $AB = l$ with acceleration $a = g\sin\alpha$.

Speed at B is

$$v = \sqrt{2al} = \sqrt{2gl\sin\alpha}$$



At this instant, it fires a shell horizontally and comes to rest. Change in momentum of the cannon + shell system along x during the duration of shoot (Δt) is

$$\Delta P = (P\cos\alpha + 0) - (mv\cos\alpha)$$
$$= P\cos\alpha - m\cos\alpha\sqrt{2gl\sin\alpha}$$

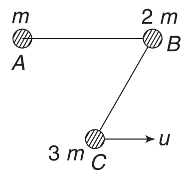
External force on the system along x-direction is

$$F_x = mg\sin\alpha$$

$$\therefore \quad F_x\,\Delta t = \Delta P$$

$$\Rightarrow \quad \Delta t = \frac{\Delta P}{F_x} = \frac{P\cos\alpha - m\cos\alpha\sqrt{2gl\sin\alpha}}{mg\sin\alpha}$$

Example 36 Three particles A, B and C of masses m, $2m$ and $3m$ respectively lie on a smooth horizontal table at the vertices of an equilateral triangle. A and B as well as B and C are connected by light inextensible strings as shown. C is imparted a velocity u parallel to AB. Find the final speed acquired by A.



Solution

Concepts

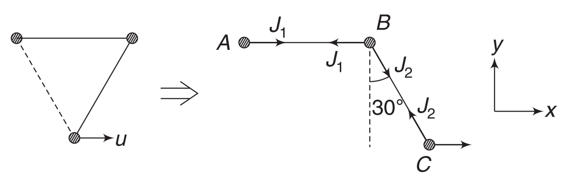
(i) Momentum of the system ($A + B + C$) will remain conserved.

(ii) When C moves to position shown in fig. below, the string BC suddenly acquires tension. It applies an impulse on C and also on B.

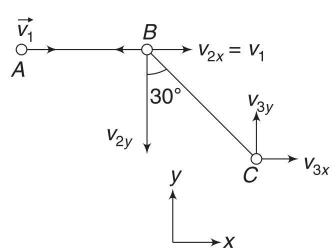
(iii) As B gets a tendency to move, the string AB also exerts an impulse on B as well as A.

A receives impulse along AB. It will have its final velocity along AB.

Let J_1 be impulse developed in string AB and J_2 be impulse developed in BC after C moves to a position shown in Fig. below.



Let final velocity of A be v_1 along AB. Let final velocity of B have two components v_{2x} and v_{2y} as shown. Velocity of C has two components v_{3x} and v_{3y}, as shown.



$$P_y = 0$$

$$\Rightarrow \quad 2m v_{2y} = 3m v_{3y}$$

$$\Rightarrow \quad 2v_{2y} = 3v_{3y} \qquad \text{...(i)}$$

$$P_x = 3mu$$

$$\Rightarrow \quad mv_1 + 2m v_{2x} + 3m v_{3x} = 3mu$$

$$\Rightarrow \quad v_1 + 2v_{2x} + 3v_{3x} = 3u$$

But $v_{2x} = v_1$. When string AB is taut

$$\therefore \quad v_1 + v_{3x} = u \qquad \text{...(ii)}$$

When BC is taut

Velocity of B along BC = Velocity of C along BC

$$\Rightarrow \quad v_{2x} \cos 60° + v_{2y} \cos 30° = -v_{3y} \cos 30° + v_{3x} \cos 60°$$

$$\Rightarrow \quad v_{2x} + \sqrt{3}\, v_{2y} = -\sqrt{3}\, v_{3y} + v_{3x}$$

$$\Rightarrow \quad v_1 + \sqrt{3}\, v_{2y} = -\sqrt{3} \cdot \frac{2}{3} v_{2y} + v_{3x}$$

$$\Rightarrow \quad v_1 - v_{3x} = -\frac{5}{\sqrt{3}} v_{2y} \qquad \text{...(iii)}$$

For C: $J_2 \cos 30° = 3m\, v_{3y}$

$$\Rightarrow \quad J_2 \frac{\sqrt{3}}{2} = 3m \cdot \frac{2}{3} v_{2y}$$

$$\Rightarrow \quad J_2 = \frac{4}{\sqrt{3}} m\, v_{2y} \qquad \text{...(a)}$$

And $3mu - J_2 \sin 30° = 3m\, v_{3x}$

$$\Rightarrow \quad J_2 = 6m(u - v_{3x}) \qquad \text{...(b)}$$

From (a) and (b)

$$\frac{4}{\sqrt{3}} v_{2y} = 6(u - v_{3x}) \qquad \text{...(iv)}$$

Solving (ii), (iii) and (iv) gives $v_1 = \dfrac{2u}{19}$

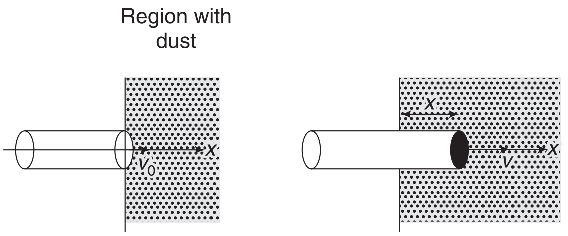
Example 37 A cylindrical solid of mass $m = 10^{-2}$ kg and cross-sectional area $A = 10^{-4}\,\text{m}^2$ is moving parallel to its axis (the x-axis) with a uniform speed of $10^3\,\text{ms}^{-1}$ in positive direction. At $t = 0$, its front face passes the plane $x = 0$. The region to the right of this plane is filled with stationary dust particles of uniform density, $\rho = 10^{-3}\,\text{kg m}^{-3}$. When a dust particle hits the face of the cylinder, it sticks to it. Assuming that the dimensions of the cylinder remains practically unchanged, and that the dust sticks only to the front face of the cylinder, find the x-coordinate of the front face of the cylinder at $t = 150$ s.

Solution

Concepts

(i) When the cylinder moves through a distance x, the dust particles present in the volume Ax will stick to the cylinder. Mass of the cylinder goes on increasing.

(ii) Momentum being conserved, the speed of cylinder decreases as more and more dust sticks to it. Using momentum conservation we can find its speed when it has moved through a distance x.



Volume of dust swept when the cylinder moves a distance x is $A \cdot x$

Mass of dust in this entire volume sticks to the cylinder. This mass is $\Delta m = \rho A \cdot x$

Momentum conservation gives

$$(m + \Delta m)v = mv_0$$

$$\Rightarrow \quad (m + \rho A x)v = mv_0$$

$$\Rightarrow \quad (m + \rho A x)\frac{dx}{dt} = mv_0$$

$$\Rightarrow \quad \int_0^x (m + \rho A x)\,dx = mv_0 \int_0^t dt$$

$$\Rightarrow \quad mx + \frac{1}{2}\rho A x^2 = mv_0 t$$

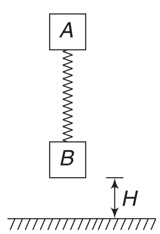
$$\Rightarrow \quad 10^{-2}x + \frac{1}{2} \times 10^{-3} \times 10^{-4}x^2 = 10^{-2} \times 10^3 \times 150$$

$$\Rightarrow \quad x^2 + (2 \times 10^5)x - 3 \times 10^{10} = 0$$

$$\text{Solving } x = \frac{-2 \times 10^5 \pm \sqrt{4 \times 10^{10} + 12 \times 10^{10}}}{2}$$

$$= 10^5 \text{ m}$$

Example 38 Two blocks A and B are connected to an ideal spring of force constant k. Masses of the blocks are m_1 and m_2 respectively. Initially, the system is kept vertical with spring relaxed and block B at a height H above the floor (see figure). The system is released from rest. Collision between B and the floor is perfectly inelastic. Find minimum value of H for which B will again bounce off the floor.



Solution

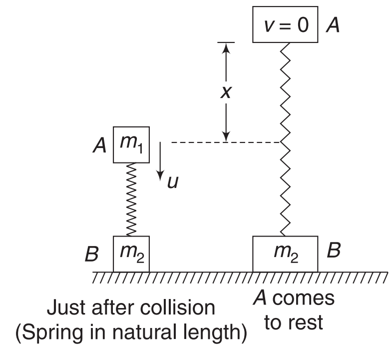
Concepts

(i) The system falls freely and speed of both A and B just before B hits the floor is $u = \sqrt{2gH}$.

(ii) B does not rebound since collision is perfectly inelastic. But A still has a velocity $u = \sqrt{2gH}$ immediately after B meets collision. A does not experience any impulse when B hits the floor.

(iii) Downward motion of A compresses the spring and A will begin to retard after some time. After A stops, it is pushed up by the spring and starts upward motion. It keeps moving up as the spring stretches. By the time it stops it must have stretched the spring to an extent such that the spring force exceeds the weight of B: then only B will bounce off the floor.

(iv) Mechanical energy is conserved for the system after the collision has taken place.



Speed of A just after B hits the floor is $u = \sqrt{2gh}$

Block A compresses the spring and then moves up. Let it stretch the spring by x when it comes to rest. B will leave the floor if

$$kx \geq m_2 g \qquad ..(i)$$

Conservation of energy gives:

$$\frac{1}{2}m_1u^2 = \frac{1}{2}kx^2 + m_1gx$$

Putting $\quad u = \sqrt{2gH} \quad$ and

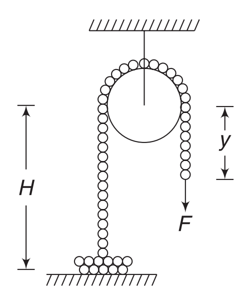
$x = \dfrac{m_2g}{k}$ from (i) for limiting case, we get

$$\frac{1}{2}m_1(2gH) = \frac{1}{2}k\left(\frac{m_2g}{k}\right)^2 + m_1g\left(\frac{m_2g}{k}\right)$$

$$\Rightarrow \quad H = \frac{m_2g}{k}\left(\frac{m_2 + 2m_1}{2m_1}\right)$$

This is the minimum value of H needed.

Example 39 A uniform chain has mass per unit length equal to λ and it is kept passing over a smooth pulley (of negligible dimension), as shown in figure. Height of pulley above the table is H and length of the chain on other side of the pulley is y. Height of heap on the table is negligible. The end of the chain is being pulled with a force F so that the chain moves at a constant speed v. Find F as a function of y.



Solution

Concepts

The heap applies force on the moving chain as small pieces are 'jerked' into motion.

An elemental length dy of the chain (in the heap) is set into motion in time dt.

Momentum imparted to the element is $dp = (\lambda dy)\cdot v$ Force applied by the vertical chain on the element to put it in motion is

$$F_{\text{th}} = \frac{dp}{dt} = \lambda v \frac{dy}{dt} = \lambda v^2$$

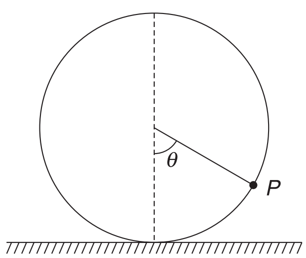
The heap applies equal force on the chain in downward direction. For moving part of the chain to have constant speed:

F + weight of length $y = F_{\text{th}}$ + weight of length H

$$\Rightarrow \quad F + \lambda yg = \lambda v^2 + \lambda Hg$$

$$\Rightarrow \quad F = \lambda v^2 + \lambda g(H - y)$$

Example 40 A ring of mass M and radius R is standing on a smooth horizontal table. A bead of mass m can slide smoothly along the ring. The system is released from rest with the bead at the top-most point. Find the velocity (u) of the centre of the ring at the instant the bead reaches a position P at an angle θ with vertical.



Solution

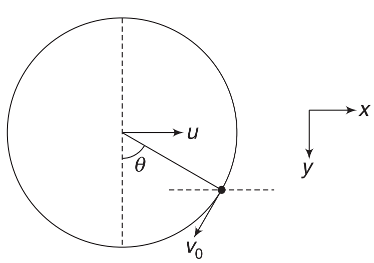
Concepts

(i) Direction of velocity of the bead relative to the ring is along the tangent on the ring.

(ii) Actual velocity of the bead is vector sum of relative velocity and velocity of the ring (u).

(iii) Horizontal momentum remains zero.

(iv) Mechanical energy is conserved.



Let v_0 be velocity of bead relative to the ring.

$$\vec{v}_0 = -v_0 \cos\theta\,\hat{i} + v_0 \sin\theta\,\hat{j}$$

Velocity of bead relative to the ground is

$$\vec{v} = \vec{v}_0 + \vec{u}$$
$$= (u - v_0\cos\theta)\hat{i} + (v_0 \sin\theta)\hat{j}$$

$$\therefore \quad v_x = u - v_0 \cos\theta;\ v_y = v_0 \sin\theta$$

Momentum is conserved in horizontal direction.

$$\therefore \quad mv_x + Mu = 0$$

$$\Rightarrow \quad m(u - v_0\cos\theta) = -Mu \qquad \text{...(i)}$$

Law of conservation of mechanical energy:

Gain in KE of ring + Gain in KE of bead = Loss in PE of bead

$$\Rightarrow \quad \frac{1}{2}Mu^2 + \frac{1}{2}m\,(v_x^2 + v_y^2) = mgR\,(1 + \cos\theta)$$

$$\Rightarrow \quad Mu^2 + m\,[(u - v_0\cos\theta)^2 + (v_0 \sin\theta)^2] = 2mgR\,(1 + \cos\theta)$$

$$\Rightarrow \quad Mu^2 + m\,[u^2 + v_0^2 - 2uv_0\cos\theta] = 2mgR\,(1 + \cos\,\theta)$$

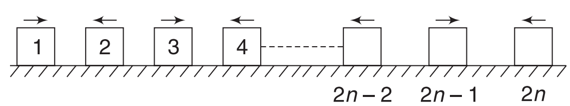
From (i) $v_0 = \dfrac{(m + M)u}{m\cos\theta}$

$$\therefore \quad u^2\left[M + m + \frac{(m + M)^2}{m\cos^2\theta} - 2(m + M)\right] = 2mgR(1 + \cos\theta)$$

Simplifying further gives:

$$u = m\cos\theta\sqrt{\frac{2gR\,(1 + \cos\theta)}{(M + m)\,(M + m\sin^2\theta)}}$$

Example 41 $2n$ number of identical elastic blocks are kept on a smooth horizontal table at uniform separation (n is an integer). All odd-numbered blocks are imparted a velocity u towards right and all even-numbered blocks are imparted a velocity u towards left. All the blocks start simultaneously. How many collision will occur in all?

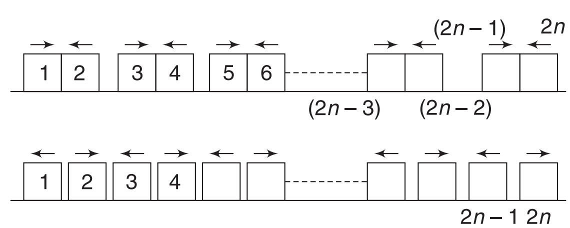


Solution

Concepts

In head-on elastic collision of equal masses, velocities get exchanged.

In first round, there will be n collisions, as shown below



After first collision, block 1 and $2n$ will leave the system and will not participate in any further collision. Now there are $(2n-2)$ blocks [Block 2 to $(2n-1)$] and they are travelling just like the initial system. There will be $(n-1)$ collisions in the next round (2 and 3, 4 and 5, ….. collide). After this, 2 and $(2n-1)$ will permanently leave and the system will have $(2n-4)$ blocks left. They will further make $(n-2)$ collisions. Process will continue and in the end, only two middle blocks will collide and then move away from each other.

$\therefore$ Total number of collisions

$$= n + (n-1) + (n-2) + ... + 2 + 1$$

= Sum of first 'n' natural numbers

$$= \frac{n(n-1)}{2}$$

Example 42 A ball moving translationally collides elastically with another stationary ball of same mass. At the instant of collision, the angle between the straight line passing through the centres of the balls and the direction of initial motion of the striking ball is $\alpha = 45°$. Assuming the balls to be smooth, find the fraction (η) of kinetic energy of the striking ball that was used up in deformation at the moment of maximum deformation.

Solution

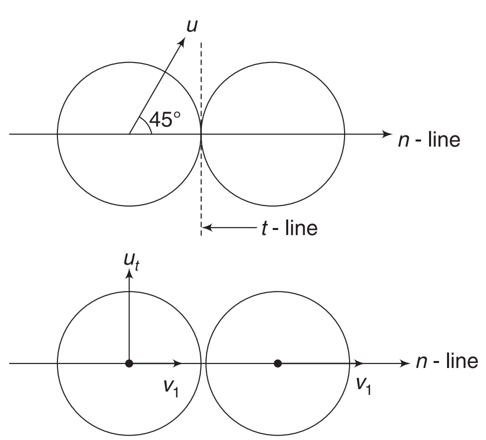
Concepts

(i) At the instant of maximum deformation, the velocity components of two balls along the LOI (i.e., n-line) will be same.

Velocity component along t-line does not change.

(ii) Using momentum conservation along n-line will give us the velocity (along n-line) of both balls when maximum deformation occurs.

(iii) Deformation PE = Loss in KE



At the instant of maximum deformation.

Components of initial velocity of moving ball are

$$u_n = u\cos 45° = \frac{u}{\sqrt{2}}$$

$$u_t = u\sin 45° = \frac{u}{\sqrt{2}}$$

When the balls are deformed by maximum amount, let the common velocity along n-line be v_1. [The second ball has no velocity along t-line]. Momentum conservation along n-line gives:

$$2mv_1 = m\frac{u}{\sqrt{2}}$$

$$\Rightarrow \qquad v_1 = \frac{u}{2\sqrt{2}}$$

At this instant speed of first ball is $\sqrt{v_1^2 + u_t^2}$

$$= \sqrt{\frac{u^2}{8} + \frac{u^2}{2}} = \frac{\sqrt{5}}{2\sqrt{2}}u$$

$\therefore$ KE at the instant of maximum deformation is

$$= \frac{1}{2}m\left(\frac{\sqrt{5}}{2\sqrt{2}}u\right)^2 + \frac{1}{2}m\left(\frac{u}{2\sqrt{2}}\right)^2$$

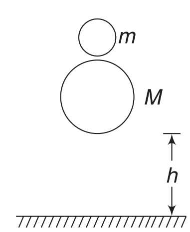
$$= \frac{3}{4}\left(\frac{1}{2}mu^2\right)$$

$\therefore$ Fraction of KE stored as deformation energy is

$$\eta = \frac{\frac{1}{2}mu^2 - \frac{3}{4}\left(\frac{1}{2}mu^2\right)}{\frac{1}{2}mu^2} = \frac{1}{4} = 0.25$$

Example 43 *Ball over ball makes a super ball!*

A small ball of mass m is held above another ball of mass M with a little gap between them. The bigger ball is at height h above the floor. The balls are released simultaneously. How high will the smaller ball rise after collision? Assume that the collisions are elastic and $M \gg m$.



Solution

Concepts

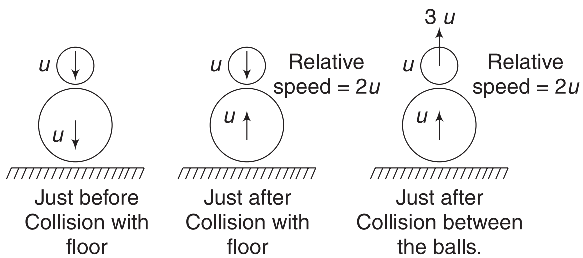
(i) Both balls fall freely to a distance h before the bigger ball hits the floor.

(ii) On elastic collision with the floor, velocity of the bigger ball gets reversed without change in magnitude.

(iii) As the velocity of bigger block becomes vertically upward, the smaller ball is still travelling downward. Thus, relative velocity of approach is high for the two balls. Relative velocity of separation will also be high.

(iv) Motion of bigger ball remains unaffected, when it hits the smaller ball, due to its large mass.

Speed of the balls just before hitting the floor, $u = \sqrt{2gh}$



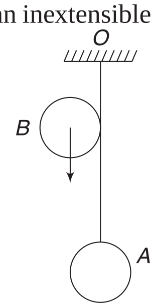
Velocity of bigger balls after colliding with floor $= u(\uparrow)$

For collision between the balls, relative speed of approach $= 2u$. Since collision is elastic, relative speed of separation after collision $= 2u$. The bigger ball keeps moving with velocity $u(\uparrow)$. Thus, velocity of smaller ball becomes $3u(\uparrow)$.

$\therefore$ Height attained by smaller ball is $H = \dfrac{(3u)^2}{2g} = 9h$

This is the reason we are calling it a super ball (!).

Example 44 A sheet ball A is suspended by an inextensible string of length $L = 1.5\,\text{m}$, from point O. Another identical ball B is projected downward such that it moves while just remaining in touch with the thread (see fig.). It collides elastically with A. Ball A just manages to complete the vertical circle after collision. Find the velocity (u) of the falling ball just before collision.



Solution

Concepts

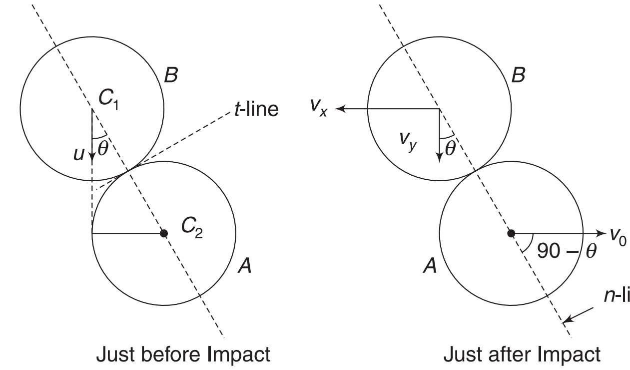
(i) LOI is the line joining the centres of the two balls. By looking at geometry, we can find the angle between initial velocity (u) of B and the LOI.

(ii) Due to impulse applied by the string (during collision), we cannot use conservation of momentum along n-line. Momentum is conserved only in horizontal direction.

(iii) We will use $-\left(\dfrac{v_{1n} - v_{2n}}{u_{1n} - u_{2n}}\right) = 1$ for elastic collision.

(iv) Velocity of A, immediately after impact is known, as it is given that it completes the circle.

(v) Velocity of B does not change along t-line of collision.



From geometry: $\sin\theta = \dfrac{r}{2r}$ [r = radius of each ball]

$\Rightarrow \qquad \theta = 30°$

Velocity of A just after impact is $v_0 = \sqrt{5gL}$, as it is just able to complete the circle.

$$\Rightarrow \qquad v_0 = \sqrt{5 \times 10 \times 1.5} = 5\sqrt{3}\ \text{ms}^{-1}$$

Let velocity components of B along horizontal and vertical directions be v_x and v_y, as shown.

Using conservation of momentum along horizontal direction

$$mv_x = mv_0 \quad \Rightarrow \quad v_x = v_0 = 5\sqrt{3}\ \text{ms}^{-1}$$

Velocity of B along n-line:

before collision is $u\cos\theta = \dfrac{\sqrt{3}}{2}u$

after collision is $v_y\cos\theta - v_x\sin\theta$

$$= \frac{\sqrt{3}}{2}v_y - \frac{1}{2} \times 5\sqrt{3}$$

Velocity of A along n-line:

before collision = 0

after collision $= v_0\cos(90 - \theta) = v_0\sin\theta$

$$= \frac{5\sqrt{3}}{2}$$

Now, $-\left(\dfrac{v_{1n} - v_{2n}}{u_{1n} - u_{2n}}\right) = 1$

$$\Rightarrow \qquad -\left(\frac{\dfrac{\sqrt{3}}{2}v_y - \dfrac{5\sqrt{3}}{2} - \dfrac{5\sqrt{3}}{2}}{\dfrac{\sqrt{3}}{2}u - 0}\right) = 1$$

$$\Rightarrow \qquad -\frac{\sqrt{3}}{2}v_y = \frac{\sqrt{3}}{2}u - 5\sqrt{3}$$

$$\Rightarrow \qquad v_y = 10 - u \qquad \text{...(i)}$$

Velocity of B will remain unchanged along t-line.

$$\therefore \qquad u\sin\theta = v_x\cos\theta + v_y\sin\theta$$

$$\Rightarrow \qquad \frac{u}{2} = 5\sqrt{3}\cdot\frac{\sqrt{3}}{2} + \frac{v_y}{2}$$

$$\Rightarrow \qquad v_y = u - 15 \qquad \text{...(ii)}$$

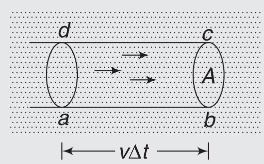
From (i) and (ii), $u = 12.5\,\text{ms}^{-1}$

Example 45 A beam of particles has each particle of mass m moving with velocity v. The beam has n number of particles per unit volume. The beam strikes against a wall at an angle θ to the normal. Collisions are elastic. Find the pressure exerted by the stream on the wall.

Solution

Concepts

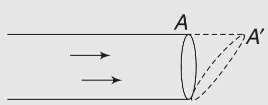
(i) If we consider a cross-sectional area A normal to the beam, then all the particles located inside cylindrical volume (abcd) will cross through the area A in time Δt.



Volume ($abcd$) = $Av\Delta t$

Number of particles crossing area A in time $\Delta t = nAv\Delta t$

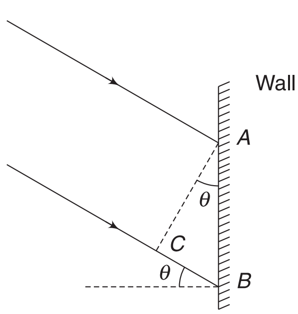
(ii) Consider an area A that is not exactly normal to the beam but its projection in a direction perpendicular to the beam is A. Number of particles crossing through A' in time Δt = number of particles crossing through A in time $\Delta t = nAv\Delta t$.



(iii) Each particle experiences change in momentum in a direction that is normal to the wall. There is no change in momentum parallel to the wall.

(iv) $F = \dfrac{\Delta P}{\Delta t}$

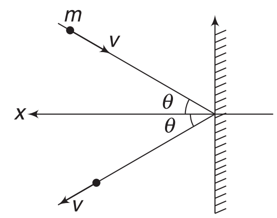
Let the area AB on the wall where the beam strikes be A. Projection of area A normal to the beam is area $(AC) = A\cos\theta$



$\therefore$ Number of particles striking the wall in time Δt = number of particles crossing area $A\cos\theta$ in time $\Delta t = n(A\cos\theta)v\Delta t$.

Change in momentum of one particle is

$$\Delta P_{\text{one}} = mv\cos\theta(\leftarrow) - mv\cos\theta\,(\rightarrow)$$
$$= 2mv\cos\theta\,(\leftarrow)$$



$\therefore$ Change in momentum of all particles hitting the wall in time Δt is

$$\Delta P = [2mv\cos\theta]\,[nA\cos\theta\cdot v\Delta t]$$

Force applied by wall on particles = force by particles on the wall

$$F = \frac{\Delta P}{\Delta t} = 2mnv^2A\cos^2\theta$$

$$\therefore \quad \text{Pressure} = \frac{F}{A} = 2mnv^2\cos^2\theta$$

Example 46 Two bodies of same mass are tied with an inelastic string of length l. They lie together on a floor. One of them is projected vertically upwards with velocity $\sqrt{6gl}$. Find the maximum height up to which the centre of mass of system of the two masses rise.

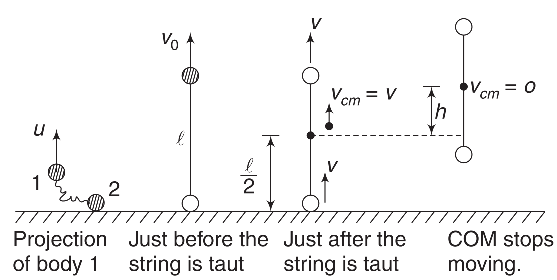
Solution

Concepts

(i) Initially, the projected body (call it body 1) moves under gravity. There is no tension in the string.

(ii) Once body 1 moves through a distance l, the string acquires tension. During the process of getting taut, the string applies an impulse on both the bodies. This sets the second body in motion. In fact, both will have same velocity after the string gets taut.

(iii) The COM of the system moves with acceleration $g(\downarrow)$ after both bodies are in air.



Let velocity of first body be v_0 immediately before the string is taut.

$$v_0 = \sqrt{u^2 - 2gl} = \sqrt{4gl}$$

COM of the system is presently at height $\dfrac{l}{2}$.

Just after the string gains tension, let velocity of two bodies be v. Momentum conservation gives

$$2mv = mv_0 \quad \Rightarrow \quad v = \frac{v_0}{2} = \sqrt{gl}$$

Velocity of the COM at this instant is $v_{CM} = \sqrt{gl}$

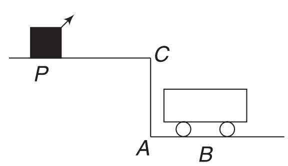
COM now moves with acceleration $g(\downarrow)$ as the only external force on the system is weight. If COM rises further by h then:

$$0^2 = v_{CM}^2 - 2g \cdot h$$

$$\Rightarrow \qquad h = \frac{v_{CM}^2}{2g} = \frac{l}{2}$$

$\therefore$ Total rise of COM $= \frac{l}{2} + \frac{l}{2} = l$

Example 47 A car P is moving with a uniform speed of $5\sqrt{3}$ ms^{-1} towards a carriage of mass 9 kg at rest kept on the rails at a point B, as shown in Fig. The height AC is 120 m. Cannon balls of 1 kg are fired from the car with an initial velocity 100 ms^{-1} at an angle 30° with the horizontal. The first cannon ball hits the stationary carriage after a time t_0 and sticks to it. Determine t_0. At t_0, the second cannon ball is fired. Assume that the resistive force between the rails and the carriage is constant and ignore the vertical motion of the carriage throughout. The second ball also hits and sticks to the carriage. What will be the horizontal velocity of the carriage just after the second impact?



Solution

Concepts

(i) Velocity of cannon balls is 100 ms^{-1} relative to the ground (and not relative to the cannon). Unless it is mentioned otherwise, we should take a given velocity as observed from the ground.

(ii) Just considering the vertical motion of the projectile can give us t_0

(iii) The cannon ball has horizontal momentum when it hits the carriage. Momentum conservation gives velocity of carriage after the ball sticks to it.

(iv) The second ball will have path (and time of flight) identical to the first one. The second ball is fired at the instant the carriage starts moving. For a hit, it is necessary that displacement of the car between two shots = displacement of carriage during time of flight of the second ball.

For cannon ball, horizontal and vertical components of initial velocity are

$$u_x = u\cos 30° = 100 \cdot \frac{\sqrt{3}}{2} = 50\sqrt{3}\,\text{ms}^{-1}$$

$$u_y = u\sin 30° = 100 \cdot \frac{1}{2} = 50\,\text{ms}^{-1}\ (\uparrow)$$

Vertical displacement of the ball by the time it hits the carriage is 120 m ($\downarrow$)

$\therefore \qquad y = u_y t + \frac{1}{2} a_y t^2$ gives

$$-120 = 50 \cdot t - \frac{1}{2} \times 10 \times t^2 \quad \Rightarrow t = 12\,\text{s}, -2\,\text{s}$$

Obivously, negative time in meaningless.

$\therefore$ time of flight, $t_0 = 12$ s

When it hits the carriage, its horizontal velocity is still u_x. Conserving momentum along horizontal direction gives:

(Mass of ball) u_x = (Mass of ball + carriage) v_1

$$\Rightarrow v_1 = \frac{1 \times 50\sqrt{3}}{10} = 5\sqrt{3}\ \text{ms}^{-1}\ [= \text{velocity of carriage}]$$

The second ball is fired at $t_0 = 12$ s after the first ball. The car moves horizontally a distance $x_0 = (5\sqrt{3}\ \text{ms}^{-1})(12\,\text{s}) = 60\sqrt{3}$ m between two shots.

The trajectory of ball 2 is identical to that of ball 1

$\therefore$ Ball 2 will have same range as 1.

Thus, ball 2 will hit the carriage at a distance $x_0 = 60\sqrt{3}$ m from the spot where first hit happened.

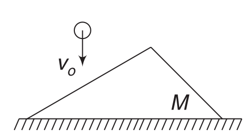
$\therefore$ Carriage travels $60\sqrt{3}$ m in 12 s. This will be possible only when resistive force on the carriage is zero as $v_1 t_0 = 5\sqrt{3} \times 12 = 60\sqrt{3}$ m. The carriage will cover a distance of $60\sqrt{3}$ m in 12 s while travelling with constant velocity $v_1 = 5\sqrt{3}$ ms^{-1}.

Again, using conservation of momentum in horizontal direction: Momentum of carriage + first ball in horizontal direction + momentum of second ball in horizontal direction = Momentum of carriage + 2 balls in horizontal direction.

$$\Rightarrow \qquad (10)(5\sqrt{3}) + (1)(50\sqrt{3}) = 11 v_2$$

$$\Rightarrow \qquad v_2 = \frac{100\sqrt{3}}{11} = 15.75\ \text{ms}^{-1}$$

Example 48 A ball of mass $m = 1$ kg in falling vertically at a velocity $v_0 = 2$ ms^{-1} when it hits a smooth inclined face of a wedge of mass $M = 2$ kg. The wedge is on a smooth table and is initially at rest. The coefficient of restitution is $e = \frac{1}{2}$. Find the velocity of the wedge and the ball after collision. Inclination angle of the wedge is $\theta = 30°$.



Solution

Concepts

(i) Conservation of momentum along horizontal direction, definition of coefficient of restitution and the fact that velocity component of the ball along *t*-line (i.e., a line parallel to wedge surface) does not change; are sufficient to solve the problem.

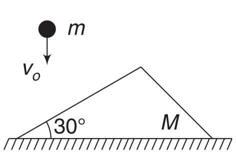
(ii) To illustrate a different line of thinking, we will solve this problem in terms of impulse between the ball and the inclined face of the wedge. However, we will still need the defining equation for coefficient of restitution.

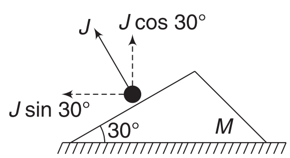
Let v_1 = velocity of wedge (towards right) after impact

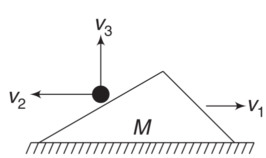
v_2 = velocity component of the ball in horizontal direction (towards left)

v_3 = velocity component of the ball in vertical direction (up) after collision.

J = impulse between the ball and the wedge (perpendicular to the incline).







Impulse = change in momentum

$$\therefore \quad J = Mv_1 = mv_2$$

$$\Rightarrow \quad J = 2v_1 = v_2 \qquad \text{...(i)}$$

Also, $J\cos 30° = mv_3 - (-mv_0) = m(v_3 + v_0)$

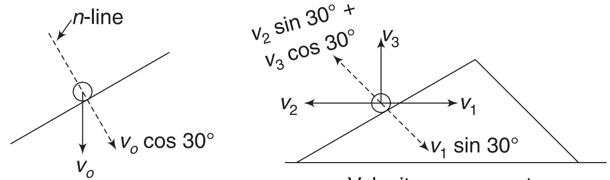
$$\Rightarrow \quad \frac{\sqrt{3}}{2} J = (v_3 + 2) \qquad \text{...(ii)}$$

$$\text{Also} \quad e = -\left(\frac{v_{1n} - v_{2n}}{u_{1n} - u_{2n}}\right)$$

$$\Rightarrow \quad \frac{1}{2} = -\left[\frac{v_1 \sin 30° - (-v_2 \sin 30° - v_3 \cos 30°)}{0 - v_0 \cos 30°}\right]$$

$$\Rightarrow \quad \frac{1}{2} v_0 \cos 30° = (v_1 + v_2)\sin 30° + v_3 \cos 30°$$

$$\Rightarrow \quad v_1 + v_2 + \sqrt{3}\, v_3 = \sqrt{3} \qquad \text{...(iii)}$$



Before collision

Velocity components along *n*-line.

Solving (i) , (ii) and (iii) gives:

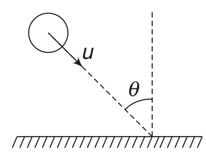
$$v_1 = \frac{1}{\sqrt{3}} \text{ ms}^{-1}$$

$$v_2 = \frac{2}{\sqrt{3}} \text{ ms}^{-1} \quad \text{and}$$

$$v_3 = 0$$

Thus, the ball moves horizontally to the left with velocity $\frac{2}{\sqrt{3}}$ ms^{-1}and the wedge moves to the right with a velocity $\frac{1}{\sqrt{3}}$ms^{-1}.

Example 49 *A ball hitting a rough surface*



A ball of mass m hits a flat horizontal surface while travelling with a velocity u. The angle of incidence is θ. Coefficient of restitution is e and coefficient of friction is μ. Find the velocity of the ball after impact.

Solution

Concepts

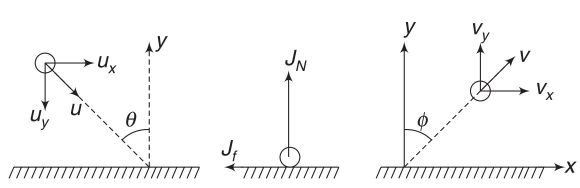
(i) Vertical component of velocity of the ball before collision is velocity of approach and vertical component of ball's velocity after collision is velocity of separation. We can use (speed of separation) = e (speed of approach).

(ii) Change in momentum of the ball in vertical direction is due to the impulse of normal reaction force that has acted on it.

(iii) Impulse of normal force $J_N = N\Delta t$

Impulse of friction force $J_f = f\Delta t = \mu N \Delta t$

(iv) Horizontal impulse by friction causes change in horizontal momentum of the ball.



Before Collision

Impulse during collision

After Collision

Speed of separation = e (speed of approach)

$$\Rightarrow \quad v_y = eu_y = eu\cos\theta \quad \text{...(i)}$$

Impulse in vertical direction = change in momentum in vertical direction

$$J_N = mv_y + mu_y \quad [J_N = N\Delta t]$$
$$= mu\cos\theta\,(e+1) \quad \text{...(ii)}$$

Impulse of friction, $J_f = \mu N\Delta t = \mu J_N$

$$= \mu mu\cos\theta\,(e+1)$$

J_f = change in horizontal momentum.

$$\mu mu\cos\theta\,(e+1)(\leftarrow) = mv_x(\rightarrow) - mu_x(\rightarrow)$$

$$\Rightarrow \quad -\mu mu\cos\theta\,(e+1) = mv_x - mu\sin\theta$$

$$\Rightarrow \quad v_x = u\sin\theta - \mu u\cos\theta\,(e+1) \quad \text{...(iii)}$$

$\therefore$ Speed of ball after impact is

$$v = \sqrt{v_x^2 + v_y^2}$$
$$= u\sqrt{[\sin\theta - \mu(e+1)\cos\theta]^2 + (e\cos\theta)^2}$$

Direction of motion makes angle ϕ with vertical

$$\tan\phi = \frac{v_x}{v_y} = \frac{\sin\theta - \mu\cos\theta\,(e+1)}{e\cos\theta} = \frac{\tan\theta}{e} - \frac{\mu(e+1)}{e}$$

Example 50 *Maximum deflection*

A mass M collides with a stationary mass m. Assume that $M > m$ and find the maximum deflection angle for M. Assume elastic collision.

Solution

Concepts

(i) If $M < m$, then it is possible for M to bounce directly backward (i.e, deflection = 180°). But for $M > m$, there is a maximum angle less than 180°.

(ii) It is possible to solve this problem in ground frame but we can save a lot of effort by considering what happens in COM frame and then shifting back to the ground frame.

(iii) COM frame is zero momentum frame. Momentum of system is zero, before and after collision, in this frame.

(iv) If u is velocity in ground frame, then velocity in COM frame is $u' = u - v_{CM}$.

Let initial velocity of M in ground frame be u_0

$$v_{CM} = \frac{Mu_0}{M+m}$$

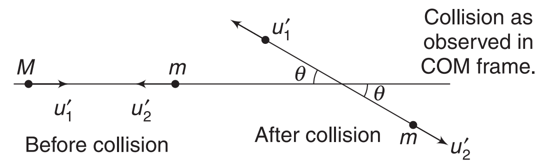
Velocity of M in COM frame (before collision) is

$$u_1' = u_0 - \frac{Mu_0}{M+m} = \frac{mu_0}{M+m}$$

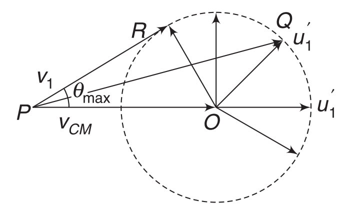
Velocity of m in COM frame (before collision) is

$$u_2' = 0 - \frac{Mu_0}{M+m} = -\frac{Mu_0}{M+m}$$

After collision, the two masses will move with their speed unchanged when observed in COM frame. Their velocities will be oppositely directed. This is necessary to conserve momentum and energy.



We do not know θ. It can have any value. Thus, velocity u_1' (velocity of M in COM frame) can have any direction. We have drawn a circle of radius $|u_1'|$ and various radii represent the various possible directions of u_1'.



In ground frame, velocity of M is $\vec{v}_1 = \vec{u}_1' + \vec{v}_{CM}$.

Take $\quad \overrightarrow{PO} = \vec{v}_{CM}$

Now, when $\overrightarrow{OQ}$ represent u_1' we have

$$\overrightarrow{PQ} = \overrightarrow{PO} + \overrightarrow{OQ}$$

$$\vec{v}_1 = \vec{v}_{CM} + \vec{u}_1'$$

Thus, $\overrightarrow{PQ}$ is velocity of M in ground frame. This way, we can construct $\vec{v}_1$ for different direction of u_1'.

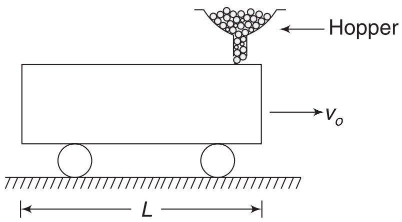
Angle between $\vec{v}_1$ and $\vec{v}_{CM}$ (which is original line of motion of M) is maximum when the vector drawn from P to the tip of u_1' becomes tangent to the circle shown.

$$\therefore \quad \sin\theta_{max} = \frac{OR}{PO} = \frac{u_1'}{v_{CM}} = \frac{m}{M}$$

Example 51 *Loading at a mine*

In a coal mine, the produced coal is fed to a hopper, which discharges it into railway wagons at a constant rate of b kgs^{-1}. An empty wagon having mass m_0 approaches the hopper while travelling at speed v_0 on a smooth track. Length of the car is L. As soon as the right end of the car is below the hopper, it begins to discharge.

Find the mass and speed of the wagon just after it moves out of the hopper.



Solution

Concepts

(i) The falling coal brings no momentum with itself in horizontal direction.

(ii) We can easily write velocity of the wagon at time t by using conservation of momentum in horizontal direction.

Let the wagon move through a distance x in time t.

Its mass at time t is $m = m_0 + bt$.

Conserving momentum:

$$mv = m_0 v_0$$

$$\Rightarrow \quad v = \frac{m_0 v_0}{m_0 + bt} \qquad \text{...(i)}$$

$$\Rightarrow \quad \frac{dx}{dt} = \frac{m_0 v_0}{m_0 + bt} \quad \Rightarrow \quad \int_0^L dx = m_0 v_0 \int_0^t \frac{dt}{m_0 + bt}$$

$$\Rightarrow \quad L = \frac{m_0 v_0}{b} \left[ln\,(m_0 + bt)\right]_0^t$$

$$\Rightarrow \quad L = \frac{m_0 v_0}{b}\, ln\left(\frac{m_0 + bt}{m_0}\right)$$

$$\Rightarrow \quad ln\left(1 + \frac{bt}{m_0}\right) = \frac{bL}{m_0 v_0}$$

$$\Rightarrow \quad 1 + \frac{bt}{m_0} = e^{\frac{bL}{m_0 v_0}}$$

$$\Rightarrow \quad m_0 + bt = m_0 e^{\frac{bL}{m_0 v_0}} = \text{mass of the wagon at time } t.$$

$$\text{From (i) } v = \frac{m_0 v_0}{m_0\, e^{\frac{bL}{m_0 v_0}}} = v_0\, e^{-\frac{bL}{m_0 v_0}}$$

Note: Using the concept of variable mass will give same result.

Worksheet 1

1. A system of N particles is free from any external forces. Which of the following is true for the magnitude of the total momentum of the system?

(a) It must be zero
(b) It could be non-zero, but it must be constant
(c) It could be non-zero, and it might not be constant
(d) The answer depends on the nature of the internal forces in the system

2. A system of N particles is free from any external forces. Which of the following must be true for the sum of the magnitudes of the momenta of the individual particles in the system?

(a) It must be zero
(b) It could be non-zero, but it must be constant
(c) It could be non-zero, and it might not be constant
(d) It could be zero, even if the magnitude of the total momentum is not zero

3. If the KE of a particle becomes four times its initial value, then the new momentum will be more than its initial momentum by

(a) 50% (b) 100%
(c) 125% (d) 150%

4. If the KE of a particle becomes 0.2% higher than its initial value, then the new momentum will be higher than its initial momentum by;

(a) 0.2% (b) 0.1%
(c) 0.125% (d) 1.50%

5. Two masses of 1 g and 4 g are moving with equal kinetic energy. The ratio of the magnitude of their linear momentum is-

(a) 1:1 (b) 1:2
(c) 1:3 (d) 1:4

6. A train of mass M is moving on a circular track of radius 'R' with constant speed v. The length of the train is half of the perimeter of the track. The magnitude of linear momentum of the train will be

(a) 0 (b) $\frac{2Mv}{\pi}$
(c) MvR (d) Mv

7. A ball of mass 50 g is dropped from a height h = 10 m. It rebounds losing 75% of its kinetic energy. If it remains in contact with the ground for Δt = 0.01 s, the impulse of the impact force is: (take g = 10 ms^{-2})

(a) 1.3 Ns (b) 1.06 Ns
(c) 1300 Ns (d) 105 Ns

8. An impulse $\vec{I}$ changes the velocity of a particle from $\vec{v}_1$ to $\vec{v}_2$. Kinetic energy gained by the particle is

(a) $\frac{1}{2}\vec{I}.(\vec{v}_1 + \vec{v}_2)$ (b) $\frac{1}{2}\vec{I}.(\vec{v}_1 - \vec{v}_2)$
(c) $\vec{I}.(\vec{v}_1 - \vec{v}_2)$ (d) $\vec{I}.(\vec{v}_1 + \vec{v}_2)$

9. A super-ball is to bounce elastically back and forth between two rigid walls, at a distance d from each other. Neglecting gravity and assuming the velocity of super-ball to be v_0 horizontal, the average force (in large time interval) being exerted by the super-ball on one wall is:

(a) $\frac{1}{2}\frac{mv_0^2}{d}$ (b) $\frac{mv_0^2}{d}$
(c) $\frac{2mv_0^2}{d}$ (d) $\frac{4mv_0^2}{d}$

10. A particle of mass $4m$, which is at rest, explodes into three fragments. Two of the fragments, each of mass m, are found to move with a speed 'v' each in mutually perpendicular directions. The minimum energy released in the process of explosion is:

(a) (2/3) mv^2 (b) (3/2) mv^2
(c) (4/3) mv^2 (d) (3/4) mv^2

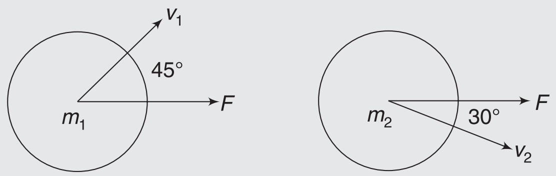
11. A 500 kg boat has an initial speed of 10 ms^{-1} as it passes under a bridge. At that instant, a 50 kg man jumps straight down into the boat from the bridge. The speed of the boat after the man and boat attain a common speed is (neglect any resistance)

(a) $\frac{100}{11}$ ms^{-1} (b) $\frac{10}{11}$ ms^{-1}
(c) $\frac{50}{11}$ ms^{-1} (d) $\frac{5}{11}$ ms^{-1}

12. A shell is fired from a canon with a velocity v at an angle θ with the horizontal direction. At the highest point in its path, it explodes into two pieces of equal masses. One of the pieces come to rest. The speed of the other piece immediately after the explosion is

(a) $3v\cos\theta$ (b) $2v\cos\theta$
(c) $\frac{3}{2}v\cos\theta$ (d) $v\cos\theta$

13. Two pucks are initially moving along a frictionless surface, as shown in the diagram. The pucks have masses $m_1 < m_2$ and have equal magnitude of momentum. A constant force F is applied to each puck directly to the right for the same amount of non-zero time. After the pushes are complete, what is the relationship for the size of the momenta of pucks (P_1 and P_2)?

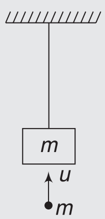


(a) $P_1 < P_2$
(b) $P_1 = P_2$
(c) $P_1 > P_2$
(d) More information about the masses, speeds, force and time are required to answer the questions

14. A spacecraft of mass M moves with velocity v in free space at first, then it explodes breaking into two pieces. If after explosion, a piece of mass m comes to rest, the other piece of space craft will have a velocity:

(a) $Mv/(M - m)$ (b) $Mv/(M + m)$
(c) $mv/(M - m)$ (d) $mv/(M + m)$

15. A bullet of mass m moving vertically upwards with velocity 'u' hits a hanging block of mass 'm' and gets embedded in it. The block is hanging with the help of a string, as shown in the figure. The height through which the block rises after collision is (assume sufficient space above block)



(a) $u^2/2g$ (b) u^2/g
(c) $u^2/8g$ (d) $u^2/4g$

16. A stationary body explodes into four identical fragments such that three of them fly off mutually perpendicular to each other, each with same kinetic energy E_0. The minimum energy of explosion will be:

(a) $6E_0$ (b) $\frac{4E_0}{3}$
(c) $4E_0$ (d) $8E_0$

17. A continuous stream of particles of mass m and velocity v, is emitted from a source at a rate of n per second. The particles travel along a straight line, collide with a body of mass M and get embedded into it. If the body of mass M was originally at rest, its velocity when it has received N particles will be

(a) $\frac{mvn}{Nm + n}$ and incresing

(b) $\frac{mvN}{Nm + M}$ and increasing

(c) $\frac{mv}{Nm + M}$ and decreasing

(d) $\frac{Nm + M}{mv}$ and increasing

18. A gun fires small balls of mass 20 gm. It is firing 20 balls per second on a smooth horizontal table. The collisions are perfectly elastic and balls strike at the centre of table with a speed 5 ms^{-1} at an angle of 60° with the vertical. The force exerted by one of the leg on ground is (assume total weight of the table is 0.2 kg and it is a four legged table):

(a) 0.5 N (b) 1 N
(c) 0.25 N (d) 0.75 N

19. Consider the following two statements:-

A: Linear momentum of a system of particles is zero.

B: Kinetic energy of a system of particles is zero.

Then-

(a) A does not imply B and B does not imply A
(b) A implies B but B does not imply A
(c) A does not imply B but B implies A
(d) A implies B and B implies A

20. A ball of mass m travelling with velocity $5v$ collides with and sticks to a ball of mass $5m$ travelling in the same direction with velocity v. Impulse applied by a ball on the other during collision is

(a) $5mv$ (b) $\frac{6mv}{5}$
(c) $\frac{8mv}{10}$ (d) $\frac{10mv}{3}$

21. Which of the following does not hold when two particles of masses m_1 and m_2 undergo elastic collision?

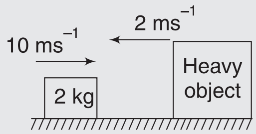
(a) When $m_1 = m_2$ and m_2 is stationary, there is maximum transfer of kinetic energy in head-on collision.
(b) When $m_1 = m_2$ and m_2 is stationary, there is maximum transfer of momentum in head-on collision.
(c) When $m_1 >> m_2$ and m_2 is stationary, after head-on collision, m_2 moves with twice the velocity of m_1.
(d) When the collision is oblique and $m_1 = m_2$, with m_2 stationary, after the collision the particle move in opposite directions.

22. A particle moves in the X-Y plane under the influence of a force such that its linear momentum is

$P(t) = A[\hat{i}\cos(kt) - \hat{j}\sin(kt)]$, where A and k are constants. The angle between the force and the momentum is:

(a) 0° (b) 30°
(c) 45° (d) 90°

23. A block of mass 2 kg is pushed towards a very heavy object moving with 2 ms^{-1} as shown. Assuming elastic collision, the velocity of 2 kg mass after collision is:



(a) 12 ms^{-1} (←) (b) 14 ms^{-1} (←)
(c) 2 ms^{-1} (←) (d) 8 ms^{-1} (→)

24. A ball is dropped on a floor from a height h. If the coefficient of restitution is e, to what height will it rise after third collision:

(a) h (b) e^2h
(c) e^3h (d) none of these

25. A sphere of mass m, moving horizontally with velocity v, enters a hanging bag of sand and stops. The bag was hanging with the help of a massless rope. If the mass of the bag is M and it is raised by height h, then the velocity of the sphere was

(a) $\dfrac{M+m}{m}\sqrt{2gh}$ (b) $\dfrac{M}{m}\sqrt{2gh}$

(c) $\dfrac{m}{M+m}\sqrt{2gh}$ (d) $\dfrac{m}{M}\sqrt{2gh}$

26. A particle of mass m moving with horizontal speed 6 ms^{-1} as shown in figure. If $m << M$, then for one-dimensional elastic collision, the speed of lighter particle after collision will be:

(a) 2 ms^{-1} in original direction
(b) 2 ms^{-1} opposite to the original direction
(c) 4 ms^{-1} opposite to the original direction
(d) 4 ms^{-1} in original direction

27. A body falling from a height of 10 m rebounds from hard floor. If it loses 20% energy in the impact, then coefficient of restitution is:

(a) 0.89 (b) 0.56
(c) 0.23 (d) 0.18

28. A body of mass M_1 collides elastically with another mass M_2 at rest. There is maximum transfer of kinetic energy when:

(a) $M_1 > M_2$
(b) $M_1 < M_2$
(c) $M_1 = M_2$
(d) same for all values of M_1 and M_2

29. A drop of liquid moving vertically down at 2 ms^{-1} it hit by another drop of half volume, moving horizontally at 1 ms^{-1}. The drops coalesce and move together. The velocity after impact will be:

(a) $\dfrac{1}{\sqrt{3}}$ ms^{-1} (b) $\dfrac{\sqrt{17}}{3}$ ms^{-1}

(c) $3\sqrt{3}$ ms^{-1} (d) $3\sqrt{17}$ ms^{-1}

30. A ball of mass m falls vertically to the ground from a height h_1 and rebound to a height h_2. The change in momentum of the ball on striking the ground is:

(a) $mg(h_1 - h_2)$ (b) $m\left(\sqrt{2gh_1} + \sqrt{2gh_2}\right)$

(c) $m\sqrt{2g(h_1 + h_2)}$ (d) $m\sqrt{2g}(h_1 + h_2)$

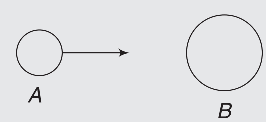
31. A spring scale is adjusted to read zero. Particles of mass 1 g fall on the pan of the scale and collide elastically and they rebound upward with the same speed. If the height of fall of particles is 2 m and their rate of collision is 100 particles per second, then the scale reading in grams will be: (g = 10 ms^{-2})

(a) 1,000 g (b) 1,100 g
(c) 120.0 g (d) 126.5 g

32. Three interacting particles of masses 100, 200 and 400 g have each a velocity 20 ms^{-1} along the positive direction of x-axis, y-axis and z-axis, respectively. Due to the force of interaction, the third particle stops moving and the velocity of second particle becomes $(10\hat{j} + 5\hat{k})$ ms^{-1}. Then the velocity of the first particle is

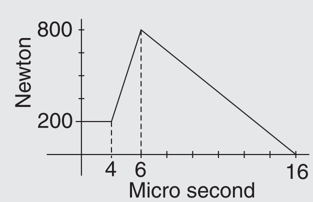
(a) $20\hat{i} + 20\hat{j} + 70\hat{k}$ (b) $10\hat{i} + 10\hat{j} + 5\hat{k}$
(c) $20\hat{i} + 70\hat{j} + 20\hat{k}$ (d) $5\hat{i} + 20\hat{j} + 70\hat{k}$

33. A solid iron ball A collides head on with another stationary solid iron ball B. If the ratio of radii of the balls is n = 2, then the ratio of their speeds just after the collision (e = 0.5) is:



(a) 3 (b) 4
(c) 2 (d) 1

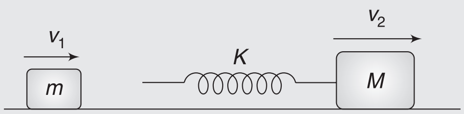
34. The magnitude of force (in Newton) acting on a body varies with time (in micro second), as shown in the figure. The magnitude of total impulse of the force on the body from $t = 4\,\mu s$ to $16\,\mu s$ is:



(a) 5×10^{-2} Ns (b) 5×10^{-3} Ns

(c) 5×10^{-4} Ns (d) 5×10^{-6} Ns

35. Two blocks of masses m and M are moving with speeds v_1 and $v_2 (v_1 > v_2)$ in the same direction on the frictionless surface, M being ahead of m. An ideal spring of force constant k is attached to the backside of M (as shown). The maximum compression of the spring when the blocks collide is:



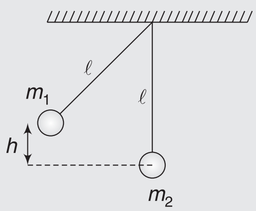
(a) $v_1 \sqrt{\frac{m}{k}}$

(b) $v_2 \sqrt{\frac{M}{k}}$

(c) $(v_1 - v_2) \sqrt{\frac{mM}{(M + m)k}}$

(d) None of above is correct.

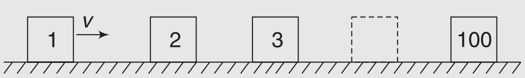
36. In the arrangement shown, the pendulum on the left is pulled aside. It is then released and allowed to collide with other pendulum, which is at rest. A perfectly inelastic collision occurs and the system rises to a height $h/4$. The ratio of the masses (m_1/m_2) of the pendulum is:



(a) 1 (b) 2

(c) 3 (d) 4

37. There are hundred identical sliders, equally spaced, on a frictionless track, as shown in the figure. Initially, all the sliders are at rest. Slider 1 is pushed with velocity v towards slider 2. In a collision, the sliders stick together. The final velocity of the set of hundred stuck sliders will be:



(a) $\frac{v}{99}$ (b) $\frac{v}{100}$

(c) zero (d) v

38. A sphere of mass m moving with a constant velocity hits another stationary sphere of the same mass. If e is the coefficient of restitution, then ratio of speed of the first sphere to the speed of the second sphere after head-on collision will be:

(a) $\left(\frac{1 - e}{1 + e}\right)$ (b) $\left(\frac{1 + e}{1 - e}\right)$

(c) $\left(\frac{e + 1}{e - 1}\right)$ (d) $\left(\frac{e - 1}{e + 1}\right)$

39. A body of mass 'm' is dropped from a height of 'h'. Simultaneously another body of mass 2 m is thrown up vertically with such a velocity v that they collide at the height $h/2$. If the collision is perfectly inelastic, the velocity at the time of collision with the ground will be:

(a) $\sqrt{\frac{5gh}{4}}$ (b) $\sqrt{gh}$

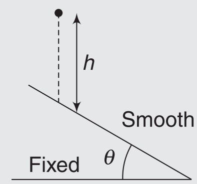
(c) $\sqrt{\frac{gh}{4}}$ (d) $\frac{\sqrt{10gh}}{3}$

40. A ball collides with a smooth and fixed inclined plane of inclination θ after falling vertically through a distance h. If it moves horizontally just after the impact, the coefficient of restitution is:

(a) $\tan^2 \theta$ (b) $\cot^2 \theta$

(c) $\tan \theta$ (d) $\cot \theta$

41. A ball of mass m strikes the fixed inclined plane after falling through a height h. If it rebounds elastically, the impulse on the ball is:



(a) $2m \cos \theta \sqrt{2gh}$ (b) $2m \cos \theta \sqrt{gh}$

(c) $\frac{2m\sqrt{2gh}}{\cos \theta}$ (d) $2m\sqrt{2gh}$

42. Two billiard balls undergo a head-on collision. Ball 1 is twice as heavy as Ball 2. Initially, Ball 1 moves with a speed v towards Ball 2, which is at rest. Immediately after the collision, Ball 1 travels at a speed of $v/3$ in the same direction. What type of collision has occured?

(a) inelastic

(b) elastic

(c) completely inelastic

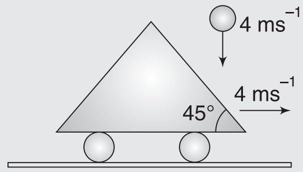
(d) Cannot be determined from the information given

43. A sphere of mass $m_1 = 2$ kg collides with a sphere of mass $m_2 = 3$ kg, which is at rest. Mass m_1 will move at right angle to the line joining centres at the time of collision, if the coefficient of restitution is:

(a) $\frac{4}{9}$ (b) $\frac{1}{2}$

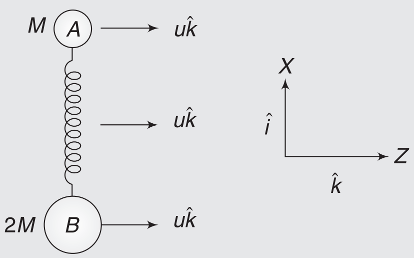
(c) $\frac{2}{3}$ (d) $\sqrt{\frac{2}{3}}$

44. A small ball falling vertically downward strikes elastically a massive inclined cart moving with velocity 4 ms^{-1} horizontally, as shown. At the instant of impact, the velocity of the ball was 4 ms^{-1}. The velocity of the rebound of the ball is



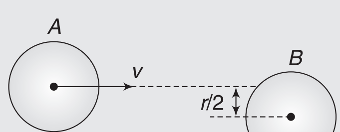
(a) $4\sqrt{2}$ ms^{-1} (b) $4\sqrt{3}$ ms^{-1}
(c) 4 ms^{-1} (d) $4\sqrt{5}$ ms^{-1}

45. Two masses A and B of mass M and $2M$ respectively are connected by a compressed ideal spring. The system is placed on a horizontal frictionless table and given a velocity $u\hat{k}$ in the z-direction, as shown in the figure. The spring also gets released. In the subsequent motion, the line from B to A is always along x-direction. At some instant of time, mass B has an x-component of velocity as $v_x\hat{i}$. The velocity $\vec{v}_A$ of mass A at that instant is



(a) $v_x\hat{i} + u\hat{k}$ (b) $-v_x\hat{i} + u\hat{k}$
(c) $-2\,v_x\hat{i} + u\hat{k}$ (d) $2v_x\hat{i} + u\hat{k}$

46. A disc A of radius r moving on perfectly smooth surface at a speed v undergoes an elastic collision with an identical stationary disc B. Find the velocity of the disc B after collision, if the impact parameter is $r/2$, as shown in the figure.



(a) $\dfrac{\sqrt{15}}{4}v$ (b) $\dfrac{v}{4}$

(c) $\dfrac{v}{2}$ (d) $\dfrac{\sqrt{3}\,v}{2}$

47. Two particles of masses m_1 and m_2 in projectile motion have velocities $\vec{v}_1$ & $\vec{v}_2$ respectively, at time $t = 0$. They collide at time t_0. Their velocities become $\vec{v}'_1$ and $\vec{v}'_2$ at time $2t_0$ while still moving in air. The value of: $[(m_1\vec{v}'_1 + m_2\vec{v}'_2) - (m_1\vec{v}_1 + m_2\vec{v}_2)]$ is:

(a) zero (b) $(m_1 + m_1)gt_0$

(c) $2(m_1 + m_2)gt_0$ (d) $\dfrac{1}{2}(m_1 + m_2)gt_0$

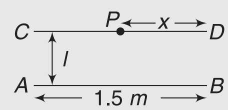
48. Two blocks of masses 10 kg and 4 kg are connected by a spring of negligible mass and placed on a frictionless horizontal surface. An impulse gives a velocity of 14 ms^{-1} to the heavier block in the direction of the lighter block. The velocity of the centre of mass is:

(a) 30 ms^{-1} (b) 20 ms^{-1}
(c) 10 ms^{-1} (d) 5 ms^{-1}

49. A projectile of mass "m" is projected from ground with a speed of 50 ms^{-1} at an angle of 53° with the horizontal. It breaks up into two equal parts at the highest point of the trajectory. One particle comes to rest immediately after the explosion. The ratio of the radii of curvatures of the moving particle just before and just after the explosion are:

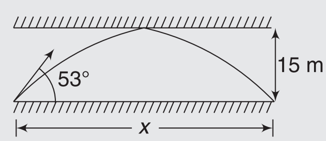
(a) 1:4 (b) 1:3
(c) 2:3 (d) 4:9

50. AB and CD are two smooth parallel wall at separation l. A child rolls a ball along ground from A towards point P. Find PD so that ball reaches point B after striking the wall CD. Given coefficient of restitution, $e = 0.5$.



(a) 0.5 m (b) 1 m
(c) 1.5 m (d) 0.8 ms

51. A ball is shot from the floor in a long hall having a roof at a height of 15 m. Initial velocity of the ball is 25 ms^{-1} at an angle of 53° with the floor. The ball lands on the floor at a distance x m from the point of projection. Assume collision with roof as elastic, if any, and find x.



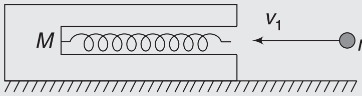
(a) 20 m (b) 40 m
(c) 30 m (d) 15 m

52. Two masses m_1 and m_2 are connected by a spring of spring constant k and are placed on a frictionless horizontal surface. Initially, the spring is stretched through a distance x_0 when the system is released from rest. Find the distance moved by the two masses before they again come to rest.

(a) $\dfrac{2m_2x_0}{m_1 + m_2}, \dfrac{2m_1x_0}{m_1 - m_2}$ (b) $\dfrac{2m_2x_0}{m_1 + m_2}, \dfrac{2m_1x_0}{m_1 + m_2}$

(c) $\dfrac{2m_2x_0}{m_1 - m_2}, \dfrac{2m_1x_0}{m_1 - m_2}$ (d) None of these

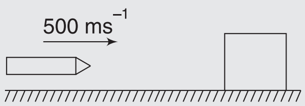
53. A ball of mass m is projected with speed v into the barrel of a spring gun of mass M, initially at rest on a frictionless surface. The mass m sticks in the barrel at the point of maximum compression of the spring. No energy is lost in friction. What fraction of the initial kinetic energy of the ball is stored in the spring?



(a) $\dfrac{2M}{M+m}$ (b) $\dfrac{M}{M+m}$

(c) $\dfrac{M}{2M+m}$ (d) None of these

54. A bullet of mass 20 g travelling horizontally with a speed of 500 ms^{-1} passes through a wooden block of mass 10.0 kg, initially at rest on a level surface. The bullet emerges with a speed of 100 ms^{-1} and the block slides 20 cm on the surface before coming to rest. Find the friction coefficient between the block and the surface:



(a) 0.16 (b) 0.26

(c) 0.36 (d) None of these

55. A ball hits the floor and rebounds after inelastic collision. In this case:

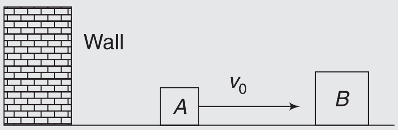
(a) The momentum of the ball just after the collision is same as that just before the collision

(b) The mechanical energy of the ball remains same in the collision

(c) The total momentum of the ball and the earth is conserved

(d) The total energy of the ball and the earth is conserved

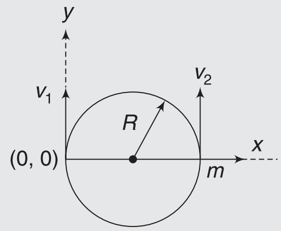
56. Block A of mass M is moving with a speed of v_0 on a frictionless surface that ends in a wall, as shown in figure. Farther from the wall is a more massive block B of mass $\alpha M(\alpha > 1)$, initially at rest. Block A undergoes elastic collision with the block B and the wall. If two blocks undergo only one collision then maximum value of α is-



(a) 1 (b) 2

(c) 3 (d) 4

57. A particle of mass m, moving in a circular path of radius R with a constant speed v_2 is located at point $(2R, 0)$ at time $t = 0$ and a man starts moving with velocity v_1 along the +ve y-axis from origin, at time $t = 0$. Calculate the linear momentum of the particle wrt the man as a function of time. $(\omega = v_2/R)$.



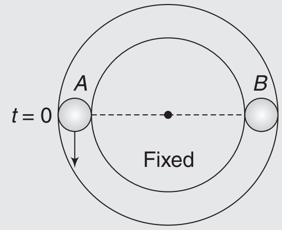
(a) $mv_2 \sin \omega t\hat{i} - m(v_2 \cos \omega t - v_1)\hat{j}$

(b) $-mv_2 \sin \omega t\hat{i} - m(v_2 \cos \omega t - v_1)\hat{j}$

(c) $-mv_2 \sin \omega t\hat{i} + m(v_2 \cos \omega t - v_1)\hat{j}$

(d) None

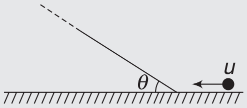
58. Particle 'A' moves with speed 10 ms^{-1} in a frictionless circular fixed horizontal pipe of radius 5 m and strikes particle 'B' having twice the mass of A. Coefficient of restitution is 1/2 and particle 'A' starts its journey at $t = 0$. The time at which second collision occurs is:



(a) $\dfrac{\pi}{2}$s (b) $\dfrac{2\pi}{3}$s

(c) $\dfrac{5\pi}{2}$s (d) 4π s

59. A particle of mass m is given initial speed u, as shown in the figure. It transfers to the fixed inclined plane without a jump. Its trajectory changes sharply from the horizontal line to the inclined line. All the surfaces are smooth and $90° \geq \theta > 0°$. Then the height to which the particle shall rise on the inclined plane (assume the length of the inclined plane to be very large)



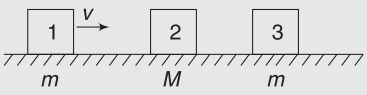
(a) increases with increase in θ

(b) decreases with increase in θ

(c) is independent of θ

(d) data insufficient

60. Three blocks are placed on smooth horizontal surface and lie on same horizontal straight line. Block 1 and block 3 have mass m each and block 2 has

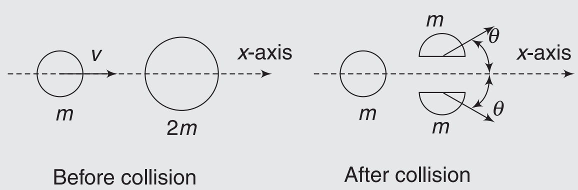


mass M ($M > m$). Block 2 and block 3 are initially stationary, while block 1 is initially moving towards block 2 with speed v, as shown. Assume that all collisions are head on and perfectly elastic. What value of $\frac{M}{m}$ ensures that Block 1 and Block 3 have the same final speed?

(a) $5 + \sqrt{2}$ (b) $5 - \sqrt{2}$

(c) $2 + \sqrt{5}$ (d) $3 + \sqrt{5}$

61. A particle of mass m is moving along the x-axis with speed v when it collides with a particle of mass $2m$ initially at rest. After the collision, the first particle has come to rest and the second particle has split into two equal-mass pieces. Which of the following statements correctly describes the speeds of the two pieces? ($\theta > 0$)



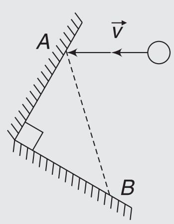
(a) Each piece moves with speed v.

(b) Each piece moves with speed $v/2$.

(c) One of the pieces moves with speed $v/2$, the other moves with speed greater than $v/2$

(d) Each piece moves with speed greater than $v/2$.

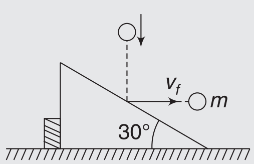
62. AB is an L-shaped obstacle fixed on a horizontal smooth table. A ball strikes it at A, gets deflected and restrikes it at B. If the velocity vector before collision is $\vec{v}$ and coefficient of restitution of each collision is 'e', then the velocity of ball after its second collision at B is



(a) $e^2\vec{v}$ (b) $-e^2\vec{v}$

(c) $-e\vec{v}$ (d) data insufficient

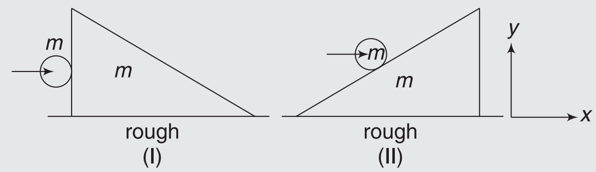
63. As shown in the figure a ball of mass m moving vertically with speed 3 ms^{-1} hits a smooth inclined plane of a wedge, which is placed on a smooth table with a back support. The ball rebounds with a velocity in the horizontal direction. Inclination of the incline is 30°. The impulse applied by the support on the wedge during collision is



(a) $3m$ (b) $\sqrt{3}\,m$

(c) $1/\sqrt{3}\,m$ (d) this is not possible

64. A ball of mass m collides horizontally with a stationary wedge on a rough horizontal surface. Consider collision in the two cases as shown. Neglect friction between ball and wedge. Two students comment on system of ball and wedge in these situations



Student 1: Momentum of system in x-direction will change by significant amount in both cases.

Student 2: There are no impulsive external forces in y-direction in both cases; hence the total momentum of system in y-direction can be treated as conserved in both cases.

(a) **Student 1** is wrong and **2** is correct

(b) **Student 1** is correct and **2** is wrong

(c) Both are correct

(d) Both are wrong

65. Two identical carts are constrained to move on a straight line. Ram and Shyam are two twins of same mass who are sitting on the two carts. The carts are moving with same velocity. At some time snow begins to pour uniformly in vertically downward direction. Ram throws off the falling snow sideways and in the second cart shyam is asleep. (Assume that friction is absent)

(a) Cart carrying Ram will speed up while cart carrying shyam will slow down

(b) Cart carrying Ram will remain at the same speed while cart carrying shyam will slow down

(c) Cart carrying Ram will speed up while cart carrying shyam will remain at the same speed

(d) Cart carrying Ram as well as shyam will slow down

66. A wagon filled with sand has a hole so that sand leaks through the bottom at a constant rate λ. An external force $\vec{F}$ acts on the wagon in the direction of motion. Assuming instantaneous velocity of the wagon to be $\vec{v}$ and initial mass of system to be m_0, the force equation governing the motion of the wagon is:

(a) $\vec{F} = m_0\frac{d\vec{v}}{dt} + \lambda\vec{v}$ (b) $\vec{F} = m_0\frac{d\vec{v}}{dt} - \lambda\vec{v}$

(c) $\vec{F} = (m_0 - \lambda t)\frac{d\vec{v}}{dt}$ (d) $\vec{F} = (m_0 - \lambda t)\frac{d\vec{v}}{dt} + \lambda\vec{v}$

67. If the thrust force on a rocket which is ejecting gases with a relative velocity of 300 ms^{-1}, is 210 N. Then the rate of combustion of the fuel will be:

(a) $10.7\ kgs^{-1}$ (b) $0.07\ kgs^{-1}$
(c) $1.4\ kgs^{-1}$ (d) $0.7\ kgs^{-1}$

68. An open water tight railway wagon of mass 5×10^3 kg coasts at an initial velocity $1.2\ ms^{-1}$ without friction on a railway track. Raindrops fall vertically downwards into the wagon. The velocity of the wagon after it has collected 10^3 kg of water will be

(a) $0.5\ ms^{-1}$ (b) $2\ ms^{-1}$
(c) $1\ ms^{-1}$ (d) $1.5\ ms^{-1}$

69. A rocket of mass 4000 kg is set for vertical firing. How much gas must be ejected per second so that the rocket may have initial upwards acceleration of magnitude $19.6\ ms^{-2}$. Exhaust speed of fuel is $980\ ms^{-1}$ and acceleration due to gravity is $9.8\ ms^{-2}$

(a) $240\ kgs^{-1}$ (b) $60\ kgs^{-1}$
(c) $120\ kgs^{-1}$ (d) None

Worksheet 2

1. In an elastic collision in absence of external force, which of the following is/are correct?

(a) The linear momentum is conserved.

(b) The potential energy is conserved in collision.

(c) The final kinetic energy is less than the initial kinetic energy.

(d) The final kinetic energy is equal to the initial kinetic energy.

2. A small ball collides with a heavy ball initially at rest. In the absence of any external impulsive force, it is possible that

(a) Both the balls come to rest.

(b) Both the balls move after collision.

(c) The moving ball comes to rest and the stationary ball starts moving.

(d) The stationary ball remains stationary, the moving ball changes its velocity.

3. A block moving in air explodes in two parts. Then, just after explosion (neglect change in momentum due to gravity)

(a) the total momentum of two parts must be equal to the momentum of the block before explosion.

(b) the total kinetic energy of two parts must be equal to that of block before explosion.

(c) the total momentum must change.

(d) the total kinetic energy must increase.

4. Two bodies of same masses collide head on elastically; then

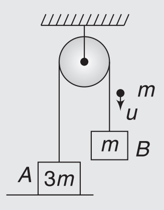
(a) their velocities are interchanged.

(b) their speeds are interchanged.

(c) their momenta are interchanged.

(d) the faster body slows down and the slower body speeds up.

5. A system of two blocks A and B are connected by an inextensible massless string, as shown. The pulley is massless and frictionless. Initially, the system is at rest with block A resting on a table and B hanging freely. A bullet of mass 'm' moving with a vertically downward velocity 'u' hits the block 'B' and gets embedded into it. The system is set into motion



(a) The impulse imparted by tension force to the block of mass $3m$ is $\frac{3mu}{5}$ during the impact of bullet.

(b) The impulse imparted by tension force to the block of mass m is $\frac{3mu}{5}$ during the impact of bullet.

(c) Momentum of B + bullet system is conserved during the impact.

(d) The impulse imparted by tension force on the ceiling is $\frac{6mu}{5}$ during the impact of bullet.

6. Two particles A and B having masses m_A and m_B respectively, start moving due to their mutual interaction only. At any time 't', $\vec{a}_A$ and $\vec{a}_B$ are their respective accelerations, $\vec{v}_A$ and $\vec{v}_B$ are their respective velocities; and W_A and W_B are the work done on A and B respectively by the mutual force upto that time. Which of the followings is/are always correct?

(a) $\vec{v}_A + \vec{v}_B = 0$ (b) $m_A\vec{v}_A + m_B\vec{v}_B = 0$

(c) $W_A + W_B = 0$ (d) $\vec{a}_A + \vec{a}_B = 0$

7. A set of n-identical cubical blocks lie at rest along a line on a smooth horizontal surface. The separation between any two adjacent blocks is L. The block at one end is given a speed v towards the next one, at time $t = 0$. All collisions are completely inelastic, then

(a) The last block starts moving at $t = n(n-1)\frac{L}{2v}$.

(b) The last block starts moving at $t = (n-1)\frac{L}{v}$.

(c) The centre of mass of the system will have a final speed v/n.

(d) The centre of mass of the system will have a final speed v.

8. A particle strikes a horizontal smooth floor with a velocity u, making an angle θ with the floor and rebounds with velocity v, making an angle ϕ with the floor. If the coefficient of restitution between the particle and the floor is e, then:

(a) the impulse delivered by the floor to the body is $mu(1+e)\sin\theta$.

(b) $\tan\phi = e\tan\theta$

(c) $v = u\sqrt{1-(1-e^2)\sin^2\theta}$

(d) the ratio of the final kinetic energy to the initial kinetic energy is $(\cos^2\theta + e^2\sin^2\theta)$

9. The figure shows a string of equally spaced beads of mass m, separated by distance d. The beads are free to slide without friction on a thin wire. A constant force F acts on the first bead initially at rest till it makes collision with the second bead. The second bead then collides with the third and so on. Suppose all collisions are elastic, then:

(a) speed of the first bead immediately before and immediately after its collision with the second bead is $\sqrt{\dfrac{2Fd}{m}}$ and zero respectively.

(b) speed of the first bead immediately before and immediately after its collision with the second bead is $\sqrt{\dfrac{2Fd}{m}}$ and $\dfrac{1}{2}\sqrt{\dfrac{2Fd}{m}}$ respectively.

(c) speed of the second bead immediately after its collision with third bead is zero.

(d) the average speed of the first bead is $\dfrac{1}{2}\sqrt{\dfrac{2Fd}{m}}$.

10. A ball moving with a velocity v hits a massive wall moving towards the ball with a velocity u. An elastic impact lasts for a time Δt.

(a) The average elastic force acting on the ball is $\dfrac{m(u+v)}{\Delta t}$.

(b) The average elastic force acting on the ball is $\dfrac{2m(u+v)}{\Delta t}$.

(c) The kinetic energy of the ball increases by $4mu(u+v)$.

(d) The kinetic energy of the ball remains the same after the collision.

11. Two blocks A and B each of mass 'm' are connected by a massless spring of natural length L and spring constant k. The blocks are initially resting on a smooth horizontal plane. A third block C, also of mass m, moves on the plane with a speed 'u' along the line joining A and B and collides elastically with A then which of the followings is/are correct:

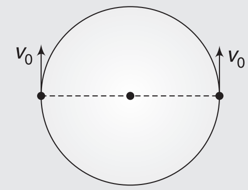
(a) KE of the AB system at maximum compression of the spring is zero.

(b) The KE of AB system at maximum compression is $(1/4)\ mu^2$.

(c) The maximum compression of spring is $u\sqrt{\dfrac{m}{k}}$.

(d) The maximum compression of spring is $u\sqrt{\dfrac{m}{2k}}$.

12. Two beads, each of mass m, are constrained to move along the circumference of a smooth circular hoop of mass m. The beads are initially located at opposite ends of a diameter and given equal velocities v_0, as shown in the figure. The entire arrangement is located in gravity-free space. Collision between the beads is elastic.



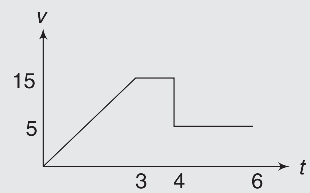
(a) The hoop will be at rest when the two beads are about to collide.

(b) Speed of the two beads immediately after collision between them is $\dfrac{2}{3}v_0$.

(c) Speed of the bead just before collision is $\dfrac{\sqrt{7}}{3v_0}$.

(d) Centre of the hoop can move in a direction inclined to the initial direction of v_0.

13. Given figure shows the velocity of an object as a function of the time. The object with mass 10 kg is being pushed along a straight line on a frictionless surface by an external force (F). At $t = 3$s, the force stops pushing and the object moves freely. It then collides head on and sticks to another object of mass 25 kg.



(a) External force F is 50 N.

(b) Velocity of the second particle just before the collision is 1 ms^{-1}.

(c) Before collision, both bodies are moving in the same direction.

(d) Before collision, bodies are moving in opposite direction with respect to each other.

14. Consider a particle at rest, which may decay into two (daughter) particles or into three (daughter) particles. Which of the following is **true in the two-body case but false in the three-body case**?

(There are no external forces and the masses of daughter particles are known.)

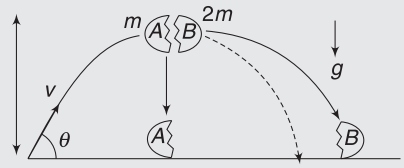
(a) Velocity vectors of the daughter particles must lie in a single plane.

(b) Given the total kinetic energy of the system, it is possible to determine the speed of each daughter particle.

(c) Given the speed(s) of all but one daughter particle, it is possible to determine the speed of the remaining particle.

(d) The total momentum of the daughter particles is zero.

15. A projectile is launched from the origin with speed v at an angle θ from the horizontal. At the highest point in the trajectory, the projectile breaks into two pieces–A and B, of masses m and $2m$, respectively. Immediately after the breakage, piece A is at rest relative to the ground. Neglect air resistance. Which of the following sentences most accurately describes what happens next?



(a) Piece B will hit the ground first, since it is more massive.

(b) Both pieces have zero vertical velocity immediately after the breakup, and therefore they hit the ground at the same time.

(c) Piece A will hit the ground first, because it will have a downward velocity immediately after the breakup.

(d) There is no way of knowing which piece will hit the ground first, because not enough information is given about the breakup.

16. A particle of mass m makes a head-on elastic collision with another particle of mass 2m initially at rest. The velocity of the first particle before and after collision is given to be u_1 and v_1 and the velocity of second particle after the collision is v_2. Which of the following statements is true in respect of this collision?

(a) For all values of u_1, v_1 will always be less than u_1 in magnitude.

(b) The fractional loss in kinetic energy of the first particle is $\frac{8}{9}$.

(c) The gain in kinetic energy of the second particle is $\left(\frac{8}{9}\right)^{\text{th}}$ of the initial kinetic energy the first particle.

(d) There is a net loss in the kinetic energy of the two-particle system in the collision.

17. Which of the following statements is/are correct?

(a) Most of the collisions on the macroscopic scale are inelastic collisions.

(b) In a perfectly inelastic collision, there is a complete loss of KE.

(c) Forces involved in elastic collision are conservative in nature.

(d) Oblique collision is that collision in which the colliding bodies do not move along the same straight-line path.

18. In a one-dimensional collision between two identical particles A and B, B is stationary and A has momentum P before impact. During the impact, B gives impulse J to A.

(A) The total momentum of the 'A plus B' system is P before and after the impact, and $(P - J)$ during the impact.

(b) During the impact, A gives impulse J to B

(c) The coefficient of restitution is $\frac{2J}{P} - 1$

(D) The coefficient of restitution is $\frac{J}{P} + 1$

19. Two trolleys A and B of equal masses M are moving in oppsite directions with velocities $\vec{v}$ and $-\vec{v}$ respectively on separate horizontal frictionless parallel tracks. When they start crossing each other, a ball of mass m is thrown from B to A and another of same mass is thrown from A to B with velocities normal to $\vec{v}$. Assume no lateral deviation of the trolleys.

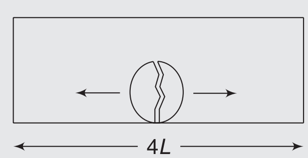
(a) If the two balls are thrown simultaneously, the event will lead to equal change in speeds of the two trolleys.

(b) When ball is thrown from A to B after the ball thrown from B reaches A, speed of B will change more than that of A.

(c) The ball thrown from A to B missed the trolley and fell on ground. Final speed of A will become smaller than B.

(d) None of the above

20. A bomb of mass 3 m is kept inside a closed box of mass 3 m and length 4 L at its centre. It explodes in two parts of masses m & $2m$. The two parts move in opposite direction and stick to the opposite walls of the box. Box is kept on a smooth horizontal surface.



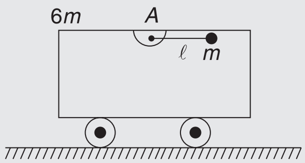
(a) Both parts of the bomb must hit the walls simultaneously.

(b) If there is friction between the parts of the bomb and the floor of the box, then the box will get displaced in the direction in which mass $2m$ travelled.

(c) The box will get displaced by $L/3$.

(d) None of the above

21. In the figure shown, the cart of mass $6m$ is intially at rest. A particle of mass m is attached to the end of the light rod, which can rotate freely about A. If the rod is released from rest in a horizontal position shown,



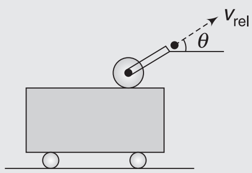
(a) The cart will be moving with speed equal to 1/6 of the speed of the particle at the instant the rod becomes vertical.

(b) The centre of mass of the cart (not including the particle) will stay at rest.

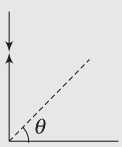
(c) The centre of mass of the cart will oscillate as the pendulum swings inside it.

(d) the velocity v_{rel} of the particle with respect to the cart when the rod is vertical is $\sqrt{\frac{7}{3}g\ell}$.

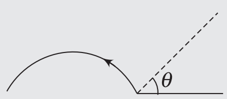
22. A cannon is mounted on a trolley. A small cannon ball is fired at an angle θ and velocity v_{rel} relative to trolley, which is free to recoil frictionlessly on a smooth surface. Which of the following trajectories (represented by dark curves) is not possible for the cannon ball?



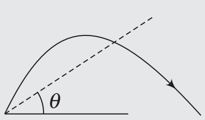
(a)



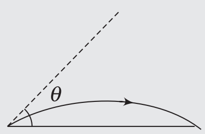
(b)



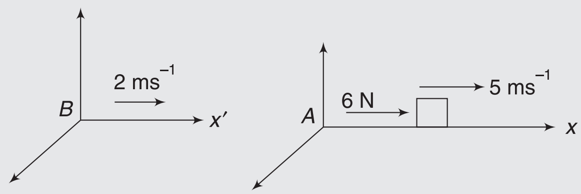
(c)



(d)



23. Consider a block of mass 10 kg, which is on a smooth surface. It is subjected to a horizontal force of 6 N for 4 s. Observer A is attached to ground and another observer B is in a reference frame moving with velocity 2 ms^{-1} as shown in figure. Which of the followings are correct?



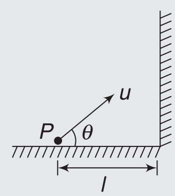
(a) The final speed of the block is 7.4 ms^{-1}, if it has initial speed of 5 ms^{-1} as measured by A.

(b) The final speed of the block is 7.4 ms^{-1}, if it has initial speed of 5 ms^{-1} as measured by B.

(c) For both the observers, change in momentum of the body is equal to impulse experienced by it.

(d) None

24. A ball of mass m is projected from a point P on the ground, as shown in the figure. It was supposed to have a horizontal range R. However, it hits a fixed vertical wall elastically at a distance l from P. Which of the following are correct?



(a) The ball will return to the point P directly if $l = R/2$.

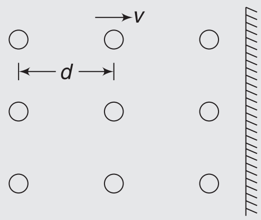
(b) The ball will fall between point P and the wall if $l < R/2$.

(c) The ball will fall between point P and the wall if $l > R/2$.

(d) If $l < R/2$, the ball spends more time in air after collision than before it.

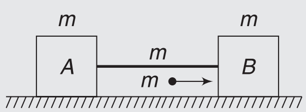
Worksheet 3

1. A beam of neutrons has all the particles travelling towards a surface with a speed v. Separation between the successive particles is d. On hitting the surface normally, the particles rebound along the original line of motion with same speed. Neglect the interaction of rebounding particles with the incoming particles. Mass per unit length of the incident beam is $\frac{m}{d}$. Find the force experienced by the surface.



2. During a hailstorm, stones of 1.0 cm diameter fall on flat roof of a house at a speed of $20\,ms^{-1}$. The stones hit the roof normally and 2000 stones fall on every square meter area of the roof in 1 second. The hailstones do not rebound. Find the force applied by the falling hailstones on the roof. Density of hailstones = $900\,kg\,m^{-3}$, and area of roof is $100\,m^2$.

3. Two identical blocks A and B have mass m each and are kept on a smooth horizontal surface connected with a rope of mass m. The rope is straight and horizontal. A bullet of mass m flying horizontally at velocity u hits blocks B and gets embedded into it. For the small time interval for which the bullet interacts with B, find

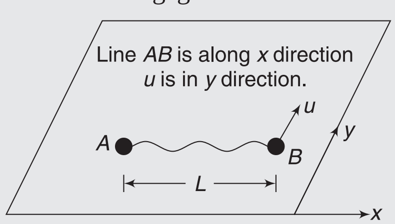


(i) Impulse applied by B on the rope.

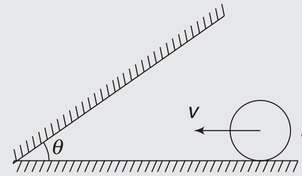
(ii) Impulse applied by the bullet on B.

(iii) Impulse applied by the rope on block A.

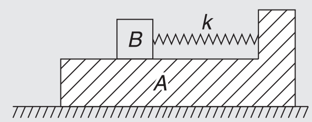
4. Two particles A and B of mass m each are placed on a smooth horizontal surface connected by a light string of length $2L$. Separation between the particles is L. Particle B is imparted a velocity u in a direction that is perpendicular to line AB (see figure). Find velocity of each particle after the string gets taut. Also find the impulse applied by the string on particle A. Express this impulse in terms of unit vectors $\hat{i}$ and $\hat{j}$.



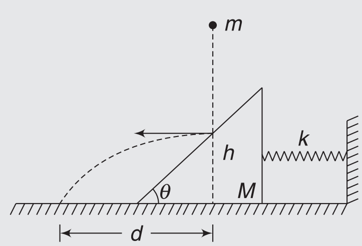
5. A ball of mass m slides with velocity v on a smooth horizontal surface. It hits an inclined wall, inclined at angle θ, and rebounds with speed $\frac{v}{2}$. Find the impulse imparted by the floor to the sphere during the time intervals for which it interacted with the wall. Wall is smooth.



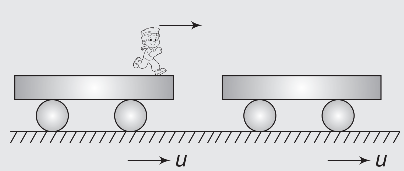
6. An L-shaped block A is placed on a smooth horizontal surface. A spring of force constant k is attached to its vertical arm as shown. A smooth block B is placed on A and is moved to right so that the spring gets compressed by $x_0 = 1\,m$. The system is released from rest. Find velocity of A when B leaves it. Take $m_A = 5\,kg$, $m_B = 1\,kg$ and $k = 100\,Nm^{-1}$.



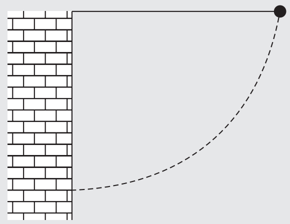
7. A triangular wedge of mass M rests on a smooth horizontal surface. It is connected to a relaxed spring of force constant k. The other end of the spring is secured to a wall and the spring is horizontal. Inclination angle of inclined surface of the wedge is θ. A ball of mass m is dropped from some height on the inclined face. It hits the inclined face at a height h above the horizontal surface. Immediately after hitting the wedge, the ball moves horizontally. It lands on the floor at a distance d from its original line of fall. Find the maximum compression in the spring.



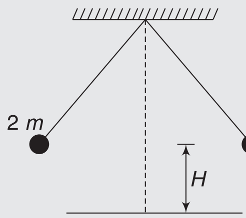
8. Two identical rail road cars (each of mass M) are moving on same smooth track with velocity u. There is a man of mass m on the rear car. He jumps horizontally with a velocity v relative to his car and lands on the front car. Find the velocities of the two cars after this jump.



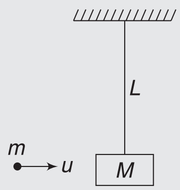
9. A simple pendulum is suspended from a peg on a vertical wall. The pendulum is pulled away from the wall to a horizontal position and released. The ball hits the wall, coefficient of restitution being $e = \frac{2}{\sqrt{5}}$. Find the minimum number of collisions after which the amplitude of oscillation becomes less than 60°.



10. Two pendulums of same length have bobs of masses m and $2m$. They are suspended from same point and raised to a height H above their lowest position (see figure). They are released from this position and they collide head on, elastically at the lowest point. To what height will the balls rise for the first time after collision?



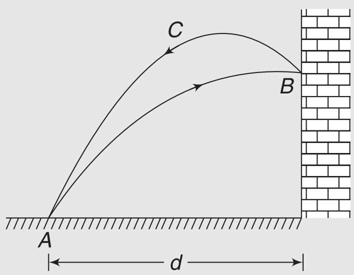
11. A wooden block of mass M is suspended like a pendulum, using a string of length L. A horizontally flying bullet has mass m and speed u. It strikes the block and gets embedded. Find



(i) Loss in mechanical energy during collision.

(ii) Minimum value of u so that the combined mass completes the vertical circle.

12. A ball is projected form point A on the ground at a velocity u, making an angle θ with horizontal, It hits a smooth vertical wall at B and returns to the projection point A along the path BCA. Distance of the wall from A is d. Find coefficient of restitution for the collision of the ball with the wall.



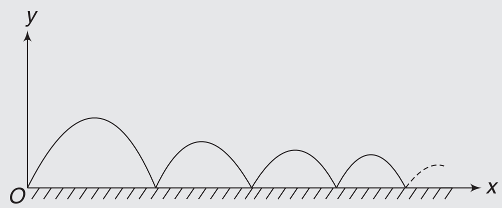
13. A 5 kg projectile is fired from ground with an initial velocity of 24 ms^{-1} making an angle of 60° with the horizontal. When at the top, it explodes into two fragments of equal mass. One of the fragments moves vertically upward after the blast. After both of them land on the ground, separation between them was found to be 45 m. Find speed of the two fragments just after explosion.

14. A projectile is launched from a point O on a floor at velocity u, making an angle θ with the horizontal. Coefficient of restitution between the projectile and the floor is e. Assuming the floor to be smooth, find:

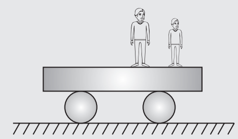
(i) Total horizontal distance travelled by the projectile before it stops.

(ii) Total time elapsed before it comes to rest.

(iii) The angle made by velocity vector of the projectile with horizontal after n^{th} impact.

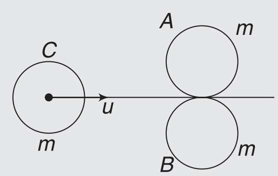


15. Two men, each of mass m, stand on the edge of a stationary flat car of mass m. The car is on a smooth horizontal surface. Calculate the velocity of the car after both men jump off with same horizontal velocity u relative to the car.

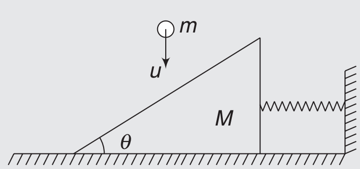


(i) Simultaneously (ii) One after the other.

16. Two identical discs of mass m are in contact on a smooth table. A third identical disc travelling with velocity u, hits them symmetrically. After collision, the third disc comes to rest. Find coefficient of restitution (e) and percentage loss in kinetic energy.

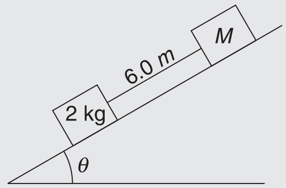


17. A ball of mass m moving downwards with a velocity v hits a wedge of mass M which is placed on a smooth horizontal table. The wedge is connected to a relaxed spring of force constant k as shown. Find maximum compression in the spring subsequent to collision. θ = 45° and collision is elastic.



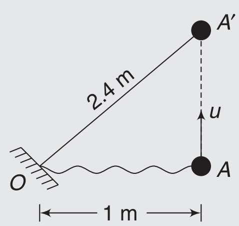
18. Two blocks of mass 2 kg and M are at rest on an inclined plane and are separated by a distance of 6.0 m as shown. The coefficient of friction between each of the blocks and the inclined plane is 0.25.

The 2 kg block is given a velocity of $10.0\,\text{ms}^{-1}$ up the inclined plane. It collides with M, comes back and has a velocity of $1.0\,\text{ms}^{-1}$ when it reaches its initial position. The other block M after the collision moves 0.5 m up and comes to rest. Calcualte the coefficient of restitution between the blocks and the mass of the block M.



[Take $\sin\theta \approx \tan\theta = 0.05$ and $g = 10\,\text{ms}^{-2}$]

19. A small ball of mass $m = 2$ kg is connected to a fixed point O by an inextensible cord of length $L = 2.4$ m. The ball is resting on a smooth horizontal table at point A at a distance of 1 m from point O. It is imparted a velocity u in a direction perpendicular to the line OA. The string gets taut when the ball reaches point A'.



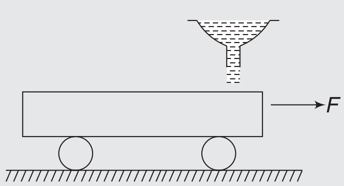
(i) Find maximum allowable u if the impulse applied by the string on the support at O is not to exceed 3 Ns.

(ii) Find loss in kinetic energy as the strings gets taut, if the ball is initially given a velocity equal to maximum allowable value calculated in part (i).

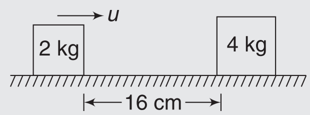
20. Two identical smooth balls are projected towards each other from two points A and B on horizontal ground with same speed of projection. The angle of projection in each case is 30°and distance AB = 100 m. The two balls collide in mid air and return to their respective projection points. Find the speed of projection of either ball if coefficient of restitution for the collision is $e = 0.7$.

21. A rocket is designed to move at constant acceleration a in space. It shoots out exhaust gases at constant speed u relative to itself. At time $t = 0$, mass of the rocket is M_0. Find its mass at a later time t.

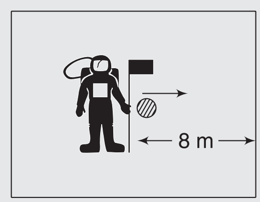
22. An empty railway wagon of mass m_0 starts moving to the right due to a constant horizontal force F. The moment force begins to act, sand begins falling on the wagon from a fixed hopper. Rate of loading is constant equal to b kgs^{-1}. Find time dependence of velocity and acceleration of the wagon in the process. Neglect friction.



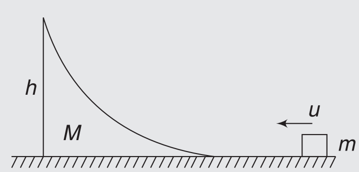
23. A block of mass 4 kg is at rest on a horizontal table. Another block of mass 2 kg is approaching it. When the separation between the blocks is 16 cm, the velocity of 2 kg block was $1\,\text{ms}^{-1}$. Coefficient of friction for both the blocks with table is 0.2 and collision is elastic. Find the final separation between the blocks.



24. An astronaut is inside a satellite. There is effectively no gravity inside the satellite (More about it in the chapter of Gravitation). The astronaut has a mass of 50 kg and is at rest inside the satellite. He throws an objects of mass 5 kg towards a wall of the satellite. He finds that the object moves away at a velocity of $12\,\text{ms}^{-1}$ relative to him. The object collides with the front wall and rebounds. How long after throwing the object the astronaut can catch the object? Assume that collision of the object with the satellite wall is elastic and that the astronaut does not meet a collision till the object is back to him. The object was thrown at a distance of 8 m from the wall.



25. A block of mass m is slid on a smooth horizontal table with a velocity u towards a wedge of mass $M = \eta \cdot m$. The wedge has a smooth curved surface on which the block climbs without any abrupt change in its velocity when it hits the wedge. Height of the wedge is h. Find minimum value of u so that the block can climb to the top of the wedge. Remember, the wedge is free to move!



Answers Sheet

Your Turn

1. The cycle
2. Truck
3. (i) $\Delta P = mu\sin\theta\ (\downarrow)$ (ii) $\Delta P = 2mu\sin\theta\ (\downarrow)$
4. (i) 10^5 N (ii) 1.5×10^5 J
5. (6.88 ± 0.37) N
6. 20 N
7. $\dfrac{2mv^2}{\pi R}$
8. $\rho A v^2$; force becomes 4 times
9. $\mu\sqrt{2gH} + \mu gt$
10. 250 N
11. Larger impact time means smaller average force in all three cases
12. 20 Ns
13. 10000 Ns
14. 20 Ns
15. $M\sqrt{2gL}\ (\downarrow)$
16. $\dfrac{2mu}{3}$
17. $(2\hat{i} + 7\hat{j})\,\text{ms}^{-1}$
18. $m\vec{u} + m\vec{g}\cdot t$
19. 3520 J
20. (i) $\dfrac{mu}{M}$ (ii) Energy released is greater than $1/2(mu^2)(1 + m/M)$
21. $\dfrac{mu}{M + m}$
22. $\dfrac{mu}{2(M + m)}$
23. $v_A = \sqrt{\dfrac{k}{6m}}\cdot x$; $v_B = 2\sqrt{\dfrac{k}{6m}}\cdot x$
24. $\sqrt{21}$ ms^{-1}
25. $\overrightarrow{BG}$
26. $\dfrac{mv}{M + m}$
27. With a velocity $\left(\dfrac{M + m}{m}\right)u$ in forward direction.
28. No change
29. $-\dfrac{mu}{M + m}$
30. (i) $\dfrac{mu}{M + m}$ (ii) $-\dfrac{1}{2}\left(\dfrac{mM}{M + m}\right)u^2$
31. $\dfrac{mu}{M + m}$
32. (i) $K_A = 0$, $K_B = \dfrac{1}{2}mu^2$ (ii) $\dfrac{1}{4}mu^2$
33. $\dfrac{2L}{u}$
34. $\dfrac{1}{\sqrt{2}}$
35. (i) 4 ms^{-1} (ii) 2.5 ms^{-1}
36. $M > \dfrac{5m}{4}$
37. 0.23 d
38. $\dfrac{x}{2}$
39. $m_2 = \sqrt{m_1 m_3}$
40. $\dfrac{3}{5}$
41. (i) $2\hat{i} - 3\hat{k}$ (ii) $2\hat{i}$ (iii) $2\hat{i} - 1.5\hat{k}$
42. $v_B = u$ in a direction opposite to original direction of motion
    $v_A = 2u$ in a direction at right angle to the original line of motion.
43. 45°
44. $e = 0.453$
45. $\dfrac{1}{3}$
46. $\dfrac{1}{e}$
47. $\dfrac{u}{\sqrt{2}}$ making 45° with x-axis
48. 2.5 kgs^{-1}
49. 3.75 kgs^{-1}
50. 0
51. 2 $ln2$

Worksheet 1

1. (b)	2. (c)	3. (b)	4. (b)	5. (b)	6. (b)	7. (b)	8. (a)	9. (b)
10. (b)	11. (a)	12. (b)	13. (a)	14. (a)	15. (c)	16. (a)	17. (b)	18. (b)
19. (c)	20. (d)	21. (d)	22. (d)	23. (b)	24. (d)	25. (a)	26. (a)	27. (a)
28. (c)	29. (b)	30. (b)	31. (d)	32. (a)	33. (c)	34. (b)	35. (c)	36. (a)
37. (b)	38. (a)	39. (d)	40. (a)	41. (a)	42. (b)	43. (c)	44. (d)	45. (c)
46. (a)	47. (c)	48. (c)	49. (a)	50. (b)	51. (c)	52. (b)	53. (b)	54. (a)
55. (c)	56. (c)	57. (c)	58. (c)	59. (b)	60. (c)	61. (d)	62. (c)	63. (b)
64. (d)	65. (d)	66. (c)	67. (d)	68. (c)	69. (c)			

Worksheet 2

1. (a,d) 2. (b,c) 3. (a,d) 4. (a,b,c,d) 5. (a,b,d) 6. (b) 7. (a,c) 8. (a,b,c,d)
9. (a,c,d) 10. (b) 11. (b,d) 12. (c) 13. (a,b,c) 14. (b,c) 15. (b) 16. (a,b,c)
17. (a,c,d) 18. (b,c) 19. (a,c) 20. (c) 21. (a,c,d) 22. (a,b,d) 23. (a,b,c) 24. (a,c,d)

Worksheet 3

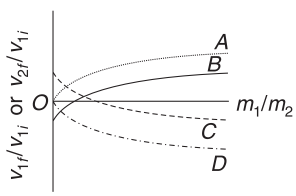
1. $2\left(\frac{m}{d}\right)v^2$
2. 1884 N
3. (i) $\frac{1}{2}mu$ $(\rightarrow)$ (ii) $\frac{3}{4}mu$ (iii) $\frac{mu}{4}$
4. $v_A = \frac{\sqrt{3}}{4}u,\ v_B = \frac{\sqrt{7}}{4}u;\ \vec{J} = \left(\frac{\sqrt{3}}{8}mu\right)\hat{i} + (\frac{3}{8}mu)\hat{j}$
5. $\frac{3}{2}mv\cot\theta$
6. $\sqrt{\frac{10}{3}}\ \text{ms}^{-1}$
7. $md\sqrt{\frac{g}{2hkM}}$
8. $v_1 = u - \frac{mv}{M+m};\ v_2 = u + \frac{Mmv}{(M+m)^2}$
9. 4
10. $\frac{25H}{9}, \frac{H}{9}$
11. (i) $\frac{Mmu^2}{2(M+m)}$ (ii) $\left(\frac{M+m}{m}\right)\sqrt{5gL}$
12. $\left(\frac{1}{\frac{u^2\sin 2\theta}{d\cdot g} - 1}\right)$
13. $24.1\,\text{ms}^{-1}$, $2.55\,\text{ms}^{-1}$
14. (i) $\frac{u^2\sin^2\theta}{g(1-e)}$ (ii) $\frac{2u\sin\theta}{g}\left(\frac{1}{1-e}\right)$ (iii) $\tan^{-1}(e^n\tan\theta)$
15. (i) $\frac{2mu}{M+2m}$ (ii) $\frac{mu}{M+2m} + \frac{mu}{M+m}$
16. 33.3%
17. $\left(\frac{2mu}{m+2M}\right)\sqrt{\frac{M}{K}}$
18. 0.84, 15.01 kg
19. (i) $1.65\,\text{ms}^{-1}$ (ii) 2.25 J
20. 37.5 ms^{-1}
21. $M = M_0 e^{-\frac{at}{u}}$
22. $v = \frac{Ft}{m_0 + bt};\ a = \frac{Fm_0}{(m_0+bt)^2}$
23. 5 cm
24. 1.6 s
25. $\sqrt{2gh\left(1 + \frac{1}{\eta}\right)}$

CHAPTER 3

Miscellaneous Problems on Chapter 1 and 2

MATCH THE COLUMN

1. A particle of mass m_2 is at rest. Another particle of mass m_1 hits it. Collision is one-dimensional elastic. Suffix 'f' represents final and 'i' represents initial in following.

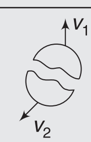
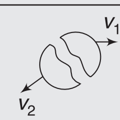
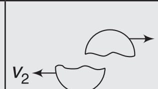
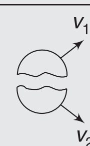


	Column I		Column II
(a)	The curve in the figure, which is the graph of the velocity ratio v_{1f}/v_{1i} versus the mass ratio m_1/m_2	(p)	A
(b)	The value of the intercept of the graph of v_{1f}/v_{1i} versus the mass ratio m_1/m_2 with the vertical axis is	(q)	C or D
(c)	The maximum value taken by the graph of v_{2f}/v_{1i} versus m_1/m_2 is	(r)	B
(d)	The curve that is the graph of the velocity ratio v_{2f}/v_{1i} versus m_1/m_2 is	(s)	0
		(t)	–1
		(u)	2

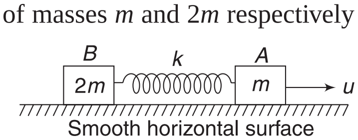
2. A smooth disk (mass M_1) moving on a smooth floor undergoes elastic collision with another smooth disk (mass M_2) at rest. Quantities in Column II refer to possible values after collision. ($v \rightarrow$ speed, $p \rightarrow$ momentum, $k \rightarrow$ kinetic energy)

	Column I		Column II
(a)	$M_1 >> M_2$, collision is head-on	(p)	$\lvert \vec{v}_{2_f} \rvert > \lvert \vec{v}_{1_f} \rvert$
(b)	$M_1 << M_2$, collision is head-on	(q)	$\lvert \vec{p}_{2_f} \rvert > \lvert \vec{p}_{1_f} \rvert$
(c)	$M_1 = M_2$, collision is head-on	(r)	$k_{2_f} > k_{1_f}$
(d)	$M_1 = M_2$, collision is oblique	(s)	$k_{2_f} < k_{1_f}$

3. A body initially moving towards right explodes into two pieces, 1 and 2. Direction of motion of the pieces is shown in Column I and possible mass ratios are shown in Column II.

	Column I		Column II
(a)		(p)	$m_1 > m_2$
(b)		(q)	$m_1 = m_2$
(c)	 , $v_1 < v_2$	(r)	$m_1 < m_2$
(d)		(s)	Impossible for any masses

4. Two blocks A and B of masses m and $2m$ respectively are connected by a massless spring of spring constant k. This system lies over a smooth horizontal surface. At $t = 0$, the



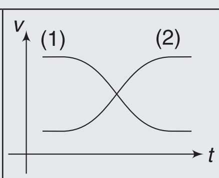
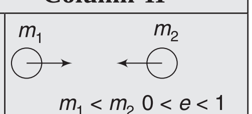
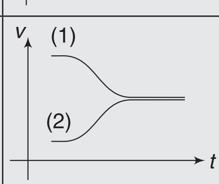
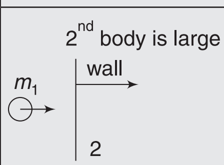
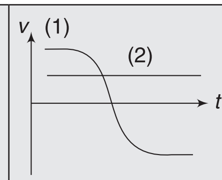
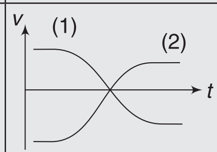
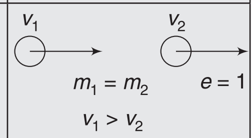
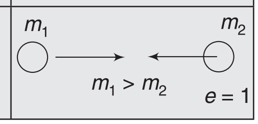
block A has velocity u towards the right, as shown, while the speed of block B is zero, and the length of spring is equal to its natural length at that instant.

Column I		Column II	
(a)	The velocity of block A	(p)	can never be zero
(b)	The velocity of block B	(q)	may be zero at certain instants of time
(c)	The kinetic energy of the system of two block	(r)	is minimum at maximum compression of spring
(d)	The potential energy of the spring	(s)	is maximum at maximum extension of spring

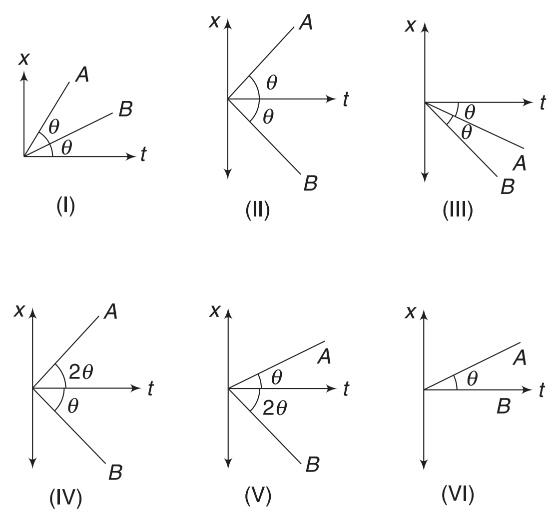
5. A truck of mass M crashes into a tempo of mass $m < M$ and the two masses stick together. For each pair of quantities below, state the relationship.

Column I			Column II	
	Quantity 1	*Quantity 2*		
(a)	Magnitude of force exerted by truck on tempo	Magnitude of force exerted by tempo on truck	(p)	Quantity 1 > Quantity 2
(b)	Magnitude of acceleration of truck during collision	Magnitude of acceleration of tempo during collision	(q)	Quantity 1 < Quantity 2
(c)	Total kinetic energy of the two-vehicle system before collision	Total kinetic energy of the two-vehicle system after collision	(r)	Quantity 1 = Quantity 2
(d)	Kinetic energy of the tempo before collision	Kinetic energy of the tempo after collision	(s)	Impossible to determine from information given

6. Assume that two bodies collide head on. The graph of their velocities with time are shown in Column I. Match them with appropriate situation in Column II.

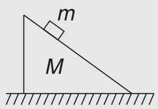
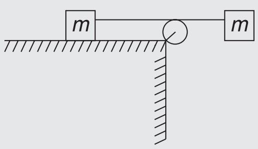
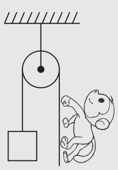
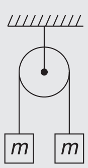
Column I		Column II	
(a)		(p)	
(b)		(q)	
(c)		(r)	
(d)		(s)	
		(t)	

7. An initially stationary box on a frictionless floor explodes into two pieces, piece A with mass m_A and piece B with mass m_B. The two pieces then move across the floor along x-axis. Possible graph of position versus time for the two pieces are given.

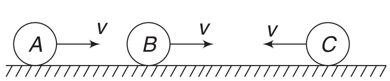


Column I		Column II	
(a)	I	(p)	Explosion could happen due to internal reasons
(b)	II	(q)	$m_A = m_B$
(c)	III	(r)	$m_A > m_B$
(d)	IV	(s)	$m_A < m_B$
(e)	V	(t)	Explosion must have happened due to an impulsive external force
(f)	VI		

8. In each situation of Column I, a system involving two bodies is given. All strings and pulleys are light and friction is absent everywhere. Initially, each body is at rest. Then match the statements in Column I with the corresponding results in Column II.

Column I		Column II	
(a)	The block plus wedge system is placed over a smooth horizontal surface. After the system is released from rest, the centre of mass of the system 	(p)	Shifts towards right
(b)	The string connecting both the blocks of mass m is horizontal. The left block is placed over smooth horizontal table, as shown. After the two-block system is released from rest, the centre of mass of the system 	(q)	Shifts downwards
(c)	The block and monkey have the same mass. The monkey starts climbing up the rope. After the monkey starts climbing up, the centre of mass of monkey + block system 	(r)	Shifts upwards
(d)	Both blocks of mass m are initially at rest. The left block is given initial velocity u downwards. Then, the centre of mass of the two-block system afterwards 	(s)	Does not shift

9. Two spheres A and B are moving along a straight line with same velocity v. A third sphere C is moving in the opposite direction on the same surface, with the same speed v. All spheres are identical and elastic. v_{CM} represents velocity of COM. Match the entries in Column I to those in Column II.

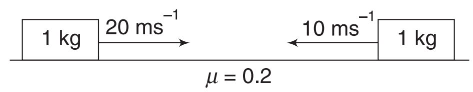


Column I		Column II	
(a)	A and B are connected by an ideal spring (which is relaxed)	(p)	$v_{CM} = \frac{v}{3}$ before and after all collisions
(b)	A and B are connected by an ideal string	(q)	There will be more than one collision.
(c)	A and B are connected by a light rigid rod	(r)	A and B always have equal and opposite momentum after all collisions.
(d)	B and C are connected by an ideal string. Gap between A and B is large.	(s)	Final mechanical energy after all collisions is less than initial.

PASSAGE-BASED PROBLEMS

Passage 1

Two blocks are far apart and are approaching each other with velocities as shown in figure. The coefficient of friction for both the blocks is μ = 0.2.

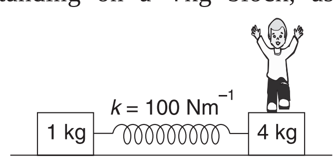


1. Linear momentum of the system is
   (a) conserved all the time
   (b) never conserved
   (c) is conserved upto 5 seconds
   (d) none of these

2. How much distance will the centre of mass travel before coming permanently to rest
   (a) 25 m   (b) 37.5 m
   (c) 42.5 m   (d) 50 m

Passage 2

A boy of mass 10 kg is standing on a 4 kg block, as shown. The whole system is on smooth level ground. Suddenly, the boy jumps to the right with a velocity of 7 ms^{-1} relative to the 4 kg block. Interaction time of the boy and the 4 kg block during the event of jump is very small and the displacement of the bock can be neglected during that period.



3. For the system, comprising the boy and the 4 kg block:
   (a) momentum is not coserved during the jump due to presence of the spring force.

(b) both blocks have finite velocities immediately after the boy jumps

(c) speed of 4 kg block immediately after the boy jumps is $5\,ms^{-1}$

(d) none of the above

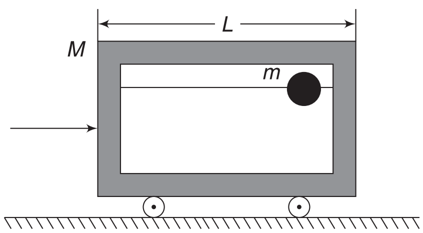
4. The maximum velocity of 1 kg block (in ms^{-1}) in the subsequent motion is

(a) 0.8 (b) 0.4

(c) 0.1 (d) 1.0

Passage 3

A horizontal frictionless rod is threaded through a bead of mass m. The rod is mounted horizontally inside a cart of mass M, as shown. The length of the cart is L and the radius of the bead, r, is very small in comparison to L ($r << L$). Initially, the bead is at the right edge of the cart. The cart is given a sharp blow (at $t = 0$) towards the right and as a result, it begins to move with velocity v_0. When the bead collides with the cart's walls, the collisions are always completely elastic. (Given: $m > M$).



5. What is the velocity of the cart after the first collision, in the centre of mass frame?

(a) $\dfrac{-mv_0}{m+M}$ (b) $\dfrac{Mv_0}{m+M}$

(c) $\dfrac{M-m}{M+m}v_0$ (d) $\dfrac{2M}{m+M}v_0$

6. The first collision takes place at time t_1 and the second collision takes place at $t = t_2$. Find $t_2 - t_1$.

(a) $\dfrac{2L}{v_0}$ (b) $\dfrac{L}{v_0}$

(c) $\dfrac{L}{2v_0}$ (d) $\dfrac{L}{3v_0}$

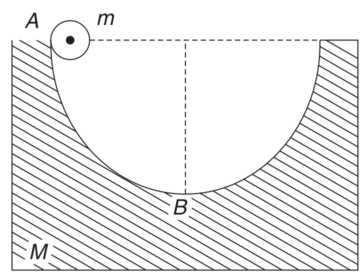
7. What is the distance the cart travels from $t = 0$ till $t = t_2$

(a) $\dfrac{2ML}{m+M}$ (b) $\dfrac{mL}{2(m+M)}$

(c) $\dfrac{2m}{m+M}L$ (d) $\dfrac{ML}{2(m+M)}$

Passage 4

A block of mass M with a semi-circular track of radius R rests on a horizontal frictionless surface. A uniform cylinder of radius 'r' and mass 'm' is released from rest at the top point A. The cylinder slips in the semi-circular frictionless track.



8. Which of the following is true?

(a) Momentum of the system of the block and the cylinder is conserved after the cylinder is released.

(b) Momentum of the system of the block and the cylinder is equal to the initial momentum only when the cylinder reaches the bottom of the track.

(c) The cylinder will never get back to the topmost position in the track from where it is released.

(d) Motion of the block is not oscillatory.

9. How fast is the block moving when the cylinder reaches the bottom of the track?

(a) $\dfrac{m}{M}\left[\dfrac{2gM(R-r)}{M+m}\right]^{1/2}$

(b) $\dfrac{m}{M}\left[\dfrac{2gM(R-r)}{2M+m}\right]^{1/2}$

(c) $\dfrac{m}{M}\left[\dfrac{gM(R-r)}{M+m}\right]^{1/2}$

(d) $\dfrac{2m}{M}\left[\dfrac{2gM(R-r)}{M+m}\right]^{1/2}$

Passage 5

A shell is flying with a velocity $u = 500\,ms^{-1}$. It bursts into three identical fragments and the kinetic energy of the system increases k times. One of the fragments continues to move along the original line of motion with speed v_1. Answer the following questions for $k = 1.5$.

10. For v_1 to be maximum

(a) the other two particles must move in opposite directions perpendicular to the original line of motion

(b) the other two particles must move in directions making equal angles with the original line of motion

(c) one of the particles should come to rest

(d) None of the above

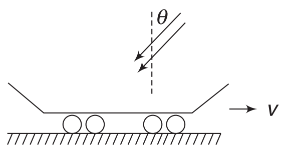
11. Maximum possible value of v_1 is

(a) 1000 ms^{-1} (b) 500 ms^{-1}

(c) 1200 ms^{-1} (d) 800 ms^{-1}

Passage 6

A freight car is moving on a smooth horizontal track with no external force acting in horizontal direction. Rain is falling with a velocity u ms^{-1} at an angle θ with the vertical. Rain drops are collected in the car at the rate of $\mu\,kgs^{-1}$. If initial mass of the car is m_0 and velocity v_0 then



12. The instantaneous acceleration (a) of the car is related to instantaneous velocity v of the car as

(a) $(m_0 + \mu t)a = -(u\sin\theta + v)\mu$

(b) $(m_0)a = -(u\cos\theta + v)\mu$

(c) $(m_0 + \mu t)a = -(u\cos\theta + v)\mu$

(d) $(m_0)a = -(u\cos\theta + v)\mu$

13. Velocity of the car as a function of time is

(a) $v = u\cos\theta\left(\dfrac{\mu t}{m_0 + \mu t}\right) + \dfrac{m_0 v_0}{(m_0 + \mu t)}$

(b) $v = u\cos\theta\left(\dfrac{\mu t}{m_0}\right) + \dfrac{m_0 v_0}{(m_0 + \mu t)}$

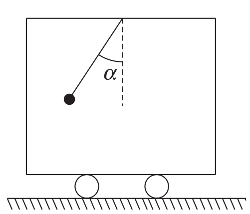
(c) $v = -u\sin\theta\left(\dfrac{\mu t}{m_0 + \mu t}\right) + \dfrac{m_0 v_0}{(m_0 + \mu t)}$

(d) $v = u\sin\theta\left(\dfrac{\mu t}{m_0}\right) + \dfrac{m_0 v_0}{(m_0 + \mu t)}$

Passage 7

A wagon of mass M can move without friction along horizontal rails. A simple pendulum consisting of a bob of mass m is suspended from the ceiling by a string of length l. At the initial moment, the wagon and pendulum are at rest and the string is deflected through an angle α from the vertical. The system is released from this position.

The velocity of wagon, when the string forms an angle β ($\beta < \alpha$) with vertical is v and u is the speed of pendulum in the frame fixed to the wagon.



Before releasing the bob

14. u and v are related as

(a) $u = \dfrac{(M + m)v}{m\cos\beta}$ (b) $u = \dfrac{(M + m)v}{m\cos(\alpha - \beta)}$

(c) $u = \dfrac{(M + m)v}{m\cos\alpha}$ (d) $u = \dfrac{mv}{(M + m)\cos\beta}$

15. The velocity of wagon (v) is

(a) $v = \sqrt{\dfrac{2mMgl}{M + m}\left[\dfrac{\cos\beta - \cos\alpha)\cos^2\beta}{M + m\sin^2\beta}\right]}$

(b) $v = \sqrt{\dfrac{2m^2gl}{M + m}\left[\dfrac{(\cos\beta - \cos\alpha)\cos^2\beta}{M + m\sin^2\beta}\right]}$

(c) $v = \sqrt{\dfrac{2mMgl}{M + m}\left[\dfrac{(\cos\beta - \cos\alpha)\cos\beta}{M + m\sin\beta}\right]}$

(d) $v = \sqrt{\dfrac{2m^2gl}{M + m}\left[\dfrac{(\cos\beta - \cos\alpha)\cos 2\beta}{M + m\sin 2\beta}\right]}$

16. The velocity of wagon when velocity of the bob is horizontal relative to the wagon is

(a) $v = 2m\sin\dfrac{\alpha}{2}\sqrt{\dfrac{gl}{mM}}$

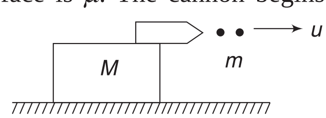
(b) $v = m\sin\dfrac{\alpha}{2}\sqrt{\dfrac{gl}{(M + m)M}}$

(c) $v = 2m\sin\dfrac{\alpha}{2}\sqrt{\dfrac{gl}{(M + m)M}}$

(d) $v = m\sin\dfrac{\alpha}{2}\sqrt{\dfrac{gl}{(M + m)m}}$

Passage 8

A cannon with shots has total mass M_0. It is kept on a rough horizontal surface. The coefficient of friction between the cannon and the horizontal surface is μ. The cannon begins to fire the shots at a uniform frequency with a velocity u relative to it. All the shots are fired horizontally.



17. If mass of each shot is m, the smallest frequency of firing, which will cause the cannon to accelerate just as the firing begins is

(a) $\dfrac{\mu M_0 g}{(mu)}$ (b) $\dfrac{2\mu M_0 g}{(mu)}$

(c) $\dfrac{\mu M_0 g}{(u)}$ (d) difficult to decide

18. Which of the following is true?

(a) If the cannon does not recoil initially, it will not move till the end of shots.

(b) If the cannon does not recoil initially, it may move a little later as the shots are being fired.

(c) If the barrel of the gun is inclined so as to have a large range, the cannon will experience a greater thrust force.

(d) None of the above.

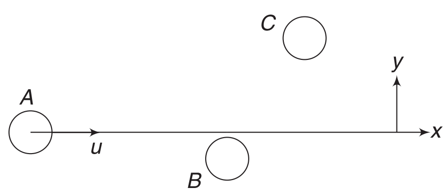
19. The velocity of the cannon when it possesses a total mass M (with the remaining shots), after time t from starting is

(a) $ut\ln(M_0/M) - \mu gt$ (b) $u\ln(M_0/M) - \mu gt$

(c) $ut\ln(M_0/M) - \mu g$ (d) $u\ln(M/M_0) - \mu gt$

Passage 9

Consider three identical discs, A, B and C, on a smooth horizontal floor (xy-plane). B and C are at rest. A is imparted a velocity u in positive x-direction. It first hits B and then goes on to hit C. Finally, A was found to be travelling along x-direction with velocity $\frac{u}{2}$. Assume all collisions to be elastic.



20. Final velocity of B is

(a) $\frac{u}{2}\hat{j}$ (b) $-\frac{u}{2}\hat{j}$

(c) $\frac{u}{2}(\hat{i} - \hat{j})$ (d) $\frac{u}{4}(\hat{i} - \hat{j})$

21. Final velocity of C is

(a) $\frac{u}{4}(\hat{i} + \hat{j})$ (b) $\frac{u}{4}(-\hat{i} + \hat{j})$

(c) $\frac{u}{2}\hat{j}$ (d) $\frac{u}{4}\hat{j}$

Passage 10

A model of a helicopter has mass M. Its fan has diameter d. When switched on, the fan collects the motionless air above it and blow it downwards. Assume that density of air, above and below the fan, is the same and equal to ρ.

22. Assuming that the speed imparted to air particles by the fan is u; volume of air that the fan must push down in 1 second, so as to just lift the helicopter is

(a) $\sqrt{\frac{2M^2g}{\rho^2u}}$ (b) $\sqrt{\frac{M^2g}{\rho^2u}}$

(c) $\frac{2Mg}{\rho u}$ (d) $\frac{Mg}{\rho u}$

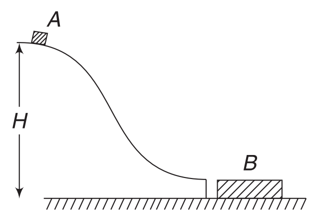
23. The minimum power needed for take-off is

(a) $\frac{(Mg)^{3/2}}{d\sqrt{\pi\rho}}$ (b) $\frac{(Mg)^{3/2}}{d\sqrt{2\pi\rho}}$

(c) $\frac{(Mg)^{3/2}}{2d\sqrt{\pi\rho}}$ (d) $\frac{(Mg)^{3/2}}{2d\sqrt{2\pi\rho}}$

Passage 11

A small block A slides down a smooth hill of height H and lands on a plank B, as shown. Mass of A is m. A quickly stops moving relative to B and stays on it. The plank B slides on the horizontal surface and the friction force acting on it is found to be directly proportional to its speed. The proportionality constant is k. That is, friction is given by



$$f = -kv$$

24. Impulse of friction that acted on ($A + B$) combined, if block B travels a distance L is

(a) $\sqrt{2g}\cdot m\cdot\frac{L}{H^2}$ (b) $\sqrt{2g}\cdot m\cdot\frac{H^2}{L}$

(c) $2kL$ (d) kL

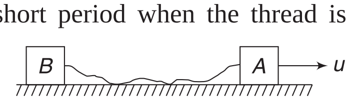
25. Minimum value of H needed for B to travel through a distance L is

(a) $\frac{k^2L^2}{2gm^2}$ (b) $\frac{k^2L^2}{gm^2}$

(c) $\frac{2k^2L^2}{gm^2}$ (d) None

Passage 12

Two blocks, A and B, of masses m_1 and m_2 respectively, are placed on a frictionless horizontal floor. They are connected by a light cord, as shown. Block A is imparted a velocity u towards right. During the very short time interaction when the thread is gaining tension, it gets slightly extended and after acquiring a maximum extension, it again goes back to relaxed state. The short period when the thread is getting stretched is called deformation period and



the period in which the thread returns to its relaxed state is called 'restitution period'. Let us define a number η as

$$\eta = \frac{\text{Impulse of tension during restitution period}}{\text{Impulse of tension during deformation period}}$$

26. Energy absorbed by the thread during its deformation period is

(a) $\frac{1}{2}\left(\frac{m_1 m_2}{m_1 + m_2}\right)u^2$ (b) $\frac{1}{4}\, m_1 u^2$

(c) $\frac{1}{4}\left(\frac{m_1 m_2}{m_1 + m_2}\right)u^2$ (d) $\frac{1}{2}\, m_1 u^2$

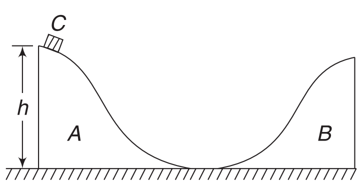
27. Final velocity of B is

(a) $\frac{m_1(1-\eta)}{m_1 + m_2}u$ (b) $\frac{m_1(1+\eta)}{m_1 + m_2}u$

(c) $\frac{m_2(1-\eta)}{m_1 + m_2}u$ (d) None

Passage 13

There are two identical smooth slides A and B, each of height h and mass M. They are placed on a smooth floor, as shown. A small disc C of mass m ($<<M$) is released from the top of slide A. The disc slides down, then slides up the other slide, reaches a maximum height and then slides down. The sequence continues till all interaction between the slides and the disc stops.



28. Maximum height attained by the disc on slide B is

(a) $\left(\frac{m}{M+m}\right)^2 h$ (b) $\left(\frac{M}{M+m}\right)^2 h$

(c) $\frac{Mm}{(M+m)^2}h$ (d) $\frac{2M}{M+m}\cdot h$

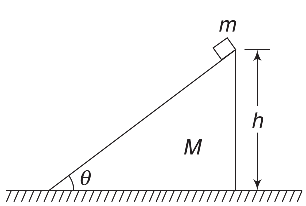
29. Final speed acquired by each slide is nearly

(a) $\sqrt{\frac{mgh}{M}}$ (b) $2\sqrt{\frac{mgh}{M}}$

(c) $\sqrt{\frac{2mgh}{M+m}}$ (d) $\sqrt{\frac{mgh}{M+m}}$

Passage 14

A triangular wedge of mass M = 4 kg is on a smooth horizontal table. Its angle of inclination is θ = 30° and its height is h = 0.5 m. A small block of mass m = 1 kg is held at rest at the top of the inclined surface of the wedge. System is released from this position. Assume no friction and take g = 10 ms^{-2}.



30. The ratio of horizontal and vertical components of velocity of the block of at any time is

(a) $\frac{4\sqrt{3}}{5}$ (b) $\frac{2\sqrt{3}}{5}$

(c) $\frac{\sqrt{3}}{5}$ (d) $\frac{4\sqrt{3}}{25}$

31. The vertical component of velocity of the block after it has come down by a distance $y(<h)$ in vertical direction is

(a) $\sqrt{\frac{10}{17}y}$ (b) $\sqrt{\frac{100}{17}y}$

(c) $\sqrt{\frac{y}{17}}$ (d) $\sqrt{\frac{5y}{17}}$

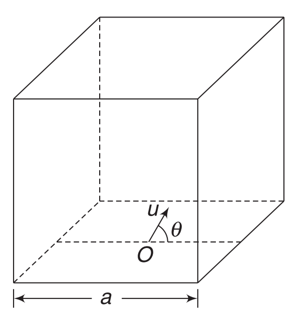
32. Time taken by the block to reach the bottom of the incline

(a) $\frac{1}{\sqrt{5}}$ s (b) $\frac{\sqrt{17}}{5}$ s

(c) $\frac{\sqrt{8.5}}{5}$ s (d) $\frac{1}{5}$ s

Passage 15

A room has a square floor of side length a. A particle is projected from the centre of the floor (O), at an angle θ with horizontal. All its collisions with the walls and the roof are elastic. The particle returns to its point of projection after time T.



33. The speed of projection, if the particle collided with two parallel walls and the roof is

(a) $\frac{a}{T\cos\theta}$ (b) $\frac{2a}{T\cos\theta}$

(c) $\frac{a}{T\sin\theta}$ (d) $\frac{a}{T\sqrt{\cos^2\theta - \sin^2\theta}}$

34. The speed of projection, if the particle just makes one collision with a wall, is

(a) $\dfrac{Tg}{\sin\theta}$ (b) $\dfrac{2Tg}{\sin\theta}$

(c) $\dfrac{a}{u\cos\theta}$ (d) None

Passage 16

A 3 kg shell moving in space explodes into three fragments of equal masses. After explosion, the fragments move apart but interact through mutual attraction with each other. At some instant after explosion, velocities of the three fragments are

$(30\hat{i} + 20\hat{j})\,\text{ms}^{-1}$, $-20\hat{i}\,\text{ms}^{-1}$ and $(50\hat{i} - 50\hat{j})\,\text{ms}^{-1}$.

35. Find the KE of the shell before explosion

(a) 150 J (b) 450 J

(c) 750 J (d) 2250 J

36. Find the minimum energy released in explosion

(a) 2600 J (b) 2150 J

(c) 1600 J (d) 450 J

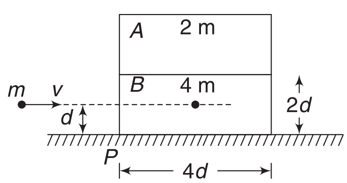
37. Find the least KE of the system that is possible at a time after explosion

(a) 750 J (b) 400 J

(c) 3500 J (d) 1600 J

Passage 17

A block A of mass $2m$ is placed on another block B of mass $4m$, which in turn is placed on a fixed table. Length of each block is $4d$ and they are placed as shown. Coefficient of friction between B and the table is μ. There is no friction between A and B. A small objects of mass m moving horizontally along a line passing through the COM of B, and perpendicular to its vertical face, with speed u collides elastically with B at a height d above the table.



38. Minimum value of u (say u_0) so as to make block A topple is

(a) $\frac{1}{2}\sqrt{6\mu gd}$ (b) $\frac{3}{2}\sqrt{6\mu gd}$

(c) $\frac{5}{2}\sqrt{6\mu gd}$ (d) $\sqrt{\mu gd}$

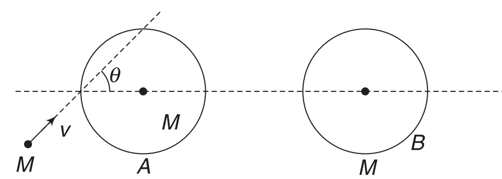
39. Assuming $u = 2u_0$, the distance measured from point P at which the mass m falls on the table after colliding with B is

(a) $6d\sqrt{3\mu}$ (b) $4d\sqrt{3\mu}$

(c) $4d\mu$ (d) $d\mu$

Passage 18

Two identical discs of mass M are lying on a smooth horizontal surface. A small ball of mass M strikes one of the discs, while moving along a line, making an angle $\theta = 37°$ with the line joining the centres of the two discs. Velocity of the ball just before collision is $v = 10\,\text{ms}^{-1}$.



Coefficient of restitution for collision between the ball and disc A is $\frac{1}{2}$ and the coefficient of restitution for collision between A and B is 1.

40. Relative velocity of separation along the line of impact for the ball and Disc A after collision is

(a) $2\,\text{ms}^{-1}$ (b) $8\,\text{ms}^{-1}$

(c) $4\,\text{ms}^{-1}$ (d) zero

41. Speed of ball after collision is

(a) $\sqrt{10}\ \text{ms}^{-1}$ (b) $3\sqrt{10}\ \text{ms}^{-1}$

(c) $4\ \text{ms}^{-1}$ (d) $2\sqrt{10}\ \text{ms}^{-1}$

42. Speed of Disc B after collision is

(a) $6\ \text{ms}^{-1}$ (b) $3\ \text{ms}^{-1}$

(c) $4\,\text{ms}^{-1}$ (d) None.

Answers Sheet

Match the Columns

1. (a) r; (b) t; (c) u; (d) p
2. (a) p, s; (b) q, s; (c) p, q, r; (d) p, q, r, s
3. (a) s; (b) s; (c) p; (d) p, q, r
4. (a) q; (b) q; (c) p, r; (d) q, s
5. (a) r; (b) q; (c) p; (d) s
6. (a) s; (b) r; (c) q; (d) p
7. (a) t; (b) p, q; (c) t; (d) p, s; (e) p, r; (f) t
8. (a) q; (b) p, q; (c) r; (d) s
9. (a) p, r, s; (b) p, q, r; (c) p; (d) p, q, s

Passage-based Problems

1. (c)	2. (b)	3. (c)	4. (a)	5. (a)	6. (b)	7. (c)	8. (b)	9. (a)
10. (d)	11. (a)	12. (a)	13. (c)	14. (a)	15. (b)	16. (c)	17. (a)	18. (b)
19. (b)	20. (c)	21. (c)	22. (d)	23. (a)	24. (d)	25. (a)	26. (a)	27. (b)
28. (b)	29. (a)	30. (a)	31. (b)	32. (c)	33. (b)	34. (c)	35. (c)	36. (a)
37. (a)	38. (c)	39. (a)	40. (c)	41. (d)	42. (a)			

CHAPTER 4

Torque and Equilibrium

"Living things have no inertia, and tend to no equilibrium."

–Thomas Henry Huxley

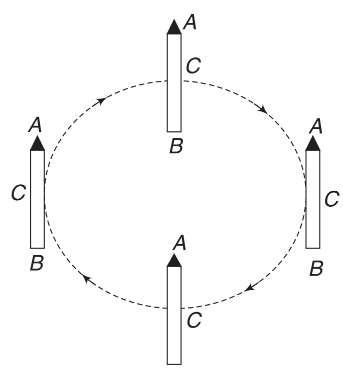
1. INTRODUCTION

We solved some problems on equilibrium of bodies in the chapter on Newton's laws of motion. Knowledge of torque will let us solve diverse kinds of equilibrium problems, which were beyond our range till now.

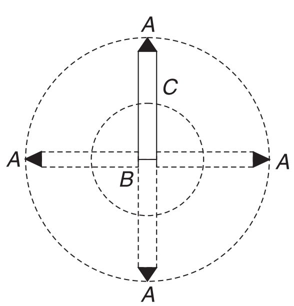
Moreover, our understanding of torque will help us in understanding the dynamics of rotational motion of rigid bodies in the chapters to follow. We will ultimately realise that torque is rotational analogue of force.

2. WHAT IS ROTATION?

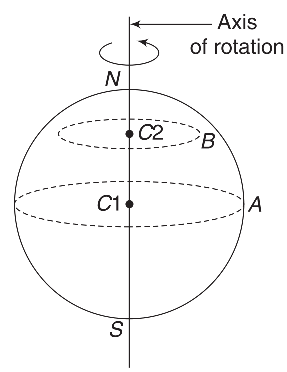
When a body moves such that all its particles undergo identical motion, we say it is translating. Think of a pencil being moved along a circle, as shown in figure. Its particles, *A*, *B* and *C*, always have the same speed and travel equal distances in equal intervals of time. The pencil is performing translational motion.



The same pencil now moves with its end *B* fixed. Tip *A* goes in a circle. Its centre *C* also goes in a circle. In fact, all points in the pencil are moving in circles, having common centre at *B*. All the points have the same angular speed (ω) but different linear speeds ($v = \omega r$). Points *A* and *C* travel different distances in equal time intervals. The pencil is said to be rotating about point *B*, or about an axis passing through point *B* and perpendicular to the plane of the figure.



Similarly, think of the spinning of the Earth. All particles are moving in circles. The centres of all such circles lie on the line passing through the poles of the Earth. This line is the rotation axis of the Earth. The Earth is rotating about its axis.



Spinning Earth

Point *A* goes in Circle with centre at *C*1
point *B* moves in circle with centre at *C*2

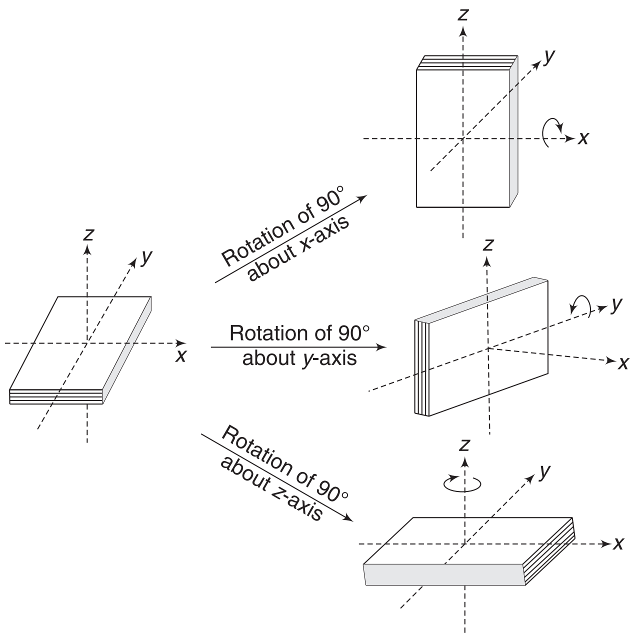
Following diagram shows the effect of rotating a book by 90° about three perpendicular axes. The diagram helps in understanding the meaning of rotation of a body about a given axis.

3. TORQUE

Torque is a physical quantity that measures the turning effect of a force.

When you switch on a fan, it begins to rotate with increasing angular speed. The fan is experiencing a torque (applied by electric motor). Since there is no acceleration of the COM of the fan, we know that the sum of all force on it is zero.

This implies that forces may have zero resultant but non-zero turning effect.



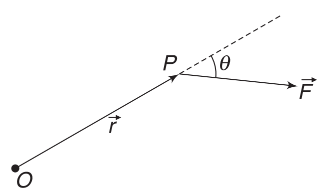
3.1 Torque of a Force about a Point

A force $\vec{F}$ is acting at point P. Torque of this force about a point O is defined as

$$\vec{\tau} = \vec{r} \times \vec{F} \qquad \text{...(1)}$$

Where $\vec{r}$ = position vector of point of application relative to O

In the figure shown, direction of τ is perpendicular to the plane of the figure. [This is the direction in which your thumb points when you curl your right-hand fingers from $\vec{r}$ to $\vec{F}$]. For ease of discussion, let us call the direction going into the plane of the figure as z-axis. Our force has a torque about O pointing along z-axis. What does this mean? If there was a body, free to rotate about point O and force $\vec{F}$ acted at some point P in the body, the torque of $\vec{F}$ about O would try to spin the body about z-axis. The direction of rotation is in the direction of curling fingers of your right hand (from $\vec{r}$ towards $\vec{F}$). With reference to the figure shown, we can say that the torque of $\vec{F}$ tries to produce a clockwise rotation about point O. So we either say that torque is directed into the plane (along z-direction), or we simply say that torque of the force about O is clockwise.

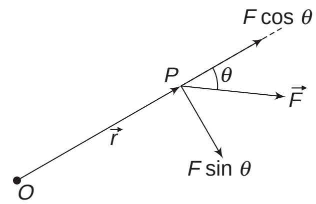


Now, we know that torque of force $\vec{F}$ about point O will try to rotate a body about an axis that is perpendicular to both $\vec{r}$ and $\vec{F}$.

Let us talk a little about the magnitude of the torque.

$$\tau = rF\sin\theta$$
$$= rF_{\perp} \qquad \text{...(2)}$$

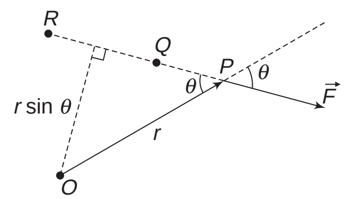
where $F_{\perp} = F\sin\theta$ is a component of $\vec{F}$ in a direction perpendicular to $\vec{r}$.



We can also interpret the magnitude of torque as

$$\tau = F(r\sin\theta)$$
$$= F r_{\perp} \qquad \text{...(3)}$$

Where $r_{\perp} = r\sin\theta$ is the perpendicular distance of line of action of force $\vec{F}$ from the point O.



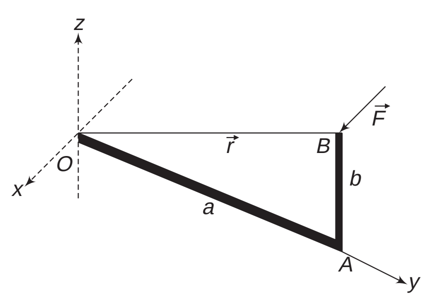
It is easy to see that torque of force $\vec{F}$ about O will not change if the point of application of the force is moved to a point like Q or R that lies on its line of action.

Obviously, unit of torque is N-m. We should avoid writing it as Joule.

3.2 Torque of a Force about an Axis

Consider an L-shaped body OAB that is free to rotate about point O. Arm OA is along y-axis and has length a. Arm AB is parallel to z-axis and has length b. A force $\vec{F} = F\hat{i}$ is applied at tip B. Torque of this force about O is

$$\tau = \vec{r} \times \vec{F} = (a\hat{j} + b\hat{k}) \times (F\hat{i})$$
$$= bF\hat{j} - aF\hat{k}$$



Actually, this torque is along an axis that is perpendicular to both $\vec{r}$ and $\vec{F}$. But in our co-ordinate system, we have two components of torque.

$$\tau_y = bF \quad \text{and} \quad \tau_z = -aF$$

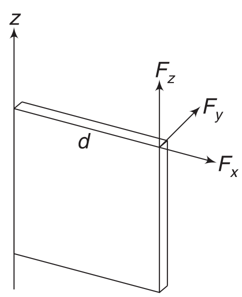
It implies that the force will try to rotate the body about y-axis as well as z-axis. However, $\tau_x = 0$ means that the force will have no rotational effect about x-axis. Think of the above-described situation practically by having an L-shaped body in your hand. You can easily see that such force will have no turning effect about x-axis but will try to turn the body about z-axis as well as y-axis.

If our body is constrained to rotate about y-axis only, then we may not be interested in τ_z. In such cases, $\tau_y = bF$ is the relevant quantity. We call this as torque about y-axis.

While calculating torque of a force about an axis, one just needs to learn the following three points:

1. If the given force is parallel to the axis, torque about the said axis is zero.
2. If the line of action of the force intersects the axis, there is no torque.
3. A force F has torque about an axis (say z) only when F is neither parallel to z-axis nor its line of action intersects z-axis. When a force is perpendicular to z-axis but its line of action does not intersect it (skewed lines), torque about z-axis is $\tau = Fd$, where d is the perpendicular distance between the line of force and z-axis.

Think of a door that can rotate about z-axis. If you apply a force F_x (See fig.) it will produce no rotation [Try with the door in your room]. Line of F_x intersects z-axis. It has no torque about this axis. If you apply a force F_z, it is also not going to rotate the door. F_z is parallel to z-axis.



When you apply F_y, the door rotates; F_y has torque about z-axis and its torque is $\tau_y = F_y \cdot d$, where d is the distance between z-axis and line of F_y.

Note: When many forces are acting on a body and we are supposed to find the resultant torque about a point (or about an axis), we find torques due to individual forces and add them.

Example 1 A force $\vec{F} = (2\hat{i} + 3\hat{j})$ N acts at a point having position vector $\vec{r} = (2\hat{i} - \hat{k})$ m.

(i) Find the torque of the force about the origin.

(ii) Find the torque of the force about z-axis.

Solution

Concepts

(i) $\vec{\tau} = \vec{r} \times \vec{F}$

(ii) z-component of τ is known as torque about z-axis

(i) $\vec{\tau} = \vec{r} \times \vec{F} = (2\hat{i} - \hat{k}) \times (2\hat{i} + 3\hat{j})$

$= (3\hat{i} - 2\hat{j} + 6\hat{k})$ N-m

(ii) $\tau_z = 6$ N-m

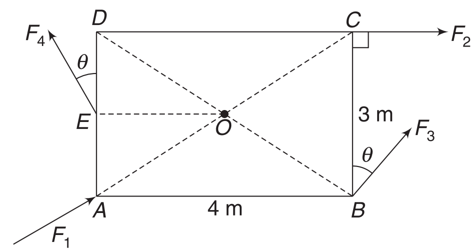
Example 2 $ABCD$ is a rectangular plate having sides of length 4 m and 3 m. Forces F_1, F_2, F_3 and F_4 are applied on the plate as shown. Each force has magnitude equal to 20 N and is in the plane of the figure. Given, $AE = ED$ and $\theta = 37°$.

(i) Find torque of all forces about centre O.

(ii) What is the direction of the torque?

(iii) What is the torque due to all forces about an axis through O that is perpendicular to the plane of the figure.

$$\left[\text{Take } \sin 37° = \frac{3}{5}\right]$$



Solution

Concepts

(i) $\tau = F\, r_\perp$ or $\tau = r\, F_\perp$

(ii) We will find torque due to individual forces and then add them to get the resultant torque

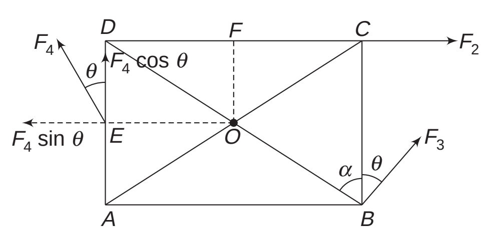
(i) Torque due to F_1 is zero as the line of action of the forces passes through O. F_2 acts along DC.

Its perpendicular distance from O is $OF = 1.5$ m

$$\therefore \quad \tau_2 = F_2 \times 1.5$$

$$= 20 \times 1.5 = 30 \text{ N-m} (\circlearrowright)$$

Torque due to F_2 tries to produce a clockwise rotation in the plate about O. ($\circlearrowright$) sign denotes clockwise direction. We can also say that this torque is directed into the plane of the figure (along $\overrightarrow{OC} \times \vec{F}_2$)



$$\tan\alpha = \frac{4}{3} \quad \Rightarrow \quad \alpha = 53°$$

Given $\quad \theta = 37°$

$\therefore$ F_3 is perpendicular to OB.

$\therefore$ Torque due to F_3 is $\tau_3 = OB \cdot F_3 = \left(\frac{5}{2}\text{ m}\right)(20\,\text{N})$

$$= 50 \text{ N-m } (\circlearrowleft)$$

This torque has anticlockwise ($\circlearrowleft$) direction. You can also say that the torque vector is out of the plane of the figure. [If you still feel confused about this direction, just curl you right-hand fingers in anti-clockwise direction; the upward-pointing thumb is the direction of torque].

Torque due to F_4 is

$$\tau_4 = (OE)\,(F_4\cos\theta) \qquad [\because \quad \tau = r \cdot F_\perp]$$

$$= 2 \times 20 \times \frac{4}{5} = 32 \text{ N-m } (\circlearrowright)$$

Resultant torque is $\tau = \tau_1 + \tau_2 + \tau_3 + \tau_4$

$$= 0 + 30\,(\circlearrowright) + 50\,(\circlearrowleft) + 32\,(\circlearrowright)$$

$$= 12 \text{ N-m } (\circlearrowright)$$

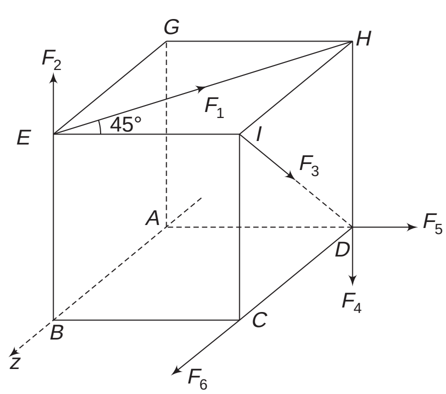
(ii) Direction of resultant torque is clockwise or we can say that the torque vector is directed into the plane of the figure.

(iii) Torque about O is directed along the given axis.

$\therefore$ Torque about the given axis is same as the torque about point O.

Note: When all $\vec{r}$ and $\vec{F}$ are in a plane (say xy plane), torque about a point O in the plane is always perpendicular to the plane (along z-axis). In such cases, it hardly makes a difference whether we say ‘torque about point O’ or ‘torque about z-axis’.

Example 3 A cube has side length a. Six forces are acting on it, as shown. Each force has magnitude equal to F. Find the torque on the cube about the side AB marked as z-axis.



Solution

Concepts

(i) Forces parallel to z-axis have no torque.

(ii) Forces having their line of action intersecting z-axis have no torque.

(iii) Forces which are along x- and y-direction and do not intersect z-axis have torque.

$\tau = F \cdot d$, where d = perpendicular distance between the force and z-axis.

F_6 is parallel to z-axis. F_2 and F_5 have their lines intersecting z-axis. These force have no torque about z-axis.

F_1 can be split into two components: $\frac{F_1}{\sqrt{2}}$ along EI and $\frac{F_1}{\sqrt{2}}$ along EG. The later component is parallel to z-axis and has no torque.

$$\therefore \qquad \tau_1 = \frac{F_1}{\sqrt{2}} \cdot (EB) = \frac{Fa}{\sqrt{2}} \quad \text{[Along negative } z]$$

Direction of this torque is along negative z-direction. Curl your right-hand fingers in the direction of rotation produced by $\frac{F_1}{\sqrt{2}}$; the thumb points along negative z-direction.

Similarly, component of F_3 along IH is parallel to z-axis and has no torque. Component along IC has torque given by

$$\tau_3 = \frac{F}{\sqrt{2}}\,(BC) = \frac{Fa}{\sqrt{2}} \qquad \text{[along negative } z]$$

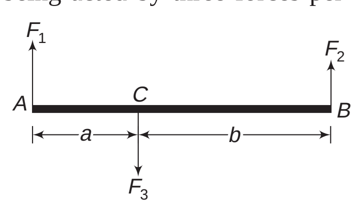
Torque due to F_4 is

$$\tau_4 = F_4(DA) = aF \qquad \text{[along negative } z]$$

$\therefore$ Resultant torque is

$$\tau = \frac{Fa}{\sqrt{2}} + \frac{Fa}{\sqrt{2}} + Fa = Fa(\sqrt{2} + 1) \text{ in negative } z \text{ direction.}$$

Example 4 A rod AB is being acted by three forces perpendicular to its length, as shown in figure. Sum of the forces is zero. Find resultant torque on AB about end A, about end B and point C.



Solution

Concepts

$$\tau = F r_\perp$$

Given, $F_3 = F_1 + F_2$ ...(i)

Torque about point A is

$$\tau_A = \tau_1 + \tau_2 + \tau_3 = 0 + F_2(a + b)(\circlearrowleft) + F_3 a(\circlearrowright)$$

$$= F_2\,(a + b) - F_3 a\,(\circlearrowleft) \qquad \text{...(ii)}$$

Torque about B

$$\tau_B = \tau_1 + \tau_2 + \tau_3$$

$$= F_1\,(a + b) - (F_1 + F_2)b\ (\circlearrowright)$$

$$= F_1\,a - F_2 b\ (\circlearrowright) \quad \text{...(iii)}$$

Using (ii) and (i) we can write:

$$\tau_A = F_2(a + b) - (F_1 + F_2)\,a\ (\circlearrowleft)$$

$$= F_2 b - F_1 a\ (\circlearrowleft)$$

$$= F_1 a - F_2 b\ (\circlearrowright)$$

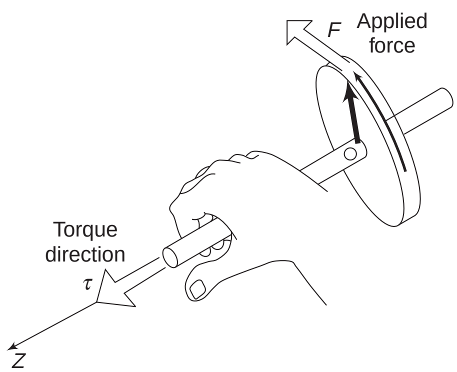
Torque about C

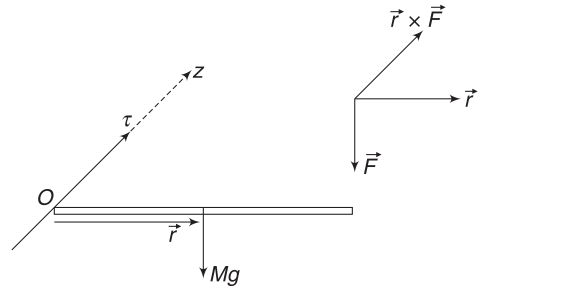
$$\tau_C = F_1 a - F_2 b\ (\circlearrowright)$$

Note: one can see that $\tau_A = \tau_B = \tau_C$. This is important. If sum of all the forces is zero, we get same torque about any point (A, B or C).

In Short:

(i) Torque is directed perpendicular to the plane containing $\vec{r}$ and $\vec{F}$.

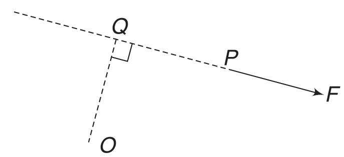




Torque of $M\vec{g}$ about O is directed into the plane of the figure. We may simply say that torque is clockwise. (Curl you right-hand fingers clockwise and the thumb points into the paper.)

(ii) In the above two figures, torque about z-axis is same as torque about point O. Torque about O has no component in x- or y-direction.

(iii) $\vec{F}$ is a force acting at point P and we wish to find its torque about O. Extend the line of $\vec{F}$ (if needed) and drop a perpendicular from O on this line. OQ is this perpendicular. Torque of F about O is $\tau = r_\perp \cdot F = (OQ)F$. In the shown figure, the force has a tendency to rotate a body clockwise about O. We can simply say that the torque is clockwise.



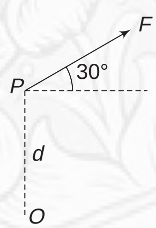
(iv) A force parallel to z-axis or intersecting z-axis will have no torque about z-axis. A force that is away from z-axis directed perpendicular to z-direction has a torque about z-axis.

(v) If resultant of all forces acting on a body is zero and the sum of their torques is zero about a point, then the resultant torque must be zero about any other point.

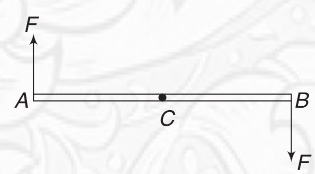
YOUR TURN

Q.1 A force $\vec{F} = 3\hat{j}$ i.e., N acts at a point (2, 2) m. Find torque of the force about the origin.

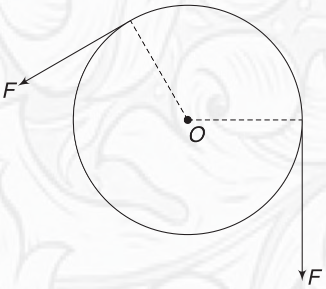
Q.2 In the figure shown, find torque of force F about O. Given $F = 20\,\text{N}$, $d = 1\,\text{m}$.



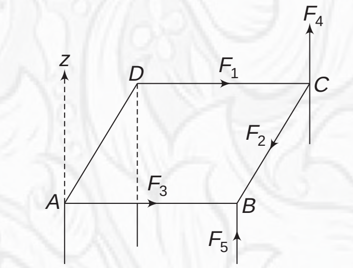
Q.3 AB is a rod. Two forces of same magnitude F act at the two ends of the rod in opposite directions. Show that torque on the rod is same about A or B or any other point C. *Such pair of equal and opposite forces are often known as a couple.*



Q.4 A pulley is free to rotate about an axis through its center O that is perpendicular to the plane of the figure. A thread passing over the pulley is being pulled at its two ends, with force F. Mass of the pulley is M. Find resultant torque on the pulley about its rotation axis.



Q.5 A table is acted upon by five forces, as shown. [Do not worry about equilibrium]. All forces have same magnitude of 20 N. Find the resultant torque about z-axis shown. Given $AB = 2$ m; $AD = 1$ m.



Q.6 A ball of mass m is projected from ground with velocity u, making an angle θ with horizontal. Find torque of weight of the ball about the point of projection when the ball is at its maximum height.

4. EQUILIBRIUM

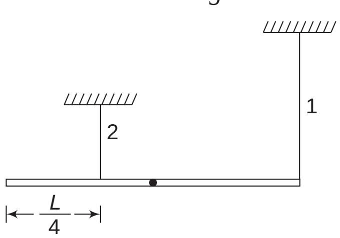
A body is said to be in equilibrium if the sum of all forces acting on it is zero and the sum of all torques acting on it about a point is also zero.

$$\left.\begin{aligned} F_{\text{net}} &= 0 \quad \text{[Translational Equilibrium]} \\ \tau_{\text{net}} &= 0 \quad \text{[Rotational Equilibrium]} \end{aligned}\right\} \text{Complete equilibrium}$$

In previous chapters, we have never actually cared about the point of application of a force. Now, we need to be careful. Torque of a force will depend on its point of application. In this context, it is very important to remember that *weight of a body must always be shown acting at the COM of the body*.

Example 5 *A stick suspended with two strings*

A uniform stick of mass M and length L is suspended horizontally with the help of two vertical strings, as shown. String 1 is tied to one end of the stick and String 2 is tied at a distance $\frac{L}{4}$ from the other end. Find tension in the two strings.



Solution

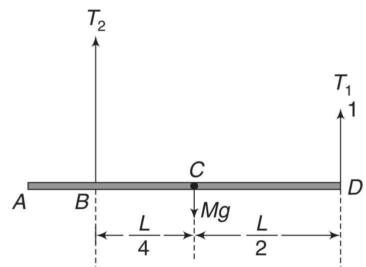
Concepts

(i) Net force on the stick is zero.
(ii) Torque on the stick about any point is zero.
(iii) The above two facts help us write two equations, which are enough to find the tension in the strings.

Weight of the rod must be shown acting at its COM.

$$F_{\text{net}} = 0$$

$$\Rightarrow \quad T_1 + T_2 = Mg \qquad \text{...(i)}$$



We will take torque on the rod about point C to be zero [You can take any other point A, B or D].

$$T_c = 0$$

Mg has no torque about this point. Tension T_1 has anti-clockwise and T_2 has clockwise torque about C. They must be of same magnitude for net torque to be zero.

$$\therefore \qquad T_1\frac{L}{2} = T_2\frac{L}{4}$$

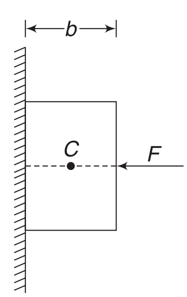
$$\Rightarrow \qquad 2T_1 = T_2 \qquad \text{...(ii)}$$

Solving (i) and (ii) gives:

$$T_1 = \frac{Mg}{3} \quad \text{and} \quad T_2 = \frac{2Mg}{3}$$

Example 6 *Line of action of normal reaction*

A uniform block of mass M is held against a vertical wall by applying a horizontal force F on it. The line of action of the force passes through its COM (C). The normal force applied by the wall on the block is acting effectively along a line that is at a distance d from C. Find d. Width of the block is b, as shown.

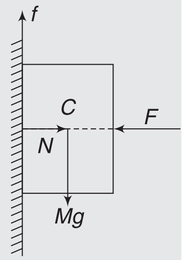


Solution

Concepts

(i) The block must be in translational as well as rotational equilibrium.

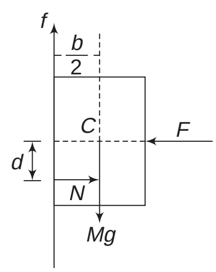
(ii) The instinctive answer to this question is that normal force must be horizontal and passing through C, but this is wrong. If N were actually passing through C, then N, F and Mg will have no torque about C. This leaves friction (f), producing an unbalanced torque about C.



It is easy to guess that N must act along a line that is below C.

For translational equilibrium,

$$N = F \quad \text{and}$$
$$f = Mg$$



For rotational equilibrium, torque about point C must be zero. In fact, torque is zero about any other point also. It is just that we have chosen point C.

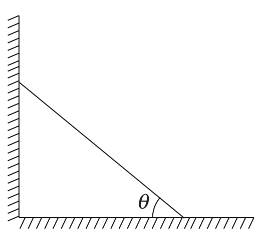
F and Mg have no torque about C. Friction (f) has a clockwise torque and normal force N has an anti-clockwise torque. For zero torque about C, these two torques must balance.

$$N \cdot d = f \cdot \frac{b}{2}$$

$$\Rightarrow \qquad F \cdot d = Mg \cdot \frac{b}{2} \quad \Rightarrow \quad d = \frac{Mgb}{2F}$$

Example 7 *Leaning ladder*

A ladder leans against a smooth wall, as shown. Coefficient of friction is μ. Find the smallest angle that the ladder can make with the floor and not slip.

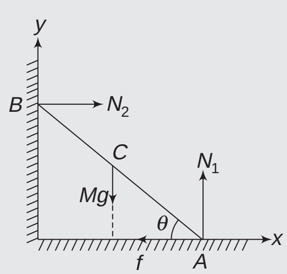


Solution

Concepts

(i) The lower end of the ladder has a tendency to slip towards right. Hence, friction on it is towards left.

(ii) We have three unknown forces: the friction (f), the normal forces N_1 and N_2 by the floor and the wall.



(iii) We also have three equations to solve for the three unknowns:

$$F_x = 0, \quad F_y = 0 \quad \text{and} \quad \tau = 0$$

(iv) Torque is zero about any point. Any point like A, B or C (or any other point) will work fine. But a certain choice can make calculation easier. The best choice, in general, is a point at which most number of forces act. In this case, $\tau = 0$ equation will have the least number of terms. In this problem, two forces act at end A (friction and N_1). We will choose A as the pivot point about which we will balance the torque.

$F_x = 0$ gives

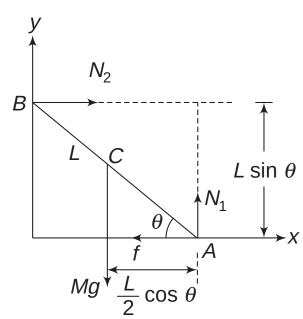
$$N_2 = f \qquad \text{...(i)}$$

$F_y = 0$ gives

$$N_1 = Mg \qquad \text{...(ii)}$$

$\tau_A = 0$ gives:

$$N_2 \, L \sin\theta = Mg \, \frac{L}{2} \cos\theta$$



[Note that line of N_2 is at a perpendicular distance $L \sin\theta$ from A and line of Mg is at a perpendicular distance $\frac{L}{2}\cos\theta$ from A]

$$\Rightarrow \qquad N_2 \tan\theta = \frac{1}{2} Mg$$

Using (i): $f \tan\theta = \frac{1}{2} Mg$

$$\Rightarrow \qquad f = \frac{Mg}{2\tan\theta}$$

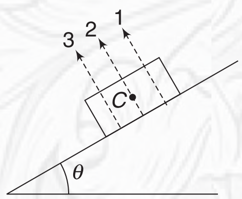
But $f \le \mu N_1 \;\Rightarrow\; f \le \mu Mg$ [using (ii)]

$$\therefore \quad \frac{Mg}{2\tan\theta} \le \mu Mg$$

$$\Rightarrow \quad \tan\theta \ge \frac{1}{2\mu}$$

Your Turn

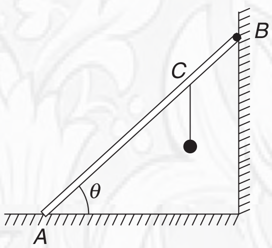
Q.7 A uniform rectangular block is placed on an incline and it stays at rest. C is the centre of the block. The normal force by incline on the table will be effectively acting along a line as indicated by 1, 2 or 3. Which is correct?



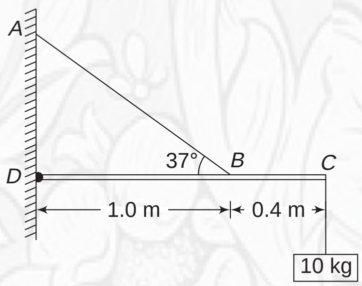
Q.8 A rod of length L = 6 m is hinged to the floor at its lower end A and leans against a smooth wall. Mass of the rod is 20 kg. A 40 kg ball is tied at point C of the rod such that AC = 4 m. Given, θ = 53°

(i) Find the force applied by the wall at B on the rod.

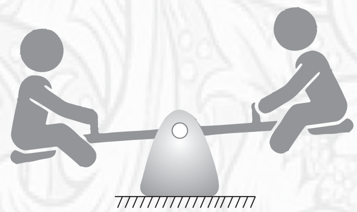
(ii) Find the vertical component of force applied by the hinge at A.



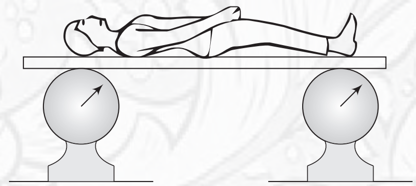
Q.9 CD is a uniform 5 kg beam. It is horizontal and is hinged at end D. Cable AB supports it with $\angle ABD$ = 37°. Find tension in the cable and horizontal and vertical components of hinge force when a 10 kg load is hanged from the end C of the beam.



Q.10 A seasaw has a length of 5.0 m and its fulcrum (support) is at its centre. Two boys weighing 10 kg and 15 kg are trying to keep it in balance. The 10 kg boy sits at one of the ends. At what distance from the centre, the other boy must sit to keep the seesaw in equilibrium?

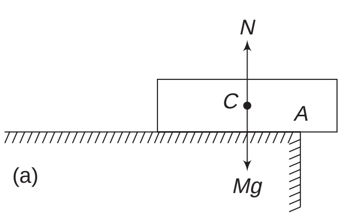


Q.11 A 160-cm tall boy is sleeping on a massless board. The board is horizontal, supported by two weighing scales – one right under the head of the boy and the other under his toe. Reading of the two scales are 30 kg and 25 kg respectively. At what distance is the centre of gravity of the boy from his toe?

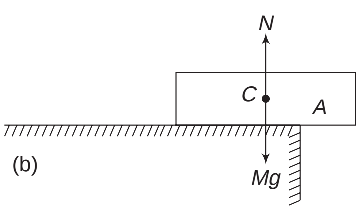


5. TOPPLING

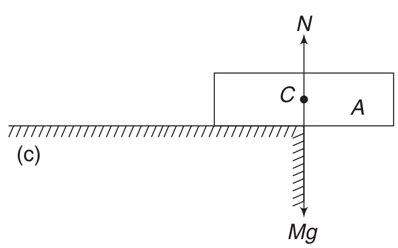
Consider a brick placed on a table. In figure (a), the brick is shown slightly protruding out of the edge of the table. Its weight (Mg) is a vertical force through its COM (C). For equilibrium, the normal force (effectively) passes through C. This makes F_{net} = 0 and τ_{net} = 0. The brick can stay in equilibrium in this position forever. The brick is pushed slightly to the left (figure b) but the vertical line of Mg is still passing through the support region. Normal force will adjust so



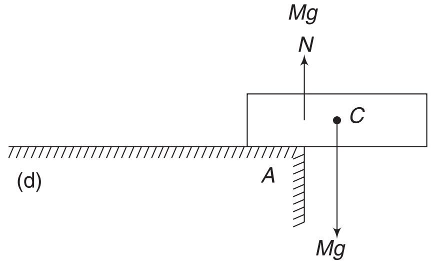
(a)



(b)



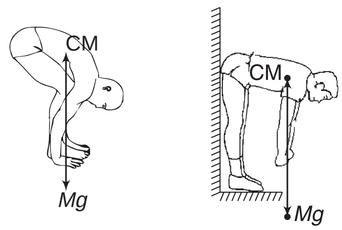
(c)



(d)

that it passes through the COM (C) and the body will stay in equilibrium. If the brick is pushed a little more towards right, such that C lies just above the edge A. It will be in critical equilibrium with normal force, adjusting itself to pass through C. Any further push to the brick will cause the line of Mg to fall outside the support region. Now, normal force cannot shift to the right of point A; at best, it can stay at A (see Fig. *d*). Clearly, there is an unbalanced torque on the brick about A. This causes it to rotate and it topples.

This result is true in general. When you bend to touch your feet, you body posture adjusts so that the vertical line through COM of your body passes through the support region between your feet. If you try doing this act while your back rests against a wall, you will never be able to do it. As you bend, your hip cannot go back (as there is wall) and COM of your body moves forward. The vertical line of Mg falls outside the support region between your feet.



5.1 A Cube on an Incline

Consider a uniform cube placed on a rough incline. When inclination angle is very small (as shown in Fig. (a)), the line of Mg passes within the support region, AB and the cube will not topple. If friction is large enough ($\mu > \tan\theta_1$), the cube will not slide also. It will stay at rest.

If inclination angle is increased to θ_2 (see Fig. (1)). Such that line of Mg passes through point A, the normal forces shifts to point A. Now all three forces (Mg, N and friction f) are passing through point A. There is no torque about A. Block will still not rotate.

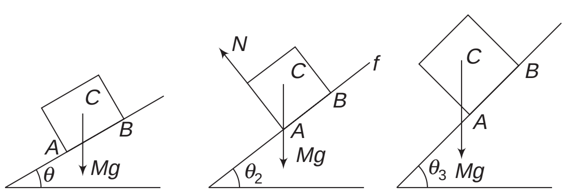
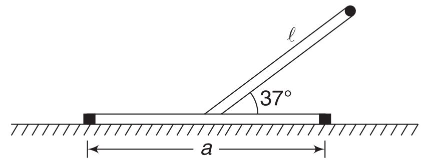


Fig. (a) Fig. (b) Fig. (c)

If coefficient of friction μ is large enough ($\mu > \tan\theta_2$), it will not slide as well. This is the critical case. If inclination is increased beyond this (see Fig (c)), line of Mg passes outside the support region AB. Normal force cannot move beyond A. Now the block has a net torque about A due to Mg. If friction coefficient is still good enough ($\mu > \tan\theta_3$), the block will not slide but it will topple.

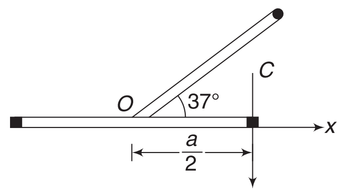
Example 8 A uniform bar has length a. It is resting on a horizontal table. Another bar made of the same material and having the same cross-sectional area is welded to the centre of the first bar so that angle between them is 37°. Find the maximum length l of the second bar so that the body can stay in equilibrium in a vertical plane, as shown.



Solution

Concepts

The vertical line through COM of the body should pass within the support region for equilibrium. In critical case, the vertical line through COM will pass through end of the horizontal bar.



Let λ = linear mass density of the two bars.

Mass of horizontal bar, $m_1 = a\,\lambda$

Mass of inclined bar, $m_2 = l\lambda$

Taking O as orgin, the x co-ordinate of COM of the system is

$$x_{CM} = \frac{m_1(0) + m_2\left(\frac{l}{2}\cos 37°\right)}{m_1 + m_2}$$

$$= \frac{(l\lambda)\left(\frac{l}{2}\cdot\frac{4}{5}\right)}{a\lambda + l\lambda} = \frac{2l^2}{5(a+l)}$$

For equilibrium, $x_{CM} \le \frac{a}{2}$

In critical case, $x_{CM} = \frac{a}{2}$

$$\Rightarrow \quad \frac{2l^2}{5(a+l)} = \frac{a}{2} \quad \Rightarrow \quad 4l^2 - 5al - 5a^2 = 0$$

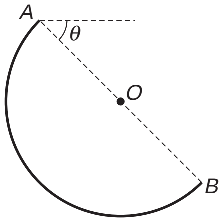
$$\Rightarrow \quad l = \frac{5a \pm \sqrt{25a^2 + 80a^2}}{8}$$

$$\Rightarrow \quad l = \left(\frac{5 + \sqrt{105}}{8}\right)a$$

If l is increased beyond this, x_{CM} will exceed $\frac{a}{2}$ and the body will topple.

Example 9 *Suspended semi-circular ring*

A semi-circular ring is suspended in a vertical plane such that it can rotate freely about its end A. Find the angle (θ) that the diameter AB makes with the horizontal in equilibrium.



Solution

Concepts

(i) The COM of the semi-circular ring is at a distance $\frac{2R}{\pi}$ from O, lying on its symmetry line.

(ii) The ring will stay in equilibrium only if torque of external force about A is zero.

This is possible only if COM of the half ring lies vertically below point A so that Mg produces no torque about A.

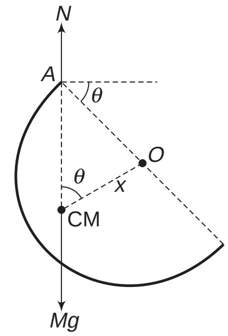
The equilibrium position is as shown in the figure.

CM is exactly below A.

Support (at A) applies a force (N)

$$N = Mg$$

Both N and Mg have zero torque about A.



$$x = \frac{2R}{\pi}$$

$$\cos\theta = \frac{x}{R} = \frac{2}{\pi}$$

$$\therefore \quad \theta = \cos^{-1}\left(\frac{2}{\pi}\right)$$

In Short:

(i) A body is in equilibrium if sum of all external forces acting on it is zero and the sum of all external torques (about any point) is also zero.

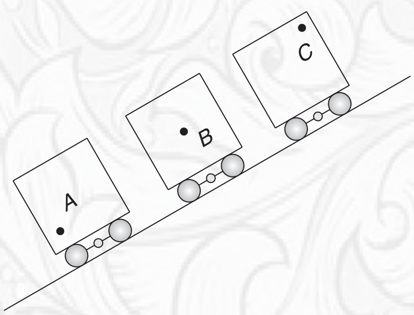
(ii) A body remains in equilibrium if vertical line through COM of the body passes through the support region. If this line falls outside the support region, torque due to Mg makes the body unstable.

(iii) A freely suspended body stays in equilibrium in a position where its COM is exactly below its point of suspension.

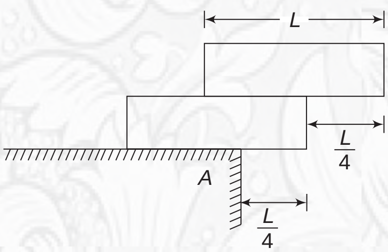
YOUR TURN

Q.12 Why must you bend forward while carrying a heavy load on your back?

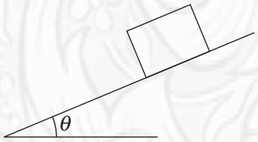
Q.13 Three trucks are parked on a hill. Their centre of gravity lies at A, B and C (see fig). Which truck will tip over?



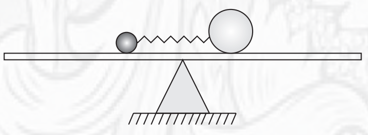
Q.14 Two identical uniform bricks have been arranged on a table, as shown. Is the system in equilibrium?



Q.15 A cubical block is kept on a rough incline. The inclination angle (θ) of the incline is increased gradually. Find the maximum value of coefficient of friction (μ) so that the block starts sliding before it topples.



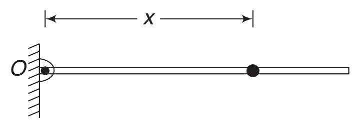
Q.16 A long smooth plank is balanced like a seesaw. It supports a small ball and a more massive ball, with compressed spring between the two. The system is balanced with the balls at the centre of the plank. Now the spring is released. The two balls move away from each other. Does the plank tip clockwise, anti-clockwise or remain in balance, as the balls move away from each other?



Miscellaneous Examples

Example 10 *Rod with a bead*

A bead of mass m is threaded on a light rod that is pivoted to a point O on a vertical wall. Initially, the rod is held in horizontal position with the bead at a distance x from O. The rod is released from this position and it rotates in vertical plane about O. Calculate the angle θ that the rod makes with horizontal at time t after its release.

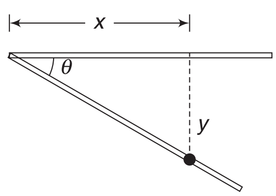


Solution

Concepts

The massless rod cannot apply a normal force on the bead. If there is such a force, then the rod will experience a finite torque (and hence infinite angular acceleration). If there is no normal force, there cannot be friction between the bead and the rod.

The bead experiences free fall. In time t, it falls through a distance given by:



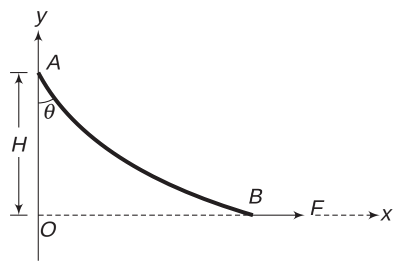
$$y = \frac{1}{2}gt^2$$

$$\therefore \quad \tan\theta = \frac{y}{x} = \frac{gt^2}{2x}$$

$$\therefore \quad \theta = \tan^{-1}\left(\frac{gt^2}{2x}\right)$$

Example 11 *Flexible rope*

One end (A) of a flexible rope is fixed to a vertical wall and its other end (B) is pulled by a horizontal force F. The rope makes an angle of $\theta = \sin^{-1}\left(\frac{3}{5}\right)$ with the wall and its top end is at a vertical height H above the other end. Find the mass of the rope and x co-ordinates of its centre of mass (in the co-ordinate system shown in the figure.)



Solution

Concepts

(1) Horizontal component of tension at $A = F$ and vertical component of tension at $A = mg$.

(2) For rotational equilibrium, the torque about any point should be zero

$$T\sin\theta = F$$

$$\Rightarrow \quad T = \frac{5F}{3}$$

Also $\quad mg = T\cos\theta$

$$\Rightarrow \quad m = \frac{5F}{3}\cdot\frac{4}{5\cdot g}$$

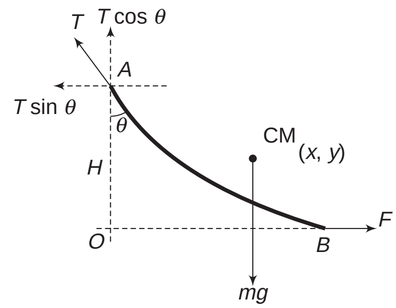
$$= \frac{4F}{3g}$$

Now consider torque about end A.

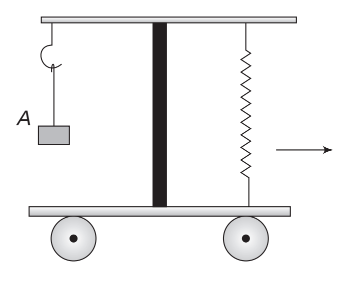
$$\tau_A = 0$$

$$\Rightarrow \quad mg\cdot x = FH$$

$$\Rightarrow \quad \frac{4F}{3}x = FH \quad \Rightarrow \quad x = \frac{3H}{4}$$



Example 12 A balance is mounted on a stationary trolley, with a weight (A) attached to its one end. The other end of the balance is linked to the floor of the trolley by a spring. The trolley is accelerated to the right and the string holding the weight gets inclined to the vertical. When the weight gets stable and is not swinging, does the tension in the spring change?



Solution

Concepts

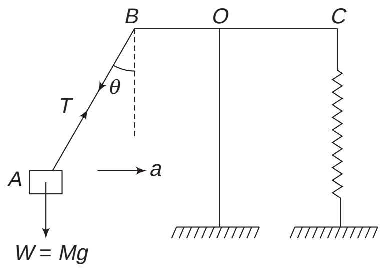
The balance will remain balanced if torque of forces at the two ends, about the pivot point, is zero.

Tension in the string increase since

$$T\cos\theta = Mg$$

$$T\sin\theta = Ma$$

$$\Rightarrow \qquad T = M\sqrt{g^2 + a^2}$$

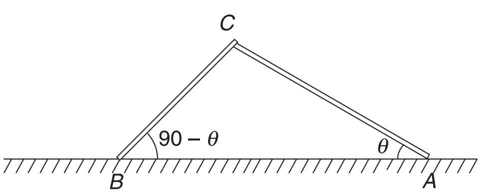


But torque of tension force acting at B (torque about O) is

$$T\cos\theta\cdot(BO) = Mg\cdot(BO)$$

Since torque is unchanged, the spring tension will also not change.

Example 13 One thin rod leans on another in vertical plane, as shown. The right rod makes an angle θ with the horizontal floor and the left rod makes an angle $(90 - \theta)$ with the floor. The left rod extends infinitesimally beyond the end of the right rod. Coefficient of friction between the two rods is μ and both sticks have same mass per unit length. The ends A and B on the ground are hinged so that the rods can rotate about A and B. Find minimum value of θ for which the rods do not fall.



Solution

Concepts

(i) The angle between the rods is 90°.

(ii) Mass of each rod is proportional to its length.

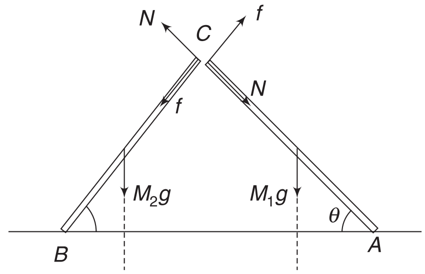
(iii) The two rods apply normal force (N) and friction (f) on one another.

(iv) To avoid hinge force in our equations, we will balance torque about A and about B, for rods AC and BC respectively.

Let λ = mass of unit length

Mass of rod AC is $M_1 = \lambda(AC)$

Mass of rod BC is $M_2 = \lambda(BC)$



Notice the direction of friction on two rods. Friction on AC must produce clockwise torque to balance the anti-clockwise torque of M_1g about A.

Forces on the two rods are as shown. We have not shown forces acting at A and B, since we plan to take torque about these points.

For rotational equilibrium of rod AC, we take

$$\tau_A = 0$$

$$f(AC) = M_1g\left(\frac{AC}{2}\cos\theta\right)$$

$$\Rightarrow \qquad f = \frac{M_1g}{2}\cos\theta \qquad \text{...(i)}$$

For rotational equilibrium of BC, let us take $\tau_B = 0$

$$N(BC) = M_2g\left(\frac{BC}{2}\cos(90 - \theta)\right)$$

$$\Rightarrow \qquad N = \frac{M_2g}{2}\sin\theta \qquad \text{...(ii)}$$

We know that

$$f \le \mu N$$

$$\Rightarrow \qquad \frac{M_1g}{2}\cos\theta \le \mu\frac{M_2g}{2}\sin\theta$$

$$\Rightarrow \qquad \frac{M_1}{M_2} \le \mu\tan\theta$$

$$\Rightarrow \qquad \frac{\lambda AC}{\lambda BC} \le \mu\tan\theta \quad \Rightarrow \quad \frac{1}{\tan\theta} \le \mu\tan\theta$$

$$\Rightarrow \qquad \tan^2\theta \ge \frac{1}{\mu} \quad \Rightarrow \quad \theta_{\min} = \tan^{-1}\left(\frac{1}{\sqrt{\mu}}\right)$$

Note: Let us check our answer in two extreme cases:

(i) When $\mu \to 0$, θ must be close to 90°. It makes sense as we need to have a very small angle between the rods, if friction between the sticks is very small.

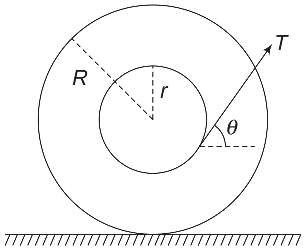
(ii) When $\mu \to \infty$ (i.e, very sticky rods), θ can be very small. This also makes sense.

Example 14 *A spool at rest*

A spool has an axle of radius r and outer disc of radius $2R$. A thread is tightly wrapped around the axle and is pulled with force T at an angle θ with horizontal. Friction between ground and the spool is sufficiently large and the spool does not moves (it neither translates nor rotates).

(i) Find θ

(ii) If coefficient of friction between the spool and the ground is μ, find the largest value of T for which the spool remains at rest.



Solution

Concepts

(i) We will write equations using followings:

$F_{\text{Horz}} = 0;\quad F_{\text{vert}} = 0;\quad \tau_{\text{centre}} = 0.$

(ii) Another quick way of solving this problem is to realise that the friction force, the normal force and the weight of the spool pass through the contact point. If line of action of T also passes through this point, then there will be no resultant torque.

(i) $F_{\text{Horz}} = 0$

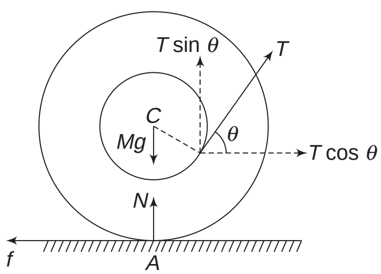
$$\Rightarrow \qquad f = T\cos\theta \qquad \text{...(i)}$$

$$F_{\text{vert}} = 0$$

$$N + T\sin\theta = Mg \qquad \text{...(ii)}$$

Torque about centre, $\tau_{\text{centre}} = 0$

$$T\cdot r = f\cdot R \qquad \text{...(iii)}$$



[Force T is tangential at a distance r from C and f is at a distance R. N and Mg pass through C]

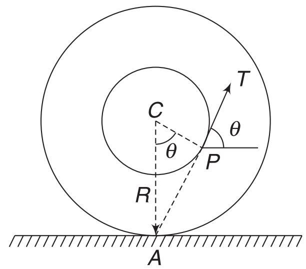
From (i) and (iii)

$$\frac{f}{T} = \cos\theta = \frac{r}{R}$$

$$\therefore \qquad \theta = \cos^{-1}\left(\frac{r}{R}\right)$$

Alternate:

Mg, N and f pass through A. If T also passes through A, net torque on spool will be zero. If line of T does not pass from A it will create a torque about A and the spool cannot remain in equilibrium.



From figure in ΔCPA: $\cos\theta = \dfrac{r}{R}$

From equation (ii),

$$N = Mg - T\sin\theta$$

Since $f \le \mu N$

$T\cos\theta \le \mu(Mg - T\sin\theta)$ $\quad [f = T\cos\theta$ from (i)]

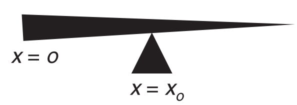
$$\Rightarrow \qquad T \le \frac{\mu Mg}{\cos\theta + \mu\sin\theta}$$

Since, $\cos\theta = \dfrac{r'}{R}$ $\qquad$ Hence, $\sin\theta = \dfrac{\sqrt{R^2 - r^2}}{R}$

$$\therefore \qquad T \le \frac{\mu Mg}{\dfrac{r}{R} + \dfrac{\mu\sqrt{R^2 - r^2}}{R}} = \frac{\mu MgR}{r + \mu\sqrt{R^2 - r^2}}$$

Example 15 A stick is semi-infinite, i.e, one end goes off to infinity. Linear mass density of the stick is given by $\lambda = \lambda e^{-x/l}$, where x is the distance from one of its ends and l is a constant. It balances on a support at a distance x_0 from its end at $x = 0$. Find x_0.

Given: $\displaystyle\int_0^\infty x e^{-ax} = \frac{1}{a^2}$



Solution

Concepts

The support must be at the COM of the stick. It means $x_{\text{CM}} = x_0$

Consider a small elemental length dx of the stick at co-ordinate x.

Mass of element $dm = \lambda dx = \lambda e^{-x/l} dx$

$$\therefore \qquad x_{\text{CM}} = \frac{\int_0^\infty x\,dm}{\int_0^\infty dm} = \frac{\lambda_0 \int_0^\infty x e^{-x/l}\,dx}{\lambda_0 \int_0^\infty e^{-x/l}\,dx}$$

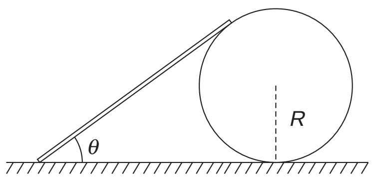
$$\int_0^\infty x e^{-\frac{x}{l}}\,dx = \frac{1}{\left(-\frac{1}{l}\right)^2} = l^2$$

And $\displaystyle\int_0^\infty e^{-x/l}\,dx = -l\left[e^{-x/l}\right]_0^\infty = -l[0-1] = l$

$\therefore \qquad x_{CM} = \dfrac{l^2}{l} = l$

Example 16 *Rod on a hoop*

A rod of linear mass density λ leans against a hoop of radius R. The rod makes an angle θ with the horizontal and is tangent to the hoop at its upper end. Friction is large enough to keep the system at rest. Find the friction force between the ground and the hoop.

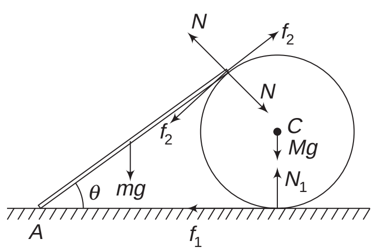


Solution

Concepts

(i) Mass of the hoop is not given. We will not write the equation of vertical equilibrium for the hoop. We will consider its horizontal equilibrium and rotational equilibrium only.

(ii) We will consider torque on the rod about its lower end. This will remove the need to consider force acting on the lower end of the rod.



Ground friction on hoop $= f_1$

Friction between the rod and the hoop $= f_2$

Normal force between the rod and the hoop $= N$

For rotational equilibrium of the hoop, net torque on it about its centre (C) must be zero.

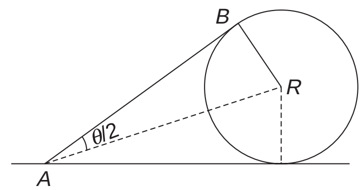
N, N_1 and Mg have no torque about C.

$\therefore$ Torque due to f_1 and f_2 must be equal and opposite

$$\Rightarrow \qquad f_1R = f_2R \quad \Rightarrow \quad f_1 = f_2 \qquad \text{...(i)}$$

Length of rod $AB = \dfrac{R}{\tan\dfrac{\theta}{2}}$

$\therefore$ Mass of rod, $m = \dfrac{\lambda R}{\tan\dfrac{\theta}{2}}$



Torque on the rod about A is zero.

$$\therefore \qquad N(AB) = mg\left(\frac{AB}{2}\cos\theta\right)$$

$$\Rightarrow \qquad N = \frac{1}{2}\frac{\lambda Rg}{\tan\dfrac{\theta}{2}}\cos\theta \qquad \text{...(ii)}$$

For horizontal equilibrium of the hoop,

$$f_1 + f_2\cos\theta = N\sin\theta$$

$$\Rightarrow \quad f_1 + f_1\cos\theta = \frac{\lambda Rg}{2\tan\dfrac{\theta}{2}}\cdot\cos\theta\cdot\sin\theta$$

$$\Rightarrow \qquad f_1 = \frac{\lambda Rg\sin\theta\cos\theta}{2\tan\dfrac{\theta}{2}(1+\cos\theta)}$$

$$\left[\because \quad \frac{\sin\theta}{1+\cos\theta} = \tan\frac{\theta}{2}\right]$$

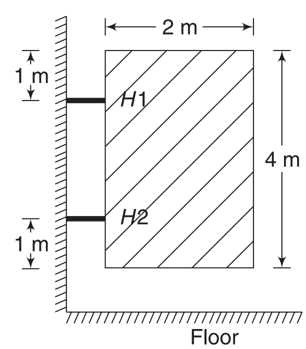
$$= \frac{1}{2}\lambda Rg\cos\theta$$

Let us check the result under extreme condition when $\theta \to \dfrac{\pi}{2}$, $\quad f_1 \to 0$

This makes perfect sense.

Example 17 *A hinged door*

A door in a showroom is 4 m tall and 2 m wide. It is hinged at two locations on the wall. Hinge $H1$ is located 1 m from the top and hinge $H2$ is located 1 m from the bottom of the door. Mass of the door is 100 kg and each hinge applies force of equal magnitude on the door. Find force applied by each hinge on the door.

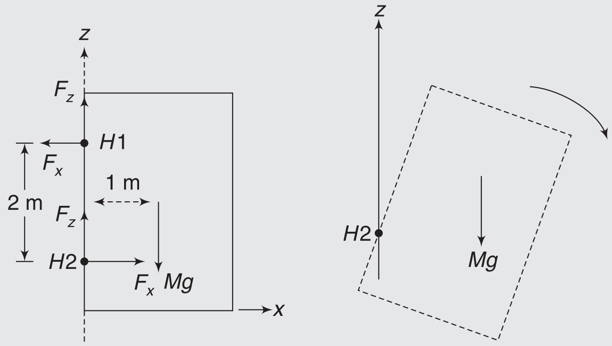


Solution

Concepts

(i) Though the door is free to rotate about vertical (z) axis, we are not concerned about the torque about z-axis in this problem.

(ii) Look at the figure. Mg has a tendency to rotate the door clockwise about $H2$. If hinge $H1$ breaks, torque of Mg will turn the door as shown in the second figure.



This is rotation about y-direction. To prevent this rotation, the horizontal force applied by $H1$ on the door must be to the left. This will cause an anti-clockwise torque on the door about $H2$.

(iii) For horizontal equilibrium of the door, the horizontal components of two hinges forces must be equal and opposite.

(iv) Since forces by two hinges are of equal magnitude and their x components have same magnitude; the z components must also be same.

$$2F_z = Mg \quad \Rightarrow \quad F_z = \frac{100 \times 10}{2} = 500\,\text{N}$$

Net torque about $H2$ is zero.

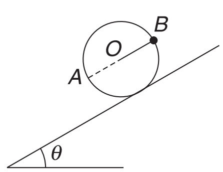
$$\therefore \qquad F_x(2\,\text{m}) = Mg(1\,\text{m}) \qquad \left[\begin{array}{r} F_z \text{ has no torque about} \\ H2.\ F_x \text{ passing through} \\ H2 \text{ also has no torque} \end{array}\right]$$

$$\Rightarrow \qquad F_x = 500\,\text{N}$$

$$\therefore \quad \text{Force by one hinge} = \sqrt{F_x^2 + F_z^2} = 500\sqrt{2}\ \text{N}$$

Example 18 *Keeping a disc at rest on an incline*

A uniform disc of mass M is kept on a rough incline of inclination angle $\theta = 37°$. A small particles of mass m is fixed on the circumference at point B. When the disc is placed on the incline with its diameter AB parallel to the incline, it stays in equilibrium. Find:



(i) m

(ii) minimum coefficient of friction between the disc and the incline.

Solution

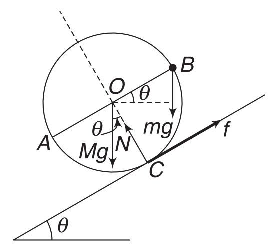
Concepts

(i) Net force and torque on the (disc + particle) combined system must be zero.

(ii) We will balance forces along the incline and perpendicular to it and will take torque to be zero about the centre of the disc.

However, choosing contact point as pivot point [for $\tau = 0$] can straightaway give the value of m. This method is being left for students to try.

One other method, is to understand that COM of the combined system must lie exactly above the contact point.



Net force along the incline is zero.

$$\therefore \qquad (Mg + mg)\sin\theta = f$$

$$\Rightarrow \qquad \frac{3}{5}(M + m)g = f \qquad \text{...(i)}$$

Net force normal to the incline is also zero.

$$\therefore \quad N = (Mg + mg)\cos\theta = \frac{4}{5}(M + m)g \qquad \text{...(ii)}$$

Torque about centre (O) of the disc = 0

$$\Rightarrow \qquad f \cdot R = mg \cdot R\cos\theta$$

[$\because$ Mg and N have no torque about O]

$$\Rightarrow \qquad f = \frac{4}{5}mg \qquad \text{...(iii)}$$

From (i) and (ii), $4m = 3(M + m) \quad \Rightarrow \quad m = 3M$

Since $\qquad f \le \mu N$

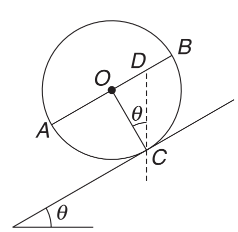
$$\therefore \qquad \frac{4}{5}mg \le \mu\,\frac{4}{5}(M + m)g$$

$$\Rightarrow \qquad 3M \le 4M\mu$$

$$\Rightarrow \qquad \frac{3}{4} \le \mu$$

Alternate for part (i)

Consider a vertical line through contact point C. This line intersects the diameter AB at point D. This point D must be the COM of the system.



From figure: $\dfrac{OD}{OC} = \tan\theta$

$$\Rightarrow \quad OD = R \times \frac{3}{4} = \frac{3}{4}R. \text{ And } DB = \frac{R}{4}$$

Mass (M) of the disc is assumed at O. For D to be COM of the system,

$$M(OD) = m(DB)$$

$$\Rightarrow \quad M\left(\frac{3}{4}R\right) = m\left(\frac{R}{4}\right) \quad \Rightarrow \quad m = 3M$$

Example 19 *Lifting a rod*

A uniform rod is lying on a table. A boy applies force at its one end so as to slowly lift the rod. He keeps the force always perpendicular to the rod. The other end of the rod does not slide and the rod was brought to a vertical position. Determine the minimum coefficient of friction between the rod and the table.

Solution

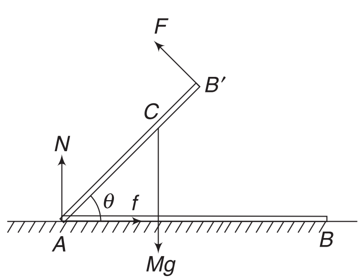
Concepts

(i) The rod is raised slowly. It means there is no acceleration and the rod is in equilibrium at any position.

(ii) To do this, the torque due to applied force must always equal to the torque due to weight of the rod.

(iii) Considering the equation for horizontal and vertical equilibrium will give us the normal force and friction force by the floor on the rod.

AB is the original position of the rod. Consider it at position AB'. End A does not slide and applied force F is perpendicular to the rod AB'.



Rod is always in equilibrium.

$$\tau_A = 0$$

$$Fl = Mg\left(\frac{l}{2}\cos\theta\right), \text{ where } l = \text{length of the rod}$$

$$\Rightarrow \quad F = \frac{1}{2}Mg\cos\theta \qquad \text{...(i)}$$

For vertical equilibrium:

$$N + F\cos\theta = Mg \quad \Rightarrow \quad N = Mg - F\cos\theta \qquad \text{...(ii)}$$

For horizontal equilibrium:

$$f = F\sin\theta \qquad \text{...(iii)}$$

But $\qquad f \le \mu N$

$$\therefore \quad F\sin\theta \le \mu(Mg - F\cos\theta)$$

$$\Rightarrow \quad \frac{F\sin\theta}{Mg - F\cos\theta} \le \mu$$

$$\Rightarrow \quad \frac{\sin\theta}{\dfrac{Mg}{F} - \cos\theta} \le \mu \quad \Rightarrow \quad \frac{\sin\theta}{\dfrac{2}{\cos\theta} - \cos\theta} \le \mu$$

[Using (i)]

$$\Rightarrow \quad \frac{\sin\theta\cos\theta}{2 - \cos^2\theta} \le \mu \quad \Rightarrow \quad \frac{\sin\theta\cos\theta}{2(\sin^2\theta + \cos^2\theta) - \cos^2\theta} \le \mu$$

$$\Rightarrow \quad \frac{\sin\theta\cos\theta}{2\sin^2\theta + \cos^2\theta} \le \mu \quad \Rightarrow \quad \frac{1}{2\tan\theta + \cot\theta} \le \mu$$

μ is minimum when $z = 2\tan\theta + \cot\theta$ is maximum.

$$\Rightarrow \quad \frac{dz}{d\theta} = 0 \quad \Rightarrow \quad 2\sec^2\theta - \operatorname{cosec}^2\theta = 0$$

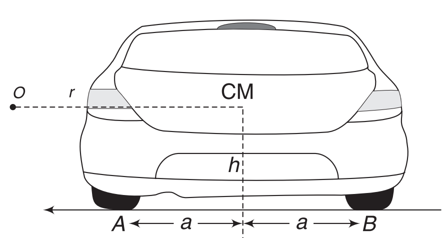
$$\Rightarrow \quad \tan\theta = \frac{1}{\sqrt{2}}$$

$$\therefore \quad \mu_{\min} = \frac{1}{2\tan\theta + \cot\theta} = \frac{1}{\dfrac{2}{\sqrt{2}} + \sqrt{2}}$$

$$\Rightarrow \quad \mu_{\min} = \frac{1}{2\sqrt{2}}$$

Example 20 *Car on a circular track*

A car is negotiating a circular track of radius r. Centre of the track is O and the car is going into the plane of the figure. In the figure, you are seeing the rear wheels. Separation between the wheels is $2a$ and height of the COM of the car above the road is h; friction is large enough to prevent slipping. Find maximum speed (v_0) of the car so that it does not overturn about the outer wheels.

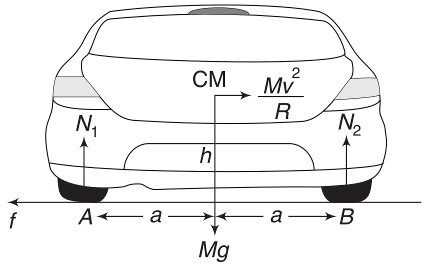


Solution

Concepts

(i) In reference frame of the car, it is a problem of equilibrium. In this frame, we need to apply the centrifugal force at the COM.

(ii) The car begins to overturn when normal contact force between the inner wheel (A) and the ground becomes zero.



For horizontal equilibrium (in *RF* attached to the car):

$$f = \frac{Mv^2}{R}$$

For vertical equilibrium:

$$N_1 + N_2 = Mg$$

For the rotational equilibrium let us take torque about *B* to be zero

$$N_1 \cdot 2a + \frac{Mv^2}{R} \cdot h = Mg\,a$$

$$\Rightarrow \quad N_1 = \frac{Mg}{2} - \frac{Mv^2 h}{2aR}$$

Car will not overturn if $N_1 \geq O$

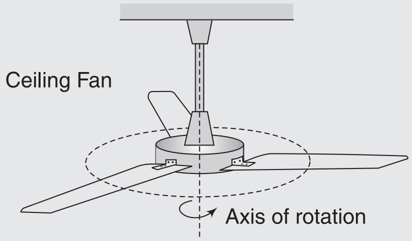
$$\Rightarrow \quad \frac{Mg}{2} \geq \frac{Mv^2 h}{2aR}$$

$$\Rightarrow \quad \sqrt{\frac{aRg}{h}} \geq v$$

$$\therefore \quad v_0 = \sqrt{\frac{aRg}{h}}$$

Worksheet 1

1. When a ceiling fan rotates,



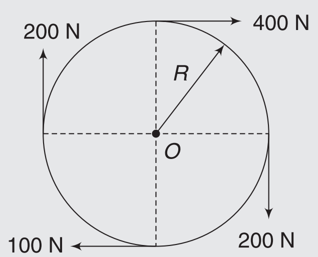
(a) all particles go in circles of same radius.

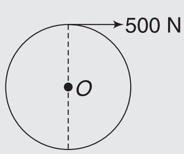
(b) its weight has a torque about its rotation axis.

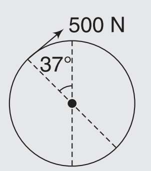
(c) all particles go in circles having their centres on the rotation axis.

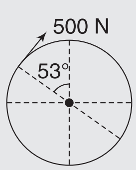
(d) None of the above.

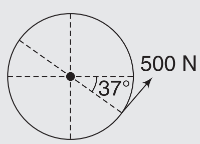
2. Four forces tangent to the circle of radius 'R' are acting on a wheel, as shown in the figure. The resultant equivalent one-force system will be:



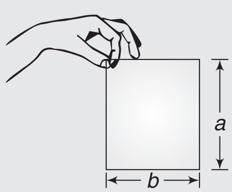
(a)


(b)


(c)


(d)


3. A person supports a book of weight W between the forefinger and the thumb, as shown. The point of grip is assumed to be at the corner of the book. Think of an axis that is passing through the grip point and is perpendicular to the plane of the figure. The person is producing a torque on the book about this axis, which is equal to:



(a) $W\frac{a}{2}$ anti-clockwise

(b) $W\frac{b}{2}$ anti-clockwise

(c) Wa anti-clockwise

(d) Wa clockwise

4. A weightless horizontal rod is acted on by upward parallel forces of 2 N and 4 N at ends A and B respectively. The total length of the rod AB = 3 m. To keep the rod in equilibrium, a force of 6 N should act in the following manner.

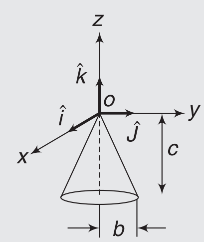
(a) Downwards, at any point between A and B.

(b) Downwards, at mid-point of AB.

(c) Downwards, at a point C such that AC = 1m.

(d) Downwards, at a point D such that BD = 1m.

5. A solid cone hangs from a frictionless pivot at the origin O, as shown. If $\hat{i}$, $\hat{j}$ and $\hat{k}$ are unit vectors, and a, b, and c are positive constants, which of the following forces F applied at a point on the lower rim of the cone results in a torque τ on the cone with a negative z component τ_z?



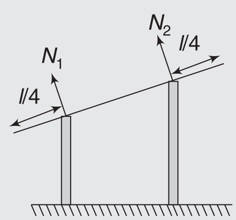
(a) $F = a\hat{k}$, at point $(0, b, -c)$

(b) $F = -a\hat{k}$, at point $(0, -b, -c)$

(c) $F = a\hat{j}$, at point $(-b, 0, -c)$

(d) None

6. A uniform rod of length l is placd symmetrically on two walls, as shown in figure. The rod is in equilibrium. If N_1 and N_2 are the normal forces exerted by the walls on the rod, then



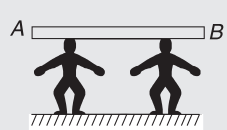
(a) $N_1 > N_2$

(b) $N_1 < N_2$

(c) $N_1 = N_2$

(d) N_1 and N_2 would be in the vertical directions

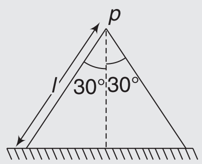
7. Two persons of equal height are carrying a long, uniform wooden beam of length l. They are at a distance $l/4$ and $l/6$ from the nearest ends of the rod. The ratio of normal reactions at their heads is



(a) 2 : 3 (b) 1 : 3

(c) 4 : 3 (d) 1 : 2

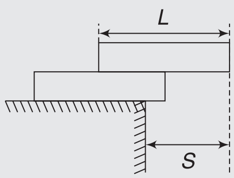
8. An inverted V is made up of two uniform boards, each weighing 200 N. Each side has the same length and makes an angle 30° with the vertical, as shown in the figure. What is the magnitude of the static frictional force that acts on each of the lower end of the V?



(a) $\frac{50}{\sqrt{3}}$ N (b) $\frac{200}{\sqrt{3}}$ N

(c) $\frac{100}{\sqrt{3}}$ N (d) $\frac{60}{\sqrt{3}}$ N

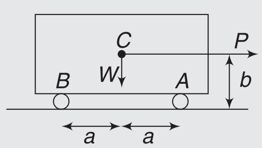
9. Two identical bricks of length L are piled one on top of the other, on a table, as shown in the figure. The maximum distance S that the top brick can overhang the table with the system still balanced is:



(a) $\frac{1}{2}L$ (b) $\frac{2}{3}L$

(c) $\frac{3}{4}L$ (d) $\frac{7}{8}L$

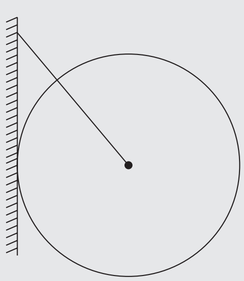
10. An automobile of weight W is standing on a smooth road as shown. Distance between the front wheels and the rear wheels is $2a$ and the height of COM is b from the road. A horizontal pull force P is applied on the automobile with its line of action passing through the COM. The reaction at the front wheel (A) is



(a) $(W/2) - (Pb/2a)$ (b) $(W/2) + (Pb/2a)$

(c) $(W/2) - (Pa/2b)$ (d) None of these

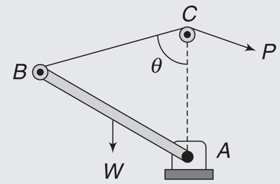
11. A uniform disk of radius R and mass m is connected to a wall by string of length $2R$. The normal reaction of wall is



(a) mg (b) $\frac{mg}{2}$

(c) $\frac{mg}{\sqrt{3}}$ (d) 2 mg

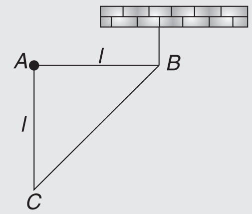
12. A uniform rod AB of weight W is movable in a vertical plane about a smooth hinge at A, and is kept in equilibrium by a force P acting on a string BCP passing over a smooth peg C, as shown. AC is vertical. If AC is equal to AB, then the force P is



(a) $W/\cos\theta$ (b) $W\cos\theta$

(c) $W/\sin\theta$ (d) $W\sin\theta$

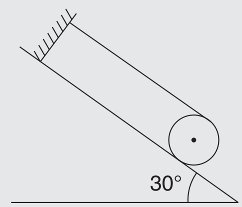
13. A right triangular plate ABC of mass m is free to rotate in the vertical plane about a fixed horizontal axis through A. It is supported by a string such that the side AB is horizontal. The reaction at the support A in equilibrium is:



(a) $\frac{mg}{3}$ (b) $\frac{2mg}{3}$

(c) $\frac{mg}{2}$ (d) mg

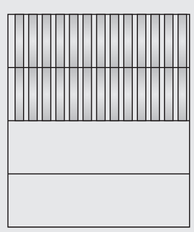
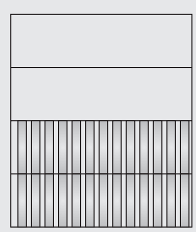
14. A thin hoop of weight 500 N and radius 1 m rests on a rough inclined plane, as shown in the figure. The minimum coefficient of friction needed for this configuration is

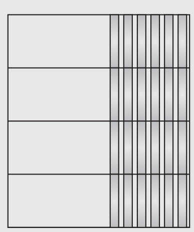
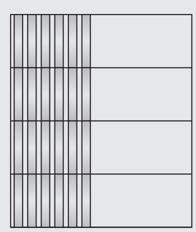


(a) $\frac{1}{3\sqrt{3}}$ (b) $\frac{1}{\sqrt{3}}$

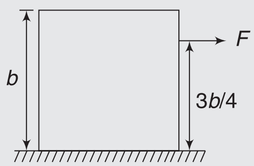
(c) $\frac{1}{2}$ (d) $\frac{1}{2\sqrt{3}}$

15. Same number of books are placed in four book cases, as shown. Which bookcase is most likely to topple forward if pulled a little at the top towards the right?

(a)  (b) 

(c)  (d) 

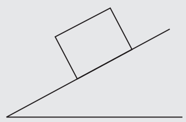
16. A uniform cube of side 'b' and mass M rests on a rough horizontal table. A horizontal force F is applied normal to one of the faces at a point, at a height $3b/4$ above the base. What should be the coefficient of friction (μ) between the cube and the table so that is will tip about an edge before it starts slipping?



(a) $\mu > \frac{2}{3}$ (b) $\mu > \frac{1}{3}$

(c) $\mu > \frac{3}{2}$ (d) none

17. A homogeneous cubical brick lies motionless on a rough inclined surface. The half of the brick, which applies greater pressure on the plane is:

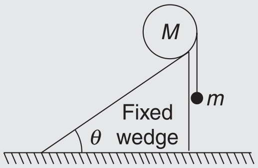


(a) left half
(b) right half
(c) both apply equal pressure
(d) the answer depends upon coefficient of friction

18. A tight-rope walker in a circus holds a long, flexible pole to help stay balanced on the rope. Holding the pole horizontally and perpendicular to the rope helps the performer

(a) by lowering the overall centre of gravity
(b) by increasing the mass
(c) in the ability to adjust the centre of gravity, to stay on the rope
(d) in providing something to hold

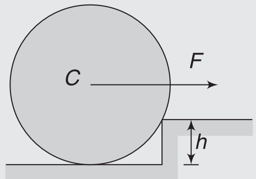
19. A uniform cylinder of mass M lies on a fixed plane inclined at an angle θ with horizontal. A light string is tied to the cylinder at the right-most point, and a mass m hangs from the string, as shown. Assume that the coefficient of friction between the cylinder and the inclined plane is sufficiently large to prevent slipping. For the cylinder to the remain static, the value of m is



(a) $M\left(\frac{\sin\theta}{1-\sin\theta}\right)$ (b) $\frac{M\cos\theta}{1+\sin\theta}$

(c) $\frac{M\sin\theta}{1+\sin\theta}$ (d) $\frac{M\cos\theta}{1-\sin\theta}$

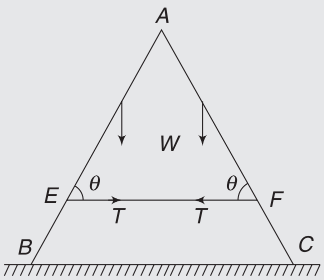
20. A cylinder of radius R and weight W is pulled against a step of height h ($< R$) by applying a horizontal force at its centre, as shown in the figure. The cylinder does not move.



(a) Maximum possible value of applied force is $F = W\frac{\sqrt{2Rh-h^2}}{R-h}$
(b) The cylinder cannot remain in rotational equilibrium
(c) Whatever be the value of F, the cylinder cannot move.
(d) None of the above.

Worksheet 2

1. Two uniform, equal ladders AB and AC, each of weight 'W' lean against each other and a string is tied between E and F. They stand on a smooth horizontal surface. Then:



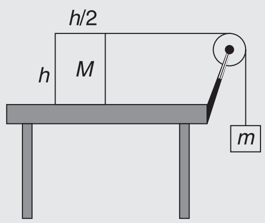
(a) the force exerted by one rod on the other at A is equal in magnitude to the tension T in the string

(b) tension $T = (W/2) \cot \theta$

(c) the normal reaction at B and C are equal

(d) the normal reaction at B or C is greater than 'W'

2. A cylinder of height h, diameter $h/2$ and mass M is placed on a horizontal table. One end of a string running over a pulley is fastened to the top of the cylinder; a body of mass m is hung from the other end and the system is released. Friction is large enough to prevent slipping of M.



(a) Maximum ratio m/M for which the cylinder will not tilt is 1/4.

(b) As m is increased (and the cylinder does not move) the line of normal force acting on the cylinder moves to the right.

(c) As m is increased (and the cylinder does not move), the line of normal force acting on the cylinder moves to left.

(d) None of the above.

3. A body is in equilibrium under the influence of a number of forces. Each force has a different line of action. The minimum number of forces required is

(a) 2, if their lines of action pass through the centre of mass of the body.

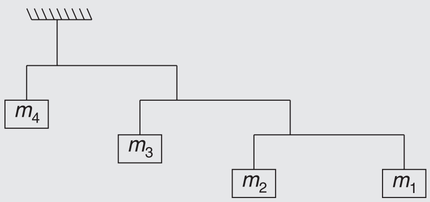
(b) 3, if their lines of action are not parallel.

(c) 3, if their lines of action are parallel.

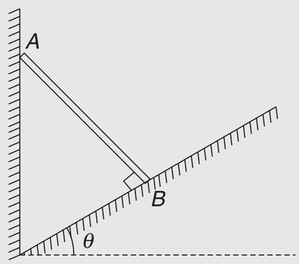
(d) 4, if their lines of action are parallel and all the forces have the same magnitude.

Worksheet 3

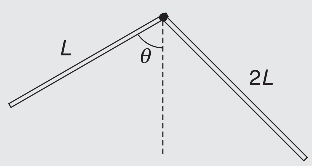
1. Four masses are suspended from a ceiling using massless rods, as shown in figure. All rods are horizontal and extend three times as far to the right of the wire supporting it, as to the left. Find mass m_1 if $m_4 = 96\,\text{kg}$.



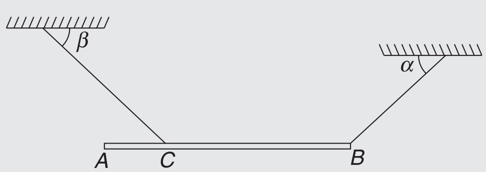
2. A uniform rod leans against a vertical wall with its lower end on a smooth incline, as shown. The rod is perpendicular to the incline and inclination angle of the incline is θ. Find minimum coefficient of friction between the rod and the incline so that the rod does not slip. Given, $\theta = 45°$.



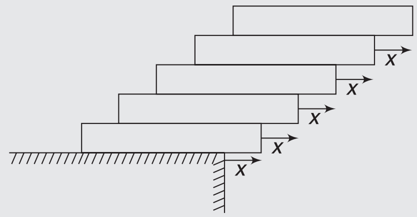
3. A rod in L-shape has two arms, having lengths L and $2L$. It rests on a peg, as shown. Find angle θ in equilibrium. Both arms have same linear density.



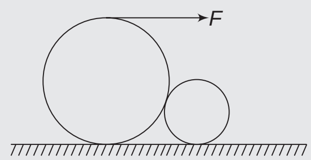
4. A uniform bar AB of length L is suspended by two light strings as shown. The string tied at end B makes an angle $\alpha = 30°$ with the horizontal. Another string is tied at point C, which is at a distance $\frac{L}{4}$ from end A. This string makes an angle β in equilibrium if the bar is horizontal. Find β.



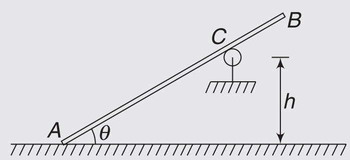
5. Five identical bricks, each of length 5 cm are arranged as shown. Find maximum value of x so that the system is in equilibrium.



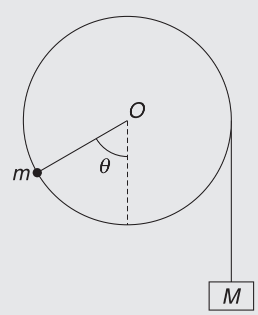
6. Two cylindrical rollers of radii R and r rest on a rough horizontal surface as shown. The larger roller has a light string tightly wrapped around it and the end of the string is being pulled horizontally with a force F. Coefficient of friction at each contact is μ. Find the smallest value of μ so that the smaller roller remains in equilibrium.



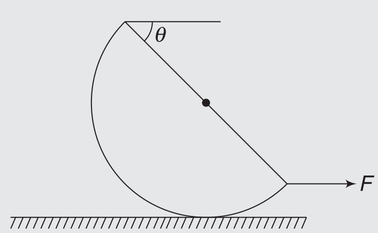
7. AB is a uniform rod of length L. Its end A lies on a rough ground and a smooth roller supports it at C. Height of C above the ground is h. Find the friction coefficient between end A and the ground, if the smallest angle that the rod can make with horizontal in equilibrium is θ.



8. A uniform disc-shaped pulley can rotate freely about an axis through its centre O. A thread is wrapped tightly on the pulley and supports a mass M at its free end. A small particle of mass m is stuck at the circumference of the pulley. The system stays in equilibrium when the radius to m makes an angle θ with the vertical as shown. Find $\frac{m}{M}$.



9. A hemisphere of mass M is being pulled by a constant horizontal force F, as shown in the figure. The hemisphere moves with constant velocity while maintaining an inclination angle θ between the horizontal and its circular base. Find the coefficient of friction between the hemisphere and the floor.



Answers Sheet

Your Turn

1. $6\hat{k}$ N-m
2. $10\sqrt{3}$ N-m (↻)
4. Zero
5. 60 N-m along negative z-direction.
6. $mu^2 \sin\theta . \cos\theta$
7. 3
8. (i) 275 N (ii) 600 N
9. Tension = 292 N, F_H = 234 N; F_v = 25.2 N
10. $\frac{5}{3}$ m
11. 87.3 cm
13. Truck having its CG at A
14. yes
15. 1
16. In balance

Worksheet 1

1. (c) 2. (c) 3. (b) 4. (d) 5. (c) 6. (c) 7. (c) 8. (c) 9. (c)
10. (d) 11. (c) 12. (b) 13. (b) 14. (d) 15. (c) 16. (a) 17. (a) 18. (a)
19. (a) 20. (a)

Worksheet 2

1. (a,c) 2. (a,b) 3. (b,c,d)

Worksheet 3

1. 2 kg
2. $\frac{1}{3}$
3. $\tan^{-1}4$
4. $\tan^{-1}\left(\frac{2}{\sqrt{3}}\right)$
5. $\frac{5}{6}$ cm
6. $\mu = \sqrt{\frac{r}{R}}$
7. $\mu \geq \frac{L\cos\theta\cdot\sin^2\theta}{2h - L\cos^2\theta\cdot\sin\theta}$
8. $\frac{1}{\sin\theta}$
9. $\mu = \frac{3\sin\theta}{8(1-\sin\theta)}$

CHAPTER 5

Kinematics of Rotation

"We live-on a spinning planet in a world of spin."

–Christopher Buckley

1. INTRODUCTION

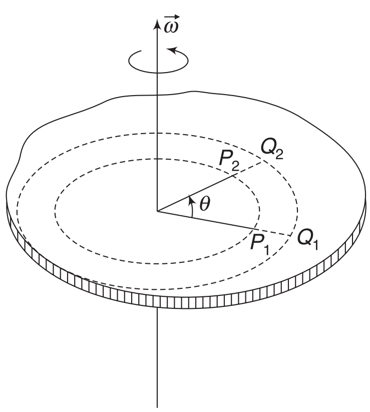
In a rotating body, all the particles do not move identically. Description of motion is quite different from that of translational motion.

The centre of mass of a body will play a pivotal role in understanding the general motion of a body.

We will consider pure rotation of rigid bodies, as well as rotation and translation, with a major thrust on understanding rolling motion. In this chapter, we will discuss the basic concepts related to kinematics of rotating bodies and then take up the dynamics parts in the next chapter. You are advised to quickly revise the chapter on *Kinematics of Circular Motion* before you move ahead in this chapter.

2. KINEMATICS OF PURE ROTATION

A body is said to be in pure rotation if it is rotating about a fixed axis. The axis may pass through the body or may lie completely outside it.



In such motion, all particles move in a circle, with their centre lying on the rotation axis. Angular displacement of each particle is the same, though their speeds may not be the same. The figure shows a rotating body. As point P_1 moves to P_2 and rotates through an angle θ, any other point (like Q) in the body also moves through the same angle θ. We say that the body has rotated by an angle θ.

Obviously, the angular velocity (ω) and angular acceleration (α) are also same for all particles in the body.

Angular velocity is a vector. Its direction is along the rotation axis. Curl your right hand fingers in the direction of rotation (trying to hold the rotation axis) and the thumb gives the direction of angular velocity vector, $\vec{\omega}$.

If ω is increasing then $\vec{\alpha}$ is directed parallel to $\vec{\omega}$ and the two vectors are oppositely directed if ω is decreasing.

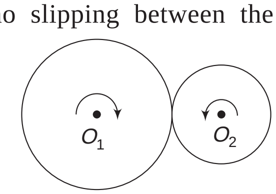
For a particle in the body, rotating in a circle of radius r, the components of acceleration are:

$$a_r = \omega^2 r \text{ [Radial acceleration]}$$

$$a_t = r\frac{d\omega}{dt} = r\alpha \text{ [Tangential acceleration]}$$

When ω is constant, α is zero.

Example 1 Two wheels can rotate about their fixed axles. The wheels have radii R and r and they rotate with their edges touching. There is no slipping between the wheels. Find the ratio of magnitudes of acceleration of points on the circumference of the two wheels. Assume that the wheels rotate with uniform angular speeds.

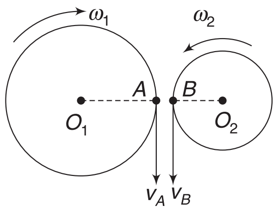


Solution

Concepts

(i) No slipping implies that the speed of particles on the circumference of the wheels is same. This gives us the ratio of angular speeds.

(ii) Acceleration of a particle on circumference = $\omega^2 r$



For no slipping, speed of point

$$A = \text{speed of point } B$$

$$\Rightarrow \quad \omega_1 R = \omega_2 r$$

$$\Rightarrow \quad \frac{\omega_1}{\omega_2} = \frac{r}{R}$$

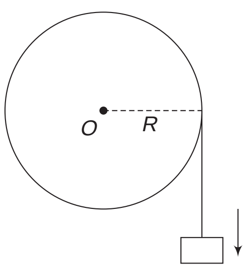
$$\therefore \quad \frac{a_A}{a_B} = \frac{\omega_1^2 R}{\omega_2^2 r} = \left(\frac{r}{R}\right)^2 \frac{R}{r} = \frac{r}{R}$$

Example 2 *Mass suspended to a thread wrapped around a pulley*

A block is hanging from a thread, which is wrapped tightly around a pulley of radius R.

(i) Find the angular speed of the pulley if the block is moving down at a speed v.

(ii) Find the angular acceleration of the pulley if the block is moving down with an acceleration a.

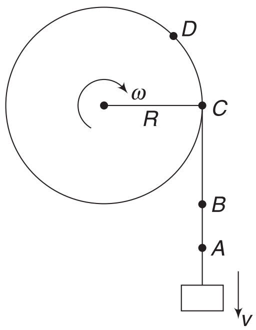


Solution

Concepts

The phrase 'thread is tightly wrapped' means that the thread will not slip on the pulley.

This implies that speed of a point on the thread is same as the speed of a point on the circumference of the pulley.



(i) Speed of block (v) = Speed of points A, B, C or D on the thread

Speed of point C (or D) = Speed of point on the circumference of the pulley = ωR

$$\therefore \quad \omega R = v \quad \Rightarrow \quad \omega = \frac{v}{R}$$

(ii) $$\omega R = v$$

$$\Rightarrow \quad \left(\frac{d\omega}{dt}\right) R = \frac{dv}{dt} \quad \Rightarrow \quad \alpha \cdot R = a$$

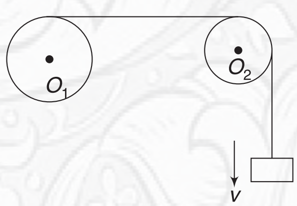
$$\Rightarrow \quad \alpha = \frac{a}{R} \qquad \left[\frac{d\omega}{dt} = \text{angular acceleration of the pulley}\right]$$

Note: One must note that if the pulley rotates through an angle θ, a thread of length $x = R\theta$ will unwind out of it and the block will descend through distance x.

Your Turn

Q.1 A fan having diameter of 36 in is rotating uniformly at an angular speed of $\omega = 20$ rad s^{-1}. Find acceleration of a point at the tip of the blade.

Q.2 The diagram shows two pulleys, which can rotate about their central axes. Radii of the pulleys are 0.2 m and 0.1 m. A thread is tightly wrapped on the bigger pulley. It passes over the smaller pulley and its free-end carries a load. The load is moving down at a speed of $v = 10$ ms^{-1}. Find the angular speed of the two pulleys. Assume that the thread does not slip on the smaller pulley.

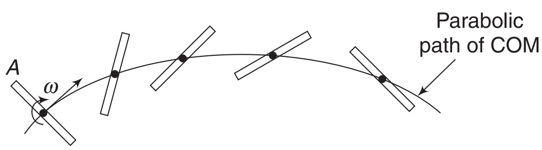


Q.3 In Example 2, how far does the block fall if the pulley makes two complete turns?

3. KINEMATICS OF ROTATION PLUS TRANSLATION

When you toss a stick in air, its motion is quite complicated. It is rotating as well as translating. Similarly, a spinning cricket ball is rotating about an axis and the axis itself is translating. Any general motion of a rigid body involves both rotation and translation. Description of such a motion becomes easier with the help of *Chasle's theorem*.

This theorem asserts that it is always possible to represent an arbitrary displacement of a rigid body by a translation of its centre of mass plus a rotation about its centre of mass. When you toss a stick in air, its COM translates along a parabolic path and all particles of the stick rotate about the COM with the same angular velocity (ω).



Any point (like A) rotates about the COM.

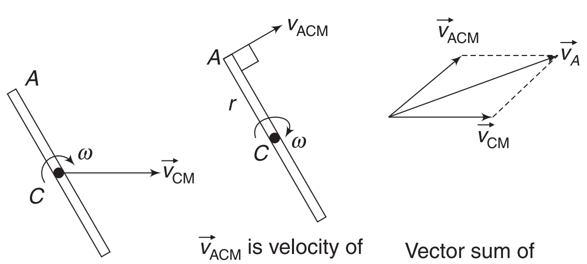
If we wish to write velocity of a point (like A) on the stick, we can do it in two steps. First we write the velocity of the point relative to COM. Relative to COM, all particles are going in circles with angular speed ω. Velocity of point A relative to COM is written as

$$v_{\text{ACM}} = \omega r$$

where r is the distance of A from the COM. Direction of $\vec{v}_{\text{ACM}}$ is normal to r.

In second step, we add the velocity of COM ($= \vec{v}_{\text{CM}}$) to $\vec{v}_{\text{ACM}}$ to get velocity of point A.

$$\vec{v}_A = \vec{v}_{\text{ACM}} + \vec{v}_{\text{CM}} \qquad ...(1)$$



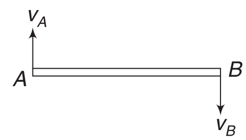
$\vec{v}_{\text{ACM}}$ is velocity of A wrt COM. In COM frame A is going in a circle of radius r (= AC)

Vector sum of $\vec{v}_{\text{ACM}}$ and $\vec{v}_{\text{CM}}$ gives $\vec{v}_A$

Similarly, for writing the acceleration of a point like A, we first write its acceleration relative to COM and then add the acceleration of COM to it.

$$\vec{a}_A = \vec{a}_{\text{ACM}} + \vec{a}_{\text{CM}} \qquad ...(2)$$

Example 3 A metre stick is tossed in air. At a certain instant, the stick is horizontal and its two end points have velocities $v_A = 6\ \text{ms}^{-1}$ upwards and $v_B = 4\ \text{ms}^{-1}$ downwards. Find the velocity of COM of the stick and its angular speed.



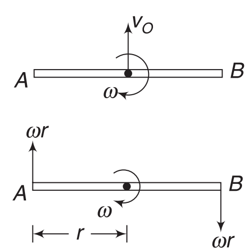
Solution

Concepts

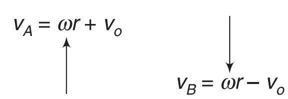
(i) If all points in a body have same velocity, it implies that there is no rotation. In this question, velocities of points A and B are different. It implies that there is rotation.

(ii) We will assume that COM is translating and the body is rotating about its COM. We will take velocity of COM (v_{CM}) and angular velocity (ω) as two unknowns.

By writing the velocities of end points (A and B) using equation (1), we will create two equations.



Motion in COM frame



Velocities of A and B in ground frame

Let the stick rotate about its COM with angular speed ω, and velocity of COM be v_0, as shown. In COM frame, speed of both A and B is ωr as shown in the second figure.

Here $\quad r = \dfrac{1}{2}$ m

$\therefore \quad v_A = \omega r + v_0\ (\uparrow)$

$$\Rightarrow \quad 6 = \omega\left(\frac{1}{2}\right) + v_0 \qquad ...(i)$$

And $\quad v_B = \omega r - v_0\ (\downarrow)$

$$4 = \omega\left(\frac{1}{2}\right) - v_0 \qquad ...(ii)$$

Adding (i) and (ii) gives

$$\omega = 10 \text{ rad s}^{-1}$$

Now, (i) gives $v_0 = 1 \text{ ms}^{-1}$

Alternate Solution

Concepts

(i) If angular speed of a body is ω about its COM, then it has same angular speed (ω) about any other point in the body.

(ii) We can easily write the velocity of A relative to B and then angular speed of A relative to B can be written as $\omega = \dfrac{v_{AB}}{AB}$. If needed, revisit the concepts learnt in the chapter of *Kinematics of Circular Motion*

Velocity of point A relative to point B is

$$v_{AB} = 10 \text{ ms}^{-1} (\uparrow).$$

This velocity is perpendicular to AB.

$\therefore$ Angular speed of A relative to B is

$$\omega = \frac{v_{AB}}{AB} = \frac{10}{1} = 10 \text{ rad s}^{-1}$$

And this is the angular speed of the body about any point like, A or C. Now, to get the velocity of COM(v_0), we again go back to equation (i) in previous method and write

$$6 = v_0 + \omega\left(\frac{1}{2}\right) \Rightarrow v_0 = 1 \text{ ms}^{-1}.$$

Your Turn

Q.4 A uniform stick moving in air has its COM moving horizontally at a velocity of 2 ms^{-1}. At the instant one of its ends is at rest, find the velocity of the other end.

Q.5 A uniform stick is tossed in air. It moves under gravity while rotating with constant angular speed $\omega = 4 \text{ rad s}^{-1}$. Find the acceleration of its top point when the stick is vertical. Length of the stick is 2 m.

3.1 Instantaneous Centre of Rotation

A rigid body is said to be undergoing plane motion if a particle in the body moves in a fixed plane. In the example given above (and also in the problems you solved just now), any particle in the stick moves in a fixed vertical plane. The stick is undergoing a plane motion. In other words, you can say that in a plane motion, the orientation of the axis of rotation of the body does not change. In the above example, the stick keeps rotating about its COM and rotation axis is horizontal, perpendicular to the plane of the figure. As the stick moves, the position of axis changes but its direction remains same.

When a body is undergoing a plane motion, we can find an axis inside or outside the body about which the body– can be considered to be in pure rotation. This axis is known as *instantaneous axis* of rotation. The axis may itself keep changing its position.

When the body is two dimensional and undergoing plane motion, we can find a point– inside or outside the body about which the body can be considered to be in pure rotation. This point is *instantaneous centre* of rotation.

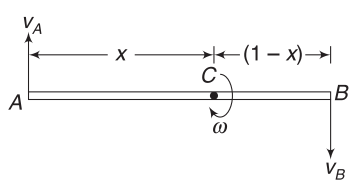
Example 4 In Example 3, find the position of instantaneous centre of rotation of the stick at the instant shown.

Solution

Concepts

Instantaneous centre is the point about which each and every particle in the stick is rotating in circle. If C is the centre of rotation, then v_A is perpendicular to AC and v_B is perpendicular to BC.

This implies that the centre of rotation is somewhere on the stick AB.



Let centre of rotation (C) be at a distance x from A and $(1 - x)$ from B.

$$v_A = \omega . x$$

$$\Rightarrow \quad 6 = \omega x \quad \text{...(1)}$$

$$\text{And} \quad v_B = \omega(1 - x)$$

$$\Rightarrow \quad 4 = \omega(1 - x) \quad \text{...(ii)}$$

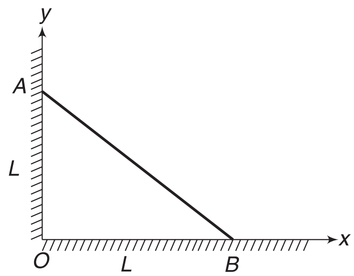
Solving (i) and (ii)

$$x = 0.6 \text{ m} \quad \text{and} \quad \omega = 10 \text{ rad s}^{-1}$$

This means that all particles in the rod are going in a circle with angular speed $\omega = 10\ \text{rad s}^{-1}$, with the centre of each circle being at C. Point C itself has zero velocity.

Example 5 *Instantaneous centre of a sliding stick*

A stick AB is leaning against a wall. It begins to slide. Find the position co-ordinates of its instantaneous centre of rotation at the instant length OA and OB both are equal to L.

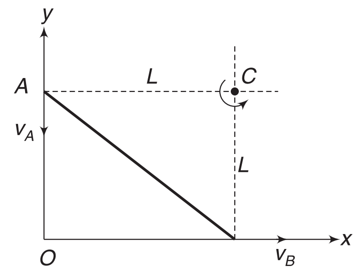


Solution

Concepts

(i) Velocity of end A is vertically downward and the velocity of end B is horizontally rightward.

(ii) Instantaneous centre is on a line that is perpendicular to $\vec{v}_A$. It is also on a line that is perpendicular to $\vec{v}_B$. Instantaneous centre is always on a line that is perpendicular to velocity of any point in the stick.



$\vec{v}_A$ and $\vec{v}_B$ are shown. Lines perpendicular to $\vec{v}_A$ and $\vec{v}_B$ intersect at C. Thus, C is the centre of rotation.

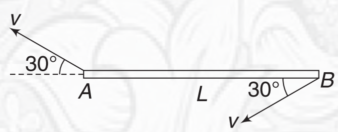
All particles in the rod are rotating about C at this instant [At a later time, centre of rotation will move to a different point.] From Fig., co-ordinates of C are (L, L).

Note: If speed of end A is v, then angular speed of the rod about C is

$$\omega = \frac{v_A}{CA} = \frac{v}{L}$$

YOUR TURN

Q.6 A stick is moving in air. Ends A and B have velocities as shown. Length of the stick is L.



(i) Find the location of instantaneous centre of rotation.

(ii) Find angular velocity of the rod.

3.2 Rolling

Rolling is possibly the most familiar kind of motion of a rigid body, which can be described as a combination of translation and rotation. Motion of a motorcycle wheel can be described as superposition of two independent motions:

(i) Translatory motion of the centre of mass

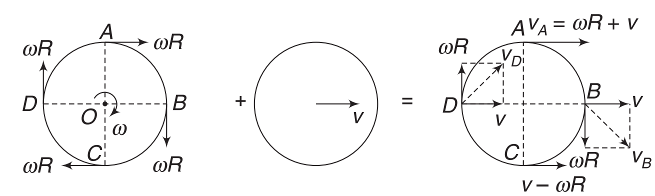
(ii) Rotation about the COM

An observer located at the centre of the wheel will find the wheel to be rotating with an angular velocity (ω). Velocity of different points on the circumference of the wheel appear to this observer as shown in fig. (a).

Velocity of any point (Like A, B, C or D) relative to ground is obtained by adding the velocity of COM to the velocity shown in Fig (a). Velocity of any point (P) relative to ground is given by

$$\vec{v}_P = \vec{v}_{\text{PCM}} + \vec{v}$$

$\vec{v}_{\text{PCM}}$ is velocity of a point P wrt COM



Velocity of different points relative to COM. An observer at COM, sees rotation. Velocity of each point is perpendicular to radius = ωR

Fig. (a)

Translation of COM.

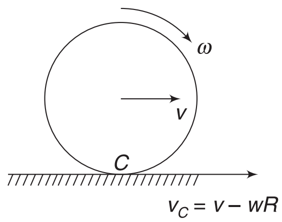
Fig. (b)

Velocity of different points in a rolling wheel

Fig. (c)

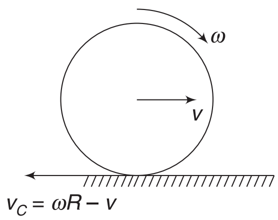
Figure (c) shows the velocity of different points as a combination of velocity of centre (= v) and the velocity due to rotation relative to centre.

Obviously, the speed of top point A is maximum and equal to $v_A = v + \omega R$. Speed of the lowest point C is the smallest and equal to $v_c = |v - \omega R|$.



If $v > \omega R$, Contact point is sliding in forward direction.

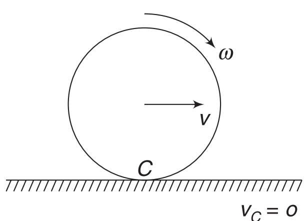
If $v > \omega R$, velocity of contact point C of the wheel is forward. It is said to be forward slipping on the ground.



If $v < \omega R$, Contact point is sliding in backward direction.

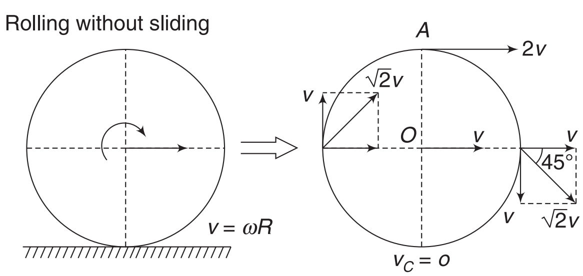
If $\omega R > v$, velocity of contact point C is backward. We say that the wheel is backward slipping on the ground.

However, the most interesting situation arises when $v = \omega R$. In this case, velocity of the contact point is zero. The contact point is not sliding (rubbing) on the ground. We say that the wheel is *rolling without sliding* or it is performing *pure rolling*.



If $v = \omega R$, Contact point is at rest.

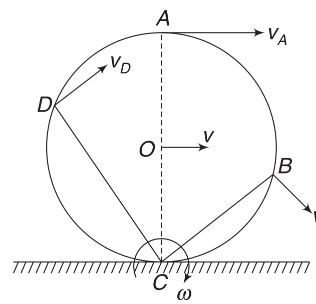
Velocities of different points on the wheel, when it is rolling without sliding, have been shown in the figure.



3.2.1 Instantaneous axis of rotation in rolling without sliding

We have learned that rolling motion is a combination of translation of COM and rotation about it. *In case of rolling without sliding*, we can view the motion in an entirely different way.

The point of contact (C) is instantaneously at rest. An axis through C, perpendicular to the plane of the figure can be regarded as instantaneous axis of rotation for the body. At each instant, there is a new point of contact and therefore, a new axis of rotation; but instantaneously, the motion of the entire body can be regarded simply as rotation about C.



Every point in the wheel can be regarded to be in pure rotation about C. This is true if $v = \omega R$.

In this model, velocity of any point, like B is perpendicular to CB and has magnitude ωr where $r = CB$.

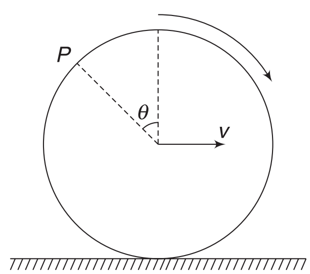
Velocity of centre O is $v = \omega R$

Velocity of top point A is $v_A = \omega(2R) = 2\omega r = 2v$

Velocity of a point written in this fashion is same as the velocity of a point predicted by considering rolling as a combination of translation and rotation.

It is important to note that in case of rolling with sliding, the contact point cannot be regarded as instantaneous centre.

Example 6 A wheel is rolling without sliding on a horizontal surface with its centre moving at a speed v. Find the speed of point P shown in figure.



Solution

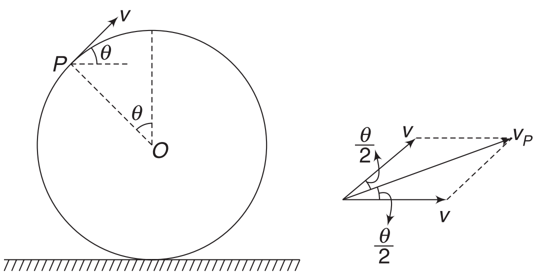
Concepts

(i) In rolling without sliding, $\omega R = v$

(ii) Relative to centre, velocity of point P is tangential and equal to $\omega R = v$. We add to this the velocity (v) of the centre and get velocity of point P.

(iii) We can also think that the wheel is instantaneously in pure rotation about the point of contact. Speed of point P is $= \omega r$, where r = distance of P from the point of contact.

Velocity of P wrt centre is $\omega r = v$ in a direction making θ with horizontal. We add velocity of centre $v(\rightarrow)$ to it. Resultant velocity v_p is directed making an angle $\dfrac{\theta}{2}$ with horizontal and has magnitude



Velocity of P wrt Centre

$$v_p = \sqrt{v^2 + v^2 + 2vv\cos\theta} = v\sqrt{2(1+\cos\theta)}$$

$$= 2v\cos\frac{\theta}{2}$$

Alternate:

We can consider the wheel to be rotating about its instantaneous centre C.

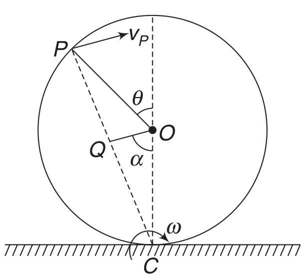
In figure: $\alpha = \dfrac{\pi - \theta}{2} = \dfrac{\pi}{2} - \dfrac{\theta}{2}$

$$CQ = R\sin\alpha = R\sin\left(\frac{\pi}{2} - \frac{\theta}{2}\right) = R\cos\frac{\theta}{2}$$

$$\therefore \quad CP = 2R\cos\frac{\theta}{2}$$

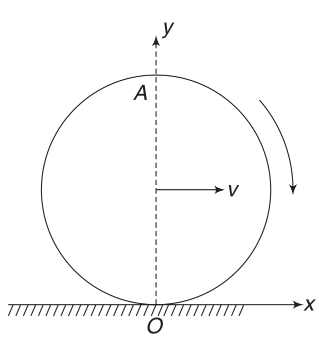
$$\therefore \quad v_P = \omega(CP) = \frac{v}{R}\cdot 2R\cos\frac{\theta}{2} = 2v\cos\frac{\theta}{2}$$

Direction of v_p is normal to CP.



Example 7 *Path of a point in a rolling wheel*

A wheel is rolling without sliding on a horizontal ground. Its centre is moving at constant velocity v. In a co-ordinate system fixed to the ground, co-ordinates of top point A are $(0, 2R)$ at time $t = 0$. Write the co-ordinates of this point at any later time t.

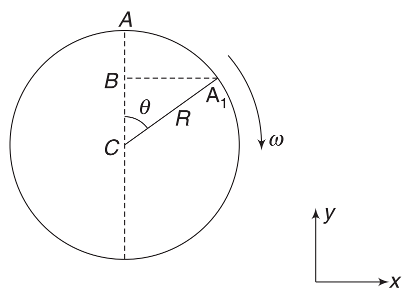


Solution

Concepts

(i) Wheel is rotating about its centre at angular speed $\omega = \dfrac{v}{R}$

(ii) We will write displacement of point A relative to centre and then add the displacement of the centre to get the displacement of A relative to the ground.



Relative to the centre, the point A rotates to position A_1 in time t.

$$\theta = \omega t = \frac{v}{R}t$$

Displacement of A in x-direction is

$$= BA_1 = R\sin\theta$$

Displacement of A in negative y-direction is

$$= AB = R(1 - \cos\theta)$$

In time t, the centre moves along x-direction by a distance $= vt$.

$\therefore$ Displacement of A wrt ground in x-direction

$$= vt + R\sin\theta$$

Displacement of A wrt ground in negative y-direction is $= R(1 - \cos\theta)$

$\therefore$ Co-ordinates of A, at time t (in ground frame), are:

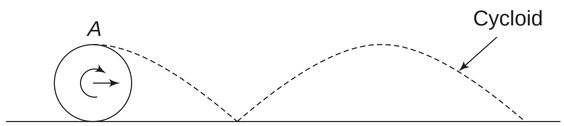
$$x = vt + R\sin\theta = vt + R\sin\left(\frac{vt}{R}\right) \qquad \text{...(i)}$$

and

$$y = 2R - R(1 - \cos\theta)$$

$$= R(1 + \cos\theta) = R\left[1 + \cos\left(\frac{vt}{R}\right)\right] \qquad \text{...(ii)}$$

Note: Equation (i) and (ii) together represent a curve known as cycloid. Path of point A (or any other point on the circumference) is a cycloid. It has been represented in the figure given below.



Path of a point on the circumference of the wheel is a cycloid.

Example 8 *Acceleration of the contact point*

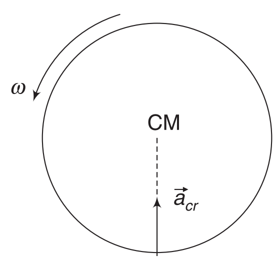
A wheel of radius R is rolling without sliding on a horizontal surface with its centre moving uniformly at speed v. Find the acceleration of the contact point.

Solution

Concepts

We will write acceleration of the contact point wrt COM. Then we will add the acceleration of COM to it.

This particular question is quite easy, as COM has no acceleration.



In a reference frame attached to the centre, the whole wheel is just rotating with angular speed

$$\omega = \frac{v}{R}$$

Acceleration of point C relative to COM is

$$a_{\text{cr}} = \omega^2 R = \frac{v^2}{R} \ (\uparrow)$$

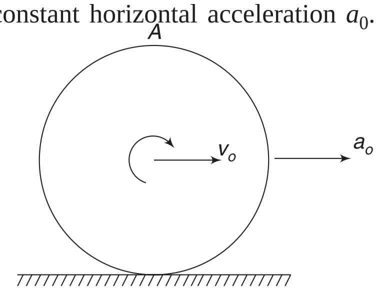
$\therefore$ Actual acceleration of point C is

$$a_c = a_{\text{cr}} + a_{\text{CM}} = \frac{v^2}{R} \ (\uparrow) \text{ since } a_{\text{CM}} = 0$$

Note: The contact point C is instantaneously at rest but its acceleration is not zero.

Example 9 *Rolling with acceleration*

A wheel of radius R is rolling without sliding on a horizontal surface. Its centre has a constant horizontal acceleration a_0. Find the acceleration of the top point A at the instant speed of the centre of the wheel is v_0.



Solution

Concepts

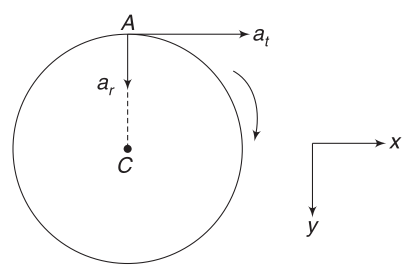
(i) For pure rolling, $v = \omega R$ condition must always be true. If v is increasing due to acceleration, ω must also increase. It means that there must be an angular acceleration (α) of the wheel as it rotates about its centre.

$$v = \omega R$$

$$\Rightarrow \quad \frac{dv}{dt} = R\frac{d\omega}{dt} \quad \Rightarrow \quad a = R\alpha$$

(ii) A point on the wheel appears to be in non-uniform circular motion for an observer, who is standing at its centre.

(iii) We will find acceleration of A relative to the centre and then add the acceleration of centre to get the resultant acceleration of A.



Acceleration of A in RF attached to C.

Acceleration of point A has radial and tangential components in a reference frame attached to the centre.

$$a_t = R\alpha$$

$$a_r = \omega^2 R = \left(\frac{v_0}{R}\right)^2 \cdot R$$

$$= \frac{v_0^2}{R}$$

$$\therefore \ \vec{a}_{AC} = (R\alpha)\hat{i} + \left(\frac{v_0^2}{R}\right)\hat{j}$$

Acceleration of C is $a_C = a_0\hat{i}$

$$\therefore \quad \vec{a}_A = \vec{a}_{AC} + \vec{a}_C = (R\alpha + a_0)\hat{i} + \left(\frac{v_0^2}{R}\right)\hat{j}$$

$$= 2a_0\hat{i} + \left(\frac{v_0^2}{R}\right)\hat{j} \qquad \left[\begin{array}{c}\text{Since } a_0 = R\alpha \text{ for} \\ \text{pure rolling}\end{array}\right]$$

Magnitude of this acceleration is

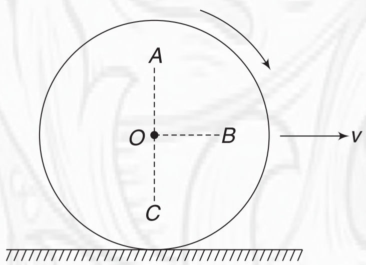
$$a_A = \sqrt{(2a_0)^2 + \left(\frac{v_0^2}{R}\right)^2}$$

YOUR TURN

Q.7 A disc is rolling on a flat ground. Its angular velocity is $\omega = 2$ rad s^{-1} and velocity of its centre is 3 ms^{-1}. Radius of the disc is 0.5 m. Is the disc slipping? Find the velocity of contact point of the disc.

Q.8 A disc is rolling without sliding with velocity $v = 4$ ms^{-1}. Radius of the disc is $R = 1.0$ m. Find speed of points A, B and C shown in figure.

$$OA = OB = OC = 0.5\text{ m}$$

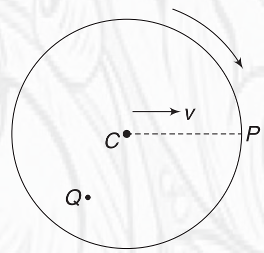


Q.9 In the last problem, find the acceleration of point A if the wheel is moving with a constant velocity v.

Q.10 A wheel is rolling uniformly with its centre moving at speed v.

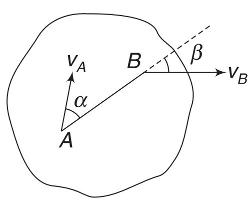
(i) Find the speed of point P that lies on the circumference on horizontal radius.

(ii) Which point – P or Q – is moving faster at the instant shown?



In Short:

(i) In a rigid body, the distance between two points never changes. Velocity component of a point A along AB = velocity component of point B along AB.



$\Rightarrow \quad v_A \cos\alpha = v_B \cos\beta$

(ii) In pure rotation, the rotation axis remains fixed and all particles of the body go in circle about the fixed axis. Speed of any particles is given by $v = \omega r$, where r is distance of the particles from the axis.

ω is same for all particles.

(iii) In pure rotation, if the angular speed of the body changes, then there is angular acceleration

$$\alpha = \frac{d\omega}{dt}$$

Now, particles in the body are performing non-uniform circular motion.

(iv) In case of a thread passing over a pulley (or wrapped on a pulley) and not slipping on the pulley, speed of the thread (v) is related to angular speed of pulley (ω) as $v = \omega R$. R is radius of the pulley. If the thread is accelerated, then $a = R\alpha$

Where α = angular acceleration of the pulley and a is rate of change of speed of the thread.

(v) Any general motion of a rigid body can be understood as translation of its centre of mass superimposed with rotation about the centre of mass. Velocity and acceleration of a point in the body can be written using this approach as

$$\vec{v}_P = \vec{v}_{PCM} + \vec{v}_{CM} \quad \text{and}$$

$$\vec{a}_p = \vec{a}_{PCM} + \vec{a}_{CM}$$

(vi) A rolling wheel is a typical example where we understand the motion as the rotation of the wheel about its centre plus the translational motion of its centre.

(vii) When the contact point of a wheel is at rest (i.e., does not rub the ground) we say it is rolling without sliding (or pure rolling). This happens when $v = \omega R$, where v is velocity of the centre and ω is the angular speed of the wheel about the centre.

If the wheel is accelerated, then $a = R\alpha$ must also be true for pure rolling to sustain. Here, a is the acceleration of the centre and α is the angular acceleration of the wheel about the centre.

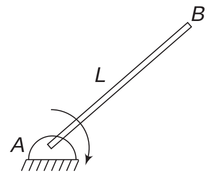
(viii) When $v \neq \omega R$, the contact point slides on the ground. It is rolling with sliding or impure rolling.

(ix) A plane motion of a rigid body can also be understood as rotation about a certain axis known as instantaneous axis of rotation.

(x) For a ball rolling without sliding, the instantaneous axis passes through its point of contact with the ground.

Miscellaneous Examples

Example 11 A rod of length L is hinged at its end A. It begins to fall from a vertical position, rotating about A. Find acceleration of its end B, at the instant its angular velocity and angular acceleration are ω and α respectively.



Solution

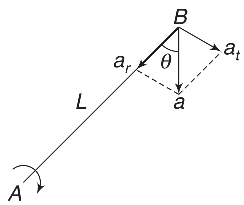
Concepts

(i) Point B is rotating in a circle of radius L, with its centre at A.

(ii) Motion of B about A is non-uniform circular motion. Point B has a radial and a tangential acceleration.

$$a_r = \omega^2 L$$

$$a_t = L\alpha$$



Acceleration of point B has magnitude

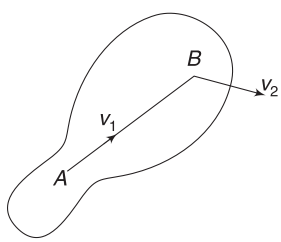
$$a = \sqrt{a_r^2 + a_t^2}$$

$$= \sqrt{(\omega^2 L)^2 + (L\alpha)^2}$$

It makes an angle θ with the rod such that

$$\tan\theta = \frac{a_t}{a_r} = \frac{\alpha}{\omega^2}$$

Example 12 A flat, rigid body is moving in the plane of the figure. At an instant, velocity of its two points A and B are as shown. Velocity of point A is $v_1 = 10\,\text{ms}^{-1}$ along AB and velocity of point B is $v_2 = 12.5\ \text{ms}^{-1}$. Find the angular speed of the body. Length $AB = 0.5\,\text{m}$.



Solution

Concepts

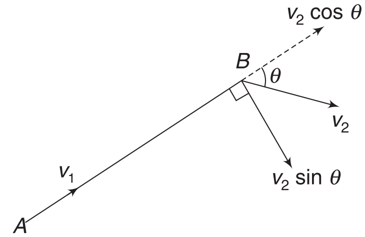
(i) Velocity component of B along AB must be equal to v_1. This is necessary as distance between two points in a rigid body cannot change.

(ii) We will find angular speed of point B wrt A. This will be our answer, as a rigid body has the same angular speed about any point in the body.

For length AB to remain unchanged, $v_2 \cos\theta = v_1$

$$\Rightarrow \quad 12.5\cos\theta = 10$$

$$\Rightarrow \quad \cos\theta = \frac{4}{5}$$



Velocity of B perpendicular to AB is

$$v_2 \sin\theta = 12.5 \times \frac{3}{5} = 7.5\,\text{ms}^{-1}$$

$\therefore$ Angular velocity of B relative to A is

$$\omega = \frac{7.5}{AB} = \frac{7.5}{0.5} = 15\ \text{rad}\,\text{s}^{-1}$$

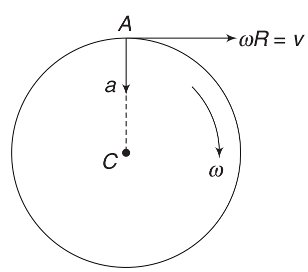
Example 13 A wheel of radius R is rolling without sliding on a horizontal surface with velocity v. As discussed in Example 7, the path of a particle on its circumference is a cycloid. Find the radius of curvature of this path at the top-most point.

Solution

Concepts

(i) If you look at the path drawn in Example 7, you will get convinced that radius of curvature is certainly larger that the radius (R) of the wheel.

(ii) We can write the acceleration of the top point and its velocity is also known. If radius of curvature of the path is r, then radial acceleration of the top point (A) can be written as $a_r = \dfrac{v_A^2}{2r}$.



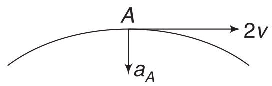
Motion of A in COM frame is uniform circular motion

If RF attached to the centre, velocity of point A is $\omega R = v$. Acceleration of A in this frame is $a_{\text{ACM}} = \omega^2 R = \dfrac{v^2}{R}$ $(\downarrow)$.

If the wheel is rolling uniformly, its COM has no acceleration. It means that acceleration of point A in ground frame is also $\dfrac{v^2}{R}(\downarrow)$. But speed of A in ground frame is

$v + \omega R = 2v$. Thus, in ground frame, acceleration of point A is $a_A = \frac{v^2}{R}$ ($\downarrow$) and its velocity is $v_A = 2v$ ($\rightarrow$).

If radius of curvature is r (i.e., we think as if point A is going in a circle of radius r at this instant), then

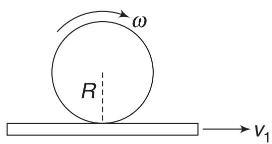


$$\frac{v_A^2}{r} = a_A$$

$$\Rightarrow \quad \frac{(2v)^2}{r} = \frac{v^2}{R} \quad \Rightarrow \quad r = 4R$$

Example 14 *Pure rolling on a moving surface*

A plank is moving horizontally with a velocity v_1. A cylinder of radius R, having angular speed ω, is on the plank. Find the velocity of the centre of the cylinder if it does not slip on the plank.



Solution

Concepts

The point of the cylinder that is touching the plank must have a velocity $v_1(\rightarrow)$. Thus, velocity of the point of contact relative to the plank will be zero and it will not slide (rub) the plank surface.

Let velocity of centre of the cylinder be $v_2(\rightarrow)$ velocity of contact point relative to the centre is $\omega R(\leftarrow)$.

$\therefore$ Velocity of the contact point of the cylinder wrt ground will be

$$v_2(\rightarrow) + \omega R\ (\leftarrow) = v_2 - \omega R(\rightarrow)$$

For no slipping this should be equal to v_1

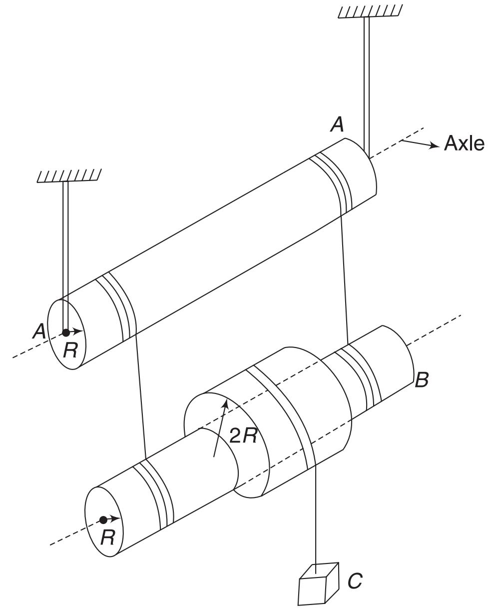
$$\therefore \quad v_2 - \omega R = v_1 \quad \Rightarrow \quad v_2 = \omega R + v_1$$

Example 15 *Constraint relation among accelerations*

In the system shown in figure, cylinder A can rotate freely about its axle. Two threads are wrapped tightly on it at symmetric locations. The two threads are also wrapped on another identical cylinder B, as shown. Both cylinders have radius R. Cylinder B can rotate about its axis and can also translate. At the centre of B, there is a sleeve of radius $2R$. The sleeve is fixed to B and does not slide on B. The sleeve has another thread tightly wrapped on it, which carries a block C at its free end. At a certain instant, angular acceleration of cylinders A and B are α_1 and α_2 respectively.

(i) Find acceleration of axle of cylinder B at this instant

(ii) Find acceleration of block C at this instant.



Solution

Concepts

(i) B falls by a distance that is equal to the length of thread that unwinds from A plus the length of the thread that unwinds from B.

(ii) C falls through a distance that is equal to the displacement of B plus the length of the thread that unwinds from the sleeve.

(i) Let A rotate by θ_1 and B rotate by θ_2 in time t. The thread that unwinds from A is $R\theta_1$ and the thread that unwinds from B is $R\theta_2$.

$\therefore$ Displacement of B in time t is

$$x_B = R\theta_1 + R\theta_2$$

$$\Rightarrow \quad \frac{dx_B}{dt} = R\frac{d\theta_1}{dt} + R\frac{d\theta_2}{dt}$$

$$\Rightarrow \quad v_B = R\omega_1 + R\omega_2$$

$$\Rightarrow \quad \frac{dv_B}{dt} = R\frac{d\omega_1}{dt} + R\frac{d\omega_2}{dt}$$

$$\Rightarrow \quad a_B = R\alpha_1 + R\alpha_2$$

(ii) Thread that unwinds from the sleeve in time t is $= 2R\theta_2$

$\therefore$ Displacement of C in time t is

$$x_C = x_B + 2R\theta_2$$

Differentiating twice wrt time, as above, gives

$$a_C = a_B + 2R\alpha_2$$

$$\therefore \quad a_C = R\alpha_1 + R\alpha_2 + 2R\alpha_2$$

$$= R(\alpha_1 + 3\alpha_2)$$

Example 16 *Distance travelled by a point on a rolling wheel*

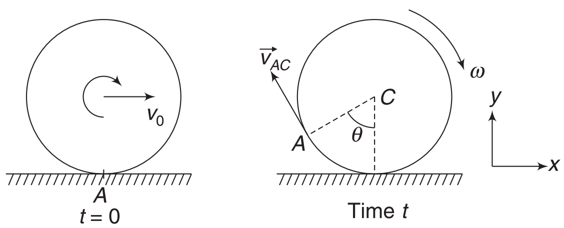
A wheel rolls without sliding on a horizontal surface. Its centre moves uniformly with velocity v_0. Consider a particle on the circumference. How much distance does this particle travel by the time the wheel makes one complete rotation about its centre? Radius of the wheel is R.

Solution

Concepts

We will write speed (v) of the particle as a function of time. Distance travelled is

$$S = \int_0^T v\,dt \text{ where } T = \text{time for one rotation}$$



Consider a point A at the bottom of the wheel, at $t = 0$.

In time t, the wheel rotates through an angle θ about the centre, where $\theta = \omega t$. Velocity of point A, relative to centre (C) is tangential and is equal to $\omega R = v_0$.

$$\therefore \quad \vec{v}_{AC} = -v_0 \cos\theta \hat{i} + v_0 \sin\theta \hat{j}$$

Velocity of A wrt ground is $\vec{v}_A = \vec{v}_{AC} + \vec{v}_C$

$$\Rightarrow \quad \vec{v}_A = (v_0 - v_0\cos\theta)\hat{i} + v_0 \sin\theta \hat{j}$$

Speed of point A is $v_A = \sqrt{v_0^2(1-\cos\theta)^2 + v_0^2 \sin^2\theta}$

$$\Rightarrow \quad v_A = v_0\sqrt{2 - 2\cos\theta} = \sqrt{2}\, v_0 \sqrt{1-\cos\theta}$$

$$= 2\, v_0 \sin\frac{\theta}{2} = 2v_0 \sin\left(\frac{\omega t}{2}\right) = 2v_0 \sin\left(\frac{v_0 t}{2R}\right)$$

Time for one rotation is $T = \dfrac{2\pi}{\omega} = \dfrac{2\pi R}{v_0}$

$\therefore$ Distance travelled by A in one full rotation is

$$S = \int_0^T v_A dt = 2v_0 \int_0^T \sin\left(\frac{v_0 t}{dt}\right) dt$$

$$= \frac{2v_0}{\frac{v_0}{2R}}\left[-\cos\left(\frac{v_0 t}{2R}\right)\right]_0^T$$

$$= -4R\left[\cos\left(\frac{v_0}{2R}\cdot\frac{2\pi R}{v_0}\right) - \cos 0\right]$$

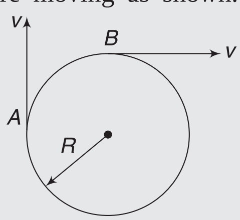
$$= -4R[\cos\pi - \cos 0] = -4R[-1-1] = 8R$$

Worksheet 1

1. A rigid body is in rotation about a fixed axis. Which of the followings must be true?
   (a) No two particles in the body can have the same speed
   (b) No two particles in the body can have the same velocity
   (c) No two particles can have the same angular speed
   (d) All particles have the same angular acceleration

2. A rigid body is in pure rotation.
   (a) You can find two points in the body, in a plane perpendicular to the axis of rotation, having the same velocity.
   (b) You can find two points in the body, in a plane perpendicular to the axis of rotation, having the same acceleration.
   (c) Speed of all the particles lying on the curved surface of a cylinder whose axis concides with the axis of rotation is same.
   (d) Relative acceleration of any two points in the body is always zero.

3. Two points of a rigid body are moving as shown. The angular velocity of the body is:

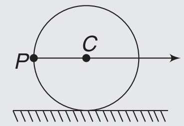


   (a) $\dfrac{v}{2R}$
   (b) $\dfrac{v}{R}$
   (c) $\dfrac{2v}{R}$
   (d) $\dfrac{2v}{3R}$

4. A ladder of length L is slipping with its ends against a vertical wall and a horizontal floor. At a certain moment, the speed of the end in contact with the horizontal floor is v and the ladder makes an angle $\alpha = 30°$ with the horizontal. Then the speed of the ladder's centre must be
   (a) $2v/\sqrt{3}$ (b) $v/2$
   (c) v (d) None

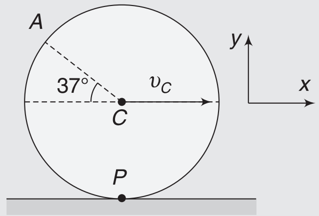
5. A sliding solid sphere, on a fixed flat surface, has a velocity (of centre of mass) v and angular velocity ω. Its contact point
   (a) must be moving forward (in direction of v)
   (b) must be moving backward (opposite to v)
   (c) cannot be at rest
   (d) data is insufficient

6. A disc of radius R is rolling purely on a flat horizontal surface, with a constant angular velocity. The angle between the velocity and acceleration vectors of point P is



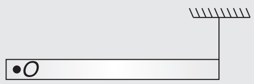
   (a) zero (b) 45°
   (c) 135° (d) $\tan^{-1}(1/2)$

7. A cylinder of radius 5 m rolls on a horizontal surface. Velocity of its centre is 25 ms^{-1}. Velocity of the point A is (in ms^{-1})



   (a) $40\hat{i} + 20\hat{j}$
   (b) $20\hat{i} + 30\hat{j}$
   (c) $20\hat{i} + 30\hat{j}$
   (d) $20\hat{i} + 30\hat{j}$

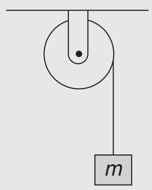
8. A rod is pivoted at its one edge about point O. Other edge of rod is suspended from the ceiling through a rope, as shown. If the rope is suddenly cut and its initial angular acceleration is found to be α.



   (a) initial acceleration of COM of the rod will be zero
   (b) initial acceleration of the COM of the rod will be more that of any other point in the rod
   (c) initial acceleration of the COM of the rod will be in vertically downward direction.
   (d) none of the above

9. A chain couples and rotates two wheels in a bicycle. The radii of bigger and smaller wheels are 0.5 m and 0.2 m respectively. The bigger wheel rotates at the rate of 200 rotations per minute, then the rate of rotation of smaller wheel will be
   (a) 1000 rpm (b) 50/3 rpm
   (c) 200 rmp (d) 500 rpm

10. In the figure shown, the block of mass m is going down with a constant velocity of 2 ms^{-1}. Radius of the pulley is 0.1 m



   (a) acceleration of a point on the circumference of the pulley has magnitude equal to 40 ms^{-2}
   (b) acceleration of a point on the circumference of the pulley is 0
   (c) acceleration of a point on the circumference of the pulley could be 10 ms^{-2}
   (d) acceleration of a point on vertical segment of the string is 40 ms^{-2}

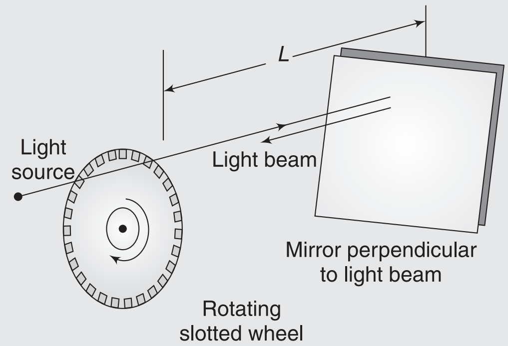
Worksheet 2

1. A wheel is rotating with angular velocity 2 rad s^{-1}. It is subjected to uniform angular acceleration 2.0 rad s^{-2}.
   (a) angular velocity of the wheel after 10 s is 22 rad s^{-1}
   (b) number of rotations made by the wheel in 10 s is close to 10
   (c) angular velocity of different points will be different in general
   (d) linear acceleration of a point in the wheel must be towards the axis of rotation

2. A ring rolls without slipping on the ground. Its centre C moves with a constant speed u. P is any point on the ring. The speed of P with respect to the ground is v.
   (a) $0 \le v \le 2u$
   (b) $v = u$, if CP is horizontal
   (c) $v = u$, is possible if CP makes an angle of 30° with the horizontal and P is below the horizontal level of C.
   (d) $v = \sqrt{2}u$, if CP is horizontal

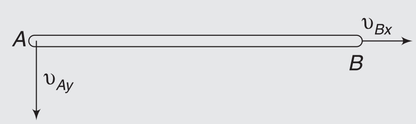
3. A wheel of radius r is rolling on a straight line, the velocity of its centre being v. At a certain instant, the point of contact of the wheel with the grounds is M and N is the highest point on the wheel (diametrically opposite to M). The correct statements are:
   (a) The velocity of any point P of the wheel is proportional to MP.
   (b) Points of the wheel moving with velocity greater than v form a larger area of the wheel than points moving with velocity less than v.
   (c) The point of contact M is instantaneously at rest.
   (d) The velocities of any two parts of the wheel, which are equidistant from the centre, are equal.

4. An early method of measuring the speed of light makes use of a rotating slotted wheel. A beam of light passes through a slot at the outside edge of the wheel, as shown in figure below, travels to a distant mirror, and returns to the wheel just in time to pass through the next slot in the wheel. One such slotted wheel has a radius of 5.0 cm and 500 slots at its edge. Measurements taken when the mirror is at a distance L = 500 m from the wheel indicate speed of light to be 3.0×10^5 kms^{-1}.



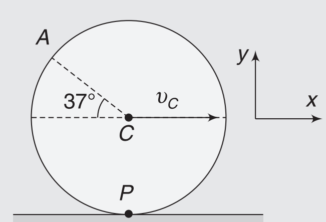
   (a) a possible value of the (constant) angular speed of the wheel is nearly 3.8×10^3 rad s^{-1}?
   (b) a possible linear speed of a point on the edge of the wheel is 1.9×10^2 ms^{-1}
   (c) speed of light is too high for such experiment to be successful
   (d) None of the above

5. A 50-cm long rod AB is in combined translation and rotation motion on a table. At an instant, velocity component of point A perpendicular the rod is 10 cm s^{-1}, velocity component of point B parallel to the rod is 6.0 cm s^{-1} and the angular velocity of the rod is 0.4 rad s^{-1} in anti-clockwise direction



   (a) the x component of velocity of A is 6.0 cm s^{-1}
   (b) the y component of velocity of B is 10.0 cm s^{-1}
   (c) the instantaneous centre of rotation is located at a distance of 15 cm from the rod.
   (d) None

6. A round body of radius 10 cm starts rolling, without sliding, on a horizontal stationary surface with uniform angular acceleration 2 rad s^{-2}.



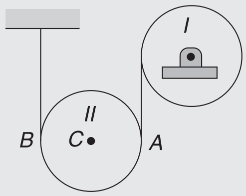
(a) initial acceleration of the centre C is 20 cms^{-2}

(b) initial acceleration of the top point A is 20 cms^{-2}

(c) initial acceleration of the top point A is 40 cms^{-2}

(d) initial acceleration of the centre C is 0

7. Two identical disks, each of radius r, are connected by a cord, as shown in the figure. The cord wraps tightly on Disc I and does not slip on Disc II. Disk I rotates with constant angular acceleration α in anti-clockwise direction.



(a) acceleration of the centre of Disk II will be $1/2r\alpha$

(b) angular acceleration of Disk II will be $1/2\alpha$

(b) angular acceleration of Disk II will be α

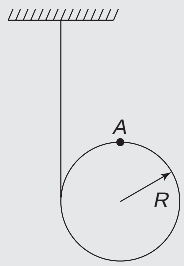
(d) acceleration of Disk II is difficult to predict unless its mass is known.

Worksheet 3

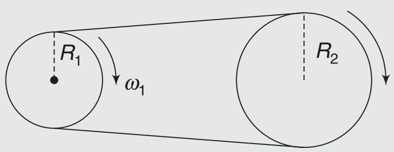
1. A thread is tightly wrapped on a disc. The free end of the thread is fixed to the ceiling and the disc is allowed to fall. Radius is $R = 0.2\,\text{m}$.

(i) Find the angular speed of the disc at the instant its centre is moving down at speed $v = 4\,\text{ms}^{-1}$.

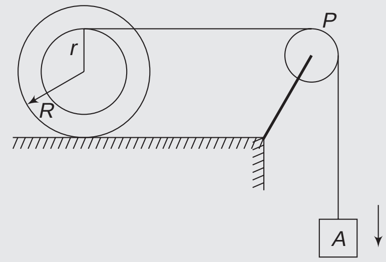
(ii) Is there any point in the disc having zero speed at the instant speed of the centre is 4 ms^{-1}? What is speed of the top point of the disc at this instant?



2. A belt moves on two cylinders without slipping. Radius of one cylinder is R_1 and it is spinning with angular velocity ω_1 about its central axis. Find the angular speed of the other cylinder, if its radius is R_2.



3. A spool has outer radius $R = 0.2\,\text{m}$. Thread is wound tightly on its inner wheel, having radius $r = 0.1\,\text{m}$. The spool is kept on a horizontal table and the free end of the thread is used to suspend a block A, as shown. Pulley P is fixed. Find acceleration of the centre of the spool when acceleration of block A is α. It is given that the spool does not slip on the table.



4. A stick of length $L = 1.0\,\text{m}$ is tossed up such that its centre moves up vertically. Initially, the stick was horizontal and it falls back to the projection point again being horizontal. During its journey, the stick, made one complete rotation. The total time of flight of the stick is 2 s. Find the speed of the ends of the stick 0.5 s after projection. Assume that the angular speed of the stick does not change during the flight.

Answers Sheet

Your Turn

1. $182.9\,\text{ms}^{-2}$
2. $50\ \text{rad s}^{-1}$, $100\ \text{rad s}^{-1}$
3. $4\pi R$
4. $4\,\text{ms}^{-1}$
5. $14\ \text{ms}^{-2}$
6. (i) C is instantaneous centre such that ΔABC is equilateral (ii) v/L
7. Yes, $2\,\text{ms}^{-1}$ in forward direction
8. $v_A = 6\,\text{ms}^{-1}$; $v_B = 2\sqrt{5}\,\text{ms}^{-1}$; $v_C = 2\text{ms}^{-1}$
9. $8\ \text{ms}^{-2}$ ($\downarrow$)
10. (i) $\sqrt{2}\ v$ (ii) P

Worksheet 1

1. (d) 2. (c) 3. (b) 4. (c) 5. (d) 6. (b) 7. (a) 8. (c) 9. (d)
10. (a)

Worksheet 2

1. (a) 2. (a,c,d) 3. (a,b,c) 4. (a,b) 5. (a,b,c) 6. (a,c) 7. (a,b)

Worksheet 3

1. (i) $\omega = 20\ \text{rad s}^{-1}$ (ii) The point of disc in contact with vertical thread has no speed. $v_A = 4\sqrt{2}\ \text{ms}^{-1}$
2. $\dfrac{\omega_1 R_1}{R_2}$
3. $\dfrac{2}{3}\,a$
4. $5.24\,\text{ms}^{-1}$

CHAPTER 6

Rotational Dynamics

"It's not magic! It's physics. The speed of the turn is what keeps you upright. It's like a spinning top."

–*Deborah Bull*

1. INTRODUCTION

An unbalanced force produces acceleration in a body and causes its momentum to change. An unbalanced torque produces angular acceleration and causes the angular momentum of a body to change. This is the basic subject matter of this chapter. We will also learn about rotational kinetic energy.

By the end of this chapter, we will be able to analyse a plane motion of a rigid body. We will pay special attention to rolling motion.

Conservation of angular momentum is a fundamental conservation principle in physics just like conservation of momentum and energy. We will learn this conservation principle towards the end of the chapter.

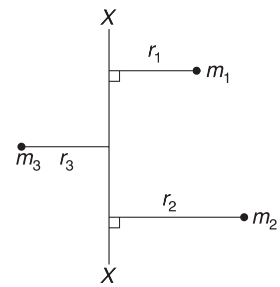
2. MOMENT OF INERTIA

When a finite force is applied on a body, its acceleration is decided by its inertia (mass). Similarly, when a finite torque acts on a body (about an axis), its angular acceleration is decided by its rotational inertia. Rotational inertia of a body about an axis is termed as its moment of inertia.

Think of two fans having same mass but different diameters. If you want to spin them with the same angular speed, the one with the larger diameter will require more effort. If both of them are rotating with the same angular speed, the one with the larger diameter will need more effort to bring it to rest. We say that the fan with larger diameter has more rotational inertia (i.e., moment of inertia) than the other fan.

Rotational inertia of a body depends on mass as well as geometry of distribution of mass around the rotation axis. Moment of inertia of a body is always defined with respect to a given rotation axis. If the axis changes, moment of inertia may also change.

Moment of Inertia (*MI*) of an n-particle system, about a fixed line (usually called as axis), is the sum of the products of the masses of the particles with square of their respective distances from the line. [You will soon realise why it has been defined this way. Right now, just learn this definition].

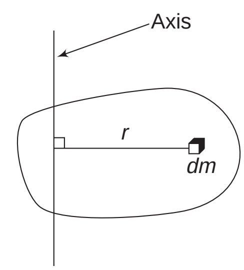


XX is a fixed line and particles of masses m_1, m_2, m_3 ... are scattered around it. Their distances from the line are r_1, r_2, r_3 ... Moment of Inertia of this collection of particles about line *XX* is

$$I = m_1r_1^2 + m_2r_2^2 + m_3r_3^2 + ... = \sum m_ir_i^2 \quad ...(1)$$

SI unit of *MI* is kg m^2.

If the distribution of mass is continuous, we consider a small elemental mass dm in the body. If distance of this element from the concerned axis is r, then *MI* of the element is



$$dI = r^2\,dm$$

MI of the entire body is obtained by summing the *MI* of every such element.

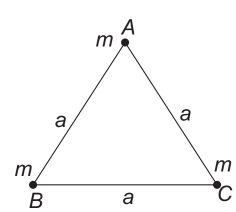
$$I = \int r^2\,dm \quad ...(2)$$

Following examples will illustrate the process.

Example 1 Three point-masses, each of mass m, are placed at the vertices of an equilateral triangle of side length a. Find the *MI* of the system about

(i) an axis through vertex *A* perpendicular to the plane of the figure.

(ii) line BC.



Solution

Concepts

(i) A line passing through A is at a perpendicular distance 'a' from both B and C.

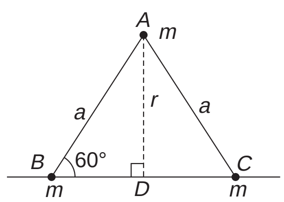
(ii) A particle that lies on the axis has no contribution to MI.

(i) MI about an axis through A and perpendicular to the plane of the figure is

$$I_A = m(0)^2 + m(a)^2 + m(a)^2$$
$$= 2ma^2$$

(ii) Distance of A from line BC is $r = a \sin 60° = \frac{\sqrt{3}}{2} a$
B and C lie on the axis itself.

$$\therefore \quad I = mr^2 = \frac{3m}{4}a^2$$



Example 2 *MI of a ring about central axis*

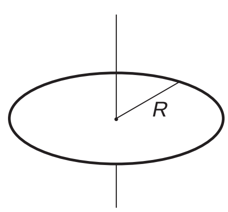
Find MI of a ring of mass M and radius R about an axis passing through its centre and perpendicular to its plane.

Solution

Concepts

Every mass point is at a distance R from the axis.

$$I = \int R^2 dm = R^2 \int dm = MR^2$$



Example 3 *MI of a thin rod about an axis perpendicular to it through its end*

A uniform thin rod has mass M and length L. Find its MI about an axis that is perpendicular to the rod and passes through its end.

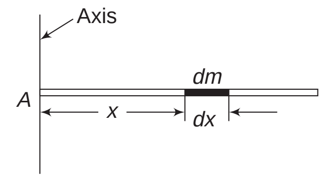
Solution

Concepts

(i) 'Thin rod' implies that mass can be assumed to be distributed on a line.

(ii) Different mass-points are at different distances from the axis.

(iii) We will consider a small element on the rod, write its MI and then integrate to find the MI of the entire rod.



Axis is as shown in figure. Consider a small segment of length dx at a distance x from the axis.

Mass of the element, $dm = \frac{M}{L} \cdot dx$

MI of the element about axis, $dI = x^2\, dm = \frac{M}{L}\, x^2 dx$

$\therefore$ MI of the rod about the axis is

$$I = \int dI = \frac{M}{L}\int_0^L x^2\, dx = \frac{M}{L}\left[\frac{x^3}{3}\right]_0^L = \frac{ML^2}{3}$$

By taking the limits of integration from $x = 0$ to $x = L$, we have considered all elements in the rod.

Example 4 *MI of disc about central axis*

A uniform disc of mass M has radius R. Find its MI about an axis passing through its centre and perpendicular to its plane.

Solution

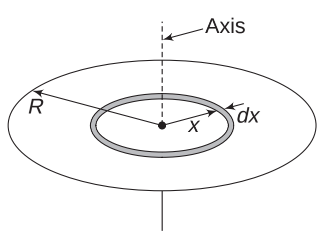
Concepts

(i) We will divide our disc into a series of thin rings.

(ii) According to the result obtained in Example 2, we can write MI of such a ring about the given axis. Integration is used to add MI of all such rings.

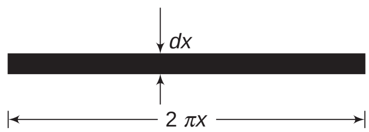
The disc and its axis is shown in the figure. Mass per unit surface area of the disc is

$$\sigma = \frac{M}{\pi R^2}$$



Consider a ring of radius x and width dx in the disc. [i.e., consider the mass lying between radius x and $x + dx$].

Area of the elemental ring = area of a rectangle of length $2\pi x$ and width dx



If you cut open the elemental ring and straighten it, it will appear like a rectangular strip.

$$\Rightarrow \quad dA = 2\pi x\, dx$$

Mass of ring element is

$$dm = \sigma dA = \frac{M}{\pi R^2} \cdot 2\pi x\, dx$$

$$= \frac{2M}{R^2} x\, dx$$

MI of this ring about the given axis is

$$dI = x^2 \cdot dm \quad \text{[Refer to Example 2]}$$

$$= \frac{2M}{R^2} x^3\, dx$$

To get *MI* of the disc, we need to add *MI* of all such rings.

$$\therefore \quad I = \int dI = \frac{2M}{R^2} \int_0^R x^3\, dx = \frac{2M}{R^2} \left[\frac{x^4}{4}\right]_0^R$$

$$\Rightarrow \quad I = \frac{MR^2}{2}$$

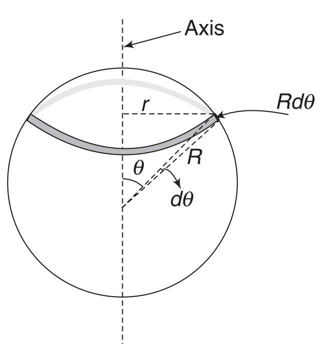
Example 5 *Uniform spherical shell*

A uniform spherical shell has mass M and radius R. Find its *MI* about a diameter. Given, $\int_0^{\pi} \sin^3 \theta d\theta = \frac{4}{3}$.

Solution

Concepts

We will divide the shell into multiple thin rings with their axes along the given diameter.



Consider a ring element between angular position θ and $(\theta + d\theta)$, as shown in figure.

Radius of ring, $r = R\sin\theta$

Width of ring $= Rd\theta$

Surface area of the ring

$$dA = (2\pi r)(Rd\theta) = 2\pi R^2 \sin\theta \cdot d\theta$$

Mass per unit surface area, $\sigma = \dfrac{M}{4\pi R^2}$

$\therefore$ Mass of the ring element, $dm = \sigma \cdot dA = \dfrac{M}{2} \sin\theta d\theta$

MI of the ring, $dI = (dm)r^2 = \dfrac{M}{2} R^2 \sin^3 \theta d\theta$

$$\therefore \quad I = \frac{MR^2}{2} \int_0^{\pi} \sin^3 \theta d\theta = \frac{2}{3} MR^2$$

Take note of the limit of integration.

Example 6 *Uniform solid sphere*

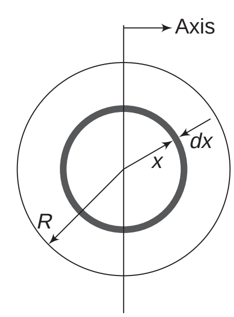
A uniform solid sphere has mass M and radius R. Find its *MI* about a diameter.

Solution

Concepts

We will consider the sphere to be made up of infinite shells, laid one over another.

To write *MI* of a shell, we will use the result obtained in the last example.



Consider an infinitesimal shell element of inner and outer radii x and $(x + dx)$ respectively.

Volume of shell element

$$= \text{(Surface area)} \times \text{(thickness)}$$

$$\therefore \quad dV = 4\pi x^2 \cdot dx$$

Density of sphere, $\rho = \dfrac{M}{\frac{4}{3}\pi R^3}$

$\therefore$ Mass of shell element is

$$dm = \rho \cdot dV = \frac{3M}{R^3} x^2\, dx$$

MI of the shell about diameter is $dI = \dfrac{2}{3}(dm) \cdot x^2$

$$\Rightarrow \quad dI = \frac{2M}{R^3} x^4\, dx$$

$$\therefore \quad I = \int dI = \frac{2M}{R^3} \int_0^R x^4\, dx = \frac{2M}{R^3}\left(\frac{R^5}{5}\right)$$

$$\therefore \quad I = \frac{2}{5} MR^2$$

2.1 Theorem of Perpendicular Axis

This theorem is applicable for a laminar (two-dimensional) body. The theorem states that:

The sum of moments of inertia of a laminar body, about any two mutually perpendicular axes lying in the plane of the body, is equal to *MI* about a third axis normal to the plane of the body and passing through the point of intersection of the two axes.

x and y-axes are in the plane of a flat body. *MI* of the body about these two axes is I_x and I_y.

z-axis is perpendicular to the plane of the body.

MI about z-axis is

$$I_z = I_x + I_y \qquad ...(3)$$

Proof: Consider the body to be made of a number of particles, specified by mass m_i and co-ordinates $(x_i, y_i, 0)$.

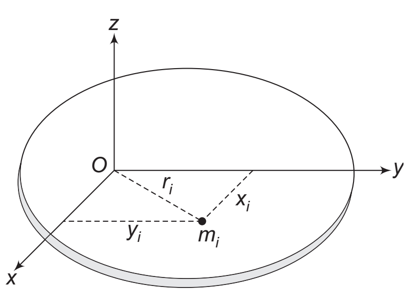
Distance of a particle from z-axis is

$$r_i = \sqrt{x_i^2 + y_i^2}$$

$$\therefore \qquad I_z = \sum m_i r_i^2 = \sum m_i (x_i^2 + y_i^2)$$

$$= \sum m_i x_i^2 + \sum m_i y_i^2$$

$$\Rightarrow \qquad I_z = I_y + I_x \qquad \begin{bmatrix} x_i = \text{distance of } m_i \text{ from } y\text{-xis} \\ y_i = \text{distance of } m_i \text{ from } x\text{-axis} \end{bmatrix}$$



Example 7 *MI of ring about its diameter*

A ring has mass M and radius R. Find its *MI* about a diameter.

Solution

Concepts

(i) We know that *MI* of a ring about its central axis perpendicular to its plane is MR^2. Let us take this axis as z-axis.

(ii) Any two perpendicular diameters will be taken as x and y-axes.

(iii) Due to symmetry, *MI* of the ring about any diameter must be same.

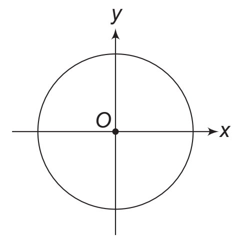
From symmetry: $I_x = I_y$

perpendicular axis theorem:

$$I_x + I_y = I_z$$

$$\Rightarrow \qquad 2I_x = MR^2$$

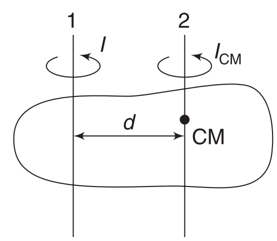
$$\Rightarrow \qquad I_x = \frac{1}{2} MR^2$$



z-axis is $\perp$ to the plane passing through O.

2.2 Parallel Axis Theorem

This theorem is valid for objects of all shapes. It states that the *MI* of a body of mass M about any axis is equal to $I_{CM} + Md^2$, where I_{CM} is *MI* of the body about another axis that passes through COM of the body and is parallel to the first axis and d is perpendicular distance between the two axes.

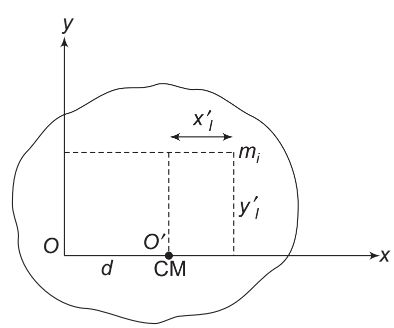


$$I = I_{CM} + Md^2 \qquad ...(4)$$

I is *MI* about axis 1 shown in figure and I_{CM} is *MI* about axis 2 that is parallel to 1 and passes through COM.

Axis 1 may be completely outside the body.

Proof: Consider a rigid body shown in the figure. Our axis is perpendicular to the plane of the figure and passes through point O (call it z-axis).



COM of the body is assumed to lie on x-axis, at a distance d from O. There is no loss in generality in assuming the COM to be on x-axis.

Consider an axis through COM and parallel to the first axis (z-axis).

Let co-ordinates of a particle of mass m_i be (x_i', y_i', z_i') in a co-ordinate system having its origin at COM.

Distance of the particle from axis through COM will be

$$r_i' = \sqrt{(x_i')^2 + (y_i')^2}$$

$\therefore$ *MI* of the body about this axis will be

$$I_{CM} = \sum m_i (r_i')^2 = \sum m_i \left[(x_i')^2 + (y_i')^2\right] \qquad ...(i)$$

But $\qquad x_i' = x_i - d \quad \text{and} \quad y_i' = y_i$

where (x_i, y_i, z_i) are co-ordinates of m_i in co-ordinate system with origin at O. MOI about z-axis through O is

$$I = \sum m_i r_i^2 = \sum m_i[(x_i)^2 + (y_i)^2]$$
$$= \sum m_i[(x_i' + d)^2 + (y_i')^2]$$
$$= \sum m_i[(x_i')^2 + (y_i')^2] + \left[\sum m_i\right] d^2 + 2d \sum m_i x_i'$$
$$I = I_{CM} + Md^2 + 0$$

The third term is zero, as x_i' refers to x co-ordinates of the particle in COM frame.

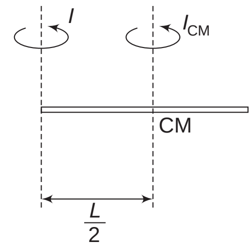
Theorems of parallel axis and perpendicular axis are two very useful tools, which help us in calculating *MI* of a body about some axis if its *MI* is known about certain other axis. We need not perform elaborate intergration all the times.

Example 8 A uniform thin rod has mass M and length L. Find its *MI* about an axis that is perpendicular to the rod and passes through it COM.

Solution

Concepts

We will use the result obtained in Example 3 and parallel axis theorem.



From Example 3, *MI* about an axis through one end and perpendicular to the rod is

$$I = \frac{ML^2}{3}$$

Other axis is parallel to this axis and passes through COM.

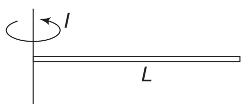
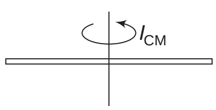
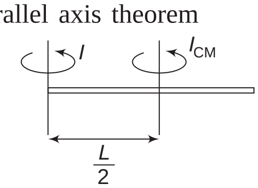
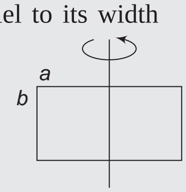
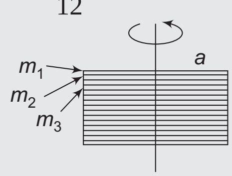
$$\therefore \qquad I = I_{CM} + Md^2 \text{ gives}$$

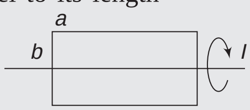
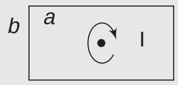
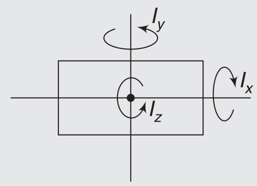
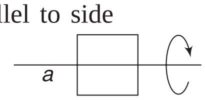
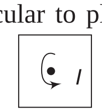
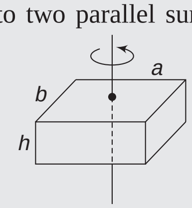
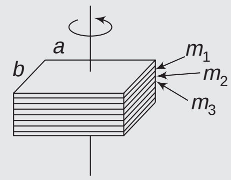
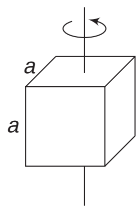
$$\frac{ML^2}{3} = I_{CM} + M\left(\frac{L}{2}\right)^2$$

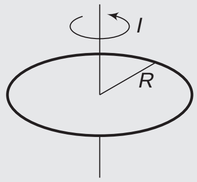
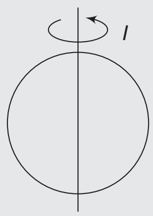
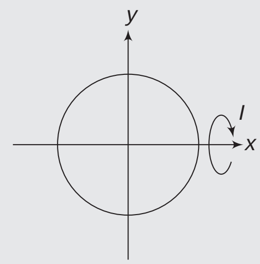
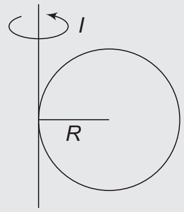
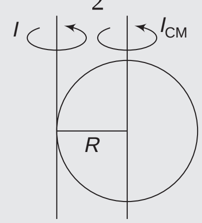
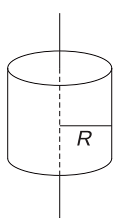
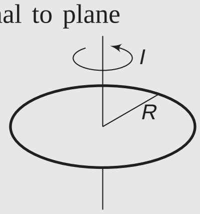
$$\Rightarrow \qquad I_{CM} = \frac{ML^2}{3} - \frac{ML^2}{4} = \frac{ML^2}{12}$$

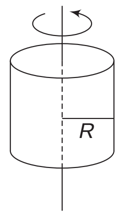
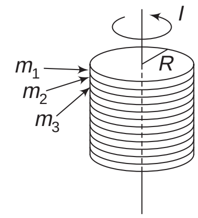
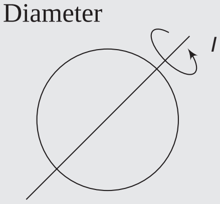
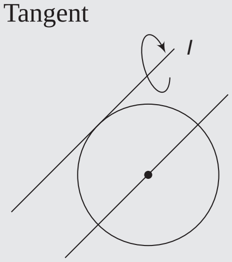
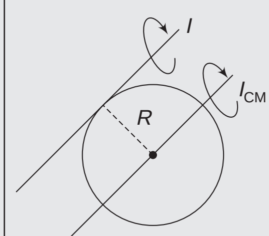
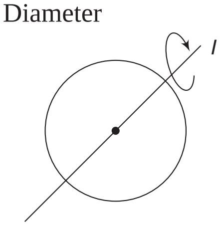
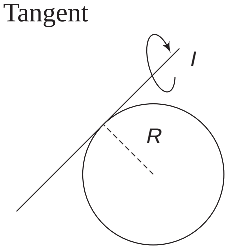
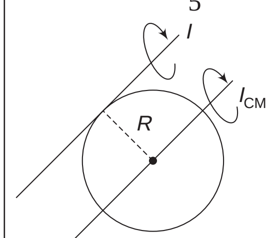
In Short:

The following table presents the *MI* of important geometrical shapes about various important axes. We have also incorporated the logic/steps involved in writing the *MI*. [All bodies are uniform.]

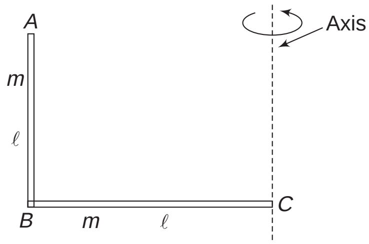
Body	*Axis*	*Moment of Inertia*	*Logic/steps*
1. Thin rod	(a) through one end perpendicular to rod 	$I = \frac{ML^2}{3}$	Direct integration. See Example 3.
	(b) through centre perpendicular to rod 	$I_{CM} = \frac{ML^2}{12}$	Parallel axis theorem  $I = I_{CM} + Md^2$
2. Rectangular plate	(a) axis through centre, parallel to its width 	$I = \frac{Ma^2}{12}$	Divide the plate into a number of thin strips. Each strip is like a rod. If masses of these strips are m_1, m_2, m_3 ..., etc. $I = \frac{m_1a^2}{12} + \frac{m_2a^2}{12} + \frac{m_3a^2}{12} + ...$ $= (m_1 + m_2 + m_3 + ...)\frac{a^2}{12}$ $\Rightarrow \quad I = \frac{Ma^2}{12}$ 

	(b) axis through centre, parallel to its length 	$I = \frac{Mb^2}{12}$	Can be concluded from above result.
	(c) axis through centre, perpendicular to plane of the plate 	$I = \frac{M}{12}(a^2 + b^2)$	Perpendicular axis theorem $I_z = I_x + I_y = \frac{M}{12}(a^2 + b^2)$  z-axis is perpendicular to plane of figure
3. Square plate	(a) passing through centre, parallel to side 	$I = \frac{Ma^2}{12}$	It can be concluded from MI of rectangular plate
	(b) passing through centre, perpendicular to plane 	$I = \frac{Ma^2}{6}$	Put $b = a$ in the formula for rectangular plate
4. Rectangular brick	Passing through centre, normal to two parallel surfaces 	$I = \frac{M}{12}(a^2 + b^2)$	Divide the brick into number of thin rectangular plates. If m_1, m_2, m_3 ... are masses of these slices, then $I = \frac{m_1}{12}(a^2 + b^2) + \frac{m^2}{12}(a^2 + b^2) + \ldots$ $= \frac{(m_1 + m_2 + \ldots)}{12}(a^2 + b^2)$ $= \frac{M}{12}(a^2 + b^2)$ 
5. Cube	Passing through the centre, normal to two parallel faces 	$I = \frac{Ma^2}{6}$	Put $b = a$ in the above expression for rectangular brick.

6. Ring	(a) passing through the centre, normal to plane 	$I = MR^2$	All points are at distance R from the axis.
	(b) a diameter 	$I = \dfrac{MR^2}{2}$	Any two diameters are similar. Consider two perpendicular diameters as x and y-axes  $I_x + I_y = I_z$ $2I = MR^2$ $I = \dfrac{1}{2} MR^2$
	(c) a tangent 	$I = \dfrac{3}{2} MR^2$	Parallel axis theorem $I = I_{CM} + MR^2$ $= \dfrac{MR^2}{2} + MR^2$ $= \dfrac{3}{2} MR^2$ 
7. Cylindrical shell	Central axis parallel to length 	$I = MR^2$	All particles are at distance R from the axis.
8. Disc	(a) passing through centre, normal to plane 	$I = \dfrac{1}{2} MR^2$	By method of integration. See Example 4.

	(b) About diameter	$I = \dfrac{MR^2}{4}$	Using perpendicular axis theorem, as in case of a ring.
9. Solid cylinder	Passing through centre, parallel to length 	$I = \dfrac{1}{2} MR^2$	Divide the cylinder into many this discs of masses m_1, m_2, m_3 ...etc.  $I = \dfrac{1}{2} m_1R^2 + \dfrac{1}{2} m_2R^2 + \dfrac{1}{2} m_3 R^2 + ...$ $= \dfrac{1}{2} (m_1 + m_2 + m_3 + ...)R^2 = \dfrac{1}{2} MR^2$
10. Thin spherical shell	(a) Diameter 	$I = \dfrac{2}{3} MR^2$	Integration. See Example 5.
	(b) Tangent 	$I = \dfrac{5}{3} MR^2$	Parallel axis theorem  $I = I_{CM} + MR^2$ $= \dfrac{2}{3} MR^2 + MR^2 = \dfrac{5}{3} MR^2$
11. Solid sphere	(a) Diameter 	$I = \dfrac{2}{5} MR^2$	Integration See Example 6
	(b) Tangent 	$I = \dfrac{7}{5} MR^2$	$I = I_{CM} + MR^2$ $= \dfrac{2}{5} MR^2 + MR^2 = \dfrac{7}{5} MR^2$ 

Example 9 *ABC* is a *L* shaped rod made of two perpendicular segments – *AB* and *BC* – each of length *l* and mass *m*. Consider an axis passing through *C* and parallel to *AB*. Write the moment of inertia of the rod about this axis.



Solution

Concepts

(i) Write the *MI* of two segments (*AB* and *BC*) separately, about the given axis, and add them.

(ii) For segment *AB*, entire mass is at a distance ℓ from the axis.

MI of *AB*, $I_1 = ml^2$ [$\because$ Each particle is at distance *l* from the axis]

MI of *BC*, $I_2 = \frac{1}{3}ml^2$

$\therefore \quad I = I_1 + I_2 = \frac{4}{3}ml^2$

Example 10 In the last problem, find the moment of inertia of the *L*-shaped rod if line *AB* itself is the axis.

Solution

Concepts

When mass lies on the axis, it does not contribute to moment of inertia.

MI of rod *AB* about the said axis = 0

MI of rod *BC* about the said axis = $\frac{1}{3}ml^2$

$\therefore \quad I = \frac{1}{3}ml^2$

Example 11 *MI of a cube about its edge*

A uniform cube has mass *M* and side length *a*. Find its *MI* about one of its edge.

Solution

Concepts

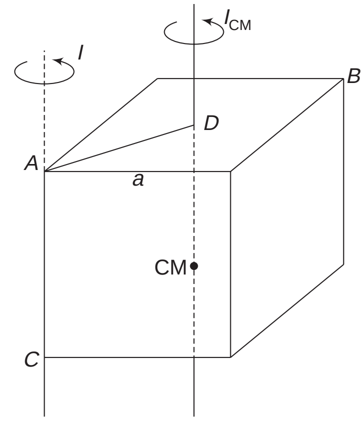
(i) *MI* of a cube about its central axis (See the table given above) is $\frac{1}{6}Ma^2$

(ii) Use parallel axis theorem.

$$I_{CM} = \frac{Ma^2}{6}$$

AC is an axis that is parallel to the axis passing through COM.

Distance between the two axes is



$$d = AD = \frac{AB}{2} = \frac{\sqrt{2}\,a}{2} = \frac{a}{\sqrt{2}}$$

$$\therefore \quad I = I_{CM} + Md^2 = \frac{Ma^2}{6} + M\left(\frac{a}{\sqrt{2}}\right)^2 = \frac{2}{3}Ma^2$$

Example 12 *MI of a square plate about diagonal*

A uniform square plate has mass *M* and side length *a*. Find its *MI* about a diagonal.

Solution

Concepts

(i) Diagonals of a square are perpendicular. The two diagonals can be regarded as *x* and *y*-axes in the plane of the square. We will use perpendicular axis theorem.

(ii) $I_z = \frac{Ma^2}{6}$. See the table.

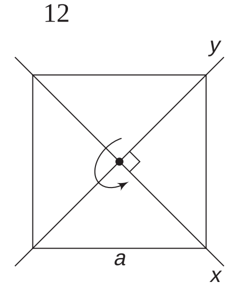
From symmetry, $I_x = I_y$

Perpendicular axis theorem:

$$I_x + I_y = I_z$$

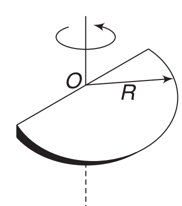
$$2I_x = \frac{Ma^2}{6}$$

$$\Rightarrow \quad I_x = \frac{Ma^2}{12}$$



Example 13 *Half disc*

A semi-circular disc has mass *M* and radius *R*. Find its moment of inertia about an axis, which is perpendicular to its plane passing through the centre of the circle.

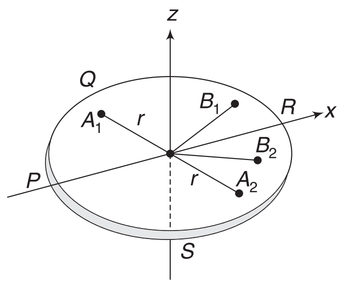


Solution

Concepts

(i) Think of a disc of mass M and radius R. Its MI about central axis perpendicular to its plane is $\frac{MR^2}{2}$.

(ii) If this disc is folded about its diameter to make it a semi-circular disc, the distance of any mass point from the axis does not change.



If half-disc PQR is flipped over to cover the other half PSR, a point like A_1 moves to a position A_2. Distance of this point from z-axis remains unchanged. A point B_1 moves to B_2. Its distance from z-axis remains unchanged.

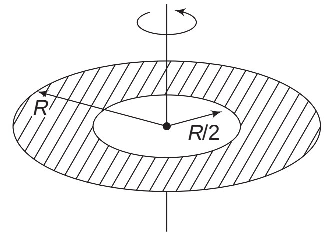
Thus, $\sum m_i r_i^2$ will not change.

$\therefore$ MI of half-disc about z-axis = MI of full disc about z-axis

$$\therefore \qquad I = \frac{1}{2}MR^2$$

Example 14 *Disc with a hole*

A disc of radius R has a concentric hole of radius $\frac{R}{2}$. 'The disc with hole' has mass M. Find its moment of inertia about its central axis normal to its plane.



Solution

Concepts

(i) Fill up the hole. Write MI of completed disc about the given axis and then subtract the MI of disc of radius $\frac{R}{2}$.

(ii) Remember that mass of the completed disc (without hole) is larger than M.

Mass per unit area is $\sigma = \dfrac{M}{\pi R^2 - \pi\left(\frac{R}{2}\right)^2} = \dfrac{4M}{3\pi R^2}$

Mass of disc without hole is, $M_1 = \sigma \cdot \pi R^2 = \dfrac{4M}{3}$

MI of disc without hole about the given axis is

$$I_1 = \frac{M_1 R^2}{2} = \frac{2M}{3}R^2$$

MI of the removed disc $\left(\text{of radius } \frac{R}{2}\right)$ about the given axis is

$$I_2 = \frac{1}{2}M_2\left(\frac{R}{2}\right)^2 = \frac{MR^2}{24}$$

$\therefore$ Required MI is $I = I_1 - I_2 = \left(\frac{2}{3} - \frac{1}{24}\right)MR^2 = \frac{5}{8}MR^2$

Your Turn

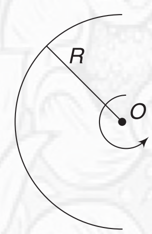
Q.1 Three particles of masses 10 g, 20 g and 30 g are kept in xy-plane at points (2, 0) cm, (0, 6) cm and (4, 3) cm respectively. Find moment of inertia of the particle system about

(a) x-axis (b) y-axis (c) z-axis

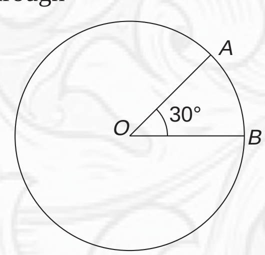
Q.2 A uniform rod has mass m and length $2l$. Two particles of mass m each, are attached to the ends of the rod. Find moment of inertia of the system about the perpendicular bisector of rod.

Q.3 A ring has mass M and radius R. One half of the ring is twice as heavy as the other half. Find the moment of inertia of the ring about its central axis perpendicular to its plane.

Q.4 A half-ring has mass M and radius R. Find its moment of inertia about an axis perpendicular to its plane passing through centre O of the circle.



Q.5 A uniform disc has radius 0.1 m and its moment of inertia about an axis perpendicular to its plane, passing through its centre is 1.2 kg m^2. A sector of angle 30° is cut and removed from the disc. Find moment of inertia of the remaining disc, axis being same.



Q.6 Two circular discs of same mass and thickness are made from metals having different densities– d and $2d$. Which disc will have larger moment of inertia about its central axis?

Q.7 Four thin sticks, each of mass m and length l, are joined to form a square. Find moment of inertia of the square about one of its side.

Q.8 Three rods, each of mass m and length l, are joined together to form an equilateral triangle. Find the moment of inertia of the triangle about an axis passing through its centre of mass and perpendicular to the plane of the triangle.

Q.9 A thin rod has its linear mass density changing with distance from one of its end (at $x = 0$) as $\lambda = \lambda_0 x$. Length of the rod is L. Find its moment of inertia about a line that is perpendicular to the rod and passes through $x = 0$.

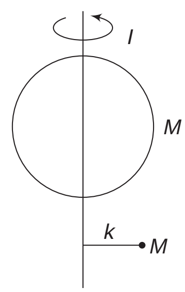
Q.10 Find percentage change in moment of inertia of a solid sphere about its diameter if its radius increases by 0.1% without change in its mass.

Q.11 Two particles of masses M and m are connected by a rod of negligible mass and length L to form a dumbbell. Find the smallest moment of inertia of this dumbbell about an axis that is perpendicular to the rod.

Q.12 A uniform sphere of radius a has concentric spherical cavity of radius b. Find the moment of inertia of this object of mass M about its diameter.

2.3 Radius of Gyration

Radius of gyration (K) of a body about an axis is the effective distance from the axis where the whole mass can be assumed to be concentrated so that the moment of inertia remains same.



A body of mass M has MI equal to I about certain axis. A point mass M kept at a distance K from the axis also has the same MI. K is known as radius of gyration of the body about the axis.

$$MK^2 = I \quad \Rightarrow \quad K = \sqrt{\frac{I}{M}} \qquad ...(5)$$

Example 15 Find radius of gyration of a uniform solid sphere about its diameter.

Solution

Concepts

$$MK^2 = I$$

$$MK^2 = \frac{2}{5}MR^2 \quad \Rightarrow \quad K = \sqrt{\frac{2}{5}} \cdot R$$

A higher value of K implies that the mass is effectively at a larger distance from the axis.

YOUR TURN

Q.13 Find the ratio of radius of gyration of a solid cylinder and a cylindrical shell about the central axis, parallel to length. Both have same radii.

3. NEWTON'S SECOND LAW FOR ROTATION

A torque can cause rotation and produce angular acceleration in a body. Angular acceleration (α) produced depends on moment of inertia (I) of the body.

Consider a rigid body, which is free to rotate about a fixed axis (say z-axis). τ_z is net external torque acting on the body. It produces an angular acceleration α in the body, which is given by

$$\tau_z = I_z \cdot \alpha \qquad ...(6)$$

where I_z is the moment of inertia of the body about z-axis.

We will use equation (6) about an axis

(i) that is fixed

(ii) that passes through COM of the body, even if COM is accelerated.

There are other axes about which equation (6) can be used directly, without thinking about the torques due to pseudo forces. But having said this, there is usually nothing to be gained by picking such axes. We will better avoid picking any such axis, which is neither stationary nor passes through COM.

COM has this remarkable property that rotational motion about it is independent of translational motion of the body. Equation (6) can be used about any axis passing through

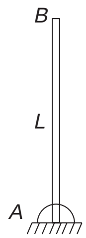
COM of the body, whatever be the state of motion of the COM itself.

Example 16 *Hinged rod released from vertical position*

A rod of mass M and length L is hinged at its lower end and held vertical. It is given a gentle push and released. The rod rotates without friction about a horizontal axis at the hinged end.

(i) Find its angular acceleration when the rod rotates through an angle θ.

(ii) Use integration to find angular speed of the rod when it becomes horizontal.



Solution

Concepts

(i) Rotation axis is a horizontal line passing through A and perpendicular to the plane of the figure.

(ii) The rod rotates about this fixed axis. We can use $\tau = I\alpha$ about this axis.

(iii) MI of the rod about rotation axis is $I = \dfrac{ML^2}{3}$

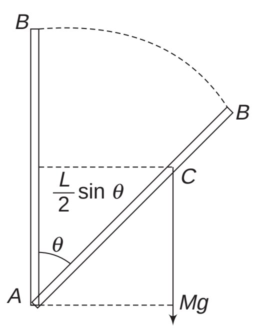
(iv) Only Mg has torque about rotation axis.

(v) $\dfrac{d\omega}{dt} = \alpha$

$$\Rightarrow \quad \frac{d\omega}{d\theta}\cdot\frac{d\theta}{dt} = \alpha \quad \Rightarrow \quad \omega\frac{d\omega}{d\theta} = \alpha$$

$$\Rightarrow \quad \omega d\omega = \alpha d\theta$$

Integrating this will give ω at any position.



(i) When the rod rotates through angle θ, torque on it is

$$\tau = Mg\frac{L}{2}\sin\theta$$

$\left[\dfrac{L}{2}\right.$ = distance of line of Mg from the axis through $A\Big]$

Hinge force has no torque about A.

Using $\quad I\alpha = \tau$

$$\frac{1}{3}ML^2\cdot\alpha = Mg\frac{L}{2}\sin\theta \quad \Rightarrow \quad \alpha = \frac{3}{2}\frac{g}{L}\sin\theta$$

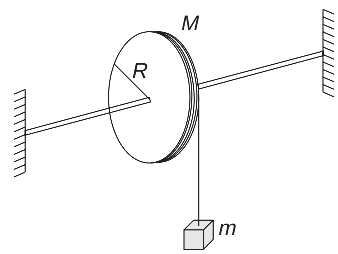
(ii) α changes with θ.

$$\omega d\omega = \alpha d\theta \quad \Rightarrow \quad \int_0^\omega \omega d\omega = \frac{3}{2}\frac{g}{L}\int_0^{90^\circ}\sin\theta d\theta$$

$$\Rightarrow \quad \frac{\omega^2}{2} = \frac{3}{2}\frac{g}{L}[-\cos\theta]_0^{90^\circ}$$

$$\Rightarrow \quad \omega = \sqrt{\frac{3g}{L}}$$

Example 17 A pulley is in the shape of a uniform disc of radius R. Its mass is M. It can rotate freely about its fixed horizontal axle. A light thread is tightly wound on it. Free end of the thread is connected to a block of mass m. System is released. Find acceleration of the block and angular acceleration of the pulley. Also find tension in the string.



Solution

Concepts

(i) If a is acceleration of the block and α is angular acceleration of the pulley, then $a = R\alpha$.

(ii) We will write equation of translational motion for the block using $F = ma$

(iii) We will write equation for rotational motion of the pulley about its fixed axis using $\tau = I\alpha$.

(iv) Solving the equations will give the desired results.

$$a = R\alpha \qquad \text{...(i)}$$

For the block: $mg - T = ma$ ...(ii)

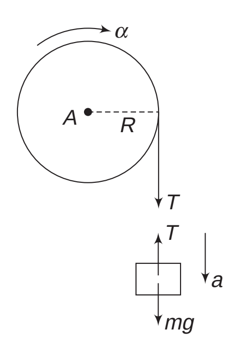
For rotation of pulley (about A)

$I\alpha = \tau$ [Weight of pulley and force by axle have no torque about A]

$$\frac{MR^2}{2}\cdot\alpha = T\cdot R$$

$$\Rightarrow \quad M(R\alpha) = 2T$$

$$\Rightarrow \quad Ma = 2T \text{ [using (i)]} \qquad \text{...(iii)}$$



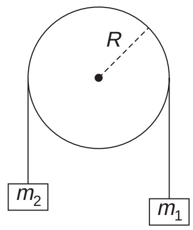
We have three equations with three unknowns: T, a and α. Eliminating T between (ii) and (iii) gives

$$a = \frac{mg}{m + \dfrac{M}{2}}$$

$$\therefore \qquad \alpha = \frac{a}{R} = \frac{mg}{R\left[m + \frac{M}{2}\right]} \quad \text{and} \quad T = \frac{Mmg}{2m + M}$$

Example 18 *Atwood's machine with a massive pulley*

A light thread, carrying two blocks of masses m_1 and m_2 ($< m_1$) at its end, passes over a pulley of radius R. The pulley can rotate about its axle without friction and has a moment of inertia I about its axle.



Thread does not slip on the pulley. Find acceleration of the blocks and tension in the threads on two sides.

Solution

Concepts

(i) Friction between the thread and the pulley is large, which does not allow the thread to slip.

(ii) Pulley rotates and transfers thread from left side to right side.

(iii) Tension on two sides of the pulley produces a torque on the pulley about O. Torque on the pulley is $\tau = T_1R - T_2R$ (See figure).

(iv) If pulley is massless, net torque on it must always be zero. Even a small torque will imply infinite angular acceleration ($\because \; I = 0$).

(v) For a massive pulley, we cannot have angular acceleration without torque. This implies that $T_1 > T_2$; otherwise, the pulley cannot accelerate.

Tension on the two sides must be different. Friction between the thread and pulley cause this difference in tension.

(vi) For no slipping, $a = R\alpha$, where a is the acceleration of the blocks and α is the angular acceleration of the pulley.

(vii) We will write $\tau = I\alpha$ for pulley and $F = ma$ for the blocks.

a = acceleration of blocks with m_1 going down and m_2 moving up

α = angular acceleration of the pulley

$a = R\alpha$ for no slipping ...(i)

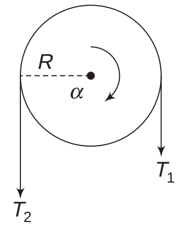
For rotation of pulley:

$$T_1R - T_2R = I\alpha$$

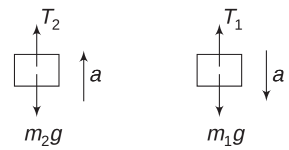
$$\Rightarrow \qquad T_1 - T_2 = \frac{I}{R}\alpha$$

$$\Rightarrow \qquad T_1 - T_2 = \frac{I}{R^2}a \qquad \text{...(ii)}$$

$$[\because \; R\alpha = a]$$



Tensions on two sides of the string are different. Friction causes this difference



For translational motion of blocks:

$$m_1g - T_1 = m_1a \qquad \text{...(iii)}$$

$$T_2 - m_2g = m_2a \qquad \text{...(iv)}$$

We have four unknowns (α, a, T_1 and T_2) and there are four equations as well. Adding (ii), (iii) and (iv)

$$m_1g - m_2g = \left(\frac{I}{R^2} + m_1 + m_2\right)a$$

$$\Rightarrow \qquad a = \frac{(m_1 - m_2)g}{m_1 + m_2 + \frac{I}{R^2}} \qquad \text{...(v)}$$

Substituting in (iii) and (iv) gives T_1 and T_2

$$T_1 = m_1g\left[\frac{2m_2 + \frac{I}{R^2}}{m_1 + m_2 + \frac{I}{R^2}}\right] \qquad \text{...(vi)}$$

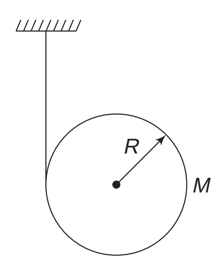
$$T_2 = m_2g\left[\frac{2m_1 + \frac{I}{R^2}}{m_1 + m_2 + \frac{I}{R^2}}\right]$$

Note: If pulley is massless, then $I = 0$ and equations (v) and (vi) give

$$a = \frac{(m_1 - m_2)g}{(m_1 + m_2)} \quad \text{and} \quad T = \frac{2m_1m_2g}{m_1 + m_2}$$

Example 19 *Combined rotation and translation of a body*

A light thread is wound tightly around a disc of mass M and radius R. Free end of the thread is fixed to ceiling of a room and the disc is released. Find the acceleration of the disc.



Solution

Concepts

(i) 'Acceleration of the disc' implies acceleration of the COM of the disc.

(ii) The disc rotates and thread unwinds. If the disc rotates by θ, a length $x = R\theta$ unwinds. The COM falls by x. Thus, displacement (x) of COM and angular displacement (θ) of the disc are related as $x = R\theta$.

$\Rightarrow \quad a = R\alpha,$

where $\quad a$ = acceleration of COM

α = angular acceleration.

(iii) Though COM is accelerated, we can use $\tau = I\alpha$ about an axis through COM.

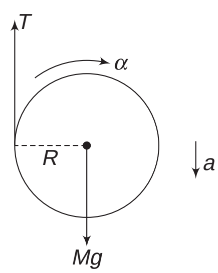
(iv) We will write $F_{ext} = Ma_{CM}$ and $\tau = I\alpha$ (about an axis through COM perpendicular to the figure) for the disc.

$$a = R\alpha \qquad \text{...(i)}$$

where $\quad a$ = acceleration of COM

α = angular acceleration about COM

Motion of the disc is a combination of translation of its COM and rotation about it.



For translatory motion, point of application of forces are not important.

$$F_{ext} = Ma_{CM}$$

$$Mg - T = Ma \qquad \text{...(ii)}$$

We can always use $\tau = I\alpha$ about an axis through COM. Mg has no torque about this axis. Only tension (T) has a torque.

$$\therefore \quad I\alpha = \tau$$

$$\frac{1}{2}MR^2 \cdot \alpha = T.R \qquad \left[I = \frac{1}{2}MR^2 \text{ for a disc about central axis}\right]$$

$$\Rightarrow \quad \frac{1}{2}Ma = T \qquad \text{...(iii)}$$

$$[\because \quad R\alpha = a]$$

From (ii) and (iii) $a = \dfrac{2g}{3}$

Note: We have considered translation of COM (Without worrying about the fact that there is rotation also) and we have considered a rotation about COM (not worrying about the fact that the COM is accelerated). Superposition of these two independent motions is the resultant motion of the body. The equation $a = R\alpha$ is the constraint relation, which relates the rotational and translational motion.

Example 20 *Hinged rod*

A uniform rod of mass M and length L is hinged at its end. It can rotate freely about this end in a vertical plane. The rod is released from horizontal position. Immediately after the rod is released, find

(i) its angular acceleration

(ii) force applied by the hinge on it

Solution

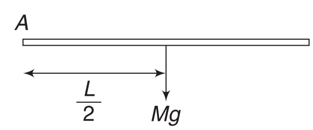
Concepts

(i) Immediately after the rod is released, it is still horizontal and has zero speed.

(ii) Using $\tau = I\alpha$ about the rotation axis is easy. Hinge force does not have any torque about the rotation axis. Only Mg has torque.

(iii) Once α is known, acceleration of COM of the rod can be calculated. The COM will go in a circle about the hinged end. Immediatedly after release, speed of COM is zero. It has no radial acceleration. It only has tangential acceleration.

(iv) $F_{ext} = Ma_{CM}$ will help us know the hinge force.



(i) Torque on rod about A is

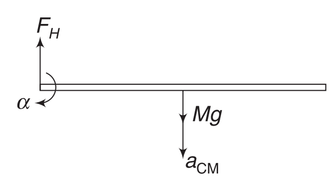
$$\tau = Mg\frac{L}{2}$$

Angular acceleration (α) is given by

$$I\alpha = \tau$$

$$\frac{ML^2}{3} \cdot \alpha = Mg\frac{L}{2} \quad \Rightarrow \quad \alpha = \frac{3g}{2L} \qquad \text{...(i)}$$

(ii) COM moves in a circle of radius $\frac{L}{2}$. Its initial speed is zero.



Radial acceleration of COM is zero.

It has a tangential acceleration (directed vertically downward)

$$a_t = \alpha\frac{L}{2} = \frac{3g}{4}$$

$$\therefore \quad a_{CM} = \frac{3g}{4} \; (\downarrow)$$

If F_H is the hinge force in vertically upward direction

$$Mg - F_H = Ma_{CM} \quad \Rightarrow \quad Mg - F_H = \frac{3Mg}{4}$$

$$\Rightarrow \qquad F_H = \frac{Mg}{4}$$

Hinge force has no horizontal component, as COM has no horizontal acceleration.

Note:

(i) Acceleration of COM depends on F_{ext} and has got nothing to do with point of application of the force

(ii) If we consider rotation about COM, we can write

$$I_{CM} \cdot \alpha = F_H \cdot \frac{L}{2}$$

$$\Rightarrow \qquad \frac{ML^2}{12} \cdot \alpha = \frac{Mg}{4} \cdot \frac{L}{2} \qquad \Rightarrow \quad \alpha = \frac{3g}{2L}$$

This is same as (i).

In Short:

(i) Use $\tau = I\alpha$ about a fixed axis or about an axis that passes through COM of the body.

(ii) For hinged objects, using $\tau = I\alpha$ about the hinge is easiest, as it does not involve writing torque due to hinge force.

(iii) For a body that is rotating as well as translating, use

$$\tau = I\alpha \text{ about an axis through COM and}$$

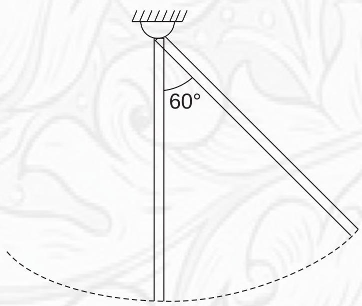
$$F_{ext} = Ma_{CM}$$

While writing the last equation, one need not worry about the point of application of the forces.

YOUR TURN

Q.14 A sphere of mass $m = 2$ kg and radius $r = 0.5$ m is free to rotate about its vertical diameter. A horizontal tangential force $F = 10$ N acts on it for $t = 3$ s. Find the angular speed gained by the sphere.

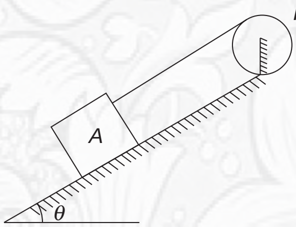
Q.15 A uniform rod of mass m and length l is swinging about its one end like a pendulum. In its extreme position, it makes an angle of 60° with vertical. Find its angular acceleration at



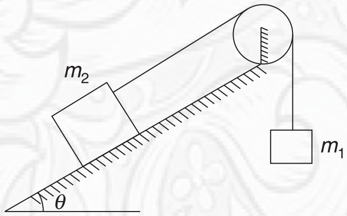
(i) Extreme position

(ii) Vertical position.

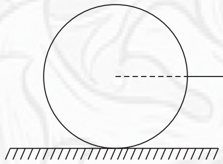
Q.16 A disc-shaped pulley (P) has mass M and radius R. It is fixed on top of a smooth incline and can rotate freely about its axle. A thread is tightly wound on the pulley. The free end of the thread is connected to a block of mass m kept on the incline. System is released. Find the acceleration of the block. Inclination angle of the incline is θ.



Q.17 In the arrangement of Atwood machine shown in figure, the pulley has radius R and moment of inertia I about its rotation axis. There is no friction between the block of mass m_2 and the incline surface. On releasing the system, m_1 goes down. Find its acceleration.



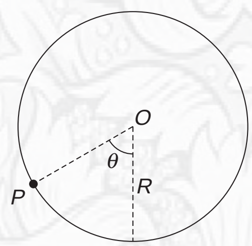
Q.18 A cylinder of mass M and radius R is kept on a smooth table. It is pulled by a horizontal force F that acts along a line passing through centre of the cylinder.



(i) Find acceleration of centre of the cylinder.

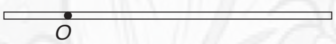
(ii) Find angular acceleration of the cylinder

Q.19 A uniform disc of mass M and radius R is free to rotate about a horizontal axis through its centre (O). A particle (P) of mass m is stuck at the circumference at angular position θ as shown. The system is released from this position.



Find the acceleration of the particle, immediately after release.

Q.20 A uniform rod of length $4l$ and mass M is free to rotate about a horizontal axis passing through a point (O) at a distance l from its one end. The rod is released from horizontal position. Immediately after release,

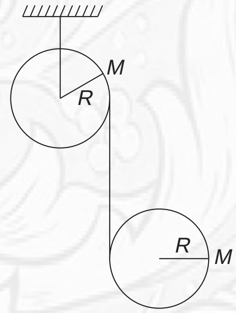


(i) Find angular acceleration of the rod

(ii) Find acceleration of COM of the rod.

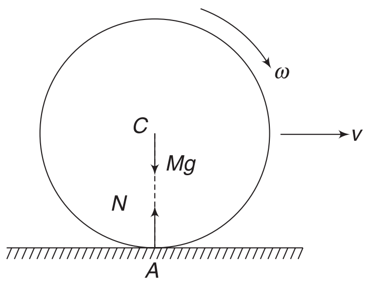
(iii) Find hinge force on the rod.

Q.21 Two identical disc, each of mass M, are connected by a massless string wrapped around them as shown. The upper disc is mounted on a frictionless horizontal axis about which it can rotate. The other disc is free to translate as well as rotate. Assume that the axis of lower disc remains horizontal, find its acceleration after the system is released.



4. ROLE OF FRICTION IN ROLLING

Think of a perfectly rigid ball rolled on a flat horizontal surface. The ball is 'rigid' implies that it touches the horizontal surface at a single point (A). Normal force (N) on the ball and its weight (Mg) have their line of action along line AC [C is the centre of the ball]. These forces are vertical, they cannot produce any horizontal acceleration. They do not have any torque [about C or A]. They cannot change angular speed (ω).



Friction force (f) will assume a direction depending on the direction of motion of contact point A. If $v = \omega R$, then the ball rolls without sliding and the contact point is instantaneously at rest. Friction will adjust itself to zero so that no sliding condition prevails. The ball will keep rolling forever! Then why does a rolling ball stops? keep pondering. We will return to this in a while.

If the ball is rolled with $v > \omega R$, then the contact point A has a velocity in forward direction. Friction does not like rubbing. It immediately takes a direction opposite to the direction of motion of contact point A. Friction causes a retardation and decreases v. It also produces a torque about C and provides an angular acceleration (α). This increases ω. Therefore, after some time, velocity (v_0) will become equal to $\omega_0 R$ and sliding will stop. Now, friction will adjust itself to zero and the ball keeps rolling forever!

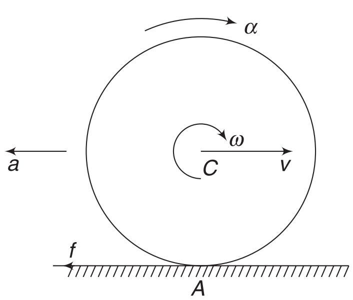
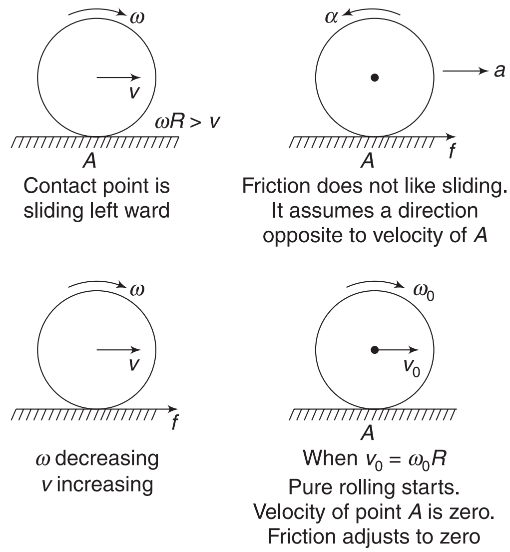


Figure below shows another situation where friction converts an imperfect rolling into perfect rolling motion. Fact is that, if it is left to friction, any motion will eventually get transformed to a motion where no sliding occurs.

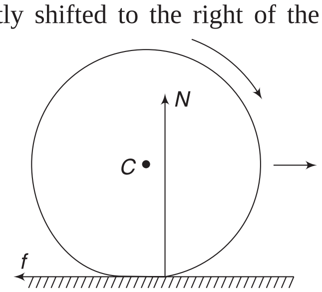


During the period, the ball rolls with sliding, friction force is kinetic in nature. Heat is produced (in rubbing) and mechanical energy is lost.

Example 21 *Why does a rolling ball stop*?

If a perfectly rigid ball is rolled on a horizontal floor with $v = \omega R$, our theoretical discussion indicates that it will keep rolling forever. In practice, no ball rolls forever. Why?

Solution In practice, no object is perfectly rigid. Objects deform under action of forces. A ball deforms under action of its weight and normal force by the floor. This results in multiple contact points of the ball with floor. The shown figure is a bit of exaggeration if you talk of a steel ball (but quite OK for a soft rubber ball). The normal force on the ball has a line of action slightly shifted to the right of the centre (C). This produces a torque about C. The torque tries to reduce the angular speed ω. Friction immediately takes leftward direction so as to reduce the speed (v) also, and to maintain the no sliding condition ($v = \omega R$). Thus, the ball will slow down and eventually stop.



Normal force passes to the right of the centre (C).

Example 22 A solid ball is imparted a velocity v (and no spin) and released on a rough horizontal surface. Find its velocity when pure rolling begins.

Solution

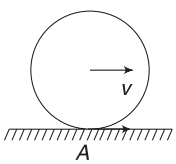
Concepts

(i) Every point (including the contact point) of the ball has a forward velociy v in the beginning. Friction will act on the ball in backward direction.

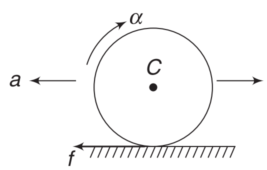
Friction decreases v and increases ω. Soon, speed will become equal to ωR and pure rolling will start

(ii) We will write $a = \frac{F}{M}$ and $\alpha = \frac{\tau}{I}$ to write the linear and angular accelerations of the ball.

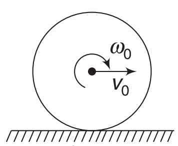
(iii) Using $v = u + at$ and $\omega = \omega_0 + \alpha t$, we will write v and ω after time t.



Contact point slides in forward direction



Friction produces linear retardation and angular acceleration. v is decreasing and ω is increasing



time t

$v_0 = \omega_0 R$

Friction will now adjust to zero.

Let f be friction force on the ball.

Linear retardation of the ball is $a = \frac{f}{M}$

Torque due to friction about C is $\tau = fR$

Moment of inertia about an axis through C is

$$I = \frac{2}{5} MR^2$$

Angular acceleration (clockwise) is

$$\alpha = \frac{\tau}{I} = \frac{5f}{2MR}$$

Velocity of the ball at time t is

$$v_0 = v - at = v - \frac{f}{M}t \quad \text{...(i)}$$

Angular velocity of the ball at time t is

$$\omega_0 = 0 + \alpha t = \frac{5ft}{2MR} \quad \text{...(ii)}$$

When pure rolling begins, $v_0 = \omega_0 R$

$$\Rightarrow \quad v - \frac{f}{M}t = \frac{5ft}{2MR}.R$$

$$\Rightarrow \quad v = \frac{7f}{2M}\cdot t$$

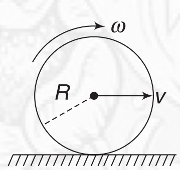
$$\Rightarrow \quad \frac{2v}{7} = \frac{ft}{M}$$

Putting this in (i) gives: $v_0 = v - \frac{2v}{7} = \frac{5v}{7}$.

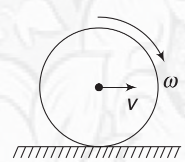
This is the speed of the ball when rolling begins.

Your Turn

Q.22 A disc is rolled on a rough horizontal surface. At an instant, its angular velocity is found to be increasing. Which is larger–velocity (v) of the centre or ωR?



Q.23 A disc is rolled on a horizontal surface with its centre moving at a velocity (v) and the disc spinning about its centre at an angular velocity $\omega = \frac{2v}{R}$. Find the velocity of the disc when slipping ceases.



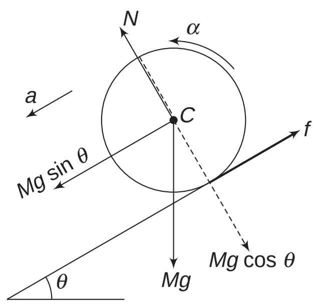
4.1 Rolling on an Incline

A round object of mass M and radius R is released on an inclined plane, set at an angle θ to the horizontal. Component of Mg, down the incline (= $Mg \sin\theta$) will try to accelerate the body. To ensure that there is no slipping, friction assumes an upward direction so as to spin the body about its COM. If friction is sufficiently strong, it will ensure that v is always equal to ωR. In fact, this will be possible if

$$a = R\alpha \quad \text{...(i)}$$

where a = acceleration of the centre and

α = angular acceleration of the body



Equation for translational motion can be written as

$$Mg\sin\theta - f = Ma \qquad ...(ii)$$

For rotation of the body about its COM, we can write

$$\tau = I\alpha \qquad [Mg \text{ and } N \text{ have no torque}]$$

$$fR = I\alpha$$

$$\Rightarrow \quad f = \frac{I}{R^2}(R\alpha) \Rightarrow f = \frac{I}{R^2}a \qquad ...(iii)$$

Adding (ii) and (iii) gives

$$a = \frac{Mg\sin\theta}{M + \dfrac{I}{R^2}} \qquad ...(7)$$

If K is radius of gyration of the body about its rotation axis through COM, then $I = MK^2$.

$$a = \frac{g\sin\theta}{1 + \dfrac{K^2}{R^2}} \qquad ...(8)$$

Writing the expression for acceleration in above form shows you that it does not depend on mass of the body. It depends on K, i.e., shape of the body.

Substituting value of a in (iii) gives

$$f = \frac{I}{R^2}a = \frac{\dfrac{I}{R^2}Mg\sin\theta}{M + \dfrac{I}{R^2}}$$

$$\Rightarrow \quad f = \frac{Mg\sin\theta}{\dfrac{MR^2}{I} + 1} = \frac{Mg\sin\theta}{\dfrac{R^2}{K^2} + 1} \qquad ...(9)$$

If friction coefficient between the incline and the object is small and friction cannot assume a value given by equation (9), the body will begin to slip.

The friction force is static friction when there is no slipping. This friction does not produce heat. Mechanical energy of the rolling body does not get dissipated.

Example 23 *Sphere is fastest, ring slowest*

A solid sphere, a cylinder, a disc, a hollow sphere and a ring are released on an incline from same height. Which body will be first to reach the bottom? Which is last?

Solution

Concepts

(i) $a = \dfrac{g\sin\theta}{1 + \dfrac{K^2}{R^2}} = \dfrac{Mg\sin\theta}{M + \dfrac{I}{R^2}}$

(ii) Acceleration does not depend on mass, it depends on shape of the body.

For solid sphere: $MK^2 = \dfrac{2}{5}MR^2 \Rightarrow \dfrac{K^2}{R^2} = \dfrac{2}{5}$

$$\therefore \quad a_{sph} = \frac{g\sin\theta}{1 + \dfrac{2}{5}} = \frac{5g\sin\theta}{7}$$

For shell: $MK^2 = \dfrac{2}{3}MR^2 \Rightarrow \dfrac{K^2}{R^2} = \dfrac{2}{3}$

$$\therefore \quad a_{shell} = \frac{g\sin\theta}{1 + \dfrac{2}{3}} = \frac{3g\sin\theta}{5}$$

For cylinder or disc: $MK^2 = \dfrac{1}{2}MR^2 \Rightarrow \dfrac{K^2}{R^2} = \dfrac{1}{2}$

$$\therefore \quad a_{cyl} = a_{disc} = \frac{g\sin\theta}{1 + \dfrac{1}{2}} = \frac{2g\sin\theta}{3}$$

For ring: $MK^2 = MR^2 \Rightarrow \dfrac{K^2}{R^2} = 1$

$$\therefore \quad a_{ring} = \frac{g\sin\theta}{1 + 1} = \frac{g\sin\theta}{2}$$

$$\therefore \quad a_{sph} > a_{disc} = a_{cyl} > a_{shell} > a_{ring}.$$

Solid sphere is fastest and the ring is slowest.

Note: Acceleration does not depend on mass or radius of the body.

Example 24 *Minimum coefficient of friction for pure rolling*

A solid uniform cylinder is released on a rough incline. Find the minimum coefficient of friction between the cylinder and the incline so that it does not slip.

Solution

Concepts

For pure rolling, friction force must be equal to the value given by equation (9).

For pure rolling, $f = \dfrac{Mg\sin\theta}{\dfrac{MR^2}{I} + 1}$

$$I = \frac{1}{2}MR^2$$

$$\therefore \qquad f = \frac{Mg\sin\theta}{3}$$

Normal force on the cylinder is $N = Mg\cos\theta$.

$$f \le \mu N$$

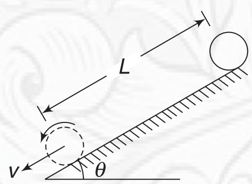
$$\Rightarrow \qquad \frac{Mg\sin\theta}{3} \le \mu Mg\cos\theta \quad \Rightarrow \quad \frac{\tan\theta}{3} \le \mu$$

$$\therefore \qquad \mu_{\min} = \frac{1}{3}\tan\theta$$

Your Turn

Q.24(a) A disc of mass M and radius R and a ring of same mass and same radius rolls down an incline. They start from the same height at same time. Which will reach the bottom of the incline travelling at higher speed? Assume that they do not slip.

(b) A small solid ball is released from top of an incline having inclination angle θ. It rolls down without sliding. It reaches the bottom of the incline after travelling a distance L. Find its speed at bottom. What is its angular speed if its radius is r?



Q.25 Find minimum coefficient of friction between a ring and a 45° incline, so that the ring can roll without sliding when released on the incline.

Q.26 A ball is released on an incline, which is smooth. Find the angular speed of the ball after it has travelled a distance S.

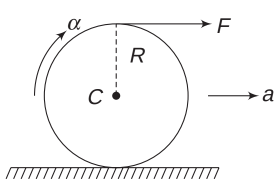
Q.27 A ball of mass M and radius R is released on a rough incline plane of inclination θ. Friction is not sufficient to prevent slipping. Coefficient of friction is μ. Find

(a) acceleration of the centre of the ball

(b) angular acceleration of the ball about its centre

4.2 Rolling with Other Forces, Apart from Friction, Producing Torque

A round object is kept on a horizontal rough surface with a light thread wound on it. The free end of the thread is pulled horizontally with a force F, as shown. This force produces linear acceleration as well as angular acceleration. Friction adjusts itself so as to ensure pure rolling (i.e., $a = R\alpha$). Slipping will occur only if friction is small and cannot assume a value that is needed to ensure $a = R\alpha$.



For a moment, assume that friction is absent. Acceleration (a) and angular acceleration (α) of the body will be given by

$$a = \frac{F}{M} \quad \text{and} \quad \alpha = \frac{F \cdot R}{MK^2} \quad \Rightarrow \quad R\alpha = \frac{FR^2}{MK^2} = \frac{R^2}{K^2}a.$$

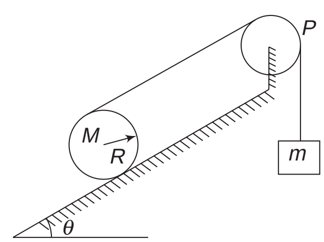
where K is radius of gyration about the rotation axis through C.

If the round object is a ring, then $MK^2 = MR^2$ and $R\alpha = \dfrac{F}{M} = a$. Any value of force F, on a ring, will ensure pure rolling in absence of friction. Friction will adjust itself to zero!

For a body having $K < R$ [all the round objects that we have seen, apart from ring, have $K < R$] $R\alpha > a$ and the contact point will have a tendency of backward slipping. Friction will act so as to prevent this slipping. It will take a forward direction. This will increase a and reduce $R\alpha$ to ensure $a = R\alpha$.

If you still find it difficult to figure out the direction of friction, do not worry. If a problem says that the body rolls without slipping, assume friction to be in any direction you want. Imposing the condition $a = R\alpha$, will ensure that mathematics corrects the direction of friction if it is wrongly chosen. On solving your equations, you may get a negative value for friction, implying the direction chosen to be wrong.

Example 25 A string is wrapped around a uniform cylinder of mass M, which rests on a fixed incline plane of inclination angle θ. The string passes over massless pulley (P) and is connected to a block of mass m, as shown. Assume that the cylinder rolls without slipping on the plane and that the string is parallel to the plane.



(a) Find acceleration of mass m.

(b) Find the ratio $\dfrac{M}{m}$ for which the cylinder accelerates down the plane.

Solution

Concepts

(i) If we assume that the cylinder rolls down, it will wrap the thread and the block will move up.

Velocity of top-point of the cylinder will be twice that of its centre ($v + \omega R = 2v$). Thus, velocity of the block will always be twice that of the centre of the cylinder. It means acceleration of the block is twice that of the centre of the cylinder.

(ii) Friction on cylinder will adjust itself so as to ensure pure rolling. Without thinking much, we will assume this friction to be in any direction, say up the plane.

(iii) We will write equation for translation and rotation of the cylinder and for translation of the block.

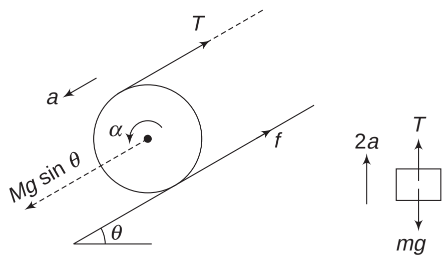


Figure shows forces on the cylinder along the incline. Using $F = Ma_{CM}$ for cylinder:

$$Mg\sin\theta - T - f = Ma \qquad \text{...(i)}$$

Using $\tau = I\alpha$ about COM:

$$fR - TR = I\alpha \quad \Rightarrow \quad fR - TR = \frac{1}{2}MR^2 \cdot \alpha$$

$$\Rightarrow \qquad f - T = \frac{1}{2}M(R\alpha) \quad \Rightarrow \quad f - T = \frac{1}{2}Ma \qquad \text{...(ii)}$$

[For pure rolling, $R\alpha = a$]

Acceleration of the block is twice that of the cylinder.

$\therefore$ For block

$$T - mg = m(2a) \qquad \text{...(iii)}$$

(i) + (ii) + 2 × (iii) gives

$$Mg\sin\theta - 2mg = \left(\frac{3}{2}M + 4m\right)a$$

$$\therefore \qquad a = \frac{(M\sin\theta - 2m)g}{\left(\frac{3}{2}M + 4m\right)a}$$

(a) Acceleration of the block $= 2a = \dfrac{2(M\sin\theta - 2m)g}{\frac{3}{2}M + 4m}$

(b) For cylinder to roll down, 'a' must be positive.

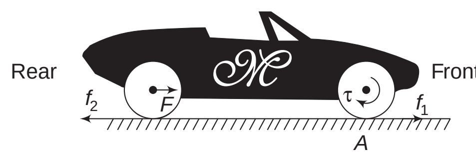
$$\therefore \quad M\sin\theta > 2m \quad \Rightarrow \quad \frac{M}{m} > \frac{2}{\sin\theta}$$

If $\theta \to 0$, this gives $\dfrac{M}{m} \to \infty$, which makes sense.

Note: You can calculate friction (f) from the above equation. It turns out to be a positive number. This means we have chosen the correct direction of the friction force.

4.3 Role of Friction in Motion of a Car

Consider a front-wheel drive car. Once you start the car, put it in gear and release the clutch, the engine imparts torque (τ) to the front axle. Rear wheel receives no power. If there is no friction between the front wheel and the road, the front wheel will spin but the car will not move! There is no external force. If front wheels are placed on a slippery surface, the car will really not move, however hard you may press the accelerator pedal.



When friction is present, it will try to prevent slipping of wheel at A. It will assume a forward direction (f_1) and will accelerate the car. If you do not press the accelerator a lot (i.e., you do not give a large angular acceleration α to the front wheel), friction will ensure that the wheels do not slip and acceleration (a) of the car is $a = R\alpha$ (R = radius of the wheel).

Due to f_1, car has a tendency to move forward. The chassis transfers a force F on the rear axle. This force will try to accelerate (and not rotate) the rear wheel. To prevent the rear wheels from slipping, friction (f_2) will act on it in backward direction.

The car will fail to move if f_1 is not greater than f_2. Engine, which is heavy, is put at the front so that normal force on front wheel is larger. This ensures that $f_1 > f_2$.

In heavy trucks or SUVs, we need more traction (forward force). In such automobiles, both axles are powered. When engine rotates both of them, friction on both front and rear wheels is in forward direction.

When you apply brakes (on all wheels), the brake shoes clamp the rotating wheels to the body of the car and rotation slows down. To maintain pure rolling (i.e., no sliding), friction takes backward direction on all wheels and decreases the velocity (v) and tries to maintain $\omega R = v$.

It is friction which accelerates, it is friction which retards a moving car!

In Short:

(i) A body rolls without sliding on a surface if its contact point does not have a velocity relative to the surface. Friction tries its best to maintain rolling without sliding.

(ii) If a body rolls without sliding (pure rolling), friction acting on it is static in nature.

(iii) If a body rolls with sliding, friction is kinetic.

(iv) On a horizontal surface, friction converts all motion into no sliding if it is the only force.

(v) Acceleration of a round object performing pure rolling on an incline is given by

$$a = \frac{g\sin\theta}{1 + \dfrac{K^2}{R^2}}$$

where K is radius of gyration about rotation axis through centre.

Friction force acting on the body is given by

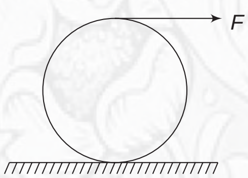
$$f = \frac{Mg\sin\theta}{1 + \dfrac{R^2}{K^2}}$$

(vi) In pure rolling, static friction does not dissipate any mechanical energy.

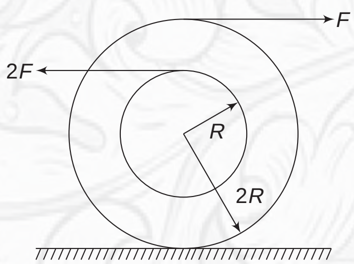
(vii) When a body is rolling without sliding, it is fine to choose any direction for friction. Solving the equations with condition $a = R\alpha$ results in correct magnitude and direction of friction.

YOUR TURN

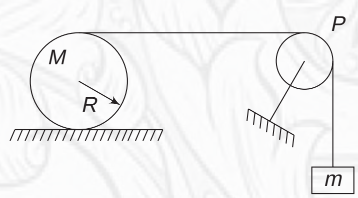
Q.28 A disc of mass M stands on a horizontal surface. A light thread is wrapped on it and its free end is pulled horizontally with a force F, as shown. The disc begins to roll without slipping. Find its acceleration.



Q.29 A spool has mass M. It has two light threads wound on it on circles of radii $2R$ and R. Free ends of the two threads are pulled horizontally with forces F and $2F$, as shown. The moment of inertia of the spool about the central axis perpendicular to the plane of the figure is $2MR^2$. Find the direction and magnitude of friction force on spool if it does not slip.



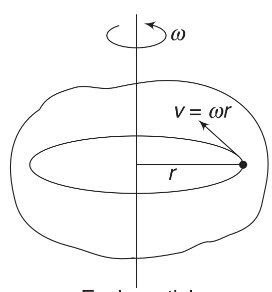
Q.30 In the figure shown, a string is wrapped around a cylinder of mass M and radius R. The string passes over a smooth pulley (P) and its free end holds a mass m. Find acceleration of m if the cylinder does not slip



5. KINETIC ENERGY OF ROTATION

A rigid body is an aggregate of particles and its kinetic energy is just equal to the sum total of kinetic energies of the individual particles.

Consider a rigid body in pure rotation about a fixed axis. Its angular speed is ω. A particle at a distance r from the rotation axis has speed $v = \omega r$. Kinetic energy of the rotating body is



Each particle has same ω.

$$k = \sum \frac{1}{2} m_i v_i^2$$

$$= \frac{1}{2}\sum m_i (r_i\omega)^2$$

$$= \frac{1}{2}\left[\sum m_i r_i^2\right]\omega^2$$

or, $$k = \frac{1}{2} I\omega^2 \qquad \text{...(10)}$$

Example 26 Find the kinetic energy of Earth related to its rotation about its axis. Given: Mass of Earth = 6×10^{24} kg, Radius of Earth = 6,400 km. Assume that the Earth is a uniform sphere.

Solution

Concepts

(i) Angular speed of Earth is

$$\omega = \frac{2\pi\,\text{rad}}{24\,\text{h}}$$

(ii) $$k = \frac{1}{2} I\omega^2$$

Angular speed $\omega = \dfrac{2\pi \text{rad}}{24 \times 60 \times 60\,\text{s}} = 7.3 \times 10^{-5}\,\text{rad s}^{-1}$

MI of Earth about its axis

$$I = \frac{2}{5}MR^2 = \frac{2}{5} \times 6 \times 10^{24} \times (6400 \times 10^3)^2$$

$$= 9.83 \times 10^{37}\ \text{kg m}^2$$

$$\therefore \quad k = \frac{1}{2}I\omega^2 = \frac{1}{2} \times 9.83 \times 10^{37} \times (7.3 \times 10^{-5})^2$$

$$= 2.62 \times 10^{29}\ \text{J}$$

5.1 Kinetic Energy in Combined Rotation and Translation

Kinetic energy of any system of particles can be written as

$$k = k_{\substack{\text{wrt}\\ \text{COM}}} + \frac{1}{2}Mv_{\text{CM}}^2$$

In context of a rigid body, the first term is the energy associated with rotation of the body with respect to the COM.

$$k_{\substack{\text{wrt}\\ \text{COM}}} = \frac{1}{2}I_{\text{CM}}\omega^2$$

Here, I_{CM} is moment of inertia of the body about rotation axis through COM. We will call this energy as rotational kinetic energy.

$$k_R = \frac{1}{2}I_{\text{CM}}\omega^2$$

The second term in the above equation is kinetic energy associated with translational motion. It is written assuming the entire mass of the body to be concentrated at COM and moving with speed of COM. We will call this energy as translational kinetic energy.

$$k_T = \frac{1}{2}Mv_{\text{CM}}^2$$

Total kinetic energy of a body that is translating plus rotating is

$$k = k_R + k_T = \frac{1}{2}I_{\text{CM}} \cdot \omega^2 + \frac{1}{2}Mv_{\text{CM}}^2 \qquad \text{...(ii)}$$

It is important to note that velocities ω and v_{CM} are independent of one another and one can give any amount of rotational kinetic energy and translational kinetic energy to a body.

If one can locate the instantaneous axis of rotation of a body, its kinetic energy can be simply written as

$k = \dfrac{1}{2}I\omega^2$ where I = moment of inertia of the body about instantaneous axis of rotation.

Example 27 *Kinetic energy of a rolling ball*

A solid uniform ball of mass M and radius R is rolling without sliding. Velocity of its centre is v. Write its kinetic energy.

Solution

Concepts

(i) $k = k_R + k_T$ or else

(ii) $k = \dfrac{1}{2}I\omega^2$ where I is *MI* about instantaneous axis. We know that for a body under pure rolling, instantaneous axis passes through the contact point.

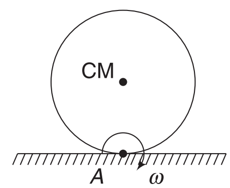
For pure rolling, $\omega = \dfrac{v}{R}$

$$\therefore \quad k = k_R + k_T = \frac{1}{2}I_{\text{CM}}\omega^2 + \frac{1}{2}Mv^2$$

$$= \frac{1}{2}\left(\frac{2}{5}MR^2\right)\left(\frac{v}{R}\right)^2 + \frac{1}{2}Mv^2 = \frac{7}{10}Mv^2$$

Alternatively: [*Refer to article 3.2.1 of previous chapter*]

The ball is under pure rotation $\left(\omega = \dfrac{v}{R}\right)$ about an axis through contact point A.



$$I_A = I_{\text{CM}} + MR^2 \qquad \text{[Parallel axis theorem]}$$

$$= \frac{2}{5}MR^2 + MR^2 = \frac{7}{5}MR^2$$

$$\therefore \quad k = \frac{1}{2}I_A\omega^2 = \frac{1}{2} \times \frac{7}{5}MR^2 \times \left(\frac{v}{R}\right)^2 = \frac{7}{10}Mv^2.$$

Example 28 A rod of mass M and length L is moving in air. At an instant, its two end points have velocities as shown. Find the *KE* of the rod.

Solution

Concepts

(i) End points have different velocities. It implies that the rod must have rotation.

(ii) We will find v_{CM} and ω and then write the *KE*

Let velocity of COM be $v_{\text{CM}}(\downarrow)$ and angular velocity of the rod be $\omega(\circlearrowright)$

$$v_A = \omega\frac{l}{2} - v_{\text{CM}}$$

$$\Rightarrow \quad v = \frac{\omega L}{2} - v_{\text{CM}} \qquad \text{...(i)}$$

$$v_B = \omega\frac{L}{2} + v_{\text{CM}}$$

$$2v = \frac{\omega L}{2} + v_{CM} \qquad \text{...(ii)}$$

Solving (i) and (ii): $\omega = \frac{3v}{L}$; $v_{CM} = \frac{v}{2}$

$\therefore \qquad k = k_R + k_T$

$$= \frac{1}{2}I_{CM}\cdot\omega^2 + \frac{1}{2}Mv_{CM}^2$$

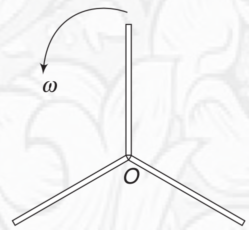
$$= \frac{1}{2}\left(\frac{ML^2}{12}\right)\left(\frac{3v}{L}\right)^2 + \frac{1}{2}M\left(\frac{v}{2}\right)^2 = \frac{1}{2}Mv^2$$

YOUR TURN

Q.31 A uniform cylinder of mass M is rolling without sliding. Its centre has velocity v. Find its kinetic energy.

Q.32 A uniform ring of mass M is rolling without sliding. Find the ratio of its translational to rotational kinetic energies.

Q.33 A fan is made of three thin sticks. Each stick has mass m and length l. The fan spins about its central axis with angular speed ω. Write its kinetic energy.



6. WORK DONE BY A TORQUE AND WORK–ENERGY THEOREM

When a torque acts on a rigid body, free to rotate about a fixed axis, it produces an angular acceleration, thereby increasing/decreasing the rotational KE.

Power delivered by torque = Rate of change of KE

$$P = \frac{dk}{dt} = \frac{d}{dt}\left(\frac{1}{2}I\omega^2\right) = I\omega\frac{d\omega}{dt} = I\,\alpha\omega$$

or, $\qquad P = \tau\cdot\omega \qquad$ ...(12)

Work done in an infinitensimal angular displacement $d\theta$ is

$$dW_\tau = P\,dt = \tau(\omega dt) = \tau d\theta$$

$\therefore$ Work done in finite angular displacement ($\Delta\theta$) is

$$W_\tau = \int_0^{\Delta\theta} \tau d\theta \qquad \text{...(13)}$$

If torque is constant; $W = \tau\Delta\theta$

When a rigid body is having a general motion, which is a combination of rotation and translation, we will consider torque about COM and its work done will be equal to change in rotational kinetic energy of the body.

The work – energy theorem divides into two parts for a rigid body in general motion:

1. Change in translational KE = Work done by forces acting on it

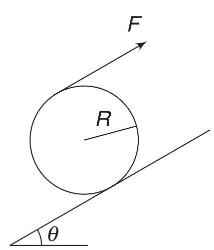
$$\Rightarrow \frac{1}{2}Mv_f^2 - \frac{1}{2}Mv_i^2 = \int_{\vec{r}_1}^{\vec{r}_2} \vec{F}\cdot\vec{dr}$$

2. Change in rotational KE = work done by torque

$$\Rightarrow \frac{1}{2}I_{CM}\,\omega_f^2 - \frac{1}{2}I_{CM}\,\omega_i^2 = \int_{\theta_1}^{\theta_2} \tau d\theta \qquad \text{...(14)}$$

When system is conservative, we can use law of conservation of energy as usual. Now the kinetic energy means sum of translational and rotational kinetic energies.

Example 29 A light tape is wrapped around a cylinder of mass M and radius R. One end of the tape is pulled along the incline so as to prevent the cylinder from falling or climbing. The incline is smooth and applied force is parallel to the incline. Find:



(a) Work done on the cylinder in time t.

(b) Length of tape unwound by the time the cylinder attains an angular speed ω.

Solution

Concepts

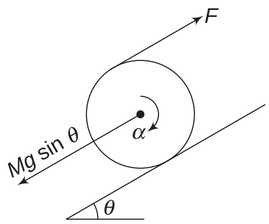
(i) Cylinder does not translate. Net force on it is zero. Work done by all forces = 0. There is no change in translational KE.

(ii) There is an unbalanced torque (about COM). This causes the cylinder to rotate. Torque performs work. This increases the rotational kinetic energy.

Net force = 0 since COM does not move.

$$F = Mg\sin\theta$$

Torque on cylinder (about its COM) is



$$\tau = FR = MgR\sin\theta$$

$\therefore$ Angular acceleration $\alpha = \dfrac{\tau}{I} = \dfrac{MgR\sin\theta}{\frac{1}{2}MR^2} = \dfrac{2g\sin\,\theta}{R}$

(a) Angular speed of the cylinder after time t is

$$\omega = 0 + \alpha t = \frac{2gt\sin\theta}{R}$$

Work–Energy theorem says:

Work done by torque = Change in rotational KE

$$W_\tau = \frac{1}{2}I\omega^2 = \frac{1}{2}\left(\frac{1}{2}MR^2\right)\frac{4g^2t^2\sin^2\theta}{R^2} = M^2g^2t^2\sin^2\theta$$

A slight alternative approach can be as follows:

Angular displacement in time t is

$$\theta = \frac{1}{2}\alpha t^2 = \frac{gt^2\sin\theta}{R}$$

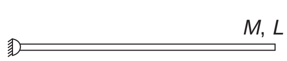
$$\therefore \quad W_\tau = \tau\theta = Mg^2t^2\sin^2\theta$$

(b) $\omega^2 = \omega_0^2 + 2\alpha\theta \Rightarrow \theta = \dfrac{\omega^2}{2\alpha} = \dfrac{\omega^2 R}{4g\sin\theta}$

Length of thread that unwinds is

$$L = 2\pi Rn = 2\pi R\frac{\theta}{2\pi} = R\theta = \frac{\omega^2R^2}{4g\sin\theta}$$

Example 30 A uniform rod of mass M and lengh L is hinged at one of its ends and can freely rotate in a vertical plane about a horizontal axis through this end. Rod is released from horizontal position. Find its angular speed at the moment it becomes vertical.



Solution

Concepts

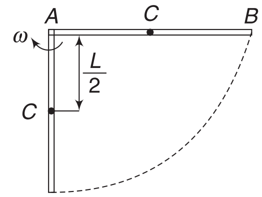
(i) The system is conservative. Only Mg does perform work. Mechanical energy of rod is conserved. It loses PE as it falls and gains KE.

(ii) Loss in $PE = Mgh$; where h = fall in height of COM.

(iii) KE of rod $= \frac{1}{2}I\omega^2$; where $I = MI$ about the fixed axis. In fixed axis rotation, $KE = \frac{1}{2}I\omega^2$

Let ω = angular speed of the rod in vertical position.

Gain in KE = Loss in PE



$$\frac{1}{2}I_A\omega^2 = Mg\frac{L}{2}$$

$$\frac{1}{2}\left(\frac{ML^2}{3}\right)\omega^2 = Mg\frac{L}{2}$$

$$\Rightarrow \quad \omega = \sqrt{\frac{3g}{L}}$$

Example 31 *Energy conservation in rolling down an incline*

A cylinder of mass M is released on an incline plane of inclination angle θ. Friction is large enough to prevent slipping and the cylinder rolls down the incline. Find its speed after it has descended through a vertical height h.

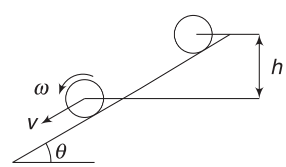
Solution

Concepts

In pure rolling, friction is static. There is no rubbing and friction does not dissipate any energy. Mechanical energy of the cylinder is conserved. Gain in its KE is equal to loss in its PE.

Note: Actually, torque due to friction (about COM) does positive work and increases the rotational KE. But “force” friction does negative work (it is opposite to displacement of the cylinder) and decreases the translational KE. Net change in KE due to friction is zero if there is no sliding.

Let velocity of cylinder be v after its centre has fallen through a height h.



$$K = k_T + k_R$$

$$= \frac{1}{2}Mv^2 + \frac{1}{2}I_{CM}\omega^2$$

$$= \frac{1}{2}Mv^2 + \frac{1}{2}\left(\frac{1}{2}MR^2\right)\omega^2$$

$$= \frac{1}{2}Mv^2 + \frac{1}{4}Mv^2 \quad [\because \quad v = \omega R]$$

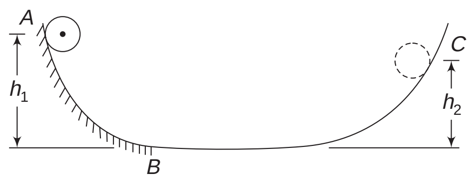
$$= \frac{3}{4}Mv^2$$

Energy Conservation:

Gain in KE = Loss in PE

$$\frac{3}{4}Mv^2 = Mgh \Rightarrow v = \sqrt{\frac{4}{3}gh}$$

Example 32 Figure shows a track in vertical plane. Part AB of the track is rough and the rest of the part is smooth.



A small cylinder is released from A and it rolls without sliding on the part AB. It climbs on the other side of the track, up to point C.

Height of A is h_1 and height of C is h_2 from the bottom of the track.

(a) Which is larger – h_1 or h_2?

(b) Express h_2 in terms of h_1

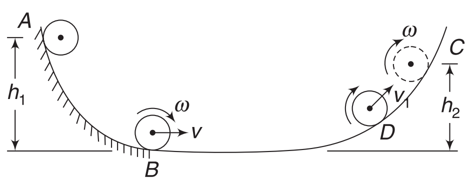
Solution

Concepts

(i) From A to B, friction is large enough to prevent slipping and the cylinder performs pure rolling.

(ii) At B, kinetic energy of the cylinder has two components–translational KE and rotational KE. Total KE is mgh_1 as energy is conserved. Friction does not dissipate energy in pure rolling.

(iii) After B, there is no torque on the cylinder about its COM. Its angular velocity cannot change now. Rotational KE does not decrease. Only translational KE decreases and the cylinder gains potential energy. When the cylinder stops at C, it is still rotating.



At B cylinder has k_R and k_T
At D, $v_1 < v$; k_T has decreased but k_R has not changed.
At C, $k_T = 0$ but k_R is still same as that at B.

(a) Mechanical energy of the cylinder is conserved throughout.

$$E_A = mgh_1$$

At C

$$E_C = mgh_2 + \text{rotational } KT$$

Since, $E_A = E_C$

$$\therefore \quad h_2 < h_1$$

(b) $k_B = mgh_1$

$$\Rightarrow \frac{3}{4}mv^2 = mgh_1$$ [Refer to Example 31. KE of a purely rolling cylinder is $\frac{3}{4}$ Mv^2]

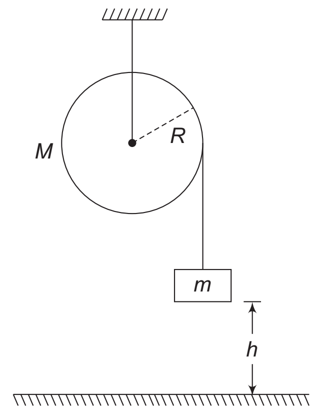
$$\Rightarrow \frac{1}{2}mv^2 = \frac{2}{3}mgh_1$$

From B to C, the translational KE gets converted into potential energy. Rotational KE does not change

$$\therefore \quad mgh_2 = \frac{2}{3}mgh_1$$

$$\Rightarrow \quad h_2 = \frac{2}{3}h_1$$

Example 33 A uniform disc-shaped pulley has mass M and radius R. A light thread is tightly wrapped around it and its free end carries a block of mass m. The block is released from a height h above the floor. Find the speed with which the block hits the floor.



Solution

Concepts

(i) You can find acceleration of the block [using $F = Ma$ and $\tau = I\alpha$] and then use $v^2 = u^2 + 2as$ to find the speed when the block has travelled through a distance s.

(ii) Just to demonstrate an altenative approach, we will use energy method.

Mechanical energy of the pulley + block system is conserved, as the thread does not slip on pulley. [A slipping thread would mean rubbing and kinetic friction will dissipate mechanical energy.]

(iii) While writing KE, remember that the pulley also has KE

Let, v = speed of the block after falling through h.
ω = angular speed of pulley at the instant the block is about to hit the floor.

Obviously, $v = \omega R$

Gain in KE of the system = Loss in gravitational PE of the block.

$$\frac{1}{2}mv^2 + \frac{1}{2}(I_{\text{pulley}})\omega^2 = mgh$$

$$\Rightarrow \quad \frac{1}{2}mv^2 + \frac{1}{2}\frac{1}{2}MR^2\omega^2 = mgh$$

$$\Rightarrow \quad \frac{1}{2}mv^2 + \frac{1}{4}Mv^2 = mgh \quad \Rightarrow \quad v^2 = \frac{4mgh}{2m + M}$$

$$\Rightarrow \quad v = \sqrt{\frac{4mgh}{2m + M}}$$

In Short:

(i) KE of a body rotating about a fixed axis is

$$k = \frac{1}{2}I\omega^2$$

(ii) In case of a general motion of a rigid body, KE is written as:

$$k = k_R + k_T = \frac{1}{2}I_{CM}\cdot\omega^2 + \frac{1}{2}Mv_{CM}^2$$

or else, if we can identify the instantaneous axis of rotation, then KE of the body can be simply written as $k = \frac{1}{2}I\omega^2$

where I = moment of inertial about instantaneous axis of rotation.

(iii) KE of a purely rolling solid ball is

$$k = k_T + k_R = \frac{7}{10}Mv^2$$

KE of a purely rolling disc or cylinder is $\frac{3}{4}Mv^2$.

KE of a purely rolling ring is Mv^2.

Here, v is the speed of the COM of the body.

(iv) Work done by a torque (τ) is given by

$$dW_\tau = \tau d\theta$$

where $d\theta$ is the angular displacement of the body on which the torque acts.

dW_τ is positive if τ and $d\theta$ are in same direction and it is negative if they are in opposite directions.

For finite displacement: $W_\tau = \int_0^{\Delta\theta} \tau d\theta$

(v) Change in rotational KE of a body = work done by torque acting on it.

(vi) Rate of work done by a torque (i.e., power of the torque) = $\tau\cdot\omega$

where ω = instantaneous angular speed of the body.

(vii) In pure rolling motion, friction does not dissipate energy.

(viii) Use conservation of energy for a conservative system. Be careful while writing KE. You must write rotational as well as translational KE.

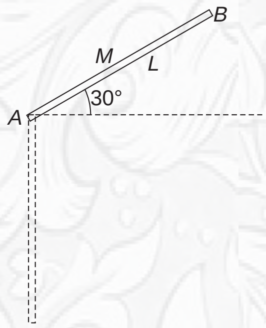
YOUR TURN

Q.34 A rigid body can rotate about a fixed axis. Its moment of inertia about the axis is $I = \pi$ kgm^2. A constant torque $\tau = 20$ Nm acts on it till it makes two full rotations. Find the KE of the body.

Q.35 In the last problem, what is the maximum power of the torque?

Q.36 When you brake your car such that it comes to rest while the wheels do not slip, where goes the kinetic energy of the car?

Q.37 A uniform rod of mass M and length L is free to rotate about its one end in a vertical plane. It is released from position shown in figure with the rod making an angle of 30° with the horizontal (see fig).

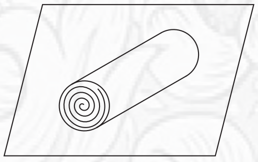


(a) Find angular speed of the rod when it becomes vertical.

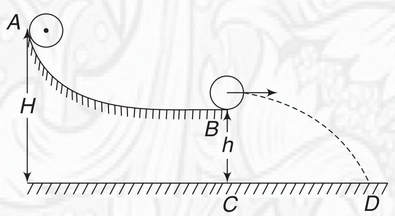
(b) Find force applied by the hinge on the rod when it becomes vertical.

Q.38 In Example 22, find work done by friction on the ball. Mass of ball is M.

Q.39 A carpet of mass M, made of inextensible material, is rolled along its length in the form of a cylinder of radius R and is kept on a rough horizontal floor. A gentle push is given to it and it begins to unroll, without sliding on the floor. Calculate the horizontal velocity of the axis of the cylindrical part of the carpet, when its radius reduces to $\frac{R}{2}$.



Q.40 A small ring rolls down without slipping from the top of a track in a vertical plane. The track has an elevated section and a horizontal part. Horizontal part is h above the ground and the top of the track is at height H. Find the distance of the point (D), where the ring lands on the ground, from point C.



7. ANGULAR MOMENTUM

We know the importance of linear momentum in dealing with the translational motion of a particle or a system of particles. Angular momentum is an analogous concept in dealing with rotational motion.

Angular momentum, as we shall see, is moment of momentum and an external torque is needed to change angular momentum just like an external force is required to change linear momentum.

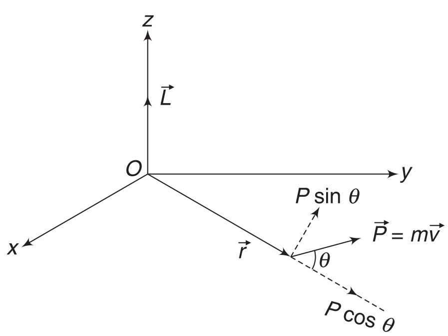
7.1 Angular Momentum of a Particle

Consider a particle moving with momentum $\vec{P}$. Position vector of the particle (at any instant) is $\vec{r}$ with respect to a point O. Angular momentum of the particle about point O is defined as

$$\vec{L} = \vec{r} \times \vec{P} = \vec{r} \times (m\vec{v}) \qquad ...(15)$$

where m and $\vec{v}$ are mass and velocity of the particle.

Consider the plane defined by $\vec{r}$ and $\vec{P}$ to be our xy-plane. Angular mometum ($\vec{L}$) is a vector directed perpendicular to $\vec{r}$ and $\vec{P}$ and it must be along z-direction. In the figure shown, $\vec{r} \times \vec{P}$ (by right hand rule) is along positive z-direction. If the particle were moving in exactly opposite direction, $\vec{r} \times \vec{P}$ would have been along negative z-direction.



Magnitude of angular momentum of the particle about O is

$$L = rP\sin\theta = rP_{\perp} \qquad ...(16)$$

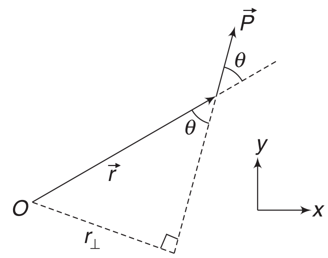
where $P_{\perp}$ = component of momentum perpendicular to $\vec{r}$. One can also interpret the above expression as

$$L = P(r\sin\theta) = Pr_{\perp} \qquad ...(17)$$

where $r_{\perp}$ is perpendicular distance of line of $\vec{P}$ from O.

Note: unit of angular momentum is kgm^2s^{-1}.

In the examples cited above, angular momentum of the particle about O is directed in z-direction. Angular momentum about point O is same as angular momentum about z-axis.



$L = Pr_{\perp}$

In this fig, $\vec{L}$ is directed out of the plane of the fig. Direction of $\vec{L}$ about O can also be conveyed simply by saying that it is anticlockwise

However, if the motion is not restricted to xy plane, angular momentum of the particle about O can be in any direction. The component of $\vec{L} = \vec{r} \times \vec{P}$ parallel to z-axis is defined as angular momentum of the particle about z-axis (L_z)

You must have noticed that the concept of angular momentum is mathematically similar to that of a torque.

$$\vec{\tau} = \vec{r} \times \vec{F} \quad \text{and} \quad \vec{L} = \vec{r} \times \vec{P}$$

To find angular momentum of a moving particle about a fixed line (axis), we need to remember the following three points [It is quite similar to calculation of torque about an axis.]

(i) If line of $\vec{P}$ intersects the given axis, there is no angular momentum about the axis.

(ii) If $\vec{P}$ is parallel to the given axis, there is no angular momentum about the axis.

(iii) When line of $\vec{P}$ is perpendicular to the axis (but not intersecting it) at a distance $r_{\perp}$ from it, the angular momentum about the axis is $Pr_{\perp}$.

Example 34 A particle has momentum $\vec{P} = P_x\hat{i} + P_y\hat{j}$ and its position vector at an instant is $\vec{r} = x\hat{i} + y\hat{j}$. Find the angular momentum of the particle

(a) about the origin (b) about the z-axis.

Solution

Concepts

$$\vec{L} = \vec{r} \times \vec{P}$$

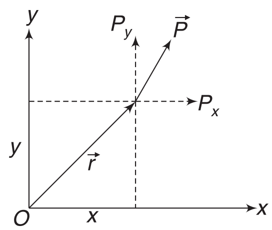
(a) $L = \vec{r} \times \vec{P} = (x\hat{i} + y\hat{j}) \times (P_x\hat{i} + P_y\hat{j})$

$$= (xP_y - yP_x)\hat{k}$$

(b) Angular momentum about z-axis = z component of $\vec{L}$

$\therefore \quad L_z = xP_y - yP_x$

Alternate: Think of the two components of $\vec{P}$ separately.



Angular momentum correponding to P_x is

= (P_x) (Perpendicular distance of line of P_x from O)

= $P_x y$ Direction is clockwise (or along negative z)

Angular momentum corresponding to P_y is

= P_y (perpendicular distance of line of P_y from O)

= $P_y x$ Direction is anti-clockwise (or along positive z)

$\therefore$ $L = (xP_y - yP_x)$ in anti-clockwise direction (or positive z).

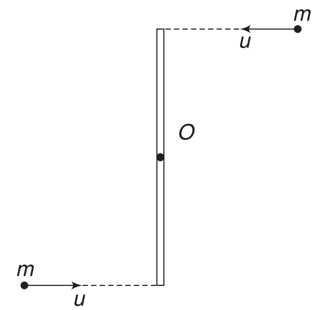
Example 35 A stick of length L is lying on a floor. Two horizontally moving particles have mass m each. They are moving perpendicular to the stick in opposite directions, as shown. Find sum of angular momentum of the two particles about centre (O) of the stick.

Solution

Concepts

(i) $L = Pr_{\perp}$.

(ii) Angular momentum in same direction will add.



Line of motion of both particles are at a distance $\frac{L}{2}$ from O.

$\therefore$ Each has angular momentum $= mu\frac{L}{2}$ about O.

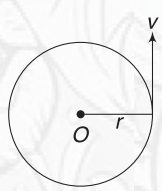
Both angular momenta are anticlockwise (or along a direction perpendicularly out of the plane of the paper). A simple observation tells you that the particles are, in a way, rotating anticlockwise about O. This judgement is good enough to write the direction of angular momentum.

$\therefore$ Required angular momentum is $2 \times mu\frac{L}{2} = muL$ (anticlockwise).

Note: If the particles strike the stick, they will cause it to rotate. This is the significance of angular momentum. Particles are having angular momentum about O, implies that they can cause rotation about O.

Your Turn

Q.41 A particle of mass m is moving in a circle of radius r with velocity v. Find its angular momentum about (a) centre O (b) about a line passing through O and perpendicular to the plane of motion.



Q.42 A pendulum has length l and bob of mass m. It is released from a position where the string is horizontal. Find the maximum angular momentum of the particle about point of suspension.

Q.43 A projectile of mass m is projected with velocity u at angle θ with horizontal. Find its angular momentum about its point of projection:

(a) When the projectile is at highest point

(b) When the projectile is about to hit the ground.

7.2 Angular Momentum of a Rotating Rigid Body

Angular momentum of a rigid body (or any system of particles) about an axis is the sum of angular momentum of individual particles.

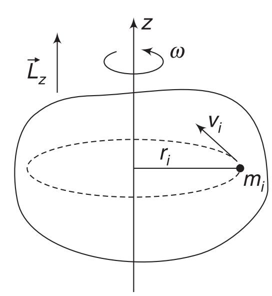
Consider a rigid body rotating with angular speed (ω) about a fixed axis (z-axis). Its angular momentum about z-axis is

$$L_z = \sum m_i v_i r_i = \sum m_i(\omega r_i) r_i$$

or $$L_z = \left[\sum m_i r_i^2\right]\omega = I\omega \quad ...(18)$$

where $I = \sum m_i r_i^2$ is the moment of inertia of the body about z-axis.

Direction of L_z is as shown in the figure. Curl your right hand fingers in the sense of rotation of the body while trying to hold the rotation axis in your hand. The thumb points in the direction of angular momentum vector.



Notice that linear momentum (P) is mass (inertia) times velocity and its rotational analogue-angular momentum (L) is rotational inertia (I) times angular velocity (ω).

7.3 Angular Momentum in Combined Rotation and Translation

Any plane motion of a rigid body can be understood as a combination of rotation about COM and translation of COM. We have already learnt how to write kinetic energy of a body using this thought. Angular momentum of the body about a point is also written using this approach.

Angular momentum of a body about some fixed point is

$$\vec{L} = \vec{L}_{\text{spin}} + \vec{L}_{\text{orbital}} \qquad ...(19)$$

where $\vec{L}_{\text{spin}}$ = angular momentum associated with spin (i.e., rotation) about COM.

and $\vec{L}_{\text{orbital}}$ = angular momentum arising due to motion of COM.

$L_{\text{spin}} = I_{\text{CM}}\,\omega$, if the body is rotating about a single axis passing through COM. I_{CM} is the moment of inertia of the body about the axis.

$$\vec{L}_{\text{orbital}} = \vec{r} \times (M\vec{v}_{\text{CM}}) \qquad ...(20)$$

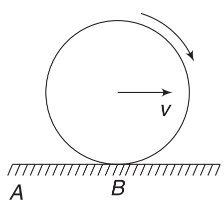
where $\vec{r}$ is position vector of COM with respect to the fixed point.

Obviously, $\vec{L}_{\text{orbital}}$ is the angular momentum of a particle (located at COM) of mass M moving with velocity $\vec{v}_{\text{CM}}$.

Example 36 *Angular momentum of a rolling disc*

A disc of mass M and radius R is rolling without sliding on a horizontal surface. Write its angular momentum about a point A on the surface.

Also write its angular momentum about contact point B.



Solution

Concepts

$$L_A = L_{\text{spin}} + L_{\text{orbital}}$$

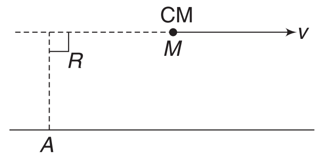
L_{spin} = Angular momentum due to rotation about an axis through COM.

$$= I_{\text{CM}} \cdot \omega = \frac{1}{2}MR^2 \cdot \omega \qquad \text{[Direction is clockwise]}$$

L_{orbital} is written considering a particle (at COM) of mass M moving with velocity v

$$L_{\text{orbital}} = (Mv)\,r_{\perp} = MvR \qquad \text{[Direction is clockwise]}$$

Angular momentum of disc about A is



$$L_A = \frac{1}{2}MR^2\omega + MvR$$

$$= \frac{1}{2}MR(R\omega) + MvR$$

$$= \frac{1}{2}MvR + MvR = \frac{3}{2}MvR \qquad \text{[Clockwise].}$$

Both L_{spin} and L_{orbital} remain same for point B also.

$$\therefore \qquad L_B = L_A = \frac{3}{2}MvR$$

Note: $L_B = I_{\text{CM}} \cdot \omega + MvR$

Let us write this in terms of ω

$$L_B = I_{\text{CM}}\,\omega + M(\omega R)R = (I_{\text{CM}} + MR^2)\,\omega$$

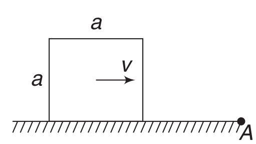
$$\Rightarrow \qquad L_B = I_B \cdot \omega$$

Where I_B is moment of inertia of the rolling body about its instantaneous axis of rotation through B.

Therefore, angular momentum of a body in pure rolling about contact point can be written assuming the body to be in rotation about instantaneous axis through the point of contact.

Example 37 *A sliding block*

A cube of Mass M and velocity v is moving on a horizontal surface. Find its angular momentum about an axis through point A, which is perpendicular to the plane of the figure.



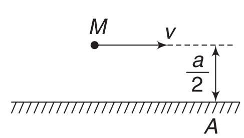
Solution

Concepts

$L_{\text{spin}} = 0$ as there is no rotation about COM. Only L_{orbital} is there.

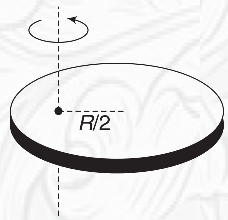
Consider a particle of mass M moving with velocity v along a line that is at a distance $\frac{a}{2}$ from A.

$$L = Mv\frac{a}{2} \text{ (clockwise)}$$

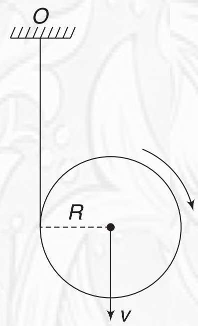


YOUR TURN

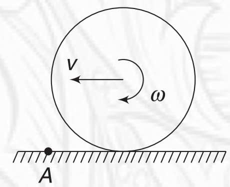
Q.44 A disc of mass M and radius R is rotating about a fixed axis at a distance of $\frac{R}{2}$ from the centre of the disc that is perpendicular to the plane of the disc. Find angular momentum of the disc about the axis if its angular speed is ω



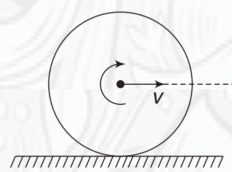
Q.45 A disc of mass M and radius R has a light thread wrapped on it. Free end of the thread is fixed at O and the disc is allowed to fall. Find the angular momentum of the disc about point O at the instant its centre has velocity v.



Q.46 A ring of mass M and radius R is placed on a horizontal surface and imparted an angular speed ω in clockwise sense apart from a velocity v towards left. Find angular momentum of the ring about A, assuming $v = 3\omega R$



Q.47 A ball of mass M radius R is rolling without sliding with velocity v. Find its angular momentum about point A shown in the figure.



In Short:

(i) Angular momentum of a particle about a point is $\vec{L} = \vec{r} \times \vec{P}$.

(ii) For a particle, $L = r_{\perp}P$ and $L = rP_{\perp}$.

(iii) Angular momentum of a particle about an axis is zero if its line of motion intersects the axis or is parallel to the axis.

$L = Pr_{\perp}$ where $r_{\perp}$ is the distance of line of motion of the particles from the axis.

(iv) For a rigid body in rotation about a fixed axis, $L = I\omega$ where I = moment of inertia about the axis.

Direction of $\vec{L}$ is along the rotation axis. Proper direction is given by right hand rule. Curl you right hand fingers in the sense of rotation and the thumb gives the direction of $\vec{L}$.

(v) For plane motion of a rigid body,

$$\vec{L} = \vec{L}_{\text{spin}} + \vec{L}_{\text{orbital}}$$

$L_{\text{spin}} = I_{\text{CM}} \cdot \omega$. This is the angular momentum of a body related to its spin in COM frame.

L_{orbital} is written using $\vec{r} \times \vec{P}$ where $\vec{P} = M\vec{v}_{\text{CM}}$ is linear momentum of COM.

(vi) For a body in pure rolling, angular momentum about point of contact can be written as $I\omega$, where I is moment of inertia about instantaneous axis through point of contact.

8. TORQUE AND ANGULAR MOMENTUM

Recall the relation $\frac{d\vec{P}}{dt} = \vec{F}_{\text{ext}}$. There is a rotational analogue to this. Just as a force causes momentum to change, a torque causes angular momentum to change. Angular momentum of a particle about a point is:

$$\vec{L} = \vec{r} \times \vec{P}$$

$$\Rightarrow \qquad \frac{d\vec{L}}{dt} = r \times \frac{d\vec{P}}{dt} + \frac{d\vec{r}}{dt} \times \vec{P}$$

$$= \vec{r} \times \frac{d\vec{P}}{dt} + \vec{v} \times (m\vec{v})$$

$$= \vec{r} \times \vec{F}_{\text{ext}} + 0 \qquad [\because \ \vec{v} \times \vec{v} = 0]$$

$$\therefore \qquad \frac{d\vec{L}}{dt} = \vec{\tau}_{\text{ext}} \qquad \text{...(21)}$$

Thus, rate of change of angular momentum of a particle about a point is equal to the external torque acting on it about the point.

For a system of particles

For a system consisting of N particles (the system can be a rigid body as well), angular momentum about a fixed point is

$$\vec{L} = \vec{L}_1 + \vec{L}_2 + \ldots + \vec{L}_N$$

where $\vec{L}_i$ is angular momentum of i^{th} particle about the fixed point.

$$\therefore \quad \frac{d\vec{L}}{dt} = \frac{d\vec{L}_1}{dt} + \frac{d\vec{L}_2}{dt} + \ldots + \frac{d\vec{L}_N}{dt}$$

$$= \vec{\tau}_1 + \vec{\tau}_2 + \ldots + \vec{\tau}_N$$

Here, $\vec{\tau}_i$ is net torque on i^{th} particle. $\vec{\tau}_i$ may have two sources–

(i) Torque due to internal interactions amongst particles

(ii) Torque due to external forces.

Torques due to internal interactions cancel out in pair and the above relation is simply

$$\frac{d\vec{L}}{dt} = \vec{\tau}_{\text{ext}} \quad \text{...(22)}$$

This relation is true for a single particle or a collection of particles – rigid or non-rigid – interacting or non-interacting.

For a rotating rigid body

For a rigid body rotating about a fixed axis, there is a much simpler way to get the above relation.

$$L = I\omega \quad \Rightarrow \quad \frac{dL}{dt} = I\frac{d\omega}{dt}$$

$$\Rightarrow \quad \frac{dL}{dt} = I\alpha \quad \Rightarrow \quad \frac{dL}{dt} = \tau_{\text{ext}}$$

Note:

(i) Equation $\frac{dL}{dt} = \tau$ shall be used in an inertial frame of reference or in a frame attached to COM of the body.

(ii) $\frac{d\vec{L}}{dt} = \vec{\tau}$ has three components

$$\frac{dL_x}{dt} = \tau_x; \quad \frac{dL_y}{dt} = \tau_y; \quad \frac{dL_z}{dt} = \tau_z$$

Example 38 A projectile of mass m is projected with velocity u at an angle θ to the horizontal. Find the rate of change of angular momentum of the particle about the point of projection when it is at the highest point.

Solution

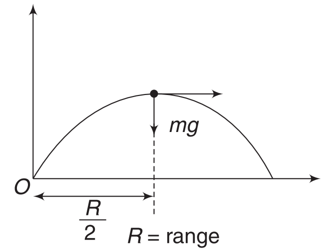
Concepts

Only force acting on the particle is its weight. Torque of weight is equal to rate of change of L.

When the particle is at the top, torque acting on it about O is

$$\tau = mg\frac{R}{2}$$

$$= mg\frac{u^2\sin^2\theta}{2g} = \frac{mu^2\sin^2\theta}{2}$$



$$\therefore \quad \frac{dL}{dt} = \tau = \frac{mu^2\sin 2\theta}{g}$$

Example 39 An amusement park ride has a big wheel having moment of inertia 10^4 kgm^2 about its rotation axis. When switched on, its electric motor sets it into motion. The wheel acquires its full speed of $4\,\text{rad s}^{-1}$ in 2 minutes. Find the average torque applied by the motor on the wheel.

Solution

Concepts

We can easily find change in angular momentum of the wheel and use $\tau_{\text{av}} = \frac{\Delta L}{\Delta t}$

$$L_i = 0; \; L_f = I\omega = 10^4 \times 4 \text{ kg m}^2\text{s}^{-1}$$

$$\Delta L = L_f - L_i = 4 \times 10^4$$

$$\therefore \quad \tau_{\text{av}} = \frac{\Delta L}{\Delta t} = \frac{4 \times 10^4 \text{ kg m}^2\text{s}^{-1}}{2 \times 60\,\text{s}} = 333.3 \text{ Nm.}$$

8.1 Angular Impulse – Momentum theorem

$$\frac{d\vec{L}}{dt} = \vec{\tau} \quad \Rightarrow \quad d\vec{L} = \vec{\tau}\,dt$$

Integrating on both sides gives

$$\int_{\vec{L}_i}^{\vec{L}_f} d\vec{L} = \int_0^{\Delta t} \vec{\tau}\,dt$$

$$\Rightarrow \quad \vec{L}_f - \vec{L}_i = \int_0^{\Delta t} \vec{\tau}\,dt \quad \text{...(23)}$$

The left side of the above equation is the change in angular momentum and right side is known as angular impulse of torque.

Recall that force × time is linear impulse and is equal to change in linear momentum.

The above equation implies that when a torque $\vec{\tau}$ acts on a body (or a system), it causes change in angular momentum ($\vec{L}$) and the change depends on torque as well the interval of time for which it acts.

If a constant torque acts on a body for time Δt, then change in angular momentum is

$$\Delta\vec{L} = \vec{\tau}\Delta\vec{t}$$

When a large torque acts on a body for a very small interval of time such that there is practically no rotation of the body in that time, we call it an *impulsive torque*.

Example 40 A fan has moment of inertia $4\,\text{kg m}^2$ about its rotation axis. When switched on, its motor applies a constant torque of $10\,\text{Nm}$. Find the angular speed of the fan $4\,\text{s}$ after it is switched on.

Solution

Concepts

$$\Delta L = \tau \Delta t$$

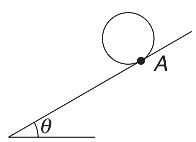
One can also find angular acceleration using $\tau = I\alpha$ and then use $\omega = \omega_0 + at$

$$\Delta L = \tau\,\Delta t \quad \Rightarrow \quad L_f - L_i = \tau\Delta t$$

$$\Rightarrow \quad I\omega_f - O = 10 \times 4$$

$$\Rightarrow \quad 4\,\omega_f = 40 \quad \Rightarrow \quad \omega_f = 10\ \text{rad s}^{-1}$$

Example 41 A disc of mass M and radius R is released on an inclined plane at point A. It rolls down without slipping. Find its angular momentum about point A on the incline after time t. Will you answer change if there is no friction and the disc slides down the incline?



Solution

Concepts

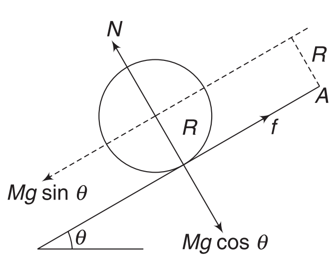
(i) One way of doing this problem is to find its velocity (v) and angular velocity (ω) after time t and then use $L = L_{\text{spin}} + L_{\text{orbital}}$ to write its angular moment about A.

But it is easier to solve the problem by using $\Delta L = \tau\,\Delta t$.

(ii) Forces on the disc are Mg, friction (f) and normal reaction (N). Torque of all these forces about point A can be calculated easily as friction has no torque and torque due to N and $Mg\cos\theta$ always cancel out.

Forces on the disc are as shown. Friction has no torque about A.

$$N = Mg\cos\theta$$



Also, line of action of N and $Mg\cos\theta$ are always same. They produce equal and opposite torque about A. Only force that has a torque about A, is $Mg\sin\theta$.

$$\tau_A = (Mg\sin\theta)R.$$

In time t, this will produce a change in angular momentum of the disc (about A) that is

$$\Delta L = \tau_A \Delta t$$

$$L_f - L_i = (MgR\sin\theta)t$$

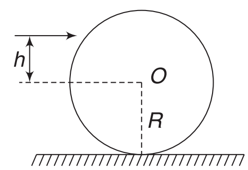
$$\Rightarrow \quad L_f - 0 = MgRt\sin\theta.$$

$$\therefore \quad L_f = MgRt\sin\theta.$$

Our answer will not change even if the disc slides.

Example 42 *Hit a ball to roll it without sliding*

A billiard ball of radius R is hit horizontally to impart a sharp impulse. At what height h above the centre shall it be hit so that it starts rolling without sliding?



Solution

Concepts

(i) There is a linear impulse on the ball, which causes its momentum to change.

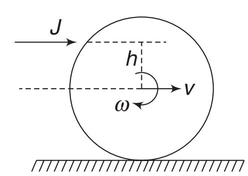
There is an angular impulse about the centre of the ball as well. This imparts a spin to the ball.

(ii) The ball begins to move with its centre having a velocity v and its angular velocity being ω. For pure rolling $v = \omega R$.

(iii) If $h = 0$, there is no torque of the applied force about the centre and angular impulse is zero. It means $\omega = 0$ if h is small, angular impulse will be small and $v > \omega R$. If h is larger, $\omega R > v$.

There is an appropriate value of h for which $v = \omega R$.

If the applied force (F) acted for a small time Δt, its impulse will be



$$J = \int_0^{\Delta t} F\,dt$$

If centre of mass of the ball acquired a velocity v then,

$$J = \Delta P \quad \Rightarrow \quad \int_0^{\Delta t} dt = Mv \qquad \text{...(i)}$$

Torque of force (F) about centre is $\tau = F\cdot h$. This torque also acted for time Δt and its angular impulse is

$$= \int_0^{\Delta t} \tau dt = h\int_0^{\Delta t} F\cdot dt = h\cdot \text{J}.$$

Using angular–impulse momentum theorem about an axis through centre gives

$$h \cdot J = \Delta L \quad \Rightarrow \quad h \cdot J = I_{CM} \cdot \omega$$

$$\Rightarrow \quad h \cdot Mv = \frac{2}{5} MR^2 \cdot \omega \qquad \text{[From (i) } J = Mv\text{]}$$

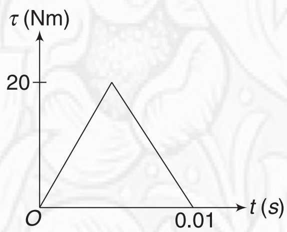
$$\Rightarrow \quad h \cdot (R\omega) = \frac{2}{5} R^2\omega \qquad \text{[For pure rolling } v = R\omega\text{]}$$

$$\Rightarrow \quad h = \frac{2}{5} R$$

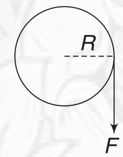
Note: It is safe to neglect impulse due to friction in the small interval for which the force was applied.

YOUR TURN

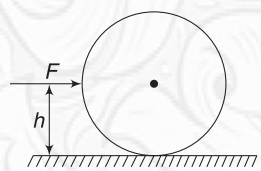
Q.48 A variable torque (τ) acts on a fan for a small time interval Δt = 0.01 s. The variation of torque with time is as shown. Moment of inertia of the fan about its rotation axis is 0.4 kgm^2. Find the final angular speed of the fan if initially it was at rest.



Q.49 A solid cylinder of mass M and radius R is free to rotate freely about the fixed horizontal axis through its centre. A light thread is wrapped around it. Its free end is pulled vertically with a force that changes with time as $F = at$. Find the angular speed of the cylinder after time T.



Q.50 A ball of mass M and radius R rests on a horizontal surface. It is given a sharp horizontal blow at a height $h(>0)$ above the floor. For what value of h will the ball acquire maximum kinetic energy? For what value of h the kinetic energy is minimum?



9. CONSERVATION OF ANGULAR MOMENTUM

We known that $\dfrac{d\vec{L}}{dt} = \vec{\tau}$

If $\vec{\tau} = 0$ then $\vec{L}$ = a constant.

When net external torque on a system is zero, the angular momentum of the system remains conserved. This is principle of conservation of angular momentum.

Like conservation of linear momentum, we can use angular momentum conservation principle in component form. If torque about z-axis on a system is zero then its angular momentum about z-axis is conserved. $L_z = a$ constant if $\tau_z = 0$ [τ_x and τ_y may not be zero].

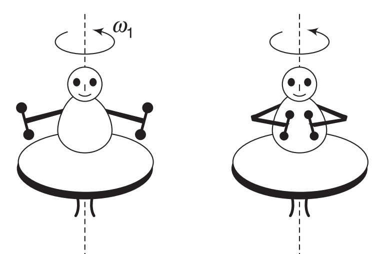
This conservation principle has wide applicability in understanding of macroscopic as well as microscopic world.

In problems in mechanics, we will use this conservation principle even when $\tau \neq 0$ but when we are convinced that the impulse of the external torque in the duration of an event is very small and can be neglected.

We will use this conservation principle about a fixed axis or about an axis passing through COM of a system.

Example 43 *Man on rotating platform*

A man is sitting on a platform that is spinning freely about its vertical axis. He has his arms stretched and is holding two dumbbells in his hands. Moment of inertia of the man – platform system is I_1 and angular speed is ω_1. Now, the man folds his arms to bring the dumbbells closer. Moment of inertia of the system about the rotation axis is now equal to I_2



(a) Find the new angular speed (ω_2) of the system

(b) What happened to kinetic energy of the system? What is change in KE? Give physical reason for this change.

Solution

Concepts

(i) As the man fold his arms and brings the dumbbells closer, the moment of inertia of the system about the rotation axis decreases. $I_2 < I_1$

(ii) Angular momentum of the system about the fixed rotation axis is conserved, as there is no external torque on it.

(a) Conservation of angular momentum (about rotation axis) gives:

$$L_f = L_i$$

$$\Rightarrow \quad I_2\omega_2 = I_1\omega_1 \quad \Rightarrow \quad \omega_2 = \left(\frac{I_1}{I_2}\right)\omega_1$$

(b) $k_i = \frac{1}{2} I_1\omega_1^2$

$$k_f = \frac{1}{2} I_2\omega_2^2 = \frac{1}{2}I_2\left[\frac{I_1}{I_2}\ \omega_1\right]^2 = \frac{I_1}{I_2}\left(\frac{1}{2}\ I_1\omega_1^2\right) = \frac{I_1}{I_2}\ k_i$$

Since, $\frac{I_1}{I_2} > 1$, the KE of the system has increased.

$$\Delta k = k_f - k_i = \left(\frac{I_1}{I_2} - 1\right)k_i = \left(\frac{I_1 - I_2}{I_2}\right)\frac{1}{2}\ I_1\omega_1^2$$

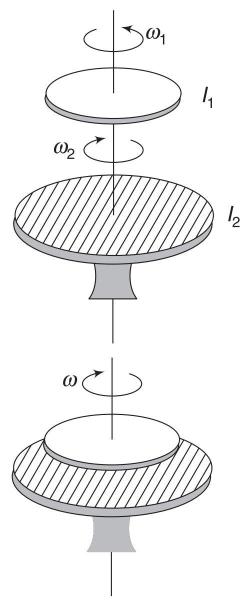
KE has increased due to work performed by the man in pulling the dumbbells inward.

Remember, internal forces can change KE of a system.

Note: A gymnast uses this principle to increase/decrease her spinning speed. When she pulls her arms close to her body, her rotational inertia decreases and angular speed increases.

Example 44 *Two spinning discs*

Two discs are rotating about their common axis of symmetry as shown. Moment of inertia of the two discs are $I_1 = 2\,\text{kg m}^2$ and $I_2 = 6\ \text{kg m}^2$ and they have angular velocities $\omega_1 = 3\ \text{rad s}^{-1}$ and $\omega_2 = 4\ \text{rad s}^{-1}$ in opposite directions as shown. The upper disc is allowed to gently fall over the lower disc. The two discs subsequently spin with one common angular speed (ω).



(a) Find ω.

(b) Find the amount of heat dissipated during the period the two discs slip on one another.

Solution

Concepts

(i) Initially, $\omega_2 > \omega_1$ and the discs slip. Kinetic friction between them develops a torque. This torque speeds up the upper disc and slows down the lower disc. Thus, the two discs eventually acquire a common angular speed.

(ii) During the process, external torque on the system of two discs about the rotation axis is zero. Angular momentum of the system is conserved.

(iii) Due to rubbing between the two discs, kinetic energy gets dissipated as heat. Heat dissipated is equal to loss in KE.

In the last problem, KE of the system increased and in this problem, it decreases. You must ponder over the working of internal forces in the two cases.

(a) Let final angular speed of the system be ω. Conservation of angular momentum gives:

$$L_f = L_i$$

$$(I_1 + I_2)\,\omega = I_1\omega_1(\uparrow) + I_2\omega_2(\downarrow)$$

[Direction of $I_1\omega_1$ is up, along the rotation axis. Curl your right-hand fingers in the sense of rotation of the upper disc, trying to hold the axis. Thumb points up. This is the direction of L for upper disc. Similarly, L for lower disc is directed down along the axis]

$$\Rightarrow \quad (2 + 6)\,\omega = 2 \times 3\ (\uparrow) + 6 \times 4\ (\downarrow)$$

$$\Rightarrow \quad 8\omega = 6(\uparrow) + 24(\downarrow) \quad \Rightarrow \quad \omega = \frac{9}{4}\ \text{rad s}^{-1}(\downarrow).$$

Final angular speed is $\frac{9}{4}\ \text{rad s}^{-1}$ in the sense of ω_2.

(b)
$$k_i = \frac{1}{2}\ I_1\,\omega_1^2 + \frac{1}{2}\ I_2\omega_2^2$$

$$= \frac{1}{2} \times 2 \times 9 + \frac{1}{2} \times 6 \times 16 = 57\,\text{J}$$

$$k_f = \frac{1}{2}(I_1 + I_2)\,\omega^2$$

$$= \frac{1}{2}(2 + 6)\left(\frac{9}{4}\right)^2 = 20.25\,\text{J}$$

Loss in $KE = k_i - k_f = 36.75\,\text{J}$

This is heat dissipated during the period the two discs slip on each other.

Example 45 *Alternative solution to Example 22*

Use conservation of angular momentum to solve example 22.

Solution

Concepts

(i) As learnt earlier, friction will eventually convert the sliding motion into a pure rolling motion.

(ii) There is no external torque on the ball about a point (A) on the horizontal surface. Line of friction passes through this point and normal force and Mg produce equal and opposite torque.

(iii) Use equation (19) while writing the angular momentum of the ball about A.

Angular momentum of the ball is conserved about a point on the horizontal surface (say point A). Initial angular momentum about A is $L_i = L_{\text{spin}} + L_{\text{orbital}} = 0 + MvR$

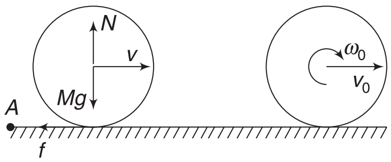
Final angular momentum about A when pure rolling starts is

$$L_f = L_{\text{spin}} + L_{\text{orbitals}} = \frac{2}{5}MR^2\cdot\omega_0 + Mv_0R$$

$$= \frac{2}{5}MRv_0 + Mv_0R = \frac{7}{5}Mv_0R.$$

$$L_f = L_i$$

$$\frac{7}{5}Mv_0R = MvR \Rightarrow v_0 = \frac{5v}{7}$$



Line of friction passes through *A*. It has no torque about *A*. *N* = *Mg* and together they have no torque about *A*

Pure rolling begins $v_0 = \omega_0 R$

Example 46 A platform of mass *M* and radius *R* is free to rotate about its central vertical axis. Initially, the platform is at rest with a man of mass *m* standing at the edge. The man starts walking along the edge with a tangential velocity *v* relative to the disc. Find angular velocity acquired by the platform. Treat man as a point object.

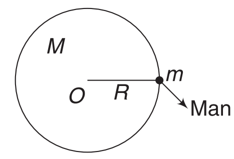
Solution

Concepts

(i) Angular momentum of (man + platform) system is zero, initially. It will continue to be zero even when the man starts walking.

(ii) If the man moves anti-clockwise, the platform will begin to rotate clockwise. This happens due to friction between the shoes of the man and the platform. But when we consider man + platform as our system, we need not worry about friction. It is an internal force.

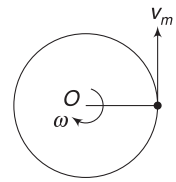
Let angular speed of the platform be ω (clockwise) after the man begins to walk.



Top view

v_m = velocity of man in tangential direction.

Velocity of a point on the circumference of the platform just beneath the shoes of the man is ωR in a direction opposite to the velocity of the man.



v = velocity of man relative to the platform

$$\Rightarrow \quad v = v_m - (-\omega R) \quad \Rightarrow \quad v_m = v - \omega R.$$

$$L_{final} = L_{initial} \quad \text{[about rotation axis through } O\text{]}$$

$$mv_m \cdot R - \frac{MR^2}{2}\omega = 0$$

$$\Rightarrow \quad \omega = \frac{2mv}{(2m + M)R}$$

Note:

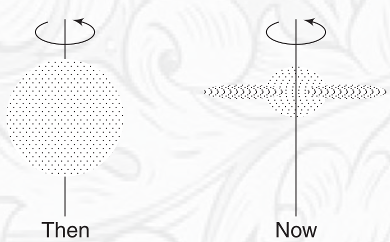
(i) In the figure shown, angular momentum of the man about the rotation axis is out of the plane of the figure (or, simply anti-clockwise). Angular momentum of the platform is into the plane of the figure (or, simply clockwise).

(ii) *KE* of the system has increased. Why? The man has performed work. When you walk on a hard ground, you work to impart *KE* to yourself. While walking on this platform, you have to work to impart *KE* to self as well as to the platform.

YOUR TURN

Q.51 If all of us – the inhabitants of Earth – move to equator, how would this affect the length of the day? And what if all of us move to the poles?

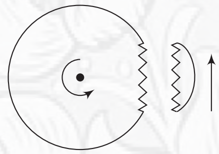
Q.52 We believe that our galaxy was formed from a huge cloud of gas. The original cloud was more or less spherical and very large in size. It was spinning about some axis. The present shape of the galaxy is somewhat like as shown in the second figure. Is it correct to say that the galaxy is spinning faster now, than when it was a large spherical cloud?



Q.53 A platform (in shape of a disc) of mass *M* and radius *R* is rotating freely about a vertical axis through its centre with an angular speed ω_0. An insect of mass *m* is sitting at the centre of the platform. The insect walks along a radius to the edge of the platform. Find the new angular speed of the platform.

Q.54 A boy is standing on the edge of a rotating platform. He drops a ball. Will the angular speed of the platform increase or decrease due to this event?

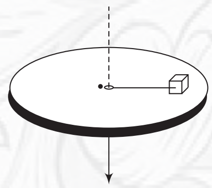
Q.55 A uniform disc of mass *M* and radius *R* is spinning about its central horizontal axis with angular speed ω. A chip of mass m breaks off the edge of the disc at the instant its (chips's) velocity is vertically up.



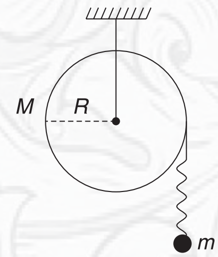
(a) What is the new angular velocity of the disc?

(b) How high does the chip rise?

Q.56 A small block of mass 4 kg is attached to a cord passing through a hole in a horizontal smooth table. The block is revolving in a circle of radius 0.5 m about the hole with a speed of 4 ms^{-1}. The cord is slowly pulled below the table so as to shorten the radius of circular path of the block. Find the radius of circular path when tension in the cord becomes 600 N.

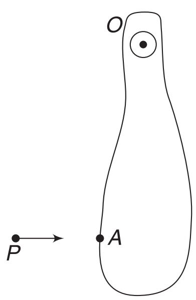


Q.57 A disc-shaped pulley has mass M and radius R. A light thread is tightly wrapped around it and its free end carries a small ball of mass m. The ball is held in a position where the thread is loose and the pulley is at rest. The ball is released from rest. It falls a distance L before the thread gets taut. Find the speed of the ball immediately after the string gets taut.



10. COLLISION OF A PARTICLE WITH A RIGID BODY

When a moving particle hits a rigid body, it can impart it a rotation apart from translation. Consider a rigid body that can rotate freely about an axis (perpendicular to the figure) through point O. A particle P strikes it at A. In general, the axis (at O) will exert a large impulsive force on the body during the collision. [The axis will be a pin or a shaft]. Therefore, the linear momentum of the (particle + body) system will not be conserved. However, the impulsive force by the axis will have no torque about itself. It means, the angular momentum of the system will remain conserved about the axis through O. It is important to understand that the angular momentum cannot be conserved about any other axis that is not passing through O. It will not be safe. The impulsive force can have torque about any such axis.



Body can rotate freely about an axis through O.

It coefficient of restitution is e, then:

Relative speed of separation of P and point A after collision

= e(relative speed of approach of P and A before collision)

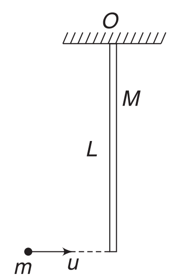
We are not concerned with velocity of any other point in the body while writing equation for e. It is point A that is important.

If a moving particle strikes a rigid body that is free (not hinged) then linear momentum is conserved. Angular momentum conservation can be applied about some fixed point or the COM of the system under consideration.

Equation for coefficient of restitution shall be written as explained above.

Example 47 *A hinged rod hit by a bullet*

A rod of mass M and length L is hinged at its one end at O such that it can freely swing in the vertical plane. A horizontally flying bullet of mass m hits its lower end and comes to rest. speed of the bullet just before impact is u.



(a) Find the angular speed of the rod immediately after the impact.

(b) Find the angle (θ) with the vertical that the rod will make in its extreme position after the impact.

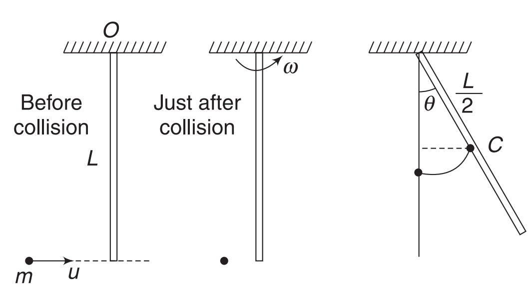
Solution

Concepts

(i) The rod can rotate about a horizontal axis through O that is perpendicular to the plane of the figure. Angular momentum of the (rod + bullet) system is conserved about the axis.

(ii) After collision, the mechanical energy of the rod remains conserved.

Remember, the kinetic energy of the bullet before collision is higher than that of rod after collision. Kinetic energy is lost during collision.



(a) Conserving angular momentum about axis through O.

$$L_{\substack{\text{before}\\\text{collision}}} = L_{\substack{\text{after}\\\text{collision}}}$$

$$\Rightarrow \quad muL = \frac{1}{3}ML^2 \cdot \omega$$

$$\Rightarrow \quad \omega = \frac{3mu}{ML}$$

(b) After impact, the rod has a *KE*. It gets converted into *PE*, as the rod rises. The rod stops after rotating through angle θ where its entire *KE* gets converted into its *PE*.

COM of the rod rises by $\frac{L}{2} - \frac{L}{2}\cos\theta$

$$= \frac{L}{2}(1 - \cos\theta)$$

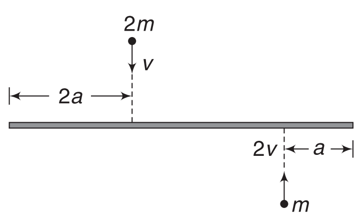
$$\therefore \quad Mg\frac{L}{2}(1 - \cos\theta) = \frac{1}{2}I\omega^2$$

$$\Rightarrow \quad Mg\frac{L}{2}(1 - \cos\theta) = \frac{1}{2}\left(\frac{ML^2}{3}\right)\left(\frac{3mu}{ML}\right)^2$$

$$\Rightarrow \quad 1 - \cos\theta = \frac{3m^2}{M^2}\frac{u^2}{gL}$$

$$\Rightarrow \quad \cos\theta = 1 - \frac{3m^2 u^2}{M^2 gL}$$

Example 48 A uniform bar of length $6a$ and mass $8m$ lies on a smooth horizontal table. Two point masses–m and $2m$–moving in opposite directions hit the bar simultaneously. The velocities of the particles are $2v$ and v, perpendicular to the bar as shown in figure. The particles stick to the bar. Find



(a) velocity of centre of the rod after collision.
(b) angular velocity of the rod after collision.
(c) Find kinetic energy of the system after collision.

Solution

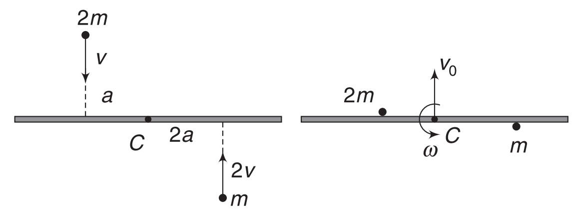
Concepts

(i) After the particles stick to the rod, the COM of the system is at the center of the rod. This is because the particle of mass m is at a distance $2a$ from the centre of the rod and the particle of mass $2m$ is at a distance a from the centre.

v_{CM} can be found using conservation of momentum.

(ii) We can use conservation of angular momentum about any point, as there is no external force on the system. We will take the COM of the system (i.e., centre of the stick) as our preferred point.

One can also think of a point attached to the table, just below the centre of the stick. This is a point in inertial frame and yields same equation as obtained by considering centre of the rod as pivot point for application of conservation of angular momentum.



(a) COM of the system is at C (centre of the stick) after collision. Let its velocity be $\vec{v}_0$. Conservation of momentum.

$$P_f = P_i$$

$$P_f = 2mv - m(2v) = 0$$

$$\therefore \quad v_0 = 0$$

(b) Angular momentum is conserved about C.

$$L_i = L_f$$

$$2mva + m(2v)(2a) = I\omega$$

$$\Rightarrow \quad 6mva = I\omega \qquad \text{...(i)}$$

I = moment of inertia of rotating system about an axis through C perpendicular to plane of the figure

$$= \frac{8m(6a)^2}{12} + 2ma^2 + m(2a)^2 = 30ma^2$$

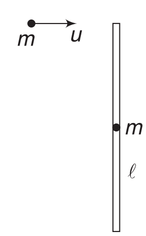
Putting in (i) gives

$$\omega = \frac{v}{5a}$$

(c) The system is just rotating about C. There is no translation.

$$\therefore \quad K = \frac{1}{2}I\omega^2 = \frac{1}{2} \times 30ma^2 \times \left(\frac{v}{5a}\right)^2 = \frac{3mv^2}{5}$$

Example 49 *Conserve angular momentum about any point!*



A particle of mass m travelling with speed u on a smooth horizontal table, hits one end of a stationary stick of mass m and length l. The stick is free. The particle strikes the stick perpendicularly and sticks to it. Find angular speed of the system after collision.

Solution

Concepts

(i) There is no external force on the system. Angular momentum is conserved about any point. To demonstrate the process, we will solve the problem by selecting three different points for conserving angular momentum.

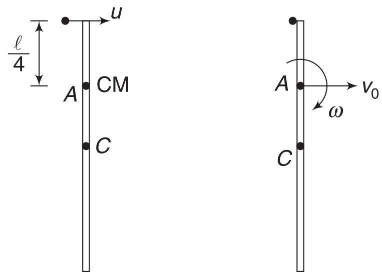
(ii) v_{CM} can be obtained using conservation of momentum.

Method 1

COM of the stick + particle system, just before the particle strikes, is at a distance $\frac{l}{4}$ from the centre of the stick. *A is a point on the table just beneath the COM. We will conserve the angular momentum about this point.*

$$L_i = L_f$$

$$mu\frac{l}{4} = L_{\text{spin}} + L_{\text{orbital}}$$



The velocity of COM $(= v_0)$ has its line passing through A. Therefore, the second term on the right side of the above equation is zero.

$$\therefore \quad mu\frac{l}{4} = I_{CM}\cdot\omega$$

$$\Rightarrow \quad mu\frac{l}{4} = \left[\frac{ml^2}{12} + m\left(\frac{l}{4}\right)^2 + m\left(\frac{l}{4}\right)^2\right]\omega$$

$$\Rightarrow \quad \frac{mu\,l}{4} = \frac{5ml^2}{24}\cdot\omega \quad\Rightarrow\quad \omega = \frac{6u}{5l}$$

Method 2

This time, we will choose the *origin to be a fixed point on the table just below centre (C) of the stick.*

$$L_i = L_f$$

$$mu\frac{l}{2} = L_{\text{spin}} + L_{\text{orbital}}$$

$$mu\frac{l}{2} = I_{CM}\cdot\omega + 2mv_0\frac{l}{4} \qquad \text{...(i)}$$

Here, $I_{CM} = \frac{5}{24}ml^2$ (as in last method) and L_{orbital} is written assuming a particle of mass $2m$ travelling at speed v_0 at a distance $\frac{l}{4}$ from the origin.

To get v_0, we need to apply conservation of momentum:

$$mu = 2mv_0 \quad\Rightarrow\quad v_0 = \frac{u}{2}$$

Putting in (i) gives

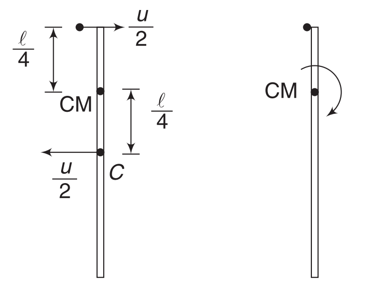
$$mu\frac{l}{2} = \left(\frac{5}{24}ml^2\right)\omega + 2m\left(\frac{u}{2}\right)\frac{l}{4}$$

$$\Rightarrow \quad \omega = \frac{6u}{5l}$$

Method 3

Let the origin (point selected for conservation of angular momentum) be the COM of the system.

Before collision, velocity of COM is $\frac{u}{2}$ towards right. In a reference frame attached to COM, the particle is seen moving with velocity $\frac{u}{2}$ to right and the centre of the rod is seen moving to left with velocity $\frac{u}{2}$.



Before collision in COM frame

After collision in COM frame

$\therefore$ Initial angular momentum of the system before collision, wrt COM of the system is

$$L_i = m\frac{u}{2}\frac{l}{4} + m\frac{u}{2}\frac{l}{4}$$

$$= mu\frac{l}{4} \; (\circlearrowright)$$

After collision, in COM frame, angular momentum is

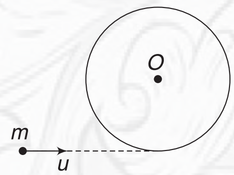
$$L_f = I_{CM}\cdot\omega = \frac{5ml^2}{24}\cdot\omega$$

$$\therefore \quad \frac{5}{24}ml^2\omega = mu\frac{l}{4}$$

$$\Rightarrow \quad \omega = \frac{6}{5}\frac{u}{l}$$

YOUR TURN

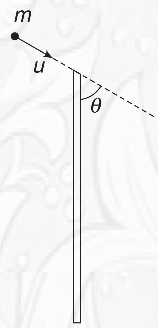
Q.58 A disc of mass M and radius R is fixed to rotate about its central vertical axis through O. A horizontally flying bullet of mass m has speed u. It kisses the disc at the circumference and loses half its kinetic energy. The bullet maintains its line of motion. Find the angular speed acquired by the disc.



Q.59 A rod of length L and mass $2m$ lies on a smooth table. A particle of mass m is moving with a velocity u, making an angle θ with the length of the rod (see fig.). The particle hits one end of the rod and stops. Find:

(a) speed of centre of the rod and

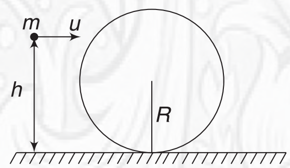
(b) angular speed of the rod after the hit.



Q.60 A ball of mass M lies on a smooth horizontal surface. A particle of mass m, travelling horizontally at a speed u, collides and sticks to the ball at a height h above the surface. Assume that $M \gg m$ so that the COM of the system practically remains at the geometrical centre of the sphere after the particle sticks to it. Find the following just after collision:

(a) Velocity of COM

(b) Angular speed of the sphere



In Short:

(i) $\dfrac{d\vec{L}}{dt} = \vec{\tau}_{ext}$

$$\Rightarrow \frac{dL_x}{dt} = \tau_x; \quad \frac{dL_y}{dt} = \tau_y; \quad \frac{dL_z}{dt} = \tau_z$$

(ii) Angular momentum of a body about a point changes only when there is an external torque on the body about that point.

(iii) When $\tau_{ext} = 0$, L = a constant.

This is known as law of conservation of angular momentum.

(iv) If a torque τ acts on a body for a time Δt, its angular impulse is defined as $\int_0^{\Delta t} \tau dt$.

If τ is constant, angular impulse is $\tau\Delta t$.

(v) Angular impulse experienced by a body is equal to the change in its angular momentum. This is known as angular impulse momentum theorem.

(vi) When you apply a force on a body, it can cause its linear momentum as well as angular momentum to change.

$$\Delta P = \int_0^{\Delta t} F\,dt \quad \text{and} \quad \Delta L = \int_0^{\Delta t} \tau dt$$

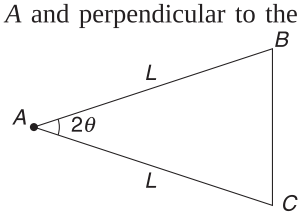
(vii) If a flying particle strikes a hinged body, use conservation of angular momentum about the hinge point.

If a particle hits a free body, angular momentum is conserved about any fixed point. We can also use conservation of angular momentum about COM of the system.

Miscellaneous Examples

Example 50 (a) An isosceles triangle has mass M, vertex angle 2θ and common side length L. Find its moment of inertia about an axis through tip A and perpendicular to the plane of the triangle.

(b) A regular polygon of N sides has mass M. Distance of any of its vertex from the centre is R. Find its moment of inertia about an axis through its centre and perpendicular to its plane.



Solution

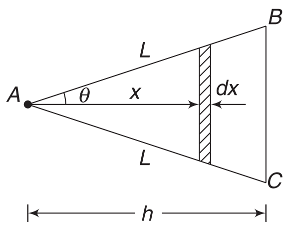
Concepts

(i) We will divide the Δ into thin strips, parallel to the base.

(ii) MI of each strip about an axis through its centre and parallel to given axis is given by formula $I = \dfrac{ml^2}{12}$ – for a rod.

(iii) Using parallel axis theorem, we can write MI of each strip about given axis.

(iv) For part (b), the polygon is made up of N isosceles triangles with $\theta = \dfrac{\pi}{N}$.



(a) $h = L\cos\theta$

Consider a strip (like a thin rod) as shown.

Length of strip $= 2x\tan\theta$

Area of strip $dA = (2x\tan\theta)\,dx$

Mass per unit area of the triangle:

$$\sigma = \frac{M}{\frac{1}{2}\cdot h(BC)} = \frac{2M}{h(2h\tan\theta)} = \frac{M}{h^2\tan\theta} \qquad ...(1)$$

Mass of the strip, $dm = \sigma dA = (2\sigma\tan\theta)\,x\,dx$

MI of the strip about an axis perpendicular to the figure and passing through centre of the strip is

$$= \frac{1}{12}(dm)(2x\tan\theta)^2$$

MI of strip about given axis through A is (using parallel axis theorem)

$$dI = \frac{1}{12}(dm)(2x\tan\theta)^2 + (dm)x^2$$

$$= (2\sigma\tan\theta\cdot x\,dx)\left[\frac{(2x\tan\theta)^2}{12} + x^2\right]$$

$$\therefore \quad I = 2\sigma\tan\theta\left(1 + \frac{\tan^2\theta}{3}\right)\int_0^h x^3\,dx$$

$$= 2\sigma\tan\theta\left(1 + \frac{\tan^2\theta}{3}\right)\frac{h^4}{4}$$

$$= \frac{M}{h^2\tan\theta}\tan\theta\left(1 + \frac{\tan^2\theta}{3}\right)\frac{h^4}{2}$$

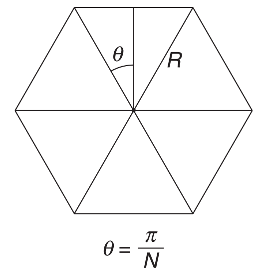
$$= \frac{Mh^2}{2}\left(1 + \frac{\tan^2\theta}{3}\right)$$

$$= \frac{M(L\cos\theta)^2}{2}\left[1 + \frac{\tan^2\theta}{3}\right]$$

$$= \frac{ML^2}{2}\left[\cos^2\theta + \frac{\sin^2\theta}{3}\right]$$

$$= \frac{ML^2}{2}\left[1 - \frac{2}{3}\sin^2\theta\right] \qquad ...(2)$$

(b) The N–gon is made up of N isosceles triangles. MI of each Δ is given by equation (2) above. When we add, masses of each Δ adds and we get.



$$I = \frac{ML^2}{2}\left[1 - \frac{2}{3}\sin^2\left(\frac{\pi}{N}\right)\right]$$

Note: For hexagon, $N = 6$

$$I = \frac{ML^2}{2}\left[1 - \frac{2}{3}\times\frac{1}{4}\right] = \frac{5}{12}ML^2$$

Example 51 A uniform cylinder has length L, radius R and mass M. Find its moment of inertia about an axis that passes through its centre and is perpendicular to its length.

Solution

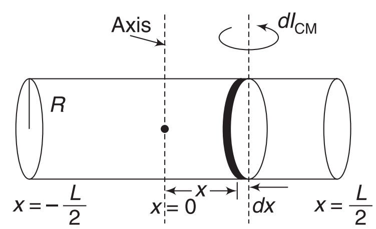
Concepts

(i) We will consider the cylinder to be made up of a number of thin discs.

(ii) MI of each disc of mass dm can be written about its diameter as $\frac{1}{4}(dm)R^2$.

(iii) Using parallel axis theorem, we can write MI of each disc about the given axis.

(iv) Adding MI of all such discs will give the desired result.



Mass of cylinder per unit length $= \dfrac{M}{L}$

Mass of disc of thickness dx is

$$dm = \frac{M}{L}dx$$

MI of this disc element about its own diameter is

$$dI_{CM} = \frac{1}{4}(dm)R^2 = \frac{1}{4}\frac{M}{L}R^2\,dx$$

MI of disc element about desired axis is

$$dI = dI_{CM} + (dm)x^2 = \frac{MR^2}{4L}dx + \frac{M}{L}x^2\,dx$$

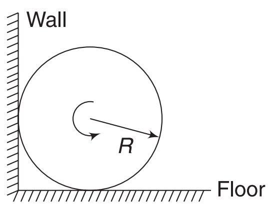
$$\therefore \qquad I = \int_{x=-\frac{L}{2}}^{L/2} dI = \frac{MR^2}{4L}\int_{-\frac{L}{2}}^{\frac{L}{2}} dx + \frac{M}{L}\int_{-\frac{L}{2}}^{\frac{L}{2}} x^2 dx$$

$$= \frac{MR^2}{4L}\left[\frac{L}{2} - \left(-\frac{L}{2}\right)\right] + \frac{M}{3L}\left[\left(\frac{L}{2}\right)^3 - \left(\frac{-L}{2}\right)^3\right]$$

$$I = \frac{MR^2}{2} + \frac{ML^2}{12}$$

Example 52 *A spinning cylinder in a corner*

A uniform cylinder of radius R is spun about its central axis at an angular speed ω_0 and then placed in a corner as shown. Coefficient of friction between the cylinder and two surfaces is μ. How many turns will the cylinder accomplish before stopping.



Solution

Concepts

(i) It is obvious that the cylinder cannot translate (due to presence of wall). Therefore, the cylinder rotates about a fixed axis through its COM.

(ii) Since $a_{CM} = 0$, net force on the cylinder must be zero.

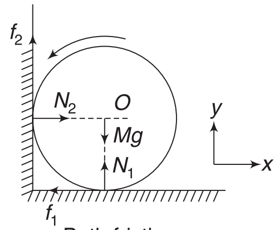
(iii) There is a net torque about COM that causes the spinning to retard.

The normal forces (due to the floor and the wall) cannot produce torque about the central axis. Their line of action intersects the axis.

Only friction produces torque.

(iv) We can use $\omega^2 = \omega_0^2 + 2\alpha\theta$ to find angular displacement of the cylinder before it stops rotating.

FBD of the cylinder is a as shown. Friction forces are directed opposite to the velocities of the rubbing contact points of the cylinder.



Both frictions are Kinetic

For $\quad a_{CMx} = 0$

$$F_x = 0$$

$$\Rightarrow \quad N_2 = f_1 \quad \Rightarrow \quad N_2 = \mu N_1 \qquad \text{...(i)}$$

For $\quad a_{CMy} = 0;\ F_y = 0$

$$\Rightarrow \quad N_1 + f_2 = Mg$$

$$\Rightarrow \quad N_1 + \mu N_2 = Mg \qquad \text{...(ii)}$$

Solving (i) and (ii) gives

$$N_1 = \frac{Mg}{1+\mu^2},\ N_2 = \frac{\mu Mg}{1+\mu^2}$$

Net torque on cylinder about its rotation axis through O is

$$\tau = f_2 R + f_1 R\ (\circlearrowright)$$

$$= \mu(N_2 + N_1)R = \frac{\mu Mg(1+\mu)R}{(1+\mu^2)}$$

Angular retardation produced in the cylinder is given by

$$I\alpha = \tau$$

$$\frac{1}{2}MR^2.\alpha = \frac{\mu Mg(1+\mu)R}{(1+\mu^2)} \quad \Rightarrow \quad \alpha = \frac{2\mu g(1+\mu)}{R(1+\mu^2)}$$

Since, α is constant, we can use

$$\omega^2 = \omega_0^2 + 2\alpha\theta$$

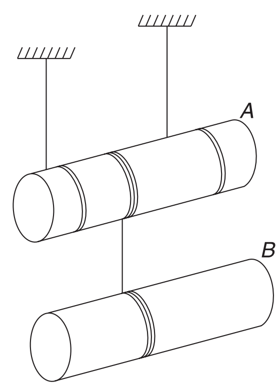
$$\Rightarrow \quad 0 = \omega_0^2 - 2\frac{2\mu g(1+\mu)}{R\,(1+\mu^2)}\cdot\theta$$

$$\Rightarrow \quad \theta = \frac{\omega_0^2\,R(1+\mu^2)}{4g\,\mu(1+\mu)}$$ = angular displacement before the cylinder stops.

$$\therefore \quad \text{Number of turns, } n = \frac{\theta}{2\pi} = \frac{\omega_0^2 R(1+\mu^2)}{8\pi g\,\mu(1+\mu)}$$

Example 53 *Watch the relation between acceleration*

A uniform cylinder A of mass M and radius R is suspended using two symmetrically wrapped strings, as shown. Another identical cylinder B is suspended from A using another string, which is tightly wrapped at the centre of two cylinders. The string holding B is wrapped on A in opposite sense to the other two strings. Axes of the cylinders remain horizontal. During fall, find acceleration of B, and acceleration of A.



Solution

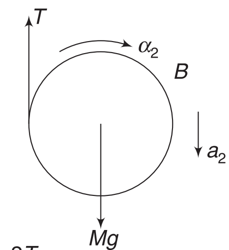
Concepts

(i) As A rotates, threads at its two ends unwind. A moves down. If acceleration of A is a_1 and its angular acceleration is α_1, then $a = R\alpha_1$.

(ii) As A rotates, thread at its centre rewinds. Note that this thread is wound in opposite sense. If A falls through x, the length of middle thread that it rewinds is also x. Thus, motion of A has no effect on motion of B.

If α_2 is angular acceleration of B and a_2 is its linear acceleration, $a_2 = R\alpha_2$

Motion of B is not affected by motion of A. From Example 19

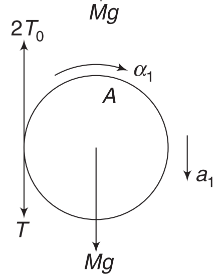


$$a_2 = \frac{2g}{3} \quad \text{and}$$

$$T = \frac{M}{2}a_2 = \frac{Mg}{3}$$

Let tension in upper string be T_0

For A:



$$T + Mg - 2T_0 = Ma_1$$

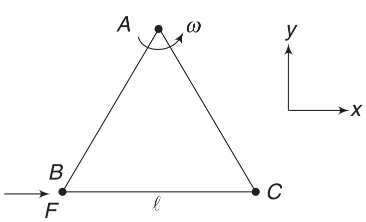
$$\frac{4Mg}{3} - 2T_0 = Ma_1 \quad \text{...(i)}$$

$$\text{And } 2T_0R - TR = \frac{1}{2}MR^2\alpha_1$$

$$2T_0 - \frac{Mg}{3} = \frac{1}{2}Ma_1 \quad \text{...(ii)}$$

Adding (i) and (ii) gives $a_1 = \dfrac{2g}{3}$

Example 54 Three particles A, B, and C each of mass m are connected to each other by three massless rigid rods to form a rigid, equilateral triangular body of side length l. This body is placed on a horizontal frictionless table (xy plane) and is hinged at point A so that it can rotate, without friction, about a vertical axis through A. The body is set into rotational motion on the table about the axis with constant angular velocity ω.



(a) Find the magnitude of horizontal force exerted by the hinge (at A) on the body.

(b) At time t, when BC is parallel to x-axis, a force F is applied on B along BC. Obtain the x and y components of the force exerted by the hinge on the body, immediately after time t.

Solution

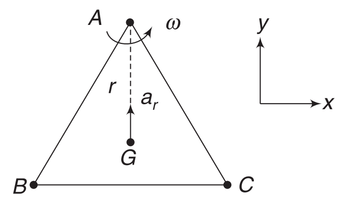
Concepts

(i) The COM of the system is at the centroid of the triangle.

(ii) COM is rotating in a circle about A. Hinge force provides the necessary acceleration to the COM.

(iii) When force F is applied, there is a torque about A. Body acquires angular acceleration. The COM gains a tangential acceleration apart from the centripetal acceleration.

In the position described, tangential acceleration of COM is in x-direction and radial acceleration is along y-direction.



(a) G is COM. Distance of G from A is

$$r = \frac{l}{\sqrt{3}}\left[= \frac{2}{3} \times \text{height}\right]$$

COM goes in a circle of radius r about A.

Force on the body must be equal to

$$F = Ma_{CM} = 3m\omega^2 \cdot r = 3m \cdot \omega^2 \cdot \frac{l}{\sqrt{3}} = \sqrt{3}\,ml\,\omega^2$$

This force is applied by the hinge on the body.

(b) Torque about A is: $\tau = F\left(\dfrac{\sqrt{3}}{2}l\right)$

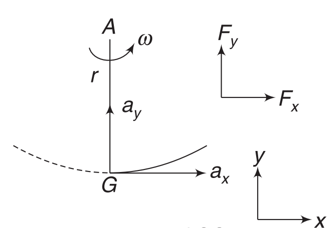
Moment of inertia about axis through A and perpendicular to the plane of the figure is

$$I = ml^2 + ml^2 + 0 = 2ml^2$$

$\therefore$ Angular acceleration produced due to F is

$$\alpha = \frac{\tau}{I} = \frac{\sqrt{3}F}{4ml}$$

Now COM is moving in a circle of radius r and at the given instant its angular speed is ω and angular acceleration is α. Radial acceleration of COM is



Accelerations of COM

$$a_y = \omega^2 r$$

$\therefore$ Radial force on it is

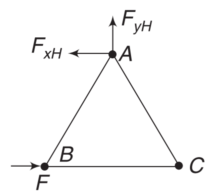
$$F_y = 3ma_y = \sqrt{3}\,ml\,\omega^2$$

[as calculated in part (a)]

Tangential acceleration of COM is

$$a_x = r\alpha = \frac{l}{\sqrt{3}}\cdot\frac{\sqrt{3}\,F}{4ml} = \frac{F}{4m}$$

$\therefore$ Force on the body in x-direction must be



$$F_x = 3ma_x = \frac{3F}{4}$$

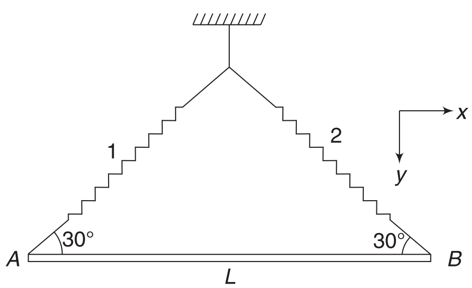
x component of hinge force must be towards left equal to

$$F_{xH} = \frac{F}{4}$$

y component of hinge force is $F_{yH} = \sqrt{3}\,ml\omega^2$

Example 55 *A bar suspended with springs*

A uniform slender bar AB of mass M is suspended using two identical springs, as shown in figure. Spring 2 is cut. Find acceleration of COM and end A of the bar immediately after this.



Solution

Concepts

(i) We can find tension in each spring when the bar is in equilibrium.

(ii) When spring 2 is cut, the extension in spring 1 cannot change suddenly. It means tension in spring 1 will remain uncharged immediately after 2 is cut.

(iii) Any motion of a rigid body can be understood as translation of COM and rotation about it. We will assume the COM has acceleration $\vec{a}_0$ and the bar has an angular acceleration about COM.

(iv) Since spring 1 applies a force that has horizontal component as well, the COM will have a horizontal component of acceleration as well.

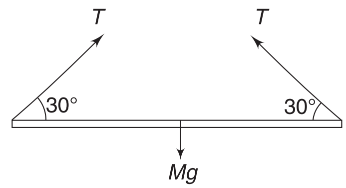
In equilibrium:

$$2T\sin 30° = Mg$$

$$\Rightarrow \quad T = Mg$$

Where spring 2 is cut, tension in 1 remains $T = Mg$

Let acceleration of COM be $\vec{a}_0 = a_x\hat{i} + a_y\hat{i}$



Let angular acceleration of the bar be α.

$$Ma_y = Mg - T\sin 30°$$

$$\Rightarrow \quad a_y = \frac{g}{2}$$

$$Ma_x = T\cos 30°$$

$$a_x = \frac{\sqrt{3}}{2}g$$

$$\therefore \quad \vec{a}_0 = \frac{\sqrt{3}}{2}g\hat{i} + \frac{g}{2}\hat{j} \quad \text{[Ans.]}$$

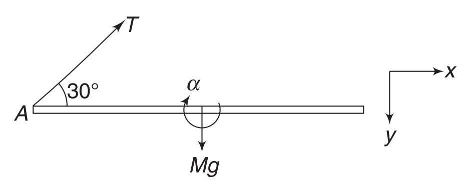
Writing $\tau = I\alpha$ about COM

$$(T\sin 30°)\left(\frac{L}{2}\right) = \frac{ML^2}{12}\alpha \quad \Rightarrow \quad \alpha = \frac{3g}{L}$$

Acceleration of end A wrt COM is

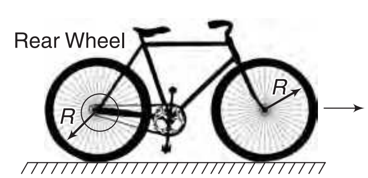
$$\vec{a}_{ACM} = \alpha\frac{L}{2}\,(\uparrow) = \frac{3g}{2}\,(\uparrow) = -\frac{3g}{2}\hat{j}$$

$$\therefore \quad \vec{a}_A = \vec{a}_{ACM} + \vec{a}_0 = \frac{\sqrt{3}}{2}g\hat{i} - g\hat{j}$$



Example 56 *Bicycle*

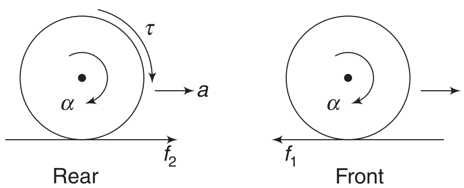
A bicycle is accelerating without slipping on a straight horizontal road. The mass of the bicycle along with the rider is M and radius of each wheel is R. The moment of inertia of each wheel about axle is I. The accelerating torque applied to the rear wheel by the pedal and gear system is τ. Find the acceleration of the bicycle.



Solution

Concepts

As the pedal–gear system applies torque on the rear wheel, it acquires angular acceleration (α). Friction force (f_2) on it must be in forward direction so as to impart a linear acceleration (a) to the wheel. The frame of the bicycle exerts a push on the axle of the front wheel. Thus, friction (f_1) on the front wheel must be in backward direction so as to provide it angular acceleration (α) and prevent the wheel from slipping.



For rear wheel, we can write

$$\tau - f_2 R = I \cdot \alpha \qquad \text{...(i)}$$

For front wheel, we can write

$$f_1 R = I . \alpha \qquad \text{...(ii)}$$

For translation of the complete bicycle, we have

$$f_2 - f_1 = Ma$$

Using (i) and (ii), we get

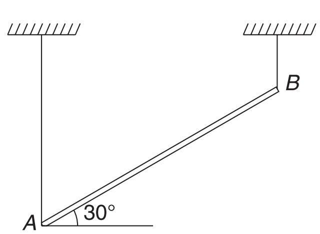
$$\frac{\tau}{R} - \frac{I\alpha}{R} - \frac{I\alpha}{R} = Ma$$

$$\Rightarrow \quad \frac{\tau}{R} = Ma + \frac{2I\alpha}{R}$$

$$\Rightarrow \quad \frac{\tau}{R} = Ma + \frac{2Ia}{R^2} \qquad [\because \ a = R\alpha]$$

$$\Rightarrow \quad a = \frac{\tau/R}{M + \dfrac{2I}{R^2}}$$

Example 57 A thin uniform bar AB of mass M and length $2L$ is held at an angle of 30° to the horizontal by means of two vertical strings at each end. The string at the right end breaks. Find tension in the left string immediately after the right string breaks.



Solution

Concepts

(i) The stick experiences only vertical forces. Its COM will have vertical acceleration only.

(ii) Acceleration of end A will be directed perpendicular to the string. End A cannot have vertically downward acceleration since string is inextensible.

Let a = Vertically downward acceleration of COM and
α = angular acceleration of the stick

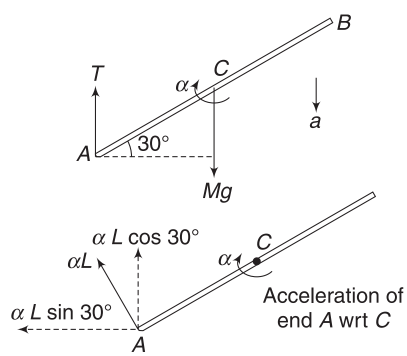
Using $\quad F = Ma_{CM}$

$$Mg - T = Ma \qquad \text{...(i)}$$

Using $\quad \tau = I\alpha$ about COM

$$T . L\cos 30° = \frac{M(2L)^2}{12} \cdot \alpha$$

$$\Rightarrow \quad T = \frac{2ML\alpha}{3\sqrt{3}} \qquad \text{...(ii)}$$



Acceleration of end A relative to COM is αL in a direction perpendicular to the rod. Vertical component of this acceleration is $\alpha L \cos 30°$.

For A to have no acceleration in ground frame

$$\alpha L \cos 30° = a \quad \Rightarrow \quad \frac{\sqrt{3}}{2}\alpha L = a \qquad \text{...(iii)}$$

Putting this in (ii) gives $T = \dfrac{4Ma}{g}$ ...(iv)

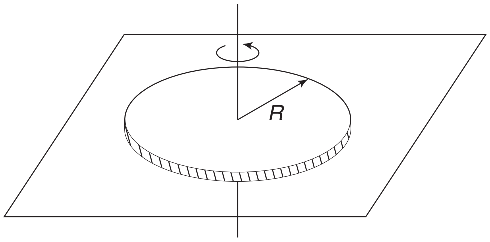
Put this in (i) $Mg - \dfrac{4Ma}{9} = Ma$

$$\Rightarrow \quad a = \frac{9g}{13}$$

$$\therefore \quad T = \frac{4M}{9} \cdot \frac{9g}{13} = \frac{4Mg}{13}$$

Example 58 *A flat spinning disc*

A uniform disc of radius R is spun to an angular velocity ω_0 about its central axis normal to its plane and is then placed carefully on a horizontal surface, as shown. How long will the disc be rotating if friction coefficient between the disc and the surface is μ? Pressure exerted by the disc on the surface can be regarded as uniform.



Solution

Concepts

(i) Friction acting on the disc produces a retarding torque. Friction is kinetic and is directed opposite to the velocity of a point.

(ii) We will consider a ring-shaped element and write torque on it due to friction. Integration to cover all such ring elements gives total torque on the disc.

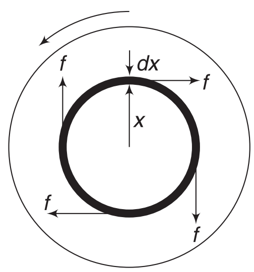
Mass per unit area $\sigma = \dfrac{M}{\pi R^2}$

Consider a ring of radius x and width dx.

Area of ring, $dA = 2\pi x\,dx$

Mass of ring, $dm = \sigma\, dA$

$$= \frac{2M}{R^2}\, x\,dx$$



Top view
Direction of friction at every point is opposite to the velocity of the point

Normal force on the ring element is

$$dN = (dm)g = \frac{2Mg}{R^2}\, x\,dx$$

Friction will have a torque on this ring element given by

$$d\tau_f = (\mu\, dN)\cdot x = \frac{2Mg\mu}{R^2}\, x^2 dx$$

Total torque on disc $\tau_f = \displaystyle\int d\tau_f = \frac{2Mg\mu}{R^2}\int_0^R x^2 dx = \frac{2Mg\mu R}{3}$

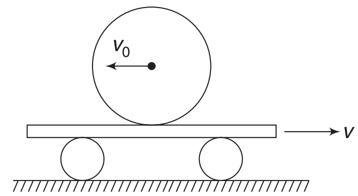
Moment of inertia of disc about rotation axis, $I = \dfrac{MR^2}{2}$

Angular retardation is $\alpha = \dfrac{\tau_f}{I} = \dfrac{4}{3}\dfrac{\mu g}{R}$

Using $\omega = \omega_0 + \alpha t$ gives

$$0 = \omega_0 - \frac{4\mu g}{3R}.t \quad \Rightarrow \quad t = \frac{3\omega_0 R}{4\mu g}$$

Example 59 A sphere of radius R and mass M has linear velocity v_0 directed to the left (and no angular velocity) as it is placed on a flat car moving to right with a constant velocity v. After sliding on the car for some time, the sphere finally stops slipping.Length of the car is sufficiently large. Coefficient of kinetic friction between the sphere and the car is μ. Find v_0 in terms of v, if the slipping ceases at the moment velocity of the sphere relative to ground is zero. Also find the time when slipping ceases.



Solution

Concepts

(i) Initially, the contact point has a velocity towards left relative to the car. Friction acts towards right.

(ii) v_0 decreases and the sphere acquires an angular speed (ω).

(iii) At some later time, speed of (centre of) the sphere is v_1 and it has an angular speed ω (anticlockwise). The contact point has a velocity $\omega R - v_1$ towards right.

(iv) Slipping stops when velocity of contact point becomes equal to the velocity of the car, i.e.,

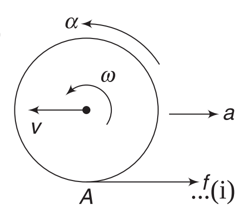
$$\omega R - v_1 = v$$

(v) Friction is kinetic during this entire period.

Acceleration of sphere is $a = \dfrac{f}{M}(\rightarrow)$

Angular acceleration (α) of the sphere is

$$\alpha = \frac{\tau}{I} = \frac{fR}{\frac{2}{5}MR^2} = \frac{5f}{2MR}\ (\circlearrowleft)$$



Velocity at time t is $v = v_0 - at$

$$v = v_0 - \frac{f}{M}t \qquad \text{...(i)}$$

Angular speed at time t is $\omega = \alpha t = \dfrac{5ft}{2MR}\ (\circlearrowleft)$

Velocity of contact point at time t is:

$$v_A = (\omega R - v)(\rightarrow) = \frac{5ft}{2M} - \left(v_0 - \frac{ft}{M}\right)$$

$$= \frac{7}{2}\frac{ft}{M} - v_0\ (\rightarrow)$$

Slipping ceases when $v_A = v$

$$\Rightarrow \qquad \frac{7}{2}\frac{ft}{M} - v_0 = v$$

But, $$f = \mu N = \mu Mg$$

$$\therefore \quad \frac{7}{2}(\mu g t) - v_0 = v \quad \Rightarrow \quad t = \frac{2(v + v_0)}{7\mu g} \qquad \text{...(ii)}$$

According to the question, slipping ceases when $v = 0$

$$\Rightarrow \quad v_0 - \frac{f}{M}t = 0 \qquad \Rightarrow \quad t = v_0\frac{M}{\mu Mg} = \frac{v_0}{\mu g}.$$

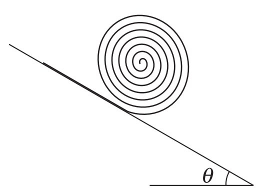
$$\therefore \quad \frac{v_0}{\mu g} = \frac{2(v + v_0)}{7\mu g} \quad \Rightarrow \quad 7v_0 = 2v + 2v_0$$

$$\Rightarrow \quad v_0 = \frac{2v}{5}$$

From (ii) $$t = \frac{2\left(v + \frac{2v}{5}\right)}{7\mu g} = \frac{2v}{5\mu g}$$

Example 60 *Unwinding tape roll on an incline*

A length L of a flexible tape is tightly wound. The tape is allowed to unwind as it rolls down an incline making an angle θ with horizontal, the free end of the tape is held fixed on the incline. In what time the tape unwinds completely?



Solution

Concepts

(i) The unrolling tape is a system of varying mass. But the outgoing mass does not apply any thrust force since its velocity (relative to contact point) is zero.

(ii) The rolling cylinder has decreasing mass and radius but its acceleration remains same. Acceleration does not depend on size or mass. It depends on shape only.

$$a = \frac{g\sin\theta}{1 + \frac{K^2}{R^2}}$$

for cylinder $$\frac{K^2}{R^2} = \frac{1}{2}$$

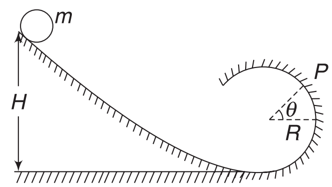
$$\therefore \quad a = \frac{2g\sin\theta}{3}$$

The tape unwinds completely when distance travelled by rolling cylinder is L.

$$L = \frac{1}{2}at^2 \quad \Rightarrow \quad t = \sqrt{\frac{2L}{a}} = \sqrt{\frac{3L}{g\sin\theta}}$$

Example 61 *Rolling ball in a vertical circle*

A small solid ball of mass m starts to roll down a loop track from height H, as shown in figure. The circular part of the track has radius R. The ball does not slip anywhere.



(a) Find kinetic energy of the ball when it reaches point P where radius makes an angle θ with horizontal.

(b) Find the radial and tangential acceleration of the COM of the ball when it is at P.

(c) Find the normal force and friction force applied by the track on the ball when it is at P if

$$H = 1\text{m},\ R = 0.1\text{m},\ \theta = 0° \text{ and } m = 0.1 \text{ kg}.$$
$[g = 9.8\,\text{ms}^{-2}]$

Solution

Concepts

(i) Mechanical energy of the ball is conserved. Friction is static (as the ball does not slip) and it does not dissipate energy.

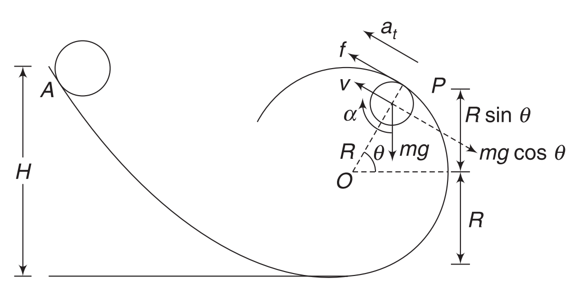
(ii) The COM of the ball is moving on a circle of radius $R - r \simeq R$ [$\because$ Radius r of the ball is very small] when the ball is on circular track.

(iii) Radial acceleration of COM is $\frac{v^2}{R}$ and its tangential acceleration is dv/dt where v is speed of the COM.

(iv) Normal force provides the radial acceleration to the COM and tangential component of mg and friction force decide the tangential acceleration of COM.

(a) Kinetic energy at P = Loss in potential energy between A and P.

$$k_P = mg\,[H - R(1 + \sin\theta)]$$



(b) At P, let the velocity of centre of the sphere be v. The ball is rolling and it has an angular speed also. Its KE is $k_P = \frac{7}{10}mv^2$ [Refer Example 27]

$$\therefore \quad \frac{7}{10}mv^2 = mg\,[H - R(1 + \sin\theta)]$$

$$\Rightarrow \quad v^2 = \frac{10g}{7}[H - R(1 + \sin\theta)] \qquad \text{...(i)}$$

COM is going in a circle of radius $R - r \simeq R$

$\therefore$ Radial acceleration is $a_r = \dfrac{v^2}{R}$

$$\Rightarrow \quad a_r = \frac{10g}{7}\left[\frac{H}{R} - (1 + \sin\theta)\right]$$

Forces acting in tangential direction are $mg\cos\theta$ and friction (f). If a_t is acceleration of COM along tangential direction:

$$ma_t = f - mg\cos\theta \qquad \text{...(ii)}$$

For rotation about COM

$$\left(\frac{2}{5}mr^2\right)\alpha = -fr$$

[Notice that for pure rolling, angular acceleration must be clockwise, if tangential acceleration is assumed up along the track. In fact, a_t is down the track (slowing the ball) and α is anti-clockwise (slowing the spin). But it hardly makes any difference to our analysis if we take them in directions indicated. Mathematics corrects the directions and gives negative values of a_t and α, as you can see below]

$$\Rightarrow \quad \frac{2}{5}m(r\alpha) = -f \quad \Rightarrow \quad \frac{2}{5}m\,a_t = -f \qquad \text{...(iii)}$$

$$[\because \; a_t = r\alpha]$$

Solving (ii) and (iii) gives

$$a_t = -\frac{5}{7}g\cos\theta$$

Alternative method to find a_t

$$v = \sqrt{\frac{10g}{7}[H - R(1 + \sin\theta)]}$$

$$a_t = \frac{dv}{dt} = \sqrt{\frac{10g}{7}}\,\frac{-R\cos\theta}{2\sqrt{H - R(1 + \sin\theta)}}\,\frac{d\theta}{dt}$$

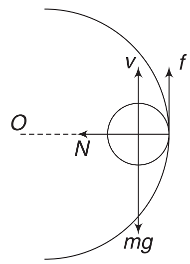
$$= -\sqrt{\frac{10g}{7}}\,\frac{1}{2}\,\frac{\cos\theta}{\sqrt{H - R(1 + \sin\theta)}}\,\frac{d}{dt}(R\theta)$$

Look at the figure: $\dfrac{d\theta}{dt}$ is angular speed of COM about O. $R\dfrac{d\theta}{dt}$ is linear speed (v) of the COM

$$\therefore \; a_t = -\sqrt{\frac{10g}{7}}\,\frac{1}{2}\,\frac{\cos\theta}{\sqrt{H - R(1 + \sin\theta)}}\,\sqrt{\frac{10g}{7}[H - R(1 + \sin\theta)]}$$

$$= -\frac{5}{7}g\cos\theta \qquad \text{...(iv)}$$

(c) At $\theta = 0°$



$$a_r = \frac{10g}{7}\left[\frac{H}{R} - 1\right]$$

Normal force is

$$N = \frac{mv^2}{R} = \frac{10}{7}mg\left[\frac{H}{R} - 1\right]$$

$$= \frac{10}{7} \times 0.1 \times 9.8\left[\frac{1}{0.1} - 1\right] = 12.6\text{N}$$

$$a_t = -\frac{5g}{7} \quad [\text{Put } \theta = 0° \text{ in (iv)}]$$

$$\therefore \qquad m\,a_t = f - mg$$

$$\Rightarrow \qquad -\frac{5}{7}mg = f - mg$$

$$\Rightarrow \qquad f = \frac{2}{7}mg = \frac{2}{7} \times 0.1 \times 9.8 = 0.28 \text{ N}$$

Example 62 *Falling rod on a smooth table*

A rod of mass M and length L, initially upright on a smooth table, starts falling when disturbed slightly. Find the speed of COM of the rod as a function of angle θ it makes with the vertical.

Solution

Concepts

(i) In absence of friction, there is no horizontal force on the rod. The COM has no horizontal acceleration. It falls on a vertical line.

(ii) The rod rotates about its COM as the COM falls. There is a fixed geometric relation between the fall of COM and angular displacement of the rod. This will helps us relate speed of COM ($= v$) and angular speed (ω).

(iii) Mechanical energy of the falling rod is conserved.

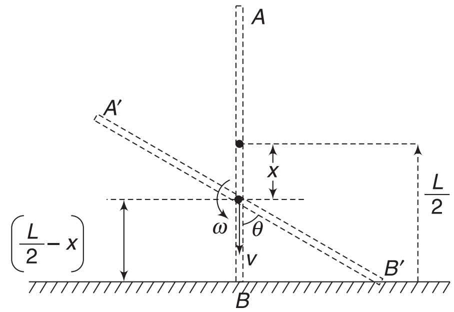
(iv) KE of the rod $= k_R + k_T$.

Figure shows the rod (AB) in its initial vertical position and any later position ($A'\,B'$) making an angle θ with the vertical. Let v be velocity of COM of the rod and ω be its angular velocity in position $A'\,B'$.

From figure: $x = \dfrac{L}{2} - \dfrac{L}{2}\cos\theta$

$$\Rightarrow \qquad \frac{dx}{dt} = \frac{L}{2}\sin\theta\,\frac{d\theta}{dt}$$

$$\Rightarrow \qquad v = \omega\frac{L}{2}\sin\theta \qquad \text{...(i)}$$



$v = \frac{dx}{dt}$ and $\omega = \frac{d\theta}{dt}$

Kinetic energy of the rod in position $A'B'$ = Loss in PE between position AB and $A'B'$

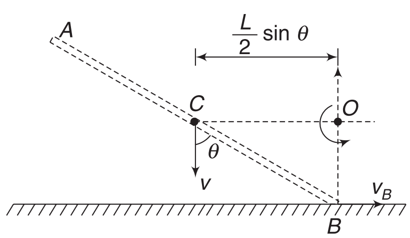
$$\frac{1}{2} Mv^2 + \frac{1}{2} I_{CM} \cdot \omega^2 = Mg\,x$$

$$\Rightarrow \qquad \frac{1}{2} Mv^2 + \frac{1}{2}\left(\frac{ML^2}{12}\right)\left(\frac{2v}{L\sin\theta}\right)^2 = Mg\,\frac{L}{2}[1 - \cos\theta]$$

$$\Rightarrow \qquad v = \sqrt{\frac{3gL(1 - \cos\theta)\,\sin^2\theta}{3\sin^2\theta + 1}}$$

Alternate way of writing KE

We can write KE of the rod by assuming it to be in pure rotation about its instantaneous axis of rotation. Velocity of point C (COM) is vertically down. The instantaneous centre of rotation will be on a line that is ⊥ to velocity of C. Similarly, point B is moving horizontally and instantaneous centre will be on a line that is ⊥ to v_B. [In fact, velocity of any point is normal to the line joining the instantaneous centre to the point]. O (see fig.) is the instantaneous centre of rotation. Axis is a line normal to the figure passing through O. MI of the rod about this axis is



$$I = \frac{ML^2}{12} + M\left(\frac{L}{2}\sin\theta\right)^2 = \frac{ML^2}{12}(1 + 3\sin^2\theta)$$

∴ KE of the rod is

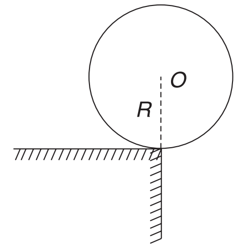
$$k = \frac{1}{2} I\omega^2 = \frac{ML^2}{24}(1 + 3\sin^2\theta)\,\omega^2$$

When you equate this KE to loss in PE

$$= Mg\,\frac{L}{2}\,(1 - \cos\,\theta], \text{ you get } \omega.$$

v can be calculated as $v = \omega\frac{L}{2}\sin\theta$

Example 63 *Rolling off an edge*



A solid homogeneous cylinder of radius R is placed horizontally at rest with its length parallel to the edge off the table. There is sufficient friction at the edge so that a small displacement causes the cylinder to roll off the edge without slipping. Determine–

(a) The angle θ_o through which the cylinder rotates before it leaves contact with the edge.

(b) The speed of COM of the cylinder just before leaving contact.

(c) The ratio of translational to rotational kinetic energy of the cylinder when its COM is in horizontal line with the edge.

Solution

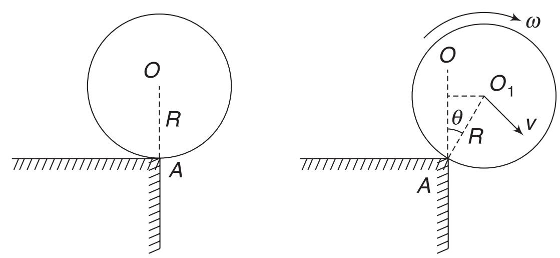
Concepts

(i) No slipping implies that friction does not perform any work. Mechanical energy of the cylinder remains conserved.

(ii) As long as the cylinder is in contact with the edge, its COM is rotating in a circle of radius R.

(iii) Writing equation for centripetal force (for motion of COM) and putting normal reaction = 0, will give us the position where the cylinder leaves contact.

(iv) After leaving contact, the rotational KE will not change as there is no torque about the COM.



Position of cylinder after it has rotated through θ

Consider the cylinder when it has rotated by an angle θ. It is in pure rotation about the edge of the table. Friction at A is preventing it from slipping. The COM of the cylinder has descended by $(R - R\cos\theta)$.

Loss in PE has been converted into KE. KE of the cylinder can be written as $k = \frac{1}{2} Mv_{CM}^2 + \frac{1}{2} I_{CM}\,\omega^2$ or,

$$k = \frac{1}{2} I_A \cdot \omega^2$$

Using the later expression,

$$k = \frac{1}{2}\left[\frac{MR^2}{2} + MR^2\right]\omega^2 = \frac{3}{4} M(R\omega)^2$$

since, speed of COM is $v = R\omega$

$\therefore \qquad k = \frac{3}{4} Mv^2$

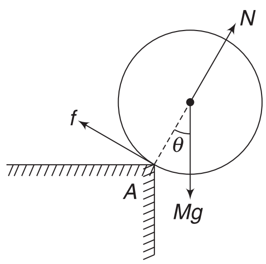
Using conservation of energy

$$\frac{3}{4}Mv^2 = MgR(1 - \cos\theta)$$

$$\Rightarrow \qquad \frac{v^2}{R} = \frac{4}{3} g(1 - \cos\theta) \qquad \text{...(i)}$$

(a) Forces on the cylinder are as shown. When it is on verge of leaving contact, $N = 0$.

COM is moving in a circle of radius R about A.



$$\therefore \quad Mg\cos\theta = \frac{Mv^2}{R}$$

Using (i) $Mg\cos\theta = \frac{4Mg}{3}(1 - \cos\theta)$

$$\Rightarrow \qquad \cos\theta = \frac{4}{7} \quad \Rightarrow \quad \theta = \cos^{-1}\left(\frac{4}{7}\right)$$

(b) From (i) Value of v when $\cos\theta = \frac{4}{7}$ is

$$v = \sqrt{\frac{4}{3}gR\left(1 - \frac{4}{7}\right)}$$

$$= \sqrt{\frac{4}{7}gR}$$

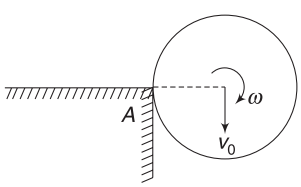
(c) At the instant the cylinder leaves contact

$$v = \sqrt{\frac{4}{7}gR} \quad \text{and} \quad \omega = \frac{v}{R} = \sqrt{\frac{4g}{7R}}$$

Rotational KE of the cylinder at this instant is

$$k_R = \frac{1}{2} I_{CM} \cdot \omega^2 = \frac{1}{2} \cdot \frac{1}{2} MR^2 \cdot \frac{4g}{7R} = \frac{MgR}{7}$$

Once the cylinder loses contact, Mg is the only force acting on it. There is no torque about the centre and k_R does not change. When COM is on a horizontal line through A:



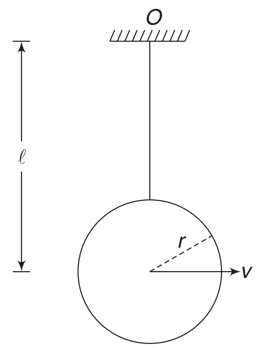
$$k_R + k_T = MgR \qquad [= \text{loss in } PE \text{ between initial and final position}]$$

$$\Rightarrow \qquad k_T = MgR - \frac{MgR}{7} = \frac{6}{7}MgR$$

$$\therefore \qquad \frac{k_T}{k_R} = 6$$

Example 64 *You may miss to see this rotation!*

A sphere of mass m and radius r is suspended from point O using a light thread. Distance of its centre from point O is l. The sphere is pushed so as to impart its centre a horizontal velocity v. What is the kinetic energy of the sphere? Can you write this KE using equation (11), i.e.,



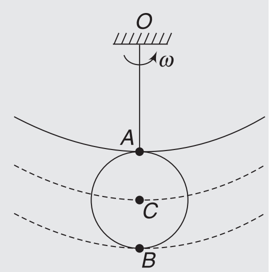
$$k = \frac{1}{2}I_{CM} \cdot \omega^2 + \frac{1}{2} mv_{CM}^2$$

Solution

Concepts

(i) KE is not equal to $\frac{1}{2} mv^2$.

All particles in the sphere do not have the same velocity. For example, point A will have velocity $\omega(l - r)$ as it moves in a circle of radius $(l - r)$. Point B will have velocity $\omega(l + r)$ as it moves on a circle of radius $(l + r)$.



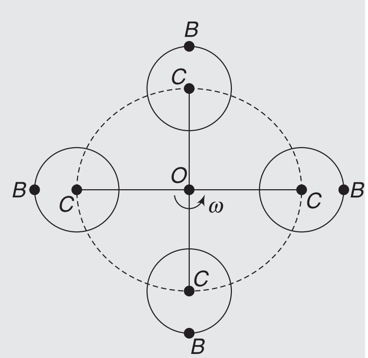
(ii) $\omega = \frac{v}{l}$ = angular speed about O.

(iii) Body is in pure rotation about an axis through O that is perpendicular to the plane of the figure.

$$\therefore \qquad k = \frac{1}{2}I_0\omega^2$$

Where, I_0 = MI about the axis through O.

(iv) An observer located at COM (C) sees that the ball is spinning about C. Look at the given figure. As the ball completes one revolution, an observer at C finds that a point (like B) spins about it. Point B is below C, then it is to right of C, then it is above C, then it is to the left of C and then again it is below C. Clearly, a point like B is spinning around C. Angular speed of this spin is same as the angular speed of the body about O.



Method I

Body is rotating about O.

$$\therefore \qquad k = \frac{1}{2}I_0\omega^2 = \frac{1}{2}\left[\frac{2}{5}mr^2 + ml^2\right]\omega^2$$

$$= \frac{1}{2}\left[\frac{2}{5}mr^2 + ml^2\right]\frac{v^2}{l^2}$$

$$= \frac{1}{2}mv^2\left[\frac{2}{5}\frac{r^2}{l^2} + 1\right]$$

Method II

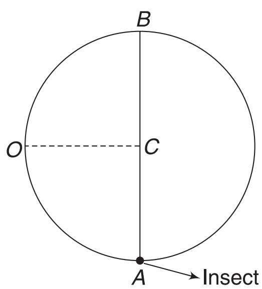
$$k = \frac{1}{2}I_{CM}\cdot\omega^2 + \frac{1}{2}mv_{CM}^2$$

$$= \frac{1}{2}\left(\frac{2}{5}mr^2\right)\omega^2 + \frac{1}{2}mv^2 \qquad \left[\because \omega = \frac{v}{l}\right]$$

$$= \frac{1}{2}mv^2\left[\frac{2}{5}\frac{r^2}{l^2} + 1\right]$$

Example 65 *Crawling insect*

A frame made of a uniform wire of linear mass density ρ is in the shape of a circle. The circle has a diameter AB made of same wire. The frame lies on a smooth table and can rotate freely about a vertical axis through point O on the circumference. An insect of mass $m_o = \frac{R\rho}{3}$ [where R is radius of the circle] is initially at rest at A and begins to crawl along the diameter AB. By what angle does the frame rotate by the time the insect reaches B?



OC is $\perp$ to AB

Solution

Concepts

(i) Angular momentum of the frame + insect system is conserved about the vertical axis through O.

(ii) As the insect moves from A to B, it generates an angular momentum about O in anti-clockwise direction. The frame must rotate clockwise so as to have equal and opposite angular momentum. This will keep the total angular momentum = 0

Mass of circle, $M = 2\pi R\rho$

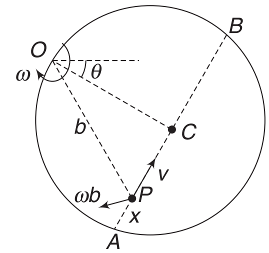
Mass of diameter AB, $m = 2R\rho$

MI of the frame about vertical axis through O is

$$I = 2MR^2 + \frac{m(2R)^2}{12} + mR^2 = 4\pi R^3\rho + 2R^3\rho\frac{4}{3}$$

$$= 2R^3\rho\left(2\pi + \frac{4}{3}\right)$$

Let the insect be at P at any time t and its velocity relative to the frame be v.



If $\qquad AP = x$, then

$$v = \frac{dx}{dt}$$

At this moment, angular speed of the frame is ω about O. If θ is the angle by which the frame has rotated by now, then

$$\omega = \frac{d\theta}{dt}$$

Angular momentum of frame $L_{frame} = I\omega(\circlearrowright)$

$$= 2R^3\rho\left(2\pi + \frac{4}{3}\right)\omega\,(\circlearrowright)$$

Actual velocity of insect (in ground frame) = v (along AB) + ωb (perpendicular to AB)

$\therefore \quad L_{insect} = L$ (Due to v along AB) + L (Due to ωb perpendicular to AB)

$$= m_o vR\,(\circlearrowleft) + m_o(\omega b)\,b\,(\circlearrowright)$$

$$= m_o vR - m_o\omega b^2\,(\circlearrowleft)$$

But $\qquad L_{insect} + L_{frame} = 0$

$$\Rightarrow \quad m_o vR - m_o\omega b^2 = 2R^3\rho\left(2\pi + \frac{4}{3}\right)\omega$$

$$\Rightarrow \quad \frac{1}{3}\rho R\left[R\frac{dx}{dt} - \omega\{R^2 + (R - x^2)\}\right] = 2R^3\rho\left(2\pi + \frac{4}{3}\right)\omega$$

$$\Rightarrow \quad \frac{1}{3}\rho R^2\frac{dx}{dt}$$

$$= \left[2R^3\rho\left(2\pi + \frac{4}{3}\right) + \frac{\rho R}{3}[R^2 + (R - x^2)]\right]\frac{d\theta}{dt}$$

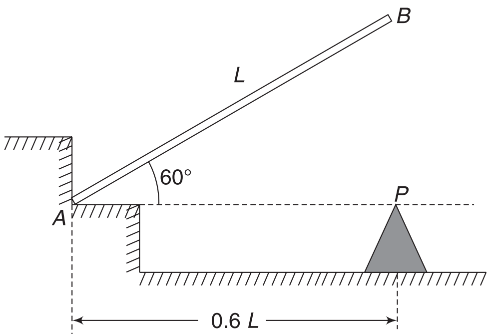
$$\Rightarrow \quad R\frac{dx}{dt} = [n^2R^2 + (R - x)^2]\frac{d\theta}{dt} \quad [\text{where } n^2 = 12\pi + 9]$$

$$\Rightarrow \quad R\,dx = [n^2R^2 + (R - x)^2]\,d\theta$$

$$\Rightarrow \quad \int_0^{\theta_0} d\theta = \int_0^{2R} \frac{R}{n^2R^2 + (R - x)^2}\,dx$$

$$\Rightarrow \quad \theta_0 = \frac{1}{n}\tan^{-1}\left(\frac{2n}{n^2 - 1}\right)$$

Example 66 A rod of mass M and length L is positioned as shown in figure. Surfaces at A are smooth. The rod is released from the shown position. The rod hits the peg P and sticks to it. Find the angular speed of the rod after it hits the peg at P.



$AP = 0.6L$ where L is length of the rod. [Note that the rod is not hinged at A]

Solution

Concepts

(i) The rod is in pure rotation about A till it hits P. Using energy conservation, we can find angular speed of the rod just before the collision.

(ii) Angular momentum about P remains unchanged during collision. The impact force does not produce any impulse about P.

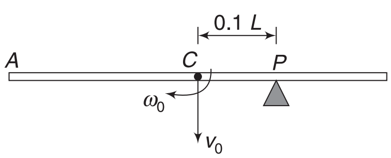
Let ω_0 = angular speed of the rod just before hitting P. Conservation of energy gives

Loss in PE = Gain in KE

$$Mg\frac{L}{2}\sin 60° = \frac{1}{2}\frac{ML^2}{3}\cdot\omega_0^2$$

[Rod is in pure rotation about A]

$$\Rightarrow \qquad \omega_0 = \sqrt{\frac{3\sqrt{3}}{2}\frac{g}{L}} \qquad \text{...(i)}$$



Angular momentum of the rod about P, just before impact is $L_{\text{before collsion}} = L_{\text{spin}} + L_{\text{orbitals}}$

$$= I_{CM}\cdot\omega_0\ (\circlearrowright) + Mv_0(0.1\,\text{L})\ (\circlearrowleft)$$

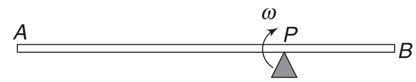
Note that motion of COM produces an anti-clockwise angular momentum about P.

Also, $\qquad v_0 = \omega_0\dfrac{L}{2}$

$$\therefore L_{\text{before collision}} = \frac{ML^2}{12}\omega_0 - M\frac{\omega_0 L}{2}[0.1L]$$

$$= ML^2\omega_0\left(\frac{1}{12} - 0.05\right) = \frac{1}{30}ML^2\omega_0\ (\circlearrowright)$$

After collision, the rod sticks at P and begins to rotate about P with angular speed ω.



Angular momentum about P after collision is

$$L_{\text{after collision}} = I_P\cdot\omega$$

$$= \left[\frac{ML^2}{12} + M(0.1L)^2\right]\omega = \frac{7}{75}ML^2\omega$$

$$L_{\text{after collision}} = L_{\text{before collision}}$$

$$\frac{7}{75}ML^2\omega = \frac{1}{30}ML^2\omega_0$$

$$\Rightarrow \qquad \omega = \frac{5}{14}\omega_0 = \frac{5}{14}\sqrt{\frac{3\sqrt{3}}{2}\frac{g}{L}}$$

Example 67 A uniform rod of mass m and length l rests on a smooth horizontal surface. One of the ends of the rod is given a sharp blow in horizontal direction at right angles to the rod. As a result, the centre of the rod beings to move with velocity v_0. Find the force with which one half of the rod will act on the other half during the course of subsequent motion.

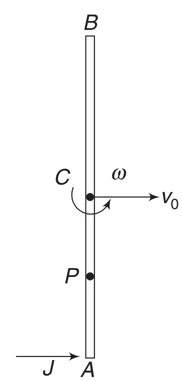
Solution

Concepts

(i) The centre moves with velocity v_0 and the entire rod spins around it with some angular velocity ω.

(ii) ω can be calculated using angular – impulse momentum theorem.

(iii) After the hit, the COM is moving with constant velocity and an RF attached to it is inertial. In this frame, each particle of the rod is seen rotating at angular speed ω. The centripetal force necessary for rotation of one half of the rod is provided by the other half.



If $\qquad J$ = impulse given, then

$$J = mv_0 \qquad \text{...(i)}$$

Angular impulse about COM is

$$J\frac{l}{2} = I_{CM}\cdot\omega$$

$$\Rightarrow \qquad mv_0\frac{l}{2} = \frac{ml^2}{12}\omega$$

$$\Rightarrow \qquad \omega = \frac{6v_0}{l}$$

In RF attached to C, segment AC is seen rotating with angular speed ω. COM of segment AC is at P.

$$C_P = \frac{l}{4}$$

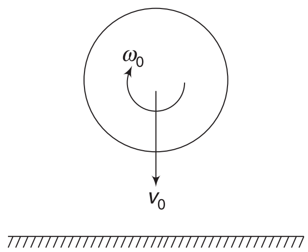
$\therefore$ Centripetal force needed by AC is

$$F = \frac{m}{2}\omega^2\frac{l}{4} = \frac{1}{8}ml\left(\frac{6v_0}{l}\right)^2 = \frac{9}{2}\frac{mv_0^2}{l}$$

This force is applied by BC on AC.

Example 68 *Spinning ball hits a rough floor*

A ball of radius r and mass m is spinning about its centre at an angular speed ω_0. It falls vertically and hits a solid floor while its centre was travelling at speed v_0. Coefficient of friction and coefficient of restitution are $\mu = 0.2$ and $e = 1$ respectively. Find the kinetic energy of the ball immediately after impact. Take $r\omega_0 = 2v_0$



Solution

Concepts

(i) $e = 1$, implies that the centre of the ball will have a vertical velocity $v_0(\uparrow)$ after the impact.

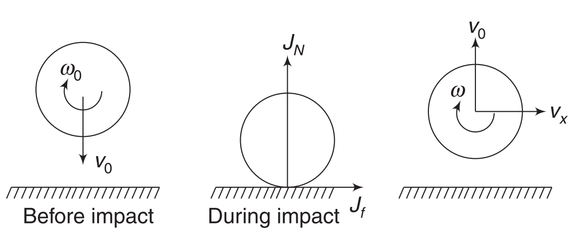
(ii) Normal force by the floor is impulsive. Since the ball is spinning, the contact point will slide on the floor and there will be a friction on the ball towards right. This friction is impulsive since N is impulsive.

(iii) Impulse of friction imparts horizontal velocity to the ball.

(iv) Friction also imparts an angular impulse. This changes the angular speed of the ball.

If J_N is impulse of normal force then

$$J_N = 2mv_0 \qquad \text{...(i)}$$



Impulse of friction $J_f = \mu J_N = 2\mu mv_0$ $\quad [\because f = \mu N]$

$$\therefore \quad mv_x = 2\mu mv_0$$

$$\Rightarrow \quad v_x = 2\mu v_0 = 0.4\, v_0 \qquad \text{...(ii)}$$

Angular impulse due to friction = $r\, J_f(\circlearrowleft)$

$$\therefore \quad I\omega_0 - rJ_f = I\omega$$

$$\Rightarrow \quad \frac{2}{5}mr^2\omega_0 - r(0.4mv_0) = \frac{2}{5}mr^2\cdot\omega$$

$$\Rightarrow \quad r\omega_0 - v_0 = \omega r \quad \Rightarrow \quad 2v_0 - v_0 = \omega r$$

$$\Rightarrow \quad \omega = \frac{v_0}{r}$$

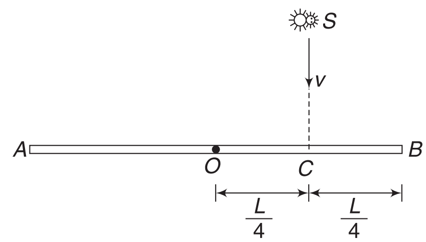
$$\therefore \quad KE = k_T + k_R$$

$$= \frac{1}{2}m(v_0^2 + v_x^2) + \frac{1}{2}I_{CM}\cdot\omega^2$$

$$= \frac{1}{2}m(v_0^2 + 0.16\, v_0^2) + \frac{1}{2}\times\frac{2}{5}mr^2\left(\frac{v_0}{r}\right)^2$$

$$= 0.78\, mv_0^2$$

Example 69 A homogeneous rod AB of length $L = 1.8\ m$ and mass M is pivoted at centre O is such a way that it can rotate freely in vertical plane. The rod is initially in horizontally position. An insect (S) of same mass M falls vertically with speed v at point C on the rod. Point C is at a distance $\frac{L}{4}$ from centre (O). Immediately after falling, the insect moves towards end B so that the rod rotates with constant angular velocity ω.



(a) Find ω in terms of v and L

(b) If the insect reaches end B when the rod has turned by 90° determine v.

Solution

Concepts

(i) Weight of the insect produces a torque about O. This will cause the angular momentum of the system to increase.

(ii) Angular momentum is increasing but the angular speed can be maintained constant by increasing moment of inertia. As the insect moves towards B, MI of the system about rotation axis increases.

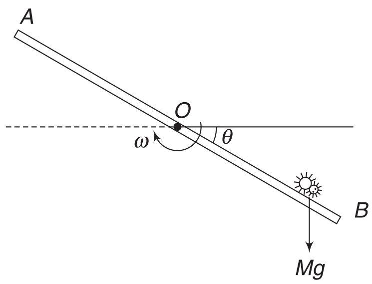
Let us denote angular momentum by A ($\because L$ is used for length of rod)

Angular momentum about O before the insect hits the rod = angular momentum just after the insect hits the rod.

$$\therefore \quad Mv\frac{L}{4} = \left[\frac{ML^2}{12} + M\left(\frac{L}{4}\right)^2\right]\omega$$

$$\Rightarrow \quad \omega = \frac{12v}{7L} \qquad \text{...(i)}$$

Let the insect be at a distance x from O when the rod has turned by an angle θ.



Torque on system about O is

$$\tau = Mgx\cos\theta$$

Angular momentum of the system about O at this moment is

$$A = \left[\frac{ML^2}{12} + Mx^2\right]\omega$$

Using $\quad \tau = \dfrac{dA}{dt}$

$$Mgx\cos\theta = 2Mx\omega\frac{dx}{dt} \qquad [\omega \text{ is a constant}]$$

$$\Rightarrow \quad dx = \left(\frac{g}{2\omega}\right)\cos\theta \cdot dt$$

$$\Rightarrow \quad \frac{dx}{d\theta}\frac{d\theta}{dt} = \frac{g}{2\omega}\cos\theta$$

$$\Rightarrow \quad \omega\frac{dx}{d\theta} = \frac{g}{2\omega}\cos\theta$$

$$\Rightarrow \quad \int_{L/4}^{L/2} dx = \frac{g}{2\omega^2}\int_0^{\pi/2}\cos\theta\, d\theta$$

$$\Rightarrow \quad \frac{L}{4} = \frac{g}{2\omega^2}[\sin\theta]_0^{\pi/2}$$

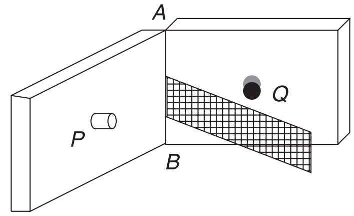
$$\Rightarrow \quad \omega = \sqrt{\frac{2g}{L}}$$

From (i) $\dfrac{12v}{7L} = \sqrt{\dfrac{2g}{L}} \quad \Rightarrow \quad v = \dfrac{7}{12}\sqrt{2gL}$

$$= \frac{7}{12}\sqrt{2 \times 10 \times 1.8}$$

$$= 3.5\,\text{ms}^{-1}$$

Example 70 Two heavy metallic plates are joined together at 90° to each other. A laminar sheet of mass 30 kg is hinged at the line AB joining the two heavy metallic plates. The hinges are frictionless. The moment of inertia of the laminar sheet about an axis parallel to AB and passing through its centre of mass is 1.2 kg m^2. Two rubber obstacles P and Q are fixed, one on each metallic plate at a distance 0.5 m from the line AB. This distance is chosen so that the reaction due to the hinges on the laminar sheet is zero during the impact. Initially, the laminar sheet hits one of the obstacles with an angular velocity 1 rad s^{-1} and turns back. If the impulse on the sheet due to each obstacle is 6 Ns.



(a) Find the location of the centre of mass of the laminar sheet from AB?

(b) At what angular velocity does the laminar sheet come back after the first impact?

(c) After how many impacts does the laminar sheet come to rest?

Solution

Concepts

(i) Hinges do not apply force. This means that impulse by an obstacle on the sheet is equal to change in linear momentum of the sheet.

(ii) We will also apply angular impulse-momentum theorem about AB.

Let $\quad r$ = distance of COM from line AB

ω = angular speed of the sheet after first collision with the rubber obstacle

Velocity of COM before and after collision are

$$v_i = r(1\text{ rad s}^{-1}) = r \quad \text{and} \quad v_f = r\omega$$

Linear impulse $(J) = \Delta P$

$$\Rightarrow \quad 6 = mv_f - (-mv_i) = m(v_f + v_i)$$

$$\Rightarrow \quad 6 = 30(r + r\omega) \quad \Rightarrow \quad r(1+\omega) = \frac{1}{5} \qquad \text{...(i)}$$

Angular impulse about $AB = \Delta L$

$$\Rightarrow \quad J \cdot r = I_{AB}\omega - (-I_{AB} \cdot 1\text{ rad s}^{-1})$$

$$\Rightarrow \quad 6 \times 0.5 = I_{AB}(\omega + 1)$$

$$\Rightarrow \quad 3 = (I_{CM} + mr^2)(\omega + 1)$$

$$\Rightarrow \quad 3 = (1.2 + 30r^2)(\omega + 1) \qquad \text{...(ii)}$$

Solving (i) and (ii) for r gives

$$r = 0.4\,m \quad \text{and} \quad r = 0.1\,m$$

For $\quad r = 0.4\,m$, $\omega < 0$ which is not possible

$\therefore \quad r = 0.1\,m$, [Ans to (a)]

Putting $\quad r = 0.1$ in (i) gives

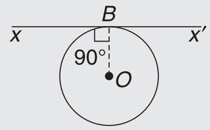
$\omega = 1\text{ rad s}^{-1}$ [Ans to (b)]

Since, the sheet returns with same angular velocity of 1 rad s^{-1}, it will make infinite number of collisions.

Worksheet 1

1. For the same total mass, which of the following will have the largest moment of inertia about an axis passing through its centre of mass and perpendicular to the plane of the body
   (a) a disc of radius a
   (b) a ring of radius a
   (c) a square lamina of side $2a$
   (d) four rods forming a square of side $2a$

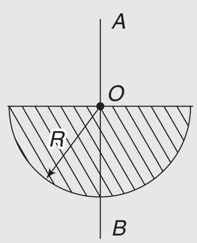
2. A thin wire of length L and uniform linear mass density ρ is bent into a circular loop with centre at O, as shown. The moment of inertia of the loop about the axis XX'



   (a) $\rho L^3/8\pi^2$ (b) $\rho L^3/16\pi^2$
   (c) $5\rho L^3/16\pi^2$ (d) $3\rho L^3/8\pi^2$

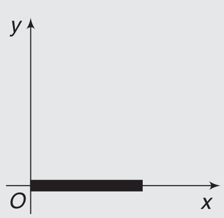
3. A thin uniform rod of mass M and length L has its moment of inertia I_1 about its perpendicular bisector. The rod is bent in the form of a semi-circular arc. Now, its moment of inertia through the centre of the semi-circular arc and perpendicular to its plane is I_2. The ratio of $I_1 : I_2$ will be
   (a) < 1 (b) > 1
   (c) $= 1$ (d) can't be said

4. Moment of inertia of a thin semicircular disc (mass = M and radius = R) about an axis AB through point O, is given by



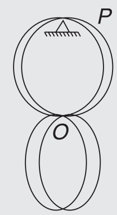
   (a) $\frac{1}{4}MR^2$
   (b) $\frac{1}{2}MR^2$
   (c) $\frac{1}{8}MR^2$
   (d) MR^2

5. The figure shows a uniform rod lying along the x-axis. Consider moment of inertia of the rod about different axes parallel to z-axis. All the axes about which the moment of inertia of the rod is same, interect the xy plane at points whose locus is



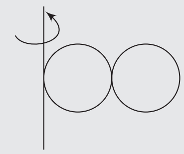
   (a) an ellipse (b) a circle
   (c) a parabola (d) a straight line

6. Two identical rings, each of mass m, with their planes mutually perpendicular, and radius R are welded at their point of contact O. If the system is free to rotate about an axis passing through the point P perpendicular to the plane of the upper ring and parallel to the plane of lower ring, the moment of inertia of the system about this axis is equal to



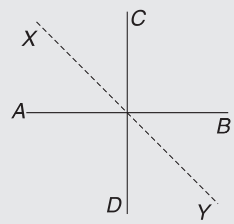
   (a) $6.5\ mR^2$ (b) $12\ mR^2$
   (c) $6\ mR^2$ (d) $11.5\ mR^2$

7. Two identical hollow spheres of mass M and radius R are joined together, and the combination is rotated about an axis tangent to one sphere and perpendicular to the line connecting their centres. The rotational inertia of the combination is



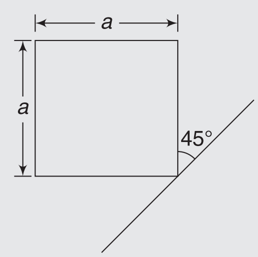
   (a) $10\ MR^2$ (b) $\frac{4}{3}MR^2$
   (c) $\frac{32}{3}\mathrm{MR}^2$ (d) $\frac{34}{3}MR^2$

8. AB and CD are two identical rods each of length L and mass M joined to form a cross. Find the MI of the system about a bisector of the angle between the rods (XY)



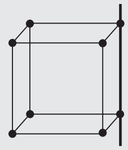
   (a) $\frac{ML^2}{12}$ (b) $\frac{ML^2}{6}$
   (c) $\frac{ML^2}{3}$ (d) $\frac{4ML^2}{3}$

9. The moment of inertia of a thin square sheet of mass M about the axis shown is



   (a) $\frac{7}{12}Ma^2$ (b) $\frac{5}{12}Ma^2$
   (c) $\frac{1}{3}Ma^2$ (d) $\frac{1}{12}Ma^2$

10. Eight point objects, each of mass M, are at the vertices of a rigid, massless frame in the form of a cube. Each side is of length a. What is the moment of inertia of this arrangement about an axis passing through one edge of the cube?

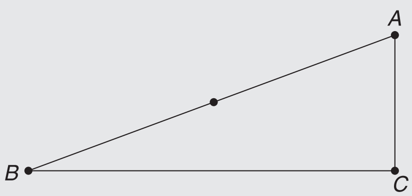


(a) $4Ma^2$ (b) $6Ma^2$
(c) $8Ma^2$ (d) $10Ma^2$

11. A rigid body has its COM on x-axis and can rotate about an axis that intersects the x-axis and is parallel to y-axis. When the axis is chosen to pass through a point on x-axis that is at a distance x from the origin of co-ordinate system, the moment of inertia is given by $I = 2x^2 - 12x + 27$. The x-coordinate of centre of mass of the body is

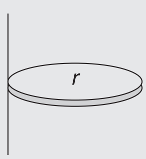
(a) $x = 2$ (b) $x = 0$
(c) $x = 1$ (d) $x = 3$

12. A flat triangular sheet of uniform material is shown in the drawing. Angle B is smaller than angle A. There are three possible axes of rotation, each perpendicular to the sheet and passing through one corner - A, B, or C. For which axis is the greatest external torque needed to impart the triangle an angular speed of 1.0 rad s^{-1} in 1.0 s, starting from rest?



(a) A (b) B
(c) C (d) All are equal

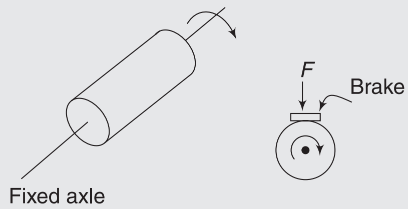
13. A solid sphere of radius R has moment of inertia I about its diameter. It is melted into a disc of radius r and thickness t. If its moment of inertia about the tangential axis (which is perpendicular to plane of the disc) is also equal to I, then the value of r and t are (respectively)



(a) $\frac{2}{15}R,\ 5R$ (b) $\frac{2}{\sqrt{5}}R,\ 2R$

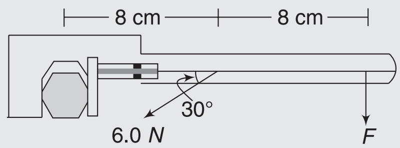
(c) $\frac{3}{\sqrt{15}}R,\ \frac{R}{2}$ (d) $\sqrt{\frac{3}{15}}R,\ \frac{R}{5}$

14. The figure shows a 15 kg solid cylinder mounted on a fixed axle, with a radius 25 cm, rotating at an angular speed of 500 rotations per minute. If a 100 N braking force is applied normal to the curved surface of the cylinder, bringing it to rest in 15 seconds, what is the approximate value of coefficient of kinetic friction between the brake shoe and the cylinder?



(a) 0.027 (b) 0.042
(c) 0.065 (d) 0.140

15. When a force of 6.0 N is exerted at 30° to a wrench at a distance of 8 cm from the nut, it is just able to loosen the nut. What force F would be sufficient to loosen it, if it acts perpendicularly to the wrench at 16 cm from the nut?

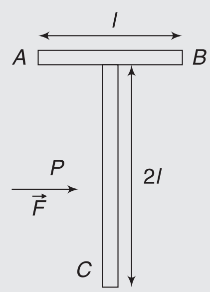


(a) 3 N (b) 1.5 N
(c) 6 N (d) 1 N

16. Two bodies have moment of inertia I and $2I$ respectively about their axis of rotation. If their kinetic energies of rotation, about fixed axes, are equal, their angular momenta will be in the ratio

(a) 2 : 1 (b) 1 : 2
(c) $\sqrt{2} : 1$ (d) $1 : \sqrt{2}$

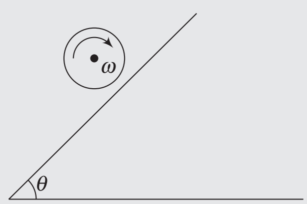
17. A T-shaped object is made by welding two rods of same linear mass density. It has dimensions shown in the figure. It is lying on a smooth floor. A force $\vec{F}$ is applied at the point P parallel to AB, such that the object has only translational motion without rotation. Find the distance of P from C



(a) $\frac{4l}{3}$ (b) l

(c) $\frac{2l}{3}$ (d) $\frac{3l}{2}$

18. A cylinder of radius R is spinned about its central axis with angular speed ω and then placed on an incline having inclination



angle θ. The cylinder continues to spin without falling. The cylinder can stay at the same location for a time

(a) $\dfrac{R\omega}{3g \sin \theta}$ (b) $\dfrac{R\omega}{2g \sin \theta}$

(c) $\dfrac{R\omega}{g \sin \theta}$ (d) $\dfrac{2R\omega}{g \sin \theta}$

19. A body rolls down an inclined plane without slipping. If its kinetic energy of rotation is 40% of its kinetic energy of translation, then the body is

(a) solid cylinder (b) solid sphere

(c) disc (d) ring

20. A body rolling without sliding has a fraction 'f' of its entire KE as rotational KE. Maximum possible value of 'f' is

(a) 1/2 (b) 1/3

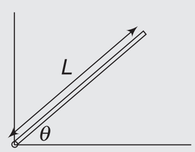
(c) 2/3 (d) 2/7

21. A particle having mass 2 kg is moving along straight line $3x + 4y = 5$ with speed 8 ms^{-1}. Find angular momentum of the particle about origin. x and y are in metres.

(a) 16 kgm^2s^{-1} (b) 8 kgm^2s^{-1}

(c) 36 kgm^2s^{-1} (d) none

22. A uniform thin pole of length L and mass M is pivoted on the ground with a frictionless hinge. The pole makes an angle θ with the horizontal. If it starts falling from the position shown in the accompanying figure, the linear acceleration of the free end of the pole immediately after release would be



(a) (2/3) $g \cos\theta$ (b) (2/3) g

(c) g (d) (3/2) $g \cos \theta$

23. If the earth shrinks to half the present radius such that mass remains constant, without any change in mass, then the duration of day and night becomes:

(a) 24 hours (b) 12 hours

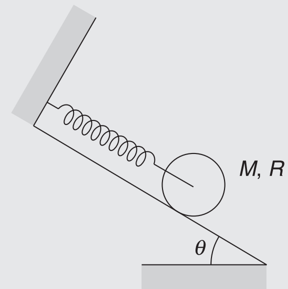
(c) 6 hours (d) 3 hours

24. At any time t, a particle of mass 0.01 kg has the position vector $\vec{r} = 3 \cos \theta\hat{i} + 4 \sin \theta\hat{j}$. Here, θ is the angle that the position vector makes with positive direction of x-axis. If its angular momentum with respect to origin is 0.6 kg $m^2s^{-1}\hat{k}$ and remain constant then angular displacement of the position vector of particle in time interval t is

(a) zero (b) $(5t)$

(c) $\sin^{-1}(0.1/t)$ (d) t

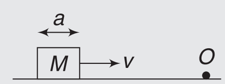
25. A uniform cylinder of mass M and radius R is placed on an incline of inclination θ. It can rotate freely about its central axle, which is connected to an ideal spring of force constant k. The cylinder is released from rest with the spring relaxed. Friction is large enough to prevent slipping. As the cylinder rolls down the inclined surface, the maximum elongation in the spring is



(a) $\dfrac{3}{4}\dfrac{Mg \sin \theta}{k}$ (b) $\dfrac{2Mg \sin \theta}{k}$

(c) $\dfrac{Mg \sin \theta}{k}$ (d) None of these

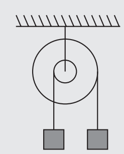
26. A cubical block of side a is moving with velocity v on a horizontal smooth plane as shown. It hits a ridge at point O. The angular speed of the block just after it hits O is



(a) $\dfrac{3v}{4a}$ (b) $\dfrac{3v}{2a}$

(c) $\dfrac{\sqrt{3}v}{\sqrt{2}a}$ (d) Zero

27. In the situation shown, a heavy wheel with a small drum attached at its centre is suspended by its frictionless axle from a ceiling. Attached to strings around the rims of the wheel and drum are two blocks of equal mass. The system is originally at rest. When the blocks are released



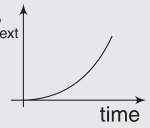
(a) Nothing moves, since the blocks have equal mass.

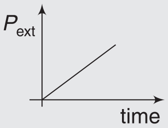
(b) The right-hand block falls and the left-hand one rises, with accelerations of the same magnitude.

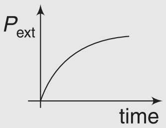
(c) While the blocks are moving, the tension in the right-hand string is less than that in the left-hand string.

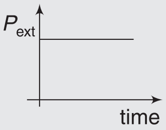
(d) Which block falls depends on the moment of inertia of the wheel-drum system.

28. A rod is hinged at its centre and rotated by applying a constant torque starting from rest. The power developed by the external torque as a function of time is

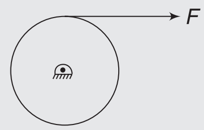
(a) 

(b) 

(c) 

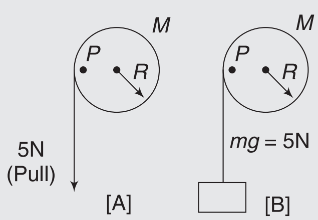
(d) 

29. A pulley is hinged at the centre and a massless thread is wrapped around it. The thread is pulled with a constant force F starting from rest. As the time increases,



(a) its angular velocity increases, but force on hinge remains constant
(b) its angular velocity remains same, but force on hinge increases
(c) its angular velocity increases and force on hinge increases
(d) its angular velocity remains same and force on hinge is constant

30. A uniform disc of mass $M = 2.50$ kg and radius $R = 0.20$ m is mounted on an axle supported on fixed frictionless bearings. A light cord wrapped around the rim is pulled with a force 5 N. On the same system of pulley and string, instead of pulling it down, a body of weight 5 N is suspended. If the first process is termed A and the second B, the tangential acceleration of point P will be



(a) equal in the processes A and B.
(b) greater in process A than in B.
(c) greater in process B than in A.
(d) independent of the two processes.

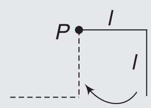
31. A cylinder rolls up an inclined plane, reaches some height, and then rolls down (without slipping throughout these motions). The direction of the frictional force acting on the cylinder is

(a) Up the incline while ascending and down the incline while descending
(b) Up the incline while ascending as well as descending
(c) Down the incline while ascending and up the incline while descending
(d) Down the incline while ascending as well as descending

32. A unifrom cylinder of mass M and radius R rolls without slipping down a slope of angle θ with horizontal. The cylinder is connected to a spring of force constant k at the centre, the other side of which is connected to a fixed support at A. The cylinder is released when the spring is unstretched. The force of friction (f)

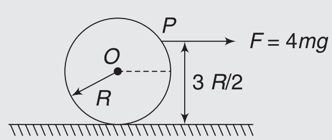
(a) is always upwards
(b) is always downwards
(c) is initially upwards and then becomes downwards
(d) is initially upwards and then becomes zero.

33. An L-shaped thin uniform rod of total length $2l$ is free to rotate in a vertical plane about a horizontal axis at P, as shown in the figure. The rod is released from rest from the position shown. Neglect friction. The angular velocity at the instant it has rotated through 90° and reached the dotted position shown is



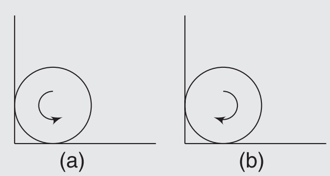
(a) zero (b) $\sqrt{\dfrac{6g}{5l}}$
(c) $\sqrt{\dfrac{3g}{5l}}$ (d) none

34. A uniform cylinder of mass m and radius R is placed on a rough horizontal surface having coefficient of friction $\mu = ½$. A horizontal force $F = 4mg$ is applied on sphere at point P (as shown in figure). The friction between cylinder and the floor is



(a) $f = 0$ (b) $f = \dfrac{mg}{4}$
(c) $f = \dfrac{mg}{2}$ (d) none of these

35. A spinning sphere is placed with its centre initially at rest in a corner as shown in figure (a) & (b). Coefficient of friction between all surfaces and the

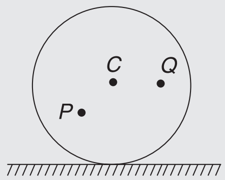


sphere is $\frac{1}{3}$. Find the ratio of the frictional force $\frac{f_a}{f_b}$ by ground in situations (a) & (b).

(a) 1 (b) $\frac{9}{10}$

(c) $\frac{10}{9}$ (d) none

36. A disc is rolling without slipping with angular velocity ω. P and Q are two points equidistant from the centre C. The order of magnitude of velocity is



(a) $v_Q > v_C > v_P$

(b) $v_P > v_C > v_Q$

(c) $v_P = v_C$, $v_Q = \frac{v_C}{2}$

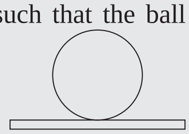
(d) $v_P < v_C > v_Q$

37. Two uniform spheres of densities $2d$ and d have radii R and $2R$ respectively. Each sphere is rotating about a fixed axis through a diameter. The rotational kinetic energies of the spheres are identical. What is the ratio of the magnitude of the angular momenta of these spheres?

(a) 4 (b) $2\sqrt{2}$

(c) 1/2 (d) $\sqrt{2}$

38. A ball is placed on a plank after being given a velocity v and angular velocity ω such that the ball rolls without sliding on a plank. The upper surface of the plank is rough enough to prevent slipping but lower surface in contact with the ground is smooth. No other force is acting on system.



(a) The plank will recoil back

(b) The plank will also move forward but with a lesser velocity than that of the ball.

(c) The plank will also move forward but with a greater velocity than that of the ball.

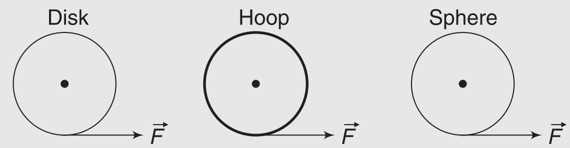
(d) The plank will remain at rest.

39. A child is standing with folded hands at the centre of a platform rotating about its central axis. Kinetic energy of the system is K. The child now stretches his arms so that moment of inertia of the system doubles.The kinetic energy of the system now is

(a) $2K$ (b) $\frac{K}{2}$

(c) $\frac{K}{4}$ (d) $4K$

40. A uniform disk, a thin hoop (ring), and a uniform sphere, all with the same mass and same outer radius, are each free to rotate about fixed axes through their centres. Assume the hoop is connected to the rotation axis by light spokes. With the objects at rest, identical forces are simultaneously applied to the rims, as shown. Rank the objects according to their angular momentum after a given time t, least to greatest.



(a) all tie (b) disk, hoop, sphere

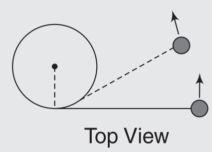
(c) hoop, disk, sphere (d) hoop, sphere, disk

41. A uniform thin rod of mass 'm' and length L is hinged about its upper end, and is free to swing in a vertical plane. A tiny ball of mass $m/4$ hits the lower end of the rod. The rod was initially vertical and the ball was moving horizontally with velocity v_0. If the ball stops just after collision, then coefficient of restitution for the collision is

(a) 0.75 (b) 0.25

(c) 0.5 (d) none

42. A ball is attached to a string that is attached to a pole. When the ball is hit, the string wraps around the pole and the ball spirals inwards, sliding on the frictionless surface. Considering angular momentum about centre of pole, what happens as the ball swings around the pole?



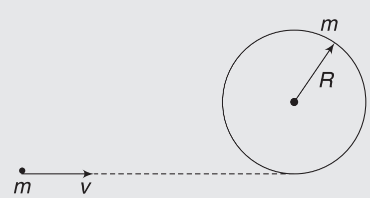
(a) The mechanical energy and angular momentum are conserved.

(b) The angular momentum of ball is conserved but the mechanical energy of the ball increases.

(c) The angular momentum of the ball is conserved and the mechanical energy of the ball decreases.

(d) The mechanical energy of the ball is conserved and angular momentum of ball decreases.

43. A circular hoop of mass m and radius R rests flat on a horizontal frictionless surface. A bullet, also of mass m and moving with a velocity v, strikes the hoop and gets embedded in it. The thickness of the hoop is much smaller than R. The angular velocity with which the system rotates after the bullet strikes the hoop is



(a) $v/(4R)$ (b) $v/(3R)$

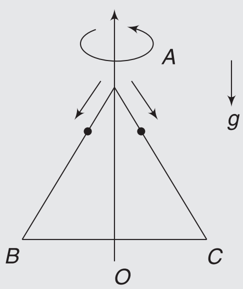
(c) $2v/(3R)$ (d) $3v/(4R)$

44. A smooth sphere A is moving on a frictionless horizontal plane with angular speed ω and centre of mass velocity v. It collides elastically and head on with an identical sphere B at rest. Neglect friction everywhere. After the collision, their angular speeds are ω_A and ω_B, respectively. Then

(a) $\omega_A < \omega_B$ (b) $\omega_A = \omega_B$

(c) $\omega_A = \omega$ (d) $\omega_B = \omega$

45. An equilateral triangle ABC formed from a uniform wire has two small identical beads initially located at A. The triangle is set rotating about the vertical axis AO. Then the beads are released from rest simultaneously and allowed to slide down, one along AB and the other along AC, as shown. Neglecting frictional effects, the quantities that are conserved as the beads slide down, are



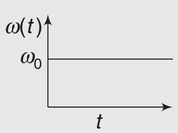
(a) Angular velocity and total energy (kinetic and potential).

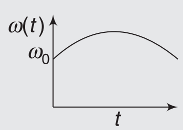
(b) Total angular momentum and total energy.

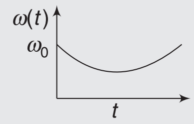
(c) Angular velocity and moment of inertia about the axis of rotation.

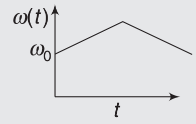
(d) Total angular momentum and moment of inertia about the axis of rotation.

46. A circular platform is free to rotate in a horizontal plane about a vertical axis passing through its centre. A tortoise is sitting at the edge of the platform. Now, the platform is given an angular velocity ω_0. When the tortoise moves along a chord of the platform with a constant velocity (with respect to the platform), the angular velocity of the platform $\omega(t)$ will vary with time t as

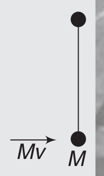
(a) 

(b) 

(c) 

(d) 

47. Two particles, each of mass M, are connected by a massless rod of length l. The rod is lying on the smooth sufrace. If one of the particles is given an impulse Mv, as shown in the figure, then angular velocity of the rod would be:



(a) v/l (b) $2v/l$

(c) $v/2l$ (d) None

48. A child with mass m is standing at the edge of a horizontal disc with moment of inertia I, radius R, and initial angular velocity ω about its central vertical axis. The child jumps off the edge of the disc with tangential velocity v with respect to the ground. The new angular velocity of the disc is

(a) $\sqrt{\dfrac{I\omega^2 - mv^2}{I}}$ (b) $\sqrt{\dfrac{(I + mR^2)\omega^2 - mv^2}{I}}$

(c) $\dfrac{I\omega - mvR}{I}$ (d) $\dfrac{(I + mR^2)\omega - mvR}{I}$

49. A uniform rod of length l and mass M is rotating about a fixed vertical axis passing through its centre, on a smooth horizontal table. It elastically strikes a particle placed at a distance $l/3$ from its axis and stops. Mass of the particle is

(a) $3M$ (b) $\dfrac{3M}{4}$

(c) $\dfrac{3M}{2}$ (d) $\dfrac{3M}{3}$

50. A uniform rod AB of mass m and length l is at rest on a smooth horizontal surface. An impulse J is applied to the end B, perpendicular to the rod in the horizontal direction. Speed of point P at a distance $\dfrac{l}{6}$ from the centre towards end A of the rod at time $t = \dfrac{\pi ml}{12J}$ after impact is

(a) $2\dfrac{J}{m}$ (b) $J/\sqrt{2}m$

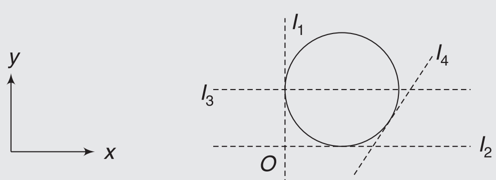
(c) $\dfrac{J}{m}$ (d) $\sqrt{2}\dfrac{J}{m}$

Worksheet 2

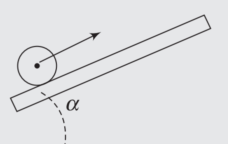
1. Suppose you are standing on the edge of a spinning platform and step off at right angle to the edge (radially outward). Now consider it the other way. You are standing on the ground next to a spinning platform and you step onto the platform at right angle to the edge (radially inward).
   (a) There is no change in rotational speed of the platform in either situation.
   (b) There is a change in rotational speed in the first situation but not the second.
   (c) There is a change in rotational speed in the second situation but not in the first.
   (d) There is a change in rotational speed in both instances.

2. A hollow spherical ball is given an initial push up an incline of inclination angle α. The ball rolls purely. Coefficient of static friction between ball and incline = μ. During its upward journey,
   (a) friction force decreases in magnitude as the ball slows down
   (b) friction force must increase in magnitude as the ball rolls down
   (c) friction force becomes zero at the top point where the ball stops
   (d) $\mu \geq 2/5 \tan \alpha$

3. A disc has radius R, mass m and has moment of inertia I_1, I_2, I_3 and I_4 about axes shown in figure. Moment of inertia of disc about z-axis passing through O will be:
   (a) $I_1 + I_2$ (b) $\frac{5}{2} mR^2$
   (c) $3mR^2$ (d) $I_1 + I_4$

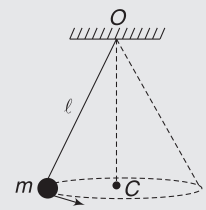


4. In the figure shown, friction coefficient between the solid sphere and the incline is μ. The centre of the sphere is given an initial upward velocity at $t = 0$, as shown, without imparting any initial angular velocity. Then, which of the following statement (s) is/are true?



   (a) The sphere will definitely stop sliding at a point during its upward journey even if μ is small.
   (b) Pure rolling will definitely begin before the sphere reaches the highest point but the sphere will continue to roll purely after that (even while coming down) only if μ is greater than a certain value.
   (c) Friction will decrease once pure rolling starts if $\mu > (2\tan\theta)/7$
   (d) None of the above

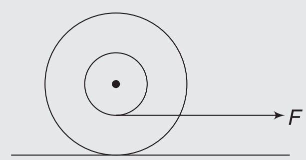
5. A small ball of mass m suspended from the ceiling at a point O by a thread of length ℓ moves along a horizontal circle with a constant angular velocity ω.



   (a) angular momentum is constant about O
   (b) angular momentum is constant about C
   (c) vertical component of angular momentum about O is constant
   (d) Magnitude of angular momentum about O is constant

6. A round body kept on a smooth horizontal surface is pulled by a constant horizontal force applied at the top point of the body.
   (a) If the body rolls purely on the surface, its shape can be that of a thin pipe
   (b) If the body rolls purely on the surface, its shape can be that of a solid uniform cylinder
   (c) If the body moves with backward sliding (i.e., its contact point having a backward velocity relative to the ground) its shape can be that of a sphere.
   (d) All bodies, irrespective of the shape, will slip

7. Inner and outer radii of a spool are r and R respectively. A thread is wound over its inner surface and placed over a rough horizontal surface. Thread is pulled by a force F, as shown in fig. Friction is large enough to prevent slipping. Which of the followings are possible?



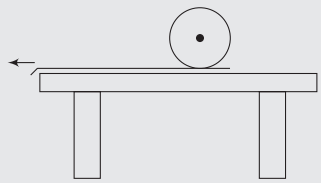
   (a) Thread unwinds, spool rotates anti-clockwise and friction acts leftwards.

(b) Thread winds, spool rotates clockwise and friction acts leftwards.

(c) Thread winds, spool moves to the right and friction acts rightwards.

(d) Thread winds, spool moves to the right and friction does not come into existence.

8. A thin tabecloth covers a horizontal table and a uniform round body lies on top of it. The tabecloth is pulled from under the body, and friction causes the body to slide and rotate. (Assume that the table is large and the body does not fall off it during its course of motion.)



(a) Body will finally roll towards left if it is a ring

(b) Body will finally roll towards right if it is a cylinder

(c) Body will finally come to rest, irrespective of its shape

(d) shape and size, both are important to decide the final state of the body.

9. A hoop and a solid cylinder have the same mass and radius. They both roll, without slipping, on a horizontal surface. If their kinetic energies are equal

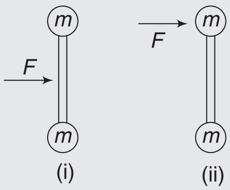
(a) the hoop has a greater translational speed than the cylinder

(b) the cylinder has a greater translational speed than the hoop

(c) the hoop and the cylinder have the same angular momentum relative to their respective COM

(d) if they begin climbing an incline, they will attain same height but the hoop will be first to reach the maximum height (assume no slipping)

10. A dumbbell is lying on a smooth horizontal table. A force F is applied to the dumbbell for a small time interval, t, first as in (i) and then as in (ii).



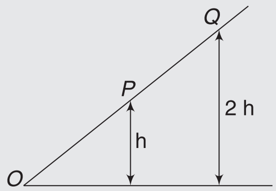
(a) In case (i), the dumbbell's COM acquires higher speed

(b) In case (ii), the dumbbell acquires higher kinetic energy

(c) In case (ii), the dumbbell acquires higher kinetic energy in its COM frame

(d) In case (ii), the dumbbell acquires higher momentum in COM frame

11. A ball rolls down an inclined plane, as shown in figure. The ball is first released from rest at P and then later from Q. Which of the following statement(s) is/are correct?



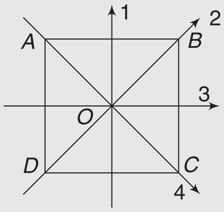
(a) The ball takes twice as much time to roll from Q to O as it does to roll from P to O.

(b) The acceleration of the ball at Q is twice as large as the acceleration at P.

(c) The ball has twice as much KE at O when rolling from Q as compared to when it starts from P.

(d) In case of the ball released from Q, angular momentum of the ball about point Q changes more during its journey from Q to P as compared to the change during its motion from P to O.

12. $ABCD$ is a square plate with centre O. The moments of inertia of the plate about the perpendicular axis through O is I and about the axes 1, 2, 3 and 4 are I_1, I_2, I_3 and I_4, respectively. It follows that:



(a) $I_2 = I_3$ (b) $I = I_1 + I_4$

(c) $I = I_2 + I_4$ (d) $I_1 = I_3$

13. A paritcle falls freely near the surface of the earth. Consider a fixed point O (not vertically below the particle) on the ground.

(a) Angular momentum of the particle about O is increasing.

(b) Torque of the gravitational force on the particle about O is decreasing.

(c) The moment of inertia of the particle about O is decreasing.

(d) The angular velocity of the particle about O is increasing.

14. The torque $\vec{\tau}$ on a body about a given point is found to be equal to $\vec{A} \times \vec{L}$ where $\vec{A}$ is a constant vector and $\vec{L}$ is the angular momentum of the body about that point. From this, it follows that

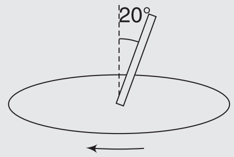
(a) $d\vec{L}/dt$ is perpendicular to $\vec{L}$ at all instants of time

(b) the components of $\vec{L}$ in the direction of $\vec{A}$ does not change with time

(c) the magnitude of $\vec{L}$ does not change with time

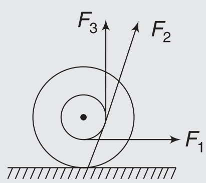
(d) $\vec{L}$ does not change with time

15. A uniform rod is fixed to a rotating turntable so that its lower end is on the axis of the turntable and it makes an angle of 20° to the vertical. (The rod is thus rotating with uniform angular velocity about a vertical axis passing through one end.) The turntable is rotating clockwise as seen from above.



(a) The rod's angular momentum vector about its lower end is directed down at 20° to the horizontal.
(b) The rod's angular momentum vector about its lower end is directed down at 80° to the horizontal.
(c) There is a torque acting on the system and it is in horizontal direction.
(d) There is a torque acting on the system and it is in vertical direction.

16. A yo-yo is resting on a rough horizontal table. Friction is large enough to prevent slipping. Forces F_1, F_2 and F_3 are applied separately on the thread as shown. Line of action of F_2 passes through the contact point. The correct statement is

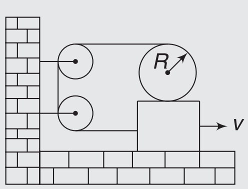


(a) when F_3 is applied, the centre of mass will move to the right.
(b) when F_2 is applied, the centre of mass will move to the left.
(c) when F_1 is applied, the centre of mass will move to the right.
(d) when F_2 is applied, the centre of mass will move to the right.

17. A plank with a uniform sphere placed on it rests on a smooth horizontal plane. Plank is pulled to right by a constant force F. If sphere does not slip over the plank, which of the following is correct?

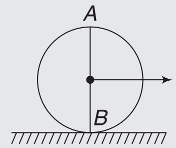
(a) Acceleration of the centre of sphere is less than that of the plank.
(b) Work done by friction acting on the sphere is equal to its total kinetic energy.
(c) Total kinetic energy of the system is equal to work done by the force F
(d) None of the above

18. In the figure shown, the plank is being pulled to the right with a constant acceleration a. If the cylinder does not slip, then



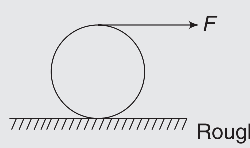
(a) the acceleration of the centre of mass of the cylinder is $2a$.
(b) the acceleration of the centre of mass of the cylinder is zero.
(c) the angular acceleration of the cylinder is a/R.
(d) the angular acceleration of the cylinder is zero.

19. A uniform disc of radius R is rolling, without sliding, on a horizontal surface. B is a point attached to the ground at the contact and A is a fixed point just above the top point of the disc.



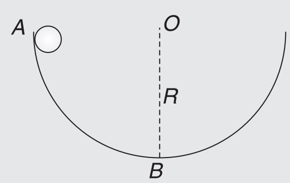
(a) The magnitude of the angular momentum of the disc about B is thrice that about A.
(b) The angular momentum of the disc about A is anti-clockwise.
(c) The angular momentum of the disc about B is clockwise
(d) The angular momentum of the disc about A is equal to that about B.

20. A solid sphere of radius R has a light thread wrapped around it. It is pulled horizontally by a force F as shown in the figure. There is no slipping and initially the sphere was at rest.



(a) work done by force F when the centre of mass moves a distance S is $2FS$.
(b) speed of the COM when COM moves a distance S is $\sqrt{\frac{20}{7}\frac{FS}{M}}$
(c) work done by the force F when COM moves a distance S is FS
(d) speed of the COM when COM moves a distance S is $\sqrt{\frac{4FS}{M}}$

21. A small sphere A of mass m and radius r rolls without slipping inside a large fixed hemispherical bowl of radius R (>> r), as shown in figure. The sphere starts from rest at the top point of the hemisphere.



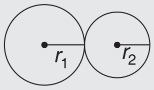
(a) the angular momentum of the sphere about O is maximum when it reaches the bottom of the bowl.

(b) the normal force exerted by the small sphere on the hemisphere when it is at the bottom B of the hemisphere is $\frac{17}{7}mg$

(c) friction force acting on the sphere is maximum just after it is released

(d) the sphere will come out of the bowl if the part of the hemisphere to the right of B is smooth

22. Two horizontal discs of different radii are free to rotate about their central vertical axes. One is given some angular velocity and the other is stationary. Their rims are now brought in contact. There is friction between the rims.



(a) The force of friction between the rims will disappear when the discs rotate with equal angular speeds.

(b) The force of friction between the rims will disappear when they have equal linear velocities at the point of contact.

(c) The angular momentum of the system about the contact point will be conserved.

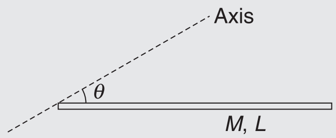
(d) The rotational kinetic energy of the system will not be conserved.

Worksheet 3

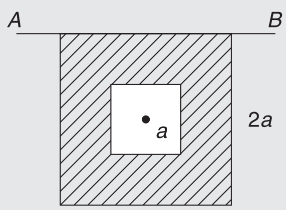
1. Particles of masses 1g, 2g, 3g ... 100g are kept at marks 1 cm, 2 cm, ... 100 cm respectively on a metre scale. Find the moment of inertia of the system of particles about a perpendicular bisector of the scale.

2. The density of a sphere of radius R varies with distance x from centre as $\rho = \rho_0\left(1 + \frac{x}{R}\right)$, where ρ_0 is a positive constant. Calculate the moment of inertia of the sphere about its diameter.

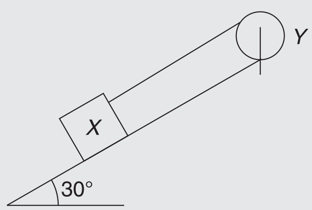
3. A uniform rod has mass M and length L. Consider an axis through its end inclined at an angle θ to the rod. Find moment of inertia of the rod about this axis.



4. A square of side length a is cut from a square plate of side length $2a$. The centre of the hole is at the centre of the square plate, as shown in figure. Mass of the square with hole is M. Find the moment of inertia of the plate about axis AB.



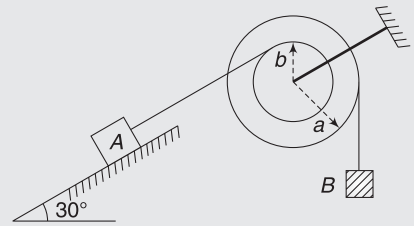
5. A block X of mass m = 0.5 kg is held by a long massless string on a smooth incline plane of inclination angle θ = 30°. The string is wound on a uniform solid cylindrical drum Y of mass M = 2 kg and radius r = 0.2 m. The drum is given an initial angular velocity, such that the block X starts moving up the plane.



   (a) Find tension in the string during motion.

   (b) At a certain instant of time, magnitude of angular velocity of Y is 10 rad s^{-1}. Calculate the distance travelled by X from that instant of time until it comes to rest.

6. In the system shown, block A and B have masses m_1 = 2 kg and $m_2 = \frac{26}{7}$ kg respectively. Pulley has moment of inertia I = 0.11 kg m^2 about its frictionless axle. Pulley can only rotate about its axis and cannot translate. The pulley has two dises of radii a = 15 cm and b = 10 cm, on which two threads are wound tightly. The inclined plane and A have coefficient of friction $\mu = \frac{\sqrt{3}}{10}$ between them. Calculate (a) tension in each theread and (b) acceleration of each block



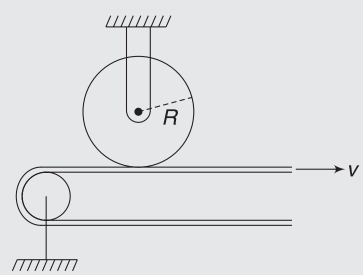
7. A uniform stick of mass M and length L is held horizontal with the help of two vertical strings as shown. The right string is cut. Immediately after the right string is cut, find:

   (a) acceleration of COM of the rod

   (b) tension in the other string.

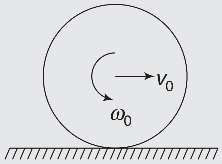
8. The strings in the previous question are replaced by two identical ideal springs. Find acceleration of end A of the rod immediately after the right spring is cut.

9. A disc of mass M = 4 kg and radius R = 75 mm can rotate feely about its central axis. The disc, originally at rest, is placed in contact with a belt moving at a constant velocity v = 18 ms^{-1}. The coefficient of friction between the belt and the disc is μ = 0.25 and normal force between the disc and the belt is equal to half the weight of the disc. Find the number of revolutions completed by the disc before it stops slipping on the belt.



10. A ring of radius R has a light thread wrapped on it. The free end of the thread is pulled so as to apply a constant horizontal force on the ring at the top. The ring is standing on a smooth horizontal surface. Find distance travelled by the centre of the ring during the time it makes one full rotation.

11. A sphere of radius R is moving with velocity v_0 and rotating with velocity ω_0 in anti-clockwise direction as shown. The surface is rough.

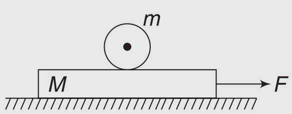


Find the ratio $\dfrac{v_0}{\omega_0 R}$ if

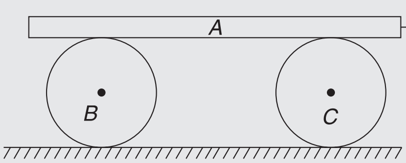
(a) the sphere stops translating and rotating simultaneously.

(b) velocity of the sphere becomes zero while it is still rotating anti-clockwise.

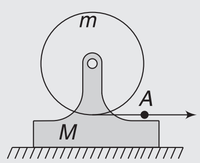
12. A plank of mass M, with a sphere of mass m placed on it, rests on a smooth horizontal surface. A constant horizontal force F is applied to the plank. Calculate the minimum coefficient of friction between the sphere and the plank so that there is no slipping between the sphere and the plank.



13. B and C are identical uniform cylinders, each having mass m and radius r. Uniform plank A has mass M and is kept symmetrically on the two cylinders. A horizontal force F pulls the plank such that there is no slipping anywhere. Find speed of A after it has moved through a distance L.



14. A uniform disc of mass m can freely rotate about a horizontal axis fixed to a mount of mass M as shown. A constant horizontal force F is applied to the free end A of the light thread wound on the disc. There is no friction between the mount and the horizontal surface. Find



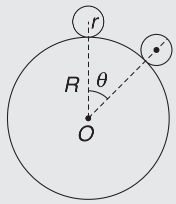
(a) the acceleration of point A on the thread.

(b) the kinetic energy of this system t seconds after the beginning of the motion.

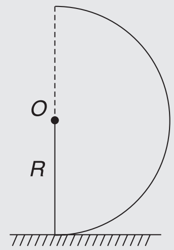
15. A uniform ball of radius r rolls without slipping down the top of a fixed sphere of radius R. Initial velocity of the ball is negligible.

(a) Find angle θ where the ball breaks off the sphere.

(b) Find angular velocity of the ball (about its centre) at the instant it breaks off the sphere.

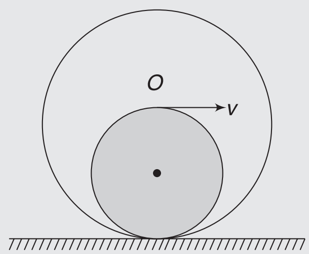


16. A uniform heavy solid hemisphere of radius R is held at rest with its base vertical and its curved surface in contact with a horizontal plane. The hemisphere is released from this position. The plane is rough enough to prevent slipping. Write the kinetic energy of the hemisphere when the base makes an angle θ with horizontal.

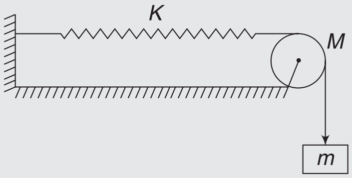


17. In the last problem, find the minimum coefficient of friction such that the hemisphere does not slip immediately after it is released.

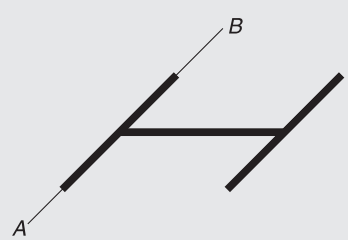
18. A uniform sphere of radius R has a spherical cavity of radius $\dfrac{R}{2}$ scooped out of it as shown in figure. Mass of the sphere with cavity is M. The sphere is rolling without sliding on a rough horizontal floor. When the cavity is at lowest position, centre of the sphere has velocity v. Write the KE of the sphere at this moment.



19. A block of mass $m = 4\,\text{kg}$ is attached to a spring of force constant $k = 32\,\text{N m}^{-1}$ by a string that passes over a pulley of mass $M = 8\,\text{kg}$. Pulley is a disc. System is released from rest with spring unstretched. Find the speed of the block after it has fallen through 1m.

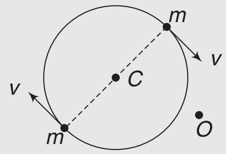


20. A rigid body is made of three identical thin rods, each having length l, fastened together to form letter 'H'. This body is free to rotate about a horizontal axis AB that runs along the length of one of the arms of H. The body is allowed to fall from rest from a position in which the plane of H is horizontal. What is the angular speed of the body when plane of 'H' becomes vertical?

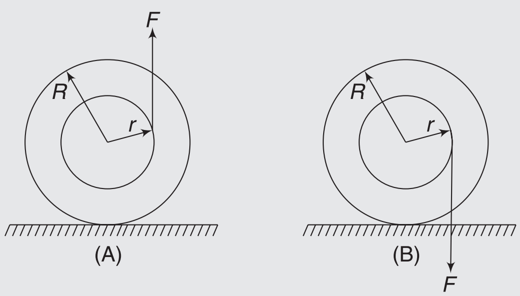


21. A stick of mass m and length l is lying flat on a horizontal table. It is struck with a hammer. The blow is made perpendicular to the stick at one end. Let the blow occur quickly, so that the stick doesn't have time to move much while the hammer is in contact. Due to the blow, the COM of the stick starts moving with a velocity v. Find velocities of the two ends of the stick immediately after the hit.

22. Two particles of same mass m move on a circle of radius r while always remaining at the ends of a diameter. Both have speed v. Find the angular momentum of the system about a point O that is at a distance R from the centre of the circle.

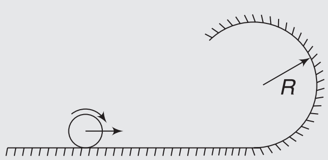


23. A spool has mass M and outer radius R. A light thread is wound on its inner disc of radius $r = \dfrac{R}{2}$. The thread is pulled vertically with a force F. Find acceleration of the centre of the spool in two cases shown in (A) and (B). The spool does not slip and its radius of gyration (K) about an axis through centre perpendicular to the figure is $\dfrac{R}{\sqrt{2}}$. Assume that the spool remains on horizontal surface.

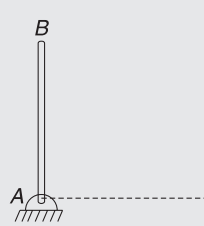


(A) (B)

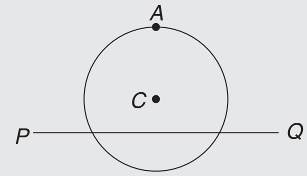
24. Figure shows a track which has a cylindrical part of radius R attached to a horizontal part. A sphere of radius r is set rolling with speed v on the horizontal part. The sphere does not slip anywhere and completes the vertical circle. Find minimum value of v.



25. A uniform slender rod AB of mass M and length L is supported by a frictionless pivot at A. It is released from rest in its vertical position (See fig.). Find the force applied by the pivot on the bar when it gets horizontal.

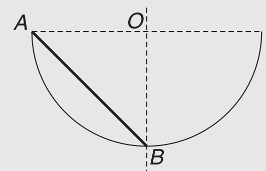


26. A uniform circular disc has radius R and mass m. A particle, also of mass m, is fixed at point A on the edge of the disc as shown. The disc can rotate freely about a fixed horizontal chord PQ that is at a distance $\dfrac{R}{4}$ from the centre C of the disc. The line AC is perpendicular to PQ. Initially, the disc is held vertical with point A at its highest position. It is then allowed to fall so that it starts rotating about PQ. Find the linear speed of the particle as it reaches its lowest position.

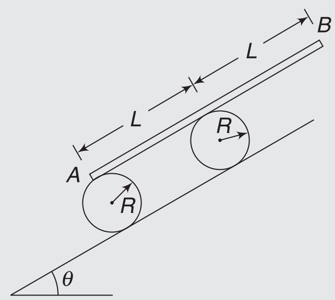


27. A particle of mass m is projected at time $t = 0$ from a point O with a speed u at an angle of 45° to the horizontal. Find the magnitude and direction of angular momentum of the particle about the point O at time $t = \dfrac{u}{g}$.

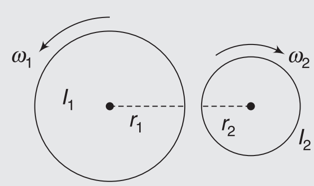
28. A homogeneous rod of mass M and length $\sqrt{2}\,R$ is released inside a smooth cylindrical surface from the position shown. Radius of the cylinder is R. The rod slides along the cylindrical surface in vertical plane. Find the velocity of COM of the rod at the instant it becomes horizontal.



29. A uniform plank of mass M and length $2L$ is placed on two identical cylindrical rollers, as shown in the figure. Each roller has mass M and radius R. Inclination of the incline is θ = 30°. Initially, the separation between centres of the two rollers is L and one end of the plank lies on roller A. System is released from rest. There is no slipping anywhere. Find the speed of the plank at the instant it is about to leave the rear roller.

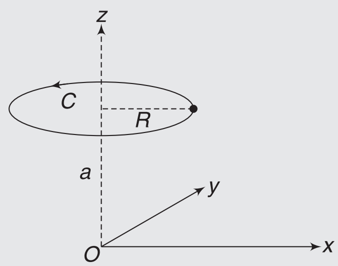


30. Figure shows two cylinders of radii r_1 and r_2 having moment of inertia I_1 and I_2 about their respective central axes. The cylinders are spinning about their axes with angular speeds ω_1 and ω_2, as shown. They are brought closer and put into contact, keeping their axes parallel. The cylinders first slip over each other but the slipping finally ceases. Find



the angular speed of the cylinders after the slipping cease.

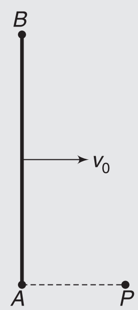
31. A particle of mass m is rotating in a circle of radius R with angular speed ω, as shown in figure. Centre of the circle (C) has co-ordinates (0, 0, a) and the plane of rotation is parallel to xy plane. At time t = 0, the particle is located at (R, 0, a).



(a) Find the angular momentum of the particle about the origin (O) at time t.

(b) Find the angular momentum of the particle about z-axis at time t.

32. Two small balls – A and B – each of mass m, are joined using a light rigid rod of length L. The system is placed on a smooth table and is translating with velocity v_0 in a direction perpendicular to the length of the rod. A particle P of mass m kept at rest on the table, sticks to the ball A, as the ball collides with it. Find, after collision,

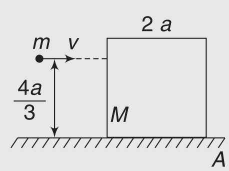


(a) velocity of COM of the system (A + B + P)

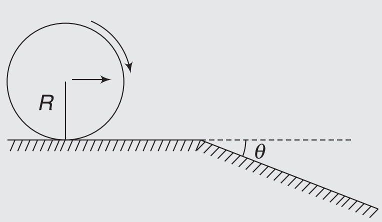
(b) the angular speed of the system about COM and

(c) linear speed of A and B.

33. A solid cube of mass M and side length $2a$ is placed on a horizontal surface. It can rotate freely about an axis passing through point A and perpendicular to the plane of the figure. Any other motion of the cube is not allowed. A bullet of mass m hits the cube while flying horizontally with a speed v at a height $\frac{4a}{3}$ above the horizontal surface. The bullet gets embedded into the block. Find minimum value of v needed to topple the cube. Assume $m << M$.



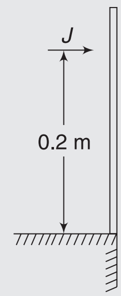
34. A uniform solid cylinder of radius R rolls over a horizontal plane passing into an inclined plane forming an angle θ with horizontal. Friction is large enough to prevent slipping. Linear velocity of the cylinder on the horizontal plane is v_0.



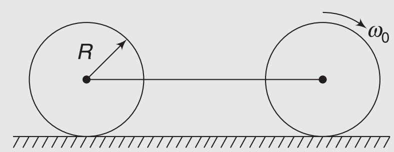
(a) Find the linear speed (i.e., speed of COM) of the cylinder immediately after it is on the inclined plane.

(b) Find maximum value of v_0, which permits the cylinder to rolls on the inclined plane without a jump.

35. A uniform thin rod with mass M = 0.6 kg and length L = 0.3 m, stands on the edge of a smooth table as shown. The rod is given a sharp below at a height of 0.2 m from the table. The impulse imparted is J = 6 Ns. Find kinetic energy of the rod t = 1.0 s after the blow.

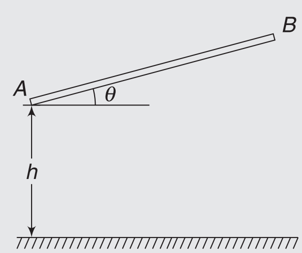


36. Two identical discs can rotate freely about their central axles. The axles of the two discs are connected using a light rod. One of the discs is given an angular speed ω_0 about its axle and the system in placed on a rough horizontal surface. Find the final linear velocity of the system when the two discs stop sliding.



37. A ball rolls without sliding on a rough horizontal floor with its centre moving towards a wall at a velocity of 14 ms^{-1}. The ball collides with the smooth wall. Coefficient of restitution is e = 0.7. Calculate the velocity of the centre of the ball a long time after collision.

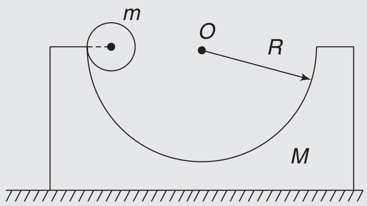
38. A rod of length L is held above a hard floor making an angle θ with the horizontal. It is released from position shown. End A of the rod makes an elastic collision with the floor after falling through a height h. Find angular velocity of the rod immediately after impact.



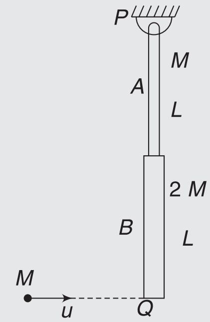
39. A block of mass M = 1 kg rests on a smooth horizontal surface. There is a semi-cylindrical track of radius R = 62.5 cm cut in the block. A cylinder of mass m = 0.5 kg and radius r = 10 cm is held at the top of the track with its axis horizontal and at the same level as the axis of the track (see fig.). System is released from this position. The cylinder does not slip on the track. When the cylinder reaches the bottom of the track

(a) Calculate velocity of its centre

(b) Calculate force applied by the block on the floor.



40. Two uniform thin rods, A and B, of length L each and masses M and $2M$ respectively, are rigidly joined end-to-end. The combination is pivoted at the lighter end P, as shown, such that it can freely rotate about point P in vertical plane. A small object of mass M, moving horizontally, hits the lower end Q of the combination. What should be velocity (u) of the object so that the system could just be raised to horizontal position, if



(a) the particle sticks to end Q.

(b) the coefficient of restitution is e.

Answers Sheet

Your Turn

1. (a) 990 gcm^2 (b) 520 gcm^2 (c) 1510 gcm^2
2. $\frac{7}{3}ml^2$
3. MR^2
4. MR^2
5. 1.1 kgm^2
6. Disc with smaller density
7. $\frac{5}{3}ml^2$
8. $\frac{ml^2}{2}$
9. $\frac{\lambda_0 L^4}{4}$
10. 0.2%
11. $I = \mu L^2$ where $\mu = \frac{mM}{m+M}$
12. $\frac{2M}{5}\left(\frac{a^5-b^5}{a^3-b^3}\right)$
13. $\frac{1}{\sqrt{2}}$
14. 75 $rad\,s^{-1}$
15. (i) $\frac{3\sqrt{3}}{4}\frac{g}{l}$ (ii) Zero
16. $a = \frac{mg\sin\theta}{m+\frac{M}{2}}$
17. $a = \frac{(m_1 - m_2\sin\theta)g}{m_1+m_2+\frac{I}{R^2}}$
18. (i) $\frac{F}{M}$ (ii) Zero
19. $\frac{2mg\sin\theta}{M+2m}$
20. (i) $\frac{3g}{7l}$ (ii) $\frac{3g}{7}$ (iii) $\frac{4Mg}{7}$
21. 4g/5
22. v is larger.
23. $\frac{4v}{3}$
24. (a) Disc (b) $v = \sqrt{\frac{10}{7}gL\sin\theta}$, $\omega = \frac{v}{r}$
25. $\mu_{min} = 0.5$
26. Zero
27. (a) $g(\sin\theta - \mu\cos\theta)$ (b) $\frac{5\mu g\cos\theta}{2R}$
28. $\frac{4F}{3M}$
29. $\frac{F}{3}$ towards right
30. $\frac{8mg}{3M+8m}$
31. $\frac{3}{4}Mv^2$
32. 1
33. $\frac{1}{2}ml^2\omega^2$
34. 80π J
35. 3,200 W
37. (a) $3\sqrt{\frac{g}{2L}}$ (b) $\frac{13}{4}Mg$
38. $-\frac{Mv^2}{7}$
39. $\sqrt{\frac{14}{3}gR}$
40. $\sqrt{2h(H-h)}$
41. (a) mvr Directed out of the plane of figure (b) mvr Directed out of the plane of figure
42. $ml\sqrt{2gl}$
43. (a) $\frac{mu^3\cos\theta\cdot\sin^2\theta}{2g}$ (b) $\frac{2mu^3\cdot\sin^2\theta\cdot\cos\theta}{g}$
44. $\frac{3}{4}MR^2\omega$
45. $\frac{3}{2}MvR$
46. $\frac{2}{3}MvR$ anti-clockwise
47. $\frac{2}{5}MRv$
48. 0.25 $rad\,s^{-1}$
49. $\frac{aT^2}{MR}$
50. $2R$, R
51. Length of day will increase if we move to the equator and will decrease if we move to the poles.
52. yes
53. $\frac{M\omega_0}{M+2m}$
54. Will not change
55. (a) ω (b) $\frac{\omega^2R^2}{2g}$
56. 3 m
57. $\frac{2mu}{2m+M}$
58. $\frac{\sqrt{2}(\sqrt{2}-1)mu}{MR}$
59. (a) $\frac{u}{2}$ (b) $\frac{3u\sin\theta}{L}$
60. (a) $\frac{mu}{M+m}$ (b) $\frac{mu(h-R)}{\left(\frac{2}{5}M+m\right)R^2}$

Worksheet 1

1. (d)	2. (d)	3. (a)	4. (a)	5. (b)	6. (d)	7. (d)	8. (a)	9. (a)
10. (c)	11. (d)	12. (b)	13. (a)	14. (c)	15. (b)	16. (d)	17. (a)	18. (b)
19. (b)	20. (a)	21. (a)	22. (d)	23. (c)	24. (b)	25. (b)	26. (a)	27. (c)
28. (b)	29. (a)	30. (b)	31. (b)	32. (c)	33. (b)	34. (a)	35. (b)	36. (a)
37. (c)	38. (d)	39. (b)	40. (a)	41. (a)	42. (d)	43. (b)	44. (c)	45. (b)
46. (b)	47. (a)	48. (d)	49. (b)	50. (d)				

Worksheet 2

1. (c)	2. (d)	3. (a,b,d)	4. (a,b,c)	5. (b,c,d)	6. (a,c)	7. (b)	8. (c)	9. (b)
10. (b,c)	11. (c,d)	12. (a,b,c,d)	13. (a,c,d)	14. (a,b,c)	15. (a,c)	16. (c)	17. (a,b,c)	18. (b,c)
19. (a,b,c)	20. (a,b)	21. (a,b,c)	22. (b,d)					

Worksheet 3

1. 0.404 kgm^2

2. $I = \frac{44}{45}\pi\rho_0 R^5$

3. $\frac{ML^2}{3}\sin^2\theta$

4. $\frac{17}{12}Ma^2$

5. (a) $\frac{5}{3}$N (b) 7.5 m

6. (a) 14.28 N, 25.22 N (b) 2.14 ms^{-2}, 3.21 ms^{-2}

7. (a) $\frac{3g}{2}$ (b) $\frac{Mg}{4}$

8. $g(\uparrow)$

9. 137.6

10. $2\pi R$

11. (a) $\frac{2}{5}$ (b) $\frac{v_0}{\omega_0 R} < \frac{2}{5}$

12. $\frac{2F}{(7M + m)g}$

13. $\sqrt{\frac{8FL}{4M + 3m}}$

14. (a) $F\left[\frac{2}{m} + \frac{1}{M + m}\right]$ (b) $F^2t^2\left[\frac{1}{m} + \frac{1}{2(M + m)}\right]$

15. (a) $\theta = \cos^{-1}\left(\frac{10}{17}\right)$ (b) $\omega = \frac{l}{r}\sqrt{\frac{10g(R + r)}{17}}$

16. $\frac{3R}{8}\cos\theta$

17. 0.27

18. $\frac{31}{40}Mv^2$

19. 2.4 ms^{-1}

20. $\omega = \frac{3}{2}\sqrt{\frac{g}{l}}$

21. The end that is hit moves with velocity $4v$ and the other end has a velocity $2v$ in opposite direction

22. $2mvr$

23. $a = \frac{F}{3M}$ in both cases. Towards left in case (A) and towards right in case (B).

24. $\sqrt{\frac{27}{7}g(R - r)}$

25. $\frac{\sqrt{37}}{4}Mg$

26. $\sqrt{5gR}$

27. $\frac{mu^3}{2\sqrt{2}g}$ directed perpendicular to the plane of the motion

28. $\frac{1}{2}\sqrt{3(\sqrt{2} - 1)gR}$

29. $v = \sqrt{\frac{4}{7}gL}$

30. $\left[\frac{I_1\omega_1 r_2 + I_2\omega_2 r_1}{I_1 r_2^2 + I_2 r_1^2}\right]r_2$ and $\left[\frac{I_1\omega_1 r_2 + I_2\omega_2 r_1}{I_1 r_2^2 + I_2 r_1^2}\right]r_1$

31. (a) $mR^2\omega\hat{k} - mRa\omega[\cos(\omega t)\hat{i} + \sin(\omega t)\hat{j}]$

32. (a) $\frac{2v_0}{3}$ (b) $\frac{v_0}{2L}$ (c) $v_A = \frac{v_0}{2}$; $v_B = v_0$

33. $v = \frac{M}{m}[3ag(\sqrt{2} - 1)^{\frac{1}{2}}]$

34. (a) $v = \sqrt{v_0^2 + \frac{4}{3}gR(1 - \cos\theta)}$ (b) $v_{0\,\max} = \sqrt{\frac{7gR}{3}\cos\theta - \frac{4}{3}gR}$

35. 70 J

36. $\frac{\omega_0 R}{6}$

37. 3 ms^{-1}.

38. $\frac{6\sqrt{2gh}\cos\theta}{L(1 + 3\cos^2\theta)}$

39. (a) 2 ms^{-1} (b) 26.67 N

40. (a) $\sqrt{\frac{11gL}{72}}$ (b) $\frac{1}{(1 + e)}\sqrt{\frac{567gL}{20}}$

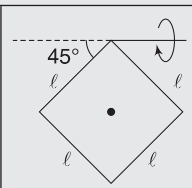
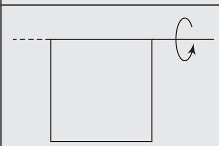
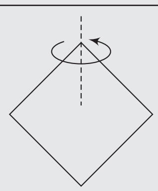
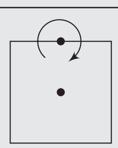
CHAPTER 7

Miscellaneous Problems on Chapter 4 to 6

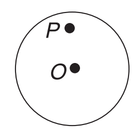
MATCH THE COLUMNS

Match the entries in Column I with those in Column II. An item in Column I can match with any number of entries in Column II. It may also happen that an item in Column I does not match with any of the entries in Column II.

1. Square frame shown in figures is made of four uniform rods. Mass of each rod is m and length is ℓ. Figures show rotation axis. Column II has moment of inertia of the square frame about an axis. Match the entries in the two columns.

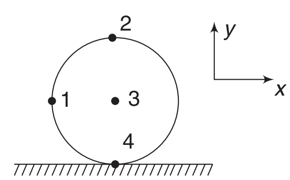
Column I		Column II	
(Axis of rotation)		*(Moment of inertia) (about axis of rotation)*	
(a)		(p)	$\frac{2}{3}m\ell^2$
(b)		(q)	$\frac{5}{3}m\ell^2$
(c)		(r)	$\frac{7}{3}m\ell^2$
(d)		(s)	$\frac{8}{3}m\ell^2$

2. A uniform disc performs pure rolling motion on rough horizontal surface with constant angular velocity. O is the centre of the disc and P is a point on the disc. Then match the statements in Column I with Column II.



Column I		Column II	
(a)	Velocity of point P on disc	(p)	Magnitude changes with time
(b)	Acceleration of point P on disc	(q)	Always directed from point P towards centre O of the disc
(c)	Tangential acceleration of point P on disc	(r)	must be directed from point P to the centre of disc at some instant, not always.
(d)	Normal acceleration of point P on disc	(s)	Non-zero and remains constant in magnitude.

3. A rigid cylinder is kept on a smooth horizontal surface, as shown. If Column I indicates velocities of various points (3-centre of cylinder, 2-top point, 4-bottom point, 1-on the level of 3 at the rim) on it, choose correct state of motion from Column II.

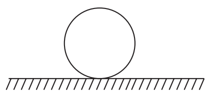


Column I		Column II	
(a)	$\vec{v}_1 = \hat{i} + \hat{j}$, $\vec{v}_2 = 2\hat{i}$	(p)	Pure rotation about centre
(b)	$\vec{v}_1 = \hat{i} + \hat{j}$, $\vec{v}_3 = -\hat{i}$	(q)	Rolling without slipping to left
(c)	$\vec{v}_2 = \hat{i}$, $\vec{v}_3 = 0$	(r)	Rolling without slipping to right
(d)	$\vec{v}_4 = 0$, $\vec{v}_1 = -\hat{i} - \hat{j}$	(s)	Not possible

4. A particle moves with position given by $\vec{r} = 3t\hat{i} + 4\hat{j}$. Where, $\vec{r}$ is measured in metres and $t(>0)$ in seconds.

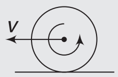
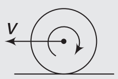
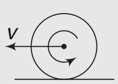
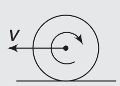
	Column I		Column II
(a)	Rate of change of distance from origin.	(p)	Increasing with time
(b)	Magnitude of linear acceleration of particle	(q)	Decreasing with time
(c)	Magnitude of angular velocity of particle about origin	(r)	Constant
(d)	Magnitude of angular momentum of particle about origin	(s)	Zero

5. Assume that a spherical ball is kept on a rough ground. Column I indicates situation related to ball and Column II indicates the effect that friction has on the ball.

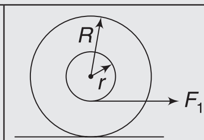
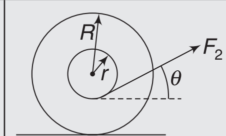
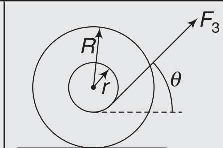
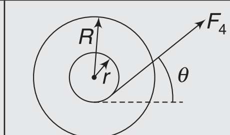


	Column I		Column II
(a)	Ball is suddenly given clockwise ω, with centre of mass initially at rest.	(p)	Increases v_{cm}
(b)	Ball is given a velocity to the right without any ω	(q)	Decreases v_{cm}
(c)	Ball is given clockwise ω and given velocity to the right such that $v_{cm} < \omega R$	(r)	Increases ω
(d)	Ball is given clockwise ω and velocity to the right such that $v_{cm} > \omega R$	(s)	Decreases ω

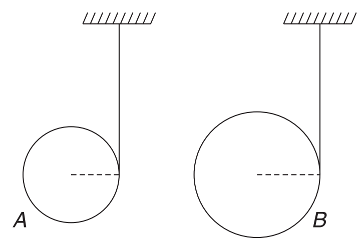
6.

	Column I		Column II
	(Initially)		*(When pure rolling begins)*
(a)	 $\omega = \dfrac{2v}{R}$	(p)	v_{cm} is towards left in case of ring
(b)	 $\omega = \dfrac{2v}{R}$	(q)	v_{cm} is towards left in case of solid sphere
(c)	 $\omega = \dfrac{v}{2R}$	(r)	v_{cm} is towards right in case of ring
(d)	 $\omega = \dfrac{v}{2R}$	(s)	v_{cm} is towards right in case of solid sphere

7. A yo-yo is resting on a rough horizontal table where friction is large enough to prevent slipping. Forces F_1, F_2, F_3 and F_4 are applied separately on the free end of the thread, as shown.

	Column I		Column II
(a)		(p)	centre of mass accelerates towards left
(b)	 $\cos\theta < (r/R)$	(q)	centre of mass accelerates towards right
(c)	 $\cos\theta > (r/R)$	(r)	friction acts towards left
(d)	 $\cos\theta = (r/R)$	(s)	friction acts towards right

8. Two solid uniform round objects of equal mass but different radii r_A and r_B, shown in figure, can rotate freely about horizontal axles passing through their respective centres. On each of them, light inextensible strings are wound. Free ends of both the threads are fixed to the ceiling and the objects are released during the subsequent motion; assume their axles to remain horizontal. ℓ represent the length of thread unwound in time t after release and a is acceleration of the centre.



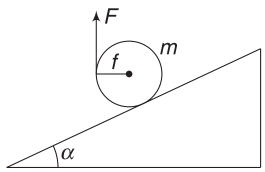
	Column I		Column II
(a)	A is disc and B is ring with $r_B > r_A$	(p)	$a_A > a_B$
(b)	Both A and B are discs with $r_B > r_A$	(q)	$l_A > l_B$
(c)	A is disc and B is ring with $r_A = r_B$	(r)	$l_A = l_B$
(d)	Both A and B are discs with $r_A = r_B$	(s)	$a_A = a_B$

PASSAGE-BASED PROBLEMS

Every passage is followed by a series of questions. Every question has four options. Choose the most appropriate option for the questions.

Passage 1

A solid uniform cylinder of mass m has a light thread wrapped on it. It is kept in equilibrium on a fixed incline of angle $\alpha = 37°$ with the help of a force F applied on the free-end of the thread.



1. What force F is required to keep the cylinder in balance when the thread is pulled vertically, as shown?

(a) $\frac{mg}{2}$ (b) $\frac{3mg}{4}$

(c) $\frac{3mg}{4}$ (d) None of these

2. If there is flexibility to apply force F in any direction, then the minimum force required to hold the cylinder is

(a) $\frac{3mg}{10}$ parallel to incline

(b) vertical $\frac{3mg}{8}$

(c) horizontal $\frac{mg}{5}$

(d) None of these

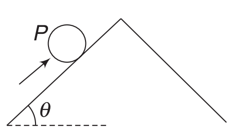
3. The values of minimum coefficient of static friction between cylinder and incline in the cases mentioned in above two questions are: _______ and _____ respectively:

(a) $\frac{4}{3}, \frac{3}{4}$ (b) $\frac{4}{3}, \frac{3}{8}$

(c) $\frac{3}{4}, \frac{3}{8}$ (d) None of these

Passage 2

In a mythological story, a worker was punished by a king. The punishment was to roll a spherical stone up a hill from where it would invariably roll down to the other side. He would run after the stone; will catch it and roll it back uphill. The sphere and the worker had the same mass. Coefficient of friction between the man and the hill and that between the sphere and the hill is the same. Neither the sphere nor the man slips.



4. At what height from the centre (measured normal to the hill surface) should the man push, in a direction parallel to the hill, to roll the sphere slowly, with minimum force.

(a) 0 (b) $R/2$

(c) $2/5R$ (d) R

5. When the sphere is rolling down, the man wants to run along with the same acceleration, without exerting any force on it.

(a) friction on the man would be backwards and equal to that on the sphere

(b) friction on the man would be forwards and greater than on the sphere

(c) friction on the man would be backwards and less than on the sphere

(d) friction on the man would be forwards and equal to that on the sphere

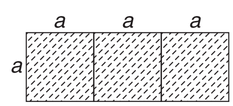
6. What is the minimum friction coefficient for the man to just push the sphere up when he pushes at the top point in a direction parallel to the hill

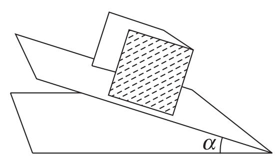
(a) $\frac{3}{2}\tan\theta$ (b) $\tan\frac{\theta}{2}$

(c) $\frac{2}{7}\tan\theta$ (d) $\frac{5}{7}\tan\theta$

Passage 3

A cardboard strip is in the shape of a rectangle of length $3a$ and width a. The extreme one-third portions are bent at right angles to the middle part so as to give the strip a shape as shown in the figure. It is put on a rough inclined plane, as shown in the figure.





7. At what minimum angle of inclination of the inclined plane will the cardboard strip topple? (assume that it does not slide).

(a) $\tan^{-1}\left(\frac{2}{3}\right)$ (b) $\tan^{-1}\left(\frac{3}{2}\right)$

(c) 45° (d) $\tan^{-1}\left(\frac{1}{3}\right)$

8. What should be the coefficient of friction so that it does not slide before toppling.

(a) 0.66 (b) 0.75

(c) 1.0 (d) 0.33

9. What should be the time taken to travel a distance a, if the inclined plane is made completely smooth?

(a) $\sqrt{\frac{a}{g\sin\alpha}}$ (b) $\sqrt{\frac{2a}{g\sin\alpha}}$

(c) $\sqrt{\frac{2a}{3g\sin\alpha}}$ (d) None of these

Passage 4

A winch is mounted on an axle, and a six-sided nut is welded to the winch. By turning the nut with a wrench, a person can rotate the winch. For instance, turning the nut clockwise lifts the block off the ground, because more and more rope gets wrapped around the winch.

Three students agree that using a longer wrench makes it easier to turn the winch. But they disagree about why. All three students are talking about the case where the winch is used, over a 10-second time interval, to lift the block one metre off the ground.

Student A

By using a longer wrench, the person decreases the average *force* he must exert on the wrench, in order to lift the block one metre in ten seconds.

Student B

Using a longer wrench reduces the *work* done by the person.

Student C

Using a longer wrench reduces the *power* that the person must exert to lift the block.

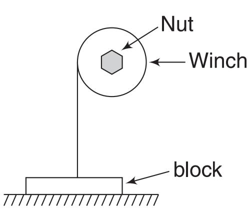


Fig. 1

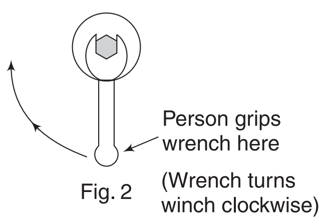


Fig. 2

10. Student A is

(a) Correct, because the torque that the wrench must exert to lift the block doesn't depend on the wrench's length

(b) Correct, because using a longer wrench decreases the torque it must exert on the winch.

(c) Incorrect, because the torque that the wrench must exert to lift the block doesn't depend on the wrench's length

(d) Incorrect, because using a longer wrench decreases the torque it must exert on the winch.

11. Which of the following is true about students B and C?

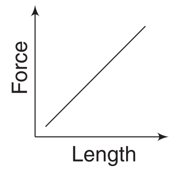
(a) Students B and C are both correct.

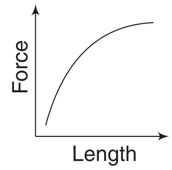
(b) Student B is correct, but student C is incorrect.

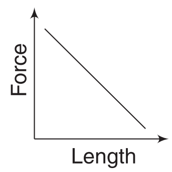
(c) Student C is correct, but student B is incorrect.

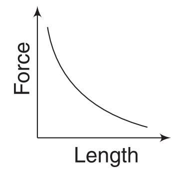
(d) Students B and C are both incorrect.

12. If several wrenches are used one after another to apply the same torque on the nut, which graph best expresses the relationship between the average force the person must apply on the wrench, and the length of the wrench?

(a) 

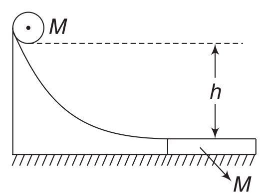
(b) 

(c) 

(d) 

Passage 5

A smooth hill has height h. A small solid cylinder of radius r and mass M slides down from rest and gets onto a long rough plank of mass M, lying on the smooth horizontal plane at the base of the hill, as shown in the figure. Due to friction between the cylinder and the plank, the cylinder slows down and starts rolling without sliding over the plank. When the cylinder starts rolling without sliding over the plank, the velocity of the plank is v_P, the velocity of the centre of mass of the cylinder is v_C and angular velocity of the cylinder is ω.



13. Which of the following is correct?

(a) $v_C = \omega r$ (b) $v_C < \omega$

(c) $v_C > \omega r$ (d) none

14. The value of v_C is

(a) constant and equals to $\frac{3}{4}\sqrt{2gh}$

(b) variable with $(v_{cm})_{max} = \frac{1}{2}\sqrt{2gh}$

(c) constant and equal to $\sqrt{2gh}$

(d) variable with $(v_{cm})_{max} = \sqrt{2gh}$

15. Which of the followings is incorrect?

(a) $2v_p + \omega r = \sqrt{2gh}$

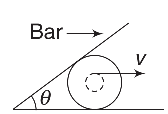
(b) $2v_C - \omega r = \sqrt{2gh}$

(c) $v_p = v_C - \omega r$

(d) none

Passage 6

A yo-yo is pulled by its string along a horizontal surface without slipping. The horizontal velocity of the end of the string



remains equal to v. A smooth bar is pivoted as shown and remains supported by the yo-yo. The outer and the inner radii of the yo-yo are R and r, respectively.

16. The velocity of the centre of mass of yo-yo is

(a) vr/R (b) $\dfrac{v(R+r)}{R}$

(c) $\dfrac{v(R-r)}{R}$ (d) $\dfrac{vR}{R+r}$

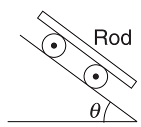
17. The angular speed of the bar as a function of θ is

(a) $\left(\dfrac{2v}{R+r}\right)\dfrac{\sin^2(\theta/2)}{\cos(\theta/2)}$ (b) $\left(\dfrac{v}{R+r}\right)\dfrac{1}{\sin^2\theta}$

(c) $\dfrac{2v}{R+r}\sin^2(\theta/2)$ (d) $\dfrac{v(R+r)}{R^2}\sin\theta$

Passage 7

A uniform rod of mass m is supported on two rollers, each of mass $\dfrac{m}{2}$ and radius r. The system is placed on an incline, as shown, and released. Assume no slipping at any contact and treat the rollers as a uniform solid cylinder.



18. If acceleration of rod down the plane is a_R and acceleration of the centre of mass of roller is a_0, then

(a) $a_R > a_0$

(b) $a_R < a_0$

(c) $a_R = a_0$

(d) Information is insufficient to decide

19. If the friction force on roller at the contact point with incline is f_1 and at the contact point with rod is f_2, then

(a) $f_1 > f_2$

(b) $f_1 < f_2$

(c) $f_1 = f_2$

(d) data is insufficient to decide

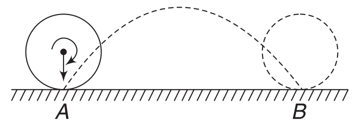
20. If the inclined surface is smooth then acceleration of roller is

(a) $2g\sin\theta$ (b) $\dfrac{12g}{11}\sin\theta$

(c) $g\sin\theta$ (d) none of these

Passage 8

In an experiment, a spinning ball of radius r and mass m is made to fall on a rough horizontal surface. While hitting the surface, the ball has a velocity v_0 and angular velocity $\dfrac{5v_0}{r}$. After hitting the surface at A, the ball moves on the path shown and hits the surface again at B. This experiment is repeated for many surfaces of varying roughness. For a certain surface 'X', the ball covers the distance $AB = \dfrac{2v_0^2}{g}$. Assume that all the surfaces used in the experiment are perfectly elastic.



21. The coefficient of friction between the surface 'X' and the ball is

(a) 0.5 (b) 0.25

(c) 0.75 (d) 1.0

22. The magnitude of impulse imparted by surface 'X' to the ball during the collision at A is

(a) mv_0 (b) $2mv_0$

(c) $2\sqrt{2}\,mv_0$ (d) $\sqrt{3}\,mv_0$

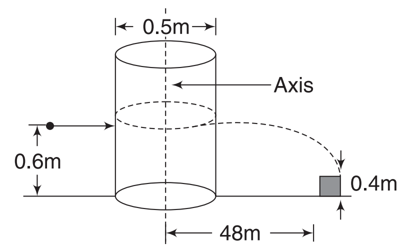
23. Change in KE of the ball during collision at A is

(a) $\dfrac{mv_0^2}{4}$ (b) $-\dfrac{9}{2}mv_0^2$

(c) $-\dfrac{mv_0^2}{4}$ (d) $\dfrac{5}{4}mv_0^2$

Passage 9

An open-top, hollow wooden cylinder of outer radius $R = 0.25\,\text{m}$ is standing on a frictionless ice surface.



The cylinder contains some ice inside it. A rifle is placed very close to the cylinder and a bullet of mass $m = 0.5$ kg is fired horizontally with a muzzle speed of $800\,\text{ms}^{-1}$ so as to pass the cylinder, just toucing it on its outer surface, at a height $0.6\,m$ above ground. Eventually, the bullet hits a target placed at a distance of 48 m from the cylinder at a height 0.4 m. After the bullet went past the cylinder, the cylinder began to rotate with a speed of one rotation per 4 s. [$g = 10\,\text{ms}^{-2}$]

Inner radius of cylinder = 0.23 m, density of wood = $600\,\text{kgm}^{-3}$ height of cylinder = 1 m, and thickness of base = 0.02 m

24. The velocity of bullet just after touching the cylinder is

(a) $480\ \text{ms}^{-1}$ (b) $240\ \text{ms}^{-1}$

(c) $120\ \text{ms}^{-1}$ (d) $960\ \text{ms}^{-1}$

25. The moment of inertia of the cylinder alonwith its content about the axis shown is

(a) $2.46\ \text{kgm}^2$ (b) $3.46\ \text{kgm}^2$

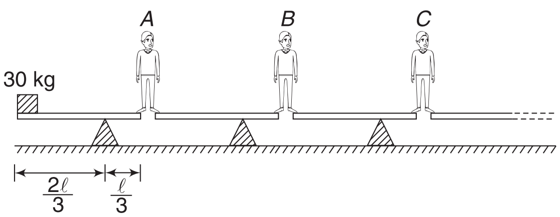
(c) $4.46\ \text{kgm}^2$ (d) $8.92\ \text{kgm}^2$

26. Use the data given and the calculated value of moment of inertia to find the mass of ice inside the cylinder

(a) 27 kg (b) 227 kg
(c) 50 kg (d) 127 kg

Passage 10

In an entertainment show, large number of light boards are placed, as shown. Each of them can rotate about a fixed fulcrum. The fulcrum of each board divides the length of the board in the ratio of 2:1. At the left end of the left-most board, there is a block of mass 30 kg and a team of joker stand keeping their feet at the ends of adjacent boards. Mass of each jokers is 80 kg. The jokers are able to maintain balance.



27. N_{LA} represents the force applied by the left foot of joker A on the board and N_{RA} is the force applied by the right foot of joker A on the board. Symbols N_{LB}, N_{RB}, etc., have similar meaning.

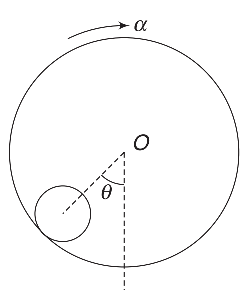
(a) $N_{LB} = N_{RB}$ (b) $N_{RA} > N_{LB}$
(c) $N_{LA} + N_{RA} < N_{LB}$ (d) $N_{RB} > N_{LB}$

28. Maximum number of jokers that can keep balance in this way is

(a) 3 (b) 4
(c) 2 (d) 5

Passage 11

A hollow cylindrical drum of radius R is kept rotating with constant angular acceleration (α) about its stationary horizontal axis, through its centre (O). A uniform solid cylinder is placed inside the drum with its axis horizontal. It was found that the solid cylinder rolls without sliding inside the hollow cylinder at angular position θ and its axis remains motionless.



29. If radius of the solid cylinder is r, then its angular acceleration about its axis is

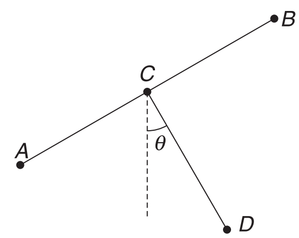
(a) α (b) $\frac{r\alpha}{R}$
(c) $\frac{R\alpha}{r}$ (d) $\frac{(R-r)\alpha}{R}$

30. Value of α, for which the solid cylinder has its axis motionless is

(a) $\frac{3}{2}\frac{g\cos\theta}{R}$ (b) $\frac{2g\sin\theta}{R}$
(c) $\frac{g\sin\theta}{R}$ (d) $\frac{3g\sin\theta}{R}$

Passage 12

Three identical light rods – AC, BC and DC – are rigidly connected to form a T-shaped structure. At the free ends of each rod are fixed three particles, each of mass m. The structure can rotate freely without friction about a fixed horizontal axis, through junction point C, perpendicular to the plane of the figure. Structure is released from the position shown with $\theta = 37°$.



31. Immediately after the structure is released

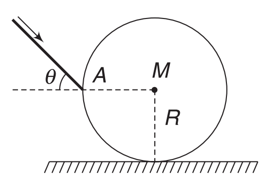
(a) acceleration of particle at A and D are perpendicular to each other.
(b) acceleration of particle at A and B are in same direction.
(c) magnitude of acceleration of COM of the system is half in magnitude to the acceleration of the particle at B.
(d) None of the above

32. Magnitude of force applied by rod CD on the particle at D is

(a) $2mg$ (b) $\frac{mg}{\sqrt{5}}$
(c) $\frac{2mg}{\sqrt{5}}$ (d) $3mg$

Passage 13

A billiard player strikes a ball (a uniform sphere with mass M and radius R) with a stick at the middle (i.e, at a height R above the table at A). The impact is along a line making an angle θ with horizontal. The impact causes the ball to move towards right as well as ''back spin'' – spin in anti clockwise direction



Coefficient of kinetic friction between the ball and the table is μ. Assume that the ball does not rebound off the table due to the strike and neglect the impulse of friction due to table on the ball during the period of impact. It was observed that the ball came to complete rest after travelling through some distance.

33. Value of θ must be

(a) $45°$ (b) $30°$

(c) $\tan^{-1}\left(\frac{2}{5}\right)$ (d) $\tan^{-1}\left(\frac{1}{5}\right)$.

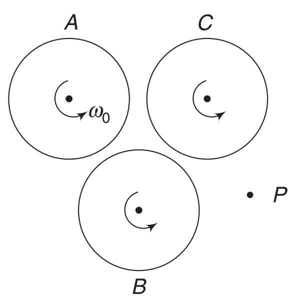
34. Distance that the ball travels before coming to rest is proportional to

(a) $\frac{1}{\mu M^2}$ (b) $\frac{R}{\mu M^2}$

(c) $\frac{R^2}{\mu M^2}$ (d) $\frac{1}{\mu M}$

Passage 14

Three identical cylinders of rough surfaces are mounted on frictionless horizontal axles passing through their centres. They are all spinning about their axles in anti-clockwise direction with same angular speed ω_0. They are brought closer into contact and their axels are held stationary till slipping between them ceases. [The cylinders are held symmetrically relative to one another as shown, so that all of them touch one another]



35. Angular momentum of the system of three cylinders is

(a) conserved about the axle of any of the three cylinders.

(b) conserved about contact point of any two cylinders.

(c) conserved about any random point P (see figure).

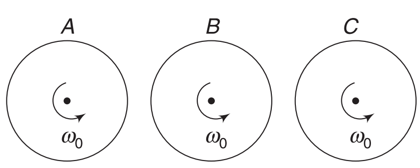
(d) not conserved about any of the above-mentioned points.

36. Final angular speed of cylinder B will be

(a) $\frac{\omega_0}{3}$ (b) $\frac{\omega_0}{2}$

(c) $\frac{\omega_0}{4}$ (d) 0

37. The arrangement of the three cylinders is changed as shown in figure. Now, what is their final angular velocity if they are brought into contact?

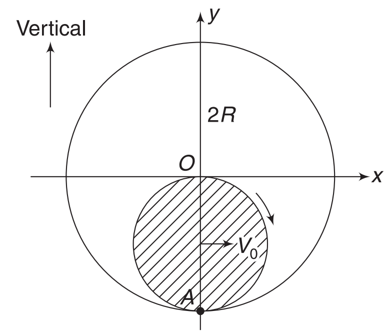


(a) $\frac{2\omega_0}{3}$ (b) $\frac{\omega_0}{3}$

(c) $\frac{\omega_0}{4}$ (d) 0

Passage 15

A solid cylinder of radius R is rolling without sliding on the inner surface of a stationary cylinder of radius $2R$. At the instant shown, velocity of the centre of the rolling cylinder is v_0. A is a massless dust particle stuck at point A on the solid cylinder.



38. Co-ordinates of the centre of the solid cylinder, when the co-ordinates of the dust particle becomes (0,0), are

(a) $(0, R)$ (b) $(-R, 0)$

(c) $(R, 0)$ (d) None

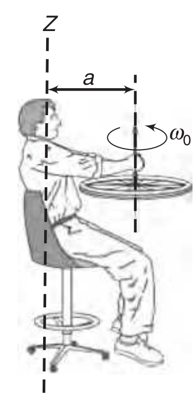
39. Speed of the centre of the solid cylinder when co-ordinates of the dust particle becomes zero is

(a) $\sqrt{v_0^2 - \frac{4}{3}gR}$ (b) $\sqrt{v_0^2 - \frac{2}{3}gR}$

(c) $\sqrt{v_0^2 - \frac{gR}{3}}$ (d) zero

Passage 16

A student is sitting on a chair that can rotate about the vertical axis (z-axis), as shown. He is holding a spinning wheel. Initially, the chair is not rotating but the wheel is rotating about its central vertical axis with angular speed ω_0. Distance of the axis of the spinning wheel from the rotation axis of the chair is a. Moment of inertia of the chair and student combined is I_1 about the z-axis shown. Moment of inertia of the wheel about its central rotation axis is I_0 and mass of the wheel is equal to $\frac{I_0}{3a^2}$. Take $I_1 = 4I_0$



[Neglect friction in bearings of the wheel and the chair]

40. Angular momentum of the entire system about z-axis is

(a) $I_0\omega_0\hat{k}$ (b) $\frac{3}{2}I_0\omega_0\hat{k}$

(c) $\frac{1}{2}I_0\omega_0\hat{k}$ (d) zero

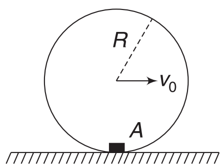
41. The student turns the spinning wheel upside down. Now the angular speed of the chair and the student about z-axis will be:

(a) $\frac{\omega_0}{4}$ (b) $\frac{\omega_0}{8}$

(c) $\frac{3\omega_0}{8}$ (d) $\frac{5\omega_0}{8}$

Passage 17

A small body A is fixed to the inside of a thin rigid hoop of radius R and mass equal to that of the body A. The hoop rolls without slipping over a horizontal plane. At the instant the body A is at lowest position, the centre of the hoop moves with velocity v_0.



42. Speed of the centre when body A is at the highest point is given by

(a) $\sqrt{\frac{v_0^2 - 2gR}{3}}$ (b) $\sqrt{\frac{v_0^2 - gR}{3}}$

(c) $\sqrt{\frac{v_0^2 - 2gR}{2}}$ (d) $\sqrt{\frac{v_0^2 - gR}{4}}$

43. Maximum value of v_0 so that the hoop remains in contact with the horizontal plane when body A is at the top, is

(a) $\sqrt{gR}$ (b) $\sqrt{2gR}$

(c) $\sqrt{4gR}$ (d) $\sqrt{8gR}$

Answers Sheet

Match the Columns

1. (a) s (b) q (c) p (d) r
2. (a) p (b) q, s (c) r (d) r
3. (a) r (b) s (c) p (d) q
4. (a) p (b) r, s (c) q (d) r
5. (a) p, s (b) q, r (c) p, s (d) q, r
6. (a) p, q (b) q, r (c) p, q (d) p, q
7. (a) q, r (b) p, r (c) q, r (d) r
8. (a) p, q (b) r, s (c) p, q (d) r, s

Passage-based Problems

1. (c)	2. (a)	3. (c)	4. (d)	5. (a)	6. (a)	7. (a)	8. (a)	9. (b)
10. (a)	11. (d)	12. (d)	13. (c)	14. (a)	15. (d)	16. (d)	17. (c)	18. (a)
19. (a)	20. (c)	21. (d)	22. (c)	23. (b)	24. (b)	25. (c)	26. (d)	27. (a)
28. (a)	29. (c)	30. (b)	31. (a)	32. (c)	33. (a)	34. (a)	35. (d)	36. (d)
37. (b)	38. (c)	39. (a)	40. (a)	41. (c)	42. (a)	43. (d)		

Fluid Mechanics

"Eureka! – I have found it!"

–*Archimedes*

1. INTRODUCTION

Fluid is a substance that can flow. Liquids and gases are fluids. A solid has definite volume and shape. A liquid has definite volume but no definite shape and an unconfined gas has neither a fixed volume nor a definite shape. Intermolecular forces are strong in a solid. These forces are relatively weak in a liquid and therefore, allow the shape to be easily changed. However, it is rather difficult to change the density of a liquid. In gases, intermolecular forces are very weak. This allows easy change in shape as well as volume.

In this chapter, we will mainly discuss the properties of a liquid, though most of the results that we obtain are applicable for gases as well. We will discuss only about an *ideal liquid*. A liquid that is *incompressible and non-viscous* is said to be ideal. Incompressibility means that density of the liquid cannot be changed even by changing pressure. Liquid is non-viscous – this implies that liquid surfaces in contact do not exert tangential force on one another. In simple language, two layers of a liquid do not exert friction on one another. They apply force that is normal to their surface.

Mechanics of fluids is important in many engineering design. It is equally important for a civil engineer, a mechanical engineer, an aeronautical engineer and even a nuclear engineer.

In this chapter, we will first study about the behaviour of a static fluid (a fluid that is at rest relative to the container). Such study is often termed as *hydrostatics* or *fluid statics*. In the later half of the chapter, we will discuss flowing fluids. Such study is called *hydrodynamics* or *fluid dynamics*.

2. DENSITY AND RELATIVE DENSITY

Density of a substance is defined as mass per unit volume.

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}} \quad \text{or,} \quad \rho = \frac{m}{v} \qquad \ldots(1)$$

Density of the most common liquid, water, at 4°C is $1\,\text{gcm}^{-3}$ or $1{,}000\ \text{kgm}^{-3}$. Though density of water changes (slightly) with temperature, we will disregard this variation in most of our discussions and assume its density to be nearly $1\ \text{gcm}^{-3}$ for a fairly wide range of temperature.

Relative density of a solid or a liquid is defined as the ratio of density of the substance to the density of water at 4°C.

$$\therefore \qquad Rd = \frac{\text{density of substance}}{\text{density of water at 4°C}} \qquad \ldots(2)$$

Relative density (Rd) is a ratio and has no unit. It is also known as *specific gravity*.

Example 1 *Gold is heavy*

Relative density of gold is 19.6. Find mass of a gold bar having volume of 1 litre.

Solution

Concepts

(i) Density of substance = (Rd) (density of water)

(ii) 1 litre = 1,000 ml = $1{,}000\,\text{cm}^3$

Density of gold = $19.6\ \text{gcm}^{-3}$ [= $19{,}600\ \text{kgm}^{-3}$]

Mass of $1{,}000\ \text{cm}^3$ gold is

$$m = \rho \cdot V = 19.6\,\frac{\text{g}}{\text{cc}} \times (1{,}000\ \text{cm}^3)$$

$$= 19600\ \text{g} = 19.6\ \text{kg}$$

Example 2 *How pure is your jewellery?*

A jewellery piece weighs 34 g and has a volume of $2\ \text{cm}^3$. It was sold to you as 22 karat gold. 22 karat gold has 91.67% gold and 8.33% other metal. Your jewellery piece has gold (Rd = 19.6) and copper (Rd = 9.0). How much gold is there in your jewellery? Do you think that you have been cheated?

Solution

Concepts

(i) Volume $= \dfrac{\text{Mass}}{\text{Density}}$

(ii) Volume of Au + Volume of Cu = 2 cm^3

Let mass of Au in the jewellery be x grams and mass of Cu be $(34 - x)$ grams.

$$\text{Volume of Au} = \frac{x\ g}{19.6\ \dfrac{g}{cm^3}} = \frac{x}{19.6}\ cm^3$$

$$\text{Volume of Cu} = \frac{(34 - x)\ g}{9\ \dfrac{g}{cc}} = \frac{34 - x}{9}\ cm^3$$

$$\therefore\ \frac{x}{19.6} + \frac{34 - x}{9} = 2$$

$$\Rightarrow\ x\left[\frac{1}{9} - \frac{1}{19.6}\right] = \frac{34}{9} - 2$$

$$\Rightarrow\ x[0.11 - 0.05] = 3.78 - 2 \quad \Rightarrow x = 29.67 \text{ g}$$

$$\text{Percentage of gold} = \frac{29.67}{34} \times 100 = 87.26\%$$

Yes, you were cheated!

Your Turn

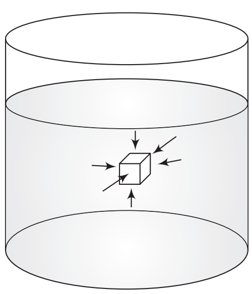
Q.1 A beaker can hold 0.5 kg water when full. It is filled with mercury. What is the mass of mercury that it holds? Relative density of mercury = 13.6.

Q.2 *Aluminium bronze* is an alloy made by adding 10% (by weight) Aluminium to Copper. Find density of the alloy in CGS unit. It is given that relative density of Cu and Al are 9.0 and 2.7 respectively.

3. PRESSURE IN A FLUID

In mechanics of solids, force was a physical quantity of immense importance. In mechanics of fluids, it is much easier to talk in terms of pressure.

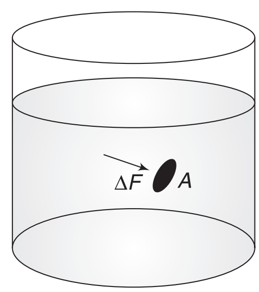
When a liquid is at rest, there are no tangential forces (or you can say that there are no shear forces). A tangential force acting on a liquid layer will cause it to flow. Therefore, liquid layers exert only normal force on one another, when there is no flow. If you submerge a solid in a static liquid, the force exterted by the liquid is always normal to the surface of the solid. The effect is to compress the solid from all sides (see fig.).



Force due to a liquid is normal to the surface of the solid and tries to compress it from all sides.

Forget about a solid; think of the cube shown in the figure as a volume of the liquid itself. The neighbouring liquid compresses it from all sides in a manner identical to it compressing a solid cube.

Therefore, a fluid at rest exerts force perpendicular to any surface in its contact – such as a container wall or a body immersed in it. This force per unit area is termed as *fluid pressure*. This arises due to molecules of the fluid hitting a surface.

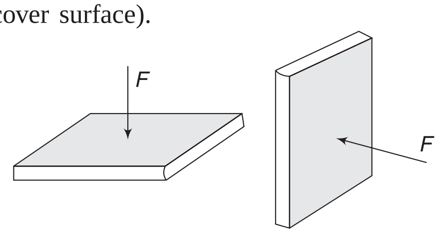


Consider a small area ΔS surrounding a point A inside a fluid. Force on this area due to liquid molecules hitting this surface from one side is ΔF. This force is normal to the surface. (The surface experiences equal and opposite forces due to molecules on other side of it. While writing pressure, you need to consider force from one side only).

Pressure at A is defined as

$$P = \frac{\Delta F}{\Delta S} \qquad \ldots(3)$$

Pressure is a scalar; it has no direction. The force that a fluid exerts on a surface has direction. Force is always normal to the surface. If you hold a book as shown in figure (a), the force on its cover surface due to atmospheric pressure is vertically down. When you hold the book as shown in figure (b), the force on cover surface due to air is horizontal. In both cases, force = (pressure due to air) × (area of cover surface).



(a) (b)

Obviously we cannot assign a unique direction to pressure.

Unit of pressure

SI unit of pressure is Nm^{-2} also known as pascal (Pa). A commonly used unit in meterology is Bar.

$$1 \text{ Bar} = 10^5 \text{ Pa}$$

3.1 Atmospheric Pressure

Pressure applied by atmospheric air is known as atmospheric pressure. Its mean value at sea level is 1.013×10^5 Pa.

Sometimes, pressure is expressed in multiplies of atmospheric pressure also. Thus,

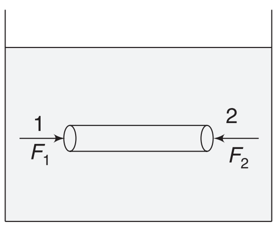
$$1 \text{ atm} = 1.013 \times 10^5 \text{ Pa}$$

3.2 Variation of Pressure in a Static Fluid

When you swim under water, you can feel the water pressure on your eardrums. The deeper you swim, the greater the pressure. When a balloon is taken deep inside water, we find that size of the balloon decreases substantially. This is due to higher water pressure. It means pressure in a fluid increase with depth. However, pressure remains same at all points on a horizontal level. Let us understand the variation in pressure inside a static fluid in much detail.

3.2.1 Pressure is Uniform on a Horizontal Plane

Consider a liquid of density ρ kept in a tank. There is no flow. Total force on the entire mass of the liquid is zero. If we consider a small volume of liquid, net force on it is also zero. Consider one such volume of the liquid, which is in the shape of a horizontal cylinder having small cross-sectional area ΔS. Net force on this element is zero as it has no acceleration. Consider horizontal forces on this cylindrical fluid element.



Force on cylindrical element of the liquid is zero

$F_1 = P_1 \, \Delta S$ = force on circular face at 1 due to surrounding liquid (Force is towards right)

$F_2 = P_2 \, \Delta S$ = force on circular face at 2 due to surrounding liquid (Force is towards left)

P_1 and P_2 are pressure inside the liquid at 1 and 2 respectively.

For horizontal equilibrium

$$F_1 = F_2 \quad \Rightarrow \quad P_1 \Delta S = P_2 \Delta S$$

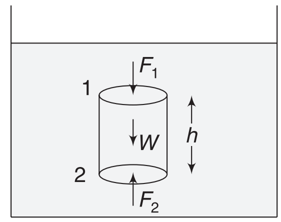
$$\Rightarrow \qquad P_1 = P_2$$

Thus, pressure at points like 1 and 2, which are on same horizontal level, is same.

Any difference in P_1 and P_2 will mean that the cylindrical element is accelerated. [Remember, our fluid is static.]

3.2.2 Pressure Increases with Depth

Consider a vertical *cylinder of liquid* inside a tank filled with liquid of density ρ. Height of the cylinder is h and its cross-sectional area is ΔS.



Force on cylindrical liquid element is zero

Pressure (P_1) is uniform on its top surface (the surface is horizontal) and downward force due to surrounding liquid on this surface is

$$F_1 = P_1 \Delta S$$

Similarly, pressure (P_2) is uniform on its bottom surface and upward force due to surrounding liquid on this surface is

$$F_2 = P_2 \, \Delta S$$

Weight of the cylindrical volume of the liquid is

$$W = \text{volume} \times \text{density} \times g = \Delta S h \rho \cdot g$$

For vertical equilibrium of the cylindrical liquid element, we must have

$$F_2 = F_1 + W \qquad \text{...(i)}$$

$$\Rightarrow \qquad P_2 \, \Delta S = P_1 \Delta S + \Delta S h \rho g$$

$$\Rightarrow \qquad P_2 = P_1 + \rho g h \qquad \text{...(4)}$$

It means that pressure increases linearly with depth if we travel down into a fluid having constant density.

Look at equation (i). It is actually the weight of fluid column, which causes the pressure to increase with depth. Heavier the fluid, higher is the change in pressure with depth.

Example 3 *You carry a lot of weight on your head*

Consider your head top to be a flat square of side length 20 cm. How much force does atmospheric air apply on your head top?

Solution

Concepts

Atmospheric Pressure, $P_0 = 1.0 \times 10^5 \text{ Nm}^{-2}$

Force due to atmosphere on a surface is $F = P_0 \Delta S$. Direction of force is normal to the surface.

$$F = P_0 \, \Delta S = \left(1.0 \times 10^5 \, \frac{\text{N}}{\text{m}^2}\right)(0.2\,\text{m} \times 0.2\,\text{m})$$

$$= 4{,}000 \text{ N}$$

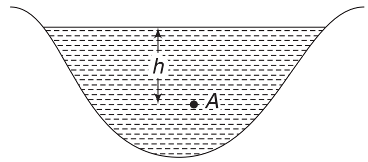
You are carrying a load of 400 kg on your head. Why don't your legs crumble? Actually, atmospheric pressure squeezes you from all sides. On every small surface of our body, force is normal to the surface. At certain locations, the force is downward and at some other location, it is upward. Net force is nearly zero. (We will shortly learn that net force is upward, known as buoyancy. However, it is small).

Atmospheric pressure acts everywhere on your body. It is normal to the surface. Net force on your body is (nearly) zero.

The next natural question that comes to our mind is-why don't our stomach or lungs get flattened? The answer is that our lungs are filled with air and stomach also is not empty. The air in lungs and stomach (or the food material in stomach) is nearly at atmospheric pressure. Therefore, there is an outward force also on your stomach muscles. Net force on our stomach wall is definitely zero.

However, if one goes deep in a lake, the increased outside pressure does cause a problem.

Example 4 Find pressure at a depth of 20 m in a lake. Atmospheric pressure is $P_0 = 1.0 \times 10^5$ Nm^{-2}.



Solution

Concepts

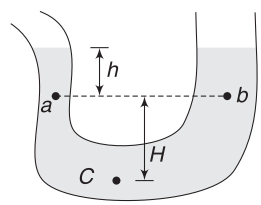
(i) Pressure on the surface of the lake = Pressure due to atmospheric air = P_0

(ii) Pressure at depth h is $P = P_0 + \rho gh$

$$P = P_0 + \rho gh$$
$$= 1 \times 10^5 \text{ Nm}^{-2} + \left(10^3 \frac{\text{kg}}{\text{m}^3}\right)\left(10 \frac{\text{m}}{s^2}\right)(20\,\text{m})$$
$$= 1 \times 10^5 + 2 \times 10^5 = 3 \times 10^5 \text{ Nm}^{-2}$$

Note: In the above calculation, shape of the lake is not important.

In fact, for the container shown in the figure $P_a = P_b = P_0 + \rho gh$, where P_0 is atmospheric pressure. [Note that a and b are points at same horizontal level in a static liquid.]



Also, $\qquad P_c = P_a + \rho gH$

Your Turn

Q.3 How does water pressure one metre below the surface of a small pond compare with water pressure one metre below the surface of a huge lake?

Q.4 Air inside the middle ear is normally at atmospheric pressure. Surface area of a man's eardrum is nearly 1 cm^2. Estimate the force exerted on his eardrum due to water, when he is swimming in a lake at a depth of 10 m.

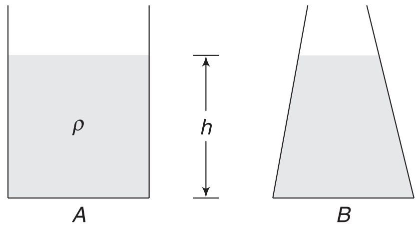
3.3 Pressure in a Gas

When we move down inside a water tank, the pressure increases according to equation (4). Is this also true for air inside our room when we move from ceiling to the floor? The air has too small a density for any significant change in pressure over a height variation of few metres. The pressure inside the room, or inside any reasonable-sized container is uniform. However, if you consider the large atmosphere the variation in pressure will be significant. Atmospheric pressure at Mt. Everest is much less than the pressure that we experience at the sea level.

4. FORCE ON CONTAINER SURFACE

4.1 Force on Horizontal Base

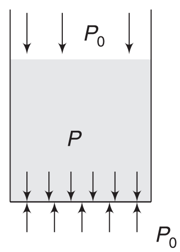
Container A and B have same area (S) of their bottom surface. They are filled to the same height (h) by a liquid of



density ρ. How much force acts on the bottom surface of the two containers?

First consider container A. Liquid pressure at the bottom is

$P = P_0 + \rho gh$, where P_0 is atmospheric pressure. Force applied by the liquid on the surface is normal to it (in vertically downward direction).



$$F_L = PS = (P_0 + \rho gh)S \quad \text{...(i)}$$

If air is present below the bottom surface (when you keep a beaker on table, air is present between the beaker and table surface), it will exert an upward force on the base of the container in upward direction. This force is

$$F_a = P_0 S \quad \text{...(ii)}$$

Thus, the resultant force that the base experiences (due to liquid + atmosphere) is

$$F = F_L - F_a = \rho ghS \quad \text{...(iii)}$$

It is important to note that the force due to atmospheric pressure is present on both sides and it cancels out.

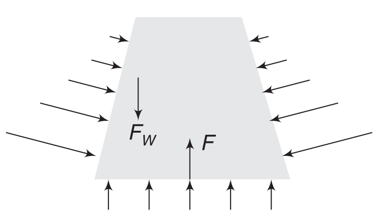
Now, $\rho(hs)g$ = density × volume × g = weight of liquid in the container. Thus,

$$F = \text{Weight of liquid in the container.}$$

This result is intuitive. Just consider the liquid as your system and study its equilibrium. Its weight is balanced by the force applied by the container base on it.

On the vertical wall of the container the liquid applies horizontal force all of which sum up to zero.

Now, consider container B. Pressure at the bottom is again $P = P_0 + \rho gh$ and equations (i), (ii) and (iii) hold true in this case also. Force on the base is still $F = \rho ghS$, which is higher than the weight of the liquid in the container (obviously, B contains less liquid than A). How is the force on the base larger than the weight of the liquid?



Force on the liquid due to container walls

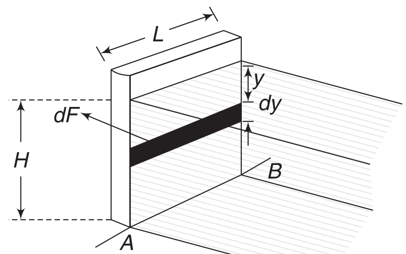
The liquid exerts force on the slant wall of the container. This force is normal to the surface and has a vertical component. The container wall also exerts same force on the liquid. As can be seen in the figure, the horizontal component will add to zero.

There is a net vertical (downward) force (F_w) on the liquid due to the walls. If F is the force due to the base on the liquid, then for vertical equilibrium of liquid

$$F = F_w + \text{weight of the liquid}$$

4.2 Force on a Vertical Wall

Consider a wall of width L. A Liquid of density ρ is filled to the right to the wall up to a height H. The force on the wall due to the liquid cannot be calculated by multiplying area of the wall twith pressure. The pressure is changing with depth. We must write force on a small patch and then integrate to find the resultant force.



Force on a strip of width dy is dF. The force is horizontal

Consider a strip of width dy at a depth y from the liquid surface.

Area of strip, $dS = L\,dy$.

Pressure of liquid at depth y is $P = \rho gy$

Force on the strip (in horizontal direction) due to the liquid is

$$dF = P \cdot dS = (\rho gy)(Ldy) = \rho gL\, y\,dy \quad \text{...(i)}$$

Total force on the wall is obtained by summing up the force on all such strips.

$$\therefore \quad F = \int dF = \rho gL \int_0^H y\,dy = \frac{1}{2}\rho gLH^2 \quad \text{...(5)}$$

There are two important things that you must notice:

1. We have deliberately left atmospheric pressure in our calculations. Atmospheric pressure acts on the wall from left to right as well. If you include atmospheric pressure in your calculations (by taking pressure as $P = P_0 + \rho gy$), then effect of atmospheric pressure on both sides just cancels out.
2. Pressure varies linearly with depth. Therefore, average pressure due to liquid over the surface of the wall is the average of the pressure at the top and the pressure at the bottom.

$$P_{av} = \frac{P_{\text{top}} + P_{\text{bottom}}}{2} = \frac{0 + \rho gH}{2} = \frac{1}{2}\rho gH$$

Force on the wall is equal to

$$F = (P_{\text{av}})(\text{Area of the wall}) = \left(\frac{1}{2}\rho gH\right)(LH) = \frac{1}{2}\rho gLH^2$$

Thinking this way, you can avoid integration.

4.2.1 Torque on the Wall

The thrust of liquid on the wall produces a torque about the base line AB (see the last figures). How can we calculate this torque?

Torque of force dF about AB is

$$d\tau = (H - y)\,dF = (H - y)\,\rho g L y\, dy$$

$$= \rho g L (Hy - y^2)\, dy$$

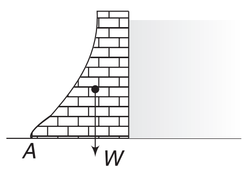
Total torque due to forces on all strips is

$$\tau = \int d\tau = \rho g L \int_0^H (Hy - y^2)\, dy$$

$$= \rho g L \left[\frac{Hy^2}{2} - \frac{y^3}{3}\right]_0^H = \rho g L \left[\frac{H^3}{2} - \frac{H^3}{3}\right]$$

$$= \frac{1}{6}\rho g L H^3 \qquad \text{...(6)}$$

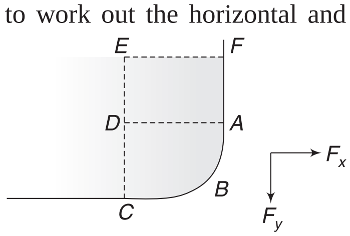
This equation is an important consideration in design of a dam wall. The torque due to water pressure tries to topple the wall about A. Weight of the wall (W) must produce a higher counter torque about A; otherwise, the wall will topple to the left, about A.



Dam wall. Torque due to weight of the wall about A must be greater than torque due to liquid force.

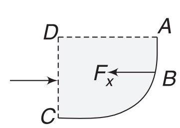
4.3 Force on a Curved Surface

Consider a curved surface ABC of a container wall, which contains a liquid. We wish to write force on this curved wall due to the liquid. It is easy to work out the horizontal and vertical components of the force, separately.



For finding horizontal component (F_x) of the force, consider the horizontal equilibrium of liquid in the volume represented by $ABCD$.

Force by curved surface on the liquid has a horizontal component (F_x) that is balanced by horizontal force on vertical wall CD due to the surrounding liquid.



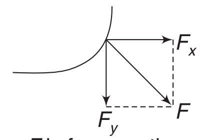
For horizontal equilibrium of liquid volume ABCD, force on face CD = F_x

$$\therefore\ F_x = \left(\frac{P_C + P_D}{2}\right)(\text{Area of face } CD)$$

Now, consider vertical equilibrium of the liquid volume $FABCDE$ (See first figure). Vertical force by the curved surface on the liquid balances the weight of liquid in volume represented by $FABCDE$.

$\therefore\ F_y$ = weight of liquid in volume $FABCDE$.

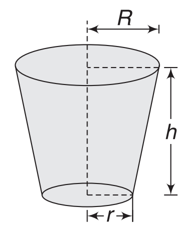
Force on the curved wall will have two components as shown in the figure. The resultant force is



F is force on the curved wall due to the liquid.

$$F = \sqrt{F_x^2 + F_y^2}$$

Example 5 A bucket is in the shape of a frustum of a cone with radii of top and bottom surfaces equal to R and r respectively. Height of the bucket is h. Water (density = ρ) is filled in the bucket to the brim. Find force on slant surface of the bucket due to water.



Solution

Concepts

(i) Volume of the frustum is $V = \frac{1}{3}\pi h (R^2 + r^2 + Rr)$

(ii) The bucket wall experiences force due to water that is normal everywhere. Obviously, the horizontal components add to zero. Vertical component is downward. From Newton's third law, the bucket wall applies the same force on water in upward direction.

(iii) By considering the equilibrium of water in the bucket, we can find the force applied by bucket wall on the water body.

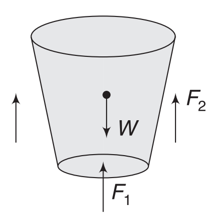
Pressure at base is

$$P = \rho g h$$

Force on base, $F_1 = \pi r^2 \cdot P$

$$= \pi r^2 \rho g h$$

Same force is applied by the base of the bucket on the water body.



Forces on water body

F_2 = Force by bucket wall on the water body.

W = weight of water = volume × density × g

$$= \frac{1}{3}\pi h (R^2 + r^2 + Rr)\rho g$$

For equilibrium of water

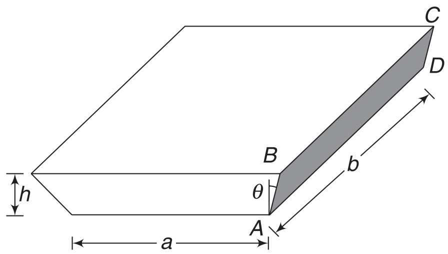
$$F_1 + F_2 = W \quad \Rightarrow \quad F_2 = W - F_1$$

$$\Rightarrow \qquad F_2 = \frac{1}{3}\pi h\,(R^2 + r^2 + Rr)\rho g - \pi r^2 \rho g h$$

$$= \frac{1}{3}\pi \rho g h [R^2 + r^2 + Rr - 3r^2]$$

$$= \frac{1}{3}\pi\rho gh[R^2 + Rr - 2r^2]$$

Example 6 A tray is of the shape shown in figure. It is completely filled with a liquid of density ρ. Find the force on its rectangular wall $ABCD$ due to liquid. The wall is inclined at an angle θ to the vertical.



Solution

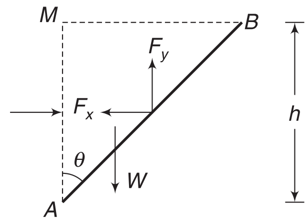
Concepts

(i) Like all perivous questions, we can forget about atmospheric pressure. Its effect is on both sides of the wall and balances out.

(ii) We will write the horizontal and vertical component of force separately. This can be done easily by considering the equilibrium of liquid contained in volume having its vertical cross-section denoted by ΔAMB (see figure below.)

We will consider equilibrium of liquid present in volume, having ΔAMB as cross-section and b as length.

Force on face AM of the considered volume due to liquid on left is equal to horizontal force by wall on the liquid.



$\therefore$ F_x = force on face AM by liquid on left

$$= (P_{av})\text{ (Area of face)}$$

$$= \left(\frac{P_M + P_A}{2}\right)(hb) = \left(\frac{0 + \rho gh}{2}\right)(hb)$$

$$= \frac{1}{2}\rho g h^2 b$$

Vertical force by the wall on liquid = Weight of liquid in volume denoted by ΔAMB

$$\Rightarrow \quad F_y = \text{volume} \times \text{density} \times g$$

$$= \frac{1}{2}b\,(AM)\cdot(BM)\,\rho\cdot g$$

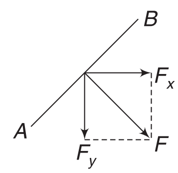
$$= \frac{1}{2}\,bh\,(h\tan\theta)\cdot\rho g = \frac{1}{2}\rho gh^2 b\,\tan\theta$$

Magnitude of resultant force on the wall is

$$F = \sqrt{F_x^{\,2} + F_y^{\,2}}$$

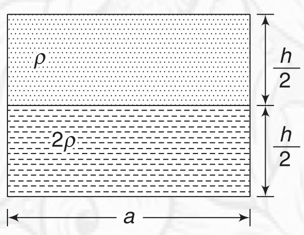
$$= \frac{1}{2}\rho gh^2 b\sqrt{1 + \tan^2\theta}$$

$$= \frac{1}{2}\rho gh^2 b\cdot\sec\theta$$



YOUR TURN

Q.5 A container has a square base of length a = 0.5 m. Its height is h = 1.0 m. Half of it is filled with a liquid of density 2ρ and upper half contains another liquid of density ρ = 2000 kgm^{-3}.

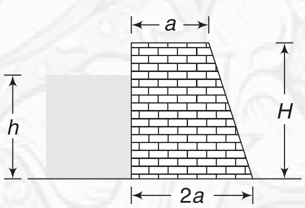


(a) Find force on base of the container due to the liquids.

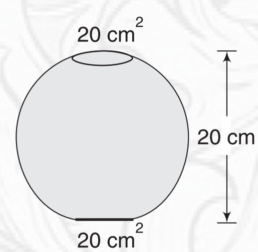
(b) Plot a graph showing variation of pressure with height (y) from the base of the container. Ignore atmospheric pressure.

Q.6 In the last problem, find the force experienced by a vertical wall of the tank due to the liquids.

Q.7 Figure shows a dam wall whose length perpendicular to the plane of the figure is L = 30 m. Height of the wall is H = 12 m and water in the dam has a depth of h = 10 m. The bricks used in construction of the wall have density of 3 gcm^{-3}. Find the width a if it is required that weight of the dam wall be 10 times the horizontal force on it due to water.



Q.8 An earthen pot has the shape shown in figure. Its base surface as well as the mouth has area 20 cm^2. Height of the pot is 20 cm. It is completely filled with half a litre water.

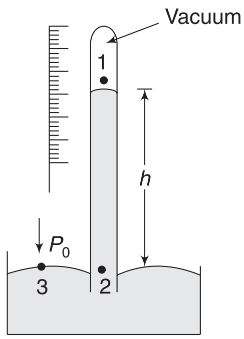


(a) Find the increase in force on the bottom surface due to filling of the container with water.

(b) Find force applied by the pot wall on water.

5. BAROMETER

Barometer is a device used to measure atmospheric pressure. Figure shows a mercury barometer, which was invented by Torricelli in 1640 s. A vertical glass tube, closed at the top, stands on an open mercury-filled basin. Lower end of the tube, which is open, is held dipped inside mercury. The top of the tube has no air. With the aid of a scale, height (h) of the mercury in the tube above the liquid surface in the basin is recorded. This gives us the value of atmospheric pressure.



Pressure at point 1 is zero as there is no fluid present. Pressure at point 2 is

$$P_2 = P_1 + \rho_{Hg} \cdot g \cdot h = 0 + \rho_{Hg} \cdot g \cdot h = \rho_{Hg} \cdot g \cdot h$$

Pressure at point 3 (on surface of mercury outside the tube) is equal to atmospheric pressure (P_0). The basin is open to air and pressure on surface of liquid in the basin is P_0.

We known that pressure at all points on same horizontal plane are equal in a static liquid.

$$\therefore \qquad P_3 = P_2$$

$$\Rightarrow \qquad P_0 = \rho_{Hg} \cdot g \cdot h \qquad \text{...(7)}$$

If you record the height of mercury column as h = 76 cm then the atmospheric pressure is

$$P_0 = (13.6 \times 10^3 \,\text{kgm}^{-3}) \times (9.81 \,\text{ms}^{-2}) \times (0.76 \,\text{m})$$
$$= 1.013 \times 10^5 \text{ Nm}^{-2}$$

Note: Density of Hg is 13.6×10^3 kgm^{-3}

Actually, barometer is a kind of balance. Atmospheric pressure pushes the liquid surface down. Inside the tube, it is weight of *Hg*-column that creates same pressure at point 2. If somehow the atmospheric pressure increases, the balance is destroyed. More liquid gets pushed into the tube so that height of *Hg*-column increases to balance the pressure.

Many a times, pressure is expressed in terms of height of *Hg*-column. A pressure equal to h height of *Hg*-column is same as $P = \rho_{Hg} \cdot g \cdot h$.

Example 7 *Why not a water Barometer*?

Mercury is poisonous and costly too. Why should not we construct a barometer using water?

Solution The primary reason is height of liquid column in the barometer tube. If water is used, its height in the tube is given by

$$P_0 = \rho_w \text{g} \cdot h \quad \Rightarrow \quad h = \frac{P_0}{\rho_w g}$$

$$h = \frac{1.013 \times 10^5}{10^3 \times 9.81} = 10.3 \,\text{m}$$

This is almost equal to height of a three-storey building! This will make the barometer impractical to use.

YOUR TURN

Q.9 Suppose we don't have air all around us but have mercury instead. How high up would that mercury atmosphere extend from ground if it were to make the same pressure that we feel?

Q.10 Assume that the atmospheric pressure is same at equator and at poles. Where will the barometer show higher reading – at equator or at poles?

Q.11 A barometer has a rigidly fixed tube that extends 1m above the mercury surface in the basin. The basin is wide. If the barometer is placed inside a freely falling lift. What reading will it show?

Q.12 A faulty barometer has some air trapped in the space in the tube above the *Hg*-column. It shows a reading of 72 cm on a day when atmospheric pressure is equal to 76 cm height of *Hg*. Find the pressure of the air in the tube in Pa.

6. MANOMETER

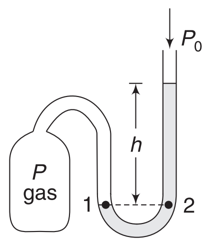
It is a device used to measure pressure difference between two points. Here, we will discuss a manometer which is commonly known as *U*-tube manometer.

A *U*-tube manometer is simply a transparent *U*-shaped tube filled with a liquid (usually mercury). Imagine one end of the tube connected to a gas cylinder and the other end left open to atmosphere. If the pressure of the gas is larger than the atmospheric pressure (P_0), then the liquid in the *U*-tube will appear as shown in the figure–elevated on right and depressed on left side of the tube.

Pressure at point 1 in the tube is same as pressure (P) of the gas in the cylinder and pressure at point 2 can be written as

$$P_2 = P_0 + \rho gh$$

Where ρ is density of manometer liquid.



'h' gives the pressure difference between two ends of the manometer tube

But 1 and 2 are two points on same horizontal level in a liquid.

$$\therefore \qquad P_1 = P_2$$

$$\Rightarrow \qquad P = P_0 + \rho gh$$

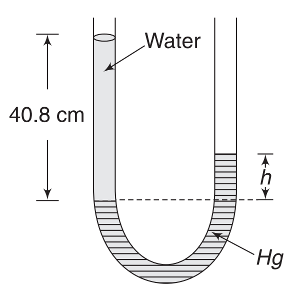
$$\Rightarrow \qquad P - P_0 = \rho gh \qquad ...(8)$$

The above equation gives pressure difference between two ends of the manometer.

The left side of the above equation is the difference in absolute pressure of the gas and the atmospheric pressure. This is sometimes known as *guage pressure* of the gas.

$$P_{\text{guage}} = P_{\text{absolute}} - P_{\text{atmospheric}}$$

Example 8 A *U*-shaped tube contains mercury. Water is poured in the left limb to a height of 40.8 cm. Now, mercury column has different surface levels on two sides of the tube. Find difference (h) in height of *Hg*-level on two sides. Relative density of Hg is 13.6.

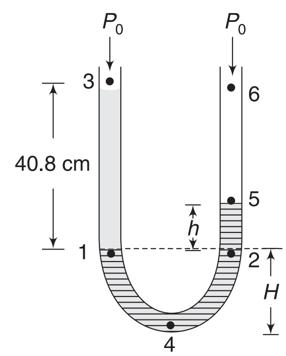


Solution

Concepts

(i) The method of writing equation (8) can be used to get value of h.

(ii) We will take a point (1) in *Hg* in left column just below the *Hg*-surface and another point (2) in right column, which is at same horizontal level as 1. Pressure at point 1 = pressure at point 2.



$$P_2 = P_1$$

$$P_0 + \rho_{\text{Hg}} \cdot g \cdot h = P_0 + \rho_w \cdot g \cdot (40.8\,\text{cm})$$

$$\Rightarrow \qquad \frac{\rho_{\text{Hg}}}{\rho_w} h = 40.8\,\text{cm}$$

$$\Rightarrow \qquad 13.6h = 40.8\,\text{cm}$$

$$\Rightarrow \qquad h = 3\,\text{cm}$$

Note: Value of h is independent of values of cross-sectional area of the two tubes. Even if the tubes on left and right have unequal cross-sectional area, value of h remains same.

Alternate way of writing the above equation:

Start from top (point 3) of the left tube and move down to point 1 and then to point 4. Keep adding ρgh terms to write pressure at point 4. Now move up from point 4 to 5. subtract ρgh terms to get pressure at 5. Pressure at 3 and 5 both are equal to atmospheric pressure.

$$\underbrace{\underbrace{P_3 + \rho_w\, g(40.8\,\text{cm})}_{\text{Pressure at 1}} + \rho_{\text{Hg}} gH}_{\text{Presssure at 4}} - \rho_{\text{Hg}}\, g(H + h) = P_5$$

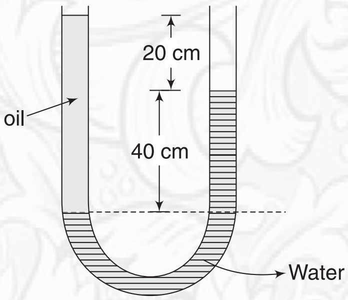
$$\Rightarrow \qquad P_0 + \rho_w g(40.8\,\text{cm}) - \rho_{\text{Hg}} gh = P_0$$

$$\Rightarrow \qquad h = (40.8\,\text{cm})\,\frac{\rho_w}{\rho_{\text{Hg}}} = \frac{40.8}{13.6} = 3\,\text{cm}$$

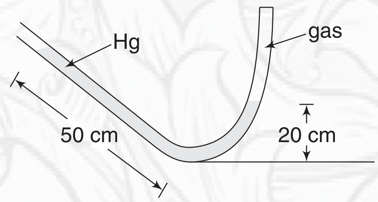
One may also chose to write pressure at 4 while moving down from 3 to 4 and then while moving down from 5 to 4. Equate the two values. You will get same equation as above.

YOUR TURN

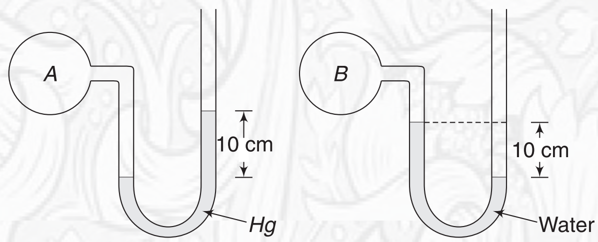
Q.13 In the arrangement shown in the figure, find the relative density of the oil. What is guage pressure at oil-water interface?



Q.14 Left arm of a U-tube is inclined at 30° to the horizontal. It contains mercury. The right end of the tube is closed and contains a gas. In equilibrium position, the length of Hg column on left and right of the tube is 50 cm and 20 cm, as shown. Find pressure of the gas in Pa. Atmospheric pressure is $P_0 = 1 \times 10^5$ Nm^{-2}. Density of Hg is 13.6×10^3 kgm^{-3}. $g = 10$ ms^{-2}



Q.15 Bulbs A and B contain gas. Bulb A is connected to mercury barometer and B is connected to a water barometer.



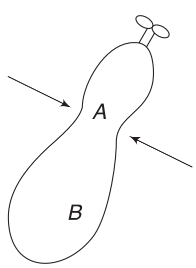
(a) In which bulb is the pressure of the gas higher?

(b) Which bulb has negative guage pressure?

(c) Find difference in pressure of A and B. Express your answer in terms of height of Hg column.

7. PASCAL'S LAW

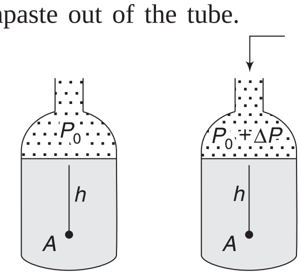
If you have a well inflated balloon and you squeeze it at some point (A), it may burst at some other point (B). The pressure inside the balloon cannot be locally increased at A, leaving all other points unaffected. Same pressure change happens everywhere inside the balloon and it brusts where the wall is weakest.



A ballon pressurised at A may burst at B.

Pascal's law states that *a change in pressure applied to an enclosed imcompressible fluid is transmitted undiminished to every point of the fluid and to the walls of the container*.

When you squeeze the lower end of the a toothpaste tube, you increase the presence everywhere inside the tube. This forces the toothpaste out of the tube.



Consider a jar filled with some water. The jar is open at the top and pressure at the water surface is equal to atmospheric pressue P_0. Pressure at a point A at depth h is

$$P_A = P_0 + \rho g h$$

Now, a lot of air in pumped inside the jar using an air pump. The pressure at the water surface now has increased and is equal to $P_0 + \Delta P$. Pressure at A now is.

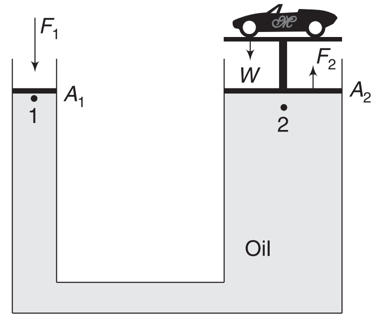
$$P_A' = (P_0 + \Delta P) + \rho g h$$

Thus, pressure at A has increased by $P_A' - P_A = \Delta P$. This is independent of h. Thus, pressure must have increased by ΔP at all points in the water. Pressure change on the surface has been transmitted throughout the water. This is a simple proof of Pascal's law.

7.1 Hydraulic Lift

A mechanic at a car workshop uses hydraulic lift to raise a car. A hydraulic lift is a forces-multiplying device that works on Pascal's principle.

Consider a U-shaped tank shown in figure. It is filled with an oil. Left side of the tank has a small cross-sectional area (A_1) and is fitted with a tight-fitting smooth piston. Right side is also fitted with a movable smooth piston. Area of right limb is A_2, which is much larger than A_1.



In a hydraulic lift, a non-corrosive oil is filled in U-shaped tank. Area $A_2 >> A_1$

The heavy object (such as a car) that is to be lifted is placed on the right piston and the person (such as mechanic) applies downward force F_1 on the piston of area A_1.

Increase in pressure at 1 is $\Delta P = \dfrac{F_1}{A_1}$

The same change in pressure occurs everywhere. Pressure at 2 increases by ΔP. Thus, the right piston experiences an extra upward force equal to

$$F_2 = \Delta P \cdot A_2 = F_1\left(\frac{A_2}{A_1}\right) \quad ...(9)$$

If F_2 is equal to weight (W) of the car, the car begins to rise.

$\dfrac{A_2}{A_1} >> 1$; hence, $F_2 >> F_1$

For example, if $\dfrac{A_2}{A_1} = 100$ then applying a force of $F_1 = 100\,\text{N}$ will result in

$$F_2 = F_1\left(\frac{A_2}{A_1}\right) = 100 \times 100 = 10{,}000\,\text{N}$$

This is good enough to lift a 1,000 kg car!

You must pay attention to the fact that there is no gain in terms of work. If the force F_1 pushes the piston down by $x_1 = 100\,\text{cm}$, the other piston rises up by only $x_2 = 1\,\text{cm}$ (volume of liquid in 100 cm length of left column = volume of liquid in 1 cm length of the right column).

Work done in pushing the left piston is $W_1 = F_1 x_1$ and the work output that we get in form of the right piston raising our car is $W_2 = F_2 x_2 = (100F_1)\left(\dfrac{x_1}{100}\right) = F_1 x_1 = W_1$

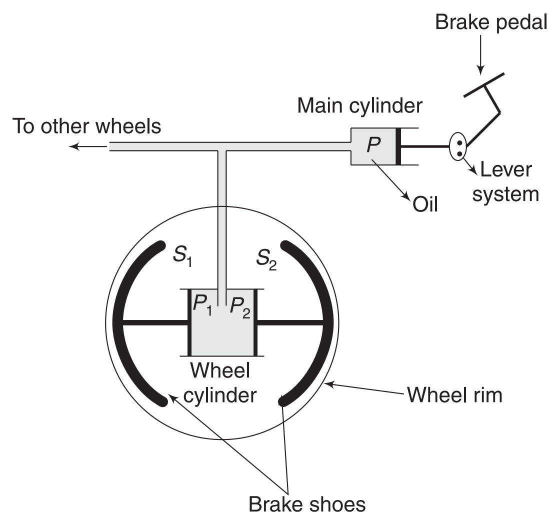
Therefore, work input = work output.

Due to various frictional losses, in practice, we will get $W_2 < W_1$.

It is easily possible to design a hydraulic lift in which a series of up and down strokes of the left piston can be made one after another, each time raising the car a little.

Example 9 *Hydraulic Brake*

Figure shows a schematic diagram of a hydraulic braking system in an automobile. Main cylinder is fitted with piston P having area of $A = 4\,\text{cm}^2$. The wheel cylinder has two pistons – P_1 and P_2 – each of area $A_1 = 36\,\text{cm}^2$. When the brake pedal is pushed, the lever system exerts a force of $F = 20\,\text{N}$ on piston P. As a result, the pistons P_1 and P_2 push the brake shoes S_1 and S_2, which in turn press against the wheel. Due to friction between the brake shoes and the wheel, the rotation of the wheel is arrested. Find the force with which each brake shoe presses the wheel.



Solution

Concepts

Pressure in the fluid increases by $\dfrac{F}{A}$

Change in pressure is same everywhere; pistons P_1 and P_2 experience outward force due to this increase in pressure.

Increase in liquid pressures when the brake pedal is pressed is

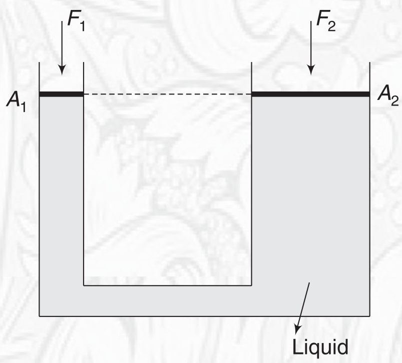
$$\Delta P = \frac{F}{A} = \frac{20\,\text{N}}{4\,\text{cm}^2}$$

Force that acts on each of S_1 and S_2 is

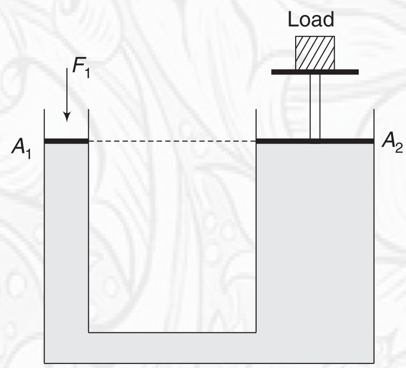
$$F_1 = \Delta P \cdot A_1 = \left(\frac{20\,\text{N}}{4\,\text{cm}^2}\right)(36\,\text{cm}^2) = 180\,\text{N}$$

YOUR TURN

Q.16 The areas of cross-section of two arms of the *U*-tube shown in figure are $A_1 = 2\,\text{cm}^2$ and $A_2 = 10\,\text{cm}^2$. A force $F_1 = 10\,\text{N}$ is applied on the piston of area A_1 as shown. The mass of the piston itself is 0.5 kg. The piston on the other side has a mass of 2 kg. How much force F_2 must be applied on this piston of area A_2 so that the entire system remains in equilibrium?



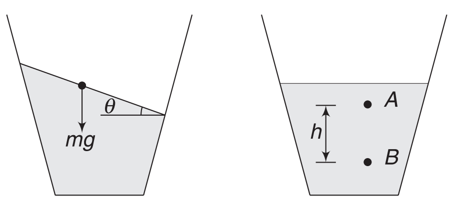
Q.17 In the hydraulic lift shown, cross-sectional areas of the two pistons are $A_1 = 10\,\text{cm}^2$, $A_2 = 100\,\text{cm}^2$



The load has a mass of 100 kg and both pistons are massless. When pistons try to move, a friction force of 20 N acts on each of them. Find the forec F_1 with which the left piston shall be pushed down so as to make the load just move up.

8. FLUID IN ACCELERATED CONTAINER

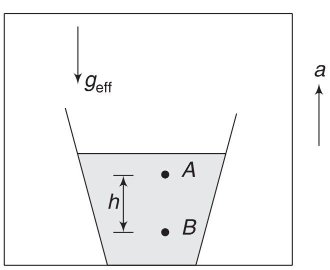
Can you keep water in a glass as shown in figure (A)? The answer is no, if the glass is at rest. But it is possible if the glass is accelerated.



(A) (B)

If water surface, in a glass at rest, were inclined (as in figure *A*), then a water particle on the surface will experience a tangential force ($mg\sin\theta$), which will cause it to slide. [Remember, there is no viscous force (i.e., friction) between layers of an ideal fluid.] The water surface will ultimately become horizontal, as shown in figure *B*. Now, a particle on the surface experiences no tangential force. We know that in this position $P_B - P_A = \rho gh$.

Imagine a glass of water kept inside an elevator that is accelerating up with acceleration a. Any observer inside the elevator will experience effective value of acceleration of free fall to be $g_{\text{eff}} = (g + a)$. The water surface remains horizontal (normal to g_{eff}) and pressure difference between A and B is given by



Liquid in accelerated lift.

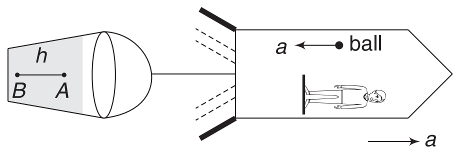
$$P_B - P_A = \rho g_{\text{eff}}\, h = \rho(g + a)h \qquad ...(10)$$

If the container is having a vertically downward acceleration of a, then $g_{\text{eff}} = (g - a)$ and

$$P_B - P_A = \rho(g - a)h \qquad ...(11)$$

Recall that, no observer can differentiate between the effect of gravity and acceleration.

To re-emphasise the above point, consider a thought experiment. A man is riding a space rocket, which is moving with acceleration a. There is a bucket tied to its tail. Bucket contains water. The orientation of the bucket and the water surface will be as shown in the figure. If the man releases a ball, it falls towards the tail of the rocket with acceleration a. Everything in this environment behaves as if there is gravity towards left. The water surface also orients itself



A bucket full of water tied to a space rocket

perpendicular to the direction of effective gravity. Note that

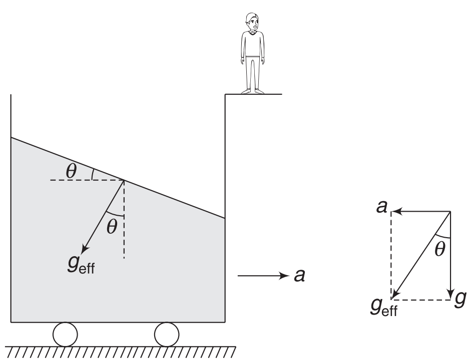
$$P_B - P_A = \rho g a \qquad ...(12)$$

Now, consider an open, water-filled container travelling horizontally with acceleration *a*. An observer attached to this container is in an environment where effective acceleration of free fall is

$$g_{\text{eff}} = \sqrt{g^2 + a^2} \qquad ...(13)$$

Making an angle θ with vertical, where

$$\tan\theta = \frac{a}{g}$$



The liquid surface will orient itself perpendicular to g_{eff}

If the observer attached to the container releases a ball, he finds it to move in the direction of g_{eff} shown in the figure. The liquid also feels g_{eff} in the shown direction and orients its surface perpendicular to it. Therefore, the liquid surface gets inclined to the horizontal at an angle

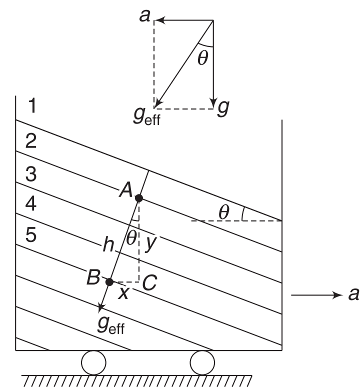
$$\theta = \tan^{-1}\left(\frac{a}{g}\right) \qquad ...(14)$$

In this position, a liquid particle on the surface will experience no tangential force. The particle will not flow. It will remain static, relative to the container.

8.1 Variation of Pressure in a Liquid in Horizontally Accelerated Container

Fluid in the container is in an environment where

$g_{\text{eff}} = \sqrt{g^2 + a^2}$ and direction of g_{eff} makes an angle $\theta = \tan^{-1}\left(\frac{a}{g}\right)$ with vertical.



Planes indicated by 1, 2, 3, 4... are equipressure planes

Consider a plane (such as 1, 2, 3, 4 ...) in liquid that is normal to g_{eff}. Pressure at all points on any such plane is same.

Consider two points *A* and *B* inside the liquid, as shown in figure. Pressure difference between the two points is

$$P_B - P_A = \rho g_{\text{eff}} \cdot h \qquad ...(15)$$

$$= \rho (g \sec\theta) . h \qquad \left[\because \frac{g}{g_{\text{eff}}} = \cos\theta\right]$$

$$= \rho (g \sec\theta)(y \sec\theta)$$

[y = vertical separation between *A* and *B*]

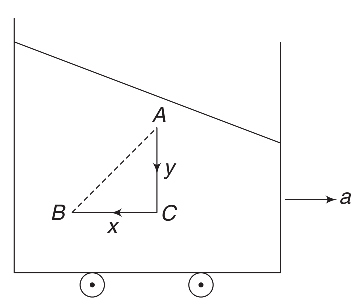
$$= \rho g y \sec^2\theta = \rho g y (1 + \tan^2\theta)$$

$$= \rho g y + \rho (g \tan\theta)\,(y \tan\theta)$$

$$\therefore \quad P_B - P_A = \rho g y + \rho \cdot a \cdot x \qquad ...(16)$$

$$\left[\because \frac{a}{g} = \tan\theta \text{ and } \frac{x}{y} = \tan\theta\right]$$

The last equation gives us a simple method of writing pressure difference between any two points in a fluid, which is in an accelerated container.



To write pressure difference between point *B* and *A*, move vertically from *A* to *C* and write

$$P_C - P_A = \rho g y \qquad ...(17)$$

Now move horizontally from *C* to *B* and write

$$P_B - P_C = \rho a x \qquad ...(18)$$

$$\Rightarrow \quad P_B - (P_A + \rho g y) = \rho a x \quad \Rightarrow \quad P_B - P_A = \rho g y + \rho a x$$

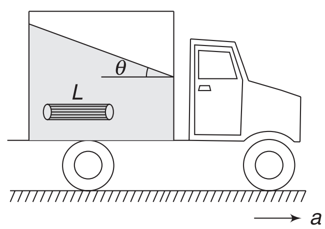
Remember that pressure increases when you move horizontally opposite to a.

Example 10 *Alternative way to get equation 14*

A liquid of density ρ is being transported in a tanker that is moving horizontally with acceleration a. Consider a horizontal cylindrical element of the liquid inside the tanker, as shown in the figure. Cross-sectional area of the cylinder is small.

(a) Write pressure difference between the ends of the cylinder by considering its accelerated motion in horizontal direction.

(b) Using the above result, find the angle θ that the liquid surface must make with horizontal.



Solution

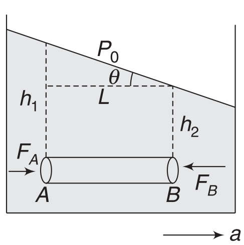
Concepts

(i) The cylindrical element of the liquid can accelerate only if there is a net horizontal force on it. This implies that pressure at the left end is higher than pressure at the right end.

(ii) Net horizontal force on cylinder due to pressure difference at two ends = (mass) × (acceleration)

(iii) The pressure at left end is high because the depth from free surface of liquid is more. The liquid surface is slant, creating more depth at left end.

(a) Let cross-sectional area of the liquid cylinder be ΔS and its length be L. Horizontal force on the cylinder is



$$F = F_A - F_B$$

$$= P_A \Delta S - P_B \Delta S$$

Using $F = ma$

$$P_A \Delta S - P_B \Delta S = (\rho L \Delta S) a$$

$$\Rightarrow \quad P_A - P_B = \rho L a \qquad ...(i)$$

This is nothing but equation 18.

(b) Look into the figure. We can write

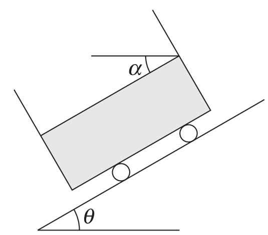
$$P_A = P_0 + \rho g h_1 \text{ and } P_B = P_0 + \rho g h_2$$

$$\therefore \qquad P_A - P_B = \rho g (h_1 - h_2)$$

From (i) $\quad \rho g (h_1 - h_2) = \rho L a$

$$\Rightarrow \quad \frac{h_1 - h_2}{L} = \frac{a}{g} \quad \Rightarrow \quad \tan\theta = \frac{a}{g}$$

Example 11 A trolley containing a liquid slides down a smooth incline of inclination angle θ. Find the angle α that the free surface of the liquid makes with horizontal



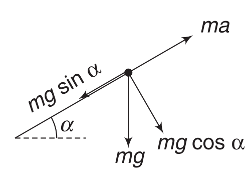
Solution

Concepts

A particle on the surface of the liquid must not have any tangential force acting on it in the reference frame of the trolley.

The trolley has an acceleration $a = g \sin\theta$ down the incline.

In *RF* attached to the trolley, forces acting on a particle on liquid surface are as shown in the figure. Tangential force will be zero, if



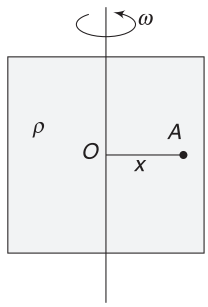
ma = pseudo force in *RF* of trolley

$$mg \sin\alpha = ma$$

$$\Rightarrow \quad g \sin\alpha = g \sin\theta \Rightarrow \alpha = \theta$$

Example 12 *Rotating tank*

A tank is filled completely with a liquid of density ρ. It is spinning about a vertical axis with angular speed ω. Find difference in pressure at a point O on the axis and a point, A located at a distance x from the axis.

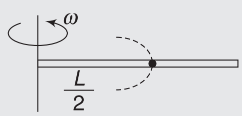


Solution

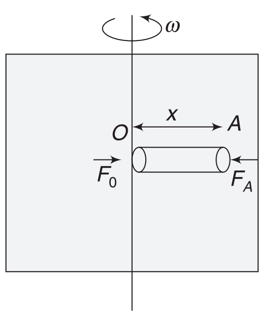
Concepts

(i) Consider a horizontal cylinder of the fluid. The centripetal force for its rotation is generated by the pressure difference at its two ends.

(ii) If a cylindrical rod of mass M and length L is rotating about an axis as shown, then centripetal force on it can be written assuming the entire mass of the rod to be at its COM.



$$F = M\omega^2 \frac{L}{2}$$



Consider a horizontal cylinder of liquid of length x and small cross-sectional area ΔS. Centripetal force needed for its rotation is $\quad F = \Delta m\omega^2 \dfrac{x}{2}$

Where $\quad \Delta m$ = mass of cylindrical element

$$= \rho \cdot \Delta \text{S} \cdot x$$

$$\therefore \qquad F = \frac{1}{2}\rho\ \Delta S\,\omega^2 x^2$$

This force is provided by the surrounding fluid. Pressure at A is larger than pressure at O and

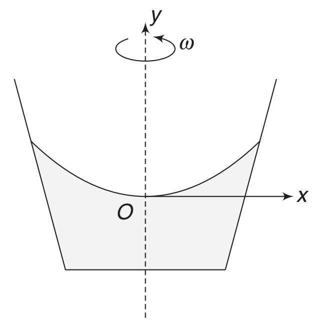
$$F = F_A - F_0$$

$$\Rightarrow \quad \frac{1}{2}\rho\Delta S\,\omega^2 x^2 = P_A \cdot \Delta S - P_0 \cdot \Delta S$$

$$\Rightarrow \qquad P_A - P_0 = \frac{1}{2}\rho\ \omega^2 x^2 \qquad \text{...(19)}$$

Example 13 *Water in spinning bucket*

Water is kept in a bucket, which is spinning about its central vertical axis with angular speed ω. The surface of the liquid forms a depression as shown. Take the lowest point of depression as origin (O) and prove that the vertical cross-section of the surface of water is a parabola. Find its equation.



Solution

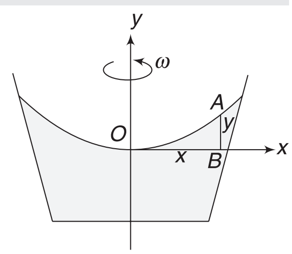
Concepts

We will solve this problem in two different ways. In the first method, we will use the result obtained in the last example.

In the second method, we will once again use the fact that a water particle on the surface must not experience a tangential force in the reference frame attached to the bucket.

Point A on the surface has co-ordinates (x, y).

From result in the last problem, we have



$$P_B - P_0 = \frac{1}{2}\rho\,\omega^2 x^2$$

$$\text{But} \qquad P_B = P_A + \rho g y$$

$$\therefore \quad P_A + \rho g y - P_0 = \frac{1}{2}\rho\,\omega^2 x^2$$

$$\Rightarrow \qquad \rho g y = \frac{1}{2}\rho\,\omega^2 x^2$$

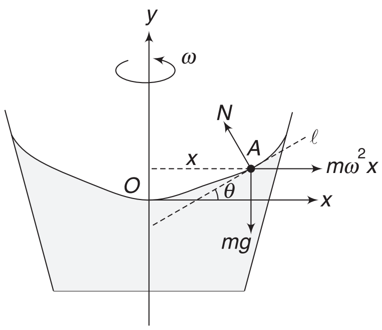
[$\because \quad P_A = P_0$ = atmospheric pressure]

$$\Rightarrow \qquad y = \left(\frac{\omega^2}{2\text{g}}\right)x^2$$

This is an equation of a parabola. If we rotate this parabola about y-axis, we get the three-dimensional surface of water. The shape of the surface (obtained by rotating a parabola) is known as paraboloid.

Alternate

Consider a particle of mass m on the surface of the liquid at point A. This particle is rotating in a circle of radius x. $m\,\omega^2 x$ is the centrifugal force on this particle in RF of the bucket.



Line l is tangent to the surface at A and makes an angle θ with x-axis. Obviously, $\tan\theta$ is the slope of the curved surface at point A.

$$\therefore \qquad \frac{dy}{dx} = \tan\theta$$

For no tangential force on the particle,

$$m\omega^2 x\cos\theta = mg\sin\theta \quad \Rightarrow \quad \tan\theta = \frac{\omega^2}{g}x$$

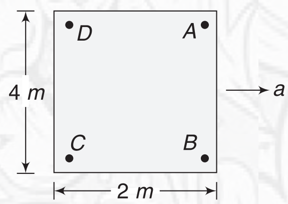
$$\therefore \qquad \frac{dy}{dx} = \frac{\omega^2}{g}x$$

$$\Rightarrow \qquad \int_{y=0}^{x} dy = \int_{x=0}^{x} \frac{\omega^2}{g}x\,dx$$

$$\Rightarrow \qquad y = \frac{\omega^2}{2g}\cdot x^2$$

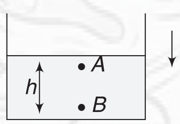
Your Turn

Q.18 A closed container is completely filled with water. It is accelerated to the right with an acceleration of a = 4ms^{-2}. A, B, C and D are points in the liquid at the four corners, as shown. Find [take g =10ms^{-2}]

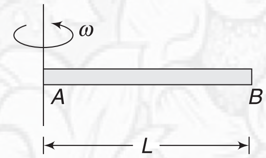


(a) $P_B - P_A$ (b) $P_D - P_A$ (c) $P_A - P_C$.

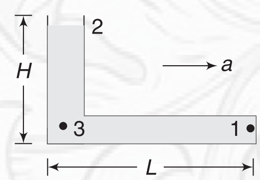
Q.19 What is pressure difference between two points, A and B, inside a freely falling water-filled container?



Q.20 A tube AB of length L = 0.1 m is rotating about its end A at an angular speed ω = 20 rad s^{-1}. The tube is horizontal and is completely filled with water. End A of the tube is open to atmosphere. Find pressure at end B. Atmospheric pressure is $P_0 = 1 \times 10^5$ Pa.



Q.21 An L-shaped tube contains a liquid of density ρ. It is moved horizontally with acceleration $a = \frac{g}{2}$. Find pressure at End 1 of the tube. End 2 is open and atmospheric pressure is P_0. Express L in terms of H, if pressure at point 1 is equal to P_0.



In Short:

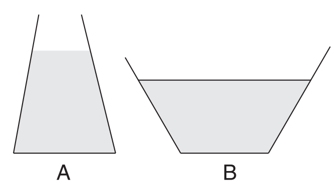
(i) A fluid exerts force on any surface, which comes into contact with it. The force is normal to the surface. This force per unit area is called pressure. Its SI unit is Nm^{-2} (also known as Pascal).

(ii) Any small volume of fluid experiences force from all sides, normal to its surface, due to surrounding fluid. Force per unit area is pressure.

(iii) In a static fluid, pressure is same at all points on a horizontal plane.

(iv) In a static fluid, pressure increases with depth. If point B is at a depth h below A, then $P_B = P_A + \rho gh$

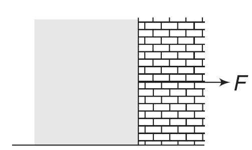
(v) A real-life container having a gas has same pressure at all points. The term ρgh becomes significant for gases when h is too large.



(vi) Force on base of container A due to liquid is more than weight of the liquid in the container. This is due to force applied by slant wall on the liquid. In container B, force on base is less than weight of the liquid in the container.

(vii) Force on vertical wall due to the liquid is

$F = (P_{av})$ (Area of wall in contact with liquid)



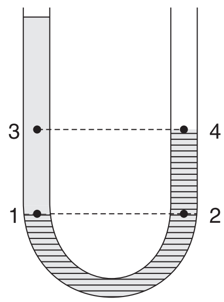
$$= \left(\frac{P_{top} + P_{bottom}}{2}\right) \text{(Area)}$$

(viii) If height of *Hg*-column in a barometer is *h*, then atmospheric pressure is $P_0 = \rho_{Hg} g h$

(ix) In the *U*-tube shown in figure

$$P_3 \neq P_4$$

Points 3 and 4 are at same horizontal level but they are not in the same liquid.



It is correct to write

$$P_1 = P_2$$

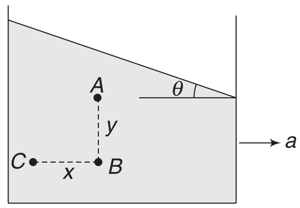
(x) Any change in pressure at a point in an enclosed fluid is transmitted throughout the fluid. This is known as Pascal's law. Using this, we can make a hydraulic lift, which is a force-multiplying device.

(xi) If a liquid is kept in a vertically accelerated container, pressure difference between two points (*B* at depth *h* below *A*) is

$$P_B - P_A = \rho g_{eff} h$$

where $g_{eff} = g + a$ for a container having upward acceleration and $g_{eff} = g - a$ for a container having downward acceleration.

(xii) In a liquid kept in a horizontally accelerated container



$$\tan\theta = \frac{a}{g}$$

$$P_B - P_A = \rho g y \text{ and}$$

$$P_C - P_B = \rho a x$$

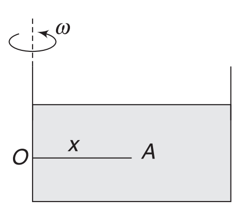
(xiii) If the container shown in the above figure is having an upward acceleration (a_0) apart from horizontal acceleration (a), then

$$\tan\theta = \frac{a}{g + a_0}$$

$$P_B - P_A = \rho(g + a_0)y$$

and $P_C - P_B = \rho a x$

(xiv) For a tank filled with liquid and rotating about a vertical axis

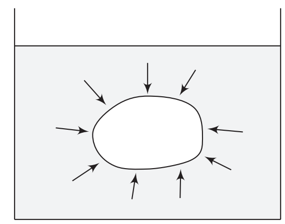


$$P_A - P_0 = \frac{1}{2}\rho\omega^2 x$$

(xv) Tangential force on a liquid particle on its surface must be zero if the liquid is static, relative to the container.

9. BUOYANCY

When a body is dipped, partially or completely, into a fluid, the fluid exerts force on the body. On any small surface area of the body, the force due to the surrounding fluid is normal to the surface. The resultant of these forces acting on all small patches on the surface of the body is known as *force of buoyancy* or *buoyant force*. The force arises because the pressure is not same everywhere in the fluid.

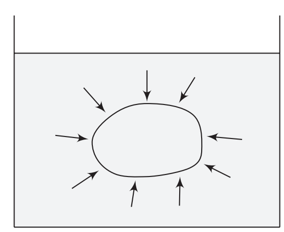


Buoyancy is resultant of forces applied by the fluid on all small patches on the solid surface.

We all have experienced this force in our life. Any floating object has its weight balanced by buoyancy. A bucket inside water in a well feels lighter due to buoyancy. It was Archimedes who first quantified the buoyancy force.

Archimedes' principle states that when a body is partially or completely immersed in a fluid at rest, the fluid exerts an upward force of buoyancy that is equal to the weight of the displaced fluid.

The principle is valid for a body of any shape. How could Archimedes conclude this? This is a bit tricky. Consider the above diagram of a solid immersed in a fluid. Surrounding liquid molecules strike its surface from all sides, producing a resultant force on the body. Now, suppose that we replace the solid body with the same fluid, occupying the volume occupied by the solid. Nothing has changed for the surrounding liquid particles. They keep on hitting the boundary in the same way as they were doing with the boundary of the solid. Therefore, the surrounding fluid exerts a force on the solid that is same as the force it exerts on the replaced fluid. It is very easy to write the force on the replaced fluid. As the entire fluid now becomes homogenous, all parts will remain in equilibrium.



Solid replaced with fluid. Volume of replacement fluid can stay in equilibrium if it experiences a net force in upward direction equal to its weight, due to surrounding fluid.

The volume of the fluid that has been used to replace the solid also remains in equilibrium. This volume of fluid can stay in equilibrium only if surrounding liquid exerts a force on it that balances its weight. Therefore, force applied by surrounding fluid is equal to weight of the volume of fluid under consideration and is in vertically upward direction. Force on submerged solid is equal to this force.

Buoyancy force on a solid dipped partially or completely in a fluid can be written as

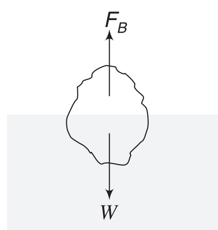
$$\begin{aligned} F_B &= \text{Weight of displaced fluid} \\ &= (\text{volume of displaced fluid}) \times (\text{density of fluid}) \times g \\ &= (\text{volume of submerged part of solid}) \times (\text{density of fluid}) \times g \end{aligned}$$

or, $$F_B = V\rho_f g \qquad \text{...(20)}$$

Where ρ_f is density of fluid and V is the volume of submerged part of the solid.

9.1 Floatation

A body is said to be floating when its weight is balanced by the buoyancy force. If you put a block of wood in water, it does not sink. A part of it gets submerged and displaces water. It stays in equilibrium in a position where the weight of displaced water ($=F_B$) balances its weight (W).



A floating body

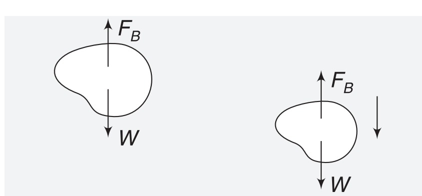
$$F_B = W$$

$$\Rightarrow \quad V_{\text{sub}}\rho_L g = V\rho_s g$$

$$\Rightarrow \quad \frac{V_{\text{sub}}}{V} = \frac{\rho_S}{\rho_L} \qquad \text{...(21)}$$

Where V_{sub} is the volume of submerged part of the solid and V is the volume of the solid. ρ_S and ρ_L are densities of the solid and the liquid respectively.

If $\rho_S = \rho_L$, then buoyancy can balance the weight of the solid only if it is completely submerged. If we place a body of relative density 1.0 inside water, it will remain at the location where it is released. It neither sinks nor comes up.



If $\rho_S = \rho_L$ body floats while remaining completely submerged

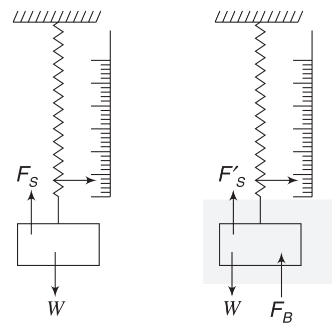
If $\rho_S > \rho_L$ the body sinks.

If $\rho_S > \rho_L$, then weight of the solid is always greater than buoyancy and the solid sinks.

The same holds true for a balloon in air. A balloon that weighs 1 ton displaces 1 ton of air, if it is hovering at a constant altitude. If it displaces more, it rises; if it displaces less, it falls.

9.2 Apparent Weight

When you suspend a solid block from a spring balance, it shows true weight of the body. The spring balance measures tension in the spring, which is equal to the weight of the body.



Tension in spring is equal to weight of the body.

Tension in the spring decreases due to buoyancy

$$F_S = W$$

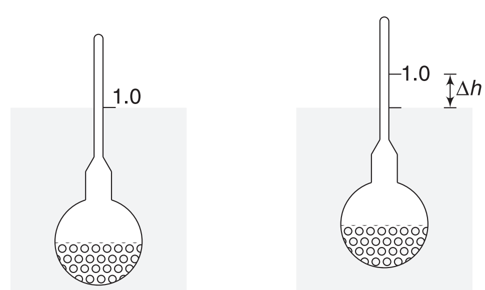
If the body is submerged in a liquid, reading of the balance decreases. This is due to buoyancy acting on the solid. Now, the tension in spring (= reading of spring balance) is

$$F'_S = W - F_B$$

Thus, Apparent weight = True weight – Buoyant force ...(22)

9.3 Hydrometer

A hydrometer is a device used to measure specific gravities of liquids. It has a glass bulb attached to a stem. The bulb is loaded with mercury or lead shots to make it float upright. [You will understand this after going through Section 9.5].



In Water

In a different liquid

When placed in water, it floats with a total volume V_0 submerged. Weight of the entire hydrometer is equal to buoyancy.

$$\therefore \quad V_0\rho_w g = W \qquad \text{...(i)}$$

A mark is made on the stem where it intersects the surface of water and 1.0 is written at the mark. Now, the hydrometer is placed in a different liquid. Figure shows the hydrometer in a liquid of specific gravity larger than 1.0. To balance the weight, a lesser volume of liquid needs to be displaced. The hydrometer moves up by Δh. We can write

$$(V_0 - S\Delta h)\rho_l g = W \qquad \text{...(ii)}$$

Where S = area of cross-section of the stem

Dividing (ii) by (i)

$$\left(\frac{V_0 - S\Delta h}{V_0}\right)\frac{\rho_l}{\rho_w} = 1$$

$\Rightarrow$ Specific gravity of liquid $= \dfrac{\rho_l}{\rho_w} = \dfrac{V_0}{V_0 - S\Delta h}$

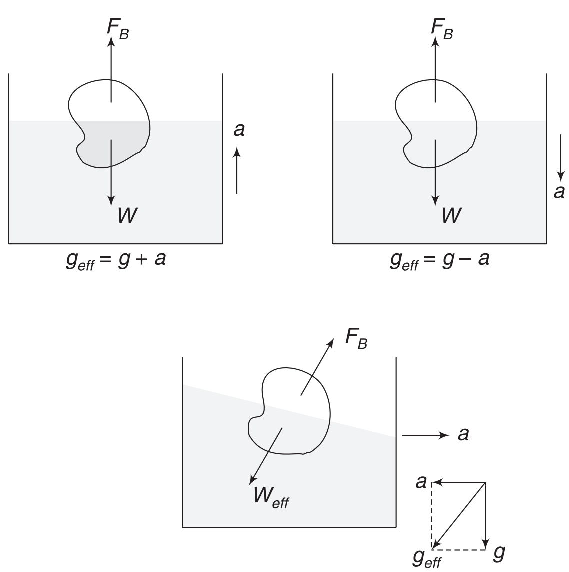
Using the above equation, marking can be done on the stem to directly give the reading of specific gravity when the hydrometer is placed in a liquid.

9.4 Buoyancy in Fluid kept in Accelerated Container

Article 8 explains the variation of pressure inside a fluid that is kept in an accelerated container. Buoyancy force arises due to liquid pressure only. Therefore, it is easy to understand that buoyancy is given by

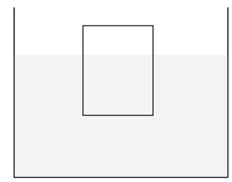
$$F_B = V_{\text{sub}}\rho_L g_{\text{eff}} \qquad \text{...(23)}$$

Where g_{eff} is the effective gravity in the environment of the fluid (i.e., in *RF* attached to the container). The figures given below are self-explanatory.



From equation (21), one can see that fraction of volume of the solid submerged in the liquid is independent of g. Therefore, in all the cases in the above diagram, the fraction of the volume of the solid that is submerged is same.

Example 14 *Ice Block in a beaker*



An ice block (relative density = 0.9) floats in water in a beaker.

(a) Find the percentage of the volume of ice block that is submerged.

(b) How will the level of water in the beaker change if the ice block melts?

Solution

Concepts

(i) Net force on ice blocks is zero. Buoyancy balances its weight. Using this gives equation (21).

Percentage of volume submerged is $\dfrac{V_{\text{sub}}}{V} \times 100$

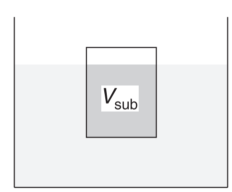
(ii) To answer part (b), we must figure out how much volume of water will be produced by melting of the ice.

The key is to realise that mass will remain same even if the ice melts. Since density of water > density of ice, the volume of water produced due to melting of ice is less than the volume of ice.

(a) From equation (21)

$$\frac{V_{\text{sub}}}{V} = \frac{\rho_{\text{ice}}}{\rho_w} = 0.9$$

$$\frac{V_{\text{sub}}}{V} \times 100 = 90\%$$



90% of the volume of the ice block remains submerged inside water. 10% of its volume is outside water.

(b) Weight of ice block = Weight of displaced water

$\Rightarrow$ Mass of ice block = Mass of displaced water

$\Rightarrow$ Mass of ice block = Mass of (V_{sub}) volume of water

Where V_{sub} = volume shown in dark shade in figure.

Thus, when the ice melts, it froms water which has volume equal to V_{sub}. The produced water exactly fills the space shown in dark shade. The level of water does not change.

Example 15 *A steel boat*

Steel is eight times heavier than water. Why doesn't a boat made of steel sinks?

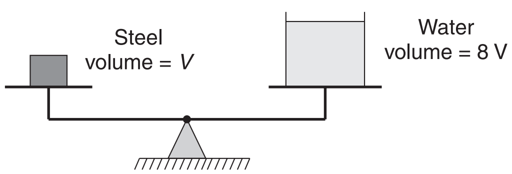
Solution

Concepts

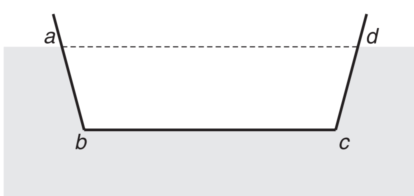
The boat is made wide enough to displace water equal to its weight before it sinks too deep in water.

Focus on understanding the meaning of "displaced water".

To balance the weight of V volume of steel, we need $8V$ volume of water. A solid iron block, if left in water, can only displace V volume of water and will definitely sink.

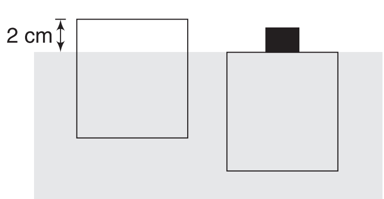


The same quantity of iron, if given the shape of a boat, can displace a lot of water. In the figure shown, (*abcd*) represents the volume of water displaced. Buoyancy can easily balance the weight.



(abcd) is volume of water displaced.

Example 16 A cube of wood is floating in water with its 2 cm height above water surface. A 200 g mass is placed on the top surface of the cube and it is now observed to float while remaining completely submerged (see figure). Find the side length of the cube.



Solution

Concepts

Initially, buoyancy = Weight of the cube

Finally, buoyancy = Weight of (the cube + Added mass)

Let side length of the cube be x and its weight be W.

Area of its face $= x^2$

Initially, $F_{B1} = W$

$$\Rightarrow \quad V_{sub}\, \rho_w g = W \qquad \text{...(i)}$$

Finally, $F_{B2} = W + mg \quad [m = 200\,\text{g}]$

$$\Rightarrow \quad V \rho_w g = W + mg \qquad \text{...(ii)}$$

[Submerged volume = total volume of cube]

(ii) – (i) gives

$$(V - V_{sub})\rho_w g = mg$$

$$\Rightarrow \quad x^2 (2\,\text{cm})\,\rho_w = 200$$

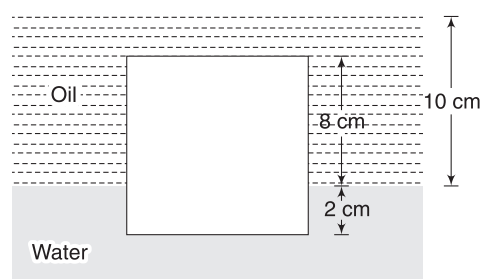
$[V - V_{sub} = x^2 \times 2$; see first figure]

$$\Rightarrow \quad x^2\ (2\ \text{cm})\ \frac{1\,\text{g}}{\text{cm}^3} = 200$$

$$\Rightarrow \quad x = 10\,\text{cm}$$

Example 17 *A solid at the interface of oil and water*

A solid cube has a side length of 10 cm. It floats at the interface of water and oil, as shown in figure. Its 2 cm height is inside water and remaining 8 cm is in oil. The layer of oil is 10 cm thick. Relative density of the oil is 0.8.



(a) Find relative density of the solid.

(b) Find pressure exerted by water on the lower surface of the cube. Atmospheric pressure is

$$P_0 = 1 \times 10^5\ \text{Nm}^{-2}$$

$$g = 10\ \text{ms}^{-2}$$

Solution

Concepts

Archimedes' principle works just fine in such situations.

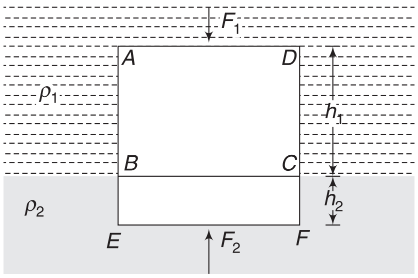
Buoyancy = Weight of displaced oil + Weight of displaced water.

Here, we will give a simple proof for this, though one can use the result directly.

Let A = area of cross-section of the solid.

Buoyancy is

$$F_B = F_2 - F_1$$
$$= P_2A - P_1A$$



where P_2 is pressure at lower face EF and P_1 is pressure at top surface AD.

$$\therefore \quad F_B = (P_2 - P_1)A = (\rho_1 g h_1 + \rho_2 g h_2)A$$
$$= (\rho_1 h_1 A)g + (\rho_2 h_2 A)g$$

= weight of volume $ABCD$ of oil + Weight of volume of water equal to $BEFC$

(a) In the present question

Weight of cube = F_B

$$\Rightarrow \quad (10^3\,\text{cm}^3)\,d\cdot g = (10 \times 10 \times 8\,\text{cm}^3)\ \rho_{oil}\ g + (10 \times 10 \times 2\,\text{cm}^3)\ \rho_w \cdot g$$

$$\Rightarrow \quad \frac{d}{\rho_w} = \frac{8}{10}\frac{\rho_{oil}}{\rho_w} + \frac{2}{10}$$

$\Rightarrow$ $R\cdot d$ of solid = $0.8 \times R\cdot d$ of oil + 0.2

$= 0.8 \times 0.8 + 0.2 = 0.84$

(b) Absolute pressure at lower surface of the cube is

$$P = P_0 + \rho_{oil}\ g(10\,\text{cm}) + \rho_w\ g(2\,\text{cm})$$
$$= 1 \times 10^5 + (0.8 \times 10^3)(10)(0.1) + (10^3)\ (10)\ (0.02)$$
$$= 1 \times 10^5 + 0.8 \times 10^3 + 0.2 \times 10^3$$
$$= 1.01 \times 10^5\ \text{Nm}^{-2}$$

Example 18 A metallic ball weighs 210 g in air, 180 g in water and 120 g in an unidentified liquid. Find the density of the metal and of liquid.

Solution

Concepts

From equation (22), loss in weight = Buoyancy force

If V is volume of ball and d is its density then

$$Vdg = 210\,g \qquad \text{...(i)}$$

Loss of weight in water = Buoyancy in water

$$\Rightarrow \quad 30\,g = V\rho_w g \qquad \text{...(ii)}$$

From (i) and (ii), $\dfrac{Vdg}{V\rho_w g} = \dfrac{210}{30}$

$$\Rightarrow \quad \frac{d}{\rho_w} = 7 \quad \Rightarrow \quad d = 7\,\text{gcm}^{-3}.$$

When dipped in liquid

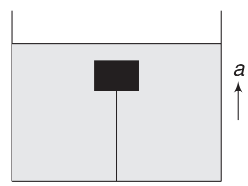
Buoyancy = loss in weight

$$V\rho_L g = 90\,g \qquad \text{...(iii)}$$

From (iii) and (ii), $\dfrac{V\rho_L g}{V\rho_w g} = \dfrac{90\,g}{30\,g}$

$$\Rightarrow \quad \frac{\rho_L}{\rho_w} = 3 \quad \Rightarrow \quad \rho_L = 3\rho_w = 3\,\text{gcm}^{-3}$$

Example 19 A solid block has density (d), which is less than the density of water (ρ). It is kept submerged in water with the help of a string, as shown in the figure. Tension in the string is T_0.



(a) Find weight of the block in terms of T_0.

(b) The container is given an upward acceleration a. Find the tension in the string in terms of T_0.

Solution

Concepts

(i) When the solid is fully submerged, buoyancy acting on it ($F_B = V\rho g$) is greater than its weight ($W = Vdg$).

The tension in the string (a downward force) is necessary for equilibrium.

(ii) When the container accelerates, $g_{eff} = g + a$.

Both the weight of the block ($= Mg_{eff}$) and buoyancy ($= V\rho g_{eff}$) change.

(a) For equilibrium of the block:

$$W + T_0 = F_B$$

But $F_B = V\rho g = \dfrac{M}{d}\rho g = \dfrac{W\rho}{d}$

$$\therefore \quad W + T_0 = W\frac{\rho}{d} \quad \Rightarrow \quad W\left[\frac{\rho}{d} - 1\right] = T_0$$

$$\Rightarrow \quad W = \frac{T_0}{\dfrac{\rho}{d} - 1}$$

(b) Initially, $W + T_0 = F_B$

$$\Rightarrow \quad Mg + T_0 = V\rho g$$

$$\Rightarrow \quad T_0 = V\rho g - Mg \qquad \text{...(i)}$$

When the container is accelerated

$$W' + T = F'_B$$

$$\Rightarrow \quad Mg_{\text{eff}} + T = V\rho g_{\text{eff}}$$

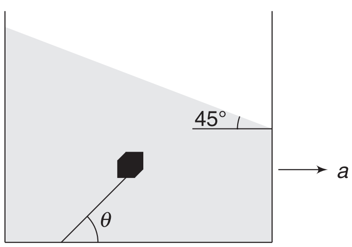
$$\Rightarrow \quad T = (V\rho - M)g_{\text{eff}} \qquad \text{...(ii)}$$

From (i) and (ii),

$$\frac{T}{T_0} = \frac{g_{\text{eff}}}{g} = \frac{g + a}{g}$$

$$\Rightarrow \quad T = T_0\left(1 + \frac{a}{g}\right)$$

Example 20 A solid of density $\frac{\rho}{2}$ (ρ = density of water) and mass m is held inside a water tank, tied to a string, as shown in the figure. The tank is accelerated uniformly in horizontal direction such that the water surface makes an angle of 45°with the horizontal.



(a) Find the angle θ that the string makes with the base of the container in equilibrium.

(b) Find the tension in the string.

Solution

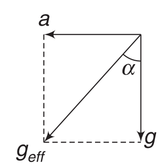
Concepts

(i) We can find acceleration of the container using equation (14). $\tan 45° = \frac{a}{g}$

(ii) In *RF* attached to the container, everything is in an environment where $g_{\text{eff}} = \sqrt{a^2 + g^2}$. g_{eff} makes an angle $\alpha = \tan^{-1}\left(\frac{a}{g}\right)$ with vertical.

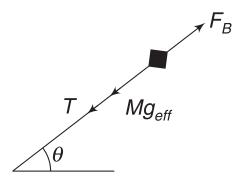
(iii) Both the effective weight and buoyancy act along the line of g_{eff}. String tension must also be along this line for equilibrium.

(a) $\frac{a}{g} = \tan 45° \quad \Rightarrow \quad a = g$



Effective free fall acceleration will be

$$g_{\text{eff}} = \sqrt{a^2 + g^2} = \sqrt{2}\, g$$



It makes an angle α with vertical

$$\tan\alpha = \frac{a}{g} = 1$$

$$\Rightarrow \quad \alpha = 45°$$

Buoyancy will act in direction opposite to g_{eff}.

String will align itself along g_{eff} as shown.

$$\therefore \quad \theta = 45°$$

(b) $T = F_B - Mg_{\text{eff}}$

$$= V\rho g_{\text{eff}} - Mg_{\text{eff}} = V\rho(\sqrt{2}\, g) - M(\sqrt{2}\, g)$$

$$= 2\sqrt{2}\, Mg - \sqrt{2}\, Mg \qquad \left[\because \quad M = V\frac{\rho}{2}g\right]$$

$$= \sqrt{2}\, Mg$$

YOUR TURN

Q.22 A metal ball weighs 0.1 N. When suspended in water, it has an apparent weight of 0.07 N. Find the density of the ball.

Q.23 A block has density ρ_1. When placed in an unidentified liquid, it floats with its three-fourth volume submerged. Find the density of the liquid.

Q.24 A block of wood is held deep inside water. Its acceleration immediately after release is 2 ms^{-2} upwards. Find the relative density of the wood [g = 10 ms^{-2}].

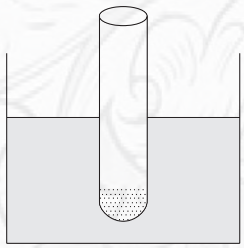
Q.25 A log of wood (density = 6 gm^{-3}) of mass 120 kg floats in water. How much weight can be placed on the log to make it just sink?

Q.26 A jewellery piece weighs 50 g in air and 46 g when dipped in water. Assuming that copper ($r \cdot d$ = 10) is mixed with gold ($r \cdot d$ = 20), find the mass of copper in the jewellery piece.

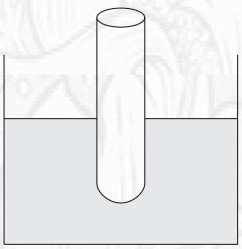
Q.27 A balloon and its contents have a total mass of 800 kg. It is floating in air and is in equilibrium. Its mass is reduced by Δm without changing its volume. Its acceleration is observed to be $2\,ms^{-2}$. Find Δm. [$g = 10\,ms^{-2}$]

Q.28 A block of ice has a piece of gold frozen inside it. The ice block is floating in water in a beaker. How will the level of water in the beaker change if the ice melts?

Q.29 A glass tube (open at top) has outer diameter of 1.6 cm. It floats in water in a large tank, as shown. Some sand is poured into the tube so as to sink it further by 2 cm. Calculate the mass of sand added.

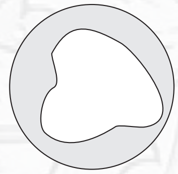


Q.30 An open top glass tube is floating in water in a beaker, as shown. Some water from the beaker is taken and poured into the tube. Will the water level in the beaker change?



Q.31 A solid ball of density half of that of water falls freely under gravity from a height of 19.6 m above the surface of water in a lake. Upto what depth will the ball go inside the lake? Neglect viscosity (i.e., friction due to water) and take $g = 9.8\ ms^{-2}$.

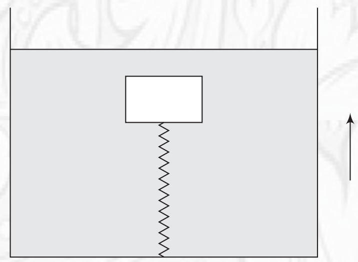
Q.32 An iron ball has cavity. The ball weighs 6000 N in air and 4000 N in water. Find the volume of cavity inside the ball.



Density of iron, $d = 7.87\,gcm^{-3}$

Density of water, $\rho = 1\,gcm^{-3}$ and $g = 9.8\ ms^{-2}$

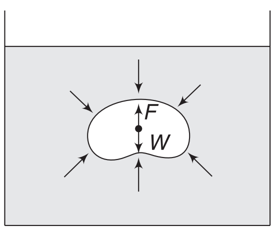
Q.33 A solid block is held by a spring inside a water tub, as shown in the figure. The spring is compressed by 3 cm in equilibrium. The container is now moved up with an acceleration $a = 2\ ms^{-2}$.



Find the compression in the spring in equilibrium. Take $g = 10\ ms^{-2}$

9.5 Centre of Buoyancy

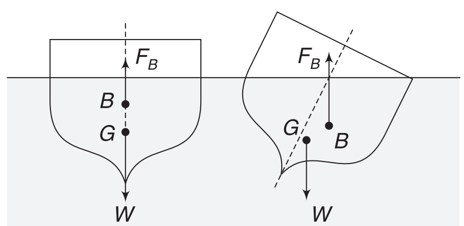
Consider a volume of liquid inside a tank full of liquid. W is the weight of this volume of liquid. It is balanced by force applied by the surrounding liquid on considered volume. This force (F) due to surrounding liquid must be considered acting at the COM of the considered volume. This will ensure rotational equilibrium. If line of F and W are separate, it will produce a torque, implying that the considered volume is rotating.



If a solid is used to replace the volume of liquid shown in the diagram, it will also be acted upon by the surrounding liquid in same fashion. Net force on the solid will be $F (= W =$ weight of displaced liquid) and it will be assumed to be acting at the COM of the displaced liquid volume.

Therefore, *the buoyancy force acting on a solid submerged in a fluid must be considered to be acting at the COM of the displaced fluid. This point is called centre of buoyancy*.

Consider a ship having its COM at G. The water displaced by it has its COM at B. Thus, B is centre of buoyancy. Weight of the ship acts at G and buoyancy force on it acts at B. If the ship tilts a little towards right, the centre of buoyancy moves to right at COM of the displaced water. This happens as displaced water has higher volume to the right of vertical line through G. F_B and W form an anti-clockwise couple that restores the position of the ship. The ship is definitely in stable equilibrium.



Example 21 A ship has it centre of buoyancy below its COM. Can it be in equilibrium for small tilts?

Solution

Concepts

It can be in equilibrium if the couple formed by buoyancy and weight is restoring.

If a small shift causes the centre of buoyancy to shift a lot (as shown in figure (b)), a restoring couple is developed. Ship will be stable.

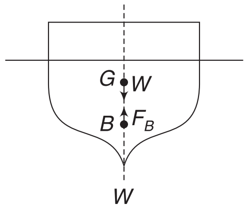
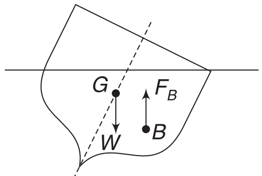
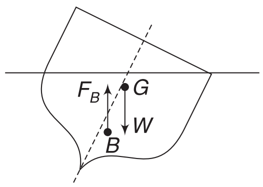


Fig (a)



ship will be restored back to equilibrium.

Fig (b)



Ship is unstable.

Fig (c)

If the tilt causes a situation as shown in figure(C), then the couple formed due to F_B and W will further tilt the ship to right. This will drown it.

Example 22 *Stick in water*

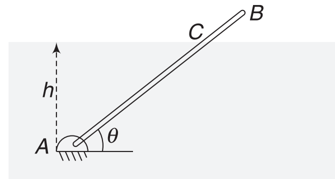
A stick (AB) is $L = 2\,\text{m}$ long. It is hinged at its lower end A inside a water tank. End A is $h = 1.0\,\text{m}$ below the surface of water. In equilibrium, the stick makes $\theta = 37°$ with horizontal. Find the relative density of the stick. Take $\tan 37° = \dfrac{3}{4}$

Solution

Concepts

(i) Buoyancy force acts at the centre of mass of displaced water, i.e., at the centre of part AC of the stick.

(ii) Torque of buoyancy balances the torque due to weight.



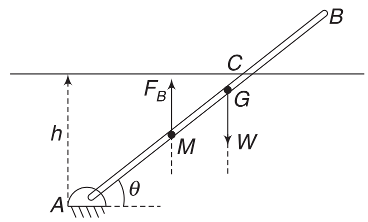
$$AC = \frac{h}{\sin 37°} = \frac{1}{\frac{3}{5}} = \frac{5}{3}\,\text{m}$$

Buoyancy acts at mid-point (M) of the submerged part AC.

$$\text{AM} = \frac{AC}{2} = \frac{5}{6}\text{ m}$$

Weight acts at mid-point (G) of AB.

$$AG = \frac{L}{2} = 1\,\text{m}$$



Buoyancy acts at midpoint (M) of submerged part AC.

Buoyancy force, $F_B = S(AC)\rho_w g = S\left(\frac{5}{3}\right)\rho_w \cdot g$

[S = cross-sectional area of the stick]

Weight, $W = SL\rho \cdot g = S(2)\rho \cdot g$

For equilibrium, torque due to F_B (about A) must be equal to torque due to W.

$$\therefore \quad \tau_W = \tau_B$$

$$W \cdot (AG\cos\theta) = F_B(AM\cos\theta)$$

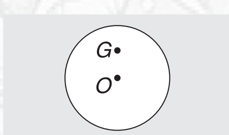
$$\Rightarrow \quad S(2)\rho g \cdot (1) = S\left(\frac{5}{3}\right)\rho_w\, g\left(\frac{5}{6}\right)$$

$$\Rightarrow \quad \frac{\rho}{\rho_w} = \frac{25}{36} = 0.69$$

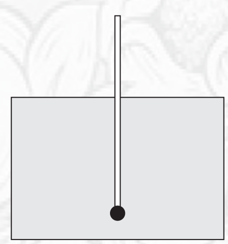
$\therefore$ Relative density = 0.69

Your Turn

Q.34 A non-uniform sphere floats in a liquid while remaining completely submerged. It is in a position where its centre of mass (G) lies above its geometrical centre (O), as shown in figure. What will happen if the sphere is disturbed a little?



Q.35 A uniform stick of mass m has a particle of negligible volume attached to its end. It floats in a liquid, as shown, with half its length submerged. Find mass of the particle if the centre of buoyancy coincides with the centre of mass of the system.



In Short:

(i) When a solid is dipped in a fluid, the resultant force acting on it due to fluid pressure from all sides is known as buoyancy.

(ii) Archimedes' principle states that buoyancy is upward. It is equal to weight of displaced fluid.

$$F_B = V_{\text{sub}}\ \rho_f g$$

Where V_{sub} = volume submerged part of the solid and ρ_f is density of fluid.

(iii) In an accelerated container, we should take g_{eff} in place of g for writing buoyancy.

(iv) When a solid is dipped in a fluid, it appears to be lighter.

$$W_{\text{apparent}} = W_{\text{true}} - F_B$$

(v) A solid having density (ρ_S) less than that of liquid (ρ_l) floats with a part of it remaining outside the liquid. The fraction of volume that is submerged is

$$\frac{V_{\text{sub}}}{V} = \frac{\rho_S}{\rho_l}$$

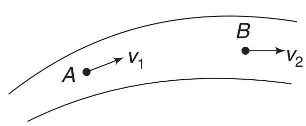
(vi) The buoyancy force acts at a point that is centre of mass of the displaced volume of the fluid. This point is known as centre of buoyancy.

10. FLOW OF FLUIDS

Study of a flowing fluid is far more complex than study of a static fluid. However, we will make many idealistic assumptions to simplify our analysis of a flowing fluid.

10.1 Steady and Turbulent Flow

A flow is said to be *steady* (or *streamline* flow) if flow conditions do not change with time. Imagine a gentle flow of water through a pipe. If you observe the fluid particles passing through a specific point A, you will find them all to have same velocity (say v_1) when they reach A. *Velocity at A does not change with time*. Similarly, all particles passing through another point B have a common velocity (say, v_2) at B. If a particle passing through point A, later passes through B, then all particles which pass through A will also pass through B. It means that all particles passing through A follow the same path. Such a flow is called steady flow.

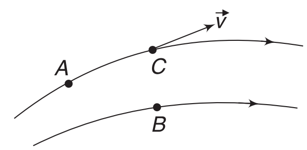


Motion of water in a fall or in a fast-flowing river is not that simple. Velocities of particles passing through the same point may be different. In fact, the flow condition at a point changes erratically with time. Such a flow is called *turbulent flow*. In general, liquid flowing at high speeds will be in turbulent flow.

10.1.1 Streamline

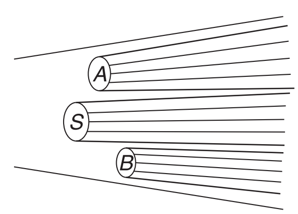
Path taken by a fluid particle is known as its *line of flow*. In case of a steady flow, a line of flow is known as a *streamline*. In this case, all particles passing through a point follow the same path and hence we have a unique line of flow passing through a point.

A tangent draw on a stream line at a point gives the direction of motion at that point. Obviously, only one streamline can pass through a point. Two streamlines can never intersect. Intersection of two streamlines will imply two different directions of motion at a point.



Steady flow. Line passing through *A* is path of all particles passing through *A*. This line is a streamline. Line passing through *B* is another streamline. Velocity of a particle at *C* is along the tangent at *C*.

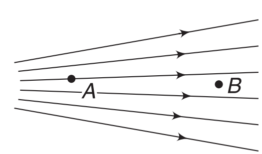
Consider water flowing steadily through a pipe. S represents an area in the cross-section of the pipe. Many streamlines are passing through this area S. These streamlines form a tube. Such a tube is often termed as a *tube of flow*. The figure shows three tubes of flow in a water pipe having cross-sections S, A and B. There is no physical wall between two tubes of flow but the liquid in tube A will not intermix with liquid in tube S. This is because two streamlines never intersect. The entire pipe itself is a tube of flow.



Fluid flowing through different tubes of flow cannot intermix

Streamline diagrams are often used to represent the nature of flow. One more important feature of such diagrams is that density of the lines give an idea of flow speed. Higher density of lines means higher speed. In the streamline diagram shown in the figure, speed of flow at A is higher than speed of flow at B.

By injecting some dye in a fluid, we can easily see the streamlines. Path followed by dye particles are the streamlines.



Speed at *A* is grater than speed at *B*.

10.2 Irrotational Flow

Irrotational flow means there is no spinning of fluid particles. Put a tiny piece of paper in a flowing fluid. If the paper does not rotate about its centre of mass, the flow is irrotational.

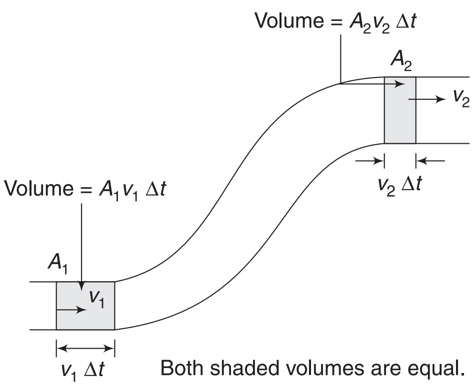
In practice, flow of viscous liquids is mostly rotational. Small fluid elements, while moving, also rotate about their own centre of mass. In a rapidly flowing river full of boulders, the flow is highly turbulent and rotational. Formation of vortex and eddies are quite common in such *flow*.

We will study only about *steady flow* of an ideal fluid. An ideal fluid is *incompressible* and *non-viscous*.

11. EQUATION OF CONTINUTIY

This important equation in fluid dynamics follows from *conservation of mass*. It states that *in a steady flow of an ideal fluid, the product of area of cross section and speed remain same at all locations in a tube of flow*.

Consider a fluid flowing through a pipe of varying cross-section. Flow speed is v_1 at a location where area of cross-section is A_1 and flow speed is v_2 at location where area of cross-section of the pipe is A_2.



Both shaded volumes are equal.

Volume of liquid entering through A_1 in time Δt is $A_1 v_1\,\Delta t$. [Actually, the shaded volume of liquid will cross section A_1 in time Δt].

Mass of liquid entering at A_1 in time Δt is

$$\Delta m_1 = \rho A_1 v_1 \Delta t \qquad [\rho_1 = \text{density of liquid at } A_1]$$

Similarly, volume of liquid flowing out through section A_2 is $A_2 v_2 \Delta t$. Mass flowing out is

$$\Delta m_2 = \rho_2 A_2 v_2\,\Delta t \qquad [\rho_2 = \text{density at } A_2]$$

But mass of liquid in the pipe between two sections at A_1 and A_2 cannot change.

$$\therefore \qquad \Delta m_1 = \Delta m_2$$

$$\Rightarrow \qquad \rho_1 A_1 v_1 = \rho_2 A_2 v_2 \qquad ...(24)$$

Since liquid is incompressible, its density cannot change.

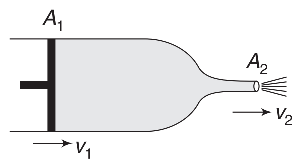
$$\therefore \qquad \rho_1 = \rho_2$$

$$\text{Hence,} \qquad A_1 v_1 = A_2 v_2 \qquad ...(25)$$

This is continuity equation. It can be applied for any tube of flow.

You must have noticed that speed of water coming out of a garden hose can be increased by partially closing the hose opening (thereby decreasing the area) with your thumb. This is continuity equation in action.

Example 23 A syringe has a piston of cross-sectional area $A_1 = 5\,\text{cm}^2$. The outlet of the nozzle has area of $A_2 = 4\,\text{mm}^2$. The piston is pushed with a speed of $v_1 = 1\,\text{cms}^{-1}$. Find the speed with which the liquid is ejected out of the nozzle.



Solution

Concepts

The liquid in contact with the piston moves with speed v_1.

$$A_1 v_1 = A_2 v_2$$

$$A_2 v_2 = A_1 v_1$$

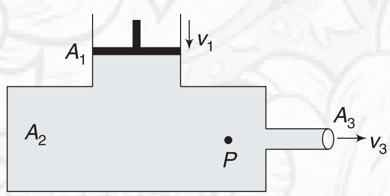
$$\Rightarrow \quad (4\text{mm}^2)\, v_2 = (5 \times 10^2\,\text{mm}^2) \times (1\ \text{cms}^{-1})$$

$$\Rightarrow \quad v_2 = 125\,\text{cms}^{-1} = 1.25\ \text{ms}^{-1}$$

An important thing to realise is that the volume swiped by the piston will be equal to liquid pushed out of the nozzle.

YOUR TURN

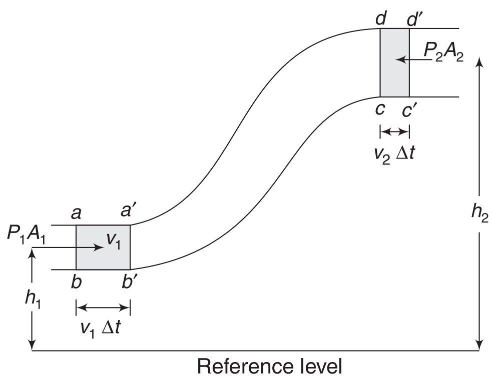
Q.36 A water tank has the shape shown in figure. The cross-sectional areas are $A_1 = 0.5\,\text{m}^2$, $A_2 = 1.0\,\text{m}^2$ and $A_3 = 0.05\,\text{m}^2$. The piston having area A_1 is pushed with a speed of $v_1 = 0.1\,\text{ms}^{-1}$. With what speed ($v_3$) the water flows out of outlet having area A_3? With what speed is a fluid particle at point P moving?



12. BERNOULLI'S EQUATION

The equation bears the name of Daniel Bernoulli, who studied fluid flow in eighteenth century. It is not something fundamental, but a form of work–energy theorem which is more suitable for use in study of flowing fluids.

Consider a fluid flowing through a tube having variable cross-section and changing height. Consider a liquid volume contained in the volume *abcd* of the pipe. This liquid volume advances to new position $a'b'c'd'$ in a small time interval Δt. We will apply work–energy theorem on considered volume of fluid, as it moves from its initial position *abcd* to its final position $a'b'c'd'$.



Volume ($aa'b'b$) = volume ($cc'd'd$) = Δv

Let A_1 and A_2 be cross sectional area of the pipe at *ab* and *cd* respectively. v_1 and v_2 are flow speeds at *ab* and *cd*. P_1 and P_2 and pressure at *ab* and *cd* respectively.

Volume $aa'b'b$ is $A_1v_1\Delta t$ as the distance $aa' = v_1\Delta t$. Similarly, volume $cc'd'd = A_2v_2\Delta t$.

From continuity equation:

$$A_1v_1 = A_2v_2$$

$$\Rightarrow \quad A_1v_1\Delta t = A_2v_2\Delta t = \Delta V \text{ (say)} \qquad \text{...(i)}$$

$$\Rightarrow \quad \text{Volume } (aa'b'b) = \text{Volume } (cc'd'd) = \Delta V$$

Change in kinetic energy (ΔK)

As the fluid moves from position (*abcd*) to new position ($a'b'c'd'$), its kinetic energy changes. This change is due to the fact that *KE* of volume ($cc'd'd$) is different from *KE* of volume ($aa'b'b$). *KE* of fluid particle in the region $a'b'cd$ is same in initial and final positions. In steady flow, speed at a point does not change with time.

$$\therefore \quad \Delta k = KE \text{ of volume } (cc'd'd) - KE \text{ of volume } (abb'a')$$

$$= \frac{1}{2}\Delta m v_2^2 - \frac{1}{2}\Delta m v_1^2$$

Where $\Delta m = \rho\Delta V$ = mass of fluid in volume ($aa'b'b$) or ($cc'd'd$)

$$\therefore \quad \Delta k = \frac{1}{2}\rho\Delta V v_2^2 - \frac{1}{2}\rho\Delta V v_1^2 \qquad \text{...(ii)}$$

Change in gravitational potential energy (ΔU)

As the fluid volume (*abcd*) shifts to ($a'b'c'd'$), change in its *PE* is

$$\Delta U = PE \text{ of volume } (cc'd'd) - PE \text{ of volume } (aa'b'b)$$

$$= \Delta m g h_2 - \Delta m g h_1$$

$$= \rho\Delta V g h_2 - \rho\Delta V g h_1 \qquad \text{...(iii)}$$

Work done by surrounding liquid (W)

The fluid to the left of boundary *ab* pushes with a force P_1A_1 and its point of application gets displaced by $v_1\Delta t$. It does positive work $P_1A_1v_1\Delta t$ on our fluid volume considered.

The fluid to the right of boundary *cd* exerts a leftward force P_2A_2 and the boundary *cd* moved by a distance $v_2\Delta t$ towards right. Work done by this force is negative equal to $-P_2A_2v_2\Delta t$.

$$\therefore \quad W = P_1A_1v_1\Delta t - P_2A_2v_2\Delta t$$

$$= P_1\Delta V - P_2\Delta V \quad \text{[using (i)]} \qquad \text{...(iv)}$$

If the fluid is ideal, there is no viscous force. Normal force by the pipe wall does no work, as it is perpendicular to the direction of motion.

Using work-energy theorem, we can write

$$W = \Delta K + \Delta U$$

$$\Rightarrow \quad P_1\Delta V - P_2\Delta V = \frac{1}{2}\rho\Delta V v_2^2 - \frac{1}{2}\rho\Delta V v_1^2 + \rho\Delta V g h_2 - \rho\Delta V g h_1$$

$$\Rightarrow \quad P_1 - P_2 = \frac{1}{2}\rho v_2^2 - \frac{1}{2}\rho v_1^2 + \rho g h_2 - \rho g h_1$$

$$\Rightarrow \quad P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2 \qquad \text{...(26)}$$

$$\Rightarrow \quad P + \frac{1}{2}\rho v^2 + \rho g h = \text{constant}$$

This equation is known as Bernoulli's equation.

12.1 Pressure as a form of Energy

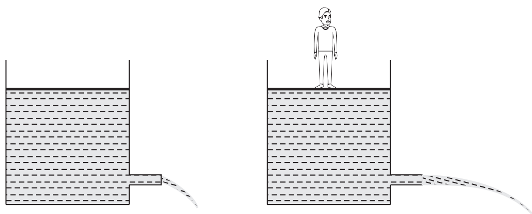
The term $\frac{1}{2}\rho v^2$ in Bernoulli's equation represents the *kinetic energy of a unit volume of the fluid*. [This is because ρ is nothing but mass of unit volume.] The unit of this term is Jm^{-3}.

Similarly, $\rho g h$ is nothing but *gravitational potential energy of unit volume* of the fluid.

The third term–pressure P – can be interpreted as pressure energy of unit volume of the fluid. Unit of pressure Nm^{-2} is same as Jm^{-3}.

It is a bit difficult to realise that pressure is a form of energy. Let us try to understand this in a simple way.

Imagine a small tank of water with a plugged outlet. If you open the plug, water flows out with some speed. Where does the water get kinetic energy from? You can argue that water in the tank is having gravitational potential energy. Ok, let us accept this argument for a moment. Now imagine the same tank having water at increased pressure.



pressurised water comes out with much higher speed. pressurised liquid has more energy.

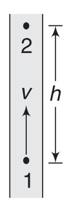
You can think that the tank is closed with a movable piston and a large weight is placed on the piston [think yourself standing on the piston]. If the plug is removed, the water flows out with a great speed. What resulted in increased kinetic energy of the outgoing water?

The increased pressure caused the water to gain more kinetic energy. Therefore, pressurised liquid has more energy. It is nothing wrong to regard pressure as a form of fluid energy.

Bernoulli's equation can be simply stated as – *Sum of kinetic energy, pressure energy and gravitational potential energy (per unit volume) is constant for a steady flow of an ideal fluid*.

Example 24 *PE increases, KE does not decrease*

A liquid of density ρ is flowing in a vertical pipe of uniform cross-section. At point 1, speed of flow is v. What is the speed of flow at point 2, which is at a height h above 1? Justify your result in terms of energy of the fluid.



Solution

Concepts

(i) Use of continuity equation will give speed at 2

(ii) A flowing liquid has energy in three forms – kinetic energy, potential energy and pressure energy.

Use of $A_1v_1 = A_2v_2$ tells us that speed at 2 is same as speed at 1 ($\because \; A_1 = A_2$).

The gravitational *PE* of the liquid increases as it moves from 1 to 2 but its *KE* does not change. A liquid has a third from of energy – pressure energy. The liquid pressure decreases at the liquid climbs up. Fall in pressure energy results in rise in *PE*.

13. APPLICATIONS OF BERNOULLI'S EQUATION

Bernoulli's equation is valid for steady flow of an *ideal fluid* but it gives us a lot of meaningful insights into real-life situations. Here, we will learn that some daily-life observation can be easily understood in terms of this principle.

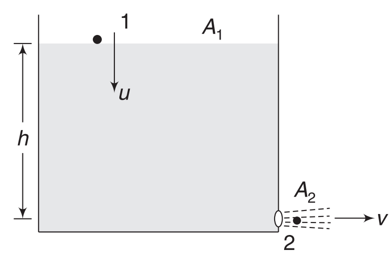
13.1 Speed of Efflux

Consider a liquid of density ρ filled in a tank of cross-sectional area A_1. The liquid is filled to a height h and the tank is open to atmosphere. There is a small hole of cross-sectional area A_2 in the side wall near the bottom of the tank. The liquid flows out of the hole and we wish to find the speed (v) with which it comes out of the hole.

The top surface of the liquid moves down at speed u such that

$$A_1u = A_2v \qquad \text{...(i)}$$

This is continuity equation.



For the flowing liquid, we will use Bernouli's equation, considering two points as 1 and 2 shown in figure. Pressure at 1 is atmospheric pressure (P_0). Pressure at 2 (just outside the hole) is also atmospheric pressure. [A jet of liquid in air is subjected to atmospheric pressure only].

$$P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2 = P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1$$

$$\Rightarrow \quad P_0 + \frac{1}{2}\rho v^2 + 0 = P_0 + \frac{1}{2}\rho u^2 + \rho g h$$

[We have chosen the level of point 2 as the reference level for measuring height. One can choose the reference level elsewhere also. It makes no difference.]

$$\Rightarrow \quad (v^2 - u^2) = 2gh$$

$$\Rightarrow \quad v^2\left[1 - \frac{A_2^2}{A_1^2}\right] = 2gh \qquad \left[\text{using (i), } u = \frac{A_2}{A_1}v\right]$$

$$\Rightarrow \quad v = \sqrt{\frac{2gh}{1 - \dfrac{A_2^2}{A_1^2}}} \qquad \text{...(27)}$$

If the tank is wide, then $A_1 \gg A_2$ and

$$\frac{A_2^2}{A_1^2} << 1$$

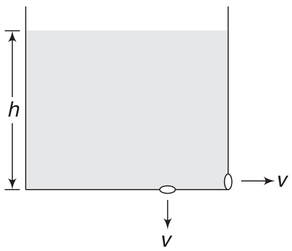
$$\therefore \qquad v \simeq \sqrt{2gh} \qquad ...(28)$$

This speed is known as *speed of efflux* and the result expressed by equation (28) is known as *Torricelli's theorem*. The theorem can be stated as:

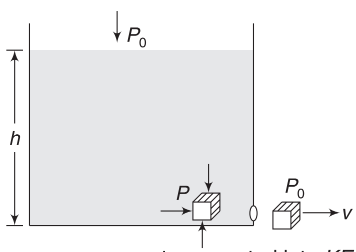
The speed of liquid coming out of a small hole at a depth h below the free surface of an open tank is same as the speed gained by a particle falling freely under gravity through a height h.

Note: Speed of efflux remains same even if the hole is made at the bottom surface of the tank.

Position of the hole is not important; it is height (h) of liquid above the hole that matters.



An alternative way of finding the speed of efflux can be as follows (though there is no difference in basic principle used):



Pressure energy gets converted into *KE* as the liquid element pops out of the tank.

Consider a small element of liquid at the bottom of the tank, just inside the hole. If tank is large ($A_1 \gg A_2$), speed everywhere inside the tank is zero. The considered element has no speed but it is subjected to large pressure.

Pressure is $P = P_0 + \rho gh$

As the element moves out, pressure on it reduces to atmospheric pressure (P_0). There is no change is gravitational *PE* as the element is essentially at same height. The loss in pressure energy manifests as gain in *KE*.

Gain in *KE* = Loss in pressure energy

$$\Rightarrow \qquad \frac{1}{2}\Delta mv^2 = (P - P_0)\text{ (volume)}$$

['Pressure' is energy per unit volume]

$$\Rightarrow \quad \frac{1}{2}\rho\text{ (Volume)}\,v^2 = (\rho gh)\text{(volume)}$$

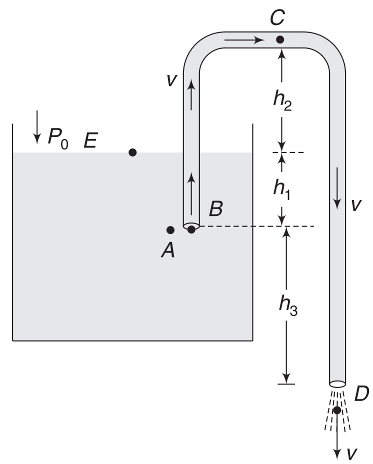
$$\Rightarrow \qquad \frac{1}{2}\rho v^2 = \rho gh$$

This equation is nothing but Bernoulli's equation used between points just inside and just outside the hole.

$$\therefore \qquad v = \sqrt{2gh}$$

13.2 Siphon

This is a simple device, which most of us have used in our daily life. It is a simple pipe used to drain liquid from a tank with the help of gravity only.



Look at the figure. All air is sucked out of the pipe and the liquid begins to flow. Let the speed at outlet D be v. If the pipe has uniform cross-section, the continuity equation requires that the speed of flow must be same everywhere in the pipe.

Let us apply Bernoulli's equation between points E and D.

$$P_D + \frac{1}{2}\rho v_D^2 + \rho gh_D = P_E + \frac{1}{2}\rho v_E^2 + \rho gh_E$$

Taking reference level (for height) to be at D and assuming that the tank is wide (which means $v_E \simeq 0$), we get

$$P_0 + \frac{1}{2}\rho v^2 + 0 = P_0 + 0 + \rho g(h_3 + h_1)$$

$$\Rightarrow \qquad v = \sqrt{2g(h_1 + h_3)} \qquad ...(29)$$

Note that the speed of outflow depends on the height difference between the liquid surface and the outlet of the pipe. It has got nothing to do with position of inlet. If the inlet end of the pipe is pushed deeper into the tank, it will not make any difference to the speed of liquid flowing out at D.

What about pressure in the pipe? Where is it minimum? At D, pressure is P_0 (atmospheric pressure). As you move up from D, the kinetic energy does not change but gravitational potential energy increase. It increases at the cost of pressure energy. Higher you move above D, lower is the pressure. Pressure is smallest at C. Using Bernoulli's equation between C and D gives:

$$P_C + \frac{1}{2}\rho v^2 + \rho g(h_1 + h_2 + h_3) = P_D + \frac{1}{2}\rho v^2 + 0$$

$$\therefore \quad P_C = P_0 - \rho g(h_1 + h_2 + h_3)$$

The siphon will stop working if pressure at C drops to zero. For successful working of the device

$$P_C > 0$$

$$\Rightarrow \quad P_0 > \rho g(h_1 + h_2 + h_3)$$

$$\Rightarrow \quad (h_1 + h_2 + h_3) < \frac{P_0}{\rho g} \quad \text{...(30)}$$

Are pressures at point A and B different? Let us look at it. Pressure at A is

$$P_A = P_0 + \rho g h_1 \qquad [\text{Speed at } A \simeq 0]$$

At B (a point inside the pipe), the liquid has kinetic energy. Height of A and B are same. It means pressure at B is less than that at A.

$$P_B + \frac{1}{2}\rho v^2 = P_A + 0$$

$$\Rightarrow \quad P_B = P_A - \frac{1}{2}\rho v^2$$

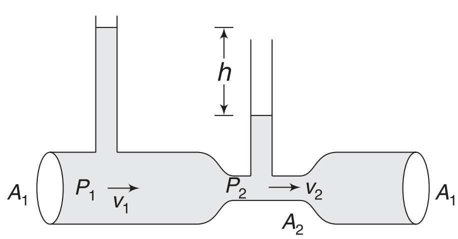
$$= P_0 + \rho g h_1 - \rho g(h_1 + h_3) \qquad [\text{using (29)}]$$

$$= P_0 - \rho g h_3$$

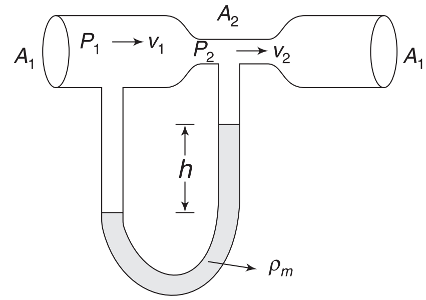
Please note that if a fluid has more speed at a point compared to another point at same horizontal level, the pressure is lower at the point of higher speed.

13.3 Venturi Meter

Venturi meter is a device used to measure the speed and flow rate of fluid through a pipe. Consider a horizontal pipe of cross-sectional area A_1 through which a liquid is flowing at some unknown speed v_1. A constriction (i.e., a throat) of cross-sectional area A_2 ($<A_1$) is attached to the pipe. This narrow zone is known as venturi. As the liquid flows through the narrow zone, its speed increases. The pressure decreases, as required by the Bernoulli's equation. By measuring the drop in pressure, we can workout the value of flow speed v_1.



For a flow of liquid in a pipe, we can insert two tubes as shown. The difference in height of liquid in the two tubes gives the pressure difference.



For a flow of gas, we attach a manometer having liquid of density ρ_m. The pressure difference between two ends of manometer is $\Delta P = \rho_m gh$.

To measure pressure difference between a point in the pipe and at constriction, we use manometers, as shown in the figures. *With regard to first figure* we can write.

$$P_1 - P_2 = \rho g h \quad [\rho = \text{density of flowing liquid}]$$

$$\Rightarrow \quad \Delta P = \rho g h \quad \text{...(i)}$$

From continuity equation:

$$A_1 v_1 = A_2 v_2 \quad \text{...(ii)}$$

Using Bernoulli's equation

$$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$$

$$\Rightarrow \quad P_1 - P_2 = \frac{1}{2}\rho(v_2^2 - v_1^2)$$

$$\Rightarrow \quad \Delta P = \frac{1}{2}\rho\left(\frac{A_1^2}{A_2^2} - 1\right)v_1^2$$

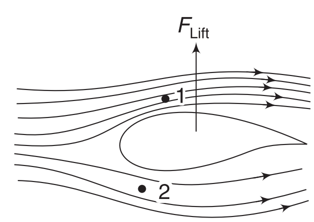
$$\Rightarrow \quad v_1 = \sqrt{\frac{2\Delta P}{\rho\left(\frac{A_1^2}{A_2^2} - 1\right)}} \quad \text{...(30)}$$

Value of ΔP is known form (i). Thus, we know v_1.

Volume flow rate of the fluid through the pipe is $A_1 v_1$ ($m^3 s^{-1}$).

13.4 Flight of Aeroplanes

One of the most interesting applications of Bernoulli's equation is flight of aeroplanes. When an aeroplane runs on a runway, it feels air blowing against it. The cross-section of the wings is designed such that the wind causes less drag (resistnace to motion) and more lift (an upward force against gravity). A term *aerofoil*



Streamlines get crowded at the top. Speed at 1 is higher than speed at 2. This results in higher pressure at 2.

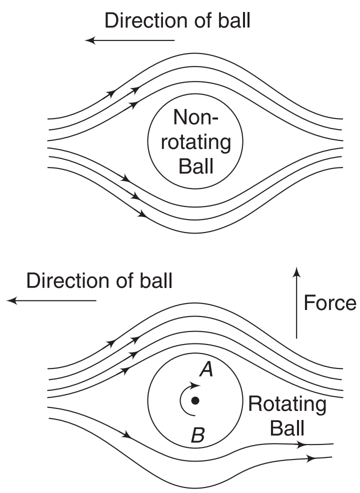
is used to describe the cross-sectional shape of objects such as aeroplane wings.

Consider an aeroplane running towards left. It feels air passing over it towards right. The aerofoil design of the cross-section of the wing causes most of the air to glide above the wing. Very less air passes beneath the wing.

The streamlines get crowded above the wing. Thus, speed of air is higher at point 1 (above the wing) compared to speed at point 2 (below the wing). There is not a considerable height difference between 1 and 2 and therefore, pressure at 2 is higher than pressure at 1. This gives an upward force to the wing, helping the aeroplane to take–off. This is a simple explanation of the phenomena of take–off, though the reality is more complex.

13.5 Magnus effect

A spinning ball travelling through air does experience a sidewise force, deflecting its path. This effect is known as Magnus effect.



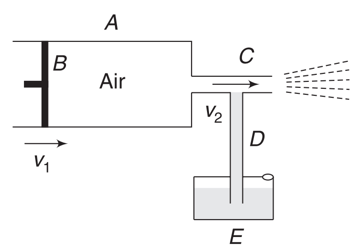
In *RF* of the ball, velocity of point *A* is in the direction of air flow. Due to this the air speed increases at *A*. At *B* speed of ball's surface is opposite to air flow. Speed of air below *B* is low.

A ball travelling through air experiences a wind in opposite direction. In case of a non-rotating ball, the streamlines separate evenly on both sides. It means speed of air is same on both sides of the ball. For a rapidly spinning ball, the situation is slightly different. At one side, the spinning surface of the ball drags air along with it and increases the flow speed. On the other side, motion of the surface of the ball is against the wind, thereby decreasing the flow speed. This difference in speed on two sides results in pressure difference in accordance with Bernoulli's equation.

In the figure shown, speed at *A* is higher than speed at *B*. Thus, pressure at *B* is higher. The ball experiences a force, due to this pressure difference that is perpendicular to its direction of motion. This causes the ball to deviate. You must have seen a curling path of football kicked by Lionel Messi, bluffing the goalkeeper.

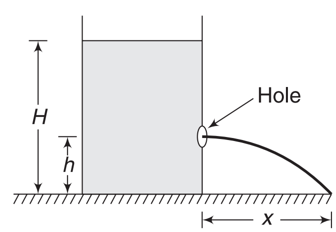
13.6 A Sprayer

Let's understand the Physics behind a hand pump used to spray insecticides. It has a cylinder *A* with a movable piston *B*. The cylinder has a narrow barrel *C*. It is connected by a thin tube *D* to a tank (*E*) containing insecticide. There may be a hole in the tank to maintain pressure equal to atmospheric pressure inside it. When the piston (*B*) is pushed with speed v_1. The air passes through tube *C* at a much higher speed (v_2) due to its small cross-sectional area. Due to high speed of air inside *C*, there is a sharp drop in pressure. The pressure inside the liquid tank (*E*) is atmospheric pressure. This high pressure causes the liquid to get pushed up in tube *D*. [It is similar to the phenomena taking place in a barometer]. The liquid rising in tube *D* gets mixed with lot of air coming from the cylinder and gets sprayed.



Example 25 *Maximum range*

A tank placed on ground has water filled in it, up to a height *H*. A small hole is punched in its side wall at a height *h* from the bottom. Water jet strikes the ground at a horizontal distance *x*.



(a) For what value of *h* is *x* maximum?

(b) For two value of *h*, we get same range (*x*). If one value of *h* is *H*/4, what is its other value?

Solution

Concepts

(i) For small hole, speed of efflux is $v = \sqrt{2gy}$ where *y* is height of water above the hole.

(ii) Range is obtained as product of *v* and time of flight (*T*). Time of flight can be obtained by treating the water jet as a horizontal projectile.

$$T = \sqrt{\frac{2h}{g}}$$

(a) Speed of efflux; $v = \sqrt{2g(H-h)}$

Time of flight; $T = \sqrt{\dfrac{2h}{g}}$

$\therefore$ Range $x = vT = 2\sqrt{h(H-h)}$ ...(i)

Range is maximum when the product $Z = h(H-h)$ is maximum. This happens when

$$\frac{dZ}{dh} = 0 \quad \Rightarrow \quad H - 2h = 0 \quad \Rightarrow \quad h = \frac{H}{2}$$

(b) From (i), it is obvious that range will be same for two values of h: h and $(H-h)$.

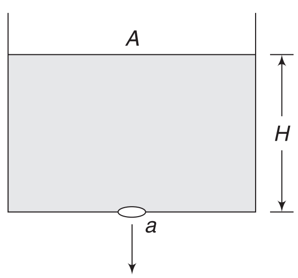
If one value of h is $\dfrac{H}{4}$, the other must be

$$H - \frac{H}{4} = \frac{3H}{4}$$

This implies that if we make two holes at an equal vertical distance from top and bottom, both jets will hit the ground at same spot.

Example 26 *Time to empty a tank*

A tank has cross-sectional area (A) and a liquid is filled in it up to a height H. A small hole of cross-sectional area a is opened at the bottom of the tank. In how much time the tank will get emptied?



Solution

Concepts

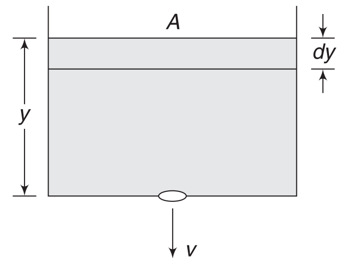
(i) Speed of efflux (v) depends on height (y) of the liquid in the tank. It is not constant. It is decreasing.

(ii) Volume flow rate (m^3s^{-1}) is given by av. Volume of liquid flowing out in time dt is given by $av\,dt$.

(iii) The hole is small, implying that speed of efflux (v) can be written using Torricelli's theorem. The water level moves down at a much slower pace compared to the speed of liquid coming out of the hole.

When height of liquid is y, speed of efflux is $v = \sqrt{2gy}$

Volume of liquid leaving the tank in a small interval dt is $avdt$.



If liquid level in the tank decreases by dy in time interval dt, we have

$$-A\,dy = av\,dt$$

Negative sign is necessary as y is decreasing.

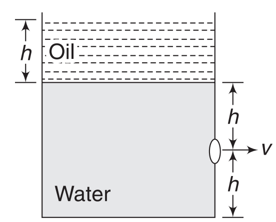
$$\therefore \qquad -A\,dy = a\sqrt{2gy}\;dt$$

$$\Rightarrow \qquad \int_{t=0}^{t} dt = -\frac{A}{a\sqrt{2g}}\int_{H}^{0} y^{-\frac{1}{2}}dy$$

$$\Rightarrow \qquad t = -\frac{2A}{a\sqrt{2g}}\left[\sqrt{y}\right]_H^0$$

$$\Rightarrow \qquad t = \frac{A}{a}\sqrt{\frac{2H}{g}}$$

Example 27 A large tank contains water (density 2ρ) filled up to height $2h$. Oil (density ρ) is poured over water to a height h. A hole is punched in the side wall at a height h from the bottom. Find speed of efflux.

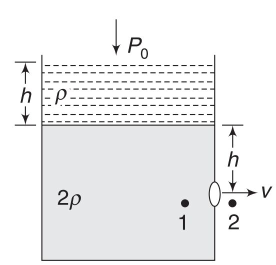


Solution

Concepts

Use Bernoulli's equation by considering two points in the same liquid.

Consider a point 1 just inside the hole and a point 2 just outside it.



Flow speed at 1 is $\simeq 0$

Pressure at 1 is $P_1 = P_0 + \rho gh + 2\rho gh$

$$= P_0 + 3\rho gh$$

Bernoulli's equation applied between 1 and 2 gives

$$P_1 + \rho g h_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho g h_2 + \frac{1}{2}\rho v_2^2$$

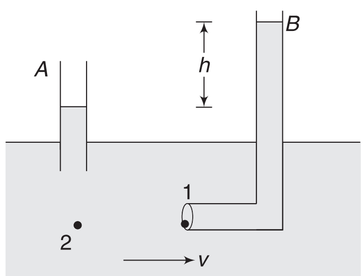
$$h_1 = h_2 \quad \text{and} \quad v_1 = 0$$

Also, density of liquid is 2ρ.

$$\therefore \quad P_0 + 3\rho g h = P_0 + \frac{1}{2}(2\rho)v^2$$

$$\Rightarrow \quad v = \sqrt{3gh}$$

Example 28 A liquid is flowing in a horizontal pipe. Two tubes, A and B, are inserted in the pipe. Tube B is L-shaped and its cross-section at 1 is held normal to the flow direction. Liquid rises in both tubes. Height of liquid column in B is h more than that in A. Find the speed of flow (v) in the pipe.



Solution

Concepts

(i) Liquid in tube B is at rest. This means that speed of flow at the mouth of tube at 1 is zero. But speed at a point like 2 on the same horizontal level is v. This implies that pressure at 1 is more than the pressure at 2.

(ii) Pressure difference between 1 and 2 is measured by h.

$$P_1 - P_2 = \rho g h \quad \text{...(i)}$$

Using Bernoulli's equation between points 1 and 2

$$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$$

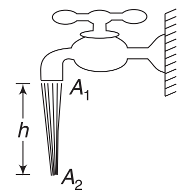
$$\Rightarrow \quad P_1 - P_2 = \frac{1}{2}\rho v^2 \qquad [\because \; v_1 = 0]$$

$$\Rightarrow \quad \rho g h = \frac{1}{2}\rho v^2$$

$$\Rightarrow \quad v = \sqrt{2gh}$$

Example 29 *Stream of water emerging from a tap "necks down"*

Open a tap discharging water at a steady rate. The stream of water narrows down as it falls.



(a) Why does this narrowing of stream happens?

(b) The cross-sectional area of the tap is $A_1 = 1.2\,\text{cm}^2$ and the area of cross-section of water stream at a depth $h = 4.5\,\text{m}$ below it is $A_2 = 0.35\,\text{cm}^2$. Calculate the volume flow rate from the tap.

Solution

Concepts

(i) From Bernoulli's equation, we can see that speed increases as water falls down.

(ii) From continuity equation, the area of cross-section must decrease with rising speed.

(a) As the water falls, its speed increases. This is due to loss in gravitational PE. Since volume flow rate is same at any cross-section, the area must be smaller if speed is higher.

(b) Bernoulli's equation

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2$$

Taking the reference level to be zero at depth h below the tap,

$$h_1 = h \quad \text{and} \quad h_2 = 0$$

$$P_1 = P_2 = P_0 \qquad \text{(atmospheric pressure)}$$

$$\therefore \quad P_0 + \frac{1}{2}\rho v_1^2 + \rho g h = P_0 + \frac{1}{2}\rho v_2^2$$

$$\Rightarrow \quad \frac{1}{2}\rho(v_2^2 - v_1^2) = \rho g h$$

$$\Rightarrow \quad v_2^2 - v_1^2 = 2gh$$

$$\text{But} \quad A_1 v_1 = A_2 v_2 \quad \Rightarrow \quad v_2 = \frac{A_1}{A_2} v_1$$

$$\therefore \quad \left(\frac{A_1^2}{A_2^2} - 1\right) v_1^2 = 2gh \quad \Rightarrow \quad v_1 = \sqrt{\frac{2gh}{\frac{A_1^2}{A_2^2} - 1}}$$

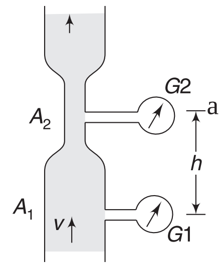
$$\Rightarrow \quad v_1 = \sqrt{\frac{2 \times 9.8 \times 0.045}{\left(\frac{1.2}{0.35}\right)^2 - 1}}$$

$$= 0.286\,\text{ms}^{-1} = 28.6\ \text{cms}^{-1}$$

Volume flow rate is $= A_1 v_1 = (1.2\,\text{cm}^2)\,(28.6\ \text{cms}^{-1})$

$$= 34\ \text{cm}^3\,\text{s}^{-1}$$

Example 30 A vertical pipe has cross-sectional area $A_1 = 0.2\,\text{m}^2$. Water (density $\rho = 10^3\ \text{kgm}^{-3}$) is flowing through it. A pressure gauge $G1$ shows reading of 200-Pa. There is a narrow zone of cross section $A_2 = 0.1\,\text{m}^2$ at a height of $h = 2\,\text{m}$ above the location of $G1$. The reading of another pressure gauge G2 located at the narrow zone is 140 kPa. Find the flow speed (v) in the pipe.



Solution

Concepts

Use of continuity equation and Bernoulli's equation.

Don't use equation (30). That equation was derived for a horizontal pipe.

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2$$

$$\Rightarrow \quad P_1 - P_2 - \rho g (h_2 - h_1) = \frac{1}{2}\rho (v_2^2 - v_1^2)$$

$$\Rightarrow \quad 200 \times 10^3 - 140 \times 10^3 - 10^3 \times 10 \times 2$$

$$= \frac{1}{2} \times 10^3 \left[\frac{A_1^2}{A_2^2} v^2 - v^2\right]$$

$$\left[A_2 v_2 = A_1 v \quad \Rightarrow \quad v_2 = \frac{A_1}{A_2} \cdot v\right]$$

$$\Rightarrow \quad 80 = \left[\left(\frac{0.2}{0.1}\right)^2 - 1\right] v^2 \quad \Rightarrow \quad 80 = 3v^2$$

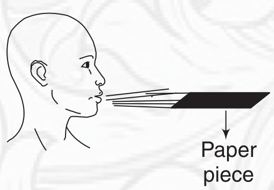
$$\Rightarrow \quad v = \sqrt{\frac{80}{3}} = 5.16\ \text{ms}^{-1}$$

YOUR TURN

Q.37 During wind storms, light roofs are blown off. Why?

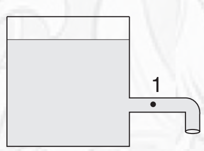
Q.38 A man is standing near a railway line. When a fast moving train crosses him, he feels as if he is being pushed towards the train. Explain.

Q.39 Hold a strip of paper at one end. By blowing air over the strip of paper, you can keep it horizontal. Explain.

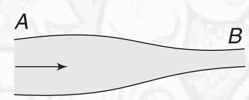


Q.40 Air is streaming past a horizontal airplane wing such that its speed is $120\,\text{ms}^{-1}$ at the upper surface of the wing and $90\ \text{ms}^{-1}$ at the lower surface. Density of air is $1.3\ \text{kg}^{-3}$. The wing is of rectangular shape; 10 m long and 2 m wide. Calculate the lift force on the wing.

Q.41 When the tap shown in figure is closed, pressure in the pipe at point 1 is $3.5 \times 10^5\,\text{Pa}$. When the tap is opened the pressure drops to $3.0 \times 10^5\,\text{Pa}$. Find the speed of water flowing through the pipe.



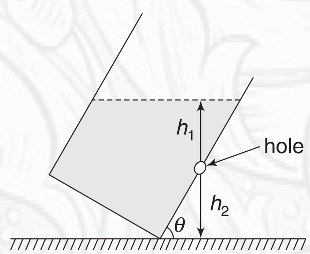
Q.42 Water flows through a horizontal tube of variable cross-section. The area of cross-section at A and B area 4 mm^2 and 2 mm^2 respectively. 1 cm^3 of water enters into the tube per second at A.



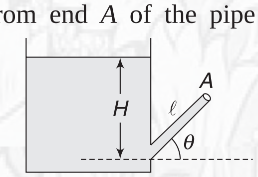
(a) Find the speed of water at B.

(b) Find pressure difference $P_A - P_B$.

Q.43 In the figure shown, find the speed of efflux from the hole.



Q.44 Find the speed of efflux from end A of the pipe shown in figure. Length of pipe is l and it is inclined at an angle θ to the horizontal. Area of cross-section of the pipe is small compared to that of the tank.



In short:

(i) A flow is said to be steady if flow conditions do not change with time.

(ii) A steady flow can be represented by a streamline diagram. A streamline represents path followed by all particles crossing through a point.

(iii) Density of streamline represents flow speed at a location and tangent on a streamline gives direction of flow.

(iv) In steady flow of an incompressible fluid, volume flow rate through any cross-section is constant.

$A_1v_1 = A_2v_2$ = volume flowing past a cross-section per unit time

(v) Pressure can be regarded as a form of energy for a fluid.

Pressure (P) is energy per unit volume.

(vi) Sum of kinetic energy, potential energy and pressure energy (per unit volume) remains constant in steady flow of an ideal fluid. This is known as Bernoulli's equation.

$$P + \frac{1}{2}\rho v^2 + \rho gh = \text{a constant}$$

(vii) In a flowing fluid the pressure at a point is higher if flow speed is lower compared to another point on same horizontal level where speed is higher.

(viii) Speed of efflux from a small hole in a large tank is $v = \sqrt{2gh}$, where h is height of water surface above the hole.

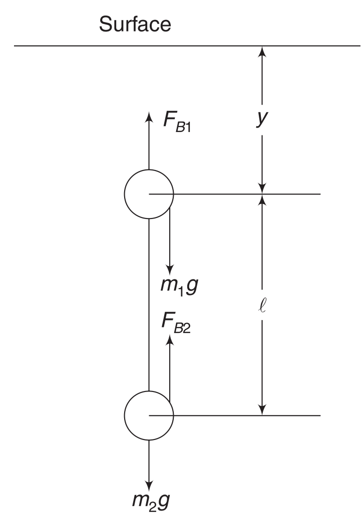
(ix) Time needed to empty an open tank of liquid through a small hole (area = a) at the bottom is

$$t = \frac{A}{a}\sqrt{\frac{2H}{g}}$$

Where A = cross-section of tank and H is initial height of liquid.

Miscellaneous Examples

Example 31 Density of liquid in a large tank increases with depth y from the surface as $\rho = a + by$, where $a = 1.0\,\text{gcm}^{-3}$ and $b = 0.01\,\text{gcm}^{-4}$. Two balls, each of volume $V = 1.0\,\text{cm}^3$ are connected by a light string of length $l = 15\,\text{cm}$ and released in the liquid. Density of two balls are $\rho_1 = 1.2\,\text{gcm}^{-3}$ and $\rho_2 = 1.4\,\text{gcm}^{-3}$. At what depth do the two balls settle in equilibrium?



Solution

Concepts

(i) Buoyancy on both balls must sum up to the combined weight of the two balls.

(ii) The heavier ball must be in lower position. Buoyancy (for same volume) increases with depth, as density of liquid in increasing. The ball in lower position experiences higher buoyancy. For string to be taut, the lower ball must have its weight greater than buoyancy acting on it.

Mass of the balls are: $m_1 = \rho_1 V = 1.2$ g

$m_2 = \rho_2 V = 1.4$ g

For equilibrium

$$F_{B1} + F_{B2} = m_1\text{g} + m_2\text{g}$$

$$V(a + by)g + V[a + b(y + l)]g = (m_1 + m_2)g$$

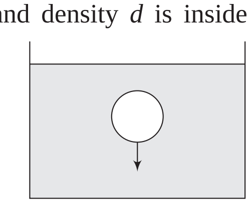
Let us write everything in CGS unit system.

$$1(1 + 0.01y) + 1[1 + 0.01(y + 15)] = 2.6$$

$\Rightarrow y = 22.5\,\text{cm}$

$\therefore$ Depth of the lighter ball is 22.5 cm and that of the heavier ball is 22.5 + 15 = 37.5 cm.

Example 32 A ball of volume V and density d is inside a liquid of density ρ. It is moved vertically down by a further distance h. Find the change in gravitational PE of the system of the ball and the liquid.



Solution

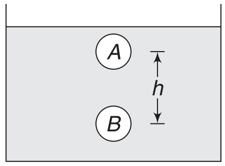
Concepts

When the ball moves down, a same volume of liquid moves up to occupy the space emptied by the ball. PE of the ball decreases and that of the liquid increases.

Ball moves from position A to B. Its PE changes by

$$\Delta U_{\text{ball}} = -mgh = -Vd\,.\,gh$$

Liquid in position B moves up to occupy the vacant position at A. change in PE of liquid.



$$\Delta U_L = m_L gh = V\rho gh$$

$$\therefore \quad \Delta U = \Delta U_L + \Delta U_{\text{ball}}$$

$$= V(\rho - d)gh$$

Example 33 In a flowing fluid, the velocity distribution in xy plane is given by

$$\vec{v} = (kx)\hat{i} - (ky)\hat{j}$$

Where k is a positive constant. Draw the streamlines in xy plane for $y > 0$ and give a possible interpretation of the pattern.

Solution

Concepts

Path of any particle is a streamline. Therefore, we need to find the path equations of various particles.

$$v_x = kx \quad \Rightarrow \quad \frac{dx}{dt} = kx \qquad \text{...(i)}$$

And

$$v_y = -ky \quad \Rightarrow \quad \frac{dy}{dt} = -ky \qquad \text{...(ii)}$$

(ii) ÷ (i)

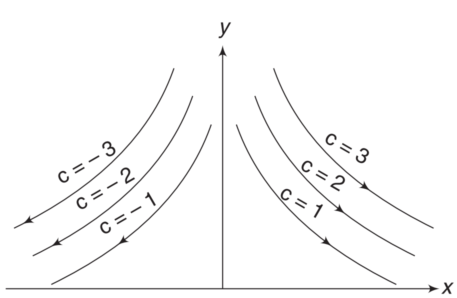
$$\frac{dy/dt}{dx/dt} = -\frac{y}{x} \quad \Rightarrow \quad \frac{dy}{dx} = -\frac{y}{x}$$

$$\Rightarrow \quad \int \frac{dy}{y} = -\int \frac{dx}{x}$$

$$\Rightarrow \quad \ln y = -\ln x + \ln c \quad \Rightarrow \quad \ln(xy) = \ln c$$

$$\Rightarrow \quad xy = c$$

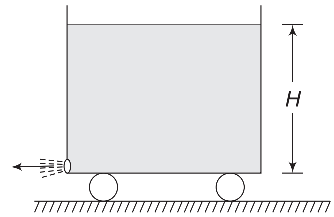
This is an equation of hyperbola. We get different hyperbolas for different values of c. These hyperbolas are streamlines. The arrowheads can be determined by looking at (i) and (ii). For example, in first quadrant, u_x is positive and v_y is negative.



The streamlines could simulate the flow of a downward stream against a flat floor.

Example 34 *Recoiling tank*

A tank of cross-sectional area A is filled with water to a height H. It stands on a smooth horizontal surface. A small orifice is made at the bottom of the vertical wall. The water flows out of the orifice and the tank recoils. Find the speed of the tank when it gets emptied. Assume mass of empty tank to be negligible.



Solution

Concepts

(i) Speed of efflux (u) with respect to the tank is given by $u = \sqrt{2gh}$ where h is instantaneous height of water in the tank.

(ii) The thrust force experienced by the water tank is $u\left|\dfrac{dm}{dt}\right|$ where $\dfrac{dm}{dt}$ is rate of leakage of mass from the tank.

[Refer to variable mass system in the chapter of momentum]

When height of water in the tank is h, let the instantaneous speed of the tank be u.

$$u = \sqrt{2gh} = \text{speed of outgoing mass wrt the tank}$$

$$\therefore \quad \left|\frac{dm}{dt}\right| = \text{rate of leakage of mass} = au \cdot \rho$$

Where a = cross-sectional area of orifice and ρ is density of liquid.

$$\therefore \quad \text{Thrust force, } F_{\text{th}} = u\left|\frac{dm}{dt}\right| = au^2\rho = 2ga\rho \cdot h$$

$$\therefore \quad m\frac{dv}{dt} = 2ga\rho \cdot h$$

But instantaneous mass of liquid in the tank can be written as

$m = Ah\rho$ where A = cross section of the tank

$$\therefore \quad \frac{dv}{dt} = 2\text{g}\left(\frac{a}{A}\right)$$

$$\Rightarrow \quad \frac{dv}{dh}\cdot\frac{dh}{dt} = 2g\frac{a}{A}$$

But volume flow rate $= a \cdot u = -A \cdot \dfrac{dh}{dt}$

$$\Rightarrow \quad \frac{dh}{dt} = -\frac{au}{A}$$

$$\therefore \quad \frac{dv}{dh}\left(-\frac{au}{A}\right) = 2g\frac{a}{A}$$

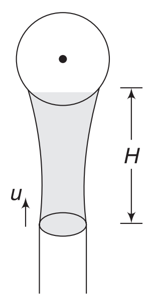
$$\Rightarrow \quad -\frac{dv}{dh}(\sqrt{2gh}) = 2g$$

$$\Rightarrow \quad \int_0^V dv = -\sqrt{2g}\int_H^0 \frac{dh}{\sqrt{h}}$$

$$\Rightarrow \quad v = 2\sqrt{2gH}$$

Example 35 *A ball balanced in a water jet*

Water jet is coming out through a pipe of cross-sectional radius r at a velocity u in upward direction. The water jet widens as it rises, though it is nearly correct to assume that velocity of water particles remain vertical. At a height H above the outlet, there is a ball supported by the water jet. Weight of the ball is W. Density of water is ρ. Find H for which the ball will remain in equilibrium. Assume that water particles striking the ball momentarily come to rest.



Solution

Concepts

(i) As the water jet rises, its speed decreases and therefore cross-sectional area widens.

(ii) Using Bernoulli's equation, we find speed at height H. Volume flow rate is same throughout the jet.

(iii) Force on ball due to impinging water jet = rate of change of momentum of the water jet due to striking the ball.

Let v = speed of water on reaching the ball.

$$P_0 + \frac{1}{2}\rho v^2 + \rho gH = P_0 + \frac{1}{2}\rho u^2 + 0$$

$$\Rightarrow \quad v^2 = u^2 - 2gH \quad \Rightarrow \quad v = \sqrt{u^2 - 2gH}$$

Volume flow rate, $Q = \pi r^2 \cdot u$

Mass of water hitting the ball per second

$$= Q\cdot\rho = \pi r^2 u\rho$$

Change in momentum per unit time = $(\pi r^2 u\rho)v$

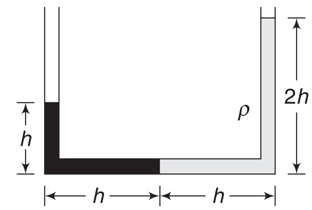
$\therefore$ Force on ball = $(\pi r^2 u\rho)v$

For equilibrium: $(\pi r^2 u\rho)v = W$

$$\Rightarrow \quad \pi r^2 u\rho\sqrt{u^2 - 2gH} = W$$

$$\Rightarrow \quad \frac{1}{2g}\left[u^2 - \frac{W^2}{\pi^2 r^4 u^2\rho^2}\right] = H$$

Example 36 A U-shaped tube has two immiscible liquids filled in it. The liquid in right arm has density ρ and the density of liquid in the left arm is not known. The equilibrium position has been shown in the diagram. Now, the tube is accelerated horizontally to the right such that liquid columns have same height on the two sides. Find the acceleration of the tube.



Solution

Concepts

(i) We can find the density of the other liquid by writing pressure at the interface of the two liquids; working from both sides and equating them.

(ii) By conserving the volume, it is easy to see that height of liquid columns in both limbs must be $\frac{3h}{2}$ when the two heights are equal.

(iii) In new position, we will again write the pressure at the interface, working from two sides and equating them.

(iv) In a horizontally accelerated container, pressure increases when we move horizontally inside liquid opposite to acceleration. Refer to equation (18).

Let us write the pressure at the interface of two liquids when the tube is at rest.

If ρ' is density of other liquid, then

$$\rho' gh = \rho g\cdot 2h \quad \Rightarrow \quad \rho' = 2\rho$$

When the tube is accelerated, height of liquid in both limbs becomes equal. This height must be $\frac{3h}{2}$ for volume to remain conserved.

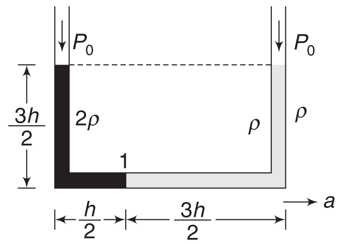
Let us write pressure at liquid interface (1), working from right.

$$P_1 = P_0 + \rho g\left(\frac{3h}{2}\right) + a\rho\left(\frac{3h}{2}\right)$$

[Refer to equation 18]

Working from left, we can write

$$P_1 = P_0 + 2\rho g\cdot\frac{3h}{2} - a2\rho\frac{h}{2}$$



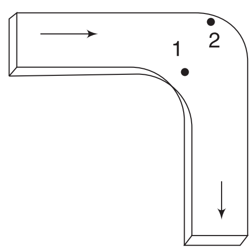
Equating the two values

$$\rho g\left(\frac{3h}{2}\right) + a\rho\left(\frac{3h}{2}\right) = 2\rho g\left(\frac{3h}{2}\right) - a2\rho\left(\frac{h}{2}\right)$$

$$\Rightarrow \qquad a = \frac{3}{5}g$$

Example 37 *Flow at a bend*

An ideal liquid is flowing through a flat horizontal tube of uniform cross-section. The flow is steady. The tube has an *L*-shaped bend as shown. Consider two points 1 and 2 at the bend. At which point the streamlines will be more crowded? At which point is the pressure higher?



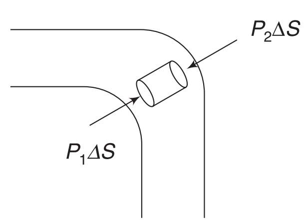
Solution

Concepts

(i) Streamlines have higher density at locations where speed of flow is high.

(ii) Pressure is lower at a point where speed is higher if the points are on same horizontal level.

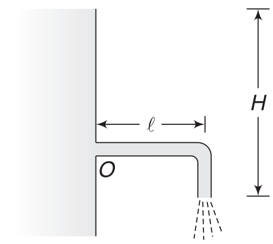
Consider a small cylindrical fluid element negotiating the bend. As the fluid element turns, it must have a radial acceleration. For this, it must have radial force acting on it.



Therefore, $P_2\Delta S > P_1\Delta S \quad \Rightarrow \quad P_2 > P_1$

Higher pressure at 2 implies lower speed at 2. Therefore, the streamlines will be more crowded near 1.

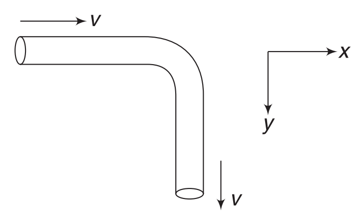
Example 38 A right-angled tube is connected to a large tank as shown. The cross-sectional area of the tube is S and its discharge end is at a depth H below the water surface. Find the torque of reaction force of flowing water, acting on the bend of the tube relative to point O. Density of water is ρ.



Solution

Concepts

As water flows out, its speed is constant throughout the tube but there is a change in direction of velocity at the bend. Thus, momentum of flowing water changes at the bend. Water experiences force due to pipe for this change is momentum. From Newton's third law, the water exerts equal force on the bend. This force causes torque.



Speed of efflux $v = \sqrt{2gH}$

Volume flow rate,

$$Q = Sv$$

Mass flow rate $= Q\cdot\rho = \rho Sv$

Change in momentum of water per unit time is force applied by the bend on it.

$$\therefore \qquad \vec{F} = (\rho Sv)v\hat{j} - (\rho Sv)v\hat{i}$$

$$= \rho Sv^2(\hat{j} - \hat{i})$$

Force applied by water on the bend is

$$\therefore \qquad \vec{F}_0 = -\vec{F} = \rho Sv^2(\hat{i} - \hat{j})$$

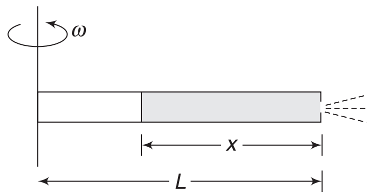
Torque about O is

$$\vec{\tau} = (l\hat{i}) \times \vec{F}_0 = (l\hat{i}) \times \rho Sv^2(\hat{i} - \hat{j})$$

$$= -l\rho Sv^2\hat{k}$$

$$\therefore \qquad |\vec{\tau}| = l\rho Sv^2 = l\rho S(2gH) = 2l\rho SgH$$

Example 39 A horizontal tube of length L is filled with a liquid of density ρ. It is rotated about a vertical axis passing through its open end. The liquid column occupies a length x in the tube, as shown. A small hole develops at the other end of the tube. Find the speed with which the liquid is initially ejected with respect to the tube.



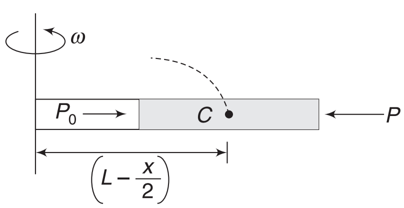
Solution

Concepts

(i) Pressure at open end is atmospheric pressure.

(ii) Pressure at the other end can be calculated by considering centripetal force on cylindrical liquid inside the tube. For writing centripetal force, we can assume the entire mass of the liquid at its COM.

(iii) After knowing pressure at the other end, we can use Bernoulli's equation between points just inside and just outside the hole to find speed of ejected liquid with respect to the tube.



C is COM of liquid

Let S = area of cross-section of the tube

Mass of liquid, $m = S \cdot x \cdot \rho$

Centripetal force needed for rotation of liquid is

$$F = m\omega^2\left(L - \frac{x}{2}\right) = S\rho x\omega^2\left(L - \frac{x}{2}\right)$$

$$\therefore \quad PS - P_0S = F$$

$$\Rightarrow \quad P = P_0 + \rho\omega^2 x\left(L - \frac{x}{2}\right)$$

Applying Bernoulli's equation between a point just inside and outside the hole

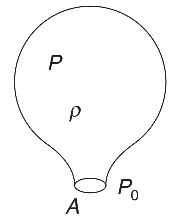
$$P + 0 + 0 = P_0 + \frac{1}{2}\rho v^2 + 0$$

$$\Rightarrow \quad \rho\omega^2 x\left(L - \frac{x}{2}\right) = \frac{1}{2}\rho v^2$$

$$\Rightarrow \quad v = \sqrt{\omega^2 x\,(2L - x)}$$

Example 40 *A balloon*

A gas-filled balloon is in equilibrium while floating in air. Pressure inside it is P, which is greater than atmospheric pressure (P_0). Density of gas in the balloon is ρ. The seal of the balloon breaks and a small hole of area A gets developed. Find the initial thrust force experienced by the balloon due to the gas escaping out of the hole.



Solution

Concepts

(i) The speed of outgoing gases relative to the balloon can be calculated using Bernoulli's equation.

(ii) Then using the concept of variable mass learnt in the chapter of momentum, we can easily write the thrust force.

OR, one can simply write the force = rate at which the outgoing gas is imparted momentum.

Let speed of outgoing gas (relative to the balloon) be v. [Since the balloon is at rest, initial speed relative to balloon will be same as speed relative to ground. You must not be very fussy about it.]

$$\frac{1}{2}\rho v^2 + P_0 = P$$

$$\Rightarrow \quad v = \sqrt{\frac{2(P - P_0)}{\rho}}$$

Volume flow rate, $Q = Av$

Mass flow rate, $\frac{dm}{dt} = Q \cdot \rho = A\rho v$

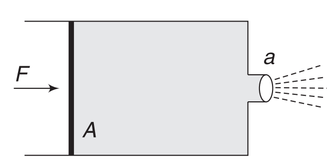
$$\therefore \quad F_{\text{thrust}} = u_{\text{rel}} \cdot \frac{dm}{dt}$$

$$= v \cdot A\rho v = A\rho v^2$$

$$= 2A(P - P_0)$$

Example 41 *A pichkari (a water gun)*

During the festival of Holi, we use a *pichkari* to drench others with colour. One such pichkari has a horizontal cylinder fitted with a piston of area A. It contains water. The opening has a small cross-section a. The piston moves slowly under action of a constant horizontal force F. Find the speed (v) at which water erupts out of the mouth.

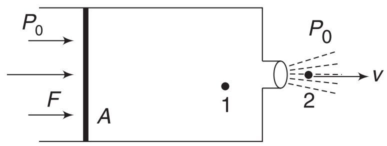


Solution

Concepts

(i) Piston is moved slowly. This implies that speed of liquid inside the cylinder is nearly zero.

(ii) The applied force increases the pressure in the liquid. This high pressure energy gets converted into kinetic energy as the liquid is ejected.



$$P_1 = P_0 + \frac{F}{A}; \qquad v_1 = 0$$

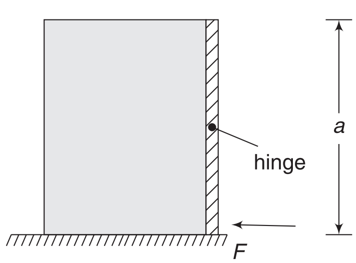
$$P_2 = p_0; \qquad v_2 = v$$

$$P_2 + \frac{1}{2}\rho v_2^2 = P_1 + \frac{1}{2}\rho v_1^2$$

$$P_0 + \frac{1}{2}\rho v^2 = P_0 + \frac{F}{A} + 0$$

$$\Rightarrow \qquad v = \sqrt{\frac{2F}{\rho A}}$$

Example 42 A square gate has dimension ($a \times a$). It is hinged at the middle and can rotate freely about a horizontal axis through its centre. It is used to retain water (density $= \rho$) on one side. For equilibrium of the gate, a horizontal force (F) is applied at its lower edge.



(a) Find F

(b) Find the force applied by the hinge on the gate.

Solution

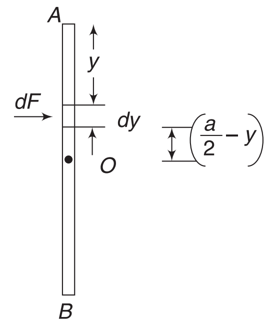
Concepts

(i) The pressure of water causes an anti-clockwise torque on the gate about the hinge. This is equal to torque on lower half minus the (clockwise) torque on upper half of the gate.

(ii) Force F is applied to ensure rotational equilibrium. Its value is obtained by making net torque about the hinge equal to zero.

(iii) We also need to find total force applied by water on the gate. This force is balanced by F plus hinge force.

(a) Consider a strip of width dy on upper half of the gate.



Area of strip $dA = a\,dy$

Pressure at strip, $P = \rho g y$

[We need not consider atmospheric pressure. It is acting on the gate from right side also. Its effect gets cancelled out.]

Force on strip,

$$dF = P\,dA = \rho g a y\,dy$$

Torque about O, $d\tau = \left(\frac{a}{2} - y\right) dF = \rho g a\, y\left(\frac{a}{2} - y\right)dy$

$\therefore$ clockwise torque on segment AO is

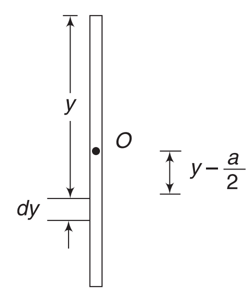
$$\tau_1 = \int_{y=0}^{a/2} d\tau = \rho g a \int_0^{a/2} \left(\frac{a}{2}y - y^2\right)dy$$

$$\therefore \qquad \tau_1 = \frac{1}{48}\rho g a^4$$

Similarly, torque on lower half can be calculated as

$$d\tau = \left(y - \frac{a}{2}\right) dF$$

$$= \rho g a\, y\left(y - \frac{a}{2}\right)dy$$



$$\tau_2 = \rho g a \int_{a/2}^{a} \left(y^2 - \frac{a}{2}y\right)dy$$

$$= \frac{5}{48}\rho g a^4$$

$$\therefore \qquad \tau_{\text{net}} = \tau_2 - \tau_1 = \frac{1}{12}\rho g a^4$$

This torque must be balanced by the torque due to applied force F.

$$\therefore \qquad F\frac{a}{2} = \frac{1}{12}\rho g a^4 \quad \Rightarrow \quad F = \frac{1}{6}\rho g a^3$$

(b) Total force on the gate due to water is

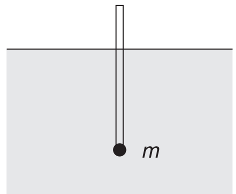
$$F_0 = \left(\frac{0 + \rho g a}{2}\right)a^2 = \frac{1}{2}\rho g a^3$$

$\therefore$ Hinge force, $F_H = F_0 - F = \left(\frac{1}{2} - \frac{1}{6}\right)\rho g\, a^3$

$$= \frac{1}{3}\rho g a^3$$

Example 43 *Stable stick*

A wooden stick of length L and radius R ($<<L$) has density ρ. A small particle of mass m is attached at its lower end and the stick floats in vertical position in water of density σ. Find smallest value of mass m for which the stick can be in stable equilibrium.

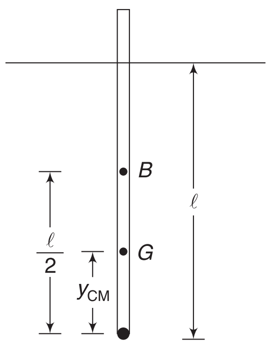


Solution

Concepts

(i) For stable equilibrium, the centre of buoyancy must be above centre of mass.

(ii) Centre of buoyancy is at geometrical centre of the submerged part.



Mass of stick, $M = \rho L \pi R^2$

Distance of COM of stick + particle system from the lower end is

$$y_{cm} = \frac{M\frac{L}{2}}{M + m}$$

Let length of submerged part be l.

$$(M + m)g = \pi R^2 l \sigma g$$

$$\therefore \quad l = \frac{(M + m)}{\pi R^2 \sigma}$$

Centre of buoyancy is at a distance $\frac{l}{2}$ from the lower end. For stable equilibrium, the centre of buoyancy (B) should be above the centre of mass (G).

$$\therefore \quad \frac{l}{2} \geq y_{cm}$$

$$\Rightarrow \quad \frac{1}{2}\left(\frac{M + m}{\pi R^2 \sigma}\right) \geq \frac{M\frac{L}{2}}{M + m}$$

$$\Rightarrow \quad (M + m)^2 \geq ML \cdot \pi R^2 \cdot \sigma$$

$$\Rightarrow \quad M + m \geq \sqrt{M \pi L R^2 \sigma}$$

$$\Rightarrow \quad m \geq \sqrt{M \pi L R^2 \sigma} - M$$

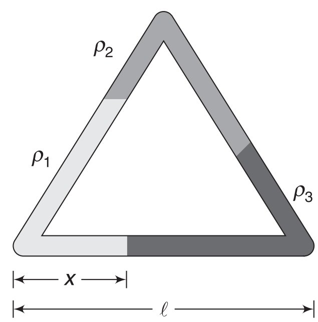
$$\Rightarrow \quad m \geq \sqrt{(\rho L \pi R^2)\ \pi L R^2 \sigma} - \rho L \pi R^2$$

$$\Rightarrow \quad m \geq \pi R^2 L \rho \left[\sqrt{\frac{\sigma}{\rho}} - 1\right]$$

$$\therefore \quad m_{\min} = \pi R^2 L \rho \left[\sqrt{\frac{\sigma}{\rho}} - 1\right]$$

Example 44 A thin triangular tube of uniform cross-section has each of its side equal to l. The tube is in vertical plane with one of its side horizontal. It contains equal volumes of three immiscible liquids having densities ρ_1, ρ_2 and ρ_3.

Find the length x of liquid of density ρ_1 in the horizontal limb.

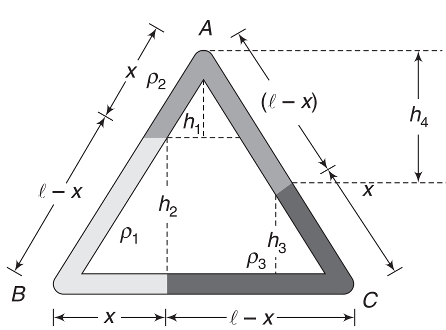


Solution

Concepts

(i) Pressure at all points in the horizontal segment will be same, otherwise a segment of fluid will get accelerated.

(ii) Each liquid covers length l along the perimeter. Thus, liquid of density ρ_1 must have a length $(l - x)$ in side AB. It implies that liquid of density ρ_2 has a length x in side AB and $(l - x)$ in AC.



$$P_B = P_C$$

$$\rho_2 gh_1 + \rho_1 gh_2 = \rho_2 gh_4 + \rho_3 gh_3$$

$$\Rightarrow \quad \rho_2 x \cdot \cos 30° + \rho_1 (l - x)\cos 30°$$

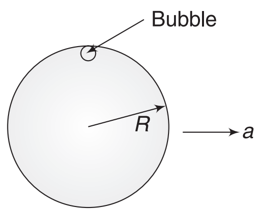
$$= \rho_2 (l - x)\cos 30° + \rho_3 x \cos 30°$$

$$\Rightarrow \quad \rho_2 x + \rho_1 l - \rho_1 x = \rho_2 l - \rho_2 x + \rho_3 x$$

$$\Rightarrow \quad x = \frac{(\rho_2 - \rho_1) l}{2\rho_2 - \rho_1 - \rho_3} = \frac{(\rho_1 - \rho_2) l}{\rho_1 + \rho_3 - 2\rho_2}$$

Example 45 *Air bubble in accelerated tank*

A closed spherical bulb of radius R is full of water. There is a small air bubble inside it. The bulb is moved with a horizontal acceleration $a = \frac{g}{2}$. Find the displacement of the air bubble as it moves to its new equilibrium position.



Solution

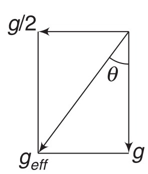
Concepts

(i) The air bubble has buoyancy force larger than its weight. It remains buoyed at the top.

(ii) As the bulb accelerates, the entire environment inside the bulb sees an effective acceleration due to gravity that makes an angle with vertical.

(iii) Buoyancy and effective weight both act along the line of g_{eff}. The bubble settles in equilibrium in a position, which is farthest from the centre along the line of buoyancy.

g_{eff} makes an angle θ with vertical, where



$$\tan\theta = \frac{1}{2}$$

The bubble moves from A to B

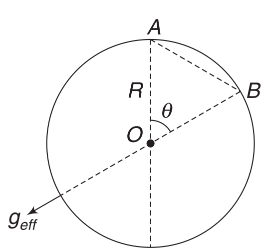
Displacement = AB

$$AB = 2R\sin\left(\frac{\theta}{2}\right) = 2R(0.23)$$

$$= 0.46\ R$$

Note: $$\tan\theta = \frac{2\tan\frac{\theta}{2}}{1 - \tan^2\frac{\theta}{2}}$$

Let $$\tan\frac{\theta}{2} = x$$



$$\Rightarrow \quad \frac{1}{2} = \frac{2x}{1 - x^2} \quad \Rightarrow \quad x^2 + 4x - 1 = 0$$

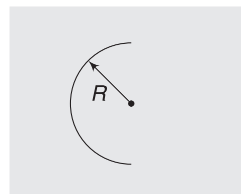
$$\Rightarrow \quad x = \frac{-4 \pm \sqrt{16 + 4}}{2} = (\sqrt{5} - 2)$$

$$\therefore \quad \tan\frac{\theta}{2} = \sqrt{5} - 2 \simeq 0.24$$

$$\therefore \quad \sin\frac{\theta}{2} = \frac{0.24}{\sqrt{1^2 + (0.24)^2}} = 0.23$$

Example 46 *A hemispherical cup*

A hemispherical cup of radius R is kept in water (density = ρ), as shown. Find the vertical thrust by the liquid on the concave surface of the cup.



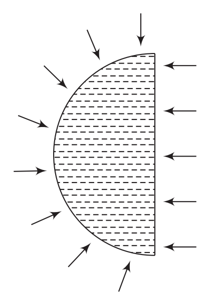
Solution

Concepts

(i) Vertical force by water on upper half is upward and on the lower half is downward. Net vertical force is downward.

(ii) We can calculate this force by considering equilibrium of a hemispherical volume of water inside the tank.

Consider a hemispherical volume of water inside the water tank. Surrounding liquid exerts pressure on it from all sides. Since the volume of water is in equilibrium, net force on it must be zero.



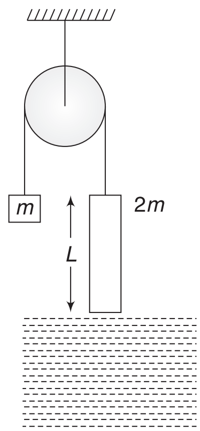
Vertically upward force on this hemispherical element due to surrounding water is equal to weight of water in the hemisphere.

$$\therefore \quad F_V = \frac{2}{3}\pi R^3 \rho \cdot g$$

Frow Newton's third law, the hemispherical volume of water applies equal vertical force on surrounding liquid in downward direction.

When the hemispherical cup is kept, the water molecules keep striking the surface in similar fashion. They do not differentiate between a cup wall and a "water wall". Thus, force on the inner surface of the cup will be $\frac{2}{3}\pi R^3 \rho g$ in downward direction.

Example 47 A uniform cylinder of mass $2m$ is connected to a block of mass m though a string passing over a smooth pulley. The system is released from position where the lower end of the cylinder just touches a liquid surface in a tank. Length and cross-section area of cylinder are L and A respectively. The cylinder just stops at the instant when its upper edge reaches the liquid surface. Find density of liquid. Neglect viscosity.

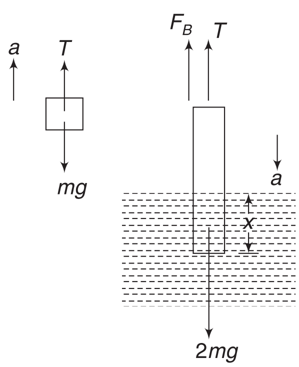


Solution

Concepts

(i) Initially, the cylinder accelerates downwards. Buoyancy force on it goes on increasing. At a point, its acceleration becomes zero (when buoyancy becomes equal to mg) and thereafter, it turns negative. Finally, the cylinder stops.

(ii) The acceleration is changing. We need to express acceleration as a function of displacement (x) of the cylinder. Then we will do integration to get the answer. Note that velocity becomes zero when $x = L$.



Consider the system when the cylinder has moved down by a distance x. Buoyancy on the cylinder is:

$$F_B = Ax\rho g$$

If acceleration is a, we can write

$$T - mg = ma \qquad \text{...(i) and}$$

$$2mg - A\rho g x - T = 2ma \qquad \text{...(ii)}$$

Adding (i) and (ii) gives

$$mg - A\rho gx = 3ma$$

$$\Rightarrow \quad a = \frac{g}{3} - \frac{A\rho g}{3m} x = k_1 - k_2 x$$

where $k_1 = \dfrac{g}{3}$ and $k_2 = \dfrac{A\rho g}{3m}$

$$\therefore \quad v\frac{dv}{dx} = k_1 - k_2 x$$

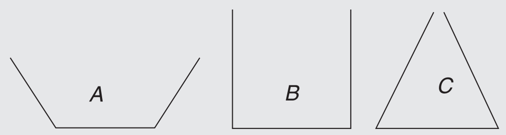
$$\Rightarrow \quad \int_o^o v\,dv = \int_L^o (k_1 - k_2 x)\,dx \quad \Rightarrow 0 = k_1 L - \frac{K_2 L^2}{2}$$

$$\Rightarrow \quad k_1 = \frac{k_2 L}{2} \quad \Rightarrow \quad \frac{g}{3} = \frac{A\rho g L}{6m}$$

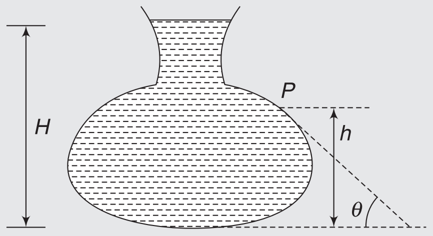
$$\Rightarrow \quad \rho = \frac{2m}{AL}$$

Worksheet 1

1. Equal masses of three liquids are kept in three identical cylindrical vessels A, B and C. The densities are ρ_A, ρ_B, ρ_C with $\rho_A < \rho_B < \rho_C$. The force on the base will be
   (a) maximum in vessel A
   (b) maximum in vessel B
   (c) maximum in vessel C
   (d) equal in all vessels
2. The three vessels shown in figure have same base area. Equal volumes of a liquid are poured in the three vessels. The force on the base will be



   (a) maximum in vessel A
   (b) maximum in vessel B
   (c) maximum in vessel C
   (d) equal in all the vessels
3. Figure shows a vertical cross-section of a vessel filled with liquid of density ρ. The normal thrust per unit area on the wall of the vessel at point P, as shown, will be



   (a) $h\rho g$ (b) $H\rho g$
   (c) $(H - h)\rho g$ (d) $(H - h)\rho g \cos\theta$
4. A beaker containing a liquid is kept inside a big closed jar. If the air inside the jar is continuously pumped out, the liquid pressure near the bottom of the jar will
   (a) increase
   (b) decrease
   (c) remain constant
   (d) first decrease and then increase
5. A piece of wood is floating in water kept in a bottle. The bottle is connected to an air pump. When more air is pushed into the bottle from the pump, the piece of wood will float with (Neglect the compressibility of water.)
   (a) larger part in water (b) lesser part in water
   (c) same part in water (d) it will sink
6. A closed cubical box is completely filled with water and is accelerated horizontally towards right with an acceleration a. The resultant normal force by the water on the top of the box-
   (a) passes through the centre of the top
   (b) passes through a point to the right of centre
   (c) passes through a point to the left of the centre
   (d) becomes zero
7. In a hydraulic lift, used at a service station, the radius of the large and small pistons are in ratio 20:1. What weight placed on the small piston will be sufficient to lift a car of mass 1,500 kg?
   (a) 3.75 kg (b) 37.5 kg
   (c) 7.5 kg (d) 75 kg
8. The pressure at the bottom of a tank of water is $3P$ where P is the atmospheric pressure. If the water is drawn out till the level of water is lowered by one fifth, the pressure at the bottom of the tank will now be
   (a) $2P$ (b) $(13/5)P$
   (c) $(8/5)P$ (d) $(4/5)P$
9. A boy carries a fish in one hand and a bucket (not full) of water in the other hand. If he places the fish in the bucket, the weight now carried by him (assume that water does not spill):
   (a) is less than before
   (b) is more than before
   (c) is the same as before
   (d) depends upon his speed
10. Two vertical cylindrical tubes contain mercury. They are connected by a horizontal tube at their bottom. The diameter of one tube is four times larger than the diameter of the other. A column of water of height 70 cm is poured into the narrower tube. How much will the mercury level rise in the other tube?
   (a) 0.3 cm (b) 4.8 cm
   (c) 5.8 cm (d) 1.2 cm
11. A body of uniform cross-sectional area floats in a liquid of density three times its own density. The portion of exposed height will be
   (a) 2/3 (b) 5/6
   (c) 1/6 (d) 1/3

12. The reading of a spring balance when a block is suspended from it in air is 60 N. Reading changes to 40 N when the block is submerged in water. The specific gravity of the block must be

(a) 3 (b) 2
(c) 6 (d) 3/2

13. A hollow sphere of volume V is floating on water surface, with half its volume remaining immersed. What should be the minimum volume of water poured inside the sphere so that the sphere now sinks into the water?

(a) $V/2$ (b) $V/3$
(c) $V/4$ (d) V

14. A and B are two metallic pieces. They are fully immersed in water and then weighed. Now they show same loss of weight. The conclusion, therefore, is

(a) A and B have same weight in air
(b) A and B have equal volumes
(c) the densities of the materials of A and B are the same
(d) A and B are immersed to the same depth inside water.

15. Two bodies are in equilibrium when suspended in water from the arms of a balance. The mass of one body is 36 g and its density is 9 gcm^{-3}. If the mass of the other is 48 g, its density in gcm^{-3} is:

(a) 4/3 (b) 3/2
(c) 3 (d) 5

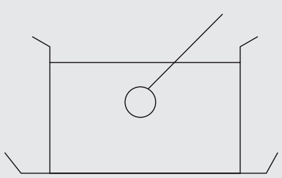
16. The mass of a balloon with its contents is 1.5 kg. It is descending with an acceleration equal to half the acceleration due to gravity. If it is to go up with the same acceleration, keeping the volume same, its mass should be decreased by:

(a) 1.2 kg (b) 1 kg
(c) 0.75 kg (d) 0.5 kg

17. A hemispherical bowl just floats without sinking in a liquid of density 1.2×10^3 kgm^{-3}. If outer diameter and the density of the bowl are 1 m and 2×10^4 kgm^{-3} respectively, then the inner diameter of the bowl will be (Given: $(0.94)^{1/3} = 0.98$)

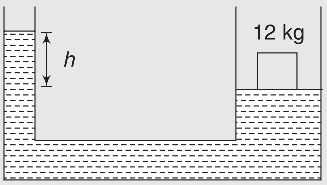
(a) 0.94 m (b) 0.97 m
(c) 0.98 m (d) 0.99 m

18. A vessel with water is placed on a weighing pan and it reads 600 g. Now a ball of mass 40 g and density 0.80 g cm^{-3} is sunk into the water. It is held sunk inside water with the help of a pin of negligible volume connected to it, as shown in figure. The weighing pan will show a reading of



(a) 600 g (b) 550 g
(c) 650 g (d) 632 g

19. The area of cross-section of the wider tube shown in figure is 800 cm^2. The left end is open and right end is covered with a massless piston. If a mass of 12 kg is placed on the piston, the difference in heights h in the level of water in the two tubes is:

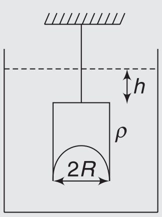


(a) 10 cm (b) 6 cm
(c) 15 cm (d) 2 cm

20. A cylindrical beaker of cross-sectional area A contains water. Pressure exerted by water on the bottom circular surface is P. A block of wood (relative density = 0.8) is put in the beaker and it floats. Its mass is M. Now, the pressure exerted by water on the bottom surface of the container is P_1. Then

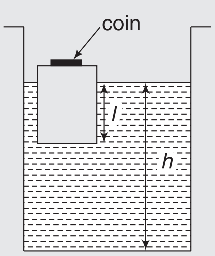
(a) $P_1 - P = 0.8Mg/A$ (b) $P = P_1$
(c) $P < P_1$ (d) $P_1 - P = Mg/A$

21. A hemispherical portion of radius R is removed from the bottom of a cylinder of radius R. The volume of the remaining cylinder is V and its mass is M. It is suspended by a string in a liquid of density ρ, where it stays vertical. The upper surface of the cylinder is at a depth h below the liquid surface. The force on the bottom of the cylinder by the liquid is



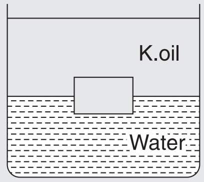
(a) Mg (b) $Mg - V\rho g$
(c) $Mg + \pi R^2 h\rho g$ (d) $\rho g(V + \pi R^2 h)$

22. A wooden block, with a metal coin placed on its top, floats in water as shown in figure. The distances l and h are as shown. After some time, the coin falls into the water. Then



(a) l decreases and h increases
(b) l increases and h decreases
(c) both l and h increase
(d) both l and h decrease

23. A cube of ice floats partly in water and partly in kerosene. Find the ratio of the volume of ice immersed in water to that in kerosene. Specific gravity of kerosene oil is 0.8 and that of ice is 0.9.

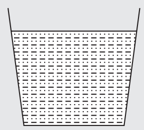


(a) 2:1 (b) 1:1
(c) 3:2 (d) None of these

24. A cubical block of iron 6 cm on each side is floating on mercury in a vessel; water is poured into the vessel so that it just covers the iron block. What is the height of water column? [Take relative density of *Hg* = 13.6 and that of *Fe* = 7.2]

(a) 4.5 cm (b) 4.75 cm
(c) 3.05 cm (d) 2.5 cm

25. A liquid of mass 1 kg is filled in a flask as shown in figure. The force exerted by the flask on the liquid is ($g = 10\,ms^{-2}$) [Neglect atmospheric pressure]



(a) 10 N
(b) greater than 10 N
(c) less than 10 N (d) zero

26. A *U*–tube having horizontal arm of length 20 cm, has uniform cross-sectional area = 1 cm^2. It is filled with water of volume 60 cm^3. What volume of a liquid of density 4 gcm^{-3} should be poured from one side into the *U*-tube so that no water is left in the horizontal arm of the tube?

(a) 60 cm^3 (b) 45 cm^3
(c) 50 cm^3 (d) 35 cm^3

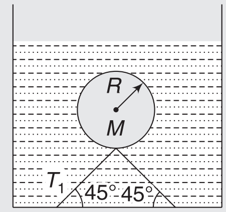
27. Two cylinders of same cross-section and length L but made of two material of densities d_1 and d_2 are cemented together to form a cylinder of length $2L$. The combination floats in a liquid of density d with a length $L/2$ above the surface of the liquid. If $d_1 > d_2$ then:

(a) $d_1 > \frac{3}{4} d$ (b) $\frac{d}{2} > d_1$
(c) $\frac{d}{4} > d_1$ (d) $d < d_1$

28. A small wooden ball of density ρ is immersed in water of density σ to depth h and then released. The height H above the surface of water up to which the ball will jump out of water is

(a) $\frac{\sigma h}{\rho}$ (b) $\left(\frac{\sigma}{\rho} - 1\right)h$
(c) h (d) zero

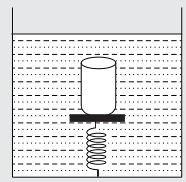
29. A hollow sphere of mass M and radius R is immersed in a tank of water (density ρ_w). The sphere would float if it were set free. The sphere is tied to the bottom of the tank by two wires, which make angle 45° with the horizontal, as shown in the figure. The tension T in each of the wires is



(a) $\dfrac{\frac{4}{3}\pi R^3 \rho_w g - Mg}{\frac{2}{3}\pi R^3 \rho_w \sqrt{2} g - Mg}$ (b)

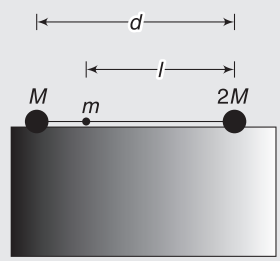
(c) $\dfrac{\frac{4}{3}\pi R^3 \rho_w g - Mg}{2}$ (d) $\frac{4}{3}\pi R^3 \rho_w g + Mg$

30. A cylindrical block of area of cross–section A and of material of density ρ is placed in a liquid of density one–third the density of block. The block compresses a spring and compression in the spring is one–third the length of the block. If acceleration due to gravity is g, the spring constant of the spring is:



(a) ρAg (b) $2\rho Ag$
(c) $2\rho Ag/3$ (d) $\rho Ag/3$

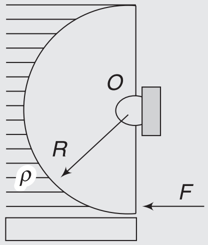
31. A dumbbell, having a light rod and two balls of masses M and $2M$ (of equal volume V), is placed in water of density ρ. It is observed that by attaching a mass m to the rod, the dumbbell floats with the rod horizontal on the surface of water and each sphere exactly half submerged, as shown in the figure. The volume of mass m is negligible. The distance l of mass from the ball of mass M is



(a) $\dfrac{d(V\rho - 3M)}{2(V\rho - 2M)}$ (b) $\dfrac{d(V\rho - 2M)}{2(V\rho - 3M)}$

(c) $\dfrac{d(V\rho + 2M)}{2(V\rho - 3M)}$ (d) $\dfrac{d(V\rho - 2M)}{2(V\rho + 3M)}$

32. A light semi-cylindrical gate of radius R is piovted at its mid-point O of the diameter as shown in the figure, supporting a liquid of density ρ. The force F required to prevent the rotation of the gate is equal to



(a) $2\pi R^3\rho g$ (b) $2\rho g R^3 l$

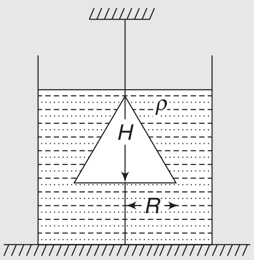
(c) $\dfrac{2R^2 l\rho g}{3}$ (d) none of these

33. A bucket contains water filled upto a height = 15 cm. The bucket is tied to a rope, which is passed over a frictionless light pulley and the other end of the rope is tied to a block which has half the mass of bucket + water. The system is released to move. The gauge pressure at the bottom of the bucket is

(a) 0.5 kPa (b) 1 kPa

(c) 5 kPa (d) None of these

34. A cone of radius R and height H, is hanging inside a liquid of density ρ by means of a string as shown in the figure. The force, due to the liquid acting on the slant surface of the cone is (Neglect atmospheric pressure)



(a) $\rho\pi g H R^2$ (b) $\pi\rho H R^2$

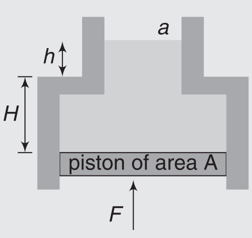
(c) $\dfrac{4}{3}\pi\rho g H R^2$ (d) $\dfrac{2}{3}\pi\rho g H R^2$

35. An open cubical tank was initially fully filled with water. When the tank was accelerated on a horizontal plane along one of its side, it was found that one-third of volume of water spilled out. The acceleration was

(a) $g/3$ (b) $2g/3$

(c) $3g/2$ (d) None

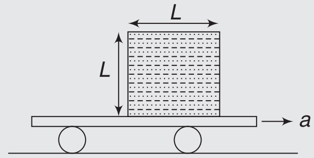
36. A liquid of density ρ is contained in an open tube of shape shown in the figure. The lower end of the tube is closed by a massless, smooth movable piston of area A and the upper end having area a is open to atmosphere. The force F needed to maintain the piston in equilibrium is



(a) $\rho g[ha-(H-h)A]$ (b) $\rho g(H+h)A$

(c) $\rho gHA + \rho gha$ (d) $\rho g(H-h)A$

37. A cubical vessel with edge L is placed on a cart, which is moving horizontally with an acceleration 'a' as shown in figure. The cube is completely filled with an ideal fluid having density ρ. It is sealed so that no air remains inside it. The pressure at the centre of the cubical vessel is



(a) $\dfrac{L}{2}\rho g$ (b) $\dfrac{L}{2}\rho(g+a)$

(c) $\dfrac{L}{2}\rho a$ (d) $\dfrac{L}{2}\rho(g-a)$

38. There is a small hole near the bottom of an open tank filled with a liquid. The speed of the water ejected does not depend on:

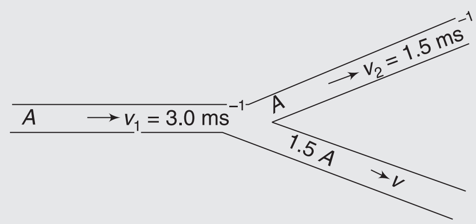
(a) area of the hole

(b) density of the liquid

(c) height of the liquid from the hole

(d) acceleration due to gravity

39. An incompressible liquid flows through a Y-shaped horizontal tube as shown in the figure. Then the velocity 'v' of the fluid in tube of cross-section 1.5 A is



(a) 3.0 ms^{-1} (b) 1.5 ms^{-1}

(c) 1.0 ms^{-1} (d) 2.25 ms^{-1}

40. Water enters through end A with a speed v_1 and leaves through end B with a speed v_2 of a cylindrical tube AB. The tube is always completely filled with water. In case I, the tube is horizontal, in case II, it is vertical with the end A upward and in case III, it is vertical with the end B upward. We have $v_1 = v_2$ for

(a) case I (b) case II

(c) case III (d) each case

41. A horizontal pipeline carries water in a steady flow. At a point along the pipe where the cross-sectional area is 10 cm^2, the water velocity is 1 ms^{-1} and the pressure is 2000 Pa. The pressure of water at another point where the cross-sectional area is 5 cm^2, is (density of water = 10^3 $kg\,m^{-3}$).

(a) 200 Pa (b) 300 Pa

(c) 400 Pa (d) 500 Pa

42. Water from a tap (at the end of a horizontal pipe) emerges vertically downwards with an initial speed of 1.0 ms^{-1}. The cross-sectional area of the tap is 10^{-4} m^2. Assume that the pressure is constant throughout the stream of water and the flow is steady. The cross-sectional area of the stream 0.15 m below the tap is:

(a) 5.0×10^{-4} m^2 (b) 1.0×10^{-5} m^2

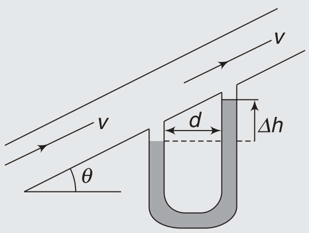
(c) 5.0×10^{-5} m^2 (d) 2.0×10^{-5} m^2

43. A large open tank has two holes in the wall. One is a square hole of side L at a depth y from the top and the other is a circular hole of radius R at a depth $4y$ from the top. When the tank is completely filled with water, the quantities of water flowing out per second from both holes are the same. Then, R is equal to:

(a) $\dfrac{L}{\sqrt{2\pi}}$ (b) $2\pi L$

(c) L (d) $\dfrac{L}{2\pi}$

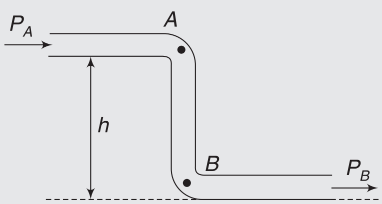
44. A pipeline is inclined at an angle θ to the horizontal. A liquid of density ρ flows through it. A mercury manometer is connected as shown in the figure. The difference in height Δh for the levels of manometer liquid is: (symbols have usual meaning) ($\rho_{Hg} >> \rho$)



(a) $\dfrac{\rho d \cot\theta}{\rho_{Hg}}$ (b) $\dfrac{\rho d \tan\theta}{\rho_{Hg}}$

(c) $\dfrac{\rho d \sin\theta}{\rho_{Hg}}$ (d) none of these

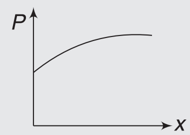
45. An ideal fluid flows through a tube of uniform cross-section. A and B are two points in the vertical section of the tube as shown. Liquid velocities are v_A & v_B and pressures are P_A & P_B at the two points. Which of the following is true?

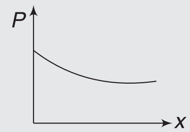


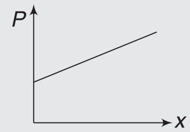
(a) $P_B > P_A$ (b) $P_B < P_A$

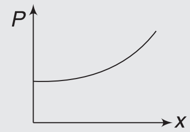
(c) $P_A = P_B$ (d) $v_A > v_B$

46. The cross-sectional area of a horizontal tube increases along its length linearly, as we move in the direction of flow. The variation of pressure, as we move along its length in the direction of flow (x-direction), is best depicted by which of the following graphs?

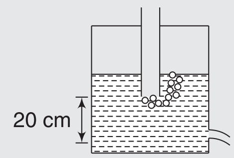
(a) 

(b) 

(c) 

(d) 

47. A vertical tube is attached to a closed vessel containing water. The vessel has rarefied air above the water surface. Lower end of the tube is held inside water in a position where the air bubbles start appearing at the tip of the tube. The velocity of water coming out from a small hole in the side wall of the container at a depth h = 20 cm from the tip of the tube is



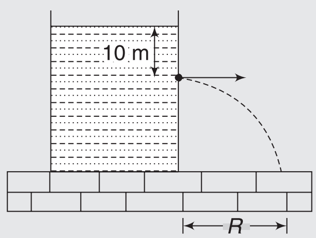
(a) $\sqrt{2}$ ms^{-1}

(b) 2 ms^{-1}

(c) depends on pressure of air inside vessel

(d) None of these

48. A large tank is filled with water (density = 10^3 kgm^{-3}). A small hole is made at a depth 10 m below water surface. The horizontal range of water issuing out of the hole is R. What extra pressure must be applied on the water surface so that the range becomes $2R$ (take 1 atm = 10^5 Pa and g = 10 ms^{-2})?



(a) 9 atm (b) 4 atm

(c) 5 atm (d) 3 atm

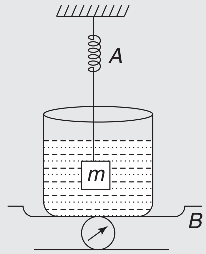
49. A cyclindrical vessel of cross-sectional area 1000 cm^2 is fitted with a frictionless piston of mass 10 kg, and filled with water completely. A small hole of cross-sectional area 10 mm^2 is opened at a point 50 cm deep from the lower surface of the piston. The velocity of efflux from the hole will be

(a) 10.51 ms^{-1} (b) 3.46 ms^{-1}

(c) 0.81 ms^{-1} (d) 0.21 ms^{-1}

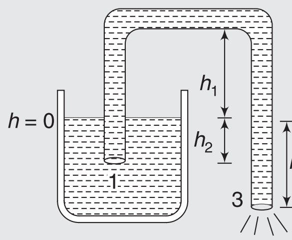
Worksheet 2

1. The spring balance A reads 2 kg with a block m suspended from it. A balance B reads 5 kg when a beaker with liquid is put on the pan of the balance. The two balances are now so arranged that the hanging mass is inside the liquid in the beaker, as shown in the figure. In this situation,



   (a) the balance A will read more than 2 kg.
   (b) the balance B will read more than 5 kg.
   (c) the balance A will read less than 2 kg and B will read more than 5 kg.
   (d) the balances A and B will read 2 kg and 5 kg respectively.

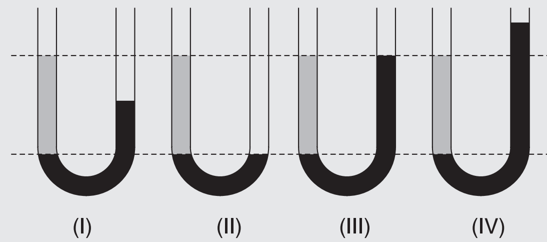
2. Figure shows a siphon. Choose the wrong statement. (P_0 = atmospheric pressure)



   (a) Siphon works when $h_3 > 0$
   (b) Pressure at point 2 is $P_2 = P_0 - \rho g h_3$
   (c) Pressure at point 3 is P_0
   (d) Siphon will always work, regardless of the value of h_1

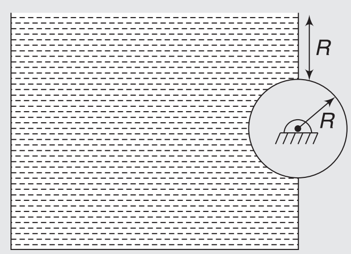
3. Water coming out of a horizontal tube at a speed v strikes normally on a vertical wall close to the mouth of the tube and falls down vertically after impact. When the speed of water is increased to $2v$,
   (a) the thrust exerted by the water on the wall will be doubled.
   (b) the thrust exerted by the water on the wall will be four times.
   (c) the energy lost per second by water striking the wall will also be four times.
   (d) the energy lost per second by water striking the wall will be increased eight times.

4. The figure below shows four situations in which a grey liquid and a black liquid (about which we known nothing) are in a U-tube.



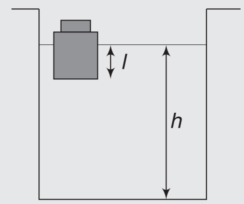
   (a) The liquids are certainly not in static equilibrium in case II only.
   (b) If the liquids are in equilibrium then the density of the grey liquid must be greater than the density of the black liquid in case IV only.
   (c) The liquids are certainly accelerated in case II.
   (d) The liquids are certainly accelerated in all cases, except III.

5. A cylinder of radius R is kept embedded along the wall of a dam as shown. Take density of water as ρ and length of the cylinder as L. Neglect atmospheric pressure



   (a) The vertical force exerted by water on the cylinder is $\dfrac{\rho \pi R^2 Lg}{2}$
   (b) The torque exerted by liquid on the cylinder about its axis is $\dfrac{\rho R^3 Lg}{2}$
   (c) The force exerted by liquid on the cylinder in horizontal direction is $4\,R^2 \rho g L$
   (d) The force exerted by liquid on the cylinder in horizontal direction is $2\,R^2 \rho g L$

6. A wooden block (having cross-sectional area A), with a coin (having volume V and density d) placed on its top floats in water, as shown in figure. Cross sectional area of the container is A_1 &



density of water is ρ. If the coin is lifted and then dropped into water.

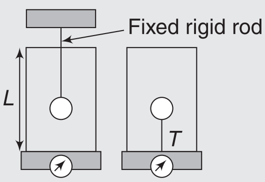
(a) change in the submerged length (l) of the block is $\Delta l = \dfrac{V\rho}{Ad}$

(b) change in the submerged length (l) of the block is $\Delta l = \dfrac{Vd}{A\rho}$

(c) change in the water level (h) in container is $\Delta h = \dfrac{V}{A_1}\left(1 - \dfrac{d}{\rho}\right)$.

(d) None of the above

7. A cylindrical container of length L is full to the brim with a liquid, which has density ρ. It is placed on a weigh-scale, which reads W. A light ball (which would float on the liquid if allowed to do so) of volume V and mass m is pushed gently down and held beneath the surface of the liquid with a rigid rod of negligible volume, as shown.



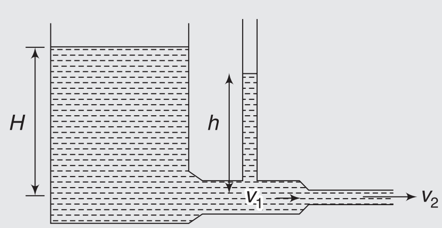
(a) the mass M of liquid, which overflowed while the ball was being pushed into the container is $2\rho V$

(b) the reading R_1 on the scale when the ball is fully immersed is W

(c) If instead of being pushed down by a rod, the ball is held in place by a fine string attached to the bottom of the container, the tension T in the string is $T = \rho Vg - mg$

(d) as described in part (c), the ball is held in place by a fine string attached to the bottom of the container. The reading R_2 on the scale is $W - \rho Vg + mg$

8. Water fills a reservoir, open to the atmosphere, to height H = 5 m above the horizontal exit pipe at the bottom of the reservoir. The first section of the pipe has radius r_1 = 2 cm, the unknown speed there is v_1 and this section of the pipe has a vertical tube conneted to it in which the water rises to an unknown height h above the the pipe.



The water leaves the second section of radius r_2 = 1 cm with speed v_2. Assume the flow is frictionless, irrotational, and steady. Also, assume the reservoir is so large compared to the pipe, that the water level in the reservoir is almost constant.

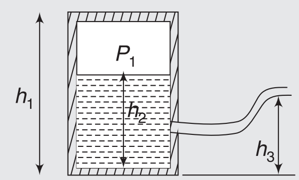
(a) the speed, v_2, of the water leaving the exit pipe is 10 ms^{-1}

(b) the speed, v_1, in the first section of the pipe is 2.5 ms^{-1}

(c) the height, h, of the water in the vertical tube is nearly 4.69 m

(d) the volume flow rate of the water flowing through the exit pipe is 3.1 ms^{-3}

9. A large tank (height h_1) of water has a hose connected to it, as shown in figure. The tank is sealed at the top and has air at atmospheric pressure between the water surface & the top. Initially, height of water in the tank is $h_2 > h_3$. Assume that the air pressure above the water decreases with expansion of air. Water begins to flow out of the hose.



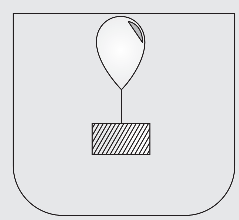
(a) flow of water will stop when height of water surface becomes equal to h_3

(b) flow of water will stop when height of water surface is more than h_3

(c) flow speed will decrease with time

(d) none of the above

10. A balloon filled with air is tied to a weight, as shown in figure. It barely floats in water while remaining completely submerged. The string is taut in the position shown. It is pushed down so that it gets submerged a short distance in water.



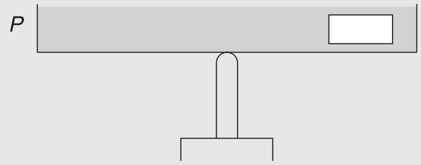
(a) If the string breaks, the balloon will move up.

(b) If the weight is pulled downward so as to take the system deep and released, the system will sink further.

(c) If the weight is pulled downward so as to take the system deep and released, the system will remain in the position it is left.

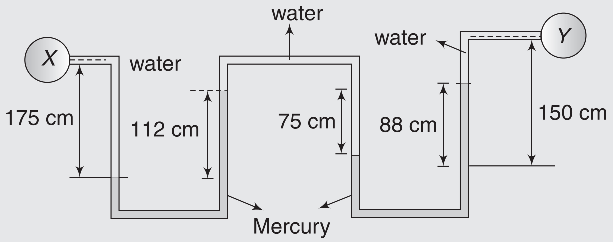
(d) None of the above

11. An open pan P filled with water (density = d_w) is placed on a vertical rod, maintaining equilibrium. A block of density d is gently placed on one side of the pan as shown. Water depth is more than height of the block.



(a) Equilibrium will be maintained only if $d < d_w$
(b) Equilibrium will be maintained in case $d = d_w$
(c) Equilibrium will be maintained for all relations between d and d_w
(d) Equilibrium will not be maintained in case $d < d_w$

12. Two bulbs X and Y contain gas at different pressures. A long tube having horizontal and vertical segments is connected to the bulbs as shown. The tube contains alternate segment of water and mercury. Water just could not enter the two bulbs. Specific gravity of mercury is 13.6

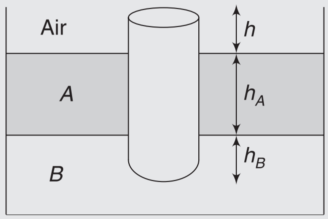


(a) Pressure in two horizontal segments of mercury must be same.
(b) Pressure difference between X and Y is nearly 248 kPa.
(c) Pressure difference between X and Y is 148 kPa.
(d) Equilibrium is not possible in the given arrangement.

13. A ship sailing from sea into a fresh water river further sinks X mm and on discharging, the cargo rises Y mm. On proceeding again into sea, the ship rises Z mm. The ship has uniform horiontal cross-sections.

(a) the specific gravity of sea–water is $\dfrac{Y}{Y - X + Z}$
(b) $X = Z$
(c) the specific gravity of sea–water is $\dfrac{Y}{Y + X + Z}$
(d) none of the above

14. A uniform solid cylinder of density 0.8 gcm^{-3} floats in equilibrium in a combination of two non–mixing liquids A and B with its axis vertical. The densities of the liquids A and B are 0.7 gcm^{-3} and 1.2 gcm^{-3}, respectively. The height of liquid A is h_A = 1.2 cm. The length of the part of the cylinder immersed in liquid B is h_B = 0.8 cm.

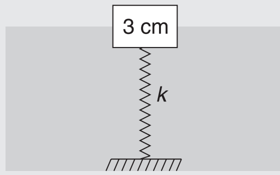


(a) the total force exerted by liquid A on the cylinder is zero
(b) the length of the part of the cylinder in air is 0.25 cm
(c) The cylinder is depressed in such a way that its top surface is just below the upper surface of liquid A and is then released. The acceleration of the cylinder immediately after it is released is $g/6$
(d) none of the above

15. A hemispherical bowl of mass m made of steel of density d is floating on the surface of water (density = ρ).
(a) volume of water displaced is m/ρ
(b) the bowl is pushed down so as to submerge it deep inside water and released. Its initial acceleration has magnitude $g\left(1 - \dfrac{\rho}{d}\right)$
(c) if the bowl is inverted, it will sink with an acceleration of $g\left(1 - \dfrac{\rho}{d}\right)$
(d) none of the above

Worksheet 3

1. A cubical block of wood of side length 3 cm floats in water. The lower surface of the cube just touches the free end of a vertical spring fixed at the bottom of the tank. Find maximum weight that can be put on the block without wetting its upper surface (it means that the block is just not completely submerged).



Density of wood $d = 0.8\,\text{gcm}^{-3}$, force constant of spring $k = 50\ \text{Nm}^{-1}$ and $g = 10\ \text{ms}^{-2}$.

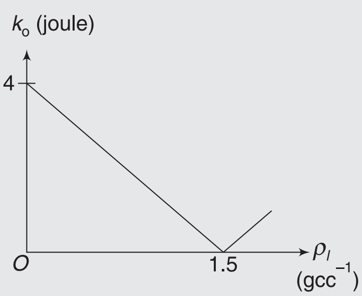
2. A piece of brass (an alloy of copper and zinc) weighs 12.9 g in air. When completely immersed in water, it weighs 11.3 g. Find the mass of copper in the allow? Specific gravities of copper and zinc are 8.9 and 7.1 respectively.

3. A cubical block of side length 5 cm is made of iron (density = 7.2 gcm^{-3}). It is floating in mercury (density = 13.6 gcm^{-3}).

(a) Find the height of block above the mercury level.

(b) Water is poured into the vessel (above mercury) so that it just covers the iron block. Find the height of water column.

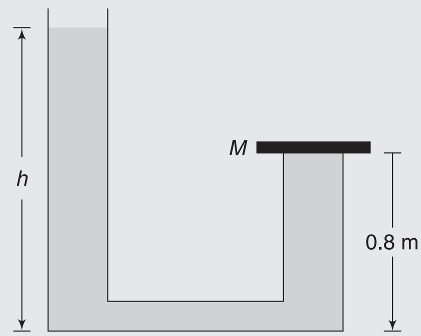
4. A small solid ball is released from rest, deep inside a liquid tank. Its kinetic energy (k_0) is measured after it travels through a distance of 8 cm in the liquid. Experiment is repeated with many non-viscous liquids. k_0 is plotted versus density (ρ_l) of liquids. Graph is shown in figure.



(a) Find mass of the ball.

(b) Find density of a liquid, which is heavier than water, in which k_0 will be equal to 4 J.[g = 10 ms^{-2}]

5. A J-shaped tube has a uniform cross-sectional area of $A = 0.01\,\text{m}^2$. Its shorter arm has length of 0.8 m and the other arm is quite tall. A steel plate of mass M = 10 kg is used to seal the opening of shorter arm. Now, water is poured into the longer arm. What should be height of water (h) in the longer arm so that the water begins to leak from the mouth of shorter arm? The area of top surface of seal is $A_0 = 0.015\,\text{m}^2$ and atmospheric pressure is $P_0 = 1 \times 10^5\,\text{Nm}^{-2}$.



6. A cubical box is constructed using iron sheets 1mm in thickness. Density of iron is = 8 gcm^{-3} and density of water is 1 gcm^{-3}.

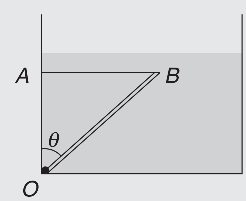
(a) Will the box sink in water if side length of the cube is large?

(b) What value of external side length of the box will you recommend if the box is supposed to float in water.

7. A concrete sphere has a cavity packed with foam. The specific gravities of the concrete and foam are 2.4 and 0.3 respectively. The sphere floats in water while remaining completely submerged. Find the ratio of mass of concrete to the mass of foam in the sphere.

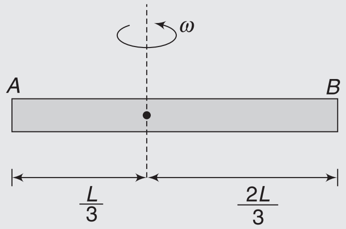
8. A 10 cm side cube weighing 5 N is immersed in a liquid of relative density 0.8 contained in a square tank of cross-section area 15 cm × 15 cm. The tank contained liquid to a height of 8 cm before immersion. Find the height of liquid surface in the tank after the cube is immersed. Is the cube touching the base of the tank? $g = 10\ \text{ms}^{-2}$.

9. A uniform rod OB is hinged at O inside water tank so that it can rotate freely in the vertical plane. Length of the rod is $L = 1.0$ m, its relative density is 2.0 and cross-sectional area is $A = 0.012\,\text{m}^2$.

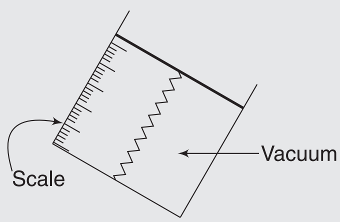


It is fully submerged in water. Its end B is tied to a horizontal sting AB, which can sustain a maximum tension of 45 N. Find the maximum angle θ that the rod can make with vertical without breaking the string.

10. A horizontal tube having length L is filled with a liquid of density ρ. It is rotating about a vertical axis as shown. Find pressure difference between point B and A.



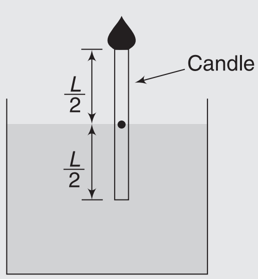
11. The device shown in the figure is a simple pressure gauge. Area of the light and smooth piston is 1.0 cm^2 and force constant of the spring is $k = 100\,Nm^{-1}$. The chamber inside which the spring lies has no air in it. The wall of the container is marked with a mm Scale. The zero of the scale is at the position of the piston when the device is kept in atmosphere.



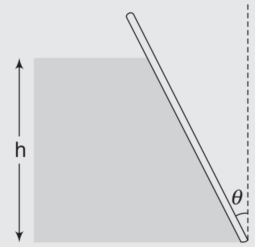
(a) Find the smallest gauge pressure that can be measured using this device.

(b) Find the displacement of the piston if the device is taken at a depth of 2m inside a swimming pool.

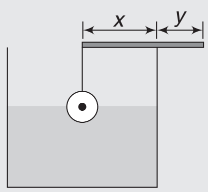
12. A candle of cross-sectional area $a = 1\,cm^2$ and length $L = 20\,cm$ is floating in equilibrium in a tank of water having cross-sectional area $A = 2\,m^2$. Half the candle is submerged in this position. The candle burns at a rate of 2 cmh^{-1}. The burnt part is completely converted into gas and goes to atmosphere. Calculate the rate of fall of top of the candle in unit of cm/hour.



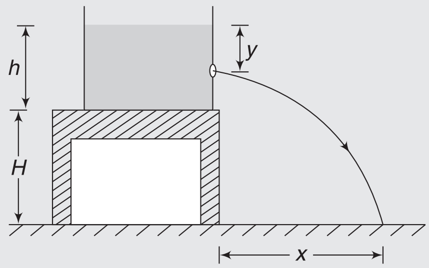
13. Find force exerted by water on the slant wall shown in figure. The wall makes an angle θ with vertical and its width perpendicular to the figure is b. Density of water is ρ. Neglect atmosphere.



14. A uniform aluminum ball of radius $r = 0.5\,cm$ is suspended on a weightless thread from the end of a uniform rod of mass $M = 4.4$ g. The rod is balanced at the edge of the container in a horizontal position. In the equilibrium position shown, half the ball is submerged. Find the ratio of length x and y. Neglect surface tension and take relative density of aluminum to be 3.0.

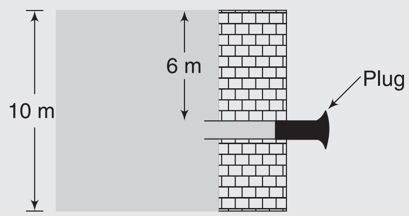


15. A water tank has water filled into it up to a height h. The tank is placed on the edge of a table that is H high. A small orifice is made at a depth y below the water surface in the side wall of the tank. For which value of y is the range x maximum, if



(a) $H = \frac{2}{3}h$ (b) $H = \frac{4}{3}h$

16. Water in a large dam is 10 m deep. There is a horizontal pipe of diameter 4.0 cm fitted in the dam wall at a depth of 6 m from the water surface. A plug in the pipe keeps the water from flowing out.

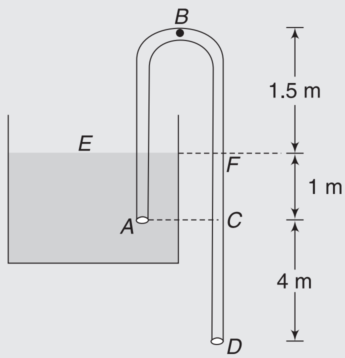


(a) Find friction force between the plug and the pipe wall.

(b) If the plug is removed, what volume of water will drain out of the pipe in 24 hours? [$g = 9.8\,ms^{-2}$]

17. A cylindrical tank of base area A has a small hole of area a at its bottom. At time $t = 0$, a tap starts to supply water into the tank. The cross-sectional area of the outlet of the tap is a_0 and water flows out from it at a speed v. Find the maximum level of water that one can have in this leaking tank.

18. A siphon tube is used to discharge a liquid of relative density 0.9 from a large reservoir as shown.



(a) Find speed of flow of liquid through the pipe

(b) Find pressure at highest point B.

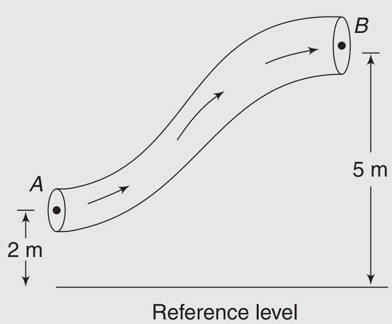
(c) Find pressure at A (just outside the tube mouth)

(d) Find pressure at A just inside the tube.

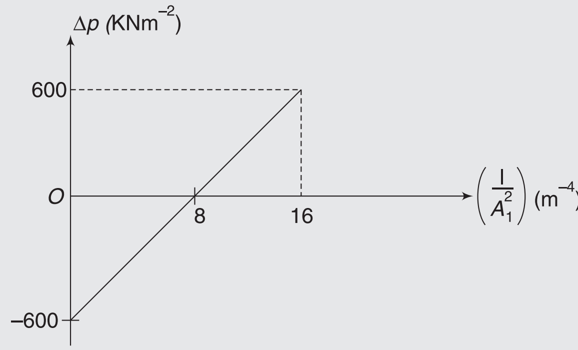
(e) Would the rate of flow be more, less or same if the liquid were water?

[Atmospheric pressure is $P_0 = 10^5$ Nm^{-2}]

19. Steady flow of water is happening through a pipe of variable cross-section. Area of cross-section of the pipe at A and B are $A_1 = 4 \times 10^{-3}$ m^2 and $A_2 = 8 \times 10^{-3}$ m^2 respectively. Height of pipe at A and B above the reference level is 2 m and 5 m. Speed of flow at A is $v_1 = 1$ ms^{-1}. Density of water is 10^3 kgm^{-3}. Find work done by surrounding liquid on a liquid element of unit volume as it moves from A to B. [$g = 10$ ms^{-2}]



20. Water flows through a horizontal pipe, which has two sections – 1 and 2. Cross-sectional area of section 2 is A_2 and is kept constant. The cross-sectional area (A_1) of section – 1 is varied during an experiment and the pressure difference between two segments ($\Delta P = P_2 - P_1$) is measured. The graph of ΔP verus $\left(\frac{1}{A_1^2}\right)$ is as shown. Volume flow rate is held constant during the experiment.

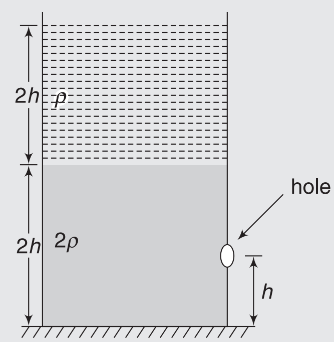


(a) Find A_2

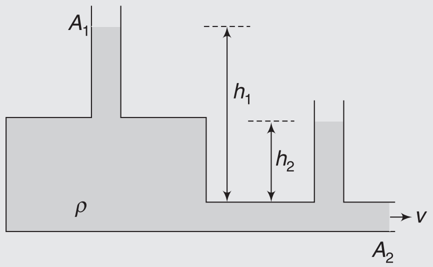
(b) Find the volume flow rate.

21. A rectangular vessel when full of water takes 10 minutes to be emptied through an orifice at its bottom. How much time will it take to be emptied if it is only half-filled?

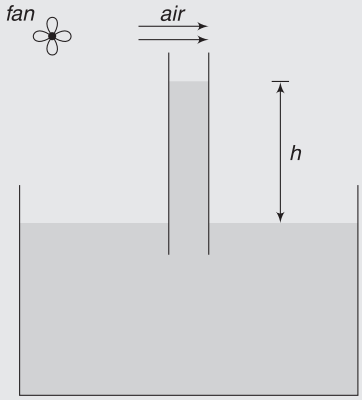
22. A cylindrical tank has cross-sectional area A. Mass of empty tank is M. It contains a liquid of density 2ρ filled upto height $2h$ and another immiscible liquid of density ρ upto a further height $2h$. A small hole of cross-section S is made in the wall of the cylinder at a height h above the horizontal table. Find the minimum coefficient of friction between the cylinder and the ground so that it does not slide immediately after the hole is opened and the liquid jet discharges horizontally.



23. A water tank has shape shown in figure. Speed of flow at the horizontal segment of cross-section A_2 is v. Height h_1 and h_2 are given. Find ratio of cross-section (A_2) of the horizontal segment to that of the vertical segment (A_1).



24. A tube is held vertically dipped in water in a beaker. A fan is put on, which causes air to blow over the tube. Density of air is 1.3 gcm^{-3}. After the fan is switched on, the water rises in the tube to a height of h = 1 cm. Find the speed of air near the mouth of the tube.



Answers Sheet

Your Turn

1. 6.8 kg	2. 7.3 gcm^{-3}	3. Both are same	4. 10 N
5. (a) 7500 N	6. 6250 N	7. a = 9.26 m	8. (a) 4N (b) 1N
9. 76 cm	10 Equator	11. 1 m	12. **
13. 0.67, 4,000 Nm^{-2}	14. 1.068×10^5 Pa	15. (a) A (b) B (c) 10.74 cm of Hg	
16. 55 N	17. 122 N	18. (a) 4×10^4 Pa (b) 8×10^3 Pa (c) -4.8×10^4 Pa	
19. 0	20. 3×10^5 Pa	21. $P_1 = P_0 + \rho g\left(H - \frac{L}{2}\right)$; $L = 2H$	
22. $\frac{1}{3} \times 10^4$ kgm^{-3}	23. $\frac{4}{3}\rho_1$	24. $\frac{5}{6}$	25. 80 kg
26. 30 g	27. 133 kg	28. Level of water falls	29. 4 g
30. No	31. 19.6 m	32. 0.12 m^3	33. 3.6 cm
34. Sphere will rotate.	35. m	36. $V_3 = 1.0\ ms^{-1}$; $V_P = 0.05\ ms^{-1}$	
40. 81,900 N	41. 10 ms^{-1}	42. (a) 50 cms^{-1} (b) 93.75 Pa	
43. $\sqrt{2gh_1}$	44. $\sqrt{2g(H - l\sin\theta)}$		

Worksheet 1

1. (d)	2. (c)	3. (c)	4. (b)	5. (c)	6. (c)	7. (a)	8. (b)	9. (c)
10. (a)	11. (a)	12. (a)	13. (a)	14. (b)	15. (c)	16. (b)	17. (c)	18. (c)
19. (c)	20. (d)	21. (d)	22. (d)	23. (b)	24. (c)	25. (a)	26. (d)	27. (a)
28. (b)	29. (a)	30. (b)	31. (b)	32. (d)	33. (b)	34. (d)	35. (b)	36. (b)
37. (b)	38. (b)	39. (c)	40. (d)	41. (d)	42. (c)	43. (a)	44. (b)	45. (a)
46. (a)	47. (b)	48. (d)	49. (b)					

Worksheet 2

1. (b, c)	2. (d)	3. (b, d)	4. (a, b, c)	5. (a, c)	6. (b, c)	7. (b, c, d)	8. (a, b, c)
9. (b, c)	10. (a, b)	11. (b, d)	12. (b)	13. (a)	14. (a, b, c)	15. (a, b, c)	

Worksheet 3

1. 0.35 N	2. 7.6 g	3. (a) 2.35 cm (b) 2.54 cm	
4. (a) 5 kg (b) 3.0 gcm^{-3}	5. 6.8 m	6. (a) No (b) Side length > 4.8 cm	7. 4
8. 97/9 cm = 10.8 cm	9. 37°	10. $\frac{1}{6}\rho\omega^2 L^2$	
11. (a) 1 kPa (b) 2 cm	12. 1 cmh^{-1}	13. $\frac{1}{2}\rho g bh^2 \sec\theta$	14. 0.46
15. (a) $\frac{5h}{6}$ (b) h	16. (a) 74 N (b) 1175 m^3		17. $\frac{a_0^2 v^2}{2ga^2}$
18. (a) 10 ms^{-1} (b) 41.5 KNm^{-2} (c) $0.9 \times 10^5\ Nm^{-2}$ (d) $4.5 \times 10^4\ Nm^{-2}$ (e) same			19. 29625 J
20. (a) 0.35 m^2 (b) 12.25 m^3s^{-1}		21. 7.07 min	22. $\frac{16\rho Sh}{M + 6hA\rho}$
23. $\sqrt{1 - \frac{2g(2h_1 - h_2)}{v^2}}$		24. 12.4 ms^{-1}	

CHAPTER 9

Surface Tension and Viscosity

"You get surface tension with water. That helps the sand grains stick together."

–Mark Mason

1. INTRODUCTION

A liquid surfaces has properties very different from its bulk. Surface of a liquid is like a stretched membrane–it is under tension. Surface tension of water–the most abundant liquid–is higher than most other liquids. Many small insects live in a world dominated by surface tension of water. Inkjet printing, design of automobile windshields, many medicals devices, rise of water from roots to leaves in a tree, etc., are just a few examples where surface tension plays a vital role.

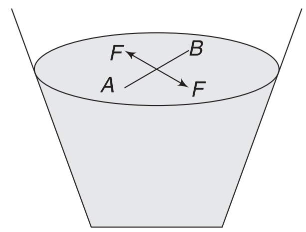
We studied Bernoulli's equation in the last chapter. It is applicable for steady flow of ideal fluids. But all real life liquids are viscous. Layers of liquids exert tangential force on one other. Viscosity is basically a measure of stickiness of a liquid. It plays an important role in deciding the nature of flow and design of mechanical systems using liquid lubricants–such as a car engine and gearbox.

2. TENSION IN A LIQUID SURFACE

Surface of liquid is like a stretched sheet and can bear load. Light insects, like mosquitores, can walk on the surface of water. A steel (relative density $\simeq 8$) needle can remain afloat on the surface of water if placed properly.

Think of a large bedsheet held at many points along its periphery by many of your friends. They all pull the sheet towards themselves. The sheet is under tension. You can go and keep a heavy ball on the sheet. A depression is formed and tension force acting on the ball from all sides has a vertical resultant, which balances the weight of the ball. A liquid surface is somewhat like this stretched sheet. If you imagine a line on the sheet dividing it into two parts, each part pulls the other one towards itself.

Consider a line *AB* of length *L* on the surface of a liquid. The two parts of liquid surface (one on right of the line and the other on left of it) pull one another with a force that is perpendicular to line *AB* and tangential to the surface. Let *F* be the magnitude of forces exerted by two sides on one another perpendicular to line *AB*. Surface tension *S* of the liquid is defined as



Surface molecules on two sides of line *AB*, pull one another. *F* is force experienced by molecules on line *AB*, due to molecules on one side of the line.

$$S = \frac{F}{L} \qquad ...(1)$$

SI unit of surface tension is Nm^{-1}.

Surface tension of few common liquids are:

Water: $0.075\ Nm^{-1}$; Soap solution: $0.03\ Nm^{-1}$;
Glycerine: $0.06\ Nm^{-1}$; Mercury: $0.465\ Nm^{-1}$.

Surface tension of water is larger than most of the common liquids but it is less than surface tension of metallic liquids like mercury.

When a solid boundary comes into contact with a liquid surface, the liquid surface pulls it tangentially-just like the stretched bedsheet pulls all your friends towards itself. The surface tension force will be obtained by multiplying length of the boundary (*L*) in contact with the surface tension (*S*).

When a liquid surface is curved, a line draw on it will be curved. Surface tension force is tangential everywhere. Consider a spherical water drop. Imagine its equator. The equator divides the entire spherical surface into two halves.

Each half exerts force on another half having magnitude.

$F = S \cdot 2\pi R$, where R is radius of drop. Figure shows surface tension force by lower half on the upper half. Force is tangential to the surface. In the given figure, it is downward. Force by upper half on the lower half is equal and opposite.

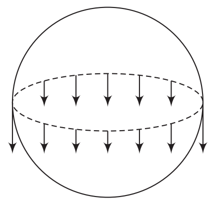


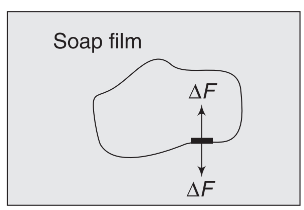
Figure shows surface tention force by lower hemispherical part on the upper part.

2.1 Demonstration of Surface Tension

A liquid surface has a property of surface tension. This fact can be demonstrated using many simple experiments. Here, we take up a few of those.

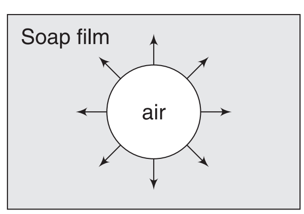
(i) Loop of thread on soap film

Take a rectangular frame of wire and dip it is soap solution. When you take out the frame from the solution, a film of soap solution bounded by the wire frame is formed. Now take a loop of light cotton thread, wet it and gently place it on the film. The loop will stay in the shape it has been placed. Now prick a hole in the film inside the loop, with the help of a needle. If done carefully, the film inside the thread loop will burst but the film outside the loop remains intact. After the inner film is bursted, the thread loop takes a circular shape.



When Soap solution is everywhere, a segment of thread experiences zero net force due to pull from both sides.

Initially, there were liquid surfaces on both side of the thread. A small segment of the thread was pulled by the force of surface tension from inside as well as outside. The two forces (shown as ΔF in figure) have equal magnitude and opposite direction–perpendicular to the segments of the thread. All segments of the thread experience zero force and stay in their original position.



When there is no soap solution inside the thread loop, every segment of the thread experiences an outward force only.

When the inner soap film is gone, every segment of the thread is pulled from outside only. Due to uniform pulling by the liquid surface from all sides, the thread loop acquires circular shape.

It is important to note that a film of liquid can survive only if it has a solid boundary to hold. The wire frame and the thread from the boundary for the liquid surface after pricking. If there were no thread and you prick the film, the entire film will collapse.

Example 1 *Be careful, there are two surfaces*

In the example cited above, after the soap film inside the thread loop is pricked, find the surface tension force acting on a small segment of thread having length Δl. Surface tension of soap solution is S.

Solution

Concepts

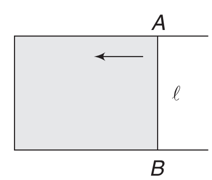
There are two surfaces in the soap film.

Surface tension force on segment of length Δl is $\Delta l S$. Since, there are two surfaces of the liquid in contact with the thread, the force will be

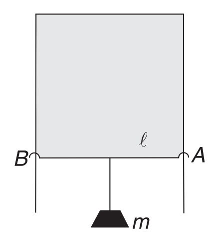
$$\Delta F = 2\Delta l S$$

(ii) Sliding wire on U-shaped frame

A U-shaped wire frame has a light wire AB mounted on it so that it can slide smoothly. If this frame is dipped in a soap solution and taken out, a film of soap solution is fomed in the rectangular region. If the frame is kept horizontal, the sliding wire moves towards the parallel arm of the frame. The force of surface tension applied by the liquid on the wire causes it to move.



Example 2 The U-frame and sliding wire system in the above discussion is held vertical. Now, a mass m is hung from the wire AB to prevent it from sliding. There is no friction, Find m if length of AB is 4 cm and surface tension of soap solution is $S = 0.03\ \text{Nm}^{-1}$. Wire AB has no mass. [$g = 10\ \text{ms}^{-2}$]



Solution

Concepts

Surface tension force on AB balances the weight

Surface tension force on the wire AB is

$$F = 2lS \text{ [There are 2 surfaces]}$$

$$= 2 \times 0.04 \times 0.03 = 2.4 \times 10^{-3}\ \text{N}$$

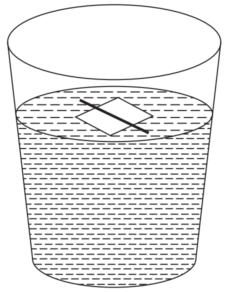
For equilibrium: $mg = 2.4 \times 10^{-3}$

$$\Rightarrow \quad m = 2.4 \times 10^{-4}\ \text{kg}$$

$$= 0.24\ \text{g}$$

(iii) A needle on surface of water

Take a glass of water and keep a small piece of tissue paper on its surface. Now place a steel needle on it. Wait for some time. The tissue paper soaks water and sinks. However, the needle remains afloat. The needle is not floating in usual sense. It is not buoyancy that prevents if from sinking. Density of steel is almost eight times that of water. It is the force of surface tension that keeps it in quilibrium.



The stretched surface of water gets depressed due to load of needle. The surface tension force (tangential to the surface is directed as shown. The force of surface tension acts on two sides. If l is length of the cylindrical needle, then

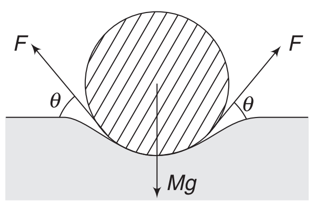
$$F = lS$$

For equilibrium,

$$2F\sin\theta = Mg$$

$$2lS\sin\theta = Mg$$

If the needle is made heavier without changing its length, then θ increases to balance the weight. It means the depression becomes deeper. In extreme case, θ can become 90°.



Cross-section of the needle.

Example 3 *Maxmium mass of needle that can stay on surface of water.*

In the above discussion, length of the needle is l. What is the maximum mass of needle that can stay on the surface of water? Take $l = 4$ cm, surface tension of water $S = 0.075$ Nm^{-1} and $g = 9.8$ ms^{-2}.

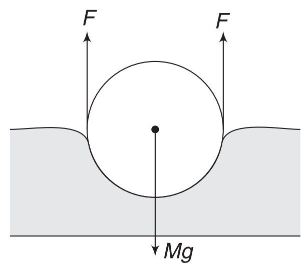
Solution

Concepts

Angle $\theta = 90°$ in extreme case.

$$2F = Mg$$

$$2lS = Mg$$



$$\Rightarrow \quad M = \frac{2lS}{g} = \frac{2 \times 0.04 \times 0.075}{9.8}$$

$$= 6.1 \times 10^{-4} \text{ kg} = 0.61 \text{ g}$$

YOUR TURN

Q.1 Water is kept in a cylindrical glass of diameter 10 cm. Consider a diameter on the circular surface of the water. Find the surface tension force by which the surface on one side of the diameter pulls the surface on the other side. Surface tension of water is $S = 0.075$ Nm^{-1}.

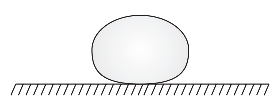
Q.2 A circular disc of radius 5 cm is placed on the surface of a liquid. It is slowly pulled upward, keeping the disc surface horizontal. In addition to the ring's weight, an upward force of 3×10^{-2} N is required to lift the disc to the point where it just breaks free of the surface. What is the surface tension of the liquid?

3. TENDENCY OF A LIQUID SURFACE TO CONTRACT

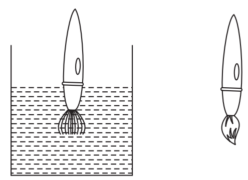
A stretched rubber sheet has a tendency to contract. Similarly, stretched surface of a liquid has a tendency to contract. A liquid always tries to keep its surface area to minimum possible. Following example demonstrates this property of a liquid surface.

(i) Small waterdrops are spherical. For a given volume, a sphere has the least surface area.

Large drops are not perfectly spherical due to effect of gravity. In a gravity-free space, all water drops (however large) will be spherical.


Waterdrop on a leaf. Large waterdrops are not spherical due to gravity.

(ii) When a printing brush (or shaving brush) is dipped in water, its hairs spread out and move freely. When it is taken out of the water, the hairs stick together. Sticking of hairs minimises the surface of water. The water particles come together to reduce their overall surface area.



Brush inside water.   Brush taken out of water.

(iii) Demonstration titled as 'loop of thread on soap film' in the last article also illustrated the fact the liquid has the tendency to minimise its surface area.

After the inner soap film is burst, the thread takes a circular shape. For a given perimeter of a plane figure, a circle has maximum area. Therefore, circular shape of the thread means that the part where there is no film (and has air) has maximum area. Thus, the area covered by the soap film is minimum possible.

(iv) 'Sliding wire on *U*-shaped film' also demonstrates that liquid will always try to reduce its surface area. The sliding wire *AB* moves to reduce the area of the film.

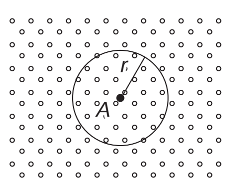
Your Turn

Q.3 An ice cube is kept in a gravity-free hall. What will be its shape when it melts?

4. MICROSCOPIC DESCRIPTION OF SURFACE TENSION

A liquid molecule that is very close to other molecule will repel it. Molecules, which are slightly away from a given molecule, will attract the molecule. And molecules which are far away (usually 10 to 20 molecular diameter away) will have no interaction with the molecule.

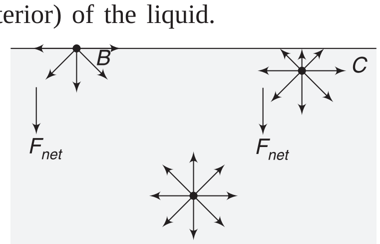
A molecule *A* interacts with other molecules, which lie in its sphere of influence. Its sphere of influences has radius of the order of 10 to 20 molecular diameter.



Molecules outside the sphere of influence do not interact with molecule *A*.

A molecule inside the bulk of a liquid (molecule *A*) feels uniform attractive forces from all sides due to liquid molecules present around it. Net force on it is zero.

A molecule on the top layer of liquid surface (like molecule *B*) feels uniform attraction due to molecules below it but above it there are air molecules or molecules of liquid vapour, which have a small density and exert almost no force on *B*. Thus, *B* experiences a net force directed into the bulk (interior) of the liquid.



Molecules like *B* and *C* experience force directed into the bulk of the liquid

A molecule like *C*, which is a few layers below the top layer, also experiences a net force towards the bulk. There are fewer molecules above *C* to attract it.

Any molecule in the top layer of 10 to 20 molecular diameter thickness will experience a net force directed into the bulk of the liquid. This zone is known as the surface of the liquid. If we go 20 molecular layers deep, a molecule finds equal force from all sides. Net force on it is zero.

Less and less number of liquid molecules will be there on the surface. They are being pulled into the bulk. For this reason, a liquid has the least surface area and a tension develops in its surface.

5. SURFACE ENERGY

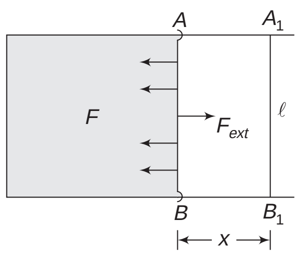
Force between a liquid molecule and other surrounding molecules is attractive (unless the molecules get extremely close). Attractive force means negative potential energy. The interior molecules have as many neighbours as they can have and thereby, have a large negative potential energy. A molecule on the surface has less number of neighbours. [A molecule like *B*, in last diagram, has no neighbor above it]. Potential energy of such molecules is higher. [A negative number but lesser magnitude] Molecules on a liquid surface have higher energy compared to molecules in the bulk.

Another way to understand this is that when a molecule moves up from inside of the liquid to its surface, it has moved up against an inward force while travelling through layers in the surface zone. It needs energy (from somewhere) to put extra molecules in the surface of a liquid (and thereby increase the surface area).

You need to work to lift a book against gravity. Similarly, molecules need energy (from somewhere) to rise to the surface against the inward pull. Molecules on the surface must have this extra energy. The extra energy that a surface layer has is called the *surface energy*.

5.1 Surface Tension as Surface Energy

Consider a *U*-shaped wire frame with a sliding arm *AB*. The frame is dipped in a soap solution and taken out. A film of soap solution is formed in the rectangular frame.



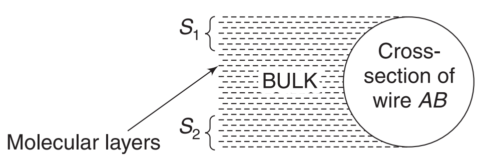
The soap film so formed appear to be thin. But on molecular scale, it may be lakhs of molecular layer thick. Nearly 20 molecular layers on both sides form two surfaces and the remaining layers of molecules form the bulk of the soap solution.

Both the surfaces pull the sliding wire *AB* with surface tension force. This force is

$$F = 2lS$$

where l = length of wire *AB*

S = Surface tension of liquid



S_1 and S_2 are the two surfaces each having a thickness of nearly 20 molecular layers.

To keep the wire in equilibrium, an external agent must apply equal and opposite force.

Now assume that you wish to stretch the film, thereby increasing the liquid's surface area. It implies that when you slide wire *AB* to the right, more and more liquid molecules will move from the bulk to the surface (the thickness of film reduces). The molecules moving from the bulk to the surface gain energy. They gain energy as you perform work in pulling the wire *AB*.

If you slide the wire *AB* slowly by a distance *x*, work done by you is

$$W = F_{ext} \cdot x = (2lS)x$$

Entire work done is used by liquid molecules in moving themselves from the bulk to the surface. We can think that the surface energy possessed by molecules on the newly created surface is equal to *W*.

New surface area created is $A = 2lx$

Thus, excess energy possessed by molecules on unit area of liquid surface is

$$\frac{W}{A} = \frac{2lSx}{2lx} = S \qquad ...(2)$$

Therefore, *surface tension of a liquid is equal to the surface energy per unit surface area.*

SI unit of surface tension is Nm^{-1} which is same as Jm^{-2}.

Molecules on $1\,m^2$ area of water surface have 0.075 J extra energy compared to the same number of molecules in the bulk.

Example 4 *Coalescing drops*

A number of little droplets of water, each of radius *r*, coalesce to form a single drop of radius *R*. Calculate the decrease in surface energy. Surface tension of water is *S*.

Solution

Concepts

(i) Using conservation of volume, we can find the number of smaller drops.

(ii) Due to decrease in surface area, the surface energy decreases.

Decrease in surface energy is

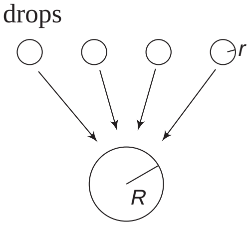
$$\Delta U = \Delta A S$$

where ΔA is decrease in surface area of liquid.

Let n = number of smaller drops

$$n \cdot \frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3$$

$$\Rightarrow \quad n = \left(\frac{R}{r}\right)^3$$



Smaller drops joining to form bigger drops decreases the surface area

Decrease in surface area of liquid

$$\Delta A = \text{surfaces area of } n \text{ smaller drops}) - (\text{surface area of bigger drop})$$

$$= n \cdot 4\pi r^2 - 4\pi R^2 = 4\pi\left[\frac{R^3}{r^3}\cdot r^2 - R^2\right]$$

$$= 4\pi R^3\left[\frac{1}{r} - \frac{1}{R}\right]$$

$\therefore$ Decrease in surface energy is

$$\Delta U = \Delta A S = 4\pi R^3\left[\frac{1}{r} - \frac{1}{R}\right]\cdot S$$

The lost surface energy goes to increase the internal kinetic energy of the water molecules. In simple language, the bigger drop will get heated. Its temperature will be higher than that of smaller drops.

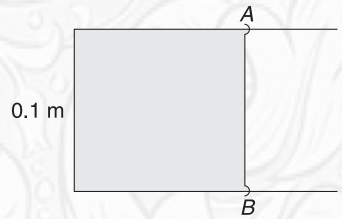
YOUR TURN

Q.4 What is the surface energy of a mercury drop of radius $r = 1$ cm? Surface tension of mercury is 0.47 Nm^{-1}.

Q.5 A waterdrop of radius 1.0 cm is sprayed into 1000 equal-sized droplets. Calculate the gain in surface energy. Surface tension of water is $S = 0.075$ Nm^{-1}.

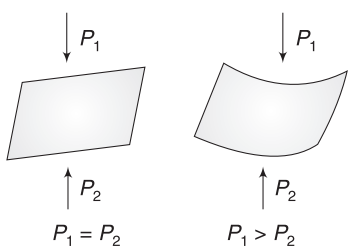
Q.6 When water is sprayed it gets cooler. Why?

Q.7 A film of soap solution is formed in a *U*-shaped frame with sliding wire *AB*. Length of *AB* is $l = 0.1$ m. Find the amount of work needed to move the wire *AB* towards right by 1 mm. Surface tension = 0.03 Nm^{-1}.



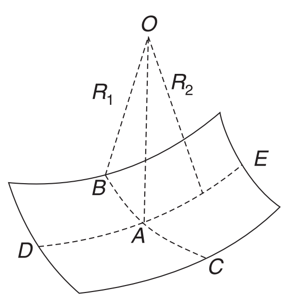
6. PRESSURE DIFFERENCE ON TWO SIDES OF A CURVED LIQUID SURFACE

Any stretched surface will remain flat if pressure is same on both sides of it. If there is a pressure difference on two sides. It will get curved. Obviously, pressure on concave side will be higher.



In a waterdrop, inside pressure must be higher than outside atmospheric pressure. Pressuer inside a soap bubble is higher than outside pressure. A water jet from a faucet is nearly cylindrical in shape. Inside the curved cylindrical surface, pressure is higher than the outside pressure.

In general, a curved surface may have different curvatures in different directions. Consider a curved surface under tension. A small patch of the surface around point *A* is as shown. If we draw arcs along the surface passing through *A*, we may get arcs of different radii in different direction. *BC* is one such arc along which radius of curvature is minimum (equal to R_1). *DE* is another arc that is perpendicular to *BC*. Radius of curvature of arc *DE* is R_2. The two radii of curvature – R_1 and R_2 – are known as principal radii of curvature of the curved surface at point *A*.



It can be proved that the pressure on concave side of such a surface is greater than the pressure on convex side by

$$\Delta P = S\left(\frac{1}{R_1} + \frac{1}{R_2}\right) \qquad ...(3)$$

Where S = surface tension

For a spherical surface, $R_1 = R_2$ (= R, say). Hence, pressure is higher on concave side of a spherical surface by an amount

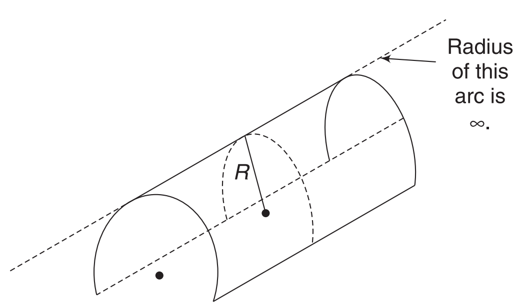
$$\Delta P = \frac{2S}{R} \qquad ...(4)$$

For a cylindrical surface,

$R_1 = R$ = radius of cylinder

$R_2 = \infty$

∴ Pressure inside a cylindrical surface is higher than outside pressure by



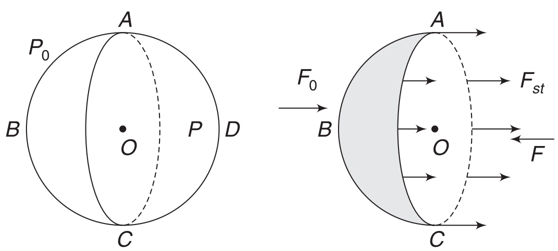
$$\Delta P = S\left(\frac{1}{R} + \frac{1}{\infty}\right)$$

$$\Rightarrow \qquad \Delta P = \frac{S}{R} \qquad ...(5)$$

Though we did not derive equation (3), we will derive equation (4) in a simpler way, considering a spherical liquid drop.

6.1 Excess Pressure inside a Spherical Liquid Drop

Consider a liquid drop. It will have spherical shape in absence of gravity. Even in presence of gravity, a small drop will be nearly spherical.



Let P be the pressure inside the drop. In absence of gravity, pressure will be same throughout (no change with depth) and in case of small drops, effects of gravity will be too small to be noticeable and we can again assume the same pressure everywhere inside the drop.

Let the outside atmospheric pressure be P_0 and surface tension of the liquid be S.

We will consider the equilibrium of half of the drop ABC. There are three forces on the hemispherical part ABC:

(i) Force due to atmospheric pressure

As we have seen in many problems, in the chapter of fluid mechanics, the force is

$F_0 = P_0$ (Projection of the area of curved hemispherical surface ABC on the flat surface represented by AOC)

$= P_0 \pi R^2$, where R is the radius of the drop

This force is towards right in our diagram.

(ii) Force due to liquid pressure

$$F = P \pi R^2$$

This force arises due to push of liquid in hemisphere ADC applied on flat circular surface AOC. This force is towards left in our diagram.

(iii) Force due to surface tension

The hemispherical surface ADC pulls the hemispherical surface ABC along the circular periphery. Force is

$$F_{st} = S \cdot 2\pi R \text{ towards right.}$$

For equilibrium of hemisphere ABC, we must have

$$F = F_0 + F_{st}$$

$$\Rightarrow \qquad P \cdot \pi R^2 = P_0 \cdot \pi R^2 + S \cdot 2\pi R$$

$$\Rightarrow \qquad P = P_0 + \frac{2S}{R}$$

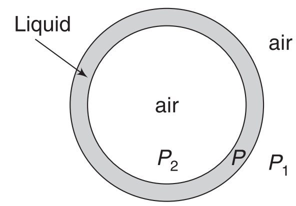
Excess pressure inside the drop is

$$\Delta P = P - P_0 = \frac{2S}{R} \qquad ...(6)$$

This is same as the result stated in equation (4).

6.2 Excess Pressure Inside a Soap Bubble

A soap bubble has apparently a very thin wall, but it may have lakhs of molecular layers in it. Therefore, it has two surfaces and a bulk of liquid between them. If the size of a bubble increases, its wall will get thinner but still, it is thick enough to have two surface layers.



A soap bubble has two surfaces. The diagram has exaggerated the thickness of wall. Both surfaces have almost equal radii.

Let pressure inside liquid wall be P and the outside atmospheric pressure be P_1. The outer surface is a spherical liquid surface of radius R. Pressure on concave side (P) is higher than the pressure on convex side (P_1).

$$\therefore \qquad P - P_1 = \frac{2S}{R} \qquad ...(i)$$

The inner surface is also concave and therefore $P_2 > P$.

$$P_2 - P = \frac{2S}{R} \qquad ...(ii)$$

Note that there is not much of difference in radii of the inner and outer surfaces. Both have same value R.

Adding (i) and (ii) gives

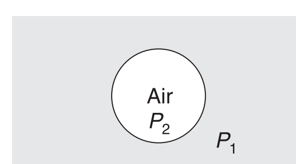
$$P_2 - P_1 = \frac{4S}{R}$$

$$\Delta P = \frac{4S}{R} \qquad ...(7)$$

This is excess pressure inside a soap bubble.

6.3 Air Bubble

Consider an air bubble inside a water tank. There is only one surface in this case. There is air on the concave side and liquid on the convex side.



An air bubble has only one surface

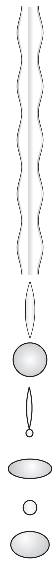
Pressure inside the air bubble is higher than pressure outside it by

$$\Delta P = P_2 - P_1 = \frac{2S}{R} \qquad ...(8)$$

Where R is radius of bubble and S is surface tension of the liquid.

6.4 Water Jet from a Faucet

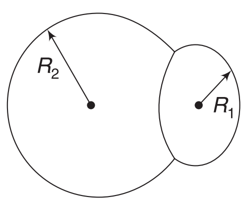
A water jet from a faucet is nearly cylindrical in shape when it begins to fall. Pressure inside the jet is $\Delta P = \frac{S}{R}$ higher than outside pressure, where R is radius of the cylinder. As R gets smaller, the pressure inside the water jet gets higher. There are always some irregularities, which cause R to decrease here and there. The higher pressure in the thinner region forces liquid away, into adjacent thicker regions. This amplifies the irregularities–making thinner region even thinner and the thick region thicker. Such a water jet is unstable and pinches itself into drops.



A cylindrical jet of water from a faucet is unstable. It breaks into pieces.

Example 5 *Double bubble*

Two soap bubbles of radii R_1 and R_2 stick together, as shown. Find the radius of curvature of the common wall.



Solution

Concepts

(i) The common wall has two surfaces.

(ii) Air inside smaller bubble is at higher pressure than that in larger bubble. Therefore, common wall is concave towards the smaller bubble.

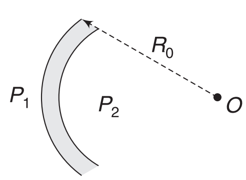
Pressure inside bubble of radius R_1 is

$$P_1 = P_0 + \frac{4S}{R_1}$$

Where P_0 is atmospheric pressure. Pressure inside bubble of radius R_2 is

$$P_2 = P_0 + \frac{4S}{R_2}$$

If $R_2 < R_1$ then, $P_2 > P_1$.



Common wall has two surfaces.

$$P_2 - P_1 = \frac{4S}{R}$$

The common wall is concave towards bubble of radius R_2. If the radius of curvature of the common wall is R_0 then

$$P_2 - P_1 = \frac{4S}{R_0}$$

[Pressure increases by $\frac{2S}{R_0}$ every time your cross to a concave side]

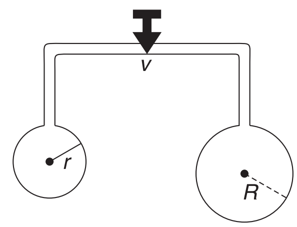
$$\therefore \quad \left(P_0 + \frac{4S}{R_2}\right) - \left(P_0 + \frac{4S}{R_1}\right) = \frac{4S}{R_0}$$

$$\Rightarrow \quad \frac{1}{R_2} - \frac{1}{R_1} = \frac{1}{R_0}$$

$$\Rightarrow \quad R_0 = \frac{R_1R_2}{R_1 - R_2}$$

Example 6 *Connected Bubbles*

Two soap bubbles are blown at the ends of a tube. The bubbles have different radii. The valve (V) is opened to connect the bubbles.



$R > r$

(a) Which bubble will grow in size after the valve is opened?

(b) What will be final state of the two bubbles?

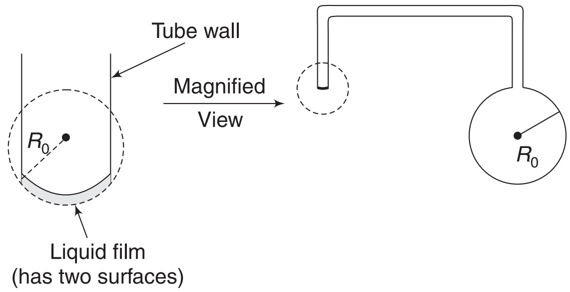
Solution

Concepts

The smaller bubble has a higher pressure inside it.

(a) As seen in the last example, the pressure inside the smaller bubble is higher. Therefore, after opening the valve, air will flow from the smaller bubble to the larger one. The larger bubble will grow in size.

(b) As the smaller bubble decreases in size, pressure inside it becomes even higher and it continues to become even smaller. Ultimately, the liquid in the smaller bubble will form a film at the end of the tube as shown in figure [something like a film one started with to blow a bubble]. The radius of curvature of the film will be equal to the radius (R_0) of the larger bubble at the other end of the tube.



This will ensure that the air inside the system is at same pressure everywhere $\left(= P_0 + \dfrac{4S}{R_0}\right)$.

Note: Our lungs contain hundreds of millions of mucus – lined sacs called *alveoli*, which are about 0.1 mm in diameter. We can exhale without muscle action by allowing surface tension of the mucus to contract the sacs (just like the surface tension caused the smaller bubble to contract). Medical patients whose breathing is aided by a pressure respirator having air blown into the lungs, but are generally allowed to exhale on their own.

YOUR TURN

Q.8 Find pressure inside an air bubble of radius 0.1 mm that is located at a depth of 10 m inside water. Surface tension of water $S = 0.072 \text{ Nm}^{-1}$, density of water $\rho = 10^3 \text{ kgm}^{-3}$, atmospheric pressure $P_0 = 1.013 \times 10^5 \text{ Nm}^{-2}$ and $g = 10.0 \text{ ms}^{-2}$.

Q.9 Soap bubble *A* has twice the mass of bubble *B*. Both the bubbles have same radii of 3 cm. Which bubble has more pressure inside it?

Q.10 A water jet from a tap has radius $r = 2$ mm at position *A* and radius $R = 4$ mm at another position *B*. What is the pressure difference inside the jet at *A* and *B*? Surface tension of water = 0.07 Nm^{-1}.

Q.11 The pressure difference inside a soap bubble and outside atmospheric pressure is P_0. Surface tension of soap solution is *S*. Find the surface energy of liquid in the bubble.

In Short:

(i) Molecules on the surface of a liquid are surrounded by less number of molecules compared to a molecule deep inside the liquid. Due to this, they experience a net inward force towards the bulk of the liquid. This causes the surface to acquire a tension and a tendency to contract.

(ii) Since molecules on the surface interact with less number of molecules, they have higher energy. [Each interaction adds negative energy].

(iii) If we draw an imaginary line on a liquid surface it divides the surface into two parts. Each part applies force on the other. This force, acting on unit length of the line, is known as surface tension of the liquid.

$$S = \frac{F}{L} \ (\text{Nm}^{-1})$$

(iv) Extra energy possessed by molecules on unit area of liquid surface, is equal to surface tension in magnitude.

(v) Pressure is higher on concave side of a curved liquid surface. For a spherical surface of radius *R*, the difference in pressure on the two sides is $\Delta P = \dfrac{2S}{R}$ and for a cylindrical liquid surface of radius R, the difference in pressure on the two sides is $\Delta P = \dfrac{S}{R}$.

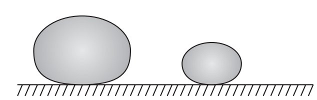
(vi) Wall of a soap bubble has two surfaces. Excess pressure inside a soap bubble is

$$\Delta P = \frac{4S}{R}$$

7. COHESIVE, ADHESIVE FORCES AND CONTACT ANGLE

Attractive force between molecules of a same substance are called *cohesive forces*. Attractive forces between molecules of different substances are known as *adhesive forces*.

When mercury drops fall on a clean glass plate they do not spread. They remain intact as cohesive force in mercury is large compared to adhesive force between mercury and glass. Mercury remains in shape of nearly spherical beads, which stand on glass surface, maintaining minimal contact with the surface. You can go and pick up a drop and no mercury will remain on the glass. We say that *mercury does not wet glass*.



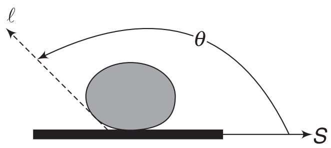
Mercury globules on glass plate. A smaller drop is less distorted due to gravity and is nearyl spherical.

When water drops fall on a clean glass surface, the liquid spreads out to form a thin, relatively uniform film over the surface. We say that *water wets glass*. This happens because of relatively stronger adhesive force between water molecules and glass.

Surface tension of a liquid is directly related to strength of cohesive force. Stronger cohesive force implies higher surface tension.

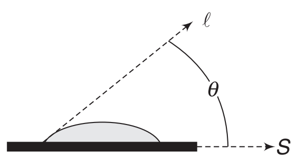
When a liquid surface touches a solid surface, the shape of liquid surface near the contact gets curved, in general. This happens due to interaction of the liquid molecules with the solid molecules. The curvature of the surface is decided by the relative strength of cohesive and adhesive forces.

The angle between the tangent planes to the soid surface and the liquid surface at the contact is called *contact angle* (θ). Tangent plane to the solid surface is drawn going towards the liquid bulk and the tangent plane to the liquid surface is drawn away from the solid. Figures show a water drop on a waxed surface and a water drop on a clean painted surface of a car.



Water drop on a waxed surface. θ is contact angle

In both the diagrams, l is tangent on the curved liquid surface at the contact, taken in direction pointing towards the solid. S is a tangent plane on solid with its direction pointing towards the liquid. Angle θ is contact angle for given pair of liquid and solid.

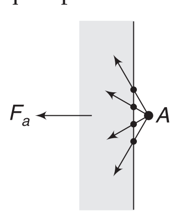


Water drop on a clean painted suface.

We all known that water does not wet a waxed surface. In this case, cohesive forces are much stronger than adhesive force between wax and water. In situations like this, contact angle is obtuse.

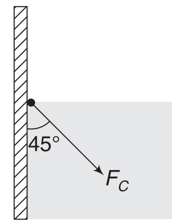
Water wets a dry and clean painted car surface. Adhesive forces are stronger here. In such cases, contact angle is acute. The larger θ is, the larger is the ratio of cohesive to adhesive forces.

Let us now see the reason behind curving of liquid surface in contact with a solid. A liquid particle in contact with solid is attracted by nearby solid molecules. Resultant of all these forces is normal to the solid surface due to symmetry of attraction from all sides. Let us call the resultant adhesive force on a small liquid particle as F_a that touches the solid surface.



Solid molecules attract liquid molecule A and the resultant adhesive force is normal to the solid surface.

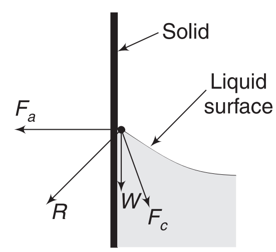
The cohesive force (due to attraction of all surrounding liquid molecules) on a liquid particle on surface is directed into the liquid and its exact direction depends on the shape of the surface itself. For a simpler case of flat liquid surface (which is not the case, usually) a liquid particle touching a solid wall will experience cohesive force in a direction making 45° with the solid surface. Let us call this force as F_c.



Cohesive force on a small liquid particle on surface near the solid. We have taken liquid surface to be flat which is usually not the case.

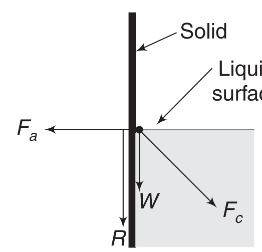
A liquid particle on surface, touching the solid boundary, will be experiencing three forces: (i) Adhesive force F_a (ii) Cohesive force F_c and (iii) it own weight W.

The direction of resultant (R) of the three forces decides the shape of the liquid surface. A liquid surface in equilibrium cannot sustain tangential force. Therefore, the surfaces curve so that R gets perpendicular to the surface. If the resultant (R) of the three forces is directed through the solid [as shown in fig (a)], the surface is concave. This is what happens when water touches glass. If resultant (R) is along the solid surface, liquid surface is flat. Contact angle is 90°. If R is directed into the liquid, the surface bulge out and contact angle is obtuse. This is what happens when mercury touches glass. [fig (c)]



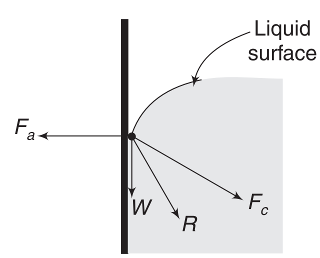
(a)

F_a is much larger than F_c. Resultant of F_a, F_c and W is towards the solid. Liquid surface is normal to R. θ is acute.



(b)

Resultant (R) of the three forces is parallel to solid wall. θ is 90°



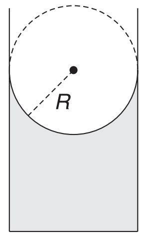
(c)

F_c is much stronger. R is towards the liquid. Surface bulges out in order to become perpendicular to R. θ is obtuse

Contact angle of pure water with glass is 0° and that of mercury with glass is close to 140°.

7.1 Shape of Meniscus

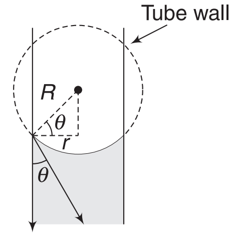
Take water in a narrow bore glass tube. The surface of water gets curved. This curved surface is known as *meniscus*. The curved surface is nearly spherical if the radius of tube is small. [If the tube is wide, the surface will not be spherical due to pronounced effect of gravity.] For contact angle of 0°, this meniscus will be a hemisphere. For contact angle $0° < \theta < 90°$, it will be a smaller part of a sphere. Radius of this (imaginary) sphere is known as radius of curvature (R) of the meniscus.



$\theta = 0°$.
Meniscus is hemispherical
Radius of curvature
(R) = tube radius (r)

As can be seen from the drawing

$$\frac{r}{R} = \cos\theta \qquad ...(9)$$



$0° < \theta < 90°$

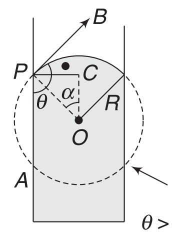
For $\theta > 90°$, the meniscus surface is convex upwards. In the diagram shown,

$$\alpha = \angle APO = \theta - 90°$$

$$\therefore \quad \frac{PC}{OP} = \sin\alpha$$

$$\Rightarrow \quad \frac{r}{R} = \sin(\theta - 90°)$$

$$\therefore \quad \frac{r}{R} = -\cos\theta \qquad ...(10)$$



$\theta > 90°$
<APB = θ (contact angle)

YOUR TURN

Q.12 Simple water (without detergent) is unable to wash grease and fat from a cloth. Why?

Q.13 A chewing gum does not stick to your teeth. What does it tell you about the relative strength of cohesive force in gum and adhesive force between gum and teeth?

Q.14 It is difficult to break a mercury drop into small droplets. Why?

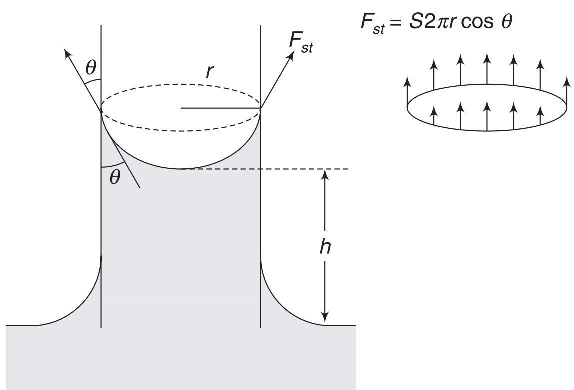
8. CAPILLARY RISE

Literal meaning of capillary is a tube, which has internal diameter of hair-like thickness.

When a glass tube of narrow bore is dipped in water, the liquid rises in the tube. Whenever the contact angle is acute, the liquid rises. This phenomena is known as *capillarity*. However, the liquid gets depressed inside the tube if contact angle is larger than 90°.

The adhesive force between water and glass is large and water spreads on a horizontal glass plate to wet it. When glass surface is vertical, water still tries to spread on glass surface but due to gravity it cannot rise much. Water surfaces forms a meniscus and water rises in the tube. Narrower the tube, higher is the rise.

Consider a capillary tube of radius r dipped in a liquid of density ρ. Let the contact angle be θ. Surface of liquid inside the tube is (nearly) spherical, having radius R. From equation (9), $\frac{r}{R} = \cos\theta$, where r is inner radius of the tube.



Surface of the liquid meets the tube wall along a circle of periphery $2\pi r$. The liquid surface pulls the tube wall tangentially (inclined at θ to the tube wall) along this circle. From Newton's third law, the tube wall must pull the liquid surface in opposite direction. The figure shows this force as F_{st}.

On a small elemental length Δl of the circumference, the force is $\Delta F_{st} = S \cdot \Delta l$, making angle θ with vertical. Vertical component of forces on all such small lengths adds to give a resultant vertical force on the liquid surface, which is equal to

$$F_{st} = \sum S \Delta l \cos\theta = S\cos\theta(2\pi r)$$

This vertically upward force by glass wall on the liquid surface supports the weight of liquid raised in the tube.

Neglecting the liquid present in the meniscus, the weight of liquid in the tube is

$$W = \pi r^2 h \cdot \rho \cdot g$$

Thus, $\quad W = F_{st}$

$$\Rightarrow \quad \pi r^2 h \rho g = S\cos\theta(2\pi r) \quad \text{...(i)}$$

$$\Rightarrow \quad h = \frac{2S\cos\theta}{r\rho g} \quad \text{...(11)}$$

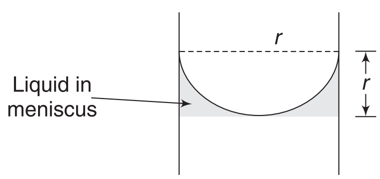
This can also be written as

$$h = \frac{2S}{R\rho g} \quad \text{...(12)}$$

where R is the radius of curvature of meniscus surface.

Correction for Weight of Liquid in Meniscus

In writing the above expression, we neglected the weight of liquid present in the meniscus. Correction for this is easy–particularly when contact angle is 0°. In this case, shape of meniscus is hemispherical and $R = r$.



Volume of liquid in meniscus is

$$= \pi r^2 \cdot r - \frac{2}{3}\pi r^3$$

$$= \frac{1}{3}\pi r^3$$

Therefore, equation (i) gets corrected as

$$\pi r^2 h \rho g + \frac{1}{3}\pi r^3 \rho \cdot g = S\cos 0°(2\pi r)$$

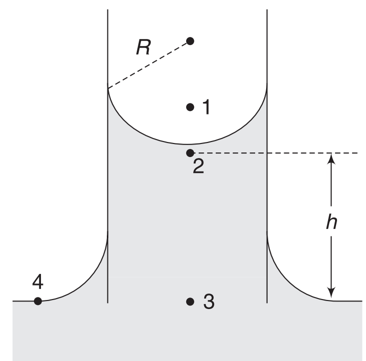
$$\Rightarrow \quad h = \frac{2S}{r\rho g} - \frac{r}{3} \quad \text{...(13)}$$

Where r is small, the correction is not very significant.

Alternative way to find capillary rise

Pressure at Point 1 above meniscus surface is atmospheric pressure.

$$P_1 = P_0.$$



Meniscus is concave upward, therefore, pressure at point 2 just below the surface is

$$P_2 = P_1 - \frac{2S}{R}$$

where R is radius of curvature of the meniscus surface

$$\Rightarrow \quad P_2 = P_0 - \frac{2S}{R}$$

Pressure at 3 is $P_3 = P_2 + \rho gh = P_0 - \frac{2S}{R} + \rho gh$

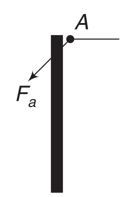
But pressure at 3 is same as pressure at Point 4, which is atmospheric pressure.

$$\therefore \quad P_0 = P_0 - \frac{2S}{R} + \rho gh$$

$$\Rightarrow \quad h = \frac{2S}{R\rho g} = \frac{2S\cos\theta}{r\rho g} \quad \text{since,} \quad \frac{r}{R} = \cos\theta$$

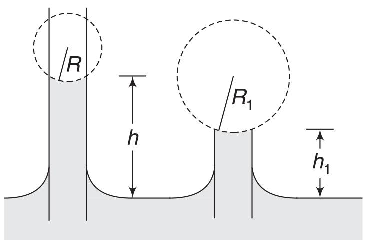
8.1 Tube of Insufficient Length

As discussed in Section 7, the force (F_a) applied by solid surface on liquid particles touching it is perpendicular to the solid surface. But this is not the case when the solid surface is just dipped in liquid. As there is no solid above the liquid surface, the adhesive force is directed towards the solid, as shown. In such situation, the angle between the solid and liquid surface may be different from the standard value of contact angle for the solid–liquid pair.



Force on a liquid particle A by the solid wall. Liquid is upto brim of the solid.

Equations (11) and (12) give us the capillary rise when a long tube is dipped in a liquid. What happens if the length of tube above the liquid surface is less than the value given by equation (11)? Will the liquid overflow out of the tube? Answer is, No. The angle between the tube wall and liquid surface (= θ) changes so as to keep surface tension force (F_{st}) equal to weight of liquid in the tube. In a much simpler language, we can say that radius of curvature (R) of the liquid surface changes.



In a tube of insufficient length, the radius of curvature of the liquid surface increases. Liquid does not overflow.

If the tube has sufficient length

$$h = \frac{2S}{R\rho g}$$

$$\Rightarrow \quad hR = \frac{2S}{\rho g} = \text{a constant}$$

When the tube has insufficient length h_1 above liquid surface, the radius of curvature (R_1) of liquid surface changes such that

$$h_1 R_1 = hR = \frac{2S}{\rho g} \quad \text{...(14)}$$

In an extreme case, when tube has no projection out of liquid surface ($h_1 = 0$), the value of R becomes infinite. It means liquid surface is flat, it does not curve.

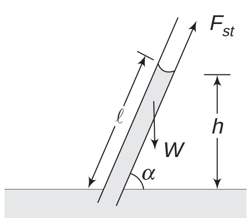
8.2 Inclined Tube

The resultant force of surface tension remains

$$F_{st} = S \cdot 2\pi r \cdot \cos\theta$$

This balances the component of weight along the tube.

$$\therefore \quad W \sin\alpha = S \cdot 2\pi r \cdot \cos\theta$$



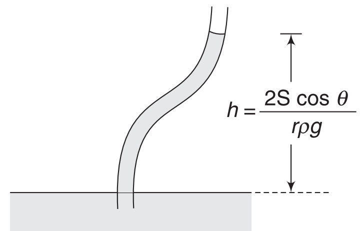
$$\Rightarrow \quad \pi r^2 \cdot l\rho \cdot g \cdot \sin\alpha = S \cdot 2\pi r \cdot \cos\theta$$

$$\Rightarrow \quad l \sin\alpha = \frac{2S\cos\theta}{r\rho g}$$

$$\Rightarrow \quad h = \frac{2S\cos\theta}{r\rho g}$$

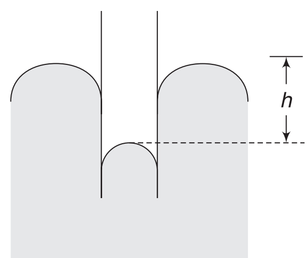
Vertical ascent of the liquid does not change even if the tube is tilted.

In fact, the vertical height of liquid in the capillary tube remains unchanged even if the tube in not straight.



8.3 Obtuse Contact Angle

In this case, the liquid gets depressed in the tube. An analysis similar to the above shows that the fall (h) of liquid level in the tube is still given by equations (11) and (12).



Liquid falls in the tube if $\theta > 90°$

cos θ is negative for $90° < \theta < 180°$ and a negative value of h in equation (11) shall be interpreted as a fall in liquid level.

8.4 Capillary Rise in Practical Life

(i) Capillary action allows the drainage of tear fluid that is constantly produced in eyes. In the inner corner of the eyes, behind the eyelids, there are two hair-like canals known as lacrymal ducts. Tear fluid from within the eyes rises into those ducts until they get expelled.

(ii) The tissue in roots, stems and branches of a tree have pores between them. These pores interconnect to form long capillary tubes. Water can rise through these tube to great heights.

(iii) Diwali 'diyas' keep burning, as oil keeps rising to the tip of burning wick due to capillary action.

(iv) Moisture in soil rises to the surface through capillaries and gets evaporated. By ploughing the field, we can break many such capillaries and reduce evaporation.

Example 7 A liquid of density 1.5 $g\,cc^{-1}$ rises 3.0 cm in a capillary tube of diameter 0.5 mm. Angle of contact for the solid–liquid pair is 0°. Calculate the excess pressure inside a spherical bubble of 1.0 cm radius blown from the same liquid, in unit of dyne cm^{-2}. g = 980 cms^{-2}

Solution

Concepts

(i) Formula for capillary rise will give the value of surface tension. We will write everything in CGS unit

(ii) Excess pressure inside bubble is $\dfrac{4S}{R}$

$$h = \frac{2S\cos 0°}{r\rho g} \Rightarrow S = \frac{hr\rho g}{2}$$

$$\Rightarrow \quad S = \frac{(3.0\,\text{cm})(0.025\,\text{cm})(1.5\,\text{gcm}^{-3})(980\,\text{cms}^{-2})}{2}$$

$$= 55 \text{ dyne cm}^{-1}$$

Excess pressure inside bubble is $\Delta P = \dfrac{4S}{R}$

$$\therefore \quad \Delta P = \frac{4 \times 55 \text{ dyne cm}^{-1}}{1 \text{ cm}} = 220 \text{ dyne cm}^{-2}$$

Example 8 A 5 cm long capillary tube with 0.1 mm internal diameter is slightly dipped in water. Find the radius of curvature of the meniscus surface in the tube. Surface tension of water is S = 75 dyne/cm and contact angle for water and glass is 0°. Take g = 981 cm s^{-2}.

Solution

Concepts

(i) We need to check whether the tube is of insufficient length.

(ii) If the tube is sufficiently long, then for θ = 0°, the radius of meniscus surface is same as the tube radius, i.e., $R = r$

If the tube is of insufficient length, then we will use equation (14) to find R.

Height of capillary rise in a tube of radius $r = \dfrac{0.1\,\text{mm}}{2}$ is

$$h = \frac{2S\cos 0°}{r\rho g} = \frac{2 \times 75 \text{ (dyne/cm)}}{(0.005 \text{ cm})\ (1\ \text{gcm}^{-3})\ (981\ \text{cms}^{-2})}$$

$$= 30.58\,\text{cm}$$

Since our tube is only 5 cm tall, it is a tube of insufficient length.

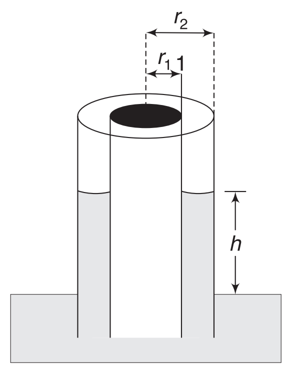
$$h_1 R_1 = hR$$

$$\Rightarrow \quad (5\,\text{cm})(R_1) = (30.58\,\text{cm})\ (0.005\,\text{cm})$$

$$\Rightarrow \quad R_1 = 0.03\,\text{cm}$$

Example 9 *A capillary tube with solid core*

A capillary tube of radius r_2 has a coaxial solid core of radius $r_1 (<r_2)$. It is dipped in water and the liquid rises in the annular space. Find the height to which water rises in the tube. Contact angle is 0° and surface tension is S.



Solution

Concepts

(i) Contact angle is 0°. This means that the force by tube wall on the water surface is vertical.

(ii) Water surface touches the tube wall along two circular peripheries of lengths $2\pi r_1$ and $2\pi r_2$.

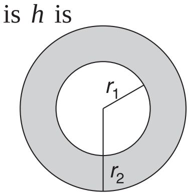
Thus, upward pulling force on water surface is

$$F_{st} = 2\pi r_1 S + 2\pi r_2 S.$$

(iii) The above force balances the weight of water in the annular space.

Volume of water in the tube when rise is h is

$$V = \pi(r_2^2 - r_1^2)\cdot h$$



For equilibrium of water in the tube:

Weight of water in tube = F_{st}

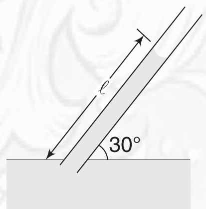
$$\pi(r_2^2 - r_1^2)h\rho g = 2\pi S(r_1 + r_2)$$

$$\Rightarrow \quad h = \frac{2S}{(r_2 - r_1)\rho g}$$

YOUR TURN

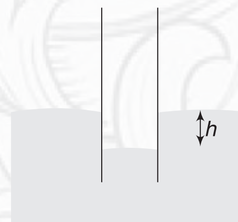
Q.15 A capillary tube of inner radius 0.5 mm is dipped in a liquid having surface tension $S = 0.07\ \text{Nm}^{-1}$. Angle of contact is 60° and density of the liquid is $10^3\ \text{kgm}^{-3}$. Find the height to which the liquid rises. $g = 9.8\ \text{ms}^{-2}$.

Q.16 A glass tube of radius $r = 0.4$ mm is dipped in water and held making an angle of 30° with the horizontal. How much length of capillary is occupied by water? Surface tension of water $S = 0.07\ \text{Nm}^{-1}$ and density of water is $\rho = 1 \times 10^3\ \text{kgm}^{-3}$. Contact angle is 0°.



Q.17 Mercury has angle of contact of 120° with wall of a capillary tube. Inner radius of the tube is 1mm. Surface tension and density of mercury are $0.5\ \text{Nm}^{-1}$ and $13.6 \times 10^3\ \text{kgm}^{-3}$ respectively. What is level of mercury in the tube when its lower end is dipped in a tank full of mercury?

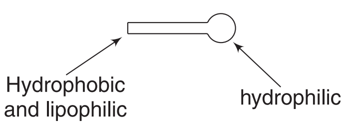
Q.18 A capillary tube dipped in mercury is shown in figure. Find the radius of curvature of the curved mercury surface. Density of mercury and its surface tension are ρ and S respectively. Depression in the tube is h. What is pressure just below mercury surface in the tube?



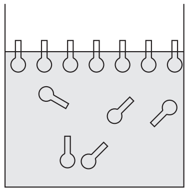
9. DETERGENTS

Washing your clothes with water does not remove grease stains. This is because water does not wet greasy stains. If water could wet grease, the flow of water could carry some grease away. However, this can be achieved by mixing detergent to water. Mixing of detergent reduces the surface tension of water.

Properties of detergents arise from their complicated molecular structure. A typical detergent molecule has a shape like a pin. Its head is hydrophilic and gets attracted to water molecules. Its tail is hydrophobic (hates water molecules) but lipophilic (gets attracted to grease, oils, fats).

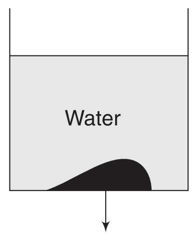


When detergent is put into water, at surface the detergent molecules have their hydrophobic ends protruding out. It is easier to pull this surface apart than it is to pull apart a surface of pure water.

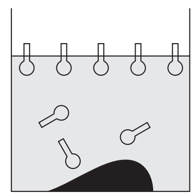


Detergent molecules in water.

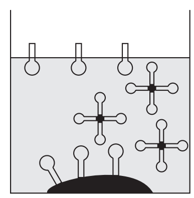
Sequence occurring during washing can be understood in simple way be looking at the following diagram.



Grease



Detergent added



Stirred

Hydrophobic ends of detergent molecules stick to grease particles and their head get attracted to water. If soapy water is removed the dirt gets washed away.

In Short:

(i) Angle between tangent plane to liquid surface (away from the solid) and tangent plane to solid surface (going towards the liquid) is known as contact angle (θ).

(ii) Value of θ is less than 90° when adhesive forces are relatively much stronger than cohesive forces. In such cases, liquid wets the solid surface.

(iii) If $\theta < 90°$, liquid rises in the capillary tube. Capillary rise is given by

$$h = \frac{2S}{R\rho g} = \frac{2S\cos\theta}{r\rho g}$$

Where R = radius of curvature of meniscus surface and r = radius of the tube

(iv) When cohesive forces are much stronger, the contact angle is obtuse ($\theta > 90°$). In such cases, the liquid does not wet the solid surface.

(v) If $\theta > 90°$, liquid level falls in capillary tube. The fall (h) is given by the above formula only. Negative value of h indicates fall.

(vi) In a capillary tube that is inclined to vertical, the vertical rise of liquid is still given by the above formula.

(vii) In a capillary tube of insufficient length, the radius of curvature of the liquid surfaces change (contact angle has no meaning in this case) and liquid does not overflow.

$$h_0 R_0 = hR$$

where,

h_0 = capillary rise in a tube of sufficient length

R_0 = radius of menicus in a tube of sufficient length

h = capillary rise in a tube of insufficient length

R = radius of meniscus in tube of insufficient length

10. VISCOSITY

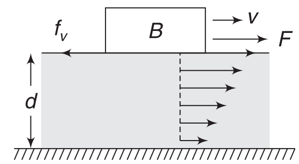
A feature which distinguishes one liquid from another is their 'thickness' or the ease with which they pour. And this property of a liquid certainly depends on how well the liquid molecules adhere to one another. If we pour water and honey on the walls of two separate funnels water flows down readily but honey takes much longer to flow down. As honey tends to flow down due to gravity, the relative motion between its layers is opposed strongly. We say honey is more viscous than water.

When a layer of fluid slips or tends to slip on another layer in contact, the two layer exert tangential force on one another. The direction of the forces are such that the relative motion between the layers is opposed. This property of a fluid to oppose relative motion between its layers is known as *viscosity*.

Before we start, let us put an important fact. In all circumstances, the velocity of a fluid is always zero (relative to the solid surface) at the surface of a solid. You must have noticed that blades of a fan collect a thin layer of dust–and it remains there even when the fan is rotated at a high speed. Dust doesn't get blown off by air. Reason is the fact that speed of air relative to the fan blade is zero, right at the surface.

10.1 Viscous Force

Consider a liquid film of thicknes d over a fixed solid surface A. A block B is placed over the liquid and dragged horizontally with velocity v. The layer of liquid in contact with the solid moves with velocity v.



Size of arrow represents the magnitude of velocity of different layers.

The lowest layer, in contact with surface A, remain at rest. Velocity of different layer increase from zero to v, as we move up from surface A. We define *velocity gradient* of fluid layers as

$$\text{Velocity Gradient} = \frac{v}{d} \qquad ...(15)$$

The force (F) needed to keep the block moving uniformly with velocity v is proportional to velocity gradient and also proportional to area of contact (A) of the block with the liquid. Thus, viscous force acting on the block is given by

$$f_v = F = \eta \cdot A\frac{v}{d} \qquad ...(16)$$

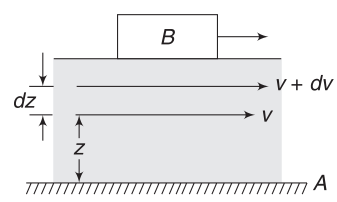
Here, η is a constant known as *coefficient of viscosity*. SI unit of coefficient of viscosity is N–s m^{-2}. Its *CGS* unit is dyne – s cm^{-2}, which is also known as poise. You can yourself prove that 1 poise = 0.1 N–s m^{-2}.

At 20°C, coefficient of viscosity for water is nearly 10^{-2} poise. At 80°C, value of η for water is 0.36×10^{-2} poise.

In fact, viscosity of liquids decrease with rise in temperature and for gases, it increases with increasing temperature.

When velocity gradient is not uniform, the force between two layers of fluids should be written as

$$f_v = \eta A\frac{dv}{dz} \qquad ...(17)$$

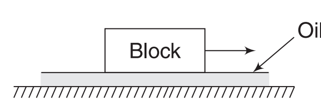


Where v = velocity of fluid layer at height z from the fixed surface A.

$v + dv$ = velocity of layer at height $z + dz$

$\frac{dv}{dz}$ = velocity gradient in the direction perpendicular to layers.

Example 10 A layer of castor oil, 1 mm thick, is spread on a horizontal floor. A block is placed on the floor and pulled horizontally at a constant speed $v = 2$ ms^{-1}. Find the force applied to the block. The face of block touching the oil has area of 20 cm^2. Coefficient of viscosity of oil at given temperature is 10 poise.



Solution

Concepts

The required force must be equal to viscous force on the block. This will ensure no acceleration

Top layer of oil moves with velocity of the block = 2 ms^{-1}. Lowest layer of oil has zero velocity as the floor is not moving.

$$\therefore \quad \text{Velocity gradient} = \frac{v}{d} = \frac{2\ \text{ms}^{-1}}{1\,\text{mm}} = 2 \times 10^{3}\,\text{s}^{-1}$$

Viscous force on the block is opposite to motion having magnitude

$$F_v = \eta A\frac{v}{d} = (10 \times 0.1\,\text{NSm}^{-2})(20 \times 10^{-4}\,\text{m}^2)(2 \times 10^{3}\,\text{s}^{-1})$$

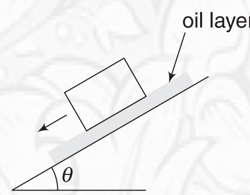
$$= 4\,\text{N}$$

External agent must apply 4 N force to keep the block moving.

Your Turn

Q.19 There is water in a bucket. It is stirred for some time and left on its own. Soon the spin of water stops. Why?

Q.20 A block of mass 50 g lies on an incline with a layer of oil 1mm thick. Area of the block in contact with oil is 20 cm^2. The block is imparted a velocity v = 2 ms^{-1} and it is found that it moves with constant velocity down the incline. Find viscosity (η) for the oil.



11. STOKES' LAW

When a solid moves through a fluid, it experiences a viscous drag. Magnitdue of the viscous force depends on the shape and size of the solid, its speed and coefficient of viscosity of the fluid.

English Physicist Stokes derived a formula for viscous force acting on a spherical body moving through a fluid. Consider a sphere of radius r moving at speed v in a fluid medium, having viscosity η. According to Stokes, the sphere experiences a viscous force given by

$$F_v = 6\pi\eta rv \qquad \text{...(17)}$$

One can find the dependence of F_v on η, r and v by the method of dimensions.

12. TERMINAL SPEED

Consider a body released from rest in a viscous medium. For example, consider a ball released from a height in atmosphere. The body first accelerates and then acquires a velocity, which remains constant during the remaining fall of the body. This constant speed (v_0) is known as terminal speed.

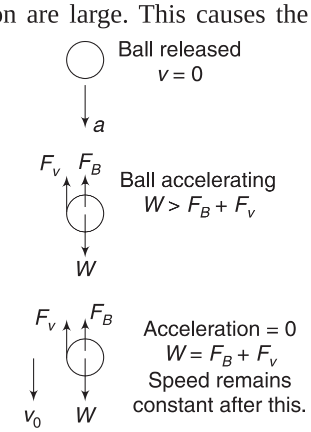
There are three forces on the ball as it falls through atmosphere:

(i) Its weight, $W = mg = Vdg$, where V is volume of the ball and d is its density.

(ii) Buoyancy, $F_B = V\rho g$, where ρ is the density of the medium (air).

(iii) Viscous force, $F_v = 6\pi\eta rv$, where r is radius of the ball and v is its speed.

The first two forces remain constant but the viscous force (F_v) increases in magnitude as the ball speeds up, since it is directly proportional to speed. Initially, when velocity is low, the viscous force (F_v) is low as well. Therefore, the net downward force and acceleration are large. This causes the speed to increase. As the speed increases, F_v also increases and in turn, the net downward force and acceleration decrease. Eventually, F_v becomes large enough to make net force on the ball equal to zero. Now, the velocity becomes constant. This constant velocity is terminal velocity (v_0).



Therefore, when velocity of the ball is v_0, net force on it is zero.

$$F_V + F_B = W$$

$$6\pi\eta rv_0 + \frac{4}{3}\pi r^3\rho g = \frac{4}{3}\pi r^3 dg$$

$$\Rightarrow \qquad v_0 = \frac{2}{9}\frac{r^2}{\eta}g(d-\rho) \qquad \text{...(18)}$$

When density of ball is large compared to density of the medium (i.e., $d >> \rho$), buoyancy can be neglected and the expression of terminal speed reduces to

$$v_0 = \frac{2}{9}\frac{r^2 g}{\eta}d \qquad \text{...(19)}$$

Example 11 A steel ball of radius $r = 5$ mm is released on the surface of a deep lake. Find the terminal velocity attained by the ball. Density of steel is 8,000 kgm^{-3} and that of water is 1,000 kgm^{-3}. Coefficient of viscosity is 10^{-3} N-sm^{-2} and acceleration due to gravity is 10 ms^{-2}.

Solution

Concepts

Direct use of equation 18

$$v_0 = \frac{2}{9}\frac{r^2 g}{\eta}(d-\rho) = \frac{2}{9} \times \frac{(5\times10^{-3})^2 \times 10}{10^{-3}}(8000-1000)$$

$$= 389\ ms^{-1}$$

This is very high! No lake will be so deep to allow the ball to reach its terminal speed.

Example 12 *Viscous force dissipates energy*

A small ball falls from rest in a viscous medium. Due to viscous force, heat is produced. Show that rate of heat generation is proportional to 5^{th} power of radius of the ball.

Solution

Concepts

Rate of loss of mechanical energy = rate of heat production = −(power developed by viscous force).

Terminal speed is $v_0 = \dfrac{2r^2 g}{9\eta}(d-\rho)$

Viscous force is $F_v = 6\pi\eta r v_0$.

Power of viscous force is

$P = -F_v v_0$ (negative sign since F_v and v_0 are in opposite direction)

$\therefore$ Rate of heat generation $= F_v \cdot v_0$

$$= 6\pi\eta r v_0^2 = \frac{8\pi g^2}{27\eta}(d-\rho)^2 r^5$$

$\therefore$ Rate of heat generation $\propto r^5$.

Your Turn

Q.21 A spherical object falls in a viscous medium and attains a terminal speed v_0. Mass of the object is m and buoyancy force on it is negligible. Find the viscous force on the object when it has attained terminal speed.

Q.22 A rain drop attains terminal speed of 3 ms^{-1} while falling through air. If the volume of rain drop were 8 times its present value, what would have been its terminal speed.

In Short:

(i) Viscous force between layers of a fluid is given by

$$F_v = \eta A\left(\frac{v}{d}\right)$$

Where A = area of contact and $\frac{v}{d}$ = velocity gradient along the thickness of the fluid.

η is coefficient of viscosity.

(ii) Viscous force on a spherical body is $F_v = 6\pi\eta r v$

(iii) When a body falls through a viscous medium, its speed becomes constant after some time. This constant speed is known as terminal speed (v_0).

For spherical bodies,

$$v_0 = \frac{2}{9}\frac{r^2 g}{\eta}(d-\rho)$$

d = density of body, ρ = density of fluid.

13. REYNOLDS' NUMBER

Osborne Reynolds studied the condition in which flow of fluid in pipes transitioned from steady flow to turbulent flow. He took a large pipe of transparent glass having water flowing through it. He introduced a small amount of dye into the centre of the flow and observed the behavior of the layer of dyed stream. When the flow speed was low, the dyed layer remained distinct and laminar throughout the length of the large pipe. When speed of flow was increased beyond a point, the dyed layer broke up and diffused throughout the tube, indicating turbulenece.

He concluded that whether the flow will be steady or turbulent depends on density (ρ), speed (v) and viscosity (η) of the fluid. It also depends on diameter (D) of the tube. He suggested a *dimensionless number*, known as Reynolds' number, which can be used to predict the nature of flow.

$$R = \frac{\rho v d}{\eta} \qquad ...(20)$$

When $R < 2000$, the flow is steady and when $R > 3000$, the flow becomes turbulent. For $2000 < R < 3000$, the flow is unstable. If may remain steady for a small interval of time and then turn turbulent for some time. R has same value in any system of units, as it is dimensionless.

Example 13 Water is flowing at a speed of 10 ms^{-1} through a pipe of dimater 0.2 m. Viscosity of water is 10^{-3} Nsm^{-2}. Is the flow steady?

Solution

Concepts

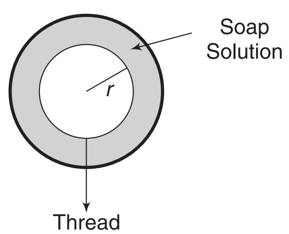
Find Reynolds' number.

$$R = \frac{\rho v d}{\eta} = \frac{10^3 \times 10 \times 0.2}{10^{-3}} = 2 \times 10^6$$

This is a large number. Flow is certainly turbulent.

Miscellaneous Examples

Example 14 There is a film of soap solution on a circular wire frame. A light thread loop is placed on the film. A pin is used to prick the film inside the loop. The thread takes a circular shape of radius r. Calculate tension developed in the thread.



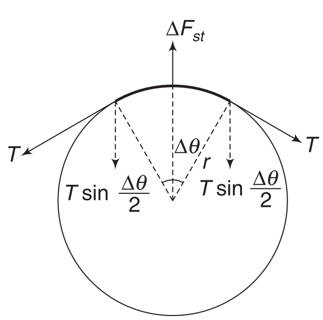
Solution

Concepts

Surface tension force on a small segment of the thread is radially outward. This is balanced by the radial component of thread tension on two sides of the segment.

Consider a segment of thread subtending a small angle $\Delta\theta$ at the centre.

Outward force on the segment due to surface tension is



$$\Delta F_{st} = 2(r\Delta\theta)\cdot S \quad \text{[There are two surfaces]}$$

Let T = Tension developed in the thread,

component of T towards centre $= T\sin\frac{\Delta\theta}{2} \simeq T\frac{\Delta\theta}{2}$

$$\left[\text{For very small } \Delta\theta, \ \sin\frac{\Delta\theta}{2} \rightarrow \frac{\Delta\theta}{2}\right]$$

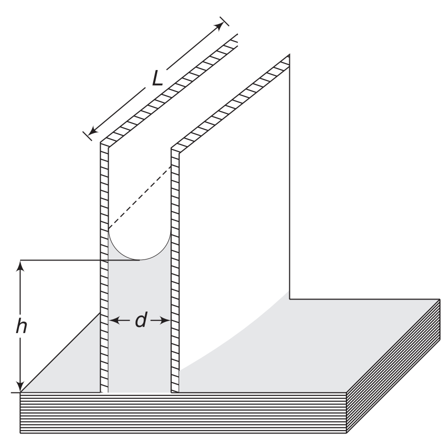
For equilibrium of the segment we must have

$$2T\,\frac{\Delta\theta}{2} = 2.(r\Delta\theta).S$$

$$\Rightarrow \qquad T = 2\cdot S\cdot r$$

Example 15 *Two parallel plates*

Two square plates have side length L each. They are kept close to each other with a small gap d between them and their lower edges are made to touch the water surface in a tank. Water rises in the gap between the plates to a height h. Surface tension of water is S and it has a contact angle of 0° with glass. Neglect edge effects, if any, and assume that the water surface between the plates has uniform curvature.



(a) Find h.

(b) Calculate the horizontal force on any one plate arising due to pressure difference on two faces.

Solution

Concepts

(i) The meniscus is cylindrical in shape.

(ii) Upward pull on water surface due to two glass plates is simply $F_{st} = 2.SL$

(iii) F_{st} balances the weight of water between the plates.

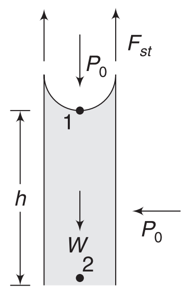
(iv) One can also find height h thinking in terms of pressure. Just below the curved menisus, pressure is less than atmospheric pressure by $\Delta P = \dfrac{S}{R}$ where $R = \dfrac{d}{2}$.

(v) Pressure on the plate due to atmospheric pressure is higher than pressure due to water. This pressure difference causes a resultant horizontal force.

(a) $$F_{st} = W$$

$$2SL = L \cdot d \cdot h \cdot \rho \cdot g$$

$$\Rightarrow \quad h = \frac{2S}{\rho d \cdot g} \qquad \text{...(i)}$$



(b) Radius of cylindrical water surface is $R = \dfrac{d}{2}$

Pressure at point 1 just below the cylindrical surface is

$$P_1 = P_0 - \frac{S}{R} = P_0 - \frac{2S}{d}$$

Pressure at Point 2 is

$$P_2 = P_1 + \rho gh = P_0 - \frac{2S}{d} + \rho gh$$

Average outward pressure on a plate is

$$P_{av} = \frac{P_1 + P_2}{2} = P_0 - \frac{2S}{d} + \frac{\rho gh}{2}$$

Outward force on a plate due to water is

$$F_1 = P_{av} \cdot L^2 = \left[P_0 - \frac{2S}{d} + \frac{\rho gh}{2}\right] L^2$$

Force due to atmospheric pressure is

$$F_2 = P_0 L^2$$

$\therefore$ Net inward force is

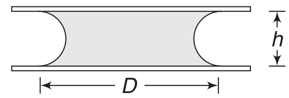
$$F = F_2 - F_1 = \left[\frac{2S}{d} - \frac{\rho gh}{2}\right] L^2$$

$$= \left[\frac{2S}{d} - \frac{S}{d}\right] L^2 \qquad \text{[iusing (i)]}$$

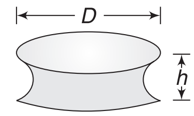
or, $$F = \frac{SL^2}{d}$$

Example 16 *Water drop between two glass plates*

A drop of water lies between a pair of horizontal flat glass plates. Figure shows the side view. The water drop spreads as a thin film having thickness h. Diameter (D) of the flattened drop is large compared to h. Contact angle is 0°. What is the extra force acting on each plate due to pressure difference on two sides?



Side view



Top view of water drop between the plates.

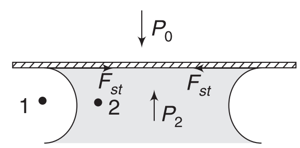
Solution

Concepts

(i) Since, $D >> h$, the surfaces of water exposed to atmosphere can be assumed to be nearly cylindrical. Pressure inside water is less than atmospheric pressure by $\dfrac{S}{R} = \dfrac{2S}{h}$

(ii) The tangential surface tension force is horizontal and has no role in present problem.

Since, contact angle is 0°, the surface tension force on glass plate at every location is horizontal. for a circular drop, resultant of this force on glass plate is zero.



Curved surface of water is nearly cylindrical with radius $R = \dfrac{h}{2}$

$$\therefore \qquad P_2 = P_1 - \frac{S}{R}$$

$$= P_0 - \frac{2S}{h}$$

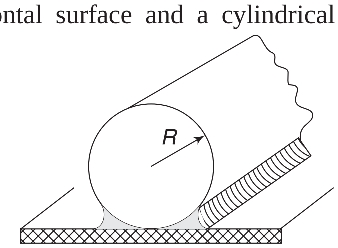
$\therefore$ Force on a plate due to pressure difference on two sides over an area $A = \dfrac{\pi D^2}{4}$ is

$$F = (P_0 - P_2)A = \frac{2S}{h} \cdot \pi \frac{D^2}{4} = \frac{\pi}{2} \frac{SD^2}{h}$$

This force is downward on upper plate and upward on the lower plate.

It will be difficult to separate the two plates by pulling them away from one another.

Example 17 A liquid having surface tension S is forming a film between a horizontal surface and a cylindrical object of radius R, as shown in figure. Cylindrical liquid surface has radius r. The contact angle is $\theta = 0°$. Find the downward force acting on unit length of the cylinder due to presence of the



liquid. Height of liquid above horizontal surface is greater than r.

Solution

Concepts

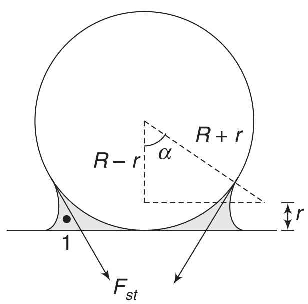
There is a surface tension force, which has vertically downward resultant and there is one force acting due to drop in pressure inside the liquid (like in last example).

From the figure

$$\cos\alpha = \frac{R - r}{R + r} \qquad \text{...(i)}$$

Consider a length L of the cylinder. Force due to surface tension acts tangentially and has a vertical resultant given by

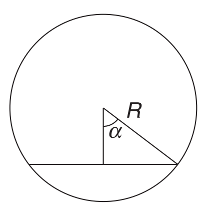
$$2SL \cdot \sin\alpha$$



Pressure inside the liquid (at 1) is less than atmospheric pressure by $\Delta P = \dfrac{S}{r}$

Pressure difference causes an additional downward force on the cylinder given by

ΔP [Rectangular area $2R\sin\alpha . L] = \dfrac{2SRL}{r}\sin\alpha.$



$\therefore$ Net downward force due to presence of liquid is

$$F_{\text{net}} = 2SL\sin\alpha + \frac{2SRL}{r}\sin\alpha$$

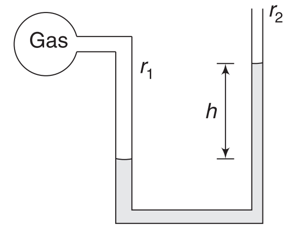
Force per unit length is

$$\frac{F_{\text{net}}}{L} = 2S\left[1 + \frac{R}{r}\right]\sin\alpha$$

$$= 2S\left[\frac{r + R}{r}\right]\sqrt{1 - \left(\frac{R - r}{R + r}\right)^2} \quad \text{[using (i)]}$$

$$= 4S\sqrt{\frac{R}{r}}$$

Example 18 A monometer contains water of density $\rho = 10^3\ \text{kgm}^{-3}$. Its arms have different radii of $r_1 = 1.44$ mm and $r_2 = 0.72$ mm. The wider tube is connected to a bulb containing a gas and the other tube is open to atmosphere. The difference in level of liquid in the two arms is $h = 0.2$ m, as shown. Find the gauge pressure in the bulb. Surface tension of water is $0.072\ \text{Nm}^{-1}$ and contact angle is 0°.



Solution

Concepts

(i) Since, radii of the two tubes are small, effect of surface tension cannot be neglected.

(ii) We will equate pressure in the two arms at the level of liquid surface in the wider arm.

Pressure in water just below the curved surface in the wider arm is

$$P_1 = P_{\text{gas}} - \frac{2S}{r_1} \qquad \left[\begin{array}{c}\text{since } \theta = 0° \\ \text{Radius of surface = radius of tube}\end{array}\right]$$

Pressure in other arm at the same horizontal level is

$$P_2 = P_0 - \frac{2S}{r_2} + \rho g h$$

$$P_1 = P_2$$

$$\therefore \quad P_{\text{gas}} - P_0 = 2S\left(\frac{1}{r_1} - \frac{1}{r_2}\right) + \rho g h.$$

$$\therefore \quad P_{\text{gas}} - P_0 = 2 \times 0.072\left[\frac{1}{1.44 \times 10^{-3}} - \frac{1}{0.72 \times 10^{-3}}\right]$$

$$+ 10^3 \times 9.8 \times 0.2$$

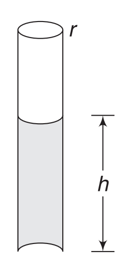
$$= 100 - 200 + 1960 = 1860\ \text{Nm}^{-2}$$

Example 19 *Capillary tube open at both ends*

A narrow capillary tube of radius $r = 0.5$ mm contain some water in it. When the tube is held vertical, the water remains inside the tube as shown. Find the radius of curvature of the upper and lower water surfaces when

(a) $h = 2$ cm

(b) $h = 3$ cm (c) $h = 4$ cm



Given: Surface tension $S = 0.075\ \text{Nm}^{-1}$ and contact angle = 0°. $g = 10\ \text{ms}^{-2}$

Solution

Concepts

(i) The upper meniscus will be hemispherical. The upward force by glass on the upper meniscus tries to balance the weight of water column. If the weight increases (i.e., h is high) the lower surface also gets concave up so as to have an upward force on it.

(ii) Working in terms of pressure is much easier.

Pressure just below the upper meniscus is

$$P_1 = P_0 - \frac{2S}{r} = P_0 - \frac{2 \times 0.075}{0.5 \times 10^{-3}}$$

$$= (P_0 - 300)\ \text{Nm}^{-2}$$

At depth h below the upper meniscus, the pressure will be

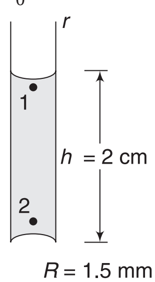
$$P_2 = P_1 + \rho g h = P_0 - 300 + 10^3 \times 10 \times h$$

$$\Rightarrow \qquad P_2 = P_0 - 300 + 10^4 h \qquad \text{...(i)}$$

(a) For $h = 2$ cm

$$P_2 = P_0 - 300 + 10^4 \times 0.02 = P_0 - 100\ \text{Nm}^{-2}$$

Since, pressure inside the lower surface is less than P_0, the surface must be convex upwards. Radius of curvature R is such that



$$\frac{2S}{R} = 100\ \text{Nm}^{-2}$$

$$\frac{2 \times 0.075}{R} = 100$$

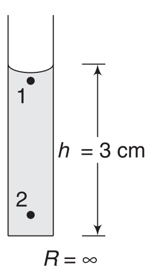
$$\Rightarrow \quad R = 1.5 \times 10^{-3}\ \text{m} = 1.5\ \text{mm}$$

(b) When $h = 3$ cm

$$P_2 = P_0 - 300 + 10^4 \times 0.03 \qquad \text{[using (i)]}$$

$$= P_0$$

$\therefore$ surface will be flat with radius of curvature $= \infty$

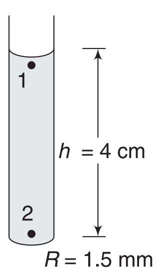


(c) When $h = 4$ cm

$$P_2 = P_0 - 300 + 10^4 \times 0.04$$

$$= P_0 + 100\ \text{Nm}^{-2}$$

Since, pressure at 2 is higher than atmospheric pressure, the lower surface will be concave up.



$$\frac{2S}{R} = 100$$

$$\Rightarrow \qquad R = 1.5\ \text{mm}$$

Example 20 Two soap bubbles are in contact. What is the angle between the tangent planes to their surfaces where they meet?

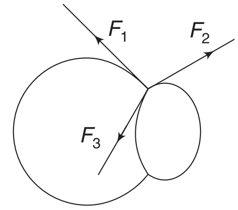
Solution

Concepts

(i) Soap solution is same, therefore surface tension is same for both bubbles.

(ii) Consider forces on a small line segment along the joint on the surface of the two bubbles.

Consider a small arc of length Δl along the circumference of the common wall. This segment experiences three surface tension forces due to three surfaces touching it. All three forces are of same magnitude $F_1 = F_2 = F_3$



For equilibrium of the segment, resultant of three forces must be zero. This is possible only if the three forces are inclined at 120° to each other.

Forces are tangential to the three surfaces. Thus, 120° is the required angle between the tangent planes on the two bubbles.

Example 21 A capillary tube of radius r is dipped in water. The liquid rises in the tube. Find heat generated due to dissipative force (like viscosity). Contact angle is 0° and surfaces tension is S.

Solution

Concepts

As soon as the capillary touches the water surfaces, the upward surface tension force pulls the water into the tube. If there is no dissipative force, the water in the tube will get accelerated. Its speed will increase till equilibrium position (where surfaces tension balances weight of water in the capillary). Having some kinetic energy, water will continue to rise. Now, it retards and stops. Now its weight is greater than upward surface tension. It will accelerate down. Thus, water column in the tube will oscillate. But this does not happen due to presence of dissipative forces. When the water moves viscous force dissipates energy.

Rise of water in the tube is $h = \dfrac{2S}{\rho g r}$

Increase in gravitational potential energy of water

$$\Delta U = (\text{mass of water is capillary})\left(\frac{h}{2}\right)g.$$

$$= (\pi r^2 h\rho)\left(\frac{h}{2}\right)g = \frac{\pi}{2}\, r^2 \rho g h^2$$

$$= \frac{2\pi S^2}{\rho g}$$

Work done by surface tension is

$W = (2\pi r\,.\,S).\,h$ [Contact angle is 0° and surface tension force is vertical]

$$= \frac{4\pi S^2}{\rho g}$$

Loss in energy (= heat generated) = $W - \Delta U = \dfrac{2\pi S^2}{\rho g}$

Example 22 *Falling raindrop*

A raindrop of radius r begins to fall from a large height. Write its speed as a function of time and plot speed (v) – time (t) graph. Assume that density of drop (d) is large compared to density of air and coefficient of viscosity is η. Take acceleration due to gravity (g) to be constant.

Solution

Concepts

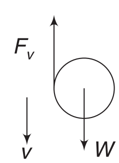
(i) Since density of drop is large compared to density of medium, we can neglect buoyancy.

(ii) The drop moves under influence of gravity and viscous force only.

(iii) We will use $F = ma$ and write $a = \dfrac{dv}{dt}$.

Integrating the obtained expression after separating the variables gives the result.

Let speed be v at time t after the drop begins to fall.



$$m\cdot a = W - F_v$$

$$\frac{4}{3}\pi r^3\cdot d\frac{dv}{dt} = \frac{4}{3}\,\pi r^3\cdot d\cdot g - 6\pi\eta r v$$

$$\Rightarrow \quad \frac{dv}{dt} = g - \frac{9}{2}\,\frac{\eta}{r^2 d}v$$

$$= g\left[1 - \frac{9\eta}{2gr^2 d}v\right]$$

$$= g\,[1 - kv] \quad \text{where} \quad k = \frac{9\eta}{2gr^2 d}$$

$$\therefore \quad \int_{v=0}^{v}\frac{dv}{1-kv} = g\int_0^t dt$$

$$\Rightarrow \quad [ln(1 - kv)]_0^v = -kgt$$

$$\Rightarrow \quad ln(1 - kv) - ln\,1 = -kgt$$

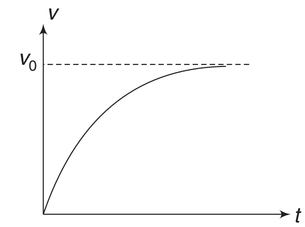
$$\Rightarrow \quad 1 - kv = e^{-kgt} \quad \Rightarrow \quad v = \frac{1}{k}\,[1 - e^{-kgt}]$$

$$\Rightarrow \quad v_0 = v_0\left[1 - e^{\frac{-gt}{v_0}}\right]$$

where $\quad v_0 = \dfrac{1}{k} = \dfrac{2gr^2 d}{9\eta}$

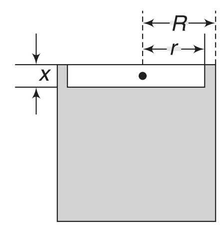
At time $\quad t \to 0;\ v = 0$

At time $\quad t \to \infty;\ v \to v_0$



v_0 is the final terminal speed. v-t graph is as shown.

Example 23 A cylinder is filled with a liquid of density ρ and viscosity η. A cylindrical piston has radius (r) slightly smaller than the inner radius (R) of the cylinder. Mass of the piston is m. The piston is released in the liquid with its axis conciding with the axis of liquid cylinder. Find the terminal speed attained by the falling piston. Thickness of the piston is x.



Solution

Concepts

(i) The fluid in contact with the piston will move with speed of piston and the fluid in contact with cylinder wall is at rest. Velocity gradient in the fluid layer between two walls is $\dfrac{v}{(R-r)}$ where v = speed of piston.

(ii) When piston moves at terminal speed, net force on it is zero.

Let v_0 = terminal speed.

Viscous force $F_V = \eta A\left(\dfrac{v_0}{R - r}\right) = \eta 2\pi r x\left(\dfrac{v_0}{R - r}\right)$

$$F_V + F_B = mg$$

$$\eta\,.\,2\pi r x\left(\frac{v_0}{R - r}\right) + \pi r^2 x\rho g = mg$$

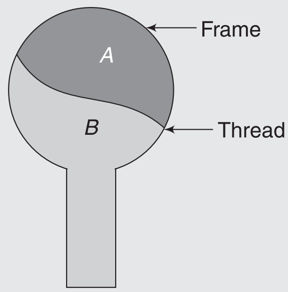
$$\Rightarrow \quad v_0 = \frac{(R - r)}{\eta\,.\,2\pi r x}\,[mg - \pi r^2 x\rho g]$$

$$= \frac{(R - r)g}{2\pi\eta r x}\,[m - \pi r^2 x\rho]$$

Worksheet 1

1. A liquid will not wet the surface of a solid if the angle of contact is
   (a) 0° (b) 45°
   (c) 60° (d) >90°

2. A thread is tied slightly loose to a wire frame as shown in the figure. And the frame is dipped into a soap solution and taken out. The frame is completely covered with the film. When the portion A is punctured with a pin, the thread:



   (a) becomes convex towards A
   (b) becomes concave towards A
   (c) remains in the initial position
   (d) either (A) or (B), depending on size of A w.r.t. B

3. A liquid rises in a capillary tube when the angle of contact is
   (a) an acute one (b) an obtuse one
   (c) $\pi/2$ radian (d) π radian

4. In an experiment, water rises to a height 0.1 m in a capillary tube. If the same experiment is repeated in an artificial satellite, which is revolving around the earth, water will rise in the capillary tube upto a height of
   (a) 0.1 m (b) 0.2 m
   (c) 0.98 m (d) full length of tube

5. Neglecting gravity, the potential energy of a molecule on the surface of a liquid compared with PE of a molecule deep inside liquid is:
   (a) greater (b) less
   (c) equal
   (d) depending on the liquid sometimes more, sometimes less

6. The surface tension of a liquid is 5 Nm^{-1}. If a film is held on a ring of area 0.02 m^2, its surface energy is about:
   (a) 5×10^{-2} J (b) 2.5×10^{-2} J
   (c) 2×10^{-1} J (d) 3×10^{-1} J

7. In a vessel, equal masses of alcohol (sp. gravity 0.8) and water are mixed together. A capillary tube of radius 1 mm is dipped vertically in it. If the mixture rises to a height 5 cm in the capillary tube, the surface tension of the mixture is (assume contact angle zero and g = 980 cms^{-2})
   (a) 218 dyne cm^{-1} (b) 234.18 dyne cm^{-1}
   (c) 107.9 dyne cm^{-1} (d) 10.79 dyne cm^{-1}

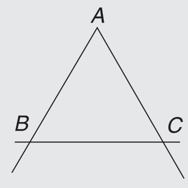
8. A glass capillary tube of inner diameter 0.28 mm is lowered vertically into water in a vessel. The gauge pressure that must be maintained above the water surface in the capillary tube so that water level in the tube is same as that in the vessel is (surface tension of water = 0.07 Nm^{-1} and atmospheric pressure = 10^5 Nm^{-2})
   (a) 500 Nm^{-2} (b) 99×10^3 Nm^{-2}
   (c) 100×10^3 Nm^{-2} (d) 101×10^3 Nm^{-2}

9. The work done in increasing the size of a rectangular soap film with dimensions 8 cm × 3.75 cm to 10 cm × 6 cm is 2×10^{-4} J. The surface tension of the film in Nm^{-1} is:
   (a) 1.65×10^{-2} (b) 3.3×10^{-2}
   (c) 6.6×10^{-2} (d) 8.25×10^{-2}

10. A capillary tube is dipped vertically in a liquid and the liquid rises up to a height of 50 cm. The tube is now tilted to an angle of 45°. Length of liquid in the tube is
   (a) 50 cm (b) $50\sqrt{2}$ cm
   (c) zero (d) none of these

11. A thin metal ring of internal radius 8 cm and external radius 9 cm is supported horizontally from the pan of a balance so that it comes in contact with water in a glass vessel. It is found that an extra weight of 7.48 g is required to pull the ring out of water. The surface tension of water is
   (a) 80×10^{-3} Nm^{-1} (b) 75×10^{-3} Nm^{-1}
   (c) 65×10^{-3} Nm^{-1} (d) 68×10^{-3} Nm^{-1}

12. A soap film is formed on a vertical equilateral triangular frame ABC. Side BC can move vertically always remaining horizontal. If m is the mass of rod BC, and S is surface tension of soap film, the length of BC in equilibrium is



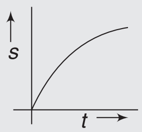
   (a) $\ell = \frac{mg}{2S}$ (b) $\ell = \frac{mg}{S}$
   (c) $\ell = \frac{2mg}{S}$ (d) $\ell = \frac{mg}{3S}$

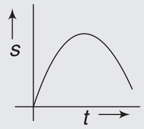
13. There is a 1mm thick layer of glycerine between a flat plate of area 100 cm^2 and a big fixed plate. If the

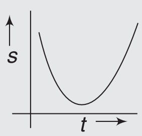
coefficient of viscosity of glycerine is 1.0 $kgm^{-1}s^{-1}$ then how much force is required to move the plate with a velocity of 7 cms^{-1}?

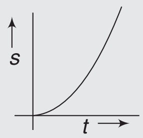
(a) 3.5 N (b) 0.7 N
(c) 1.4 N (d) None

14. The displacement of a ball falling from rest in a viscous medium is plotted against time. Choose a possible option.

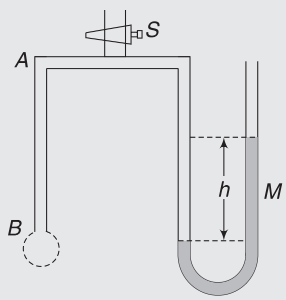
(a)


(b)


(c)


(d)


15. A tube of fine bore *AB* is connected to a manometer *M* as shown. The stop cork *S* controls the flow of air. *AB* is dipped into and taken out of a liquid whose surface tension is σ. A film of liquid is formed, which closes the end *B*. On opening the stop cork for a while, air is forced into the tube and a bubble is formed at *B*. The manometer level is recorded, showing a difference *h* in the levels in the two arms. If ρ be the density of manometer liquid and *r* the radius of curvature of the bubble, then the surface tension ρ of the liquid is given by



(a) $rhrg$ (b) $2rhgr$
(c) $4rhrg$ (d) $\dfrac{rh\rho g}{4}$

16. A viscous fluid fills the clearance between a shaft and a sleeve. When a force of 800 N is applied to the shaft, parallel to the sleeve, the shaft attains a speed of 1.5 cms^{-1}. If a force of 2.4 kN is applied instead, the shaft would move with a speed of

(a) 1.5 cms^{-1} (b) 13.5 cms^{-1}
(c) 4.5 cms^{-1} (d) None

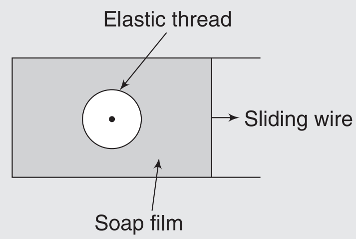
17. A solid metallic sphere of radius *r* is allowed to fall freely through air. If the frictional resistance due to air is proportional to the cross-sectional area and to the square of the velocity, then the terminal velocity of the sphere is proportional to which of the following?

(a) r^2 (b) r
(c) $r^{3/2}$ (d) $r^{1/2}$

18. Two drops of same radius are falling through air with steady velocity of v cms^{-1}. If the two drops coalesce, the new terminal velocity will be

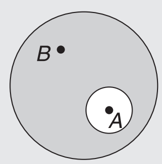
(a) $4v$ (b) $(4)^{1/3}\,v$
(c) $2v$ (d) $64\,v$

19. The figure shows a soap film in which a closed elastic thread is lying. The film inside the thread is pricked. Now the sliding wire is moved out so that the surface area increases. The radius of the circle formed by elastic thread will



(a) increase (b) decrease
(c) remains same (d) data insufficient

20. There is an air bubble of radius *R* inside a drop of water of radius 3*R*. The ratio of gauge pressure at point *A* to the gauge pressure at point *B* is



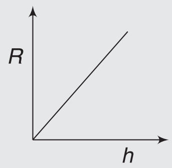
(a) 4 (b) 1
(c) 3 (d) 2

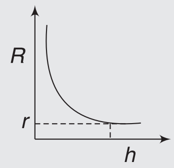
21. A long vertical capillary of radius *R* is dipped in a liquid having surface tension *S*. The gauge pressure at the mid-point of the liquid column in the capillary is (take angle of contact = θ)

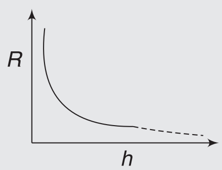
(a) $\dfrac{3S\cos\theta}{R}$ (b) $-\dfrac{S\cos\theta}{2R}$

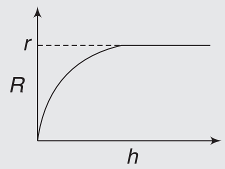
(c) $-\dfrac{S\cos\theta}{R}$ (d) $\dfrac{2S\cos\theta}{R}$

22. A long capillary tube of radius '*r*' is vertically immersed inside a liquid such that its top end is at the surface. Angle of contact for the solid - liquid pair is 0°. If the tube is slowly raised then relation between radius of curvature of meniscus inside the capillary tube and displacement (*h*) of tube can be represented by

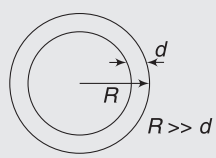
(a) 

(b) 

(c) 

(d) 

23. A soap bubble has radius R and thickness d ($<< R$), as shown. It collapses into a spherical drop.



The ratio of excess pressure in the drop to the excess pressure inside the bubble is

(a) $\left(\frac{R}{3d}\right)^{\frac{1}{3}}$ (b) $\left(\frac{R}{6d}\right)^{\frac{1}{3}}$

(c) $\left(\frac{R}{24d}\right)^{\frac{1}{3}}$ (d) None

24. When a ball is released from rest in a very long column of viscous liquid, its downward acceleration is '*a*'(just after release). Find its acceleration when it has acquired two third of the maximum velocity:

(a) $\frac{a}{3}$ (b) $\frac{2a}{3}$

(c) $\frac{a}{6}$ (d) none of these

Worksheet 2

1. When an air bubble rises from the bottom of a deep lake to a point just below the water surface, the pressure of air inside the bubble

(a) is greater than the pressure outside it

(b) is less than the pressure outside it

(c) increases as the bubble moves up

(d) decreases as the bubble moves up

2. A sphere of mass m is released from rest in a stationary viscous medium. In addition to the gravitational force of magnitude mg, the sphere experiences a retarding force of magnitude bv, where v is the speed of the sphere and b is a constant. Assume that the buoyant force is negligible. Which of the following statement/s about the sphere is correct?

(a) Its kinetic energy begins to decrease sometime after it begins to fall.

(b) Its kinetic energy increases to a maximum, then decreases to zero.

(c) Its speed increases monotonically, approaching a terminal speed that depends on b but not on m.

(d) Its speed increases monotonically, approaching a terminal speed that depends on both b and m.

3. Two balls of same material (density ρ) but radii r_1 and r_2 are joined by a light inextensible vertical thread and released from a large height in a medium of coefficient of viscosity η. Neglect buoyancy and assume acceleration due to gravity to be a constant.

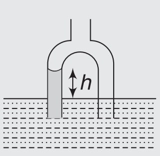
(a) the terminal velocity acquired by the balls is $\frac{2}{9}\frac{\rho g}{\eta}\left[r_1^2 - r_1 r_2 + r_2^2\right]$

(b) tension in the string when balls are moving with terminal velocity is $\frac{4}{3}\pi\rho g\left|r_1^2 r_2 - r_2^2 r_1\right|$

(c) just after the release, both balls will have same acceleration

(d) the string will get taut some time after the release.

4. The figure shows an inverted U-shaped tube with straight limbs of unequal radii $r_1 = 0.25$ mm and $r_2 = 0.50$ mm. Both the open ends of the tube are immersed below the free surface of water. There is an opening in the tube at the top. Air is pumped into the upper part of the tube so that the water in the wider tube is at level with the water outside.



Take angle of contact as zero and surface tension of water = 7×10^{-2} Nm^{-1}.

(a) height h of water in the narrower limb is nearly 0.028 m.

(b) if more air is pumped in, level of water in both limbs will not get affected.

(c) if more air is pumped in, level of water in wider limb will go below the free surface of water in the tank and water level in other limbs will not get affected.

(d) if more air is pumped in, level of water in wider limb will not change but it will go down in the other limb.

5. A sphere of radius 'a', is moving in a medium having coefficient of viscosity η. A force F starts acting on it along the line of motion. There is no gravity. (K is a numerical constant)

(a) the sphere acquires a constant velocity equal to $K\frac{F}{a\eta}$, if force F is a constant.

(b) the sphere acquires a constant velocity equal to $K\frac{F}{a\eta}$, if the force F remains constant in magnitude and changes its direction once.

(c) the sphere acquires a constant velocity equal to $K\eta\frac{a}{F}$, if force F is a constant.

(d) none of the above

6. A metallic sphere of radius 1.0×10^{-3} m and density 1.0×10^4 kgm^{-3} enters a tank of water, after a free fall through a height h in the earth's gravitational field. Its velocity remains unchanged after entering water. Given: coefficient of viscosity of water = 1.0×10^{-3} Nsm^{-2}, $g = 10$ ms^{-2} and density of water = 1.0×10^3 kgm^{-3}.

(a) Value of h is 20 m

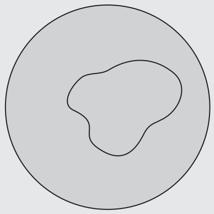
(b) If water is replaced with a liquid having double the coefficient of viscosity of water but same density as water, value of h will be 5 m

(c) If the experiment is repeated on moon (assuming we have water there), value of h will be 10 m (acceleration due to gravity on moon is 10/6 ms^{-2})

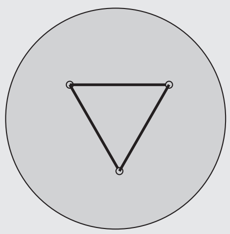
(d) None of the above

Worksheet 3

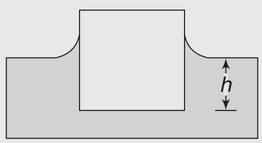
1. A thread ring is lying on surface of water in irregular shape. Some detergent in added to the surface inside the loop. What happens?



2. Three matchsticks are floating on the surface of water, forming a triangle. Some soap solution is poured gently inside the triangle. What happens?

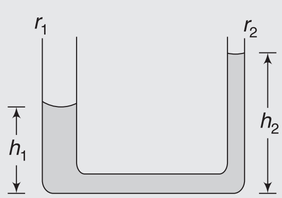


3. A cube of side length a and mass m floats in water. Water perfectly wets the material of the cube (it means that contact angle is 0°). Find the distance h between the lower face of the cube and water surface. Surface tension of water = S.



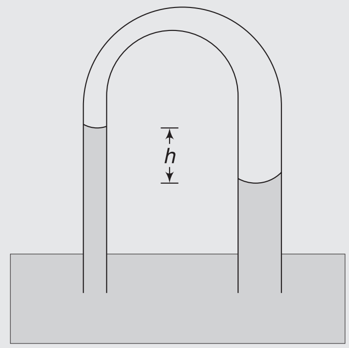
4. A thin ring of mass m and radius r is resting on a liquid surface. Surface tension of the liquid is S. The ring is slowly lifted while keeping its plane horizontal. Find the force needed to separate it from the liquid surface.

5. A U-shaped tube has two vertical arms of small cross-sections. Radii of the two arms are r_1 and r_2 ($< r_1$). A liquid is poured in the tube and its height in two columns is h_1 and h_2. Surface tension of the liquid is S and its contact angle with the tube wall is $\theta = 60°$. Find $h_2 - h_1$.



6. A plate of length 20 cm, width 0.77 cm and thickness 2 mm has mass of 8.2 gram. It is held vertically with long side horizontal and lower-half under water. Find the apparent weight of the plate. Surface tension of water = 73 dyne cm^{-1}, contact angle = 0° and g = 980 cm s^{-2}.

7. A U-shaped glass tube has two limbs of different radii. The radii are r_1 = 1.5 mm and r_2 = 3.0 mm. The tube is inverted so that both ends are dipped inside water. What is the difference in heights(h) to which water rises in the two limbs? Surface tension of water S = 0.07 Nm^{-1} and contact angle with glass is 0°.



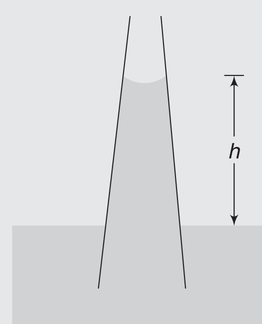
8. A water drop of radius R splits into n identical drops. Calculate the increase in surface energy. Surface tension is S.

9. A narrow capillary tube of radius r = 0.5 mm contains some water. The tube is held vertical. What is the height of water column h in the tube if the water is about to leak from the lower end?

Surface tension S = 0.075 Nm^{-1}; contact angle = 0°; g = 10 ms^{-2}.

10. Two soap bubbles of same size come into contact. Find the radius of curvature of the common wall.

11. A capillary tube has conical shape. Its radius decreases with height. The cone wall makes an angle α with vertical. The lower end of the tube is dipped in a liquid, which rises in it to a height h. At the location of liquid surface in the tube, radius of circular section of the tube is r. Contact angle is 90°. Find h if density of liquid is ρ.



12. A capillary tube is dipped in water and the liquid rises to a height h. In an experiment, the capillary tube is dipped in water and the whole experimental arrangement is inside an elevator. Length of the tube above the water surface is $2h$.

(a) With what acceleration the elevator must be moved so that water rises to the brim in the capillary?

(b) With what acceleration the elevator must be moved so that water rises only to a height $\frac{h}{2}$?

(c) What is the radius of curvature of the water surface in the tube if the elevator falls with acceleration g?

13. With what terminal velocity will an air bubble 0.8mm in diameter rise in liquid of viscosity 0.15 Nsm^{-2} and relative density 0.9? Density of air is 1.29 kgm^{-3}.

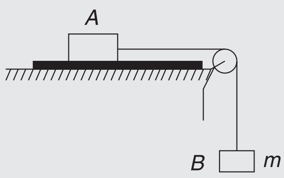
14. A metal sphere has radius r = 1 mm and mass m = 50 mg. It falls vertically in glycerine. Find

(a) Viscous force on the sphere when its speed is 0.5 cms^{-1}

(b) The terminal speed of the sphere

Density of glycerine ρ = 1260 kgm^{-3} and coefficient of viscosity = 8.0 poise.

15. A block (A) is placed on a table. There is a 3.0 mm thick layer of oil between the table and the block. Face of the block in contact with oil has an area 0.1 m^2. The block is connected to another block (B) of mass m = 0.01 kg by massless string passing over a smooth pulley (see figure). The system is released from rest. It accelerates and finally attains a constant speed of 0.085 ms^{-1}. Find coefficient of viscosity of the liquid. Neglect viscous force and buoyancy force due to air on the blocks. g = 9.8 ms^{-2}.



16. Spherical dust particles are shaken up in water and allowed to settle. Depth of water in the container is 2 cm. Estimate the diameter of the largest particles remaining in suspension 10 minute later. Density of dust particles, is $d = 1.8 \times 10^3$ kgm^{-3}. Assume that the dust particles quickly attain terminal speed.

Answers Sheet

Your Turn

1. 7.5 mN
2. 0.096 Nm^{-1}
3. Spherical
4. 5.9 μJ
5. 0.85 mJ
6. Increase in surface area requires energy.
7. 6 μJ
8. 2.0274×10^5 Nm^{-2}
9. Both have same pressure
10. 17.5 Nm^{-2}
11. $128\pi \dfrac{S^3}{P_0^2}$
15. 1.43 cm
16. 7.14 cm
17. Mercury gets depressed by 3.75 mm
18. $R = \dfrac{2S}{pgh}$; $P_0 + pgh$
19. Due to viscous force between different layers
20. 0.6 poise
21. mg
22. 12 ms^{-1}

Worksheet 1

1. (d)	2. (b)	3. (a)	4. (d)	5. (a)	6. (c)	7. (a)	8. (a)	9. (b)
10. (b)	11. (d)	12. (a)	13. (b)	14. (d)	15. (d)	16. (c)	17. (d)	18. (b)
19. (c)	20. (a)	21. (c)	22. (b)	23. (c)	24. (a)			

Worksheet 2

1. (a, d)
2. (d)
3. (a,b,c,d)
4. (a)
5. (a, b)
6. (a, b)

Worksheet 3

1. Thread becomes circular.
2. Matchsticks get separated.
3. $\dfrac{mg}{a^2\rho g} + \dfrac{4S}{a\rho g}$
4. $mg + 4\pi r S$
5. $\dfrac{S(r_1 - r_2)}{\rho g r_1 r_2}$
6. 9.67 gm. wt.
7. 4.76 mm
8. $4\pi R^2 \left[n^{\frac{1}{3}} - 1\right] S$
9. 6 cm
10. Infinite
11. $\dfrac{2S\sin\alpha}{r\rho g}$
12. (a) $\dfrac{g}{2}$ $(\downarrow)$ (b) g $(\uparrow)$ (c) ∞
13. 0.21 cms^{-1}
14. (a) 7.5×10^{-5} N (b) 2.9 cms^{-1}
15. $\eta = 0.034$ Nsm^{-2}
16. 8.8 μm

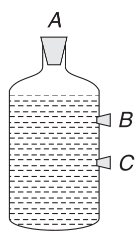
CHAPTER 10

Miscellaneous Problems on Chapters 8 and 9

MATCH THE COLUMNS

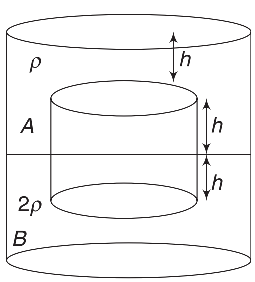
Match the entries in Column I with those in Column II. An item in Column I can match with any number of entries in Column II. It may also happen that an item in Column I does not match with any of the entries in Column II.

1. A bottle is filled with water, above which a little air at atmospheric pressure is present. Plugs are blocking the three small holes (A, B and C).



Column I		Column II	
(a)	A & C are opened	(p)	Air comes into the bottle from upper-most open hole
(a)	B & C are opened	(q)	Water flows out of holes B and C
(c)	Only C is opened	(r)	Some water comes out and the flow stops
(d)	All the holes are opened	(s)	Pressure of air inside bottle above the water remains unchanged.

2. Shown below is a cylinder of radius R floating in vessel containing liquids A and B. Neglecting atmospheric pressure, match the quantities mentioned in Column I with corresponding expression in Column II.

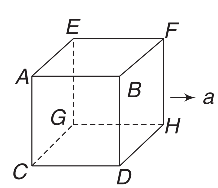


Column I		Column II	
(a)	Net force exerted by liquid A of density ρ on the cylinder.	(p)	$9\,\rho gRh^2$
(b)	Net force exerted by liquid B of density 2ρ on the cylinder.	(q)	$\pi\rho gR^2h$
(c)	Net force exerted by liquids A and B on the left half of the curved part of cylinder.	(r)	$4\pi\rho gR^2h$
(d)	Net force exerted by liquid A and B on the cylinder.	(s)	$3\pi\rho gR^2h$

3. Bucket A contains only water. An identical bucket B contains water, but also contains a solid object in the water. Consider the following four situations given in column I. Which bucket weighs more?

Column I		Column II	
(a)	The object floats in bucket B, and the buckets have the same water level	(p)	Bucket A
(b)	The object floats in bucket B, and the buckets have the same volume of water	(q)	Bucket B
(c)	The object sinks completely in bucket B, and the buckets have the same water level	(r)	Both buckets have the same weight
(d)	The object sinks completely in bucket B, and the buckets have the same volume of water	(s)	The answer cannot be determined from the information given.

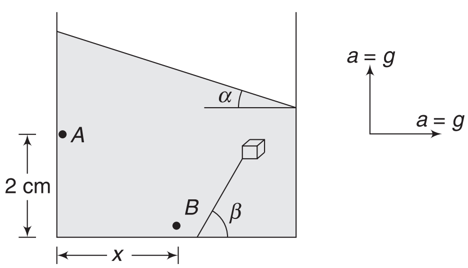
4. A cubical box is completely filled with mass m of a liquid and is given horizontal acceleration a, as shown. Match the force due to fluid



pressure on the faces of the cube with their appropriate values (assume zero pressure at the point of minimum pressure inside the tank.)

	Column I		Column II
(a)	force on face *ABFE*	(p)	$\frac{ma}{2}$
(b)	force on face *BFHD*	(q)	$\frac{mg}{2}$
(c)	force on face *ACGE*	(r)	$\frac{ma}{2} + \frac{mg}{2}$
(d)	force on face *CGHD*	(s)	$\frac{ma}{2} + mg$
		(t)	$\frac{mg}{2} + ma$

5. A container contains a liquid of density ρ. It is accelerated so as to have a horizontal component of acceleration equal to g and a vertically upward component also equal to g.



Match the enteries in Column I with respect to entries in Column II.

	Column I		Column II
(a)	Liquid surface makes an angle α with horizontal. $\tan\alpha =$	(p)	4
(b)	*A* wooden block is held under water with the help of a string tied to the base. The string makes an angle β with horizontal. $\tan\beta =$	(q)	2
(c)	A point *A* on the left wall is at a height 2 cm above the base. *B* is another point on the base at a distance *x* from the left wall. Pressure at *A* and *B* are equal. Value of *x* in cm is	(r)	$\frac{1}{2}$
(d)	Angle between line *AB* and free surface of the liquid is ϕ. $\tan\phi =$	(s)	0

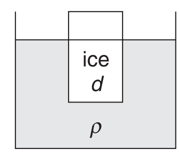
6. A liquid is filled in a cylindrical container up to a height *H* and a small hole is made at a depth *h* below the free surface of the liquid. Cross-section of the tank is *A* and that of the hole is A_o ($<< A$).

	Column I		Column II
(a)	The time taken by the liquid surface to reach level of orifice	(p)	Is independent of density of liquid
(b)	The horizontal range of liquid jet ejected from the tank	(q)	Is same for $h = y$ and $h = H - y$
(c)	Speed of efflux	(r)	Will change if the container is accelerated vertically up
(d)	Time needed to empty the tank for $h = H$	(s)	Is same if A_o is doubled.

7. Match Column I with entries in Column II

	Column I		Column II
(a)	Surface tension	(p)	Conservation of energy
(b)	Viscous force in liquid	(q)	Conservation of mass
		(r)	Terminal velocity
(c)	Continuity equation	(s)	Surface energy
(d)	Bernoulli's equation	(t)	Reynold's number

8. A block of ice floats in a liquid with a part of the ice block remaining outside the liquid. Density of ice is *d* and that of the liquid is ρ. Also, density of water is ρ_w. Match the situation described in Column I with results in Column II.



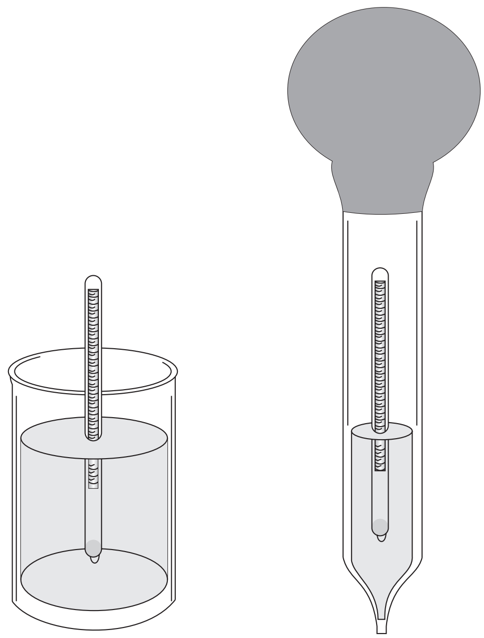
	Column I		Column II
(a)	$\rho = \rho_w$ and the ice melts	(p)	Level of liquid in the container falls
(b)	$\rho = \rho_w$ and half the ice melts	(q)	Level of liquid in the container goes up
(c)	$\rho > \rho_w$ and the ice melts	(r)	Level of liquid in the container does not change
(d)	$\rho < \rho_w$ and the ice melts	(s)	No prediction can be made about change in level of liquid

PASSAGE-BASED PROBLEMS

Every passage is followed by a series of questions. Every question has four options. Choose the most appropriate option for the questions.

Passage 1

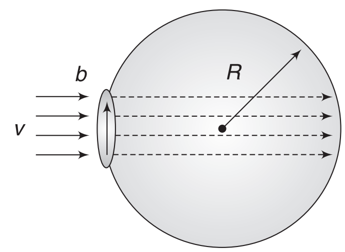
Hydrometer is used to measure the density of liquids. A calibrated tube sinks into the liquid until the weight of the liquid it displaces is exactly equal to its own weight. It is weighted at its bottom end so that the upright position is stable, and a scale in the top stem permits direct density readings. Figure (b) shows a typical type of hydrometer that is commonly used to measure the density of battery acid or antifreeze. A large tube has a calibrated hydrometer inside it. The large tube is fitted with a rubber bulb at the top and its lower end has a small opening. The bottom of the large tube is immersed in the battery liquid. The rubber bulb is squeezed to expel air and is then released, like a giant medicine dropper. The liquid rises into the tube. The hydrometer floats in this sample of the liquid.



1. For the case shown in figure (a):
   (a) the depth of the hydrometer submerged in denser liquid is more
   (b) the apparent weight of hydrometer in denser liquid is more
   (c) the depth of hydrometer in denser liquid is lesser
   (d) the apparent weight of hydrometer in denser liquid is less
2. In figure (b), pressure of air above the hydrometer is
   (a) less than atmospheric
   (b) atmospheric
   (c) more than atmospheric
   (d) cannot be said
3. The condition of stability of hydrometer in any fluid will be obtained if
   (a) buoyant force is equal to weight and centre of buoyancy coincides with the centre of mass
   (b) buoyant force is more than weight and centre of buoyancy coincides with the centre of mass
   (c) buoyant force is more than weight and centre of buoyancy lies above the centre of mass
   (d) buoyant force is equal to weight and centre of buoyancy lies above the centre of mass

Passage 2

A soap bubble is blown at the end of a tube of radius b. The liquid has surface tension S. Bubble having radius $R(b << R)$ is formed at the end of the tube. Assume that the bubble is essentially spherical in shape and the mouth of the tube is a small circle intersecting the spherical bubble. Air is blown inside the tube with velocity v, as shown, to grow the size of the bubble. Density of air is d. Assume that the air molecules entering the bubble travel straight and strike the wall of the bubble. The air molecules collide perpendicularly with the wall of the bubble and stop. The force applied by the air molecules striking the wall of the bubble pull the bubble away from the tube. When this force exceeds the force applied by the tube wall on the bubble, the bubble separates. Disregard gravity.



4. The force applied by the air molecules striking the wall of the bubble is

   (a) $2\pi b^2 \rho v^2$ (b) $\frac{1}{2}\pi b^2 \rho v^2$

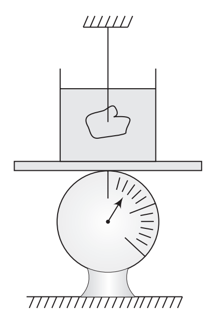
   (c) $\pi b^2 \rho v^2$ (d) None
5. The radius R at which the bubble separates from the ring is

   (a) $\frac{2S}{\rho v^2}$ (b) $\frac{S}{\rho v^2}$

   (c) b (d) None

Passage 3

A beaker containing water is placed on the pan of a balance. The balance shows a reading of M kg. A lump of sugar having mass m is suspended by a thread in such a way that it get completely submerged in water. The upper end of the thread is fixed to the ceiling. After time t_1, half the mass of sugar lump gets dissolved and at time t_2, the complete sugar lump get dissolved. Density of sugar (d) is greater than that of water (ρ). The thread has negligible volume and mass.



6. Reading of the balance at time t_1 is

   (a) $Mg + \frac{mg}{2}\frac{\rho}{d}\left[\frac{d}{\rho} + 1\right]$

   (b) $Mg + \frac{mg}{2}$

(c) $Mg - \dfrac{mg}{2}$

(d) $Mg + \dfrac{mg}{2}\dfrac{\rho}{d}\left[\dfrac{d}{\rho} - 1\right]$

7. Reading of the balance at time t_2 is

(a) $Mg + mg\dfrac{\rho}{d}$ (b) $Mg + mg\dfrac{d}{\rho}$

(c) $Mg + mg$ (d) $Mg + \dfrac{mg}{2}$

8. Reading of the balance

(a) Continuously increases from $t = 0$ (when sugar was put into water) to $t = t_2$

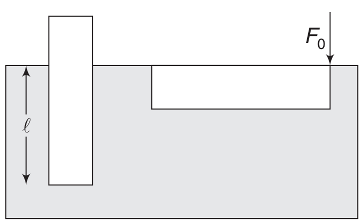
(b) Increases in the interval $t = 0$ to $t = t_1$ and then decreases in the interval t_1 to t_2

(c) Decreases first and then increases

(d) Nothing can be said

Passage 4

A cylindrical piece of wood has non-uniform density and its length and cross-sectional area are L and A respectively. It floats in water in vertical position in stable equilibrium with its length l submerged. To make the wood piece float in horizontal position while remaining completely submerged, we need to apply a vertically downward force F_0 at its end (see figure). Density of water is ρ_0.



9. Centre of mass of the wood is

(a) at the geometrical centre, when it is floating vertically

(b) above the geometrical centre, when it is floating vertically

(c) closer to the end at which F_0 is not applied in horizontal position

(d) above the water surface in vertical position

10. The distance of centre of mass of the wood piece from the end to which F_0 is applied is

(a) $\dfrac{F_0L}{2[AL\rho_0 g - F_0]}$ (b) $\dfrac{F_0L}{AL\rho_0 g + F_0}$

(c) $\dfrac{F_0L}{2[AL\rho_0 g + F_0]}$ (d) $\dfrac{F_0L}{2AL\rho_0 g}$

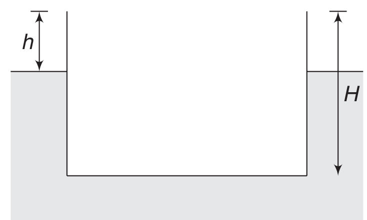
11. Value of l is

(a) $L - \dfrac{2F_0}{A\rho_0 g}$ (b) $L - \dfrac{F_0}{A\rho_0 g}$

(c) $L - \dfrac{F_0}{2A\rho_0 g}$ (d) $L - \dfrac{3F_0}{2A\rho_0 g}$

Passage 5

A thin-walled open box of height H floats in water with its height h lying outside water. The cross-section of the tank is rectangular with dimension $a \times b$. A small hole of cross-sectional area A develops in the base of the box and water begins to enter the box.



12. The height difference between level of water outside and inside the box

(a) remains constant till the brim of the box reaches the outside water level

(b) increases till the brim reaches the outside water level

(c) decreases till the brim reaches the outside water level

(d) increases or decreases depending on the relative density of material of the box

13. Speed at which water enters the hole when water inside the box collects to a height $\dfrac{h}{2}$ is

(a) $\sqrt{g(H-h)}$ (b) $\sqrt{g\left(H - \dfrac{h}{2}\right)}$

(c) $\sqrt{2g(H-h)}$ (d) $\dfrac{1}{2}g(H-2h)$

14. Time t in which the box will sink is

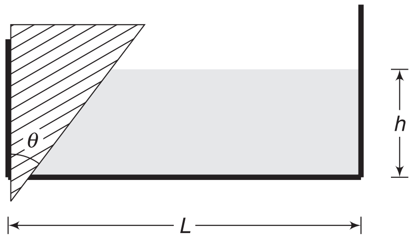
(a) $\dfrac{abh}{\sqrt{gH}}$ (b) $\dfrac{abh^2}{A^2 g(H-h)}$

(c) $\dfrac{abh}{A\sqrt{2g(H-h)}}$ (d) $\dfrac{abH}{A\sqrt{g(H-h)}}$

Passage 6

A rectangular tank has length L and width b. A slit is made at the bottom of the tank at its left edge. A wooden wedge is used to close the slit, as shown in the figure. The width of the wedge (perpendicular to the figure) is exactly b and it fits perfectly in the slit. Wedge has mass m and apex angle θ.

Its vertical surface is in contact with the dry left wall of the container. Coefficient of friction between the container wall and the wedge is μ. Density of water is ρ and it is filled to a height h in the tank. Neglect atmospheric pressure.



15. Normal contact force between the wedge and tank wall is

(a) $\frac{1}{2}\rho gbh^2$ (b) ρgbh^2

(c) $\frac{1}{2}\rho gbh^2\tan^2\theta$ (d) $\rho gbh^2\cot^2\theta$

16. Vertical component of force applied by water on the wedge is

(a) $\frac{1}{2}\rho gbh^2$ (b) ρgbh^2

(c) $\frac{1}{2}\rho gbh^2\tan^2\theta$ (d) $\rho gbh^2\cot^2\theta$

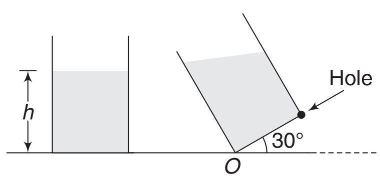
17. Maximum height h to which water can be filled in the tank without leakage from the slit is

(a) $\sqrt{\dfrac{m}{\rho b(\tan\theta-\mu)}}$ (b) $\sqrt{\dfrac{2m}{\rho b(\tan\theta-\mu)}}$

(c) $\sqrt{\dfrac{m\tan\theta}{\rho g\,(1-\mu)}}$ (d) $\sqrt{\dfrac{2m\tan\theta}{\rho b(1-\mu)}}$

Passage 7

Water is filled in a container having square base (2 m × 2 m). Height of water in the container is $h = 2$ m. Now the container is tilted so that its base makes an angle of 30° with the horizontal. A small hole is made at the bottom of the right wall of the container. Water ejects from the hole in horizontal direction [$g = 10\,\text{ms}^{-2}$]



18. The water surface in the tilted container makes angle θ with horizontal. Value of θ is

(a) 30° (b) 60°

(c) 15° (d) 0

19. Distance from point O where the water jet hits the horizontal ground is nearly

(a) 5.24 m (b) 6.27 m

(c) 4.90 m (d) 3.95 m

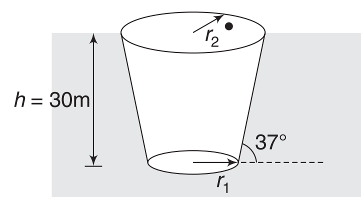
20. The tilted container is accelerated to right with a horizontal acceleration of $a = 2\,\text{ms}^{-2}$. Now the water surface makes an angle α with horizontal. Value of α is

(a) $\tan^{-1}\left(\frac{1}{8}\right)$ (b) $\tan^{-1}\left(\frac{1}{2\sqrt{2}}\right)$

(c) 15° (d) $\tan^{-1}\left(\frac{1}{5}\right)$

Passage 8

An object is in the shape of a truncated cone. It is floating in a lake with its upper surface just exposed to the atmosphere. The radii of upper and lower surfaces are r_2 and r_1 respectively and the height of the object is $h = 30$ m. The slant surface of the object makes 37° with the horizontal. It was observed that the magnitude of force applied by water on the slant (curved) surface is equal to weight of the object.



Density of water, $\rho = 10^3\,\text{kgm}^{-3}$, Atmospheric pressure $P_0 = 10^5\,\text{Nm}^{-2}$.

21. Magnitude of force applied by water on the slant curved surface of the object

(a) will not change if the object is inverted upside down

(b) will increase if the object is tilted upside down

(c) will decrease if the object is tilted upside down

(d) None of the above

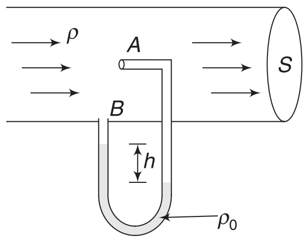
22. Value of r_1 is

(a) 10 m (b) 15 m

(c) 30 m (d) 40 m

Passage 9

Pitot tube is used in wind tunnel experiments and in airplanes to measure flow speed. The diagram shows a very simple pitot tube mounted along the axis of a gas pipeline having cross-sectional area S. Flowing gas has density ρ. The pitot tube has a liquid of density ρ_0. The mouth A of the tube is held normal to the direction of flow. The gas inside the tube is essentially at rest. Difference in height of liquid column in two arms of the pitot tube is h.



23. P_A is pressure just inside the mouth of the tube at A and P_B is pressure inside the mouth of the tube at B. Then,

(a) $P_A > P_B$ (b) $P_A = P_B$
(c) $P_A < P_B$ (d) cannot be said

24. Volume of gas flowing across the section of pipe per unit time is

(a) $S\sqrt{\dfrac{\rho_0 gh}{\rho}}$ (b) $S\sqrt{\dfrac{2\rho_0 gh}{\rho}}$

(c) $S\sqrt{\dfrac{\rho gh}{2\rho_0}}$ (d) $S\sqrt{\dfrac{\rho_0 gh}{2\rho}}$

Passage 10

Consider a fluid flowing through a narrow tube in steady flow. Because of viscosity, the layer in contact with the wall of the tube remains at rest and the layers away from the wall move fast. If pressure difference across the ends of a tube of length l is ΔP, then $\dfrac{\Delta P}{l}$ is defined as pressure gradient along the tube. It has been found that the volume of fluid flowing through the tube in unit time (call it volume flow rate Q) is dependent on pressure gradient, radius of the tube (r) and coefficient of viscosity (η) of the fluid.

In one experiment, the speed of flow along the axis of tube of radius r_0 was measured to be v_0. Length of the tube was l_0 and viscosity of the fluid used was η. Velocity gradient from the tube wall to its axis is assumed to be linear.

25. The volume flow rate (Q_0) for the tube mentioned above is

(a) $\pi v_0 r_0^2$ (b) $\dfrac{1}{2}\pi v_0 r_0^2$

(c) $\dfrac{1}{3}\pi v_0 r_0^2$ (d) $\dfrac{1}{4}\pi v_0 r_0^2$

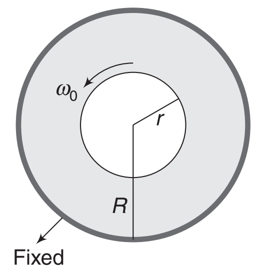
26. The pressure difference across the tube mentioned above can be expressed as (R is a dimensionless constant)

(a) $\dfrac{Q_0 l_0 \eta_0}{k r_0^4}$ (b) $\dfrac{Q_0 l_0 \eta_0}{r_0^3}$

(c) $\dfrac{Q_0 l_0^2 \eta_0}{k r_0^5}$ (d) $\dfrac{Q_0 l_0^2 \eta_0}{k r_0}$

Passage 11

A hollow cylinder has inner radius R and is held fixed. A co-axial solid cylinder of radius r and length l is inserted inside it. The solid cylinder has mass m and is free to rotate about its axis. Space between the two cylinders is filled with a liquid of viscosity η. The solid cylinder is imparted an angular speed ω_0 and released. Assume that velocity gradient in the liquid between the walls of the two cylinders is uniform. Also, assume the viscous force to act only on the curved surface of the cylinder.



27. Retarding torque acting on the cylinder when its angular speed reduces to ω is given be $\tau = k\omega$. Value of k is

(a) $\dfrac{2\pi\eta L r^3}{(R-r)}$ (b) $\dfrac{\pi\eta L r^3}{(R-r)}$

(c) $\dfrac{2\pi\eta L r^2 k}{(R-r)}$ (d) $\dfrac{\pi\eta L R^2 r}{(R-r)}$

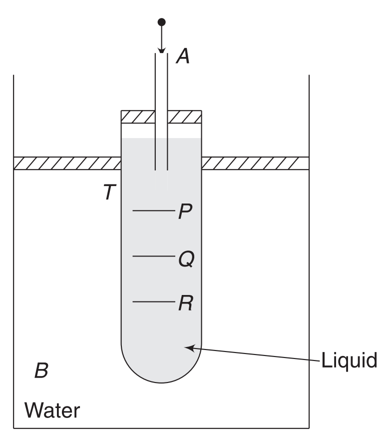
28. Angular speed at time t after the start of motion is

(a) $\omega_0 e^{\frac{kt}{mr^2}}$ (b) $\omega_0 e^{-\frac{2kt}{mr^2}}$

(c) $\omega_0\left[1-\dfrac{kt}{mr^2}\right]$ (d) $\omega_0\left[1-\dfrac{2kt}{mr^2}\right]$

Passage 12

Viscosity of a highly viscous liquid can be measured with an arrangement shown in the figure. A test tube (T) contains experimental liquid and is fitted into a water bath (B). A thermometer is used to measure the temperature of the water bath (thermometer has not been shown in the diagram). A tube A is fitted in the cap of test tube T. There are three marks – P, Q and R – on the test tube well below the lower end of tube A. Distance between the marks is such that $PQ = QR = d\cdot$ A spherical metal ball of mass m and radius r is dropped in tube A. With the help of a stop watch, time interval taken by the ball to cross the length PQ and QR are recorded. Let these time intervals be t_1 and t_2 respectively. ρ is the density of the experimental liquid. Different balls are tried till we get $t_1 = t_2$.



29. Which of the following is true in the context of the experiment?

(a) We can do away with the step of measuring the temperature of the water bath. This step is not relevant.

(b) If the time interval $t_1 \neq t_2$ then we should try a different ball of slightly smaller radius.

(c) We can have $t_1 > t_2$ for some specific balls.

(d) Highly viscous liquid ensures that the balls never attain terminal speed while falling.

30. For a ball of radius r_0 and mass m_0, it was recorded that $t_1 = t_2$ (= t_0, say). The coefficient of viscosity of the liquid at the temperature of experiment is

(a) $\dfrac{2}{9}\dfrac{t_0 g r_0^2}{d}\left[\dfrac{3m}{4\pi r_0^3} - \rho\right]$

(b) $\dfrac{2}{9}\dfrac{t_0 g r_0^2}{d}\left[\dfrac{3m}{2\pi r_0^3} - \rho\right]$

(c) $\dfrac{2}{9}\dfrac{t_0 g r_0}{d}\left[\dfrac{3m}{4\pi r_0^2} - \rho\right]$

(d) $\dfrac{t_0 g r_0^2}{d}\left[\dfrac{m}{6\pi r_0^3} - \rho\right]$

Answers Sheet

Match the Columns

1. (a) p, s (b) p, q (c) r (d) p, q, s
2. (a) q (b) r (c) p (d) s
3. (a) r (b) q (c) q (d) q
4. (a) p (b) q (c) t (d) s
5. (a) r (b) q (c) p (d) r
6. (a) p, r (b) p, q, s (c) p, r, s (d) p, r
7. (a) s (b) r, t (c) q (d) p
8. (a) r (b) r (c) q (d) p

Passage-based Problems

1. (c)	2. (a)	3. (d)	4. (c)	5. (a)	6. (a)	7. (c)	8. (a)	9. (c)
10. (a)	11. (b)	12. (a)	13. (c)	14. (c)	15. (a)	16. (c)	17. (b)	18. (d)
19. (d)	20. (d)	21. (b)	22. (d)	23. (a)	24. (b)	25. (c)	26. (a)	27. (a)
28. (b)	29. (b)	30. (a)						

CHAPTER 11

Gravitation

"Pick a flower on Earth and you move the farthest star."

–*Paul Dirac.*

1. INTRODUCTION

It will not be wrong to say that Newton did not discover gravitation. What Newton discovered was that gravitation is universal – that it is not special to Earth, as others of his time assumed. Newton published his brilliant theory in 1687.

Before 1687, a large amount of data had been collected regarding motions of the moon and the planets. Tycho Brahe (1546–1601) and Johannes Kepler (1571–1630) need special mention. This teacher – disciple pair studied the planetary motion in great detail. Kepler formulated three laws of planetary motion, based on large amount of data collected. He stated that all planets go around the sun in elliptical orbits. [We will take up the three laws later in this chapter.] But clear understanding of the motion of the planets and moon was not available till Newton gave his law of universal gravitiation.

In this chapter, we will study the law of universal gravitation. We will realise that it is the same force that causes an apple to fall downwards and also causes the moon to revolve around earth. It is the same universal gravitational force that is holding a galaxy of stars together and is trying to hold together the entire universe (which is expanding).

We will learn about the motion of satellites and planets and see how Kepler's findings could be explained in terms of Newton's law of gravitation.

2. NEWTON'S LAW OF GRAVITATION

Newton was aware that the Earth – Moon distance is close to r = 3, 85,000 km and it takes T = 27.3 days for the moon to make a revolution around the earth. Acceleration of the moon (directed towards the earth) is

$$a = \omega^2 r = \left(\frac{2\pi}{T}\right)^2 \cdot r$$

$$= \frac{4\pi^2 \times 3.85 \times 10^8 \text{m}}{(27.3 \times 24 \times 60 \times 60\,\text{s})^2}$$

$$\simeq 0.0027\,\text{ms}^{-2}$$

From his laws of motion, Newton knew that acceleration can be produced only if there is a force and direction of acceleration is in the direction of the force. A natural guess is that the Earth is attracting the Moon.

According to popular legend, he was sitting under an apple tree when an apple fell from the tree. The falling apple (and the preoccupied mind of Newton, thinking about the acceleration of the Moon) sparked the idea that the Earth attracts all objects towards itself, including the Moon.

The prevailing notion, at that point of time, was that there were two sets of natural laws: one for earthly objects, and another, altogether different, for heavenly objects. Newton's intuition that the force between an apple and the earth is the same force that pulls the moon towards earth was a revolutionary breakthrough.

Acceleration of a body (such as an apple) falling near the earth's surface is 9.8 ms^{-2}. Therefore,

$$\frac{a_{\text{apple}}}{a_{\text{moon}}} = \frac{9.8}{0.0027} = 3600 \qquad \text{...(i)}$$

To understand how the acceleration is affected by distance, let us calculate the ratio of distance of the apple and the moon from the centre of the earth.

$$\frac{r_{\text{apple}}}{r_{\text{moon}}} = \frac{6400\ \text{km}}{3.85 \times 10^5\,\text{km}} \simeq \frac{1}{60} \qquad \text{...(ii)}$$

Note that the apple is close to the surface of the earth and its distance from the centre of the earth is equal to the radius of the earth = 6,400 km.

From (i) and (ii), it can be concluded that

$$\frac{a_{\text{apple}}}{a_{\text{moon}}} = \left(\frac{r_{\text{moon}}}{r_{\text{apple}}}\right)^2$$

Newton guessed that acceleration of a body towards earth is inversely proportional to the square of distance from the centre of the earth.

$$a \propto \frac{1}{r^2}$$

Since force is mass times acceleration, force on a body of mass *m* due to the Earth is

$$F \propto \frac{m}{r^2}$$

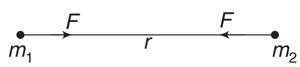
From Newton's third law, force applied by the earth on a body must be equal to the force applied by the body on earth. Therefore, this force must be proportional to the mass of earth as well.

$$\therefore \qquad F \propto \frac{mM}{r^2}$$

Newton generalised the law by saying that all material bodies in the universe attract one another. His famous law of gravitation can be stated as:

Any two particles in this universe attract each other along the line joining them. Force between them is proportional to the product of their masses and inversely proportional to the square of the distance between them.

$$F = G\frac{m_1 m_2}{r_2} \qquad \text{...(1)}$$



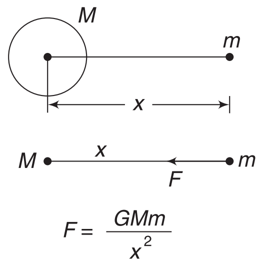
G is a constant, called *Universal Constant of Gravitation* and its value is found to be $6.673 \times 10^{-11}\ \text{Nm}^2\,\text{kg}^{-2}$.

It is important to remember the following points:

(i) Newton's law of gravitation gives the force between two point-masses. For writing gravitational force between two pens, you cannot directly use equation (1). The pens are not point masses. If the two pens are 1km apart, then they can be treated as point masses, as distance between them is large compared to their size. There is nothing wrong is treating earth to be a point mass while writing the force on it due to the sun.

(ii) A spherically symmetrical body can be replaced by a point particle of equal mass placed at its centre for the purpose of calculating gravitational force on an outside object. We will return to this point later in this chapter. It needs a good amount of calculus to prove this. What this essentially means is that all celestial objects (which are spherical) can be treated as point masses at their centres for the purpose of calculating the mutual force between them.

Force applied by the earth on an apple can be calculated by assuming the entire mass of earth to be located at its centre, like a point mass.



A spherically symmetric object can be replaced with a point object of equal mass placed at its centre

(iii) Force between two objects is independent of the medium between the particles.

(iv) Gravitational force is conservative in nature.

Example 1 *Gravitation is a weak force*

Two small bodies, each of mass 1 kg, are placed 1m apart. Find the gravitational force they exert on one another. Assuming that the only force acting on the particles is mutual gravitation, find their initial acceleration.

Solution

Concepts

$$F = \frac{Gm_1 m_2}{r_2}$$

Gravitational force applied by one particle on another is

$$F = \frac{Gm_1 m_2}{r_2}$$

$$= \frac{6.67 \times 10^{-11}\ \dfrac{\text{Nm}^2}{\text{kg}^2} \times (1\,\text{kg}) \times (1\,\text{kg})}{(1\,\text{m})^2}$$

$$= 6.67 \times 10^{-11}\,\text{N}$$

$$\text{Acceleration,} \qquad a_1 = \frac{F}{m_1} = \frac{6.67 \times 10^{-11}}{1}$$

$$= 6.67 \times 10^{-11}\,\text{ms}^{-2}$$

Acceleration of the other particle is also of same magnitude.

Note: Gravitational forces between objects of daily life are too small to be noticeable. Gravitational force is a weak force. It becomes significant only when at least one of the two masses is huge. For example, the force between your body and the earth or the force between the sun and the earth is large and meaningful. There is no point in discussing the gravitational force between a proton and an electron in an atom.

YOUR TURN

Q.1 Distance between two masses is halved. How does the gravitational force between them change?

Q.2 There is attractive force between all objects. Why do we not feel ourselves gravitating towards massive buildings in our close proximity?

Q.3 Heavier bodies are attracted more strongly by Earth. Why don't they fall faster than lighter bodies?

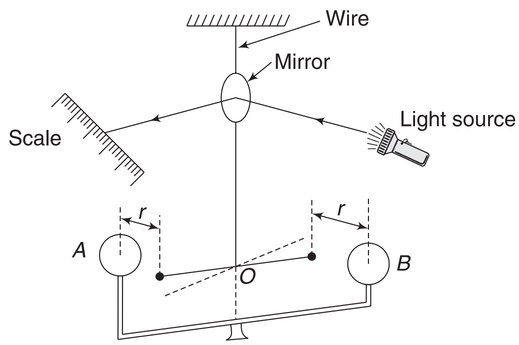
Q.4 Find gravitational force between the earth and the sun.

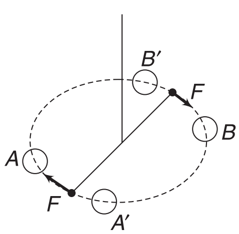
$$M_{sun} = 2.0 \times 10^{30}\,\text{kg}; \; M_{earth} = 6 \times 10^{24}\,\text{kg};$$

Earth–Sun distance, $r = 1.5 \times 10^{11}$ m

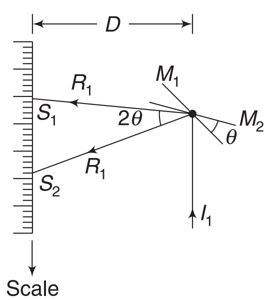
3. MEASUREMENT OF GRAVITATIONAL CONSTANT *G*

Value of G was first measured in 1798 by a gifted English scientist, Henry Cavendish (1731–1810). Cavendish made many contributions to science but his measurement of G was the most prolific of all. The experiment required a very delicate set up.





Pull of larger balls on the smaller balls produces a couple



When mirror rotates from position M_1 to M_2 (by θ), the reflected ray rotates from R_1 to R_2 (by angle 2θ). The spot of light on the scale moves from S_1 to S_2. θ can be calculated by measuring the distance S_1S_2

Two small balls, each of mass m, are attached to the ends of a light rod to form a dumb bell. The rod is suspended by a fine fibre or a thin metal wire. There is a small mirror attached to the wire, which reflects a sharp beam of light incident on it on to the scale fixed at some distance D from the mirror. Initially, there is no twist in the wire.

Two heavy spheres (A and B), of mass M each, are brought near the smaller spheres such that the centres of the four spheres fall on a horizontal circle. Let distance between the centres of a heavy ball and the smaller ball near it be r.

Gravitational pull of larger ball on the smaller ball is

$$F = \frac{GMm}{r^2}$$

Torque on the dumb bell due to gravitational force is

$$\tau = 2Fl \qquad [2l = \text{length of the dumb bell}]$$

$$= \frac{2GMml}{r^2}$$

This torque causes the dumb bell to rotate and the wire gets twisted. The twisted wire produces a counter-torque. After some time, an equilibrium is established with torque produced by wire balancing the gravitational torque.

If a wire is twisted by θ, the torque it develops is given by $\tau = k\theta$, where k is a constant for the given wire, known as its torsional constant.

If equilibrium is attained with wire twisted by θ,

$$\frac{2GMml}{r^2} = k\theta$$

$$\Rightarrow \qquad G = \frac{kr^2\theta}{2Mml} \qquad \text{...(i)}$$

How do we measure θ? If the wire turns by θ, the mirror attached to it also turns by θ. The spot of the light on the scale moves from S_1 to S_2. We measure S_1S_2. If D is the distance between the mirror and the scale, then $\theta \simeq \dfrac{S_1S_2}{D}$.

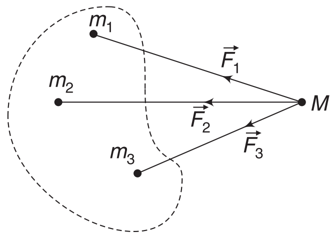
[Note that when mirror rotates by θ, the reflected ray rotates by 2θ. We will learn more about it in optics]

Since θ is a very small angle, its measurement is going to be difficult. In an actual experiment, we allow the dumb bell to settle and then move the larger balls so as to place them on opposites sides of the smaller balls, at same distance (r). See the figure. Larger balls are moved from position A, B to A', B'. As the heavy balls are shifted to a new position, the dumb bell rotates. It will settle at angle θ from mean position on the other side. Thus, the mirror rotates through 2θ (and reflected light rotates by 4θ) as the heavier balls are moved to a new position. The spot of light moves a larger distance and hence 4θ can be easily measured.

Putting all the values in (i) gives us the value of universal gravitational constant, G.

4. PRINCIPLE OF SUPERPOSITION

Consider fixed particles having masses m_1, m_2, m_3 ... We wish to write net force due to all of them on another particle of mass M.



Let $\vec{F}_1$ = force on M due to m_1, assuming that only m_1 and M are present (m_2, m_3 etc. are not there)

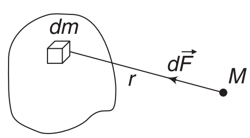
$\vec{F}_2$ = force on M due to m_2 when only M and m_2 are there and similarly, $\vec{F}_3$, $\vec{F}_4$ ... are forces due to individual particles on M when all other particles are missing.

The principle of superposition says that net gravitational force on M is obtained by vector sum of $\vec{F}_1$, $\vec{F}_2$, $\vec{F}_3$ ... etc.

$$\therefore \qquad \vec{F}_{net} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + ... \qquad ...(2)$$

In other words, the force between m_1 and M is independent of the presence or absence of other particles (m_2, m_3 ... etc)

If we wish to find force due to an extended body on a point mass M, then we divide the body into parts small enough to treat as particles. We need to write force due to each small part on M and add the forces vectorially.



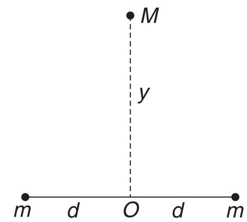
dm is a small mass in the body which applies force $d\vec{F}$ on M.

$$\vec{F} = \int d\vec{F}$$

Integration is performed so as to cover all the particles in the extended body.

Things would have been very difficult if force between two particles changed when a third particle was brought closer.

Example 2 Two point-masses, m each, are kept fixed at a separation $2a$. There is another point mass M placed on perpendicular bisector of the line joining two masses, at a distance y.



(a) Write net gravitational force (F) on M.

(b) Approximate the value of F for the condition $y \gg d$. Does the result convince you?

Solution

Concepts

(i) We will write force due to each m on M and add them. We must add the forces as vectors.

(ii) Here, we will resolve both the forces into two perpendicular components and then add them.

(iii) $y \gg d$ does not mean that $y \to \infty$.

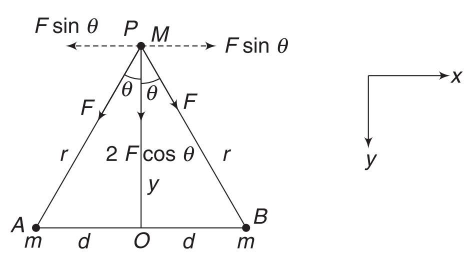
It only means that y is large compared to d. We can use approximations like $y \pm d \simeq y$, $y^2 \pm d^2 \simeq y^2$ etc.

(a) Force due to particle at A on the particle at P is

$$F = \frac{GMm}{r^2} \quad \text{(along } PA\text{)}$$

Force due to particle at B is

$$F = \frac{GMm}{r^2} \quad \text{(along } PB\text{)}$$



We resolve the two forces along x- and y-direction shown. The x-components cancel out and the y-components add up.

$\therefore$ Resultant force is $F_0 = 2F\cos\theta$ (in y-direction towards O)

$$\therefore \qquad F_0 = \frac{2GMm}{r^2}\cdot\frac{y}{r} = \frac{2GMmy}{r^3}$$

But $\qquad r = \sqrt{d^2 + y^2}$

$$\therefore \qquad F_0 = 2GMm \frac{y}{(d^2 + y^2)^{3/2}}$$

(b) For $\quad y \gg d$

$$y^2 + d^2 \simeq y^2$$

$$\therefore \quad F_0 \simeq 2GMm \frac{y}{(y^2)^{3/2}} = \frac{2GMm}{y^2}$$

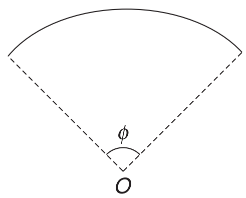
When M is placed at a large distance (y) compared to d, the two particles will appear to be very close, as seen from M. They will appear like a single point of mass $2m$, kept at a distance y. Thus, the force they will exert on M is

$$F_0 = \frac{GM(2m)}{y^2}$$

Therefore, the result obtained is absolutely convincing.

Example 3 *A circular arc*

A circular arc of radius r subtends an angle ϕ at the centre (O). The linear mass density of the arc is λ kgm^{-1}. Find the gravitational force it exerts on a point mass m, kept at O.



Solution

Concepts

(i) We will consider a small element on the arc subtending a very small angle $d\theta$ at the centre. We can write force on m due to this element.

(ii) We need to find vector sum of forces due to all small elements. These forces have varying directions.

(iii) Considering another symmetrical element (as shown below) helps us to visualise that direction of resultant force is along the line of symmetry (y-axis).

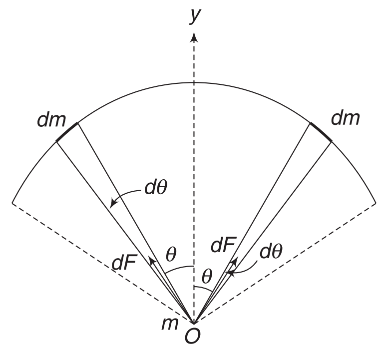
We need to take component of force due to each element in y-direction and add them. Addition will be done using integration.

Consider two elements at angle θ from y-axis as shown. Angular width of each element is $d\theta$.

Length of each element = $rd\theta$

Mass of each element = $\lambda rd\theta$

Force due to each element on mass m at O is



$$dF = \frac{Gm(\lambda r\,d\theta)}{r^2} = \frac{Gm\lambda}{r}\,d\theta$$

Direction of the two forces are as shown.

Obviously, the x-components of these two forces cancel out and y-components will add.

x-component of forces by any two symmetrical elements will cancel out. Resultant force is in y-direction.

y-component of force due to one element is

$$dF_y = dF\cos\theta = \frac{Gm\lambda}{r}\cos\theta\,d\theta$$

$\therefore$ Resultant force is

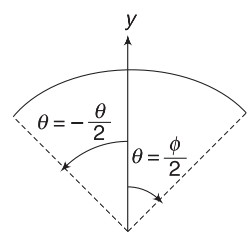
$$F = \int dF_y$$

$$= \frac{Gm\lambda}{r}\int_{-\phi/2}^{\phi/2}\cos\theta\,d\theta$$

$$\therefore \quad F = \frac{Gm\lambda}{r}[\sin\theta]_{-\phi/2}^{\phi/2}$$

$$= \frac{Gm\lambda}{r}\left[\sin\frac{\phi}{2} - \sin\left(-\frac{\phi}{2}\right)\right]$$

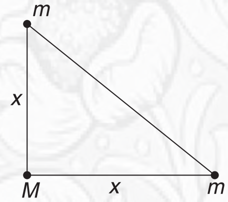
$$F = \frac{2Gm\lambda}{r}\sin\frac{\phi}{2}$$



θ is measured from y-direction. We have taken clockwise angle as positive

YOUR TURN

Q.5(a) Three particle of masses m, m and M are placed at the vertices of an isosceles right-angled triangle, as shown in figure. Equal sides of the triangle have length x. Find gravitational force on M.



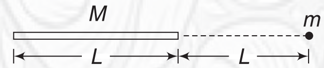
(b) Three point masses, m each, are placed at the vertices of an equilateral triangle of side a. Find net gravitational force on any one mass.

Q.6 Each of the five vertices of a regular pentagon has a point mass m kept at them. A sixth particle of mass M is placed at the centre. Distance of each vertex from the centre is x. Find

(a) Gravitational force on M

(b) magnitude of gravitational force on M, if a particle is removed from one of the vertices.

Q.7 A point mass m is at a distance L from one end of a uniform rod of mass M and length L. Find gravitational force on the point mass due to the rod.



5. GRAVITATIONAL FIELD

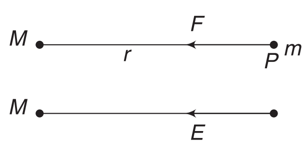
The sun applies force on the earth. They are not in contact. Similarly, two magnets, which are at some distance, apply force on one another. These are not contact forces (like tension in a string or normal reaction). These are *action at a distance*. In physics, such action at a distance forces are often called as *field forces*. Such forces are described in terms of something known as field.

The gravitational force between two masses is assumed to be set up in two steps – (i) a body A creates a gravitational field in the space around it. (Think of it as some sort of invisible change that the body creates around itself). (ii) when a body B is placed near A, this fields exerts a force on it.

If a point mass m placed at a point P experiences a gravitational force $\vec{F}$, then we define *intensity of gravitational field* ($\vec{E}$) at point P as

$$\vec{E} = \frac{\vec{F}}{m} \qquad \text{...(3)}$$

In a much simpler language, the intensity of gravitational field at a point is the gravitational force experienced by a unit mass placed at that point. Quite often, the intensity of gravitational field is abbreviated as *gravitational field*. Its SI units is N kg^{-1} (which is same as ms^{-2}).

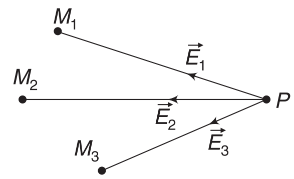


Consider a fixed point mass M. It produces field around itself. What is the intensity of field at a point P, which is at distance r from M? A mass m placed at P will experience a force $F = \frac{GMm}{r^2}$ towards M. Thus, gravitational field at P due to M is

$$E = \frac{F}{m} = \frac{GM}{r^2} \text{ directed towards } M \qquad \text{...(3a)}$$

This is the gravitational field due to a point mass at a distance r from it. It is always directed towards the point mass.

The field obeys the principle of superposition. If M_1, M_2, M_3 .... etc., are fixed point masses, then together, they produce a field at P, given by



Field at a point is superposition (addition) of field due to individual masses

$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \ldots$$

Where $\vec{E}_1$, $\vec{E}_2$, $\vec{E}_3$, etc., are fields due to individual point masses.

Example 4 What is gravitational field intensity due to the earth at a point close to its surface?

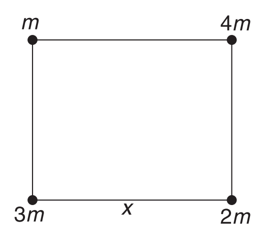
Solution Force on a particle of mass m near the surface of the Earth is

$F = mg$ towards the centre of the earth (i.e., vertically downward direction)

$\therefore$ Gravitational field, $E = \frac{F}{m} = g$

Thus, intensity of gravitational field near the surface of the earth is equal to the acceleration due to gravity.

Example 5 A square of side length x has four particles placed at its vertices, as shown in the figure. Find gravitational field at the centre of the square. How much force does point mass M, placed at the centre of the square, experience?



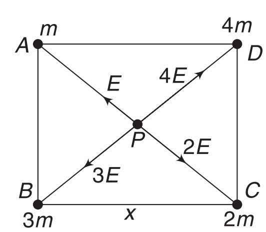
Solution

Concepts

We will write field due to individual masses and then add them vectorially.

Let $AP = r = \frac{x}{\sqrt{2}}$

Field at P due to m is



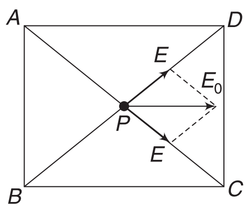
$$E = \frac{GM}{r^2} \text{ (along } \overrightarrow{PA}\text{)}$$

Field at P due to $2m$, $3m$ and $4m$ are

$$\frac{G2m}{r^2} = 2E \text{ (along } \overrightarrow{PC}\text{)}$$

$$\frac{G3m}{r^2} = 3E \text{ (along } \overrightarrow{PB}\text{)}$$

$$\frac{G4m}{r^2} = 4E \text{ (along } \overrightarrow{PD}\text{)}$$



Resultant of E and $2E$ is E along $\overrightarrow{PC}$ and resultant of $4E$ and $3E$ is E along $\overrightarrow{PD}$.

Resultant of these two fields is parallel to $\overrightarrow{BC}$ and has magnitude

$$E_0 = \sqrt{2}\,E = \frac{\sqrt{2}\,Gm}{r^2} = \frac{\sqrt{2}\,Gm}{\left(\frac{x}{\sqrt{2}}\right)^2} = \frac{2\sqrt{2}\,Gm}{x^2}$$

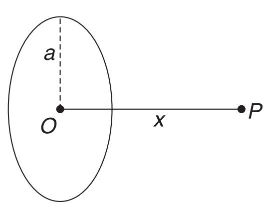
If we place a unit mass at P, it will experience a net gravitational force equal to E_0 in the direction parallel to $\overrightarrow{BC}$. If a mass M is kept at P, force on it is

$$F = ME_0 = \frac{2\sqrt{2}\,GmM}{x^2} \text{ along } \overrightarrow{BC}$$

Note: Take a note of how we are thinking. We said that the four masses at A, B, C and D created a field at P and now the field applies a force on a mass (M) placed at P.

Example 6 *Gravitational field on the axis of a ring*

A uniform ring has mass M and radius a. Calculate the gravitational field due to this ring at a point lying on its axis, at a distance x form its centre.



Solution

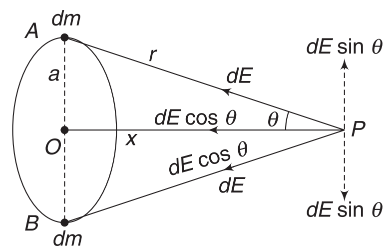
Concepts

(i) Due to symmetry, the field at P will be along the axis directed towards O. This can be seen easily by considering two identical mass elements on the ring at diametrically opposite ends.

(ii) Summing up the component of field along PO due to various element on the ring gives the resultant field.

Consider a small element of mass dm at A. Field due to this small mass at P is

$$dE = \frac{Gdm}{r^2} \text{ directed along } PA$$



Consider identical element at B. Field due to this element at P is also dE, directed along PB.

Components of dE in direction perpendicular to the axis cancel out. This is true for all pairs of masses at diametrically opposite ends.

Resultant field is obtained by adding components along the axis due to each elemental mass on the ring.

$$\therefore \quad E = \int dE \cos\theta = \int G\frac{dm}{r^2}\cdot\frac{x}{r} = \frac{Gx}{r^3}\int dm$$

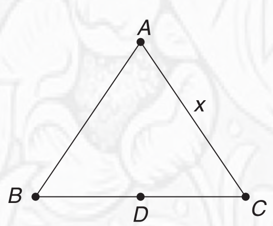
$$= \frac{GMx}{r^3} = \frac{GMx}{(a^2+x^2)^{3/2}} \text{ along } PO$$

Notice that r is same for all elemental masses on the ring.

Note: If you place a mass m at P, it will experience a gravitational force mE towards O.

YOUR TURN

Q.8 An equilateral triangle has point mass placed at its each vertex. Each mass is m and side length of the triangle is x. Find gravitational field intensity at mid-point (D) of side BC.



Q.9 Find field at the centre of a uniform ring of mass M and radius R.

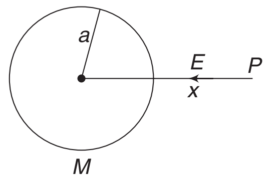
Q.10 Find field at the centre of a uniform semi-circular ring of mass M and radius R.

5.1 Gravitational Field Due to Spherical Bodies

Celestial bodies are spherical in shape. In gravitation, spherical geometry is most important. Here are two results related to spherical geometry, without proof.

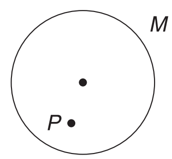
We will learn how to prove this in chapter of Electrostatics.

1. A thin uniform spherical shell can be treated as a point particle of same mass, placed at its centre, for calculation of gravitation field at an external point. Field at Point P due to shell of mass M shown in the figure is



$$E = \frac{GM}{x^2}$$

2. Gravitational field inside a uniform spherical shell is zero due to the mass of the shell. If a mass is placed inside a spherical shell at point P, it will experience no force.

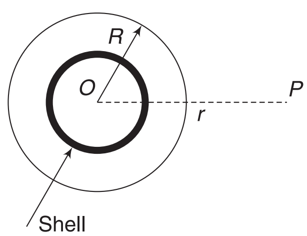


5.1.1 Gravitational field due to uniform solid sphere

Using the above two results, we can easily find the gravitational field indensity due to a uniform solid sphere.

Field at an external point

Consider a solid sphere of mass M and radius R. We need to find the gravitational field at an outside point P at a distance r from centre O.



The sphere can be divided into numerous thin shells centred at O. One such shell is shown in the figure. Field at P due to any such shell can be obtained by replacing the shell with a point mass at O. Thus, all such shells can be replaced with a point mass at O. It implies that the entire sphere of mass M can be replaced with a point mass M placed at O.

$\therefore$ Field at P is

$$E = \frac{GM}{r^2} \qquad \text{...(4)}$$

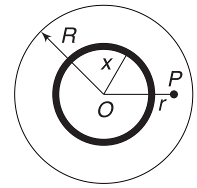
Field on the surface of the sphere is

$$E_{\text{surface}} = \frac{GM}{R^2} \qquad \text{...(5)}$$

Field at an internal point

P is a point inside a solid sphere of mass M and radius R. Distance of P from centre (O) is r.

Once again, we will think of the sphere as composed of many concentric shells. Suppose mass of one such shell is dm and its radius is x.



If $x < r$, the point P is outside the shell. For writing field at P due to the shell, we can replace the shell with a point mass at O.

If $x > r$, the point P is inside the shell. Field at P due to such a shell is zero.

Thus, field at P is contributed by shells having radii $0 < x < r$. Mass of the all these shells can be through to be at O. Combined mass of these shells is equal to the mass of solid sphere of radius r and is given by

$$m = \text{(density)(volume)}$$

$$= \frac{M}{\frac{4}{3}\pi R^3}\cdot\frac{4}{3}\pi r^3 = \frac{Mr^3}{R^3}$$

Field at P = Field due to point mass m at O

$$\therefore \qquad E = \frac{Gm}{r^2} = \frac{GM}{R^3}r \qquad \text{...(6)}$$

Direction of the field is towards the centre of the sphere.

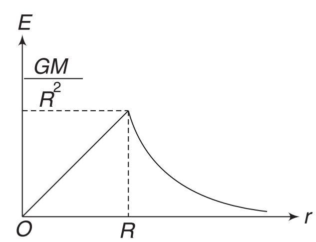
Example 7 E is gravitation field at a distance r from the centre of a uniform solid sphere of mass M and radius R. Plot a graph showing variation of E with r, for $0 \le r < \infty$.

Solution

Concepts

For inside points: $E = \dfrac{GM}{R^3}r$

For outside points: $E = \dfrac{GM}{r^2}$



For $\quad 0 \le r \le R$: $E = \left(\dfrac{GM}{R^3}\right)r$

$\Rightarrow \quad E \propto r$

At $\quad r = R$ (i.e., on surface)

$$E_{\text{surface}} = \frac{GM}{R^3}\cdot R = \frac{GM}{R^2}$$

For $\quad r \ge R$; $E = \dfrac{GM}{r^2}$

$$\Rightarrow \qquad E \propto \frac{1}{r^2}$$

Graph is as shown.

YOUR TURN

Q.11 E is gravitational field at a distance r from the centre of a uniform thin spherical shell of mass M and radius R. Plot a graph showing variation of E versus r, for $0 \le r < \infty$.

Q.12 Mass of the Moon is 7.36×10^{22} kg and its radius is 1.74×10^6 m. Assuming it to be a uniform sphere, find gravitational field on its surface.

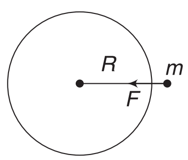
Q.13 Consider Earth to be a uniform sphere. At what depth from the surface of the earth, the gravitational field intensity is half its value on the surface? Radius of the earth is R.

6. ACCELERATION DUE TO GRAVITY (g)

The gravitational pull of earth on a body is often referred to as gravity. Acceleration produced by this force is called acceleration due to gravity. When a body of mass m is near the surface of Earth, force of gravity acting on it is

$$F = \frac{GMm}{R^2} \text{ towards the centre of earth}$$

Thus, acceleration due to gravity (denoted by g) near the surface of Earth is



$$g = \frac{F}{m} = \frac{GM}{R^2} \quad \text{...(7)}$$

The above expression has been written assuming Earth to be a uniform sphere.

Measured value of g near the surface of Earth is nearly 9.8 ms^{-2}.

Acceleration due to gravity (g) has the same value as gravitational field intensity due to Earth.

6.1 "Weighing" The Earth

Value of acceleration due to gravity (g) can be measured easily with the help of simple experiments. Radius of Earth (R) is also known. The day Cavendish measured the value of universal constant G, the mass of the Earth became known. It is often said that Cavendish was the first person to weigh the Earth.

Putting $g = 9.8\,ms^{-2}$, $G = 6.67 \times 10^{-11}$ Nm^2kg^{-2},

$R = 6.4 \times 10^6$ m in equation (7) gives the mass of Earth as

$$M \simeq 6 \times 10^{24}\,\text{kg}$$

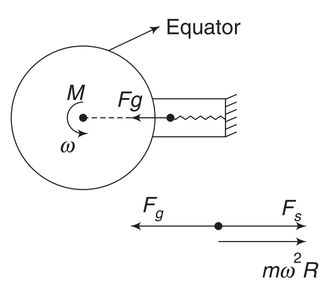
6.2 Variation in '*g*' on the surface of Earth

Acceleration due to gravity at a point on the surface of the Earth differs from the value predicted by equation (7), due to various reasons. Prominent reasons are:

1. **Non-uniform Earth:** Earth is not a uniform sphere. On the surface, we have mountains, valleys, seas, etc. Moreover, the density of Earth is not uniform. The density in the inner core is as high as 14 kgm^{-3} and the density of the mantle ranges from 3 $kg\,m^{-3}$ to 6 $kg\,m^{-3}$. The crust has density varying from region to region over the surface of Earth. Thus, g varies from region to region over the surface.

2. **Non-spherical Earth:** Equation (7) assumes that Earth is a sphere. We known that it is not. Earth is flattened at the poles and bulging at the equator. Its equatorial radius is larger than its polar radius by nearly 21 kilometres. A point on the pole is closer to the denser core of Earth as compared to a point on the equator. Accordingly, value of g is higher at poles than at the equator.

3. **Rotation of the Earth:** Earth is rotating about an axis running through its poles. Consider a ball of mass m suspended from the ceiling of a room using a spring balance, at a place on the equator.



Gravitational pull on the ball is

$$F_g = \frac{GMm}{R^2} = mg$$

If Earth were not rotating, the reading of the spring balance (i.e., tension in the spring) would have been mg.

However, in the reference frame of the rotating Earth, there is a centrifugal force ($= m\omega^2 R$) acting on the ball in a radially outward direction. Thus, the force applied by the spring balance on the ball is

$$F_S = F_g - m\omega^2 R$$

$$F_S = \text{Weight recorded by the balance} = mg'$$

Where g' is the effective acceleration due to gravity at the place.

$$\therefore \quad mg' = mg - m\omega^2 R$$

$$\Rightarrow \quad g' = g - \omega^2 R \qquad ...(8)$$

The above equation gives the value of *free fall acceleration* at the equator. It is less than the gravitational acceleration (g).

This effect is less pronounced at higher latitudes, mainly because the radius of the latitude circle decreases as one moves from the equator towards the pole. At poles, there is no effect of rotation of the Earth, as a point on the pole is rotating on a circle of zero radius.

Note that due to non-sphericity of the Earth also, value of g is higher at poles. In general, measured value of g is highest at poles and smallest at the equator.

You can also refer to our discussion in this regard in the chapter of circular motion.

Example 8 *Effect of rotation is small*

Find the percentage change in value of acceleration of free fall at a place on the equator due to rotation of Earth. Value of $\frac{GM}{R^2} = 9.80\ \text{ms}^{-2}$. Radius of the Earth $R = 6.4 \times 10^6\,\text{m}$.

Solution

Concepts

(i) Percentage change is $\frac{g' - g}{g} \times 100$

(ii) Angular speed of Earth is $\omega = \frac{2\pi \text{rad}}{24h}$

Change in g due to rotation

$$\Delta g = g' - g = (g - \omega^2 R) - g = -\omega^2 R$$

$$\text{Fractional change} = \frac{\Delta g}{g}$$

$$\text{Percentage change} = \frac{\Delta g}{g} \times 100 = -\frac{\omega^2 R}{g} \times 100$$

$$= -\left(\frac{2\pi}{24 \times 60 \times 60}\right)^2 \times \frac{6.4 \times 10^6}{9.8} \times 100 = -0.34\%$$

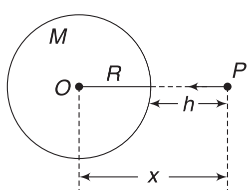
Negative sign indicates decrease.

YOUR TURN

Q.14 Suppose you could increase the angular speed of rotation of Earth. You go on increasing the speed and a man on the equator feels lighter and lighter. What will be the length of the day if he starts feeling weightlessness?

6.3 Acceleration due to Gravity at a Height from the Surface of the Earth

Consider a body of mass m located at a point P at a distance x ($>R$) from the centre of the earth. Height of point P from the surface is h.



$$x = R + h \qquad ...(i)$$

Gravitational force on m due to earth (mass = M) is

$$F = \frac{GMm}{x^2}$$

$\therefore$ Acceleration produced due to this force is

$$g' = \frac{GM}{x^2} \qquad ...(9)$$

Note that acceleration due to gravity has the same value as the gravitational field intensity due to earth. Also note that we have reserved the symbol g for $\frac{GM}{R^2}$, value of acceleration due to gravity on the surface. Any changed value of g is being represented by g'.

Using (i) in equation (9) gives

$$g' = \frac{GM}{(R+h)^2} = \frac{GM}{R^2\left[1 + \frac{h}{R}\right]^2}$$

$$\Rightarrow \quad g' = \frac{g}{\left(1 + \frac{h}{R}\right)^2} \qquad ...(10)$$

When $h \ll R$, we can approximate the above expression in the following manner:

$$g' = g\left[1 + \frac{h}{R}\right]^{-2}$$

Using binomial expansion:

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \ldots$$

$$g' = g\left[1 - \frac{2h}{R} + 3\left(\frac{h}{R}\right)^2 + \ldots\right]$$

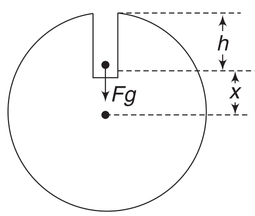
The terms having $\left(\frac{h}{R}\right)^2, \left(\frac{h}{R}\right)^3$, etc., are very small and can be neglected.

$$\therefore \qquad g' \simeq g\left(1 - \frac{2h}{R}\right) \qquad \ldots(11)$$

Note that equation (10) is the true equation and can be used for any value of h. Equation (11) is an approximated version of (10) *for the ease of calculation* and should be used only when $h \ll R$.

6.4 Acceleration Due to Gravity Inside the Earth

Imagine a particle kept inside a tunnel, at a depth h from the surface of the earth. Distance of the particle from the centre of the earth is



$$x = R - h$$

Acceleration experienced by the particle due to gravity will be same as the gravitational field at the point, which is given by equation (6).

$$\therefore \qquad g' = \frac{GM}{R^3}\cdot x \qquad \ldots(12)$$

$$\text{or,} \qquad g' = \frac{GM}{R^2}\frac{(R-h)}{R} = \frac{GM}{R^2}\left(1 - \frac{h}{R}\right)$$

$$\Rightarrow \qquad g' = g\left(1 - \frac{h}{R}\right) \qquad \ldots(13)$$

At the centre of the earth, $h = R$ and $g' = 0$. The variation of acceleration due to gravity with distance from the centre of the earth is same as depicted in the graph of Example 7.

Example 9 *Be careful while using equation (11)*

Acceleration due to gravity on the surface of Earth is g. Find its value at a height:

(a) 3,200 km (b) 32 km

from the surface of Earth. Radius of Earth is $R = 6{,}400$ km.

Solution

Concepts

(i) $g' = \dfrac{g}{\left(1 + \frac{h}{R}\right)^2}$

(ii) When $h \ll R$, $\quad g' \simeq g\left(1 - \frac{2h}{R}\right)$

(a) $g' = \dfrac{g}{\left(1 + \frac{3200}{6400}\right)^2} = \dfrac{g}{\left(1 + \frac{1}{2}\right)^2} = \dfrac{4}{9}g$

For $h = 3{,}200$ km, one should not use equation (11), as h is not small.

(b) 32 km $\ll$ 6400 km

$$\therefore \qquad g' \simeq g\left(1 - \frac{2h}{R}\right) = g\left(1 - \frac{2 \times 32}{6400}\right)$$

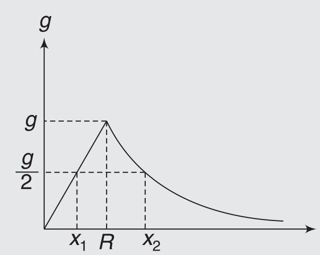
$$= g(1 - 0.01) = 0.99g$$

Example 10 At what distance from the centre of the earth is acceleration due to gravity half its value on the surface? Radius of Earth is R.

Solution

Concepts

Acceleration due to gravity changes with distance from the centre, as shown in the graph. There are two points (one inside the earth and the other outside it) where g is half its value on the surface.



For a point inside Earth, equation (12) gives

$$g' = g\frac{x}{R}$$

$$\Rightarrow \qquad \frac{g}{2} = g\frac{x_1}{R}$$

$$\Rightarrow \qquad x_1 = \frac{R}{2}$$

For a point outside Earth, equation (9) gives

$$g' = \frac{GM}{x^2} = \frac{GM}{R^2}\cdot\frac{R^2}{x^2}$$

$$\Rightarrow \qquad g' = g\cdot\frac{R^2}{x^2}$$

$$\therefore \qquad \frac{g}{2} = g\frac{R^2}{(x_2)^2} \quad \Rightarrow \quad x_2 = \sqrt{2}R$$

Your Turn

Q.15 At what depth below the surface of Earth is the value of acceleration due to gravity same as its value at a height $h = R$, where R is radius of Earth?

Q.16 At what height from the surface of Earth will the value of g get reduced by

(a) 36% (b) 0.36%

from its value at the surface? Radius of Earth = 6,400 km.

Q.17 Assuming Earth to be spherical, at what height above the south pole, acceleration of free fall is same as that on Earth's surface at the equator?

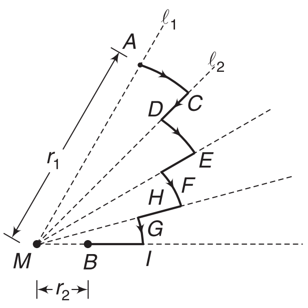
Q.18 Value of g on the surface of Earth is $9.8\,\text{ms}^{-2}$. Find its value on the surface of a planet whose density and radius both are twice that of Earth.

7. GRAVITATIONAL POTENTIAL ENERGY

Gravitational force is conservative. We can define a potential energy corresponding to it.

7.1 Gravitational Force is Conservative

Consider a fixed mass M. A point mass m is at A, at a distance r_1 from M. It is moved from A to B, where B is a point at a distance r_2 from M. The point mass is moved along a path $ACDEFGHIB$, where AC, DE, FG and HI are circular arcs centred at M and CD, EF. GH and IB are radial displacements towards M.



Obviously, no work is done by gravitational pull of M on m, in the segments AC, DE, FG and HI. Gravitational force is perpendicular to these displacements. Gravitational force does work in the segments CD, EF, GH and IB only.

Note that $CD + EF + GH + IB = r_1 - r_2$. Work done will remain same for any other path between A and B. Any path can be approximated as number of arcs and radial displacements. If the path is a smooth curve, it can be approximated as infinite number of radial and tangential (circular arc) displacements. Work is done by gravitational force only when there is a radial movement and this work done will depend on the distance of initial and final positions. It will not depend on whether the movement has taken place along radial line l_1 or l_2.

Therefore, we can say with certainty that work done by gravitational force between A and B is path independent and the force is conservative.

In Short:

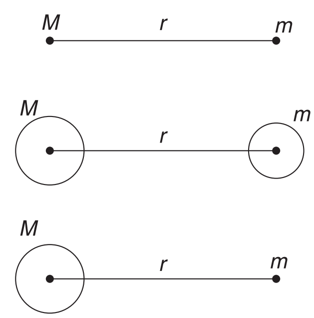
(i) Any two masses in the universe attract each other by a force known as gravitational force. Force between two masses is given by

$$F = \frac{GMm}{r^2}$$

when (a) both masses are point masses,

(b) both masses are spherical

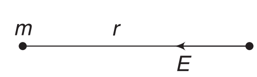
(c) one mass is spherical and other is point mass.



(ii) Gravitational force obeys principle of superposition. If force due to m_1 on M is $\vec{F}_1$ and that due to m_2 on M is $\vec{F}_2$, then net force on M in the presence of m_1 and m_2 both is $\vec{F}_1 + \vec{F}_2$.

(iii) Gravitational field at a point is defined as the force experienced by a unit mass placed at that point.

(iv) Field due to a point mass is given by $E = \dfrac{Gm}{r^2}$ Field is directed towards m.



(v) Gravitational field obeys the principle of superposition. If $\vec{E}_1$, $\vec{E}_2$, $\vec{E}_3$ are fields at a point P due to individual masses m_1, m_2 and m_3, then resultant field at P in presence of all three masses is

$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3$$

(vi) Field due to a uniform spherical shell at an inside point is zero. Field at an outside point ($r \geq R$) is

$$E = \frac{GM}{r^2}$$

(vii) Field at an inside point of a uniform solid sphere is

$$E = \frac{GM}{R^3}\cdot r \quad \text{where } r \leq R$$

At an outside point, $E = \dfrac{GM}{r^2}$ where $r \geq R$.

(viii) Acceleration due to gravity has the same value as gravitational field intensity at a point.

(ix) Acceleration due to gravity at various points is given by

(a) on surface of the Earth: $g = \dfrac{GM}{R^2}$

(b) At height h above the surface:

$$g' = \frac{g}{\left(1 + \frac{h}{R}\right)^2}$$

$$\simeq g\left[1 - \frac{2h}{R}\right] \quad \text{when} \quad h \ll R$$

(c) At depth h from the surface: $g' = g\left[1 - \dfrac{h}{R}\right]$

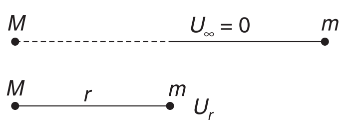
(x) Due to rotation of the earth, the value of free fall acceleration that we feel is slightly less than acceleration due to gravity. At equator:

$$g' = g - \omega^2 R$$

At poles, there is no effect of rotation.

7.2 Gravitational Potential Energy of Two Point-Masses

Gravitational *PE* of two point-masses is assumed to be zero when they are far apart (i.e., at infinite separation).



Let *PE* of the system be U_r when the masses are at separation r.

Based on your discussion in the chapter of work, energy and power, we know that change in *PE* is defined as negative of work done by the conservative force (here the gravitational force).

$$\therefore \qquad U_\infty - U_r = -W_g^{r\to\infty}$$

$$\Rightarrow \qquad 0 - U_r = -W_g^{r\to\infty}$$

$$\Rightarrow \qquad U_r = W_g^{r\to\infty} \qquad \text{...(14)}$$

In words, the PE of the system of two point-masses at separation r is equal to work done by the gravitational force in moving the masses to infinite separation from present separation r.

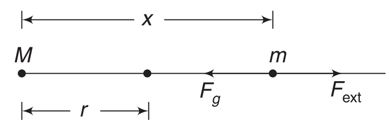
One can also define it as

$$U_r = W_g^{r\to\infty} = -W^{r\to\infty}_{\substack{\text{ext Agt}\\ \text{slowly}}} \left[= W^{\infty\to r}_{\substack{\text{ext Agt}\\ \text{slowly}}}\right] \qquad \text{...(15)}$$

In words, *the PE of a pair of point masses at separation r is negative of work done by external agent in slowly moving the masses from present separation to infinite separation*.

While calculating the work done, we can assume one of the masses (say M) to be fixed. It makes no difference to the *PE*, as it will depend only on separation r and not on the fact whether the masses are located in Delhi or in London.

Keeping M to be fixed, let us assume that m is moved away starting from a separation r.



When separation is x, the gravitational force is

$$F_g = \frac{GMm}{x^2}$$

Work done by gravitational force when m is further moved by dx is

$$dW_g = -F_g \cdot dx = -\frac{GMm}{x^2}\,dx$$

Work done by gravitational force when m moves from r to ∞ is

$$W_g^{r\to\infty} = -GMm\int_{x=r}^{\infty}\frac{dx}{x^2} = GMm\left[\frac{1}{x}\right]_r^{\infty}$$

$$= GMm\left[\frac{1}{\infty} - \frac{1}{r}\right] = -\frac{GMm}{r}$$

From equation (14)

$$U_r = W_g^{r\to\infty} = -\frac{GMm}{r} \qquad \text{...(15)}$$

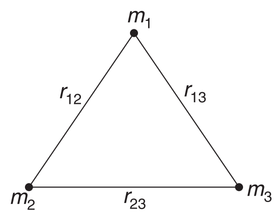
Note: 1. We wrote the gravitational *PE* of a body near the surface of Earth as *mgh*, where h was the height measured from a reference level. While writing the expression $U = mgh$, we considered the gravitational force to be constant, as h was small.

The expression (15) is very general in nature. It is not exclusive for Earth and another object. It is true in general for any pair of point masses in this universe.

2. When two particles are brought closer, the *PE* decreases (just like $U = mgh$ decreases when a mass falls towards Earth). Since $U = 0$ for $r = \infty$, the *PE* is negative (i.e., less than zero) for any finite separation and becomes progressively more negative as the particles move closer.

7.3 Gravitational PE of a System of Multiple Particles

When a system has more than two particles, the *PE* of the system is the sum of *PE* of each pair considered separately.



For a system having three particles of masses m_1, m_2 and m_3

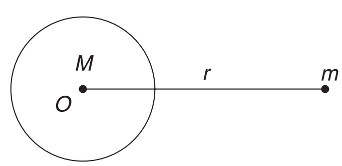
$$U = U_{12} + U_{13} + U_{23}$$

$$= -\left(\frac{Gm_1m_2}{r_{12}} + \frac{Gm_1m_3}{r_{13}} + \frac{Gm_2m_3}{r_{23}}\right) \qquad ...(16)$$

7.4 Spherical Bodies

How will we write the potential energy of a system comprising the spherical Earth (mass *M*) and a particle (mass *m*) kept at a distance r ($\geq R$) from the centre of Earth?

PE of this system is the work done by gravitational force in moving the particle from distance r to ∞, while keeping Earth fixed. The expression of *PE* will be same as equation (15), as the spherical Earth applies the same force on m as a point mass placed at the centre (*O*).



PE of a spherical body and a point mass is $-\frac{GMm}{r}$

$$\therefore \qquad U = -\frac{GMm}{r} \qquad ...(17)$$

Note: A ball of mass m is placed at a distance r from the centre of the earth. *PE* of the system (Earth + ball) is $-\frac{GMm}{r}$. However, in such cases, where one mass is fixed, we can take the liberty of saying that this *PE* belongs to the movable object. If the ball is allowed to fall, *PE* of the system decreases and *KE* increases. Owing to its large mass, Earth partically remains unmoved. The entire *KE* belongs to the ball. Any decrease (or increase) in *PE* of the system results in an increase (or decrease) in *KE* of the ball only. We can say that the *PE*, $U = -\frac{GMm}{r}$ belongs to the ball.

Example 11 *Work done in separating particles from one another*

Three particles, each of mass m, are kept at the vertices of an equilateral triangle of side length x. Find the work needed to separate the particles and place them far away from one another.

Solution

Concepts

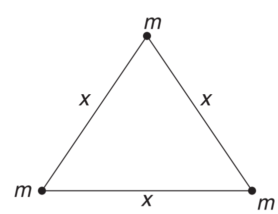
(i) We can write *PE* of the system using equation (16).

(ii) *PE* of the system is negative. When all the particles are placed at infinite separation, the *PE* becomes zero.

To increase the *PE*, from a negative number to zero, an external agent has to do work.

(iii) Amount of work done is equal to change in *PE* of the system.

$$W_{ext} = U_f - U_i$$



PE of the system is

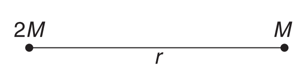
$$U_i = -\frac{Gmm}{x} - \frac{Gmm}{x} - \frac{Gmm}{x}$$

$$= -\frac{3Gm^2}{x}$$

Final *PE* at ∞ separation is: $U_f = 0$

$$\therefore \qquad W_{ext} = U_f - U_i = \frac{3Gm^2}{x}$$

Example 12 Two particles are in space, separated by a distance r. Masses are $2M$ and M. With the particle of mass $2M$ held stationary, the other particle is released. It moves under gravitational pull of mass $2M$. Find the speed of M when separation between the particles reduces to $\frac{r}{2}$.

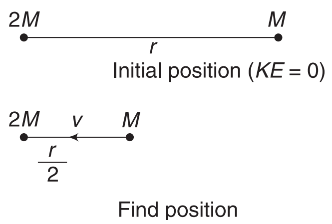


Solution

Concepts

(i) Gravitational force is conservative. Mechanical energy of M is conserved.

(ii) We can take the liberty of calling the PE as PE of mass M, since $2M$ is fixed.



$$K_i + U_i = K_f + U_f$$

$$0 + \left(-\frac{G\,2M\cdot M}{r}\right) = \frac{1}{2}Mv^2 + \left(-\frac{G\,2M\cdot M}{\frac{r}{2}}\right)$$

$$\Rightarrow \qquad \frac{1}{2}v^2 = \frac{2GM}{r}$$

$$\Rightarrow \qquad v = 2\sqrt{\frac{GM}{r}}$$

Example 13 In the last example, both particles are released from separation r. Find speed of mass M when separation reduces to $\frac{r}{2}$.

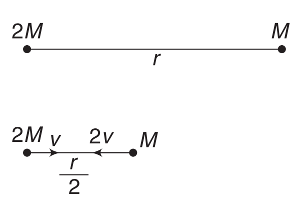
Solution

Concepts

(i) Mechanical energy is conserved. In this case, PE is strictly said to belong to the system of the particles. Any change in PE affects the KE of both particles.

(ii) The system of particles move under mutual interaction. There is no external force. Momentum of the system is conserved. Since initial momentum is zero, momentum at separation $\frac{r}{2}$ is also zero.

If velocity of $2M$ is $v(\rightarrow)$ then the velocity of M must be $2v(\leftarrow)$ for momentum of the system to remain zero.



Conservation of energy:

$$K_i + U_i = K_f + U_f$$

$$0 - \frac{G\,2M\cdot M}{r} = \frac{1}{2}(2M)v^2 + \frac{1}{2}(M)(2v)^2 - \frac{G\,2M\cdot M}{\frac{r}{2}}$$

$$\Rightarrow \qquad 3v^2 = \frac{2GM}{r}$$

$$\Rightarrow \qquad v = \sqrt{\frac{2}{3}\frac{GM}{r}}$$

Thus, speed of M is $2v = 2\sqrt{\frac{2}{3}\frac{GM}{r}}$

Note: Speed of M obtained in this example is less than the speed of M obtained in last example. Now, the same loss in PE has been shared between two particles as KE. In last example, mass M was the only beneficiary.

Example 14 *Reaching a great height*

A body is projected from the surface of Earth (mass = M, radius = R) with a speed $u = \sqrt{gR}$ ($\simeq 8\,\text{kms}^{-1}$) in vertically upward direction. To what height (above the surface) will the body reach before it begins to fall down? Neglect atmospheric resistance.

Solution

Concepts

As the projectile moves up, it slows down. Its KE decreases and gravitational PE increases. At the top point, its KE is zero.

We will use conservation of energy to solve the problem. Particle is projected from A, it stops at B.

Let $\qquad OB = x$

$$K_A + U_A = K_B + U_B$$

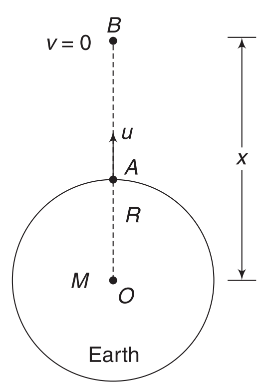
$$\frac{1}{2}mu^2 - \frac{GMm}{R} = 0 - \frac{GMm}{x}$$

$$\Rightarrow \qquad \frac{1}{2}(gR) - \frac{GM}{R} = -\frac{GM}{x}$$

$$\Rightarrow \qquad \frac{1}{2}\frac{GM}{R} - \frac{GM}{R} = -\frac{GM}{x} \qquad \left[\because \quad g = \frac{GM}{R^2}\right]$$

$$\Rightarrow \qquad -\frac{GM}{2R} = -\frac{GM}{x}$$

$$\Rightarrow \qquad x = 2R$$



$\therefore$ Height attained above the surface is $h = x - R = R$

Example 15 Two particles, each of mass m, are at a separation r. They are moved apart from one another. On reaching infinite separation, both of them have speed v. How much work was done by the external agent?

Solution

Concepts

(i) Potential energy of the system is $-\dfrac{GMm}{r}$

If an external agent delivers $+\dfrac{GMm}{r}$ energy, the total energy of the system will become zero. The two particles can be just separated by large distance if $+\dfrac{GMm}{r}$ work is done on them.

If we want them to have KE when at ∞, we need to do more work on them.

(ii) W_{ext} = Change in mechanical energy of the system.

$$\therefore \quad W_{ext} = (K_f + U_f) - (K_i + U_i)$$

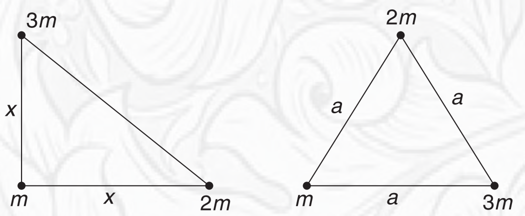
$$= \left(\frac{1}{2}mv^2 + \frac{1}{2}mv^2 + 0\right) - \left(0 - \frac{Gmm}{r}\right)$$

$$[\because \quad U_f = U_\infty = 0]$$

$$\therefore \quad W_{ext} = mv^2 + \frac{Gm^2}{r}$$

Your Turn

Q.19 Three particles of masses m, $2m$ and $3m$, are placed on the vertices of a right-angled isosceles triangle, as shown. They are moved so as to bring them on the vertices of an equilateral triangle of side length a. Find the work done by gravitational force in the process.



Q.20 Two particles of mass m each are initially at rest at very large separation. They are gently pushed and they begin to move towards one another due to mutual gravitational pull. Find their speed when separation between them becomes r.

Q.21 A body is dropped from a height R above the surface of Earth. Find its speed on reaching the surface. R is the radius of Earth and g is the acceleration due to gravity on its surface.

7.5 Escape Speed

If a body is projected up, it will slow down, stop and then fall back to Earth. If we go on increasing the projection speed, it will go higher and higher. There is a certain minimum initial speed that will cause it to move upward forever, theoretically coming to rest only at infinity. This minimum speed is called the escape speed from the surface of Earth.

A particle of mass m, on the surface of Earth, has gravitational PE equal to

$$U = -\frac{GMm}{R}$$

To make it reach infinity, we must make its total energy equal to or greater than zero. Thus, the minimum energy that we need to impart to the particles to free it from the gravity is $+\dfrac{GMm}{R}$. This energy $\left(= \dfrac{GMm}{R}\right)$ is known as *binding energy* of the Earth-particle system.

If the particle is given a speed v, its total energy becomes

$$E = K + U = \frac{1}{2}mv^2 - \frac{GMm}{R}$$

For it to escape forever, $E \geq 0$

$$\Rightarrow \quad \frac{1}{2}mv^2 - \frac{GMm}{R} \geq 0$$

$$\Rightarrow \quad v \geq \sqrt{\frac{2GM}{R}} = \sqrt{\frac{2GMR}{R^2}} = \sqrt{2gR}$$

Minimum speed to escape the gravity is known as escape speed

$$v_e = \sqrt{\frac{2GM}{R}} = \sqrt{2gR} \qquad \text{...(18)}$$

Putting $g = 9.8\,\text{ms}^{-2}$ and $R = 6.4 \times 10^6\,\text{m}$ gives $v_e \simeq 11.3\ \text{kms}^{-1}$.

Mahabharata's Bhima could throw elephants at speed large enough to make them escape Earth's gravitational pull. The elephants never returned!

Equation (18) is applicable for any celestial object, not just for Earth. Substituting mass of the moon $M = 7.36 \times 10^{22}\,\text{kg}$ and radius of the moon $R = 1.74 \times 10^6\,\text{m}$ in equation (18) gives escape speed from the surface of the moon. It is close to $2.38\ \text{kms}^{-1}$.

Escape speed on the Jupiter is close to $60\,\text{kms}^{-1}$.

7.6 Black Holes

The great Indian scientist Subrahmanyan Chandrasekhar predicted that a star having a mass more than few times the mass of the sun would collapse under its own gravity, after its fuel is exhausted, to become a superdense object of vanishingly small size. Such objects are plentiful in the space. Equation (18) predicts that escape speed from such objects will be extraordinarily high due to small radius (R).

$$v_e = \sqrt{\frac{2GM}{R}} \geq c \text{ for such objects.}$$

Where $c = 3 \times 10^8\,\text{ms}^{-1}$ is the speed of light. According to theory of relativity (Einstein, 1905), no material object can have speed greater than that of light. Therefore, no object can escape the gravity of such superdense stars. Even light cannot escape from such an object. Such objects were called Black Holes in 1969 by American scientist, John Wheeler. Chandrasekhar was awarded Nobel prize in 1983.

Example 16 A particle is located at a height $h = \frac{R}{2}$ above the surface of Earth. With what minimum speed should it be projected up, so as to escape the gravity forever? R = Radius of Earth.

Solution

Concepts

For a particle to just escape gravity, its total energy must be zero.

$$K + U = 0 \quad \text{for escape}$$

$$\Rightarrow \quad \frac{1}{2}mv^2 - \frac{GMm}{\frac{3R}{2}} = 0$$

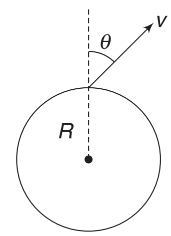
$$\left[\text{Distance from centre} = R + \frac{R}{2} = \frac{3R}{2}\right]$$

$$\Rightarrow \quad v = \sqrt{\frac{4GM}{3R}} = 2\sqrt{\frac{GM}{3R^2}\cdot R}$$

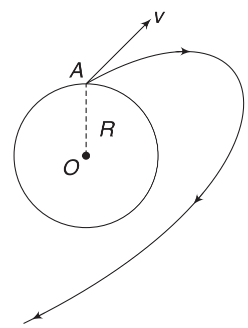
$$= 2\sqrt{\frac{gR}{3}}$$

Example 17 *Direction of projection will not matter if $v \geq v_e$*

A particle is projected from the surface of a planet, making an angle θ with vertical. Projection speed is equal to escape speed. Will the particle escape? Discuss qualitatively.



Solution If projection speed is $v_e = \sqrt{\frac{GM}{R}}$, the total energy of the projectile is zero. It is not bound to the planet. It will escape to infinity. It takes a curved path, as shown, and moves to infinity. However, depending on the angle of projection, it is possible that the planet may intercept its path and the particle hits the planet.



YOUR TURN

Q.22 Potential energy of a rock piece on the surface of a planet is $-E_0$. Due to a blast, the rock piece acquires a kinetic energy of $2E_0$.

(a) Will the rock piece escape out of the gravitational pull of the planet?

(b) At some height, its KE is $1.6E_0$. What is its PE there?

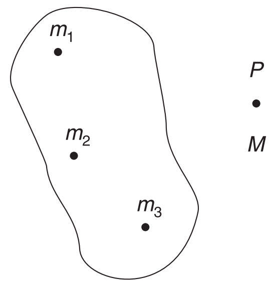
(c) What will be KE of the particle when it is far away from the planet?

Q.23 Acceleration due to gravity on surface of planets A and B are σ and 2σ respectively. Radii of A and B are R and $2R$ respectively. Find the ratio of escape velocities from the surface of the two planets.

Q.24 A rocket is shot vertically upward from Earth's surface. Find the distance from the centre of Earth that the rocket reaches if the launcher gives it half the KE that is needed to escape the gravity forever? R = Radius of Earth.

8. GRAVITATIONAL POTENTIAL

A system of particles has fixed masses: m_1, m_2, m_3 ... etc. In presence of these fixed masses, work needed to slowly bring a mass M from infinity to P is defined as PE of the mass M. [Refer to equation (15). We can also say that PE of M at P is negative of work done by external agent in slowly moving it from P to ∞].



$$U_P = W_{\substack{\text{ext Agt}\\ \text{slowly}}}^{\infty \to P}$$

Gravitational potential (V) at point P due to fixed masses is defined as

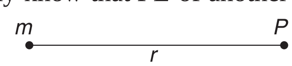
$$V_P = \frac{U_P}{M} \qquad ...(19)$$

Its unit is J kg^{-1}. It is a scalar.

Thus, the potential at a point may be defined as *the work done per unit mass by an external agent in slowly bringing a particle from infinity to the given point*.

8.1 Potential Due to a Point Mass

m is a fixed point mass. We already know that PE of another mass M placed at a point P (at a distance r from m) is



$$U = -\frac{GmM}{r}$$

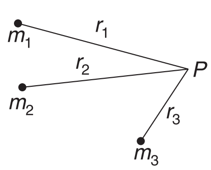
$\therefore$ Potential at P due to m is $V = \dfrac{U}{M}$

$$\Rightarrow \qquad V = -\frac{Gm}{r} \qquad ...(20)$$

This is the work done by an external agent in slowly moving a unit mass from ∞ to P.

Note: (i) Potential obeys principle of superposition. Potential at P in the shown figure is:

$$V = -\frac{Gm_1}{r_1} - \frac{Gm_2}{r_2} - \frac{Gm_3}{r_3}$$

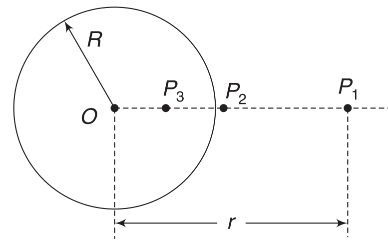


(ii) If we place a mass M at P, then its PE is given by

$$U = MV$$

8.2 Potential due to a Uniform thin Spherical Shell

Consider a shell of mass m and radius R.



What is the potential due to the shell at point P_1 (at a distance $r > R$ from centre)? It is the work done in slowly shifting a unit mass from ∞ to P_1 in the presence of the shell. Will it make a difference (to the work done) if the shell is replaced by a point mass M at its centre? Answer is no. The gravitational force due to a shell is same as that due to an equal point mass at its centre. Thus, potential at P_1 remains unchanged if shell is replaced with a point mass.

$$\therefore \qquad V_{P1} = -\frac{Gm}{r} \quad (\text{for } r \geq R) \qquad ...(21)$$

Exactly on the surface (at P_2)

$$V_{P2} = -\frac{Gm}{R} \qquad ...(22)$$

Now, what is the potential at P_3 (a point inside the shell)? By definition

$$V_{P3} = W_{\substack{\text{ext Agt}\\ \text{slowly}}}^{\infty \to P_3} \quad (\text{for a unit mass})$$

$$= W_{\substack{\text{ext Agt}\\ \text{slowly}}}^{\infty \to P_2} + W_{\substack{\text{ext Agt}\\ \text{slowly}}}^{P_2 \to P_3} \qquad (\text{for unit mass})$$

$$= V_{P2} + W_{\substack{\text{ext Agt}\\ \text{slowly}}}^{P_2 \to P_3}$$

But no work is needed to move a mass from P_2 to P_3, as there is no field inside the shell [Refer to article 5.1]. A mass inside the shell will experience no force due to the shell.

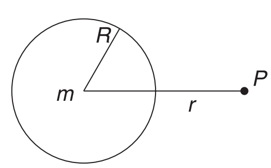
$$\therefore \qquad W_{\substack{\text{ext Agt}\\ \text{slowly}}}^{P_2 \to P_3} = 0$$

$$\therefore \qquad V_{P3} = V_{P2} = -\frac{Gm}{R} \qquad ...(23)$$

Potential at all inside points is same as surface potential.

8.3 Potential Due to a Solid Uniform Sphere

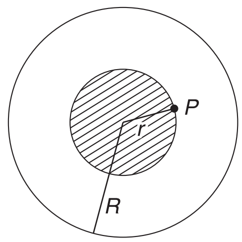
Arguing exactly as above, we can easily say that



$$V = -\frac{Gm}{r} \quad (\text{for } r \geq R) \qquad ...(24)$$

And $$V_{\text{surface}} = -\frac{Gm}{R} \qquad ...(25)$$

For getting potential at an inside point, we need to find the work done in bringing a unit mass from ∞ to that inside point. This can be calculated with the help of equation (6). However, we will take a different approach. We will use principle of superposition for this calculation.



Let us assume P is a point at a distance r from the centre where we wish to find potential.

Let us divide the sphere into two parts

(i) a solid sphere of radius r (shown as shaded part).

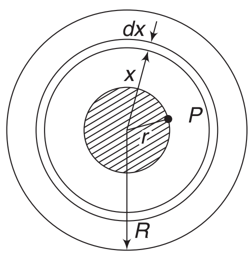
(ii) annular region between $r \leq x \leq R$.

Mass of the first part is: $m' = \dfrac{m}{\frac{4}{3}\pi R^3} \cdot \dfrac{4}{3}\pi r^3 = \dfrac{mr^3}{R^3}$

Potential at P due to solid sphere of mass m' is given by equation (25) as

$$V_1 = -\frac{Gm'}{r} = -\frac{Gm}{R^3}r^2$$

Now, we will find potential (V_2) at P due to the mass in the region $r \leq x \leq R$. For this, we divide this mass into concentric shells. Consider one such shell between radius x and $x + dx$.



Mass of the shell is

$$dm = \frac{m}{\frac{4}{3}\pi R^3} \cdot 4\pi x^2\,dx = \frac{3mx^2\,dx}{R^3}$$

Potential at P due to this shell is same as potential due to the shell at its own surface (equation (22)).

$$\therefore \qquad dV = -\frac{Gdm}{x} = -\frac{3Gm}{R^3} \cdot x \cdot dx$$

Potential due to all the shells lying between $x = r$ to $x = R$ is

$$V_2 = -\frac{3Gm}{R^3}\int_r^R x\,dx = -\frac{3Gm}{R^3}\left[\frac{x^2}{2}\right]_r^R$$

$$= -\frac{3Gm}{R^3}\left[\frac{R^2}{2} - \frac{r^2}{2}\right]$$

∴ Potential at P is

$$V = V_1 + V_2$$

$$V = -\frac{Gmr^3}{R^3} - \frac{3Gm}{2R^3}(R^2 - r^2)$$

$$V = -\frac{Gm}{R^3}\left(\frac{3}{2}R^2 - \frac{1}{2}r^2\right) \qquad ...(26)$$

Potential at the centre is obtained by putting $r = 0$ in the above equation.

$$V_{\text{centre}} = -\frac{3}{2}\frac{Gm}{R} \qquad ...(27)$$

Example 18 Four particles, each of mass m, are kept at the vertices of a square of side length a. Find the gravitational field and potential at the centre of the square.

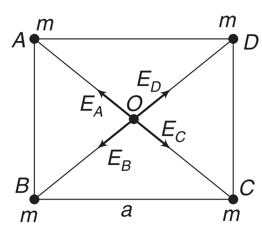
Solution

Concepts

(i) Both field and potential obey principle of superposition.

(ii) We must be careful that field is a vector, whereas potential is a scalar.

Field at O due to masses at A and C cancel out. Similarly, field due to masses at B and D also cancel out.



Resultant field at O is zero.

$$AO = r = \frac{a}{\sqrt{2}}$$

Potential due to each mass, at O is $= -\dfrac{Gm}{r}$

$$= -\frac{\sqrt{2}\,Gm}{a}$$

∴ Total potential at O is

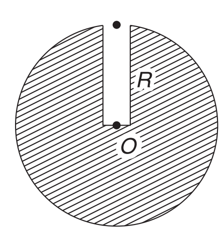
$$V = -\frac{4\sqrt{2}\,Gm}{a}$$

Note: If we place a mass M at O, then its PE in presence of the four fixed masses will be

$$U = MV = -\frac{4\sqrt{2}\,GmM}{a}$$

Example 19 *A tunnel to the centre of the Earth*

A tunnel is dug all along the radius of the Earth – from its surface to its centre. A ball is dropped in the tunnel. With what speed will it reach the centre?



Neglect any friction and assume that Earth is a uniform solid sphere of mass M and radius R.

Solution

Concepts

(i) Assuming Earth to be a uniform solid sphere allows us to use equations (26) or (27) for writing potential at a point inside it.

(ii) Potential energy of the ball (mass = m) at a point where potential is V is given by $U = mV$.

(iii) Mechanical energy is conserved.

Potential at the surface of the Earth: $V_{surf} = -\dfrac{GM}{R}$

PE of the ball at the surface: $U_{surf} = -\dfrac{GMm}{R}$

Potential at the centre of Earth:

$$V_{cent} = -\frac{3}{2}\frac{GM}{R}$$

PE of the ball at the centre: $U_{cent} = -\dfrac{3}{2}\dfrac{GMm}{R}$

Energy conservation:

$$KE_{cent} + U_{cent} = KE_{surf} + U_{surf}$$

$$\frac{1}{2}mu^2 - \frac{3}{2}\frac{GMm}{R} = 0 - \frac{GMm}{R}$$

$$\Rightarrow \quad u^2 = \frac{GM}{R} \quad \Rightarrow \quad u = \sqrt{\frac{GM}{R}}$$

Your Turn

Q.25 Gravitational potential due to Earth at a point is $-1.0 \times 10^7\ \text{J kg}^{-1}$

(a) Is the point inside or outside the Earth

(b) What is the PE of a 5 kg mass placed at this point?

Q.26 Is it right to say that PE of a particle at the centre of Earth is more than the PE of a similar particle on its surface?

Q.27 Find the gravitational potential at the centre of a ring of mass M and radius R.

In Short:

(i) Gravitational PE of a system of two point-masses is assumed to be zero when they are at ∞ separation.

(ii) Gravitational PE of a system of two point-masses, separated by distance r, can be defined as:

– (Work done by gravitational force when the masses are moved from ∞ to separation r) =

work done by gravitational force when masses are moved from r to ∞ =

work done by external agent when masses are moved slowly from ∞ to r =

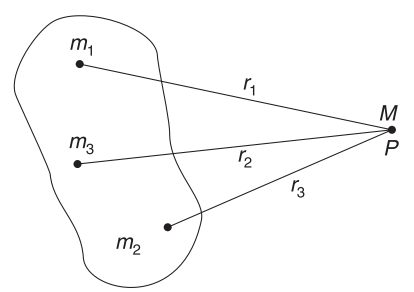
– (Work done by external agent when masses are moved slowly from r to ∞)

(iii) PE for a two-particle system is $U = -\dfrac{Gm_1m_2}{r}$

(iv) PE of a multi-particle system is obtained by adding PE of each pair.

$$U = -\frac{Gm_1m_2}{r_{12}} - \frac{Gm_1m_3}{r_{13}} - \frac{Gm_2m_3}{r_{23}} \ldots$$

(v) Let m_1, m_2, m_3 be fixed masses and M is free to move under influence of gravitational force due to fixed masses. In such cases, it is not necessary to worry about energy of pair like (m_1, m_2), (m_1, m_3) or (m_2, m_3). The PE of these pairs does not change as they are fixed. And, in solving a problem, the change in PE is important, not its absolute value. Thus, we need not consider PE of interactions of m_1, m_2 and m_3. What changes when M moves? It is energy of pairs (m_1, M), (m_2, M) and (m_3 M).



In such cases, concept of potential becomes important.

Potential at P is PE of unit mass kept at P. Here, we are taking into account PE of pairs (m_1, M), (m_2, M) and (m_3, M) with M as unity.

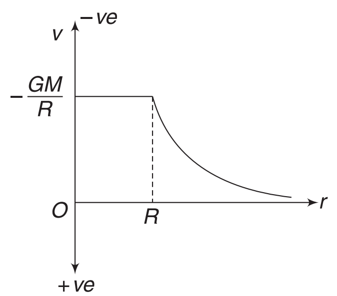
$$\therefore \quad V_P = -\frac{Gm_1}{r_1} - \frac{Gm_2}{r_2} - \frac{Gm_3}{r_3}$$

PE of a mass M at P is $U = MV_P$

(vi) Potential due to a uniform spherical shell of mass M and radius R is

$$V = -\frac{GM}{r} \quad \text{for } r \geq R$$

$$= -\frac{GM}{R} \quad \text{for } r \leq R$$

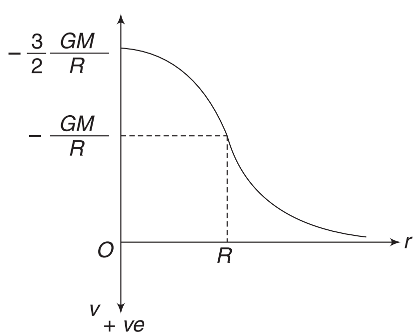


The potential is taken to be +*ve* in downward direction on the vertical axis.

(vii) Potential due to a uniform solid sphere of mass M and R is

$$V = -\frac{GM}{r} \quad \text{for } r \geq R$$

$$= -\frac{GM}{R^3}\left(\frac{3}{2}R^2 - \frac{1}{2}r^2\right) \quad \text{for } r \leq R$$



Potential due to a solid sphere. Part of the graph between $r = 0$ to $r = R$ is parabola. For $r > R$, it is hyperbola.

(viii) Gravitational PE is always negative. It shows binding. Two masses are said to be separated if they are far away from one another and their PE is zero. It means, we need to increase the energy (i.e., do positive work) on a system of two masses to separate them.

(ix) Amount of energy needed to separate a particle from another and move them to infinity, is known as binding energy. Binding energy of a particle of mass m on the surface of Earth (mass M, radius R) is $\frac{GmM}{R}$.

(x) Minimum speed of projection that takes a projectile to infinity is known as escape speed. Escape speed on the surface of a spherical celestial body is given by

$$v = \sqrt{\frac{2GM}{R}} = \sqrt{2gR}$$

Where g is acceleration due to gravity on the surface.

(xi) Earth has atmosphere. This is because the average speed of gas molecules in the atmosphere of Earth is much smaller than escape speed (11.3 kms^{-1}).

Note: The concept of field and potential may be a bit difficult to understand at this stage. We will come back to it in greater detail in the chapter of electrostatics.

9. SATELLITE

Moon is the only natural satellite of the Earth. However, there are thousands of artificial satellites going around the Earth. They are used for telecommunication, weather forecasting, spying, GPS (global positioning system) and many other applications.

To launch a satellite, a rocket is used to lift it to a desired height and then another rocket is fired to impart it a velocity parallel to the Earth's surface. Depending on the speed given, a satellite goes around the Earth in a circular orbit or an elliptical orbit.

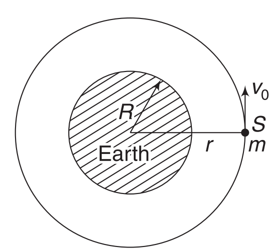
Here, we will assume a satellite in circular orbit and derive the equations for its orbital speed, period of revolution and energy.

Orbital Speed

A satellite of mass m is orbiting around Earth, in a circular orbit of radius r. Its orbital speed is v_0. There is no engine or rocket attached to the satellite. Once it is imparted a speed v_0, the gravitational pull of Earth provides it the necessary centripetal force and it keeps revolving.

$$\frac{mv_0^2}{r} = \frac{GMm}{r^2}$$

$$\Rightarrow \quad v_0 = \sqrt{\frac{GM}{r}} \quad \text{...(28)}$$



A satellite (S) going around the Earth in a circular orbit of radius r.

If h is height of the satellite above the surface of Earth, $r = R + h$.

$$\therefore \qquad v_0 = \sqrt{\frac{GM}{R+h}} \qquad ...(29)$$

To establish a satellite in a circular orbit above the surface of Earth, it must be imparted a speed given by the above equation.

Orbital speed (v_0) is independent of the mass of the satellite. It is only the radius of the circular orbit that matters. Larger the radius of circular path, smaller is the speed.

Time period of revolution

Time needed to complete one revolution is the time needed by the satellite to travel a distance $2\pi r$ with speed v_0.

$$\therefore \qquad T = \frac{2\pi r}{v_0} = \frac{2\pi r}{\sqrt{\frac{GM}{r}}} = 2\pi\sqrt{\frac{r^3}{GM}} \qquad ...(30)$$

$$\text{or,} \qquad T^2 = 4\pi^2 \frac{r^3}{GM} \qquad ...(31)$$

Square of time period is proportional to cube of the radius of the circular path. Farther the satellite, higher is its time period.

Energy of a satellite

Energy of a satellite is the sum of its kinetic and potential energies.

$$U = -\frac{GMm}{r} \qquad ...(32)$$

$$\text{and} \qquad K = \frac{1}{2} m v_0^2 = \frac{1}{2}\frac{GMm}{r} \qquad ...(33)$$

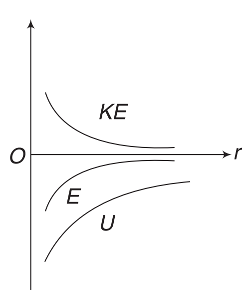
Total energy of the satellite is:

$$E = U + K = -\frac{1}{2}\frac{GMm}{r} \qquad ...(34)$$

Total energy is negative. It should be. The satellite is bound to Earth.

Total energy is equal to half the potential energy and KE is same as magnitude of total energy.

The above analysis is valid for circular orbit of any satellite of any planet. It makes no difference whether the satellite is natural or artificial.

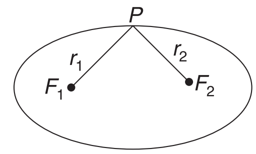


Variation of KE, U and total energy of a satellite with radius of path

9.1 Ellipitical Orbit

If the speed given to the satellite is different from v_0 (but not high enough to make it escape), the path is an ellipse.

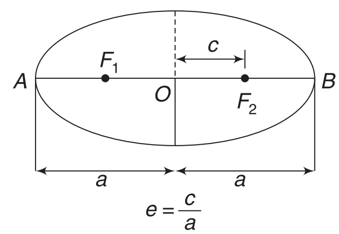
Ellipse is a path traced by a point, which moves such that sum of its distance from two fixed points (F_1 and F_2) remains constant. $r_1 + r_2$ = a constant.



$r_1 + r_2$ = a constant

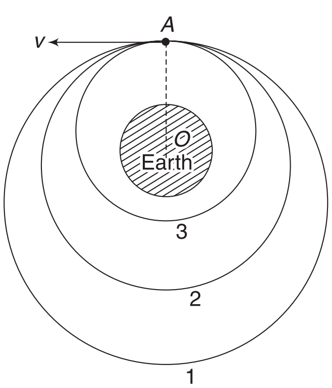
The fixed points, F_1 and F_2, are known as foci (plural of focus).

Distance of each vertex (A and B) from the geometrical centre (O) of the ellipse is called its semi-major axis (a).



If distance of a focus (F_1 or F_2) from O is c, then $\frac{c}{a} = e$ is known as eccentricity of the ellipse. Value of e ranges from 0 to 1; when $e = 0$, the ellipse is a circle and when $e = 1$, it is a straight line.

Suppose a satellite is lifted above the surface of Earth to point A and is given a velocity v parallel to the surface of Earth. If the speed is exactly equal to v_0 given by equation (28), the satellite will revolve in a circular path (shown as path 2 in figure). If $v > v_0$, it will go in an elliptical path with the centre of Earth (O) at a focus. If v is too large, the satellite will move in an open path (like a parabola) and move to infinity.



When $v = v_0 = \sqrt{\frac{GM}{r}}$

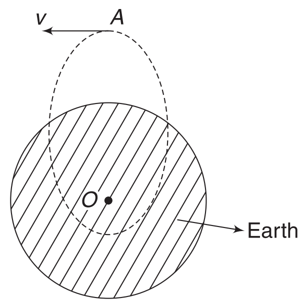
Path is circular (2).

For $v > v_0$, path is elliptical (1).

For $v < v_0$, path is elliptical (3).

If $v < v_0$, once again, the satellite follows an elliptical path (shown as path 3). If the speed is too small, the eccentricity of the path increases and the path of the satellite will intersect that of Earth. It will fall on Earth.

It can be proved that for elliptical orbit of a satellite, time period and energy is still given by equations (31) and (34) respectively, with r replaced by semi-major axis (a) of the ellipse.



If v is too small, the satellite will fall on the Earth

$$T^2 = 4\pi^2 \frac{a^3}{GM} \qquad ...(35)$$

and $$E = -\frac{1}{2}\frac{GMm}{a} \qquad ...(36)$$

The last equation tells us that the total energy of a satellite depends only on the semi-major axis (a) and not on eccentricity (e).

Example 20 *The fastest satellite*

(a) Find the maximum possible speed of a satellite of the Earth.

(b) What is the smallest time in which a satellite can complete a revolution around the Earth?

Solution

Concepts

(i) Speed is maximum for smallest radius of circular path. Smallest radius possible for an Earth's satellite is R (= radius of the Earth). Such satellites may be called as near-surface satellites.

(ii) Time period is also smallest for a near-surface satellite, as $T^2 \propto r^3$.

(a) $$v_0 = \sqrt{\frac{GM}{r}}$$

$$v_{\max} = \sqrt{\frac{GM}{R}} = \sqrt{\frac{GM}{R^2}\cdot R} = \sqrt{gR} \simeq 8\,\text{kms}^{-1}$$

(b) $$T = 2\pi\sqrt{\frac{r^3}{GM}}$$

T is minimum if $r = R$

$$T = 2\pi\sqrt{\frac{R^3}{GM}} = 2\pi\sqrt{\frac{R^2}{GM}\cdot R} = 2\pi\sqrt{\frac{R}{g}}$$

$$\simeq 84.6 \text{ minutes}$$

It means that no satellite of earth can go around the earth in a time less than 84.6 minutes.

Example 21 Distance of moon form the centre of the earth is about $r_M = 4 \times 10^5$ km and its time period of revolution is $T_M = 27$ days. Find the time period of revolution of an earth's satellite that is in a circular orbit of radius $r = 8{,}000$ km.

Solution

Concepts

Equation (31), $T^2 \propto r^3$ is valid for moon also.

$$T^2 = 4\pi^2 \frac{r^3}{GM}$$

M is the mass of earth.

Thus, $\frac{4\pi^2}{GM}$ is same for an artificial satellite as well as moon.

$$\therefore \quad T^2 \propto r^3$$

$$\frac{T^2}{T_M^2} = \frac{r^3}{r_M^3} \quad \Rightarrow \quad T^2 = (27d)^2\left(\frac{8\times 10^3}{4\times 10^5}\right)^3$$

$$\therefore \quad T = 27 \times \frac{2\sqrt{2}}{1000}\text{ day} = 1.83\,\text{h}$$

Example 22 A satellite of earth has an orbital speed v_0, in a circular orbit. A rocket, on board, is fired to increase its speed to v_1 in a quick time. Find the minimum value of v_1 for which the satellite will escape to infinity.

Solution

Concepts

The satellite will just escape if its total energy is made zero.

$$v_0 = \sqrt{\frac{GM}{r}}$$

For the satellite to just escape, its total energy must be made zero.

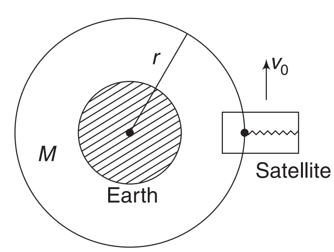
$$K + U = 0$$

$$\frac{1}{2}mv_1^2 - \frac{GMm}{r} = 0 \quad \Rightarrow \quad v_1 = \sqrt{\frac{2GM}{r}} = \sqrt{2}\,v_0$$

It implies that KE of the satellite must be doubled.

Example 23 *Weightlessness in a satellite*

Inside an earth's satellite, there is a ball suspended with the help of a spring balance. What is the reading of the spring balance if mass of the ball is m?



Solution

Concepts

Orbital speed of a satellite is independent of its mass.

If the satellite is in a circular orbit of radius r, its orbital speed is $v_0 = \sqrt{\frac{GM}{r}}$.

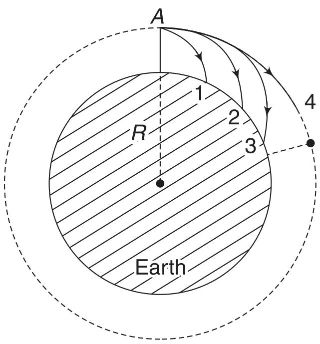
The ball, inside the satellite, is also having speed v_0. Imagine for a moment, that all the walls of the satellite (and the spring balance) are removed without disturbing the ball. What will happen to the ball? It will keep revolving like a satellite on its own! It has just the right speed for being a satellite. The gravitational pull of Earth on the ball is just equal to the necessary centripetal force, at speed v_0.

It means that the ball, inside the satellite, is moving in a circle without any force from the spring. Tension in the spring is zero. The spring balance records zero weight.

Note: An astronaut inside a satellite feels weightless.

Example 24 The moon is continuously falling towards Earth. Explain.

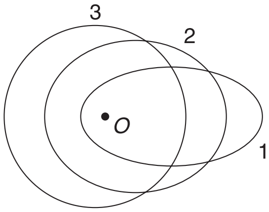
Solution Imagine yourself at Mt. Everest (A). You throw a ball horizontally and it lands on ground, at 1. Now you fire a bullet and it lands at point 2. If you fire from a high-speed cannon, the shell may reach as far as point 3. Note that with increasing speed of projection, the radius of curvature of the path (near A) is increasing.



Now, if you fire a projectile using some powerful launcher so as to impart it a speed $v = \sqrt{\frac{GM}{R}}$, the radius of curvature of the path becomes equal to radius (R) of the earth. The projectile moves along path 4. When it reaches a point B, it is still having same speed and follows the path of same curvature. The projectile will move in a circle. It is a satellite. It is continuously falling towards the Earth, just like the ball (1), bullet (2) or shell (3), but it fails to hit Earth, as the Earth curves away from its path. Radius of Earth is smaller than (or nearly equal to) the radius of curvature of the path of the satellite.

Every satellite (including the moon) is falling towards Earth (under gravity) but fails to hit Earth for the reason explained above.

Example 25 In the figure shown, O is the centre of the earth. Three identical satellites are going in paths shown in the figure. Path 3 is circular and the other two paths are elliptical. The distance between vertices of the elliptical paths 1 and 2 is equal to the diameter of the circular path 3. Eccentricity of path 1 is 0.9 and that of path 2 is 0.5.



Find the ratio of time periods and total energies of the three satellites.

Solution

Concepts

Equations (35) and (36) tell you that the time period and energy depend only on semi-major axis and are independent of eccentricity.

Distance between vertices of an ellipse = $2a$

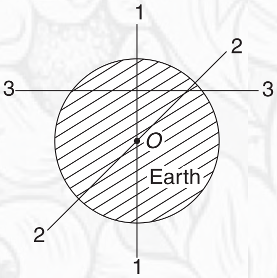
$\therefore$ All three paths have same semi-major axis.

$$\therefore \quad T_1 : T_2 : T_3 = 1 : 1 : 1$$

$$\text{And} \quad E_1 : E_2 : E_3 = 1 : 1 : 1$$

YOUR TURN

Q.28 In the figure shown, lines 1–1, 2 – 2 and 3–3 are diameters of circular paths around the earth. O is the centre of the earth. Which circular path cannot be a path for a satellite and why?



Q.29 If orbital radius of a satellite is increased, what happens to its (a) potential energy (b) kinetic energy?

Q.30 A satellite at a height of $6R$ from the surface of the earth has a time period of 24h. Find the time period of another satellite at a distance of $3.5R$ from the centre of the earth. R is the radius of the earth.

Q.31 An artificial satellite of Earth has an orbital speed equal to half the escape speed from the surface of Earth [R = Radius of Earth]

(a) Determine the height of the satellite above the surface of Earth.

(b) An alien demon stops the satellite suddenly and releases it, allowing it to fall freely. Find the speed with which it hits the Earth.

Q.32 A satellite of mass m = 2000 kg is lifted from the surface of Earth and placed in a circular orbit of radius $2R$. How much energy was given to the satellite in the process? Given, g = 10 ms^{-2} and R = 6,400 km.

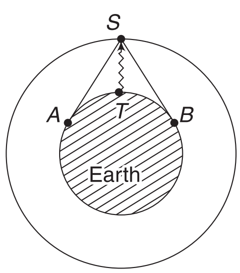
Q.33 A satellite is raised to a height R above the surface of Earth and given a velocity $v = \sqrt{\frac{2}{3}\frac{GM}{R}}$ parallel to the surface of Earth.

(a) Will the satellite escape the gravity of Earth?

(b) What kind of path will the satellite follow? R is the radius of Earth.

9.2 Geostationary Satellites

Satellites are an important component of modern-day communication. A TV station relays its signal to a satellite, which in turn broadcasts (spreads) it over a large area on the surface of Earth. T is a TV station, which sends its signal to satellite S. The satellite receives the signal, amplifies it and re-emits it. Signal coming from satellite S can be received on surface area ATB [SA and SB are tangents].



Signal emitted by satellite S can reach at all locations from A to T to B

Higher the satellite, larger is the area that it can cover.

A more important consideration is the fact that the earth is rotating from west to east and its time period is 24 hours. If the satellite has a different time period, telecasting a cricket match is going to be very difficult. At a time the satellite will be in a position so as to throw signal over the entire Indian sub-continent, and at a different time, it may be in position to provide signal to South America but not to India.

To solve this problem, we need a satellite, which remains at rest, relative to a person on Earth. Such a satellite is possible only in *equatorial plane* (think why?) and should have a time period of T = 24 hours. It must revolve from west to east. Such a satellite is called a geostationary or geosynchronous satellite.

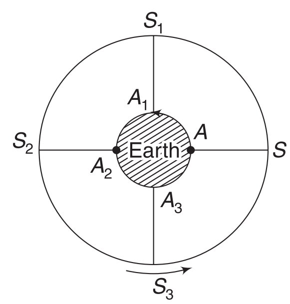
$$T^2 = \frac{4\pi^2}{GM} r^3 = \frac{4\pi^2}{g} \frac{r^3}{R^2}$$

Putting T = 24 h, $R = 6.4 \times 10^6$ m and g = 9.8 ms^{-2}, gives $r \simeq 42{,}000$ km.

Thus, height of a geostationary satellite above Earth's surface is

$$h = r - R \simeq 36{,}000 \text{ km}$$

Syncom-2 was the first geosynchronous communication satellite launched by NASA, in 1963. The first Indian geo-synchronous satellite was APPLE, launched in 1981. Since then, India has launched many INSAT, IRNSS and GSAT series of geosynchronous satellites.



Geostationary satellite. Satellite (S) is always above city (A). As S moves to S_1, S_2, S_3 the city A moves to A_1, A_2, A_3.

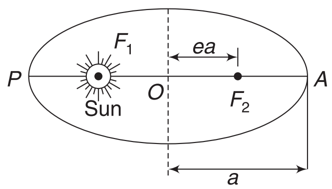
10. PLANETS AND KEPLER'S LAWS

Everything discussed in article 9 is applicable for planetary motion also. In all our discussions, M will represent mass of the sun and m will represent the mass of a planet. The gravitational pull of the sun provides the necessary centripetal force to the planets.

Tycho Brahe, a great astronomer, made years of observations of the planetary motion (without using a telescope) and compiled extensive data. Kepler made three deductions, looking carefully at this data for years. These three empirical laws are now known as Kepler's laws. All the three laws can be derived using Newton's law. The three laws hold equally well for satellites or any other body orbiting around the sun.

1. The law of orbits: All planets move in elliptical orbits with the sun at one focus.

The eccentricities of the planetary orbits are not large and the orbits are nearly circular. The eccentricity of Earth's orbit around the sun in only e = 0.0167.



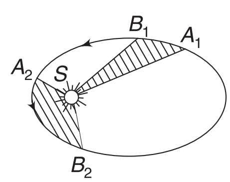
The figure is a bit exaggerated. Planetary paths are nearly circular and F_1 and F_2 are very close to O. e is very small

Point A, where the planet is farthest from the sun, is known as *Aphelion* and point P, where the planet is closest to the sun in called *Perihelion*. For Earth, we use special terms *apogee* and *perigee* respectively.

2. The law of areas: The radius vector, drawn from the centre of the sun to the planet, sweeps out equal area (in the plane of the orbit) in equal time intervals.

$$\frac{dA}{dt} = \text{a constant}$$

Where A = area swept by radius vector of the planet.



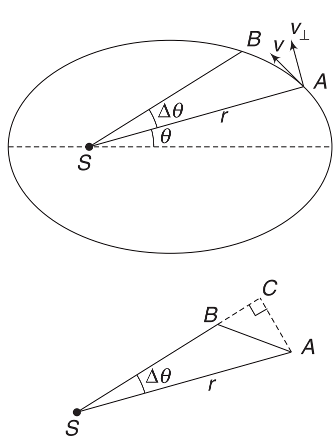
A planet moves from A_1 to B_1 in a time interval Δt and its position vector sweeps area $SA_1B_1 = \Delta A_1$. The same planet moves from A_2 to B_2 in the same interval of time Δt and its position vector sweeps area $SA_2B_2 = \Delta A_2$.

$$\Delta A_1 = \Delta A_2$$

Clearly, arc length A_2B_2 > arc length A_1B_1.

This implies that planet is moving at a higher speed in the region A_2B_2 compared to its speed in region A_1B_1. This implies that the planet is slowest when it is farthest from the sun (at aphelion) and it will be fastest when it is closest to the sun (at perihelion).

Kepler's second law is a direct outcome of law of conservation of angular momentum. Here is how we can prove this.



Consider a planet at point A in its elliptical orbit around the sun (S). Velocity of the planet is v (Tangential to the path) and its distance from the sun is r. In a small interval of time, it moves to B, rotating by an angle $\Delta\theta$.

Let velocity component perpendicular to r (= SA) be $v_\perp$. Angular velocity of the planet about S is

$$\omega = \frac{\Delta\theta}{\Delta t} = \frac{v_\perp}{r} \qquad \text{...(i)}$$

Area swept in time Δt is

$$\Delta A = \text{area (SAB)} \simeq \text{area (SAC)} = \frac{1}{2}(r)(AC)$$

$$= \frac{1}{2}r\,(r\Delta\theta)$$

$$= \frac{1}{2}r\,(v_\perp \Delta t)$$

[The approximation, area (SAB) $\simeq$ area (SAC) will give very accurate result, as $\Delta t \to 0$ and $\Delta\theta$ is extremely small.]

$$\therefore \text{ Areal velocity, } \frac{\Delta A}{\Delta t} = \frac{1}{2} r v_\perp \qquad \text{...(ii)}$$

Let us write the angular momentum of the planet (mass = m) about the sun (S).

$$L = (mv_\perp)r$$

$$\Rightarrow \qquad r v_\perp = \frac{L}{m} \qquad \text{...(iii)}$$

From (ii) and (iii),

$$\frac{\Delta A}{\Delta t} = \frac{L}{2m} \qquad \text{...(37)}$$

Angular momentum of the planet about the sun must be a constant, as the only force acting on it (the gravitational pull of the sun) has no torque about S. Hence, right side of equation (37) is a constant, implying that the areal velocity is a constant.

3. The law of periods: The square of the time-period of any planet is proportional to the cube of the semi-major axis of its elliptical orbit.

This law is nothing but equation (35).

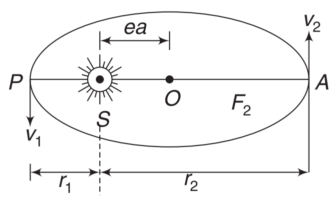
Example 26 A planet going around a star in an elliptical orbit has a semi-major axis equal to a and eccentricity $e = \frac{1}{2}$. Speed of the planet when it is nearest to the star is v_1. Find its speed when it is farthest from the star.

Solution

Concepts

(i) At farthest and nearest points (i.e., at vertices of the ellipse), velocity is perpendicular to the position vector.

(ii) Angular momentum of the planet about the star is conserved.



At perihelion (nearest point), distance of the planet from the star is

$$r_1 = a - ea = (1 - e)a$$

At aphelion (farthest point), distance of the planet from the star is

$$r_2 = a + ea = (1 + e)a$$

Angular momentum at A = Angular momentum at P

$$mv_2r_2 = mv_1r_1$$

$$\Rightarrow \quad v_2(1 + e)a = v_1(1 - e)a$$

$$\Rightarrow \quad v_2 = \left(\frac{1 - e}{1 + e}\right)v_1$$

Example 27 *Comet Halley*

Since its first predicted return in 1759, Halley's comet has come back to us three times, in 1835, 1910 and most recently in 1986. It is orbiting around the Sun with a time period of 76 years. In 1986, it was nearest to the Sun (at its perihelion) at a distance of $r_1 = 8.9 \times 10^{10}$ m. Mass of the Sun = 2×10^{30} kg.

(a) Find the comet's farthest distance from the Sun.

(b) What is eccentricity of the orbit?

[76 years = 2.4×10^9 s, $(19.4)^{1/3} \simeq 2.7$]

Solution

Concepts

(i) If r_1 and r_2 are the nearest and farthest distances of the comet from the sun

$$r_1 + r_2 = 2a$$

(ii) From the last example:

$$r_1 = (1 - e)a \text{ and } r_2 = (1 + e)a$$

(a) $T^2 = \dfrac{4\pi^2}{GM}a^3$

$$\Rightarrow \quad a = \left[\frac{GMT^2}{4\pi^2}\right]^{\frac{1}{3}}$$

$$= \left[\frac{6.67 \times 10^{-11} \times 2 \times 10^{30} \times (2.4 \times 10^9)^2}{4 \times (3.14)^2}\right]^{1/3}$$

$$= [19.4 \times 10^{36}]^{1/3} = 2.7 \times 10^{12}\,\text{m}$$

Now, $r_1 + r_2 = 2a$

$$\Rightarrow \quad r_2 = 2a - r_1 = 2.7 \times 10^{12} \times 2 - 8.9 \times 10^{10}$$

$$\simeq 5.3 \times 10^{12}\,\text{m}$$

(b) $r_1 = (1 - e)a \quad \Rightarrow \quad e = 1 - \dfrac{r_1}{a}$

$$\Rightarrow \quad e = 1 - \frac{8.9 \times 10^{10}}{2.7 \times 10^{12}} \simeq 0.97$$

Eccentricity is close to 1.0. The orbits is a long, thin ellipse.

Example 28 *From a circular to an elliptical orbit*

A satellite of mass m is going around Earth in a circular orbit of radius r. It has time period T_1. A small rocket is fired from the satellite, which decreases its speed by 4%, almost instantaneously. Direction of motion remains unchanged. Now the satellite moves in an elliptical orbit.

(a) Find the semi-major axis in terms of r.

(b) Find time period in terms of T_1.

Solution

Concepts

(i) In circular orbits, energy of the satellite is $E = -\dfrac{GMm}{2r}$.

In an elliptical orbit, energy is $E = -\dfrac{GMm}{2a}$

(ii) The square of time period is proportional to cube of semi-major axis. Here, if T_1 is period in circular orbit and T_2 is period in elliptical orbit, then

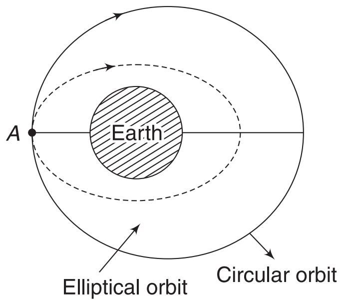
$$\frac{T_2^2}{T_1^2} = \frac{a^3}{r^3}$$

(a) Rocket is fired at A to decrease the speed by 4%. Speed before firing of rocket

$$v_0 = \sqrt{\frac{GM}{r}}$$

Energy of satellite at A is

$$E_1 = -\frac{GMm}{2r}$$



Just after the rocket is fired, speed is, $v = 0.96 v_0$

KE after rocket is fired is

$$K_2 = \frac{1}{2}m\,(0.96 v_0)^2 = \left(\frac{1}{2}\,mv_0^2\right) \times 0.92$$

$$= 0.92\left(\frac{GMm}{2r}\right)$$

PE just after rocket is fired is (the satellite is still at *A*)

$$U_2 = -\frac{GMm}{r}$$

$\therefore$ Total energy after the rocket is fired is

$$E_2 = K_2 + U_2 = \frac{-0.54\,GMm}{r}$$

If the semi-major axis of elliptical orbit is a then

$$E_2 = -\frac{GMm}{2a}$$

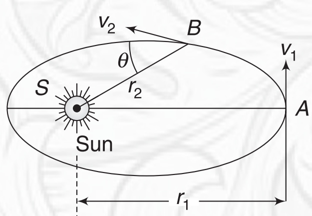
$$\Rightarrow \quad -0.54\,\frac{GMm}{r} = -\frac{GMm}{2a} \quad \Rightarrow \quad a = 0.93\,r$$

(b) $\dfrac{T_2^2}{T_1^2} = \dfrac{a^3}{r^3} \quad \Rightarrow \quad T_2 = T_1(0.93)^{3/2} = 0.89\,T_1$

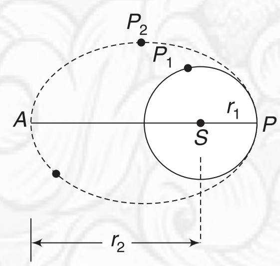
Your Turn

Q.34 "Angular momentum of a planet about the sun remains conserved but its *KE* does change." Is this statement true? Explain your answer.

Q.35 A planet is going around the sun. Its speed at aphelion (A) is v_1. Its distance from the sun in this position is r_1. At some other point *B*, it is at a distance r_2 from the sun and its velocity vector makes an angle θ with the line BS, as shown. Find speed (v_2) at *B*.



Q.36 Two planets (P_1 and P_2) have equal mass. They revolve around a common massive star *S*. P_1 goes in a circular orbit of radius $r_1 = 10^8$ km and has a time period of 2 years. Planet P_2 moves in an elliptical path with nearest and farthest distances from the star equal to $r_1 = 1 \times 10^8$ km and $r_2 = 1.8 \times 10^8$ km (see figure).



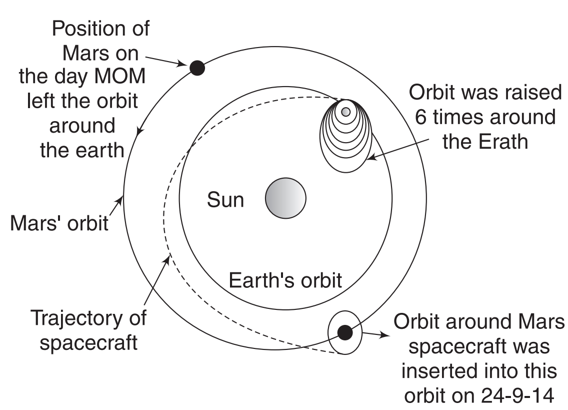
(a) Find time period of P_2.

(b) Which planet has greater speed at *P*?

(c) Which planet has greater energy at *P*?

11. MARS ORBITER MISSION (MANGALYAAN)

Mass Orbiter Mission (MOM) is a space probe orbiting Planet Mars since 24 september 2014. This was launched by ISRO (Indian Space Research Organisation) on 5 November 2013. MOM was put into an orbit around Earth. Over a span of nearly one month, its orbit was raised six times by firing rockets. The rockets were fired when MOM was nearest to Earth in its path (at Perihelion). This was done to impart maximum increase in *KE* of the spacecraft while using as little fuel as possible.



Let the speed of spacecraft be v when a rocket is fired to impart a thrust in the direction of its motion. By burning a certain quantity of fuel, the rocket can cause the velocity to increase by some fixed amount Δv.

$$k = \frac{1}{2} mv^2$$

$$\Delta k = \frac{1}{2}m\,(2v)\Delta v = mv\Delta v$$

Thus, for a given Δv, change in KE is maximum when v is maximum and speed of the spacecraft it maximum at perihelion.

Finally, on 01 December 2013, a rocket was fired to impart enough energy for the probe to leave the gravity of earth. MOM took a hyperbolic trajectory under influence of the sun's gravity. The flight path of MOM intersected the orbit of Mars at the moment when Mars was too close. At this moment, its speed was very large and it would have escaped Mars gravity too. A rocket was fired to slow it down to an adequate speed. It got captured into an orbit around Mars. Since then, it has sent many revealing photographs of Martian surface.

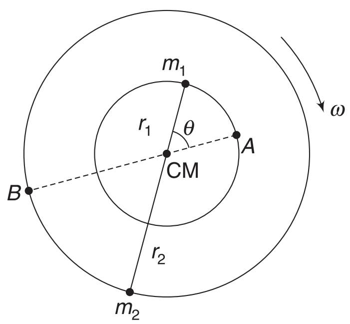
YOUR TURN

Q.37 Do you think that timing the escape of MOM from the gravity of Earth was important? Why?

12. BINARY STARS

There are many pairs of stars, which revolve about their common centre of mass, under mutual gravitational attraction. Many a times, observation of a star reveals that it is revolving but the companion star is invisible. Well, the companion is a black hole! Fact of the matter is that both – the earth and the moon – are revolving about their common COM. The COM of the Earth – Moon system is located inside the earth due to its large mass. The centre of the earth revolves on a circle of relatively small radius.

Consider a binary star system comprising two stars, of masses m_1 and m_2, revolving around their common COM. Both have the same time period and are always located on a diameter.



Binary stars. As m_1 moves to A, the other star of mas m_2 moves to B.

r = separation between stars

r_1 and r_2 are distances of the two stars from their COM.

$$m_1r_1 = m_2r_2 \qquad \text{...(i)}$$

and $$r_1 + r_2 = r \qquad \text{...(ii)}$$

Solving the above two equations,

$$r_1 = \frac{m_2r}{m_1 + m_2} \text{ and } r_2 = \frac{m_1r}{m_1 + m_2} \qquad \text{...(iii)}$$

Equation for centripetal force:

$$m_1r_1\omega^2 = m_2r_2\omega^2 = \frac{Gm_1m_2}{r^2} \qquad \text{...(iv)}$$

$$\Rightarrow \quad \omega^2 = \frac{Gm_2}{r_1r^2} = \frac{Gm_2}{\dfrac{m_2r^3}{m_1 + m_2}} = \frac{Gm}{r^3} \text{ where } M = m_1 + m_2$$

$$\Rightarrow \quad \omega = \sqrt{\frac{GM}{r^3}}$$

Time period of revolution, $T = \dfrac{2\pi}{\omega} = \dfrac{2\pi r^{3/2}}{\sqrt{GM}}$ ...(38)

Kinetic energy of the system of binary stars:

$$k = \frac{1}{2} m_1(r_1\omega)^2 + \frac{1}{2} m_2(r_2\omega)^2$$

$$= \frac{1}{2}m_1\left(\frac{m_2r\omega}{m_1 + m_2}\right)^2 + \frac{1}{2}m_2\left(\frac{m_1r\omega}{m_1 + m_2}\right)^2$$

$$= \frac{1}{2}\left(\frac{m_1m_2}{m_1 + m_2}\right)r^2\omega^2 = \frac{1}{2}\mu r^2\omega^2 \qquad \text{...(39)}$$

where, $\mu = \dfrac{m_1m_2}{m_1 + m_2}$ = reduced mass of the system

Angular momentum of the system about COM:

$$L = (I_1 + I_2)\,\omega = (m_1r_1^2 + m_2r_2^2)\,\omega$$

Substituting for r_1 and r_2 from (iii) and simplifying gives

$$L = \mu r^2 \cdot \omega \qquad \text{...(40)}$$

Effectively, we can replace this two-body problem with a single-body problem in which a body of mass μ is revolving in a circular orbit of radius r. The body experiences a force $\dfrac{Gm_1m_2}{r^2}$ toward the centre.

Miscellaneous Examples

Example 29 A thin uniform rod has mass M and length $2L$. A particle of mass M is kept at a distance a from the rod on its perpendicular bisector. Find gravitational force on the particle due to the rod.

Solution

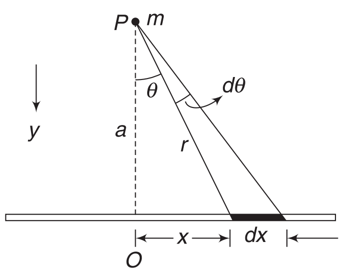
Concepts

(i) We will need to consider an infinitesimally small mass on the rod and write force on m due to that

(ii) Adding forces due to all small masses making the rod will give the answer.

(iii) Considering identical elements on two sides of centre (O) of the rod helps us to see that the resultant force is along the perpendicular bisector.

(iv) Below, we have expressed our integration in terms of θ. This keeps the integration simple, though one can always express the entire thing in terms of x.



Mass per unit length of the rod is $\lambda = \dfrac{M}{L}$

Consider an element of angular width $d\theta$ as shown.

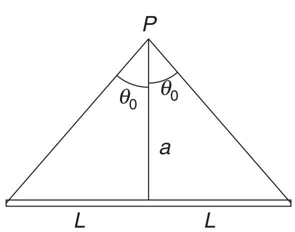
Length (dx) of the element can be expressed in terms of θ as follows.

[Note that it is wrong to write $dx = rd\theta$. Why?]

$$x = a\tan\theta \quad \Rightarrow \quad \frac{dx}{d\theta} = a\sec^2\theta$$

$$\Rightarrow \qquad dx = a\sec^2\theta \cdot d\theta$$

Mass of element, $dM = \lambda dx = \dfrac{Ma}{L}\sec^2\theta\, d\theta$



Force on m due to dM is

$$dF = \frac{GmdM}{r^2} = \frac{Gm\left(\dfrac{Ma}{L}\sec^2\theta d\theta\right)}{(a\sec\theta)^2}$$

$$\Rightarrow \qquad dF = \frac{GmM}{aL}d\theta.$$

Component of this force along the line PO is

$$dF_y = dF\cos\theta = \frac{GmM}{aL}\cos\theta \cdot d\theta$$

By considering an identical element to the left of O, one can easily show that the resultant force on m will be along PO.

Resultant force is obtained by adding dF_y due to all elements.

$$\therefore \qquad F_y = \int dF_y = \frac{GmM}{aL}\int_{-\theta_0}^{\theta_0}\cos\theta_0\, d\theta$$

$$= \frac{GmM}{aL}[\sin\theta]_{-\theta_0}^{\theta_0}$$

$$= \frac{2GmM}{aL}\sin\theta_0 = \frac{2GmM}{aL}\frac{L}{\sqrt{a^2 + L^2}}$$

$$\therefore \qquad F_y = \frac{2GmM}{a\sqrt{L^2 + a^2}}$$

Example 30 Potential energy of a particle of mass m at infinite distance from the earth (mass M) is assumed to be zero. Its PE on the surface of earth is $-\dfrac{GMm}{R}$ where R is radius of the earth. Find the change in PE of the particle when it is raised to a height $h(\ll R)$ above the surface of the earth.

Solution

Concepts

PE at a height h is $\left(-\dfrac{GMm}{R+h}\right)$

$$U_0 = -\frac{GMm}{R};\ U_h = -\frac{GMm}{(R+h)}$$

$$\Delta U = U_h - U_0 = -GMm\left[\frac{1}{R+h} - \frac{1}{R}\right]$$

$$= \frac{GMmh}{R(R+h)} = \frac{GMmh}{R^2\left(1 + \dfrac{h}{R}\right)}$$

$$= \frac{gmh}{1 + \dfrac{h}{R}}$$

when $\quad h \ll R, \frac{h}{R} \ll 1$

$\therefore \quad \Delta U \simeq mgh.$

The result is as expected.

Example 31 Distance between the centres of two stars is $10a$. The masses of these stars are M and $16M$ and their radii are a and $2a$ respectively. A body is fired straight from the surface of the larger stars towards the surface of the smaller star. What should be its minimum initial speed to reach the surface of the smaller star?

Solution

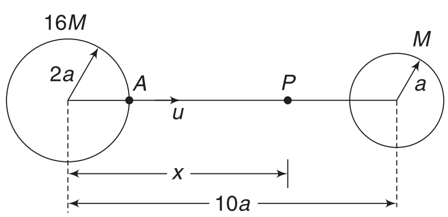
Concepts

(i) Mistake that one can do in this problem is to apply energy conservation, taking the speed of the body on the surface of the smaller star to be zero.

The body can never reach the smaller star with zero speed.

(ii) At points close to the larger star, its gravitational pull is more powerful. Similarly, at points very close to the smaller star, gravitational pull of the smaller star will be much stronger than the pull of the larger star. There is a point between them where field is zero. Force on a body kept there will be zero. Once the projected body crosses this point, it will be automatically pulled by the gravity of the smaller star.

(iii) While writing potential energy of the body, we must consider both the stars.



Let P be the point where a body experiences no force. To the left of P, a body gets attracted towards the bigger star and to the right of P, a body will fall towards the smaller star.

$$\frac{G(16M)m}{x^2} = \frac{G(M)m}{(10a - x)^2}$$

$$\Rightarrow \quad \frac{10a - x}{x} = \frac{1}{4} \quad \Rightarrow \quad x = 8a$$

A body projected from point A slows down till it reaches P. If it just manges to reach P (with almost zero speed), it will definitely hit the smaller star. To the right of P, pull of smaller star is stronger.

Energy conservation,

$$K_A + U_A = K_P + U_P$$

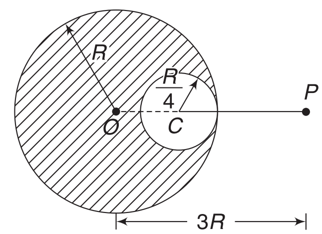
$$\frac{1}{2}mu^2 + \left[-\frac{G16Mm}{2a} - \frac{GMm}{8a}\right]$$

$$= 0 + \left[-\frac{G16Mm}{8a} - \frac{GM \cdot m}{2a}\right]$$

Simplifying this gives

$$u = \frac{3\sqrt{5}}{2}\sqrt{\frac{GM}{a}}$$

Example 32 The earth is a uniform sphere of mass M and radius R. Assume that a massive spherical cavity of radius $\frac{R}{4}$ is dug out, as shown in the figure. Find acceleration due to gravity and gravitational potential at a point P shown in the figure.



Solution

Concepts

(i) Both field and potential obey the principle of superposition.

(ii) Field (or potential) at P is equal to field (or potential) due to complete Earth (mass M) minus the field (or potential) due to the spherical mass in the cavity.

Mass of volume $\frac{4}{3}\pi R^3$ is M.

Therefore, mass of volume $\frac{4}{3}\pi\left(\frac{R}{4}\right)^3$ is $m = \frac{M}{64}$

Acceleration due to gravity is equal to the field at a point.

Field at P due to mass M is $E_1 = \frac{GM}{(3R)^2}$ (Towards PO)

Field at P due to mass m is $E_2 = \frac{GM}{(CP)^2}$

$$= \frac{GM/64}{(9R/4)^2} \text{ (towards } PO)$$

$\therefore$ Field at P due to the cavitied sphere is

$$E = E_1 - E_2 = \frac{GM}{9R^2} - \frac{GM}{4 \times 81R^2} = \frac{35}{324}\frac{GM}{R^2}$$

Potential can be calculated in similar manner, as

$$V = V_1 - V_2 = -\frac{GM}{3R} - \left(-\frac{GM/64}{9R/4}\right) = -\frac{47}{144}\frac{GM}{R}$$

Example 33 A body is projected from the surface of the earth in vertically upward direction. Velocity given to it is just sufficient to take it to infinity. Calculate the time that is taken to reach a height R above the surface. R is the radius of the earth.

Solution

Concepts

(i) Speed of projection is escape speed = $\sqrt{2gR}$

(ii) Using energy conservation, we can find speed (v) at a distance r from the centre.

(iii) Now, putting $v = \frac{dr}{dt}$, separating the variables and integrating gives the answer.

Use of energy conservation greately simplifies the mathematics.

Projection speed is $u = \sqrt{2gR}$

Let v be the speed at distance r from the centre.

Conservation of energy gives:

$$\frac{1}{2}mv^2 - \frac{GMm}{r} = \frac{1}{2}mu^2 - \frac{GMm}{R}$$

$$\Rightarrow \quad v^2 = \frac{2GM}{r} + 2gR - 2gR$$

$$\left[\because \; u^2 = 2gR \text{ and } \frac{GM}{R} = gR\right]$$

$$\Rightarrow \quad v^2 = \frac{2GM}{r} = \frac{2GM}{R^2}\frac{R^2}{r} = 2g\frac{R^2}{r}$$

$$\Rightarrow \quad v = \sqrt{2g}\cdot R\cdot\frac{1}{\sqrt{r}} \quad \Rightarrow \quad \frac{dr}{dt} = \sqrt{2g}\cdot R\cdot\frac{1}{\sqrt{r}}$$

$$\Rightarrow \quad \int_0^t dt = \frac{1}{R\sqrt{2g}}\int_R^{2R}\sqrt{r}\,dr$$

$$\Rightarrow \quad t = \frac{2}{3}\frac{1}{R\sqrt{2g}}\left[r^{3/2}\right]_R^{2R}$$

$$= \frac{2}{3R}\frac{1}{\sqrt{2g}}\left[(2R)^{3/2} - (R)^{3/2}\right]$$

$$= \frac{\sqrt{2}}{3}(2\sqrt{2} - 1)\sqrt{\frac{R}{g}}$$

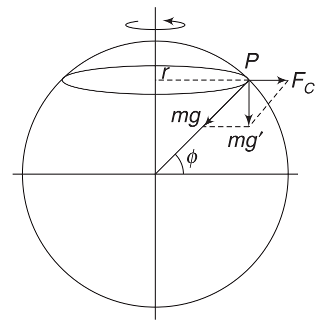
Example 34 Considering the angular speed of Earth to be ω about its rotation axis, find the fractional change in value of acceleration of free fall due to rotation of Earth at a place located at latitude ϕ. Radius of the Earth is R.

Solution

Concepts

(i) A body (mass m) on the surface of Earth feels the gravitational pull of Earth and a centrifugal force. If $\vec{F}$ is the resultant of these two forces, then acceleration of free fall is $\vec{g}' = \frac{\vec{F}}{m}$

(ii) Latitude of a place is the angle that position vector of the place relative to the centre of the earth makes with equatorial plane.



Consider a particle of mass m at place P (at latitude ϕ). Particle rotates (with the Earth) in a circle of radius

$$r = R\cos\phi$$

Centrifugal force on it is

$$F_c = m\omega^2 r$$

Resultant of gravitational force mg and F_c has been shown as mg'

$$mg' = \sqrt{(mg)^2 + F_c^2 + 2\cdot mg(F_c)\cdot\cos(180 - \phi)}$$

$$\Rightarrow \quad g' = \sqrt{g^2 + (\omega^2 r)^2 - 2g\omega^2 r\cdot\cos\phi}$$

The term $\omega^4 r^2$ is extremely small due to small angular speed of Earth $\left(\omega = \frac{2\pi}{24 \times 3600}\frac{\text{rad}}{\text{s}}\right)$

$$\therefore \quad g' \simeq \sqrt{g^2 - 2g\omega^2 r\cos\phi} = \left[g^2 - 2g\omega^2 R\cos^2\phi\right]^{\frac{1}{2}}$$

$$= g\left[1 - \frac{2\omega^2 R}{g}\cos^2\phi\right]^{\frac{1}{2}}$$

Using binomial expansion and neglecting higher order terms

$$g' \simeq g\left[1 - \frac{R\omega^2\cos^2\phi}{g}\right] = g - R\omega^2\cos^2\phi$$

$$\text{Fractional change} = \frac{\Delta g}{g}$$

$$= -\frac{R\omega^2\cos^2\phi}{g}$$

Example 35 Acceleration due to gravity at the poles of Earth is $g = 9.80000\,\text{ms}^{-2}$. Find its value at a height of 32 km above the north pole. Radius of Earth, R = 6,400 km(exact).

Solution

Concepts

(i) Value of g has been gives up to 6 significant digits. We must give answer up to 6 digits.

(ii) $g' = g\left[1 - \dfrac{2h}{R}\right]$ may not be useful, as we have neglected higher-order terms, which may affect the answer if we need 6 significant digits. We must include other higher terms as well.

$$g' = \frac{g}{\left[1 + \dfrac{h}{R}\right]^2} = g\left[1 + \frac{h}{R}\right]^{-2}$$

We know that $(1 + x)^n = 1 + nx + \dfrac{n(n-1)}{2!}x^2 + \dfrac{n(n-1)(n-2)}{3!}x^3 + \ldots$

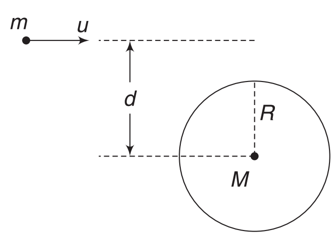
$$\therefore \quad g' = g\left[1 - \frac{2h}{R} + \frac{(-2)(-2-1)}{2}\left(\frac{h}{R}\right)^2 + \frac{(-2)\,(-2-1)\,(-2-2)}{1 \times 2 \times 3}\left(\frac{h}{R}\right)^3 + \ldots\right]$$

$$= g[1 - 2 \times 0.005 + 3 \times (0.005)^2 - 4(0.005)^3 + \ldots]$$

$$= g[1 - 0.010 + 0.000075 + \text{very small terms}]$$

$$= 9.80000 \times 0.990075 = 9.70274$$

Example 36 An asteroid of mass m is approaching a planet of mass $M (\gg m)$ from a large distance. The speed of the asteroid is u and the impact parameter is $d = 2R$, where R is the radius of the planet. The asteroid just misses the planet. Find u.



Solution

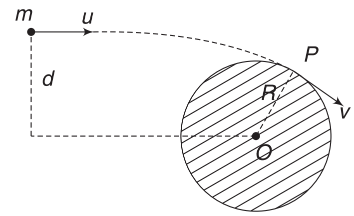
Concepts

(i) Since $M \gg m$, we can treat the planet to remain fixed.

(ii) Due to gravitational pull of the planet, the asteroid takes a curved path. It just misses the planet. This means it passes tangentially to the planet.

(iii) The angular momentum of the asteroid about the centre of the planet is conserved as the only force that it experiences passes through the centre.

(iv) Energy is conserved.



The asteroid passes tangentially to the planet at P.

Let its speed at P be v.

Angular momentum conservation about O:

$$mvR = mud \quad \Rightarrow \quad vR = u(2R) \quad \Rightarrow \quad v = 2u \qquad \ldots\text{(i)}$$

Energy conservation:

$$K_\infty + U_\infty = K_P + U_P$$

$$\Rightarrow \quad \frac{1}{2}mu^2 + 0 = \frac{1}{2}mv^2 - \frac{GMm}{R}$$

$$\Rightarrow \quad \frac{u^2}{2} = \frac{(2u)^2}{2} - \frac{GM}{R} \quad \Rightarrow \quad \frac{3}{2}u^2 = \frac{GM}{R}$$

$$\Rightarrow \quad u = \sqrt{\frac{2GM}{3R}} = \sqrt{\frac{2}{3}gR}$$

Example 37 A satellite is suddenly stopped in its circular orbit and released. Show that the time in which it will fall on the planet is $\dfrac{1}{4\sqrt{2}}$ times its original time period of revolution. Assume that the original radius of circular path of the satellite is large compared to the radius of the planet.

Solution

Concepts

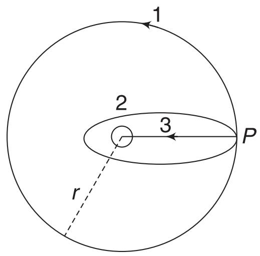
(i) For $v_0 = \sqrt{\dfrac{GM}{r}}$ path is circular. If $v < v_0$, path becomes elliptical. If v is made very small, the path will be an extended flat ellipse (eccentricity close to 1). If $v = 0$, we can think that the straight line path (on which the satellite falls) is an ellipse with eccentricity $e = 1$.

(ii) Since planet is small, we can treat it like a point.

(iii) After the satellite is stopped, its path (a straight line) is an ellipse with major axis equal to r. Here, r is the radius of its original circular path.

Path of the satellite is straight line path 3. It can be regarded as an extreme case of elliptical path, with $e = 1$. Semi-major axis of this ellipse is

$$a = \frac{r}{2}$$



Path 1 is original circular path of radius r. If speed of satellite is made very smal, it will follow elliptic path 2. It speed is made zero, it will take path 3.

If T_1 is the original time period in circular orbit and T_2 is time period in path 3, then

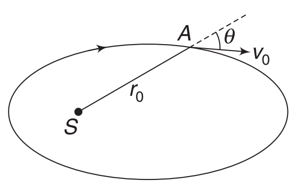
$$\left(\frac{T_2}{T_1}\right)^2 = \left(\frac{r/2}{r}\right)^3$$

$$\Rightarrow \quad T_2 = \frac{T_1}{\sqrt{8}}$$

Actually, the satellite will never complete ellipse 3 (which is a straight line). It will hit the planet after completing half the ellipse. Hence, required time is

$$t = \frac{T_2}{2} = \frac{T_1}{2\sqrt{8}} = \frac{T_1}{4\sqrt{2}}$$

Example 38 A planet is going around the sun (S). At a certain instant, distance of the planet from the sun is r_0 and its speed is $v_0 = \sqrt{\frac{GM}{2r_0}}$. Velocity vector of the planet makes an angle $\theta = 30°$ with the radius vector. Calculate maximum and minimum distance of the planet from the sun. M = mass of the sun.



Solution

Concepts

(i) Angular momentum of the planet about the sun is conserved.

(ii) Angular momentum at aphelion and perihelion can be written as mvr since v is $\perp$ to r.

In the position shown, angular momentum is $m(v_0 \sin\theta)r$.

(iii) Energy of the planet is conserved.

Let v be the velocity when it is perpendicular to the radius vector (r). At this instant, the planet is either at maximum or minimum separation from the sun.

Conservation of angular momentum:

$$mvr = m(v_0 \sin\theta) r_0$$

$$\Rightarrow \quad vr = \frac{v_0 r_0}{2} \quad \Rightarrow \quad v = \frac{1}{2r}\sqrt{\frac{GMr_0}{2}} \qquad \text{...(i)}$$

Conservation of energy gives:

$$\frac{1}{2}mv^2 - \frac{GMm}{r} = \frac{1}{2}mv_0^2 - \frac{GMm}{r_0}$$

$$\Rightarrow \quad \frac{1}{16}\frac{GMr_0}{r^2} - \frac{GM}{r} = \frac{GM}{4r_0} - \frac{GM}{r_0}$$

$$\Rightarrow \quad \frac{r_0}{16r^2} - \frac{1}{r} = -\frac{3}{4r_0} \quad \Rightarrow \quad 12r^2 - 16r_0 \cdot r + r_0^2 = 0$$

$$\Rightarrow \quad r = \left(\frac{4 \pm \sqrt{13}}{6}\right) r_0$$

$$\therefore \quad r_{\min} = \left(\frac{4 - \sqrt{13}}{6}\right) r_0 \quad \text{and} \quad r_{\max} = \left(\frac{4 + \sqrt{13}}{6}\right) r_0$$

Example 39 A satellite of Earth is revolving in a circular orbit of radius x. A gun in the satellite in aimed directly towards Earth. It fires a bullet with a velocity equal to half the orbital speed of the satellite, relative to the satellite. The satellite is massive and its recoil can be neglected. Find the maximum and minimum distance of the bullet from the centre of the earth during its subsequent motion.

Solution

Concepts

(i) Velocity of bullet wrt the satellite is $\frac{v_0}{2}$ in radial direction, where v_0 is the orbital speed of the satellite. $v_0 = \sqrt{\frac{GM}{x}}$.

(ii) Velocity of bullet (wrt earth) is the vector sum of satellite's velocity v_0 (tangential) and $\frac{v_0}{2}$ (radial).

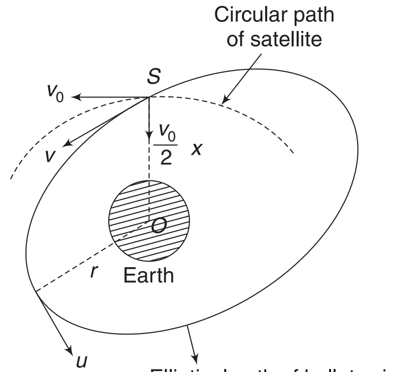
(iii) Angular momentum of the bullet about the centre of the earth and its energy is conserved.

$$v_0 = \sqrt{\frac{GM}{x}} \qquad \text{...(i)}$$

Speed of the bullet immediately after it leaves the satellite is

$$v = \sqrt{v_0^2 + \left(\frac{v_0}{2}\right)^2} = \frac{\sqrt{5}}{2} v_0$$

Let the speed of the bullet be u when its velocity is perpendicular to the position vector wrt the centre of the earth. Let its distance from the centre of the earth be r at this instant.



Elliptical path of bullet. v is velocity of bullet. It is vector sum of v_0 and $\frac{v_0}{2}$.

Conservation of angular momentum:

$$mur = mv_0 \cdot x$$

$$\Rightarrow \quad ur = v_0 x = \sqrt{GMx} \qquad \text{...(i)}$$

Energy conservation:

$$\frac{1}{2}mu^2 - \frac{GMm}{r} = \frac{1}{2}m\left(\frac{\sqrt{5}}{2}v_0\right)^2 - \frac{GMm}{x}$$

$$\Rightarrow \quad \frac{5}{8}v_0^2 - \frac{GM}{x} = \frac{v^2}{2} - \frac{GM}{r}$$

Substituting $v_0 = \sqrt{\frac{GM}{x}}$ and $u = \sqrt{\frac{GMx}{r}}$

$$\frac{5}{8}\frac{GM}{x} - \frac{GM}{x} = \frac{x^2}{r^2}\frac{GM}{2x} - \frac{GM}{r}$$

$$\Rightarrow \quad 3r^2 - 8x \cdot r + 4x^2 = 0$$

$$\Rightarrow \quad r = \frac{8x \pm \sqrt{64\,x^2 - 48x^2}}{6} = 2x \text{ and } \frac{2x}{3}$$

$$\therefore \quad r_{\max} = 2x,\ r_{\min} = \frac{2x}{3}$$

Worksheet 1

1. The force of gravitation is

(a) repulsive (b) electrostatic
(c) conservative (d) non-conservative

2. Which of the following is the evidence to show that there must be a force acting on earth and directed towards the sun?

(a) Deviation of the falling bodies towards east
(b) Revolution of the earth round the sun
(c) Phenomenon of day and night
(d) Apparent motion of sun round the earth

3. The atmosphere is held to the earth by:

(a) winds (b) gravity
(c) clouds (d) None of these

4. If the earth stops rotating, the value of free fall accelaration at the equator will:

(a) increase (b) remain same
(c) decrease (d) None of these

5. If the earth rotates faster than its present speed, the weight of an object will

(a) increase at the equator but remain unchanged at the poles
(b) decreases at the equator but remain unchanged at the poles
(c) remain unchanged at the equator but decreae at the poles
(d) remain unchanged at the equator but increase at the poles

6. If R is the radius of the earth and g is the acceleration due to gravity on the earth's surface, the mean density of earth is:

(a) $4\pi G/3gR$ (b) $3\pi R/4gG$
(c) $3g/4\pi RG$ (d) $\pi RG/12G$

7. Acceleration due to gravity on moon is 1/6 of the acceleration due to gravity on earth. If the ratio of densities of earth (ρ_e) and moon (ρ_m) is $\left(\frac{\rho_e}{\rho_m}\right) = \frac{5}{3}$ then radius of moon R_m in terms of radius of the earth R_e will be:

(a) $\frac{5}{18} R_e$ (b) $\frac{1}{6} R_e$
(c) $\frac{3}{18} R_e$ (d) $\frac{1}{2\sqrt{3}} R_e$

8. If the mass of earth is 80 times of that of a planet and diameter is double that of planet and 'g' on earth is 9.8 ms^{-2}, then the value of 'g' on that planet is:

(a) 4.9 ms^{-2} (b) 0.98 ms^{-2}
(c) 0.49 ms^{-2} (d) 49 ms^{-2}

9. The distance of the centres of moon and earth is D. The mass of earth is 81 times the mass of the moon. At what distance from the centre of the earth, the gravitational force on a partcile will be zero:

(a) $\frac{D}{2}$ (b) $\frac{2D}{3}$
(c) $\frac{4D}{3}$ (d) $\frac{9D}{10}$

10. The value of 'g' at a particular point is 9.8 ms^{-2}. Suppose the earth suddenly shrinks uniformly to half its present size without losing any mass. The value of 'g' at the same point (assuming that the distance of the point from the centre of earth does not shrink) will now be

(a) 4.9 ms^{-2} (b) 3.1 ms^{-2}
(c) 9.8 ms^{-2} (d) 19.6 ms^{-2}

11. Weight of a body of mass m decreases by 1% when it is raised to height h above the earth's surface. If the body is taken to a depth h in a mine, change in its weight is:

(a) 2% decrease (b) 0.5 decrease
(c) 1% increase (d) 0.5% increase

12. At what depth below the surface of the earth, acceleraton due to gravity g will be half its value 1,600 km above the surface of the earth [radius of the earth = 6,400 km]

(a) 4,352 km (b) 3,192 km
(c) 1,563 km (d) None of these

13. R is the radius of the earth and ω is its angular velocity and g_p is the value of g at the poles. The effective value of g at the latitude λ = 60° will be equal to:

(a) $g_p - \frac{1}{4} R\omega^2$ (b) $g_p - \frac{3}{4} R\omega^2$
(c) $g_p - R\omega^2$ (d) $g_p + \frac{1}{4} R\omega^2$

14. If both the mass and the radius of the earth decrease by 1%, the value of the acceleration due to gravity will

(a) decrease by 1% (b) increase by 1%
(c) increase by 2% (d) increase by 0.5%

15. If the radius of earth is decreased by 4% and its density remains same, then escape velocity from its surface will

(a) remain same (b) incerease by 4%
(c) decrease by 4% (d) increase by 2%

16. The escape speed for a projectile in the case of earth is 11.2 kms^{-1}. A body is projected from the surface of the earth with a velocity which is euqal to twice the escape speed. The velocity of the body when it is at infinite distance from the centre of the earth is:
(a) 11.2 kms^{-1} (b) 22.4 kms^{-1}
(c) $11.2\sqrt{3}$ kms^{-1} (d) $11.2\sqrt{2}$ kms^{-1}

17. The gravitational potential energy of a body of mass 'm' at the earth's surface is $-mgR_e$. Its gravitational potential energy at a height R_e from the earth's surface will be (Here, R_e is the radius of the earth)
(a) $-2mgR_e$ (b) $2mgR_e$
(c) $\frac{1}{2}mgR_e$ (d) $-\frac{1}{2}mgR_e$

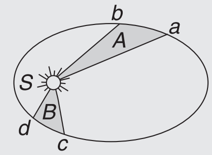
18. A satellite of earth is moving in its orbit with a constant speed v. If the gravity of earth suddenly vanishes, then this satellite will
(a) continue to move in the orbit with velocity v.
(b) start moving with velocity v in a direction tangential to the orbit.
(c) fall down with increased velocity
(d) be lost in outer space

19. A satellite is moving around the earth. In order to make it move to infinity, its velocity must be increased by
(a) 20%
(b) it is impossible to do so
(c) 82.8% (d) 41.4%

20. A planet of mass m is moving in an elliptical orbit about the sun (mass of sun = M). The maximum and minimum distances of the planet from the sun are r_1 and r_2 respectively. The period of revolution of the planet will be proportional to
(a) $r_1^{3/2}$ (b) $r_2^{3/2}$
(c) $(r_1 - r_1)^{3/2}$ (d) $(r_1 + r_2)^{3/2}$

21. A ball is dropped from a spacecraft revolving around the earth at a height of 120 km. What will happen to the ball?
(a) It will continue to move with velocity v along the original orbit of spacecraft
(b) It will move with the same speed tangentially to the spacecraft
(c) It will fall down to the earth gradually
(d) It will go very far in the space

22. The figure shows the motion of a planet around the sun in an elliptical orbit with sun at the focus. The shaded areas A and B are also shown in the figure, which can be assumed to be equal. If t_1 and t_2 represent the time for the planet to move from a to b and d to c respectively, then



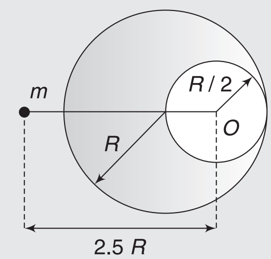
(a) $t_1 < t_2$ (b) $t_1 > t_2$
(c) $t_1 = t_2$ (d) $t_1 \leq t_2$

23. A satellite is revolving around earth in a circular orbit. The radius of orbit is half of the radius of the orbit of moon. Satellite will complete one revolution in
(a) $2^{-3/2}$ lunar month (b) $2^{-2/3}$ lunar month
(c) $2^{3/2}$ lunar month (d) $2^{2/3}$ lunar month

24. A particle falls on earth: (i) from infinity (ii) from a height 10 times the radius of earth. The ratio of the velocities gained on reaching the earth's surface is
(a) $\sqrt{11}:\sqrt{10}$ (b) $\sqrt{10}:\sqrt{11}$
(c) 10 : 11 (d) 11 : 10

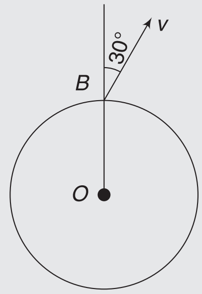
25. Two satellites of same mass m are launched in the same circular orbit of radius r around the earth (mass M) so as to rotate opposite to each other. If they collide inelastically and stick together as wreckage, the total energy of the system just after collision is:
(a) $-\frac{2GMm}{r}$ (b) $-\frac{GMm}{r}$
(c) $\frac{GMm}{2r}$ (d) $\frac{GMm}{4r}$

26. A solid sphere of radius $R/2$ is cut out of a solid sphere of radius R such that the spherical cavity so formed touches the surface on one side and the centre of the sphere on the other side, as shown. The initial mass of the solid sphere was M. A particle of mass m is placed at a distance $2.5R$ from the centre of the cavity O, as shown. What is the gravitational attraction on the mass m?



(a) $\frac{GMm}{R^2}$ (b) $\frac{GMm}{2R^2}$
(c) $\frac{GMm}{8R^2}$ (d) $\frac{23}{100}\frac{GMm}{R^2}$

27. A body B is fired with a velocity of magnitude $\sqrt{gR} < v < \sqrt{2gR}$ at an angle of 30° with the radius vector of earth. If at the highest point, the speed of the body is $v/4$, the maximum height attained by the body is equal to:



(a) $R/2$
(b) R
(c) $\sqrt{2}R$
(d) none of these

28. Suppose the gravitational force varies inversely as the n[th] power of distance. Then the time period of a planet in circular orbit of radius R around the sun will be proportional to

(a) $R^{\left(\frac{n+1}{2}\right)}$ (b) $R^{\left(\frac{n-1}{2}\right)}$
(c) R^n (d) $R^{\left(\frac{n-2}{2}\right)}$

29. A satellite is launched into a circular orbit of radius R around the earth. A second satellite is launched into an orbit of radius $(1.01)R$. The period of the second satellite is larger than that of the first one by approximately

(a) 0.5% (b) 1.0%
(c) 1.5% (d) 3.0%

30. If the distance between the earth and the sun becomes half its present value, the number of days in a year would have been

(a) 64.5 (b) 129
(c) 182.5 (d) 730

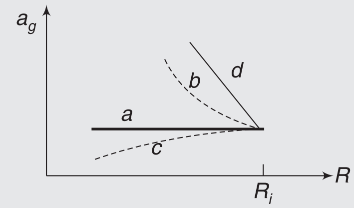
31. A spherical uniform planet is rotating about its axis. The velocity of a point on its equator is v. Due to the rotation of planet about its axis, the acceleration due to gravity g at equator is 1/2 of g at poles. The escape velocity of a particle on the pole of the planet in terms of v is

(a) $v_e = 2v$ (b) $v_e = v$
(c) $v_e = v/2$ (d) $v_e = \sqrt{3}\,v$

32. The escape velocity for a planet is v_e. A tunnel is dug along a diameter of the planet and a small body is dropped into it at the surface. When the body reaches the centre of the planet, its speed will be

(a) v_e (b) $\frac{v_e}{\sqrt{2}}$
(c) $\frac{v_e}{2}$ (d) zero

33. A (non-rotating) star collapses onto itself from an initial radius R_i, with its mass remaining unchanged. Which curve in the figure best gives the gravitational acceleration a_g on the surface of the star, as a function of the radius of the star, during the collapse?



(a) a (b) b
(c) c (d) d

34. A satellite revolves in the geostationary orbit but in a direction east to west. The time interval between its successive passing about a point on the equator is:

(a) 48 h (b) 24 h
(c) 12 h (d) never

35. A satellite of mass $5M$ orbits the earth in a circular orbit. At one point in its orbit, the satellite explodes into two pieces, one of mass M and the other of mass $4M$. After the explosion, the mass M ends up travelling in the same circular orbit, but in opposite direction. After explosion, the mass $4M$ is

(a) in a circular orbit
(b) unbound
(c) elliptical orbit
(d) data is insufficient to determine the nature of the orbit.

36. A satellite can be in a geostationary orbit around earth at a distance r from the centre. If the angular velocity of earth about its axis doubles, a satellite can now be in a geostationary orbit around earth if its distance from the centre is

(a) $\frac{r}{2}$ (b) $\frac{r}{2\sqrt{2}}$
(c) $\frac{r}{(4)^{1/3}}$ (d) $\frac{r}{(2)^{1/3}}$

37. Satellites A and B are orbiting around the earth in orbits of radii R and $4R$ respectively. The ratio of their areal velocities is:

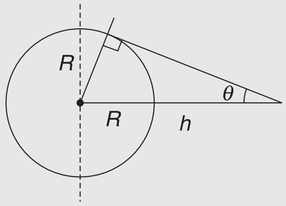
(a) 1 : 2 (b) 1 : 4
(c) 1 : 8 (d) 1 : 16

Worksheet 2

1. The magnitudes of the gravitational force on a particle at distances r_1 and r_2 from the centre of a uniform sphere of radius R and mass M are F_1 and F_2 respectively. Then
   (a) $\dfrac{F_1}{F_2} = \dfrac{r_1}{r_2}$ if $r_1 < R$ and $r_2 < R$
   (b) $\dfrac{F_1}{F_2} = \dfrac{r_2^2}{r_1^2}$ if $r_1 > R$ and $r_2 > R$
   (c) $\dfrac{F_1}{F_2} = \dfrac{r_1}{r_2}$ if $r_1 > R$ and $r_2 > R$
   (d) $\dfrac{F_1}{F_2} = \dfrac{r_2^2}{r_1^2}$ if $r_1 < R$ and $r_2 < R$

2. Inside a hollow isolated spherical shell,
   (a) gravitational potential is zero everywhere.
   (b) gravitational field is zero everywhere.
   (c) gravitational potential is same everywhere.
   (d) gravitational field is same everywhere.

3. A geostationary satellite is at a height h above the surface of earth. If earth's radius is R (colatitude at a place is 90°-latitude angle)



   (a) The minimum colatitude on earth upto which the satellite can be used for communication is $\sin^{-1}(R/(R + h))$.
   (b) The maximum colatitudes on earth upto which the satellite can be used for communication is $\sin^{-1}(R/(R + h))$.
   (c) The area on earth's surface that cannot receive communication from this satellite is given as $2\pi R^2(1 + \sin\theta)$.
   (d) The area on earth's surface that can receive signal from this satellite is given as $2\pi R^2(1 + \cos\theta)$.

4. When a satellite in a circular orbit around the earth enters the atmospheric region, it encounters small air resistance to its motion. Then
   (a) its kinetic energy increases
   (b) its kinetic energy decreases
   (c) its angular momentum about the earth decreases
   (d) its period of revolution around the earth increases

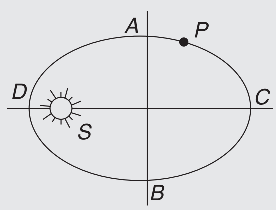
5. A communication satellite
   (a) goes round the earth from east to west
   (b) can be in the equatorial plane only
   (c) can be vertically above any place on the earth
   (d) goes round the earth from west to east

6. A satellite S is moving in an elliptical orbit around the earth. The mass of the satellite is very small compared to the mass of the earth
   (a) the acceleration of S is always directed towards the centre of the earth
   (b) the angular momentum of S about the centre of the earth changes in direction, but its magnitude remains constant
   (c) the total mechanical energy of S varies periodically with time
   (d) the linear momentum of S remains constant in magnitude

7. For a satellite to orbit around the earth, which of the following must be true?
   (a) It must be above the equator at some time
   (b) It cannot pass over the poles at any time
   (c) Its height above the surface cannot exceed 36,000 km
   (d) Its period of rotation must be $> 2\pi\sqrt{R/g}$ where R is the radius of earth

8. Two satellites s_1 & s_2 of equal masses revolve in the same sense around a heavy planet in co-planar circular orbit of radii R & $4R$
   (a) the ratio of period of revolution of s_1 & s_2 is 1 : 8.
   (b) their velocities are in the ratio 2 : 1
   (c) their angular momentum about the planet are in the ratio 2 : 1
   (d) the ratio of angular velocities of s_2 and s_1 is 2 : 5.

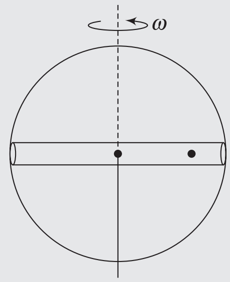
9. Figure shows the orbit of a planet P around the sun S. AB and CD are the minor and major axes of the ellipse.



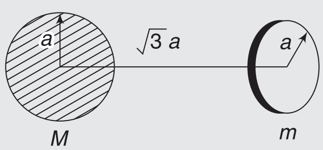
   (a) Time needed for the planet to travel half the ellipse ACB is same as the time needed to travel the other half BDA.
   (b) If U is the potential energy and K is the kinetic energy, then $|U| > |K|$ at both D & C
   (c) If U is the potential energy and K is the kinetic energy, then $|U| > |K|$ at D but not at C
   (d) If U is the potential energy and K is the kinetic energy, then $|U| > |K|$ at C but not at D

Worksheet 3

1. A planet has uniform mass density ρ and is rotating about its axis with an angular speed ω. A straight smooth tunnel is dug across it, perpendicular to its axis. An object placed inside the tunnel, at any location, does not experience acceleration. Find ω.



2. A uniform ring of mass m lies at a distance of $\sqrt{3}a$ from the centre of a solid sphere of mass M. Axis of the ring is the diameter of the sphere. Both objects have same radius a. Find the force applied by the ring on the sphere.



3. A body of mass m is raised to a height h above the surface of the earth and you use the equation $\Delta U = mgh$ for calculating change in PE. At what value of h will the error in value of ΔU be 1%?

4. Three stars, each of mass m, are located at the vertices of an equilateral triangle of side length a. They revolve in circular orbit under mutual gravitational force while preserving the equilateral triangle. Find the speed of each star. Also find the mechanical energy of the system.

5. A scientist found that a planet of radius $3R$ is made of two different materials. The inner core of radius R is a uniform dense liquid. Mass of the core is M. The remaining part is solid with density half that of the liquid core. Find the gravitational acceleration of a particle at points

   (a) R and (b) $3R$ from the centre of the planet.

6. A satellite is revolving around the earth in an equatorial plane from west to east direction. Its time period is 8 hours. On 17 January 2018, at 8 AM, the satellite was exactly above a city Pontianak in Indonesia [Pontianak is a city, which lies on both the hemispheres of Earth]. At what time and on which date the satellite will once again be overhead Pontianak? Just write the time and date nearest to 17/01/2018, 8 AM.

7. 1 astronomical unit (AU) is average Earth – sun distance. It was estimated that within a space of radius $r = 3 \times 10^9$ AU, centred at the centre of a galaxy, all celestical bodies have a total mass of 2.8×10^{11} times that of our sun. Stars located at a distance r from the centre of the galaxy were found to be orbiting the centre with a period $T = 3 \times 10^8$ years. Assume that the entire mass of the galaxy within radius r can be assumed as a point mass at its centre for calculating the time period of a star and estimate the mass that must be present in the galaxy in a space of radius r.

   How much is this mass different from the estimate made by visual observations as given in this problem? This missing mass is dark matter.

8. The speed of a point on the equator of a planet is u due to rotation of the planet about its own axis. Due to rotation, the weight of a body at the equator is half of its weight at the poles of the planet. Calculate the speed with which a body on a pole shall be projected from the pole so that it just manages to escape the gravity of the planet.

9. A 'moon' of a planet (mass = M) is orbiting in a circular orbit of radius r. It suddenly explodes into three pieces in mass ratio 1:1:4. Immediately after the explosion, one of the smaller fragments starts orbiting the planet in reverse direction in the same orbit. The other smaller piece has zero initial velocity after the explosion and falls straight towards the planet.

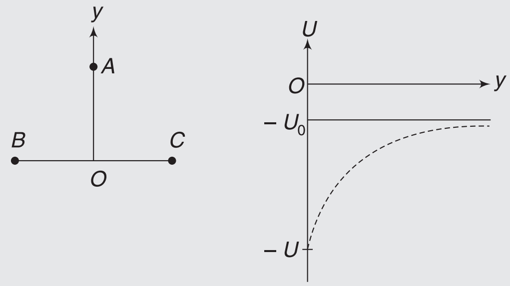
   (a) Find the speed of the smaller piece when it is about to hit the surface of the planet. Radius of the planet is $\frac{r}{2}$.

   (b) What happened to the third piece? Is is still a satellite?

10. Two particles A and B have masses m and $2m$ respectively. They are held at separation r_0 in space. A is given a velocity v_0 along the line joining the two masses (away from B) and B is released simultaneously. Find the range of velocity v_0 for which the two particles would remain bound under their mutual gravitation.

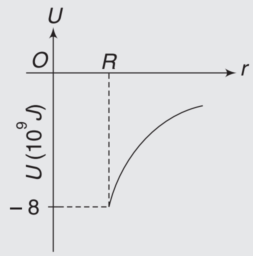
11. Two identical particles are fixed at points B and C. A third particle A is moved along perpendicular bisector of the line BC. It is moved from $y = 0$ to $y = \infty$. Mass of the third particle is twice the mass of particle at B (or C). The potential energy of the three-particle system changes with distance (y) of A

from O as shown in the graph. Express value of U in terms of U_0.



12. A projectile is fired from the surface of the earth at an angle $\theta = 60°$ to the vertical. Its projection speed is $u = \sqrt{\frac{GM}{R}}$. How high does the projectile rise? Neglect rotation of the earth. Mass of the earth is M and its radius is R.

13. Gravitational PE of a particle of mass $m = 1\,\text{kg}$ varies with distance (r) from the centre of a planet, as shown in the figure. Radius of the planet is R. The particle is projected from the surface of the planet with a KE of 6×10^9 J. The maximum height attained by the particle above the surface is R. At what angle to the vertical was the projectile fired?



14. At what angular speed of rotation will the surface material on the equator of a planet be on verge of flying off the surface of the planet, if the planet is spherical with radius 1,800 km and mass 3×10^{24} kg?

15. A planet of mass M and a star of mass $2M$ are revolving around their common COM under influence of mutual gravitational pull. Distance between them is r. Find the total mechanical energy of the system.

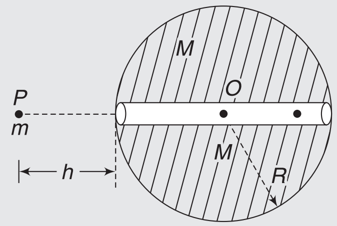
16. A ring has mass M and radius R. Write gravitational potential due to the ring at

(a) its centre

(b) a point on its axis at a distance x from the centre.

16a. A satellite of mass m is revolving in a circular orbit of radius r around the earth (Mass M and radius R). Due to atmospheric drag, it loses energy at a constant rate w. It follows a spiral path and falls to earth. Estimate the time in which it will fall to earth.

17. A tunnel is dug along the diameter of the earth. A particle of mass m is released from point P at a height $h = R$. The particle enters the tunnel and moves without friction.



(a) What is the maximum displacement of the particle after it is released?

(b) What is the maximum speed of the particle after it is released?

18. The minimum and maximum distance of a satellite from the centre of the earth are $2R$ and $4R$ respectively. Mass of the earth is M and its radius is R. Find:

(a) The minimum speed of the satellite in its path.

(b) The radius of curvature of the path when it is nearest to the earth.

19. A satellite is moving in an elliptical orbit of semi-major axis a around a planet of mass M. Find its speed when it is at a distance r from the centre of the planet.

Answers Sheet

Your Turn

1. It becomes 4 times
2. Forces between building and us are relatively small. These tiny forces are not meaningful when we are overwhelmed by the huge attraction of Earth.
3. They have same acceleration.

4. 3.56×10^{22} N

5(a). $\frac{\sqrt{2}\ GMm}{x^2}$ (b) $\frac{\sqrt{3}\ Gm^2}{a^2}$

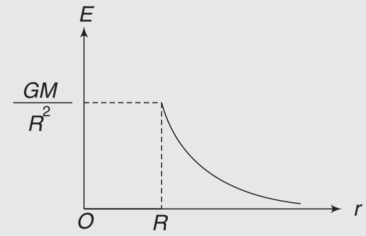
6. (a) 0 (b) $\frac{GMm}{x^2}$

7. $\frac{GMm}{2L^2}$

8. $(4GM)/(3x^2)$ along DA

9. zero

10. $\frac{2GM}{\pi R^2}$

11. 

12. $1.62\,\text{N kg}^{-1}$

13. $\frac{R}{2}$

14. $2\pi\sqrt{\frac{R}{g}} \simeq 84.6$ minutes

15. $\frac{3}{4}R$

16. (a) 1600 km (b) 11.5 km

17. Nearly 11 km

18. $39.2\,\text{ms}^{-2}$

19. $\frac{6Gm^2}{a}\left(1 - \frac{1}{\sqrt{2}}\right)$

20. $\sqrt{\frac{Gm}{r}}$

21. $\sqrt{gR}$

22. (a) Yes (b) $-0.4E_0$ (c) E_0

23. $\frac{v_A}{v_B} = \frac{1}{2}$

24. $2R$

25. (a) Outside (b) $-3.14 \times 10^8 J$

26. No

27. $-\frac{GM}{R}$

28. 3-3

29. (a) Increases (b) Decreases

30. 8.48 h

31. (a) $h = R$ (b) $\sqrt{gR}$

32. 9.6×10^{10} J

33. (a) No (b) Elliptical

34. Yes

35. $\frac{v_1 r_1}{r_2 \sin\theta}$

36. (a) 3.3 years (b) P_2 (c) P_2

37. Yes, so that MOM intercepts the Mars orbit when the planet was there.

Worksheet 1

1. (c)	2. (b)	3. (b)	4. (a)	5. (b)	6. (c)	7. (a)	8. (c)	9. (d)
10. (c)	11. (b)	12. (a)	13. (a)	14. (b)	15. (c)	16. (c)	17. (d)	18. (b)
19. (d)	20. (d)	21. (a)	22. (c)	23. (a)	24. (a)	25. (a)	26. (d)	27. (b)
28. (a)	29. (c)	30. (b)	31. (a)	32. (b)	33. (b)	34. (c)	35. (b)	36. (c)
37. (a)								

Worksheet 2

1. (a,b)	2. (b,c,d)	3. (a,c)	4. (a,c)	5. (b,d)	6. (a)	7. (a,d)	8. (a,b)
9. (b)							

Worksheet 3

1. $\omega = \sqrt{\frac{4}{3}\pi GP}$

2. $\frac{\sqrt{3}\ GMm}{8a^2}$

3. 64 km

4. $\sqrt{\frac{GM}{a}}, \frac{-3Gm^2}{2a}$

5. (a) $\frac{GM}{R^2}$ (b) $\frac{5}{9}\frac{GM}{R^2}$

6. 8 PM, 17/01/2018

7. 2×10^{10} times mass of the sun

8. $2u$

9. (a) $\sqrt{\frac{2GM}{r}}$ (b) will escape

10. $v_0 < \sqrt{\frac{6Gm}{r_0}}$

11. $9U_0$

12. $\frac{R}{2}$

13. $\sin^{-1}\left(\sqrt{\frac{2}{3}}\right)$

14. 5.85×10^{-2} rad s^{-1}

15. $-\frac{GM^2}{r}$

16. (a) $-\frac{GM}{R}$ (b) $-\frac{GM}{\sqrt{R^2 + x^2}}$

16a. $\frac{GMm}{2W}\left(\frac{1}{R} - \frac{1}{r}\right)$

17. (a) $4R$ (b) $\sqrt{2gR}$

18. (a) $\sqrt{\frac{GM}{6R}}$ (b) $\frac{8R}{3}$

19. $v = \sqrt{GM\left(\frac{2}{r} - \frac{1}{a}\right)}$

CHAPTER 12

Elasticity

"The finest steel has to go through the hottest fire"

–*Richard M. Nixon*

1. INTRODUCTION

While dealing with solid bodies, we assumed them to be perfectly rigid. However, no object in real world is perfectly rigid. When an object is subjected to external forces, it undergoes changes in size, shape, or both. The subject of elasticity deals with the behaviour of those substances which have the property of recovering their size and shape when the forces producing deformations are removed. This elastic property is present in all solids to some extent. The extent to which the shape of a body is restored when the deforming forces are removed depends on microscopic structure of the solid.

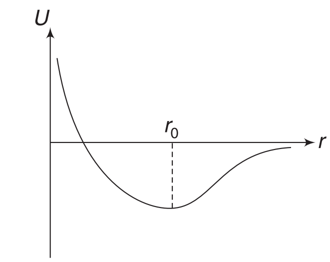
Study of elastic properties of material is of immense importance for engineering designs. When you ride a lift, you often come across messages like – "carrying capacity 400 kg''. The message may have a lot to do with the strength and elastic properties of the rope holding the elevator car.

2. MICROSCOPIC REASON FOR ELASTICITY

The property of a body to restore the original shape or to oppose the deformation is known as elasticity. Steel is able to completely regain its natural shape after deforming forces are removed (provided deformation is small). We say that steel is *perfectly elastic*. If a body has no tendency to regain its shape after deforming forces are removed, it is said to be perfectly *inelastic* or *plastic*.

Clay, putty and dough are very close to being perfectly inelastic. Most of the real-life objects are somewhere in between. They try to regain their shapes after forces are removed but cannot regain their original shape completely. Some deformation remains.

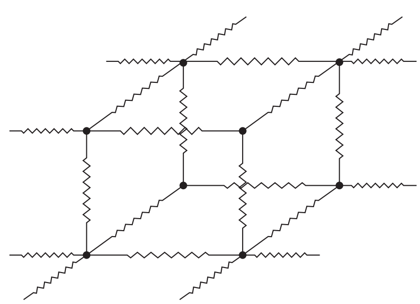
The potential energy of interaction of two neighbouring molecules in a solid varies with separation (r) between them in a typical fashion, as shown in the figure.



r_0 is equilibrium separation.

At some specific separation (r_0), the potential energy is minimum $\left(\frac{dU}{dr} = 0\right)$ and the molecules are in equilibrium. All molecules in a solid will try to settle down at this separation (r_0). If, by applying some external forces, we try to increase the separation between the molecules, the force turns attractive $\left(\frac{dU}{dr}\right.$ is positive and force $F = -\frac{dU}{dr}$ is negative$\left.\right)$, trying to restore the original position of the molecules. When external forces decrease the inter-molecular separation, the forces amongst the molecules becomes repulsive ($\frac{dU}{dr}$ is negative for $r < r_0$); again trying to push back the molecules to their original separation.

A large number of atoms come together and settle into their equilibrium position in a three-dimensional lattice, to form a metallic solid. The interatomic forces can be *modelled as* being produced by tiny (and stiff) springs holding these atoms. Increasing the atomic separation will create a restoring attractive force and decreasing the distance between the atoms will cause a repulsive force between the atoms.



Atom in a metallic solid form a repetitive lattice. Springs represent the behaviour of interatomic forces.

You may note that, in materials not so rigid, like a rubber pipe, the molecules make flexible chains, each chain being loosely bound to its neighbour.

3. STRESS AND STRAIN

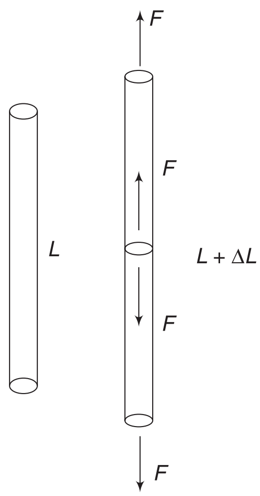
When a body is subjected to deforming forces, three kinds of deformations can be seen. Associated with each of these deformations, there are internal restoring stresses.

Stress characterises the strength of force that is causing deformation and strain is a measure of the amount of deformation produced with respect to the size of the body.

Let us take a look at three types of stresses and strains.

3.1 Longitudinal Stress and Strain

Consider a metal rod subjected to two forces (F) of equal magnitude, acting at the opposite ends, so as to keep the rod in equilibrium. The rod does not move but gets stretched. We say that the rod is subjected to a *longitudinal tensile stress*. If we consider an intermediate cross-section of the rod, the two sides pull each other with force F. This is internal restoring force. The *longitudinal stress* in the rod is defined as



Tensile stress

$$\sigma_l = \frac{F}{A} \quad \text{...(1)}$$

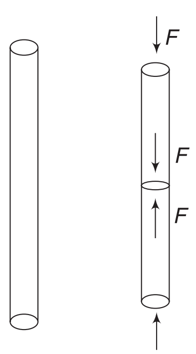
where A is the area of cross-section of the rod.

Obviously, force F applied to the ends of a thick rod and a thin rod will not have the same deforming effects.

Force per unit area gives us a better idea of the deformation-producing ability of the force. Unit of stress is Nm^{-2}.

If the forces applied to the ends of the rod are so as to compress it (see fig), we say that the rod is subjected to *longitudinal compressive stress*. At any cross-section, the two parts of the rod push against each other. Stress in the rod is still defined by equation (1).

When the rod is stretched, it elongates. Its length (L) increase by ΔL. When the rod is subjected to a compressive stress, its length decreases by ΔL. In both cases, we define *longitudinal strain* developed in the rod as fractional change in its length.



Compressive stress

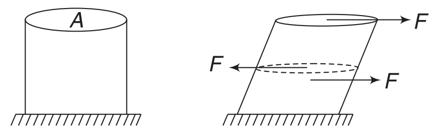
$$\text{Longitudinal strain, } \in = \frac{\Delta L}{L} \quad \text{...(2)}$$

One may use ± sign to differentiate between tensile and compressive strains.

Strains is a dimensionless quantity. It has no unit.

3.2 Shear Stress and Strain

Consider a solid cylinder of cross-sectional area A, fixed rigidly to the ground. A force F is applied on the top surface, in a direction tangential to the surface. Here, the force is trying to cause a deformation in which two parallel surfaces have a tendency to slide over one another. Internal forces develop to prevent this sliding.



Shear stress

In the figure shown, at any horizontal cross-section of the cylinder, the two parts apply a tangential force on one another.

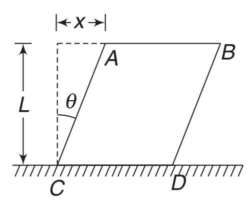
Note that the cylinder shown in figure above is in equilibrium. There are forces acting on it due to the ground as well.

Shear stress is defined as tangential force per unit area. In the figure shown, shear stress is

$$\sigma_s = \frac{F}{A} \quad \text{...(3)}$$

Unit of shear stress is also Nm^{-2}.

Shear strain is a measure of deformation produced when a body is subjected to shear stress. We can define it as displacement of a layer AB divided by its distance from the fixed layer CD.



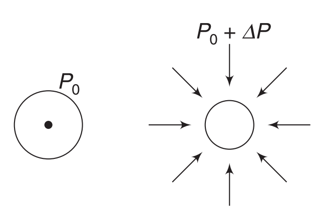
Deformation is small
Shear strain = tan θ

$$\in_s = \frac{x}{L} = \tan\theta \quad \text{...(4)}$$

Shear strain is dimensionless.

3.3 Volume Stress and Strain

All bodies on the surface of Earth are subjected to atmospheric pressure (P_0). When pressure on a body is changed uniformly from all sides, we say that it is subjected to *Volume stress* or *Bulk stress*. Change in pressure (ΔP) is defined as volume stress.



Increase is pressure is stress

$$\text{Volume stress, } \sigma_v = \Delta P \quad \text{...(5)}$$

For example, if you submerge a ball deep inside water, the volume stress on it is $\Delta P = \rho g h$, where ρ is the density of water and h is the depth.

When subjected to volume stress, the volume of a body changes. *Volume strain* is defined as the fractional change in volume. It has no unit.

$$\text{Volume strain, } \in_V = \frac{\Delta V}{V} \qquad ...(6)$$

Example 1 A steel wire is 1 m long and another wire is 2 m long. Both are stretched so as to elongate them by 1 mm. Can we say that the both wires are equally deformed? Give reasons.

Solution Strain is correct measure of deformation.

For first wire, longitudinal strain is

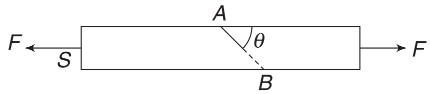
$$\in_1 = \frac{\Delta l_1}{l_1} = \frac{1\,\text{mm}}{1\text{m}} = 10^{-3}$$

$$\text{For second wire, } \in_2 = \frac{\Delta l_2}{l_2} = \frac{1\,\text{mm}}{2\text{m}} = 0.5 \times 10^{-3}$$

First wire is more deformed.

Intermolecular distance have incrased more in the first wire. You can visualise this by considering two segments of 1 m length in the second wire. Each 1 m segment is elongated by 0.5 mm only.

Example 2 A uniform rod of cross-sectional area S is pulled at its two ends by applying forces of equal magnitude acting along its length. Consider a cross-section (AB) of the rod that is inclined at an angle θ to the length of the rod. Write longitudinal and shear stress developed at the section AB.



Solution

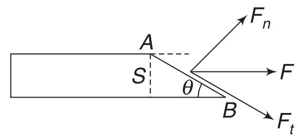
Concepts

$$\text{Stress} = \frac{\text{Force}}{\text{Area}}$$

For longitudinal stress, we consider force that is normal to the area and for shear stress, we consider component of forces tangential to the area.

Area of section AB is

$$S' = \frac{S}{\sin\theta}$$



Normal and tangential components of F are

$$F_n = F\sin\theta;\ F_t = F\cos\theta$$

$\therefore$ Longitudinal stress at section AB is

$$\sigma_l = \frac{F_n}{S'} = \frac{F\sin\theta}{\frac{S}{\sin\theta}} = \frac{F}{S}\sin^2\theta$$

Shear stress at section AB is

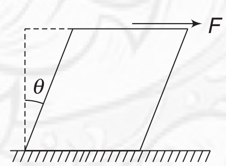
$$\sigma_s = \frac{F_t}{S'} = \frac{F\cos\theta}{\frac{S}{\sin\theta}} = \frac{F}{S}\sin\theta\cdot\cos\theta$$

YOUR TURN

Q.1 A wire of length L = 1.0 m is pulled by applying force of equal magnitudes at its two ends. Magnitude of each forces is 100 N and area of cross section of the wire is $A = 10\,\text{mm}^2$. The wire elongates by 2 mm. Find

(i) Longitudinal strain (ii) Longitudinal stress

Q.2 A cuboidal block is fixed to the ground. A tangential force applied to the top surfaces deforms the block as shown, with $\theta = 1°$. Find shear strain.



Q.3 A ball is taken deep inside the ocean to a depth of 2 km. Find volume stress on it.

[Density of sea water = 1.0×10^3 kgm^{-3}, g = 9.8 ms^{-2}]

4. HOOKE'S LAW

For small deformations, stress in a body is proportional to strain. This is known as Hooke's law. This is true for all three types of stresses and the corresponding strains. This law was discovered by English scientist Robert Hooke, in the seventeenth century.

Stress $\propto$ Strain

$$\text{Stress} = E(\text{Strain}) \qquad ...(7)$$

Proportionality constant (E) is known as *modulus of elasticity*.

We define three moduli of elasticity for three types of stresses and strains.

4.1 Young's Modulus (Y)

Consider a rod of length L and cross-section A, stretched by force F.

Longitudinal Stress, $\sigma_l = \dfrac{F}{A}$

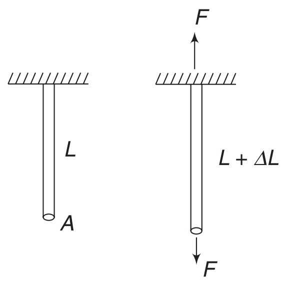
If the rod elongates by ΔL, then longitudinal strain is

$$\in_l = \frac{\Delta L}{L}$$

From Hooke's law (for small deformations)

Longitudinal Stress ∝ Longitudinal Strain

$$\Rightarrow \quad \frac{\sigma_l}{\in_l} = Y \qquad \text{...(8)}$$



Rod stretched by force F. Ceiling applies force F at the other end.

Proportionality constant Y is known as *Young's modulus of elasticity*. It is a constant for a given material.

We can write

$$Y = \frac{\sigma_l}{\in_l} = \frac{FL}{A\Delta L} \qquad \text{...(9)}$$

If the rod is subjected to compressive stress, value of Y does not change.

Example 3 One end of a wire 2 m long and 0.4 cm^2 in cross-section is fixed to the ceiling of a room and a 5 kg load is attached to the free end. Find the extension in the wire. Young's modulus of the materials of the wire is

$$Y = 1.2 \times 10^{11}\ \text{Nm}^{-2}.\ \text{Take } g = 10\ \text{ms}^{-2}$$

Solution

Concepts

Direct use of equation (9)

Tension in wire $F = Mg = 5 \times 10 = 50\,\text{N}$

Using (9), $\Delta L = \dfrac{FL}{AY} = \dfrac{(50\,\text{N})(2\,\text{m})}{(0.4\times10^{-4}\,\text{m}^2)(1.2\times10^{11}\,\text{Nm}^{-2})}$

$$= 2.08 \times 10^{-5}\ \text{m}$$

Example 4 Two wires of equal cross-sections have lengths L_1 and L_2. They are made of steel and copper respectively. They are joined end-to-end and the combination is held under a tension such that elongation in steel wire is same as that in the copper wire. Find the ratio of the original lengths of the two wires. Young's modulus of steel and copper are

$$Y_s = 2.0 \times 10^{11}\,\text{Nm}^{-2} \quad \text{and}$$

$$Y_{cu} = 1.1 \times 10^{11}\,\text{Nm}^{-2}$$

Solution

Concepts

Since, both wires have the same cross-section and tension, stress is same in both of them.

$$(\text{Stress})_s = Y_s(\text{Strain})_s = Y_s\frac{\Delta L}{L_1}$$

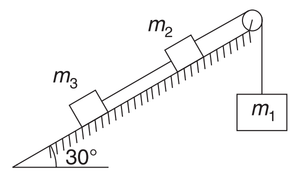
$$(\text{Stress})_{cu} = Y_{cu}(\text{Strain})_{cu} = Y_{cu}\frac{\Delta L}{L_2}$$

$$\therefore \quad (\text{Stress})_s = (\text{Stress})_{cu}$$

$$\therefore \quad Y_s\frac{\Delta L}{L_1} = Y_{cu}\frac{\Delta L}{L_2}$$

$$\Rightarrow \quad \frac{L_1}{L_2} = \frac{Y_s}{Y_{cu}} = \frac{2}{1.1} = \frac{20}{11}$$

Example 5 Three blocks of masses $m_1 = 4\,\text{kg}$, $m_2 = 2\,\text{kg}$, $m_3 = 2\,\text{kg}$ are connected with steel wires of cross-sectional area $A = 0.05\,\text{cm}^2$, as shown in figure. The incline is smooth and the pulley has no mass. Length of the wire connecting m_1 and m_2 is 2 m where as length of the other wire is 1 m. Find the strain in the two wires when the system is released to move. Take $g = 10\,\text{ms}^{-2}$ and $Y_{\text{steel}} = 2 \times 10^{11}\ \text{Nm}^{-2}$



Solution

Concepts

We need to find tension in both wires using Newton's law of motion. Then, using equation (9) separately for the two wires will give the answer.

Acceleration of the system is

$$a = \frac{(m_1 - m_2\sin30° - m_3\sin30°)g}{m_1 + m_2 + m_3}$$

$$\therefore \quad a = \frac{\left(4 - 2\times\frac{1}{2} - 2\times\frac{1}{2}\right)\times 10}{4+2+2} = 2.5\ \text{ms}^{-2}$$

Let T_1 = tension is string connecting m_1 and m_2.

$$m_1g - T_1 = m_1a \quad \Rightarrow \quad T_1 = m_1(g - a)$$

$$\Rightarrow \quad T_1 = 4(10 - 2.5) = 30\,\text{N}.$$

$$T_2 = \text{Tension in other string}$$

$$T_2 - m_3g\sin30° = m_3a$$

$$\Rightarrow \quad T_2 = 2(2.5 + 5) = 15\,\text{N}.$$

For wire having tension T_1

$$\Delta L_1 = \frac{T_1L_1}{AY} = \frac{30\times2}{(0.05\times10^{-4})\times2\times10^{11}} = 6\times10^{-5}\text{m}$$

For wire having tension T_2

$$\Delta L_2 = \frac{T_2L_2}{AY} = \frac{15\times1}{0.05\times10^{-4}\times2\times10^{11}} = 1.5\times10^{-5}\text{m}.$$

YOUR TURN

Q.4 A metal wire is subjected to a tension, and the extension observed is x. If a wire of double the radius, made of same material, is subjected to double the tension, find extension.

Q.5 A steel cable is used to suspend an elevator car. Its length is 20 m when lift is at rest and diameter is 0.05 m. Three people, having a total mass of 238 kg, enter the elevator. Find the

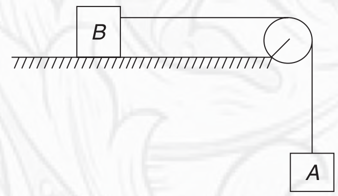
(a) Stretch in the cable when lift is at rest

(b) Stretch in the cable when lift is moving up with a constant speed of $2\,\text{ms}^{-1}$ [$Y_s = 2 \times 10^{11}\,\text{Nm}^{-2}$]

Q.6 Your leg bones have a cross-sectional area of about $9.5\,\text{cm}^2$ and experience a force of 855 N when you walk briskly. Find the fractional amount your leg bones are compressed by walking. Young's modulus for bone is

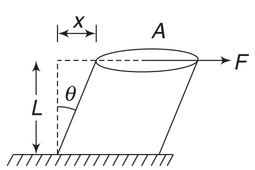
$$Y = 1 \times 10^{10}\ \text{Nm}^{-2}.$$

Q.7 Mass of block A in figure is 3 kg and that of block B is 6 kg. The cross-sectional area and Young's modulus of the wire connecting the blocks are $0.005\,\text{cm}^2$ and $2 \times 10^{11}\ \text{Nm}^{-2}$ respectively. Neglect friction and find strain developed in the wire when the system is allowed to move. Take $g = 10\,\text{ms}^{-2}$.



4.2 Shear Modulus or Modulus of Rigidity (η)

Shear Stress ∝ Shear Strain. Proportionality constant (η) is known as shear modulus of the material.



$$\eta = \frac{\text{Shear stress}}{\text{Shear strain}} = \frac{F/A}{x/L} = \frac{F}{A\theta} \qquad ...(10)$$

[For small deformations]

Note that unit of η is also Nm^{-2} (or, Pascal)

Example 6 A cubical metal block of side length $L = 25$ cm is fixed on a hard concrete floor. A force of 4000 N is applied tangentially on the upper face. Shear modulus for the metal is $\eta = 80$ GPa. Find shear strain in the block and displacement of upper surface.

Solution

Concepts

Direct use of equation (10)

$$1\,\text{GPa} = 10^9\ \text{Nm}^{-2}$$

$$\text{Shear strain} = \theta = \frac{x}{L}$$

$$\text{Shear strain} = \frac{\text{Shear stress}}{\eta} = \frac{F}{A\eta}$$

$$= \frac{4000}{(0.25 \times 0.25) \times 80 \times 10^9}$$

$$\Rightarrow \quad \text{Shear strain} = 8 \times 10^{-7}$$

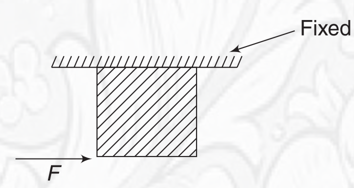
$$\Rightarrow \quad \frac{x}{L} = 8 \times 10^{-7}$$

$$\Rightarrow \quad x = 8 \times 10^{-7} \times 25\,\text{cm} = 2 \times 10^{-5}\ \text{cm}$$

This is displacement of upper surface.

YOUR TURN

Q.8 A cube of side length 5 cm has its lower face displaced by 0.2 mm when a tangential force of 8 N acts on its lower face, as shown. Find shear stress and modulus of rigidity of the cube.



4.3 Bulk Modulus (B)

Volume Stress $\propto$ Volume Strain

$$\Rightarrow \qquad \frac{\text{Volume stress}}{\text{Volume strain}} = B$$

Constant B is known as Bulk modulus of the material. If volume of a body changes by ΔV when pressure on it changes uniformly from all sides by ΔP, then

$$B = -\frac{\Delta P}{\frac{\Delta V}{V}} = -V\frac{\Delta P}{\Delta V} \qquad ...(11)$$

Negative sign is to ensure that we get a positive value of B. Actually, change in volume (ΔV) is negative (i.e., volume decrease) when pressure on a body is increased (i.e., ΔP is positive).

For infinitesimally small changes, we can write

$$B = -V\frac{dP}{dV} \qquad ...(12)$$

Unit of B is also Nm^{-2}.

It is important to note that Bulk Modulus is a term that is meaningful for solids, liquids, as well as gases. It is something that relates to elasticity of volume. All three forms of matter have volume elasticity (i.e., tendency to regain original volume).

Young's modulus and Shear modulus have no meaning for gases and liquids.

Compressibility (K) of a material is defined as the reciprocal of the Bulk Modulus (B)

$$K = \frac{1}{B} = -\frac{dV}{V\,dP} \qquad ...(13)$$

Example 7 *Bulk modulus of water*

A rigid tank contains $1 m^3$ water. It is covered with a tight fitting, light movable piston such that water in the tank is at atmospheric pressure $P_0 = 1 \times 10^5\ Nm^{-2}$. By placing weights on the piston, the pressure on water is increased to twice the atmospheric pressure. Volume of water decreases by 50 cc.

(a) Find the bulk modulus of water.

(b) Find compressibility of water.

Solution

Concepts

(i) $B = -V\frac{\Delta P}{\Delta V}$

While doing numerical calculations, we can always do away with negative sign and take positive value for everything. Ultimately, what we want is a positive value for B.

(ii) Bulk stress is not pressure. It is the change in pressure (ΔP)

(a) Volume stress, ΔP = 1 atmosphere = $1 \times 10^5\ Nm^{-2}$

$$\text{Volume strain, } \frac{\Delta V}{V} = \frac{50\,cc}{1\,m^3} = \frac{50 \times 10^{-6}}{1}$$

$$\therefore \qquad B = \frac{\Delta P}{\frac{\Delta V}{V}} = \frac{1 \times 10^5}{50 \times 10^{-6}} = 2 \times 10^9\ Nm^{-2}$$

(b) $$K = \frac{1}{B} = \frac{1}{2 \times 10^9} = 5 \times 10^{-10}\, m^2 N^{-1}$$

Note:

(i) Compressibility of mercury is much less than water. It is only $3.7 \times 10^{-11} m^2 N^{-1}$

(ii) Gases do not have fixed value of Bulk Modulus. Value of B for a gas depends on something known as thermodynamic process.

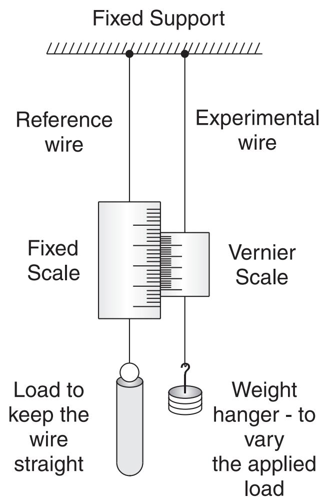
Your Turn

Q.9 Compressibility of water is $5 \times 10^{-10} m^2 N^{-1}$. Find change in volume of 1 litre water when pressure on it is increased by 15 MPa.

Q.10 Bulk modulus of a certain kind of plastic is $9.8 \times 10^8 Nm^{-2}$. To what depth should a plastic cube be taken in a lake so that its volume decreases by 0.1%?

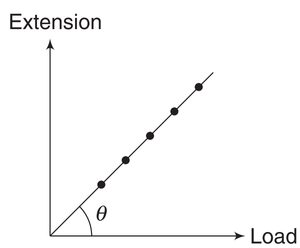
5. MEASURMENT OF YOUNG'S MODULUS

The figure shows an experimental set-up to measure Young's modulus of a wire. A long references wire, nearly 2 m long. is suspended from a fixed support. It has a scale attached to it and there is a heavy load below it. The load serves the purpose of keeping the wire straight. The experimental wire, having nearly the same length as the reference wire is suspended from the fixed support, close to the reference wire. A vernier scale attached to the end of the experimental wire can slide freely on the scale attached to the reference wire. There is a hanger attached to the vernier scale, which can carry weights.



Radius of the experimental wire is measured using a screw gauge. You will do better to measure it at several locations and take their average. Some initial weight (may be 1 kg) is placed on the hanger, to make the experimental wire straight.

Readings of the main scale and vernier scale are recorded. A known weight (say 0.5 kg) is added to the hanger. The set-up is left for about a minute to allow the elongation to happen. Reading of scale is recorded. The difference in this reading and the previous reading gives the extension in the wire. We go on adding weights and recording the extension. One should not exceed the breaking strength of the wire. For better accuracy, one can repeat the experiment in reverse order by decreasing the weight (0.5 kg at a time) and noting down the extension.



From the data obtained, a graph of extension versus load is plotted. The graph is a straight line passing through the origin.

Slope of the graph (= $\tan\theta$) is measured. It gives

$$\tan\theta = \frac{\text{Extension}}{\text{Load}} = \frac{\Delta L}{Mg}$$

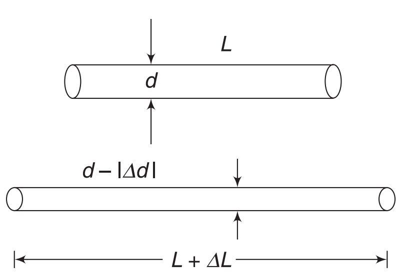
Young's modulus of the wire is calculated as

$$Y = \frac{\text{Stress}}{\text{Strain}} = \frac{Mg}{\pi r^2 \cdot \frac{\Delta L}{L}} = \frac{MgL}{\pi r^2 \Delta L}$$

$$\Rightarrow \quad Y = \frac{L}{\pi r^2 \tan\theta} \qquad \text{...(13)}$$

6. POISSON'S RATIO

When a cylindrical rod is stretched, its length increases. At the same time, its diameter decreases. The ratio of lateral strain and longitudinal strain is known as poisson's ratio.



$$\text{Lateral strain} = \frac{\text{Change in diameter}}{\text{Original diameter}} = \frac{\Delta d}{d}$$

$$\text{Longitudinal strain} = \frac{\text{Change in length}}{\text{Original length}} = \frac{\Delta L}{L}$$

Poisson's ratio is

$$\sigma = -\frac{\Delta d/d}{\Delta L/L} \qquad \text{...(14)}$$

The minus sign is for convenience so that σ comes out positive.

For most metals, $\sigma \simeq 0.25$ to 0.35. For cork, $\sigma \simeq 0$. It means the cork shows little lateral expansion when it is compressed.

Example 8 *Incompressible material*

A material whose volume does not change on (small) compression is called *incompressible*. What will be poisson's ratio for an incompressible material?

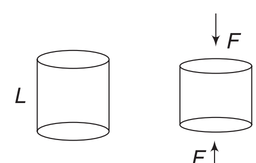
Solution

Concepts

Volume= $A \cdot L = \pi r^2 L$ for a cylindrical solid of radius r and length L. It is compressed, L decreases but r increases. If volume does not change, material is incompressible.

$$V = \pi r^2 L$$

Let a compressive force be applied on the material and its length reduce by ΔL and radius increase by Δr.



Radius increases length decreases

$$\frac{\Delta V}{V} = 2\frac{\Delta r}{r} + \frac{\Delta L}{L}$$

For $\quad \Delta V = 0;\ 2\dfrac{\Delta r}{r} = -\dfrac{\Delta L}{L}$

$$\therefore \quad \sigma = -\frac{\Delta r/r}{\Delta L/L} = \frac{1}{2} = 0.5$$

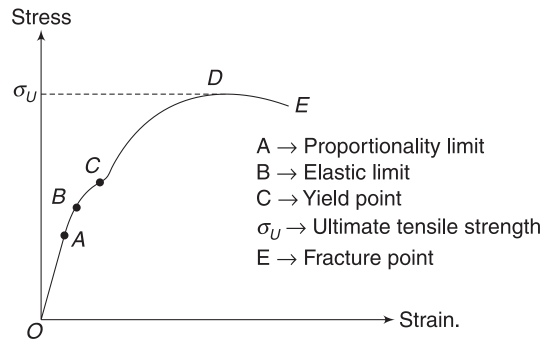
Note: For rubber, $\sigma \simeq 0.5$

7. LONGITUDINAL STRESS–STRAIN CURVE FOR A METAL WIRE

A metal wire of cross-sectional area A_0 is stretched by applying load to it. As the load is increased, it stretches more and more. We define stress as

$\sigma = \dfrac{\text{Load}}{A_0}$ and a graph of stress versus strain is plotted. A typical graph appears as shown in the figure.

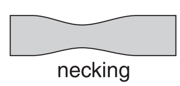
When the strain is small, the stress is proportional to strain. Stress–strain graph is a straight line and Hooke's



law is being obeyed. This is true till point A, which is known as *proportionality limit*.

Beyond A, stress is no longer proportional to strain. Graph is not a straight line. However, the material is elastic up to B. This means that the material is able to regain its shape completely if the load is removed. Point B is the *elastic limit*. Beyond B, when the load is removed, the wire is unable to regain its original length. Some permanent deformation remains. Since it is often difficult to pin-point the elastic limit, we define a point known as *yield point* (C) that is close to elastic limit. Yield point is the point at which a permanent deformation of 0.2% [i.e., strain = 0.002] remains in the wire if deforming load is removed.

If the load is increased even further, the strain increases at a much rapid pace. Beyond point D, ‘‘necking’’ is seen in the material. After necking, all subsequent deformation takes place in the neck and the material fails at E, which is known as fracture point (or breaking point).



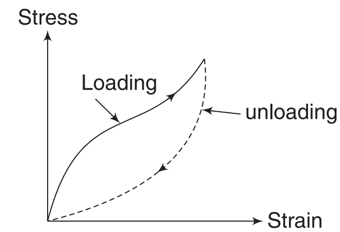
The stress corresponding to point D is known as ultimate tensile stress (σ_U) of the material. It is most common number used to judge the strength of a material in tension.

The graph takes a dip after D, indicating that the wire continues to elongate even if the load decreases a bit. Yes, this is true. The wire elongates even if the load is decreased a bit. But it does not mean that wire elongates even if stress is decreased a bit. By now, the cross-sectional area of the wire has decreased substantially and the true stress is higher even if the load is decreased slightly. The graph takes a dip because we are still calculating stress as $\left(\dfrac{\text{load}}{A_0}\right)$ when A_0 is the original cross-sectional area.

If a large deformation takes place between elastic limit and fracture point, material is called *ductile*. If it breaks soon after the elastic limit is crossed, it is called *brittle*.

8. LONGITUDINAL STRESS–STRAIN CURVE FOR RUBBER

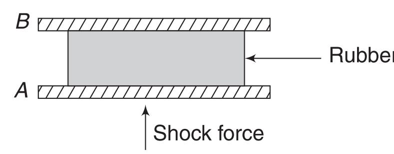
If we load a rubber cord, it stretches a lot. Strain is high but the cord remains elastic even when it has extended over few times its original length. If the load is removed, it is able to regain its original length. If the deforming force is decreased gradually, the original stress–strain curve is not retraced. The work done by rubber cord when it is returning to original length is less than the work that was done to deform it. The difference appears as heat. The phenomenon is known as *elastic hysteresis*.



Loading-represents the stress-strain curve when stretching force is being increased. Unloading represents the curve when force is being reduced.

Example 9 Why is rubber padding common in shock absorbers?

Solution Suppose a plate A is subjected to rhythmic vibration due to variable shock forces acting on it. Plate B is not in direct contact with A, but there is a rubber padding inbetween. The rubber pad is compressed and released in every cycle of vibration. In each cycle, amount of energy transmitted to B is less than amount of energy received from A. A large part of energy is dissipated by rubber due to elastic hysteries. B will have substantially less energy of vibration compared to A.

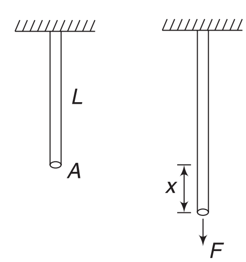


9. ELASTIC POTENTIAL ENERGY

In natural state of a body, the molecules settle in their equilibrium position and the potential energy corresponding to this position is minimum (Refer to the graph given in section 2 of this chapter). When deformed, the molecular separation changes and potential energy increases. This happens due to appearance of an internal forces, against whom work has to be performed to deform a body. The increase in potential energy when body is deformed is known as *elastic potential energy*.

Let us calculate the elastic potential energy of a wire, which is stretched by applying a force .

Consider a wire of length L and cross-section A. We pull it so as to stretch it slowly. When extension is x, the tension force is such that



$$Y = \frac{F/A}{x/L} \qquad \text{[Assuming } A \text{ to remain constant]}$$

$\Rightarrow \qquad F = \left(\frac{YA}{L}\right)x = kx$

Where $\qquad k = \frac{YA}{L}$ is a constant

Tension (F) changes with stretch (x) just like spring force. [Fact is that we assumed that an ideal spring is one which obeys Hooke's law; implying that $F \propto x$].

Therefore, the elastic potential energy of the wire, when its length is increased by ΔL is given as

$$U = \frac{1}{2}k(\Delta L)^2 = \frac{1}{2}\left(\frac{YA}{L}\right)(\Delta L)^2 \qquad ...(15)$$

This may be written as

$$U = \frac{1}{2}\left(Y\frac{\Delta L}{L}\right)\left(\frac{\Delta L}{L}\right)(LA)$$

$$\Rightarrow \quad U = \frac{1}{2}(\text{stress})(\text{strain})(\text{volume of wire}) \quad ...(16)$$

Energy stored in unit volume of the wire is known as *elastic energy density* (u)

$$\therefore \quad u = \frac{1}{2} \times \text{Stress} \times \text{Strain} \qquad ...(17)$$

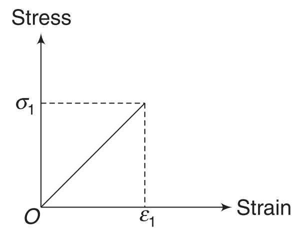
Unit of energy density (u) is Jm^{-3}.

Equation (15) can also be written as

$$U = \frac{1}{2}\left(AY\frac{\Delta L}{L}\right)\Delta L$$

$$= \frac{1}{2}(\text{maximum value of stretching force})(\text{extension}) \qquad ...(18)$$

Example 10 A steel wire has length L and cross-sectional area A. It is stretched by gradually increasing force till the strain becomes $\in_1$. The stress–strain graph for the entire process is a straight line, as shown. Find the elastic potential energy in the wire when strain is $\in_1$.



Solution

Concepts

Energy density, $u = \frac{1}{2} \times \text{Stress} \times \text{Strain}$

Area under the given graph $\left(= \frac{1}{2}\sigma_1 \in_1\right)$ gives u.

$$u = \text{Area under graph} = \frac{1}{2}\sigma_1 \in_1$$

$$\therefore \quad U = u \times \text{Volume} = \frac{1}{2}\sigma_1 \in_1 AL$$

YOUR TURN

Q.11 A metal wire is stretched by a small amount so as to produced a strain $\in$ in it. Y is Young's modulus of the material. Write the elastic energy density in the wire.

Q.12 A steel wire stretches by 2.0 mm when a load of 800 N is used to stretch it. Calculate the elastic potential energy stored in the wire.

Miscellaneous Examples

Example 11 A solid sphere of radius R, made of a material of bulk modulus B, is surrounded by a liquid in a cylindrical container. A massless piston of area A floats on the surface of the liquid. A mass M is placed on the piston to compress the liquid. Find fractional change in radius of the sphere, assuming it to be small.

Solution

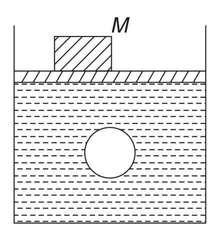
Concepts

Change in pressure due to weight placed on the piston is $\Delta P = \frac{Mg}{A}$.

As per Pascal's law, this change in pressure is transmitted uniformly throughout the liquid. Therefore, ΔP is the volume stress on the solid sphere.

Volume of sphere, $V = \frac{4}{3}\pi R^3$

If radius changes by ΔR, the corresponding change in volume is



$$\Delta V = \frac{4}{3}\pi(3R^2\Delta R)$$

$$\therefore \quad \frac{\Delta V}{V} = \frac{3\Delta R}{R}$$

Now, $\quad B = \frac{\Delta P}{-\frac{\Delta V}{V}}$

$$\Rightarrow \quad \frac{3\Delta R}{R}B = -\frac{Mg}{A}$$

$$\Rightarrow \quad \frac{\Delta R}{R} = -\frac{Mg}{3B \cdot A}$$

Negative sign indicates that R decreases.

Example 12 A wire of length $L = 10$ m is suspended from the ceiling. The radius of cross-section of the wire changes linearly from $a = 5 \times 10^{-4}$ m at the top to $b = 9.8 \times 10^{-4}$ m at the bottom. A load of mass m = 3.14 kg is attached to the lower end of the wire. Find increase in length of the wire. Young's modulus for material of the wire is $Y = 2 \times 10^{11}\,\text{Nm}^{-2}$.

Solution

Concepts

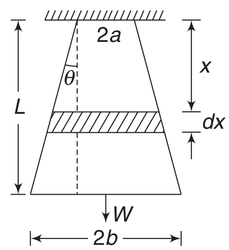
Tension is same at each cross-section but area is changing. Therefore, stress is different at different locations in the wire.

We will consider a small length dx of the wire, write extension in it and then integrate to get the overall extension.

Radius of wire at a distance x from the top is

$$r = a + x\tan\theta$$

$$= a + \left(\frac{b-a}{L}\right)x$$



Area of cross section

$$A = \pi r^2$$

$$= \pi\left[a + \left(\frac{b-a}{L}\right)x\right]^2$$

Stress $= \dfrac{W}{A}$

If extension is dl in an element of length dx, then,

$$\text{Strain} = \frac{dl}{dx}$$

$$\therefore \quad \frac{dl}{dx} = \frac{W}{AY} \Rightarrow dl = \frac{W}{AY}dx$$

$$\therefore \quad dl = \frac{mg}{Y\pi}\frac{dx}{\left[a + \left(\frac{b-a}{L}\right)x\right]^2}$$

$\therefore$ Total extension is $\int dl$

$$\Delta L = \frac{mg}{\pi Y}\int_0^L \frac{dx}{\left[a + \left(\frac{b-a}{L}\right)x\right]^2}$$

For integration, we can use substitution method.

Let $t = a + \left(\dfrac{b-a}{L}\right)x$ [at $x = 0$, $t = a$ at $x = L$, $t = b$]

$$\Rightarrow \quad dt = \left(\frac{b-a}{L}\right)dx$$

$$\therefore \quad L = \frac{mg}{\pi Y}\left(\frac{L}{b-a}\right)\int_a^b \frac{dt}{t^2}$$

$$= -\frac{mg}{\pi Y}\left(\frac{L}{b-a}\right)\left[\frac{1}{t}\right]_a^b$$

$$= -\frac{mgL}{\pi Y(b-a)}\left[\frac{1}{b} - \frac{1}{a}\right] = \frac{mgL}{\pi Yab}$$

$$= \frac{3.14 \times 9.8 \times 10}{3.14 \times 2 \times 10^{11} \times 5 \times 10^{-4} \times 9.8 \times 10^{-4}} = 10^{-3}\,\text{m}$$

Example 13 A thin ring of radius R is made of a material of density ρ and Young's modulus Y. The ring is rotated rapidly about its centre in its own place at an angular speed ω. Find the small increase in its radius.

Solution

Concepts

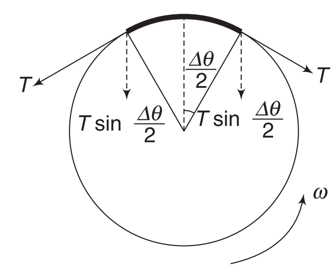
We have learnt how to find tension in the ring in the chapter on circular motion. However, we will reproduce the method here.

Tension divided by cross-sectional area gives stress. This stress is uniform throughout.

Consider an element of the ring that subtends a small angle $\Delta\theta$ at the centre.

Length of element $= R\Delta\theta$

Mass per unit length $= (A \cdot 1)\cdot\rho = A\rho$



$\therefore$ Mass of element, $\Delta m = (A\rho)(R\Delta\theta)$

Tension on either side of the element $= T$

Components of T towards centre provide the necessary centripetal force.

$$2T\sin\Delta\frac{\theta}{2} = \Delta m\omega^2 \cdot R$$

$$\Rightarrow \quad 2T\cdot\frac{\Delta\theta}{2} = A\rho R\Delta\theta\cdot\omega^2 R$$

$$\left[\text{For small } \Delta\theta,\ \sin\frac{\Delta\theta}{2} \simeq \frac{\Delta\theta}{2}\right]$$

$$\Rightarrow \quad T = A\rho\omega^2 R^2$$

$$\therefore \quad \text{Stress in the ring is } \frac{T}{A} = \rho\omega^2 R^2$$

$$\Rightarrow \quad (\text{Strain})\,Y = \rho\omega^2 R^2$$

$$\Rightarrow \quad \frac{\Delta l}{l}Y = \rho\omega^2 R^2$$

$$\Rightarrow \quad \frac{2\pi\Delta R}{2\pi R}Y = \rho\omega^2 R^2$$

$$\Rightarrow \quad \Delta R = \frac{\rho\omega^2 R^3}{Y}$$

Example 14 A block of mass 1 kg is fastened to one end of a wire of cross-sectional area $A = 3\,\text{mm}^2$ and is rotated in a vertical circle of radius 0.2 m. The speed of the block at the bottom of the circle is $2\,\text{ms}^{-1}$. Find the elongation of the wire (in cm) when the block is at the bottom of the circle. Young's modulus of the material of the wire $= 2 \times 10^{11}\ \text{Nm}^{-2}$.

Solution

Concepts

Change in length of the wire will be far less than 0.2 m because Young's modulus is high. In such a scenario, we need not worry about change in radius and assume that the block essentially swings in a circle of radius 0.2 m.

The equation of centripetal force gives the tension in the wire.

Let the tension in the wire be T and W be the weight of the block.

At the lowest point, the resultant force is $T–W$ towards the centre of the circle and it must be equal to centripetal force $m\upsilon^2/r$.

Thus, $T - W = m\upsilon^2/r$

or,
$$T = W + m\upsilon^2/r$$
$$= (1\,\text{kg})(9.8\,\text{ms}^{-2}) + \frac{(1\,\text{kg})(2\text{ms}^{-2})}{0.2\,\text{m}} = 30\,\text{N}$$

We have, $Y = \dfrac{T/A}{\Delta L/L}$ or, $\Delta L = \dfrac{TL}{AY}$

$$= \frac{30\,\text{N} \times (20\,\text{cm})}{(3 \times 10^{-6}\text{m}^2) \times (2 \times 10^{11}\text{Nm}^{-2})}$$
$$= 5 \times 10^{-5} \times 20\,\text{cm} = 10^{-3}\,\text{cm}$$

Example 15 Find the depth of lake at which density of water is 1% greater than at the surface. Given compressibility of water is $K = 50 \times 10^{-6}\text{atm}^{-1}$.

Solution

Concepts

At high pressure, volume decreases and density increases. Definition of Bulk modulus (or compressibility) will give us change in volume when pressure is increased.

For calculation of pressure, we can assume the density of water to remain constant as the variation is not much.

$$B = \frac{\Delta P}{-\Delta V/V} \Rightarrow \frac{\Delta V}{V} = -\frac{\Delta P}{B}$$

We know $P = P_{atm} + h\rho g$

Mass of a volume V of water is

$$m = \rho V = \text{constant}$$

$$d\rho V + dV\rho = 0 \Rightarrow \frac{d\rho}{\rho} = -\frac{dV}{V} \quad \text{i.e.,} \quad \frac{\Delta\rho}{\rho} = \frac{\Delta P}{B}$$

$$\Rightarrow \frac{1}{100} = \frac{h\rho g}{B} \Rightarrow h\rho g = \frac{B}{100} = \frac{1}{100K}$$

$$\Rightarrow h\rho g = \frac{1 \times 1 \times 10^5}{100 \times 50 \times 10^{-6}}$$

$$\Rightarrow h = \frac{10^5}{5000 \times 10^{-6} \times 1000 \times 10}$$

$$= \frac{100 \times 10^3}{50}\ \text{m} = 2\ \text{km}$$

Example 16 *Increment in length due to own weight*

A heavy rope of mass M and length L is hanged vertically. Find elongation in the rope due to its own weight. Assume the cross-section to remain uniform at A.

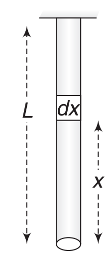
Solution

Concepts

Tension is different at different points in the rope. Stress, as well as strain, will be different at different points. A small segment at the lower end will suffer less extension compared to an identical segment at upper end.

Consider a small element of length dx in the rope at x distance from the lower end.

Tension at the location of segment is



$$T = \left(\frac{M}{L}\right)xg$$

$$\text{Stress} = \frac{T}{A} = \left(\frac{M}{L}\right)\frac{xg}{A}$$

Let increase in length of the element, be dy.

Strain in element is $= \dfrac{dy}{dx}$

Young's modulus of elasticity $Y = \dfrac{\text{Stress}}{\text{Strain}} = \dfrac{\frac{M}{L}\frac{xg}{A}}{dy/dx}$

$$\Rightarrow \left(\frac{M}{YL}\right)\frac{xg}{A}dx = dy$$

Extension in the entire wire is the summation of extension in each such element.

Total extension in the rope, $\displaystyle\int_0^{\Delta l} dy = \frac{Mg}{YLA}\int_0^L xdx = \frac{MgL}{2AY}$

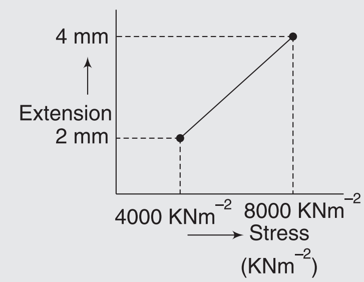
Note: Since the stress is varying linearly, we may take the average stress as

Average stress = 1/2 (Stress at top + Stress at bottom).

Strain = Average Stress/Y

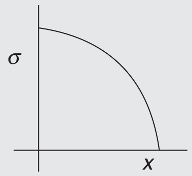
Worksheet 1

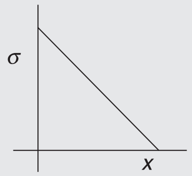
1. Two wires A and B are made of same material. The wire A has a length l and diameter r while the wire B has a length $2l$ and diameter $r/2$. If the two wires are stretched so as to produce the same strain in them, the ratio of forces applied to A and B is
   (a) 1/8 (b) 1/4
   (c) 4 (d) 8
2. A heavy, uniform rod is hanging vertically from a fixed support. It is stretched by its own weight. The diameter of the rod is
   (a) smallest at the top and gradually increases down the rod
   (b) largest at the top and gradually decreases down the rod
   (c) uniform everywhere
   (d) maximum in the middle
3. The breaking stress of a wire depends on the
   (a) material of the wire
   (b) length of the wire
   (c) radius of the wire
   (d) shape of the cross-section
4. A rope, 1 cm in diameter, breaks if the tension in it exceeds 500 N. The maximum tension that may be given to a similar rope of diameter 2 cm is
   (a) 500 N (b) 250 N
   (c) 1,000 N (d) 2,000 N
5. A wire can sustain a weight of 20 kg before breaking. If the wire is cut into two equal parts, each part can sustain a weight of
   (a) 10 kg (b) 20 kg
   (c) 40 kg (d) 80 kg
6. A wire elongates by 1.0 mm when a load W is hanged from it. If this wire goes over a pulley and two weights W each are hung at the two ends, the elongation of the wire will be
   (a) 0.5 m (b) 1.0 m
   (c) 2.0 mm (d) 4.0 mm
7. A heavy mass is attached to a thin wire and is whirled in a vertical circle. The wire is most likely to break
   (a) when the mass is at the highest point
   (b) when the mass is at the lowest point
   (c) when the wire is horizontal
   (d) at an angle of $\cos^{-1}(1/3)$ from the upward vertical
8. The length of a metal wire is l_1 when the tension in it is T_1 and is l_2 when the tension is T_2. The natural length of the wire is
   (a) $\dfrac{l_1 + l_2}{2}$ (b) $\sqrt{l_1 l_2}$
   (c) $\dfrac{l_1T_2 - l_2T_1}{T_2 - T_1}$ (d) $\dfrac{l_1T_2 + l_2T_1}{T_2 + T_1}$
9. A wire suspended vertically from one of its ends is stretched by attaching a weight of 200 N to the lower end. The weight stretches the wire by 1 mm. Then, the elastic energy stored in the wire is
   (a) 0.2 J (b) 10 J
   (c) 20 J (d) 0.1 J
10. In determination of Young's modulus of elasticity of wire, a force is applied and extension is recorded. Initial length of wire is 1 m. The curve between extension and stress is depicted. Then Young's modulus of wire will be

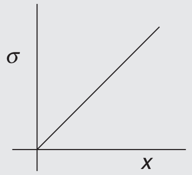


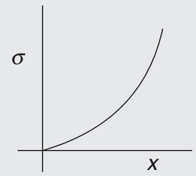
   (a) 2×10^9 Nm^{-2} (b) 1×10^9 Nm^{-2}
   (c) 2×10^{10} Nm^{-2} (d) 1×10^{10} Nm^{-2}
11. Overall changes in volume and radii of a uniform cylindrical steel wire are 0.2% and 0.002% respectively, when subjected to some suitable force. Longitudinal tensile stress acting on the wire is ($Y = 2.0 \times 10^{11}$ Nm^{-2})
   (a) 3.2×10^9 Nm^{-2}
   (b) 3.2×10^7 Nm^{-2}
   (c) 3.6×10^9 Nm^{-2}
   (d) 4.08×10^8 Nm^{-2}
12. A cylindrical wire of radius 1 mm, length 1 m, Young's modulus = 2×10^{11} Nm^{-2}, poisson's ratio $\mu = \pi/10$ is stretched by a force of 100 N. Its radius will become
   (a) 0.99998 mm (b) 0.99999 mm
   (c) 0.99997 mm (d) 0.99995 mm
13. A uniform rod is rotating in gravity-free region with certain constant angular velocity. The variation of

tensile stress with distance x from axis of rotation is best represented by which of the following graphs?

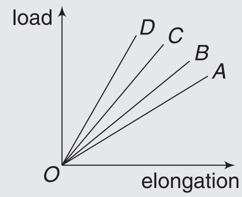
(a) 

(b) 

(c) 

(d) 

14. The load versus elongation graph for four wires of same length and made of the same material is shown in the figure. The thickest wire is represented by the line



(a) OB (b) OA

(c) OD (d) OC

Worksheet 2

1. A composite rod consists of a steel rod of length 25 cm and area 2*A* and a copper rod of length 50 cm and area *A*. The composite rod is subjected to an axial load *F*. If the Young's modulus of steel and copper are in the ratio 2 : 1.

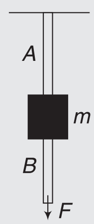
(a) The extension produced in copper rod will be more.

(b) The extension in copper and steel parts will be in the ratio 2 : 1.

(c) The stress applied to the copper rod will be more.

(d) No extension will be produced in the steel rod.

2. The wires *A* and *B*, shown in the figure, are made of the same material and have radii r_A and r_B respectively. The block between them has a mass *m*. When the force *F* is *mg*/3, one of the wires breaks.



(a) *A* breaks if $r_A = r_B$

(b) *A* breaks if $r_A < 2r_B$

(c) It is difficult to predict which wire will break if $r_A = 2r_B$

(d) The lengths of *A* and *B* must be known to predict which wire will break

3. Four rods *A*, *B*, *C*, *D* of same length and material but of different radii r, $r\sqrt{2}$, $r\sqrt{3}$ and $2r$ respectively are held between two rigid walls. The rods just fit between the walls without any stress in them.The two walls are brought slightly closer. If the rods do not bend, then

(a) the stress in the rods are in the ratio 1 : 2 : 3 : 4.

(b) the force on the rod exerted by the wall are in the ratio 1 : 2 : 3 : 4.

(c) the energy stored in the rods due to elasticity are in the ratio 1 : 2 : 3 : 4.

(d) the strains produced in the rods are in the ratio 1 : 2 : 3 : 4.

4. A body of mass *M* is attached to the lower end of a metal wire, whose upper end is fixed. The elongation of the wire is *l*.

(a) Loss in gravitational potential energy of *M* is Mgl

(b) The elastic potential energy stored in the wire is Mgl

(c) The elastic potential energy stored in the wire is 1/2 Mgl

(d) Heat produced is 1/2 Mgl.

Worksheet 3

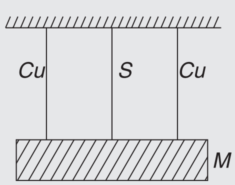
1. A person's femur (a long bone) has a diameter of 3.0 cm and a hollow centre of 0.8 cm diameter. Length of the bone is 50 cm. It is supporting a load of 600 N. How much will it be shortened by this load? Young's modulus of the bone is $Y = 1.6 \times 10^{10}$ Nm^{-2}.

2. A catapult consists of two parallel rubber strings, each of length 10 cm and cross-sectional area 1 mm^2. It is stretched by 6 cm and then released to project a stone of mass 5.0 g. Assume that the entire elastic energy is used to impart kinetic energy to the stone and find the speed of projection of the stone. Young's modulus for rubber is $Y = 5 \times 10^8$ Nm^{-2}.

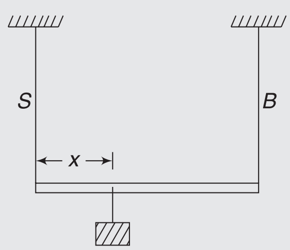
3. A wire of length 1m has a circular cross section of radius 1 mm. When subjected to a load it stretches by x. The wire is melted and is then drawn into a wire of square cross section having side length 2 mm. Find the extension is the new wire when same load is applied.

4. A wire has length L and cross-sectional area A. It is suspended at one of its ends from a ceiling. Density and Young's modulus of the material of the wire are ρ and Y respectively. Find the elastic strain energy in the wire due to its own weight

5. A uniform block of mass M is suspended on three vertical wires of equal lengths arranged symmetrically, as shown in figure. Middle wire is of steel and the other two wires are of copper. All wires have the same cross-section. Young's modulus of steel is twice that of copper. Find tension in steel wire.



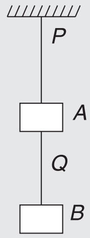
6. A light rod of length $L = 2$ m is suspended from a ceiling by means of two vertical wires of same length tied to the ends of the rod. One of the wire is made of steel and has cross-sectional area A. The other wire is made of brass and has cross-sectional area $2A$. A weight is hung from the rod at a distance x from the steel wire end. Young's modulus of steel is twice that of brass.



   (a) Find x such that both wires have equal strain.

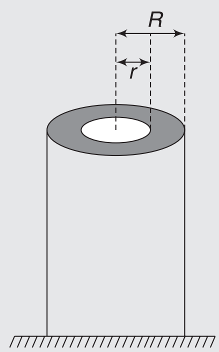
   (b) Find x such that the both wires have equal stress.

7. Wires P and Q are made of same material and have cross-sectional radii of r_1 and r_2 respectively. Two blocks A and B are suspended from the wires as shown. Masses of A and B are $3m$ and m respectively. It is observed that one of the two wires break. Find relation between r_1 and r_2 if



   (a) Wire P breaks (b) wire Q breaks.

8. A cylindrical pillar is made of two materials. The inner core, having radius r, has Young's modulus Y_1 and the outer layer (between radius r and R) has Young's modulus Y_2. A load of weight W is placed on the cylinder. Find the fraction of load supported by the outer layer.



9. The breaking stress of aluminium is 7.5×10^8 dyne cm^{-2}. Find the greatest length of aluminium wire that can hang vertically without breaking. Density of aluminium is 2.7 g cm^{-3}. [Given: g = 980 cm s^{-2}]

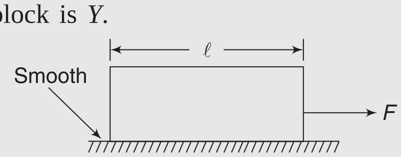
10. A thin, uniform metallic rod of length 0.5 m and radius 0.1 m rotates with an angular velocity 400 rad s^{-1} in a horizontal plane about a vertical axis passing through one of its ends. Calculate

    (a) tension in the rod at a distance r from the rotation axis and

    (b) the elongation of the rod.

    The density of material of the rod is 10^4 kgm^{-3} and the Young's modulus is 2×10^{11} Nm^{-2}.

11. A block of mass m and length l has cross-sectional area A. It is placed on a smooth horizontal surface and pulled by applying a force F at its one end. Find out the elongation in block. Young's modulus of block is Y.



Answers Sheet

Your Turn

1. (i) 0.002 (ii) 10^7 Nm^{-2}
2. 0.0174
3. 1.96×10^7
4. $\frac{x}{2}$
5. (a) 1.17×10^{-4} m (b) 1.17×10^{-4} m
6. 9×10^{-5}
7. 2×10^{-4} 8.8×10^5 Nm^{-2}
9. 7.5 ml
10. 100 m
11. $\frac{1}{2} Y\in^2$
12. 0.8 J

Worksheet 1

1. (c) 2. (a) 3. (a) 4. (d) 5. (b) 6. (b) 7. (b) 8. (c) 9. (d)
10. (a) 11. (d) 12. (d) 13. (a) 14. (c)

Worksheet 2

1. (a, c) 2. (a, b, c) 3. (b, c) 4. (a, c, d)

Worksheet 3

1. 2.9×10^{-5} m
2. 84.9 ms^{-1}
3. $\frac{\pi^2}{16} x$
4. $\frac{\rho^2 g^2 AL^3}{6Y}$
5. $Mg/2$
6. (a) 1m (b) $\frac{4}{3}$ m
7. (a) $r_2 > \frac{r_1}{2}$ (b) $r_2 < \frac{r_1}{2}$
8. $\frac{(R^2 - r^2)Y_2}{r^2 Y_1 + (R^2 - r^2)Y_2}$
9. 2.834 km
10. (a) $8\pi \times 10^6 \left[\frac{1}{4} - r^2\right]$N (b) $\frac{1}{3} \times 10^{-3}$m
11. $\frac{F\ell}{2AY}$

CHAPTER 13

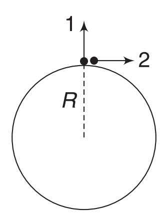
Miscellaneous Problems based on Chapters 11 and 12

MATCH THE COLUMN

1. A star of mass M is a uniform sphere of radius R. It is spinning about its axis with angular speed ω_0. It begins to contract while maintaining its spherical shape. The effect of contraction on quantities in column I has been given in Column II. Match them.

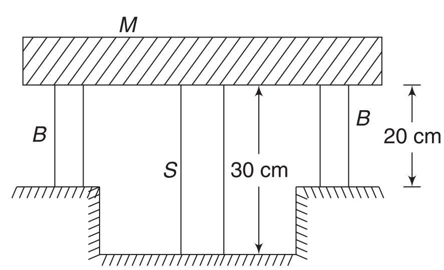
	Column I		Column II
(a)	Acceleration due to gravity at poles	(p)	Increases
(b)	Free fall acceleration at poles	(q)	Decreses
(c)	Free fall acceleration at equator	(r)	Remains unchanged
(d)	Binding energy of a particle on pole	(s)	May increase or decreases

2. Two identical particles are projected from the surface of the Earth with same speed. Particle 1 is projected vertically up and Particle 2 is projected tangentially. Neglect atmospheric resistance.



	Column I		Column II
(a)	Particles stop momentarily before falling back to the ground	(p)	Particle 1
(b)	Attains more height	(q)	Particle 2
(c)	Has higher angular momentum about the centre of the Earth	(r)	Both 1 and 2
(d)	May hit the surface of the Earth normally.	(s)	Neither 1 or 2

3. A steel rod of cross-sectional area $16\,\text{cm}^2$ and two identical brass rods of cross-sectional area $10\,\text{cm}^2$ each, support a mass of $M = 5{,}000\,\text{kg}$, as shown. The steel rod is at the centre and the other two rods are located symmetrically, bearing equal load. Given:



$$Y_S = 2 \times 10^6 \text{ kgcm}^{-2} \quad \text{and} \quad Y_B = 10^6 \text{ kgcm}^{-2}$$

Match entries in Column I with those in Column II

	Column I		Column II
(a)	Ratio of stress in steel to that in brass is close to	(p)	1
(b)	Ratio of strain in steel to that in brass is	(q)	$\frac{4}{3}$
(c)	Ratio of compressive force in steel to that in one brass rod is	(r)	$\frac{2}{3}$
(d)	Ratio of change in length of steel to that in brass is	(s)	$\frac{32}{15}$

4. In this problem, M refers to the mass of the Earth and R is its radius. Match the items in Column I to the items in Column II given below:

	Column I		Column II
(a)	Escape velocity from a height R above earth's surface	(p)	$\sqrt{\frac{4GM}{R}}$

(b)	Minimum horizontal velocity required for a particle at a height R above earth's surface to avoid falling on earth	(q)	$\sqrt{\frac{GM}{R}}$
(c)	Escape velocity on the surface of earth if earth's diameter were shrunk to half of its present value	(r)	$\sqrt{\frac{GM}{3R}}$
(d)	Velocity of a body thrown from the earth's surface to reach a height of $\frac{R}{3}$ from the surface of the earth	(s)	$\sqrt{\frac{GM}{2R}}$

PASSAGE-BASED PROBLEMS

Passage 1

Saturn's ring has aroused curiosity since Christiaan Huygens observed it in 1655 using a self-designed telescope. In 1859, James clerk Maxwell demonstrated that a non-uniform solid ring would not be stable and the ring must be composed of numerous small particles, all independently orbiting Saturn. The dense main ring extends from 8,000 km to 80,000 km away from saturn's equator, whose radius is nearly 60,000 km. The ring is composed of 99.9% water and ice with estimated thickness of 10 m to 1 km. The ring contains particles of radius ranging from 1 cm to 10 m. However, the total mass of the ring has been estimated to be very small compared to the mass of Saturn.

Acceleration due to gravitational pull of Saturn on its surface at equator is $10\,\text{ms}^{-2}$. Density of ice $\simeq$ Density of water = $10^3\,\text{k gm}^{-3}$.

1. Estimate the maximum KE of a particle in the ring of Saturn

(a) 10^{15} J (b) 10^8 J

(c) 10^4 J (d) 10^{22} J

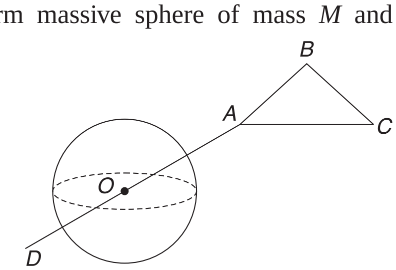
2. The ratio of smallest time period of revolution of a particle around Saturn to the largest time period of revolution of a particle around Saturn is nearly

(a) 0.8 (b) 0.1

(c) 0.7 (d) 0.35

Passage 2

Consider a fixed uniform massive sphere of mass M and radius R. O is the centre of the sphere and AD is a straight line passing through O. OA = OD. ABC is an equilateral triangle having side length equal to R. Work done in slowly moving a particle of mass m from A to B, along straight path AB is W_1 and work done in moving the same particle from A to B along the path ACB is W_2.

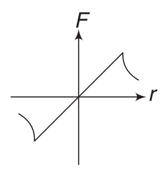


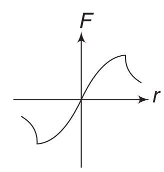
3. The differences in gravitational potential at A and B due to the sphere can be expressed as

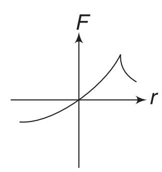
(a) $-\left(\frac{W_1 + W_2}{2m}\right)$ (b) $\frac{W_1 + W_2}{2m}$

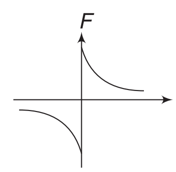
(c) $\frac{W_1}{m}$ (d) $\frac{2(W_1 + W_2)}{m}$

4. The particle is slowly transferred from A to D along the straight line AOD (assume a smooth tunnel through the sphere). Take the direction of force (F) applied by the external agent on m to be positive, when it is directed along $\overrightarrow{OA}$. Which of the following is the best representation of F versus r, where r is distance from O. r is taken to be positive from O to A and negative from O to D.

(a)


(b)


(c)


(d)


Passage 3

A meteorite of mass m collides with a satellite, which was orbiting around a planet in a circular orbit of radius R. The meteorite sticks to the satellite, which has mass $10m$. The satellite now orbits the planet in an elliptical orbit. Mass of the planet is M and the meteorite was moving towards the centre of the earth before collision.

5. What could be the maximum speed of the meteorite?

(a) $\sqrt{\frac{142\,GM}{R}}$ (b) $\sqrt{\frac{3\,GM}{R}}$

(c) $\sqrt{\frac{63\,GM}{R}}$ (d) $\sqrt{\frac{GM}{R}}$

6. If the satellite has a minimum distance from the centre of the planet = $\frac{R}{2}$ after collision, the speed of meteorite before collision was:

(a) $\sqrt{\frac{GM}{R}}$ (b) $\sqrt{\frac{58\,GM}{R}}$

(c) $\sqrt{\frac{3\,GM}{R}}$ (d) $\sqrt{\frac{2GM}{R}}$

Passage 4

Two stars rotate about their common centre of mass as a binary system. Both move in circular orbits and their time period is 1 year. Mass of one star is double that of the other and the lighter star has mass equal to one-third of the mass of our sun. Distance between the Earth and the sun is R and mass of the sun is M.

7. Distance between the two stars is

(a) R (b) $1.2R$

(c) $1.5R$ (d) $3R$

8. Kinetic energy of the smaller star is

(a) $\frac{GM}{R}$ (b) $\frac{2GM}{R}$

(c) $\frac{2}{9}\frac{GM}{R}$ (d) $\frac{GM}{9R}$

Passage 5

Two satellites S_1 and S_2 revolve round a planet in co-planar circular orbits in the same sense. There periods of revolution are 1 h and 8 h respectively. The radius of orbit of S_1 is 10^4 km. At an instant, the two satellites are nearest to each other. Answer the following questions with regard to this instant.

9. Speed of S_2 relative to S_1 is

(a) 10^4 kmh^{-1} (b) 2×10^4 kmh^{-1}

(c) $\pi \times 10^4$ kmh^{-1} (d) $2\pi \times 10^4$ kmh^{-1}

10. Angular speed of S_2 as observed by an astronaut is S_1 is

(a) π rad h^{-1} (b) $\frac{2\pi}{3}$ rad h^{-1}

(c) $\frac{\pi}{3}$ rad s^{-1} (d) $\frac{\pi}{3}$ rad h^{-1}.

Passage 6

A body is released from a height R above the north pole of the Earth. Another identical body is released from a height R above a city at latitude 30° N. There is no resistance to motion of the bodies and consider no rotation of the Earth [R is radius of the Earth and M is mass of the Earth]

11. Relative speed of the two bodies at the instant they have travelled a distance $\frac{R}{2}$ each, is

(a) $\sqrt{\frac{GM}{R}}$ (b) $\sqrt{\frac{2GM}{R}}$

(c) $\sqrt{\frac{3GM}{R}}$ (d) zero

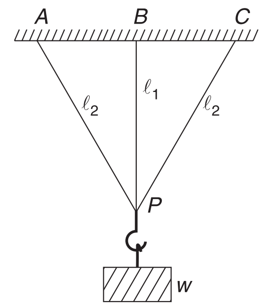
12. Distance between the two bodies at the instant they have travelled a distance $\frac{R}{2}$ is

(a) $\frac{3R}{2}$ (b) $\frac{R}{2}$

(c) $2R$ (d) zero

Passage 7

Three wires AP, BP and CP are made of same materials and have same cross sectional area. They are fixed to points A, B and C on a ceiling and their free ends are knotted together at P. Lengths of the three wires are $AP = CP = l_2$ and $BP = l_1$. A weight W is suspended at P and the wires get taut. Assume the deformations to be small.



13. Which of the following is true?

(a) Extension in wire AP is higher than extension in BP.

(b) Extension in wire BP is more than extension in AP.

(c) Stress in wire AP is more than stress in BP.

(d) None of the above.

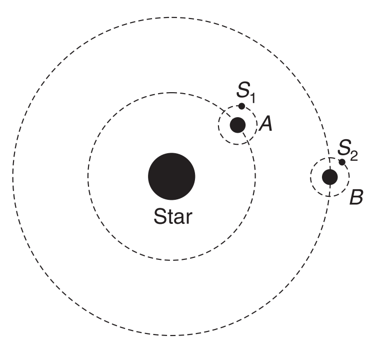
14. Tension in wire BP is

(a) $\frac{W}{1+2\left(\frac{l_1}{l_2}\right)^3}$ (b) $\frac{W}{1+\left(\frac{l_1}{l_2}\right)^2}$

(c) $\frac{W}{1+2\left(\frac{l_2}{l_1}\right)^2}$ (d) $\frac{W}{1+\left(\frac{l_2}{l_1}\right)^2}$

Passage 8

Consider a star and two planet system. The star has mass M. The planets A and B have the same mass m, radius a and they revolve around the star in circular orbits of radius r and $4r$ respectively ($M >> m$, $r >> a$). Planet A has intelligent people living on it and they have achieved a very high degree of technological advancement. They wish to shift a geostationary satellite of their own planet to a geostationary orbit of planet B. They achieve this through a series of high-precision maneuvers in which the satellite never has to apply brakes and not a single joule of energy is wasted. S_1 is a geostationary satellite of planet A and S_2 is a geostationary satellite of planet B. Neglect interactions between A and B.



15. If the time period of the satellite in geostationary orbit of planet A is T, then its time period in geostationary orbit of planet B is:

(a) T (b) $4T$

(c) $8T$ (d) Data insufficient

16. If the radius of the geostationary orbit in planet A is given by $r_G = r\left(\frac{m}{M}\right)^{1/3}$, then the time in which the geostationary satellite will complete one revolution is

I. 1 planet year = time in which planet revolves around the star

II. 1 planet day = time in which planet revolves about its axis

(a) I (b) II

(c) both I and II (d) neither I nor II.

17. If planet A and B, both complete one revolution about their own axis in the same time, then the energy needed to transfer the satellite of mass m_0 from planet A to planet B is

(a) $\frac{Gmm_0}{4r}$ (b) $\frac{GMm_0}{4r}$

(c) $\frac{3GMm_0}{8r}$ (d) Zero

Answers Sheet

Match the Columns

1. (a) p (b) p (c) q (d) p
2. (a) p (b) p (c) q (d) p
3. (a) q (b) r (c) s (d) p
4. (a) q (b) r (c) p (d) s

Passage-based Problems

1. (a)	2. (d)	3. (a)	4. (a)	5. (a)	6. (b)	7. (a)	8. (c)	9. (c)
10. (d)	11. (b)	12. (a)	13. (b)	14. (a)	15. (d)	16. (c)	17. (c)	

CHAPTER 14

Past Years' Questions

CENTRE OF MASS AND MOMENTUM

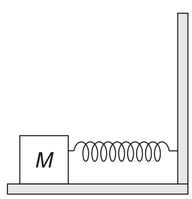
AIEEE/JEE Main Questions

1. A mass 'm' moves with a velocity 'v' and collides inelastically with another identical mass at rest. After collision, the first mass moves with velocity $v/\sqrt{3}$ in a direction perpendicular to the initial direction of motion. Find the speed of the second mass after collision. **[AIEEE 2005]**

(a) $(2/\sqrt{3})v$ (b) $v/\sqrt{3}$

(c) v (d) $\sqrt{3}\,v$

2. The block of mass M, moving on the frictionless horizontal surface, collides with a spring of spring constant k and compresses it by length L. The maximum momentum of the block after collision is **[AIEEE 2005]**



(a) Zero (b) $\dfrac{ML^2}{k}$

(c) $\sqrt{Mk}L$ (d) $\dfrac{kL^2}{2M}$

3. A bomb of mass 16 kg, at rest, explodes into two pieces of masses 4 kg and 12 kg. The velocity of the 12 kg mass is 4 ms^{-1}. The kinetic energy of the other mass is **[AIEEE 2006]**

(a) 144 J (b) 288 J

(c) 192 J (d) 96 J

4. Consider a two-particle system with particles having masses m_1 and m_2. If the first particle is pushed towards the centre of mass through a distance d, by what distance should the second particle be moved, so as to keep the centre of mass at the same position? **[AIEEE 2006]**

(a) $\dfrac{m_2}{m_1}d$ (b) $\dfrac{m_1}{m_1 + m_2}d$

(c) $\dfrac{m_1}{m_2}d$ (d) d

5. A circular disc of radius R is removed from a bigger circular disc of radius $2R$, such that the circumferences of the discs coincide. The centre of mass of the new disc is at a distance aR from the centre of the bigger disc. The value of a is **[AIEEE 2007]**

(a) 1/3 (b) 1/2

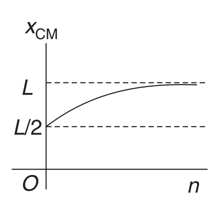
(c) 1/6 (d) 1/4

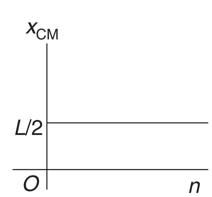
6. A body of mass m = 3.513 kg is moving along the x-axis, with a speed of 5.00 ms^{-1}. The magnitude of its momentum is recorded as **[AIEEE 2008]**

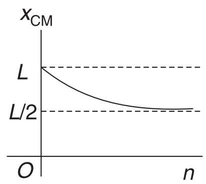
(a) 17.6 kg ms^{-1} (b) 17.565 kg ms^{-1}

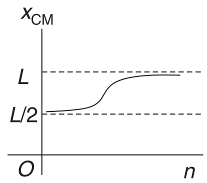
(c) 17.56 kg ms^{-1} (d) 17.57 kg ms^{-1}

7. A thin rod of length 'L' is lying along the x-axis, with its ends at x = 0 and x = L. Its linear density (mass/length) varies with x as $k(x/L)^n$, where n can be zero or any positive number. If the position x_{CM} of the centre of mass of the rod is plotted against 'n', which of the following graphs best approximates the dependence of x_{CM} on n? **[AIEEE 2008]**

(a)


(b)


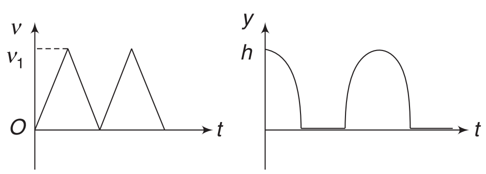
(c) 

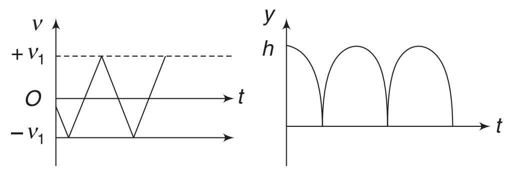
(d) 

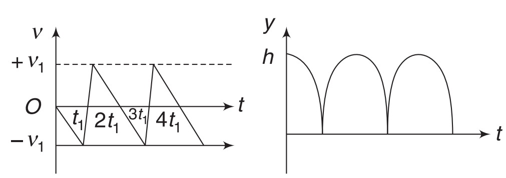
8. A block of mass 0.50 kg is moving with a speed of 2.00 ms^{-1} on a smooth surface. It strikes another mass of 1.00 kg and then they move together as a single body. The energy loss during the collision is **[AIEEE 2008]**

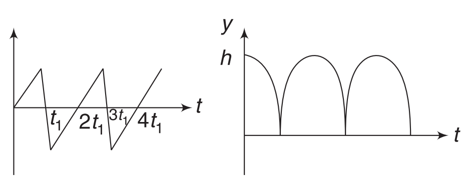
(a) 0.16 J (b) 1.00 J
(c) 0.67 J (d) 0.34 J

9. Consider a rubber ball freely falling from a height h = 4.9 m onto a horizontal elastic plate. Assume that the duration of collision is negligible and the collision with the plate is totally elastic. Then the velocity as a function of time and the height as a function of time will be **[AIEEE 2009]**

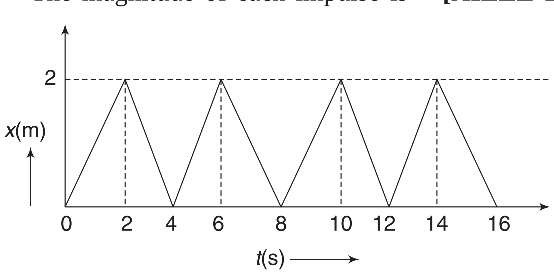
(a) 

(b) 

(c) 

(d) 

10. The figure shows the position–time (x-t) graph of the one-dimensional motion of a body of mass 0.4 kg. The magnitude of each impulse is **[AIEEE 2010]**



(a) 0.2 Ns (b) 0.4 Ns
(c) 0.8 Ns (d) 1.6 Ns

11. **Statement-1:** Two particles moving in the same direction do not lose all their energy in a completely inelastic collision.

Statement-2: Principle of conservation of momentum holds true for all kinds of collisions. **[AIEEE 2010]**

(a) Statement–1 is true, Statement–2 is false
(b) Statement–1 is true, Statement–2 is true; Statement–2 is the correct explanation of Statement–1
(c) Statement–1 is true, Statement–2 is true; Statement–2 is not the correct explanation of Statement–1
(d) Statement–1 is false, Statement–2 is true

12. This question has Statement I and Statement II. Of the four choices given after the statements, choose the one that best describes the two statements.

Statement-1: A point particle of mass m moving with speed υ collides with a stationary point-particle of mass M. If the maximum energy loss possible is given as $f\left(\frac{1}{2\,m\upsilon^2}\right)$, then $f = \left(\frac{m}{M+m}\right)$.

[JEE-Main 2013]

Statement-2: Maximum energy loss occurs when the particles get stuck together as a result of the collision.

(a) Statement-1 is true, Statment-2 is true, Statement-2 is the correct explanation of Statement-1.
(b) Statement-1 is true, Statment-2 is true, Statement-2 is not the correct explanation of Statement-1.
(c) Statement-1 is true, Statment-2 is false.
(d) Statement-1 is false, Statment-2 is true.

13. A particle of mass m moving in the x direction with speed $2v$ is hit by another particle of mass $2m$ moving in the y-direction with speed v. If the collision is perfectly inelastic, the percentage loss in the energy during the collision is close to **[JEE Main 2015]**

(a) 62% (b) 44%
(c) 50% (d) 56%

14. Distance of the centre of mass of a solid uniform cone from its vertex is z_0. If the radius of its base is R and its height is h then z_0 is equal to:

[JEE Main 2015]

(a) $\frac{3h^2}{8R}$ (b) $\frac{h^2}{4R}$

(c) $\frac{3h}{4}$ (d) $\frac{5h}{8}$

15. In a collinear collision, a particle with an initial speed v_0 strikes a stationary particle of the same mass. If the final total kinetic energy is 50% greater than the original kinetic energy, the magnitude of the relative velocity between the two particles, after collision, is: **[JEE Main 2018]**

(a) $\frac{v_0}{\sqrt{2}}$ (b) $\frac{v_0}{4}$

(c) $\sqrt{2}\, v_0$ (d) $\frac{v_0}{2}$

IIT JEE (Advanced) Questions

1. STATEMENT-1: In an elastic collision between two bodies, the relative speed of the bodies after collision is equal to the relative speed before the collision. **[2007]**

because

STATEMENT-2: In an elastic collision, the linear momentum of the system is conserved

(a) Statement-1 is true, Statement-2 is true; Statement-2 **is** a correct explanation for Statement-1

(b) Statement-1 is true, Statement-2 is true; Statement-2 is NOT a correct explanation for Statement-1

(c) Statement-1 is true, Statement-2 is false

(d) Statement-1 is false, Statement-2 is true.

(e) Statement-1 is false, Statement-2 is false

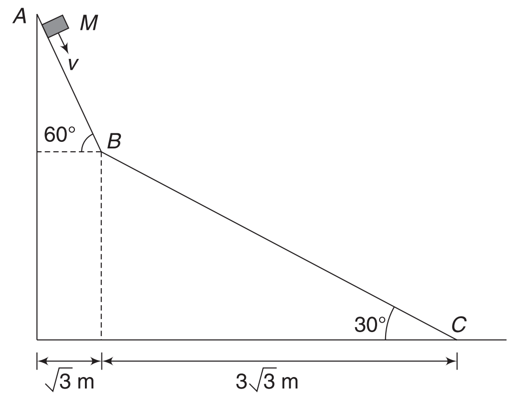
2. Two balls, having linear momenta $\vec{p}_1 = p\hat{i}$ and $\vec{p}_2 = -p\hat{i}$, undergo a collision in free space. There is no external force acting on the balls. Let $\vec{p}'_1$ and $\vec{p}'_2$ be their final momenta. Which of the following option(s) is(are) **NOT ALLOWED** for any non-zero value of p, a_1, a_2, b_1, b_2, c_1 and c_2. **[2008]**

(a) $\vec{p}'_1 = a_1\hat{i} + b_1\hat{j} + c_1\hat{k}$, $\vec{p}'_2 = a_2\hat{i} + b_2\hat{j}$ (b) $\vec{p}'_1 = c_1\hat{k}$, $\vec{p}'_2 = c_2\hat{k}$

(c) $\vec{p}'_1 = a_1\hat{i} + b_1\hat{j} + c_1\hat{k}$, $\vec{p}'_1 = a_1\hat{i} + b_1\hat{j} - c_1\hat{k}$ (d) $\vec{p}'_1 = a_1\hat{i} + b_1\hat{j}$, $\vec{p}'_2 = a_2\hat{i} + b_1\hat{j}$

3. Following three questions are based on the given passage

A small block of mass M moves on a frictionless surface of an inclined plane, as shown in figure. The angle of the incline suddenly changes from 60° to 30° at point B. The block is initially at rest at A. Assume that collision between the block and the incline is totally inelastic ($g = 10\,\text{ms}^{-2}$) **[2008]**



(i) The speed of the block at point B immediately after it strikes the second incline is

(a) $\sqrt{60}$ ms^{-1} (b) $\sqrt{45}$ ms^{-1}

(c) $\sqrt{30}$ ms^{-1} (d) $\sqrt{15}$ ms^{-1}

(ii) The speed of the block at point C, immediately before it leaves the second incline is

(a) $\sqrt{120}$ ms^{-1} (b) $\sqrt{105}$ ms^{-1}

(c) $\sqrt{90}$ ms^{-1} (d) $\sqrt{75}$ ms^{-1}

(iii) If collision between the block and the incline is completely elastic, then the vertical (upward) component of the velocity of the block at point B, immediately after it strikes the second incline is

(a) $\sqrt{30}$ ms^{-1} (b) $\sqrt{15}$ ms^{-1}

(c) 0 (d) $-\sqrt{15}$ ms^{-1}

4. If the resultant of all the external forces acting on a system of particles is zero, then from an inertial frame, one can surely say that: **[2009]**

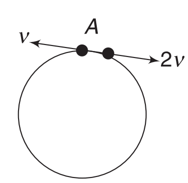
(a) Linear momentum of the system does not change in time.

(b) Kinetic energy of the system does not change in time.

(c) Angular momentum of the system does not change in time.

(d) Potential energy of the system does not change in time.

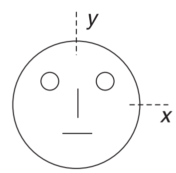
5. Two small particles of equal masses start moving in opposite directions from a point A, in a horizontal circular orbit. Their tangential velocities are v and $2v$ respectively, as shown in the figure. Between collisions, the particles move with constant speeds. After making how many elastic collisions, other than that at A, will these two particles again reach point A? **[2009]**



(a) 4 (b) 3

(c) 2 (d) 1

6. Look at the drawing given in the figure, which has been drawn with ink of uniform line thickness. The mass of ink used to draw each of the two inner circles, and each of the two line segments is m. The mass of the ink used to draw the outer circle is $6m$. The co-ordinates of the centres of the different parts are: outer circle (0, 0); left inner circle $(-a, a)$; right inner circle (a, a); vertical line (0, 0); and horizontal line $(0, -a)$. The y-co-ordinate of the centre of mass of the ink in this drawing is **[2009]**

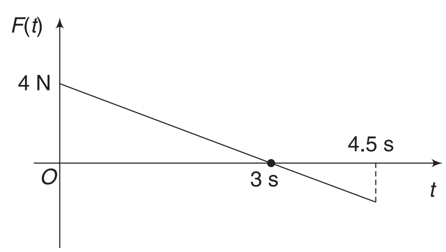


(a) $a/10$ (b) $a/8$

(c) $a/12$ (d) $a/3$

7. Three objects, A, B and C, are kept in a straight line on a frictionless horizontal surface. These have masses m, $2m$ and m, respectively. The object A moves towards B with a speed $9\,\text{ms}^{-1}$ and makes an elastic collision with it. Thereafter, B makes completely inelastic collision with C. All motions occur on the same straight line. Find the final speed (in ms^{-1}) of the object C. **[2010]**

8. A block of mass 2 kg is free to move along the x-axis. It is at rest and from $t = 0$ onwards, it is subjected to a time-dependent force $F(t)$ in the x-direction. The force $F(t)$ varies with t, as shown in the figure. The kinetic energy of the block after 4.5 seconds is **[2011]**



(a) 4.50 J (b) 7.50 J

(c) 5.06 J (d) 14.06 J

9. A point mass of 1 kg collides elastically with a stationary point mass of 5 kg. After their collision, the 1 kg mass reverses its direction and moves with a speed of $2\,\text{ms}^{-1}$. Which of the following statement (s) is (are) correct for the system of these two masses? **[2010]**

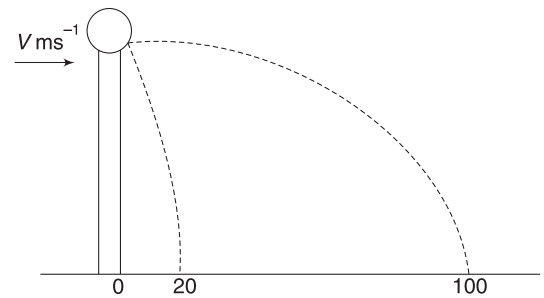
(a) Total momentum of the system is 3 $\text{kg}\,\text{ms}^{-1}$

(b) Momentum of 5 kg mass after collision is 4 $\text{kg}\,\text{ms}^{-1}$

(c) Kinetic energy of the centre of mass is 0.75 J

(d) Total kinetic energy of the system is 4 J

10. A ball of mass 0.2 kg rests on a vertical post of height 5m. A bullet of mass 0.01 kg, travelling with a velocity $v\,\text{ms}^{-1}$ in a horizontal direction, hits the centre of the ball. After the collision, the ball and the bullet travel independently. The ball hits the ground at a distance of 20 m and the bullet at a distance of 100 m from the foot of the post. The initial velocity v of the bullet is **[2011]**

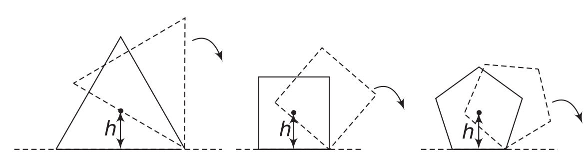


(a) $250\,\text{ms}^{-1}$ (b) $250\sqrt{2}\,\text{ms}^{-1}$

(c) $400\,\text{ms}^{-1}$ (d) $500\,\text{ms}^{-1}$

11. A bob of mass m, suspended by a string of length l_1, is given a minimum velocity required to complete a full circle in the vertical plane. At the highest point, it collides elastically with another bob of mass m suspended by a string of length l_2, which is initially at rest. Both the strings are mass-less and inextensible. If the second bob, after collision, acquires the minimum speed required to complete a full circle in the vertical plane, find the ratio $\frac{l_1}{l_2}$. **[2013]**

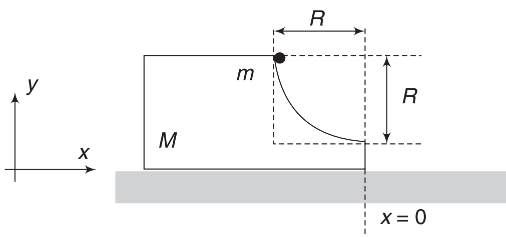
12. Consider regular polygons with number of sides $n = 3, 4, 5$ ..., as shown in the figure. The centre of mass of all the polygons is at height h from the ground. They roll on a horizontal surface about the leading vertex without slipping and sliding as depicted. The maximum increase in height of the locus of the centre of mass for each polygon is Δ. Then Δ depends on n and h as **[2017]**



(a) $\Delta = h\sin^2\left(\frac{\pi}{n}\right)$ (b) $\Delta = h\left(\frac{1}{\cos\left(\frac{\pi}{n}\right)} - 1\right)$

(c) $\Delta = h\sin\left(\frac{2\pi}{n}\right)$ (d) $\Delta = h\tan^2\left(\frac{\pi}{2n}\right)$

13. A block of mass M has a circular cut with a frictionless surface, as shown. The block rests on the horizontal frictionless surface of a fixed table. Initially, the right edge of the block is at $x = 0$, in a co-ordinate system fixed to the table. A point mass m is released from rest at the top-most point of the path, as shown, and it slides down. When the mass loses contact with the block, its position is x and the velocity is v. At that instant, which of the following options is/are correct? **[2017]**



(a) The position of the point mass m is

$$x = -\sqrt{2}\,\frac{mR}{M+m}$$

(b) The velocity of the point mass m is

$$v = \sqrt{\frac{2gR}{1+\frac{m}{M}}}$$

(c) The x component of displacement of the centre of mass of the block M is $-\dfrac{mR}{M+m}$

(d) The velocity of the block M is $V = -\dfrac{m}{M}\sqrt{2gR}$

14. A solid horizontal surface is covered with a thin layer of oil. A rectangular block of mass $m = 0.4$ kg is at rest on this surface. An impulse of 1.0 Ns is applied to the block at time $t = 0$ so that it starts moving along the x-axis with a velocity $v(t) = v_0 e^{-t/\tau}$, where v_0 is a constant and $\tau = 4\,$s. The displacement of the block, in meters, at $t = \tau$ is _________. Take $e^{-1} = 0.37$. **[JEE Adv. 2018]**

ROTATIONAL MOTION

AIEEE/JEE Main Questions

1. An annular ring with inner and outer radii R_1 and R_2 is rolling without slipping, with a uniform angular speed. The ratio of the forces experienced by the two particles situated on the inner and outer parts of the ring, F_1/F_2 is: **[AIEEE 2005]**

(a) 1 (b) R_1/R_2

(c) R_2/R_1 (d) $(R_1/R_2)^2$

2. A round uniform body of radius R, mass M and moment of inertia I rolls down (without slipping) on an inclined plane making an angle θ with the horizontal. Then its acceleration is: **[AIEEE 2007]**

(a) $\dfrac{g\sin\theta}{1 - MR^2/I}$ (b) $\dfrac{g\sin\theta}{1 + I/MR^2}$

(c) $\dfrac{g\sin\theta}{1 + MR^2/I}$ (d) $\dfrac{g\sin\theta}{1 - I/MR^2}$

3. Angular momentum of the particle rotating with a central force is constant due to: **[AIEEE 2007]**

(a) constant torque
(b) constant force
(c) constant linear momentum
(d) zero torque

4. Consider a uniform square plate of side 'a' and mass 'm'. The moment of inertia of this plate about an axis perpendicular to its plane and passing through one of its corners is: **[AIEEE 2008]**

(a) $5/6\ ma^2$ (b) $1/12\ ma^2$

(c) $7/12\ ma^2$ (d) $2/3\ ma^2$

5. A thin horizontal circular disc is rotating about a vertical axis passing through its centre. An insect is at rest at a point near the rim of the disc. The insect now moves along diameter of the disc to reach its other end. During the journey of the insect, the angular speed of the disc **[AIEEE 2011]**

(a) continuously increases
(b) first increases and then decreases
(c) remains unchanged
(d) continuously decreases

6. A pulley of radius 2m is rotated about its axis by a force $F = (20t - 5t^2)$ N (where t is measured in seconds) applied tangentially. If the moment of inertia of the pulley about its axis of rotation is 10 kg m^2, the number of rotations made by the pulley before its direction of motion is reversed, is **[AIEEE 2011]**

(a) more than 6 but less than 9
(b) more than 9
(c) less than 3
(d) more than 3 but less than 6

7. A mass m hangs with the help of a string wrapped around a pulley on a frictionless bearing. The pulley has mass m and radius R. Assuming pulley to be a perfect uniform circular disc, the acceleration of the mass m, if the string does not slip on the pulley, is: **[AIEEE 2011]**

(a) $(2/3)g$ (b) $g/3$

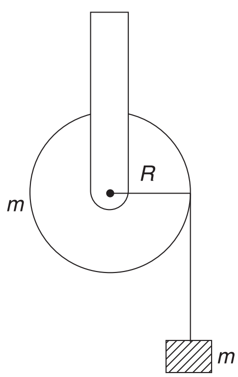
(c) $(3/2)g$ (d) g

8. A hoop of radius R and mass m, rotating with an angular velocity ω_0, is placed on a rough horizontal surface. The initial velocity of the centre of the hoop is zero. What will be the velocity of the centre of the hoop when it ceases to slip? **[JEE-Main 2013]**

(a) $\dfrac{R\omega_0}{3}$ (b) $\dfrac{R\omega_0}{2}$

(c) $R\omega_0$ (d) $\dfrac{R\omega_0}{4}$

9. A mass m is supported by a massless string wound around a uniform hollow cylinder of mass m and radius R. If the string does not slip on the cylinder, with what acceleration will the mass fall on release? **[JEE-Main 2014]**



(a) $\dfrac{2g}{3}$ (b) $\dfrac{g}{2}$

(c) $\dfrac{5g}{6}$ (d) g

10. A bob of mass m, attached to an inextensible string of length l, is suspended from a vertical support. The bob rotates in a horizontal circle with an angular speed ω rad s^{-1} about the vertical. About the point of suspension: **[JEE-Main 2014]**

(a) angular momentum is conserved.

(b) angular momentum changes in magnitude but not in direction

(c) angular momentum changes in direction but not in magnitude

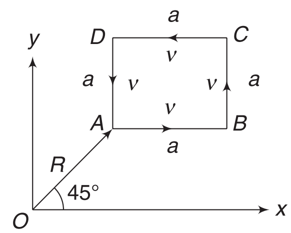
(d) angular momentum changes both in direction and magnitude

11. From a solid sphere of mass M and radius R, a cube of maximum possible volume is cut. Moment of inertia of cube about an axis passing through its centre and perpendicular to one of its faces is: **[JEE-Main 2015]**

(a) $\dfrac{4MR^2}{3\sqrt{3}\pi}$ (b) $\dfrac{MR^2}{32\sqrt{2}\pi}$

(c) $\dfrac{MR^2}{16\sqrt{2}\pi}$ (d) $\dfrac{4MR^2}{9\sqrt{3}\pi}$

12. A particle of mass m is moving along the side of a square of side 'a', with a uniform speed v in the $x-y$ plane, as shown in the figure Which of the following statements is false for the angular momentum $\vec{L}$ about the origin? **[JEE-Main 2016]**



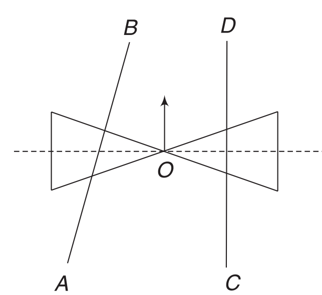
(a) $\vec{L} = -\dfrac{mv}{\sqrt{2}} R\hat{k}$ when the particle is moving from A to B.

(b) $\vec{L} = mv\left[\dfrac{R}{\sqrt{2}} - a\right]\hat{k}$ when the particle is moving from C to D

(c) $\vec{L} = mv\left[\dfrac{R}{\sqrt{2}} + a\right]\hat{k}$ when the particle is moving from B to C

(d) $\vec{L} = \dfrac{mv}{\sqrt{2}} R\hat{k}$ when the particle is moving from D to A

13. A roller is made by joining two cones at their vertices, O. It is kept on two rails, AB and CD,which are placed asymmetrically (see figure), with its axis perpendicular to CD and its centre O at the centre of the line joining AB and CD. It is given a light push so that it starts rolling with its centre O moving parallel to CD in the direction shown. As it moves, the roller will tend to: **[JEE Main 2016]**



(a) turn left (b) turn right

(c) go straight

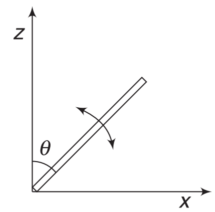
(d) turn left and right alternately

14. The moment of inertia of a uniform cylinder of length l and radius R about its perpendicular bisector is I. What is the ratio l/R such that the moment of inertia is minimum? **[JEE-Main 2017]**

(a) $\sqrt{\frac{3}{2}}$ (b) $\frac{\sqrt{3}}{2}$

(c) 1 (d) $\frac{3}{\sqrt{2}}$

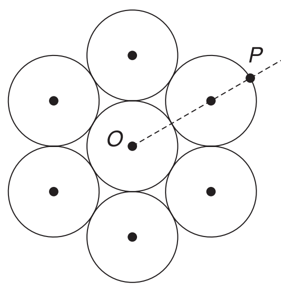
15. A slender uniform rod of mass M and length l is pivoted at one end so that it can rotate in a vertical plane (see figure). There is negligible friction at the pivot. The free end is held vertically above the pivot and then released. The angular acceleration of the rod, when it makes an angle θ with the vertical is: **[JEE-Main 2017]**



(a) $\frac{3g}{2l}\sin\theta$ (b) $\frac{2g}{3l}\sin\theta$

(c) $\frac{3g}{2l}\cos\theta$ (d) $\frac{2g}{3l}\cos\theta$

16. Seven identical circular planar disks, each of mass M and radius R are welded symmetrically as shown. The moment of inertia of the arrangement about the axis normal to the plane and passing through the point P is: **[JEE-Main 2018]**

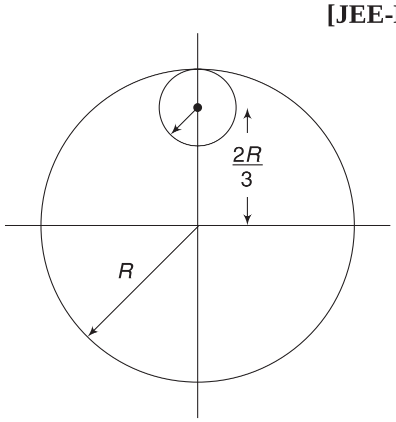


(a) $\frac{181}{2}MR^2$ (b) $\frac{19}{2}MR^2$

(c) $\frac{55}{2}MR^2$ (d) $\frac{73}{2}MR^2$

17. From a uniform circular disc of radius R and mass $9M$, a small disc of radius $\frac{R}{3}$ is removed as shown in the figure. The moment of inertia of the remaining disc about an axis perpendicular to the plane of the disc and passing through centre of disc is:

[JEE-Main 2018]



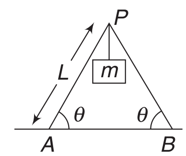
(a) $\frac{37}{9}MR^2$ (b) $4\ MR^2$

(c) $\frac{40}{9}\ MR^2$ (d) $10\ MR^2$

IIT JEE (Advanced) Questions

1. A solid cylinder of mass m and radius R rolls down an inclined plane of inclination θ. Calculate the linear acceleration of the axis of cylinder. **[2005]**

2. Two identical ladders, each of mass M and length L are resting on a rough horizontal surface, as shown in the figure. A block of mass m hangs from P. If the system is in equilibrium, find the magnitude and the direction of frictional force at A and B. **[2005]**



3. If a particle is confined to rotate in a circular path with decreasing linear speed, then which of the following is correct? **[2005]**

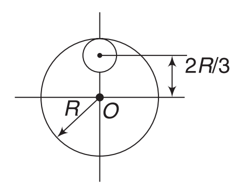
(a) $\vec{L}$ (angular momentum) is conserved about the centre

(b) Only direction of angular momentum $\vec{L}$ is conserved

(c) It spirals towards the centre.

(d) Its acceleration is towards the centre.

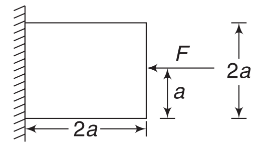
4. From a circular disc of radius R and mass $9M$, a small disc of radius $R/3$ is removed. The moment of inertia of the remaining disc about an axis perpendicular to the plane of the disc and passing through O is: **[2005]**



(a) $4MR^2$ (b) $\frac{40}{9}MR^2$

(c) $10MR^2$ (d) $\frac{37}{9}MR^2$

5. A block of mass m is held fixed against a wall by applying a horizontal force F. Line of action of F passes through the centre of the block. Which of the following option is *incorrect*? **[2005]**



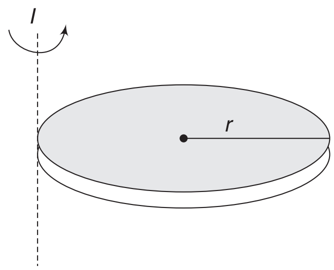
(a) friction force on the block = mg

(b) normal force will not produce a torque on the block about its centre

(c) F will not produce torque on the block about its centre

(d) normal reaction on the block = F

6. A solid sphere of radius R has moment of inertia I about its geometrical axis. It is melted into a disc of radius r and thickness t. If its moment of inertia about the tangential axis (which is perpendicular to plane of the disc), is also equal to I, then the value of r is equal to:

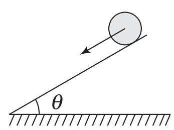


[2006]

(a) $\frac{2}{\sqrt{15}}R$ (b) $\frac{2}{\sqrt{5}}R$

(c) $\frac{3}{\sqrt{15}}R$ (d) $\frac{\sqrt{3}}{\sqrt{15}}R$

7. A solid sphere is in pure rolling motion on an inclined surface having inclination θ. **[2006]**



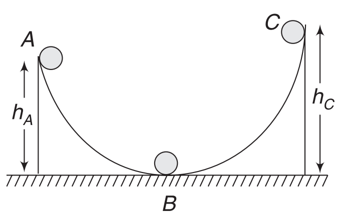
(a) frictional force acting on sphere is $f = \mu mg\cos\theta$

(b) f is dissipative force

(c) friction will increase its angular velocity and decreases its linear velocity.

(d) If θ decreases, friction will decrease

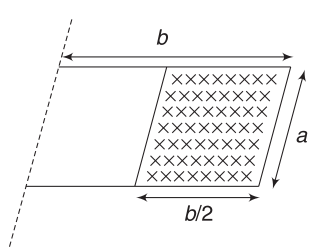
8. A ball moves over a fixed track, as shown in the figure. From A to B the ball rolls without slipping. If surface BC is frictionless and K_A, K_B and K_C are the kinetic energies of the ball at A, B and C respectively, then: **[2006]**



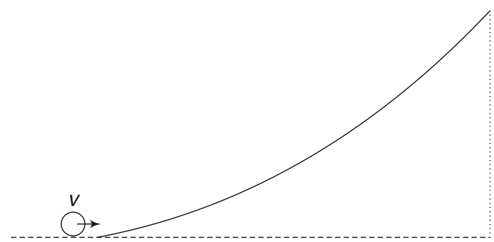
(a) $h_A > h_C$; $K_B > K_C$ (b) $h_A > h_C$; $K_C > K_A$

(b) $h_A = h_C$; $K_B = K_C$ (d) $h_A < h_C$; $K_B > K_C$

9. A rectangular plate of mass $M = 3$ kg of dimensions ($a \times b = 1\,\text{m} \times 2\,\text{m}$) is hinged along one edge. The plate is maintained in horizontal position by balls moving in vertically upward direction and colliding with the plate. Each ball has mass $m = 0.01$ kg and $n = 100$ balls hit the plate per second, per square metre of plate area on the right half of the plate (shown as shaded area). The balls are distributed uniformly over the shaded area of the plate and all collisions are elastic. Find the required speed of the balls just before they hit the plate, assuming all of them to be moving with the same speed. **[2006]**



10. A small object of uniform density rolls up a curved surface with an initial velocity v. It reaches up to a maximum height of $\frac{3v^2}{4g}$ with respect to the initial position. The object is **[2007]**



(a) ring (b) solid sphere

(c) hollow sphere (d) disc

11. **Statement–1:** If there is no external torque on a body about its centre of mass, then the velocity of the centre of mass remains constant. **[2007]**

Statement–2: The linear momentum of an isolated system remains constant.

(a) Statement-1 is true, Statement-2 is true; Statement-2 **is** a correct explanation for Statement-1

(b) Statement-1 is true, Statement-2 is true; Statement-2 **is NOT** a correct explanation for Statement-1

(c) Statement-1 is true, Statement-2 is false

(d) Statement-1 is false, Statement-2 is true.

12. **Following three questions are based on the given passage:** **[2007]**

Two discs, A and B, are mounted coaxially on a vertical axle. The discs have moments of inertia I and $2I$ respectively, about the common axis. Disc A is imparted an initial angular velocity 2ω, using the entire potential energy of a spring compressed by a distance x_1. Disc B is imparted an angular velocity ω by a spring having the same spring constant and compressed by a distance x_2. Both the discs rotate in the clockwise direction.

(i) The ratio of x_1/x_2 is:

(a) 2 (b) 1/2

(c) $\sqrt{2}$ (d) $1/\sqrt{2}$

(ii) When disc B is brought in contact with disc A, they acquire a common angular velocity in time t. The average frictional torque on one disc by the other during this period is:

(a) $\frac{2I\omega}{3t}$ (b) $\frac{9I\omega}{2t}$

(c) $\frac{9I\omega}{4t}$ (d) $\frac{3I\omega}{2t}$

(iii) The loss of kinetic energy during the above process is:

(a) $\frac{I\omega^2}{2}$ (b) $\frac{I\omega^2}{3}$

(c) $\frac{I\omega^2}{4}$ (d) $\frac{I\omega^2}{6}$

13. **Statement-1:** Two cylinders, one hollow (metal) and the other solid (wood), with the same mass and identical dimensions, are simultaneously allowed to roll without slipping down an inclined plane from the same height. The hollow cylinder will reach the bottom of the inclined plane first. **[2008]**

and

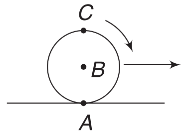
Statement-2: By the principle of conservation of energy, the total kinetic energies of both the cylinders are identical when they reach the bottom of the incline.

(a) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is true, Statement-2 is true; Statement-2 is NOT a correct explanation for Statement-1
(c) Statement-1 is true, Statement-2 is false
(d) Statement-1 is false, Statement-2 is true.

14. If the resultant of all the external forces acting on a system of particles is zero, then from an inertial frame, one can surely say that **[2009]**

(a) Linear momentum of the system does not change in time.
(b) Kinetic energy of the system does not changes in time.
(c) Angular momentum of the system does not change in time.
(d) Potential energy of the system does not change in time.

15. A sphere is rolling without slipping on a fixed horizontal plane surface. In the figure, A is the point of contact, B is the centre of the sphere and C is its top-most point. Then, **[2009]**

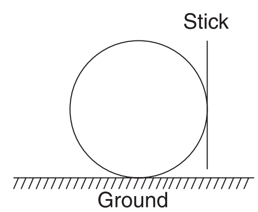


(a) $\vec{v}_C - \vec{v}_A = 2(\vec{v}_B - \vec{v}_C)$
(b) $\vec{v}_C - \vec{v}_B = \vec{v}_B - \vec{v}_A$
(c) $\left|\vec{v}_C - \vec{v}_A\right| = 2\left|\vec{v}_B - \vec{v}_C\right|$
(d) $\left|\vec{v}_C - \vec{v}_A\right| = 4\left|\vec{v}_B\right|$

16. A block of base 10 cm × 10 cm and height 15 cm is kept on an inclined plane. The coefficient of friction between them is $\sqrt{3}$. The inclination θ of this inclined plane from the horizontal plane is gradually increased from 0°. Then **[2009]**

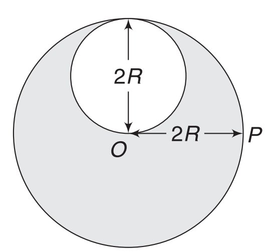
(a) at $\theta = 30°$, the block will start sliding down the plane
(b) the block will remain at rest on the plane up to certain θ and then it will topple
(c) at $\theta = 60°$, the block will start sliding down the plane and continue to do so at higher angles
(d) at $\theta = 60°$, the block will start sliding down the plane and on further increasing θ, it will topple at certain θ

17. A boy is pushing a ring of mass 2 kg and radius 0.5 m with a stick, as shown in the figure. The stick applies a force of 2N on the ring and rolls it without slipping, with an acceleration of 0.3 ms^{-2}. The coefficient of friction between the ground and the ring is large enough that rolling always occurs and the coefficient of friction between the stick and the ring is (P/10). The value of P is: **[2010]**



18. Four solid spheres, each of diameter $\sqrt{5}$ cm and mass 0.5 m are placed with their centres at the corners of a square of side 4 cm. The moment of inertia of the system about the diagonal of the square is $N \times 10^{-4}$ kg m^2, then N is. **[2011]**

19. A lamina is made by removing a small disc of diameter $2R$ from a bigger disc of uniform mass density and radius $2R$, as shown in the figure. The moment of inertia of this lamina about axes passing through O and P is I_0 and I_P, respectively. Both these axes are perpendicular to the plane of the lamina. The ratio I_P/I_O to the nearest integer is: **[2012]**



20. Two solid cylinders P and Q of same mass and same radius start rolling down a fixed inclined plane, from the same height, at the same time. Cylinder P has most of its mass concentrated near its surface, while

Q has most of its mass concentrated near the axis. Which statement (s) is (are) correct? **[2012]**

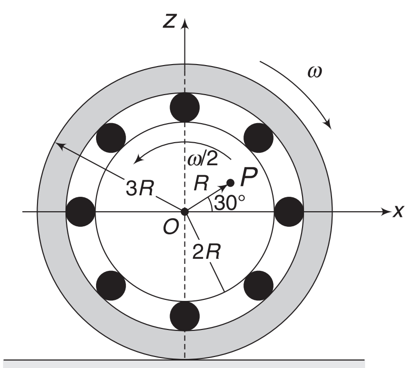
(a) Both cylinders P and Q reach the ground at the same time.

(b) Cylinder P has a larger linear acceleration than cylinder Q.

(c) Both cylinders reach the ground with the same translational kinetic energy.

(d) Cylinder Q reaches the ground with larger speed.

21. The figure shows a system consisting of (i) a ring of outer radius $3R$ rolling clockwise without slipping on a horizontal surface with angular speed ω and (ii) an inner disc of radius $2R$ rotating anti-clockwise with angular speed $\omega/2$. The ring and disc are separated by frictionless ball bearings. The system is in the x–z plane. The point P on the inner disc is at a distance R from the origin, where OP makes an angle of 30° with the horizontal. Then with respect to the horizontal surface, **[2012]**



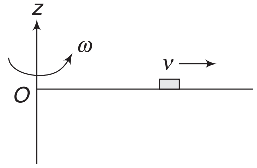
(a) the point O has a linear velocity $3R\omega\hat{i}$

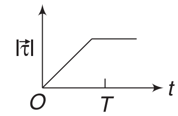
(b) the point P has a linear velocity $\frac{11}{4}R\omega\hat{i}+\frac{\sqrt{3}}{4}R\omega\hat{k}$

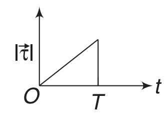
(c) the point P has a linear velocity $\frac{13}{4}R\omega\hat{i}-\frac{\sqrt{3}}{4}R\omega\hat{k}$

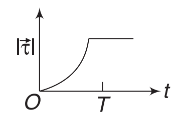
(d) the point P has a linear velocity $\left(3-\frac{\sqrt{3}}{4}\right)R\omega\hat{i}+\frac{1}{4}R\omega\hat{k}$

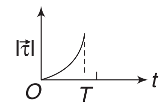
22. A thin uniform rod, pivoted at O, is rotating in the horizontal plane with constant angular speed ω, as shown in the figure. At time $t = 0$, a small insect starts from O and moves with constant speed v with respect to the rod towards the other end. It reaches the end of the rod at $t = T$ and stops. The angular speed of the system remains ω throughout. The magnitude of the torque $(|\vec{\tau}|)$ on the system about O, as function of time, is best represented by which plot? **[2012]**



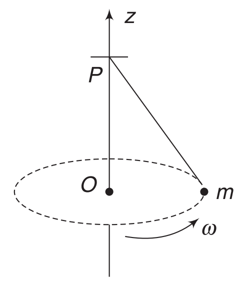
(a) 

(b) 

(c) 

(d) 

23. A small mass m is attached to a massless string, whose other end is fixed at P, as shown in the figure. The mass is under going circular motion in the x–y plane with centre at O and constant angular speed ω. If the angular momentum of the system, calculated about O and P are denoted by $\vec{L}_0$ and $\vec{L}_P$ respectively, then **[2012]**



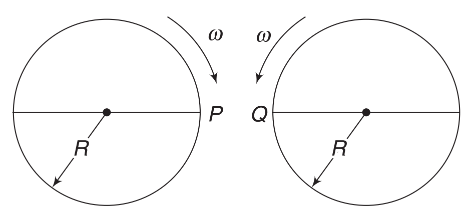
(a) $\vec{L}_0$ and $\vec{L}_P$ do not vary with time.

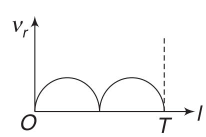
(b) $\vec{L}_0$ varies with time while $\vec{L}_P$ remains constant.

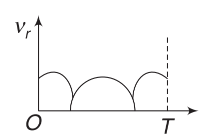
(c) $\vec{L}_0$ and $\vec{L}_P$ both vary with time.

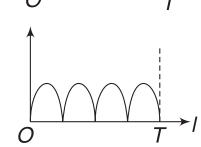
(d) $\vec{L}_0$ remains constant while $\vec{L}_P$ varies with time.

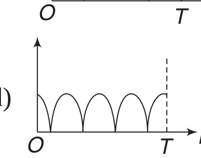
24. Two identical discs of same radius R are rotating about their axes in opposite directions with the same constant angular speed ω. The discs are in the same horizontal plane. At time $t = 0$, the points P and Q are facing each other, as shown in the figure. The relative speed between the two points P and Q is v. In one time period (T) of rotaion of the discs, v as a function of time is best represented by **[2012]**



(a) 

(b) 

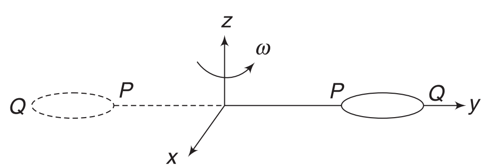
(c) 

(d) 

25. Following two questions are based on the given passage

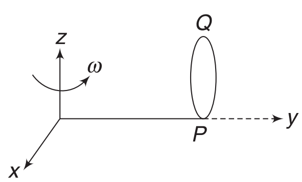
The general motion of a rigid body can be considered to be a combination of (i) a motion of its centre of mass about an axis, and (ii) its motion about an instantaneous axis passing through the centre of mass. These axes need not be stationary. Consider, for example, a thin uniform disc welded (rigidly fixed) horizontally at its rim to a massless stick, as shown in the figure.

When the disc-stick system is rotated about the origin on a horizontal frictionless plane with angular speed ω, the motion at any instant can be taken as a combination of (i) a rotation of the centre of mass of the disc about the z-axis, and (ii) a rotation of the disc through an instantaneous vertical axis passing through its centre of mass (as is seen from the changed orientation of points P and Q). Both these motions have the same angular speed ω in this case.

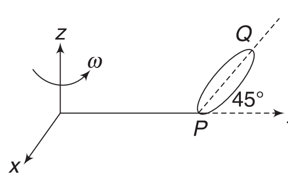


Now consider two similar systems, as shown in the figure: Case (a): the disc with its face vertical and parallel to x–z plane; Case (b): the disc with its face making an angle of 45° with x–y plane and its horizontal diameter parallel to x-axis. In both the cases, the disc is welded at point P, and the systems are rotated with constant angular speed ω about the z-axis. **[2012]**

Case (a)



Case (b)



(i) Which of the following statements regarding the angular speed about the instantaneous axis (passing through the centre of mass) is correct?

(a) It is $\sqrt{2}\,\omega$ for both the cases

(b) It is ω for case (a); and $\dfrac{\omega}{\sqrt{2}}$ for case (b)

(c) It is ω for case (a); & $\sqrt{2}\,\omega$ for case (b)

(d) It is ω for both the cases

(ii) Which of the following statements about the instantaneous axis (passing through the centre of mass) is correct?

(a) It is vertical for both the cases (a) and (b)

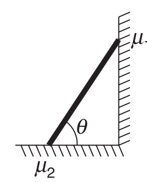
(b) It is vertical for case (a). and is at 45° to the x–z plane and lies in the plane of the disc for case (b)

(c) It is horizontal for case (a); and is at 45° to the x–z plane and is normal to the plane of the disc for case (b)

(d) It is vertical for case (a); and is at 45° to the x–z plane and is normal to the plane of the disc for case (b)

26. A uniform circular disc of mass 50 kg and radius 0.4 m is rotating with an angular velocity of 10 rad s^{-1} about its own axis, which is vertical. Two uniform circular rings, each of mass 6.25 kg and radius 0.2 m, are gently placed symmetrically on the disc in such a manner that they are touching each other along the axis of the disc and are horizontal. Assume that the friction is large enough such that the rings are at rest relative to the disc and the system rotates about the original axis. The new angular velocity (in rad s^{-1}) of the system is. **[2013]**

27. In the figure, a ladder of mass m is shown leaning against a wall. It is in static equilibrium, making an angle θ with the horizontal floor.



The coefficient of friction between the wall and the ladder is μ_1 and that between the floor and the ladder is μ_2. The normal reaction of the wall on the ladder is N_1 and that of the floor is N_2. If the ladder is about to slip, then **[2014]**

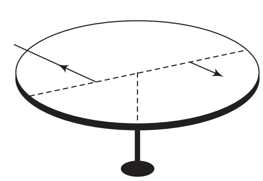
(a) $\mu_1 = 0$ $\mu_2 \neq 0$ and $N_2 \tan\theta = \dfrac{mg}{2}$

(b) $\mu_1 \neq 0$ $\mu_2 = 0$ and $N_1 \tan\theta = \dfrac{mg}{2}$

(c) $\mu_1 \neq \mu_2 \neq 0$ and $N_2 = \dfrac{mg}{1 + \mu_1\mu_2}$

(d) $\mu_1 = 0$ $\mu_2 \neq 0$ and $N_1 \tan\theta = \dfrac{mg}{2}$

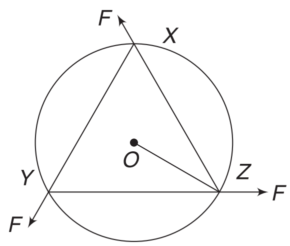
28. A horizontal circular platform of radius 0.5 m and mass 0.45 kg is free to rotate about its axis. Two massless spring toy-guns, each carrying a steel ball of mass 0.05 kg, are attached to the platform, at



a distance 0.25 m from the centre on its either sides, along its diameter (see figure).

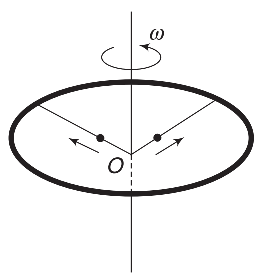
Each gun simultaneously fires the balls horizontally and perpendicular to the diameter in opposite directions. After leaving the platform, the balls have horizontal speed of $9\,\text{ms}^{-1}$ with respect to the ground. The rotational speed of the platform, in rad s^{-1}, after the balls leave the platform is **[2014]**

29. A uniform circular disc of mass 1.5 kg and radius 0.5 m is initially at rest on a horizontal frictionless surface. Three forces of equal magnitude $F = 0.5$ N are applied simultaneously along the three sides of an equilateral triangle *XYZ* with the vertices on the perimeter of the disc (see figure). One second after applying the forces, the angular speed of the disc in $\text{rad}\,\text{s}^{-1}$ is **[2014]**



30. The densities of two solid spheres *A* and *B* of the same radii *R*, vary with radial distance *r*, as $\rho_A(r) = k\left(\frac{r}{R}\right)$ and $\rho_B(r) = k\left(\frac{r}{R}\right)^5$, respectively, where *k* is a constant. The moments of inertia of the individual spheres about axes passing through their centres are I_A and I_B, respectively. If $\frac{I_B}{I_A} = \frac{n}{10}$, the value of *n* is **[2015]**

31. A ring of mass *M* and radius *R* is rotating with angular speed ω about a fixed vertical axis passing through its centre *O* with two point masses, each of mass *M*/8, at rest at *O*. These masses can move radially outwards along two massless rods fixed on the ring, as shown in the figure. At some instant, the angular speed of the system is 8/9 ω and one of the masses is at a distance of 3/5 *R* from *O*. At this instant the distance of the other mass from *O* is: **[2015]**



(a) $\frac{2}{3}R$ (b) $\frac{1}{3}R$

(c) $\frac{3}{5}R$ (d) $\frac{4}{5}R$

32. The position vector $\vec{r}$ of a particle of mass *m* is given by the following equation $r(t) = \alpha t^3\hat{i} + \beta t^2\hat{j}$, where $\alpha = 10/3\,\text{ms}^{-3}$, $\beta = 5\,\text{ms}^{-2}$ and $m = 0.1$ kg. At $t = 1$ s, which of the following statement(s) is (are) true about the particle? **[2016]**

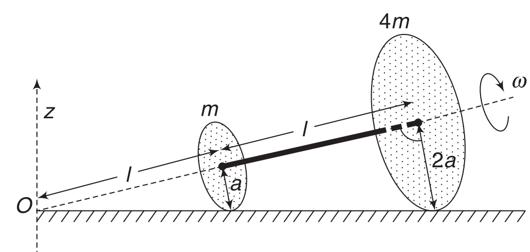
(a) The velocity $\vec{v}$ is given by $\vec{v} = (10\hat{i} + 10\hat{j})\,\text{ms}^{-1}$

(b) The angular momentum $\vec{L}$ with respect to the origin is given by $\vec{L} = -(5/3)\hat{k}$ N m s

(c) The force $\vec{F}$ is given by $\vec{F} = (\hat{i} + 2\hat{j})$ N

(d) The torque $\vec{\tau}$ with respect to the origin is given by $\vec{\tau} = -(20/3)$ k N m

33. Two thin circular discs of mass *m* and 4*m*, having radii of a and 2*a*, respectively, are rigidly fixed by a massless, rigid rod of length $l = \sqrt{24}\,a$ through their centres. This assembly is laid on a firm and flat surface and set rolling without slipping on the surface so that the angular speed about the axis of the rod is ω. The angular momentum of the entire assembly about the point '*O*' is $\vec{L}$ (see the figure). Which of the following statement(s) is (are) true? **[2016]**



(a) The magnitude of angular momentum of centre of mass of the assembly about the point *O* is $81\,ma^2\omega$

(b) The centre of mass of the assembly rotates about the *z*-axis with an angular speed of $\omega/5$

(c) The magnitude of the *z*-component of $\vec{L}$ is $55\,ma^2\omega$

(d) The magnitude of the angular momentum of the assembly about its centre of mass is $17\,ma^2\,\omega/2$

34. A uniform wooden stick of mass 1.6 kg and length *l* rests in an inclined manner on a smooth, vertical wall of height $h(<l)$ such that a small portion of the stick extends beyond the wall. The reaction force of the wall on the stick is perpendicular to the stick. The stick makes an angle of 30° with the wall and the bottom of the stick is on a rough floor. The reaction of the wall on the stick is equal in magnitude to the reaction of the floor on the stick. The ratio *h*/*l* and the frictional force *f* at the bottom of the stick are ($g = 10\,\text{ms}^{-2}$) **[2016]**

(a) $\frac{h}{l} = \frac{\sqrt{3}}{16}$, $f = \frac{16\sqrt{3}}{3}$ N

(b) $\frac{h}{l} = \frac{3}{16}$, $f = \frac{16\sqrt{3}}{3}$ N

(c) $\dfrac{h}{l} = \dfrac{3\sqrt{3}}{16}$, $f = \dfrac{8\sqrt{3}}{3}$ N

(d) $\dfrac{h}{l} = \dfrac{3\sqrt{3}}{16}$, $f = \dfrac{16\sqrt{3}}{3}$ N

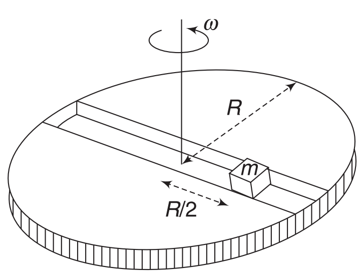
35. Following two questions are based on the given passage

A frame of reference that is accelerated with respect to an inertial frame of reference is called a non-inertial frame of reference. A co-ordinate system, fixed on a circular disc, rotating about a fixed axis, with a constant angular velocity ω is an example of a non-inertial frame of reference. The relationship between the force $\vec{F}_{rot}$ experienced by a particle of mass m moving on the rotating disc and the force $\vec{F}_{in}$ experienced by the particle in an inertial frame of reference is

$$\vec{F}_{rot} = \vec{F}_{in} + 2m(\vec{v}_{rot} \times \vec{\omega}) + m(\vec{\omega} \times \vec{r}) \times \vec{\omega}$$

where $\vec{v}_{rot}$ is the velocity of the particle in the rotating frame of reference and $\vec{r}$ is the position vector of the particle with respect to the centre of the disc.

Now, consider a smooth slot along a diameter of a disc of radius R rotating counter-clockwise, with a constant angular speed ω about its vertical axis through its centre. We assign a co-ordinate system with the origin at the centre of the disc, the x-axis along the slot, the y-axis perpendicular to the slot and the z-axis along the rotation axis ($\vec{\omega} = \omega\hat{k}$). A small block of mass m is gently placed in the slot at $\vec{r} = (R/2)\hat{i}$ at $t = 0$ and is constrained to move only along the slot. **[2016]**



(i) The distance r of the block at time t is

(a) $\dfrac{R}{2}\cos\omega t$ (b) $\dfrac{R}{4}(e^{\omega t} + e^{-\omega t})$

(c) $\dfrac{R}{2}\cos 2\omega t$ (d) $\dfrac{R}{4}(e^{2\omega t} + e^{-2\omega t})$

(ii) The net reaction of the disc on the block is

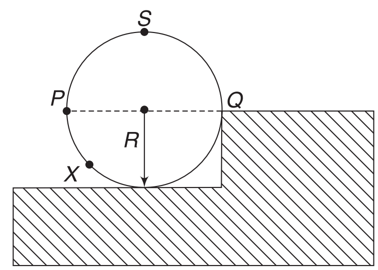
(a) $\dfrac{1}{2}m\omega^2 R(e^{\omega t} - e^{-\omega t})\hat{j} + mg\hat{k}$

(b) $\dfrac{1}{2}m\omega^2 R(e^{2\omega t} - e^{-2\omega t})\hat{j} + mg\hat{k}$

(c) $m\omega^2 R\sin\omega t\,\hat{j} - mg\hat{k}$

(d) $-m\omega^2 R\cos\omega t\,\hat{j} - mg\hat{k}$

36. A wheel of radius R and mass M is placed at the bottom of a fixed step of height R, as shown in the figure. A constant force is continuosly applied on the surface of the wheel so that it just climbs the step without slipping. Consider the torque τ about an axis normal to the plane of the paper passing through the point Q. Which of the following options is/are correct? **[2017]**



(a) If the force is applied at point P tangentially, then τ decreases continuously as the wheel climbs.

(b) If the force is applied normal to the circumference at point X, then τ is constant

(c) If the force is applied normal to the circumference at point P then τ is zero

(d) If the force is applied tangentially at point S then $\tau \neq 0$ but the wheel never climbs the step.

37. Following two questions are based on the given passage

One twirls a circular ring (of mass M and radius R) near the tip of one's finger, as shown in Figure 1. In the process, the finger never loses contact with the inner rim of the ring. The finger traces out the surface of a cone, shown by the dotted line. The radius of the path traced out by the point where the ring and the finger is in contact is r. The finger rotates with an angular velocity ω_0. The rotating ring rolls without slipping on the outside of a smaller circle described by the point where the ring and the finger are in contact (Figure 2). The coefficient of friction between the ring and the finger is μ and the acceleration due to gravity is g. **[2017]**

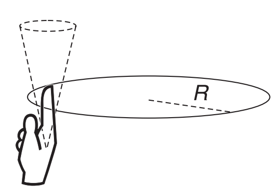


Fig. 1

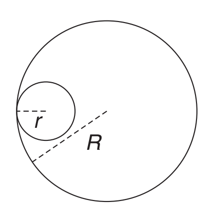


Fig. 2

(i) The total kinetic energy of the ring is

(a) $M\omega_0^2 R^2$ (b) $\frac{1}{2} M\omega_0^2 (R - r)^2$

(c) $M\omega_0^2 (R - r)^2$ (d) $\frac{3}{2} M\omega_0^2 (R - r)^2$

(ii) The minimum value of ω_0 below which the ring will drop down is

(a) $\sqrt{\frac{g}{\mu(R - r)}}$ (b) $\sqrt{\frac{2g}{\mu(R - r)}}$

(c) $\sqrt{\frac{3g}{2\mu(R - r)}}$ (d) $\sqrt{\frac{g}{2\mu(R - r)}}$

38. The potential energy of a particle of mass m at a distance r from a fixed point O is given by $V(r) = kr^2/2$, where k is a positive constant of appropriate dimensions. This particle is moving in a circular orbit of radius R about the point O. If v is the speed of the particle and L is the magnitude of its angular momentum about O, which of the following statements is (are) true? **[JEE Adv. 2018]**

(a) $v = \sqrt{\frac{k}{2m}}R$ (b) $v = \sqrt{\frac{k}{m}}R$

(c) $L = \sqrt{mk}R^2$ (d) $L = \sqrt{\frac{mk}{2}}R^2$

39. Consider a body of mass 1.0 kg at rest at the origin at time $t = 0$. A force $\vec{F} = (\alpha t\hat{i} + \beta\hat{j})$ is applied on the body, where $\alpha = 1.0\ \text{Ns}^{-1}$ and $\beta = 1.0$ N. The torque acting on the body about the origin at time $t = 1.0$s is $\vec{\tau}$. Which of the following statements is (are) true? **[JEE Adv. 2018]**

(a) $|\vec{\tau}| = \frac{1}{3}$Nm

(b) The torque $\vec{\tau}$ is in the direction of the unit vector $+\hat{k}$

(c) The velocity of the body at $t = 1s$ is $\vec{v} = \frac{1}{2}(\hat{i} + 2\hat{j})\ \text{ms}^{-1}$

(d) The magnitude of displacement of the body at $t = 1$ s is $\frac{1}{6}$m

40. A ring and a disc are initially at rest, side by side, at the top of an inclined plane which makes an angle 60° with the horizontal. They start to roll without slipping at the same instant of time along the shortest path. If the time difference between their reaching the ground is $(2 - \sqrt{3})/\sqrt{10}\ s$, then the height of the top of the inclined plane, in *metres*, is __________. Take $g = 10\ \text{ms}^{-2}$. **[JEE Adv. 2018]**

41. In the List-I below, four different paths of a particle are given as functions of time. In these functions, α and β are positive constants of appropriate dimensions and $\alpha \neq \beta$. In each case, the force acting on the particle is either zero or conservative. In List-II, five physical quantities of the particle are mentioned: $\vec{p}$ is the linear momentum, $\vec{L}$ is the angular momentum about the origin, K is the kinetic energy, U is the potential energy and E is the total energy. Match each path in List-I with those quantities in List-II, which **are conserved for that path**.

[JEE Adv. 2018]

LIST–I	LIST–II
P. $\vec{r}(t) = \alpha t\hat{i} + \beta t\hat{j}$	1. $\vec{p}$
Q. $\vec{r}(t) = \alpha\cos\omega t\hat{i} + \beta\sin\omega t\hat{j}$	2. $\vec{L}$
R. $\vec{r}(t) = \alpha(\cos\omega t\hat{i} + \sin\omega t\hat{j})$	3. K
S. $\vec{r}(t) = \alpha t\hat{i} + \frac{\beta}{2}t^2\hat{j}$	4. U
	5. E

(a) **P → 1, 2, 3, 4, 5; Q → 2, 5; R → 2, 3, 4, 5; S → 5**

(b) **P → 1, 2, 3, 4, 5; Q → 3, 5; R → 2, 3, 4, 5; S → 2, 5**

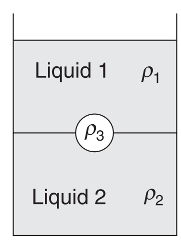
(c) **P → 2, 3, 4; Q → 5; R → 1, 2, 4; S → 2, 5**

(d) **P → 1, 2, 3, 5; Q → 2, 5; R → 2, 3, 4, 5; S → 2, 5**

FLUID MECHANICS

AIEEE/JEE Main Questions

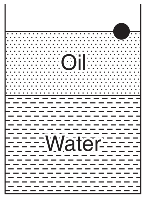
1. A jar is filled with two non–mixing liquids 1 and 2, having densities ρ_1 and ρ_2 respectively. A solid ball, made of a material of density ρ_3, is dropped in the jar. It comes to equilibrium in the position shown in the figure. Which of the following is true for ρ_1, ρ_2 and ρ_3? **[AIEEE 2008]**

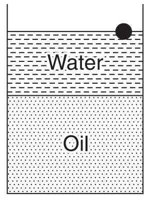


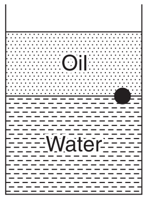
(a) $\rho_3 < \rho_1 < \rho_2$ (b) $\rho_1 < \rho_3 < \rho_2$

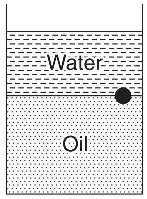
(c) $\rho_1 < \rho_2 < \rho_3$ (d) $\rho_1 < \rho_3$, ρ_2

2. A ball is made of a material of density ρ, where $\rho_{oil} < \rho < \rho_{water}$ with ρ_{oil} and ρ_{water} representing the densities of oil and water, respectively. The oil and water are immiscible. If the above ball is in equilibrium in a mixture of this oil and water, which of the following pictures represents its equilibrium position? **[AIEEE 2010]**

(a) 

(b) 

(c) 

(d) 

3. Water is flowing continuously from a tap having an internal diameter 8×10^{-3} m. The water velocity as it leaves the tap is $0.4\,\text{ms}^{-1}$. The diameter of the water stream at a distance 2×10^{-1} m below the tap is close to: **[AIEEE 2011]**

(a) 9.6×10^{-3} m (b) 3.6×10^{-3} m

(c) 5.0×10^{-3} m (d) 7.5×10^{-3} m

4. A uniform cylinder of length L and mass M, having cross-sectional area A is suspended, with its length vertical, from a fixed point by a massless spring, such that it is half submerged in a liquid of density σ at equilibrium position. The extension x_0 of the spring when it is in equilibrium is: (Here k is spring constant) **[JEE Main 2013]**

(a) $\dfrac{Mg}{k}\left(1 - \dfrac{LA\sigma}{M}\right)$ (b) $\dfrac{Mg}{k}\left(1 - \dfrac{LA\sigma}{2M}\right)$

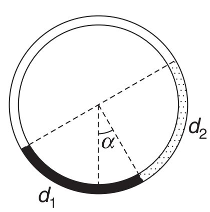
(c) $\dfrac{Mg}{k}\left(1 + \dfrac{LA\sigma}{M}\right)$ (d) $\dfrac{Mg}{k}$

5. An open glass tube is immersed in mercury in such a way that a length of 8 cm extends above the mercury level. The open-end of the tube is then closed and sealed and the tube is raised vertically up by an additional 46 cm. What will be length of the air column above mercury in the tube now? (Atmospheric pressure = 76 cm of Hg) **[JEE Main 2014]**

(a) 16 cm (b) 22 cm

(c) 38 cm (d) 6 cm

6. There is a circular tube in a vertical plane. Two liquids, which do not mix and of densities d_1 and d_2, are filled in the tube. Each liquid subtends 90° angle at the centre. Radius joining their interface makes an angle α with the vertical. Ratio (d_1/d_2) is: **[JEE Main 2014]**

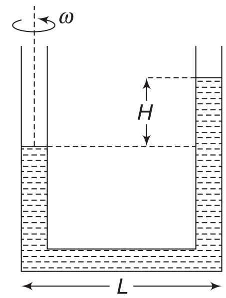


(a) $\dfrac{1 + \sin\alpha}{1 - \sin\alpha}$ (b) $\dfrac{1 + \cos\alpha}{1 - \cos\alpha}$

(c) $\dfrac{1 + \tan\alpha}{1 - \tan\alpha}$ (d) $\dfrac{1 + \sin\alpha}{1 - \cos\alpha}$

IIT JEE (Advanced) Questions

1. A U-tube is rotated about one of it s limbs with an angular velocity ω. Find difference in height H of the liquid (density ρ) level, where diameter of the tube $d \ll L$. **[2005]**



(a) $H = \dfrac{\omega^2 L^2}{2g}$ (b) $H = \dfrac{\omega L^2}{2g}$

(c) $H = \dfrac{\omega^2 L}{2g}$ (d) None

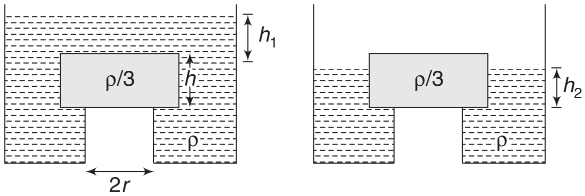
2. Water is filled in a container upto height 3m. A small hole of area 'a' is punched in the wall of the container, at a height 52.5 cm from the bottom. The cross-sectional area of the container is A. If $a/A = 0.1$, then v^2 is: (where v is the velocity of water coming out of the hole. Take $g = 10\,\text{ms}^{-2}$) **[2005]**

(a) 50 (b) 51

(c) 48 (d) 51.5

3. Following three questions are based on the given passage

A cylindrical tank has a hole of diameter $2r$ in its bottom. The hole is covered with a wooden cylindrical block of diameter $4r$, height h and density $\rho/3$, as shown. Initially, the tank is filled with water of density ρ to a certain height. The level of water is decreased gradually and it was found that the block begins to move up when the height of water above the top of the block is h_1 (measured from the top of the block). An external agent applies force to keep the block in place and the level of water is reduced further. When the level of water is reduced to a point where its height from the base of the block is h_2 (top surface of the block is exposed to atmosphere) the external force is no longer needed to keep the block in place. **[2006]**



(i) Value of height h_1 is

(a) $\frac{2h}{3}$ (b) $\frac{5h}{4}$

(c) $\frac{5h}{3}$ (d) $\frac{5h}{2}$

(ii) Value of h_2 is

(a) $\frac{4h}{9}$ (b) $\frac{5h}{9}$

(c) $\frac{h}{3}$ (d) $\frac{2h}{3}$

(iii) If height of water is lowered below h_2, then:

(a) cylinder will not move up and remains at its original position

(b) for $h_2 = h/3$, cylinder again starts moving up

(c) for $h_2 = h/4$, cylinder again starts moving up

(d) for $h_2 = h/5$ cylinder again starts moving up

4. **Statement-1:** The stream of water flowing at high speed from a garden hose pipe tends to spread like a fountain when held vertically up, but tends to narrow down when held vertically down.

Statement-2: In any steady flow of an incompressible fluid, the volume flow rate of the fluid remains constant. **[2008]**

(a) Statement–1 is true, Statement–2 is true; Statement–2 is a correct explanation for Statement–1

(b) Statement–1 is true, Statement–2 is true; Statement–2 is not a correct explanation for Statement–1

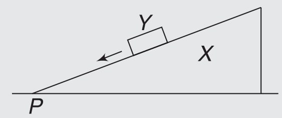
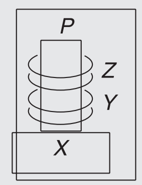
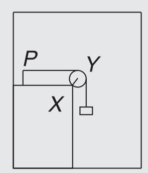
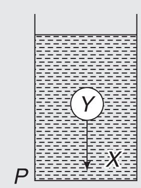
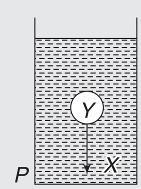
(c) Statement–1 is true, Statement–2 is false

(d) Statement–1 is false, Statement–2 is true

5. A cylindrical vessel of height 500 mm has an orifice (small hole) at its bottom. The orifice is initially closed and water is filled in it up to height H. Now the top is completely sealed with a cap and the orifice at the bottom is opened. Some water comes out from the orifice and the water level in the vessel becomes steady with height of water column being 200 mm. Find the fall in height (in mm) of water level due to opening of the orifice.

[Take atmospheric pressure 1.0×10^5 Nm^{-2}, density of water 1000 kgm^{-3} and $g = 10$ ms^{-2}. Neglect any effect of surface tension.] **[2009]**

6. **Column-II** shows five systems in which two objects are labelled as X and Y. Also, in each case, a point P is shown. **Column-I** gives some statements about X and/or Y. Match these statements to the appropriate system(s) from **Column-II**. **[2009]**

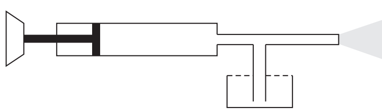
	Column–I		Column–II
(a)	The force exerted by X on Y has a magnitude Mg.	(p)	Block Y of mass M left on a fixed inclined plane X, slides on it with a constant velocity. 
(b)	The gravitational potential energy of X is continuously increasing.	(q)	Two ring magnets Y and Z, each of mass M, are kept in frictionless vertical plastic stand so that they repel each other. Y rests on the base X and Z hangs in air in equilibrium. P is the top-most point of the stand on the common axis of the two rings. The whole system is in a lift that is going up with a constant velocity. 
(c)	Mechanical energy of the system $X + Y$ is continuously decreasing.	(r)	A pulley Y of mass m_0 is fixed to a table through a clamp X. A block of mass M hangs from a string that goes over the pulley and is fixed at point P of the table. The whole system is kept in a lift that is going down with a constant velocity. 
(d)	The torque of the weight of Y about point P is zero.	(s)	A sphere Y of mass M is put in a non-viscous liquid X, kept in a container at rest. The sphere is released and it moves down in the liquid. 
		(t)	A sphere Y of mass M is falling with its terminal velocity in a viscous liquid X kept in a container. 

7. A thin uniform cylindrical shell, closed at both ends, is partially filled with water. It is floating vertically in water in half-submerged state. If ρ_C is the relative density of the material of the shell with respect to water, then the correct statement is that the shell is: **[2012]**

(a) more than half-filled if ρ_C is less than 0.5
(b) more than half-filled if ρ_C is more than 1.0
(c) half-filled if ρ_C is more than 0.5
(d) less than half-filled if ρ_C is less than 0.5

8. Following two questions are based on the given passage

A spray gun is shown in the figure where a piston pushes air out of a nozzle. A thin tube of uniform cross-section is connected to the nozzle. The other end of the tube is in a small liquid container. As the piston pushes air through the nozzle, the liquid from the container rises into the nozzle and is sprayed out. For the spray gun shown, the radii of the piston and the nozzle are 20 mm and 1 mm respectively. The upper end of the container is open to the atmosphere. **[JEE Adv. 2014]**



(i) If the piston is pushed at a speed of $5\,\text{mms}^{-1}$, the air comes out of the nozzle with a speed of

(a) $0.1\,\text{ms}^{-1}$ (b) $1\,\text{ms}^{-1}$
(c) $2\,\text{ms}^{-1}$ (d) $8\,\text{ms}^{-1}$

(ii) If the density of air is ρ_a and that of the liquid ρ_1, then for a given piston speed, the rate (volume per unit time) at which the liquid is sprayed will be proportional to

(a) $\sqrt{\dfrac{\rho_a}{\rho_1}}$ (b) $\sqrt{\rho_a \rho_1}$
(c) $\sqrt{\dfrac{\rho_1}{\rho_a}}$ (d) ρ_1

9. A person in a lift is holding a water jar, which has a small hole at the lower end of its side. When the lift is at rest, the water jet coming out of the hole hits the floor of the lift at a distance d of 1.2 m from the person. In the following, state of the lift's motion is given in List I and the distance where the water jet hits the floor of the lift is given in List II. Match the statements from List I with those in List II and select the correct answer using the code given below the lists. **[JEE Adv. 2014]**

	List I		List II
P.	Lift is accelerating vertically up	**1.**	$d = 1.2\,\text{m}$
Q.	Lift is accelerating vertically down with an acceleration less than the gravitational acceleration.	**2.**	$d > 1.2\,\text{m}$
R.	Lift is moving vertically up with constant speed.	**3.**	$d < 1.2\,\text{m}$
S.	Lift is falling freely.	4.	No water leaks out of the jar

Code:

(a) P-2, Q-3, R-2, S-4 (b) P-2, Q-3, R-1, S-4
(c) P-1, Q-1, R-1, S-4 (d) P-2, Q-3, R-1, S-1

SURFACE TENSION AND VISCOSITY

AIEEE/JEE Main Questions

1. A 20 cm long capillary tube is dipped in water. The water rises upto 8 cm. If the entire arrangement is put in a freely falling elevator, the length of water column in the capillary tube will be **[AIEEE 2005]**

(a) 8 cm (b) 10 cm
(c) 4 cm (d) 20 cm

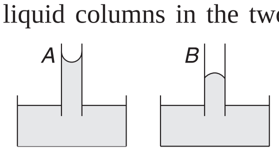
2. If the terminal speed of a sphere of gold (density = $19.5 \times 10^3\ \text{kg m}^{-3}$) is $0.2\,\text{ms}^{-1}$ in a viscous liquid (density = $1.5 \times 10^3\ \text{kg m}^{-3}$), find the terminal speed of a sphere of silver (density = $10.5\,\text{kgm}^{-3}$) of the same size in the same liquid. **[AIEEE 2006]**

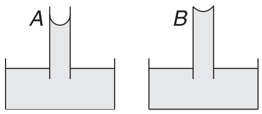
(a) $0.4\,\text{ms}^{-1}$ (b) $0.133\,\text{ms}^{-1}$
(c) $0.1\,\text{ms}^{-1}$ (d) $0.2\,\text{ms}^{-1}$

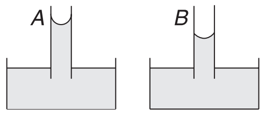
3. A spherical solid ball of volume V is made of a material of density ρ_1. It is falling through a liquid of density $\rho_2 (\rho_2 < \rho_1)$. Assume that the liquid applies a viscous force on the ball that is proportional to the square of its speed v, i.e., $F_{\text{viscous}} = kv^2 (k > 0)$. The terminal speed of the ball is **[AIEEE 2008]**

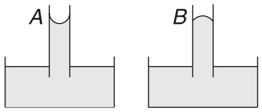
(a) $\sqrt{\dfrac{Vg(\rho_1 - \rho_2)}{k}}$ (b) $\dfrac{Vg\rho_1}{k}$
(c) $\sqrt{\dfrac{Vg\rho_1}{k}}$ (d) $\dfrac{Vg(\rho_1 - \rho_2)}{k}$

4. A capillary tube (A) is dipped in water. Another identical tube (B) is dipped in a soap-water solution. Which of the following shows the relative nature of the liquid columns in the two tubes? **[AIEEE 2008]**

(a) 

(b) 

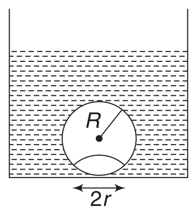
(c) 

(d) 

5. Work done in increasing the size of a soap bubble from a radius of 3 cm to 5 cm is nearly (surface tension of soap solution = 0.03 Nm^{-1}): **[AIEEE 2011]**

(a) 2π mJ (b) 0.4π mJ
(c) 4π mJ (d) 0.2π mJ

6. On heating water, bubbles being formed at the bottom of the vessel detach and rise. Take the bubbles to be spheres of radius R and making a circular contact of radius r with the bottom of the vessel. If $r \ll R$ and the surface tension of water is T, value of r just before bubbles detach is: (density of water is ρ_w)



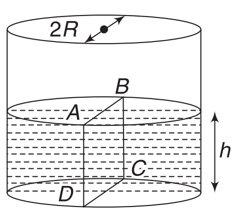
[JEE Main 2014]

(a) $R^2\sqrt{\frac{\rho_w g}{3T}}$ (b) $R^2\sqrt{\frac{\rho_w g}{6T}}$

(c) $R^2\sqrt{\frac{\rho_w g}{T}}$ (d) $R^2\sqrt{\frac{3\rho_w g}{T}}$

IIT JEE (Advanced) Questions

1. Water is filled up to a height h in a beaker of radius R as shown in the figure. The density of water is ρ, the surface tension of water is T and the atmospheric pressure is P_0. Consider a vertical section ABCD of the water column through a diameter of the beaker. The force on water on one side of this section by water on the other side of this section has magnitude: **[2007]**



(a) $\left|2P_0Rh + \pi R^2\rho gh - 2RT\right|$
(b) $\left|2P_0Rh + R\rho gh^2 - 2RT\right|$
(c) $\left|P_0\pi R^2 + R\rho gh^2 - 2RT\right|$
(d) $\left|P_0\pi R^2 + R\rho gh^2 - 2RT\right|$

2. Two soap bubbles A and B are kept in a closed chamber where the air is maintained at pressure 8 Nm^{-2}. The radii of bubbles A and B are 2 cm and 4 cm, respectively. Surface tension of the soap-water used to make bubbles is 0.04 Nm^{-1}. Find the ratio n_B/n_A, where n_A and n_B are the number of moles of air in bubbles A and B, respectively. [Neglect the effect of gravity.] **[2009]**

3. Following three questions are based on the given passage

When liquid medicine of density ρ is to be put in the eye, it is done with the help of a dropper. As the bulb on the top of the dropper is pressed, a drop forms at the opening of the dropper. We wish to estimate the size of the drop. We first assume that the drop formed at the opening is spherical because that requires a minimum increase in its surface energy. To determine the size, we calculate the net vertical force due to the surface tension T when the radius of the drop is R. When this force becomes smaller than the weight of the drop, the drop gets detached from the dropper. **[2010]**

(i) If the radius of the opening of the dropper is r, the vertical force due to the surface tension on the drop of radius R (assuming $r \ll R$) is:

(a) $2\pi rT$ (b) $2\pi RT$

(c) $\frac{2\pi r^2 T}{R}$ (d) $\frac{2\pi R^2 T}{r}$

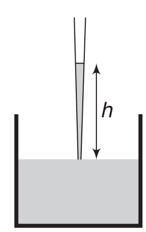
(ii) If $r = 5 \times 10^{-4}$ m, $\rho = 10^3\ \text{kg m}^{-3}$, $g = 10\ \text{ms}^{-2}$ $T = 0.11\ \text{Nm}^{-1}$, the radius of the drop when it detaches from the dropper is approximately

(a) 1.4×10^{-3} m (b) 3.3×10^{-3} J
(c) 2.0×10^{-3} m (d) 4.1×10^{-3} m

(iii) After the drop detaches, its surface energy is

(a) 1.4×10^{-3} J (b) 2.7×10^{-6} J
(c) 5.4×10^{-6} J (d) 8.1×10^{-6} J

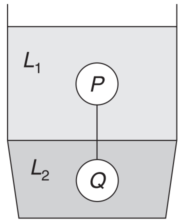
4. A glass capillary tube is of the shape of a truncated cone with an apex angle α so that its two ends have cross-sections of different radii. When dipped in water vertically, water rises in it to a height h, where the radius of its cross-section is b. If the surface tension of water is S, its density is ρ, and its contact angle with glass is θ, the value of h will be (g is the acceleration due to gravity) **[2014]**



(a) $\frac{2S}{b\rho g}\cos(\theta - \alpha)$ (b) $\frac{2S}{b\rho g}\cos(\theta + \alpha)$

(c) $\frac{2S}{b\rho g}\cos\left(\theta - \frac{\alpha}{2}\right)$ (d) $\frac{2S}{b\rho g}\cos\left(\theta + \frac{\alpha}{2}\right)$

5. Two spheres P and Q of equal radii have densities ρ_1 and ρ_2, respectively. The spheres are connected by a massless string and placed in liquids L_1 and L_2 of densities σ_1 and σ_2 and viscosities η_1 and η_2, respectively. They float in equilibrium with the sphere P in L_1 and sphere Q in L_2 and the string being taut (see figure). If sphere P alone in L_2 has terminal velocity $\vec{v}_P$ and Q alone in L_1 has terminal velocity $\vec{v}_Q$, then



[2015]

(a) $\dfrac{|\vec{v}_P|}{|\vec{v}_Q|} = \dfrac{\eta_1}{\eta_2}$ (b) $\dfrac{|\vec{v}_P|}{|\vec{v}_Q|} = \dfrac{\eta_2}{\eta_1}$

(c) $\vec{v}_P \cdot \vec{v}_Q > 0$ (d) $\vec{v}_P \cdot \vec{v}_Q < 0$

6. Consider two solid spheres P and Q each of density 8 gm cm^{-3} and diameters 1 cm and 0.5 cm, respectively. Sphere P is dropped into a liquid of density 0.8 gm cm^{-3} and viscosity η = 3 poiseulles. Sphere Q is dropped into a liquid of density 1.6 gm cm^{-3} and viscosity η = 2 poiseulles. The ratio of the terminal velocities of P and Q is **[2016]**

7. A drop of liquid of radius $R = 10^{-2}$ m having surface tension $S = \dfrac{0.1}{4\pi}$ Nm^{-1} divides itself into K identical drops. In this process, the total change in the surface energy is $\Delta U = 10^{-3}$ J. If $K = 10^{\alpha}$ then the value of α is **[2017]**

8. A uniform capillary tube of inner radius r is dipped vertically into a beaker filled with water. The water rises to a height h in the capillary tube above the water surface in the beaker. The surface tension of water is σ. The angle of contact between water and the wall of the capillary tube is θ. Ignore the mass of water in the meniscus. Which of the following statements is (are) true? **[JEE Adv. 2018]**

(a) For a given material of the capillary tube, h decreases with increase in r

(b) For a given material of the capillary tube, h is independent of σ

(c) If this experiment is performed in a lift going up with a constant acceleration, then h decreases

(d) h is proportional to contact angle θ

9. Consider a thin square plate floating on a viscous liquid in a large tank. The height h of the liquid in the tank is much less than the width of the tank. The floating plate is pulled horizontally with a constant velocity u_0. Which of the following statements is (are) true? **[JEE Adv. 2018]**

(a) The resistive force of liquid on the plate is inversely proportional to h

(b) The resistive force of liquid on the plate is independent of the area of the plate

(c) The tangential (shear) stress on the floor of the tank increases with u_0

(d) The tangential (shear) stress on the plate varies linearly with the viscosity η of the liquid

GRAVITATION

AIEEE/JEE Main Questions

1. A particle of mass 10 g is kept on the surface of a uniform sphere of mass 100 kg and radius 10 cm. Find the work to be done against the gravitational force between them to take the particle far away from the sphere (you may take $G = 6.67 \times 10^{-11}$ Nm^{2}kg^{-2} **[AIEEE 2005]**

(a) 6.67×10^{-9} J (b) 6.67×10^{-10} J

(c) 13.34×10^{-10} J (d) 3.33×10^{-10} J

2. A planet in a distant solar system is 10 times more massive than earth and its radius is 10 times smaller. Given that the escape velocity from the earth is 11 km s^{-1}, the escape velocity from the surface of the planet would be **[AIEEE 2008]**

(a) 1.1 km s^{-1} (b) 11 km s^{-1}

(c) 110 km s^{-1} (d) 0.11 km s^{-1}

3. The height at which the acceleration due to gravity becomes $g/9$ (where g = the acceleration due to gravity on the surface of the earth) in terms of R, the radius of the earth, is: **[AIEEE 2009]**

(a) $2R$ (b) $\dfrac{R}{\sqrt{2}}$

(c) $R/2$ (d) $\sqrt{2}R$

4. The mass of a spaceship is 1000 kg. It is to be launched from the earth's surface, out into free space. The value of 'g' and 'R' (radius of earth) are 10 ms^{-2} and 6400 km respectively. The required energy for this work will be: **[AIEEE 2012]**

(a) 6.4×10^{10} Joules (b) 6.4×10^{11} Joules

(c) 6.4×10^{8} Joules (d) 6.4×10^{9} Joules

5. What is the minimum energy required to launch a satellite of mass m from the surface of a planet of mass M and radius R in a circular orbit at an altitude of $2R$? **[JEE Main 2013]**

(a) $\dfrac{2GmM}{3R}$ (b) $\dfrac{GmM}{2R}$

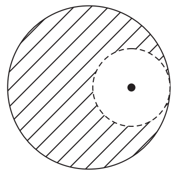
(c) $\dfrac{GmM}{3R}$ (d) $\dfrac{5GmM}{6R}$

6. Four particles, each of mass M are at the vertices of a square and move along a circle of radius R, under the action of their mutual gravitational attraction. The speed of each particle is: **[JEE Main 2014]**

(a) $\sqrt{\dfrac{GM}{R}}$ (b) $\sqrt{2\sqrt{2}\,\dfrac{GM}{R}}$

(c) $\sqrt{\dfrac{GM}{R}\,(1 + 2\sqrt{2})}$ (d) $\dfrac{1}{2}\sqrt{\dfrac{GM}{R}\,(1 + 2\sqrt{2})}$

7. From a solid sphere of mass M and radius R, a spherical portion of radius $R/2$ is removed, as shown in the figure. Taking gravitational potential $V = 0$ at $r = \infty$, the potential at the centre of the cavity thus formed is: (G gravitational constant)



[JEE Main 2015]

(a) $\dfrac{-2GM}{R}$ (b) $\dfrac{-GM}{2R}$

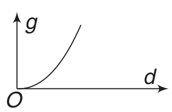
(c) $\dfrac{-GM}{R}$ (d) $\dfrac{-2GM}{3R}$

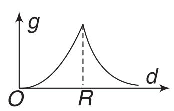
8. A satellite is revolving in a circular orbit at a height 'h' from the earth's surface (radius of earth R; $h << R$). The minimum increase in its orbital velocity required, so that the satellite could escape from the earth's gravitational field, is close to: (Neglect the effect of atmosphere). **[JEE Main 2016]**

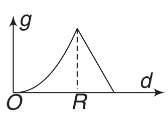
(a) $\sqrt{2gR}$ (b) $\sqrt{gR}$

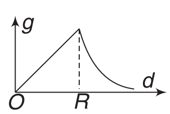
(c) $\sqrt{gR/2}$ (d) $\sqrt{gR}\,(\sqrt{2} - 1)$

9. The variation of acceleration due to gravity g with distance d from the centre of earth is best represented by (R = Earth's radius): **[JEE Main 2017]**

(a) 

(b) 

(c) 

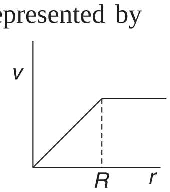
(d) 

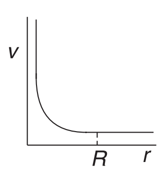
IIT JEE (Advanced) Questions

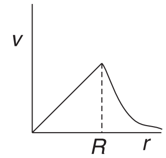
1. A spherically symmetric gravitational system of particles has a mass density **[2008]**

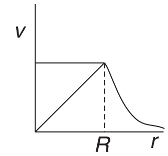
$$\rho = \begin{cases} \rho_0 & \text{for} \quad r \le R \\ 0 & \text{for} \quad r > R \end{cases}$$

Where ρ_0 is a constant. A test mass can undergo circular motion under the influence of the gravitational field of particles. Its speed v as a function of distance $r\,(0 < r < \infty)$ from the centre of the system is represented by

(a) 

(b) 

(c) 

(d) 

2. **Statement-1:** An astronaut in an orbiting space station above Earth experiences weightlessness.

Statement-2: An object moving around Earth under the influence of Earth's gravitational force is in a state of "free-fall". **[2008]**

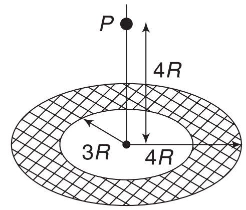
(a) Both Assertion and Reason are true and Reason is the correct explanation of Assertion

(b) Both Assertion and Reason are true and Reason is not the correct explanation of Assertion'

(c) Assertion is true but Reason is false

(d) Assertion is false but Reason is true

3. A thin uniform annular disc (see figure) of mass M has outer radius $4R$ and inner radius $3R$. The work required to take a unit mass from point P on its axis to infinity is: **[2010]**



(a) $\dfrac{2GM}{7R}\,(4\sqrt{2} - 5)$ (b) $-\dfrac{2GM}{7R}\,(4\sqrt{2} - 5)$

(c) $\dfrac{GM}{4R}$ (d) $\dfrac{2GM}{5R}\,(\sqrt{2} - 1)$

4. Gravitational acceleration on the surface of a planet is $\sqrt{6}/11g$, where g is the gravitational acceleration on the surface of the earth. The average mass density of the planet is 2/3 times that of the earth. If the escape speed on the surface of the earth is taken to be 11 km s^{-1}, the escape speed on the surface of the planet in km s^{-1} will be **[2010]**

5. A satellite is moving with a constant speed 'v' in a circular orbit about the earth. An object of mass 'm' is ejected from the satellite such that it just escapes from gravitational pull of the earth. At the time of its ejection, the kinetic energy of the object is:

[2011]

(a) $(1/2)mv^2$ (b) mv^2

(c) $(3/2)mv^2$ (d) $2mv^2$

6. Two bodies, each of mass M, are kept fixed with a separation $2L$. A particle of mass m is projected from the mid-point of the line joining their centres, perpendicular to the line. The gravitational constant is G. The correct statement(s) is (are) **[2013]**

(a) The minimum initial velocity of mass m to escape the gravitational field of the two bodies is $4\sqrt{GM/L}$

(b) The minimum initial velocity of the mass m to escape the gravitational field of the two bodies is $2\sqrt{GM/L}$

(c) The minimum initial velocity of the mass m to escape the gravitational field of the two bodies is $\sqrt{\dfrac{2GM}{L}}$

(d) The energy of the mass m remains constant

7. A planet of radius $R = \dfrac{1}{10} \times$ (radius of Earth) has the same mass density as Earth. Scientists dig a well of depth $R/5$ on it and lower a wire of the same length and of linear mass density 10^{-3} kg m^{-1} into it. If the wire is not touching anywhere, the force applied at the top of the wire by a person holding it in place is (take the radius of Earth = 6×10^6 m and the acceleration due to gravity on Earth is 10 ms^{-2}) **[2014]**

(a) 96 N (b) 108 N

(c) 120 N (d) 150 N

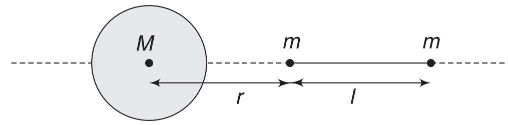
8. A spherical body of radius R consists of a fluid of constant density and is in equilibrium under its own gravity. If $P(r)$ is the pressure at $(r < R)$, then the correct option (s) is (are) **[2015]**

(a) $P(r = 0) = 0$ (b) $\dfrac{P(r = 3R/4)}{P(r = 2R/3)} = \dfrac{63}{80}$

(c) $\dfrac{P(r = 3R/5)}{P(r = 2R/5)} = \dfrac{16}{21}$ (d) $\dfrac{P(r = R/2)}{P(r = R/3)} = \dfrac{20}{27}$

9. A bullet is fired vertically upward with velocity v from the surface of a spherical planet. When it reaches its maximum height, its acceleration due to the planet gravity is one-fourth of its value at the surface of the planet. If the escape velocity from the planet is $v_{ese} = v\sqrt{N}$, then the value of N is (ignore energy loss due to atmosphere) **[2015]**

10. A large spherical mass M is fixed at one position and two identical point masses m are kept on a line passing through the centre of M (see figure). The point masses are connected by a rigid massless rod of length l and this assembly is free to move along the line connecting them. All three masses interact only through their mutual gravitational interaction. When the point mass nearer to M is at a distance $r = 3l$ from M, the tension in the rod is zero for $m = k\left(\dfrac{M}{288}\right)$. The value of k is **[2015]**



11. A rocket is launched normal to the surface of the Earth, away from the Sun, along the line joining the Sun and the Earth. The Sun is 3×10^5 times heavier than the Earth and is at a distance 2.5×10^4 times larger than the radius of the Earth. The escape velocity from Earth's gravitational field is $v_e = 11.2$ km s^{-1}. The minimum initial velocity (v_s) required for the rocket to be able to leave the Sun–Earth system is closest to (Ignore the rotation and revolution of the Earth and the presence of any other planet) **[2017]**

(a) $v_s = 22$ km s^{-1} (b) $v_s = 42$ km s^{-1}

(c) $v_s = 62$ km s^{-1} (d) $v_s = 72$ km s^{-1}

12. A planet of mass M, has two natural satellites with masses m_1 and m_2. The radii of their circular orbits are R_1 and R_2 respectively. Ignore the gravitational force between the satellites. Define v_1, L_1, K_1 and T_1 to be, respectively, the orbital speed, angular momentum, kinetic energy and time period of revolution of satellite 1; and v_2, L_2, K_2 and T_2 to be the corresponding quantities of satellite 2. Given $m_1/m_2 = 2$ and $R_1/R_2 = 1/4$, match the ratios in List-I to the numbers in List-II. **[JEE Adv. 2018]**

LIST–I	LIST–II
P. $\dfrac{v_1}{v_2}$	**1.** $\dfrac{1}{8}$
Q. $\dfrac{L_1}{L_2}$	**2.** 1
R. $\dfrac{K_1}{K_2}$	**3.** 2
S. $\dfrac{T_1}{T_2}$	**4.** 8

(a) **P → 4; Q → 2; R → 1; S → 3**

(b) **P → 3; Q → 2; R → 4; S → 1**

(c) **P → 2; Q → 3; R → 1; S → 4**

(d) **P → 2; Q → 3; R → 4; S → 1**

ELASTICITY

AIEEE/JEE Main Questions

1. If S is stress and Y is Young's modulus of material of a wire, the energy stored in the wire per unit volume is **[AIEEE 2005]**

(a) $2S^2Y$ (b) $S^2/2Y$

(c) $2Y/S^2$ (d) $S/2Y$

2. A wire elongates by l when a load w is hanged from it. If the wire goes over a pulley and two weights, w each, are hung at the two ends, the elongation of the wire will be (in mm) **[AIEEE 2006]**

(a) l (b) $2l$

(c) zero (d) $l/2$

3. Two wires are made of the same material and have the same volume. However, wire 1 has cross-sectional area A and wire 2 has cross-sectional area $3A$. If the length of wire 1 increases by Δx on applying force F, how much force is needed to stretch wire 2 by the same amount? **[AIEEE 2009]**

(a) F (b) $4F$

(c) $6F$ (d) $9F$

4. A man grows into a giant such that his linear dimensions increase by a factor of 9. Assuming that his density remains same, the stress in the leg will change by a factor of: **[JEE Main 2017]**

(a) 9 (b) $\frac{1}{9}$

(c) 81 (d) $\frac{1}{81}$

5. A solid sphere of radius r made of a soft material of bulk modulus K is surrounded by a liquid in a cylindrical container. A massless piston of area a floats on the surface of the liquid, covering entire cross section of cylindrical container. When a mass m is placed on the surface of the piston to compress the liquid, the fractional decrement in the radius of the sphere, $\left(\frac{dr}{r}\right)$, is: **[JEE Main 2018]**

(a) $\frac{mg}{Ka}$ (b) $\frac{Ka}{mg}$

(c) $\frac{Ka}{3mg}$ (d) $\frac{mg}{3Ka}$

IIT JEE (Advanced) Questions

1. When temperature of a gas is 20°C and pressure is changed from $P_1 = 1.01 \times 10^5$ Pa to $P_2 = 1.165 \times 10^5$ Pa then the volume changed by 10%. The bulk modulus is: **[2005]**

(a) 1.55×10^5 Pa (b) 0.115×10^5 Pa

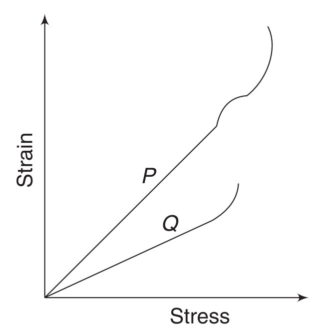
(c) 1.4×10^5 Pa (d) 1.01×10^5 Pa

2. One end of a horizontal thick copper wire of length $2L$ and radius $2R$ is welded to an end of another horizontal thin copper wire of length L and radius R. When the arrangement is stretched by applying forces at two ends, the ratio of the elongation in the thin wire to that in the thick wire is **[2013]**

(a) 0.25 (b) 0.50

(c) 2.00 (d) 4.00

3. In plotting stress versus strain curves for two materials P and Q, a student by mistake puts strain on the y-axis and stress on the x-axis, as shown in the figure. Then the correct statement(s) is (are) **[2015]**



(a) P has more tensile strength than Q

(b) P is more ductile than Q

(c) P is more brittle than Q

(d) The Young's modulus of P is more than that of Q

Answers Sheet

Centre of Mass and Momentum

AIEEE/JEE (Main) Questions

1. (a) 2. (c) 3. (b) 4. (a) 5. (a) 6. (a) 7. (a) 8. (c) 9. (c)
10. (c) 11. (b) 12. (d) 13. (d) 14. (c) 15. (c)

IIT JEE (Advanced) Questions

1. (a) 2. (a) 3. (i) (b) (ii) (b) (iii) (c) 4. (a) 5. (c) 6. (a) 7. (4)
8. (c) 9. (a, c) 10. (d) 11. (5) 12. (b) 13. (b, c) 14. (6.30 m)

Rotational Motion

AIEEE/JEE (Main) Questions

1. (b) 2. (b) 3. (d) 4. (d) 5. (b) 6. (d) 7. (a) 8. (b) 9. (b)
10. (c) 11. (d) 12. (b,d) 13. (a) 14. (a) 15. (a) 16. (a) 17. (b)

IIT JEE (Advanced) Questions

1. $\frac{2}{3}g \sin\theta$ 2. $\frac{1}{2}(M + m)g \cot\theta$ 3. (b) 4. (a) 5. (b) 6. (a) 7. (c, d)
8. (b) 9. $10\,\text{ms}^{-1}$ 10. (d) 11. (d) 12. (i) (c) (ii) (a) (iii) (b) 13. (d) 14. (a)
15. (b,c) 16. (b) 17. (4) 18. (9) 19. (3) 20. (d) 21. (a,b) 22. (b) 23. (d)
24. (a) 25. (i) (d) (ii) (a) 26. 8 rad s^{-1} 27. (c,d) 28. 4 rad s^{-1} 29. 2 rad s^{-1} 30. (6) 31. (d)
32. (a,b,d) 33. (b,d) 34. (d) 35. (i) (b) (ii) (a) 36. (c) 37. (i) none (ii) (a) 38. (b,c)
39. (a,c) 40. (0.75 m) 41. (a)

Fluid Mechanics

AIEEE/JEE (Main) Questions

1. (d) 2. (c) 3. (b) 4. (b) 5. (a) 6. (c)

IIT JEE (Advanced) Questions

1. (a) 2. (a) 3. (i) (c) (ii) (a) (iii) (a) 4. (a) 5. 6 mm
6. (a) (p, t) (b) (q, s, t) (c) (p, r, t) (d) (q) 7. (a) 8. (i) (c) (ii) (a) 9. (c)

Surface Tension and Viscosity

AIEEE/JEE (Main) Questions

1. (d) 2. (c) 3. (a) 4. (c) 5. (b) 6. none

IIT JEE (Advanced) Questions

1. (b) 2. (6) 3. (i) (c) (b) (a) (iii) (b) 4. (d) 5. (a,d) 6. (3) 7. (6)
8. (a,c) 9. (a,c,d)

Gravitation

AIEEE/JEE (Main) Questions

1. (b) 2. (c) 3. (a) 4. (a) 5. (d) 6. (d) 7. (c) 8. (d) 9. (d)

IIT JEE (Advanced) Questions

1. (c) 2. (a) 3. (a) 4. (3) 5. (b) 6. (b, d) 7. (b) 8. (b, c) 9. (2)
10. (7) 11. (b) 12. (b)

Elasticity

AIEEE/JEE (Main) Questions

1. (b) 2. (a) 3. (d) 4. (a) 5. (d)

IIT JEE (Advanced) Questions

1. (a) 2. (c) 3. (a, b)

Solutions

Chapter 1 Centre of Mass

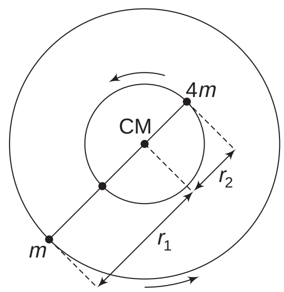
Your Turn

1. The two stars move in circles of radii r_1 and r_2 about the COM, as shown in figure.

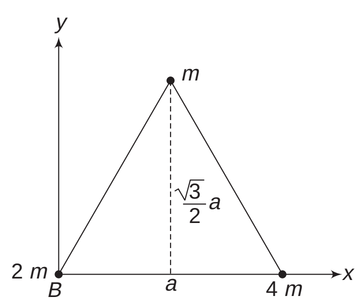
   Radius of circular path of m

   $$= \text{distance of } m \text{ from COM } (r_1)$$

   $$= \frac{4md}{m + 4m} = \frac{4d}{5}$$



2. Consider the co-ordinate system as shown, with B as origin.



   x co-ordinate of mass m is $\frac{a}{2}$ and its y co-ordinate is $= \frac{\sqrt{3}}{2} a$

   $$x_{CM} = \frac{2m \times 0 + 4m \times a + m \times \frac{a}{2}}{2m + 4m + m} = \frac{9}{14}a = 0.64a$$

   $$y_{CM} = \frac{2m \times 0 + 4m \times 0 + m \times \frac{\sqrt{3}}{2} a}{7m} = 0.12a$$

   Distance of COM from origin is

   $$r = \sqrt{(0.64a)^2 + (0.12a)^2} = 0.65a = 0.65 \times 10$$

   $$= 6.5m$$

3. Let the line joining the particles be x-axis

   $$\Delta x_{CM} = \frac{m_1 \Delta x_1 + m_2 \Delta x_2}{m_1 + m_2}$$

   $$= \frac{m(6) + 3m(-2)}{4m} = 0$$

4. (i) $x_{CM} = 0$

   $\Rightarrow m_1 x_1 + m_2 x_2 + m_3 x_3 + m_4 x_4 = 0$

   $\Rightarrow 1 \times 4 + 2 \times 6 + 3 \times (-1) + 2x_4 = 0$

   $\Rightarrow x_4 = -6.5$ cm

   (ii) The fourth particle shall be placed at the COM of the system of m_1, m_2 and m_3.

   $$\Rightarrow x = \frac{1 \times 4 + 2 \times 6 + 3 \times (-1)}{1 + 2 + 3} = \frac{13}{6} \text{ cm}$$

5. One can find x_{CM}, y_{CM} and z_{CM} separately or can directly use

   $$\vec{r}_{CM} = \frac{1}{M} \Sigma m_i \vec{r}_i$$

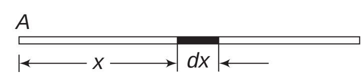
   $$\vec{r}_{CM} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2 + m_3 \vec{r}_3}{m_1 + m_2 + m_3}$$

   $$= \frac{1(\hat{i} + 4\hat{j} + \hat{k}) + 2(\hat{i} + \hat{j} + \hat{k}) + 3(2\hat{i} - \hat{j} - 2\hat{k})}{1 + 2 + 3}$$

   $$= \frac{9\hat{i} + 3\hat{j} - 3\hat{k}}{6} = \left(\frac{3}{2}\hat{i} + \frac{1}{2}\hat{j} - \frac{1}{2}\hat{k}\right) \text{m}$$

6. Consider origin at end A and x-axis along the rod.

   An element of length dx has mass given by



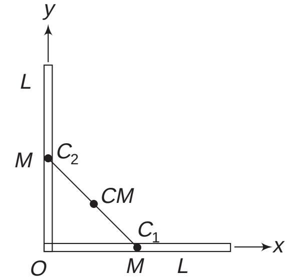
   $$dm = \lambda dx = \lambda_0 \left[1 + \frac{x}{L}\right] dx$$

   Mass of the rod is the sum of masses of all such elements.

   $$\therefore M = \int dm = \lambda_0 \int_0^L \left(1 + \frac{x}{L}\right) dx = \lambda_0 L + \frac{\lambda_0 L}{2} = \frac{3}{2} \lambda_0 L$$

   $$x_{CM} = \frac{1}{M} \int x \, dm = \frac{2}{3\lambda_0 L} \int_0^L x \lambda_0 \left(1 + \frac{x}{L}\right) dx = \frac{5L}{9}$$

7. O is the common end. Both the rods can be replaced with a point mass placed at their respective centres (at C_1 and C_2). Co-ordinates of COM of this two-particle system are



$$x_{CM} = \frac{M \cdot \frac{L}{2} + M(0)}{2M} = \frac{L}{4}$$

$$y_{CM} = \frac{M(0) + M\left(\frac{L}{2}\right)}{2M} = \frac{L}{4}$$

Distance of the COM of the system from point O is

$$d = \sqrt{\left(\frac{L}{4}\right)^2 + \left(\frac{L}{4}\right)^2} = \sqrt{2}\frac{L}{4} = \frac{L}{2\sqrt{2}}$$

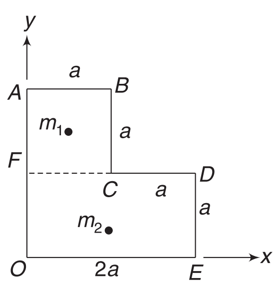
8. Let mass per unit area of the plate be σ.

Mass of segment ABCF is

$$m_1 = \sigma \cdot a^2$$

Mass of segment of OFDE is

$$m_2 = \sigma(2a^2)$$

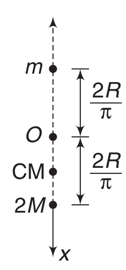


We can assume a particle of mass m_1 at the COM of ABCF at $\left(\frac{a}{2}, \frac{3a}{2}\right)$ and another particle of mass m_2 at COM of FDEO at $\left(a, \frac{a}{2}\right)$.

$$\therefore \quad x_{CM} = \frac{(\sigma a^2)\left(\frac{a}{2}\right) + (2\sigma a^2)(a)}{3\sigma a^2} = \frac{5}{6}a$$

Similarly, $y_{CM} = \frac{5}{6}a$

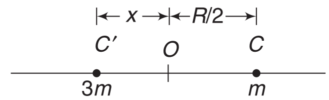
9. Let mass of ADC be m. Mass of ABC is $2m$. COM of a half-ring is at a distance $\frac{2R}{\pi}$ from centre O [Refer to Example 5]. We can assume two point masses as shown. Taking O as origin and downward direction as positive x-direction, we can write



$$x_{CM} = \frac{(2m)\left(\frac{2R}{\pi}\right) + m\left(-\frac{2R}{\pi}\right)}{2m + m} = \frac{2R}{3\pi}$$

Thus, COM is below the origin O, at a distance $\frac{2R}{3\pi}$ from it.

10. Let mass of removed part be m. (Area of the removed part is $\frac{\pi R^2}{4}$). Mass of the original plate (i.e., plate without hole) will be $4m$ ($\because$ its area is πR^2)



$\therefore$ Mass of remaining plate $= 3m$

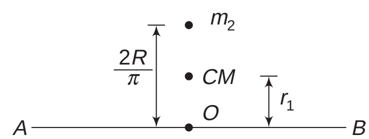
Let C' be the COM of the remaining plate. We can assume a point mass m at C and another point mass $3m$ at C'. These two masses together make a system, whose COM is at O.

Refer to Example 9 and the note given thereafter.

$$3m \cdot x = m \cdot \frac{R}{2} \quad \Rightarrow \quad x = \frac{R}{6}$$

11. $2R + \pi R = L \quad \Rightarrow \quad R = \frac{L}{2 + \pi}$

Let linear mass density of the wire be λ. Mass of straight wire AB is $m_1 = \lambda(2R)$. Mass of curved segment ACB is $m_2 = \lambda(\pi R)$. We can replace AB with a point mass m_1, placed at its centre O. ACB can be replaced with a point mass m_2, kept at a distance $\frac{2R}{\pi}$ from O.



Distance of COM from O is

$$r_1 = \frac{0 + m_2\left(\frac{2R}{\pi}\right)}{m_1 + m_2} = \frac{\lambda \cdot \pi R. \frac{2R}{\pi}}{\lambda \cdot 2R + \lambda . \pi R}$$

$$= \frac{2R}{2 + \pi}$$

13. COM is closer to the heavier side.

15. (i) $\vec{v}_1 = \frac{d\vec{r}_1}{dt} = 3\hat{i} + 4t\hat{j}$

At $\quad t = 1.0\,\text{s}, \vec{v}_1 = 3\hat{i} + 4\hat{j}\ \text{cms}^{-1}$

$$\vec{v}_2 = \frac{d\vec{r}_2}{dt} = -6\hat{i}\ \text{cms}^{-1}.$$

$$\therefore \quad \vec{v}_{CM} = \frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1 + m_2} = \frac{2(3\hat{i} + 4\hat{j}) + 3(-6\hat{i})}{2 + 3}$$

$$= \frac{-12\hat{i} + 8\hat{j}}{5} = (-2.4\hat{i} + 1.6\hat{j})\ \text{cms}^{-1}$$

(ii) $\vec{a}_1 = \frac{d\vec{v}_1}{dt} = 4\hat{j}\ \text{cms}^{-2}$

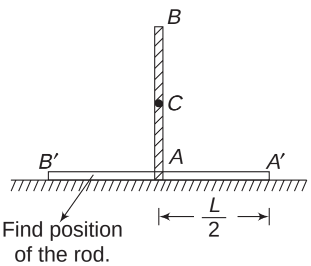
$$\vec{a}_2 = \frac{d\vec{v}_2}{dt} = 0$$

$$\vec{a}_{CM} = \frac{m_1\vec{a}_1 + m_2\vec{a}_2}{m_1 + m_2} = \frac{8\hat{j} + 0}{5} = 1.6\,\hat{j}\ \text{cms}^{-2}$$

(iii) $\vec{F}_{ext} = (m_1 + m_2)\vec{a}_{CM} = (5)(1.6)\,\hat{j}g\,\text{cms}^{-2}$

$$= 8\,\hat{j}\ \text{dyne}$$

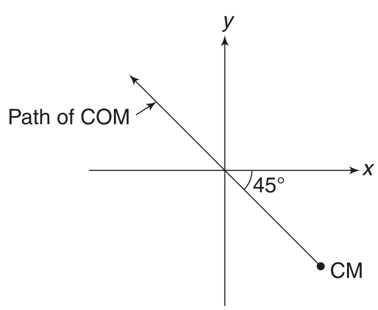
16. Refer to Example 17. The COM of the rod will fall on a vertical line. The centre of the rod will finally be at the initial position of lower end A. Lower end moves to A', such that $AA' = \frac{L}{2}$.



17. On the system of five blocks, there is no external force. Hence, motion of the COM does not change due to collision. Velocity of the COM before and after collision remains the same.

$$v_{CM} = \frac{mu + 0}{5m} = \frac{u}{5}$$

18. $\vec{v}_{CM} = \dfrac{-mv\hat{i} + mv\hat{j}}{2m} = \dfrac{v}{2}(-\hat{i} + \hat{j})$ collision will not change the motion of the COM of the system, as there is no external force.



19. Length of base of the wedge is

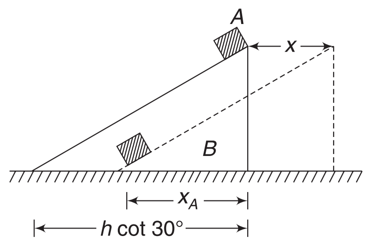
$$h\cot 30° = 10\sqrt{3}\cdot\sqrt{3} = 30\,m.$$

If block B moves by x to the right, then horizontal displacement of A to the left is $h\cot 30° - x = 30 - x$

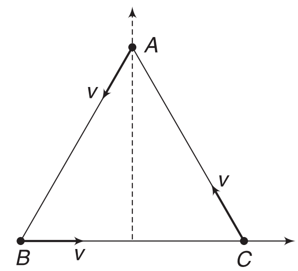
Since the system ($A + B$) has no horizontal force acting or it, the COM will have no horizontal displacement.

$$m_A x_A = m_B x_B$$

$\Rightarrow \quad m(30 - x) = 5m\cdot x \quad \Rightarrow \quad x = 5m$



20. $\vec{v}_{CM} = \dfrac{m(\vec{v}_A + \vec{v}_B + \vec{v}_C)}{3m} = 0$



Since $\vec{v}_A$, $\vec{v}_B$ and $\vec{v}_C$ are vectors of same magnitude at separation of 120° from one another, their resultant is zero.

The COM will not move.

21. Initial co-ordinates of COM are (0, 0, 0). COM has initial velocity $\vec{v}_0 = (120\,\text{ms}^{-1})\hat{i}$ and it continues to move with an acceleration $-g\hat{j} = -(10\,\text{ms}^{-2})\,\hat{j}$

After $\Delta t = 3$ s

$$x_{CM} = 120 \times 3 = 360\,\text{m}$$

$$y_{CM} = -\frac{1}{2}gt^2 = -\frac{1}{2} \times 10 \times 3^2 = -45\,\text{m}$$

$$z_{CM} = 0,$$

Since there is no movement of COM in z-direction.

$$\vec{r}_{CM} = \frac{\frac{2}{5}M\vec{r}_A + \frac{3}{5}M\vec{r}_B}{M}$$

$$360\hat{i} - 45\hat{j} + 0\hat{k} = \frac{2}{5}(300\hat{i} + 24\hat{j} - 48\hat{k}) + \frac{3}{5}\vec{r}_B$$

$$\Rightarrow \quad \vec{r}_B = \frac{5}{3}(360\hat{i} - 45\hat{j}) - \frac{2}{3}(300\hat{i} + 24\hat{j} - 48\hat{k})$$

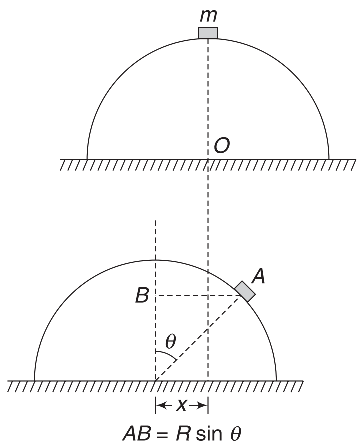
$$= (400\hat{i} - 91\hat{j} + 32\hat{k})\,\text{m}$$

22. If the hemisphere moves to the left by a distance x, the horizontal displacement of the block will be $(R\sin\theta - x)$.

Since COM of the system will not experience any horizontal displacement,

$$m(R\sin\theta - x) = M\cdot x$$

$$\Rightarrow \qquad x = \frac{mR\sin\theta}{m + M}$$



23. Mass of chain, $M = \lambda \pi R$

Height of the COM from O is $y_{CM} = \dfrac{2R}{\pi}$

$$\therefore \quad U = Mg y_{CM} = \lambda . \pi R \cdot g \cdot \frac{2R}{\pi}$$

$$= 2\lambda g R^2$$

24. $$v_{rel} = 5 - 2 = 3\,\text{ms}^{-1}$$

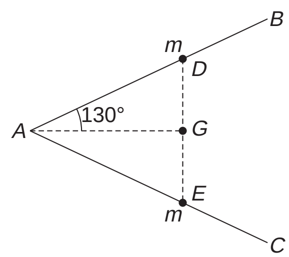
$$\mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{4 \times 2}{4 + 2} = \frac{4}{3}\text{ kg}$$

$$\therefore \quad k_{\text{wrt CM}} = \frac{1}{2}\mu v_{rel}^2 = \frac{1}{2}\left(\frac{4}{3}\right)(3)^2 = 6\,\text{J}$$

25. Momentum of a system in COM frame is always zero.

Worksheet 1

1. Particles can be located in any direction from the origin, hence $r_{CM} \le r$. For example, if a large number of particles are spread uniformly on the surface of a sphere, $r_{CM} = 0$.
2. AB is replaced by a point mass (m), placed at its centre D. AC is replaced by a point mass (m), placed at its centre E.



$\therefore$ COM of the system is at G, equidistant from D and E.

$$AB = \frac{l}{2};\ AD = \frac{l}{4};\ AG = \frac{l}{4}\cos 30° = \frac{\sqrt{3}l}{8}$$

3. Replace each brick with a point mass at its centre.

$$x_{CM} = \frac{m\left(\frac{l}{2}\right) + m\left(\frac{l}{2}+\frac{l}{2}\right) + m\left(\frac{l}{2}+\frac{l}{4}+\frac{l}{2}\right) + m\left(\frac{l}{2}+\frac{l}{4}+\frac{l}{6}+\frac{l}{2}\right)}{4m}$$

$$= \frac{25l}{24}$$

4. COM lies on the x-axis, as x-axis is the line of symmetry. Let σ = mass per unit area.

Mass of circle $= \pi r^2 \sigma$; Mass of square $= 4\ r^2 \cdot \sigma$

$$x_{CM} = \frac{(\pi r^2 \cdot \sigma) r + (4r^2 \cdot \sigma)(3r)}{\pi r^2 \sigma + 4r^2 \cdot \sigma}$$

$$= \left(\frac{\pi + 12}{\pi + 4}\right) r$$

5. $$x_{CM} = \frac{m\left(\frac{1}{2}\right) + m\left(\frac{1}{2}\right) + m\left(\frac{1}{2}\right) + m\left(\frac{3}{2}\right)}{4m} = \frac{3}{4}$$

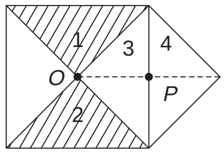
From symmetry, $x_{CM} = y_{CM} = z_{CM}$

6. Distance of COM for H end is

$$= \frac{0 + 1.27 \times (35.5\,m)}{(35.5 + 1)\,m}\ \text{Å}$$

This is a number slightly less than 1.27 Å

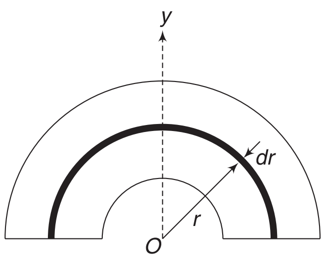
7. Mass of each quarter piece – 1, 2, 3 and 4 – is m. COM of (1 + 2) is at O and COM of (3 + 4) is at P. Thus, we can assume a point mass $2m$ at O and another mass $2m$ at P.



Distance of COM from O is

$$x_{CM} = \frac{0 + 2m(OP)}{2m + 2m} = \frac{OP}{2} = \frac{L}{4}$$

9. Let σ = mass per unit area. Mass of the ring element shown is $dm = \sigma.\pi r dr$.



Mass of complete disc is

$$M = \int dm = \sigma\pi \int_{R_1}^{R_2} r dr = \frac{\sigma\pi}{2}(R_2^2 - R_1^2)$$

$$y_{CM} = \frac{\int y\,dm}{M} = \frac{1}{M}\int_{R_1}^{R_2} \frac{2r}{\pi}\cdot dm$$

$$= \frac{1}{\frac{\sigma\pi}{2}(R_2^2 - R_1^2)} 2\sigma \int_{R_1}^{R_2} r^2 dr$$

$$= \frac{4(R_2^3 - R_1^3)}{3\pi(R_2^2 - R_1^2)}$$

10. The COM of the original rectangle and the COM of the removed part (unshaded rectangle) are joined. Extend this line. The COM of the remaining piece will fall on this line.

11. Let mass of each removed circle be m.

Mass of the complete circle (without hole) = $16m$; since its area is 16 times that of each small circle.

Mass of remaining piece = $14m$.

Now, $x_{CM} = 0$ for the complete disc. The complete disc is made of a $14m$ mass having x-co-ordinate x_0, a mass m having x co-ordinate $3R$ and a mass m having x co-ordinate 0.

$\therefore \quad 14m\cdot x_0 + m(3R) + m(0) = 0.$

$\Rightarrow \quad x_0 = -\frac{3R}{14}$

Similarly, for y co-ordinate.

12. If mass per unit length be λ, then

$$h_{CM} = \frac{\pi R\lambda(0) + \pi R\lambda(0) + \pi R\lambda\left(\frac{2R}{\pi}\right) + \pi R\lambda\left(\frac{2R}{\pi}\right)}{4\cdot\pi R\lambda} = \frac{R}{\pi}$$

14. $$x_{CM} = \frac{m\left(\frac{L}{2}\right) + m\left(\frac{L}{5} + \frac{L}{2}\right) + m\left(\frac{L}{5} + \frac{L}{5} + \frac{L}{2}\right) + m\left(\frac{L}{5} + \frac{L}{2}\right) + m\left(\frac{L}{2}\right)}{5m}$$

$$= \frac{23L}{50}$$

15. $x\cdot(7m) = \frac{\sqrt{3}l}{2}\cdot m \quad \Rightarrow \quad x = \frac{\sqrt{3}l}{14}$

16. Just before breaking, velocity of the projectile is horizontal. Just after breaking the COM will still have horizontal velocity only. It means the two pieces have no vertical motion after the projectile breaks. They will hit the ground at same time.

17. Since 60 kg mass (i.e, heavier mass) moves to the left, the plank (along with B) must move to the right so as to keep COM at the original position.

Let the displacement of plank (and B) be x towards the right.

Displacement of 40 kg mass = $(x + 4)m$ towards the right.

Displacement of 60 kg mass = $(4 - x)m$ towards the left

$$m_1\Delta x_1 + m_2\Delta x_2 + m_3\Delta x_3 = 0$$

$\therefore \quad (50 + 90)x + 40(x + 4) - 60(4 - x) = 0$

$\Rightarrow \quad x = \frac{1}{3}\ m$

18. $y_{CM} = 0$

$\frac{m}{4} \times 15 + \frac{3m}{4} \times y = 0 \quad \Rightarrow y = -5\,\text{cm}$

19. Mutual interaction cannot change velocity of COM.

20. Range $= \frac{v^2 \sin 2\theta}{g} = \frac{10^2 \times 1}{10} = 10\,\text{m}$

COM of the entire system (Platform + boy + stone) will not get displaced horizontally.

$\therefore \quad (100\,\text{kg})(x) = (1\,\text{kg})(10\,\text{m})$

$\Rightarrow \quad x = \frac{1}{10}m = 10\ \text{cm}$

21. $\vec{v}_{CM} = \dfrac{2\hat{i} + 2\hat{j}}{2} = (\hat{i} + \hat{j})\,\text{ms}^{-1}$.

$$\vec{a}_{CM} = \frac{\hat{i} + \hat{j} + 0}{2} = \frac{1}{2}(\hat{i} + \hat{j})\,\text{ms}^{-2}.$$

Since $\vec{v}_{CM}$ and $\vec{a}_{CM}$ are in same direction, the COM will continue to move along straight line.

22. $\vec{a}_{CM} = \dfrac{\vec{F}_{net}}{M} = \dfrac{M\vec{g} + \vec{R}}{M}$

24. B remains fixed.

$\therefore$ A must move 60 cm to the right, to meet B. To keep COM at a fixed position, the plank moves 60 cm to the left, as masses of B and plank are same.

Thus, the right end of the plank comes at the location of B.

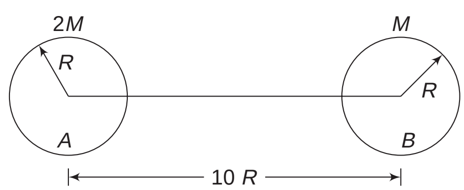
25. Horizontal force = 0.

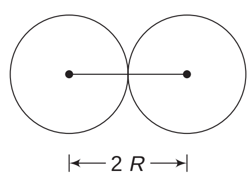
$\therefore$ There is no motion of the COM in horizontal direction. The block has acceleration in vertical direction. Thus, COM has a vertical component of acceleration.

Note that $(M + m)g > N$, Where N is normal force by the ground and $(M + m)$ is the total mass of the two-block system.

26. Mass of $B = M$

Mass of $A = 2M$





Let displacement of B be $x(\leftarrow)$, then displacement of A will be $(8R - x)$ towards right, for COM to remain at fixed position

$$M \cdot x = 2M(8R - x)$$

$$\Rightarrow \quad x = \frac{16R}{3}$$

27. $3M \cdot x = M(2 - x) \quad \Rightarrow x = 0.5\,\text{m}$

28. A completely filled shell as well as a completely empty shall have their COM at the geometrical centres.

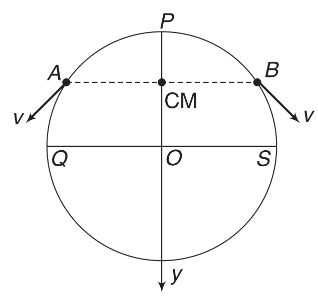
29. COM is at a distance $\dfrac{R}{2}$ from the centre of the earth.

$$x_{cg} = \frac{w_1(0) + w_2(R)}{w_1 + w_2} = \frac{mg \cdot R}{0 + mg} = R$$

30. In COM frame, momentum is always zero.

$\therefore$ Momentum of the other particle must be $-\vec{P}$

31. As the car A moves from P to Q (and B moves from P to S) the COM moves from P to O.



$$v_{CM} = \frac{mv_y + mv_y}{2m} = v_y$$

Where v_y = velocity component of A and B along y-direction.

certainly $v_y < v$.

32. Friction on wedge will act rightward. Hence, COM will move rightward. In vertical direction, there is an unbalanced force on the system (as the block has a vertical acceleration) Thus, COM has a vertical motion also.

Worksheet 2

1. If mass density continuously increases or decreases, then COM will be on the heavier side.
2. $F \neq O$ implies that $a_{CM} \neq O$
3. $\underset{\text{Rest}}{A} \longrightarrow \underset{\overset{\longleftarrow}{P}}{B} + \underset{\overset{\longrightarrow}{P}}{C}$

   B and C have equal and opposite momentum.

   Thus $\quad m_B v_B = m_C v_C \qquad$ ...(i)

   If speed of either B or C is known, we can easily find the speed of the other.

   And $\quad \frac{1}{2} m_B v_B^2 + \frac{1}{2} m_C v_C^2 = Q$ (Given) $\qquad$ ...(ii)

   Solving (i) and (ii), will give v_B and v_C if Q is given. If the decay is like $A \to B + C + D$

   Then, $\quad \vec{P}_B + \vec{P}_C + \vec{P}_D = O$

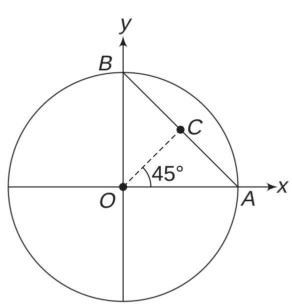
   The above equation has too many unknowns – directions of $\vec{P}_B$ and $\vec{P}_C$ apart from their magnitudes will decide the value of $\vec{P}_D$

   All unknowns cannot be found even if total KE is known.
4. C is the COM of rod AB. Its co-ordiantes are $\left(\frac{R}{2}, \frac{R}{2}\right)$.

   Mass of ring is assumed at O and mass of rod is a point mass at C. COM of the system is between O and C on the line OC.

   Also, $\quad x_{CM} = y_{CM}$

   Thus, (A) and (B) are possible co-ordinates of COM, depending on masses of the ring and the rod.

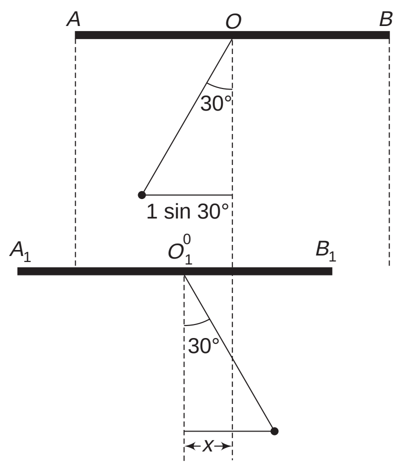


5. When the ball stops, the plank must also stop so that the velocity of COM remains zero, in horizontal direction. From conservation of energy, it is easy to see that the ball will stop when the string makes 30° with vertical, as shown in second figure.

   Let displacement of the plank be $x(\leftarrow)$.

   $$Mx = m(1 \cdot \sin 30° + 1 \cdot \sin 30° - x)$$

   $$\Rightarrow \quad 4x = 1(1 - x) \quad \Rightarrow \quad x = \frac{1}{5} = 0.2\,\text{m}$$



6. In all cases, $F_{ext} = 0$

   $\therefore$ COM will not move.
7. Net force on the (man + balloon) system is zero. Gravity is balanced by forces due to air (buoyancy). COM of the system will remain unmoved even when the man climbs.
8. According to the question, the COM of the system moves in a circle of radius R.

   $\therefore \quad F_{ext} = Ma_{CM} = 3m \cdot \omega^2 R$
9. $a_{CM} = g(\downarrow)$

   If $v_1 \cos\theta_1 = v_2 \cos\theta_2$, Then x component of velocity of the COM will be zero. COM will rise and fall on a vertical line. COM can never move in horizontal line since there is a vertical acceleration.
10. $K_o = K_{CM} + \frac{1}{2} m v_{CM}^2$

    Since v_{cm} does not change in absence of external force, the system always has $\frac{1}{2} M v_{CM}^2$ $(= K_o - K_{CM})$ amount of energy in ground frame. Also, K_{CM} change due to internal interactions.

Worksheet 3

1. The COM lies on y-axis. An element, subtending an angle $d\phi$ has length $Rd\phi$ and its mass is

$$dm = \lambda R d\phi$$

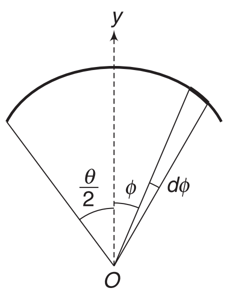
where λ = linear mass density. Y co-ordinate of this element is $y = R\cos\phi$

$$\therefore \quad y_{CM} = \frac{\int_{-\theta/2}^{\theta/2} y\,dm}{\lambda \cdot R\theta} = \frac{\lambda R^2}{\lambda R\theta}\int_{-\theta/2}^{\theta/2}\cos\phi\,d\phi$$

$$= \frac{R}{\theta}\cdot 2\sin\frac{\theta}{2} = \frac{R\sin\theta/2}{\theta/2}$$

Check: when $\frac{\theta}{2} \to O$; $\quad y_{CM} \to R$

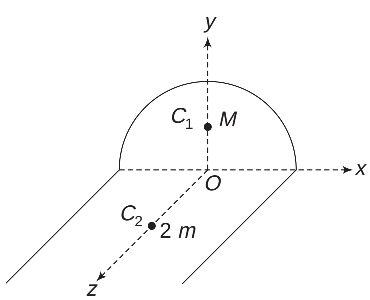
when $\frac{\theta}{2} \to \frac{\pi}{2}$; $\quad y_{CM} \to \frac{2R}{\pi}$



2. Mass of semicircle is $M = \lambda\pi R$

It can be replaced with a point mass at C_1 such that $OC_1 = \frac{2R}{\pi}$. Mass of each straight segment

$$m = \lambda 2R$$



We can assume $2m$ mass at C_2 (COM of the two straight segments) such that $OC_2 = R$.

Obviously, $x_{CM} = O$

$$y_{CM} = \frac{M\cdot\frac{2R}{\pi}}{M + 2m} = \frac{\lambda\cdot\pi R\,\frac{2R}{\pi}}{\lambda\pi R + 4\lambda R}$$

$$= \frac{2R}{\pi + 4}$$

$$z_{CM} = \frac{2m\cdot R}{M + 2m} = \frac{2(2\lambda R)R}{\lambda\pi R + 4\lambda R}$$

$$= \frac{4R}{\pi + 4}$$

$$\therefore \quad r_{CM} = \sqrt{\left(\frac{2R}{\pi + 4}\right)^2 + \left(\frac{4R}{\pi + 4}\right)^2} = \frac{2\sqrt{5}\,R}{\pi + 4}$$

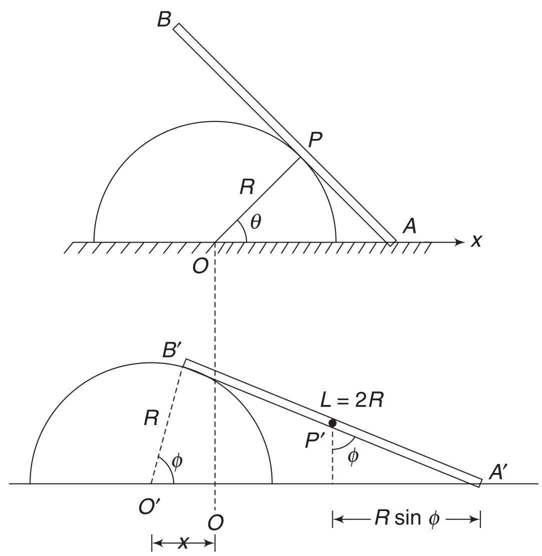
3. Initially, $AP = R\tan 45° = R$

$\Rightarrow$ COM of the rod is at P. Taking O at origin, the COM of the system of rod plus cylinder has x co-ordinates given by

$$x_{CM} = \frac{m(0) + mR\cos\theta}{2m}$$

$$= \frac{R}{2\sqrt{2}} \qquad \text{...(i)}$$

In position when the end B touches the cylinder, let $< B'O'A'$ be ϕ; clearly, $\tan\phi = \frac{L}{R} = 2$. COM of the rod (at P') is at a horizontal distance $R\sin\phi$ from end A'.



x co-ordinate of P' (with O as origin) is

$O'A' - x - R\sin\phi$ (where x is displacement of cylinder)

$$= R\sec\phi - x - R\sin\phi$$

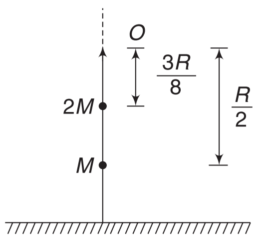
$$= \sqrt{5}\,R - x - \frac{2R}{\sqrt{5}} = \frac{3R}{\sqrt{5}} - x$$

COM of the system is still given by (i)

$$\therefore \quad \frac{R}{2\sqrt{2}} = \frac{m\left(\frac{3R}{\sqrt{5}} - x\right) - mx}{2m}$$

$$\Rightarrow \quad x = \frac{(3\sqrt{2} - \sqrt{5})R}{2\sqrt{10}}$$

4. Hemisphere (shell) can be replaced with a point mass M, kept at a distance $\frac{R}{2}$ from O on the vertical symmetry line.



The liquid is like a solid hemisphere and it can be replaced with a point mass $2M$ at a distance $\frac{3R}{8}$ from O.

$\therefore$ Distance of COM of the system from O is

$$x_{CM} = \frac{2M\left(\frac{3R}{8}\right) + M\left(\frac{R}{2}\right)}{2M + M} = \frac{5}{12}R$$

Height of COM from the floor is $= R - \frac{5R}{12} = \frac{7R}{12}$

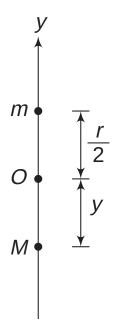
5. Let σ = mass per unit area

Mass of removed square, $m = \sigma l^2 = \frac{\sigma r^2}{2}$

Mass of remaining disc, $M = \sigma\left(\pi r^2 - \frac{r^2}{2}\right)$

$$= \sigma\left(\pi - \frac{1}{2}\right)r^2$$

Let COM of remaining disc be at a distance y from O, as shown.



If the square hole is refilled, the two masses m and M make a complete disc having its COM at O.

$$\therefore \quad My = m\frac{r}{2}$$

$$\Rightarrow \quad \sigma\left(\pi - \frac{1}{2}\right)r^2 \cdot y = \frac{\sigma r^2}{2} \cdot \frac{r}{2}$$

$$\Rightarrow \quad y = \frac{r}{4\left(\pi - \frac{1}{2}\right)} = \frac{r}{2(2\pi - 1)}$$

$\therefore$ Co-ordinates of COM are $\left(0, -\frac{r}{2(2\pi - 1)}\right)$

6. (i) COM has initial velocity $u_{CM} = 4\,\text{ms}^{-1}$ and it continues to move with this velocity.

Displacement of man relative to platform = 4 m.

Time taken to move from one end to another is

$$t = \frac{4\text{ m}}{2\,\text{ms}^{-1}} = 2\,\text{s}$$

$\therefore$ Displacement of COM of the system is

$$\Delta x_{CM} = u_{CM} \cdot t = 4 \times 2 = 8 \text{ m.}$$

(ii) Let displacement of the platform be Δx

Displacement of the man will be $(\Delta x + 4)$

$$\Delta x_{CM} = \frac{m_1 \Delta x_1 + m_2 \Delta x_2}{m_1 + m_2}$$

$$\Rightarrow \quad 8 = \frac{25(\Delta x) + 75(\Delta x + 4)}{25 + 75}$$

$$\Rightarrow \quad \Delta x = 5 \text{ m}$$

Displacement of the man = 5 + 4 = 9 m

(iii) $\Delta x = 5$ m

7. COM of the whole system will stay at rest. Let x be the leftward displacement of the car when all balls have been collected into the bucket.

Mass of man + car = M; mass of all balls = m.

Displacement of balls = $(L - x)$ to right.

$$\therefore \quad m(L - x) = M \cdot x \quad \Rightarrow \quad x = \frac{mL}{M + m} = \frac{L}{1 + \frac{M}{m}}$$

Obviously, $x < L$.

8. Let the displacement of wedge be x towards right. Displacement of $B = (x - l)$ towards right. Displacement of A relative to wedge (on the incline) is l.

$\therefore$ Displacement of A in horizontal direction $= l\cos\theta + x$

The COM of the entire system suffers no displacement in horizontal direction.

$$\therefore \quad \Delta x_{CM} = 0$$

$$\Rightarrow \quad m_1\Delta x_1 + m_2\Delta x_2 + m_3\Delta x_3 = 0$$

$$\Rightarrow \quad M \cdot x + m(x - l) + m(l\cos\theta + x) = 0$$

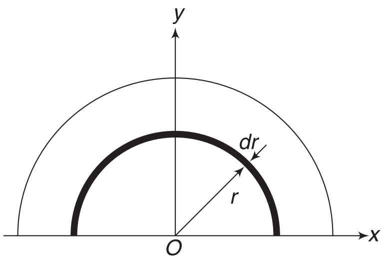
$$\Rightarrow \quad x = \frac{ml(1 - \cos\theta)}{M + 2m}$$

9. Consider a hemispherical shell of radius r and thickness dr.

Mass of shell, $dm = (2\pi r^2 \cdot dr)\rho$

$$= (2\pi r^2 dr)\rho_o \frac{r}{R}$$

$$= \frac{2\pi\rho_o}{R} r^3 \cdot dr$$



COM of the shell is on y-axis at a distance $\frac{r}{2}$ from O. We can replace the shell with a point a mass on y-axis at $y = \frac{r}{2}$. The entire hemisphere is made of infinite such shells. We can replace them all by point masses on y-axis.

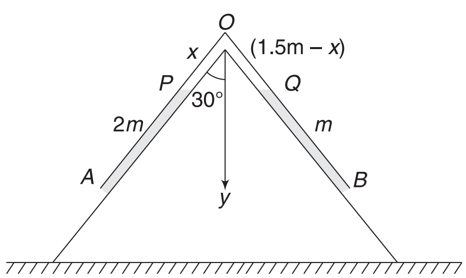
$$\therefore \quad y_{CM} = \frac{\int_0^R \frac{r}{2} dm}{\int_0^R dm} = \frac{\frac{2\pi\rho_o}{2 \cdot R}\int_0^R r^4 dr}{\frac{2\pi\rho_o}{r}\int_0^R r^3 dr} = \frac{2}{5} R$$

10. Let the mass x of the mid-segment lie on left slope and mass $(1.5m - x)$ lie on the right slope.

Equilibrium is possible only when mass of rope on both sides is same.

$$2m + x = m + 1.5\, m - x$$

$$\Rightarrow \qquad x = 0.25m$$



Length OP has mass $0.25m$, which is $\left(\frac{1}{6}\right)$th the mass of mid segment.

$$\therefore \; OP = \frac{L}{6}$$

and $\qquad OQ = \frac{5L}{6}$

$$y_{CM} = \frac{0.25m\left(\frac{1}{2}\cdot\frac{L}{6}\cos 30°\right) + 1.25m\left(\frac{1}{2}.\frac{5L}{6}\cos 30°\right) + 2m\left(\frac{L}{6}+\frac{L}{2}\right)\cos 30° + m\left(\frac{5L}{6}+\frac{L}{2}\right)\cos 30°}{2m + 1.5m + m}$$

$$= \frac{79\sqrt{3}}{216} L \simeq 0.63\, L$$

11. COM of liquid in the cone is at a height $\frac{3h}{4} = \frac{3}{4} \times 4R = 3R$ from the apex.

Height of COM from the base of cylindrical container is $h_1 = 3R + 2R = 5R$.

Mass of liquid, $m = \frac{1}{3}\pi R^2(4R)\cdot d = \frac{4}{3}\pi R^3 \cdot d$

Taking base of cylinder as reference level, the PE of liquid is

$$U_1 = mgh_1 = \frac{4}{3}\pi R^3 \cdot d \cdot g \cdot 5R = \frac{20\pi}{3} R^4 \cdot d \cdot g$$

Height of liquid column in cylinder will be given by

$$\pi R^2 \cdot h_o = \frac{1}{3}\pi R^2(4R)$$

$$\Rightarrow \qquad h_o = \frac{4R}{3}$$

$\therefore$ Height of COM above base, $h_2 = \frac{h_o}{2} = \frac{2R}{3}$. Final PE of the liquid is

$$U_2 = mgh_2 = \frac{4}{3}\pi R^3.d.g.\frac{2R}{3} = \frac{8\pi}{9} R^4 \cdot d \cdot g$$

Work done by gravity = Negative of change in PE

$$= -(U_2 - U_1) = U_1 - U_2$$

$$= \left(\frac{20}{3} - \frac{8}{9}\right)\pi R^4 \cdot d \cdot g$$

$$= \frac{52\pi}{9} R^4 \cdot d \cdot g$$

Your Turn

1. $P_{\text{bullet}} = \frac{50}{1000} \times 500 = 25\,\text{kg ms}^{-1}$

$P_{\text{cycle}} = 80 \times 2.5 = 200\ \text{kg ms}^{-1}$

$[\because\ 9\,\text{kmh}^{-1} = 2.5\,\text{ms}^{-1}]$

$\therefore\ P_{\text{cycle}} > P_{\text{bullet}}$

2. $K = \frac{1}{2}mv^2 = \frac{1}{2m}(mv)^2$

$\Rightarrow\quad mv = \sqrt{2mK}$

$\Rightarrow\quad P = \sqrt{2mK}$

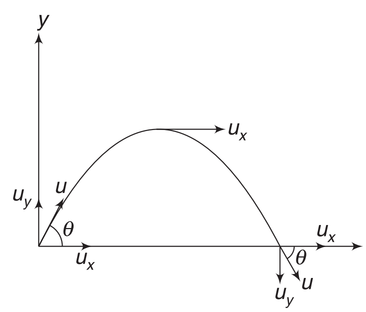
Since, K is same for both but m is larger for the truck, the truck will have higher momentum.

3. (i) $\vec{P}_i = m(u\cos\theta)\hat{i} + m(u\sin\theta)\hat{j}$

At top, $\vec{P}_f = m(u\cos\theta)\hat{i}$

$\therefore\quad \overrightarrow{\Delta P} = \vec{P}_f - \vec{P}_i = -mu\sin\theta\,\hat{j}$

$= mu\sin\theta\ (\downarrow)$



(ii) $\vec{P}_i = m(u\cos\theta)\hat{i} + m(u\sin\theta)\hat{j}$

Just before landing, $\vec{P}_f = m(u\cos\theta)\hat{i} - m(u\sin\theta)\hat{j}$

$\therefore\quad \overrightarrow{\Delta P} = \vec{P}_f - \vec{P}_i = -2mu\sin\theta\cdot\hat{j}$

$= 2mu\sin\theta\ (\downarrow)$

4. (i) $P_i = (1500\,\text{kg}) \times (15\,\text{ms}^{-1}) = 22500\ \text{kg ms}^{-1}$

$P_f = (1500\,\text{kg}) \times (5\,\text{ms}^{-1}) = 7500\ \text{kg ms}^{-1}$

$|\Delta P| = 22500 - 7500 = 15000\ \text{kg ms}^{-1}$

$F_{av} = \frac{|\Delta P|}{\Delta t} = \frac{15000}{0.15} = 10^5\,\text{N}$

(ii) Loss in $KE = \frac{1}{2} \times 1500(15^2 - 5^2) = 1.5 \times 10^5\,\text{J}$

5. Speed before hitting the floor,

$$u = \sqrt{2gH} = \sqrt{2 \times 10 \times 5} = 10\ \text{ms}^{-1}$$

Speed just after hitting the floor,

$$v = \sqrt{2gh} = \sqrt{2 \times 10 \times 1.25} = 5\,\text{ms}^{-1}$$

Change in momentum of the ball is

$$|\Delta P| = mu + mv = \frac{50}{1000}(10 + 5) = \frac{3}{4}\ \text{kg ms}^{-1}$$

The smallest time for which the ball could have been in contact with the floor is

$\Delta t_1 = 0.11 - 0.01 = 0.10\ \text{s}$

$\therefore$ Maximum average force $F_{\text{max}} = \frac{\Delta P}{\Delta t_1}$

$= \frac{0.75}{0.1} = 7.50\,\text{N}$

The largest time for which the ball could have been in contact with the floor is

$\Delta t_2 = 0.11 + 0.01 = 0.12\,\text{s}$

$\therefore$ Minimum average force, $F_{\text{min}} = \frac{\Delta P}{\Delta t_2} = \frac{0.75}{0.12}$

$= 6.25\ \text{N}$

$\therefore$ We can say that average force on the ball was in range $6.25\,\text{N} \le F \le 7.50\,\text{N}$.

or , $F = (6.88\,\text{N} \pm 0.37\,\text{N})$

$\left[\text{where } \frac{6.25 + 7.50}{2} = 6.88\right]$

6. $F = \frac{dP}{dt}$ = slope of P–t graph.

7. $\Delta P = mv(\rightarrow) = -mv(\leftarrow) = 2mv(\rightarrow)$

$\Delta t = \frac{\pi R}{v}$

$\therefore\quad F_{aV} = \frac{\Delta P}{\Delta t} = \frac{2mv^2}{\pi R}$

8. Mass of air incident per second = ρ (volume incident per second) = ρAv

Change in momentum of the incident mass = $(\rho Av)\cdot v$ per second. Thus force $F = \frac{dp}{dt} = \rho Av^2$

If speed is doubled, the force will become 4 times.

9. Refer to Example 4. Mass of water accumulated in the container at time t is μt.

$\therefore$ Weight of water in container, $W = \mu t\cdot g$

Apart from this weight, the falling water also exerts a thrust force on the container.

Speed of water hitting the container, $v = \sqrt{2gH}$

Mass of water hitting the container in unit time = μ.

Water particle does not bounce back.

$\therefore$ F = rate of change of momentum

$$= \mu\sqrt{2gH}$$

$\therefore$ Reading of scale = $\mu\sqrt{2gH} + \mu gt$

10. Gun exters force on bullets, causing their momentum to change. Bullets exert equal force on the gun in opposite direction. The army person has to apply same force on the gun to hold it.

Force = (number of bullets fired per second) × (change in momentum of each bullet)

$$= \frac{20}{4} \times \frac{50}{1000} \times 1000 = 250\,\text{N}$$

12. $F_{\text{net}} = mg\sin 30° = 10\,\text{N}$

$\therefore$ $J = F_{\text{net}} \cdot t = 10 \times 2 = 20$ Ns

13. $J = \Delta P = P_f - P_i = mv - mu = 1000(10 - 20)$

$$= -10000 \text{ Ns}$$

Negative sign indicates that impulse is opposite to the direction of motion.

14. Let the speed of the block immediately after the hit be v.

Friction on the block, when it is moving is $f = \mu mg$,

$\therefore$ Retardation is $a = \frac{f}{m} = \mu g = 2\,\text{ms}^{-2}$

Using $v^2 = u^2 + 2as$

$$0 = v^2 - 2 \times 2 \times 1 \quad \Rightarrow \quad v = 2\,\text{ms}^{-1}$$

It means that the stick imparts a velocity of $2\,\text{ms}^{-1}$ to the block.

$\therefore$ $J = mv = 10 \times 2 = 20$ Ns

[During the interaction time of the stick and the block, the impulse of friction is negligible.]

15. Change in momentum of the chain

$$\Delta P = P_f - P_i = 0 - 0 = 0$$

Final state refers to the position when the complete chain falls to the table.

Time of fall $t = \sqrt{\frac{2L}{g}}$

Impulse of gravitational force on the chain is

$$Jg = Mgt = M\sqrt{2gL}\ (\downarrow)$$

If J_t = impulse applied by table on the chain, then

$$J_t + J_g = 0 \quad \Rightarrow \quad J_t = -M\sqrt{2gL}\ (\downarrow)$$
$$= M\sqrt{2gL}\ (\uparrow)$$

Same impulse is applied by the chain on the table in downward direction.

16. Let the string apply an impulse J to both the balls so that both of them acquire same velocity after the string is taut. Let the final velocity of the two balls be v.

For ball of mass m:

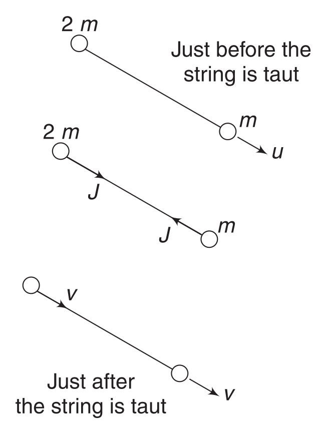
$$mv(\rightarrow) - mu(\rightarrow) = J(\leftarrow)$$

$\Rightarrow$ $mv - mu = -J$ ...(i)

For ball of mass $2m$:

$2mv\ (\rightarrow) = J(\rightarrow) \quad \Rightarrow \quad 2mv = J$ ...(ii)

From (i) and (ii): $J = \frac{2mu}{3}$



17. Impulse, $\vec{J} = \int_0^2 \vec{F}\,dt = \left[2\int_0^2 t\,dt\right]\hat{j} = 4\hat{j}$ Ns

$$\vec{P}_f - \vec{P}_i = \vec{J} \quad \Rightarrow \quad \vec{P}_f = \vec{P}_i + \vec{J}$$

$$\Rightarrow \quad 1(\vec{v}_f) = 1(2\hat{i} + 3\hat{j}) + 4\hat{j}$$

$$\Rightarrow \quad \vec{v}_f = (2\hat{i} + 7\hat{j})\,\text{ms}^{-1}$$

18. $\vec{P}_f = \vec{P}_i + \vec{J} = m\vec{u} + (m\vec{g})t$

19. Impulse J = Area under $F - t$ graph $= \frac{1}{2} \times 8 \times 20$

$$= 80 \text{ Ns}$$

$$P_i = mu = 4\ \text{kg ms}^{-1}$$

$$P_f = P_i + J = 84\ \text{kg ms}^{-1}$$

$$K_i = \frac{P_i^2}{2m} = \frac{4^2}{2 \times 1} = 8\,\text{J}$$

$$K_f = \frac{P_f^2}{2m} = \frac{84^2}{2 \times 1} = 3528\,\text{J}$$

$$W = K_f - K_i \quad \text{(WE theorem)}$$

$$= 3520\ \text{J}$$

20. (i) Let the gun recoil with velocity v. Momentum of gun + bullet system is conserved and remains zero.

$$\therefore \quad Mv = mu \quad \Rightarrow \quad v = \frac{mu}{M}$$

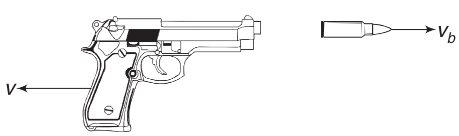
(ii) Energy released = KE of gun + KE of bullet + Heat + light + sound

$$KE \text{ of gun + bullet} = \frac{1}{2}M\left(\frac{mu}{M}\right)^2 + \frac{1}{2}mu^2$$

$$= \frac{1}{2}mu^2\left[1 + \frac{m}{M}\right]$$

$$\therefore \quad \text{Energy released} > \frac{1}{2}mu^2\left[1 + \frac{m}{M}\right]$$

21. Let velocity of bullet be $v_b(\rightarrow)$ and that of the gun be $v(\leftarrow)$.



Given $u(\rightarrow) = v_b(\rightarrow) - v(\leftarrow)$

$$\Rightarrow \quad u = v_b + v \quad \Rightarrow \quad v_b = (u - v)$$

Conservation of momentum gives

$$m(u - v) = Mv \quad \Rightarrow \quad v = \frac{mu}{M + m}$$

22. Velocity of shell wrt the gun is

$$\vec{u} = u\cos 60°\hat{i} + u\sin 60°\hat{j} = \frac{u}{2}\hat{i} + \frac{\sqrt{3}}{2}u\hat{j}$$

Let recoil velocity of the canon be $v_0(-\hat{i})$. Velocity of shell relative to the ground is

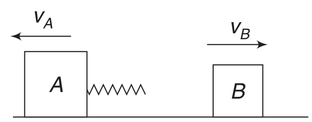
$$\vec{v} = \vec{u} + v_0(-\hat{i}) = \left(\frac{u}{2} - v_0\right)\hat{i} + \frac{\sqrt{3}}{2}u\hat{j}$$

Applying conservation of momentum in x-direction, we get $m\left(\frac{u}{2} - v_0\right) = Mv_0$

$$\Rightarrow \quad v_0 = \frac{m\dfrac{u}{2}}{M + m} = \frac{mu}{2(M + m)}$$

Note that x component of velocity of the bullet (wrt ground) is less than $u\cos 60°$. Relative to the ground, the bullet comes out of the barrel at an angle higher than 60° with horizontal.

23. Let the velocity of two blocks be v_A $(\leftarrow)$ and v_B $(\rightarrow)$ after the spring opens completely.



Momentum conservation gives:

$$2mv_A = mv_B$$

$$\Rightarrow \quad 2v_A = v_B \qquad \text{...(i)}$$

Energy is also conserved.

$$\frac{1}{2}(2m)v_A^2 + \frac{1}{2}mv_B^2 = \frac{1}{2}kx^2$$

$$\Rightarrow \quad \frac{1}{2}(2m)v_A^2 + \frac{1}{2}m(2v_A)^2 = \frac{1}{2}kx^2$$

$$\Rightarrow \quad 6mv_A^2 = kx^2 \quad \Rightarrow \quad v_A = \sqrt{\frac{k}{6m}}\cdot x$$

And $\quad v_B = 2\sqrt{\dfrac{k}{6m}}\cdot x$

24. Total momentum after explosion = 0

$$\Rightarrow \quad m(2\hat{i} + 5\hat{j} - 6\hat{k}) + m(-4\hat{i} + 3\hat{j} + 2\hat{k}) + 2m\vec{v} = 0$$

$$\Rightarrow \quad \vec{v} = (-\hat{i} - 4\hat{j} + 2\hat{k})\,\text{ms}^{-1}$$

$$\therefore \quad v = \sqrt{1^2 + 4^2 + 2^2} = \sqrt{21}\ \text{ms}^{-1}$$

25. Sum of momentum of the three particles is zero.

26. Let velocity of the man relative to the ground be v_m and the velocity of the car relative to the ground be v_c.

$$v(\leftarrow) = v_m(\leftarrow) - v_c(\rightarrow)$$

$$\Rightarrow \quad v = v_m + v_c \quad \Rightarrow \quad v_m = v - v_c(\leftarrow)$$

Momentum conservation gives:

$$Mv_c - mv_m = (M + m)u$$

$$\Rightarrow \quad Mv_c - m(v - v_c) = (M + m)u.$$

$$\Rightarrow \quad v_c = \frac{(M + m)u + mv}{M + m} = u + \frac{mv}{M + m}$$

$\therefore$ Change in velocity of the car is $\dfrac{mv}{M+m}$

27. $mv_m = (M+m)u \Rightarrow v_m = \left(\dfrac{M+m}{m}\right)u$

28. Refer to Example 15.

There is no effect along the line of motion.

29. $(M+m)v = Mu$ [Conserving momentum in horizontal direction]

$$\Rightarrow \quad v = \frac{Mu}{M+m}$$

Change in velocity $\Delta v = v - u = \dfrac{Mu}{M+m} - u$

$$= -\frac{mu}{M+m}$$

30. (i) Let final velocity of the two objects, when slipping stops, be v. Momentum of the system is conserved

$$\therefore (M+m)v = mu \Rightarrow v = \frac{mu}{M+m}$$

(ii) $KE_i = \dfrac{1}{2}mu^2$; $KE_f = \dfrac{1}{2}(M+m)\left(\dfrac{mu}{M+m}\right)^2$

$$= \frac{1}{2}\frac{m^2u^2}{M+m}$$

Loss in $KE = k_i - k_f = \dfrac{1}{2}mu^2\left[1 - \dfrac{m}{M+m}\right]$

$$= \frac{1}{2}\left(\frac{mM}{M+m}\right)u^2$$

$\therefore$ Work done by friction is

$$W_f = -\frac{1}{2}\left(\frac{mM}{M+m}\right)u^2$$

31. When the smaller block is at P, the horizontal component of velocity of m is same as horizontal velocity of M. Using conservation of momentum in horizontal direction, we get

$$(M+m)v = mu \Rightarrow v = \frac{mu}{M+m}$$

32. (i) In head on elastic collision of two bodies, velocities get exchanged.

$\therefore \quad v_A = 0; \quad v_B = u$

$\therefore \quad K_A = 0; \quad K_B = \dfrac{1}{2}mu^2$

(ii) If collision is perfectly inelastic, final velocity of both block is same (v)

$$2mv = mu; \quad \Rightarrow v = \frac{u}{2}$$

$\therefore \quad K_i = \dfrac{1}{2}mu^2$; $K_f = \dfrac{1}{2}(2m)\left(\dfrac{u}{2}\right)^2 = \dfrac{1}{4}mu^2$

$\therefore$ Loss in $KE = K_i - K_f = \dfrac{1}{4}mu^2$

33. The ball moves towards B and the cart remains stationary. Ball hits the cart (wall at B) and exchange of velocity takes place. The ball comes to rest and the cart begins to move towards right with velocity u. Now the wall at A comes from behind and hits the ball.

Time = time for ball to travel from A to B (Cart not moving) +

Time for cart to travel a distance L (ball not moving)

$$= \frac{L}{u} + \frac{L}{u} = \frac{2L}{u}$$

34. Let v_1 and v_2 be final velocities after collision.

$$mv_1 + mv_2 = mv$$

$$\Rightarrow \quad v_1 + v_2 = v \quad \text{...(i)}$$

And $\quad -\left(\dfrac{v_1 - v_2}{v - 0}\right) = e$

$$\Rightarrow \quad v_2 - v_1 = ev \quad \text{...(ii)}$$

Solving (i) and (ii) $v_2 = \dfrac{(1+e)v}{2}$ and

$$v_1 = \frac{(1-e)v}{2}$$

Given $\dfrac{1}{2}mv_1^2 + \dfrac{1}{2}mv_2^2 = \dfrac{3}{4}\cdot\dfrac{1}{2}mv^2$

$$\frac{(1-e)^2}{4}v^2 + \frac{(1+e)^2v^2}{4} = \frac{3}{4}v^2$$

$$\Rightarrow \quad (1-e)^2 + (1+e)^2 = 3$$

$$\Rightarrow \quad 2 + 2e^2 = 3 \Rightarrow e^2 = \frac{1}{2}$$

$$\Rightarrow \quad e = \frac{1}{\sqrt{2}}$$

35. (i) Relative speed of approach = 2 + 1 = 3 ms^{-1}. Let velocity of ball be $v(\leftarrow)$ after collision. Velocity of heavy wall does not change.

Relative speed of separation = Relative speed of approach

$$v - 1 = 3 \Rightarrow v = 4\,ms^{-1}$$

(ii) (Relative speed of separation) = 0.5 (Relative speed of approach)

$$\Rightarrow \quad v - 1 = 0.5 \times 3$$

$$\Rightarrow \quad v = 2.5\,ms^{-1}$$

36. Using equation (13), velocity of first ball after collision is (Note: $u_2 = 0$)

$$v_1 = \left(\frac{m - eM}{m+M}\right)u$$

For v_1 to be negative, $m < eM$

$$\Rightarrow \quad \frac{m}{e} < M$$

$$\Rightarrow \quad \frac{5m}{4} < M$$

37. Let initial velocity of m be u. Using equation (13), velocities of the two balls after collision are

$$v_1 = \left(\frac{m - e(2m)}{m + 2m}\right)u = \frac{u}{6}$$

$$v_2 = \left(\frac{m(1 + e)}{m + 2m}\right)u = \frac{5u}{12}$$

Time needed by second ball to hit the wall is

$$t_1 = \frac{d}{v_2} = \frac{12d}{5u}$$

In this time, the first ball gets closer to the wall by

$$x_1 = v_1 t_1 = \frac{2d}{5}$$

Now, after colliding with the wall velocity of second ball becomes $v_2' = 0.25\,v_2 = \dfrac{5u}{48}$

Balls approach with relative speed

$$= v_1 + v_2' = \frac{u}{6} + \frac{5u}{48}$$

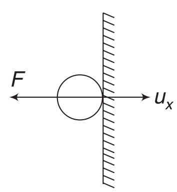
Time needed to cover distance x_1 is

$$t_2 = \frac{\frac{2d}{5}}{\frac{u}{6} + \frac{5u}{48}} = \frac{2.22d}{u}$$

$\therefore$ Distance from the wall where the collision occurs is

$$= v_2' \cdot t_2 = \frac{5u}{48} \cdot \frac{2.22d}{u} = 0.23\,d$$

38. The ball is at the top point of its trajectory while it hits the wall. It is travelling horizontally and the collision is head on. There is no impulse on the ball in vertical direction during impact. Its vertical motion has no change. Therefore, time of flight from A to the wall (top of the trajectory = time of flight from the wall to the ground).



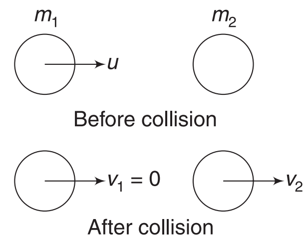
Velocity after impact $= e u_x = 0.5\,u_x$

$\therefore$ Horizontal ground covered after impact $= \dfrac{x}{2}$

39. For collision between m_1 and m_2

$$u_1 = u; \qquad u_2 = 0$$

and $\quad v_1 = 0$



From conservation of momentum:

$$m_2 v_2 = m_1 u$$

$$\Rightarrow \quad v_2 = \left(\frac{m_1}{m_2}\right)u$$

$$\therefore \quad e = -\left(\frac{v_1 - v_2}{u_1 - u_2}\right) = -\left(\frac{0 - v_2}{u - 0}\right)$$

$$\Rightarrow \quad eu = v_2 \quad \Rightarrow \quad eu = \left(\frac{m_1}{m_2}\right)u$$

$$\Rightarrow \quad e = \frac{m_1}{m_2} \qquad \text{...(i)}$$

Similarly for collision between m_2 and m_3

$$e = \frac{m_2}{m_3} \qquad \text{...(ii)}$$

From (i) and (ii), $\dfrac{m_1}{m_2} = \dfrac{m_2}{m_3}$; Hence $m_2 = \sqrt{m_1 m_3}$

40. For B to rise upto a height of $h = 3.2$ m, the minimum speed of B must be

$$v_B = \sqrt{2gh} = \sqrt{2 \times 10 \times 3.2} = 8\,\text{ms}^{-1}$$

Using equation (13) with $u_2 = 0$

$$v_B = \frac{(1 + e)m_1 u_1}{m_1 + m_2} \quad \Rightarrow \quad 8 = \frac{(1 + e)m \times 10}{2m}$$

$$\Rightarrow \quad e = \frac{3}{5}$$

41. The line of impact (n-line) is z-axis. Velocity component parallel to the wall does not change.

$$\therefore \quad v_x = 2\,\text{ms}^{-1}$$

The z-component of velocity changes direction and its magnitude is $|v_z| = e\,|u_z|$

(i) $|v_z| = 3$ $\qquad [\because e = 1]$

$\therefore \; \vec{v} = 2\hat{i} - 3\hat{k}$

(ii) $|v_z| = 0$ $\qquad [\because e = 0]$

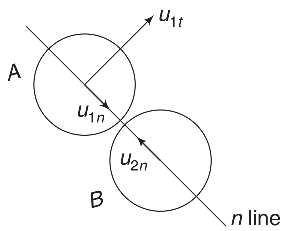
$\therefore \; \vec{v} = 2\hat{i}$

(iii) $|v_z| = 0.5 \times 3 = 1.5\,\text{ms}^{-1}$ $\qquad [\because e = 0.5]$

$\therefore \; \vec{v} = 2\hat{i} - 1.5\hat{k}$

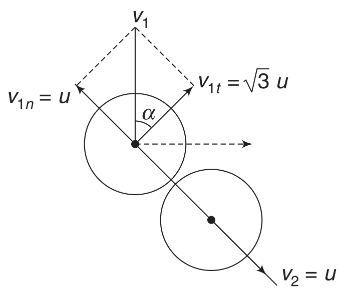
42. $u_{1n} = 2u\sin 60° = u$; $u_{1t} = 2u\sin 60° = \sqrt{3}u$

$u_{2n} = u$



Along the n-line, we have a head-on collision of two equal masses. Velocities get exchanged.

$\therefore$ Velocity of B after impact is u along n-line, as shown.



Component of velocity of A are $v_{1t} = u_{1t} = \sqrt{3}u$

$$v_{1n} = u \text{ (in direction shown)}$$

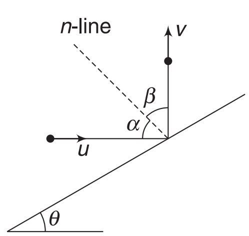
$\therefore$ Velocity of A is

$$v_1 = \sqrt{u^2 + (\sqrt{3}\,u)^2} = 2u$$

$$\tan\alpha = \frac{u}{\sqrt{3}\,u} = \frac{1}{\sqrt{3}} \Rightarrow \alpha = 30°$$

$\therefore$ Direction of v_1 makes 90° with original direction of motion

43. In elastic collision, angle of incidence (α) = angle of reflection (β)



Since $\alpha + \beta = 90°$

$\therefore \quad \alpha = \beta = 45°$

$\therefore \quad \theta = 45°$

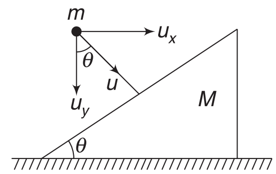
44. The ball will move along the same line (n-line). Let its velocity be v after impact, as shown. Momentum conservation along horizontal direction gives:

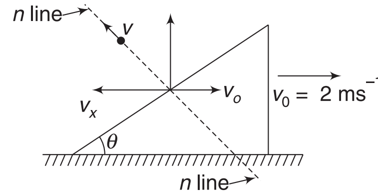
$$mu_x = Mv_0 - mv_x$$

$\Rightarrow \quad 1(10\sin 37°) = 4 \times 2 - (1)(v_x)$

$\Rightarrow \quad v_x = 2\,\text{ms}^{-1}$

But $\quad v\sin\theta = v_x$





$\Rightarrow \quad v \times \frac{3}{5} = 2 \quad = \quad v = \frac{10}{3}\text{ms}^{-1}$

Component of v_0 along n line is

$$v_0\sin\theta = 2 \times \frac{3}{5} = 1.2\,\text{ms}^{-1}$$

$\therefore \quad e = -\left(\frac{1.2 - \left(-\frac{10}{3}\right)}{0 - 10}\right) = 0.453$

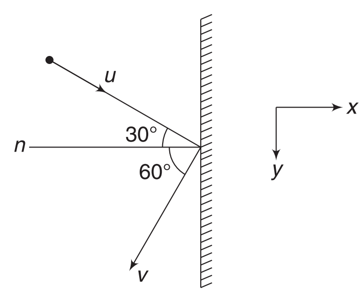
45. Y component of velocity does not change.

$\therefore \quad v_y = u\sin 30°$

And $\quad |v_x| = e|u_x| = eu\cos 30°$

$\therefore \quad \frac{v_y}{v_x} = \frac{1}{e}\tan 30°$

$\Rightarrow \quad \tan 60° = \frac{1}{e}\tan 30° \quad \Rightarrow \quad e = \frac{1}{3}$



46. Horizontal velocity of the projectile remains unchanged throughout. The vertical component of velocity gets reduced at B.

$u_y = |\text{vertical component of velocity at } A|$

$\quad = |\text{vertical component at } B \text{ just before collision}|$

After collision at B; $v_y = eu_y$

Range: $L_1 = \frac{2u_x u_y}{g}$ and $L_2 = \frac{2u_x v_y}{g} = \frac{2eu_x u_y}{g}$

$\therefore \quad \frac{L_1}{L_2} = \frac{1}{e}$

47. Let velocity of the second piece be v, making an angle θ with x-direction as shown.

$$P_x = mu$$

$$\Rightarrow m\frac{u}{2} + mv\cos\theta = mu$$

$$\Rightarrow \quad v\cos\theta = \frac{u}{2} \quad \text{...(i)}$$

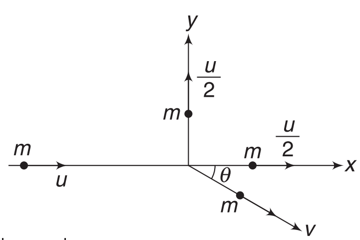
And $\quad P_y = 0$

$$\Rightarrow \quad mv\sin\theta = m\frac{u}{2} \Rightarrow v\sin\theta = \frac{u}{2} \quad \text{...(ii)}$$

Squaring (i) and (ii) and adding them

$$v^2 = 2\left(\frac{u}{2}\right)^2 \Rightarrow v = \frac{u}{\sqrt{2}}$$

Dividing (ii) by (i) gives $\tan\theta = 1 \Rightarrow \theta = 45°$



48. $F_{\text{th}} = \left|u\dfrac{dm}{dt}\right|$

Rocket just lifts if $F_{\text{th}} = mg$

$$\Rightarrow \left|u\frac{dm}{dt}\right| = mg$$

$$\Rightarrow \left|\frac{dm}{dt}\right| = \frac{500 \times 10}{2000} = 2.5\,\text{kgs}^{-1}$$

49. $$m\cdot\frac{dv}{dt} = F_{\text{th}} - mg$$

$$\Rightarrow 500 \times 5 = F_{\text{th}} - 500 \times 10$$

$$\Rightarrow F_{\text{th}} = 7500\,\text{N}$$

$$\therefore \left|u\frac{dm}{dt}\right| = 7500 \Rightarrow \left|\frac{dm}{dt}\right| = \frac{7500}{2000} = 3.75\ \text{kgs}^{-1}$$

50. Velocity of falling sand in horizontal direction = u (due to inertia).

$\therefore$ Relative velocity of the mass getting separated from the cart = 0.

$\therefore$ Thrust force on the cart = 0

$\therefore$ Acceleration = 0

51. Refer to Example 31

$$m\frac{dv}{dt} = -u\frac{dm}{dt}$$

$$\Rightarrow dv = -u\frac{dm}{m}$$

$$\Rightarrow \int_0^v dv = -u\int_m^{m/2}\frac{dm}{m}$$

$$\Rightarrow v = -u\,ln\frac{1}{2} = u\,ln2 = 2ln2$$

Worksheet 1

1. Law of conservation of momentum
2. Read the question carefully. It says "sum of magnitudes of momenta ..."

When $\vec{F}_{\text{ext}} = 0;\ \vec{P}_1 + \vec{P}_2 + \vec{P}_3 + \ldots = 0$

But $|\vec{P}_1| + |\vec{P}_2| + |\vec{P}_3|$ ... will be a positive number if particles are moving. This sum may change also.

3. $K = \dfrac{1}{2}mv^2 = \dfrac{P^2}{2m}$

If KE is made 4 times, P becomes twice. It means P increase by 100 %

4. $$K = \frac{P^2}{2m}$$

$$\Rightarrow \frac{dK}{dP} = \frac{P}{m}$$

For small changes $\Delta K \simeq \dfrac{P}{m}\Delta P$

$$\therefore \frac{\Delta K}{K} = \frac{\frac{P\Delta P}{m}}{\frac{P^2}{2m}} = \frac{2\Delta P}{P}$$

$$\therefore \frac{\Delta P}{P} \times 100 = \frac{1}{2}\left(\frac{\Delta K}{K} \times 100\right) = \frac{1}{2} \times (0.2\%)$$

$$= 0.1\%$$

5. $$K = \frac{P^2}{2m} \Rightarrow P = \sqrt{2mK}$$

$$\therefore \frac{P_1}{P_2} = \frac{\sqrt{2 \times 1 \times K}}{\sqrt{2 \times 4 \times K}} = \frac{1}{2}$$

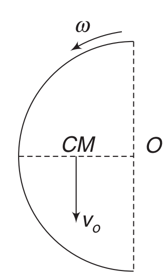
6. Angular speed $\omega = \dfrac{v}{R}$.

Distance of COM from O is $\dfrac{2R}{\pi}$

Linear momentum of train

$$P = Mv_0 = M\left(\frac{2R}{\pi}\omega\right)$$

$$= M\left(\frac{2R}{\pi}\cdot\frac{v}{R}\right) = \frac{2Mv}{\pi}$$



7. Speed just before hitting the floor,

$$u = \sqrt{2gh} = \sqrt{2 \times 10 \times 10}$$

$$= 10\sqrt{2}\ \text{ms}^{-1}$$

On collision, it loses $\dfrac{3}{4}$th of its KE. It means it has only $\dfrac{1}{4}$th of its original KE. This is possible if speed halves.

$\therefore \quad v = \dfrac{u}{2} = 5\sqrt{2}\ \text{ms}^{-1}$

Impulse = change in momentum

$= mv(\uparrow) - mu(\downarrow)$

$= (mv + mu)\uparrow$

$= \dfrac{50}{1000}[5\sqrt{2} + 10\sqrt{2}] = \dfrac{3\sqrt{2}}{4} = 1.06$ Ns

8. $\Delta K = \dfrac{1}{2}m\,(v_2^2 - v_1^2) = \dfrac{1}{2}m\,(\vec{v}_2 - \vec{v}_1)(\vec{v}_2 + \vec{v}_1)$

$= \dfrac{1}{2}\,\Delta\vec{P}\cdot(\vec{v}_2 + \vec{v}_1) = \dfrac{1}{2}I\cdot(\vec{v}_1 + \vec{v}_2)$

9. After a collision, the ball travels a distance $2d$ for next collision with the same wall. Hence, time intervals between two successive collision with a wall is

$$T = \frac{2d}{v_0}$$

Frequency of collision with one wall, $n = \dfrac{1}{T} = \dfrac{v_0}{2d}$

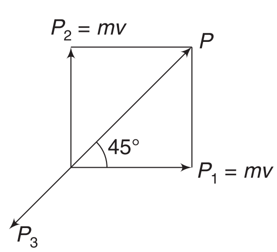
$\therefore$ Force = (Change in momentum in one collision) × (frequency of collision)

$$= 2mv_0\,\frac{v_0}{2d} = \frac{mv_0^2}{d}$$

10. Sum of momentum of particles moving in perpendicular direction is

$$P = \sqrt{(Mv)^2 + (mv)^2} = \sqrt{2}\ mv$$

Momentum of the third particle is exactly equal and opposite to P so as to keep the total momentum zero.



Minimum energy released = sum of KE of the three particles

$$= \frac{1}{2}mv^2 + \frac{1}{2}mv^2 + \frac{P^2}{2(2m)}$$

$$= mv^2 + \frac{2m^2v^2}{4m} = \frac{3}{2}mv^2$$

11. Momentum conservation along horizontal direction:

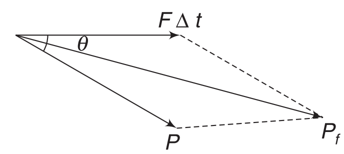
$(500 + 50)v = 500 \times 10 \Rightarrow v = \dfrac{100}{11}\ \text{ms}^{-1}$

12. Speed at top point, before explosion $= v\cos\theta$. Momentum conservation gives:

$$mv\cos\theta = 0 + \frac{m}{2}v_1$$

$\Rightarrow \quad v_1 = 2v\cos\theta$

13. Impulse $(F\Delta t)$ = change in momentum. When impulse is added vectorially to initial momentum, we get the final momentum.



Sum of two vectors will be larger if angle between them is smaller.

$\therefore \quad P_2 > P_1$

14. $\quad Mv = 0 + (M - m)v_1$

$\Rightarrow \quad v_1 = \dfrac{Mv}{M - m}$

15. Speed of the (block + bullet) combined system after the hit is given by momentum conservation.

$2mv = mu \quad \Rightarrow \quad v = \dfrac{u}{2}$

$\therefore \quad H = \dfrac{v^2}{2g} = \dfrac{u^2}{8g}$

16. Let the fragments have mass m each. Momentum of each of the three fragments has magnitude.

$$P = \sqrt{2mE_o}$$

If they move along x-, y- and z-direction then

$P\hat{i} + P\hat{j} + P\hat{k} + \vec{P}_4 = 0 \quad \Rightarrow \vec{P}_4 = -P(\hat{i} + \hat{j} + \hat{k})$

$\therefore \quad P_4 = \sqrt{3}P = \sqrt{6mE_o}$

$KE, \quad K_4 = \dfrac{P_4^2}{2m} = 3E_0$

$\therefore$ Sum of KE of all fragments $= 6E_0$

17. Momentum conservation:

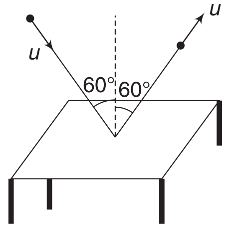
$$(M + Nm)v_0 = Nmv$$

$\Rightarrow \quad v_0 = \dfrac{Nmv}{M + Nm}$

18. Change in momentum of each ball is

$$\Delta P = 2mu\cos 60°(\uparrow)$$

$$= 2 \times \frac{20}{1000} \times 5 \times \frac{1}{2} = \frac{1}{10}\ \text{Ns}$$



Force on table $F = n \cdot \Delta P$

$$= 20 \times \frac{1}{10} = 2\,\text{N}$$

where n = number of balls hitting per second

If N = normal reactions of each leg

$$4N = F + W \quad \Rightarrow \quad 4N = 2 + 0.2 \times 10$$

$$\Rightarrow \qquad N = 1\,\text{N}$$

19. When momentum of a system is zero, the particles in it may be moving. The system can have KE. Your ceiling fan has no momentum but it has KE.

When $KE = 0$, it means all the particles in the system are at rest.

20. Final common velocity (v_1) is given by

$$6mv_1 = m(5v) + 5m(v)$$

$$\Rightarrow \qquad v_1 = \frac{5v}{3}$$

Impulse applied by m on $5m$ = change in momentum of the ball of mass $5m$

$$\Rightarrow \qquad J = 5m(v_1 - v) = 5m\frac{2v}{3} = \frac{10\,mv}{3}$$

21. Refer to Example 25

22. $$\vec{P} = A\hat{i}\cos kt - A\hat{j}\sin kt$$

$$\vec{F} = \frac{d\vec{p}}{dt} = -Ak\hat{i}\sin kt - Ak\hat{j}\cos kt$$

Now $\vec{F}\cdot\vec{P} = -A^2k\sin kt\cdot\cos kt + A^2k\sin kt\cdot\cos kt = 0$

23. Velocity of the heavy object will not change much. Relative speed of approach = $10 + 2 = 12$ ms^{-1}

Relative speed of separation = $12\,\text{ms}^{-1}$ $\quad [\because e = 1]$

$$\therefore \qquad v - 2 = 12$$

$$\Rightarrow \qquad v = 14\,\text{ms}^{-1}$$

24. Velocity before first collision $u = \sqrt{2gh}$

Velocity after first collision, $v_1 = eu(\uparrow)$

The ball will return back for second collision while moving with velocity v_1 $(\downarrow)$

Velocity just after second impact, $v_2 = ev_1 = e^2u$

Similarly, Velocity after third impact, $v_3 = e^3u$

$\therefore$ Height gained after third impact,

$$h_3 = \frac{v_3^2}{2g} = \frac{e^6\,u^2}{2g}$$

$$\Rightarrow \qquad h_3 = \frac{e^6\cdot 2gh}{2g} = e^6\cdot h$$

25. Let v_0 = speed of (bag + sphere) combined system after impact.

Energy conservation gives:

$$\frac{1}{2}(M+m)v_0^2 = (M+m)gh \quad \Rightarrow v_0 = \sqrt{2gh}$$

Momentum conservation for the collision event gives:

$$mv = (M+m)v_0 \quad \Rightarrow \quad v = \left(\frac{M+m}{m}\right)\sqrt{2gh}$$

26. Speed of M remains practically unchanged. Speed of separation = speed of approach

$$\Rightarrow \qquad 4 - v_1 = 6 - 4$$

$$\Rightarrow \qquad v_1 = 2\,\text{ms}^{-1}$$

27. KE before impact $= \frac{1}{2}\,mu^2$

KE after impact $= 0.8 \times \frac{1}{2}mu^2$

$$\Rightarrow \qquad \frac{1}{2}\,mv^2 = 0.8 \times \frac{1}{2}mu^2$$

$$\Rightarrow \qquad v = \sqrt{0.8}\cdot u = 0.89\,u$$

$$\therefore \qquad e = 0.89$$

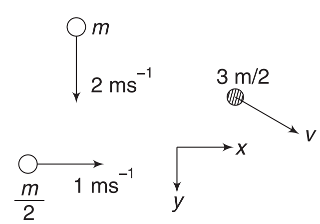
28. In head-on elastic collision of two equal masses, there is exchange of velocity. If M_2 is at rest, it will get entire KE of M_1 and M_1 will come to rest.

29. Momentum conservation:

$$\frac{3m}{2}\,\vec{v} = 2m\,\hat{j} + \frac{m}{2}\hat{i}$$

$$\Rightarrow \qquad v = \frac{4}{3}\hat{j} + \frac{1}{3}\hat{i}$$

$$\Rightarrow \qquad v = \sqrt{\frac{16}{9} + \frac{1}{9}} = \frac{\sqrt{17}}{3}\ \text{ms}^{-1}$$



30. Speed before collision is $u = \sqrt{2gh_1}$

Speed after collision is $v = \sqrt{2gh_2}$

$$\Delta P = mv - (-mu) = mv + mu = m\left(\sqrt{2gh_2} + \sqrt{2gh_1}\right)$$

31. Force applied by falling particles on the scale is

F = (number of particles hitting per second) (change in momentum of each particles)

$$= 100 \times 2mu = 200 \times \frac{1}{1000}u$$

$$= \frac{1}{5}\sqrt{2gh} = \frac{1}{5} \times \sqrt{2 \times 10 \times 2} = 1.265\,\text{N}$$

$\therefore$ Reading of the scale $= \frac{1.265}{10}\text{kg} = 0.1265\,\text{kg}$

$= 126.5\,\text{g}$

32. $m_1\vec{v}_1 + m_2\vec{v}_2 + m_3\vec{v}_3 = m_1\vec{u}_1 + m_2\vec{u}_2 + m_3\vec{u}_3$

$$\Rightarrow v_1 = \vec{u}_1 + \frac{m_2}{m_1}\vec{u}_2 + \frac{m_3}{m_1}\vec{u}_3 - \frac{m_2}{m_1}\vec{v}_2 - \frac{m_3}{m_1}\vec{v}_3$$

$$= 20\hat{i} + 2(20\hat{j}) + 4(20\hat{k}) - 2(10\hat{j} + 5\hat{k}) - 0$$

$$= 20\hat{i} + 20\hat{j} + 70\hat{k}$$

33. $r_B = 2r_A$

$\therefore$ Volume $V_B = 8V_A$

$\therefore$ Mass $m_B = 8m_A$

If $m_A = m$ then $m_B = 8m$

Let velocity of A be v_1 and that of B be v_2 after collision

$$mu = mv_1 + 8mv_2 \quad \Rightarrow \quad u = v_1 + 8v_2 \qquad \text{...(i)}$$

$$\text{And } -\left(\frac{v_1 - v_2}{u - 0}\right) = 0.5 \quad \Rightarrow v_2 - v_1 = 0.5u \qquad \text{...(ii)}$$

From (i) and (ii)

$$v_1 + 8v_2 = 2v_2 - 2v_1 \quad \Rightarrow \quad 3v_1 = -6v_2$$

$$\Rightarrow \quad \frac{v_1}{v_2} = -2$$

Negative sign tells us that v_1 and v_2 are oppositely directed.

$$\left|\frac{v_1}{v_2}\right| = 2$$

34. Impulse $= \int F dt$ = area under the graph.

35. At maximum compression, both the blocks have same velocity (Say v).

Momentum conservation gives:

$$(m + M)v = mv_1 + Mv_2$$

$$\Rightarrow \quad v = \frac{mv_1 + Mv_2}{m + M}$$

Spring PE = loss in KE

$$\frac{1}{2}kx^2 = \frac{1}{2}mv_1^2 + \frac{1}{2}Mv_2^2 - \frac{1}{2}(m + M)\left(\frac{mv_1 + mv_2}{m + M}\right)^2$$

Simplifying gives: $x = (v_1 - v_2)\sqrt{\frac{mM}{(M + m)k}}$

Alternate:

$$K = \frac{1}{2}(M + m)v_{CM}^2 + \frac{1}{2}\mu v_{rel}^2$$

During maximum compression, kinetic energy is simply $\frac{1}{2}(M + m)v_{CM}^2$.

$$\therefore \quad \frac{1}{2}kx^2 = \frac{1}{2}\mu v_{rel}^2$$

$$\Rightarrow \quad kx^2 = \frac{mM}{m + M}\cdot(v_1 - v_2)^2$$

$$\Rightarrow \quad x = \sqrt{\frac{mM}{k(m + M)}}\,(v_1 - v_2)$$

36. Speed of m_1 before collision is $u = \sqrt{2gh}$

Speed of combined mass after collision is

$$v = \sqrt{2g\frac{h}{4}} = \sqrt{\frac{gh}{2}}$$

Momentum conservation for collision:

$$m_1 u = (m_1 + m_2)v$$

$$\Rightarrow \quad m_1\sqrt{2gh} = (m_1 + m_2)\sqrt{\frac{gh}{2}}$$

$$\Rightarrow \quad 2m_1 = m_1 + m_2 \quad \Rightarrow \quad m_1 = m_2$$

37. $mv = (100\,\text{m})v_0 \quad \Rightarrow \quad v_0 = \frac{v}{100}$

39. Time required by the body of mass m to fall through $\frac{h}{2}$ is

$$t = \sqrt{\frac{2\left(\frac{h}{2}\right)}{g}} = \sqrt{\frac{h}{g}}$$

The second body of mass $2m$ is projected with a velocity so as to reach height $\frac{h}{2}$ in time t

$$\therefore \quad x = ut + \frac{1}{2}at^2 \text{ gives}$$

$$\frac{h}{2} = u\sqrt{\frac{h}{g}} - \frac{1}{2}g\left(\frac{h}{g}\right) \quad \Rightarrow u = \sqrt{gh}$$

Speed of this body when it is at height $\frac{h}{2}$ is given by

$$u_2^2 = u^2 - 2g\frac{h}{2} = gh - gh = 0$$

$$\Rightarrow \quad u_2 = 0$$

Speed of first body at the instant of collision is

$$u_1 = \sqrt{2g\frac{h}{2}} = \sqrt{gh}$$

Velocity of combined mass after collision is given as:

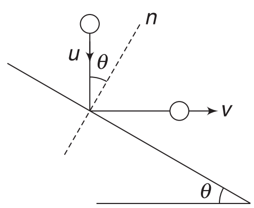
$$3mv = mu_1 \quad \Rightarrow \quad v = \frac{1}{3}\sqrt{gh}$$

Now this mass falls through a height $\frac{h}{2}$ with acceleration g. Its speed before hitting the ground is

$$v_0^2 = v^2 + 2g\frac{h}{2} = \frac{gh}{9} + gh = \frac{10\,gh}{9}$$

$$\therefore \qquad v_0 = \frac{\sqrt{10gh}}{3}$$

40. Velocity of approach = component of u along n-line $= u\cos\theta$



Velocity of separation = component of v along n-line $= v\sin\theta$

$$\therefore \qquad \frac{v\sin\theta}{u\cos\theta} = e \qquad \text{...(i)}$$

Now, the velocity component along the incline does not change.

$$\therefore \qquad u\sin\theta = v\cos\theta \quad \Rightarrow \quad \frac{v}{u} = \tan\theta$$

$$\therefore \qquad e = \tan^2\theta$$

41. In elastic collision, speed will not change. Taking help from the figure in last solution,

$$\Delta P = 2mu\cos\theta \text{ [along } n\text{-line]}$$

$$= 2m\sqrt{2gh}\cdot\cos\theta$$

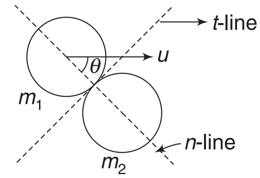
42. Let velocity of ball 2 be v_2 after collision

$$2mv = 2m\frac{v}{3} + mv_2 \quad \Rightarrow v_2 = \frac{4}{3}v$$

Now $\qquad e = -\left(\frac{v_1 - v_2}{u_1 - u_2}\right) = -\left(\frac{\frac{v}{3} - \frac{4v}{3}}{v - 0}\right) = 1$

$\therefore$ collision is elastic

43. m_1 will move along t-line if it loses all its momentum along n-line. Conservation of momentum along n-line



$$2u_n = 3v_n$$

Where $u_n = u\cos\theta$ = velocity of 2 kg mass before collision along n-line.

v_n = velocity of 3 kg mass after collision along n-line.

$$\Rightarrow \qquad v_n = \frac{2}{3}u_n$$

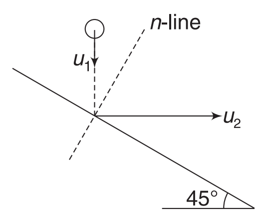
$\Rightarrow$ (Speed of separation after collision)

$= \frac{2}{3}$ (speed of approach before collision)

$$\therefore \qquad e = \frac{2}{3}$$

44. u_1 = velocity of ball before collision

u_2 = velocity of car



Relative speed of approach

$$= \frac{u_1}{\sqrt{2}} + \frac{u_2}{\sqrt{2}} = 4\sqrt{2}$$

There is no change in velocity of the massive cart.

If v_{1n} = velocity component of the ball along n-line after collision, then

$$v_{1n} - \frac{u_2}{\sqrt{2}} = e\cdot(4\sqrt{2})$$

$$\Rightarrow \qquad v_{1n} = 4\sqrt{2} + 2\sqrt{2} = 6\sqrt{2}$$

Velocity component along the incline has not changed for the ball

$$\therefore \qquad v_{1t} = \frac{u}{\sqrt{2}} = 2\sqrt{2}$$

$$\therefore \qquad v_1 = \sqrt{(6\sqrt{2})^2 + (2\sqrt{2})^2} = 4\sqrt{5}\ \text{ms}^{-1}$$

45. z component of velocity has no reason to change. Momentum conservation along x direction gives

$$Mv_{Ax} + 2Mv_x = 0 \quad \Rightarrow \quad v_{Ax} = -2v_x$$

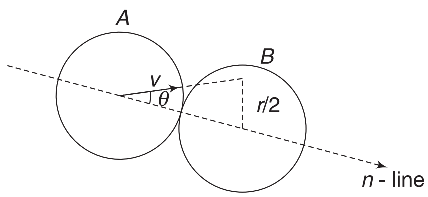
$$\therefore \qquad \vec{v}_A = -2v_x\hat{i} + u\hat{k}$$

46. $$\sin\theta = \frac{r/2}{2r} = \frac{1}{4}$$

Exchange of velocity takes place along n-line.

$\therefore$ Velocity of $B = u_{An}$

$$= v\cos\theta = v\sqrt{\frac{15}{4}}$$



47. Change in momentum of system of two particles = Impulse of external forces

$= (m_1 + m_2)g\ (2t_0)$

48. $v_{CM} = \dfrac{10 \times 14 + 4 \times 0}{10 + 4} = 10\,\text{ms}^{-1}$

49. Velocity just before explosion $= u_x = 50\cos 53°$ Radius of curvature of the path (R_1) is given by

$$\frac{mu_x^2}{R_1} = mg$$

$$\Rightarrow \quad R_1 = \frac{u_x^2}{g}$$

After explosion, the velocity of moving fragment $= 2u_x$

$\therefore$ Radius of curvature of its path

$$R_2 = \frac{(2u_x)^2}{g} = 4R_1$$

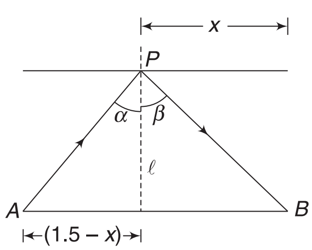
50. From Example 26

$$\tan\beta = \frac{1}{e}\tan\alpha$$

$$\Rightarrow \quad \frac{x}{l} = \frac{1}{0.5}\,\frac{(1.5 - x)}{l}$$

$$\Rightarrow \quad x = 3 - 2x \quad \Rightarrow \quad 3x = 3$$

$$\Rightarrow \quad x = 1.0 \text{ m}$$



51. $u_x = 25\cos 53° = 15\,\text{ms}^{-1}$ and $u_y = 25\sin 53° = 20\,\text{ms}^{-1}$

Time taken to reach the roof is given by

$$y = u_y t + \frac{1}{2}a_y t^2 \quad \Rightarrow \quad 15 = 20t - \frac{1}{2} \times 10 \times t^2$$

$$\Rightarrow \quad t^2 - 4t + 3 = 0 \quad \Rightarrow \quad t^2 - 3t - t + 3 = 0$$

$$\Rightarrow \quad t(t-3) - 1(t-3) = 0$$

$$\Rightarrow \quad t = 1\,\text{s},\ t = 3\,\text{s}$$

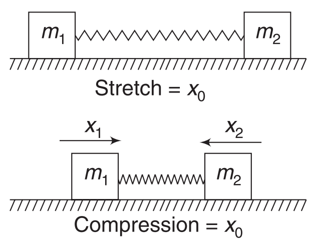
Obviously, our time is $t = 1$ s. (The projectile would have been at same height at $t = 3$ s had there been no roof.)

$\therefore$ total time of flight = 2 s

$$\therefore \quad x = u_x \times 2 = 30\,\text{m}$$

52. Initial momentum = 0.

$\therefore$ Both masses must come to rest simultaneously so that momentum is zero. When they come to rest (together), the compression in the spring must be x_0 for energy to remain conserved.



Since, the COM stays at rest

$$\therefore \quad m_1x_1 = m_2x_2 \quad \text{...(i)}$$

$$\text{And} \quad x_1 + x_2 = 2x_0 \quad \text{...(ii)}$$

Solving (i) and (ii) gives x_1 and x_2

53. At the point of maximum compression, both M and m will have same velocity.

$$\therefore \quad (M + m)v = mu \quad \Rightarrow \quad v = \frac{mu}{M + m}$$

$$K_i = \frac{1}{2}mu^2;\ K_f = \frac{1}{2}(M + m)v^2$$

$$= \frac{1}{2}(M + m)\left(\frac{mu}{M + m}\right)^2$$

$$= \frac{m^2u^2}{2(M + m)}$$

Loss in KE = Spring PE (U) $= K_i - K_f$

$$= \frac{1}{2}mu^2\left[1 - \frac{m}{M + m}\right]$$

$$= \frac{1}{2}\left(\frac{mM}{M + m}\right)u^2$$

$$\therefore \quad \frac{U}{K_i} = \frac{M}{M + m}$$

54. Momentum conservation:

$$10v + \frac{20}{1000} \times 100 = \frac{20}{1000} \times 500$$

$$\Rightarrow \qquad v = 0.8\,\text{ms}^{-1}$$

The block travels 0.2 m and comes to rest. It's retardation is given by

$$0^2 = (0.8)^2 - 2a \times 0.2 \quad \Rightarrow a = \frac{0.64}{0.4} = 1.6\,\text{ms}^{-2}$$

$\therefore$ Friction $= ma \Rightarrow \mu mg = ma$

$$\Rightarrow \mu = \frac{a}{g} = 0.16$$

55. If B is very massive (i.e., α is large), A will rebound with speed nearly equal to v_0. It will hit the wall and rebound back to hit B again. If, after colllison with B, A has speed larger than B, then it will rebound back from the wall and will chase B with speed greater than B; hitting it again.

In the critical case, speed of A, say $v(\leftarrow)$ is equal to speed of B (equal to $v(\rightarrow)$). In this case, A will not be able to catch B after rebounding from the wall.

For this critical case

$$\alpha Mv - Mv = Mv_0 \quad \Rightarrow v = \frac{v_0}{\alpha - 1}$$

Since $e = 1$

$\therefore$ speed of separation = speed of approach

$$\Rightarrow \qquad 2v = v_0$$

$$\Rightarrow \qquad \frac{2v_0}{\alpha - 1} = v_0 \qquad \Rightarrow \alpha - 1 = 2 \quad \Rightarrow \alpha = 3$$

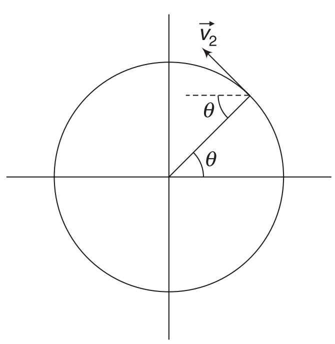
57. In time t, $\theta = \omega t$

Velocity of the particle is

$$\vec{v}_2 = -v_2 \sin\theta \hat{i} + v_2 \cos\theta \hat{j}$$

$$= -v_2 \sin(\omega t)\hat{i} + v_2 \cos(\omega t)\hat{j}$$

Velocity of the man $\vec{v}_1 = v_1 \hat{j}$



$\therefore$ Velocity of the particle relative to the man is

$$\vec{v}_{21} = \vec{v}_2 - \vec{v}_1 = -v_2 \sin(\omega t)\hat{i} + [v_2 \cos(\omega t) - v_1]\hat{j}$$

$\therefore$ Momentum of particle as observed by the man is

$$\vec{P}_{21} = m\vec{v}_{21} = -mv_2 \sin(\omega t)\hat{i} + m[v_2 \cos(\omega t) - v_1]\hat{j}$$

58. Time of first collision is $t_1 = \dfrac{\pi R}{u_1} = \dfrac{\pi \times 5}{10} = \dfrac{\pi}{2}$ s

For collision between A and B:

Speed of approach = 10 ms^{-1}

After collision relative speed of separation $= e \times 10 = 5\,\text{ms}^{-1}$

$\therefore$ Time (after first collision) for second collision is

$$t_2 = \frac{2\pi R}{5} = 2\pi\,\text{s}$$

$\therefore$ Required answer is $t = t_1 + t_2 = \dfrac{5\pi}{2}$ s

59. Due to impact, velocity normal to the incline is lost; velocity component parallel to incline does not change. Velocity after impact $= v\cos\theta$ (up the incline) Height to which particle rises is proportional to its KE after impact; which decrease with increasing θ.

60. Using equation (11), we get velocity of 1 and 2 after collision as

$$v_1 = \left(\frac{m - M}{m + M}\right)v = -\left(\frac{M - m}{m + M}\right)v$$

and $$v_2 = \frac{2mv}{m + M}$$

After collision of 2 and 3, velocity of 3 is

$$v_3 = \left(\frac{2M}{m + M}\right)v_2$$

$$\Rightarrow \qquad v_3 = \left(\frac{2M}{m + M}\right)\left(\frac{2mv}{m + M}\right) = \frac{4mMv}{(m + M)^2}$$

According to the question $|v_1| = |v_3|$

$$\Rightarrow \quad \frac{4mMv}{(m + M)^2} = \left(\frac{M - m}{m + M}\right)v \quad \Rightarrow 4Mm = M^2 - m^2$$

$$\Rightarrow \quad \left(\frac{M}{m}\right)^2 - 4\left(\frac{M}{m}\right) - 1 = 0$$

$$\Rightarrow \qquad \frac{M}{m} = \frac{4 \pm \sqrt{16 + 4}}{2} = 2 \pm \sqrt{5}$$

$$\therefore \qquad \frac{M}{m} = 2 + \sqrt{5}$$

61. For conservation of momentum along x-axis

$2mv_0 \cos\theta = mv$ [v_0 = speed of each piece]

$$\Rightarrow \quad v_0 = \frac{v}{2\cos\theta}$$

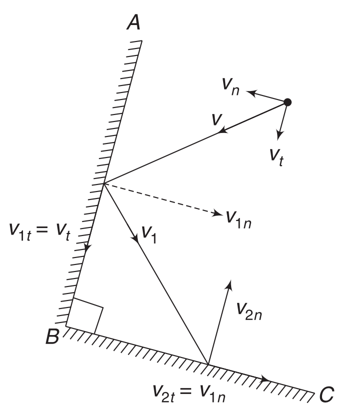
$$\because \quad \cos\theta < 1$$

$$\therefore \quad v_0 > \frac{v}{2}$$

62. v_n and v_t are velocity components perpendicular and parallel to AB. After collision, velocity is v_1. Its components are:

$$v_{1n} = e\,v_n$$

$$v_{1t} = v_t$$



Now v_{1n} is actually parallel to BC. During second collision, it remains unchanged.

v_{1t} is normal to BC.

$\therefore$ After collision,

$$v_{2n} = e\,v_{1t} = e\,v_t$$

And $\quad v_{2t} = v_{1n} = e\,v_n$.

$\therefore$ Both v_{2t} and v_{2n} have become e times their original value, and their directions is reversed.

$\therefore$ Final velocity $= -e\vec{v}$

63. v_f = find speed of ball after collision. Since, velocity component parallel to the incline does not change

$$\therefore \quad 3\cos 60° = v_f \cos 30° \quad \Rightarrow v_f = \sqrt{3}\,\text{ms}^{-1}$$

Horizontal component of impulse received by the ball $= m\sqrt{3}$. Same horizontal impulse is applied by the ball on the wedge towards left. The back support applies equal impulse on the wedge towards right to prevent it from moving.

64. In first case, momentum will not change much in x-direction during collision. Impulse of friction is negligible.

In second case, there is impulse on the system in both x- and y-directions. The ground will impart a significant normal impulse during collision. Friction will also be impulsive in nature as it depends on the normal force.

65. As mass (snow) gets added to each cart, their velocity will decrease to conserve momentum, in horizontal direction. When Ram removes snow and throws it sideways, it will not make any difference to velocity of his trolley. [Refer to Example 14 and 15]

66. Relative velocity of the mass getting separated (in horizontal direction) is zero.

$$\therefore \quad \text{Thrust force, } F_{\text{th}} = u\frac{dm}{dt} = 0$$

Mass of wagon at time t is $m_0 - \lambda t$

$$\therefore \quad (m_0 - \lambda t)\frac{d\vec{v}}{dt} = \vec{F}$$

67. $$F_{\text{th}} = u\,\frac{dm}{dt} \quad \Rightarrow 210 = 300\,\frac{dm}{dt}$$

$$\Rightarrow \quad \frac{dm}{dt} = 0.7\,\text{kgs}^{-1}$$

68. $(5 \times 10^3 + 10^3)v = 5 \times 10^3 \times 1.2 \quad \Rightarrow v = 1\,\text{ms}^{-1}$

69. $$F_{\text{th}} = u\frac{dm}{dt} = 980\,\frac{dm}{dt}$$

$$M\frac{dv}{dt} = F_{\text{th}} - Mg$$

$$\Rightarrow \quad 4000 \times 19.6 = 980\left(\frac{dm}{dt}\right) - 4000 \times 9.8$$

$$\Rightarrow \quad \frac{dm}{dt} = 120\,\text{kgs}^{-1}$$

Worksheet 2

2. If both come to rest it will violate law of conservation of momentum. Hence, (*A*) is wrong.

   If the heavy ball remains still and the moving ball changes its velocity then also law of conservation of momentum gets violated. Hence, (D) is wrong

3. In explosion, chemical energy will get converted into *KE*, sound, heat, etc. Thus, final *KE* will be greater than initial *KE*.

5. Let v be speed of the two blocks immediately after the bullets hits block *B*.

   For B: $2mv - mu = -J$

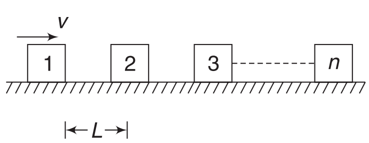
   For A: $3mv = J$

   Solving the above two equation gives $v = \frac{u}{5}$ and $J = \frac{3mu}{5}$

   Impulse on ceiling is $2J = \frac{6mu}{5}$ because tension in the string connected to ceiling is twice the tension in the string connecting the blocks.

6. Momentum is always zero.

7. Time when block 1 hit 2 is $t_1 = \frac{L}{v}$



   The two blocks stick and the common velocity becomes:

   $$v_1 = \frac{v}{2}$$

   Time, after first collision, when second collision takes place is

   $$t_2 = \frac{L}{v_1} = \frac{2L}{v}$$

   $\therefore$ Total time when $(n-1)^{\text{th}}$ collision takes place is

   $$T = t_1 + t_2 + \ldots + t_{n-1}$$

   $$= \frac{L}{v}[1 + 2 + 3 + \ldots + (n-1)] = \frac{L(n-1)n}{2v}$$

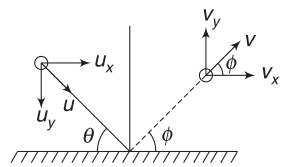
   $$v_{\text{CM}} = \frac{mv}{nm} = \frac{v}{n}$$

   This remains constant throughout.

8. $\Delta P = mv_y + mu_y$

   $$= meu_y + mu_y$$

   $$= (e+1)mu_y = (e+1)mu\sin\theta$$



   This is impulse applied by the floor on the ball.

   $$\tan\phi = \frac{v_y}{v_x} = \frac{eu_y}{u_x} = \frac{e \cdot u\sin\theta}{u\cos\theta} = e\tan\theta$$

   Speed, $v = \sqrt{v_x^2 + v_y^2} = \sqrt{(u\cos\theta)^2 + (eu\sin\theta)^2}$

   $$= u\sqrt{\cos^2\theta + \sin^2\theta + e^2\sin^2\theta - \sin^2\theta}$$

   $$= u\sqrt{1-(1-e^2)\sin^2\theta}$$

   $$\frac{KE_f}{KE_i} = \frac{\frac{1}{2}mv^2}{\frac{1}{2}mu^2} = \frac{(u\cos\theta)^2 + (eu\sin\theta)^2}{u^2}$$

   $$= \cos^2\theta + e^2\sin^2\theta$$

9. For first bead, we will use work-energy theorem to find its speed just before collision with second bead.

   $$\frac{1}{2}mu^2 = Fd \quad \Rightarrow u = \sqrt{\frac{2Fd}{m}}$$

   Since collision is head-on elastic and masses are equal, there will be exchange of velocities. The first bead will come to rest after collision.

   Second bead begins to move at speed u, hits the third bead and comes to rest. Now bead 2 is at a distance d from its original position and is at rest. In the mean time, bead 1 gets accelerated due to force F and again hits bead 2. Just before collision, speed of bead 1 is again u. It will again come to rest. After this, it will once again get accelerated for a distance d.

   Thus, bead 1 travels a distance d as its speed increases uniformly from 0 to u.

   Average Speed $= \frac{0+u}{2} = \frac{1}{2}\sqrt{\frac{2Fd}{u}}$

10. Speed of approach $= u + v$

    Speed of separation $= v_1 - u$

    where v_1 = velocity of the ball after the impact.

    Since impact is elastic

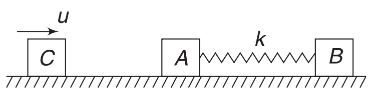
    $\therefore \quad v_1 - u = u + v \quad \Rightarrow v_1 = 2u + v$

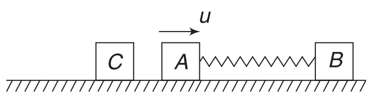
    $$\Delta P_{\text{ball}} = mv_1 + mv = 2m(u+v)$$

    $\therefore \quad F_{\text{av}} = \frac{2m(u+v)}{\Delta t}$

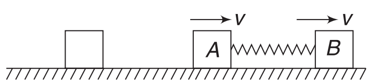
$$\Delta K = K_f - K_i = \frac{1}{2}mv_1^2 - \frac{1}{2}mu^2$$
$$= \frac{1}{2}m[4u^2 + v^2 + 4uv - u^2]$$
$$= \frac{1}{2}m[3u^2 + v^2 + 4uv]$$

11. When C hits A, exchange of velocity takes place; C comes to rest and A acquires velocity u. At maximum compression of the spring, both A and B will have same velocity.





Immediately after collision



Maximum compression

$$(m + m)v = mu$$
$$\Rightarrow \quad v = \frac{u}{2}$$
$$\therefore \quad KE = \frac{1}{2}(2m)\left(\frac{u}{2}\right)^2 = \frac{1}{4}mu^2$$

Loss in $KE = \frac{1}{2}mu^2 - \frac{1}{4}mu^2 = \frac{1}{4}mu^2$

This energy is stored in the spring as PE

$$\therefore \quad \frac{1}{2}kx^2 = \frac{1}{4}mu^2 \quad \Rightarrow \quad x = \sqrt{\frac{m}{2k}}\cdot u$$

12. $v_{CM} = \dfrac{mv_0 + mv_0}{3m} = \dfrac{2v_0}{3}$

Let u be the velocity of beads relative to the hoop just before collision.

In y-direction, the beads and the hoop will have same velocity equal to $v_{CM} = \dfrac{2v_0}{3}$.

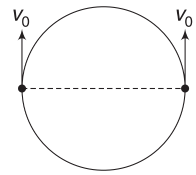
In ground frame, speed of each bead is

$$= \sqrt{u^2 + v_{CM}^2} = \sqrt{u^2 + \frac{4v_0^2}{9}}$$

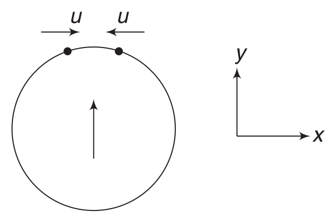
Energy conservation gives:

$$\frac{1}{2}m\left(u^2 + \frac{4v_0^2}{9}\right) \times 2 + \frac{1}{2}m\left(\frac{2v_0}{3}\right)^2 = \frac{1}{2}mv_0^2 \times 2$$

$$\Rightarrow \quad u = \frac{v_0}{\sqrt{3}}$$



Initial condition



Just before collision

$\therefore$ Speed of each bead $= \sqrt{u^2 + \dfrac{4v_0^2}{9}} = \dfrac{\sqrt{7}}{3}v_0$

13. Acceleration for first 3 s is,

$$a = \frac{15\,\text{ms}^{-1}}{3\,\text{s}} = 5\,\text{ms}^{-2}$$
$$\therefore \quad F = ma = 10 \times 5 = 50\,\text{N}$$

According to the question, a body of mass 10 kg moving at 15 ms^{-1} collides head-on with a 25 kg mass and sticks to it. Final velocity of combined mass is 5 ms^{-1}.

Let velocity of 25 kg mass before collision be v.

$$10 \times 15 + 25 \times v = (10 + 25) \times 5$$
$$\Rightarrow \quad v = 1\,\text{ms}^{-1}$$

14. (A) the sum of momenta of three particles must be zero. In three-body case also, sum of momenta will be zero if the three momenta are in same plane.
(B) Momentum of conservation and sum of KE gives two equations. These can be solved in two-body case but cannot be solved in three-body case, as number of unknowns will be 3.
(C) In three-body case, particle can move in any direction in a plane. If we do not know direction of motion of some of these particles, there will be too many unknowns.

16. Refer to equation (12)

$$v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)u_1$$
$$= \left(\frac{m - 2m}{m + 2m}\right)u_1 = -\frac{u_1}{3}$$
$$\therefore \quad |v_1| < |u_1|$$

Loss in KE of first particle $= \frac{1}{2} mu_1^2 - \frac{1}{2} m\left(\frac{u_1}{3}\right)^2$

$= \frac{1}{2} mu_1^2 \times \frac{8}{9}$

Fractional loss $= \frac{8}{9}$

KE lost by first particle must be equal to KE gained by the second particle, as the collision is elastic.

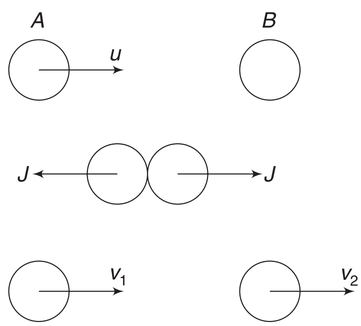
18. For A: $mv_1 = mu - J$

$\Rightarrow \quad v_1 = u - \frac{J}{m}$

For B: $mv_2 = J$

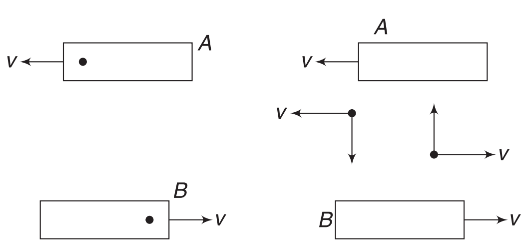
$\Rightarrow \quad v_2 = \frac{J}{m}$

$$e = \frac{v_2 - v_1}{u} = \frac{\frac{2J}{m} - u}{u} = \frac{2J}{mu} - 1 = \frac{2J}{P} - 1$$

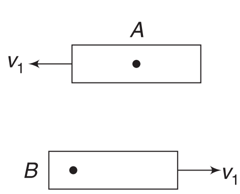


19. When a ball falls on a trolley, it brings a momentum with it which is opposite to that of the trolley. Thus, the speed of the trolley decreases when a ball lands on it.

When the balls are projected simultaneously, they have equal magnitude of momentum and will cause equal change in velocity of trolleys.



Balls projected simultaneously



Velocities decrease after balls land into trolleys

When ball from B to A reaches first, it decreases the speed of trolley A. Now the ball projected from A will carry less momentum (as it is travelling at smaller speed) and will cause less change in speed of B.

When ball from A to B misses the trolley, speed of B remains unchanged. But speed of A decreases due to ball landing on it.

20. The COM of the system of box + bomb will not get displaced.

21. When the rod is vertical, let the velocity of cart be u ($\rightarrow$). From conservation of momentum, in horizontal direction, velocity of bob must be $6u$ ($\leftarrow$).

Energy conservation:

$$\frac{1}{2} 6mu^2 + \frac{1}{2} m(6u)^2 = mgl$$

$$\Rightarrow \quad u = \sqrt{\frac{gl}{21}}$$

$\therefore$ Relative speed $= 6u + u = 7u = \sqrt{\frac{7gl}{3}}$

22. Due to recoil of the cannon, the horizontal component of velocity of the cannon ball is less than $v_{rel} \cos\theta$. But its vertical velocity is $v_{rel} \sin\theta$. Thus, actual angle of projection is larger than θ. [Refer to Example 22]. Thus, the trajectory shown in (C) is possible.

23. Both observers are in inertial frames. They can use work energy theorem and impulse-momentum equation in their usual form.

24. For $l = \frac{R}{2}$, the projectile hits the wall while moving horizontally. Its velocity gets reversed and it retraces its path. For $l < \frac{R}{2}$, the projectile will spend less time in air before hitting the wall and will spend more time in air after hitting the wall.

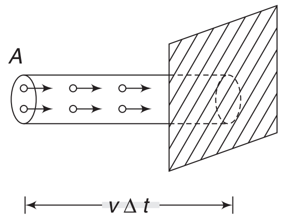
The overall time period does not change as collision does not change the vertical component of velocity. For $l > \frac{R}{2}$, the projection will spend more time while moving towards the wall.

$\therefore$ while coming back it will travel a horizontal distance $< l$.

$\therefore$ It will fall between point P and the wall.

Worksheet 3

1. Let the cross-sectional area of the beam be A. In time Δt, all the particles inside the cylindrical volume shown in figure will hit the surface.



Mass of particles inside the volume shown $= \frac{m}{d} \cdot v\ \Delta t = \Delta m$ (say). This Δm mass suffers a momentum change of $2\Delta mv$ in time interval Δt.

$$\therefore \quad F = \frac{\Delta P}{\Delta t} = \frac{2\Delta mv}{\Delta t} = \frac{2 \cdot \frac{m}{d} v \Delta t \cdot v}{\Delta t} = 2\left(\frac{m}{d}\right)v^2$$

2. Refer to Example 3

Force on the roof = (Number of stones hitting per second) × (Change in momentum of one stone)

Number of stones hitting the roof in 1 second, $n = 2000 \times 100 = 2 \times 10^5$

Mass of one stone,

$$m = \frac{4}{3}\pi r^3 \cdot d = \frac{4}{3} \times 3.14 \times \left(\frac{0.5}{100}\right)^3 \times 900$$

$$= 4.71 \times 10^{-4}\text{kg}$$

Change in momentum of one stone on hitting the roof is

$$\Delta P_{\text{one}} = mv = 4.71 \times 10^{-4} \times 20$$

$$= 9.42 \times 10^{-3}\ \text{kgms}^{-1}$$

$$\therefore \quad \text{Force, } F = n\Delta P_{\text{one}} = 2 \times 10^5 \times 9.42 \times 10^{-3} = 1884\ \text{N}$$

3. Let the final velocity of the system be v. From conservation of linear momentum

$$4mv = mu \quad \Rightarrow \quad v = \frac{u}{4}$$

Block A is set into motion due to the impulse it receives from the rope.

$$\therefore \quad J_A = mv = \frac{mu}{4} \qquad \text{[Ans. to (iii)]}$$

Impulse applied by B on the bullet is reason behind change in momentum of the bullet.

$$\Delta P_{\text{bullet}} = mv - mu = m\frac{u}{4} - mu = -\frac{3}{4}mu.$$

$$= \frac{3}{4}mu\ (\leftarrow)$$

B exerts an impulse on the bullet and the bullet exerts same impulse on the block towards right. This impulse is

$$J = \frac{3}{4}mu \qquad \text{[Ans. to (ii)]}$$

For block B

Impulse due to rope + Impulse due to bullet $= m\frac{u}{4}$

$$\Rightarrow \quad J_B + \frac{3}{4}mu = \frac{mu}{4}$$

$$\Rightarrow \quad J_B = -\frac{1}{2}mu = \frac{1}{2}mu\ (\leftarrow)$$

$\therefore$ Impulse applied by B on rope $= \frac{1}{2}mu\ (\rightarrow)$

4. Only B moves (along y-direction) till the string becomes taut. Position of A and B is as shown when the string is about to get taut (Fig. a) After the string gets taut, both the particles move such that they have same velocity component along the length of the string.

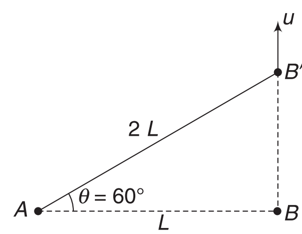


Fig. (a)

A will move along AB' since it receives impulse in this direction and originally it was at rest. Let its velocity be v. Component of velocity of B along thread = v. Let velocity of B perpendicular to the thread be v'.

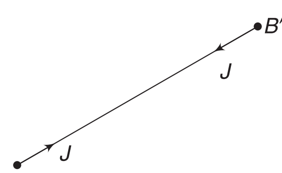


Fig. (b)

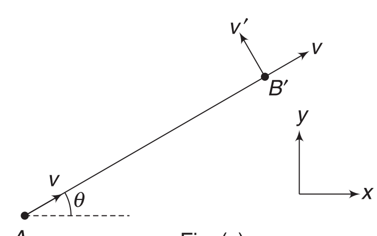


Fig. (c)

Since B does not receive any impulse perpendicular to the string, its velocity will not change in that direction.

$$\therefore \quad v' = u\cos\theta = \frac{u}{2}$$

Applying momentum conservation along the thread.

$$mv + mv = mu\sin\theta \quad \Rightarrow \quad 2v = u\sin 60°$$

$$\Rightarrow \quad v = \frac{\sqrt{3}}{4}u$$

Velocity of A is $v_A = \dfrac{\sqrt{3}}{4}u$ in a direction making 60° with x.

Velocity of B is $v_B = \sqrt{v^2 + (v')^2}$

$$= \sqrt{\left(\frac{\sqrt{3}}{4}\right)^2 + \left(\frac{1}{2}\right)^2} \cdot u = \frac{\sqrt{7}}{4}u$$

Impulse on A is $J = mv_A = \dfrac{\sqrt{3}}{4}mu$ making 60° with x direction.

$$\vec{J} = \left(\frac{\sqrt{3}}{4}mu\cos 60°\right)\hat{i} + \left(\frac{\sqrt{3}}{4}mu\sin 60°\right)\hat{j}$$

$$= \left(\frac{\sqrt{3}}{8}mu\right)\hat{i} + \left(\frac{3}{8}mu\right)\hat{j}$$

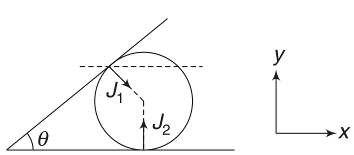
5. Let impulses by the wall and the floor be J_1 and J_2 respectively.

$$J_x = J_1\sin\theta\ (\rightarrow)$$

$$\Rightarrow \quad m\frac{v}{2}(\rightarrow) - mv(\leftarrow) = J_1\sin\theta\ (\rightarrow)$$

$$\Rightarrow \quad m\frac{v}{2} + mv = J_1\sin\theta$$

$$\Rightarrow \quad \frac{3}{2}\frac{mv}{\sin\theta} = J_1$$



Net impulse in y-direction should be zero.

$$\therefore\ J_2 = J_1\cos\theta = \frac{3}{2}\frac{mv}{\sin\theta}\cos\theta = \frac{3}{2}mv\cot\theta$$

6. Let velocity of A be $v(\rightarrow)$ and that of B be $u(\leftarrow)$ at the moment they get separated. Momentum conservation gives:

$$5v = 1\cdot u \quad \Rightarrow \quad 5v = u \qquad \text{...(1)}$$

Energy conservation gives:

$$\frac{1}{2}(5)v^2 + \frac{1}{2}(1)u^2 = \frac{1}{2}kx_0^2$$

$$\Rightarrow \quad 5v^2 + 25v^2 = 100(1)^2 \quad \Rightarrow \quad 30v^2 = 100$$

$$\Rightarrow \quad v = \sqrt{\frac{10}{3}}\ \text{ms}^{-1}$$

7. Time of flight of the ball after impact is

$$T = \sqrt{\frac{2h}{g}}$$

If velocity of ball after impact is $v_1(\leftarrow)$, then $d = v_1T$

$$\Rightarrow \quad v_1 = \frac{d}{T} = d\sqrt{\frac{g}{2h}}$$

Let $v_2(\rightarrow)$ be velocity of wedge after impact. Momentum of ball + wedge system is conserved in horizontal direction during the collision. Note that spring is relaxed and during the short interval of collision, it does not apply any impulse.

$$\therefore \quad Mv_2 = mv_1$$

$$\Rightarrow \quad v_2 = \frac{md}{M}\sqrt{\frac{g}{2h}}$$

The wedge compresses the spring and comes to rest when the spring is fully compressed. Let the maximum compression be x. Conservation of energy gives:

$$\frac{1}{2}Mv_2^2 = \frac{1}{2}kx^2$$

$$\Rightarrow \quad x = \sqrt{\frac{M}{k}}\cdot v_2 = md\sqrt{\frac{g}{2hkM}}$$

8. Refer to Example 16

After the man jumps out, velocity of the rear car will be $v_1 = u - \dfrac{mv}{M+m}$

Total momentum of the two car + man system is conserved. If v_2 is velocity of front car + man after he lands into it, then

$$(M+m)v_2 + Mv_1 = (2M+m)u.$$

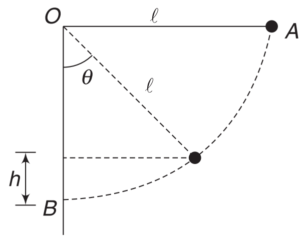
$$\Rightarrow \quad (M+m)v_2 = (2M+m)u - Mu + \frac{Mmv}{M+m}$$

$$\Rightarrow \quad v_2 = u + \frac{Mmv}{(M+m)^2}$$

9. The bob of the pendulum hits the wall at B with a horizontal velocity (v) given by conservation of energy.

$$\frac{1}{2}mv^2 = mgl \quad \Rightarrow \quad v = \sqrt{2gl}$$

The bob rebounds with speed



$$v_1 = ev = e\sqrt{2gl}$$

Now the bob rises to some height h and then again comes back to hit the wall at B. Before hitting, its velocity is $v_1(\leftarrow)$ and after the second hit, it becomes

$$v_2 = ev_1 = e^2\sqrt{2gl}\ (\rightarrow)$$

The process continues and the speed of bob after n^{th} hit is

$$v_n = e^n\sqrt{2gl}$$

If the bob rises to a height h_n after n collisions, then

$$mgh_n = \frac{1}{2}mv_n^2$$

$$\Rightarrow \qquad h_n = \frac{v_n^2}{2g} = \frac{e^{2n}\cdot 2gl}{2g} = l\cdot e^{2n}$$

$$\therefore \qquad h_n = \left(\frac{2}{\sqrt{5}}\right)^{2n} l = \left(\frac{4}{5}\right)^n\cdot l$$

From geometry, $\dfrac{l-h}{l} = \cos\theta$

$$\Rightarrow \qquad 1 - \frac{h}{l} = \cos\theta$$

$$\Rightarrow\ 1 - \left(\frac{4}{5}\right)^n = \cos\theta$$

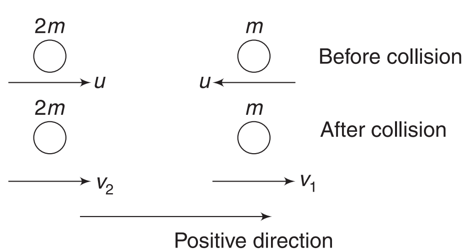
For $\theta < 60°$, $\cos\theta > \dfrac{1}{2}$

$$\Rightarrow\ 1 - \left(\frac{4}{5}\right)^n > \frac{1}{2} \quad \Rightarrow \quad \left(\frac{4}{5}\right)^n < \frac{1}{2}$$

This condition is satisfied for $n \geq 4$

$\therefore$ Minimum number of collisions needed = 4

10. Speed of the two balls just before collision is $u_1 = u_2 = \sqrt{2gH}$



Let velocities be v_1 and v_2

After collision in positive direction (see figure).

$$v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)u_1 + \left(\frac{2m_2}{m_1 + m_2}\right)u_2$$

$$= \left(\frac{m - 2m}{m + 2m}\right)(-\sqrt{2gH}) + \left(\frac{4m}{m + 2m}\right)(\sqrt{2gH})$$

$$= \frac{5}{3}\sqrt{2gH}$$

Similarly $v_2 = \left(\dfrac{2m_1}{m_1 + m_2}\right)u_1 + \left(\dfrac{m_2 - m_1}{m_2 + m_1}\right)u_2$

$$= -\frac{1}{3}\sqrt{2gH}$$

Negative sign tells us that v_2 is directed to left. Height raised by bob of mass m is given as

$$mgH_1 = \frac{1}{2}mv_1^2$$

$$\Rightarrow \qquad H_1 = \frac{25}{9}H$$

Similarly, $H_2 = \dfrac{H}{9}$

11. (i) Let velocity of combined mass (bullet + block) be v just after the collision.

Conservation of momentum along horizontal direction gives

$$(M + m)v = mu \quad \Rightarrow \quad v = \frac{mu}{M + m}$$

Loss in mechanical energy during collision is same as loss in KE in the context of the question.

$$\therefore \qquad \text{loss} = \frac{1}{2}mu^2 - \frac{1}{2}(M + m)v^2$$

$$= \frac{1}{2}mu^2 - \frac{1}{2}(M + m)\left(\frac{mu}{M + m}\right)^2$$

$$= \frac{Mmu^2}{2(M + m)}$$

(ii) For completing the circle

$$v \geq \sqrt{5gL}$$

$$\Rightarrow \quad \frac{mu}{M + m} \geq \sqrt{5gL} \quad \Rightarrow \quad v \geq \left(\frac{M + m}{m}\right)\sqrt{5gL}$$

12. The vertical component of velocity does not change due to impact. Hence, total time of flight from A to B to C to A is $T = \dfrac{2u_y}{g} = \dfrac{2u\sin\theta}{g}$

While travelling from A to B, the horizontal component of velocity of the ball is $u\cos\theta$ and while moving back from B to A it is $eu\cos\theta$.

$\therefore \qquad t_{A\to B} + t_{B\to A} = T$

$$\frac{d}{u\cos\theta} + \frac{d}{eu\cos\theta} = \frac{2u\sin\theta}{g}$$

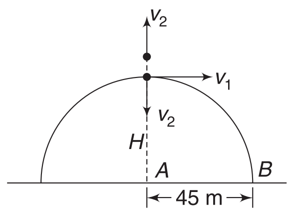
$$\Rightarrow \qquad \frac{d}{eu\cos\theta} = \frac{2u\sin\theta}{g} - \frac{d}{u\cos\theta}$$

$$\Rightarrow \qquad e = \frac{1}{\dfrac{u^2\sin 2\theta}{d\cdot g} - 1}$$

13. Velocity at top is

$$u_x = 24\cos 60° = 12\,\text{ms}^{-1}$$

Velocity of one fragment after explosion is vertically up. Let it be v_2.



For conservation of momentum in vertical direction, the other fragment must have a downward component of velocity equal to v_2.

Let horizontal component of velocity of the second fragment after explosion be v_1.

For momentum to be conserved along horizontal direction

$$mv_1 = 2mu_x \quad \Rightarrow \quad v_1 = 24\,\text{ms}^{-1}$$

According to the problem $AB = 45\,\text{m}$

$$\Rightarrow \qquad 24t = 45$$

$$\rightarrow \qquad t = \frac{15}{8}\text{s} = 1.875\,\text{s}$$

$$H = \frac{u^2\sin^2 60°}{2g} = \frac{24^2 \times 3}{2\times 10\times 4} = 21.6\ \text{m}$$

For second fragment, time of flight is $t = 1.875\,\text{s}$

$$\therefore\ v_2 \times 1.875 + \frac{1}{2}\times 10 \times 1.875^2 = 21.6$$

$$\Rightarrow \qquad v_2 = 2.55\,\text{ms}^{-1}$$

$\therefore$ Velocity of 1st particle $= v_2 = 2.55\,\text{ms}^{-1}$

Velocity of second particle

$$= \sqrt{v_1^2 + v_2^2} = \sqrt{24^2 + 2.55^2} = 24.1\,\text{ms}^{-1}$$

14. It will be useful to refer to Example 23.

t-line is line along the floor and n-line for each impact is vertical (along y).

Horizontal component of velocity remains constant $= u_x = u\cos\theta$. Vertical component of velocity:

After first collision, $v_1 \quad = eu_y \quad = eu\sin\theta$

After second collision, $v_2 \quad = ev_1 \quad = e^2u\sin\theta$

After n^{th} collision, $v_n = e^n u\sin\theta$

Total time of flight

$$T = \frac{2u\sin\theta}{g} + \frac{2eu\sin\theta}{g} + \frac{2e^2u\sin\theta}{g} + \ldots\ \infty$$ terms

$$= \frac{2u\sin\theta}{g}[1 + e + e^2 + \ldots + \infty \text{ terms}]$$

$$= \frac{2u\sin\theta}{g}\left(\frac{1}{1-e}\right) \qquad \text{[Ans. to (ii)]}$$

Total horizontal displacement is

$$x = u_x\cdot T = \frac{u^2\sin 2\theta}{g}\left(\frac{1}{1-e}\right) \qquad \text{[Ans. to (i)]}$$

Launch angle after n$^{\text{th}}$ collision is given as:

$$\tan\theta_n = \frac{v_n}{u_x} \quad \Rightarrow \quad \tan\theta_n = e^n\tan\theta$$

$$\Rightarrow \qquad \theta_n = \tan^{-1}(e^n\tan\theta)$$

15. (i) Let velocity of men after jump be v_m $(\rightarrow)$ and that of car be v_c $(\leftarrow)$

$u(\rightarrow) = v_m(\rightarrow) - v_c(\leftarrow) \Rightarrow v_m = u - v_c$.

Momentum conservation gives:

$$2m(u - v_c) = Mv_c \quad \Rightarrow \quad v_c = \frac{2mu}{2m + M}$$

(ii) When only one man has jumped, let velocity of car be $v_1(\leftarrow)$.

Velocity of man (who has jumped) is $v_m = u - v_1$

Momentum conservation gives:

$$m(u - v_1) = (M + m)v_1 \quad \Rightarrow \quad v_1 = \frac{mu}{M + 2m}\ (\leftarrow)$$

Now, the car is moving with velocity v_1. The second man jumps out and its velocity changes to v_2 $(\leftarrow)$.

$$v'_m = u - v_2\ (\rightarrow)$$

Momentum conservation gives:

$$(M + m)v_1(\leftarrow) = Mv_2(\leftarrow) + m(u - v_2)(\rightarrow)$$

$$\Rightarrow (M + m)v_1 = Mv_2 - m(u - v_2)$$

$$\Rightarrow (M+m)v_2 = mu + (M+m)\left(\frac{mu}{M+2m}\right)$$

$$\Rightarrow \qquad v_2 = \frac{mu}{M+m} + \frac{mu}{M+2m}$$

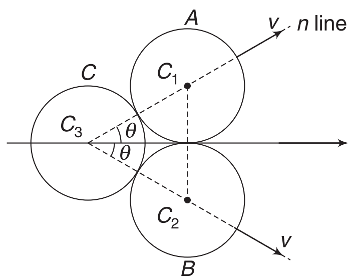
16. Collision of C with A and that with B is identical. Both A and B will move along LOI with same speed.

Momentum (for the system of three discs) is conserved along any direction. We will conserve it along x direction.

From simply geometry, we can find angle between u and LOI.

$\Delta C_1C_2C_3$ is equilateral

$\therefore \qquad \theta = 30°$



Momentum conservation along x direction gives

$$2mv\cos\theta = mu$$

$$\Rightarrow \qquad v = \frac{u}{2\cos\theta} = \frac{u}{2\frac{\sqrt{3}}{2}} = \frac{u}{\sqrt{3}}$$

For A and C: $e = -\left(\frac{v_{Cn}-v_{An}}{u_{Cn}-u_{An}}\right)$

$$= -\left(\frac{0-v}{u\cos\theta - 0}\right) = \frac{\frac{u}{\sqrt{3}}}{u\frac{\sqrt{3}}{2}} = \frac{2}{3}$$

$$\text{Loss in } KE = \frac{1}{2}mu^2 - 2\left(\frac{1}{2}mv^2\right)$$

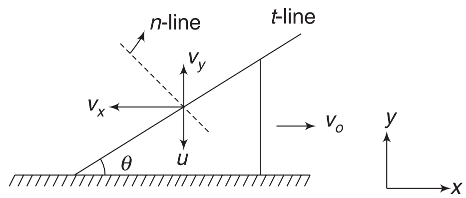
$$= \frac{1}{2}mu^2 - m\left(\frac{u}{\sqrt{3}}\right)^2 = \frac{1}{6}mu^2$$

$$\%\text{ loss} = \frac{\frac{1}{6}mu^2}{\frac{1}{2}mu^2} \times 100 = \frac{100}{3} = 33.3\%$$

17. Let velocity of wedge be $v_0(\rightarrow)$ after collision.

Horizontal and vertical velocity components of the ball after collision are v_x and v_y as shown.

Momentum conservation along x direction gives:



$$Mv_0 = mv_x \qquad \text{...(i)}$$

Velocity component of the ball along the t-line (parallel to incline does not change).

$$\therefore\ v_x\cos\theta - v_y\sin\theta = u\sin\theta \qquad \text{...(ii)}$$

And $\qquad e = -\left(\frac{v_{1n}-v_{2n}}{u_{1n}-u_{2n}}\right)$

$$\Rightarrow \qquad 1 = -\left(\frac{v_x\sin\theta + v_y\cos\theta + v_0\sin\theta}{-u\sin\theta - 0}\right) \qquad \text{...(iii)}$$

Solving (i), (ii) and (iii) gives: $v_0 = \frac{2mu}{m+2M}$

From energy conservation:

$$\frac{1}{2}kx^2 = \frac{1}{2}Mv_0^2$$

$$\Rightarrow \qquad x = \sqrt{\frac{M}{k}}\,v_0 = \left(\frac{2mu}{m+2M}\right)\sqrt{\frac{M}{k}}$$

18. Retardation of 2 kg block while going up is

$$a_1 = g(\sin\theta + \mu\cos\theta) = 10(0.05 + 0.25\times 1)$$

$$= 3\,\text{ms}^{-2} \qquad [\because \cos\theta \simeq 1 \text{ for small angle}]$$

Velocity of 2kg mass just before collision is given by

$$v_0^2 = 10^2 - 2a_1s = 100 - 2\times 3\times 6 = 64$$

$$\therefore\ \ v_0 = 8\,\text{ms}^{-1}$$

Its acceleration while coming down is

$$a_2 = g(\sin\theta - \mu\cos\theta) = 10(0.05 - 0.25) = -2\,\text{ms}^{-2}$$

Negative sign shows that the block retards.

Let its velocity after collision be v_1 (down the plane).

Then $\qquad 1^2 = v_1^2 + 2a_2\cdot s$

$$\Rightarrow \qquad v_1^2 = 1 + 2\times 2\times 6 \ \Rightarrow\ v_1 = 5\,\text{ms}^{-1}$$

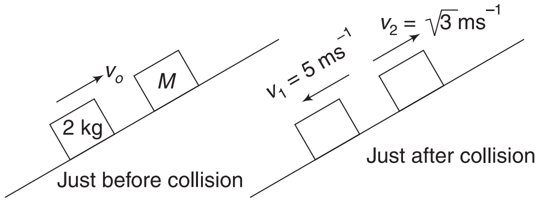
Let velocity of M after collision be v_2 (up the plane).

$$0^2 = v_2^2 - 2\times a_1\times(0.5) \Rightarrow v_2^2 = 2\times 3\times 0.5 = 3$$

$$\Rightarrow v_2 = \sqrt{3}\ \text{ms}^{-1}$$

Relative speed of approach $= v_0 = 8\,\text{ms}^{-1}$

Relative speed of separation $= (5+\sqrt{3})\,\text{ms}^{-1}$

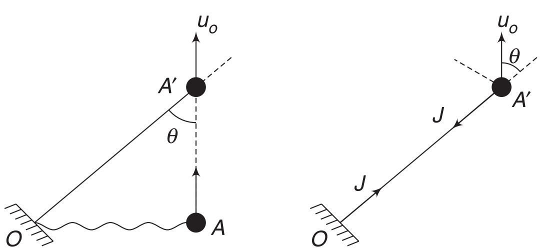


$$\therefore \qquad e = \frac{5 + \sqrt{3}}{8}$$

Momentum conservation gives:

$$M\sqrt{3} - 2 \times 5 = 2 \times 8 \quad \Rightarrow \quad M = \frac{26}{\sqrt{3}} \text{ kg}$$

19. (i) In figure, $\sin\theta = \dfrac{1}{2.4} = 0.42$. Let the critical allowed velocity be u_0. As the string gets taut, the velocity component of ball along the length of thread becomes zero.



This happens due to impulse J applied by the string on the ball.

$$\therefore \qquad J = mu_0 \cos\theta$$

$$\Rightarrow \qquad u_0 = \frac{J}{m\cos\theta} = \frac{3}{2 \times 0.9}$$

$$[\because \quad \cos\theta = \sqrt{1 - (0.42)^2} = 0.9]$$

$$\simeq 1.65 \text{ ms}^{-1}$$

(ii) Velocity component perpendicular to OA' remains unchanged.

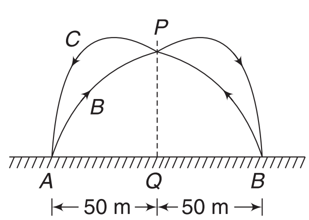
$\therefore$ After the string gets taut, velocity of ball is perpendicular to OA' and is equal to

$$u_0 \sin\theta = 1.65 \times 0.42 = 0.69 \text{ ms}^{-1}$$

$$\therefore \text{ Loss in } KE = \frac{1}{2} m(u_0^2 - u_0^2 \sin^2\theta)$$

$$= \frac{1}{2} \times 2 \times ((1.65)^2 - (0.69)^2) \simeq 2.25 \text{ J}$$

20. From symmetry the two balls will collide at P such that $AQ = BQ = 50$ m. Ball projected from A follows the trajectory $ABPCA$. Due to collision, there will be no change in vertical motion of the balls. Their time of flight will not change.



$$\therefore \qquad \frac{2u\sin 30°}{g} = \frac{50}{u\cos 30°} + \frac{50}{eu\cos 30°}$$

$$\Rightarrow \qquad u = 37.5 \text{ ms}^{-1}$$

21. $$M\frac{dv}{dt} = -u\frac{dM}{dt}$$

$$\Rightarrow \quad Ma = -u\frac{dM}{dt} \quad \Rightarrow \quad \int_{M_0}^{m} \frac{dM}{M} = -\frac{a}{u}\int_0^t dt$$

$$\Rightarrow \quad ln\left(\frac{M}{M_0}\right) = -\frac{at}{u}$$

$$\Rightarrow \quad M = M_0 e^{-\frac{at}{u}}$$

22. Let v = velocity of wagon at time t

m = mass of wagon at $t = m_0 + bt$

Relative velocity of mass being added; $u = 0 - v(\rightarrow) = v(\leftarrow)$

$$F_{th} = u\frac{dm}{dt} = bv(\leftarrow) \left[\frac{dm}{dt} = b \text{ is positive}\right]$$

$\therefore$ Net force on the wagon $= F + F_{th}$

$$m\frac{dv}{dt} = F - bv \qquad \text{...(i)}$$

$$\Rightarrow \quad (m_0 + bt)\frac{dv}{dt} = F - bv$$

$$\Rightarrow \quad \int_0^v \frac{dv}{F - bv} = \int_0^t \frac{dt}{m_0 + bt}$$

$$\Rightarrow -\frac{1}{b}[ln(F - bv)]_0^v = \frac{1}{b}[ln(m_0 + bt)]_0^t$$

$$\Rightarrow \quad ln\left(\frac{F}{F - bv}\right) = ln\left(\frac{m_0 + bt}{m_0}\right)$$

$$\Rightarrow \quad \frac{F}{F - bv} = \frac{m_0 + bt}{m_0}$$

$$\Rightarrow \quad v = \frac{Ft}{m_0 + bt}$$

$$\text{From (i) } \frac{dv}{dt} = \frac{F - bv}{m} = \frac{F - \dfrac{bFt}{m_0 + bt}}{m_0 + bt}$$

$$\therefore \quad a = \frac{Fm_0}{(m_0 + bt)^2}$$

23. Retardation of 2 kg block $a = \mu g = 2\,ms^{-2}$

Velocity of 2 kg block just before collision is given by $v_0^2 = u^2 - 2ax$

$$\Rightarrow \quad v_0^2 = 1^2 - 2 \times 2 \times 0.16 \Rightarrow v_0 = 0.6 \text{ ms}^{-1}$$

Let v_1 and v_2 be velocities of 2 kg and 4 kg blocks immediately after collision.

$$2v_1 + 4v_2 = 2 \times 0.6$$

$$\Rightarrow \quad v_1 + 2v_2 = 0.6 \quad \text{...(i)}$$

$$\text{And} \quad -\left(\frac{v_1 - v_2}{0.6 - 0}\right) = 1 \Rightarrow v_2 - v_1 = 0.6 \quad \text{...(ii)}$$

Solving (i) and (ii) gives: $v_2 = 0.4\text{ ms}^{-1}$; $v_1 = -0.2\text{ ms}^{-1}$.

Retardation after collision $= \mu g = 2\text{ ms}^{-2}$

$\therefore$ Displacement (x_1) of 2 kg block toward left after collision is given as

$$0^2 = 0.2^2 - 2 \times 2 \times x_1 \Rightarrow x_1 = 0.01\,\text{m}$$

Displacement (x_2) of 4 kg block towards right after collision is given as:

$$0^2 = 0.4^2 - 2 \times 2 \times x_2 \Rightarrow x_2 = 0.04\text{ m}$$

$\therefore$ Final separation $= x_1 + x_2 = 0.05\,m = 5\,\text{cm}$

24. Let the astronaut recoil with speed v_1 and speed of the object be v_2.

$$\text{Given} \quad v_2 + v_1 = 12$$

$$\Rightarrow \quad v_2 = 12 - v_1$$

Momentum conservation gives:

$$5v_2 = 50\,v_1 \quad \Rightarrow \quad 12 - v_1 = 10\,v_1$$

$$\Rightarrow \quad v_1 = \frac{12}{11}\text{ ms}^{-1}$$

$$\text{And} \quad v_2 = 12 - \frac{12}{11} = \frac{120}{11}\text{ ms}^{-1}$$

Let the object and astronaut meet after time t. Displacement of astronaut $x_1 = v_1 t = \frac{12t}{11}$

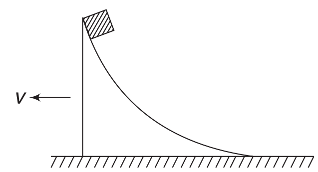
In this time the object travels a distance = 8 + 8 + x_1 with a constant speed v_2.

$$\therefore \quad 16 + x_1 = v_2 t$$

$$\Rightarrow \quad 16 + \frac{12t}{11} = \frac{120}{11}t \quad \Rightarrow \quad 16 = \frac{108}{11}t$$

$$\Rightarrow \quad t = \frac{16 \times 11}{108} = 1.6\,\text{s}$$

25. The block will stop climbing at the instant its velocity relative to the wedge becomes zero.



Let the velocity of the system (wedge + block) at this instant be $v(\leftarrow)$.

Conservation of momentum gives

$$(M + m)v = mu \quad \text{...(i)}$$

Energy conservation gives:

$$\frac{1}{2}(M + m)v^2 + mgh = \frac{1}{2}mu^2 \quad \text{...(ii)}$$

Solving (i) and (ii) for u and putting $M = \eta m$ gives

$$u = \sqrt{2gh\left(1 + \frac{1}{\eta}\right)}$$

Match the Column

1. (A) If $\frac{m_1}{m_2} \to 0 \quad \Rightarrow m_1 \to 0$

Then $v_{2f} \to 0$ and $v_{1f} = -v_{1i}$

$\therefore \quad \frac{v_{1f}}{v_{1i}} = -1$

when $\frac{m_1}{m_2} \to \infty \quad \Rightarrow \quad m_1 \to \infty$

then $v_{2f} \to 2v_{1i}$ and $v_{1f} = v_{1i}$

$\therefore \quad \frac{v_{1f}}{v_{1i}} = 1$

$\therefore$ Correct graph is (B)

(B) From above discussion, the answer is -1

(C) When $\frac{m_1}{m_2} \to \infty, \frac{v_{2f}}{v_{1i}} \to 2$

(D) $\frac{v_{2f}}{v_{1i}} \to 0$ if m_2 is massive $\Rightarrow$ when $\frac{m_1}{m_2} \to 0$

$\frac{v_{2f}}{v_{1i}} \to 2$ if m_2 is light $\Rightarrow$ when $\frac{m_1}{m_2}$ is large

3. (A) Sum of momentum of two pieces cannot be towards right

(B) Here also, sum of momentum cannot be towards right

(C) For momentum to be towards right,

$m_1v_1 > m_2v_2 \quad \Rightarrow m_1 > m_2$ [Since $v_1 < v_2$]

(D) In this case, resultant momentum can be towards right for $m_1 = m_2$ or $m_1 > m_2$ or $m_1 < m_2$.

4. Both energy and momentum are conserved. A has maximum speed in the initial state, as neither B nor the spring are having energy at this instant. Entire energy is KE of A.

At some point, A will come to rest, as it is slowing down. Its minimum speed is zero. At maximum compression, both blocks are having equal velocity of $\frac{u}{3}$.

KE can never be zero for the system, as it will mean that momentum has become zero.

5. Both the truck and the tempo apply equal force on one another. Due to large mass of the truck, its acceleration is smaller.

KE of the system decrease in collision.

KE of the temo cannot be determined using the given information. KE of the tempo can even increase if the initial velocity of the truck was very high.

6. (A) Exchange of velocity has taken place.

(B) Both objects stick and move together

(C) Velocity of body 2 is unchanged and that of body 1 has reversed.

(D) Speed of both gets reduced and direction of motion is reversed. There is loss in KE.

7. Slope of x-t graph gives velocity.

(A) Both fragments are moving in the same direction. Net momentum is not zero. An external impulsive force must have acted.

(B) The two pieces have equal and opposite velocities. If $m_A = m_B$, then momentum is conserved.

(C) Same as (A)

(D) $v_A > v_B$

For conservation of momentum, $m_Av_A = m_Bv_B$

Therefore, $m_A < m_B$

(E) Similar to (D)

(F) Net momentum is not zero.

8. In (C), the monkey and the block have the same mass. If the monkey accelerates up, the block will also accelerate up, as the same force acts on both of them.

In (D), both the blocks will always move with equal and opposite velocities.

Passage-based Problems

Passage 1

1. The friction forces on the two blocks are equal and opposite. Their resultant is zero. Thus, linear momentum of the system of two blocks is conserved.

When one of the blocks stops, friction stops acting on it. The right block stops after 5 s. After this, friction acts only on the left block and is directed leftward. Now, momentum of the system is not conserved.

2. COM stops moving after both the blocks stop.

Retardation of each block, $a = 2\,\text{ms}^{-2}$

Right blocks stops at $t = 5$ s and the left block stops at $t = 10$ s.

Initial velocity of COM,

$$u = \frac{20 \times 1 - 10 \times 1}{2} = 5\,\text{ms}^{-1}\ (\rightarrow)$$

Displacement of COM in 5 s is $x_1 = 5 \times 5 = 25$ m.

Retardation of COM between $t = 5$ and $t = 10$ s is

$$a_{CM} = \frac{1\times 2 + 0}{2} = 1\,\text{ms}^{-2}$$

Further displacement of COM before it stops is given by

$$x_2 = 5 \times 5 - \frac{1}{2} \times 1 \times 5^2 = 12.5\,\text{m}$$

$\therefore$ Total displacement of COM = 25 + 12.5 = 37.5 m.

Passage 2

3. Let leftward direction be positive

$$v_b - v_4 = -7$$

$$v_b = v_4 - 7$$

Spring does not exert any force, as it is relaxed. During the event of jump, there is no compression in the spring. Momentum of the 4 kg block and the boy remains zero during the jump. Thus,

$$0 = 10(v_4 - 7) + 4v_4$$

$$v_4 = \frac{70}{14} = 5 \text{ and } v_b = -2.0\ \text{ms}^{-1}$$

4. When velocity of 1 kg block is maximum, its acceleration will be zero.

$\Rightarrow$ spring will be in its natural length.

Conservation of momentum:

$$4 \times 5 = 1v_1 + 4v_4 \quad \Rightarrow v_4 = \frac{20 - v_1}{4} \rightarrow$$

Conservation of energy:

$$\frac{1}{2} \times 4 \times (5)^2 = \frac{1}{2} \times 1 \times v_1^2 + \frac{1}{2} \times 4 \times v_4^2$$

$$100 = v_1^2 + \frac{1}{4}(400 + v_1^2 - 4v_1) \quad \Rightarrow \quad \frac{5v_1^2}{4} = v_1$$

$$\Rightarrow v_1 = 0.8\,\text{ms}^{-1}$$

Passage 3

5. Initially, the cart moves to right with velocity v_0 and the bead is at rest. Velocity of COM of the system is

$$v_c = \frac{Mv_0}{m + M}$$

First collision occurs when the cart moves through a distance L. The bead hits the left wall.

Conservation of momentum, for first collision, gives:

$$Mv_0 = Mv_1 + mv_2 \qquad \text{...(1)}$$

Here, v_1 and v_2 are velocities of the cart and the bead immediately after collision.

Coefficient of restitution is 1: $\dfrac{v_2 - v_1}{v_0} = 1$ ...(2)

Solving (1) and (2) gives:

$$v_1 = \frac{M - m}{M + m}v_0 = -\frac{m - M}{M + m}v_0;$$

$$v_2 = \frac{2M}{m + M}v_0$$

$$\therefore \quad v_{1c} = v_1 - v_c = -\frac{mv_0}{m + M} \quad \text{and}$$

$$v_{2c} = v_2 - v_c = \frac{Mv_0}{m + M}$$

6. Relative speed after collison = Relative speed before collision

$$\therefore \quad t_2 - t_1 = \frac{L}{v_0}$$

7. Distance travelled by the cart till first collision = L

Distance travelled between first and second collision = $(t_2 - t_1)\ |v_1|$

$$= \frac{L}{v_0}\ \frac{m - M}{M + m}v_0 = \frac{m - M}{M + m}L$$

Total distance is obtained by adding the above two distances.

Passage 4

8. As the cylinder moves down, the vertical component of the momentum of the system changes. At the bottom, the cylinder only has horizontal momentum. And momentum is conserved in horizontal direction.

9. Initial energy of the system = mgR

final energy when the cylinder reaches the bottom of the track = $\frac{1}{2}mv^2 + \frac{1}{2}Mv_1^2$

where v (towards right) is the velocity of 'm' and v_1 (towards left) is the velocity of 'M' relative to the ground.

$$\therefore \quad mg(R - r) = \frac{1}{2}mv^2 + \frac{1}{2}Mv_1^2 \qquad \text{...(i)}$$

Initial momentum of the system = 0

Final momentum when cylinder has reached bottom of the track $B = mv - Mv_1$

$$\therefore \quad mv - Mv_1 = 0 \qquad \text{...(ii)}$$

$\therefore \qquad v_1 = \dfrac{mv}{M}$ Substituting in (i) gives

$$mg(R - r) = \frac{1}{2}[m][v]^2 + \frac{1}{2}[M]\left[\frac{mv}{M}\right]^2$$

$$= \frac{1}{2}mv^2 + \frac{1}{2}\frac{m^2v^2}{M} = \frac{v^2}{2}\left[m + \frac{m^2}{M}\right]$$

or,
$$v^2 = \frac{2g(R-r)}{\left[1+\dfrac{m}{M}\right]} = \left[\frac{2gM(R-r)}{M+m}\right]$$

$$v = \left[\frac{2gM(R-r)}{M+m}\right]^{1/2}$$

Substituting in (ii) gives:

$$v_1 = \frac{m}{M}\left[\frac{2gM(R-r)}{M+m}\right]^{1/2} = m\left[\frac{2g(R-r)}{M(M+m)}\right]^{1/2}$$

Passage 5

10. Let the mass of the shell be $3m$. The mass of each fragment is m.

The particle with maximum velocity must be moving in the forward direction.

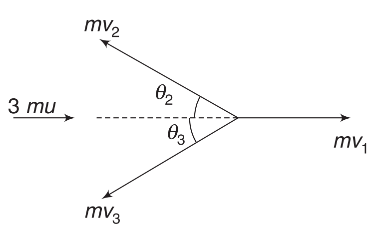
By law of conservation of momentum,

$$3mu = mv_1 - mv_2\cos\theta_2 - mv_3\cos\theta_3$$

$$3u = v_1 - v_2\cos\theta_2 - v_3\cos\theta_3$$

$$v_1 = 3u + v_2\cos\theta_2 + v_3\cos\theta_3 \qquad \text{...(i)}$$

Also, $mv_2\sin\theta_2 = mv_3\sin\theta_3$ ...(ii)



If v_1 is to be maximum,

$$\theta_2 = \theta_3 = 0$$

11. From (ii), if $\theta_2 = \theta_3$

$$v_2 = v_3 = v\,(\text{say})$$

Equation (i) becomes

$$v_1 = 3u + 2v$$

$$v = (v_1 - 3u)/2 \qquad \text{...(iii)}$$

from question,

$$k\frac{1}{2}(3m)u^2 = \left(\frac{1}{2}mv_1^2 + 2\times\frac{1}{2}mv^2\right)$$

$$3ku^2 = v_1^2 + 2v^2 \qquad \text{...(iv)}$$

Substituting for v from (iii)

$$3ku^2 = v_1^2 + \frac{1}{2}(v_1^2 + 9u^2 - 6v_1u)$$

Solving for v_1

$$v_1 = u[1 + \sqrt{2(k-1)}]$$

For $u = 500\,\text{ms}^{-1}$ and $k = 1.5$

$$v_1 = 500\,[1 + \sqrt{2(1.5-1)}] = 1000\,\text{ms}^{-1}$$

Passage 6

Sol: **12,13.** After time t mass of the car with water is $m_t = (m_0 + \mu t)$ kg. Let at that momentum speed of the car be v.

$$\therefore \qquad m_t\frac{dv}{dt} = F_{ext} + v_{rel}\frac{dm}{dt}$$

As $\quad (m_0 + \mu t)\dfrac{dv}{dt} = 0 - (u\sin\theta + v)\mu$

$$\Rightarrow \int_{v_0}^{v}\frac{dv}{(u\sin\theta + v)} = -\int_0^t \frac{dt}{\left(\dfrac{m_0}{\mu} + t\right)}$$

$$\Rightarrow \ln\left(\frac{u\sin\theta + v}{u\sin\theta + v_0}\right) = -\ln\left[\frac{\left(\dfrac{m_0}{\mu} + t\right)}{m_0/\mu}\right]$$

$$\Rightarrow \frac{u\sin\theta + v}{u\sin\theta + v_0} = \frac{m_0}{(m_0 + \mu t)}$$

$$\Rightarrow v = u\sin\theta\left(\frac{m_0}{m_0 + \mu t} - 1\right) + \frac{m_0v_0}{(m_0 + \mu t)}$$

$$\Rightarrow v = -u\sin\theta\left(\frac{\mu t}{m_0 + \mu t}\right) + \frac{m_0v_0}{(m_0 + \mu t)}$$

Passage 7

Sol: **14.** Let v be the leftward velocity of wagon (relative to earth). Let u be the velocity of pendulum in a frame fixed to the wagon. Then, $u\cos\beta$ is the relative horizontal velocity of the bob and $u\sin\beta$ is its vertical velocity. Let v_x and v_y be the absolute horizontal and vertical downward velocities of the bob.

$$\Rightarrow \qquad v_x = u\cos\beta - v \quad \text{and} \quad u\sin\beta = v_y$$

There is no external force on the system in the horizontal direction.

Therefore, by the principle of conservation of momentum,

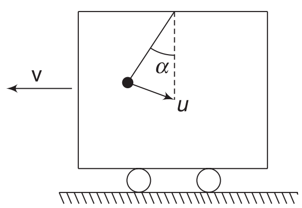
$$0 = m(u\cos\beta - v) - Mv$$

$$\Rightarrow \quad u\cos\beta - v = \frac{M}{m}\cdot v$$

$$\Rightarrow \quad u = \frac{(M+m)v}{m\cos\beta}$$

15. Kinetic energy of bob $= \frac{1}{2}m(v_x^2 + v_y^2)$

By the conservation of energy,



After releasing the bob

$$mgl(1-\cos\alpha) = mgl(1-\cos\beta) + \frac{1}{2}Mv^2 + \frac{1}{2}m\,[u\cos\beta - v)^2 + u^2\sin^2\beta]$$

or, $\quad 2mgl(\cos\beta - \cos\alpha)$

$$= Mv^2 + m\frac{M^2v^2}{m^2} + \frac{m\sin^2\beta(M+m)^2v^2}{m^2\cos^2\beta}$$

$$= Mv^2\left\{1+\frac{M}{m}\right\} + \frac{(M+m)^2v^2}{m\cos^2\beta}\sin^2\beta$$

$$= \frac{M(M+m)}{m}v^2 + \frac{(M+m)^2v^2\sin^2\beta}{m\cos^2\beta}$$

$$\Rightarrow \quad v^2 = \frac{2m^2gl}{M+m}\left[\frac{(\cos\beta-\cos\alpha)\cos^2\beta}{M+m\sin^2\beta}\right]$$

$$\therefore \quad v = \sqrt{\frac{2m^2gl}{M+m}\left[\frac{(\cos\beta-\cos\alpha)\cos^2\beta}{M+m\sin^2\beta}\right]}$$

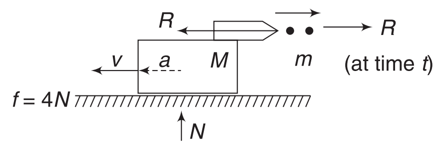
16. In this particular case, when $\beta = 0$,

$$v = \sqrt{\frac{2m^2gl}{M+m}\,\frac{(1-\cos\alpha)}{M}} = \sqrt{\frac{2m^2gl}{M+m}\,\frac{2\sin^2\frac{\alpha}{2}}{M}}$$

$$v = 2m\sin\frac{\alpha}{2}\sqrt{\frac{gl}{(M+m)M}}.$$

Passage 8

17. Each shot of mass m (say) leaves the cannon with a relative velocity u and a frequency n; the rate of loss of mass of the system is given as



$$r = \frac{dm}{dt} = mn$$

$\Rightarrow$ the thrust force, $R = u\frac{dm}{dt} = mnu$

For the cannon to accelerate, this force shall be larger than the friction

$$mnu > \mu M_0 g$$

$$\Rightarrow \quad n > \mu M_0 g/(mu)$$

18. The canon can recoil later, as its mass goes on decreasing and therefore, limiting friction force on it also goes on decreasing.

19. The net force acting on the system after a time t

$$\Rightarrow \quad F = R - f \quad \Rightarrow \quad m'a = mnu - \mu N$$

where f is the force of friction

where $\quad N = m'g$

Thus, $\quad m'a = mnu - \mu m'g \qquad$ ...(i)

m' = mass of the cannon with the shots remaining inside it after a time t

$m' = M - mnt \qquad$ ...(ii)

using (i) and (ii), $a = \dfrac{mnu}{M_0 - mnt} - \mu g$

Integrating both sides

$$\int_0^v dv = mnu\int_0^t \frac{dt}{M_0 - mnt} - \mu g\int_0^t dt$$

$$v = mnu\left[-\frac{1}{mn}ln\,|M_0 - mnt|\right]_0^t - \mu gt$$

$$\Rightarrow \quad v = -u\ln\left(\frac{M_0 - mnt}{M_0}\right) - \mu gt$$

$$= u\ln\left(\frac{M_0}{M_0 - mnt}\right) - \mu gt$$

Since after a time t, the cannon + remaining shots has mass M

$$\Rightarrow \quad M_0 - mnt = M$$

$$\Rightarrow \quad v = u\,ln(M_0/M) - \mu gt$$

Passage 9

20,21. Refer to Example 25 in the chapter.

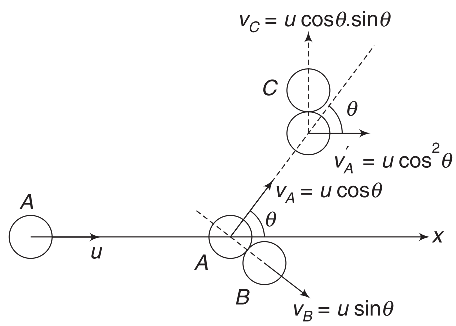
Let A get deflected by angle θ due to collision with B.

Velocity of B = velocity component of A along LOI, which gets imparted to B

$$v_B = u\sin\theta$$

Velocity of A = velocity component of $A \perp$ to LOI before collision

$$v_A = u\cos\theta$$



After hitting C, the disc A moves along x-direction. During collision, it will impart its velocity, along LOI, to C and retain its velocity perpendicular to LOI. It means that LOI is along y-direction. C begins to move in y-direction.

$$v_C = v_A\sin\theta = u\cos\theta\sin\theta$$

and $\quad v_A' = v_A\cos\theta = u\cos^2\theta$

Given, $\quad v_A' = \dfrac{u}{2} \Rightarrow \cos^2\theta = \dfrac{1}{2} \Rightarrow \cos\theta = \dfrac{1}{\sqrt{2}}$

$\therefore \quad \vec{v}_C = \dfrac{u}{2}\hat{j}$

$$\vec{v}_B = \frac{u}{\sqrt{2}}\left[\frac{1}{\sqrt{2}}\hat{i} - \frac{1}{\sqrt{2}}\hat{j}\right] = \frac{u}{2}\hat{i} - \frac{u}{2}\hat{j}$$

Passage 10

22. Lift force on helicopter = momentum imparted to blown air in unit time

$\therefore \quad Mg = V\rho\cdot u \quad \begin{bmatrix} V = \text{volume of air pushed} \\ \text{down per second} \end{bmatrix}$

$\Rightarrow \quad V = \dfrac{Mg}{\rho u}$

23. Volume of air pushed down in unit time can also be written as

$$V = \pi\left(\frac{d}{2}\right)^2\cdot u$$

For take off:

Momentum imparted in unit time = Mg

$$(\rho V)u = Mg$$

$$\Rightarrow \quad \frac{\rho\pi d^2}{4}u^2 = Mg \qquad \text{...(i)}$$

Power = KE imparted to air

$$= \frac{1}{2}(\rho V)u^2 = \frac{1}{2}\rho\cdot\frac{\pi d^2}{4}u^3$$

$$= \frac{\rho\pi d^2}{8}u^3 = \frac{\rho\pi d^2}{8}\left(\frac{4Mg}{\rho\pi d^2}\right)^{3/2}$$

$$= \frac{(Mg)^{3/2}}{d\sqrt{\pi\rho}}$$

Passage 11

24,25. Impulse of friction is

$$J_f = k\int v\,dt = k\cdot L$$

Speed of A when it reaches the bottom of the slide is $u = \sqrt{2gH}$

Momentum = $mu = m\sqrt{2gH}$

For the system to stop

$$J_f = m\sqrt{2gH}$$

$$kL = m\sqrt{2gH}$$

$$\Rightarrow \quad H = \frac{k^2L^2}{2gm^2}$$

Passage 12

Sol. The entire process is like a head-on collision of two blocks with coefficient of restitution = η

[Refer to Example 24 in the chapter]

$\therefore \quad v_B = \dfrac{m_1(1+\eta)}{m_1+m_2}u \quad$ [Ans. to Q.27]

At the instant of maximum deformation,

$$v_A = v_B = v\text{(Say)}$$

$$(m_1+m_2)v = m_1u \Rightarrow v = \frac{m_1u}{m_1+m_2}$$

Energy absorbed by the thread is

$$= \frac{1}{2}m_1u^2 - \frac{1}{2}(m_1+m_2)\left(\frac{m_1u}{m_1+m_2}\right)^2$$

$$= \frac{1}{2}m_1u^2\left[1 - \frac{m_1}{m_1+m_2}\right] = \frac{1}{2}\left(\frac{m_1m_2}{m_1+m_2}\right)u^2$$

Passage 13

Sol. Every time the disc climbs a slide (and slides down it), it imparts a KE to the slide. Height attained on B will be maximum when it climbs on it for the first time.

Let velocity of A be v_1 ($\leftarrow$) when C reaches the bottom. Let velocity of C be u_1 ($\rightarrow$) at this instant.

Conservation of momentum gives $Mv_1 = mu_1$ ...(i)

Conservation of energy: $\frac{1}{2}Mv_1^2 + \frac{1}{2}mu_1^2 = mgh$

$$\Rightarrow \quad Mv_1^2 + m\left(\frac{Mv_1}{m}\right)^2 = 2mgh$$

$$\Rightarrow \quad v_1 = \sqrt{\frac{2m^2gh}{M(M+m)}} \quad \text{and} \quad u_1 = \sqrt{\frac{2Mgh}{M+m}}$$

Now C climbs on B. At maximum height, C stop moving on B. It means both have same velocity; say v_2 ($\rightarrow$).

Momentum conservation: $(M+m)v_2 = mu_1$

$$\Rightarrow \quad v_2 = \left(\frac{m}{M+m}\right)\sqrt{\frac{2Mgh}{M+m}}$$

Energy conservation: $mgH + \frac{1}{2}(M+m)v_2^2 = \frac{1}{2}mu_1^2$

$$\Rightarrow \quad mgH = -\frac{1}{2}(M+m)\left(\frac{m}{M+m}\right)^2\frac{2Mgh}{M+m} + \frac{1}{2}m\left(\frac{2Mgh}{M+m}\right)$$

$$= \frac{Mmgh}{M+m}\left[1 - \frac{m}{(M+m)}\right]$$

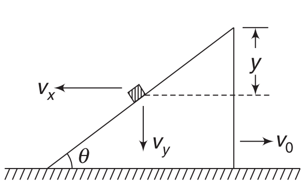
$$\Rightarrow \quad H = \frac{M^2}{(M+m)^2}\cdot h$$

Finally, the disc will slow down such that its speed will becomes slightly less than the wedge it is chasing. It will no longer be able to climb the wedge. At this instant, the entire energy will be shared between A and B as disc will have negligible KE (due to $m \ll M$).

$$\therefore \quad \frac{1}{2}Mv^2 + \frac{1}{2}Mv^2 = mgh \quad \Rightarrow \quad v = \sqrt{\frac{mgh}{M}}$$

Passage 14

Sol. Let velocity components of the block, after it has descended a distance y, be v_x and v_y, as shown. Velocity of wedge is v_0. We know that (from our discussion in constraint-related problems in the chapter of Newton's law of motion)



$$\frac{v_y}{v_x + v_0} = \tan\theta \quad \Rightarrow \quad v_y = \frac{1}{\sqrt{3}}(v_x + v_0) \quad \text{...(i)}$$

Conservation of momentum in horizontal direction gives

$$Mv_0 = mv_x \quad \Rightarrow \quad 4v_0 = v_x \quad \text{...(ii)}$$

From (i) and (ii),

$$v_0 = \frac{\sqrt{3}}{5}v_y \quad \text{and} \quad v_x = \frac{4\sqrt{3}}{5}v_y$$

Conservation of mechanical energy gives

$$\frac{1}{2}Mv_0^2 + \frac{1}{2}m(v_x^2 + v_y^2) = mgy$$

$$\Rightarrow \quad 4\left(\frac{\sqrt{3}}{5}v_y\right)^2 + \left(\frac{48}{25}v_y^2 + v_y^2\right) = 2gy$$

$$\Rightarrow \quad v_y^2 = \frac{10}{17}gy$$

$$\Rightarrow \quad v_y = \sqrt{\frac{10}{17}gy} \quad \Rightarrow \quad \frac{dy}{dt} = \sqrt{\frac{10}{17}gy}$$

$$\Rightarrow \quad \int_0^h \frac{dy}{\sqrt{y}} = \sqrt{\frac{100}{17}}\int_0^t dt$$

$$\Rightarrow \quad \sqrt{h} = \frac{5}{\sqrt{17}}\cdot t \quad \Rightarrow \quad t = \frac{\sqrt{0.5\times 17}}{5} = \frac{\sqrt{8.5}}{5}\text{ s}$$

Passage 15

There is no change is magnitude of horizontal component of velocity due to impacts.

33. $\therefore \quad T = \frac{2a}{u_x} \quad \Rightarrow \quad u_x = \frac{2a}{T}$

$$\Rightarrow \quad u\cos\theta = \frac{2a}{T} \quad \Rightarrow \quad u = \frac{2a}{T\cos\theta}$$

34. In this case, $T = \frac{a}{u_x} \quad \Rightarrow \quad u_x = \frac{a}{T}$

$$\Rightarrow \quad u = \frac{a}{T\cos\theta}$$

Passage 16

35. $\vec{v}_{CM} = \dfrac{30\hat{i} + 20\hat{j} - 20\hat{i} + 50\hat{i} - 50\hat{j}}{3}$

$$= (20\hat{i} - 10\hat{j})\text{ ms}^{-1}$$

$\vec{v}_{CM}$ will not change due to internal interactions.

$\therefore$ Velocity of shell before explosion = $\vec{v}_{CM}$

$\therefore$ KE of shell before explosion

$$= \frac{1}{2}Mv_{CM}^2 = \frac{1}{2} \times 3 \times (20^2 + 10^2) = 750\,\text{J}$$

36. KE at given instant is

$$K = \frac{1}{2} \times 1 \times (30^2 + 20^2) + \frac{1}{2} \times 1 \times (20^2) + \frac{1}{2} \times 1 \times (50^2 + 50^2)$$

$$= 3{,}350\,\text{J}$$

KE at given instant is higher than initial KE by $3340 - 750 = 2600\,\text{J}$.

37. $$\text{KE} = \text{KE}_{\text{wrt COM}} + \frac{1}{2}Mv_{CM}^2$$

$\text{KE}_{\text{wrt COM}}$ can become zero if all the particles stop moving due to attraction. But, $\frac{1}{2}\text{Mv}_{CM}^2$ will remain unchanged

$\therefore$ Minimum KE possible after explosion = $\frac{1}{2}\text{Mv}_{CM}^2$

$$= 750\text{ J}$$

Passage 17

38. Let v_1 and v_2 be velocities of the flying object and block B just after collision.

$$mu = mv_1 + 4mv_2 \Rightarrow u = v_1 + 4v_2 \qquad \text{...(i)}$$

Since, collision is elastic

$$\therefore \quad -\left(\frac{v_1 - v_2}{u - 0}\right) = 1 \Rightarrow v_2 - v_1 = u \qquad \text{...(ii)}$$

Solving (i) and (ii); $v_1 = -\frac{3u}{5}$, $v_2 = \frac{2u}{5}$ ...(iii)

After collision, B begins to move with velocity v_2. The upper block A does not move. There is no horizontal force on it!

B retards due to friction from the floor.

Friction on $B = \mu(4m + 2m)g = 6\,\mu mg$

Retardation of $B = \frac{6\mu mg}{4m} = \frac{3}{2}\mu g$

$\therefore$ Displacement of B till it stops is given by

$$v^2 = u^2 + 2ax$$

$$\Rightarrow \quad 0 = \left(\frac{2u}{5}\right)^2 - 2\left(\frac{3}{2}\mu g\right)x$$

$$\Rightarrow \quad x = \frac{4}{75}\frac{u^2}{\mu g}$$

For A to topple, $x > 2d$

$$\Rightarrow \quad \frac{4}{75}\frac{u^2}{\mu g} > 2d$$

$$\Rightarrow \quad u > \frac{5}{2}\sqrt{6\mu gd}$$

$$\therefore \quad u_{\min} = u_0 = \frac{5}{2}\sqrt{6\mu gd}$$

39. When $u = 2u_0 = 5\sqrt{6\mu gd}$, the flying object will have velocity towards left (after collision) given by (iii) as

$$v_1 = \frac{3u}{5} = 3\sqrt{6\mu gd}$$

It will fly like a horizontal projectile and its range will be

$$x = v_1\sqrt{\frac{2d}{g}} = 3\sqrt{6\mu gd}\sqrt{\frac{2d}{g}} = 6d\sqrt{3\mu}$$

Passage 18

Sol. Velocity of approach $= v\cos\theta = 10 \times \frac{4}{5} = 8\,\text{ms}^{-1}$

Relative speed of separation $= e \times 8 = 4\,\text{ms}^{-1}$

Let velocity of Disc A (after collision with the ball) be v_1 along LOI and velocity of ball be v_2 along LOI.

$$v_1 - v_2 = 4 \qquad \text{...(i)}$$

Conserving momentum along LOI

$$Mv_1 + Mv_2 = M \times 8$$

$$\Rightarrow \quad v_1 + v_2 = 8 \qquad \text{...(ii)}$$

Solving (i) and (ii) $v_1 = 6\,\text{ms}^{-1}$, $v_2 = 2\,\text{ms}^{-1}$

Velocity of ball along t-line remains unchanged

$$v_t = v\sin 37° = 10 \times \frac{3}{5} = 6\,\text{ms}^{-1}$$

$$\therefore \quad v_{\text{ball}} = \sqrt{2^2 + 6^2} = \sqrt{40}\ \text{ms}^{-1}$$

Disc A hits head-on, disc B with a velocity of $v_1 = 6\,\text{ms}^{-1}$ and the collision is elastic.

$\therefore$ Disc B acquires a velocity of $6\,\text{ms}^{-1}$ and A comes to rest.

Chapter 4 Torque and Equilibrium

Your Turn

1. $\vec{r} = (2\hat{i} + 2\hat{j})m;\ \vec{F} = 3\hat{j}$

   $\vec{\tau} = \vec{r} \times \vec{F} = 6\hat{k}$ N-m.

2. Component of force perpendicular to OP is $F\cos 30° = 10\sqrt{3}$ N.

   $\therefore\quad \tau = dF_{\perp} = 1 \times 10\sqrt{3} = 10\sqrt{3}$ N-m $(\circlearrowright)$

   OR, $\vec{r} = \overrightarrow{OP}$

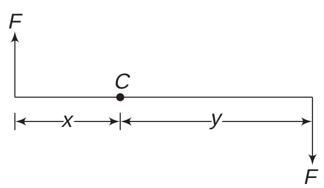
   Angle between $\vec{r}$ and $\vec{F}$ is 60°.

   $\therefore\quad \tau = rF\sin 60° = 1 \times 20 \times \dfrac{\sqrt{3}}{2} = 10\sqrt{3}$ N-m

3. Torque about C:

   $$\tau = Fx + Fy\ (\circlearrowright)$$
   $$= F(x + y) = FL\ (\circlearrowright)$$

   Where L is the length of the rod. Thus, value of torque is independent of choice of point C.

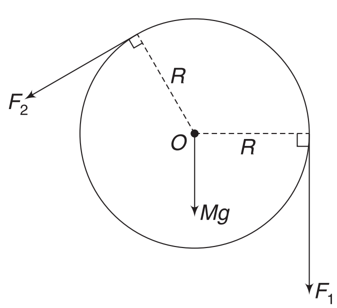


4. Torque due to string tension F_1 is $\tau_1 = F_1R\ (\circlearrowright)$

   Torque due to F_2 is $\tau_2 = F_2R\ (\circlearrowleft)$

   Torque due to Mg is zero as line of Mg intersects the axis at O. The axis (the support) at O must by applying a force on the pulley. But this force is acting at O. Whatever be its direction, it has no torque.

   $\therefore$ Total Torque = 0.



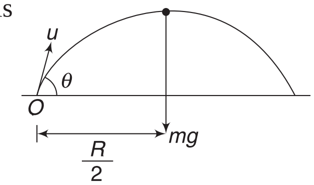
5. $\tau_1 = F_1(AD) = 20$ N-m, directed along negative z [This is same as saying clockwise if the observer is looking downwards]

   $$\tau_2 = F_2(AB) = 40 \text{ N-m along negative } z.$$
   $$\tau_3 = 0;\ \tau_4 = 0;\ \tau_5 = 0$$

   $\therefore\quad \tau = 60$ N-m along negative z.

6. Torque of mg about O is

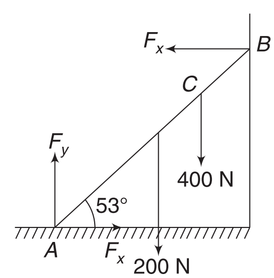
   $$\tau = mg \cdot \frac{R}{2}$$
   $$= mg\,\frac{u^2 \sin 2\theta}{2g}$$
   $$= mu^2 \sin\theta \cos\theta$$



7. Friction produces an anti-clockwise torque about C. To balance this, N must produce a clockwise torque about C. The correct line for N is 3.

8. For horizontal equilibrium, it is necessary that horizontal force at A = force at B $(=F_x)$. For vertical equilibrium $F_y = 600$ N

   $$\tau_A = 0$$
   $$F_x(6\sin 53°) = 200(3\cos 53°) + 400(4\sin 53°)$$
   $$\Rightarrow\quad F_x = 275\text{ N}$$



9. For horizontal equilibrium,

   $$F_H = T\cos 37° \qquad ...(i)$$

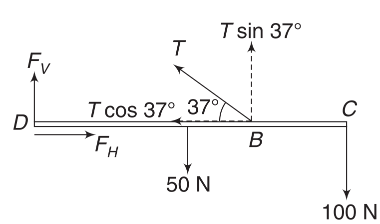
   For vertical equilibrium,

   $$F_V + T\sin 37° = 150 \qquad ...(ii)$$

   Taking torque about D to be zero

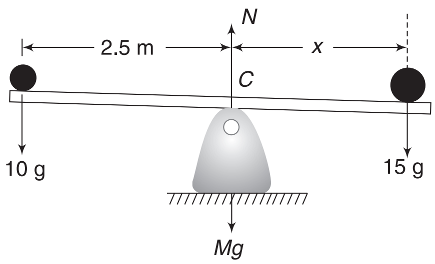
   $$(T\sin 30°) \times 1 = 50 \times 0.7 + 100 \times 1.4 \qquad ...(iii)$$

   Solving (i), (ii) and (iii) gives the answer.



10. Balancing torque about the centre point C,

    $$15g(x) = 10g(2.5)$$
    $$\Rightarrow\quad x = \frac{5}{3}\text{ m}$$



11. $N_1 + N_2 = Mg$

$$\Rightarrow \quad 30g + 25g = Mg$$

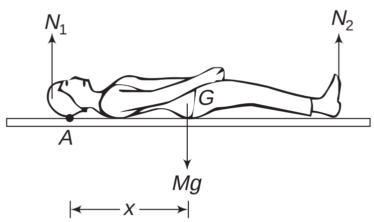
$$\Rightarrow \quad 55g = Mg$$

$$\tau_A = 0$$

$$\Rightarrow \quad Mg \cdot x = N_2(160\,\text{cm})$$

$$\Rightarrow \quad x = \frac{25}{55} \times 160\,\text{cm} = 72.7\,\text{cm}$$

$\therefore$ Required answer is $160 - 72.7 = 87.3\,\text{cm}$

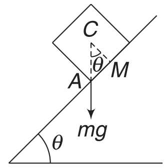


14. Taking the edge of the table to be origin, the COM of the system of two bricks will have its x co-ordinate given by

$$x_{CM} = \frac{M\left(-\frac{L}{4}\right) + M(0)}{2M} = -\frac{L}{8}$$

It means COM is well within the support region. The system will be in equilibrium.

15. Block will slide if $\tan\theta > \mu$. The cube will be on the verge of toppling if vertical line through its COM (C) passes through its edge A. In this case:



$$\tan\theta = \frac{AM}{CM} = 1$$

$$\Rightarrow \quad \theta = 45°$$

$\therefore$ Block will slide before toppling if $\tan\theta \geq \mu$ for $\theta < 45°$.

$$\Rightarrow \quad \mu \leq 1$$

16. Displacement of the two balls are related as:

$$m_1x_1 = m_2x_2$$

Since the COM of the system will not move in absence of external forces.

The COM of the entire system remains exactly above the pivot point.

Worksheet 1

1. Weight acts at COM. Its line is the rotation axis. It has no torque.
2. Resultant force is

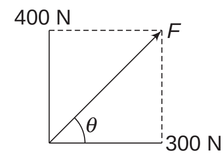
$$F = \sqrt{(300)^2 + (400)^2} = 500\,\text{N}$$

And $\tan\theta = \dfrac{4}{3} \Rightarrow \theta = 53°$

Torque about O is

$$\tau = (400R + 200R + 100R)(\circlearrowright) + 200R(\circlearrowleft)$$

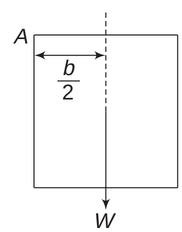
$$= 500R\ (\circlearrowright)$$



Force shown in option (C) will have exactly the same effect on the wheel as the given system of forces.

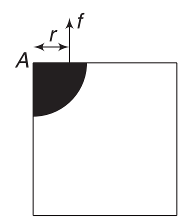
3. Torque applied by the person balances the torque due to weight about the corner.

$$\therefore \quad \tau = W\frac{b}{2}$$



Torque due to weight (about A) is clockwise. The person must produce an anti-clockwise torque.

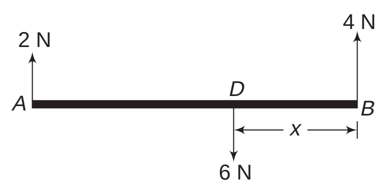
Note: Actually, the fingers apply large friction on the book and this friction causes a torque since size of fingers is not exactly zero. Shaded portion in the figure is thumb. It exerts friction (f), which produces a torque fr about A. The book has a tendency to rotate about a point that is slightly inside from corner A – somewhere around centre of the thumb.



4. For equilibrium, torque about any point = 0.

Let us take torque about B

$2 \times 3 = 6 \times x \Rightarrow x = 1\text{m}$



5. In option C:

$$\vec{r} = -b\hat{i} + o\hat{j} - c\hat{k}$$

$$\vec{F} = a\hat{j}$$

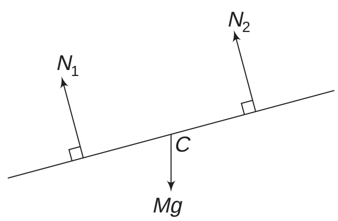
$$\therefore \quad \vec{\tau} = \vec{r} \times \vec{F} = -ab\hat{k} + ac\hat{i}$$

$$\therefore \quad \tau_z = -ab$$

6. Torque about the centre of the rod = 0

$$N_1 \frac{l}{4} = N_2 \frac{l}{4}$$

$$\Rightarrow \quad N_1 = N_2$$



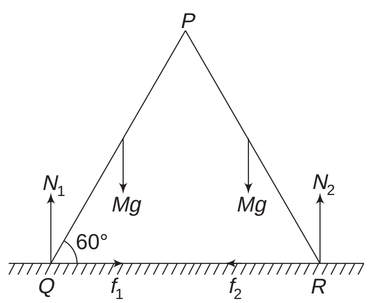
7. Balancing torque about COM of the beam

$$N_1\left(\frac{l}{2} - \frac{l}{4}\right) = N_2\left(\frac{l}{2} - \frac{l}{6}\right) \quad \Rightarrow \quad \frac{N_1}{N_2} = \frac{4}{3}$$

8. Due to symmetry: $N_1 = N_2$ and $f_1 = f_2$

$$N_1 + N_2 = 2Mg$$

$$\Rightarrow \quad N_1 = N_2 = Mg = 200\,\text{N}$$

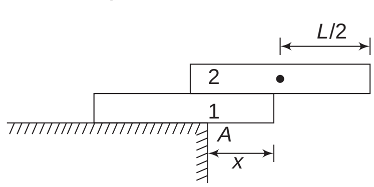


Considering rotational equilibrium of board PQ about P, gives

$$N_1 l\cos 60° = Mg\frac{l}{2}\cos 60° + f_1\, l\sin 60°$$

$$Mg\cdot\frac{1}{2} = \frac{Mg}{4} + f_1\frac{\sqrt{3}}{2} \quad \Rightarrow \quad f_1 = \frac{100}{\sqrt{3}}\ \text{N}$$

9. For Brick 2 to remain in equilibrium above Brick 1, the maximum overhang of 2 over 1 can be $\frac{L}{2}$. For the entire system to remain stable and not topple about A, COM of the two-brick system shall be to the left of edge A. In critical case, when COM is at A, taking A to be the origin, we can write



$$M\left(\frac{L}{2} - x\right) = M\cdot x \quad \Rightarrow \quad x = \frac{L}{4}$$

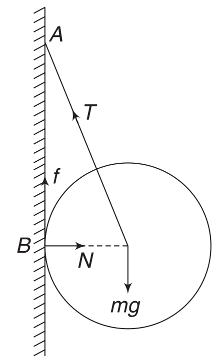
$$\therefore \quad S_{max} = \frac{L}{4} + \frac{L}{2} = \frac{3L}{4}$$

10. The COM will accelerate. We will take $\tau = 0$ about COM.

$$N_B\cdot a = N_A\cdot a \quad \Rightarrow \quad N_B = N_A$$

$$\therefore \quad N_B = N_A = \frac{W}{2}$$

11. Balancing torque about A gives



$$N(AB) = mgR$$

$$N\sqrt{(2R)^2 - R^2} = mgR$$

$$N = \frac{mg}{\sqrt{3}}$$

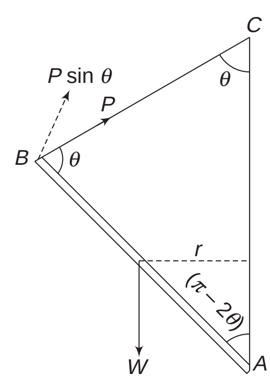
12. About A:

Torque due to W = torque due to P

$$W\cdot r = P\sin\theta\,(AB)$$

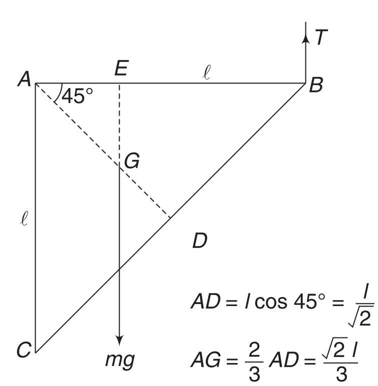
$$W\frac{AB}{2}\cdot\sin(\pi - 2\theta) = P\sin\theta\,(AB)$$

$$\Rightarrow \quad P = W\frac{\sin 2\theta}{2\sin\theta} = W\cos\theta$$



13. Torque about A is zero

$$mg\cdot(AE) = T\cdot(AB)$$



$$mg \cdot (AG \cos 45°) = T \cdot l$$

$$\Rightarrow \quad mg\left(\frac{\sqrt{2}l}{3} \cdot \frac{1}{\sqrt{2}}\right) = T.l$$

$$\Rightarrow \quad T = \frac{mg}{3}$$

For translational equilibrium, hinge at A must apply a vertical force (F) such that

$$F + T = mg \quad \Rightarrow \quad F = \frac{2mg}{3}$$

14. For translational equilibrium:

$$N = Mg \cos 30° \qquad \text{...(i)}$$

and $\quad T + f = Mg \sin 30° \qquad$ ...(ii)

Torque about $C = 0$

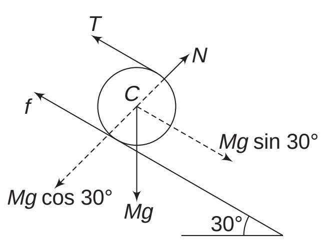
$$\Rightarrow \quad TR = fR \quad \Rightarrow \quad T = f$$

From (ii), $\quad f = \frac{1}{2} Mg \sin 30°$

But $\quad f \leq \mu N$

$$\therefore \frac{1}{2} Mg \sin 30° \leq \mu \, Mg \cos 30°$$

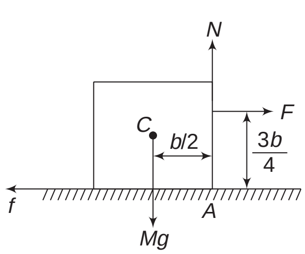
$$\Rightarrow \frac{1}{2} \tan 30° \leq \mu \quad \Rightarrow \quad \frac{1}{2\sqrt{3}} \leq \mu$$



15. Vertical line through COM will fall outside the base region if the system shown in (C) is tilted slightly towards right.

16. Just when the cube is about to rotate about A, the normal force will pass through point A. In critical case, torque of F about A is equal to torque due to Mg about A. [Friction has no torque about A]

$$\therefore \quad F \cdot \frac{3b}{4} = Mg \frac{b}{2} \quad \Rightarrow \quad F = \frac{2Mg}{3}$$



If the cube is not to slide when force $F = \frac{2}{3} Mg$ is applied,

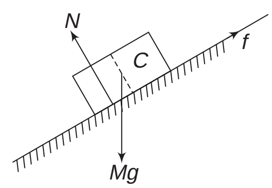
$$f = F$$

$$\Rightarrow \quad f = \frac{2}{3} Mg$$

But $\quad f \leq \mu N$

$$\therefore \quad \frac{2}{3} Mg \leq \mu \, Mg \quad \Rightarrow \quad \frac{2}{3} \leq \mu$$

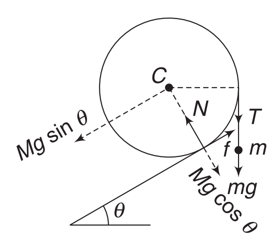
17. To balance torque due to friction (f) about C, the effective normal force must pass through left of C. Thus, the left half exerts more force on the incline compared to its right half.



18. Lowering the centre of gravity is the basic physical principle that is used in the show. If the pole is heavy and flexible, the COM of the system of the man + pole can be below the rope as well.

19. String tension $T = mg$

Component of 'T' parallel to incline plane will be $T \sin \theta = mg \sin \theta$.



[This is not shown in the figure]

Net force on cylinder along the incline is zero.

$$\therefore \quad f = Mg \sin \theta + mg \sin \theta \qquad \text{...(i)}$$

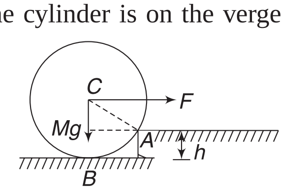
For zero torque about C

$$T \cdot R = f \cdot R \quad \Rightarrow \quad T = f (= mg)$$

Substituting in (i) gives; $mg = Mg \sin \theta + mg \sin \theta$

$$\Rightarrow \quad m = \frac{M \sin \theta}{1 - \sin \theta}$$

20. When the torque due to F about A just becomes equal to the torque due to Mg, the cylinder is on the verge of moving. Normal force at $B = 0$ and normal force and friction at A are of no concern if we are considering torque about A.

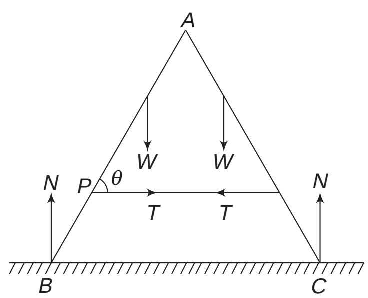


Worksheet 2

1. For vertical equilibrium of entire system, $2N = 2W$

   $\Rightarrow \qquad N = W$

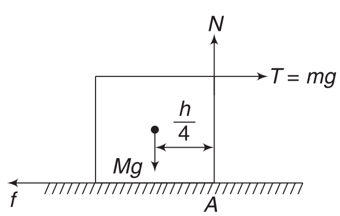
   Considering equilibrium of part AB only, we can easily see that there will be a horizontal force on it at A and it will be equal to T acting towards left. As we have already concluded that $N = W$, there cannot be a vertical force on AB at A. Value of T will depend on the distance of Point P from A. Larger the distance AB, smaller will be T.



2. $T = mg$.

   As mg increases, tension produces more and more torque about A. Normal force moves to right and in extreme case (when the block is about to topple), the contact is only at A. Normal force acts at A. Cylinder will not tilt as long as:

$$Mg\,\frac{h}{4} \geq T\cdot h \quad \Rightarrow \quad \frac{Mg}{4} \geq mg \quad \Rightarrow \quad \frac{M}{4} \geq m$$



3. For equilibrium, $F_{net} = 0$ and $\tau_{net} = 0$

Worksheet 3

1. $m_1(3l) = m_2 l \quad \Rightarrow \quad 3m_1 = m_2$

   $(m_1 + m_2) = m_3 l \quad \Rightarrow \quad (4m_1)(3l) = m_3 l$

   $\Rightarrow \quad 12m_1 = m_3$

   And $(m_1 + m_2 + m_3)(3l) = m_4 l$

   $\Rightarrow (m_1 + 3m_1 + 12m_1)(3l) = m_4 l$

   $\Rightarrow \qquad 48m_1 = m_4$

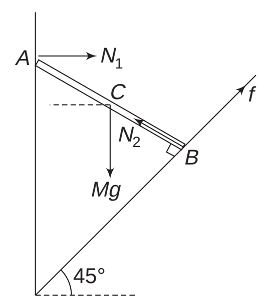
   $\Rightarrow \qquad m_1 = \dfrac{96}{48} = 2\,\text{kg}$

2. $\tau_A = 0$

$$f(AB) = Mg(AC\sin 45°)$$

$$f(AB) = Mg\left(\frac{AB}{2}\cdot\frac{1}{\sqrt{2}}\right)$$

$$\Rightarrow \qquad f = \frac{Mg}{2\sqrt{2}} \qquad \text{...(i)}$$



Net force in vertical direction = 0

$\Rightarrow N_2\cos 45° + f\sin 45° = Mg$

$$\Rightarrow \qquad \frac{N_2}{\sqrt{2}} + \frac{Mg}{4} = Mg$$

$$\Rightarrow \qquad N_2 = \frac{3Mg}{2\sqrt{2}}$$

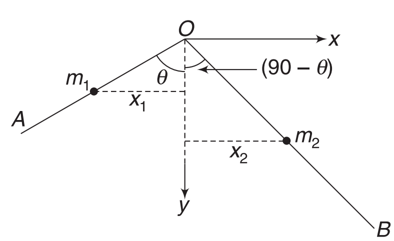
Since $\qquad f \leq \mu N_2$

$$\therefore \qquad \frac{Mg}{2\sqrt{2}} \leq \mu\cdot\frac{3Mg}{2\sqrt{2}} \quad \Rightarrow \quad \frac{1}{3} \leq \mu$$

3. The COM lies vertically below the peg. If linear mass density $= \lambda$, then

   mass of OA is $m_1 = \lambda L$

   mass of OB is $m_2 = 2\lambda L$



For COM to lie on y axis:

$$m_1x_1 = m_2x_2$$

$$(\lambda L)\left(\frac{L}{2}\sin\theta\right) = (\lambda 2L)\left(\frac{2L}{2}\cdot\cos\theta\right)$$

$\Rightarrow$ $\tan\theta = 4$

$\Rightarrow$ $\theta = \tan^{-1}4$

4. For horizontal equilibrium:

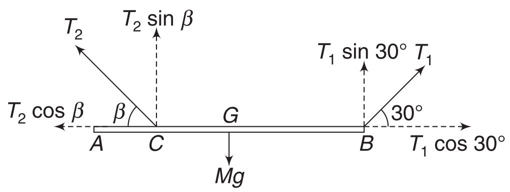
$$T_2\cos\beta = T_1\cos 30°$$

$\Rightarrow$ $T_2\cos\beta = \frac{\sqrt{3}}{2}T_1$ ...(i)

For vertical equilibrium:

$$T_1\sin 30° + T_2\sin\beta = Mg$$

$\Rightarrow$ $\frac{T_1}{2} + T_2\sin\beta = Mg$ ...(ii)



Taking torque about C to be zero:

$$(T_1\sin 30°)\left(\frac{3L}{4}\right) = Mg\cdot\frac{L}{4}$$

$\Rightarrow$ $T_1 = \frac{2Mg}{3}$ ...(iii)

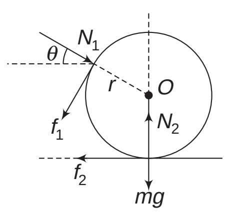
Equation (i) becomes $T_2\cos\beta = \frac{Mg}{\sqrt{3}}$ ...(iv)

Equation (ii) becomes $T_2\sin\beta = \frac{2Mg}{3}$ ...(v)

Dividing (v) by (iv) gives

$$\tan\beta = \frac{2}{\sqrt{3}}$$

6. Fig. shows forces on the smaller roller.



In critical case, frictions f_1 and f_2 are

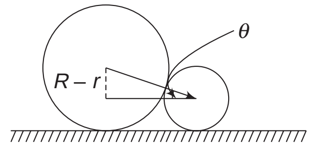
$$f_1 = \mu N_1 \quad \text{and} \quad f_2 = \mu N_2$$

For horizontal equilibrium:

$N_1\cos\theta - \mu N_2 - \mu N_1\sin\theta = 0$ ...(i)

Taking torque about centre (O) to be zero

$\mu N_1 r - \mu N_2 r = 0 \quad \Rightarrow \quad N_1 = N_2$ ....(ii)



$$\sin\theta = \frac{R-r}{R+r}$$

From (i) and (ii): $\mu N_1 + \mu N_1\sin\theta = N_1\cos\theta$

$\Rightarrow$ $\mu(1 + \sin\theta) = \cos\theta$ ...(iii)

From geometry: $\sin\theta = \frac{R-r}{R+r}$

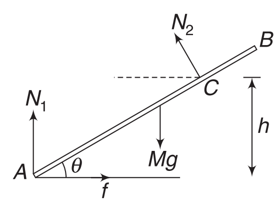
$\therefore$ $\cos\theta = \sqrt{1-\sin^2\theta} = \frac{2\sqrt{Rr}}{R+r}$

$\therefore$ Equation (iii) becomes

$$\mu\left(1 + \frac{R-r}{R+r}\right) = \frac{2\sqrt{Rr}}{R+r}$$

$\Rightarrow$ $\mu = \sqrt{\frac{r}{R}}$

7. There is no friction between roller and the rod. The normal force by the roller on the rod is perpendicular to the rod. Write equation of horizontal and vertical equilibrium



$f = N_2\sin\theta$ ...(i)

$N_1 + N_2\cos\theta = Mg$ ...(ii)

$\boldsymbol{\tau_A = 0}$ gives

$$N_2(AC) = Mg\frac{L}{2}\cos\theta$$

$$N_2\frac{h}{\sin\theta} = Mg\frac{L}{2}\cos\theta$$

$\Rightarrow$ $N_2 = \frac{MgL}{2h}\sin\theta\cdot\cos\theta$

Putting this value in (i) and (ii) gives:

$$f = \frac{MgL}{2h}\sin^2\theta\cos\theta \quad \text{and}$$

$$N_1 = Mg\left[1 - \frac{L}{2h}\sin\theta\cos^2\theta\right]$$

Since $f \le \mu N_1$

$\therefore$ $\frac{L}{2h}\sin^2\theta\cdot\cos\theta \le \mu\left(1 - \frac{L}{2h}\sin\theta\cdot\cos^2\theta\right)$

$\Rightarrow$ $\mu \ge \frac{L\sin^2\theta\cdot\cos\theta}{2h - L\sin\theta\cdot\cos^2\theta}$

8. For equilibrium of M,

$$T = Mg$$

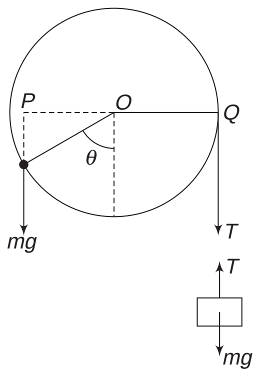
For torque on pulley to be zero about its centre O,

$$T(OQ) = mg(OP)$$

$$TR = mg\,R\sin\theta$$

$$\Rightarrow \quad Mg = mg\sin\theta$$

$$\Rightarrow \quad \frac{m}{M} = \frac{1}{\sin\theta}$$



9. There is no horizontal or vertical acceleration.

$$\therefore \quad N = Mg$$

and $\quad f = F$

$$\Rightarrow \quad \mu N = F$$

$$\Rightarrow \quad \mu Mg = F$$

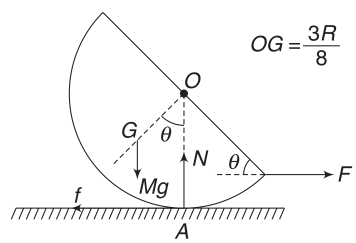
For rotational equilibrium: torque about $A = 0$

$$\Rightarrow \quad Mg(OG\sin\theta) = F(R - R\sin\theta)$$

$$\Rightarrow \quad Mg\frac{3R}{8}\sin\theta = \mu MgR(1 - \sin\theta)$$

$$\Rightarrow \quad 3\sin\theta = 8\mu - 8\mu\sin\theta$$

$$\Rightarrow \quad \mu = \frac{3\sin\theta}{8(1 - \sin\theta)}$$



Chapter 5 Kinematics of Rotation

Your Turn

1. $a = \omega^2 R = (20)^2 \left(\dfrac{18 \times 2.54}{100}\text{ m}\right) = 182.9\text{ ms}^{-2}$

2. Let ω_1 and ω_2 be the angular speed of larger and smaller pulleys respectively.

   For no slipping, speed of a point on the circumference of the pulling = speed of the load.

   $\Rightarrow \quad \omega_1 r_1 = \omega_2 r_2 = v$

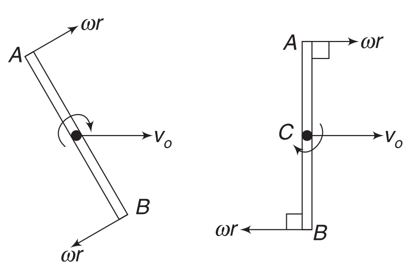
   $\Rightarrow \quad \omega_1(0.2) = \omega_2(0.1) = 10$

   $\Rightarrow \quad \omega_1 = 50\text{ rad s}^{-1}; \quad \omega_2 = 100\text{ rad s}^{-1}.$

3. $\theta = 2$ turns $= 4\pi$ radian

   $x = R\theta = 4\pi R$ meter

4. The stick must be vertical at the instant mentioned. If it is not so (as shown in first figure), then velocity of A or B cannot be zero since velocity of A or B is the vector sum of v_0 and ωr.



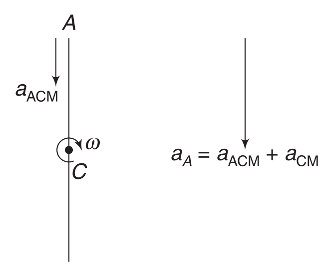
   $v_B = 0$ is possible if situation is as shown in the second figure. For v_B to be zero, $\omega r = v_0 = 2\text{ ms}^{-1}$

   $\therefore \quad v_A = v_0 + \omega r = 4\text{ ms}^{-1}$

5. In COM frame, the stick is rotating uniformly. Point A is going in a circle of radius $r (= CA = 1\text{m})$.

   $\therefore$ Acceleration of A wrt COM is

   $$a_{\text{ACM}} = \omega^2 r = 4^2 \times 1 = 4\text{ ms}^{-2}\ (\downarrow)$$



In COM frame A is performing uniform circular motion

   Acceleration of the COM is $g(\downarrow)$ as only external force on the stick is Mg and a_{CM} depends only on external force.

   $\therefore \quad \vec{a}_A = \vec{a}_{\text{ACM}} + \vec{a}_{\text{CM}}$

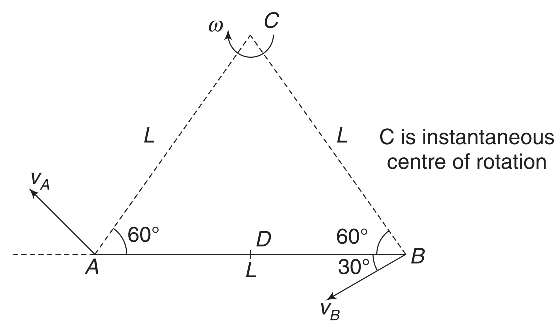
   $\Rightarrow \quad a_A = 4(\downarrow) + 10(\downarrow) = 14\text{ ms}^{-2}(\downarrow).$

6. Instantaneous centre of rotation lies at the point of intersection of perpendiculars drawn to $\vec{v}_A$ and $\vec{v}_B$. Obviously ΔABC is equilateral Δ.

   $$v_A = \omega(AC)$$

   $$v = \omega L \quad \Rightarrow \quad \omega = \frac{v}{L}$$

Note: Velocity of the centre of the rod (D) is $\omega(CD)$ along a direction perpendicular to CD; i.e., along BA.



7. $\omega R = 2 \times 0.5 = 1\text{ ms}^{-1}$

   $v > \omega R$

   Hence, the disc is slipping.

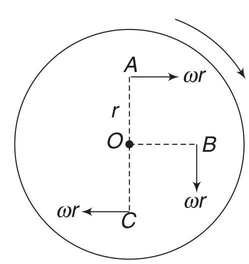
   Velocity of the contact point is $v - \omega R = 2\text{ ms}^{-1}$ in forward direction.

8. $\omega = \dfrac{v}{R} = \dfrac{4}{1} = 4\text{ rad s}^{-1}$

   $v_A = \omega r + v = 4 \times 0.5 + 4 = 6\text{ ms}^{-1}$

   $v_B = \sqrt{(\omega r)^2 + v^2} = \sqrt{2^2 + 4^2} = 2\sqrt{5}\text{ ms}^{-1}$

   $v_C = v - \omega r = 4 - 2 = 2\text{ ms}^{-1}.$



Velocities relative to O

9. Relative to centre, point A is performing uniform circular motion.

$$a_{AC} = \omega^2 r\ (\downarrow) = 4^2 \times 0.5 = 8\,\text{ms}^{-2}(\downarrow)$$

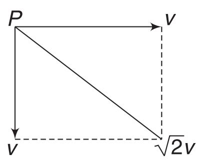
Acceleration of the centre is zero

$\therefore\quad a_A = 8\,\text{ms}^{-2}\ (\downarrow)$

10. (i) $v_{pc} = \omega R = v\ (\downarrow)$

$\vec{v}_p = \vec{v}_{pc} + \vec{v}_c$

$\therefore\quad v_P = \sqrt{2}\ v$



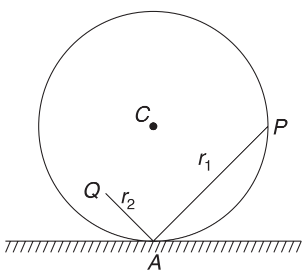
(ii) The wheel may be considered to be in pure rotation about the contact point (A).

Speed of P is $v_P = \omega r_1$

Speed of Q is $v_Q = \omega r_2$

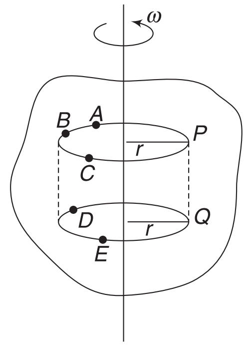
Since $r_1 > r_2$

$\therefore\quad v_P > v_Q$



Worksheet 1

1. Particles like P, A, B, C ... have the same speed. Particles like P and Q have the same velocity.



2. Refer to the figure in previous solution. In the plane represented by circle $ABCP$, no two particles can have the same velocity (direction and magnitude both same). Speed of all the particles on the cylinder $PABDEQP$ is same.
3. All points have velocity along tangents. Thus, body is in pure rotation about its centre. $\omega = \dfrac{v}{R}$
4. l is a line perpendicular to v_B and m is line perpendicular to v_A. Intersection of l and m at O is the instantaneous centre of rotation.

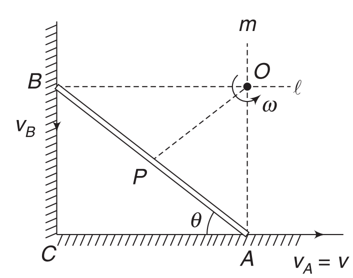
$$\omega(OA) = v_A$$

$$\omega\frac{L}{2} = v \quad\Rightarrow\quad \omega = \frac{2v}{L}$$

Speed of the centre (P) is $v_p = \omega(OP)$

$$= \frac{2v}{L}\cdot\frac{L}{2} = v$$

Note: Velocity of P is $\perp$ to OP.



5. It depends on the values of v and ωR.

If $v = \omega R$ there is no slipping.

If $v < \omega R$ then the contact point moves backwards, etc.

6. Velocity of P wrt centre (C) is $v_{pc} = \omega R = v$ in tangential direction Velocity of P in ground frames is

$$\vec{v}_P = \vec{v}_{PC} + \vec{v}_C$$

$$\therefore \quad v_P = \sqrt{v^2 + v^2} = \sqrt{2}v$$

And $\quad \tan\theta = \dfrac{v}{v} = 1 \quad \Rightarrow \quad \theta = 45°$

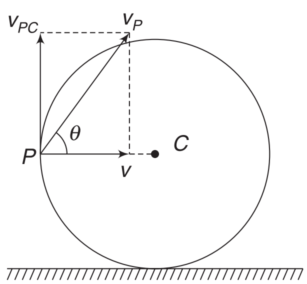
Acceleration of P wrt C is

$$a_{PC} = \omega^2 R = \frac{v^2}{R} \text{ towards centre } C.$$

Acceleration of C itself is zero

$$\therefore \quad \vec{a}_P = \vec{a}_{PC} + \vec{a}_C = \vec{a}_{pc}$$

$\therefore \quad \vec{a}_P$ is horizontally along PC.



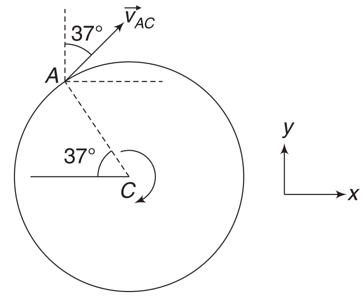
7. $v_{AC} = \omega R = v = 25\,\text{ms}^{-1}$

Direction of v_{AC} is as shown.

$$\vec{v}_{AC} = (25\sin 37°)\hat{i} + (25\cos 37°)\hat{j}$$
$$= 15\hat{i} + 20\hat{j}$$

Velocity of point A wrt ground is

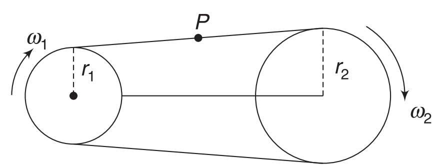
$$\vec{v}_{AC} = \vec{v}_{AC} + \vec{v}_C = 15\hat{i} + 20\hat{j} + 25\hat{i}$$
$$= (40\hat{i} + 20\hat{j})\,\text{ms}^{-1}.$$



8. COM will move in a circle with centre at O. Immediately after the string is cut, there is no speed. Thus, COM has no radial acceleration. COM has tangential acceleration, which is directed vertically down.

9. Speed of any point (like P) on the chain is $\omega_1 r_1$ or $\omega_2 r_2$

$$\therefore \quad \omega_1 r_1 = \omega_2 r_2$$
$$\omega_1(0.2) = (200 \text{ rpm})(0.5)$$
$$\Rightarrow \quad \omega_1 = 500 \text{ rpm}$$



10. Speed of a point on the circumference is $v = 2\ \text{ms}^{-1} = \omega R$. There is no acceleration of the block. It means there is no angular acceleration of the pulley. A point on the circumference has radial acceleration only.

$$a = \frac{v^2}{R} = \frac{2^2}{0.1} = 40\ \text{ms}^{-2}$$

Worksheet 2

1. $$\omega = \omega_0 + \alpha t$$

$$\Rightarrow \quad \omega = 2 + 2 \times 10 = 22 \text{ rad s}^{-1}$$

$$\theta = \omega_0 t + \frac{1}{2}\alpha t^2$$

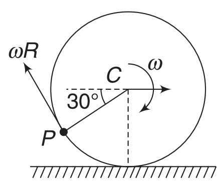
$$\Rightarrow \quad \theta = 2 \times 10 + \frac{1}{2} \times 2 \times 10^2 = 120 \text{ rad}$$

$$= \frac{120}{2\pi} \text{ rotations} \simeq 20$$

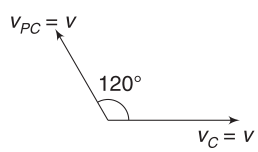
Angular velocity is same for all points.

Acceleration of a point is the vector sum of radial and tangential accelerations. It is not in radial direction.

2. Smallest speed is for the particle in contact with the ground. It is 0 in pure rolling. Highest speed is for the top point. It is $v + \omega R = 2v$ for a point on the top. Velocity of any point relative to C is ωR in tangential direction. For the point shown in figure $\vec{v}_{PC}$ is equal to $\omega R = v$ in the direction shown.



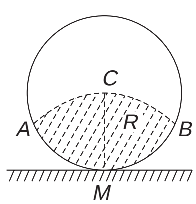
WR is velocity of *P* wrt C.



Velocity of P wrt ground is obtained by adding velocity of centre ($\vec{v}_C = \vec{v}$) to $\vec{v}_{PC}$. Obviously, speed of P is v.

3. The wheel can be thought to be in pure rotation about it contact point M. Speed of any point is ωr, where r is the distance from M.

ACB is an arc of radius R with centre at M. All particles in the shaded region will have speed less than $\omega R = v$.



4. Time taken by light to travel from the wheel to the mirror and back is

$$\Delta t = \frac{2L}{C} = \frac{2 \times 500}{3 \times 10^8} = \frac{1}{3} \times 10^{-5} \text{ s}$$

In this time interval, the wheel rotates through an angle

$$\Delta\theta = \frac{2\pi}{500} \text{ rad}$$

$$\therefore \quad \omega = \frac{\Delta\theta}{\Delta t} = \frac{2\pi}{500 \times \frac{1}{3} \times 10^{-5}}$$

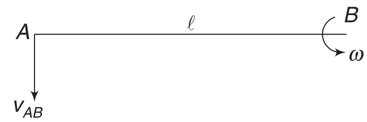
$$= 3.77 \times 10^3 \text{ rad s}^{-1} \simeq 3.8 \times 10^3 \text{ rad s}^{-1}$$

Linear speed of a point on the edge is

$$v = \omega R = 3.8 \times 10^3 \times 0.05$$

$$= 1.9 \times 10^2 \text{ ms}^{-1}$$

5. Separation between points A and B cannot change.



$$\therefore \quad v_{Ax} = 6 \text{ cms}^{-1}$$

Let velocity of A wrt $B = v_{AB}$

Then, $\omega = \dfrac{v_{AB}}{l}$

$$\Rightarrow \quad 0.4 = \frac{v_{AB}}{0.5} \quad \Rightarrow \quad v_{AB} = 0.2 \text{ ms}^{-1}.$$

$$v_{AB} = v_{Ay} - v_{By}$$

$$\Rightarrow 0.2\,(\downarrow) = 0.1 \text{ ms}^{-1}(\downarrow) - v_{By}$$

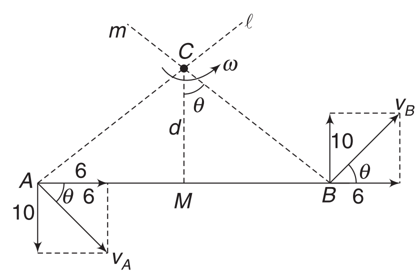
$$\Rightarrow \quad v_{By} = 0.1 \text{ ms}^{-1}\,(\uparrow) = 10 \text{ cms}^{-1}\,(\uparrow)$$

Velocities of A and B are as shown. $\tan\theta = \dfrac{10}{6} = \dfrac{5}{3}$

l is a line $\perp$ to $\vec{v}_A$ and m is a line $\perp$ to $\vec{v}_B$ and their intersection point C is instantaneous centre of rotation.

$$\frac{BM}{d} = \tan\theta$$

$$\Rightarrow \frac{25 \text{ cm}}{d} = \frac{5}{3} \quad \Rightarrow \quad d = 15 \text{ cm}.$$



6. For pure rolling, $a = R\alpha$.

$\therefore$ Acceleration of centre is $a = (10 \text{ cm})(2 \text{ rad s}^{-2}) = 20 \text{ cms}^{-2}$. Acceleration of top point A wrt centre is $R\,\alpha(\rightarrow) = 20 \text{ cms}^{-2}(\rightarrow)$

$\therefore$ Acceleration of A is

$$a_A = 20 + 20 = 40 \text{ cms}^{-2}(\rightarrow)$$

7. If disc I rotates by an angle θ, it wraps $x = R\theta$ length of the thread. Disc II will move up by

$$\frac{x}{2} = \frac{R\theta}{2}$$

Worksheet 3

1. (i) If disc rotates through θ, a length $x = R\theta$ unwinds and the centre falls through x.

$$x = R\theta$$

$$\Rightarrow \quad \frac{dx}{dt} = R\frac{d\theta}{dt} \quad \Rightarrow \quad v = R\omega$$

$$\Rightarrow \quad 4 = 0.2\ \omega \quad \Rightarrow \quad \omega = 20 \text{ rad s}^{-1}$$

(ii) Vertical part of thread has no speed. Point C of the disc must also be at rest.

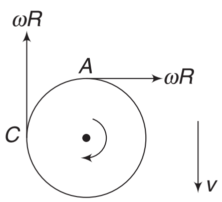
$$v_c = \omega R(\uparrow) + v(\downarrow) = 0$$

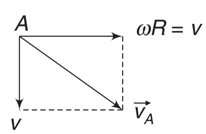
Velocity of A wrt centre is $\vec{v}_{\text{ACM}} = \omega R(\rightarrow) = v(\rightarrow)$.

Actual velocity of A is obtained by adding velocity of the centre $v(\downarrow)$ to the velocity $\vec{v}_{\text{ACM}}$

$$\therefore \quad v_A = \sqrt{v^2 + v^2} = \sqrt{2}\ v = 4\sqrt{2}\ \text{ms}^{-1}$$

Direction of $\vec{v}_A$ makes 45° with horizontal.





2. Speed of belt = Speed of a point on circumference of the cylinders

$$\Rightarrow \quad v = \omega_1 R_1 = \omega_2 R_2$$

$$\Rightarrow \quad \omega_2 = \frac{\omega_1 R_1}{R_2}$$

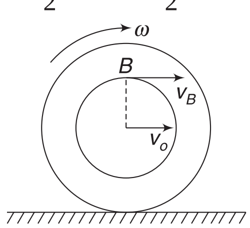
3. Let velocity of the centre of the spool be v_0 and its angular velocity be ω.

$$v_0 = \omega R$$

Velocity of point B is

$$v_B = \omega r + v_0 = \frac{v_0}{R} r + v_0 = \frac{v_0}{2} + v_0 = \frac{3v_0}{2}$$

It means that the thread at B is moving at a speed $\frac{3v_0}{2}$. Block A must also be moving at this speed.



$$\therefore \quad v_A = \frac{3v_0}{2}$$

$$\Rightarrow \quad \frac{dv_A}{dt} = \frac{3}{2}\frac{dv_0}{dt}$$

$$\Rightarrow \quad a = \frac{3}{2} a_0 \quad [a_0 = \text{acceleration of the centre}]$$

$$\Rightarrow \quad a_0 = \frac{2}{3} a$$

4. Initial velocity of centre $= u(\uparrow)$

$$\frac{2u}{g} = 2 \text{ s} \quad \Rightarrow \quad u = 10 \text{ ms}^{-1}.$$

In two seconds, the stick rotated once. Hence, angular speed of the stick is

$$\omega = \frac{\Delta\theta}{\Delta t} = \frac{2\pi \text{rad}}{2 \text{s}} = \pi \text{ rad s}^{-1}.$$

In 0.5 seconds, the stick rotates by $\frac{\pi}{2}$ radian. It is vertical in this position. Velocity of ends A and B relative to the centre of the stick is $\frac{\omega L}{2} = \pi \times \frac{1}{2} = \frac{\pi}{2}$ rad s^{-1} in the directions shown. Velocity of A and B relative to ground is obtained by adding the velocity of centre to this.

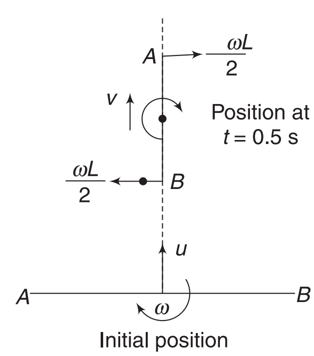
The COM moves with acceleration $g(\downarrow)$.

$\therefore$ Velocity of COM at $t = 0.5$ s is

$$v = u - gt = 10 - 10 \times 0.5 = 5 \text{ ms}^{-1}(\uparrow)$$

$$\therefore \quad v_A = v_B = \sqrt{v^2 + \left(\frac{\omega L}{2}\right)^2} = \sqrt{5^2 + \left(\frac{\pi}{2}\right)^2}$$

$$= 5.24 \text{ ms}^{-1}$$

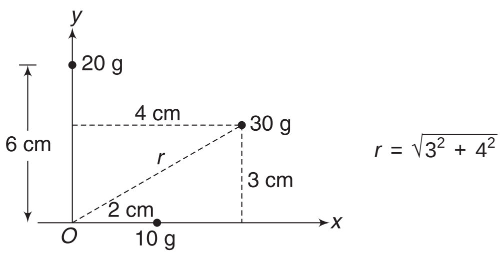


Your Turn

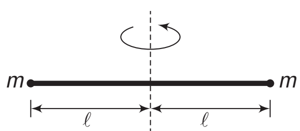
1. (a) $I_x = m_1r_1^2 + m_2r_2^2 + m_3r_3^2$
$= 0 + 20 \times 6^2 + 30 \times 3^2$
$= 990 \text{ gcm}^2$

(b) $I_y = 10 \times 2^2 + 0 + 30 \times 4^2$
$= 520 \text{ gcm}^2$

(c) $I_z = 10 \times 2^2 + 20 \times 6^2 + 30 \times (3^2 + 4^2)$
$= 1510 \text{ gcm}^2$



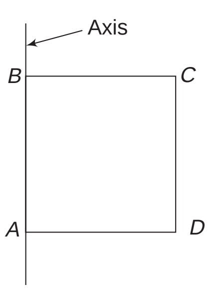
2. $I = I_{\text{rod}} + I_{\text{particles}}$
$= \dfrac{m(2l)^2}{12} + ml^2 \times 2$
$= \dfrac{7}{3}ml^2$



3. All particles are at a distance R from the axis.
4. All particles are at distance R from the axis.
5. The disc can be divided into 12 sectors of 30° each. Each sector has identical mass distribution about the given axis. Hence, each sector has same *MI*.

$$\therefore \quad I = \frac{11}{12} \times I_{\text{disc}} = \frac{11}{12} \times 1.2 = 1.1 \text{ kgm}^2.$$

6. Disk with smaller density will have larger radius for the two disks to have same mass. Thus, it's *MI* will be larger.
7. $I = I_{AB} + I_{BC} + I_{CD} + I_{DA}$
$= 0 + \dfrac{ml^2}{3} + ml^2 + \dfrac{ml^2}{3}$
$= \dfrac{5ml^2}{3}$



8. Distance of centre of rod BC from the COM of the triangle is

$$r = BD \cdot \tan 30° = \frac{1}{2\sqrt{3}}$$

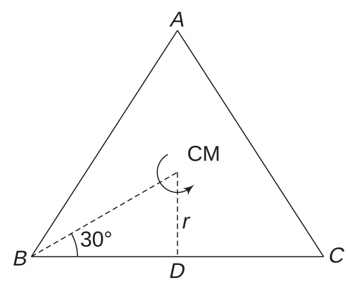
D is COM of rod BC. *MI* of rod BC about an axis through D and perpendicular to the plane of the figure is

$$I_{\text{CM}} = \frac{ml^2}{12}$$

MI of rod BC about a parallel axis through COM of the triangle is given by parallel axis theorem

$$I = I_{\text{CM}} + mr^2 = \frac{ml^2}{12} + m\left(\frac{l}{2\sqrt{3}}\right)^2 = \frac{ml^2}{6}.$$

MI of all three rods will be thrice of this $= \dfrac{ml^2}{2}$.

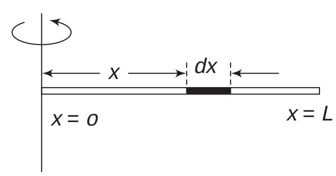


9. Consider an element of length dx as shown. Mass of the element is

$$dm = \lambda dx = \lambda_0 x\, dx$$

$$\therefore \quad dI = x^2 dm = \lambda_0 x^3 dx$$

$$\therefore \quad I = \int dI = \lambda_0 \int_0^L x^3 dx = \lambda_0 \frac{L^4}{4}$$



10. $$I = \frac{2}{5}MR^2$$

$$\frac{\Delta I}{I} = 2\frac{\Delta R}{R}$$

$$\frac{\Delta I}{I} \times 100 = 2\left(\frac{\Delta R}{R} \times 100\right) = 2 \times 0.1 = 0.2\%$$

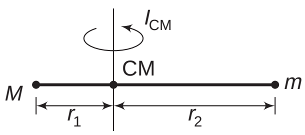
11. *MI* will be minimum about an axis passing through COM of the system.

[About any other axis $I = I_{\text{CM}} + Md^2$ $(> I_{\text{CM}})$]

$$r_1 = \frac{mL}{m+M} \quad \text{and} \quad r_2 = \frac{ML}{m+M}$$

$$\therefore \quad I_{CM} = Mr_1^2 + mr_2^2 = \frac{MmL^2}{(M+m)^2}(M+m)$$

$$= \left(\frac{Mm}{M+m}\right)L^2$$



12. Let ρ = density of sphere

$$M = \rho\,\frac{4}{3}\pi\,(a^3 - b^3) \quad \Rightarrow \quad \rho = \frac{3M}{4\pi(a^3-b^3)}$$

$$I = I_{\substack{\text{sphere of}\\ \text{radius a}}} - I_{\substack{\text{sphere of}\\ \text{radius b}}}$$

$$= \frac{2}{5}\left[\rho\cdot\frac{4}{3}\,\pi a^3\right]a^2 - \frac{2}{5}\left[\rho\cdot\frac{4}{3}\,\pi b^3\right]b^2$$

$$= \frac{2}{5}\left(\frac{4}{3}\,\pi\rho\right)(a^5 - b^5) = \frac{2M}{5}\left(\frac{a^5-b^5}{a^3-b^3}\right)$$

13. For solid cylinder $MK_1^2 = \frac{1}{2}\,MR^2 \quad \Rightarrow \quad K_1 = \frac{1}{\sqrt{2}}\,R$

For cylindrical shell $MK_2^2 = MR^2 \quad \Rightarrow \quad K_2 = R$

$$\therefore \qquad \frac{K_1}{K_2} = \frac{1}{\sqrt{2}}$$

14. $I = \frac{2}{5}mr^2 = \frac{2}{5}\times 2\ (0.5)^2 = 0.2\ \text{kgm}^2$

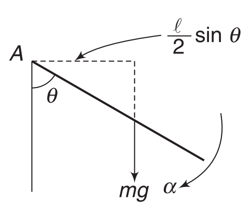
$$\tau = F\cdot r = 10\times 0.5 = 5\,\text{Nm}$$

$$\therefore \qquad \alpha = \frac{\tau}{I} = \frac{5}{0.2} = 25\ \text{rad}\,\text{s}^{-2}$$

Using $\omega = \omega_0 + \alpha t$; $\omega = 0 + 25\times 3 = 75\ \text{rad}\,\text{s}^{-1}$

15. (i) Torque on the rod about it's rotation axis through A is

$$\tau = mg\frac{l}{2}\,\sin\theta$$



$\therefore$ Using $I\alpha = \tau$ gives

$$\frac{ml^2}{3}\cdot\alpha = mg\frac{l}{2}\,\sin\theta$$

$$\Rightarrow \qquad \alpha = \frac{3}{2}\,\frac{g}{l}\,\sin 60° = \frac{3\sqrt{3}}{4}\,\frac{g}{l}$$

(ii) In vertical position, line of mg passes through A. There is no torque about A.

$$\therefore \qquad \alpha = 0$$

16. $$a = R\alpha \qquad \text{...(i)}$$

For block: $mg\sin\theta - T = ma$ ...(ii)

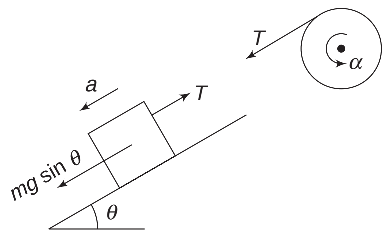
For pulley: $TR = I\alpha$

$$TR = \frac{1}{2}MR^2\cdot\alpha$$

$$T = \frac{1}{2}Ma \qquad \text{...(iii)}$$

Adding (ii) and (iii) gives

$$a = \frac{g\sin\theta}{m + \frac{M}{2}}$$



17. The relevant equations are:

$$a = R\alpha \qquad \text{...(i)}$$

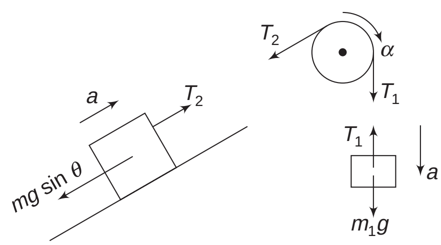
$$m_1g - T_1 = m_1a \qquad \text{...(ii)}$$

$$T_2 - m_2g\sin\theta = m_2a \qquad \text{...(iii)}$$

$$T_1R - T_2R = I\cdot\alpha$$

$$T_1 - T_2 = \frac{I}{R^2}a \qquad \text{...(iv)}$$

Add (ii), (iii) and (iv) to get the desired result.



19. Net torque on (disk + particle) system is

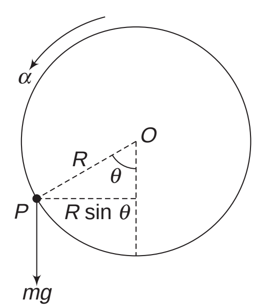
$$\tau = mgR\sin\theta$$

MI about horizontal axis through O is

$$I = \frac{1}{2}MR^2 + mR^2$$

$$\therefore \qquad \alpha = \frac{\tau}{I} = \frac{mgR\sin\theta}{\left(\frac{M}{2}+m\right)R^2}$$

$$= \frac{2mg\sin\theta}{(M+2m)R}$$



Speed of particle immediately after release = 0.

$\therefore$ Radial acceleration = 0

Tangential acceleration $a_t = R\alpha = \dfrac{2mg\sin\theta}{(M+2m)}$

20. (i) Torque about O

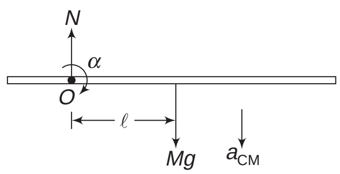
$$\tau = Mgl$$

Moment of inertia about rotation axis through O is

$$I = I_{CM} + Md^2 = \frac{1}{12}M(4l)^2 + Ml^2$$

$$= \frac{7}{3}Ml^2$$

$$\therefore \quad \alpha = \frac{\tau}{I} = \frac{Mgl}{\frac{7}{3}Ml^2} = \frac{3g}{7l}$$



(ii) COM will move in a circle of radius l about O. It will have no speed immediately after release, hence no radial acceleration.

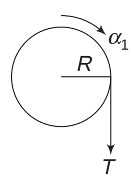
Tangential acceleration of COM is

$$a_t = l\cdot\alpha = \frac{3g}{7}\ (\downarrow)$$

(iii) $Mg - N = M\cdot a_{CM}$

$$\Rightarrow \quad N = Mg - \frac{3Mg}{7} = \frac{4Mg}{7}.$$

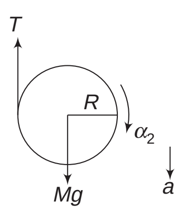
21. For upper disc:



$$TR = \frac{1}{2}MR^2\cdot\alpha_1$$

$$\Rightarrow \quad T = \frac{1}{2}M(R\alpha_1) \quad \text{...(i)}$$

For lower disc [about its COM]



$$TR = \frac{1}{2}MR^2\cdot\alpha_2$$

$$T = \frac{1}{2}M(R\alpha_2) \quad \text{...(ii)}$$

From (i) and (ii) $\alpha_1 = \alpha_2$.

Linear acceleration (a) of the disc results due to unwinding of thread from both the discs.

$$\therefore \quad a = R\alpha_1 + R\alpha_2 = 2R\alpha_2 \quad \text{...(iii)}$$

For translation: $Mg - T = Ma$ ...(iv)

Equation (ii) can be written as $T = \dfrac{1}{2}M\dfrac{a}{2} = \dfrac{Ma}{4}$

Substituting in (iv) gives:

$$Mg - \frac{Ma}{4} = Ma \quad \Rightarrow \quad a = \frac{4g}{5}$$

22. When $v > \omega R$, contact point has velocity in forward direction. Friction is in backward direction. Torque due to friction increases ω.

23. $\omega R = 2v$

Contact point has a velocity towards left. Friction acts towards right.

$$a = \frac{f}{M}$$

$\therefore$ Velocity at time t is

$$v_0 = v + \frac{f}{M}t \quad \text{...(i)}$$

$$\alpha = \frac{\tau}{I} = \frac{fR}{\frac{1}{2}MR^2} = \frac{2f}{MR}$$

$\therefore$ Angular velocity at t is

$$\omega_0 = \omega - \alpha t = \omega - \frac{2ft}{MR}$$

$$= \frac{2v}{R} - \frac{2ft}{MR} \quad \text{...(ii)}$$

When pure rolling (i.e, no sliding) begins

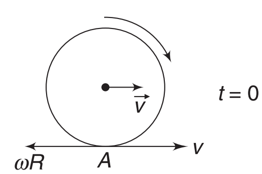
$$v_0 = \omega_0 R$$

$$\Rightarrow \quad v + \frac{f}{M}t = 2v - \frac{2ft}{M}$$

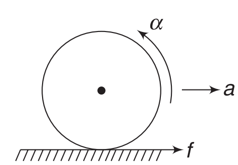
$$\Rightarrow \quad \frac{3ft}{M} = v$$

Putting this value of time in (i) gives

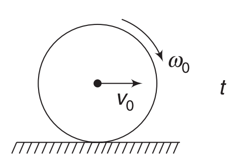
$$v_0 = v + \frac{f}{M}t = v + \frac{v}{3} = \frac{4v}{3}$$



Velocity of contact point A is $(\omega R - v)$ towards left



Friction increases v and decreases ω.



Pure rolling starts when $v_0 = \omega_0 R$

24. (a) Refer to Example 23. $a_{disc} > a_{ring}$.

(b) $a = \dfrac{g\sin\theta}{1 + \dfrac{k^2}{R^2}} = \dfrac{g\sin\theta}{1 + \dfrac{2}{5}} = \dfrac{5g\sin\theta}{7}$

$\therefore \quad v^2 = 0^2 + 2aL \quad \Rightarrow \quad v = \sqrt{\dfrac{10}{7}\, gL\sin\theta}$

$\omega = \dfrac{v}{r} = \dfrac{1}{r}\sqrt{\dfrac{10}{7}\, gL\sin\theta}$

25. $f = \dfrac{Mg\sin\theta}{\dfrac{MR^2}{I} + 1} = \dfrac{Mg\sin\theta}{1 + 1} = \dfrac{Mg\sin\ \theta}{2}$

$f \le \mu N$

$\therefore \quad \dfrac{Mg\sin\theta}{2} \le \mu Mg\cos\theta \quad \Rightarrow \quad \dfrac{1}{2}\tan\theta \le \mu$

$\Rightarrow \quad \dfrac{1}{2} \le \mu$

26. There is no friction. There is no torque about centre. There is no rotation.

27. Friction is kinetic.

$f = \mu N = \mu Mg\cos\theta$

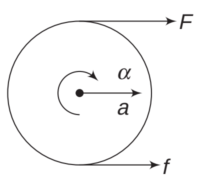
$\therefore \quad Ma = Mg\sin\theta - \mu Mg\cos\theta$

$\Rightarrow \quad a = g(\sin\theta - \mu\ \cos\theta)$

And $\quad \alpha = \dfrac{\tau}{I} = \dfrac{f\cdot R}{\dfrac{2}{5}MR^2} = \dfrac{5\mu Mg\cos\theta}{2M\cdot R} = \dfrac{5\mu g\cos\ \theta}{2R}$

28. Let friction (f) be in forward direction.

$F + f = Ma$ ...(i)



And $FR - fR = \dfrac{1}{2}MR^2\cdot\alpha$

$\Rightarrow \quad F - f = \dfrac{1}{2}M(R\alpha)$

$\Rightarrow \quad F - f = \dfrac{1}{2}Ma$ ...(ii)

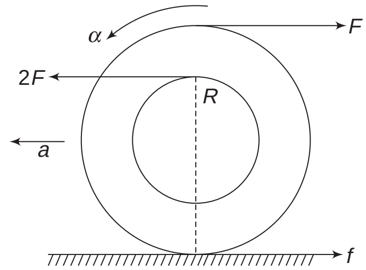
Solving (i) and (ii) $a = \dfrac{4F}{3M}$

29. Net torque due to two forces is $2FR(\circlearrowleft) + F(2R)(\circlearrowright) = 0$. The spool will have a linear acceleration towards left. It must have an anticlockwise angular acceleration to avoid slipping. Thus friction is to the right.

$2F - F - f = Ma$

$F - f = Ma$ ...(i)

$f\cdot 2R = I\alpha$



$f\cdot 2R = 2MR^2\cdot\alpha$

$\Rightarrow \quad 2f = M(2R\alpha)$

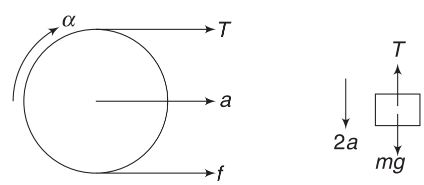
$\Rightarrow \quad 2f = Ma \quad [\because \quad a = \alpha(2R)]$ ...(ii)

Solving (i) and (ii) $f = \dfrac{F}{3}$

30. Let acceleration of centre of the cylinder be a and its angular acceleration be α.

For no slipping $a = R\alpha$ ...(i)

Speed of block = speed of top point of cylinder
= twice the velocity of centre of the cylinder



Thus, acceleration of m must be twice that of the cylinder = $2a$.

$T + f = Ma$ ...(i)

$TR - fR = \dfrac{1}{2}MR^2\ \alpha \quad \Rightarrow \quad T - f = \dfrac{1}{2}Ma$ ....(iii)

And $\quad mg - T = m(2a)$ ...(iv)

Solving the above equations gives

$$a = \frac{4mg}{3M + 8m}$$

31. $K = \dfrac{1}{2}I_{CM}\omega^2 + \dfrac{1}{2}Mv_{CM}^2 = \dfrac{1}{2}\left(\dfrac{1}{2}MR^2\right)\omega^2 + \dfrac{1}{2}Mv^2$

$= \dfrac{1}{4}M(R\omega)^2 + \dfrac{1}{2}Mv^2 = \dfrac{3}{4}Mv^2.$

32. $\dfrac{k_T}{k_R} = \dfrac{\dfrac{1}{2}Mv^2}{\dfrac{1}{2}I_{CM}\cdot\omega^2} = \dfrac{Mv^2}{MR^2\omega^2} = 1 \qquad [\because \quad R\omega = v]$

33. $I = \left(\dfrac{1}{3}ml^2\right) \times 3 = ml^2$

$\therefore \quad k = \frac{1}{2} I\omega^2 = \frac{1}{2}(ml^2)\,\omega^2$

34. $W_\tau = \tau\Delta\theta = 20 \times (2 \times 2\pi) = 80\pi$ J.

$\therefore \quad KE = W_\tau = 80\pi$ J

35. $P = \tau \cdot \omega$

P is maximum when ω is maximum, i.e, at the end of two rotations.

Final ω can be calculated as:

$$\frac{1}{2}I\omega^2 = 80\pi \quad \Rightarrow \quad \omega = \frac{160\pi}{\pi} = 160 \text{ rad s}^{-1}$$

$\therefore \quad P_{\max} = (20 \text{ Nm})(160 \text{ rad s}^{-1}) = 3200$ watt.

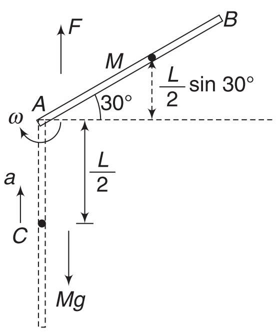
37. (a) Energy conservation.

gain in KE = Loss in PE

$$\frac{1}{2} I_A \omega^2 = Mg\left[\frac{L}{2} + \frac{L}{2}\sin 30°\right]$$

$$\Rightarrow \quad \frac{1}{2}\frac{ML^2}{3}\omega^2 = \frac{3}{4}MgL$$

$$\Rightarrow \quad \omega = \sqrt{\frac{g}{2}\frac{g}{L}} = 3\sqrt{\frac{g}{2L}}$$



(b) In vertical position, COM of the rod has a radial acceleration (towards A) given by

$$a = \omega^2\frac{L}{2} = \frac{9}{4}g$$

If hinge force is F, then

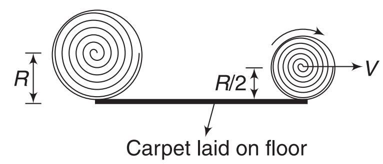
$$F - Mg = Ma \quad \Rightarrow \quad F = Mg + \frac{9}{4}Mg = \frac{13}{4}Mg$$

38. Work done by friction = change in KE of the ball

$\therefore \quad W_f = k_f - k_i$

$$= \frac{7}{10}M\left(\frac{5v}{7}\right)^2 - \frac{1}{2}Mv^2 = -\frac{1}{7}Mv^2$$

39. As the carpet unrolls, it loses PE and gains KE. The part that unrolls, lies on the floor without any KE. There is no sliding (rubbing) and friction is static. It does not dissipate mechanical energy. Volume of cylinder with radius $\frac{R}{2}$ becomes $\left(\frac{1}{4}\right)^{\text{th}}$ the original. Thus, mass of the unrolled cylinder is $\frac{M}{4}$.



Loss in $PE = MgR - \frac{M}{4}g\frac{R}{2} = \frac{7}{8}MgR$.

KE of a purely rolling cylinder is $\frac{3}{4}mv^2$

$\therefore \quad KE$ of rolling part $= \frac{3}{4}\frac{M}{4}\cdot v^2 = \frac{3}{16}Mv^2$

Energy conservation: $\frac{3}{16}Mv^2 = \frac{7}{8}MgR$

$$\Rightarrow \quad v = \sqrt{\frac{14gR}{3}}$$

40. At B, KE of the ring $= Mg(H - h)$

$$\Rightarrow \quad \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2 = Mg(H-h)$$

$$\Rightarrow \quad \frac{1}{2}Mv^2 + \frac{1}{2}MR^2\omega^2 = Mg(H-h)$$

$$\Rightarrow \quad Mv^2 = Mg(H-h)$$

$$\therefore \quad v = \sqrt{g(H-h)}$$

Now, the ring is a horizontal projectile projected from B at speed v. [Remember that rotational KE of the ring will not change once it is in air. There is no torque on it]

$$\therefore \quad CD = v\sqrt{\frac{2h}{g}} = \sqrt{g(H-h)}\sqrt{\frac{2h}{g}} = \sqrt{2h(H-h)}$$

42. Distance of line of $\vec{P}$ from point of suspension is always l. Hence, L is maximum when P (i.e, speed) is maximum. Speed is maximum at lowest position.

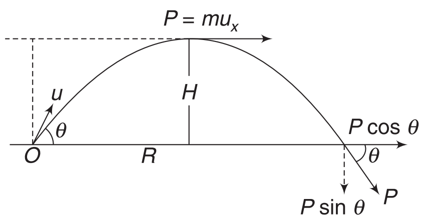
$$\frac{1}{2}mu^2 = mgl \quad \Rightarrow \quad u = \sqrt{2gl}$$

$$\therefore \quad L_{\max} = mu \cdot l = ml\sqrt{2gl}$$

43. (a) At highest point, momentum (P) is horizontal. Perpendicular distance of line of P from O is equal to maximum height (H).

$\therefore \quad L = mu_x \cdot H \quad [\text{using } L = r_\perp P]$

$$= m(u\cos\theta)\frac{u^2\sin^2\theta}{2g} = \frac{mu^3\cos\theta\cdot\sin^2\theta}{2g}$$



(b) When the projectile is about to hit the ground its velocity is u directed at an angle θ to the horizontal.

Using $L = P_{\perp} \cdot r$

$$L = (P\sin\theta) \cdot R$$

where R = horizontal range.

$$= (mu\sin\theta)\,\frac{u^2\sin 2\theta}{g}$$

$$= \frac{2mu^3\sin^2\theta \cdot \cos\theta}{g}$$

44. This is a fixed axis rotation.

$$\therefore \quad L = I \cdot \omega = \left[\frac{MR^2}{2} + M\left(\frac{R}{2}\right)^2\right]\omega = \frac{3}{4}MR^2\omega$$

45. $L = L_{\text{spin}} + L_{\text{orbitals}} = \frac{1}{2}MR^2\omega + MvR$

$$= \frac{1}{2}MR(\omega R) + MvR = \frac{1}{2}MvR + MvR = \frac{3}{2}MvR$$

$$[\because \omega R = v]$$

46. $L = L_{\text{spin}} + L_{\text{orbitals}} = I_{\text{CM}} \cdot \omega(\circlearrowright) + MvR(\circlearrowleft)$

$$= MR^2\omega(\circlearrowright) + MvR(\circlearrowleft) = \frac{MvR}{3}(\circlearrowright) + MvR(\circlearrowleft)$$

$$= \frac{2}{3}MvR(\circlearrowleft)$$

47. $L = L_{\text{spin}} + L_{\text{orbital}} = \frac{2}{5}MR^2\omega + 0 = \frac{2}{5}MR \cdot v$

The line of velocity of COM passes through A.

48. Angular impulse $= \int_0^{\Delta t} \tau dt$ = area under the graph

$$= \frac{1}{2} \times 0.0\ 1 \times 20 = 0.1 \text{ N.m.s}$$

$$\Delta L = 0.1$$

$$\Rightarrow \quad I\omega = 0.1 \quad \Rightarrow \quad \omega = \frac{0.1}{0.4}$$

$$= 0.25 \text{ rad s}^{-1}.$$

49. Torque $\tau = RF = aRt$

Angular impulse $= \int_0^T \tau dt = aR\int_0^T t\,dt = \frac{aRT^2}{2}$.

$$\therefore \quad \Delta L = \frac{aRT^2}{2}$$

$$\Rightarrow \quad I\omega = \frac{aRT^2}{2} \quad \Rightarrow \quad \frac{MR^2}{2} \cdot \omega = \frac{aRT^2}{2}$$

$$\Rightarrow \quad \omega = \frac{aT^2}{MR}$$

50. Velocity (v) acquired by the COM is independent of h. Angular speed (ω) about centre will be maximum when the angular impulse is maximum. This happens when line of force is at maximum distance from the centre.

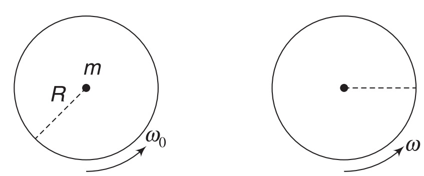
$\omega = 0$ (and hence KE is minimum) when line of action of the force passes through COM.

51. Moving to equator means moving farther from the rotation axis of the Earth. This will increase the moment of inertia of the Earth. Angular momentum of the Earth associated with its spin about its rotation axis cannot change due to internal interactions. This implies that angular speed (ω) will decrease and hence the Earth will take more time to complete one rotation.

53. $I_0\omega_0 = I\omega$

$$\frac{MR^2}{2}\omega_0 = \left[\frac{MR^2}{2} + mR^2\right]\omega$$

$$\Rightarrow \quad \omega = \left(\frac{M}{M + 2m}\right)\omega_0$$



54. Due to inertia, Velocity of ball before and after the release are equal. Angular momentum of the ball wrt the rotation axis is same before and after the release. Therefore, ω for boy + platform system will also not change in order to conserve angular momentum.

55. (a) This is similar to previous problem. Angular speed of the disc will not change.

(b) Vertical velocity of the chip just after separation, $u = \omega R\ (\uparrow)$.

$$\therefore \quad h = \frac{u^2}{2g} = \frac{\omega^2R^2}{2g}$$

56. Tension in the cord does not produce any torque on the block about the centre of the circle.

Angular momentum of the block about centre remains conserved.

$$mv_1r_1 = mv_2r_2 \qquad \text{...(i)}$$

Tension in the cord when radius is r_2 is

$$T_2 = \frac{mv_2^2}{r_2} \qquad \text{...(ii)}$$

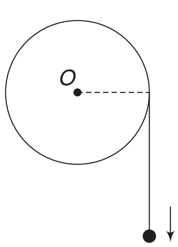
From (i) and (ii); $mv_1 r_1 = m\left[\frac{T_2 r_2}{m}\right]^{\frac{1}{2}} \cdot r_2$

$$\Rightarrow \quad r_2 = \left[\frac{mv_1^2 r_1^2}{T_2}\right]^{\frac{1}{3}} = \left[\frac{4 \times 16 \times 0.25}{600}\right]^{\frac{1}{3}} = 3\,\text{m}.$$

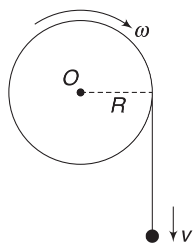
57. Speed of the ball after falling through L is

$$u = \sqrt{2gL}$$

Angular momentum of the ball + pulley system about O just before and after the string gets taut remains same.



Just before the string is taut



Just after the string is taut

[***Note:*** Weight of the ball (mg) has a torque about O. But this torque has negligible impulse during the short duration in which the string gains tension]

$$muR = mvR + \frac{1}{2}MR^2 \cdot \omega$$

$$mu = mv + \frac{1}{2}M(R\omega)$$

$$mu = mv + \frac{1}{2}Mv \qquad [\because \; v = R\omega]$$

$$\therefore \quad v = \frac{2mu}{2m + M}$$

58. Conservation of angular momentum about the rotation axis of the disc gives:

$$m\frac{u}{\sqrt{2}}R + \frac{1}{2}MR^2\omega = muR$$

[KE of the bullet is halved. This means its speed becomes $\frac{u}{\sqrt{2}}$]

$$\Rightarrow \quad \omega = \frac{\sqrt{2}(\sqrt{2} - 1)mu}{MR}$$

59. Conservation of momentum tells us that COM of the rod will move parallel to u with a velocity v given by

$$2mv = mu \quad \Rightarrow \quad v = \frac{u}{2}$$

Let angular speed of the rod be ω.

We will conserve angular momentum about a point fixed to the table which is just beneath the centre C of the rod.

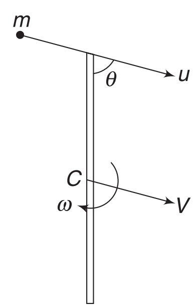
$$L_i = L_f$$

$$m(u\sin\theta)\frac{L}{2} = L_{\text{spin}} + L_{\text{orbital}}$$

$$\Rightarrow \quad mu\sin\theta\,\frac{L}{2} = \frac{(2m)(L^2)}{12}\omega + 0$$

$$\Rightarrow \quad \omega = \frac{3u\sin\theta}{L}$$

[v_{CM} is along the line passing through the origin (just below C)]



60. (a) Conservation of momentum:

$$(M + m)v = mu \quad \Rightarrow \quad v = \frac{mu}{M + m}$$

(b) We will conserve angular momentum about COM of the combined system.

$$L_i = L_f$$

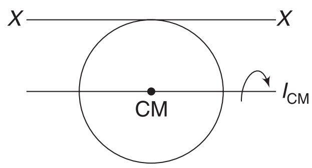
$$mu(h - R) = \left(\frac{2}{5}MR^2 + mR^2\right)\omega$$

$$\Rightarrow \quad \omega = \frac{mu(h - R)}{\left(\frac{2}{5}M + m\right)R^2}$$

Worksheet 1

1. Square of side $2a$ has its mass particles placed at largest distance from the axis.

2. Mass of loop, $m = \rho L$



Radius of loop, $r = \dfrac{L}{2\pi}$

$$I_{CM} = \frac{1}{2}mr^2$$

$$I_{xx} = \frac{1}{2}mr^2 + mr^2 = \frac{3}{2}mr^2 = \frac{3}{2}\rho L\left(\frac{L}{2\pi}\right)^2 = \frac{3\rho L^3}{8\pi^2}$$

3. $$I_1 = \frac{ML^2}{12};\ I_2 = Mr^2 = M\left(\frac{L}{\pi}\right)^2$$

$$\therefore \quad \frac{I_1}{I_2} = \frac{\pi^2}{12} < 1$$

4. If a disc of mass M is folded about its diameter (perpendicular to AB), distances of particles from line AB do not change.

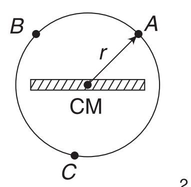
$\therefore$ $I = MI$ of a disc about diameter (AB)

$$= \frac{1}{4}MR^2$$

5. If I_{CM} is MI abut a line through centre of the rod that is parallel to z-axis, then MI about any other parallel axis is

$$I = I_{CM} + Mr^2$$

where r = distance of the axis from the line through COM. If I is to remain constant, then r must remain constant. Thus the locus is a circle with its centre at the centre of the rod



$$I_A = I_B = I_C = I_{CM} + Mr^2$$

6. MI of upper ring $I_1 = mR^2 + mR^2 = 2mR^2$

MI of lower ring

$$I_2 = \frac{1}{2}mR^2 + m(3R)^2 = 9.5\ mR^2$$

$$\therefore \quad I = I_1 + I_2 = 11.5\,mR^2$$

7. $$I = \left(\frac{2}{3}MR^2 + MR^2\right) + \left(\frac{2}{3}MR^2 + (3R)^2\right) = \frac{34}{3}MR^2$$

8. MI about an axis through junction point and perpendicular to the figure is

$$I_z = \frac{ML^2}{12} + \frac{ML^2}{12} = \frac{ML^2}{6}$$

Consider the given axis and one more axis perpendicular to it in the plane of the figure. MI about both these axes must be same due to symmetry.

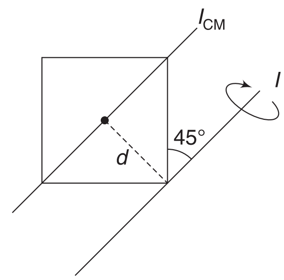
Using perpendicular axis theorem

$$I + I = I_z \quad \Rightarrow \quad I = \frac{I_z}{2} = \frac{ML^2}{12}$$

9. $$I_{CM} = \frac{Ma^2}{12}$$

$$\therefore \quad I = \frac{Ma^2}{12} + Md^2$$

$$= \frac{Ma^2}{12} + M\left(\frac{a}{\sqrt{2}}\right)^2 = \frac{7}{12}Ma^2$$



10. Two masses lie on axis. They do not contribute to MI. 4 masses are at separation a from the axis and 2 masses are at separation $\sqrt{2}\ a$.

$$\therefore \quad I = 4\cdot ma^2 + 2\cdot m(\sqrt{2}\ a)^2 = 8\,ma^2.$$

11. MI is minimum about an axis through COM.

$$I = 2x^2 - 12x + 27$$

I is minimum if $\dfrac{dI}{dx} = 0$

$$\Rightarrow \quad 4x - 12 = 0 \quad \Rightarrow \quad x = 3$$

12. COM is farthest from B

$\therefore$ I_B is maximum

14. $$\omega = 500\ \frac{\text{rot}}{\text{min}} = \frac{500 \times 2\pi}{60} = \frac{50\pi}{3}\ \text{rad}\,\text{s}^{-1}.$$

Angular retardation produced due to braking can be calculated as

$$\omega = \omega_0 + \alpha t \quad \Rightarrow \quad 0 = \frac{50\pi}{3} - \alpha \times 15$$

$$\Rightarrow \quad \alpha = \frac{10\pi}{9}\ \text{rad}\,\text{s}^{-2}$$

Friction, $f = \mu N = 100\mu$.

Torque $\tau_f = \mu N \cdot r = 100\mu \times 0.25 = 25\mu$ Nm

Moment of inertia, $I = \dfrac{1}{2}mr^2 = \dfrac{1}{2} \times 15 \times (0.25)^2$

$= \dfrac{15}{32}$ kg m^2

Using $\tau = I\alpha$

$$25\mu = \frac{15}{32} \times \frac{10\pi}{9} \quad \Rightarrow \quad \mu = 0.065$$

15. To loosen the nut we need a certain minimum torque.

$6 \times 8 \times \sin 30° = F \times 16 \quad \Rightarrow \quad F = 1.5\,\text{N}$

16. $k = \frac{1}{2} I\omega^2 = \frac{1}{2I}(I\omega)^2 = \frac{L^2}{2I}$

$\therefore \quad \frac{L_1^2}{2I_1} = \frac{L_2^2}{2I_2} \quad \Rightarrow \quad \frac{L_1}{L_2} = \sqrt{\frac{I_1}{I_2}} = \frac{1}{\sqrt{2}}$

17. For no rotation, the force must be applied at the COM of the body. Thus P is COM.

Distance of COM from C is

$$x = \frac{2ml + m\cdot 2l}{3m} = \frac{4l}{3}$$

18. For no rotation $f = mg\sin\theta$

$$\therefore \quad \tau_f = fR = mgR\sin\theta$$

$\therefore$ Angular retardation, $\alpha = \frac{\tau_f}{I}$

$$= \frac{mgR\sin\theta}{\frac{1}{2}mR^2} = \frac{2g\sin\theta}{R}$$

Using, $\quad \omega = \omega_0 + \alpha t$

$$0 = \omega - \left(\frac{2g\sin\theta)}{R}\right)t \quad \Rightarrow \quad t = \frac{R\omega}{2g\sin\theta}$$

Once the cylinder stops spinning, it will begin to roll down the incline.

19. $\frac{1}{2} I_{\text{CM}}\cdot\omega^2 = 0.4\,\frac{1}{2}Mv^2$

$\Rightarrow \quad I_{\text{CM}}\,\omega^2 = \frac{2}{5}\cdot M(\omega R)^2 \quad \Rightarrow \quad I_{\text{CM}} = \frac{2}{5}MR^2$

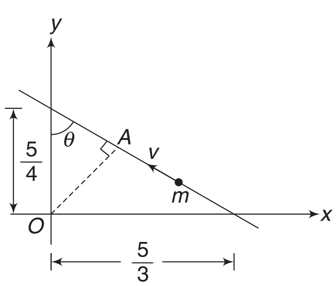
20. $f = \dfrac{\frac{1}{2} I_{\text{CM}}\,\omega^2}{\frac{1}{2} I_{\text{CM}}\,\omega^2 + \frac{1}{2}Mv^2} = \dfrac{1}{1 + \frac{MR^2}{I_{\text{CM}}}} \qquad [\because v = \omega R]$

f is maximum when I_{CM} is maximum.

Maximum value of I_{CM} for a ring is equal to MR^2

$\therefore \quad f_{\max} = \frac{1}{2}$

21. The line of motion is as shown. Perpendicular distance of line of motion from O is



$$OA = \frac{5}{4}\cdot\sin\theta = \frac{5}{4}\times\frac{4}{5} = 1$$

$$\left[\because \tan\theta = \frac{5/3}{5/4} = \frac{4}{3} \quad \Rightarrow \quad \sin\theta = \frac{4}{5}\right]$$

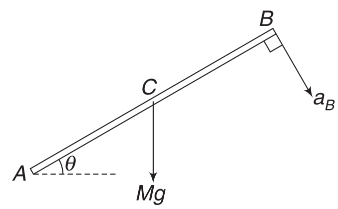
$\therefore$ Angular momentum of the particle about O is

$$L = mv(OA) = 2 \times 8 \times 1 = 16\ \text{kg}\,\text{m}^2\,\text{s}^{-1}.$$

22. $\tau_A = Mg\,\frac{L}{2}\cos\theta$

$$\frac{ML^2}{3}\cdot\alpha = Mg\,\frac{L}{2}\cos\theta$$

$$\Rightarrow \qquad \alpha = \frac{3g}{2L}\cos\theta$$



Acceleration of end B immediately after release is tangential acceleration perpendicular to the rod

$$a_B = \alpha L = \frac{3g}{2}\cos\theta.$$

23. $I_2\omega_2 = I_1\omega_1 \quad \Rightarrow \quad \frac{2}{5}M\left(\frac{R}{2}\right)^2\omega_2 = \frac{2}{5}MR^2\,\omega_1$

$$\Rightarrow \qquad \omega_2 = 4\omega_1$$

Angular speed becomes 4 times. It means it will take $\left(\frac{1}{4}\right)^{\text{th}}$ of 24 hr to complete one rotation.

24. $\vec{r} = 3\cos\theta\,\hat{i} + 4\sin\theta\,\hat{j}$

$$\vec{v} = \frac{d\vec{r}}{dt} = -3\sin\theta\,\frac{d\theta}{dt}\,\hat{i} + 4\cos\theta\,\frac{d\theta}{dt}\,\hat{j}$$

$$\frac{d\theta}{dt} = \omega = \text{angular speed about } O.$$

$$\therefore \quad \vec{v} = (-3\sin\theta\,\hat{i} + 4\cos\theta\,\hat{j})\,\omega$$

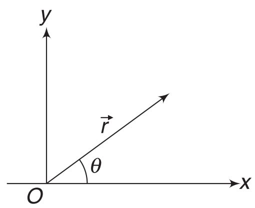
$$\vec{L} = \vec{r} \times (m\vec{v})$$

$$= m(12\cos^2\theta\,\hat{k} + 12\sin^2\theta\,\hat{k})\,\omega = 12\,m\omega\,\hat{k}$$

$$\therefore \quad 0.6 = 12 \times 0.01 \times \omega$$

$$\Rightarrow \quad \omega = 5\ \text{rad}\,\text{s}^{-1}$$

$$\therefore \qquad \theta = \omega t = 5t\ \text{rad}$$



25. Friction does not dissipate energy in pure rolling when the cylinder stops it has no *KE*.

$$\frac{1}{2}kx^2 = Mgx\sin\theta \quad\Rightarrow\quad x = \frac{2Mg\sin\theta}{k}$$

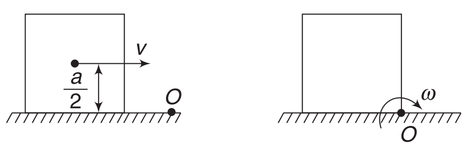
26. Angular momentum is conserved about *O*.

$$L_i = L_f$$

$$Mv\,\frac{a}{2} = I_0\omega$$

$$Mv\,\frac{a}{2} = \frac{2}{3}Ma^2\cdot\omega$$

$$\therefore \qquad \omega = \frac{3v}{4a}$$



27. $$mg - T_1 = ma_1$$

$$\therefore \qquad T_1 = mg - ma_1 \qquad \text{...(i)}$$

And $T_2 - mg = ma_2$

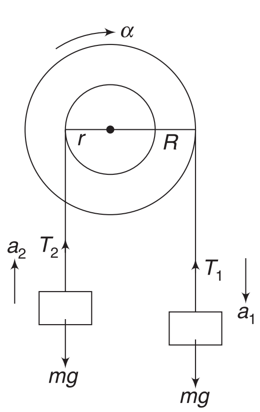
$$T_2 = mg + ma_2 \qquad \text{...(ii)}$$

From (i), T_1 is less than mg and from (ii), T_2 is larger than *mg*.

$$\therefore \qquad T_2 > T_1$$

Note: Torque due to T_1 is larger than torque due to T_2. Thus, pulley rotates clockwise

$$[T_1R > T_2r]$$



28. τ is constant.

$\therefore$ α is constant.

$$\therefore \qquad \omega = \alpha t$$

$$P_\tau = \tau\cdot\omega = \tau\alpha t$$

Thus power $\propto t$. Graph of *P* vs *t* is a straight line passing through origin.

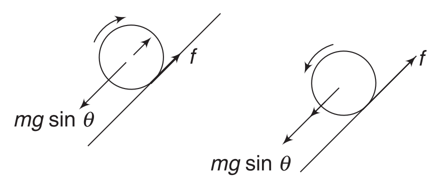
29. Hinge force is equal and opposite to applied force (*F*) as

$$F_{\text{net}} = 0 \qquad [\because a_{\text{CM}} = 0]$$

Hinge force has no torque about centre. Applied force has a torque. Thus ω increases.

30. Process *B* has more inertia, therefore less acceleration.

31. When the cylinder moves up its velocity (*v*) decreases. To maintain pure rolling, ω must also decrease. For this friction is up providing an anti clockwise torque. We already know that friction is up the plane when moving down.



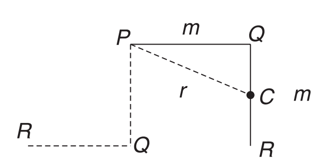
32. Initially (when stretch in spring is small), $mg\sin\theta$ is larger than spring force *kx* and the cylinder accelerates down. Friction must be up the plane to provide the necessary angular acceleration for rolling. As the spring stretches beyond equilibrium position, spring forces exceeds $mg\sin\theta$. Now the cylinder retards. It must also experience angular retardation. For this the friction assumes a direction down the incline.

33. *MI* about rotation axis through *P* is

$$I = I_{PQ} + I_{QR}$$

$$= \frac{1}{3}ml^2 + \left[\frac{1}{12}ml^2 + mr^2\right]$$

$$= \frac{1}{3}ml^2 + \frac{1}{12}ml^2 + m\left(l^2 + \frac{l^2}{4}\right) = \frac{5}{3}ml^2$$



Energy conservation:

$$\frac{1}{2}I\omega^2 = mg\frac{l}{2} + mg\frac{l}{2} \quad\Rightarrow\quad \frac{1}{2}\,\frac{5}{3}\,ml^2\omega^2 = mgl$$

$$\Rightarrow \qquad \omega = \sqrt{\frac{6}{5}\,\frac{g}{l}}$$

34. $$F - f = ma \quad \text{...(1)}$$

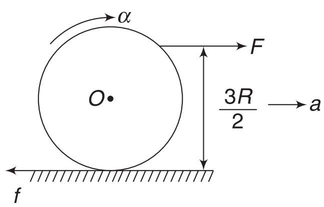
$$\frac{FR}{2} + f \cdot R = \frac{1}{2} mR^2 \cdot \alpha$$

$$\frac{1}{2}F + f = \frac{1}{2}ma \quad \text{...(2)}$$

$[\because R\alpha = a]$

Eliminating a between (1) and (2) gives

$$F - f = F + 2f \quad \Rightarrow \quad f = 0.$$



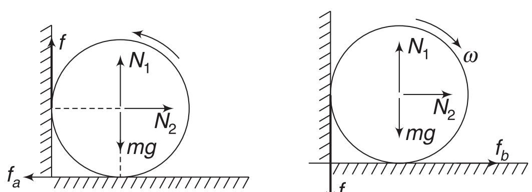
Note: The applied force itself ensures $a = R\alpha$ and therefore friction will adjust itself to zero.

35. Friction at both surfaces is kinetic and directed opposite to the contact point.

In case (a)

$$f_a = N_2 \quad \Rightarrow \quad \mu N_1 = N_2 \quad \text{...(i)}$$

And $f + N_1 = mg \quad \Rightarrow \quad \mu N_2 + N_1 = mg \quad$ ...(ii)



Case (a) Case (b)

Solving (i) and (ii) $N_1 = \dfrac{mg}{\mu^2 + 1}$

$$\therefore \quad f_a = \mu N_1 = \frac{\mu mg}{\mu^2 + 1}$$

Figure shown for case (b) clearly shows that the sphere will definitely move towards right and there will be no contact with the wall.

$$N_2 = f = 0$$

$$\therefore \quad f_b = \mu N_1 = \mu mg$$

$$\therefore \quad \frac{f_a}{f_b} = \frac{1}{\mu^2 + 1} = \frac{9}{10}$$

36. Since, the disc is rolling without sliding we can consider it to be in pure rotation about contact point.

Speed of point Q is

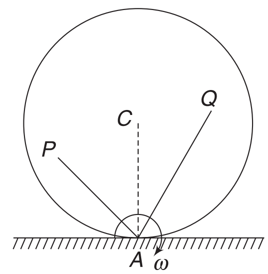
$$v_Q = (AQ)\,\omega$$

For C: $\quad v_C = (AC)\,\omega$

For P: $\quad v_P = (AP)\,\omega$

Since, $\quad AQ > AC > AP$

$\therefore \quad v_Q > v_C > v_P$



37. $$M_1 = \frac{4}{3}\pi R^3 \cdot (2d)$$

$$M_2 = \frac{4}{3}\pi\,(2R)^3 (d) = 4M_1$$

$$k_1 = k_2$$

$$\Rightarrow \quad \frac{L_1^2}{2M_1} = \frac{L_2^2}{2M_2} \quad \Rightarrow \quad \frac{L_1}{L_2} = \sqrt{\frac{M_1}{M_2}} = \sqrt{\frac{1}{4}} = \frac{1}{2}$$

38. Since, the ball begins its motion with pure rolling, friction on it is zero. Thus there is no friction on the plank. It will stay at rest.

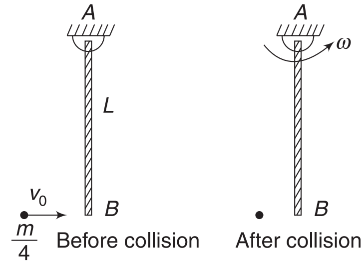
39. Angular momentum is conserved.

$$K = \frac{L^2}{2I}$$

After folding of arms, new KE is $K' = \dfrac{L^2}{2(2I)} = \dfrac{K}{2}$

40. On each object, angular impulse is F.R.t. Thus change in angular momentum is same for all of them.

41. Conservation of angular momentum (about A) gives:



$$L_i = L_f$$

$$\frac{m}{4} v_0 L = \frac{1}{3} mL^2 \cdot \omega$$

$$\Rightarrow \qquad \omega = \frac{3}{4}\frac{v_0}{L}$$

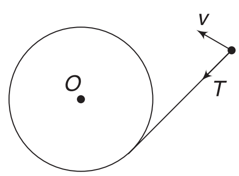
Velocity of end B after collision is:

$$v_B = \omega L = \frac{3}{4}v_0$$

$$e = \frac{\text{relative speed of separation of end } B \text{ and the particle after collision}}{\text{relative speed of approach of the particle before collision}}$$

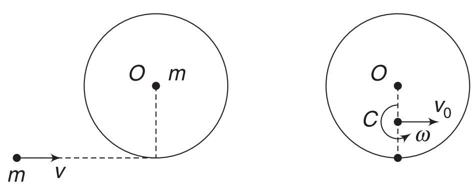
$$= \frac{\frac{3}{4}v_0}{v_0} = \frac{3}{4}$$

42. Tension is always perpendicular to the direction of velocity of the ball. Hence, tension does not perform work on the ball. KE of the ball does not change. But tension has a torque about O. Thus angular momentum of the ball about O does change.



43. COM of the (hoop + particle) system is at a distance $\frac{R}{2}$ from O (at C). After collision COM moves with velocity v_0 given by:

$$2mv_0 = mv \quad \Rightarrow \quad v_0 = \frac{v}{2}$$



We will conserve angular momentum about a point attached to the table that is just below C.

$$L_i = L_f \quad \Rightarrow \quad mv\frac{R}{2} = I_C \cdot \omega$$

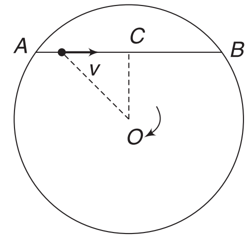
$$\Rightarrow mv\frac{R}{2} = \left[mR^2 + m\left(\frac{R}{2}\right)^2 + m\left(\frac{R}{2}\right)^2\right]\omega$$

$$\Rightarrow \qquad \omega = \frac{v}{3R}$$

44. Due to impact, linear motion of A will get transferred to B(head on elastic collision of equal masses). There will be no change in angular speeds of the two spheres as there is no angular torque on them.

45. There is no torque about rotation axis, hence angular momentum is conserved.

Only force that performs work is gravity, thus mechanical energy is conserved.



46. When tortoise moves from A to C the MI of the system about rotation axis through O decreases. As it moves from C to B the MI increases. During the entire course of motion angular momentum of the system is conserved. Thus, angular speed first increases when tortoise moves from A to C and then it decreases as MI increase when it moves from C to B. It is easy to see that the variation is not linear.

47. A massless rod will not apply a force perpendicular to its length.

$\therefore$ Velocity of the particle immediately after impact = v. And velocity of the other particle immediately after impact = 0. Angular velocity of moving end wrt the stationary end is $\omega = \frac{v}{l}$

48. $L_i = L_f$ (about centre of disc)

$$(I + mR^2)\,\omega = I\omega_0 + mvR$$

$$\Rightarrow \qquad \omega_0 = \frac{(I + mR^2)\,\omega - mvR}{I}$$

49. $L_i = L_f$ (about C)

$$\frac{Ml^2}{12}\cdot\omega = mv\frac{l}{3}$$

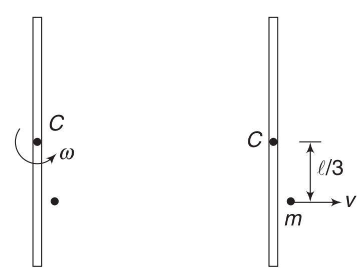
$$\Rightarrow \quad Ml\omega = 4mv \qquad \text{...(i)}$$

Since, collision is elastic

$\therefore$ Velocity of approach = velocity of separation

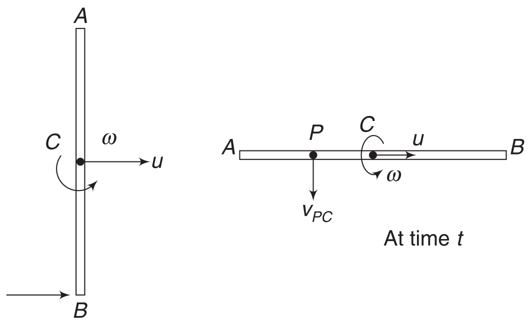
$$\omega\frac{l}{3} = v$$

Putting this in (i) gives, $m = \frac{3}{4}M$.



50. Immediately after impact, the velocity of the centre is

$$mu = J \quad \Rightarrow \quad u = \frac{J}{m}$$



Immediately after impact

Angular speed about the centre is given by angular-impulse momentum theorem.

$$\frac{ml^2}{12}\cdot\omega = J\cdot\frac{l}{2}$$

$$\Rightarrow \qquad \omega = \frac{6J}{ml}$$

Angle by which the rod will rotate in time $t = \frac{\pi ml}{12J}$ is

$$\theta = \omega t = \frac{6J}{ml}\cdot\frac{\pi ml}{12J} = \frac{\pi}{2}$$

Velocity of point P wrt C at this time is

$$v_{PC} = \omega\frac{l}{6} = \frac{J}{m}$$

$\therefore$ Velocity of P relative to ground is

$$v_P = \sqrt{(v_{PC})^2 + u^2} = \sqrt{2}\,\frac{J}{m}$$

Worksheet 2

1. In first case the man will continue to have tangential speed (ωr) due to inertia.

   Thus angular momentum of the man wrt centres does not change. Hence angular speed of the platform will also not change. In second case, after the man steps in, (due to friction) he will acquire rotation. Since *MI* of the system increases, ω must decrease

2. Friction remains constant throughout.

   Using equation (9)

   $$f = \frac{Mg\sin\theta}{\frac{R^2}{K^2}+1} = \frac{Mg\sin\theta}{\frac{3}{2}+1} = \frac{2Mg\sin\theta}{5}$$

   But $\quad f \le \mu N \;\Rightarrow\; \frac{2Mg\sin\theta}{5} \le \mu Mg\cos\theta$

   $$\Rightarrow \frac{2}{5}\tan\theta \le \mu$$

3. *MI* about any tangent will be same.

   $\therefore \qquad I_1 = I_2 = I_4$

   From perpendicular axes theorem:

   $$I_z = I_1 + I_2 = I_1 + I_4$$

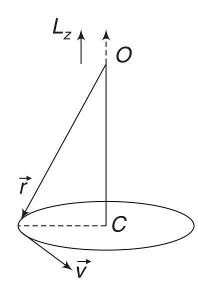
   Distance of centre from O is:

   $$r = \sqrt{R^2 + R^2} = \sqrt{2}R$$

   $$\therefore \qquad I_z = I_{CM} + mr^2 = \frac{mR^2}{2} + m(2R^2) = \frac{5}{2}mR^2$$

4. During uphill motion v decrease and ω increases. Thus there will be a point where $v = \omega R$. During sliding, friction is kinetic. Its value is μN. Once sliding ceases, static friction adjusts to a value given by equation (9).

5. Angular momentum about O is $\vec{L} = \vec{r} \times (m\vec{v})$

   As the particle rotates, the plane containing $\vec{r}$ and $\vec{v}$ changes. Thus direction of $\vec{L}$ changes.

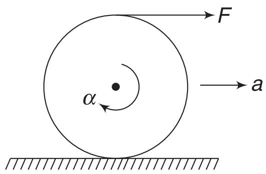
   However, magnitude of $\vec{L}$ about O does not change.

   Vertical component of $\vec{L}$ about O is same as angular momentum about line OC.

   $L_z = mvr$ (Directed up in the diagram)

   L_c is also mvr directed along L_z.

6. Refer to Section 4.2.



For a ring (or a hollow cylinder) the force produces acceleration $a = \dfrac{F}{M}$

And angular acceleration produced is

$$\alpha = \frac{FR}{MR^2} = \frac{F}{MR} \quad \Rightarrow \quad \alpha R = \frac{F}{M}$$

$$\Rightarrow \qquad a = R\alpha.$$

For any other object: $\alpha = \dfrac{FR}{MK^2} \;\Rightarrow\; R\alpha = \dfrac{FR^2}{MK^2}$

$$\Rightarrow \quad R\alpha = a\left(\frac{R^2}{K^2}\right)$$

$$\therefore \quad R\alpha > a$$

7. Assume the situation shown in figure.

$$F - f = Ma \qquad \text{...(i)}$$

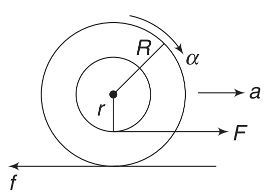
$$fR - Fr = MK^2 \cdot \alpha$$

$$f - F\left(\frac{r}{R}\right) = \frac{MK^2}{R^2} \cdot a \qquad \text{...(ii)}$$

(i) + (ii) gives

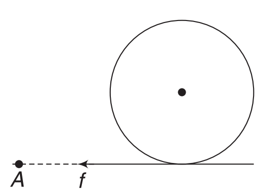
$$F\left[1 - \frac{r}{R}\right] = M\left[1 + \frac{K^2}{R^2}\right]a$$

a is positive as $1 > \dfrac{r}{R}$.



$\therefore$ rotation is clockwise and the thread will wind up if rotation is clockwise.

8. Friction applied by the table cloth will provide a linear as well as angular velocity to the body. But the friction force has no torque about a point like A. Thus angular momentum of the body about A is zero [$L_{\text{spin}} + L_{\text{orbital}} = 0$]. Angular momentum about A will be conserved after the table cloth is removed (i.e., it will remain zero). Thus, the body will finally come to rest while having no spin.



9. For hoop: $\qquad k_1 = k_T + k_R = Mv_1^2$

For cylinder: $\quad k_2 = k_T + k_R = \dfrac{3}{4} Mv_2^2$

Since, $k_1 = k_2$

$$\therefore \quad v_1^2 = \frac{3}{4} v_2^2 \;\Rightarrow\; v_1 = \frac{\sqrt{3}}{2} v_2 \;\Rightarrow\; v_1 < v_2$$

$$L_{\text{spin}}^{\text{hoop}} = I_{\text{CM}} \cdot \frac{v_1}{R} = MR^2 \cdot \frac{\sqrt{3}v_2}{2R} = \frac{\sqrt{3}}{2} Mv_2R$$

$$L_{\text{spin}}^{\text{cylinder}} = I_{\text{CM}} \frac{v_2}{R} = \frac{1}{2} MR^2 \frac{v_2}{R} = \frac{1}{2} Mv_2R$$

$$\therefore \quad L_{\text{spin}}^{\text{hoop}} \neq L_{\text{spin}}^{\text{cylinder}}$$

Since, both have same KE, they will attain same height on an incline. But they will have different retardations.

$$a_1 = \frac{g\sin\theta}{1 + \dfrac{k^2}{R^2}} = \frac{g\sin\theta}{2}$$

$$a_2 = \frac{g\sin\theta}{1 + \dfrac{1}{2}} = \frac{2}{3} g\sin\theta$$

Time needed by hoop to stop is given by

$$t_1 = \frac{v_1}{a_1} = \frac{\sqrt{3}\,v_2}{2 \cdot \dfrac{g\sin\theta}{2}} = \frac{\sqrt{3}\,v_2}{g\sin\theta}$$

For cylinder, $t_2 = \dfrac{v_2}{a_2} = \dfrac{v_2}{\dfrac{2}{3} g\sin\theta} = \dfrac{3}{2} \dfrac{v_2}{g\sin\theta}$

$$\therefore \qquad t_1 > t_2$$

10. In case (i) there is no rotation about COM.

In case (ii) there is rotation. In both cases speed acquired by COM is same.

12. $I_1 + I_3 = I_z$ But $I_1 = I_3$

$$\therefore \quad I_1 = I_3 = \frac{I_z}{2}$$

Similarly, $I_2 = I_4 = \dfrac{I_z}{2}$

14. $\dfrac{d\vec{L}}{dt} = \vec{\tau}$

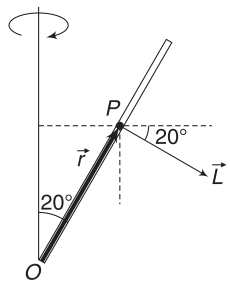
Since $\vec{\tau}$ ($= \vec{A} \times \vec{L}$) is perpendicular to $\vec{L}$, thus $\dfrac{d\vec{L}}{dt}$ is perpendicular to $\vec{L}$.

$\dfrac{d\vec{L}}{dt}$ is perpendicular to both $\vec{L}$ and $\vec{A}$.

It means $d\vec{L}$ is perpendicular to $\vec{A}$.

It means component of $\vec{L}$ along $\vec{A}$ does not change.

15. Consider any particle on the rod. Its angular momentum is $\vec{L} = \vec{r} \times (m\vec{v})$. Direction of $\vec{L}$ is perpendicular to $\vec{r}$ as well as $\vec{v}$.

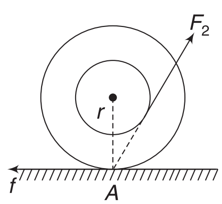


In the position shown, point P has velocity $(\vec{v})$ directed out of the plane of figure. Thus $\vec{L}$ is in the direction shown.

As the rod rotates, vertical component of L does not change but horizontal component of L is rotating. Horizontal component of vector L is changing (in direction). There must be a torque in the direction of change of L.

16. As seen in Q.7 of this worksheet, when F_1 is applied, acceleration will be towards right.

When F_3 is applied, it gives an anticlockwise torque. There is no tendency of translation. Friction will act towards left to oppose the tendency of slippage. Thus, the yo–yo will move towards left.

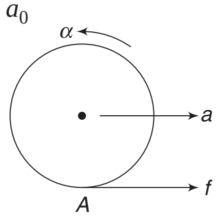


Line of F_2 passes through point of contact A. There is no torque about A. Friction will adjust to a value equal to horizontal component of F_2 and there will be no motion.

17. If a_0 is acceleration of the plank, then for no slipping, tangential acceleration of the contact point must be a_0.

$$a + r\alpha = a_0 \quad \Rightarrow \quad a < a_0$$

Friction does perform work on the sphere. The contact point (A) when the friction acts is not at rest. Work done by friction on the sphere is equal to change in KE of the sphere. Friction does negative work on the plank.



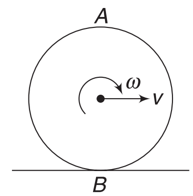
On the system (sphere + plank) as a whole, static friction does not perform work.

19. $L_B = L_{spin} + L_{orbital}$

$$= \frac{1}{2} MR^2\omega + MvR$$

$$= \frac{3}{2} MvR\ (\circlearrowright) \qquad [\because R\omega = v]$$

And $L_A = L_{spin} + L_{orbital} = \frac{1}{2} MR^2\omega\ (\circlearrowright) + MvR\ (\circlearrowleft)$



$$= \frac{1}{2} MvR\ (\circlearrowright) + MvR\ (\circlearrowleft) = \frac{1}{2} MvR\ (\circlearrowleft)$$

$$\therefore \quad |L_B| = 3|L_A|$$

20. When COM moves by S, the point of application of F moves by $2S$.

$$\therefore \qquad W_F = F(2S)$$

KE of rolling sphere is $\frac{7}{10} Mv^2$

$$\therefore \quad \frac{7}{10} Mv^2 = 2FS \quad \Rightarrow \quad v = \sqrt{\frac{20FS}{7M}}$$

21. $L = L_{spin} + L_{orbital}$

Both L_{spin} and $L_{orbitals}$ are maximum when the sphere reaches the lowest position.

Friction on incline plane is $f = \dfrac{mg\sin\theta}{1 + \dfrac{R^2}{K^2}}$

Friction is maximum where $\sin\theta$ is maximum.

Slope is maximum at the top, hence friction is maximum at the top.

When the sphere reaches the bottom, its speed is given by

$$\frac{7}{10} mv^2 = mgR \quad \Rightarrow \quad \frac{mv^2}{R} = \frac{10\,mg}{7}$$

For circulation motion COM: $N = mg = \dfrac{mv^2}{R}$

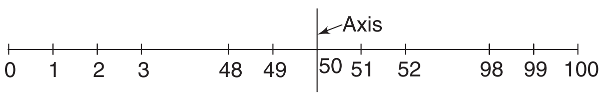
$$\Rightarrow \qquad N = \frac{17}{7} mg$$

22. Friction is kinetic and it dissipates KE.

The force applied by the axes on the discs do not have zero torque about the point of contact.

Worksheet 3

1. 1g and 99 g particles are at a distance of 49 cm from the axis. 2 g and 98 g particles are at a distance of 48 cm from the axis; and so on.



$\therefore\ I = (1 + 99) \times 49^2 + (2 + 98) \times 48^2 + (3 + 97) \times 47^2 + \ldots + (49 + 51) \times 1^2 + 50 \times 0^2$

$= 100\,[49^2 + 48^2 + 47^2 + \ldots + 2^2 + 1^2]$

$= 100 \times \dfrac{49(49+1)(2 \times 49 + 1)}{6}$

$\left[\text{Sum of squares of first } n \text{ natural numbers is } S = \dfrac{n(n+1)(2n+1)}{6}\right]$

$= 4.04 \times 10^6$ g-cm^2

$= 0.404$ kgm^2

2. Consider a shell of radius x and thickness dx.

Volume of shell, $dv = 4\pi x^2\,dx$

Mass of shell, $dm = \rho \cdot dv$

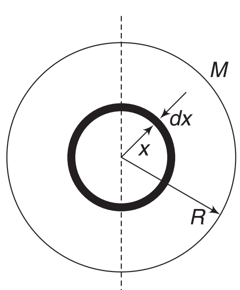
$= \rho_0\left(1 + \dfrac{x}{R}\right) 4\pi x^2\,dx$

$dI_{\text{shell}} = \dfrac{2}{3}(dm)x^2 = \dfrac{2}{3}\rho_0\left(1 + \dfrac{x}{R}\right)4\pi x^4 dx$

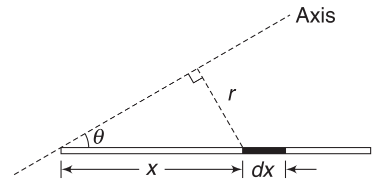
$= \dfrac{8\pi\rho_0}{3}x^4 dx + \dfrac{8\pi\rho_0}{3R}x^5 dx$

$\therefore\ I = \int_{x=0}^{R} dI_{\text{shell}} = \dfrac{8\pi\rho_0}{3}\int_0^R x^4 dx + \dfrac{8\pi\rho_0}{3R}\int_0^R x^5 dx$

$= \dfrac{44}{45}\pi\rho_0 R^5$



3. Consider an element as shown. Mass of element,



$dm = \dfrac{M}{L}dx.$

Perpendicular distance of the element from the axis is

$r = x\sin\theta$

$\therefore\quad dI = \left(\dfrac{M}{L}dx\right)(r^2) = \dfrac{M}{L}\sin^2\theta\, x^2 dx$

$\therefore\quad I = \dfrac{M}{L}\sin^2\theta\int_0^L x^2 dx = \dfrac{1}{3}ML^2\sin^2\theta$

4. Area of plate $= (2a)^2 - a^2 = 3a^2$

Mass per unit area, $\sigma = \dfrac{M}{3a^2}$

Mass of removed part, $m = \sigma . a^2 = \dfrac{M}{3}$

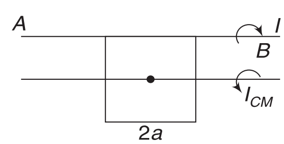
Mass of plate before making of hole $= M + \dfrac{M}{3}$

$M_0 = \dfrac{4M}{3}$

For a square plate of mass M_0, moment of inertia about its side is

$I = I_{\text{CM}} + M_0 a^2 = \dfrac{M_0(2a)^2}{12} + M_0 a^2$

$= \dfrac{4}{3}M_0 a^2 = \dfrac{4}{3}\cdot\dfrac{4M}{3}a^2 = \dfrac{16Ma^2}{9}$



For square of side a [the hole] and mass m, moment of inertia about AB will be

$I_{\text{hole}} = I_{\text{CM}} + md^2$

$= \dfrac{ma^2}{12} + ma^2 = \dfrac{13}{12}ma^2 = \dfrac{13}{12}\cdot\dfrac{M}{3}a^2$

$= \dfrac{13Ma^2}{36}$

$\therefore$ *MI* of the plate with hole about *AB* is

$= I - I_{\text{hole}} = \left(\dfrac{16}{9} - \dfrac{13}{36}\right)Ma^2 = \dfrac{17}{12}Ma^2.$

5. (a) As the drum rotates, it winds the string and X moves up.

a = retardation of X; α = angular retardation of Y.

Both X and Y retard and eventually stop.

For X: $mg\sin\theta - T = ma$ ...(i)

For Y: $T\cdot r = \frac{1}{2}Mr^2\cdot\alpha$

$\Rightarrow$ $T = \frac{1}{2}M(r\alpha)$

$\Rightarrow$ $T = \frac{1}{2}Ma$ ...(ii)

Solving (i) and (ii) gives

$$a = \frac{mg\sin\theta}{m + \frac{M}{2}} = \frac{0.5 \times 10 \times \frac{1}{2}}{0.5 + \frac{2}{2}} = \frac{5}{3}\ \text{ms}^{-2}$$

and $T = \frac{5}{3}N$

(b) Initial velocity of x is $u = \omega r = 10 \times 0.5 = 5\ \text{ms}^{-1}$ up the plane.

Using $v^2 = u^2 + 2as$

$0 = 5^2 - 2 \times \frac{5}{3} \times s$

$\Rightarrow$ $s = 7.5\,m$

6. Let α be angular acceleration of the pulley in clockwise sense.

Acceleration of B will be $a_2 = a\alpha\ (\downarrow)$

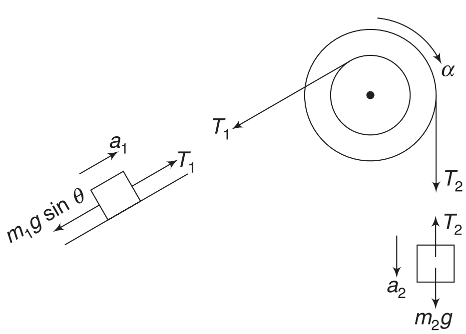
Acceleration of A will be $a_1 = b\alpha$ up the incline.

For pulley:

$T_2a - T_1b = I\alpha$

$0.15\,T_2 - 0.1\,T_1 = 0.11\,\alpha$

$\Rightarrow$ $15\,T_2 - 10\,T_1 = 11\,\alpha$ ...(i)



For A:

$T_1 - m_1g\sin\theta = m_1a_1$

$T_1 - 2 \times 10 \times \frac{1}{2} = 2(0.1\,\alpha)$

$\Rightarrow$ $T_1 - 10 = 0.2\,\alpha$ ...(ii)

For B:

$m_2g - T_2 = m_2a_2 \Rightarrow \frac{26}{7}g - T_2 = \frac{26}{7}(0.15\,\alpha)$

$\Rightarrow$ $260 - 7\,T_2 = 3.9\,\alpha$ ...(iii)

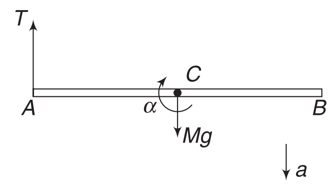
Solving (i), (ii) and (iii) gives

$\alpha = 21.4\ \text{rad s}^{-2}$; $T_1 = 14.28\,N$; $T_2 = 25.22\,N$

$a_1 = 0.1 \times \alpha = 2.14\,\text{ms}^{-2}$;

$a_2 = 0.15 \times \alpha = 3.21\ \text{ms}^{-2}$.

7. After the right string is cut, forces on the rod are-its weight (Mg) and tension at end A. Both the forces are vertical. Hence, COM can only have vertical acceleration.



Let acceleration of COM be $a(\downarrow)$ and angular acceleration of the rod be α.

$Mg - T = Ma$ ...(i)

$I_{CM}\cdot\alpha = \tau_{CM}$

$\frac{ML^2}{12}\cdot\alpha = T\cdot\frac{L}{2}$

$\Rightarrow$ $\frac{ML\alpha}{6} = T$ ...(ii)

Acceleration of end A in COM frame $= \frac{L}{2}\alpha\ (\uparrow)$.

String is inextensible and acceleration of end A in vertical direction shall be zero, in ground frame

$\therefore$ $a_A = 0 \Rightarrow a(\downarrow) + \frac{L}{2}\alpha\ (\uparrow) = 0.$

$\Rightarrow$ $\frac{L\alpha}{2} = a$

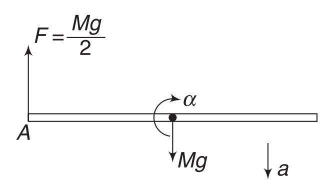
put in (ii) $\frac{Ma}{3} = T$

put in (i) $a = \frac{3g}{4}$

And $T = \frac{M}{3}a = \frac{Mg}{4}$.

8. In equilibrium both the springs apply a force

$F = \frac{Mg}{2}$ on the rod.



When right spring is cut, the spring force at A cannot change suddenly. [String tension can change suddenly but not the spring tension].

$$\therefore \quad Ma = Mg - \frac{Mg}{2}$$

$$\Rightarrow \quad a = \frac{g}{2}$$

Writing $\tau = I\alpha$ about COM,

$$\frac{ML^2}{12}\cdot \alpha = F\cdot\frac{L}{2}$$

$$\Rightarrow \quad \frac{ML^2}{12}\alpha = \frac{Mg}{2}\cdot\frac{L}{2} \quad \Rightarrow \quad \alpha = \frac{3g}{L}$$

Acceleration of the end, in COM frame is

$$\alpha\frac{L}{2} = \frac{3g}{2}\ (\uparrow).$$

Acceleration of COM is $\frac{g}{2}\ (\downarrow)$

$$\therefore \quad a_A = \frac{3g}{2}(\uparrow) + \frac{g}{2}(\downarrow) = g\ (\uparrow).$$

9. Slipping ceases when velocity of a point on the circumference of the disc becomes v.

$$\omega R = v \quad \Rightarrow \quad \omega = \frac{18}{0.075} = 240 \text{ rad s}^{-1}$$

Friction on disc $f = \mu N = \frac{\mu Mg}{2}$

$$= 0.25 \times 4 \times \frac{10}{2} = 5N$$

MI of disc $I = \frac{1}{2}MR^2 = \frac{1}{2} \times 4\,(0.075)^2$

$$= 0.01125 \text{ kg-m}^2$$

$\therefore$ Angular acceleration, $\alpha = \frac{\tau}{I} = \frac{f\cdot R}{I} = \frac{5 \times 0.075}{0.01125}$

$$= 33.33 \text{ rad s}^{-2}$$

Using $\omega^2 = \omega_0^2 + 2\alpha\theta$

$$(240)^2 = 0 + 2 \times (33.33)\,\theta$$

$$\Rightarrow \quad \theta = 864 \text{ rad.}$$

$\therefore$ Number of turns $= \frac{864}{2\pi} = 137.6$

10. Refer to Article 4.2 in the text. The ring will roll without sliding even if there is no friction.

11. The contact point has velocity in forward direction. Hence, friction is towards left.

Retardation: $a = \frac{f}{M}$

Angular retardation $\alpha = \frac{\tau}{I} = \frac{fR}{\frac{2}{5}MR^2} = \frac{5f}{2MR}$

After time t: $v = v_0 - at = v_0 - \frac{ft}{M}$

$$\omega = \omega_0 - \alpha t = \omega_0 - \frac{5ft}{2MR}$$

v becomes zero at time given by $v_0 - \frac{ft}{M} = 0$.

$$\Rightarrow \quad \frac{ft}{M} = v_0 \qquad \text{...(i)}$$

(a) At this time ω is also zero

$$\Rightarrow \quad \omega_0 - \frac{5ft}{2MR} = 0 \quad \Rightarrow \quad \omega_0 - \frac{5}{2}\frac{v_0}{R} = 0.$$

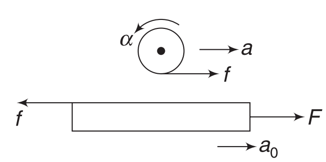
$$\Rightarrow \quad \frac{v_0}{\omega_0 R} = \frac{2}{5}.$$

(b) For sphere to be rotating $\omega > 0$.

$$\Rightarrow \quad \omega_0 > \frac{5ft}{2MR} \quad \Rightarrow \quad \omega_0 > \frac{5}{2}\frac{v_0}{R} \quad \text{[using (i)]}$$

$$\Rightarrow \quad \frac{2}{5} > \frac{v_0}{\omega_0 R}$$

12. a_0 = acceleration of plank.
a = acceleration of sphere
α = angular acceleration of sphere



For no slipping

$$R\alpha + a = a_0 \qquad \text{...(i)}$$

For sphere: $f = ma$ ...(ii)

$$fR = \frac{2}{5}mR^2\cdot\alpha$$

$$\Rightarrow \quad f = \frac{2}{5}m\,(R\alpha) \qquad \text{...(iii)}$$

For plank: $F - f = Ma_0$ ...(iv)

Calculating a, $R\alpha$ and a_0 from (ii), (iii) and (iv) and putting into (i) gives:

$$\frac{5f}{2m} + \frac{f}{m} = \frac{F}{M} - \frac{f}{M}$$

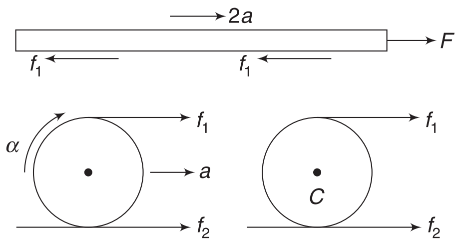
$$\Rightarrow f\left[\frac{5}{2m} + \frac{1}{m} + \frac{1}{M}\right] = \frac{F}{M} \quad \Rightarrow \quad f = \frac{2Fm}{7M + m}$$

But $f \le \mu mg$

$$\therefore \quad \frac{2Fm}{7M + m} \le \mu mg \quad \Rightarrow \quad \frac{2F}{(7M + m)g} \le \mu$$

13. Let acceleration of two cylinders be a.

Acceleration of plank A = horizontal acceleration of top point of the two cylinders = $2a$.



Friction forces at various surfaces are assumed to be f_1 and f_2 as shown.

For A: $F - 2f_1 = M(2a)$ ...(i)

For B: $f_1 + f_2 = ma$ ...(ii)

and $f_1 r - f_2 r = \frac{1}{2}mr^2 \cdot \alpha$

$\Rightarrow \quad f_1 - f_2 = \frac{1}{2}ma \quad [\because r\alpha = a]$ ...(iii)

Adding (i), (ii) and (iii) $F = \left(2M + \frac{3}{2}m\right)a.$

$$\Rightarrow \quad a = \frac{2F}{4M + 3m}$$

$\therefore$ Acceleration of A is $= \frac{4F}{4M + 3m}$

Using $\quad v^2 = u^2 + 2as$

$$v^2 = 0 + \frac{8FL}{4M + 3m} \Rightarrow v = \sqrt{\frac{8FL}{4M + 3m}}$$

14. (a) Torque on disc about its central rotation axis is

$$\tau = F \cdot r$$

$$\therefore \quad \alpha = \frac{\tau}{I} = \frac{F \cdot r}{\frac{1}{2}mr^2} = \frac{2F}{mr}$$

We have used $\tau = I\alpha$ about an axis through COM and it is perfectly alright even if COM is accelerated.

Linear acceleration of the (disc + mount) system is

$$a = \frac{F}{M + m}$$

Acceleration of point A wrt centre of the disc is

$$r\alpha = \frac{2F}{m}$$

$\therefore$ Acceleration of point A relative to the ground is

$$a_A = \frac{2F}{m} + \frac{F}{M + m}$$

(b) Angular speed of disc at time t is

$$\omega = \alpha t = \frac{2Ft}{mr}$$

It's rotational KE is: $k_R = \frac{1}{2}\left(\frac{1}{2}mr^2\right)\omega^2 = \frac{F^2t^2}{m}$

Linear speed of system at time t:

$$v = at = \frac{Ft}{M + m}$$

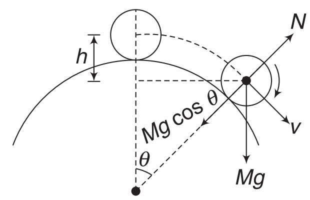
Translational KE at t is:

$$k_T = \frac{1}{2}(M + m)\left(\frac{Ft}{M + m}\right)^2$$

$$= \frac{1}{2}\frac{F^2t^2}{M + m}$$

$$\therefore \quad k = k_R + k_T = F^2t^2\left[\frac{1}{m} + \frac{1}{2(M + m)}\right]$$

15. (a) The COM moves in a circle of radius $(R + r)$.



Let v be speed of the COM of the ball at angular position θ.

$$\frac{7}{10}mv^2 = mgh$$

$$\frac{7}{10}v^2 = g(R + r)[1 - \cos\theta]$$

$$\Rightarrow \quad v^2 = \frac{10}{7}g(R + r)[1 - \cos\theta] \quad ...(i)$$

If the ball loses contact at this position, then $N = 0$

$$\Rightarrow \quad mg\cos\theta = \frac{mv^2}{R + r}$$

$$\Rightarrow \quad g\cos\theta = \frac{10g}{7}[1 - \cos\theta] \text{ [using (i)]}$$

$$\Rightarrow \quad 17\cos\theta = 10 \quad \Rightarrow \quad \cos\theta = \frac{10}{17}$$

(b) From (i) $v^2 = \frac{10}{7}g(R + r)\left[\frac{17 - 10}{17}\right] = \frac{10}{17}g(R + r)$

$$\therefore \quad v = \sqrt{\frac{10g(R + r)}{17}}$$

$$\therefore \quad \omega = \frac{v}{r} = \frac{1}{r}\sqrt{\frac{10g(R + r)}{17}}$$

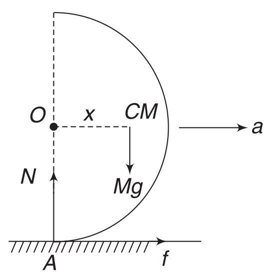
16. Height of point O from the surface cannot change. Mechanical energy is conserved.

17. Distance of COM from O is $x = \dfrac{3R}{8}$

Torque about A is

$$\tau_A = Mg \cdot x = \frac{3MgR}{8}$$

Moment of inertia about the instantaneous axis through A is



$$I_A = \frac{7}{5}MR^2$$

[It makes no difference whether it is a complete sphere or half sphere when we are writing MI about axis through A. Why?]

$$\therefore \qquad \alpha = \frac{\tau_A}{I_A} = \frac{15}{56}\frac{g}{R}$$

COM has a vertically downward acceleration given by $x\alpha$ [you can think of COM rotating about O. Or else, think of COM rotating about A. Acceleration of COM will be $\perp$ to line joining A to COM. Component of this acceleration in vertical direction is $x\alpha$]

$$\therefore \quad Mg - N = M(x\alpha)$$

$$\Rightarrow \qquad N = Mg - M\left(\frac{3R}{8}\right)\left(\frac{15}{56}\frac{g}{R}\right)$$

$$= \frac{403}{308}Mg.$$

Horizontal acceleration is $a = \dfrac{f}{M}$ (this is also acceleration of point O).

For pure rolling $a = R\alpha$.

$$\therefore \qquad \frac{f}{M} = \frac{15g}{56} \quad \Rightarrow \quad f = \frac{15Mg}{56}$$

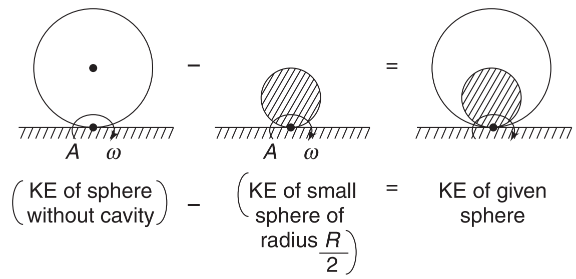
$$\text{But} \qquad f \le \mu N \quad \Rightarrow \quad \frac{15Mg}{56} \le \mu\frac{403}{408}Mg.$$

$$\Rightarrow \quad 0.27 \le \mu$$

18. Density $\rho = \dfrac{M}{\dfrac{4}{3}\pi R^3 - \dfrac{4}{3}\pi\left(\dfrac{R}{2}\right)^3} = \dfrac{8M}{7\left(\dfrac{4}{3}\pi R^3\right)}$

Mass of removed part $\left(\text{i.e., mass of a sphere of radius } \dfrac{R}{2}\right)$ is

$$m = \frac{4}{3}\pi\left(\frac{R}{2}\right)^3\rho = \frac{4}{3}\pi\frac{R^3}{8}\,\frac{8M}{7\left(\frac{4}{3}\pi R^3\right)} = \frac{M}{7}$$



Mass of cavity filled sphere, $M_0 = M + m = \dfrac{8M}{7}$

KE of given sphere = (kinetic energy of cavity filled sphere) – (kinetic energy of sphere of radius $\dfrac{R}{2}$ occupying the cavity space)

For writing KE we will consider pure rotation about instantaneous axis through contact point A.

$$\therefore \qquad KE = \frac{1}{2}\left(\frac{7}{5}M_0R^2\right)\omega^2 - \frac{1}{2}\left(\frac{7}{5}m\left(\frac{R}{2}\right)^2\right)\omega^2$$

$$= \frac{7}{10}\cdot\frac{8M}{7}v^2 - \frac{7}{40}\cdot\frac{M}{7}v^2 \text{ [where } R\omega = v]$$

$$= \frac{31}{40}Mv^2.$$

19. Speed of the rim of the pulley = speed of the block.

$$\Rightarrow \qquad \omega R = v$$

where ω = angular speed of the pulley and
v = speed of the block.

When the block falls by a distance x, its PE decreases by mgx. PE in the spring increases by $\dfrac{1}{2}kx^2$ and the block and the pulley both gain kE.

$$\therefore \qquad \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 + \frac{1}{2}kx^2 = mgx$$

$$\Rightarrow \quad \frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{1}{2}MR^2\right)\left(\frac{v}{R}\right)^2 + \frac{1}{2}kx^2 = mgx$$

Put the values of m, M, k and x and simplify to get

$$v = 2.4\,\text{ms}^{-1}.$$

20. MI of the body about rotation axis is

$$I = 0 + ml^2 + \frac{ml^2}{3} \text{ where } m = \text{mass of each rod.}$$

$$= \frac{4}{3}ml^2.$$

Loss in PE when the body becomes vertical is

$$= mgl + mg\frac{l}{2} = \frac{3}{2}mgl$$

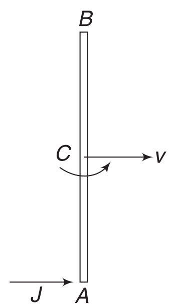
$\left[\text{Centre of one rod falls through } l \text{ and the centre of the other rod falls through } \dfrac{l}{2}\right]$

Conservation of energy:

$$\frac{1}{2}I\omega^2 = \frac{3}{2}mgl$$

$$\Rightarrow \frac{1}{2}\left(\frac{4}{3}ml^2\right)\omega^2 = \frac{3}{2}mgl \quad \Rightarrow \quad \omega = \frac{3}{2}\sqrt{\frac{g}{l}}.$$

21. Let the hammer impart the blow at end A.



Impulse = ΔP

$\therefore \quad J = mv$ ...(i)

Angular impulse about COM is $J\cdot\frac{l}{2}$

$$J\cdot\frac{l}{2} = \Delta L$$

$$\Rightarrow \quad J\frac{l}{2} = \frac{ml^2}{12}\omega \quad \Rightarrow \quad \omega = \frac{6J}{ml} = \frac{6mv}{ml} = \frac{6v}{l}$$

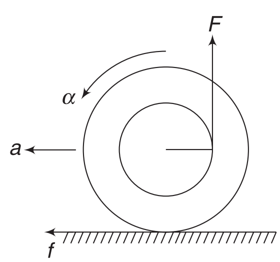
$$\therefore \quad v_A = \omega\frac{l}{2} + v = 4v;\ v_B = \omega\frac{l}{2} - v = 2v(\leftarrow).$$

22. L_{spin} = Angular momentum about COM (C)

$= mvr + mvr = 2mvr.$

$L_{\text{orbital}} = 0$ since COM is not moving.

23. In case (A) the applied force provides an anticlockwise torque. To ensure pure rolling friction will act towards left so as to provide linear acceleration.



$f = Ma$ ...(i)

$$F\frac{R}{2} - fR = Mk^2\cdot\alpha$$

$$\Rightarrow \quad \frac{F}{2} - f = \frac{Ma}{2}\left[\because k^2 = \frac{R^2}{2} \text{ and } R\alpha = a\right] \text{...(ii)}$$

Adding (i) and (ii): $\frac{F}{2} = \frac{3Ma}{2} \quad \Rightarrow \quad a = \frac{F}{3M}$

Similarly, one can solve for case B.

24. COM of the sphere moves in a circle of radius $(R - r)$ and it will complete the circle if it has a minimum speed of $v_0 = \sqrt{g(R-r)}$ at the top.

KE of a sphere rolling with speed v_0 is $\frac{7}{10}mv_0^2$.

$\therefore$ Minimum KE of the sphere at the top of the path is $\frac{7}{10}mg(R-r)$.

Mechanical energy of the sphere is conserved as it rolls up.

$$\therefore \frac{7}{10}mv^2 = \frac{7}{10}mg(R-r) + mg\,2(R-r).$$

$$\Rightarrow \quad v^2 = \frac{27}{7}g(R-r) \quad \Rightarrow \quad v = \sqrt{\frac{27}{7}g(R-r)}$$

25. Energy conservation gives ω when the rod is horizontal.

$$\frac{1}{2}I_A\omega^2 = Mg\frac{L}{2} \quad \Rightarrow \quad \frac{1}{2}\frac{ML^2}{3}\omega^2 = Mg\frac{L}{2}$$

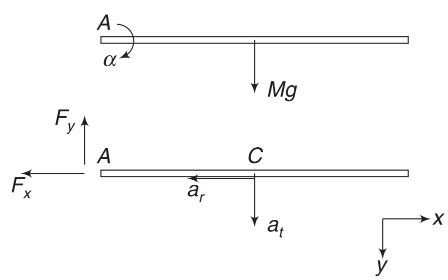
$$\Rightarrow \quad \omega = \sqrt{\frac{3g}{L}}$$

Now we will find angular acceleration of the rod in horizontal position.

$$I_A\cdot\alpha = \tau_A$$

$$\frac{ML^2}{3}\cdot\alpha = Mg\frac{L}{2}$$

$$\Rightarrow \quad \alpha = \frac{3}{2}\frac{g}{L}$$



COM of the rod is moving in circle of radius $\frac{L}{2}$ with angular acceleration α and angular speed ω. Radial acceleration of COM is horizontal.

$$a_r = \omega^2\frac{L}{2} = \frac{3g}{2}$$

Horizontal component of hinge force is responsible for this acceleration $F_x = Ma_r = \frac{3Mg}{2}$

Tangential acceleration of COM is vertically downwards, equal to

$$a_t = \frac{L}{2}\cdot\alpha = \frac{3g}{4}$$

$$\therefore \quad Mg - F_y = \frac{3Mg}{4}$$

[F_y = vertical component of hinge force]

$$\Rightarrow \quad F_y = \frac{Mg}{4}$$

$\therefore$ Hinge force is $F = \sqrt{F_x^2 + F_y^2} = \frac{\sqrt{37}}{4}Mg.$

26. During the fall, the system gains *KE* at the expense of potential energy.

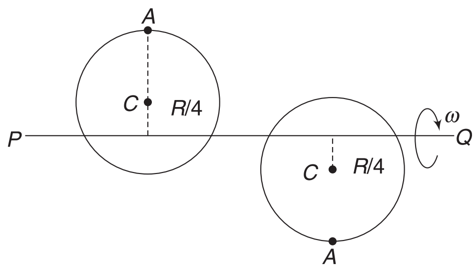


Figure shows initial and final positions.

Loss in *PE* is

$$\Delta U = mg\left(\frac{R}{4} + \frac{R}{4}\right) + mg\left(\frac{5R}{4} + \frac{5R}{4}\right)$$

$$= 3\ mgR$$

$\left[\because\right.$ Centre of the disc falls through $\frac{R}{2}$ and particle at A falls through $\left.\frac{5R}{4} + \frac{5R}{4} = \frac{5R}{2}\right]$

Moment of inertia of the disc-mass system about line *PQ* is

$$I = I_{\text{disc}} + I_{\text{particle}} = \left[\frac{mR^2}{4} + m\left(\frac{R}{4}\right)^2\right] + m\left(\frac{5R}{4}\right)^2$$

$$= \frac{15}{8}\ mR^2$$

Gain in *KE* = Loss in *PE*

$$\frac{1}{2}\ I\omega^2 = 3mgR$$

where ω = angular speed in final position

$$\Rightarrow \quad \frac{15}{16}\ mR^2\omega^2 = 3mgR \quad \Rightarrow \quad \omega = \sqrt{\frac{16g}{5R}}$$

Speed of particle at *A* is $v = \omega\left(\frac{5R}{4}\right) = \sqrt{5gR}$

$\left[\right.$Since, the particle is rotating in a circle of radius $\left.\frac{5R}{4}\right]$

27. $u_x = \frac{u}{\sqrt{2}};\ u_y = \frac{u}{\sqrt{2}}$

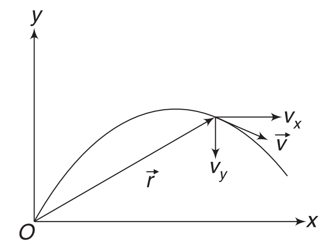
At $t = \frac{u}{g};$

$$v_y = \frac{u}{\sqrt{2}} - g.\frac{u}{g} = -u\left(1 - \frac{1}{\sqrt{2}}\right)$$

$$\therefore \quad \vec{v} = \frac{u}{2}\ \hat{i} - u\left(1 - \frac{1}{\sqrt{2}}\right)\hat{j}$$

Co-ordinates at time $t = \frac{u}{g}$ are

$$x = \frac{u}{\sqrt{2}}\cdot t = \frac{u^2}{\sqrt{2}\,g}$$



$$y = \frac{u}{\sqrt{2}}t - \frac{1}{2}gt^2 = \frac{u^2}{\sqrt{2}\,g} - \frac{1}{2}g\frac{u^2}{g^2} = \frac{u^2}{2g}\ (\sqrt{2} - 1)$$

$\therefore$ Position vector relative to *O* is

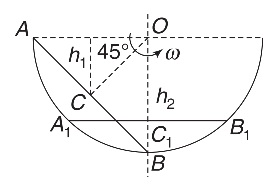
$$\vec{r} = \frac{u^2}{\sqrt{2}\,g}\ \hat{i} + \frac{u^2}{2g}\ (\sqrt{2} - 1)\hat{j}$$

$$\vec{L} = \vec{r} \times \vec{P} = m(\vec{r} \times \vec{v})$$

$$= m\left[\frac{u^2}{\sqrt{2}\,g}\ \frac{u}{\sqrt{2}}\ (1 - \sqrt{2}) - \frac{u^2}{2g}(\sqrt{2} - 1)\frac{u}{\sqrt{2}}\right]\hat{k}$$

$$= -\frac{mu^3}{2\sqrt{2}\,g}\ \hat{k}$$

28. The entire rod is rotating about point *O*. Note that any particle in the rod remains at a fixed distance from *O* when the rod moves. Let angular speed of the rod be ω when it becomes horizontal.



$$OC = OC_1 = R\cos 45°$$

$$h_1 = OC\cdot\sin\ 45°$$

Loss in *PE* of the rod is

$$\Delta U = Mg\,(h_2 - h_1)$$

$$= Mg\,[R\cos 45° - R\cos 45°\cdot\sin 45°]$$

$$= MgR\left[\frac{1}{\sqrt{2}} - \frac{1}{2}\right] = \frac{MgR\,(\sqrt{2} - 1)}{2}$$

MI of the rod about an axis passing through *O* perpendicular to the plane of the figure is

$$I = \frac{M(\sqrt{2}\,R)^2}{12} + M(OC_1)^2$$

$$= \frac{MR^2}{6} + M\left(\frac{R}{\sqrt{2}}\right)^2 = \frac{2}{3}\ MR^2$$

KE of the rod when it is horizontal is $k = \frac{1}{2}I\omega^2 = \frac{1}{3}\ MR^2\,\omega^2$

Gain in *KE* = loss in *PE*

$$\frac{1}{3}MR^2\omega^2 = MgR\left(\frac{\sqrt{2}-1}{2}\right)$$

$$\Rightarrow \qquad \omega = \sqrt{\frac{3(\sqrt{2}-1)}{2}\frac{g}{R}}$$

Velocity of centre: $v = (OC_1)\omega = \frac{R}{\sqrt{2}}\cdot\omega$

$$= \frac{1}{2}\sqrt{3(\sqrt{2}-1)gR}$$

29. Top point of roller moves with a speed that is twice that of the centre.

Each roller moves by L when the plank moves by $2L$.

Let speed of centre of each roller be v.

KE of each roller $= \frac{3}{4}Mv^2$.

Speed of plank $= 2v$.

KE of plank + rollers $= \frac{1}{2}M(2v)^2 + 2 \times \frac{3}{4}Mv^2 = \frac{7}{2}Mv^2$

Loss in PE = Loss in PE of plank + (loss in PE of each roller) × 2

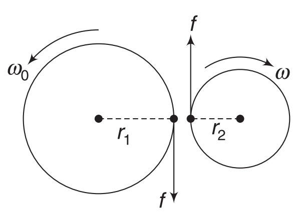
$$= Mg(2L\sin 30°) + [Mg(L\sin 30°)] \times 2$$

$$= MgL + MgL = 2MgL.$$

From conservation of energy: $\frac{7}{2}Mv^2 = 2MgL$

$$\Rightarrow \qquad v = \sqrt{\frac{4}{7}gL}$$

30. Let kinetic friction between the two cylinders be f and t be the time in which they stop slipping. Let final angular speed of first cylinder be ω_0 and that of the second cylinder be ω.



When there is no slipping

$$\omega_0 r_1 = \omega r_2 \qquad \text{...(i)}$$

Using angular-impulse momentum theorem for both cylinders, we can write:

$$-fr_1t = I_1\omega_0 - I_1\omega_1$$

$$\Rightarrow \qquad -ft = \frac{I_1\omega_0}{r_1} - \frac{I_1\omega_1}{r_1} \qquad \text{...(ii)}$$

And $\qquad fr_2t = I_2\omega - I_2\omega_2$

$$ft = \frac{I_2\omega_0 r_1}{r_2^2} - \frac{I_2\omega_2}{r_2} \quad \text{[using (i)]} \qquad \text{...(iii)}$$

Adding (ii) and (iii)

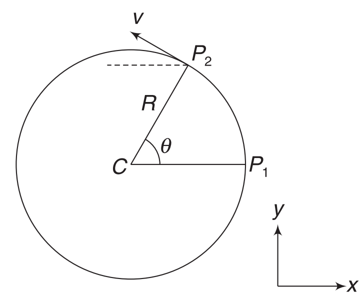
$$\omega_0\left[\frac{I_1}{r_1} + \frac{I_2 r_1}{r_2^2}\right] = \frac{I_1\omega_1}{r_1} + \frac{I_2\omega_2}{r_2}$$

$$\therefore \qquad \omega_0 = \left[\frac{I_1\omega_1 r_2 + I_2\omega_2 r_1}{I_1 r_2^2 + I_2 r_1^2}\right]r_2$$

And $\qquad \omega = \frac{\omega_0 r_1}{r_2} = \left[\frac{I_1\omega_1 r_2 + I_2\omega_2 r_1}{I_1 r_2^2 + I_2 r_1^2}\right]r_1$

31. Particle moves from P_1 to P_2 in time t.

$$\theta = \omega t.$$



Co-ordinates of the particle are:

$$x = R\cos\omega t$$

$$y = R\sin\theta = R\sin\omega t$$

$$z = a.$$

$$\therefore \qquad \vec{r} = R(\cos\omega t)\hat{i} + (R\sin\omega t)\hat{j} + a\hat{k}$$

Velocity: $\vec{v} = -v\sin\theta\,\hat{i} + v\cos\theta\,\hat{j}$

$$= R\omega[-\sin(\omega t)\hat{i} + \cos(\omega t)\hat{j}]$$

(a) Angular momentum about O is

$$\vec{L} = \vec{r} \times (m\vec{v}) = m(\vec{r} \times \vec{v})$$

$$= mR^2\omega[(\cos\omega t)\cdot\cos\omega t + \sin^2\omega t]\hat{k} - mRa\omega\sin(\omega t)\hat{j} - mRa\omega\cos(\omega t)\hat{i}$$

$$= mR^2\omega\hat{k} - mRa\omega[\cos(\omega t)\hat{i} + \sin(\omega t)\hat{j}]$$

(b) $L_z = mR^2\omega$

L_z can be written by simple observation.

Note: (i) Magnitude of $\vec{L}$ is constant (ii) $\vec{L}$ vector has a fixed z component and its component in xy plane rotates with angular velocity ω.

32. (a) Conservation of momentum gives:

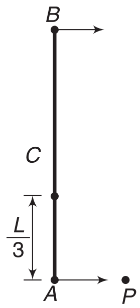
$$3mv = 2mv_0 \quad \Rightarrow \quad v = \frac{2v_0}{3}$$

(b) The COM of the system is at a distance $\frac{L}{3}$ from end A. We will conserve angular momentum about a point fixed on the table which is just beneath the COM of the system (at C).

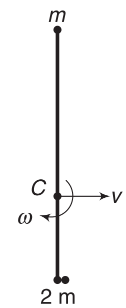
$$L_i = L_B + L_A + L_P$$

$$= mv_0 \frac{2L}{3}(\circlearrowright) + mv_0 \frac{L}{3}(\circlearrowleft) + 0$$

$$= mv_0 \frac{L}{3}\ (\circlearrowright)$$



COM is at C just before collision



Just after collision the COM of the system translates along a line that passes through C.

After collision, angular momentum about our origin is

$$L_f = L_{\text{spin}} + L_{\text{orbital}} = I\omega + 0$$

$$= \left[2m\left(\frac{L}{3}\right)^2 + m\left(\frac{2L}{3}\right)^2\right]\omega$$

Using $L_i = L_f$ gives $\omega = \dfrac{v_0}{2L}\ (\circlearrowright)$

(c) v_A = velocity of COM + velocity wrt COM

$$= v - \frac{\omega L}{3} = \frac{2v_0}{3} - \frac{v_0}{2L}\frac{L}{3} = \frac{v_0}{2}$$

$$v_B = v + \omega\frac{2L}{3} = v_0$$

Alternate: A massless rod will not exert any force on A in the direction of motion ($\perp$ to rod), during collision.

$\therefore$ Momentum of $(A + P)$ system is separately conserved.

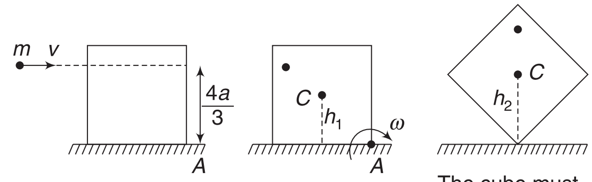
$$\therefore \quad mv_0 = 2mv_A \quad \Rightarrow \quad v_A = \frac{v_0}{2}$$

Velocity of B remains unchanged, i.e, $v_B = v_0$.

33. Angular momentum is conserved about rotation axis through A, during collision.

$$L_{\text{before collision}} = L_{\text{after collision}}$$

$$mv\frac{4a}{3} = \frac{2}{3}M(2a)^2 \cdot \omega$$



Just before impact Just after impact The cube must be able to reach this position before its angular velocity becomes zero.

[***Note:*** $m << M$, hence we are neglecting contribution of m in moment of inertia. MI of cube about its edge is $\dfrac{2}{3}ma^2$]

$$\therefore \qquad \omega = \frac{mv}{2Ma} \qquad \text{...(i)}$$

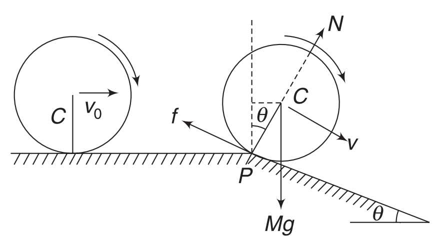
After collision, the cube is set into rotation about A and its mechanical energy remains conserved. For topping it must be able to reach the position shown in 3rd figure (In this position COM is at maximum height. PE is maximum).

$$\therefore \qquad \frac{1}{2}I_A \cdot \omega^2 = Mg(h_2 - h_1)$$

$$\Rightarrow \quad \frac{1}{2}\left(\frac{8}{3}Ma^2\right)\left(\frac{mv}{2Ma}\right)^2 = Mg(\sqrt{2} - 1)a$$

$$\Rightarrow \qquad v = \frac{M}{m}\left[3ag(\sqrt{2} - 1)\right]^{\frac{1}{2}}$$

34. (a) KE of rolling cylinder $= \dfrac{3}{4}mv_0^2$. Let its speed be v immediately after it is on incline.



Gain in KE = loss in PE

$$\frac{3}{4}mv^2 - \frac{3}{4}mv_0^2 = mgR(1 - \cos\theta)$$

$$\Rightarrow \quad v = \sqrt{v_0^2 + \frac{4}{3}gR(1 - \cos\theta)} \qquad \text{...(1)}$$

(b) The COM of the cylinder is instantaneously rotating about P with speed v.

$$\therefore \qquad \frac{mv^2}{R} = mg\cos\theta - N$$

$$\Rightarrow \qquad N = mg\cos\theta - \frac{mv^2}{R}$$

$$= mg\cos\theta - \frac{mv_0^2}{R} - \frac{4}{3}mg + \frac{4}{3}mg\cos\theta$$

[using (i)]

For no jump, $N \geq 0$

$$\Rightarrow \quad \frac{7}{3}mg\cos\theta - \frac{4}{3}mg - \frac{mv_0^2}{R} \geq 0$$

$$\Rightarrow \quad v_0 \leq \sqrt{\frac{7gR}{3}\cos\theta - \frac{4}{3}gR}$$

35. Speed of COM is given by $Mu = J$

$$\Rightarrow \quad 0.6u = 6 \quad \Rightarrow \quad u = 10 \text{ ms}^{-1}.$$

Angular speed about COM is given by

$$I_{CM}\cdot\omega = \text{angular impulse about COM}$$

$$\frac{1}{12}ML^2\cdot\omega = J\cdot(0.05\,\text{m})$$

$$\Rightarrow \frac{1}{12}\times 0.6\times 0.3^2\cdot\omega = 6\times 0.05$$

$$\Rightarrow \quad \omega = \frac{200}{3}\text{ rad s}^{-1}$$

As the rod is kicked off the edge, its angular speed (ω) becomes constant. There is no torque about COM.

Velocity of COM increases in vertical direction due to Mg.

After 1.0 s the vertical component of velocity of COM is

$$v = gt = 10\,\text{ms}^{-1}.$$

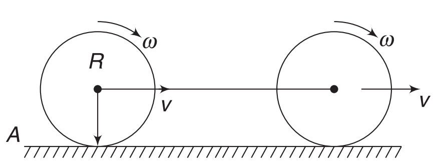
$$\therefore \quad v_{CM} = \sqrt{v^2 + u^2} = 10\sqrt{2}\text{ ms}^{-1}.$$

$$KE = \frac{1}{2}Mv_{CM}^2 + \frac{1}{2}I_{CM}\cdot\omega^2$$

$$= \frac{1}{2}0.6\times(10\sqrt{2})^2 + \frac{1}{2}\times\frac{1}{12}\times 0.6\times 0.3^2\times\left(\frac{200}{3}\right)^2$$

$$= 70 \text{ J}$$

36. Consider a point A on the surface that is to the left of both discs. Angular momentum of the system is conserved about this point. [Refer to example 45]



$$L_i = L_f$$

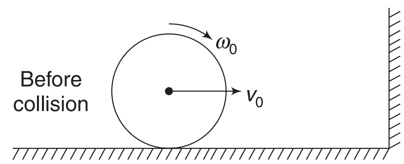
$$\frac{1}{2}mR^2\omega_0 = \left[\frac{1}{2}mR^2\omega + mvR\right]\times 2$$

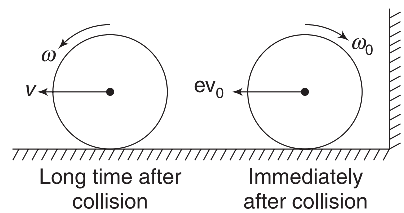
$$\Rightarrow \quad \frac{1}{2}mR^2\omega_0 = \left[\frac{1}{2}mR^2\frac{v}{R} + mvR\right]\times 2$$

[When pure rolling starts, $v = \omega R$]

$$\Rightarrow \quad v = \frac{\omega_0 R}{6}$$

37. Before collision, $\omega_0 = \frac{v_0}{R}$. After collision, velocity of the ball is $ev_0(\leftarrow)$ and its angular speed is $\omega_0(\circlearrowright)$.





Note that ω_0 do not change during collision.

Friction converts this rolling with sliding motion into pure rolling. When pure rolling starts $v = \omega R$.

Conserving angular momentum about point A:

$$L_i = L_f$$

$$m(ev_0)R\ (\circlearrowleft) + \frac{2}{5}mR^2\omega_0(\circlearrowright) = mvR\ (\circlearrowleft) + \frac{2}{5}mR^2\omega\ (\circlearrowleft)$$

$$\Rightarrow \quad ev_0 - \frac{2}{5}v_0 = v + \frac{2}{5}v \qquad \left[\begin{array}{l}\because R\omega_0 = v_0\\ \text{and } R\omega = v\end{array}\right]$$

$$\Rightarrow \quad 0.7\times 14 - \frac{2}{5}\times 14 = \frac{7}{5}v$$

$$\Rightarrow \quad 1.4 - 0.8 = 0.2v$$

$$\Rightarrow \quad 3\,\text{ms}^{-1} = v$$

38. Just before impact, velocity of COM is $v_0 = \sqrt{2gh}$

Let v be the velocity of COM and ω be angular velocity of the rod after impact.

J = impulse applied by the floor.

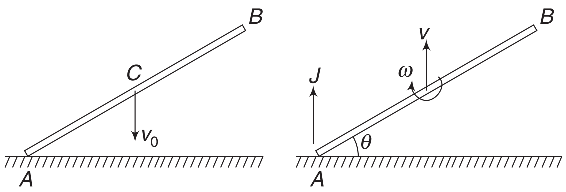
$$J = P_f - P_i$$

$$J = mv - (-mv_0) = m(v + v_0) \qquad \text{...(i)}$$

Angular impulse about COM = Change in angular momentum about COM

$$\Rightarrow \quad J\frac{L}{2}\cos\theta = \frac{mL^2}{12}\omega; \quad \text{using (i)}$$

$$\Rightarrow \quad (v + v_0)\cos\theta = \frac{L\omega}{6} \qquad \text{...(ii)}$$



Before impact After impact

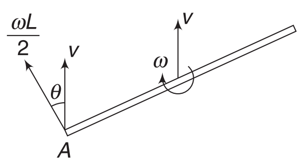
Since, collision is elastic,

Relative speed of approach of point A wrt floor = relative speed of separation of A wrt the floor.

$$\Rightarrow \qquad v_0 = \frac{\omega L}{2}\cos\theta + v \qquad \text{...(iii)}$$

Solving (ii) and (iii) gives

$$\omega = \frac{6v_0\cos\theta}{L(1+3\cos^2\theta)} = \frac{6\sqrt{2gh}\,\cos\theta}{L(1+3\cos^2\theta)}$$

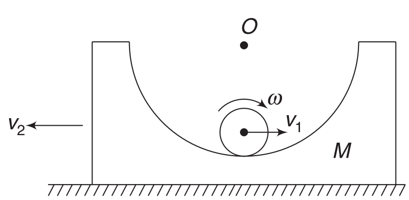


Velocity of point A is vector sum of v and $\frac{\omega L}{2}$

Note: Alternatively one can get equation (ii) by applying conservation of angular momentum about A.

39. (a) Let velocity of the centre of the cylinder and the block be $v_1(\rightarrow)$ and $v_2(\leftarrow)$ when the cylinder reaches the bottom. ω is angular velocity of cylinder.

For no slipping: $\omega r - v_1 = v_2$ ...(i)



Momentum conservation along horizontal direction: $mv_1 = Mv_2$ ...(ii)

Energy conservation:

$$mg(R-r) = \frac{1}{2}mv_1^2 + \frac{1}{2}\left(\frac{1}{2}mr^2\right)\omega^2 + \frac{1}{2}Mv_2^2 \qquad \text{...(iii)}$$

Solving the above equation gives

$v_1 = 2\,\text{ms}^{-1}$. and $v_2 = 1.5\,\text{ms}^{-1}$.

(b) In RF attached to the block, the COM of the cylinder is moving in a circle of radius $(R-r)$ at a speed of $v_1 + v_2$. At this point in time, the block has no acceleration and the RF attached to it is inertial.

$$\therefore\ N - mg = \frac{m(v_1+v_2)^2}{R-r}$$

where N = normal force by block on cylinder.

$$N = 0.5\times 10 + \frac{0.5\times(2+1.5)^2}{0.525} = 16.67\,\text{N}.$$

(c) Considering vertical equilibrium of the block.

$N_1 = Mg + N$ [N_1 = normal reaction of floor]

$= 1\times 10 + 16.67 = 26.67\,\text{N}.$

40. MI of the combination of rods about rotation axis through P is

$$I = \frac{ML^2}{3} + \frac{2ML^2}{12} + 2M\left(\frac{3L}{2}\right)^2 = 5ML^2$$

(a) Conservation of angular momentum about P gives

$$Mu(2L) = [5ML^2 + M(2L)^2]\,\omega$$

where ω = angular speed just after impact.

$$\Rightarrow \qquad \omega = \frac{2u}{L}$$

Now, the KE of the system just sufficient to raise it to horizontal position.

Loss in KE = gain in PE

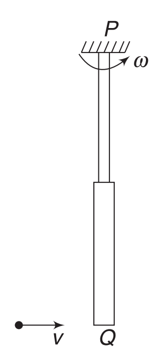
$$\frac{1}{2}[5ML^2 + M(2L)^2]\,\omega^2 = Mg\left(\frac{L}{2}\right) + 2Mg\cdot\left(\frac{3L}{2}\right) + Mg\cdot(2L)$$

$$\Rightarrow \qquad \frac{9}{2}ML^2\,\frac{4u^2}{L^2} = \frac{11}{4}MgL$$

$$\Rightarrow \qquad u = \sqrt{\frac{11}{72}gL}$$

(b) Let velocity of the object be v after impact.

Angular momentum conservation about P gives:



$$Mu(2L) = Mv(2L) + I\omega$$

$$\Rightarrow \qquad 2MuL = 2MvL + 5ML^2\omega$$

$$\Rightarrow \qquad 2u = 2v + 5\omega L \qquad \text{...(i)}$$

Relative speed of separation of end Q and the object after collision = e (relative speed of approach of the object and end Q before collision)

$\Rightarrow \omega(2L) - v = eu$ ...(ii)

Eliminating v between (i) and (ii) gives:

$$\omega = \frac{2u(1+e)}{9L} \quad \text{...(iii)}$$

After collision:

Loss in KE = gain in PE

$$\Rightarrow \quad \frac{1}{2} I\omega^2 = Mg\frac{L}{2} + 2Mg\left(\frac{3L}{2}\right)$$

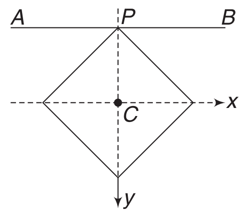
$$\Rightarrow \quad \frac{1}{2}5ML^2\left[\frac{2u(1+e)}{9L}\right]^2 = \frac{7}{2}MgL$$

$$\Rightarrow \quad u = \frac{1}{(1+e)}\sqrt{\frac{567gL}{20}}$$

Chapter 7 Miscellaneous Problems on Chapters 4–6

Match the Column

1. (A) MI about axis through C and perpendicular to the plane of the frame is



$$I_z = \left[\frac{ml^2}{12} + m\left(\frac{l}{2}\right)^2\right] \times 4$$

$$= \frac{4}{3}\, ml^2$$

Now, $\quad I_x + I_y = I_z$

$$\Rightarrow \quad 2I_x = \frac{4}{3}\, ml^2 \quad \Rightarrow \quad I_x = \frac{2}{3}\, ml^2$$

$$\therefore \quad I_{AB} = I_x + (4m)(PC)^2 = \frac{2}{3}ml^2 + 4m \cdot \left(\frac{l}{\sqrt{2}}\right)^2$$

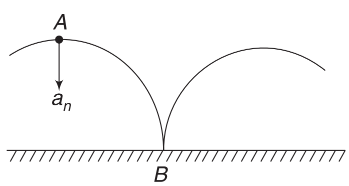
$$= \frac{8}{3}\, ml^2$$

(B) $I = 0 + \dfrac{1}{3}\, ml^2 + \dfrac{1}{3}\, ml^2 + ml^2 = \dfrac{5}{3}\, ml^2$

(C) As found in (A), $I_x = \dfrac{2}{3}\, ml^2$

(D) $I = I_z + (4m)\left(\dfrac{l}{2}\right)^2 = \dfrac{7}{3}\, ml^2$

2. Total acceleration is always directed towards the centre of the disc and has a magnitude $a = \dfrac{v^2}{R}$.



Path of Point P (in ground frame) is a cycloid. At Point A, its normal acceleration is directed vertically down, i.e, towards the centre of the disc.

At Point B, when it is touching the ground, its tangential acceleration is vertically up. Note that at this point, normal acceleration must be zero as speed of the point is zero.

3. (A) $\vec{v}_1 = v\hat{i} + \omega R\,\hat{j}$

Where v is velocity of centre and ωR is velocity of point 1 relative to the centre.

It is given that $\vec{v}_1 = \hat{i} + \hat{j}$

$\therefore \qquad v = 1 \quad \text{and} \quad \omega R = 1$

Also, $\qquad v_2 = (v + \omega R)\hat{i} = 2\hat{i}$

Thus the cylinder is in pure rolling motion.

(B) $\vec{v}_1 = \hat{i} + \hat{j}$ and $\vec{v}_3 = -\hat{i}$, implies that the point 1 and 3 are getting closer (one is moving to right and the other is moving to left). Distance between two points cannot change in a rigid body.

(C) and (D) can be understood on similar lines.

4. (A) $\vec{r} = 3t\hat{i} + 4\hat{j}$

$$\vec{v} = \frac{d\vec{r}}{dt} = 3\hat{i}$$

Distance (r) from origin is increasing at a constant rate of $3\,\text{ms}^{-1}$.

(B) Acceleration, $\vec{a} = \dfrac{d\vec{v}}{dt} = 0$

(C) Angle between $\vec{r}$ and $\vec{v}$ given by $\cos\theta = \dfrac{\vec{r}\,.\,\vec{v}}{r\;v}$

$$\therefore \quad \cos\theta = \frac{9t}{\left(\sqrt{9t^2 + 16}\right)3} = \frac{3t}{\sqrt{9t^2 + 16}}$$

Component of velocity perpendicular to $\vec{r}$ is

$$v_\perp = v\sin\theta = 3\sqrt{1 - \cos^2\theta} = \frac{12}{\sqrt{9t^2 + 16}}$$

$$\therefore \quad \text{Angular velocity is, } \omega = \frac{v_\perp}{r} = \frac{12}{9t^2 + 16}$$

$\therefore \quad \omega$ decreases with time.

(D) $\vec{L} = m(\vec{r} \times \vec{v}) = -(12m)\hat{k}$

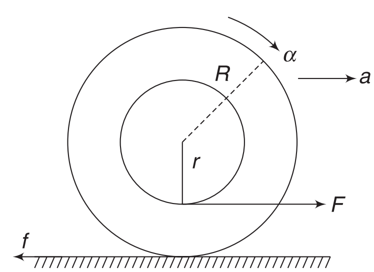
5. First, write the direction of motion of the contact point. Friction will be directed opposite to this.

7. (a) $$F - f = Ma \qquad \text{...(i)}$$

And $\quad fR - F \cdot r = I \cdot \alpha$

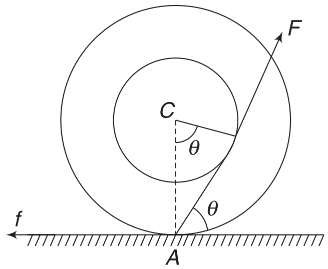
$$f - \frac{Fr}{R} = \frac{I}{R^2} \cdot (R\,\alpha)$$

$$f - \frac{Fr}{R} = \frac{I}{R^2} \cdot a \qquad \text{...(ii)}$$



(i) + (ii) gives: $F\left(1-\frac{r}{R}\right)=\left(M+\frac{I}{R^2}\right)a.$

Since $\frac{r}{R}<1$, a is positive. Yo-yo moves to the right. Putting the value of a in (i) gives friction f. It is also positive, which implies that friction is correctly shown in the above diagram.



$\cos\theta = \frac{-r}{R}$

When line of F passes through the contact point (A), it will product no torque about A. Friction also does not have torque about A. Thus, there is no rotation. For no sliding, there should not be any translation as well. Hence, f is to left and is equal to $F\cos\theta$.

When angle θ is greater than that shown in the above figure, it means line of F passes from right of A. This produces an anti-clockwise torque about A. The body has anti-clockwise angular acceleration. It must have leftward acceleration for pure rolling. Hence, friction is to the left.

When θ is less than that shown in the above figure, line of F passes from left of A. This produces a clockwise torque. Body has clockwise angular acceleration. It must move to the right for pure rolling.

8. $Mg - T = Ma$ ...(i)

and $T\cdot r = I\cdot\alpha \Rightarrow T = \frac{I}{R^2}\cdot(R\alpha)$ ...(ii)

From (i) and (ii), $Mg = \left(M+\frac{I}{R^2}\right)a \quad [\because \ R\alpha = a]$

For disc, $I = \frac{MR^2}{2}$ and for ring, $I = MR^2$

$\therefore\ a_{\text{ring}} < a_{\text{disc}}$, irrespective of the radius.

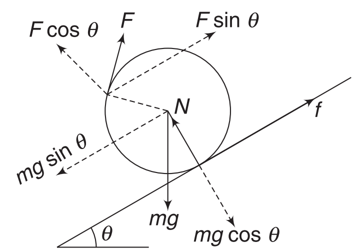
If both the objects are disc (or ring) then $a_A = a_B$

Higher the acceleration, higher will be the distance through which the object falls. It means higher acceleration implies higher length of thread unwound.

Passage-based Problems

Passage 1

1.



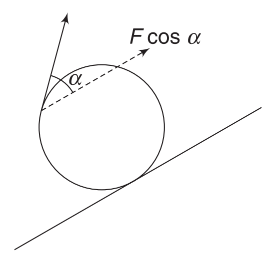
For rotational equilibrium: $F\cdot r = f\cdot r \quad\Rightarrow\quad F = f.$ ...(i)

For translational equilibrium along the incline:

$$f + F\sin\theta = mg\sin\theta \qquad \text{...(ii)}$$

$$\Rightarrow\quad F\left[1+\frac{3}{5}\right] = mg\times\frac{3}{5} \quad\Rightarrow\quad F = \frac{3}{8}mg.$$

2. The above equation (i) holds true for any direction of applied force F.



For translational equilibrium:

$$F_{//} + f = mg\sin\theta$$

Where $F_{//} = F\cos\alpha$ is component of F parallel to the incline.

$\therefore\quad F\cos\alpha + F = mg\sin\theta$

Obviously, F is minimum when $\cos\alpha = 1$ ($\Rightarrow\ \alpha = 0°$)

$$\therefore\qquad F_{\min} = \frac{mg\sin\theta}{2} = \frac{3mg}{10}$$

3. In first question: $N + F\cos\theta = mg\cos\theta$

$$\Rightarrow\qquad N = \frac{4}{5}mg - \frac{3}{8}mg\cdot\frac{4}{5} = \frac{1}{2}mg$$

And $\quad f = F = \frac{3}{8}mg$

Since $\quad f \le \mu N \quad\Rightarrow\quad \frac{3}{8}mg \le \mu.\frac{1}{2}mg \quad\Rightarrow \frac{3}{4} \le \mu$

Similarly, find for second question.

Passage 2

4. For slowly rolling the sphere (i.e., with no acceleration), we have

$$F + f = mg\sin\theta \qquad \text{...(i)}$$

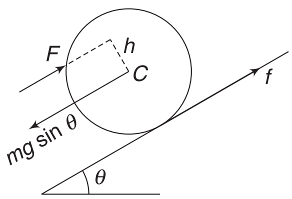
And $\quad Fh = fR$

$$\Rightarrow \quad f = \frac{Fh}{R} \qquad \text{...(ii)}$$

Substituting in (i) gives

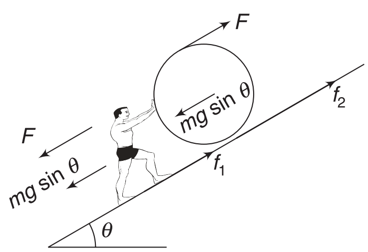
$$F + \frac{Fh}{R} = mg\sin\theta \quad \Rightarrow \quad F = \frac{mg\sin\theta}{1 + \frac{h}{R}}$$

F is minimum when h is maximum, i.e., equal to R.



5. Both have the same mass. For same acceleration, friction on both will be the same.

6. For slow motion of the man:



$$mg\sin\theta + F = f_1 \qquad \text{...(i)}$$

For sphere:

$$F + f_2 = mg\sin\theta \qquad \text{...(ii)}$$

And $\quad FR = f_2R$

$$\Rightarrow \quad F = f_2 \qquad \text{...(iii)}$$

Solving the above equation gives

$$F = \frac{mg\sin\theta}{2}$$

$$\therefore \quad f_2 = \frac{mg\sin\theta}{2} \quad \Rightarrow \quad \mu mg\cos\theta \geq \frac{mg\sin\theta}{2}$$

$$\Rightarrow \quad \mu \geq \frac{1}{2}\tan\theta$$

And from (i), $f_1 = \frac{3}{2}mg\sin\theta$

$$\therefore \quad \frac{3}{2}mg\sin\theta \leq \mu mg\cos\theta$$

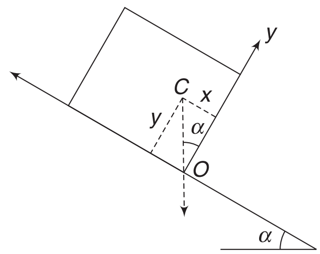
$$\Rightarrow \quad \frac{3}{2}\tan\theta \leq \mu \text{ for the man to not slip.}$$

$\therefore$ For both to move without slipping

$$\mu \geq \frac{3}{2}\tan\theta$$

Passage 3

7. In the co-ordinate system shown, the co-ordinates of the COM of the card board are



$$y_{cm} = \frac{a}{2}$$

$$x_{cm} = \frac{\frac{a}{2} \times m + \frac{a}{2} \times m + 0 \times m}{3m} = \frac{a}{3}$$

The board is on verge of toppling if the vertical line through the COM passes through O.

$$\tan\alpha = \frac{x}{y} = \frac{2}{3}$$

8. For no sliding, $\mu \geq \tan\alpha$

For no sliding before toppling $\mu \geq \frac{2}{3}$

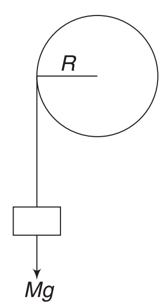
9. Acceleration $= g\sin\alpha$

Using $\quad S = ut + \frac{1}{2}at^2$

$$a = \frac{1}{2}g\sin\alpha \cdot t^2 \quad \Rightarrow \quad t = \sqrt{\frac{2a}{g\sin\alpha}}$$

Passage 4

Torque needed to lift the block is independent of choice of wrench. It is $Mg\,R$ to just lift the block. To accelerate the block, torque should be higher.



Torque produced by wrench on the nut is τ = force × arm length.

For a fixed τ, force is inversely proportional to length.

Passage 5

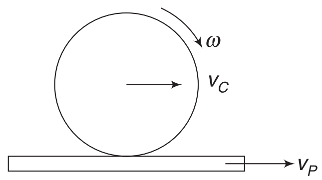
When the cylinder reaches the bottom of the hill, its velocity is v (it has no spin as the hill is smooth) such that

$$\frac{1}{2}Mv^2 = Mgh \quad \Rightarrow \quad v = \sqrt{2gh}$$

Friction slows dows the cylinder and imparts it a spin. It also accelerates the plank. Finally, pure rolling starts.

$$v_C - \omega r = v_P \qquad \text{...(i)}$$

$$\therefore \qquad v_C > \omega r \qquad \text{[Ans to 13]}$$



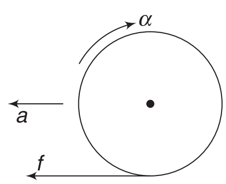
Conservation of linear momentum gives

$$Mv = Mv_C + Mv_P \Rightarrow v = v_C + v_P \qquad \text{...(ii)}$$

From (i) and (ii), $2v_C = v + \omega r$ ....(iii)

Adding (i) and (ii) gives $2v_P + \omega r = v = \sqrt{2gh}$.

[The above relations give answer to Q.15]



When the cylinder is sliding, friction on it is leftwards.

Retardation, $a = \dfrac{f}{M}$

Angular acceleration, $\alpha = \dfrac{fr}{\frac{1}{2}Mr^2}$

$$\Rightarrow \qquad \alpha = \frac{2f}{Mr}$$

Speed after time t is, $v_C = v - at = v - \dfrac{ft}{M}$

Angular speed at time t is, $\omega = \alpha t = \dfrac{2ft}{Mr}$

For pure rolling condition, (iii) must hold.

$$\therefore \qquad 2\left(v - \frac{ft}{M}\right) = v + \frac{2ft}{M}$$

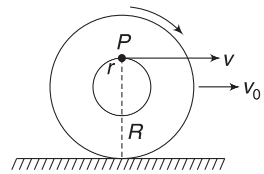
$$\Rightarrow \quad v = \frac{4ft}{M} \Rightarrow \frac{ft}{M} = \frac{v}{4}$$

$$\therefore \quad v_C = v - \frac{ft}{M} = v - \frac{v}{4} = \frac{3}{4}v = \frac{3}{4}\sqrt{2gh} \ \text{[Ans to 14]}$$

Passage 6

16. Let speed of centre be v_0 and angular velocity of the yo-yo be ω.

Horizontal velocity of the string is same as velocity of Point P.



$$\therefore \qquad v = v_0 + \omega r = v_0 + \frac{v_0}{R}\cdot r$$

[$\because$ For pure rolling $v_0 = \omega R$]

$$\Rightarrow \qquad v = v_0\left(\frac{R+r}{R}\right)$$

17. $\tan\left(\dfrac{\theta}{2}\right) = \dfrac{R}{x}$

$$\therefore \qquad x = R\cot\left(\frac{\theta}{2}\right)$$

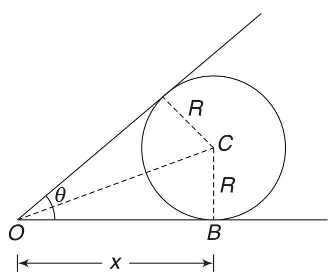
$$\Rightarrow \qquad \frac{dx}{dt} = -R\,\text{cosec}^2\left(\frac{\theta}{2}\right)\left[\frac{1}{2}\frac{d\theta}{dt}\right]$$

$$\Rightarrow \qquad -\frac{d\theta}{dt} = \frac{2v_0}{R}\sin^2\left(\frac{\theta}{2}\right)$$

$$-\frac{d\theta}{dt} = \text{angular speed of the bar.}$$

$\dfrac{d\theta}{dt}$ is negative, as θ is decreasing.

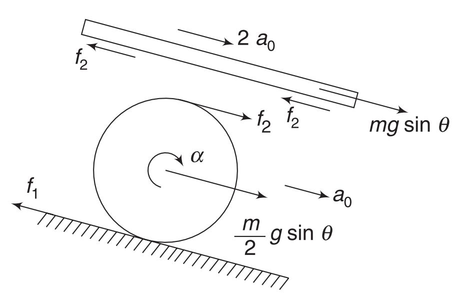
$$\therefore \qquad \omega = \left(\frac{2v}{R+r}\right)\sin^2\frac{\theta}{2}$$



Passage 7

18. If acceleartion of centre of a roller is a_0, then acceleration of the rod will be $a_R = 2a_0$.

19. One can find f_1 and f_2 by solving the following set of equations:



$$mg\sin\theta - 2f_2 = m(2a_0) \qquad \text{...(i)}$$

$$\frac{m}{2}\,g\sin\theta + f_2 - f_1 = \frac{m}{2}\,a_0 \qquad \text{...(ii)}$$

$$f_1 r + f_2 r = \frac{1}{2}\left(\frac{m}{2}\right) r^2 \cdot \alpha$$

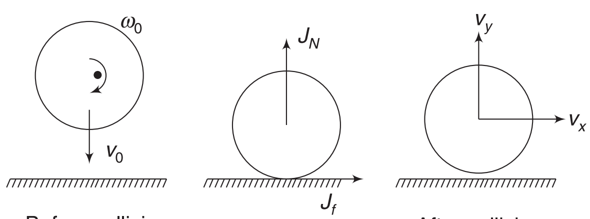
$$\Rightarrow \qquad f_1 + f_2 = \frac{m}{4}\,a_0 \qquad \text{...(iii)}$$

20. For no slipping, both must slide with an acceleration of $g\sin\theta$. The cylinders do not spin.

Passage 8

21. Vertical component of velocity after impact is

$$v_y = v_0 \quad [\because \text{ collision is elastic}]$$



Before collision   After collision

Impulse of normal force is $J_N = 2mv_0 \quad \left[= \int_0^{\Delta t} N\,dt\right]$

Impulse of friction is

$$J_F = \mu J_N \left[= \int_0^{\Delta t} \mu N\,dt\right]$$

$$\therefore \qquad mv_x = J_f \;\Rightarrow\; v_x = \frac{\mu J_N}{m} = 2\mu v_0$$

Range of the ball after impact is

$$R = \frac{2v_x v_y}{g} = \frac{2\mu v_0^2}{g}$$

If $R = \dfrac{2v_0^2}{g}$ then $\mu = 1$.

22. Impulse $J = \sqrt{J_N^2 + J_f^2} = \sqrt{J_N^2 + \mu^2 J_N^2}$

$$= \sqrt{2}J_N = 2\sqrt{2}\,mv_0$$

23. Angular impulse about centre is $= J_f \cdot r$

$$\therefore \quad I\omega - I\omega_0 = -\mu J_N \cdot r$$

$$\frac{2}{5}mr^2(\omega - \omega_0) = -2mv_0 r \qquad [\because \mu = 1;\ J_N = 2mv_0]$$

$$\Rightarrow \qquad \omega = \omega_0 - \frac{5v_0}{r} = \frac{5v_0}{r} - \frac{5v_0}{r} = 0$$

Change in $KE = \dfrac{1}{2}\,m\,(v_x^2 + v_y^2) - \left[\dfrac{1}{2}mv_0^2 + \dfrac{1}{2}I\omega_0^2\right]$

$$= \frac{1}{2}\,m(2v_0^2) - \frac{1}{2}\,mv_0^2 - \frac{1}{2}\left(\frac{2}{5}mr^2\right)\left(\frac{5v_0}{r}\right)^2$$

$$= -\frac{9}{2}\,mv_0^2$$

Passage 9

24 to 26.

After touching the cylinder, the bullet moves down by 0.2 m while it travels 48 m horizontally

Time of fall $\rightarrow 0.2 = \dfrac{1}{2}\,gt^2$

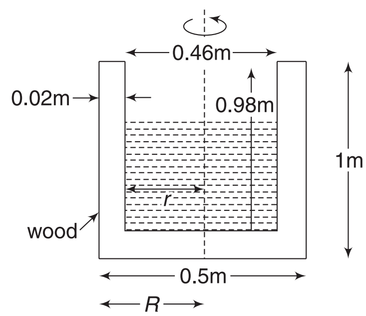
$$t = 0.2 \text{ s}$$

For horizontal motion

$$vt = 48 \quad \Rightarrow \quad v = 240 \text{ ms}^{-1} \text{ [Ans to (24)]}$$

Angular momentum of bullet before hitting

$$L = mvR = 0.05 \times 800 \times 0.25 = 10 \text{ kg ms}^{-2}$$



Angular momentum conservation

$$mvR + I\omega = 10$$

$$\Rightarrow \qquad 0.05 \times 240 \times 0.25 + I \times \frac{2\pi}{4} = 10$$

$$\Rightarrow \qquad I = 4.46 \text{ kg-m}^2 \quad [\textbf{Ans} \text{ to 25}]$$

Moment of inertia of the cylinder with content can be written as

I = (MI of solid wooden cylinder of radius R)
– (MI of solid wooden cylinder of radius r)
+ (MI of ice)

$$\therefore \quad 4.46 = \frac{1}{2}m_{\text{solid}}R^2 - \frac{1}{2}m_{\text{hole}}r^2 + \frac{1}{2}m_{\text{ice}}r^2$$

$$4.46 = \frac{1}{2} \times 600 \times \pi \times (0.25)^2 \times 1 \times (0.25)^2$$

$$- \frac{1}{2} \times 600 \times \pi \times (0.25)^2 \times 0.98 \times (0.23)^2$$

$$+ \frac{1}{2}m_{\text{ice}} \times (0.23)^2$$

Solving, $m_{\text{ice}} = 127$ kg **[Ans.** to 26]

Passage 10

27and28. For a board to be in equilibrium, weight on left and right ends are related as:

$$W_L\left(\frac{2l}{3}\right) = W_R\left(\frac{l}{3}\right) \quad \Rightarrow \quad 2W_L = W_R$$

$\therefore$ Joker A must apply 60 kg force on the first board (i.e, $N_{LA} = 60$ kg).

Since weight of joker is 80 kg

$\therefore \qquad N_{RA} = 20$ kg

For balance of second board

$$N_{LB} = 2 \times 20 = 40\,\text{kg}$$

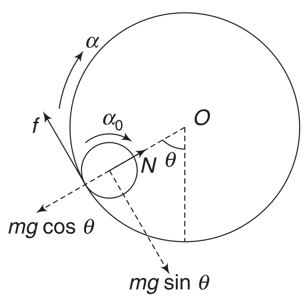
$\therefore \qquad N_{RB} = 40$ kg

For balance of third board, $N_{LC} = 2N_{RB} = 80$ kg

Thus, joker C will have to put his entire weight on the third board.

Passage 11

29and30. For no slipping, rate of increase of speed of a point on the circumference of the drum must be equal to the rate of increase of speed of a point on the circumference of solid cylinder.



$\therefore \qquad R\alpha = r\alpha_0 \qquad$ [α_0 = angular acceleration of solid cylinder]

$$\Rightarrow \qquad \alpha_o = \frac{R\alpha}{r}$$

Forces on the solid cylinder are as shown.

For no translation: $f = mg\sin\theta$

For rotation about its COM:

$$\tau = I\alpha_0$$

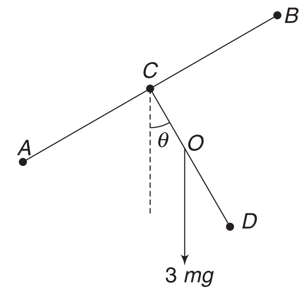
$$f\cdot r = \frac{1}{2}mr^2\left(\frac{R\alpha}{r}\right)$$

$$\Rightarrow \quad mg\sin\theta\cdot r = \frac{1}{2}mr^2\frac{R\alpha}{r} \quad \Rightarrow \quad \alpha = \frac{2g\sin\theta}{r}$$

Passage 12

31&32. COM of the structure is at O.

If $\qquad AC = CB = CD = l$, then



$$CO = \frac{2m(0) + m(l)}{3m} = \frac{l}{3}$$

Torque on the structure (about C)

$$\tau = 3mg\,\frac{l}{3}\sin\theta = mgl\sin\theta$$

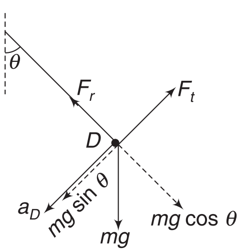
Using $\qquad \tau = I\alpha$

$$mgl\sin\theta = (3ml^2)\cdot\alpha$$

$$\Rightarrow \qquad \alpha = \frac{1}{3}\frac{g}{l}\sin\theta = \frac{g}{3l}\times\frac{3}{5} = \frac{g}{5l}$$

Immediately after release, speed of each particle = 0.

Each particle has tangential acceleration only.



Acceleration of A is $\perp$ to CA and acceleration of D is $\perp$ to CD.

$$a_D = \alpha\cdot l = \frac{g}{5}$$

Radial force by the rod

$$F_r = mg\cos\theta = \frac{4}{5}mg$$

Tangential force ($\perp$ to the rod) is given by

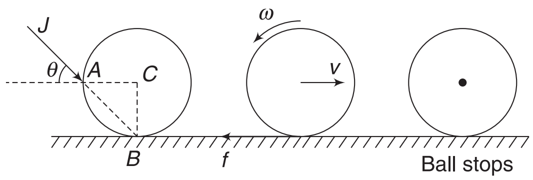
$$mg\sin\theta - F_t = ma_D$$

$$\Rightarrow \qquad F_t = \frac{3}{5}mg - \frac{mg}{5} = \frac{2mg}{5}$$

$$\therefore \quad F = \sqrt{F_r^2 + F_t^2} = \frac{2mg}{5}\sqrt{2^2 + 1^2} = \frac{2}{\sqrt{5}}mg$$

Passage 13

33. Angular momentum of the ball about B (a point fixed to the table) will remain conserved, as it moves on the table.



$$L_f = 0, \text{ since the ball stops.}$$

$$\therefore \quad L_i = 0$$

This will be true if line of impact passes through B. The impulse given to the ball has no angular impulse about B. Thus $\theta = 45°$.

34. Let the ball acquire a velocity u due to impact.

Friction is kinetic, $f = \mu N = \mu Mg$

Retardation, $a = \mu g$

Using $\quad v^2 = u^2 - 2as$

$$O = u^2 - 2\,\mu g \cdot s$$

$$s = \frac{u^2}{2\mu g}$$

But the initial speed u is given by $Mu = J$

$$\Rightarrow \quad u = \frac{J}{M}$$

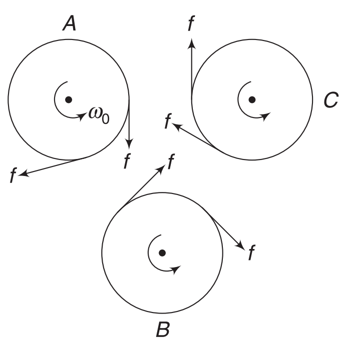
$$\therefore \quad s = \frac{J^2}{2M^2\mu g}$$

Passage 14

35. To keep the axles stationary, forces must be applied on them. Those external forces can produce torque about a selected point.

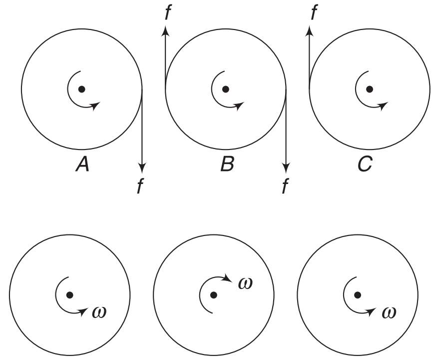
Hence answer is (D).

36. Friction forces acting on each cylinder are as shown. Each cylinder experiences the same torque ($\tau = 2fR$) about its axle. Thus, each of them has the same angular retardation.



Slipping will cease when each comes to rest.

37. In this case, torque on B is twice that of torque on A or C.



In final situation Contact points have same velocities

Let angular retardation of $A = \alpha$. Then, retardation of $B = 2\alpha$.

Slipping ceases when angular speed of A (and C) becomes $\omega(\circlearrowleft)$ and that of B becomes $\omega(\circlearrowright)$

For A: $\quad \omega = \omega_0 - \alpha t$

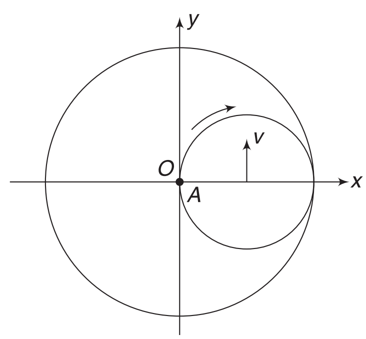
For B: $\quad -\omega = \omega_0 - 2\alpha t$

Solving: $\quad \alpha t = \dfrac{2\omega_0}{3}$

$$\therefore \quad \omega = \omega_0 - \alpha t = \frac{\omega_0}{3}$$

Passage 15

38. Position when dust particle (A) reaches origin (O) is as shown.



39. Kinetic energy of rolling cylinder is $\dfrac{3}{4}mv^2$.

Therefore,

$$\frac{3}{4}mv^2 + mgR = \frac{3}{4}mv_0^2$$

$$\Rightarrow \quad v = \sqrt{v_0^2 - \frac{4gR}{3}}$$

Passage 16

40. $L = L_{\substack{\text{chair}\\\text{student}}} + L_{\text{wheel}}$

$= 0 + L_{\text{spin}} + L_{\text{orbital}}$

$= 0 + I_0\omega_0 + 0 = I_0\omega_0$ along positive z-direction.

41. After the wheel is turned upside down, its spin angular momentum reverses direction. To conserve angular momentum, the system of student and chair begins to rotate with angular velocity ω, such that

$$\vec{L}_{\substack{\text{chair}\\\text{student}}} + \vec{L}_{\text{wheel}} = (I_0\omega_0)\hat{k}$$

$$\Rightarrow \quad I_1\omega\hat{k} + [\vec{L}_{\text{spin}} + \vec{L}_{\text{orbital}}] = (I_0\omega_0)\hat{k}$$

$$\Rightarrow \quad I_1\omega\hat{k} + [-(I_0\omega_0)\hat{k} + (I_0 + ma^2)\omega\hat{k}] = (I_0\omega_0)\hat{k}$$

$$\Rightarrow \quad (I_1 + I_0 + ma^2)\,\omega = 2I_0\omega_0$$

$$\Rightarrow \quad \omega = \frac{2I_0\omega_0}{I_1 + I_0 + ma^2} = \frac{2I_0\,\omega_0}{4I_0 + I_0 + \dfrac{I_0}{3}} = \frac{6\omega_0}{16}$$

$$= \frac{3\omega_0}{8}$$

Passage 17

42&43. Let v = velocity of centre when the body is at highest point.

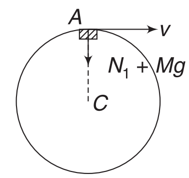
$2v$ = Velocity of body at this instant. Apply conservation of energy:

[Energy of rolling hoop = Mv^2 and speed of A when it is at lowest point is 0]

$$Mv_0^2 = Mv^2 + \frac{1}{2}M(2v)^2 + Mg(2R).$$

$$\Rightarrow \quad 3v^2 = v_0^2 - 2gR$$

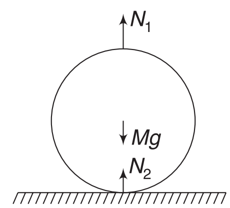
$$\Rightarrow \quad v = \sqrt{\frac{v_0^2 - 2gR}{3}} \qquad \text{...(i)}$$



When body A is at the top, centre C is not having any acceleration. Relative to C, A is moving in circle at speed v.

If N_1 = Normal force between A and C

$$N_1 + Mg = \frac{Mv^2}{R} \qquad \text{...(ii)}$$



Forces on hoop are as shown.

For $\quad N_2 > 0$

$$Mg > N_1$$

$$\Rightarrow \quad Mg > \frac{Mv^2}{R} - Mg \qquad \text{[Using (ii)]}$$

$$\Rightarrow \quad 2Mg > \frac{Mv^2}{R} \quad \Rightarrow \quad 2gR > v^2$$

$$\Rightarrow \quad 2gR > \frac{v_0^2}{3} - \frac{2gR}{3}$$

$$\Rightarrow \quad \sqrt{8gR} > v_0$$

Chapter 8 Fluid Mechanics

Your Turn

1. Hg is 13.6 times heavier than water.

   $\therefore$ Mass of Hg in the beaker $= 0.5 \times 13.6 = 6.8$ kg

2. Consider a volume V (in cm^3) of the alloy, which has mass m (in g).

   Mass of Al in alloy is $m_1 = 0.1\,m$

   Volume of Al in alloy is $V_1 = \dfrac{M_1}{\rho_1} = \dfrac{(0.1\ m)\ g}{(2.7)\ \text{g/cm}^3}$

   $$= \frac{m}{27}\ \text{cm}^3$$

   Mass of Cu in alloy is $m_2 = 0.9\,m$

   Volume of Cu in alloy is $V_2 = \dfrac{m_2}{\rho_2} = \dfrac{0.9\ m}{9} = \dfrac{m}{10}$ cm^3

   Volume of alloy, $V = V_1 + V_2$

   $$\Rightarrow \qquad V = \frac{m}{27} + \frac{m}{10}$$

   $$\Rightarrow \qquad \rho = \frac{m}{V} = \frac{1}{\dfrac{1}{27} + \dfrac{1}{10}} = \frac{270}{37}$$

   $$= 7.3\,\text{gcm}^{-3}$$

4. Pressure at depth h is $P = P_0 + \rho g h$ where P_0 is atmospheric pressure.

   The eardrum experiences force due to water that is inward. The air inside the ear exerts a force on the eardrum in outward direction. Net force on eardrum is difference of the two forces.

   Pressure at 10 m depth is

   $$P = P_0 + \rho g h = P_0 + 10^3 \times 10 \times 10$$
   $$= P_0 + 10^5\ \text{Nm}^{-2}$$

   Inward force due to water

   $$F_1 = P\Delta S = (P_0 + 10^5)\Delta S$$

   Outward force due to air in the middle air (behind the eardrum) is

   $$F_2 = P_0 \Delta S$$
   $$\therefore \quad F_{\text{net}} = F_1 - F_2 = (10^5)(\Delta S)$$
   $$= 10^5 \times (1 \times 10^{-4}) = 10\ \text{N}$$

5. (a) Pressure at bottom is

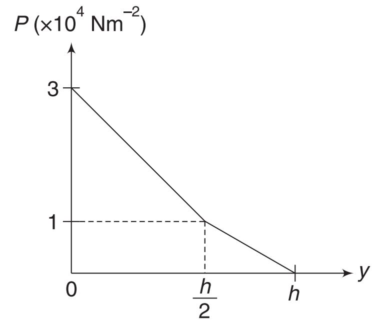
   $$P = \rho g \frac{h}{2} + 2\rho \cdot g \frac{h}{2}$$
   $$= \frac{3}{2}\rho g h = \frac{3}{2} \times 2000 \times 10 \times 1$$
   $$= 3 \times 10^4\ \text{Nm}^{-2}$$

   $\therefore$ Force on base is

   $$F = PS = 3 \times 10^4 \times (0.5 \times 0.5) = 7{,}500\,\text{N}$$

   (b) Pressure at the interface of the two liquids is

   $$P_1 = \rho g \frac{h}{2} = 10^4\ \text{Nm}^{-2}$$



   Pressure decrease linearly from 3×10^4 to $1 \times 10^4\,\text{Nm}^{-2}$ when one travels a height 0.5 m through the lower liquid. Then it decrease linearly from $1 \times 10^4\ \text{Nm}^{-2}$ to zero when one moves from $y = \dfrac{h}{2}$ to $y = h$ in upper liquid.

6. Average pressure on upper half of the wall is (see solution of last Q)

   $$P_{\text{av}} = \frac{0 + P_1}{2} = \frac{0 + 10^4}{2} = 5{,}000\,\text{Pa}$$

   Average pressure on lower half of the wall is

   $$P'_{\text{av}} = \frac{1 \times 10^4 + 3 \times 10^4}{2} = 2 \times 10^4\ \text{Pa}$$

   Area of half wall is, $S = \dfrac{h}{2}a = 0.5 \times 0.5 = 0.25\ \text{m}^2$.

   $\therefore$ Required force is

   $$F = P_{\text{av}} \cdot S + P'_{\text{av}} \cdot S = (5{,}000 + 20{,}000) \times 0.25$$
   $$= 6250\,\text{N}$$

7. Horizontal force due to water is

   $F = (P_{\text{av}})$ (area of wall in contact with water)

   $$= \left(\frac{0 + \rho g h}{2}\right)(hL) = \frac{10^3 \times 10 \times 10^2 \times 30}{2}$$
   $$= 1.5 \times 10^7\,\text{N}$$

   Weight of the wall, W = volume $\times$ density $\times$ g

   $$\Rightarrow \qquad W = \frac{1}{2}H\,(a + 2a) \cdot L \times 3000 \times 10$$
   $$= \frac{1}{2} \times 12(3a) \times 30 \times 3000 \times 10$$
   $$= 162\,a \times 10^5\,\text{N}$$

As per the question: $W = 10F$

$\Rightarrow \quad 162a \times 10^5 = 15 \times 10^7 \quad \Rightarrow \quad a = 9.26\,\text{m}$

8. (a) $P = \rho gh = 10^3 \times 10 \times 0.2 = 2{,}000\ \text{Nm}^{-2}$

Force on base due to water is

$$F_1 = P \cdot S = 2{,}000 \times (20 \times 10^{-4}) = 4\,\text{N}$$

(b) Consider equilibrium of water in the pot.

Base surface exerts a force of $F_1 = 4\text{N}(\uparrow)$ on water.

Weight of water is $W = (0.5\ \text{l})\ (1\ \text{kgl}^{-1}) \times g = 5\,\text{N}\ (\downarrow)$

The curved wall must exert an upward force of 1N on the water body for its equilibrium.

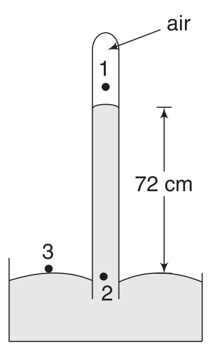
10. g is smaller at equator; therefore, h will be higher at equator.

11. Inside a freely falling lift, $g_{\text{eff}} = 0$

From equation (7): $h = \dfrac{P_0}{\rho_{\text{Hg}} \cdot g_{\text{eff}}} = \infty$

Hg will rise as much as the tube permits.

12. Let us express pressure in unit of height of Hg–column. Pressure of air in the tube is P_1.



$$P_2 = P_3$$

$$P_1 + 72\,\text{cm} = 76\,\text{cm}$$

$\Rightarrow \quad P_1 = 4\,\text{cm}$ height of Hg

$$= (0.04\,\text{m}) \times (13600\,\text{kgm}^{-3}) \times (9.81\,\text{ms}^{-2})$$

$$= 5337\ \text{Nm}^{-2}$$

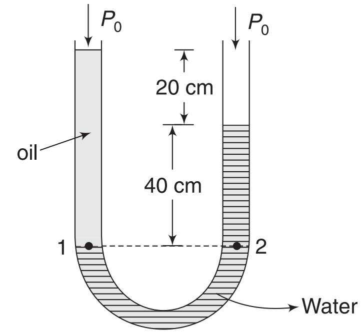
13. $P_1 = P_2$

$$P_0 + \rho_{\text{oil}}\, g(60\,\text{cm}) = \rho_0 + \rho_w\, g(40\,\text{cm})$$

$\Rightarrow \quad \dfrac{\rho_{\text{oil}}}{\rho_w} = \dfrac{2}{3}$

Pressure at 1 is

$$P_1 = P_0 + \rho_{\text{oil}}\, g(60\,\text{cm})$$

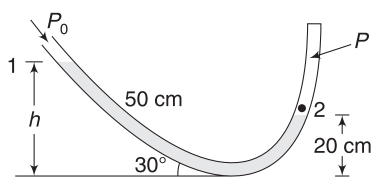


Guage pressure at 1 is $P_1 - P_0 = \rho_{\text{oil}}\, g(60\,\text{cm})$

$$= \left(\frac{2}{3} \times 10^3\ \text{kgm}^{-3}\right)(10\ \text{ms}^{-2})(0.6\,\text{cm})$$

$$= 4{,}000\ \text{Nm}^{-2}$$

14. Let us write pressure at the bottom of the tube. It can be written starting from point 1 and it can also be written starting from point 2. The two values should be same.



$$P_0 + \rho_{\text{Hg}} \cdot g \cdot h = P + P_{\text{Hg}}\, g(20\,\text{cm})$$

$\therefore \quad P_0 + \rho_{\text{Hg}} \cdot g(50 \sin 30°\ \text{cm}) = P + \rho_{\text{Hg}}\, g(20\ \text{cm})$

$\Rightarrow \quad P = P_0 + \rho_{\text{Hg}}\, g(25 - 20\,\text{cm})$

$$= 1 \times 10^5 + 13600 \times 10 \times \left(\frac{5}{100}\text{m}\right) = 1.068 \times 10^5\ \text{Nm}^{-2}$$

15. $P_A = (P_0 + 10\,\text{cm})$ of Hg column

$P_B = P_0 - 10\,\text{cm}$ of water column

$= P_0 - \dfrac{10}{13.6}$ cm of Hg column

$\therefore \quad P_A - P_B = 10 + \dfrac{10}{13.6}$ cm of Hg

$= 10.74\,\text{cm}$ of Hg

16. Pressure below left piston = pressure below right piston

$$P_0 + \frac{0.5g + F_1}{A_1} = P_0 + \frac{2g + F_2}{A_2}$$

$\Rightarrow \quad \dfrac{5 + 10}{2} = \dfrac{20 + F_2}{10}$

$\Rightarrow \quad F_2 = 55\,\text{N}$

17. Force on larger piston needed to just move the load is

$$F_2 = Mg + f = 100 \times 10 + 20 = 1020\,\text{N}$$

Pressure in the liquid must increase by

$$\Delta P = \frac{F_2}{A_2} = \frac{1020\,\text{N}}{100\,\text{cm}^2}$$

$\therefore$ Force on smaller piston is given by

$$\frac{F_1 - f}{A_1} = \Delta P$$

$$F_1 - 20 = \left(\frac{1020\,\text{N}}{100\,\text{cm}^2}\right)(10\,\text{cm}^2) = 102\,\text{N}$$

$$\Rightarrow \quad F_1 = 122\,\text{N}$$

18. (a) $P_B - P_A = \rho g y = 10^3 \times 10 \times 4 = 4 \times 10^4$ Pa

(b) $P_D - P_A = \rho a x = 10^3 \times 4 \times 2 = 8 \times 10^3$ Pa

(c) $P_C - P_A = \rho g y + \rho a x = 4 \times 10^4 + 8 \times 10^3$

$$= 4.8 \times 10^4 \text{ Pa}$$

$$\therefore \quad P_A - P_C = -4.8 \times 10^4\,\text{Pa}$$

20. Using result obtained in example 12:

$$P_B - P_A = \frac{1}{2}\rho\omega^2 \cdot L^2$$

$$\Rightarrow \quad P_B = P_A + \frac{1}{2}\rho\omega^2 L^2$$

$$= 1 \times 10^5 + \frac{1}{2} \times 10^3 \times 20^2 \times 1^2$$

$$= 3 \times 10^5 \text{ Pa}$$

21. $P_3 = P_2 + \rho g H = P_0 + \rho g H$

And $\quad P_3 - P_1 = \rho a L$

$$\Rightarrow \quad P_0 + \rho g H - P_1 = \rho \frac{g}{2} L$$

$$\Rightarrow \quad P_1 = P_0 + \rho g \left(H - \frac{L}{2}\right)$$

for $P_1 = P_0$, we must have $L = 2H$

22. Buoyancy = Loss in weight

$$V\rho_w g = 0.03 \qquad \text{...(i)}$$

And $\quad Vd \cdot g = 0.1\,\text{N} \qquad$ ...(ii)

$$\therefore \quad \frac{Vdg}{V\rho_w g} = \frac{0.1}{0.03}$$

$$\Rightarrow \quad d = \frac{10}{3}\rho_w = \frac{1}{3} \times 10^4 \text{ kgm}^{-3}$$

23. $\frac{3}{4}V\rho_L g = V\rho_1 g \quad \Rightarrow \quad \rho_L = \frac{4}{3}\rho_1$

24. $ma = F_B - mg$

$$\Rightarrow \quad Vda = V\rho_w g - Vd \cdot g$$

where d = density of wood.

$$\Rightarrow \quad d(g + a) = \rho_w g \quad \Rightarrow \quad \frac{d}{\rho_w} = \frac{g}{g + a}$$

$$\Rightarrow \quad r \cdot d = \frac{10}{10 + 2} = \frac{5}{6}$$

25. When the log is completely submerged, buoyancy on it is $F_B = V\rho g = \left(\frac{m}{d}\right)\rho g$

$$\Rightarrow \quad F_B = \left(\frac{120\,\text{kg}}{600\,\text{kgm}^{-3}}\right)\left(10^3\ \frac{\text{kg}}{\text{m}^3}\right) \cdot g = 200 \cdot \text{g}$$

According to the question

Weight of log + Additional weight = F_B

$$\Rightarrow \quad 120 \cdot g + mg = 200\,\text{g}$$

$$\Rightarrow \quad m = 80\,\text{kg}$$

26. Let x = mass of copper in jewellery. (in gram)

$(50 - x)$ = mass of gold (in gram)

Volume of jewellery = volume of copper + volume of gold

$$\Rightarrow \quad V = \left(\frac{x}{10} + \frac{50 - x}{20}\right) \text{ in cm}^3$$

Buoyancy = Loss in weight

$$\Rightarrow \quad V\rho_w g = 4 \cdot g$$

$$\Rightarrow \quad \left(\frac{x}{10} + \frac{50 - x}{20}\right) \times (1) \times g = 4 \cdot g \ [\rho_w = 1\,\text{gcm}^{-3}]$$

$$\Rightarrow \quad \frac{x}{10} + \frac{50 - x}{20} = 4 \quad \Rightarrow \quad x = 30 \text{ g}$$

27. For equilibrium of the balloon,

$$F_B = 800 \text{ g} \qquad \text{...(i)}$$

Let the mass of balloon be m after it is reduced. Buoyancy does not change as volume has not changed and it displaces same amount of air as earlier.

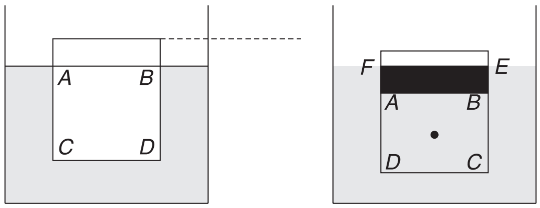
$$ma = F_B - mg \quad \Rightarrow \quad m(a + g) = 800\,\text{g}$$

$$\Rightarrow \quad m = \frac{800 \times 10}{2 + 10} = 667\,\text{kg}$$

$$\therefore \quad \Delta m = 800 - 667 = 133\,\text{kg}$$

28. Consider an ice block (without gold) floating in water.

It displaces ($ABCD$) volume of water. When it melts, is makes water, which has volume exactly equal to ($ABCD$).



ice block with gold displaces more water

When the ice block has gold embedded in it, it will have to displace more water to balance the weight of the gold. Now volume of water displaced is ($CDFE$). In fact, the volume ($ABEF$) is 19.6 times the volume of gold piece as relative density of gold is 19.6. It means we need 19.6 cm^3 water to balance the weight of 1cm^3 gold.

When ice melts, it forms water to fill the volume $ABCD$. The volume $ABEF$ will remain unoccupied. Thus, the level of water will fall.

29. Area of cross-section of tube $A = \pi r^2 = 3.14 \times 0.8^2$

$$= 2.0\,\text{cm}^2$$

Weight added = Increase in buoyancy due to displacement of extra water

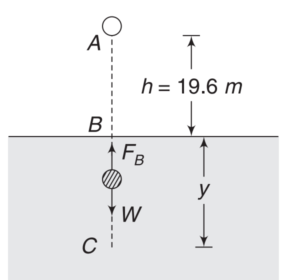
$$\Rightarrow \quad \Delta mg = (A \times 2\,\text{cm})\,\rho_w \cdot g$$

$$\Rightarrow \quad \Delta m = (2 \times 2\,\text{cm}^3) \times \left(1 \frac{\text{g}}{\text{cm}^3}\right) = 4\ \text{g}$$

30. If m mass of water is removed and poured into the tube, the tube will sink further. The extra mass of water that it will displace will be equal to m (as increase in mass of tube is m). It will make no difference to level of water in the beaker.

If you are rowing a boat and you drink some water from the lake, has the level of water in the lake changed? Obviously, no.

31. Speed of the ball when it reaches B (at water surface) is



$$v = \sqrt{2gh}$$

$$= \sqrt{2 \times 9.8 \times 19.6}$$

$$= 19.6\ \text{ms}^{-1}$$

As the ball enters the liquid, it experiences buoyancy (F_B) apart from its weight (W).

Now acceleration (a) of the ball is

$$a = \frac{F_B - W}{m}\ \text{(upward)}$$

$$= \frac{V\rho_w g - V\dfrac{\rho_w}{2}\cdot g}{V\dfrac{\rho_w}{2}} = g(\uparrow)$$

The ball retards and stops at C.

For motion from B to C: $v^2 = u^2 + 2ax$

$$\Rightarrow \quad 0^2 = (19.6)^2 - 2 \times 9.8 \times y \quad \Rightarrow \quad y = 19.6\,\text{m}$$

32. Volume of iron (the shaded part in diagram) is

$$V_1 = \frac{\text{mass of iron}}{\text{density of iron}} = \frac{6000/9.8}{7.87 \times 10^3} = 0.078\,\text{m}^3$$

When placed in water

Buoyancy = loss in weight

$(V_1 + V_2)\rho g = 2000$ [V_2 = volume of cavity]

$$\Rightarrow \quad (0.078 + V_2) \times 10^3 \times 9.8 = 2{,}000$$

$$\Rightarrow \quad V_2 \simeq 0.12\,\text{m}^3$$

33. Initially, $F_B + F_S = W$

$$\Rightarrow \quad V\rho g + kx_1 = Vdg$$

[d = density of solid and ρ = density of liquid]

$$\therefore \quad kx_1 = V(d - \rho)g \qquad \text{...(i)}$$

When the container is accelerated, $g_{\text{eff}} = g + a$

In equilibrium: $F'_B + F'_S = W'$ [In RF of container]

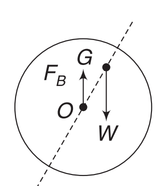
$$\Rightarrow \quad V\rho\, g_{\text{eff}} + kx_2 = Vd\,g_{\text{eff}}$$

$$\Rightarrow \quad kx_2 = V(d - \rho)\, g_{\text{eff}} \qquad \text{...(ii)}$$

Dividing (ii) by (i) gives: $\dfrac{x_2}{x_1} = \dfrac{g_{\text{eff}}}{g}$

$$\Rightarrow \quad \frac{x_2}{3\,\text{cm}} = \frac{10 + 2}{10} \quad \Rightarrow \quad x_2 = 3.6\,\text{cm}$$

34. Centre of buoyancy is O. A slight tilt in clockwise sense will cause the two forces (F_B and W) to form a clockwise couple. This will further rotate the sphere.



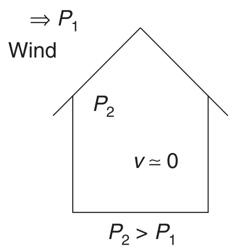
35. Centre of buoyancy is at mid-point of the submerged part. For COM to be at the same location, mass of the particle shall be *m*.

36. $A_1 v_1 = A_2 v_P = A_3 v_3$

$$0.5 \times 0.1 = 1 \times v_P = 0.05 \times v_3$$

$$\Rightarrow \quad v_P = 0.05\,\text{ms}^{-1};\ v_3 = 1\,\text{ms}^{-1}$$

37. Due to high speed of air outside the house, pressure is substantially lower than inside pressure. The difference in pressure causes an upward force on the roof.



38. Air, in immediate contact with the train, is moving at speed equal to the speed of the train. In the narrow space between man's body and the train, speed of air is high. This results in drop in pressure and the atmospheric pressure from behind pushes the man towards the train.

40. P_1 and P_2 are pressure at the top and bottom of the wing. Height difference between two points is negligible.

$$\therefore \quad P_2 + \frac{1}{2}\rho v_2^2 = P_1 + \frac{1}{2}\rho v_1^2$$

$$\Rightarrow \quad P_2 - P_1 = \frac{1}{2}\rho(v_1^2 - v_2^2)$$

$$\Rightarrow \quad \Delta P = \frac{1}{2} \times 1.3 \times (120^2 - 90^2) = 4{,}095\,\text{Nm}^{-2}$$

Lift force is $F = \Delta P A$

$$= 4{,}095 \times (10 \times 2) = 81{,}900\,\text{N}$$

41. Loss in pressure energy (per unit volume) = gain in *KE* per unit volume

$$\Rightarrow \quad (3.5 - 3) \times 10^5 = \frac{1}{2}\rho v^2$$

$$\Rightarrow \quad 0.5 \times 10^5 = \frac{1}{2} \times 10^3 \times v^2$$

$$\Rightarrow \quad v = 10\ \text{ms}^{-1}$$

42. $A_1 v_1 = A_2 v_2 = 1\ \text{cm}^3\text{s}^{-1}$

$$\therefore \quad v_1 = \frac{1\,\text{cm}^3\text{s}^{-1}}{4\ \text{mm}^2} = \frac{1\,\text{cm}^3\text{s}^{-1}}{4 \times 10^{-2}\,\text{cm}^2} = 25\ \text{cms}^{-1}$$

$$\text{and} \quad v_2 = \frac{1\,\text{cm}^3\text{s}^{-1}}{2\,\text{mm}^2} = 50\ \text{cms}^{-1}$$

Using Bernoulli's equation

$$P_A + \frac{1}{2}\rho v_1^2 = P_B + \frac{1}{2}\rho v_2^2$$

$$\Rightarrow \quad P_A - P_B = \frac{1}{2}P(v_2^2 - v_1^2)$$

$$= \frac{1}{2} \times 10^3\ (0.5^2 - 0.25^2) = 93.75\ \text{Pa}$$

44. Speed of efflux will depend on height difference between surface of water and end *A*.

$$h = H - l\sin\theta$$

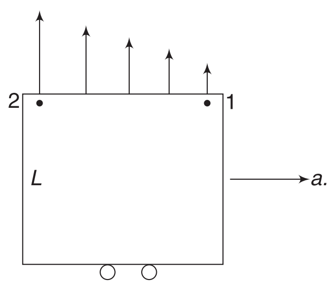
$$\therefore \quad v = \sqrt{2gh} = \sqrt{2g(H - l\sin\theta)}$$

Worksheet 1

1. In a cylindrical vessel, force on base is equal to weight of liquid in it.
2. In case (C), the slant container wall exerts downward force on the liquid, thus force on the base increases. Or, height of liquid will be maximum in vessel (C). Therefore, pressure at bottom is maximum in (C).
3. Pressure is normal thrust per unit area.

   $\therefore \quad P = \rho g(H - h)$
4. When air is pumped out, pressure at the top decreases. Pascal's law says that pressure decreases by same amount everywhere.
5. Reason is same as in Question 4.

   If atmospheric pressure increases, buoyancy force on a floating wood piece will not change.
6. Pressure increases as one moves horizontally from point 1 to 2. Thus, force on a small segment of the top surface close to point 2 is higher than force on a segment of same size that is close to 1. Thus resultant force is nearer to 2.

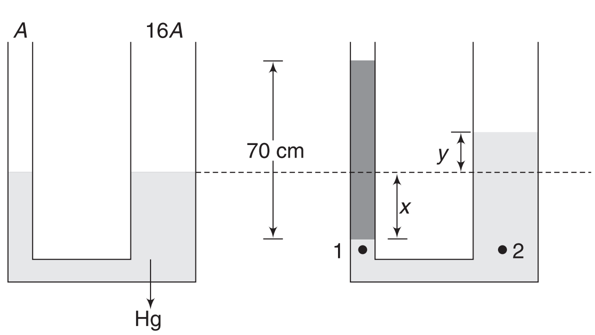


7. Force gets multiplied by $\frac{A_2}{A_1} = \left(\frac{20}{1}\right)^2 = 400$

   $\therefore \quad mg \times 400 = 1500\,\text{g} \quad \Rightarrow \quad m = 3.75\,\text{kg}$
8. $3P = P + \rho gH \quad \Rightarrow \quad \rho gH = 2P$

   When the height of water is lowered by a fifth, the new height is $h = \frac{4H}{5}$. Pressure at bottom of the tank is

   $P' = P + \rho gh = P + \frac{4}{5}\rho gH = P + \frac{8P}{5} = \frac{13P}{5}$
10. If area of one side is A then the area of other limb is $16A$. When water is added, let the mercury level fall by x in the thinner tube. Mercury level moves up by y in the other limb.



Volume conservation gives:

$$Ax = 16Ay \quad \Rightarrow \quad y = \frac{x}{16}$$

Now, $\quad P_1 = P_2$

$$P_0 + \rho_w\, g(70\,\text{cm}) = P_0 + \rho_{Hg} \cdot g \cdot (x + y)$$

$$\Rightarrow \quad 70\,\text{cm} = \frac{\rho_{Hg}}{\rho_w}(16y + y)$$

$$\Rightarrow \quad \frac{70}{13.6 \times 17}\,\text{cm} = y \quad \Rightarrow \quad y = 0.3\text{ cm}$$

11. $V_{\text{sub}} \cdot \rho_l \cdot g = V_{\text{solid}}\, \rho_s \cdot g$

    $\Rightarrow \quad V_{\text{sub}} = V_{\text{solid}} \times \frac{1}{3}$

    $\therefore$ One-third of its volume is submerged and two third is exposed
12. Loss in weight = Buoyancy

    $60 - 40 = V\rho_w g$

    $\Rightarrow \quad 20 = V\rho_w g$ ...(i)

    Also, $\quad 60 = V\rho_s g$ ...(ii)

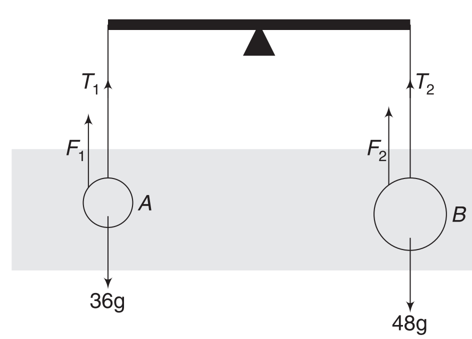
    Dividing (ii) by (i) gives $\frac{\rho_s}{\rho_w} = \frac{60}{20} = 3$
13. Let V = volume of the sphere

    Weight of the sphere = weight of $\frac{V}{2}$ volume of water.

    When the sphere is about to sink (it is completely submerged); Weight of sphere + Weight of water in it = Weight of V volume water.

    $\therefore$ Weight of water in sphere = weight of $\frac{V}{2}$ Volume of water.
14. Weight loss = Buoyancy

    For same loss in weight buoyancy must be same
15. Let mass of A be 36 g and that of B be 48 g.



F_1 and F_2 are buoyancy forces on the two objects. For equilibrium of the balance

Tension T_1 = tension T_2

$\therefore \quad 36\,\text{g} - F_1 = 48\,\text{g} - F_2$

$\Rightarrow \quad 36\text{ g} - \dfrac{36}{9} \times 1 \times g = 48\text{ g} - \dfrac{48}{d} \times 1 \times g$

[$F_1 = V_1 \rho_w g = \dfrac{36\,\text{gram}}{9\,\text{gcm}^{-3}} \times 1\text{gcm}^{-3} \times g$;

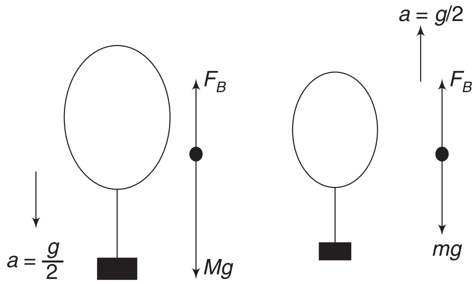
similarly F_2 is written. d = density of second object]

$\Rightarrow \quad d = 3\,\text{gcm}^{-3}$

16. $M = 1.5\,\text{kg}$

Initially, $Mg - F_B = Ma$

$\Rightarrow \quad F_B = Mg - \dfrac{Mg}{2} = \dfrac{Mg}{2}$



After the mass is decreased to m, the buoyancy force has not changed as volume remain same.

$\therefore \quad F_B - mg = ma \quad \Rightarrow \quad \dfrac{Mg}{2} - mg = m\dfrac{g}{2}$

$\Rightarrow \quad m = \dfrac{M}{3} = 0.5\,\text{kg}$

$\therefore$ Decrease in mass = 1.0 kg

17. Weight of the hemispherical bowl = Buoyancy

$\Rightarrow \quad W = \dfrac{2}{3}\pi R^3 \cdot \rho_l \cdot g \qquad [R = \text{radius} = 0.5\,\text{m}]$

$\therefore$ Mass of bowl, $M = \dfrac{2}{3}\pi R^3 \rho_l$

Volume of material in the bowl is

$$V = \frac{M}{\rho_s} = \frac{2}{3}\pi R^3 \frac{\rho_l}{\rho_s}$$

$$= \frac{2}{3}\pi (0.5)^3 \times \frac{1.2}{20}$$

If inner radius of the bowl is r, then volume of material is

$$V = \frac{2}{3}\pi (R^3 - r^3)$$

$\therefore \quad \dfrac{2}{3}\pi (R^3 - r^3) = \dfrac{2}{3}\pi (0.5)^3 \times \dfrac{1.2}{20}$

$\Rightarrow \quad 0.5^3 - r^3 = 0.5^3 \times 0.06$

$\Rightarrow \quad r^3 = 0.5^3 \times 0.94 \qquad \Rightarrow \quad r = 0.5 \times (0.94)^{1/3}$

$\qquad = 0.5 \times 0.98$

$\therefore$ Inner diameter = $2r$ = 0.98 m

18. Reading = weight of vessel and water + weight of ball + vertical force needed to keep the ball sunk (F)

$= 600\,\text{g} + 40\,\text{g} + F = 640\,\text{g}\cdot + F$

If the ball is released it will float with 20% of its volume outside water and 80% immersed in water if volume of ball is V then

$0.8\,V\rho_w g = 40\,g$ ...(i)

To submerge the remaining ball the pin must apply a force that is equal to additional buoyancy when remaining 20% is pushed into water.

$\therefore \quad F = 0.2\,V\rho_w g = 10\,g$ [from (i)]

$\therefore$ Reading = $640\,g + 10\,g = 650\,g$

= 650 gram wt.

19. Pressure just below the piston is

$P_1 = P_0 + \dfrac{12.g}{A}$ $[A = 800\text{ cm}^2]$

Pressure at same horizontal level in the left limb is

$P_2 = P_0 + \rho gh$

Since, $P_2 = P_1$

$\therefore \quad \rho gh = \dfrac{12.g}{A}$

$\Rightarrow \quad h = \dfrac{12\text{ kg}}{\left(10^3\,\dfrac{\text{kg}}{\text{m}^3}\right) \times (800 \times 10^{-4}\text{ m}^2)} = 0.15\text{ m}$

20. In a cylindrical container, if P is pressure at bottom and A is area of base then

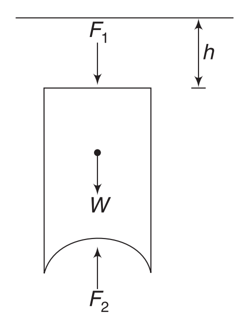
PA = Weight of content in the cylinder

$\therefore \quad P_1A$ = Weight of water

P_2A = Weight of water + block.

$\therefore \quad (P_2 - P_1)A = Mg$

21. Let F_1 = downward force on top surface due to liquid pressure.



F_2 = upward force on curved lower face due to liquid.

Buoyancy = $F_2 - F_1$

$V\rho g = F_2 - \pi R^2.(\rho gh)$

$\Rightarrow \quad F_2 = V\rho g + \pi R^2 \rho gh$

22. Metal coin has high density compared to water. To balance its weight, a large volume of water is displaced by the wooden block. When the metal coin is removed, the cube moves up and level of water goes down. If the coin is placed inside water, it will displace volume of liquid equal to its own volume, which is not much.

23. Let volume of ice in kerosene be V_1 and that in water be V_2.

$$\text{Buoyancy} = \text{Weight}$$

$$V_1\rho_k g = V_2\rho_w g = (V_1 + V_2)\,\rho_{\text{ice}}\,g$$

$$\Rightarrow \quad \frac{\rho_k}{\rho_w} + \frac{V_2}{V_1} = \left(1 + \frac{V_2}{V_1}\right)\frac{\rho_{\text{ice}}}{\rho_w}$$

$$\Rightarrow \quad 0.8 + \frac{V_2}{V_1} = \left(1 + \frac{V_2}{V_1}\right) \times 0.9$$

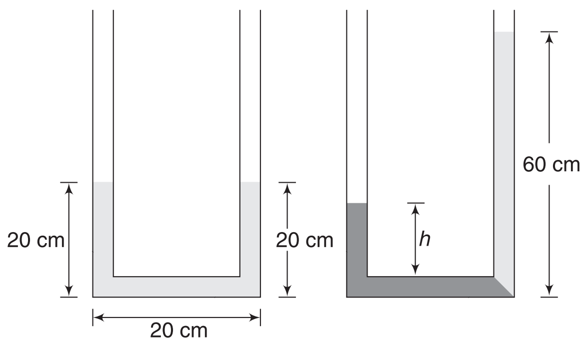
$$\Rightarrow \quad \frac{V_2}{V_1}(1 - 0.9) = 0.1 \quad \Rightarrow \quad \frac{V_2}{V_1} = 1$$

24. Proceed as in last question.

25. Considering equilibrium of the liquid, we can easily see that force on it due to the container balances its weight.

26. The situation has been shown in the figure. Initially, 3 segments of the tube contain 20 cm^3 volume of water in each of them.

When the new liquid is poured in the left column, the water moves to the right. Final situation is as shown in second figure.



$$\rho g h = \rho_w g \cdot (60\text{ cm})$$

$$\Rightarrow \quad 4\rho_w g h = \rho_w g\ (60\text{ cm})$$

$$\Rightarrow \quad h = 15\text{ cm}$$

$\therefore$ Length of new liquid in the tube = 15 + 20 = 35 cm

$\therefore$ Volume = 35 cm^3

27. $\frac{3}{4}$ of the volume of the combined cylinder is submerged.

$$\text{Buoyancy} = \text{Weight}$$

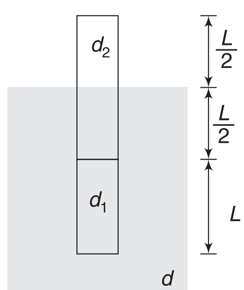
$$\frac{3}{4}Vdg = \frac{V}{2}d_1 g + \frac{V}{2}d_2 g$$

$$\Rightarrow \quad d_1 + d_2 = \frac{3d}{2}$$

Since, $\quad d_1 > d_2$

$$\therefore \quad 2d_1 > \frac{3d}{2}$$

$$\Rightarrow \quad d_1 > \frac{3d}{4}$$



28. Upward acceleration of the ball inside water is given by

$$ma = F_B - W$$

$$V\rho a = V\sigma g - V\rho g \quad \Rightarrow \quad a = \left(\frac{\sigma}{\rho} - 1\right)g$$

Speed gained when it reaches the surface is given by

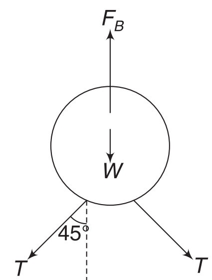
$$v^2 = 0^2 + 2ah \quad \Rightarrow \quad v^2 = 2ah \qquad \text{...(i)}$$

Outside the water, the ball moves under gravity. Height attained above surface of water is

$$H = \frac{v^2}{2g} = \frac{ah}{g} = \left(\frac{\sigma}{\rho} - 1\right)h$$

29. For equilibrium

$$2 \cdot T\cos 45° + W = F_B$$



$$\Rightarrow \quad \sqrt{2}\,T + Mg = \frac{4}{3}\pi R^3\,\rho_w \cdot g$$

$$\Rightarrow \quad T = \frac{\frac{4}{3}\pi R^3 \rho_w g - Mg}{\sqrt{2}}$$

30. Let length of the block be L. Compression in the spring $= \frac{L}{3}$

For equilibrium: $k\frac{L}{3} + (AL)\left(\frac{\rho}{3}\right)g = AL \cdot \rho \cdot g$

$$\Rightarrow \quad k = 2\rho A g$$

31. For equilibrium, force on dumbbell = 0

$$\therefore \quad 2\left(\frac{V}{2}\rho g\right) = (3M + m)g$$

$$\Rightarrow \quad V\rho g = (3M + m)g \qquad \text{...(i)}$$

For rotational equilibrium, let us make torque about the centre of right ball to be zero.

$$mg \cdot l + Mg \cdot d = \frac{V}{2}\rho g d$$

$$\Rightarrow \quad (V\rho - 3M)l = \frac{V}{2}\rho d - Md \qquad \text{[Using (i)]}$$

$$\Rightarrow \quad l = \frac{(V\rho - 2M)d}{2(V\rho - 3M)}$$

32. Force due to liquid pressure on any small element on the surface of the cylinder is normal to the cylinder. It is radial and will have no torque about O.

    Therefore, $F = 0$.

    [Applying any force F will cause the gate to rotate]. Which force balances the horizontal force on the gate due to liquid pressure? Obviously, hinge force.

33. Bucket falls down with an acceleration

$$a = \frac{2mg - mg}{3\text{m}} = \frac{g}{3}$$

[m = mass of block]

$\therefore$ Gauge pressure at the bottom of the bucket is

$$P = \rho \cdot g_{\text{eff}} h = 10^3 \times \left(g - \frac{g}{3}\right) \times 0.15$$

$$= 10^3 \times \frac{2 \times 10}{3} \times 0.15 = 10^3 \text{ Pa}$$

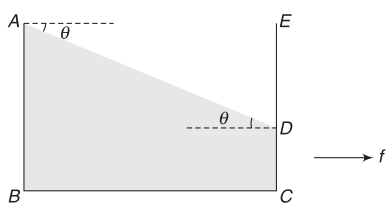
$$= 1 \text{kPa}$$

34. Conceptually similar to Q.21.

35. Let side length of cube be a.

$$\text{Volume } ADE = \frac{1}{2} \times AE \times ED \times a$$

$$= \frac{1}{2}\, a \cdot (a \tan\theta) \cdot a = \frac{1}{2} a^3 \tan\theta$$



Given volume, $ADE = \frac{1}{3}a^3$

$$\therefore \quad \frac{1}{2}\tan\theta = \frac{1}{3}$$

$$\Rightarrow \quad \tan\theta = \frac{2}{3}$$

If f is acceleration of the tank then,

$$\Rightarrow \quad \frac{f}{g} = \frac{2}{3} \quad \Rightarrow \quad f = \frac{2g}{3}$$

36. The question has been created to test your understanding of the contribution made by atmospheric pressure.

    Consider the equilibrium of the piston.

$$F + P_0 A = PA$$

$$\Rightarrow \quad F + P_0 A = [P_0 + \rho g (H + h)]\, A$$

$$\Rightarrow \quad F = \rho g (H + h) A$$

37. Pressure at top-right corner = 0

    This is because there is no air and the liquid tends to move backward.

$$\therefore \quad P_{\text{centre}} = \rho g \frac{L}{2} + \rho a \frac{L}{2}$$

39.

$$A_1 v_1 = A_2 v_2 + A_3 v_3$$

$$A \times 3 = A \times 1.5 + 1.5\, A \cdot v$$

$$\Rightarrow \quad v = 1 \text{ ms}^{-1}$$

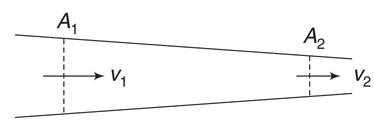
40. Continuity equation requires $v_1 = v_2$.

41.

$$A_1 v_1 = A_2 v_2$$

$$10 \times 1 = 5 \times v_2$$

$$\Rightarrow \quad v_2 = 2 \text{ ms}^{-1}$$



Bernoulli's equation:

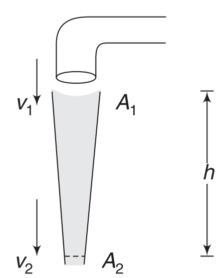
$$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$$

$$2000 + \frac{1}{2} \times 10^3 \times 1^2 = P_2 + \frac{1}{2} \times 10^3 \times 2^2$$

$$\Rightarrow \quad P_2 = 500 \text{ Pa}$$

42. Bernoulli's equation

$$\frac{1}{2}\rho v_2^2 + P_0 + 0 = \frac{1}{2}\rho v_1^2 + P_0 + \rho g h$$



$$\therefore \quad v_2^2 = v_1^2 + 2gh = 1^2 + 2 \times 10 \times 0.15$$

$$\therefore \quad v_2 = 2 \text{ ms}^{-1}$$

Continuity equation:

$$A_2 v_2 = A_1 v_1$$

$$A_2 \times 2 = 10^{-4} \times 1 \quad \Rightarrow A_2 = 5 \times 10^{-5} \text{ m}^2$$

43. Speed of efflux at square hole is $v_1 = \sqrt{2gy}$

    Volume flow rate from square hole is

$$Q_1 = A_1v_1 = L^2\sqrt{2gy}$$

Speed of efflux at the circular hole is $v_2 = \sqrt{2g(4y)}$

$$= 2\sqrt{2gy}$$

Volume flow rate from circular hole is

$$Q_2 = A_2v_2 = \pi R^2 2\sqrt{2gy}$$

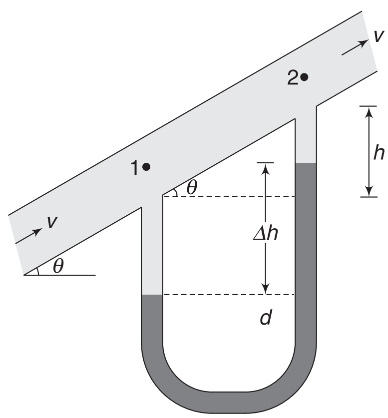
Given $\quad Q_1 = Q_2$

$\Rightarrow \quad L^2 = 2\pi R^2 \quad \Rightarrow \quad R = \dfrac{L}{\sqrt{2\pi}}$

44. $P_1 - P_2 = \rho_{Hg}\cdot g\cdot\Delta h$

$$\Delta P = \rho_{Hg}\cdot g\cdot\Delta h$$

Note that flowing liquid that fills up some space in manometer will have no effect in ΔP as $\rho_{Hg} >> \rho$.



Speed does not change as cross section is uniform. Height difference between point 2 and 1 is

$$h = d\cdot\tan\theta$$

$$P_1 + \frac{1}{2}\rho v^2 + 0 = P_2 + \frac{1}{2}\rho v^2 + \rho gh$$

$\therefore \quad P_1 - P_2 = \rho g(d\tan\theta)$

$$\rho_{Hg}\cdot g\cdot\Delta h = \rho g\cdot d\cdot\tan\theta \Rightarrow \Delta h = \frac{\rho d}{\rho_{Hg}}\tan\theta$$

45. Fluid has more potential energy at A and more pressure energy at B.

47. Bubbles start forming at the tip when pressure there is equal to atmospheric.

Therefore, pressure in the liquid at the level of hole is

$$P = P_0 + \rho gh$$

Applying Bernoulli's equation between points just inside the hole and outside it, gives

$$P + 0 + 0 = P_0 + \frac{1}{2}\rho v^2 + 0$$

$\Rightarrow \quad P_0 + \rho gh = P_0 + \dfrac{1}{2}\rho v^2$

$\Rightarrow \quad v = \sqrt{2gh} = \sqrt{2\times 10\times 0.2} = 2\ \text{ms}^{-1}$

48. Initially, $\quad v = \sqrt{2gh}$

Range $\quad R \propto v$

For range to double, the speed of efflux must double. Let extra pressure on water surface be ΔP for speed of efflux to be $2v$.

$$P_0 + \Delta P + \rho gh = P_0 + \frac{1}{2}\rho(2v)^2$$

$\Rightarrow \quad \Delta P = 2\rho\left(\sqrt{2gh}\right)^2 - \rho gh = 3\rho gh$

$$= 3\times 10^3\times 10\times 10 = 3\times 10^5\ \text{Nm}^{-2}$$

49. Pressure just beneath the piston is

$$P = P_0 + \frac{Mg}{A} = P_0 + \frac{10\times 10}{1000\times 10^{-4}}$$

$$= (P_0 + 10^3)\ \text{Nm}^{-2}$$

Using Bernoulli's equation,

$$P + \rho gh = P_0 + \frac{1}{2}\rho v^2$$

$\Rightarrow \quad P_0 + 10^3 + 10^3\times 10\times 0.5 = P_0 + \dfrac{1}{2}\times 10^3\times v^2$

$\Rightarrow \quad v^2 = 12 \quad \Rightarrow \quad v = 3.46\ \text{ms}^{-1}$

Worksheet 2

1. When the block is submerged, the liquid applies a buoyancy (F) on the block. Due to this reading of the spring balance will decrease.

   From Newton's third law, the block applies F on the liquid in downward direction. Thus, reading of balance B will increase.

2. Siphon works only when discharge end is below the level of water in the tank. It means $h_3 > 0$.

   Pressure at 2 can be calculated using Bernoulli's equation

$$P_2 + \frac{1}{2}\rho v^2 + \rho g h_3 = P_3 + \frac{1}{2}\rho v^2 + 0$$

$$\Rightarrow \qquad P_2 = P_0 - \rho g h_3$$

   With increase in h_1, pressure decrease at the top. Siphon will not work if h_1 is too large such that pressure at the top becomes zero.

3. A = area of tube

   Volume flow rate = Av

   Momentum of water particles that hit the wall in time Δt is $= (Av\Delta t\rho)v = A\rho\Delta t v^2$

   Force = rate of change of momentum $= \dfrac{A\rho\Delta t\, v^2}{\Delta t}$

$$= A\rho v^2$$

$$\therefore \quad F \propto v^2$$

   If v is doubled, F becomes four times.

   Kinetic energy of water coming out in Δt time is

$$\Delta k = \frac{1}{2}(Av\Delta t\rho)v^2$$

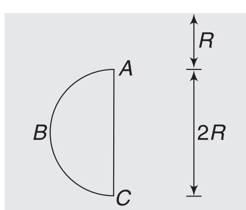
$$\frac{\Delta k}{\Delta t} = \frac{1}{2}A\rho v^3$$

   If v is doubled, rate of loss of KE becomes 8 times

5. Average pressure between the top and bottom of the cylinder is

$$P_{av} = \rho g R + \frac{\rho g \cdot 3R}{2} = 2\rho g R$$

   Consider a semi-cylindrical water element (as shown) inside a tank full of water. The element of water is in equilibrium.



   Horizontal force on face ABC due to surrounding liquid = force on flat rectangular face represented by AC due to surrounding liquid.

$$F_H = 2R \cdot L \cdot \rho_{av} = 2R.L\; 2\rho g R = 4\rho g L R^2$$

   Our solid cylinder will also experience same force on curved face. Vertical force due to surrounding liquid on semi-cylindrical liquid element is

$$F_v = \text{Weight of semi-cylindrical liquid}$$

$$= \frac{\pi R^2}{2} L\rho \cdot g$$

   Same vertical force will act on solid cylinder also.

   Torque is zero since force due to liquid on any surface element of the cylinder is radial.

6. Weight of coin = Vdg

   Volume of water having same weight as the coin is

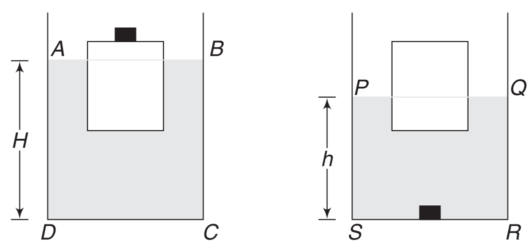
$$\Delta V = \frac{Vdg}{\rho g} = V\frac{d}{\rho}$$

   When the coin is removed, the block will rise up so as to reduce the volume of submerged part by Δv.

$$\therefore \quad \Delta l A = V\frac{d}{\rho} \quad \Rightarrow \quad \Delta l = \frac{Vd}{\rho A}$$

   To find change in level of water, let us assume that volume of submerged part of the block is V_1 and V_2 before and after the coin is removed. Also, assume that volume of water in the container is V_0.

   Before coin is removed



   Volume $ABCD = V_0 + V_1$

$$\Rightarrow \quad A_1 H = V_0 + V_1 \qquad \text{...(i)}$$

   After the coin is removed from the block and placed in the tank

   Volume $PQRS = V_0 + V_2 + V$

$$\Rightarrow \quad A_1 h = V_0 + V_2 + V \qquad \text{...(ii)}$$

   (i) – (ii) gives $A_1 (H - h) = V_1 - V_2 - V$

$$\Rightarrow \quad A_1 \Delta h = (V_1 - V_2) - V$$

$$= \frac{\text{Weight of coin}}{\rho g} - V$$

$$= \frac{Vdg}{\rho g} - V = V\left(\frac{d}{\rho} - 1\right)$$

$$\therefore \quad \Delta h = \frac{V}{A_1}\left(\frac{d}{\rho} - 1\right)$$

7. Volume of liquid that overflows = volume of the ball (V). When ball is submerged, it experiences buoyancy equal to $F_B = Vrg$. It applies same force on the liquid in downward direction. Thus, the ball applies a force $F_B = Vrg$ on the liquid downward. But same weight of the liquid has overflown also. Thus, reading will not change.

   In second case, reading is

$$R_2 = W + mg - \text{Weight of liquid that overflows}$$
$$= W + mg - Vrg$$

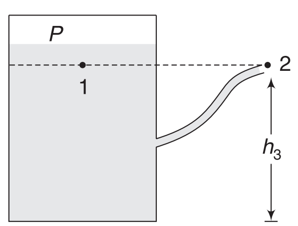
8. $v_2 = \sqrt{2gH} = \sqrt{2 \times 10 \times 5} = 10 \text{ ms}^{-1}$

$$v_1 = \frac{v_2 A_2}{A_1} = v_2 \frac{r_2^2}{r_1^2} = 10 \times \frac{1}{4} = 2.5 \text{ ms}^{-1}$$

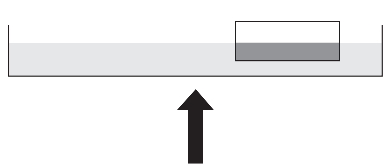
   Pressure at the location of vertical tube can be calculated using Bernoulli's equation.

$$P + \frac{1}{2}\rho v_1^2 + 0 = P_0 + 0 + \rho gH$$
$$\Rightarrow \quad P = P_0 + \rho gH - \frac{1}{2}\rho v_1^2$$
$$\Rightarrow \quad \rho gh + P_0 = P_0 + \rho gH - \frac{1}{2}\rho v_1^2$$
$$\Rightarrow \quad h = H - \frac{v_1^2}{2g} = 5 - \frac{2.5^2}{2 \times 10} = 4.69 \text{ m}$$

10. As the level of water falls, air inside the tank expands and pressure at the surface of water decreases. Flow will stop when ($P_1 = P_2$) ($= P_0$).



11. At large depths, due to large hydrostatic pressure, the size of balloon with decrease substantially. It will displace less water and buoyancy will fail to balance the weight of the system. It will sink further.
12. Equilirbium is maintained for $d \le d_w$. As long as the block floats, it applies a force on the liquid that is equal to weight of displaced liquid.



   The block exerts downward force that is equal to weight of dark shaded volume of liquid (= Buoyancy). So removing the block and filling the dark shaded portion with liquid will not change the situation.

13. $P_x + \rho_w g(175 \text{ cm}) - \rho_{Hg} g(112 \text{ cm}) + \rho_w g(75 \text{ cm})$

$$-\rho_{Hg} g(88 \text{ cm}) - \rho_w g(150 - 88 \text{ cm}) = P_y$$
$$\Rightarrow P_x - P_y = \rho_{Hg} g(200 \text{ cm}) - \rho_w g(188 \text{ cm})$$
$$= 13600 \times 9.8 \times 2 - 1000 \times 9.8 \times 1.88$$
$$= 266560 - 18424 = 248136 \text{ Nm}^{-2}$$
$$\simeq 248 \text{ Nm}^{-2}$$

14. Let M be mass of ship and m be mass of cargo. Let the volume of sea water displaced initially be V_0.

   Cross-sectional area of the ship is A.

   ρ_s = density of sea water, ρ_r = density of river water

$$(M + m)g = V_0 \rho_s g \Rightarrow M + m = V_0 \rho_s \quad \text{...(i)}$$

   When cargo is removed in river, the ship rises by Y, it means

$$mg = AY\rho_r \quad \Rightarrow \quad m = AY\rho_r \quad \text{...(ii)}$$

   Putting in (i) gives $\quad M = V_0 \rho_s - AY\rho_r \quad$ ...(iii)

   After unloading of cargo, when the ship is back in sea water

$$Mg = (V_0 + AX - AY - AZ)\rho_s g$$
$$\Rightarrow V_0 \rho_s - AY\rho_r = V_0 \rho_s + (AX - AY - AZ)\rho_S \quad \text{[using (iii)]}$$
$$\Rightarrow \quad Y\rho_r = \rho_s (Y + Z - X)$$
$$\Rightarrow \quad \frac{\rho_s}{\rho_r} = \frac{Y}{Y + Z - X}$$

15. Liquid A applies force normal to the cylinder wall which cancels out. Liquid A does not apply force on cylinder. But it does not mean that liquid A has no role to play. Its pressure increases the pressure in liquid B. For finding h, consider equilibrium.

$$\text{Buoyancy} = \text{weight}$$
$$Ah_B \cdot \rho_B \cdot g + Ah_A \rho_A \cdot g = A(h + h_A + h_B) d \cdot g$$
$$\Rightarrow \quad 0.8 \times 1.2 + 1.2 \times 0.7 = (h + 1.2 + 0.8) \times 0.8$$
$$\Rightarrow \quad h = 0.25 \text{ cm}$$

   When cylinder is depressed by h, the resultant force on it is upward and is equal to change in buoyancy. Length of cylinder inside A does not change but its length inside B increases by h.

$$\therefore \quad F = Ah\rho_B g$$
$$\Rightarrow \quad ma = Ah\rho_B g$$
$$\Rightarrow \quad A(h + h_A + h_B) d \cdot a = Ah\rho_B \cdot g$$

$$\Rightarrow \quad a = \frac{0.25 \times 1.2}{0.8(0.25 + 1.2 + 0.8)} g = \frac{g}{6}$$

16. Volume of water displaced must have weight equal to the bowl

$$\therefore \quad V\rho g = mg$$

$$\Rightarrow \quad V = \frac{m}{\rho}$$

Volume of steel in bowl $V_0 = \dfrac{m}{d}$

When submerged completely, volume of water displaced is V_0

$$\therefore \quad ma = mg - F_B$$

$$ma = mg - V_0 \rho g = mg - \frac{m\rho g}{d}$$

$$\therefore \quad a = g\left(1 - \frac{\rho}{d}\right)$$

Worksheet 3

1. Initially, spring is relaxed. Buoyancy balances the weight.

$\therefore$ Fraction of volume submerged $= \dfrac{d}{\rho_w} = 0.8$

20% Volume of the cube is outside water.

$\Rightarrow$ 3 cm × 0.2 = 0.6 cm height is outside water.

When the cube is completely submerged, it compresses the spring by 0.6 cm = 0.006 m

For equilibrium:

Additional weight (W) + weight of block = Buoyancy + spring force

$$W + (3 \times 10^{-2})^3 \times 800 \times 10$$
$$= (3 \times 10^{-2})^3 \times 1000 \times 10 + 50 \times 0.006$$

$$\Rightarrow \quad W = 0.35\,\text{N}$$

2. Let mass of $Cu = x$ g

Mass of $Zn = (12.9 - x)$ g

$$\text{Volume of alloy} = \frac{x}{\rho_{cu}} + \frac{12.9 - x}{\rho_{zn}} = \frac{x}{8.9} + \frac{12.9 - x}{7.1}$$

Loss of weight in water = Buoyant force

$$\therefore \quad (12.9 - 11.3)g = \left(\frac{x}{8.9} + \frac{12.9 - x}{7.1}\right) \rho_w \cdot g$$

$$\Rightarrow \quad 1.6 = \frac{x}{8.9} + \frac{12.9 - x}{7.1} \qquad [\because \quad \rho_w = 1\,\text{gcm}^{-3}]$$

Solving gives $x = 7.6g$

3. (a) $V_{\text{sub}} \cdot \rho_{\text{Hg}} \cdot g = V \rho_{\text{Fe}} \cdot g$

$$\Rightarrow \quad \frac{V_{\text{sub}}}{V} = \frac{\rho_{\text{Fe}}}{\rho_{\text{Hg}}} = \frac{7.2}{13.6} = 0.53$$

$\Rightarrow$ 53% of volume is submerged and 47% is outside Hg.

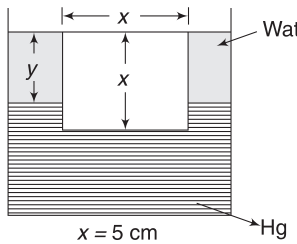
$\therefore$ Height of block above Hg is 0.47 × 5 = 2.35 cm

(b) Let y = height of water level for equilibrium. Weight = Buoyancy

$\Rightarrow$ W = weight of displaced water + weight of displaced Hg.

$$\Rightarrow \quad x^2 \cdot \rho_{\text{Fe}} g = x^2 y \rho_w g + (x - y)x^2 \rho_{\text{Hg}} \cdot g$$

$$\Rightarrow \quad \frac{\rho_{\text{Fe}}}{\rho_w} = \frac{y}{x} + \left(1 - \frac{y}{x}\right) \frac{\rho_{\text{Hg}}}{\rho_w}$$



$$\Rightarrow \quad 7.2 = \frac{y}{x} + 13.6 - 13.6\frac{y}{x} \quad \Rightarrow \quad \frac{y}{x} = \frac{6.4}{12.6}$$

$$\Rightarrow \quad y = 5 \times \frac{6.4}{12.6} = 2.54\,\text{cm}$$

4. (a) Weight and buoyancy are forces on the ball.

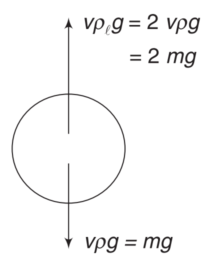
$\rho_l \to 0$ implies that there is no buoyancy and the ball is falling freely.

KE gained when ball falls through 8 cm is given to be 4 J.

$\therefore \quad KE = mgh$

$\Rightarrow \quad 4 = m \times 10 \times 0.08 \quad \Rightarrow \quad m = 5\,\text{kg}$

(b) Density of the ball is $1.5\,\text{gcm}^{-1}$. This is because it experiences no net force when placed in the liquid of density $\rho_l = 1.5$. The graph shows that its KE is nearly zero even after it has travelled through 8 cm. [It travelled with a speed nearly equal to zero and never accelerated]

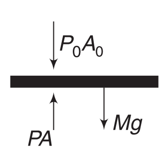


Situation when density of liquid (ρ_l) is twice that of the ball (ρ).

If density of liquid = $3.0\,\text{gcm}^{-3}$, net force on the ball will be mg upwards. It will move up with acceleration g.

$\therefore$ After moving a distance 8 cm in upward direction, its KE will be 4 J (as in (a).)

5. For water to leak, the seal must move up. It means force due to water on the seal must become greater than its weight. In critical case-



$$PA = Mg + P_0A_0$$

$$\Rightarrow \quad P \times 0.01 = 1 \times 10^5 \times 0.015 + 10 \times 10$$

$$\Rightarrow \quad P = 1{,}60{,}000\,\text{Nm}^{-2} = 1.6 \times 10^5\,\text{Nm}^{-2}$$

$$\Rightarrow \quad P_0 + \rho g(h - 0.8) = 1.6 \times 10^5$$

$$\Rightarrow \quad 10^3 \times 10(h - 0.8) = 1.6 \times 10^5 - 1.0 \times 10^5$$

$$= 0.6 \times 10^5$$

$$\Rightarrow \quad h = 6.8\,\text{m}$$

6. (a) If side length is x, volume of iron used to make the box is

$$V_{\text{Fe}} = 6x^2 \cdot t \qquad [t = 1\text{mm} = \text{thickness}]$$

Mass of box $m = \rho_{\text{Fe}} V_{\text{Fe}} = 6\rho_{\text{Fe}} x^2 t$

With increasing x, mass of box increase as $m \propto x^2$. But with increasing x, its overall volume increases as $V \propto x^3$. Therefore, larger box will have a large volume so as to displace sufficient quantity of water needed for floating.

A large box will not sink.

(b) For the box to float while remaining completely Submerged: Buoyancy = weight

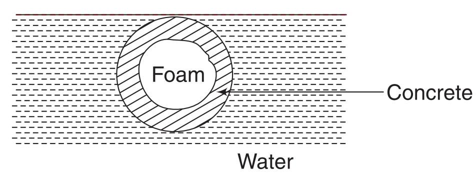
$$x^3 \cdot \rho_w g = 6\,\rho_{\text{Fe}} x^2 t \cdot g$$

$$\Rightarrow \quad x = 6t \cdot \frac{\rho_{\text{Fe}}}{\rho_w} = 6 \times (1\,\text{mm}) \times 8$$

$$= 48\,\text{mm} = 4.8\,\text{cm}$$

For $x > 4.8$ cm, the box floats with its some part remaining outside water.

7. Let volume of sphere be V_0 and Volume of foam in it be V.



Volume of concrete = $V_0 - V$

For floatation: Buoyancy = weight

$$V_0 \rho_w g = (V_0 - V)\rho_C g + V\rho_F g$$

$$\Rightarrow \quad V_0 = (V_0 - V)\frac{\rho_C}{\rho_w} + V\frac{\rho_F}{\rho_w}$$

$$\Rightarrow \quad V_0 = (V_0 - V) \times 2.4 + V \times 0.3 \quad \Rightarrow V = \frac{2}{3}V_0$$

Thus, $\frac{2}{3}$rd volume is foam and remaining $\frac{1}{3}$rd is concrete.

$$\therefore \quad \frac{mc}{m_F} = \frac{\rho_c \cdot V_c}{\rho_F V_F} = \frac{2.4 \times V_0/3}{0.3 \times 2V_0/3} = 4$$

8. Let density of cube be d.

$$Vdg = W$$

$$\Rightarrow \quad (0.1\,\text{m})^3 \times d \times 10 = 5$$

$\Rightarrow \quad d = 0.5 \times 10^3 \,\text{kgm}^{-3}$

Density of cube is less than that of liquid. It can float in the liquid such that

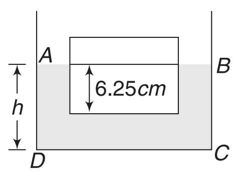
$$V_{sub}\rho_l g = Vd \cdot g \quad \Rightarrow \quad V_{sub} = V \cdot \frac{d}{\rho_l} = V \times \frac{0.5}{0.8} = \frac{5}{8}V$$

Thus, the cube will float with its vertical edge remaining submerged up to length $= \frac{5}{8} \times 10 = 6.25\,\text{cm}$

The tank contains liquid up to much greater depth. Thus the cube is floating without touching the base of the tank.

If height of liquid in tank is h, then,

Volume ($ABCD$) = Volume of liquid in the tank + volume of submerged part of cube

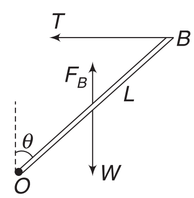


$\therefore \quad 15 \times 15 \times h = 15 \times 15 \times 8 + 6.25 \times 10 \times 10$

$\Rightarrow \quad h = \frac{97}{9}\text{ cm} = 10.8\text{ cm}$

9. Both buoyancy and weight act at the centre of the rod.

For rotational equilibrium of the rod, torque on it about O must be zero.



$$T \cdot L\cos\theta = \frac{L}{2}\sin\theta \cdot (W - F_B)$$

$$\Rightarrow \quad \tan\theta = \frac{2T}{W - F_B}$$

θ is maximum when T is maximum.

$$\therefore \quad \tan\theta_{max} = \frac{2 \times 45}{0.012 \times 1 \times 2000 \times 10 - 0.012 \times 1 \times 1000 \times 10} = \frac{3}{4}$$

$\Rightarrow \quad \theta_{max} = 37°$

10. Refer to Example 12.

$$P_B - P_0 = \frac{1}{2}\rho\omega^2\left(\frac{2L}{3}\right)^2$$

where P_0 = pressure at rotation axis

$$\Rightarrow \text{ And } \quad P_A - P_0 = \frac{1}{2}\rho\omega^2\left(\frac{L}{3}\right)^2$$

Taking difference of the two equations:

$$P_B - P_A = \frac{1}{2}\rho\omega^2 L^2\left[\frac{4}{9} - \frac{1}{9}\right] = \frac{1}{6}\rho\omega^2 L^2$$

11. Initially spring is compressed by x_0 such that

$$kx_0 = P_0 A$$

Where P_0 = atmospheric pressure, A = area of piston.

If the device is kept at a location where pressure is P then the spring will get further compressed by x such that

$$PA = k(x_0 + x) \quad \Rightarrow \quad (P - P_0)A = kx$$

$$\Rightarrow \quad \text{gauge pressure} = P - P_0 = \frac{kx}{A}$$

(a) Least count of $x = 1\,\text{mm}$

$\therefore$ Smallest gauge pressure that can be recorded is

$$= \frac{kx}{A} = \frac{100 \times 1 \times 10^{-3}}{1 \times 10^{-4}} = 1{,}000\,\text{Nm}^{-2}$$

$$= 1\,\text{kPa}$$

(b) At depth of 2 m, the gauge pressure is

$= \rho gh = 10^3 \times 10 \times 2 = 2 \times 10^4\,\text{Nm}^{-2}$

$= 20$ kPa

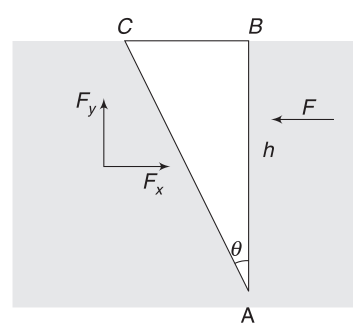
$\therefore$ Further compression in spring (= displacement of piston) = 20 mm = 2 cm

12. $A \gg a$, therefore level of water in the tank will practically remain unchanged.

At every moment, half the candle will be submerged when length becomes 18 cm, after 1 hour half the length (= 9 cm) will be above water. Thus, the top of the candle has moved down by 1 cm only.

13. Consider a water volume represented by ABC inside a tank full of water. Horizontal force by surrounding water on face AC = Horizontal force by surrounding water on face AB.

$$\therefore \quad F_x = \left(\frac{0 + \rho gh}{2}\right)hb = \frac{1}{2}\rho gbh^2$$



Vertical force by surrounding water on face AC = Weight of volume ABC.

$$\Rightarrow \quad F_y = \frac{1}{2}h \cdot (h\tan\theta)b\rho g = \frac{1}{2}\rho ghb^2\tan\theta$$

$\therefore$ Resultant force on AC is

$$F = \sqrt{F_x^2 + F_y^2} = \frac{1}{2}\rho g b h^2 \sqrt{1 + \tan^2\theta}$$

$$= \frac{1}{2}\rho g b h^2 \sec\theta$$

Same force will be applied by water on the wall.

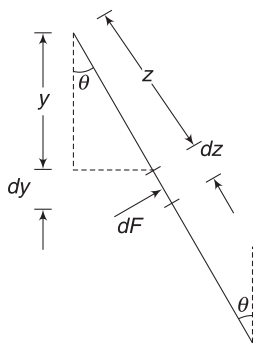
Alternate (by direct integration)

Consider a strip of width dz in the wall, as shown.

Area of the strip $dA = b \cdot dz$

Depth, $y = z\cos\theta \quad \Rightarrow \quad dy = dz \cdot \cos\theta$ ...(i)

Pressure at the strip, $P = \rho g y$



Force on the strip (normal to wall) is

$$dF = P \cdot dA$$

$$= \rho g y \cdot b dz = \rho g b \sec\theta \cdot y dy \quad \text{[using (i)]}$$

$$\therefore \quad F = \rho g b \sec\theta \int_0^h y dy = \frac{1}{2}\rho g b h^2 \sec\theta$$

14. Tension in thread is

$$T = W - F_B = V(3\rho_0)g - \frac{V}{2}\rho_0 g$$

$$= \frac{5}{2}V\rho_0 g$$

where ρ_0 = density of water and $3\rho_0$ is density of the ball. $V = \frac{4}{3}\pi r^3$ = volume of the ball

COM of the rod is at a distance $\left(\frac{y-x}{2}\right)$ from the brim on right side of it.

For equilibrium, torque about the brim (the support) must be zero.

$$\frac{5}{2}V\rho_0 g \cdot x = mg\left(\frac{y-x}{2}\right)$$

$$\Rightarrow \quad 5 \times \frac{4}{3} \times \pi \times (0.5 \times 10^{-2})^3 \times 1 \times 10^3 \times x$$

$$= 4.4 \times 10^{-3}(y - x)$$

$$\Rightarrow \quad 5.22x = 4.4(y - x) \quad \Rightarrow \quad \frac{x}{y} = \frac{4.4}{9.62} = 0.46$$

15. (a) Refer to Example 25. Range is maximum when

$$y = \frac{H + h}{2} = \frac{5h}{6}$$

(b) Range is maximum if $y = \dfrac{H+h}{2} = \dfrac{7h}{6}$

But y cannot be larger than h.

$\therefore$ Hole is at the bottom of the tank for range to be maximum

$$\therefore \quad y = h$$

16. (a) Area of cross-section of the pipe (plug) is

$$A = \pi r^2 = 3.14 \times (0.02)^2 = 1.26 \times 10^{-3}\,\text{m}^2$$

Outward force on the plug due to water pressure is

$$F = PA = \rho g h \cdot A = 10^3 \times 9.8 \times 6 \times 1.26 \times 10^{-3}$$

$$= 74\,\text{N}$$

Friction must be equal to 74 N to prevent the plug from moving.

(b) Speed of efflux $v = \sqrt{2gh} = \sqrt{2 \times 9.8 \times 6}$

$$= 10.84\ \text{ms}^{-1}$$

Volume of water flowing out per second

$$= \text{Av} = 1.26 \times 10^{-3} \times 10.84$$

$$= 1.36 \times 10^{-2}\ \text{m}^3\text{s}^{-1}$$

Water flowing out in 24 h is

$$= 1.36 \times 10^{-2} \times 24 \times 60 \times 60 = 1{,}175\ \text{m}^3$$

17. When height of water is h, rate of outflow is $a\sqrt{2gh}$. When the rate of outflow becomes equal to the rate of inflow, the level of water becomes constant.

$$a\sqrt{2gh} = a_0 v \quad \Rightarrow \quad h = \frac{a_0^2 v^2}{2g a^2}$$

18. (a) Apply Bernoulli's equation between E and D

$$P_0 + \frac{1}{2}\rho v^2 + 0 \quad \Rightarrow \quad P_0 + 0 + \rho g(5\,\text{m})$$

$$\Rightarrow \quad v^2 = 2 \times 10 \times 5 \quad \Rightarrow \quad v = 10\,\text{ms}^{-1}$$

(b) Speed at all points inside the pipe is same.

Using Bernoulli's equation between B and D gives

$$P_B + \frac{1}{2}\rho v^2 + \rho g(6.5\,\text{m}) = P_0 + \frac{1}{2}\rho v^2 + 0$$

$$\Rightarrow \quad P_B = 1 \times 10^5 - 900 \times 10 \times 6.5$$

$$= 41.5 \times 10^3\,\text{Nm}^{-2}$$

$$= 41.5\,\text{kNm}^{-2}$$

(c) $P_A = P_E - \rho g(1\,\text{m}) = 1 \times 10^5 - 10^3 \times 10 \times 1$

$$= 0.9 \times 10^5\ \text{Nm}^{-2}$$

(d) $P_A' + \frac{1}{2}\rho v^2 = P_A$

$\Rightarrow \quad P_A' = 0.9 \times 10^5 - \frac{1}{2} \times 900 \times 10^2$

$= 4.5 \times 10^4 \text{ Nm}^{-2}$

(e) From solution of part (a) it is evident that v is independent of ρ.

19. $W = \Delta U + \Delta K = \rho g(h_2 - h_1) + \frac{1}{2}\rho(v_2^2 - v_1^2)$

Since, $A_2 v_2 = A_1 v_1 \quad \therefore\ v_2 = \frac{v_1}{2} = 0.5\,\text{ms}^{-1}$

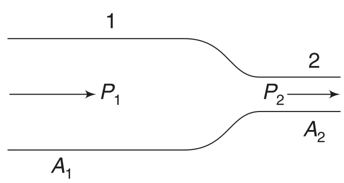
$\therefore\ W = 10^3 \times 10 \times (5 - 2) + \frac{1}{2} \times 10^3 \times (0.5^2 - 1^2)$

$= 3 \times 10^4 - 0.375 \times 10^3 = 29625\,\text{J}$

20. (a) $P_1 + \frac{1}{2}\rho v_1^2 = \rho_2 + \frac{1}{2}\rho v_2^2$

$$\Rightarrow \quad \Delta P = P_2 - P_1 = \frac{1}{2}\rho(v_1^2 - v_2^2)$$

$$= \frac{1}{2}\rho\left(\frac{Q^2}{A_1^2} - \frac{Q^2}{A_2^2}\right)$$



where $Q = A_1 v_1 = A_2 v_2$ = volume flow rate (m^3/s)

$$\therefore \quad \Delta P = \left(\frac{1}{2}\rho Q^2\right)\frac{1}{A_1^2} - \left(\frac{1}{2}\frac{\rho Q^2}{A_2^2}\right)$$

$$\Delta P = a\frac{1}{A_1^2} - b$$

where $\quad a = \frac{1}{2}\rho Q^2$ and $b = \frac{1}{2}\frac{\rho Q^2}{A_2^2}$

(a) $\Delta P = 0$ when $A_1 = A_2$

$\therefore \quad \frac{1}{A_2^2} = 8 \quad \Rightarrow \quad A_2 = \frac{1}{2\sqrt{2}} = 0.35\,\text{m}^2$

(b) From graph, slope is $= \frac{600 \times 10^3}{8} = 75 \times 10^3$

$\therefore \quad \frac{1}{2}\rho Q^2 = 75 \quad \Rightarrow \quad Q^2 = \frac{75 \times 10^3 \times 2}{10^3}$

$Q = 12.25\,\text{m}^3\text{s}^{-1}$

21. Refer to Example 26

$$t \propto \sqrt{H} \quad \Rightarrow \quad \frac{t}{10\,\text{min}} = \sqrt{\frac{H/2}{H}}$$

$$\Rightarrow \quad t = \frac{10\,\text{min}}{\sqrt{2}} = 7.07\,\text{min}$$

22. Let speed of efflux be v.

Applying Bernoulli's equation between a point just outside the hole and a point just inside it, we get

$(P_0 + \rho g \cdot 2h + 2\rho \cdot g \cdot h) + 0 = P_0 + \frac{1}{2}\rho v^2$

$\Rightarrow \quad v = \sqrt{8gh}$

Volume of liquid flowing out per second = Sv

Mass of liquid flowing out per second = $2\rho(Sv)$

Momentum gained by the outgoing liquid per second = $(2\rho Sv)v$

$\therefore$ Force experienced by the tank horizontally towards left is

$$F_{\text{thrust}} = 2\rho S v^2 = 16\rho Sgh$$

Mass of the tank and its content

$= M + (2hA)\rho + (2hA)(2\rho)$

$= M + 6h\rho A$

For friction to be able to prevent sliding

$$\mu N \geq F_{\text{thrust}}$$

$\mu(M + 6h\rho A)g \geq 16\rho Sgh$

$$\Rightarrow \quad \mu \geq \frac{16\rho Sh}{M + 6hA\rho}$$

23. $A_1 v_1 = A_2 v_2 \quad \Rightarrow \quad v_1 = \frac{A_2}{A_1}v$

$P_1 = P_0$; and $P_1 - P_2 = \rho g(h_2 - h_2)$

$\Rightarrow \quad P_2 = P_0 - \rho g(h_1 - h_2)$

$\therefore \quad P_0 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + 0$

$$\Rightarrow \quad P_0 + \frac{1}{2}\rho\left(\frac{A_2}{A_1}v\right)^2 + \rho g h_1$$

$$= P_0 - \rho g(h_1 - h_2) + \frac{1}{2}\rho v^2$$

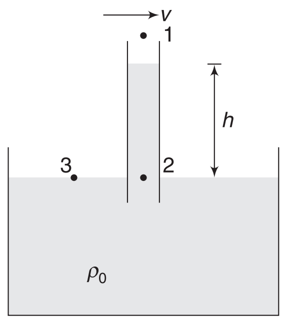
$$\Rightarrow \quad \rho g(2h_1 - h_2) = \frac{1}{2}\rho v^2\left[1 - \left(\frac{A_2}{A_1}\right)^2\right]$$

$$\Rightarrow \quad 2g\frac{(2h_1 - h_2)}{v^2} = 1 - \left(\frac{A_2}{A_1}\right)^2$$

$$\Rightarrow \quad \frac{A_2}{A_1} = \sqrt{1 - \frac{2g(2h_1 - h_2)}{v^2}}$$

24. $P_2 = P_3$

$P_1 + \rho_0 g h = P_0 \quad \Rightarrow \quad P_1 = P_0 - \rho_0 g h$



Bernoulli's equation

$$\frac{1}{2}\rho v^2 + P_1 = P_0$$

$$\Rightarrow \quad \frac{1}{2}\rho v^2 + P_0 - \rho_0 gh = P_0 \quad \Rightarrow \quad \frac{1}{2}\rho v^2 = \rho_0 gh$$

$$\Rightarrow \quad v = \sqrt{\frac{2\rho_0 gh}{\rho}}$$

$$= \sqrt{\frac{2 \times 10^3 \times 10 \times 10^{-2}}{1.3}} \simeq 12.4 \text{ ms}^{-1}$$

In simple language, the loss in pressure = gain in *KE* per unit volume by air

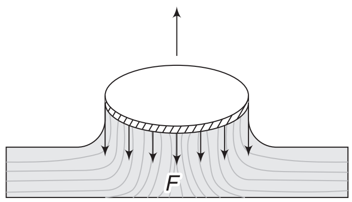
$$\therefore \quad \Delta P = \frac{1}{2}\rho v^2$$

where ΔP = amount by which pressure at 1 is less than atmospheric pressure

$$\therefore \quad \rho_0 gh = \frac{1}{2}\rho v^2$$

Your Turn

1. $F = SL = (0.075\ \text{Nm}^{-1})\ (0.1\ \text{m}) = 7.5 \times 10^{-3}\ \text{N}$
2. As the disc is pulled up, the water surface clings to the outer periphery and the surface tension force acts downward on the disc. In the extreme case the water surface becomes tangential (i.e., vertical) to the disc. If the disc is moved up further, the liquid surface leaves it.



$\therefore$ Downward force on disc due to surface tension is

$$F = 2\pi R \cdot S = 2 \times 3.14 \times 0.05 \times S$$

$$\Rightarrow\ 3 \times 10^{-2} = 2 \times 3.14 \times 0.05 \times S$$

$$\Rightarrow\ S = 0.096\ \text{Nm}^{-1}$$

3. Spherical
4. $A = 4\pi r^2 = 4 \times 3.14 \times (0.01)^2 = 1.256 \times 10^{-5}\text{m}^2$

$$\therefore\ U = S \cdot A = \left(0.47\ \frac{\text{J}}{\text{m}^2}\right)(1.256 \times 10^{-5})$$

$$= 5.9 \times 10^{-6}\,\text{J}$$

5. $R = 1\,\text{cm}$ = radius of original drop

r = radius of each of the smaller droplets formed

$n = 1000$

$$n \cdot \frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3 \quad \Rightarrow \quad r = \left(\frac{R^3}{1000}\right)^{\frac{1}{3}} = \frac{R}{10} = 0.1\ \text{cm}$$

Increase in surface energy is

$$\Delta A = n \cdot 4\pi r^2 - 4\pi R^2 = 36\pi \times 10^{-4}\ \text{m}^2$$

$\therefore$ Increase in surface energy is

$$\Delta U = S \cdot \Delta A = 0.075 \times 36 \times 3.14 \times 10^{-4}$$

$$= 0.85 \times 10^{-3}\,\text{J}$$

7. Work done = Increase in surface energy

$$= S.\Delta A = S \cdot (2lx)$$

[There are two surfaces]

$$\therefore\ W = 0.03 \times 2 \times 0.1 \times 10^{-3} = 6 \times 10^{-6}\,\text{J}$$

8. Pressure at a depth of 10 m inside water is

$$P = P_0 + \rho g h = 1.013 \times 10^5 + 10^3 \times 10^1 \times 10$$

$$= 2.013 \times 10^5\ \text{Nm}^{-2}$$

Excess pressure inside air bubble is

$$\Delta P = \frac{2S}{r} = \frac{2 \times 0.072}{0.1 \times 10^{-3}} = 1440\ \text{Nm}^{-2}$$

$\therefore$ Pressure inside the bubble

$$= 2.013 \times 10^5 + 1440$$

$$= 2.0274 \times 10^5\ \text{Nm}^{-2}$$

9. Thickness of wall of A is more, so that it contains more liquid. But thickness of wall is always much smaller compared to radius of the bubble and is of no concern in deciding the pressure inside a bubble.

10. Excess pressure at A: $\Delta P_A = \dfrac{S}{r}$

Excess pressure at B: $\Delta P_B = \dfrac{S}{R}$

$\therefore$ Difference in pressure at A and B is

$$= S\left(\frac{1}{r} - \frac{1}{R}\right) = 0.07 \times \left(\frac{1}{2 \times 10^{-3}} - \frac{1}{4 \times 10^{-3}}\right)$$

$$= 70 \times \frac{1}{4} = 17.5\ \text{Nm}^{-2}.$$

11. $P_0 = \dfrac{4S}{R} \quad \Rightarrow \quad R = \dfrac{4S}{P_0}$

Surface energy = Surface area $\times\ S$

$= 2 \times 4\pi R^2 \times S$ [There are two surfaces]

$$= 8\pi\left(\frac{4S}{P_0}\right)^2 \cdot S = 128\pi\frac{S^3}{P_0^2}$$

15. $h = \dfrac{2S\cos\theta}{r\rho g} = \dfrac{2 \times 0.07 \times \cos 60°}{0.5 \times 10^{-3} \times 10^3 \times 9.8} = 1.43 \times 10^{-2}\ \text{m}$

$$= 1.43\,\text{cm}$$

16. Vertical rise, $h = \dfrac{2S\cos\theta}{r\rho g} = 3.57\ \text{cm}$

$$l = \frac{h}{\sin 30°} = 7.14\ \text{cm}$$

17. $h = \dfrac{2S\cos\theta}{r\rho g} = \dfrac{2 \times 0.5 \times \cos 120°}{1 \times 10^{-3} \times 13.6 \times 10^{3} \times 9.8}$

$= -3.75 \times 10^{-3}$ m $= -3.75$ mm

Negative sign indicates that mercury level falls inside the tube by 3.75 mm

18. Pressure just below mercury surface is $P_0 + \dfrac{2S}{R}$.

Pressure at the same level can also be written as $P_0 + \rho gh$.

$$\therefore \quad P_0 + \frac{2S}{R} = P_0 + \rho gh \quad \Rightarrow \quad R = \frac{2S}{\rho gh}$$

19. Due to viscous forces between different layers.

20. Velocity of top layer of oil in contact with the block is $v = 2$ ms^{-1}.

Velocity of lowest layer touching the incline = 0

$\therefore$ Velocity gradient $= \dfrac{2\,\text{ms}^{-1}}{10^{-3}\,\text{m}} = 2000$ s^{-1}

For constant velocity

$$F_v = mg\sin\theta$$

$$\Rightarrow \quad \eta A\left(\frac{v}{d}\right) = 50 \times 10^{-3} \times 10 \times \frac{1}{2}$$

$$\Rightarrow \quad \eta \cdot (20 \times 10^{-4})(2000) = 0.25$$

$$\Rightarrow \quad \eta = 0.06 \text{ Nsm}^{-2} = 0.6 \text{ poise}$$

21. At terminal speed, acceleration = 0 $\Rightarrow$ $F_v = mg$

22. $v_0 \propto r^2$

If volume is made 8 times, radius doubles

$\left(\because \quad v = \dfrac{4}{3}\pi r^3\right)$

$\therefore$ Terminal speed becomes 4 times.

Worksheet 1

2. The thread will get pulled by liquid on side B. It will become convex towards B and concave towards A.

4. $h = \dfrac{2S}{R\rho g} \quad \Rightarrow \quad h \propto \dfrac{1}{g}$

Inside a satellite, $g_{\text{eff}} = 0$

$\therefore \qquad h = \infty$

For a tube of finite length, water will rise to full length.

6. There are two surfaces

$\therefore \qquad U = 2 \times 0.02 \times 5 = 0.2$ J

7. Density of mixture is $\rho = \dfrac{\text{mass}}{\text{volume}}$

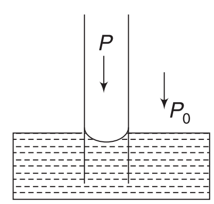
$$\therefore \qquad \rho = \frac{m + m}{\dfrac{m}{0.8} + \dfrac{m}{1.0}} = \frac{2 \times 0.8}{1.8} = 0.89 \text{ gcm}^{-3}$$

$$h = \frac{2S}{r\rho g} \quad \Rightarrow \quad S = \frac{hr\rho g}{2}$$

$$\therefore \qquad S = \frac{(5\text{ cm})(0.1\text{ cm})\left(0.89\,\dfrac{\text{g}}{\text{cm}^3}\right)\left(980\,\dfrac{\text{cm}}{\text{S}^2}\right)}{2}$$

$$= 218 \text{ dyne cm}^{-1}$$

8. $P - \dfrac{2S}{r} = P_0$



$\Rightarrow \quad P - P_0 = \dfrac{2S}{r}$

$= \dfrac{2 \times 0.07}{0.28 \times 10^{-3}} = 500$ Nm^{-2}

9. Work done = Increase in surface energy

or, W = (increase in surface area)S

or, $2 \times 10^{-4} = 2 \times [0.1 \times 0.06 - 0.08 \times 0.0375] \times S$

$\Rightarrow \qquad S = 3.3 \times 10^{-2}$ Nm^{-1}.

10. Vertical rise still remains 50 cm.

$\therefore$ Length of liquid column in tube $= 50\sqrt{2}$ cm

11. As the ring tries to move up the water surface, two surfaces are formed – one touching the inner circumference and the other touching the outer circumference. Extra weight is needed to overcome the surface tension force on the ring.

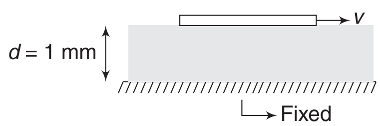
$\therefore \quad S(2\pi r + 2\pi R) = mg$.

$$S = \frac{mg}{2\pi(r+R)} = \frac{7.48 \times 10^{-3} \times 9.8}{2 \times 3.14[8 \times 10^{-2} + 9 \times 10^{-2}]}$$

$$= 0.068 \text{ Nm}^{-1}$$

12. $2lS = mg \Rightarrow l = \frac{mg}{2S}$

13. Velocity gradient $= \frac{v}{d} = \frac{0.07}{0.001} = 70 \text{ s}^{-1}$



$$F = \eta A\left(\frac{v}{d}\right) = 1 \times (100 \times 10^{-4}) \times 70 = 0.7\,\text{N}$$

14. Velocity first increases and then becomes constant. Therefore, slope of *s*-*t* graph increases and then becomes constant.

    Option (*D*) appear to be the best choice.

15. Pressure of air in the tube is higher than atmospheric pressure by

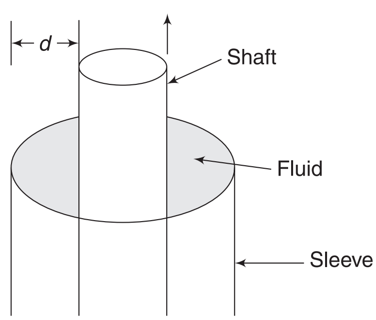
    $$\Delta P = \rho g h$$

    Excess pressure inside bubble is $\Delta P = \frac{4\sigma}{r}$

    The pressure of air inside the bubble and the entire tube is same.

    $$\therefore \quad \frac{4\sigma}{r} = \rho g h \Rightarrow \sigma = \frac{\rho g h r}{4}$$

16. Let gap between the shaft and the sleeve be *d*.



Viscous force

$$F_V = \eta A \frac{v}{d}$$

Where A = area of shaft in contact with fluid.

To maintain constant speed, external force must be equal to F_v in magnitude.

$\therefore \quad F \propto v$

$\therefore$ Force is tripled if v is tripled.

17. Density of metal >> density of air.

    $\therefore$ Let us neglect buoyancy. [Even if you consider buoyancy, the result will remain unchanged]

    Net force on the sphere is zero when it travels at terminal speed (v_0).

    Air friction is $f \propto \pi r^2 \cdot v^2$

    $$\therefore \quad f = k r^2 v^2$$

    At terminal speed, $f = mg$

    $$\Rightarrow \quad k r^2 v_0^2 = \frac{4}{3}\pi r^3 \cdot d \cdot g$$

    $$\Rightarrow \quad v_0 \propto r^{1/2}$$

18. Let radius of new drop be R and that of original drops be r each.

    $$\frac{4}{3}\pi R^3 = 2 \cdot \frac{4}{3}\pi r^3 \Rightarrow R = (2)^{1/3} \cdot r$$

    Terminal speed $v \propto$ (radius)^2

    $$\therefore \quad \frac{v_0}{v} = \frac{R^2}{r^2} = (2)^{2/3} = (4)^{1/3}$$

    $$\therefore \quad v_0 = (4)^{1/3}\, v$$

19. Surface tension does not change

20. Gauge pressures at B and A are

    $$P_B = \frac{2S}{3R}$$

    and $$P_A = P_B + \frac{2S}{R} = \frac{2S}{3R} + \frac{2S}{R} = \frac{8S}{3R}$$

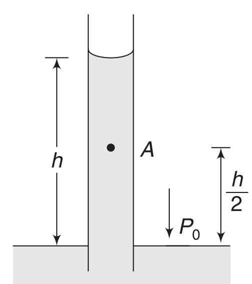
    $$\therefore \quad \frac{P_A}{P_B} = \frac{4}{1}$$

21. $P_A = P_0 - \rho g \frac{h}{2}$

    $$\therefore \quad P_A - P_0 = -\rho g \frac{h}{2}$$

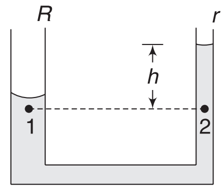
    $$= -\rho g \frac{2S\cos\theta}{2(R\rho g)}$$

    $$= -\frac{S\cos\theta}{R}$$



Negative sign indicates that pressure at A is less than atmospheric pressure.

22. $P_2 = P_1$



$$P_0 - \frac{2S}{r} + \rho gh = P_0 - \frac{2S}{R}$$

$$\Rightarrow \quad h = \frac{2S}{\rho g}\left[\frac{1}{r} - \frac{1}{R}\right]$$

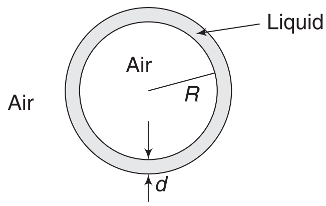
$$= \frac{2 \times 7.5 \times 10^{-2}}{10^3 \times 10}\left[\frac{1}{1 \times 10^{-3}} - \frac{1}{1.5 \times 10^{-3}}\right]$$

$$= 5 \times 10^{-3} \text{ m} = 5 \text{ mm}$$

23. Initially, the meniscus if flat, i.e., radius of curvature is ∞. As the tube is pulled out, R decreases (i.e., surface gets curved). After the tube length becomes sufficient, the radius of curvature becomes $R = r\cos 0° = r$ and remains constant.

24. Excess pressure in bubble, $\Delta P_1 = \dfrac{4S}{R}$

Let the radius of drop formed be r. Conserving the volume of liquid, we get



Volume of liquid in the bubble wall is $4\pi R^2.d$

$$\frac{4}{3}\pi r^3 = 4\pi R^2 \cdot d$$

$$\Rightarrow \quad r = (3R^2 d)^{1/3}$$

Excess pressure inside the drop is

$$\Delta P_2 = \frac{2S}{r} = \frac{2S}{(3R^2 d)^{1/3}}$$

$$\therefore \quad \frac{\Delta P_2}{\Delta P_1} = \frac{2S}{(3R^2 d)^{1/3}} \times \frac{R}{4S} = \left(\frac{R}{24d}\right)^{1/3}$$

25. $ma = mg - F_B$ [as viscous force is zero initially] ...(i)

Maximum velocity is terminal velocity (v_0).

$$6\pi\eta r v_0 + F_B = mg$$

$$\Rightarrow \quad 6\pi\eta r v_0 = mg - F_B = ma \qquad \text{[using (i)]}$$

when velocity is $\dfrac{2}{3}v_0$

$$ma_1 = (mg - F_B) - \left(6\pi\eta r\frac{2}{3}v_0\right)$$

$$= ma - \frac{2}{3}ma$$

$$\therefore \quad a_1 = a - \frac{2}{3}a = \frac{1}{3}a$$

Worksheet 2

1. When air bubble rises, the surrounding hydrostatic pressure decreases. The bubble expands. Excess pressure inside it decreases with increasing radius. Therefore, pressure inside the bubble will decrease.

2. Speed increases and becomes constant after some time.

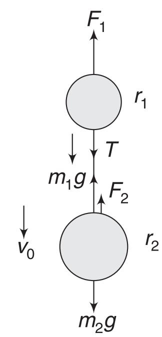
$\therefore$ KE increases and attains a constant value.

If v_0 is terminal speed, then

$$bv_0 + F_B = mg \quad \Rightarrow \quad v_0 = \frac{mg}{b} - \frac{F_B}{b}$$

$\therefore$ Terminal speed depends on m as well as b.

3. When the spheres are moving at terminal speed (v_0), net force on the system = 0



$$\Rightarrow \quad m_1 g + m_2 g = F_1 + F_2$$

Where F_1 and F_2 are viscous forces

$$\therefore \quad \frac{4}{3}\pi r_1^3 \rho g + \frac{4}{3}\pi r_2^3 \rho g = 6\pi\eta r_1 v_0 + 6\pi\eta r_2 v_0$$

$$\Rightarrow \quad v_0 = \frac{2}{9}\frac{\rho g}{\eta}[r_1^2 - r_1 r_2 + r_2^2]$$

$$\left[\because r_1^3 + r_2^3 = (r_1 + r_2)(r_1^2 - r_1 r_2 + r_2^2)\right]$$

Considering the motion of lower ball:

$$T + F_2 = m_2 g \Rightarrow \quad T = m_2 g - F_2$$

$$\Rightarrow \quad T = \frac{4}{3}\pi r_2^3 \rho g - 6\pi\eta r_2 v_0$$

$$= \frac{4}{3}\pi r_2^3 \rho g - 6\pi\eta r_2 \frac{2}{9}\frac{\rho g}{\eta}\left[r_1^2 - r_1 r_2 + r_2^2\right]$$

$$= \frac{4}{3}\pi\rho g\left[r_2^3 - r_2 r_1^2 + r_1 r_2^2 - r_2^3\right]$$

$$= \frac{4}{3}\pi\rho g r_1 r_2 [r_2 - r_1]$$

Sphere of large radius will be at lower position. If $r_1 > r_2$, then

$$T = \frac{4}{3}\pi\rho g r_1 r_2 [r_1 - r_2]$$

4. If pressure of air in the tube is P_0 then

$$P_0 - \frac{2S}{r_1} + \rho gh = P_0 - \frac{2S}{r_2}$$

$$\Rightarrow \quad h = \frac{2S}{\rho g}\left[\frac{1}{r_1} - \frac{1}{r_2}\right] \simeq 0.028 \text{ m}$$

Note that h does not depend on P_0.

5. Terminal speed is given by

$$6\pi\eta a v_0 = F \quad \Rightarrow \quad v_0 = \frac{F}{6\pi\eta a} = k\,\frac{F}{\eta a}.$$

Even if the force change direction, ultimately the sphere will move in the direction of force with speed v_0.

6. Terminal speed in water is given by

$$v_0 = \frac{2}{9}\,\frac{r^2 g}{\eta}(d - \rho)$$

$$= \frac{2 \times (10^{-3})^2 \times 10}{9 \times 10^{-3}}\,[1 \times 10^4 - 10^3] = 20\ \text{ms}^{-1}$$

The sphere speed $v_0 = 20\ \text{ms}^{-1}$ on reaching the surface of water.

$$\therefore \quad \sqrt{2gh} = 20 \quad \Rightarrow \quad h = 20\,\text{m}$$

If η is doubled $v_0 = 10\ \text{ms}^{-1}$

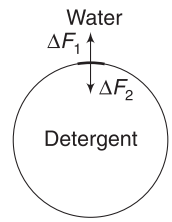
$$\therefore \quad \sqrt{2gh_1} = 10 \quad \Rightarrow \quad h_1 = 5\,\text{m}$$

If g is made $\dfrac{g}{6}$

$$v_0 = \frac{20}{6} = \frac{10}{3}\ \text{ms}^{-1}$$

$$\therefore \quad \sqrt{2.\frac{10}{6}.h_2} = \frac{10}{3} \quad \Rightarrow \quad h_2 = \frac{10}{3}\ \text{m}$$

Worksheet 3



1. Surface tension of water inside the loop decrease due to adding of detergent.

   Force due to pure water on an element of thread (ΔF_1) is larger than force on the element due to detergent mixed water. Therefore, thread becomes circular.

2. Same as in Q.1

3. $$F_B = W + F_{\text{st}}$$

$$(a^2 \cdot h)\rho g = mg + 4a \cdot S$$

$$\Rightarrow \quad h = \frac{mg}{a^2 \rho g} + \frac{4S}{a\rho g}$$

4. The liquid forms two surfaces as the ring is lifted. One surface touches its outer edge and the other surface touches its inner circumference. In extreme case, when the ring is just about to leave the liquid surface, the tangential force of surface tension is vertical.

$$\therefore \quad F = mg + (2\pi r \cdot S) \times 2$$

5. Radius of meniscus in two arms are

$$R_1 = \frac{r_1}{\cos 60°} = 2r_1 \text{ and } R_2 = \frac{r_2}{\cos 60°} = 2r_2$$

Pressure at the bottom of the tube can be written as:

$$P_0 - \frac{2S}{R_1} + \rho g h_1 = P_0 - \frac{2S}{R_2} + \rho g h_2$$

$$\Rightarrow \quad \rho g(h_2 - h_1) = S\left(\frac{1}{r_2} - \frac{1}{r_1}\right)$$

$$\Rightarrow \quad h_2 - h_1 = \frac{S(r_1 - r_2)}{\rho g r_1 r_2}$$

6. Volume of submerged part of plate is

$$V = 20 \times \frac{0.77}{2} \times 0.2 = 1.54\ \text{cm}^3$$

Buoyancy force, $F_B = V\rho g$

$= (1.54\ \text{cm}^3)\ (1\ \text{g/cm}^3)\ (980\ \text{cms}^{-2}) = 1509$ dyne

Water touches the plate along a rectangular periphery having total length $L = 2(20 + 0.2) = 40.4$ cm

$\therefore$ Downward pull due to surface tension is

$$F_{\text{st}} = 73 \times 40.4 = 2949\ \text{dyne}$$

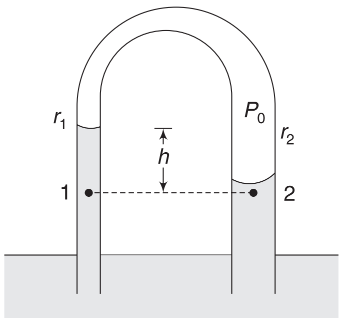
Apparent weight $= (8.2)g + F_{\text{st}} - F_B$

$$= 8.2 \times 980 + 2949 - 1509 = 9476 \text{ dyne}$$

$$= \frac{9476}{980} \text{ gm wt} = 9.67 \text{ gm wt.}$$

7. $$P_1 = P_2$$

$$P_0 - \frac{2S}{r_1} + \rho g h = P_0 - \frac{2S}{r_2}$$



$$\Rightarrow \quad h = \frac{2S}{\rho g}\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$$

$$= \frac{2 \times 0.07}{10^3 \times 9.8}\left(\frac{1}{1.5 \times 10^{-3}} - \frac{1}{3 \times 10^{-3}}\right)$$

$$= 4.73 \times 10^{-3} \text{ m} = 4.73 \text{ mm}$$

8. r = radius of each small drop

$$n \cdot \frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3 \quad \Rightarrow \quad r = n^{-\frac{1}{3}} \cdot R$$

$\therefore$ increase in surface energy is given by

$$\Delta U = [n \cdot 4\pi r^2 - 4\pi R^2]S$$

$$= 4\pi\left[n \cdot n^{-\frac{2}{3}} R^2 - R^2\right]S$$

$$= 4\pi R^2\left[n^{\frac{1}{3}} - 1\right]S$$

9. Refer to Example 15

In extreme case, the lower surface will become hemispherical and its radius will be $r = 0.5$ mm.

Pressure just inside the lower surface will be

$$P_2 = P_0 + \frac{2S}{r} = P_0 + 300 \text{ Nm}^{-2}$$

Pressure below upper surface is

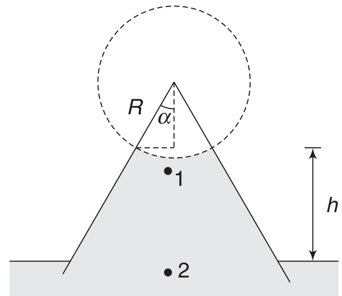
$$P_1 - \frac{2S}{r} = P_0 - 300 \text{ Nm}^{-2}$$

But $P_2 - P_1 = \rho g h$

$$\Rightarrow \quad 600 = 10^3 \times 10 \times h \Rightarrow h = \frac{6}{100} \text{m} = 6 \text{cm.}$$

10. The common wall will be flat as the pressure on boths sides is same.

11. Contact angle is 90°. This means that the tube wall is conical with its apex at the centre of curvature of the spherical meniscus.



From figure: $\frac{r}{R} = \sin\alpha$

$$\Rightarrow \quad R = \frac{r}{\sin\alpha}$$

$$\therefore \quad P_1 = P_0 - \frac{2S}{R}$$

$$P_2 = P_1 + \rho g h = P_0 - \frac{2S}{R} + \rho g h$$

But $\quad P_2 = P_0$

$$\therefore \quad h = \frac{2S}{R\rho g} = \frac{2S\sin\alpha}{r\rho g}$$

12. Capillary rise $= \frac{2S}{R\rho g} = h$

(a) Rise will become $2h$ if $g_{\text{eff}} = \frac{g}{2}$

The elevator must move down with acceleration $a = \frac{g}{2}\ (\downarrow)$

(b) For rise to be $\frac{h}{2}$, $g_{\text{eff}} = 2g$

The elevator must move with an acceleration $g(\uparrow)$

(c) In this case, $g_{\text{eff}} = 0$

$\therefore$ Capillary rise $\to \infty$

Water will rise to the top and its surface will become flat ($R = \infty$).

13. Here, buoyancy is larger than weight. The air bubble goes up. Since density of air (in bubble) is very small compared to density of liquid, the weight of bubble can be neglected. Terminal velocity is attained when buoyancy gets balanced by viscous force.

$$6\pi\eta r v_0 = \frac{4}{3}\pi r^3 \rho \cdot g$$

$$\Rightarrow \quad v_0 = \frac{2r^2\rho g}{9\eta} = \frac{2}{9} \times \frac{(0.4 \times 10^{-3})^2 \times (0.9 \times 10^3) \times 9.8}{0.15}$$

$$= 2.1 \times 10^{-3}\ \text{ms}^{-1} = 0.21\ \text{cms}^{-1}$$

14. (a) $F_v = 6\pi\eta r v = (6 \times 3.14) \times (0.8) \times (1 \times 10^{-3}) \times (0.5 \times 10^{-2})$

$$= 7.5 \times 10^{-5}\ \text{N}$$

(b) Buoyancy; $F_B = \frac{4}{3}\pi r^3 \rho g$

$$= \frac{4}{3} \times 3.14 \times (1 \times 10^{-3})^3 \times 1260 \times 9.8$$

$$= 5.2 \times 10^{-5}\ \text{N}$$

At terminal speed,

$$F_V + F_B = mg$$

$$\Rightarrow \quad 6\pi\eta r v_0 + 5.2 \times 10^{-5} = 50 \times 10^{-6} \times 9.8$$

$$\Rightarrow \quad 6 \times 3.14 \times 0.8 \times 10^{-3} \times v_0 = 43.8 \times 10^{-5}$$

$$\Rightarrow \quad v_0 = 0.029\ \text{ms}^{-1} = 2.9\ \text{cms}^{-1}$$

15. When speed is constant (i.e., $a = 0$), we must have

$$mg = T \text{ and } T = F_v$$

$$\therefore \quad F_V = mg \Rightarrow \eta . A\frac{v}{d} = mg$$

$$\Rightarrow \quad \eta \times 0.1 \times \frac{0.085}{3 \times 10^{-3}} = 0.01 \times 9.8$$

$$\Rightarrow \quad \eta = 0.034\ \text{Nsm}^{-2}$$

16. A particle will be seen suspended if it has not travelled 2 cm in 10 minute, Speed for such particles must be

$$v \le \frac{2\ \text{cm}}{10\ \text{minute}} = \frac{2 \times 10^{-2}}{600} = \frac{1}{3} \times 10^{-4}\ \text{ms}^{-1}$$

Since, particle quickly attains terminal speed, we can assume them to travel at constant terminal speed.

$\therefore$ Particle having terminal speed $\le \frac{1}{3} \times 10^{-4}\ \text{ms}^{-1}$ will remain suspended.

$$v_0 \le \frac{1}{3} \times 10^{-4}$$

$$\frac{2}{9}r^2\frac{g}{\eta}(d - \rho) \le \frac{1}{3} \times 10^{-4}$$

$$\Rightarrow \quad r^2 \le \frac{1}{3} \times 10^{-4} \times \frac{9 \times 10^{-3}}{2 \times 9.8 \times (1.8 - 1) \times 10^3}$$

$$= 0.191 \times 10^{-10}$$

$$\Rightarrow \quad r \le 0.44 \times 10^{-5}\ \text{m}$$

$$\therefore \quad r_{max} = 4.4\ \mu m.$$

$$D_{max} = 2 \times 4.4 = 8.8\ \mu m$$

Chapter 10 Miscellaneous Problems on Chapters 8 and 9

Match the Column

1. (A) When A and C are opened, water flows out through C and pressure above water surface remains atmospheric pressure

(C) Water flows out initially and air in the bottle expands. Pressure of air drops. When pressure just inside the hole at C drops to atmospheric pressure the flows stops.

2. (A) $F_A = (\rho gh)\cdot \pi R^2$ = Force on top surface

(B) $F_B = [\rho g(2h) + 2\rho gh]\pi R^2 = 4\pi\rho gR^2 h$

= Force on bottom surface

(C) Force by A + Force by B

$$= (P_{av})_1(h\cdot 2R) + (P_{av})_2(h\cdot 2R)$$

$$= \left(\frac{\rho gh + \rho g\cdot 2h}{2}\right)(h\cdot 2R)$$

$$+ \left(\frac{\rho g2h + (\rho g\cdot 2h + 2\rho\cdot g\cdot h)}{2}\right)(h\cdot 2R)$$

$$= 9\rho\cdot gRh^2$$

(D) $F_B - F_A = 3\pi\ \rho gR^2 h$

3. (A) In bucket B, the volume of water displaced (= extra water that is there is bucket A) is equal to weight of the block.

(B) Clearly, the block is an additional weight in B.

(C) Weight of the object is more than the water displaced.

4. Pressure varies with depth as: $P = \rho gh$

Pressure increases in horizontal direction opposite to acceleration, as $P = \rho\ell a$

At the right extreme top, the pressure is zero, but it increases as one moves to the left.

Average pressure on $ABFE$ is $\left[\frac{0 + \rho\ell a}{2}\right]$. Force on this surface is

$$\left[\frac{0 + \rho\ell a}{2}\right][\ell^2] = \frac{\ell\rho a}{2}(\ell^2) = \frac{(\rho\ell^3)}{2}a = \frac{mg}{2}$$

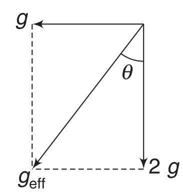
Average pressure on $BDHF$ is $\left[\frac{0 + \rho\ell a}{2}\right]$. Force on this surface is

$$\left[\frac{0 + \rho g\ell}{2}\right][\ell^2] = \frac{\rho g}{2}(\ell^3) = \frac{\rho(\ell)^3}{2}(g) = \frac{mg}{2}$$

Similarly, for the other part.

5. Effective acceleration of free fall in an RF attached to the container is $\sqrt{g^2 + (2g)^2} = \sqrt{5}\,g$.

g_{eff} makes angle θ with vertical, where $\tan\theta = \frac{1}{2}$



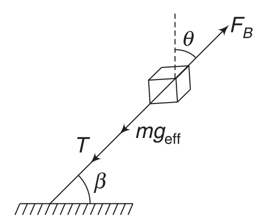
(A) Water surface is perpendicular to g_{eff}.

$$\therefore\quad \alpha = \theta = \tan^{-1}\left(\frac{1}{2}\right)$$

(B) Effective buoyancy is in a direction opposite to g_{eff}. String will be along g_{eff}.

$$\therefore\quad \beta = 90° - \theta$$

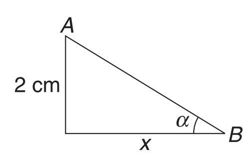
$$\tan\beta = \cot\theta = 2$$



(C) Points A and B lie on a equi-pressure line. This line is parallel to liquid surface.

$$\tan\alpha = \frac{2}{x}$$

$$\Rightarrow\quad \frac{1}{2} = \frac{2}{x} \quad\Rightarrow\quad x = 4\,\text{cm}$$



(D) As explained above, $\phi = \alpha = \tan^{-1}\left(\frac{1}{2}\right)$

6. Speed of efflux $v = \sqrt{2gh}$ it is not dependent on density but depends on g. Range of liquid jet will change if container is accelerated since time of flight will depend on g but v will depend on g_{eff}

Time needed to empty the tank is given by

$$t = \frac{A}{A_o}\sqrt{\frac{2H}{g}}$$

It depends on A_o. It does not depend on density but will change if g_{eff} changes.

8. (A) Volume of displaced liquid $V_{sub} = V \cdot \dfrac{d}{\rho_w}$

Mass water formed due to melting of ice = mass of ice = Vd.

$\therefore$ Volume of water formed due to melting ice is = $V \dfrac{d}{\rho_w}$

$\therefore$ Volume of water formed fits exactly in the space occupied by ice block inside water.

(B) It can be argued as above.

(C) If $\rho > \rho_w$ then volume of submerged part will be less compared to V_{sub} in water. Melting of ice will produce larger volume of water than the volume occupied by ice block in liquid. Level will rise.

(D) It can be argued as above.

Passage-based Problems

Passage 1

2. When air is expelled out of the large tube, pressure inside it drops. This is the reason why the liquid rises in the tube. The hydrometer floats in this sample of liquid

3. For rotational equilibrium it is necessary that the COM lies below the centre of buoyancy.

Passage 2

4. Volume of air striking the bubble wall per second is = vA

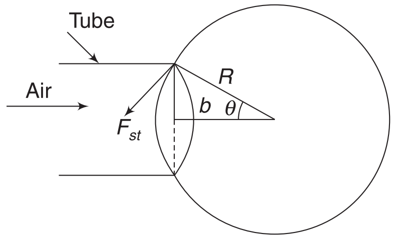
Mass of air striking the wall per second = $vA\rho$

Rate of change of momentum of air particles per second = $(vA\rho)v = A\rho v^2$

$\therefore$ Force $= A\rho v^2$ $[\because A = \pi b^2]$

$= \pi b^2 \rho v^2$

5. Surface tension force on bubble is acting tangentially over the circumference of length $2\pi b$.



Resultant surface tension force is towards right in the diagram.

$$F_{st} = S \cdot 2\pi b . \sin\theta = S \cdot 2\pi b \cdot \frac{b}{R} = 2\pi S \frac{b^2}{R}$$

Bubble detaches when force due to striking air molecules exceed this surface tension force.

$$\Rightarrow \quad \pi b^2 \rho v^2 \geq 2\pi S \cdot \frac{b^2}{R}$$

$$\Rightarrow \quad R \geq \frac{2S}{\rho v^2}$$

Passage 3

Sol. Reading of balance immediately after the sugar lump is put into it is $R_0 = Mg + F_B$

[Water applies buoyancy on sugar lump. The lump pushes the water with the same force in downward direction]

$$\therefore \quad R_0 = Mg + V_{sugar} \cdot \rho g = Mg + \frac{m}{d}\rho g \quad \text{...(i)}$$

At time t_1, the reading is

$$R_1 = \left(M + \frac{m}{2}\right)g + V_{sugar}\,\rho g$$

$$= \left(M + \frac{m}{2}\right)g + \frac{m}{2d}\rho g$$

$$= Mg + \frac{mg}{2} + \frac{1}{2}\frac{m}{d}\rho g \quad \text{...(ii)}$$

$$= Mg + \frac{mg\rho}{d}\left[\frac{1}{2}\frac{d}{\rho} + \frac{1}{2}\right] \quad \text{...(iii)}$$

Since $\dfrac{d}{\rho} > 1$, therefore quantity inside the bracket $\left[= \dfrac{1}{2}\dfrac{d}{\rho} + \dfrac{1}{2}\right]$ in the above equation is greater than 1.

Thus, comparing (i) and (ii) gives: $R_1 > R_0$.

At time t_2, reading is

$$R_2 = (M + m)g$$

$$= Mg + \frac{mg}{2} + \frac{mg}{2} \quad \text{...(iv)}$$

Comparing (ii) and (iv) gives $R_2 > R_1$ $\left[\text{Since } \dfrac{\rho}{d} < 1\right]$

$$\therefore \quad R_2 > R_1 > R_0$$

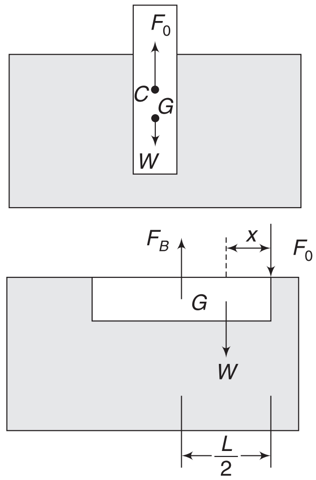
The reading of the balance must have increased continuously.

Passage 4

9. For stable equilibrium, centre of mass must be below the centre of buoyancy (C).

C is at mid-point of submerged part and centre of mass (G) is below it.

When the wood piece is horizontal and complelety submerged, centre of buoyancy shifts to geometrical centre of the wood piece.



Both F_B and F_o have clockwise torque about G. Equilibrium is not possible

For translational and rotational equilibrium, the point of action of weight (W) of the wood piece must be to the left of F_B and F_0, as shown. Hence, (c) is correct.

10. $$F_B = F_0 + W \qquad \text{...(i)}$$

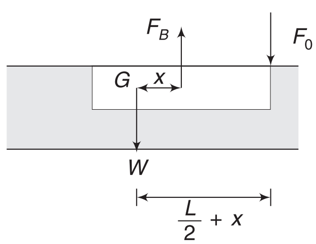
Balancing torque about G gives

$$F_0\left(x + \frac{L}{2}\right) = F_B \cdot x$$

$$\Rightarrow \quad F_0 x + F_0 \frac{L}{2} = AL\rho_0 g \cdot x$$

$$\Rightarrow \quad (AL\rho_0 g - F_0)x = \frac{F_0 L}{2}$$

$$\Rightarrow \quad x = \frac{F_0 L}{2(AL\rho_0 g - F_0)}$$



F_B and F_o have opposite torques about G. Equilibrium is possible.

11. In horizontal position: $W = F_B - F_0 = AL\rho_0 g - F_0$

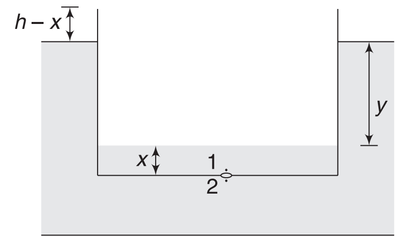
In vertical position: Buoyancy = W

$$\Rightarrow \quad Al\rho_0 g = AL\rho_0 g - F_0$$

$$\Rightarrow \quad l = L - \frac{F_0}{A\rho_0 g}$$

Passage 5

12. Initially, weight of the box is balanced by buoyancy. When water level inside the box becomes x, the box must sink further by x to balance the additional weight of water inside it.



The difference in height (y) between the level of water outside and inside the tank remains unchanged.

$$x + y + (h - x) = H \quad \Rightarrow \quad y = H - h$$

13. Consider two points 1 and 2, which are just above and below the hole respectively. Using Bernoulli's equation:

$$P_1 + \frac{1}{2}\rho v^2 = P_2 + 0$$

$$\Rightarrow \quad P_0 + \rho g x + \frac{1}{2}\rho v^2 = P_0 + \rho g(x + y)$$

$$\Rightarrow \quad v^2 = 2gy = 2g(H - h)$$

$$\therefore \quad v = \sqrt{2g(H - h)}$$

This speed is independent of x.

14. Volume flow rate $= vA = A\sqrt{2g(H - h)}$ = a constant

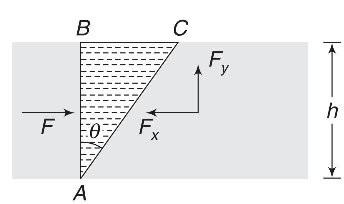
The box will sink when $x = h$

Volume of water inside the box $= abh$

$$\therefore \quad \text{Time needed, } t = \frac{abh}{A\sqrt{2g(H - h)}}$$

Passage 6

Sol. To know the force applied by water on the slant face of the wooden wedge, consider a volume ABC of water inside a water tank. Force applied by surrounding liquid on face AC will be same as force by water on the given wedge.



Horizontal force on AC = Force on AB by surrounding liquid.

$$F_x = F$$

$$F_x = (P_{av})(hb) = \left(\frac{0 + \rho g h}{2}\right)hb = \frac{1}{2}\rho g b h^2$$

Vertical force

$$F_y = \text{weight of water in volume ABC}$$

$$= \frac{1}{2} b\,(h \cdot h \tan\theta)\rho g = \frac{1}{2}\rho g b h^2 \tan\theta$$

15. $N = F_x = \frac{1}{2}\rho g b h^2$

16. $F_y = \frac{1}{2}\rho g b h^2 \tan\theta$

17. Friction on left wall, $f = \mu N(\downarrow)$

In critical case, when the wedge is about to slide up:

$$f + mg = F_y$$

$$\mu \frac{1}{2}\rho g b h^2 + mg = \frac{1}{2}\rho g b h^2 \tan\theta$$

$$\Rightarrow \quad h = \sqrt{\frac{2m}{\rho b(\tan\theta - \mu)}}$$

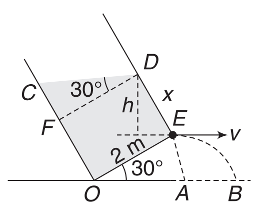
Passage 7

18. Water surface has to remain horizontal, as acceleration due to gravity is downwards.

19. Volume of water in the tank is $8\,\text{m}^3$.

Let $\quad DE = x$

Then $\quad OC = x + 2\tan 30° = x + \frac{2}{\sqrt{3}}$



Volume of water can be expressed in terms of x as:

$$V = 2\ (\text{area of trapezium OCDE})$$

$$= 2 \cdot \frac{1}{2} \cdot 2\left(x + x + \frac{2}{\sqrt{3}}\right) = 2\left(2x + \frac{2}{\sqrt{3}}\right)$$

$$\therefore \quad 2\left(2x + \frac{2}{\sqrt{3}}\right) = 8$$

$$\Rightarrow \quad x + \frac{1}{\sqrt{3}} = 2 \quad \Rightarrow \quad x = 1.42\,\text{m}$$

Vertical height difference between D and E is

$$h = x\cos 30° = 1.42 \times \frac{\sqrt{3}}{2} = 1.23\,\text{m}$$

$\therefore$ Speed of efflux: $v = \sqrt{2gh} = \sqrt{2 \times 10 \times 1.23}$

$$= 4.96\,\text{ms}^{-1}$$

$$AE = 2\sin 30° = 1\,\text{m}$$

$$\therefore \quad \text{Range } AB = v\sqrt{\frac{2(AE)}{g}} = 4.96\sqrt{\frac{2 \times 1}{10}}$$

$$= 2.22\,\text{m}$$

$$\therefore \quad OB = 2\cos 30° + 2.22 = 3.95\,\text{m}$$

20. $\tan\alpha = \frac{a}{g} = \frac{2}{10} = \frac{1}{5}$

Passage 8

Sol. Object is floating while remaining completely submerged. Its density is equal to that of the water.

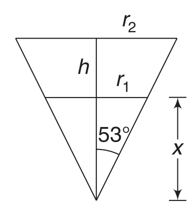
Force on slant surface = Buoyancy [As told in the question]

$\Rightarrow$ Downward force on top surface due to atmospheric pressure = upward force on bottom surface due to water.

$$\Rightarrow \quad P_0 \cdot \pi r_2^2 = (P_0 + \rho g h)\pi r_1^2$$

$$\Rightarrow \quad 10^5\, r_2^2 = (10^5 + 10^3 \times 10 \times 30)\, r_1^2$$

$$\Rightarrow \quad r_2^2 = 4r_1^2 \quad \Rightarrow \quad r_2 = 2r_1 \qquad \text{...(i)}$$



From geometry: $\tan 53° = \frac{r_1}{x} = \frac{r_2}{x + 30}$

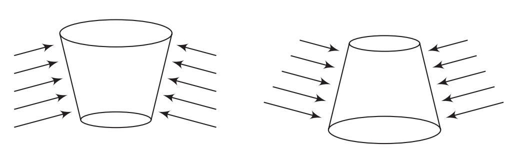
$$\Rightarrow \quad \frac{r_1}{x} = \frac{4}{3} \quad \Rightarrow \quad x = \frac{3}{4}r_1$$

And $\quad \frac{r_1}{x} = \frac{r_2}{x + 30}$

$$\Rightarrow \quad r_1 x + 30 r_1 = r_2 x$$

$$\Rightarrow \quad \frac{3}{4}r_1^2 + 30 r_1 = (2r_1)\left(\frac{3}{4}r_1\right) \quad \Rightarrow \quad \frac{3}{4}r_1 = 30$$

$$\Rightarrow \quad r_1 = 40\,\text{m}$$



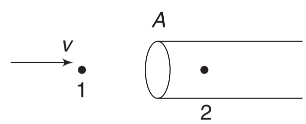
Increasing size of arrow represents increasing pressure.

If the body is inverted, the force on two flat surfaces are no longer equal. Force on slant surface will increase in magnitude. This can be understood simply by observing that pressure increase with depth and the inverted object has more surface area exposed to higher pressure.

Passage 9

23. Just outside the mouth of the pitot tube, at A, the gas has speed and just inside, there is no speed. Therefore, pressure at a point inside the tube (at A) is higher than outside pressure. This is the reason the liquid is pushed down in the right arm.

24. Apply Bernoulli's equation between points just outside the tube and just inside it (at A).



$$P_1 + \frac{1}{2}\rho v^2 = P_2 + 0$$

$$\Rightarrow \qquad \frac{1}{2}\rho v^2 = P_2 - P_1$$

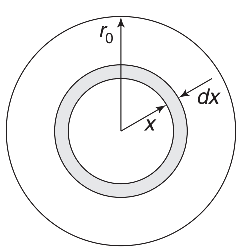
$$\Rightarrow \qquad \frac{1}{2}\rho v^2 = \rho_0 gh \quad \Rightarrow \quad v = \sqrt{\frac{2\rho_0}{\rho}gh}$$

$\therefore$ Volume flow rate is

$$Q = Sv = S\sqrt{\frac{2\rho_0 gh}{\rho}}$$

Passage 10

25. Cross-section of the tube has been shown. Speed of flow along the axis is v_0 and speed of fluid at tube wall is zero.



Speed increases linearly from 0 to v_0 over a distance r_0.

$\therefore$ Velocity gradient $= \dfrac{v_0}{r_0}$

Speed at a distance x from the axis is

$$u = \frac{v_0}{r_0}(r_0 - x) = v_0\left(1 - \frac{x}{r_0}\right)$$

$\therefore$ Volume flow rate through shaded portion of the cross-section is

$$dQ_0 = dA \cdot u = (2\pi x dx)v_0\left(1 - \frac{x}{r_0}\right)$$

$\therefore$ Volume flow rate through entire cross-section is

$$Q_0 = 2\pi v_0 \int_0^{r_0}\left(x - \frac{x^2}{r_0}\right)dx = 2\pi v_0\left[\frac{r_0^2}{2} - \frac{r_0^2}{3}\right]$$

$$= \frac{1}{3}\pi v_0 r_0^2$$

26. Q depends on $\dfrac{\Delta P}{l}$, η and r.

Let $Q = k\left(\dfrac{\Delta P}{l}\right)^a \eta^b r^c$; $\qquad k$ is a constant

Taking dimension;

$$[L^3\ T^{-1}] = \left[\frac{ML^{-1}T^{-2}}{L}\right]^a [ML^{-1}T^{-1}]^b \cdot L^c$$

$$[L^3T^{-1}] = [M^{a+b}\ L^{-2a+c-b}\ T^{-2a-b}]$$

Equating the dimension of M, L and T, we get

$$0 = a + b;\ 3 = -2a - b + c;\ -1 = -2a - b$$

On solving, $a = 1$, $b = -1$ and $c = 4$

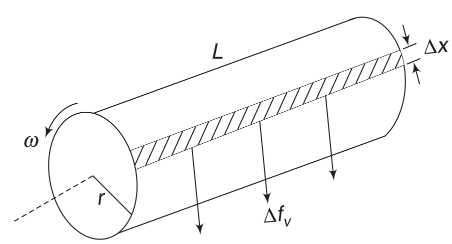
$\therefore \qquad Q = k\dfrac{\Delta P r^4}{l\eta}$ [In fact, this equation is known as Poiseuille's equation]

$\therefore \qquad \Delta P = \dfrac{Q_0 l_0 \eta_0}{k r_0^4}$

Passage 11

Sol. Let angular speed be ω after time t.

Speed of fluid layer in contact with inner cylinder is $v = \omega r$.



Speed of layer in contact with outer cylinder = 0.

$\therefore$ Velocity gradient is $= \dfrac{\omega r - 0}{R - r} = \dfrac{\omega r}{R - r}$

Viscous force on a strip of width Δx on the surface of inner cylinder is

$$\Delta f_v = \eta \Delta A\left(\frac{\omega r}{R - r}\right)$$

$$= \eta L \Delta x\left(\frac{\omega r}{R - r}\right)$$

Torque due to this force (about rotation axis) is

$$\Delta\tau = \Delta f_v \cdot r = \eta L\left(\frac{\omega r^2}{R-r}\right)\Delta x$$

Total torque due to force on all such element is

$$\tau = \sum \Delta\tau = \eta L\left(\frac{\omega r^2}{R-r}\right)\sum \Delta x = \eta L\left(\frac{\omega r^2}{R-r}\right)2\pi r$$

$$= 2\pi\eta L\,\frac{\omega r^3}{(R-r)}$$

$$= k\omega$$

Where, $k = \dfrac{2\pi\eta L r^3}{(R-r)}$

Angular retardation of the cylinder is

$$\alpha = \frac{\tau}{I} = \frac{k\omega}{\frac{1}{2}mr^2} = \frac{2k}{mr^2}\cdot\omega$$

$$\therefore \quad \frac{d\omega}{dt} = -\frac{2k}{mr^2}\cdot\omega$$

$$\Rightarrow \quad \int_{\omega_0}^{\omega}\frac{d\omega}{\omega} = -\frac{2k}{mr^2}\int_0^t dt$$

$$\Rightarrow \quad ln\,\omega - ln\,\omega_0 = -\frac{2k}{mr^2}\cdot t \quad \Rightarrow \quad \frac{\omega}{\omega_0} = e^{-\frac{2k}{mr^2}\cdot t}$$

$$\Rightarrow \quad \omega = \omega_0\, e^{-\frac{2k}{mr^2}\cdot t}$$

Passage 12

29. Viscosity depends on temperature. We must know the temperature at which we are measuring it.

When the ball is not able to attain terminal speed before reaching P, $t_1 > t_2$ [Ball is still accelerated]. We want to ensure that $t_1 = t_2$. For this, the ball should quickly attain terminal speed after it is dropped. A ball of smaller radius has smaller terminal speed and it is more likely to attain terminal speed before reaching P. Highly viscous liquid ensures that terminal speed is attained quickly.

30. Terminal speed, $v_0 = \dfrac{d}{t_0}$

$$\Rightarrow \quad \frac{2}{9}\,\frac{g r_0^2(\sigma-\rho)}{\eta} = \frac{d}{t_0} \qquad \left[\sigma = \frac{m}{\frac{4}{3}\pi r_0^3} = \text{density of ball}\right]$$

$$\Rightarrow \quad \eta = \frac{2}{9}\,\frac{t_0 g r_0^2}{d}\,(\sigma-\rho)$$

$$= \frac{2}{9}\,\frac{t_0 g {r_0}^2}{d}\left[\frac{m}{\frac{4}{3}\pi r_0^3} - \rho\right]$$

Chapter 11 Gravitation

Your Turn

1. $F \propto \frac{1}{r^2}$

   If you put $\frac{r}{2}$ in place of r, value of F will become four times.

4. $$F = \frac{G M_{\text{sun}} M_{\text{earth}}}{r^2}$$

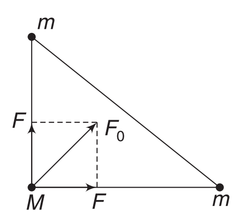
   $$= \frac{6.67 \times 10^{-11} \times 2 \times 10^{30} \times 6 \times 10^{24}}{(1.5 \times 10^{11})^2}$$

   $$= 3.56 \times 10^{22} \text{ N}$$

5(a). $F = \frac{GMm}{x^2}$

   Resultant, $F_0 = \sqrt{F^2 + F^2} = \sqrt{2}\, F$

   $$= \frac{\sqrt{2}\, GMm}{x^2}$$

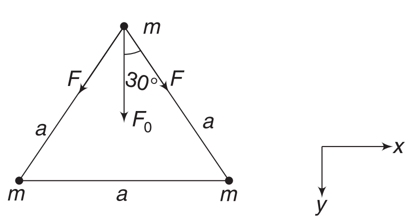


(b) $F = \frac{Gm^2}{a^2}$

   Resultant is along y-direction as x component cancel out.

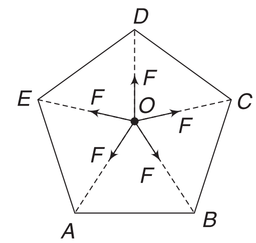
   $$F_0 = 2F\cos 30° = 2 \cdot \frac{Gm^2}{a^2} \cdot \frac{\sqrt{3}}{2}$$

   $$= \frac{\sqrt{3}\, Gm^2}{a^2}$$



6. (i) The five force acting on the particle at O are in direction shown. Each has a magnitude

   $$F = \frac{GmM}{x^2}$$

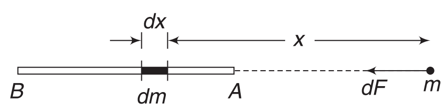


We have studied in the chapter of vector that resultant of five such symmetrical forces is zero.

If particle at A is removed, the force along $\overrightarrow{OA}$ gets removed.

Resultant of other four forces must be F along $\overrightarrow{AO}$.

7. A common mistake is to replace the rod with a point mass at its centre.



There is no such principle in physics. Only *spherical* masses behave like point masses placed at their centres.

Consider an element of length dx as shown.

Mass of this element is $dm = \frac{M}{L}\, dx$

Force applied by this element on m is

$$dF = \frac{G(dm)m}{x^2} = \frac{GMm}{L}\frac{dx}{x^2}$$

$$\therefore \quad F = \frac{GMm}{L}\int_L^{2L} \frac{dx}{x^2}$$

[Note: at A $x = L$, at B $x = 2L$]

$$= -\frac{GMm}{L}\left[\frac{1}{x}\right]_L^{2L} = -\frac{GMm}{L}\left[\frac{1}{2L} - \frac{1}{L}\right] = \frac{GMm}{2L^2}$$

8. Field due to the mass at B and C cancel out. Field at D is only because of mass at A.

   $$AD = \frac{\sqrt{3}}{2}x$$

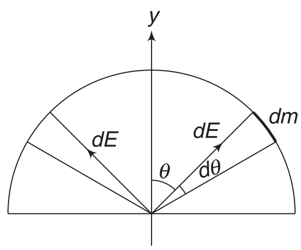
   $$\therefore \quad E = \frac{GM}{\left(\frac{\sqrt{3}}{2}x\right)^2} = \frac{4}{3}\frac{GM}{x^2} \text{ along } DA$$

9. Field due to elements at diametrically opposite ends cancels out.

10. From symmetry, field is along y-direction.

For element shown,

$$dm = \frac{M}{\pi R}(R\,d\theta)$$



$$= \frac{M}{\pi}\,d\theta$$

$$dE = \frac{Gdm}{R^2}$$

$$= \frac{GM}{\pi R^2}\,d\theta$$

Resultant field is

$$E_0 = \int_{-\pi/2}^{\pi/2} dE\cos\theta = \frac{GM}{\pi R^2}\int_{-\pi/2}^{\pi/2}\cos\theta\,d\theta$$

$$= \frac{2GM}{\pi R^2}$$

12. $$E = \frac{GM}{R^2} = \frac{6.67\times10^{-11}\times7.36\times10^{22}}{(1.74\times10^6)^2}$$

$$= 1.62\,\text{N kg}^{-1}$$

13. On surface, $E_{\text{surface}} = \dfrac{GM}{R^2}$

At distance r from centre (inside Earth)

$$E = \frac{GM}{R^3}\cdot r$$

$$E = \frac{1}{2}E_{\text{surface}}$$

$$\therefore\quad \frac{GM}{R^3}\cdot r = \frac{1}{2}\frac{GM}{R^2} \quad\Rightarrow\quad r = \frac{R}{2}$$

Depth of such a point from the surface of Earth is

$$h = R - r = \frac{R}{2}$$

14. Man on the equator will feel no weight if the effective free fall acceleration (g') becomes zero, putting $g' = 0$ in equation (8) gives

$$\omega^2 R = g \quad\Rightarrow\quad \omega = \sqrt{\frac{g}{R}} \quad\Rightarrow\quad \frac{2\pi}{T} = \sqrt{\frac{g}{R}}$$

$$T = 2\pi\sqrt{\frac{R}{g}}$$

Note that time period of rotation of the Earth is the length of the day.

Putting $R = 6.4\times10^6$ m and $g = 9.8\,\text{ms}^{-2}$ in the above expression gives $T \simeq 84.6$ minutes.

15. Value of acceleration due to gravity at a depth d is $g\left(1-\dfrac{d}{R}\right)$ and its value at height h is $\dfrac{g}{\left(1+\dfrac{h}{R}\right)^2}$.

Given: $$g\left(1-\frac{d}{R}\right) = \frac{g}{\left(1+\dfrac{R}{R}\right)^2}$$

$$\Rightarrow\quad d = \frac{3}{4}R$$

16. (a) $$\frac{g}{\left(1+\dfrac{h}{R}\right)^2} = 0.64g \quad\Rightarrow\quad 1+\frac{h}{R} = \frac{5}{4}$$

$$\Rightarrow\quad h = \frac{R}{4} = 1600\ \text{km}$$

(b) For small height

$$g' \simeq g\left(1-\frac{2h}{R}\right)$$

$$\therefore\quad g(1-0.0036) = g\left(1-\frac{2h}{R}\right)$$

$$\Rightarrow\quad \frac{2h}{R} = 0.0036$$

$$\Rightarrow\quad h = \frac{6400\times0.0036}{2} = 11.5\ \text{km}$$

17. At height h above a pole, $g' = \dfrac{g}{\left(1+\dfrac{h}{R}\right)^2}$

Effective acceleration of free fall at equator is

$$g_{\text{eq}} = g - \omega^2 R$$

As per the question

$$\frac{g}{\left(1+\dfrac{h}{R}\right)^2} = g - \omega^2 R$$

$$\Rightarrow\quad g\left[1-\frac{2h}{R}\right] = g - \omega^2 R$$

[Since effect of rotation is small at equator, the value of h must also be small]

Substituting data in above expression gives $h \simeq 11$ km.

Note: Just by going through Example 8 and Question 16 one can guess the answer.

18. $$g = G\frac{M}{R^2} = G\cdot\frac{\rho\cdot\frac{4}{3}\pi R^3}{R^2} = \frac{4}{3}\pi G\,\rho\cdot R$$

$\therefore \quad g \propto \rho\cdot R$

Since ρ and R for planet are twice that of Earth, value of g on the surface is four times that on Earth.

19. $W_g = -\Delta U = U_i - U_f$

$$= \left(-\frac{Gm\cdot 2m}{a} - \frac{Gm\cdot 3m}{a} - \frac{G\,2m\cdot 3m}{\sqrt{2}\;a}\right) -$$

$$\left(-\frac{Gm\cdot 2m}{a} - \frac{Gm\cdot 3m}{a} - \frac{G\,2m\,.\,3m}{a}\right)$$

$$= 6\left(1 - \frac{1}{\sqrt{2}}\right)\frac{Gm^2}{a}$$

20. Both must have equal speed in order to conserve momentum. Let their speed be r.

$$K_i + U_i = K_f + U_f$$

$$0 + 0 = \frac{1}{2}mv^2 + \frac{1}{2}mv^2 - \frac{Gm\cdot m}{r}$$

$$\Rightarrow \qquad v = \sqrt{\frac{Gm}{r}}$$

21. $$K_i + U_i = K_f + U_f$$

$$0 - \frac{GmM}{2R} = \frac{1}{2}\,mv^2 - \frac{GmM}{R}$$

[Initial distance of the body from the centre of the Earth is R]

$$\therefore \qquad v = \sqrt{\frac{GM}{R}} = \sqrt{\frac{GM}{R^2}\cdot R} = \sqrt{gR}$$

22. (a) Total mechanical energy $= -E_0 + 2E_0 = E_0$.

This is positive. Hence the rock piece will escape out of the gravitational pull.

(b) Mechanical energy is conserved.

$$K + U = E_0$$

$$\Rightarrow \quad 1.6E_0 + U = E_0 \quad \Rightarrow \quad U = -0.4E_0$$

(c) At infinity, $U = 0$

$\therefore K =$ total energy $= E_0$

23. $$v = \sqrt{2gR}$$

$$\therefore \qquad \frac{v_A}{v_B} = \sqrt{\frac{2\sigma R}{2(2\sigma)(2R)}} = \frac{1}{2}$$

24. PE on surface $= -\dfrac{GMm}{R}$

KE needed to escape $= \dfrac{GMm}{R}$

KE imparted $= \dfrac{1}{2}\dfrac{GMm}{R}$

$\therefore$ Total energy of rocket $= -\dfrac{GMm}{R} + \dfrac{1}{2}\dfrac{GMm}{R}$

$$= -\frac{1}{2}\frac{GMm}{R}$$

When at maximum distance (x) from the centre, its energy is purely potential energy.

$$\therefore \quad -\frac{GMm}{x} = -\frac{1}{2}\frac{GMm}{R} \quad \Rightarrow x = 2R$$

25. (a) Potential at the surface,

$$V_{\text{surf}} = -\frac{GM}{R} = -\frac{GM}{R^2}\cdot R = -gR$$

$$= -9.8 \times 6.4 \times 10^6$$

$$= -6.27 \times 10^7\ \text{J kg}^{-1}$$

At a distance x from the centre (outside the Earth)

$$V = -\frac{GM}{x}$$

Since $\left|-1 \times 10^7\right| < \left|-6.27 \times 10^7\right|$

$\therefore \qquad x > R$

(b) $U = mV = -5 \times 6.27 \times 10^7 = -3.14 \times 10^8\,\text{J}$

27. Potential due to a small mass dM on the ring is

$$dV = -\frac{GdM}{R}$$

All elemental masses are at a distance R from the centre of the ring.

$$\therefore \quad V = \int dV = -\frac{G}{R}\int dM = -\frac{GM}{R}$$

28. Force on a satellite always acts towards the centre of the Earth. If the path has 3-3 as diameter, this force is not towards centre of the circular path.

29. $U = -\dfrac{GMm}{r} \quad \Rightarrow \quad U \propto -\dfrac{1}{r}$

$\therefore$ U will have decreased magnitude with negative sign if r is increased.

$\therefore$ U increases

$$k = \frac{GMm}{2r} \Rightarrow k \text{ will decrease}$$

30. $$\left(\frac{T_2}{T_1}\right)^2 = \left(\frac{r_2}{r_1}\right)^3$$

$$\Rightarrow \quad T_2 = \left(\frac{r_2}{r_1}\right)^{3/2} \cdot T_1 = \left(\frac{3.5R}{7R}\right)^{3/2} \cdot (24\,\text{hr})$$

$$\therefore \quad T_2 = 8.48\,\text{hr}$$

31. (a) $$v_0 = \sqrt{\frac{GM}{r}} = \sqrt{\frac{GM}{R+h}}$$

Given $v_0 = \dfrac{v_e}{2}$

$$\Rightarrow \sqrt{\frac{GM}{R+h}} = \frac{1}{2}\sqrt{\frac{2GM}{R}} \Rightarrow R + h = 2R$$

$$\Rightarrow \quad h = R$$

(b) Energy conservation gives

$$\frac{1}{2}mv^2 + \left(-\frac{GMm}{R}\right) = 0 + \left(-\frac{GMm}{2R}\right)$$

$$\Rightarrow \quad v^2 = \frac{GM}{R} \Rightarrow v = \sqrt{gR}$$

32. Energy given = (Energy of the satellite in the orbit) – (Energy of the object when it was on the Earth)

$$= -\frac{GMm}{2(2R)} - \left(-\frac{GMm}{R}\right) = \frac{3}{4}\frac{GMm}{R}$$

$$= \frac{3}{4}mgR = \frac{3}{4} \times 2000 \times 10 \times 6.4 \times 10^6$$

$$= 9.6 \times 10^{10}\,\text{J}$$

33. (a) Velocity for circular orbit, $v_0 = \sqrt{\dfrac{GM}{r}} = \sqrt{\dfrac{GM}{2R}}$

Velocity for escape (Refer to example 22)

$$v_1 = \sqrt{2}\,v_0 = \sqrt{\frac{GM}{R}}$$

Since $v = \sqrt{\dfrac{2}{3}\dfrac{GM}{R}}$ is less than v_1, the satellite will not escape.

(b) Speed $v > v_0$

$\therefore$ Path is elliptical.

34. Only force on planet is gravitational pull of the sun. It has no torque about the centre of the sun. Hence, angular momentum is conserved.

Force is not perpendicular to the velocity of the planet (except at perihelion and aphelion). Thus, work done by force is not zero in a finite displacement. Thus, KE keeps changing.

35. Angular momentum conservation

$$mv_1r_1 = m(v_2 \sin\theta)r_2 \Rightarrow v_2 = \frac{v_1r_1}{r_2\sin\theta}$$

36. (a) For P_2, $a = \dfrac{r_1 + r_2}{2} = 1.4 \times 10^8$ km

$$\left(\frac{T_2}{T_1}\right)^2 = \left(\frac{1.4 \times 10^8}{1 \times 10^8}\right)^3 \Rightarrow T_2 = (2\,\text{yr})(1.4)^{3/2}$$

$$\simeq 3.3\,\text{years}$$

(b) Speed of P_2 is larger

(c) PE of both P_1 and P_2 are same at P but KE of P_2 is larger.

$\therefore$ $E_2 > E_1$

Worksheet 1

1. Gravitational force is conservative. Work done by it does not depend on path.
2. Gravitational force provides the necessary centripetal force
3. In absence of gravity, the gas molecule will escape.
4. Due to rotation, acceleration of free fall is $g' = g - \omega^2 R$. If ω becomes zero g' increases.
5. $g' = g - \omega^2 R$ at equator but there is no effect of rotation at poles
6. $g = \dfrac{GM}{R^2} \Rightarrow g = \dfrac{G\cdot\frac{4}{3}\pi R^3\rho}{R^2}$

$$\Rightarrow \qquad \rho = \frac{3g}{4\pi GR}$$

7. $g = \dfrac{G\cdot\frac{4}{3}\pi R^3\rho}{R^2} \Rightarrow g \propto R\rho$

$$\frac{g_m}{g_e} = \frac{R_m\rho_m}{R_e\rho_e} \Rightarrow R_m = R_e\left(\frac{g_m}{g_e}\right)\left(\frac{\rho_e}{\rho_m}\right)$$

$$= R_e\left(\frac{1}{6}\right)\left(\frac{5}{3}\right) = \frac{5}{18}R_e$$

8. $g = \dfrac{GM}{R^2} = \dfrac{4GM}{D^2}$ [D = diameter]

$$\therefore \quad g \propto \frac{M}{D^2} \Rightarrow \frac{g_p}{g_e} = \frac{M_p}{D_p^2}\cdot\frac{D_e^2}{M_e} = \left(\frac{M_p}{M_e}\right)\left(\frac{D_e}{D_p}\right)^2$$

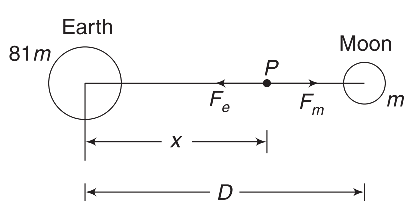
$$= \frac{1}{80}\times 2^2 = \frac{1}{20}$$

$$\therefore \quad g_P = \frac{g_e}{20} = \frac{9.8}{20} = 0.49\,\text{ms}^{-2}$$

9. Let P be the point where force (on a body of mass M) due to Earth and the moon are equal and opposite.

$$\frac{G(81m)M}{x^2} = \frac{Gm\cdot M}{(D-x)^2}$$

$$\Rightarrow \qquad \frac{D-x}{x} = \frac{1}{9} \Rightarrow x = \frac{9D}{10}$$



10. The whole mass of Earth can be assumed to be concentrated at its centre. Thus, shrinking of Earth will not make any difference.

11. At height h, $g_1 = g\left[1 - \dfrac{2h}{R}\right]$

Decrease in g is $\Delta g_1 = \left(\dfrac{2g}{R}\right)h$ [This is given as 1%]

At depth h, $g_2 = g\left[1 - \dfrac{h}{R}\right]$

Decrease in g is $\Delta g_2 = \left(\dfrac{g}{R}\right)h$

Obviously, the answer is 0.5%

12. $$g\left[1 - \frac{h}{R}\right] = \frac{1}{2}\frac{g}{\left[1 + \frac{1600}{6400}\right]^2}$$

$$1 - \frac{h}{6400} = \frac{8}{25} \Rightarrow h = \frac{17}{25}\times 6400\,\text{km}$$

$$= 4352\,\text{km}$$

13. From Example 34

$$g' = g - R\omega^2\cos^2\phi = g - \frac{R\omega^2}{4}$$

14. $g = \dfrac{GM}{R^2} \Rightarrow \dfrac{\Delta g}{g} = 1\cdot\dfrac{\Delta M}{M} - 2\cdot\dfrac{\Delta R}{R}$

$$\therefore \quad \frac{\Delta g}{g}\times 100 = \frac{\Delta M}{M}\times 100 - 2\,\frac{\Delta R}{R}\times 100$$

$$= (-1\%) - 2(-1\%) = +1\%$$

15. $v = \sqrt{2\dfrac{GM}{R}} = \sqrt{\dfrac{2G\,\frac{4}{3}\pi R^3\rho}{R}}$

$$\therefore \quad v \propto R$$

16. If v is escape speed, then $\dfrac{1}{2}mv^2 - \dfrac{GMm}{R} = 0$

$$\Rightarrow \qquad \frac{1}{2}mv^2 = \frac{GMm}{R} \qquad \text{...(i)}$$

Let speed at ∞ be u when the body is projected with speed $2v$.

$$\frac{1}{2}mu^2 = \frac{1}{2}m(2v)^2 - \frac{GMm}{R}$$

$$\Rightarrow \qquad \frac{1}{2}mu^2 = \frac{4}{2}mv^2 - \frac{1}{2}mv^2 \qquad \text{[Using (i)]}$$

$$\Rightarrow \qquad u = \sqrt{3}\,v$$

17. Given, $U = -mgR_e \Rightarrow -\dfrac{GMm}{R_e} = -mgR_e$

$$\therefore \quad \text{At height } h = R_e,\ U_h = -\frac{GMm}{2R_e} = -\frac{1}{2}mgR_e$$

18. Switching off the gravity means that there will be no centripetal force on the satellite. It will move along a straight line.

19. The satellite will just manage to escape if its speed is increased so as to make its total energy zero.

$$\frac{1}{2}mv^2 - \frac{GMm}{r} = 0$$

$$\Rightarrow \qquad v = \sqrt{\frac{2GM}{r}} = \sqrt{2}\,v_0 = 1.414\,v_0$$

Where $v_0 = \sqrt{\frac{GM}{r}}$ is the orbital speed of the satellite.

Percentage increase in speed needed is 41.4%

20. Semi-major axis of elliptical path, $a = \frac{r_1 + r_2}{2}$

$$\therefore \qquad T^2 \propto a^3 \Rightarrow T \propto a^{3/2}$$

$$\therefore \qquad T \propto (r_1 + r_2)^{3/2}$$

21. The ball is moving with the satellite. After its release, due to its inertia, it will have the same velocity as the satellite. It will continue to revolve around the planet under gravitational pull of the planet.

23. 1 lunar month = time period of revolution of the moon.

$$\therefore \quad \left(\frac{T}{1\text{ lunar month}}\right)^2 = \left(\frac{r/2}{r}\right)^3$$

$$\Rightarrow \quad T = \left(\frac{1}{8}\right)^{\frac{1}{2}} \text{lunar month} = 2^{-3/2} \text{ lunar month}$$

24. When a particle falls from ∞, its KE on reaching the surface of Earth is

$$k_1 = \text{Loss in } PE = 0 - \left(-\frac{GMm}{R}\right) = \frac{GMm}{R} \qquad \text{...(i)}$$

When it falls from a height $h = 10R$, its KE on reaching Earth is

$$k_2 = \text{Loss in } PE = -\frac{GMm}{11R} - \left(-\frac{GMm}{R}\right)$$

$$= \frac{10}{11}\frac{GMm}{R} \qquad \text{...(ii)}$$

$$\therefore \qquad \frac{k_1}{k_2} = \frac{11}{10} \Rightarrow \frac{\frac{1}{2}mv_1^2}{\frac{1}{2}mv_2^2} = \frac{11}{10}$$

$$\Rightarrow \qquad \frac{v_1}{v_2} = \sqrt{\frac{11}{10}}$$

25. Both have same speed, as they are in the same orbit. After collision, the combined mass will have zero velocity to conserve momentum.

$$\therefore \quad \text{Total energy} = PE = -\frac{2GMm}{r}$$

26. Volume of solid sphere $= \frac{4}{3}\pi R^3 = V$

Volume of removed part $= \frac{4}{3}\pi\left(\frac{R}{2}\right)^3 = \frac{V}{8}$

$\therefore$ Mass of removed part $= \frac{M}{8}$

Force on m = Force due to solid sphere – Force due to mass that was present inside the cavity region

$$= \frac{GMm}{(2R)^2} - \frac{G\frac{M}{8}m}{(2.5R)^2} = \frac{23}{100}\frac{GMm}{R^2}$$

27. Conservation of angular momentum gives

$$\frac{mv}{4}\cdot r = m(v\sin 30°)R.$$

$$\Rightarrow \qquad r = 2R$$

$\therefore$ Height attained, $h = 2R - R = R$.

28. $m\omega^2 R = \frac{GMm}{R^n} \quad \Rightarrow \quad \omega = \left(\frac{GM}{R^{n+1}}\right)^{\frac{1}{2}}$

$$\therefore \quad \omega \propto \frac{1}{R^{\left(\frac{n+1}{2}\right)}} \quad \Rightarrow \quad T \propto R^{\left(\frac{n+1}{2}\right)}$$

29. $T \propto r^{3/2}$

$$\therefore \quad \frac{\Delta T}{T} = \frac{3}{2}\frac{\Delta r}{r} \quad \Rightarrow \quad \frac{\Delta T}{T} \times 100 = \frac{3}{2}\left(\frac{\Delta r}{r} \times 100\right)$$

$$= \frac{3}{2}(1\%) = 1.5\%$$

30. $$\frac{T}{365\,\text{days}} = \left(\frac{r/2}{r}\right)^{3/2}$$

$$\therefore \qquad T = \frac{365}{2\sqrt{2}}\text{ days} = 129\,\text{days}$$

31. $$\frac{g}{2} = g - \omega^2 R$$

$$\Rightarrow \quad \omega^2 R = \frac{g}{2} \quad \Rightarrow \quad \omega = \sqrt{\frac{g}{2R}}$$

Velocity of a point on the equator is

$$v = \omega R = \sqrt{\frac{gR}{2}}$$

$$v_e = \sqrt{2gR} = 2v$$

32. $v_e = \sqrt{\frac{2GM}{R}}$

Let speed of the body at the centre of the tunnel be v. From conservation of energy, we get:

$$\frac{1}{2}mv^2 - \frac{3}{2}\frac{GMm}{R} = -\frac{GMm}{R}$$

$$\left[\text{Potential at the centre} = -\frac{3}{2}\frac{GM}{R}\right]$$

$$\Rightarrow \quad v^2 = \frac{GM}{R} \quad \Rightarrow \quad v = \sqrt{\frac{GM}{R}} = \frac{v_e}{\sqrt{2}}$$

33. $a_g = \dfrac{GM}{R^2} \quad \Rightarrow \quad a_g \propto \dfrac{1}{R^2}$

34. Time period of satellite = 24 h

Earth and the satellite are orbiting in opposite directions. Both will rotate by 180° (in opposite direction) in 12 hours. The satellite will once again be above the same point on Earth.

35. From momentum conservation, we can find the speed of the fragment of mass $4M$.

$$5Mv_0 = -Mv_0 + 4Mv \quad \Rightarrow \quad v = \frac{3}{2}v_0$$

If speed of a satellite is made more than $\sqrt{2}$ times its orbital speed, it will escape.

36. Angular speed of Earth doubles. This implies that its time period becomes half (= 12 h).

$$T^2 \propto r^3$$

$$\therefore \quad \left(\frac{T/2}{T}\right)^2 = \left(\frac{R}{r}\right)^3 \quad \Rightarrow \quad R = \frac{r}{(4)^{1/3}}$$

37. Areal velocity $\dfrac{dA}{dt} = \dfrac{L}{2m} = \dfrac{mv_0r}{2m} = \dfrac{1}{2}v_0r$

$$= \frac{1}{2}\sqrt{\frac{GM}{r}}\cdot r = \frac{1}{2}\sqrt{GMr}$$

$\therefore$ areal velocity $\propto \sqrt{r}$

$\therefore$ Required ratio $= \sqrt{\dfrac{R}{4R}} = \dfrac{1}{2}$

Worksheet 2

1. If a particle is inside the sphere

$$F \propto r$$

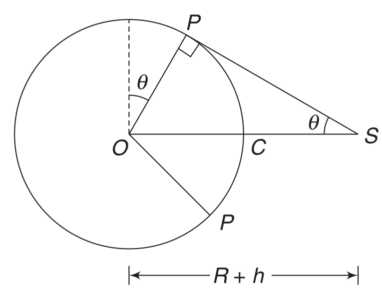
$\therefore$ (A) is correct

If a particle is outside the sphere $F \propto \dfrac{1}{r^2}$

$\therefore$ (B) is correct

3. S is satellite. Any signal sent from it can reach up to a point P on the surface of Earth. Colatitude at P is θ.

$$\sin\theta = \frac{R}{R+h}$$



Area of the spherical cap PCP is given by

$$A = 2\pi R^2[1 - \cos(90 - \theta)]$$
$$= 2\pi R^2[1 - \sin\theta]$$

The area that cannot be captured is

$$A' = 4\pi R^2 - 2\pi R^2\,[1 - \sin\theta] = 2\pi R^2\,[1 + \sin\theta]$$

4. Due to air resistance, the mechanical energy of the satellite decreases. But its KE increases, as orbital speed increases for smaller radius of path.

5. A satellite that is not above the equator cannot remain continuously over a point on Earth. Similarly, a satellite that is rotating opposite to the Earth's spin cannot remain above a point. A communication satellite, must always remain above a particular point.

6. Angular momentum is directed perpendicular to the plane of rotation and is fixed in magnitude and direction.

7. Centre of the Earth is focus of any path of satellite. Such a path will definitely pass over the equator. The smallest time period is for a satellite, which is near to the surface of Earth.

$$T_{\min} = 2\pi\sqrt{\frac{R}{g}}$$

8. (A) $\dfrac{T_1}{T_2} = \left(\dfrac{R}{4R}\right)^{3/2} = \dfrac{1}{8}$

(B) $\dfrac{v_1}{v_2} = \sqrt{\dfrac{4R}{R}} = \dfrac{2}{1}$

(C) $L = mvr = m\sqrt{\dfrac{GM}{r}} \cdot r = m\sqrt{GMr}$

$\therefore \quad L \propto \sqrt{r}$

$\therefore \quad \dfrac{L_1}{L_2} = \sqrt{\dfrac{1}{4}} = \dfrac{1}{2}$

(D) $\omega = \dfrac{2\pi}{T}$

$\Rightarrow \quad \dfrac{\omega_1}{\omega_2} = \dfrac{T_2}{T_1} = \dfrac{8}{1}$

9. Along ACB, the planet is relatively farther (from sun) as compared to its distance when it travels along BDA. It means along ACB the planet is slower.

   Time needed in ACB > time needed in BDA

   Since planet is bound to the sun, its total energy is negative.

   $\therefore \quad |U| > k$ at all points.

Worksheet 3

1. Consider the object (mass = m) at a distance x from the centre inside the tunnel. For it to experience no acceleration, net force on it (in RF of the planet) must be zero.

   $\Rightarrow$ Gravitational force = centrifugal force

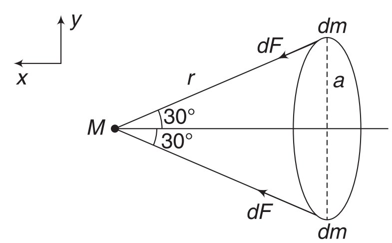
   $\Rightarrow \quad mg' = m\omega^2 x$

   $\Rightarrow \quad g' = \omega^2 x$

   $\Rightarrow \quad \left(\dfrac{GM}{R^3}\right)x = \omega^2 x \quad \Rightarrow \quad \omega = \sqrt{\dfrac{GM}{R^3}}$

   $\Rightarrow \quad \omega = \sqrt{\dfrac{G \cdot \frac{4}{3}\pi R^3 \cdot \rho}{R^3}} = \sqrt{\dfrac{4}{3}\pi G\rho}$

2. Force applied by the ring on the sphere = force applied by the sphere on the ring. For finding force on the ring due to the sphere, we can replace the latter by a point mass placed at its centre.



   Consider two small masses, dm each, on the ring at diametrically opposite ends. Force on each element due to M is

   $$dF = \frac{GMdm}{r^2}$$

   Components of the two forces along x-direction will add and their components perpendicular to it will cancel out

   $\therefore$ Total force on the ring

   $$F = \int dF \cos 30° = \frac{GM}{r^2}\cos 30° \int dm$$

   [Note that r and cos 30° are same for every element selected on the ring]

   $$\therefore \quad F = \frac{GM}{r^2}\cos 30° \cdot m$$

   $$= \frac{GMm}{(a^2 + 3a^2)} \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{3}\ GMm}{8a^2}$$

3. The equation $\Delta U = mgh$ assumes a constant value of g and therefore, it gives a value of ΔU that is larger than the general value (i.e., correct value).

$$\Delta U_{\text{correct}} = -\frac{GMm}{R+h} - \left(-\frac{GMm}{R}\right)$$

$$= GMm\left[\frac{1}{R} - \frac{1}{R+h}\right] = \frac{GMmh}{R(R+h)}$$

According to the problem,

$$\Delta U = 1.010\ (\Delta U_{\text{correct}})$$

$$\Rightarrow \quad mgh = 1.010 \cdot \frac{GMmh}{R(R+h)}$$

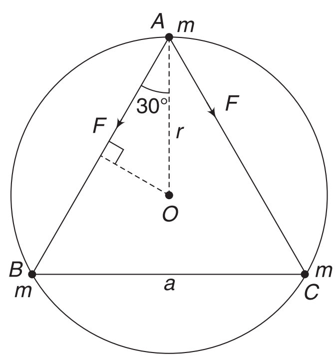
$$\Rightarrow \quad g = 1.010\ g\left(\frac{R}{R+h}\right) \Rightarrow \frac{R+h}{R} = 1.010$$

$$\Rightarrow \quad 1 + \frac{h}{R} = 1.010 \qquad \Rightarrow \frac{h}{R} = 0.010$$

$$\Rightarrow \quad h = 0.010R = 0.010 \times 6400\,\text{km} = 64\,\text{km}$$

4. All the stars move in a common circle of radius r about the centroid O.

$$r = \frac{a}{2\sec 30°} = \frac{a}{\sqrt{3}}$$



Force on the star at A is

$$F_A = 2F\cos 30° \quad (\text{along } AO)$$

$$= 2\ \frac{Gm \cdot m}{a^2} \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{3}\,Gm^2}{a^2}$$

This force provides the necessary centripetal force to the star A.

$$\Rightarrow \quad \frac{mv^2}{r} = \frac{\sqrt{3}\,Gm^2}{a^2} \quad \Rightarrow \quad \frac{v^2}{\frac{a}{\sqrt{3}}} = \frac{\sqrt{3}\,Gm}{a^2}$$

$$\Rightarrow \quad v = \sqrt{\frac{GM}{a}}$$

Mechanical energy of the system

$$E = U + K$$

$$= -\frac{3Gm \cdot m}{a} + 3 \cdot \left(\frac{1}{2}\,mv^2\right)$$

$$= -\frac{3Gm^2}{a} + \frac{3}{2}m\left(\frac{Gm}{a}\right) = -\frac{3}{2}\,\frac{Gm^2}{a}$$

5. (a) Acceleration due to gravity has the same value as gravitational field intensity.

At $r = R$, field is contributed by the core only. The solid part can be divided into many shells and each shell creates zero field at the point (as the point lies inside each shell).

$$\therefore \qquad g_1 = \frac{GM}{R^2}$$

(b) In this case, the entire mass can be assumed to be at the centre.

Volume of core $= \frac{4}{3}\pi R^3 = V(\text{say})$

Volume of solid part $= \frac{4}{3}\pi(3R)^3 - \frac{4}{3}\pi R^3$ $= 8V$.

$\therefore$ Mass of solid part $= 4M$

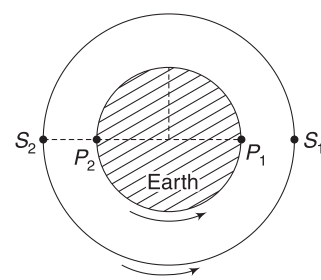
[$\because$ density is half that of core]

Total mass $= 5M$

$$\therefore \qquad g_2 = \frac{G5M}{(3R)^2}$$

$$\Rightarrow \qquad g_2 = \frac{5}{9}\,\frac{GM}{R^2}$$

6. P_1 is the city and S_1 is the location of the satellite at given time.



In the next 8 hours, the satellite will make one full rotation and will be back to S_1. But the city (P) will have rotated by 120° during this period $\left(\frac{8}{24} \times 360 = 120\right)$.

In next t hours, the satellite rotates through $\left(\frac{360}{8}t\right)^\circ$ and P rotates through $\left(\frac{360}{24}\ t\right)^\circ$.

For S to be above P –

$$120° + \left(\frac{360}{24}t\right)^\circ = \left(\frac{360}{8}t\right)^\circ$$

$$\Rightarrow \qquad t = 4\,\text{h}$$

$\therefore$ Satellite will be exactly overhead P at position S_2.

Total time elapsed = 8 + 4 = 12 h

$\therefore$ Time in clock = 8 PM on 17 January 2018.

7. For the Earth: $(1\,\text{yr})^2 = \dfrac{4\pi^2}{6\,M_s}(1\,\text{AU})^3$ ...(i)

M_s is mass of the sun.

For the star in the galaxy:

$$(3 \times 10^8\,\text{yr})^2 = \frac{4\pi^2}{GM}(3 \times 10^9\,\text{AU})^3 \quad \text{...(ii)}$$

From (i) and (ii): $(3 \times 10^8)^2 = \dfrac{M_s}{M}\, 27 \times 10^{27}$

$$\Rightarrow \quad M = (3 \times 10^{11})M_s$$

$$\Delta M = 3 \times 10^{11} M_s - 2.8 \times 10^{11} M_s = 0.2 \times 10^{10} M_s = 2 \times 10^{10} M_s$$

8. Effective value of 'g' at equator is

$$g' = g - \omega^2 R$$

$$\Rightarrow \quad \frac{g}{2} = g - \omega^2 R \quad \Rightarrow \quad \omega = \sqrt{\frac{g}{2R}}$$

$$\therefore \quad u = \omega R = \sqrt{\frac{1}{2}gR}$$

Escape speed is $v_e = \sqrt{2gR} = 2u$.

9. Orbital speed, $v_0 = \sqrt{\dfrac{GM}{r}}$ ...(i)

Masses of the fragments are

$$m_1 = m,\ m_2 = m,\ m_3 = 4m$$

After explosion: $v_1 = -v_0$, $v_2 = 0$, $v_3 = ?$

Momentum conservation: $6mv_0 = -mv_0 + 4mv_3$

$$\Rightarrow \quad v_3 = \frac{7}{4}v_0$$

A satellite will escape if its velocity is increased to $v_1 = \sqrt{2}\,v_0$.

In this case $v_3 > \sqrt{2}\,v_0$

Therefore, the third piece will escape the gravity of the planet.

Energy conservation for m_2:

$$-\frac{GMm}{r/2} + \frac{1}{2}mv^2 = -\frac{GMm}{r}$$

$$v = \sqrt{\frac{2GM}{r}} = \sqrt{2}\,v_0 \quad \text{[From (i)]}$$

10. Momentum of the system will remain conserved.

A will slow down and B will speed up. At a point they will have the same velocity (at separation r). After this they will start getting closer. Thus, the system will be bound if r is not infinity.

$$3mv = mv_0 \quad \Rightarrow \quad v = \frac{v_0}{3}$$

$$-\frac{Gm(2m)}{r_0} + \frac{1}{2}mv_0^2 = \frac{1}{2}(3m)\left(\frac{v_0}{3}\right)^2 - \frac{G(m)(2m)}{r}$$

$$\Rightarrow \frac{Gm(2m)}{r} = \frac{2Gm^2}{r_0} - \frac{1}{3}mv_0^2$$

For $r < \infty$, $\dfrac{2Gm^2}{r_0} > \dfrac{1}{3}mv_0^2$

$$\Rightarrow \quad v_0 < \sqrt{\frac{6Gm}{r_0}}$$

11. When A is at ∞, $U_0 = -\dfrac{GMM}{2a}$

Where M is mass of particles at B and C and $2a$ is separation between them.

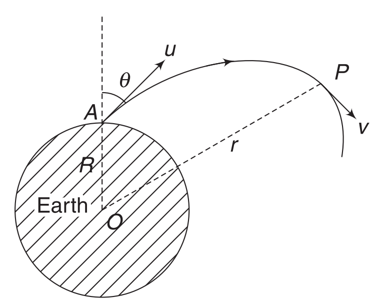
When A is at $y = 0$,

$$U = -\frac{GMM}{2a} - \frac{GM(2M)}{a} - \frac{GM(2M)}{a}$$

$$= \frac{-9\,GMM}{2a}$$

$$= 9U_0$$

12. At highest point, velocity (v) will be perpendicular to the position vector OP.



At farthest point (P), Velocity v is perpendicular to position vector OP

[When v is $\perp r$ to r, it implies that $\dfrac{dr}{dt} = 0$ and r has stopped growing]

Angular momentum of the projectile about O is conserved.

$$\therefore \quad L_A = L_P$$

$$(mu\sin\theta)R = mvr$$

$$\Rightarrow \quad (u\sin 60°)R = vr \quad \text{...(i)}$$

Energy conservation gives:

$$-\frac{GMm}{r} + \frac{1}{2}mv^2 = -\frac{GMm}{R} + \frac{1}{2}mu^2$$

$$\Rightarrow \quad -\frac{GMm}{r} + \frac{1}{2}m\left(\frac{\sqrt{3}}{2}\frac{uR}{r}\right)^2 = -\frac{GMm}{R} + \frac{1}{2}mu^2$$

$$\Rightarrow \quad -\frac{GMm}{r} + \frac{3}{8}\frac{GMm}{r^2}R = -\frac{GMm}{R} + \frac{1}{2}\frac{GMm}{R}$$

$$\left[\because u = \sqrt{\frac{GM}{R}}\right]$$

$$\Rightarrow \quad -\frac{1}{r} + \frac{3}{8}\frac{R}{r^2} = -\frac{1}{2R}$$

$$\Rightarrow \quad 4r^2 - 8Rr + 3R^2 = 0$$

$$\Rightarrow \quad r = \frac{R}{2} \text{ and } \frac{3R}{2}$$

Obvious answer is $\frac{3R}{2}$. Height above surface $= \frac{R}{2}$

13. $U_{\text{surface}} = -8 \times 10^9 \text{ J} = -\frac{GMm}{R}$

PE at height R is $U = -\frac{GMm}{2R} = -4 \times 10^9$ J

Look into the figure of last solution

$$K_P + U_P = K_A + U_A$$

$$\frac{1}{2}mv^2 - 4 \times 10^9 = 6 \times 10^9 - 8 \times 10^9$$

$$\therefore \quad v^2 = 2 \times 10^9 \quad \Rightarrow \quad v = \sqrt{2 \times 10^9} \text{ ms}^{-1}$$

At surface, $\frac{1}{2}mu^2 = 6 \times 10^9 \quad \Rightarrow u = \sqrt{12 \times 10^9}$ ms^{-1}

Conservation of angular momentum:

$$mu\sin\theta \cdot R = mv(2R)$$

$$\sqrt{12 \times 10^9}\,\sin\theta = 2\sqrt{2 \times 10^9}$$

$$\Rightarrow \quad \sin\theta = \sqrt{\frac{2}{3}}$$

$$\Rightarrow \quad \theta = \sin^{-1}\left(\sqrt{\frac{2}{3}}\right)$$

14. $g' = g - \omega^2 R = 0$

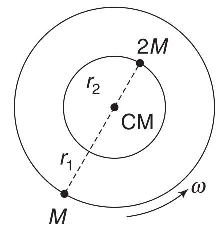
$$\Rightarrow \quad \omega = \sqrt{\frac{g}{R}} = \sqrt{\frac{GM}{R^3}} = \sqrt{\frac{6.67 \times 10^{-11} \times 3 \times 10^{24}}{(1.8 \times 10^6)^3}}$$

$$= 5.85 \times 10^{-2} \text{ rad s}^{-1}$$

15. $r_1 = \frac{2Mr}{2M + M} = \frac{2r}{3}$; $r_2 = \frac{r}{3}$

$$Mr_1\omega^2 = 2Mr_2\omega^2 = \frac{G(M)(2M)}{r^2} \qquad \text{...(i)}$$

And $\quad \omega^2 = \frac{2GM}{r_1 r^2} = \frac{2GM}{\frac{2r}{3}\cdot r^2} = \frac{3GM}{r^3}$



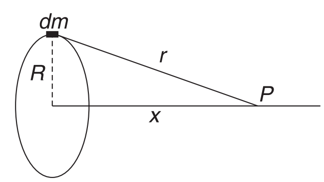
$$\therefore \quad E = -\frac{G(M)(2M)}{r} + \frac{1}{2}Mr_1^2\omega^2 + \frac{1}{2}(2M)r_2^2\omega^2$$

$$= -\frac{2GM^2}{r} + \frac{1}{2}\left(\frac{2GM^2}{r}\right)\left(\frac{2r}{3}\right) + \frac{1}{2}\left(\frac{2GM^2}{r^2}\right)\left(\frac{r}{3}\right)$$

$$= -\frac{2GM^2}{r} + \frac{2}{3}\frac{GM^2}{r} + \frac{1}{3}\frac{GM^2}{r} = -\frac{GM^2}{r}$$

16. (a) All particles in the ring are at distance R from the centre.

$$\therefore \quad V = -\int\frac{G\,dm}{R} = -\frac{G}{R}\int dm = -\frac{GM}{R}$$

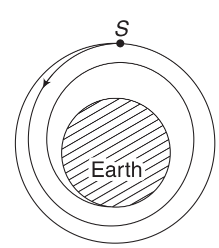


(b) All particles in the ring are at distance $r = \sqrt{R^2 + x^2}$ from the given point

$$\therefore \quad V = -\int\frac{G\,dm}{r} = -\frac{GM}{r}$$

$$= -\frac{GM}{\sqrt{R^2 + x^2}}$$

16a. Energy of the satellite in its orbit is $E_1 = -\frac{GMm}{2r}$



Its energy when it gets close to Earth (i.e., it is revolving in a circle of radius R) is

$$E_2 = -\frac{GMm}{2R}$$

$\therefore$ Loss in energy $\Delta E = E_1 - E_2$

$$= \frac{GMm}{2}\left(\frac{1}{R} - \frac{1}{r}\right)$$

$$\therefore \quad t = \frac{\Delta E}{w} = \frac{GMm}{2w}\left(\frac{1}{R} - \frac{1}{r}\right).$$

17. (a) From conservation of energy, it follows that the particle will come out of the tunnel from the other side and reach height h.

$\therefore$ Maximum displacement = $2R + 2R = 4R$

(b) Particle has maximum speed when it reaches the centre (O). After this, the particle retards.

$$k_P + U_P = k_0 + U_0$$

$$0 - \frac{GMm}{2R} = \frac{1}{2}mv^2 - \frac{3}{2}\frac{GMm}{R}$$

$$\left[\textit{Note:} \text{ Potential at centre} = -\frac{3}{2}\frac{GM}{R}\right]$$

$$\Rightarrow \quad v = \sqrt{\frac{2GM}{R}} = \sqrt{2gR}$$

18. (a) Let v_1 be maximum and v_2 be minimum speeds.

$$mv_1(2R) = mv_2(4R)$$

$$\Rightarrow \quad v_1 = 2v_2 \quad \text{...(i)}$$

$$\frac{1}{2}mv_1^2 - \frac{GMm}{2R} = \frac{1}{2}mv_2^2 - \frac{GMm}{4R} \quad \text{...(ii)}$$

Solving (i) and (ii) gives $v_2 = \sqrt{\frac{GM}{6R}}$ and

$$v_1 = \sqrt{\frac{2GM}{3R}}$$

(b) If r is the radius of curvature of point A, then

$$\frac{mv_1^2}{r} = \frac{GMm}{(2R)^2} \Rightarrow r = \frac{4v_1^2R^2}{GM} = \frac{8R}{3}$$

19. Total energy of a satellite in elliptical orbit is $-\frac{GMm}{2a}$

$$\therefore \quad \frac{1}{2}mv^2 - \frac{GMm}{r} = -\frac{GMm}{2a}$$

$$\Rightarrow \quad v = \sqrt{GM\left(\frac{2}{r} - \frac{1}{a}\right)}$$

Your Turn

1. (i) $\in_l = \dfrac{\Delta L}{L} = \dfrac{2\,mm}{10^3\,mm} = 0.002$

(ii) $\sigma_l = \dfrac{F}{A} = \dfrac{100\,\text{N}}{10 \times 10^{-6} m^2} = 10^7\,\text{Nm}^{-2}$

2. $\in_s = \dfrac{x}{L} = \tan\theta = \tan 1° = \tan\left(\dfrac{\pi}{180}\right) \simeq \dfrac{\pi}{180}$

$= 0.0174$

[for small angles $\tan\theta \simeq \theta$]

3. Volume stress = increase in pressure = ρgh

$= 10^3 \times 9.8 \times 2000 = 1.96 \times 10^7\,\text{Nm}^{-1}$

4. $x = \dfrac{FL}{AY}$

On doubling the radius, A becomes 4 times.

Therefore, doubling F and radius will make value of x half.

5. Area of cross-section of the cable, $A = \pi\dfrac{d^2}{4}$

$\therefore \quad A = 3.14\left(\dfrac{0.05}{2}\right)^2 = 0.002\,\text{m}^2$

(a) Additional tension in the cable when people enter the elevator is

$F = (238\,\text{kg}) \times (9.8\,\text{ms}^{-2}) = 2332\,\text{N}$

$\Delta L = \dfrac{FL}{AY} = \dfrac{2332 \times 20}{0.002 \times 2 \times 10^{11}}$

$= 1.17 \times 10^{-4}$ m.

(b) Tension remains unchanged if the lift moves with constant speed.

6. Strain = fractional change in length

$= \dfrac{\Delta L}{L} = \dfrac{\text{Stress}}{Y}$

$= \dfrac{855}{(9.5 \times 10^{-4}) \times (1 \times 10^{10})} = 9 \times 10^{-5}$

7. $3g - T = 3a$ and

$T = 6a$

Solving, $T = 20\,N$

Strain $= \dfrac{\text{Stress}}{Y}$

$= \dfrac{(20\,N)}{(0.005 \times 10^{-4}\,m^2)2 \times 10^{11}\,\text{Nm}^{-2}} = 2 \times 10^{-4}$

8. Shear strain $= \dfrac{x}{L} = \dfrac{0.2 \times 10^{-1}\,\text{cm}}{5\,\text{cm}} = 0.004$

Shear stress $= \dfrac{F}{A} = \dfrac{8\,\text{N}}{0.05 \times 0.05\,\text{m}^2} = 3{,}200\,\text{Nm}^{-2}$

$\therefore \quad \eta = \dfrac{\text{Shear stress}}{\text{Shear strain}} = \dfrac{3200}{0.004} = 8 \times 10^5\,\text{Nm}^{-2}$

9. $K = \dfrac{1}{B} = \dfrac{\Delta V}{VP}$

$\Rightarrow \quad \Delta V = KVP$

$= (5 \times 10^{-10}\,\text{m}^2\,\text{N}^{-1})(1000\text{ ml})\,(15 \times 10^6\,\text{Nm}^{-2})$

$= 7.5$ ml

10. $B = \dfrac{\Delta P}{\dfrac{\Delta V}{V}} \Rightarrow \dfrac{\Delta V}{V} = \dfrac{\Delta P}{B}$

$\dfrac{\Delta V}{V} = \dfrac{0.1}{100} = 1 \times 10^{-3}$

$\therefore \quad \dfrac{\Delta P}{B} = 1 \times 10^{-3} \Rightarrow \Delta P = 9.8 \times 10^5$

$\Rightarrow \quad \rho gh = 9.8 \times 10^5$

$\Rightarrow \quad 10^3 \times 9.8 \times h = 9.8 \times 10^5$

$\Rightarrow \quad h = 100\,\text{m}$

11. $u = \dfrac{1}{2} \times \text{Stress} \times \text{Strain}$

$= \dfrac{1}{2} \times (Y \cdot \text{strain}) \cdot \text{strain} = \dfrac{1}{2} Y\in^2$

12. $U = \dfrac{1}{2} \times$ (maximum stretching force) (extension)

$= \dfrac{1}{2} \times 800 \times 2 \times 10^{-3} = 0.8\,\text{J}$

Worksheet 1

1. Same material implies same Y.

$$(\text{Strain})_A = (\text{Strain})_B$$

$$\frac{(\text{Strain})_A}{Y} = \frac{(\text{Stress})_B}{Y} \Rightarrow (\text{Stress})_A = (\text{Stress})_B$$

$$\Rightarrow \frac{F_A}{A} = \frac{F_B}{A/4} \Rightarrow \frac{F_A}{F_B} = \frac{4}{1}$$

2. Tension (and hence, stress) is maximum at the top. Longitudinal strain is maximum at the top.

4. A material fails if breaking stress is applied to it. What it means is that it is stress (and not the force), which decides the breakage of the rope. Both ropes fail when stress reaches breaking stress.

$$\frac{F}{4A} = \frac{500\,\text{N}}{A} = F = 2{,}000\,\text{N}$$

5. Solution to the previous question has the clue.

6. Tension in the wire in both cases is W.

7. Tension is maximum at the lowest position.

8. Let natural length be l_0.

$$\text{Strain} = \frac{\text{Stress}}{Y} \Rightarrow \frac{l_1 - l_0}{l_0} = \frac{T_1}{AY}$$

$$\Rightarrow l_1 - l_0 = \frac{T_1 l_0}{AY} \quad \text{...(a)}$$

Similarly, $l_2 - l_0 = \frac{T_2 l_0}{AY}$ ...(b)

From (a) and (b): $\frac{l_1 - l_0}{T_1} = \frac{l_2 - l_0}{T_2}$

$$\Rightarrow l_0 = \frac{T_2 l_1 - T_1 l_2}{T_2 - T_1}$$

9. $U = \frac{1}{2}$

(Final value of force stretching the wire) × (extension)

$$= \frac{1}{2} \times 200 \times 0.001 = 0.1\,\text{J}$$

10. $\text{Strain} = \frac{\text{Stress}}{Y} \Rightarrow \frac{\Delta l}{l} = \frac{\text{Stress}}{Y}$

$$\therefore \Delta l = (\text{Stress})\frac{l}{Y}$$

Graph between Δl and (stress) is a straight line with slope $= \frac{l}{Y}$

$$\Rightarrow \frac{4\,\text{mm} - 2\,\text{mm}}{(8 \times 10^6 - 4 \times 10^6)\,\text{Nm}^{-2}} = \frac{1\text{m}}{Y}$$

$$\Rightarrow Y = 2 \times 10^9\ \text{Nm}^{-2}$$

11. $V = \pi r^2 \cdot l$

$$\frac{\Delta V}{V} = 2\frac{\Delta r}{r} + \frac{\Delta l}{l}$$

$$\frac{\Delta V}{V} \times 100 = -2\frac{\Delta r}{r} \times 100 + \frac{\Delta l}{l} \times 100$$

[Radius decrease]

$$\Rightarrow 0.2 = -2 \times 0.002 + \frac{\Delta l}{l} \times 100$$

$$\Rightarrow \frac{\Delta l}{l} = 0.204$$

$$\text{Stress} = Y(\text{strain})$$

$$= 2 \times 10^{11} \times 0.204 = 4.08 \times 10^8\ \text{Nm}^{-2}$$

12. $\frac{\Delta l}{l} = \frac{F}{AY} = \frac{100}{\pi(10^{-3})^2 \times 2 \times 10^{11}} = \frac{1}{2\pi} \times 10^{-3}$

$$\text{Passion's Ratio} = \frac{\Delta r/r}{\Delta l/l}$$

$$\Rightarrow \frac{\Delta r}{r} = \frac{\pi}{10}\cdot\frac{1}{2\pi} \times 10^{-3} = 0.5 \times 10^{-4}$$

$$\Rightarrow \Delta r = 0.00005\,\text{mm.}$$

$$\therefore \text{New radius} = 1\text{mm} - 0.00005\,\text{mm}$$

$$= 0.99995\ \text{mm}$$

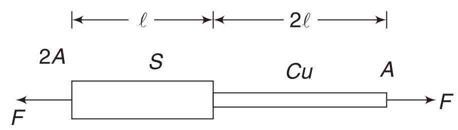
13. Go through solution given to Q.9 in Worksheet 3. Look at equation (i) in the solution.

14. Thickest wire will have the highest load for a given strain.

Worksheet 2

1. Tension is the same everywhere.

$\therefore$ Stress in *Cu* wire is more. Obviously, strain is higher for *Cu* wire, as it has a smaller Young's modulus as well.



Cu wire has more strain and longer original length. Therefore, it has more extension.

2. Tension in A > Tension in B

$$T_A = mg + \frac{mg}{3} = \frac{4}{3}mg$$

And $\quad T_B = \frac{1}{3}mg \quad \Rightarrow \quad T_A = 4T_B$

When $r_A = r_B$, Stress in A is more. Therefore, it breaks

$$(\text{Stress})_A = \frac{T_A}{\pi(r_A)^2}; \ (\text{Stress})_B = \frac{T_B}{\pi(r_B)^2}$$

$(\text{Stress})_A > (\text{Stress})_B$ if

$$\frac{T_A}{r_A^2} > \frac{T_B}{r_B^2} \quad \Rightarrow \quad \frac{T_A}{T_B} > \left(\frac{r_A}{r_B}\right)^2$$

$$\Rightarrow \qquad 4 > \left(\frac{r_A}{r_B}\right)^2 \quad \Rightarrow \quad 2r_B > r_A$$

When $r_A = 2r_B$, both wires have the exact same stress.

3. Strain in all rods are same. Therefore, stress is same in all of them.

Cross-sectional areas are in ratio

$$A_A : A_B : A_C : A_D = 1:2:3:4$$

$$\frac{F_A}{A_A} = \frac{F_B}{A_B} = \frac{F_C}{A_C} = \frac{F_D}{A_D}$$

$$\therefore \quad F_A : F_B : F_C : F_D = 1:2:3:4$$

Energy stored, $\quad U = \frac{1}{2} F \cdot \Delta l \quad \Rightarrow \quad U \propto F$

$$\therefore \quad U_A : U_B : U_C : U_D = 1:2:3:4$$

4. When a block attached to a spring is released from its unstretched position, it will go down to stretch the spring by x such that

$$\frac{1}{2}kx^2 = mgx \quad \Rightarrow \quad x = \frac{2mg}{k}$$

Ultimately, due to air friction, etc., the block will settle down in its equilibrium position, stretching the spring by

$$x_0 = \frac{mg}{k} \left(= \frac{x}{2}\right)$$

Thus, energy equal to $mg\frac{x}{2}$ is lost forever. This energy is mainly lost as heat in overcoming viscosity of air.

In case of the wire (which is just like a very stiff spring), similar thing happens. We may not be able to observe oscillations as in the case of a spring due to small amplitude. Ultimately, the system settles in equilibrium where half the loss in gravitational PE of the block is lost as heat.

Worksheet 3

1. Cross-sectional area supporting the load is

$$A = \pi\left[\left(\frac{3}{2}\right)^2 - \left(\frac{0.8}{2}\right)^2\right]\text{cm}^2$$

$$\Delta L = \frac{FL}{AY}$$

Substituting the values (in SI unit) will give the answer

2. Force constant of each cord is $k = \dfrac{YA}{l}$

$$\Rightarrow \quad k = \frac{5\times10^8\times1\times10^{-6}}{0.1} = 5\times10^3\ \text{Nm}^{-1}$$

Elastic energy stored in both cords is

$$E = \left(\frac{1}{2}k\Delta l^2\right)\times 2$$

$$\Rightarrow \quad E = 5\times10^3\times(0.06)^2 = 18\,\text{J}$$

$$\therefore \quad \frac{1}{2}mv^2 = 18 \quad \Rightarrow \quad \frac{1}{2}\times\frac{5}{1000}\times v^2 = 18$$

$$\Rightarrow \quad v = 84.9\,\text{ms}^{-1}$$

3. Let the length of original wire be l_1

$$\text{Strain} = \frac{\text{Stress}}{Y};\ \text{or, } \frac{x}{l_1} = \frac{F}{A_1Y}$$

$$\Rightarrow \quad x = \frac{Fl_1}{A_1Y} = \frac{Fl_1}{\pi r^2 Y} \qquad \text{...(i)}$$

When the wire is re-drawn to have square section of side length a, its length (l_2) will be given by volume conservation

$$l_2 . a^2 = \pi r^2\cdot l_1 \quad \Rightarrow l_2 = \frac{\pi r^2}{a^2}\cdot l_1 \qquad \text{...(ii)}$$

$$\text{Strain} = \frac{\text{Stress}}{Y} \quad \Rightarrow \frac{x_2}{l_2} = \frac{F}{A_2Y}$$

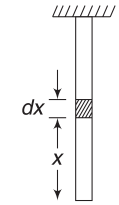
$$\Rightarrow \quad x_2 = \frac{Fl_2}{A_2Y} = \frac{Fl_2}{a^2\cdot Y}$$

$$= \frac{F\left(\dfrac{\pi r^2}{a^2}\,l_1\right)}{a^2Y} = \frac{\pi F r^2 l_1}{a^4 Y}$$

But $\quad a = 2r$

$$\therefore \quad x_2 = \frac{\pi}{16}\,\frac{Fl_1}{r^2Y} = \frac{\pi^2}{16}x \qquad \text{[Using (i)]}$$

4. Stress at distance x above the lower end is



$$\text{Stress} = \frac{xA\rho g}{A} = x\rho g$$

Elastic potential energy stored in an element of length dx is

$$dU = \frac{1}{2}\,(\text{stress})(\text{strain})(\text{volume})$$

$$= \frac{1}{2}\,(x\rho g)\left(\frac{x\rho g}{Y}\right)(A\,dx) = \frac{1}{2}\,\frac{\rho^2g^2A}{Y}\,x^2dx$$

$$\therefore \quad U = \int dU = \frac{1}{2}\,\frac{\rho^2g^2A}{Y}\int_0^L x^2dx = \frac{\rho^2g^2AL^3}{6Y}$$

5. Strain is same in all wires.

$$\therefore \quad (\text{stress})_s = Y_s\cdot(\text{strain})$$

$$(\text{stress})_{\text{cu}} = Y_{\text{cu}}\cdot(\text{strain})$$

$$\therefore \quad \frac{(\text{stress})_s}{(\text{stress})_{\text{cu}}} = \frac{Y_s}{Y_{\text{cu}}} = \frac{2}{1}$$

$$\Rightarrow \quad \frac{T_s}{T_{\text{cu}}} = \frac{2}{1} \quad \Rightarrow \quad T_{\text{cu}} = \frac{1}{2}T_s$$

Also $\quad 2T_{\text{cu}} + T_s = Mg$

$$T_s + T_s = Mg$$

$$T_s = \frac{Mg}{2}$$

6. Let tension in two wires be T_s and T_B. For rotational equilibrium:

$$T_S\cdot x = T_B = T_B\,(2-x) \qquad \text{...(i)}$$

(a) $\quad (\text{Strain})_S = (\text{Strain})_B$

$$\Rightarrow \quad \frac{(\text{Stress})_S}{Y_S} = \frac{(\text{Stress})_B}{Y_B}$$

$$\Rightarrow \quad \frac{T_S}{2A} = \frac{T_B}{2A} \quad [\because\ Y_S = 2Y_B \text{ and } A_B = 2A_S]$$

$$\Rightarrow \quad T_S = T_B \qquad \text{...(ii)}$$

Solving (i) and (ii) gives: $x = 1$m

(b) $\quad (\text{Stress})_S = (\text{Stress})_B$

$$\frac{T_S}{A} = \frac{T_B}{2A} \quad \Rightarrow \quad T_S = \frac{T_B}{2} \qquad \text{...(iii)}$$

Solving (i) and (iii) gives: $x = \dfrac{4}{3}$ m

7. $T_P = 4mg$; $T_Q = mg$

A wire will break if stress in it exceeds the breaking stress. Since both wires are made of the same material, they have same breaking stress.

(a) For P to fail:

$$(\text{Stress})_P > (\text{Stress})_Q$$

$$\frac{4mg}{\pi r_1^2} > \frac{mg}{\pi r_2^2} \quad \Rightarrow \quad \frac{r_2}{r_1} > \frac{1}{2}$$

(b) For Q to fail:

$$(\text{Stress})_Q > (\text{Stress})_P$$

$$\Rightarrow \qquad \frac{r_2}{r_1} < \frac{1}{2} \quad \Rightarrow \quad r_2 < \frac{r_1}{2}$$

8. $$A_1 = \pi r^2;\ A_2 = \pi(R^2 - r^2)$$

$$\sigma_1 A_1 + \sigma_2 A_2 = W \qquad \text{...(i)}$$

Strain in both sections is same.

$$\frac{\sigma_1}{Y_1} = \frac{\sigma_2}{Y_2} \qquad \text{...(ii)}$$

Solving (i) and (ii) gives $\sigma_2 = \dfrac{WY_2}{A_1Y_1 + A_2Y_2}$

Required fraction is: $\dfrac{\sigma_2 A_2}{W} = \dfrac{A_2Y_2}{A_1Y_1 + A_2Y_2}$

$$= \frac{(R^2 - r^2)Y_2}{r^2 Y_1 + (R^2 - r^2)Y_2}$$

9. Let L be the greatest length of the wire that can hang vertically without breaking.

Mass of wire m = cross-sectional area (A) × length (L) × density (d),

Weight of wire = $mg = AL\,dg$

The top of the wire experiences maximum tension equal to weight of the wire. Wire is likely to break at the top.

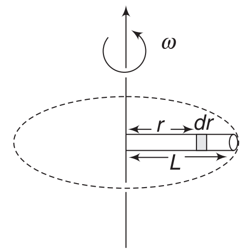
Breaking stress = $\dfrac{LA\,dg}{A} = Ldg$

$$\Rightarrow \quad 7.5 \times 10^8 = L \times 2.7 \times 980$$

$$\Rightarrow \qquad L = \frac{7.5 \times 10^8}{2.7 \times 980}\text{ cm} = 2.834 \times 10^5\text{ cm}$$

$$= 2.834\text{ km}$$

10. (a) Tension in the rod at a distance r from the axis of rotation will be equal to the centripetal force needed by all elements between $x = r$ and $x = L$. Consider an element of length dr, at a distance r from the axis of rotation, as shown in figure. The centripetal force acting on this element will be $= dmr\omega^2 = (\rho A dr) r\omega^2$.



Tension in the rod at distance r from axis is

$$T = \int_r^L \rho A\omega^2 r dr = \frac{1}{2}\rho A\omega^2 [L^2 - r^2] \qquad \text{...(i)}$$

[One can also assume the entire mass of the segment of the rod between r and L to be located at its COM.

$$T = \frac{1}{2} \times 10^4 \times \pi \times 10^{-2} \times (400)^2 \left[\left(\frac{1}{2}\right)^2 - r^2\right]$$

$$= 8\pi \times 10^6 \left[\frac{1}{4} - r^2\right]\text{N}$$

(b) Now, if dy is the elongation in the element of length dr, at position r, then strain

$$\frac{dy}{dr} = \frac{\text{stress}}{Y} = \frac{T}{AY} = \frac{1}{2}\frac{\rho\omega^2}{Y}[L^2 - r^2]$$

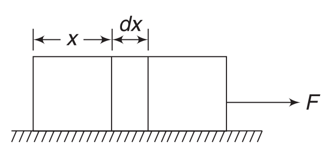
So the elongation of the whole rod

$$\Delta y = \frac{\rho\omega^2}{2Y}\int_0^L (L^2 - r^2)\,dr = \frac{1}{3}\frac{\rho\omega^2 L^3}{Y}$$

$$= \frac{1}{3} \times \frac{10^4 \times (400)^2 (0.5)^3}{2 \times 10^{11}} = \frac{1}{3} \times 10^{-3}\text{m}$$

11. Acceleration of the block is $a = \dfrac{F}{m}$

Tension at a distance x from the rear end is



$T = m'a$, where $m' = \dfrac{m}{\ell}x$ = mass of segment of length x.

$$\Rightarrow \qquad T = \frac{Fx}{\ell}$$

Elongation in element of length dx is $= \dfrac{Tdx}{AY}$,

Total elongation, $d = \displaystyle\int_o^\ell \frac{Tdx}{AY} = \int_o^\ell \frac{Fxdx}{A\ell Y} = \frac{F\ell}{2AY}$

Match the Column

1. $I_0\omega_0 = I\omega \quad \Rightarrow \quad \frac{2}{5} MR^2 \cdot \omega_0 = \frac{2}{5} Mr^2 \cdot \omega$

$\Rightarrow \quad \omega = \left(\frac{R}{r}\right)^2 \cdot \omega_0.$ Thus, ω decreases.

Free-fall acceleration at equator is

$$g' = g - \omega^2 r = \frac{GM}{r^2} - \left[\left(\frac{R}{r}\right)^2 \omega_0\right]^2 \cdot r$$
$$= \frac{GM}{r^2} - \frac{R^4\omega_0^2}{r^3}$$

As the planet contracts, r decreases. The second term in above expression $\left(\alpha \frac{1}{r^3}\right)$ increases more rapidly compared to the first term $\left(\alpha \frac{1}{r^2}\right)$. Thus, g' decreases.

There is no effect of rotation at poles.

$\therefore$ Acceleration due to gravity at poles = free fall acceleration at poles $= \frac{GM}{r^2}$.

With decrease in r, g_{pole} will increase.

Binding energy $= \frac{GMm}{r}$

With decrease in r, binding energy will increase

2. At maximum height, Particle 2 will have a velocity in a direction perpendicular to its radius vector.

Particle 2 has angular momentum = muR, where u is its speed of projection. It cannot hit the surface of the Earth normally, as this will violate the law of conservation of angular momentum. [Angular momentum of a radially moving body about the centre of the Earth will be zero]

3. Change in length of steel = Change in length of brass

$\therefore \quad (\text{Strain})_S = \frac{\Delta l}{30}; \ (\text{Strain})_B = \frac{\Delta l}{20}$

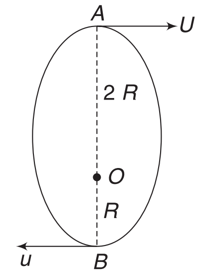
$$\Rightarrow \quad \frac{(\text{Strain})_S}{(\text{Strain})_B} = \frac{2}{3}$$

$$\Rightarrow \quad \frac{\frac{(\text{Stress})_S}{Y_S}}{\frac{(\text{Stress})_B}{Y_B}} = \frac{2}{3}$$

$$\Rightarrow \quad \frac{(\text{Stress})s}{(\text{Stress})_B} = \frac{2}{3}\frac{Y_S}{Y_B} = \frac{2}{3} \times 2 = \frac{4}{3}$$

$$\frac{F_S}{F_B} = \frac{(\text{Stress})_S \cdot A_S}{(\text{Stress})_B \cdot A_B} = \frac{4}{3} \times \frac{16}{10} = \frac{32}{15}$$

4. O is centre of the Earth. Particle is projected from A and its smallest distance from O is R (when the particles reaches B). If smallest distance is less than R, the particle will hit the Earth.



$$mv \cdot 2R = muR \quad \Rightarrow \quad 2v = u \qquad \text{...(i)}$$

$$\frac{1}{2} mv^2 - \frac{GMm}{2R} = \frac{1}{2} mu^2 - \frac{GMm}{R}$$

$$\Rightarrow \quad (2v)^2 - v^2 = \frac{GM}{R}$$

$$\Rightarrow \quad v = \sqrt{\frac{GM}{3R}}$$

Passage-based Problems

Passage 1

1. Each particle is like a satellite of the Saturn. Every particle experiences gravitational pull of the planet and force on it due to the particles in the ring can be neglected, as the ring has a very small mass.

The particles closest to the planet have highest speed. It is given by

$$v_0 = \sqrt{\frac{GM}{R+h}} = \sqrt{\frac{GM}{R\left[1 + \frac{h}{R}\right]}}$$

$$= \sqrt{\frac{gR}{1 + \frac{h}{R}}} = \sqrt{\frac{10 \times 6 \times 10^7}{1 + \frac{8}{60}}}$$

$$= 2.3 \times 10^4 \text{ ms}^{-1}.$$

Mass of heaviest particle = mass of an ice ball of radius 10 m

$$m_0 = \frac{4}{3}\pi r^3 \cdot d = \frac{4}{3} \times 3.14 \times (10)^3 \times 10^3$$
$$= 4.19 \times 10^6 \text{ kg}$$

$$\therefore \quad KE_{\max} = \frac{1}{2} m_0 v_0^2$$

$$= \frac{1}{2} \times 4.19 \times 10^6 \times (2.3 \times 10^4)^2$$

$$= 1.1 \times 10^{15} \text{ J}$$

2. The nearest particles are going in a circle of radius $r_1 = 68{,}000$ km and the farthest ones are in orbit of radius $r_2 = 60{,}000 + 80{,}000 = 1{,}40{,}000$ km.

$$\therefore \qquad \left(\frac{T_1}{T_2}\right)^2 = \left(\frac{r_1}{r_2}\right)^3 = \left(\frac{68}{140}\right)^3 = 0.12$$

$$\therefore \qquad \frac{T_1}{T_2} = 0.35$$

3. Work done in moving the mass from A to B along any path will be same.

$$\therefore \qquad W_1 = W_2$$

$$W_1 = W^{A\to B}_{\substack{\text{ext Agt}\\ \text{slowly}}} = U_B - U_A = m(V_B - V_A)$$

$$\therefore \quad V_A - V_B = -\frac{W_1}{m} = -\left(\frac{W_1 + W_2}{2m}\right)$$

4. From A to O, the gravitational force is along $\overrightarrow{AO}$. Force by external agent is equal and opposite. Thus, F is positive. F decreases as $\frac{1}{r^2}$ when m is outside the sphere. Inside the tunnel, $F \propto r$. At centre, $F = 0$. From O to D, gravitational force is in positive direction. Force by external agent is in negative direction.

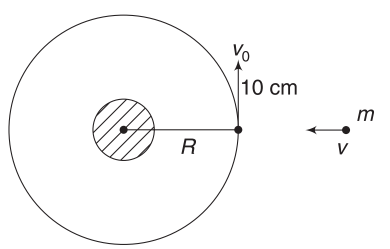
5. Momentum conservation gives:

$$11m\,u_x = mv$$

$$\Rightarrow \qquad u_x = \frac{v}{11}$$

$$11m\,u_y = 10m \cdot v_0$$

$$\Rightarrow \qquad u_y = \frac{10}{11}\,v_0 = \frac{10}{11}\sqrt{\frac{GM}{R}}$$



Before Collision

Speed after collision is

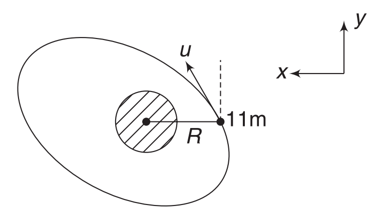
$$u = \sqrt{u_x^2 + u_y^2}$$

$$= \frac{1}{11}\sqrt{v^2 + 100v_0^2}$$

For the satellite to avoid escaping

$$u < \sqrt{2}\ v_0$$

$$\Rightarrow \qquad u^2 < 2v_0^2$$



Just after collision

$$\Rightarrow \qquad \frac{1}{121}\,(v^2 + 100\,v_0^2) < 2v_0^2$$

$$\Rightarrow \qquad v^2 < 142\,v_0^2 \quad \Rightarrow \quad v < \sqrt{142}\,v_0 = \sqrt{142}\,\sqrt{\frac{GM}{R}}$$

6. Let speed at minimum distance be u_1

$$mu_1\,\frac{R}{2} = mu_y \cdot R \quad \Rightarrow \quad \frac{u_1}{2} = \frac{10}{11}\sqrt{\frac{GM}{R}}$$

$$\Rightarrow \qquad u_1 = \frac{20}{11}\sqrt{\frac{GM}{R}}$$

Conservation of energy

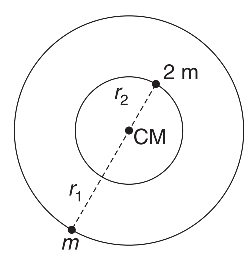
$$-\frac{GMm}{R/2} + \frac{1}{2}\,mu_1^2 = -\frac{GMm}{R} + \frac{1}{2}mu^2$$

Substituting for u_1 and u we get an equation in v. solving this gives: $v = \sqrt{\frac{58GM}{R}}$

7. $$r_1 + r_2 = r$$

$$m_1 r_1 = 2m r_2$$

Solving, $r_1 = \frac{2r}{3}$ and $r_2 = \frac{r}{3}$



For star of mass m:

$$m\omega^2 r_1 = \frac{Gm(2m)}{r^2}$$

$$\Rightarrow \qquad \frac{2r}{3}\,\omega^2 = \frac{2Gm}{r^2}$$

$$\Rightarrow \qquad \omega^2 = \frac{3Gm}{r^3} \qquad \text{...(i)}$$

For the Earth $m_e \omega_e^2 R = \frac{Gm_e M}{R^2}$

$$\Rightarrow \qquad \omega_e^2 = \frac{GM}{R^3} \qquad \text{...(ii)}$$

According to the problem ω for the star system is same as that of the Earth.

$\therefore$ From (i) and (ii)

$$\frac{3Gm}{r^3} = \frac{GM}{R^3}$$

Given $\quad M = 3m$

$$\therefore \quad r = R$$

8. From (i) $\omega^2 = \dfrac{3Gm}{r^3} = \dfrac{GM}{R^3}$

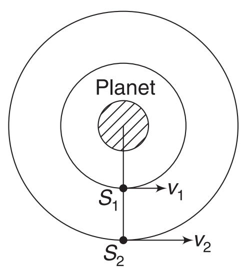
$\therefore$ *KE* of the star of smaller mass is

$$K_1 = \frac{1}{2}\,m(\omega r_1)^2$$

$$= \frac{1}{2}\cdot\frac{M}{3}\cdot\left(\frac{2r}{3}\right)^2\cdot\frac{3GM}{r^3}$$

$$= \frac{2}{9}\,\frac{GM}{r} = \frac{2}{9}\,\frac{GM}{R}$$

9. $T^2 \propto r^3$



$$\therefore \quad \left(\frac{T_2}{T_1}\right)^2 = \left(\frac{r_2}{r_1}\right)^3$$

$$\Rightarrow \quad \left(\frac{8}{1}\right)^2 = \left(\frac{r_2}{10^4}\right)^3$$

$$\Rightarrow \quad r_2 = 4 \times 10^4\,\text{km}$$

Speed of S_1 is $v_1 = \omega_1 r_1 = \left(\dfrac{2\pi}{1\,\text{h}}\right)(10^4\,\text{km})$

$$= 2\pi \times 10^4 \text{ kmh}^{-1}.$$

Speed of S_2 is $v_2 = \omega_2 r_2 = \left(\dfrac{2\pi}{8\,\text{h}}\right)(4 \times 10^4\,\text{km})$

$$= \pi \times 10^4 \text{ kmh}^{-1}.$$

$\therefore$ The position of the satellites when they are nearest is as shown in the figure.

Relative speed $= |v_1 - v_2| = \pi \times 10^4$ kmh^{-1}.

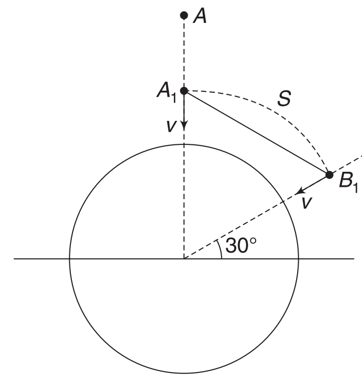
10. Angular speed

$$\omega = \frac{\text{Relative speed}}{\text{Distance } S_1S_2}$$

$$= \frac{\pi \times 10^4}{3 \times 10^4} = \frac{\pi}{3}\,\text{rad}\,\text{h}^{-1}.$$

11. Speed of each body at height $\dfrac{R}{2}$ can be calculated using conservation of energy as

$$\frac{1}{2}mv^2 - \frac{GMm}{R/2} = -\frac{GMm}{R}$$



$$\Rightarrow \quad v = \sqrt{\frac{2GM}{R}}$$

Angle between velocities of two bodies is 60° at this instant.

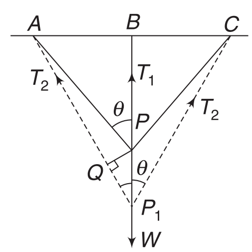
$\therefore$ Relative speed is

$$v_r = \sqrt{v^2 + v^2 + 2v\,v\cos 120^\circ} = v = \sqrt{\frac{2GM}{R}}$$

12. Distance between the bodies = chord length A_1B_1

$$= \left(\frac{3R}{2}\sin 30^\circ\right) \times 2 = \frac{3R}{2}.$$

13. After the weight is added, the joint P moves to P_1 Extension in BP is $PP_1 = \Delta l_1$ (say)



Drop a perpendicular from P on the line AP_1. Extension of wire AP is

$$QP_1 = \Delta l_2 \text{ (say)}$$

[This is because $AQ \simeq AP$]

Since extension are small,

$$\angle APB \simeq \angle AP_1B = \theta \text{ (say)}$$

$$P_1Q = PP_1\cos\theta \quad \Rightarrow \quad \Delta l_2 = \Delta l_1 \cos\theta \quad \text{...(i)}$$

$\therefore$ $\Delta l_1 > \Delta l_2$. It means extension in BP is more.

14. For BP, Stress = Y strain

$$\therefore \quad \frac{T_1}{A} = Y\cdot\frac{\Delta l_1}{l_1} \quad \Rightarrow \quad T_1 = YA\,\frac{\Delta l_1}{l_1}$$

For AP, $\dfrac{T_2}{A} = Y\dfrac{\Delta l_2}{l_2} \quad \Rightarrow \quad T_2 = YA\,\dfrac{\Delta l_2}{l_2}$

$$\therefore \qquad \frac{T_1}{T_2} = \frac{\Delta l_1}{\Delta l_2}\frac{l_2}{l_1} = \frac{1}{\cos\theta}\cdot\frac{1}{\cos\theta}$$

$$\Rightarrow \qquad T_2 = T_1\cos^2\theta \qquad ...(ii)$$

Therefore, $T_1 > T_2$

For vertical equilibrium

$$T_1 + 2T_2\cos\theta = W \qquad ...(iii)$$

$$\Rightarrow \quad T_1 + 2\cdot T_1\cos^3\theta = W \qquad \text{[using (ii)]}$$

$$\Rightarrow \qquad T_1 = \frac{W}{1 + 2\cos^3\theta}$$

$$= \frac{W}{1 + 2\left(\frac{l_1}{l_2}\right)^3}$$

15. The time in which the planet rotates about its axis is not given for either planet.

16. For geostationary satellite, time period = 1 planet day

Let $T = 1$ planet day

$T_0 = 1$ planet year

Now $T_2 = \frac{4\pi^2}{Gm}r_G^3 = \frac{4\pi^2}{Gm}\left(\frac{m}{M}\right)r^3 = T_0^3 \quad \Rightarrow T = T_0$

17. The energy of any geostationary satellite is the sum of kinetic energy of satellite, interaction energy of satellite and its own planet and interaction energy of satellite and star. Both planets have same mass and same length of day. Geostationary satellite–planet system will have same interaction energy in either planet. Also, kinetic energy of both satellites will be same. But the satellite-star system will account for the energy difference.

$$U_i = -\frac{GMm_0}{2r} + U_{\text{satellite}-\text{planet}}$$

$$U_f = -\frac{GMm_0}{2(4r)} + U_{\text{satellite}-\text{planet}}$$

$$E_{\min} = U_f - U_i = \frac{3GMm_0}{8r}$$

Centre of Mass and Momentum

AIEE/JEE Main Questions

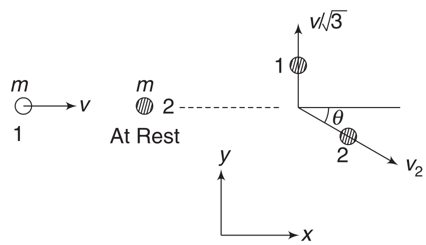
1. Momentum conservation along x and y directions gives:

$$mv = mv_2 \cos\theta$$

$$\Rightarrow \quad v = v_2 \cos\theta \quad ...(a)$$

And $$m\frac{v}{\sqrt{3}} = mv_2 \sin\theta$$

$$\Rightarrow \quad \frac{v}{\sqrt{3}} = v_2 \sin\theta \quad ...(b)$$



Squaring and adding the two equation gives

$$v_2^2(\sin^2\theta + \cos^2\theta) = v^2 + \frac{v^2}{3}$$

$$\Rightarrow \quad v_2 = \frac{2}{\sqrt{3}}v$$

2. The block compresses the spring and then again returns to its original position (where the spring is relaxed). At this position, the KE of the block is maximum (equal to its original KE). This is when the momentum of the block is maximum too.

KE when spring is relaxed = PE when spring has maximum compression.

$$\therefore \quad \frac{1}{2}Mv^2 = \frac{1}{2}KL^2$$

$$\Rightarrow \quad v = \sqrt{\frac{K}{M}}L \quad \Rightarrow \quad Mv = \sqrt{MK}\cdot L$$

3. The pieces must have equal and opposite momenta

$$\therefore \quad 4v = 12 \times 4 \quad \Rightarrow \quad v = 12\,\text{ms}^{-1}$$

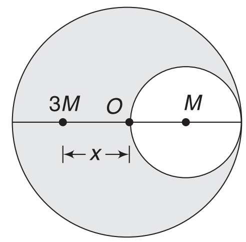
$$\therefore \quad KE = \frac{1}{2} \times 4 \times 12^2 = 288\,\text{J}$$

4. $m_1\Delta x_1 = m_2\Delta x_2$

With Δx_1 and Δx_2 in opposite directions.

$$\therefore \quad \Delta x_2 = \frac{m_1}{m_2}\cdot d$$

5. Mass of removed part = M [Area = πR^2]



Mass of remaining part = $3M$ [Area = $3\pi R^2$]

$$\therefore \quad 3M\cdot x = M\cdot R \quad \Rightarrow \quad x = \frac{R}{3}$$

6. We must pay attention to significant digits.

$$P = m\cdot v$$

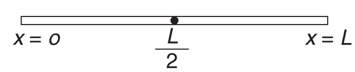
$$M = 3.513\,\text{kg (4 significant digits)}$$

$$v = 5.00\,\text{ms}^{-1}\text{ (3 significant digits)}$$

$$P = 17.565\,\text{kg m s}^{-1} \simeq 17.6\ \text{kgms}^{-1}$$

The final answer must be rounded off to keep three significant digits.

7. For $n = 0$, linear density is a constant. The COM lies at $x = \frac{L}{2}$.



For any other value of n, the rod is heavier towards its end at $x = L$. The COM moves to the right of $x = \frac{L}{2}$.

As n becomes larger and larger, the linear density becomes higher and higher. The COM moves to the right for every rise is value of n. In extreme case of $n \to \infty$, the rod becomes extremely massive at its right end and has near zero density at locations close to $x = 0$. The COM moves very close to $x = L$.

8. $(0.5 + 1)v = 0.5 \times 2 \quad \Rightarrow \quad v = \frac{2}{3}\,\text{ms}^{-1}$

$$\text{Energy loss} = \frac{1}{2} \times 0.5 \times 2^2 - \frac{1}{2} \times 1.5 \times \left(\frac{2}{3}\right)^2$$

$$= 0.67\,\text{J}$$

9. In all options, h is positive. This implies that upward direction has been considered positive. When the ball begins to fall, its speed increases linearly. Velocity is downward (negative). As soon as it hits the floor, its velocity becomes upward (positive) without change in magnitude. There is no time lag in velocity becoming $+ve$ from $-ve$, as the duration of collision is negligible.

By doing this much analysis only we can choose the answer (*C*).

However, it is recommended that you plot h versus t graph on your own.

10. Slope of $x-t$ graph is velocity.

Velocity just before $t = 2\text{s}$ is v_1 = slope of graph

$$= \frac{2}{2} = 1\,\text{ms}^{-1}$$

Velocity just after $t = 2\text{s}$ is v_2 = slope = $-1\,\text{ms}^{-1}$

Impulse acted at $t = 2\text{s}$ and changed the velocity from to v_1 to v_2.

$$\therefore \quad |\text{Impulse}| = |m(v_2 - v_1)| = 0.4 \times 2 = 0.8\,\text{Ns}$$

Similar impulse acted on the body at $t = 4, 6, 8\text{s}$ ...

11. To conserve momentum, there must be some velocity after collision. This implies that there cannot be a complete loss of kinetic energy.

12. Maximum energy is lost in perfectly inelastic collision.

$$\text{Loss} = \frac{1}{2}\mu v_{\text{rel}}^2 = \frac{1}{2}\left(\frac{mM}{m+M}\right)v^2 = f\left(\frac{1}{2}mv^2\right)$$

where $\quad f = \dfrac{M}{M+m}$

13. Let velocity after collision be $v = v_x\hat{i} + v_y\hat{j}$

$$3mv_x = m(2v) \quad \Rightarrow \quad v_x = \frac{2}{3}v$$

And $\quad 3mv_y = 2mv \quad \Rightarrow \quad v_y = \dfrac{2}{3}v$

$$\therefore \quad v = \sqrt{v_x^2 + v_y} = \frac{2\sqrt{2}}{3}v$$

Loss in energy

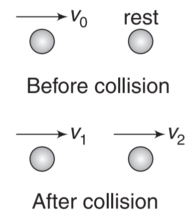
$$= \frac{1}{2}m(2v)^2 + \frac{1}{2}m(v)^2 - \frac{1}{2}(3m)\left(\frac{2\sqrt{2}}{3}v\right)^2$$

$$= \frac{5}{3}mv^2$$

$$\%\ \text{loss} = \frac{\frac{5}{3}mv^2}{\frac{1}{2}m(2v)^2 + \frac{1}{2}2m(v)^2} \times 100 \simeq 56\%$$

15. $$\frac{1}{2}m(v_1^2 + v_2^2) = 1.5\frac{1}{2}mv_0^2$$

or, $\quad v_1^2 + v_2^2 = 1.5\,v_0^2 \quad$ ...(i)



Momentum conservation gives

$$mv_1 + mv_2 = mv_0$$

$\Rightarrow \quad v_1 + v_2 = v_0 \quad$ ...(ii)

Now, $\quad (v_1 + v_1)^2 = v_0^2$

or $\quad 1.5v_0^2 + 2v_1v_2 = v_0^2 \quad \Rightarrow 2v_1v_2 = -0.5v_0^2 \quad$ ...(iii)

Now, $\quad (v_1 - v_2)^2 = v_1^2 + v_2^2 - 2v_1v_2$

$$= 1.5\,v_0^2 + 0.5v_0^2$$

or $\quad v_1 - v_2 = \sqrt{2}\,v_0$

IIT JEE (Advanced) Questions

1. Both statements are true but only statement 2 cannot lead to conclusion in statement 1.

2. $\vec{P}_1' + \vec{P}_2' = 0$

This is possible in (*b*) and (*c*) but not possible in (*a*) and (*d*).

3. (i) Velocity of the block just before reaching *B* can be calculated using energy conservation.

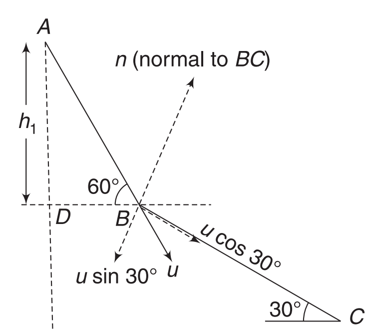
$$\frac{1}{2}mu^2 = mgh_1$$

$$u^2 = 2gh_1 = 2g\,(BD\tan 60°)$$

$$= 2 \times 10 \times \sqrt{3} \times \sqrt{3} = 60$$

$$\therefore \quad u = \sqrt{60}\ \text{ms}^{-1}$$

Component of u along BC is $u\cos 30° = \sqrt{45}\ \text{ms}^{-1}$



This remains unchanged and the velocity component normal to *BC* becomes zero after collision.

(ii) Height of *B* (above *C*) is

$$h_2 = 3\sqrt{3}\tan 30° = 3\ \text{m}$$

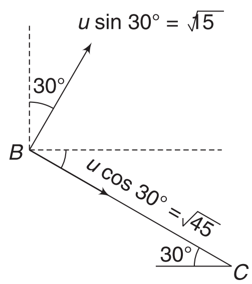
Energy conservation between B and C gives

$$\frac{1}{2} mv^2 = mgh_2 + \frac{1}{2}m(\sqrt{45})^2$$

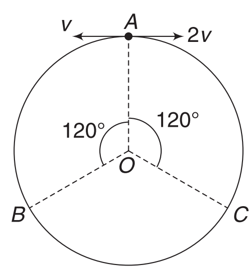
$$\Rightarrow \quad v^2 = 2 \times 10 \times 3 + 45 = 105$$

$$\therefore \quad v = \sqrt{105}\ \text{ms}^{-1}$$

(iii) If collision is elastic, velocity component $u \sin 30°$ gets inverted and component $u \cos 30°$ down the incline remains unchanged. Vertical component of velocity is $\sqrt{15}\cos 30° - \sqrt{45}\sin 30° = 0$.



5. First collision occurs at B, as the particle having speed $2u$ covers double the distance travelled by the particle having speed v.



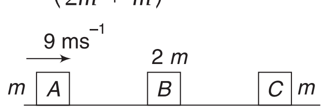
At B, the velocities get exchanged and the next collision will occur at C. Once again, velocities are exchanged and the particles meet at A.

6. y_{CM}

$$= \frac{6m(0) + m(a) + m(a) + m(0) + m(-a)}{6m + m + m + m + m} = \frac{a}{10}$$

7. Speed of B after elastic collision between A and B is

$$v = \left(\frac{2m}{2m + m}\right) \times 9 = 6\,\text{ms}^{-1}$$



After inelastic collision of B and C, speed is

$$v_0 = \frac{2m \cdot v}{2m + m} = 4\,\text{ms}^{-1}$$

8. Area under $F-t$ graph = Impulse = Change in momentum

$$\therefore \quad P = \frac{1}{2} \times 4 \times 3 - \frac{1}{2} \times 1.5 \times 2 = 4.5\,\text{kg m s}^{-1}$$

[Note that the straight-line graph tells you that F is 2N at 4.5 s]

$$\therefore \quad k = \frac{P^2}{2m} = \frac{4.5^2}{2 \times 2} = 5.06\,\text{J}$$

9. Let initial speed of 1 kg mass be u before collision and speed of 5 kg mass be v after collision.

$$1 \cdot u = -1(2) + 5v$$

$$\Rightarrow \quad u = -2 + 5v \qquad \text{...(i)}$$

Speed of approach = speed of separation

$$\Rightarrow \quad u = 2 + v \qquad \text{...(ii)}$$

Solving (i) and (ii) gives: $v = 1\,\text{ms}^{-1}$; $u = 3\,\text{ms}^{-1}$

$$\therefore \; P_{\text{initial}} = 1 \times 3 = 3\,\text{kg m s}^{-1}$$

$$v_{CM} = \frac{1 \times 3}{1 + 5} = 0.5\,\text{ms}^{-1}$$

$$\frac{1}{2} M v_{CM}^2 = \frac{1}{2} \times 6 \times 0.5^2 = 0.75\,\text{J}$$

10. For ball: $20 = v_1 \sqrt{\dfrac{2h}{g}}$

$$\Rightarrow \quad 20 = v_1 \sqrt{\frac{2 \times 5}{10}} \quad \Rightarrow \quad v_1 = 20\,\text{ms}^{-1}$$

For bullet: $100 = v_2 \sqrt{\dfrac{2h}{g}} \quad \Rightarrow \quad v_2 = 100\,\text{ms}^{-1}$

Momentum conservation

$$0.01v = 0.2 \times 20 + 0.01 \times 100 \Rightarrow v = 500\,\text{ms}^{-1}$$

11. Speed of first bob at top of its trajectory is $\sqrt{gl_1}$ Exchange of velocity takes place during collision.

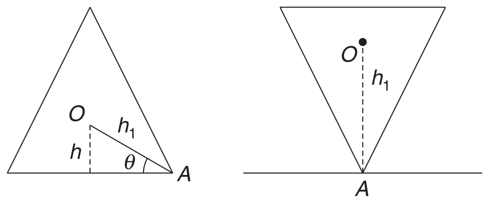
$\therefore$ Second bob acquires speed $\sqrt{gl_1}$ after collision. According to the question: $\sqrt{gl_1} = \sqrt{5gl_2}$

$$\Rightarrow \quad \frac{l_1}{l_2} = 5$$

12. In the figure, a triangle has been shown, but it can by any polygon. Sum of all interior angles

$$= (n - 2)\pi$$

$$\therefore \quad \theta = \frac{1}{2}\frac{(n-2)\pi}{n} = \left(\frac{\pi}{2} - \frac{\pi}{n}\right)$$



Clearly (from first figure): $h_1 = \dfrac{h}{\sin\theta} = \dfrac{h}{\sin\left(\dfrac{\pi}{2} - \dfrac{\pi}{n}\right)}$

$$= \frac{h}{\cos\dfrac{\pi}{n}}$$

$$\therefore \qquad \Delta h = h_1 - h = h\left[\frac{1}{\cos\dfrac{\pi}{n}} - 1\right]$$

13. COM of the system does not move. Let displacement of the block be $d(\leftarrow)$.

Displacement of the point mass = $(R-d)(\rightarrow)$

$$Md = m(R-d) \quad \Rightarrow \quad d = \frac{mR}{M+m} \ (\leftarrow)$$

$\therefore$ Displacement of COM of the block = $-\dfrac{mR}{M+m}$

x co-ordinate of point mass is same at x co-ordinate of the right edge of the block.

For finding velocities, first we use conservation of momentum.

$$mv + mV = 0 \qquad \text{...(i)}$$

Energy conservation gives:

$$\frac{1}{2}mv^2 + \frac{1}{2}MV^2 = mgR \qquad \text{...(ii)}$$

Solving (i) and (ii) gives velocity (v) of particle as well as velocity of the block (V).

14. Speed gained by the black due to the impulse is

$$v_o = \frac{J}{m} = \frac{1}{0.4} = 2.5 \text{ m/s}$$

Given, $\quad v = v_o\, e^{-t/\tau}$ [v_o = velocity at $t = 0$]

or, $\quad \dfrac{dx}{dt} = 2.5\, e^{-t/\tau}$

or, $\quad \displaystyle\int_o^x dx = 2.5 \int_o^\tau e^{-t/\tau}\, dt$

or, $\quad x = -2.5\tau\,[e^{-t/\tau}]_o^t$

or, $\quad x = -2.5 \times 4\,[e^{-1} - e^o]$

$$= 10[1 - 0.37] = 6.30 \text{ m}$$

Rotational Motion

AIEE/JEE Main Questions

1. In an RF attached to the centre, both particle are performing circular motion with speeds

$$v_1 = \omega R_1 \quad \text{and} \quad v_2 = \omega R_2$$

Accelerations are: $a_1 = \dfrac{v_1^2}{R_1} = \omega^2 R_1$

$$a_2 = \omega^2 R_2$$

Accelerations in ground frame are also a_1 and a_2 because the centre has no acceleration.

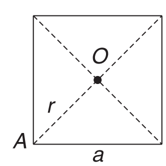
$$\therefore \quad \frac{F_1}{F_2} = \frac{a_1}{a_2} = \frac{R_1}{R_2}$$

(Assuming the particles to be of equal masses)

4. $$I_0 = \frac{Ma^2}{6}$$

$$I_A = I_0 + Mr^2 = \frac{Ma^2}{6} + M\left(\frac{a}{\sqrt{2}}\right)^2$$

$$= \frac{2}{3}Ma^2$$



5. As insect moves towards centre, MI decreases and hence angular speed increases. When the insect moves away from the centre, MI increases and angular speed decreases.

6. Torque, $\tau = rF = 2(20t - 5t)^2$

$\therefore$ Angular acceleration, $\alpha = \dfrac{\tau}{I} = 4t - t^2$

$$\therefore \qquad \frac{d\omega}{dt} = 4t - t^2 \quad \Rightarrow \quad \int_0^\omega d\omega = \int_0^t (4t - t^2)\,dt$$

$$\Rightarrow \qquad \omega = 2t^2 - \frac{t^3}{3}$$

$$\omega = 0 \quad \text{when} \quad t = 6\text{ s}$$

Now $\quad \omega = \dfrac{d\theta}{dt}$

$$\therefore \qquad \frac{d\theta}{dt} = 2t^2 - \frac{t^3}{3}$$

$$\Rightarrow \qquad \int_0^\theta d\theta = \int_0^6 \left(2t^2 - \frac{t^3}{3}\right)dt$$

$$\theta = 36 \text{ rad}$$

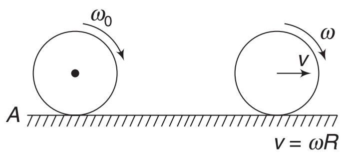
$$\Rightarrow \quad 2\pi n = 36 \quad \Rightarrow \quad n = \frac{36}{2\pi} < 6$$

8. Angular momentum conservation about A gives:

$$mR^2\omega_0 = mvR + mR^2 \cdot \omega$$

$$\Rightarrow \quad mR^2\omega_0 = mvR + mvR \qquad [\because \quad v = \omega R]$$

$$\Rightarrow \quad v = \frac{\omega_0 R}{2}$$



9. $mg - T = ma$ ...(i)

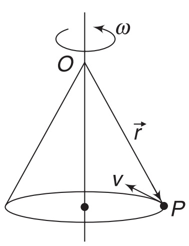
For cylinder: $TR = mR^2 \cdot \alpha$

$\Rightarrow \quad T = ma \quad [\because R\alpha = a]$ ...(ii)

From (i) and (ii); $a = g/2$

10. Particle (P) moves in a circle forming a conical pendulum. Angular momentum about O is

$$\vec{L} = \vec{r} \times (m\vec{v})$$



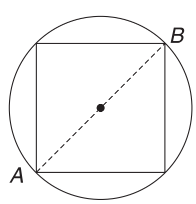
The magnitude of $\vec{r}$, $\vec{v}$ and angle between $\vec{r}$ and $\vec{v}$ does not change with time. Hence, magnitude of $\vec{L}$ is constant.

But plane containing $\vec{r}$ and $\vec{v}$ is changing. Hence, direction of $\vec{L}$ is changing.

11. Diagonal of cube = $2R$

$$\sqrt{3}\,a = 2R$$

$$a = \frac{2R}{\sqrt{3}}$$



Density, $\rho = \dfrac{M}{\frac{4}{3}\pi R^3}$

Mass of Cube $m = a^3 \cdot \rho = \left(\dfrac{2R}{\sqrt{3}}\right)^3 \left(\dfrac{M}{\frac{4}{3}\pi R^3}\right) = \dfrac{2M}{\sqrt{3} \cdot \pi}$

Momenta of inertia is $I = \dfrac{ma^2}{6} = \dfrac{4MR^2}{9\sqrt{3} \cdot \pi}$

12. For the particle moving along AB

$$\vec{L} = m(\vec{R} \times \vec{v}) = m\left(\frac{R}{\sqrt{2}}\,\hat{i} + \frac{R}{\sqrt{2}}\,\hat{j}\right) \times (v\hat{i})$$

$$= -\frac{mvR}{\sqrt{2}}\,\hat{k}$$

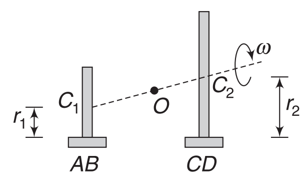
Similarly, one can find angular momentum for other segments of the path.

Though the question was intended to have one correct answer, it has two correct answers, viz (b) and (d)

13. For a moment, assume that the double cone does not deviate and its centre O continues on a straight line. After moving some distance, the left cone will be touching rail AB on a circle of radius r_1 and right cone will be touching the rail CD on a circle of radius r_2 with $r_2 > r_1$. If angular speed of the cone is ω and it is rolling, then speeds of the center C_1 and C_2 of the circles touching the rails will be

$$v_1 = \omega r_1 \quad \text{and } v_2 = \omega r_2$$

Clearly $\quad v_2 > v_1$



The double cone will steer towards left. [What will happen if the centre of right wheel of your car moves faster than the centre of left wheel? The car steers to the left. If the left tire of the car bursts suddenly, the car will steer to the left on its own causing serious accident].

14. Assuming the mass to be constant,

$$\pi R^2 l\rho = m \quad \Rightarrow R^2 = \frac{m}{\pi\rho l}$$

Now, the said MI is

$$I = \frac{ml^2}{12} + \frac{mR^2}{4} = \frac{m}{4}\left(\frac{l^2}{3} + R^2\right)$$

$$= \frac{m}{4}\left(\frac{l^2}{3} + \frac{m}{\pi l\rho}\right) \qquad \text{...(i)}$$

For minium I, $\dfrac{dI}{dl} = 0$

$$\Rightarrow \quad \frac{m}{4}\left(\frac{2l}{3} - \frac{m}{\pi l^2\rho}\right) = 0 \quad \Rightarrow \quad \frac{2l}{3} = \frac{m}{\pi l^2\rho}$$

$$\Rightarrow \quad \frac{2l}{3} = \frac{\pi R^2 l\rho}{\pi l^2\rho}$$

or, $\frac{l^2}{R^2} = \frac{3}{2} \Rightarrow \frac{l}{R} = \sqrt{\frac{3}{2}}$

One may choose to express equation (i) in terms of R, instead of l.

15. $I\alpha = \tau$

$$\frac{Ml^2}{3}\cdot\alpha = Mg\,\frac{l}{2}\sin\theta \Rightarrow \alpha = \frac{3}{2}\frac{g}{l}\sin\theta$$

16. Moment of inertia of the entire system about an axis passing through O is (O is COM of the system)

$$I_{CM} = \frac{1}{2}MR^2 + 6\left[\frac{1}{2}MR^2 + M(2R)^2\right] = \frac{55}{2}MR^2$$

$$I_P = I_{CM} + 7M(3R)^2$$

$$= \frac{55}{2}MR^2 + 63MR^2 = \frac{181}{2}MR^2$$

17. Area of removed part is $\frac{1}{9}$ of the area of the original disc

$\therefore$ mass of removed part = M

$$I = I_{\text{completed disc}} - I_{\text{removed disc}}$$

$$= \frac{1}{2}(9M)R^2 - \left[\frac{1}{2}M\left(\frac{R}{3}\right)^2 + M\left(\frac{2R}{3}\right)^2\right] = 4\,MR^2$$

IIT JEE (Advanced) Questions

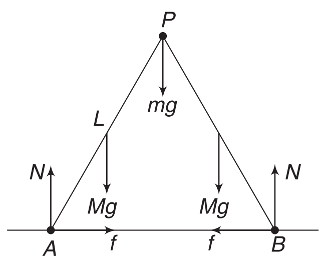
1. $a = \dfrac{g\sin\theta}{k^2/k^2 + 1} = \dfrac{g\sin\theta}{\frac{1}{2}+1} = \dfrac{2g\sin\theta}{3}$

2. Normal force at A and B are same due to symmetry. Friction at A is towards right and at B it is towards left. Both friction forces must have same magnitude for horizontal equilibrium of the system.

For vertical equilibrium

$$2N = 2Mg + mg$$

$$\Rightarrow \quad N = Mg + \frac{mg}{2}$$



Consider rotational equilibrium of segment AP and taking torque about P to be zero gives:

$$Mg\,\frac{L}{2}\cdot\cos\theta + f\cdot L\sin\theta = N\cdot \mathrm{L}\cdot\cos\theta$$

$$\Rightarrow \quad f\sin\theta = \left(Mg + \frac{mg}{2}\right)\cos\theta - \frac{1}{2}Mg\cos\theta$$

$$\Rightarrow \quad f = \frac{1}{2}(M+m)g\cdot\cot\theta$$

Note that by considering torque about P, we have to eliminate the need to think about force applied by BP on AP.

4. Area of removed part = $\frac{\pi R^2}{9}$

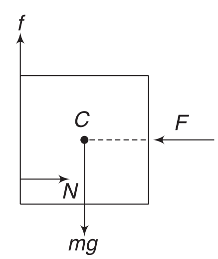
$\therefore$ Mass of removed part = M

$$I = I_{\text{whole}} - I_{\text{removed}}$$

$$= \frac{1}{2}(9M)(R)^2 - \left[\frac{1}{2}(M)\left(\frac{R}{3}\right)^2 + M\left(\frac{2R}{3}\right)^2\right]$$

$$= 4MR^2$$

5. For horizontal and vertical equilibrium $F = N$ and $f = mg$. For rotational equilibrium about the centre C, line of normal force (N) must pass below C. This will ensure that torque due to f gets balanced by torque due to N.



6. $\frac{1}{2}Mr^2 + Mr^2 = \frac{2}{5}MR^2$

$$\Rightarrow \quad r = \frac{2}{\sqrt{15}}R$$

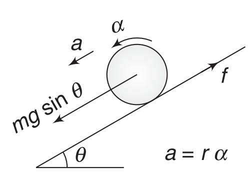
Thickness has no role to play.

7. Friction is not dissipative in pure rolling. Friction provides torque about centre and increases angular speed.

$$mg\sin\theta - f = ma \qquad \text{...(i)}$$

And $\quad f\cdot r = \frac{2}{5}mr^2\cdot\alpha \Rightarrow f = \frac{2}{5}ma$ ...(ii)

Solving (i) and (ii) gives: $f = \frac{2}{7}mg\sin\theta$ ...(iii)



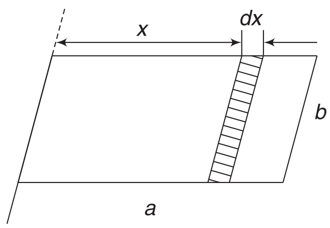
If θ decreases, friction will decrease. Static friction force is given by (iii). It is not equal to μN.

8. Energy of the ball is conserved. From A to B, rolling is pure and friction is static. There is no dissipation of energy. From B to C, there is no friction.

   KE of the ball is maximum at B since PE is least there. At B, the total KE of the ball is not translational. A part of it is rotational also. This rotational energy does not change from B to C, as there is no friction. At C, the ball has same rotational energy as at B, though its linear speed is zero. Thus, total energy at C is not potential energy.

$$h_C < h_A$$

9. Consider a strip of width dx on the plate as shown. Area of strip = $a\,dx$. Number of balls hitting the strip per second = $(a\,dx)\,n$.



Change in momentum of each ball = $2mv$

Force on the strip, $dF = (2mv)(an\,dx)$

Torque due to this force about the hinged side is

$$d\tau = x \cdot dF = 2mv\,an\,x\,dx$$

Total torque $\tau = 2mv\,an \int_{\frac{b}{2}}^{b} x\,dx = \frac{3}{4}\,mv\,ab^2 n$

This torque balances the torque due to weight

$$\therefore \quad \frac{3}{4}\,mv\,ab^2 n = Mg \cdot \frac{b}{2}$$

$$\Rightarrow \quad v = \frac{2}{3}\,\frac{Mg}{m\,abn} = \frac{2}{3} \times \frac{3 \times 10}{0.01 \times 1 \times 2 \times 100}$$

$$= 10\,\text{ms}^{-1}$$

10. Energy is conserved in pure rolling

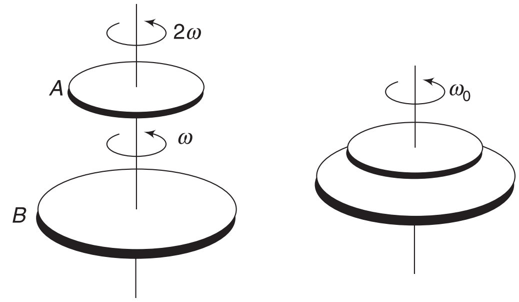
$$\frac{1}{2}\,mv^2 + \frac{1}{2} I \left(\frac{v}{R}\right)^2 = mg\left(\frac{3v^2}{4g}\right)$$

$$\Rightarrow \quad I = \frac{1}{2}\,mR^2$$

$\therefore$ Body is disc.

12. (i) $\frac{1}{2}\,I(2\omega)^2 = \frac{1}{2}\,kx_1^2$ and $\frac{1}{2}\,(2I)\,\omega^2 = \frac{1}{2}\,kx_2^2$

$$\therefore \quad \frac{x_1}{x_2} = \sqrt{2}$$



(ii) Let final common angular velocity be ω_0

$$(I + 2I)\,\omega_0 = I \cdot 2\omega + 2I \cdot \omega$$

$$\Rightarrow \quad \omega_0 = \frac{4}{3}\,\omega$$

For disc A:

Change in angular momentum is

$$|\Delta L| = I \cdot 2\omega - I\left(\frac{4}{3}\,\omega\right) = \frac{2}{3}\,I\omega$$

Since angular impulse = ΔL

$$\therefore \quad \tau \cdot t = \frac{2}{3}\,I\omega \quad \Rightarrow \quad \tau = \frac{2I\omega}{3t}$$

(iii) Loss in $KE = k_i - k_f$

$$= \left[\frac{1}{2}\,I(2\omega)^2 + \frac{1}{2}(2I)\,\omega^2\right] - \frac{1}{2}\,(3I)\left(\frac{4}{3}\,\omega\right)^2$$

$$= \frac{1}{3}\,I\omega^2$$

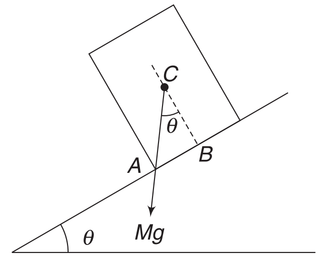
13. On an incline, $a_{\text{solid}} > a_{\text{cylinder}}$

    Thus, statement I is false.

15. If $v_B = v$ then $v_C = 2v$ and $v_A = 0$

16. The block will begin to slide when $\tan\theta > \mu$

$$\Rightarrow \quad \tan\theta > \sqrt{3} \quad \Rightarrow \quad \theta > 60°$$



The block will topple, if vertical line through COM passes from outside the base. Critical case has been shown in the figure.

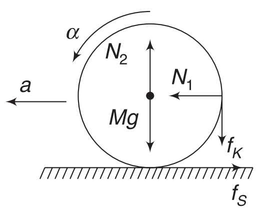
$$\tan\theta = \frac{AB}{BC} = \frac{10/2}{15/2} = \frac{2}{3}$$

This angle is much less than 60°. Thus, the block will topple even before it begins to slide.

17. Though not stated explicity, in the context of the problem, we assume that 2 N is the normal force applied by the stick on the ring.

$$N_1 = 2 \text{ N}$$

As the ring rolls, the stick applies a kinetic frciton force on it (if we assume that the stick is having no vertical motion). However, there is no slipping at the ground surface.



$$N_1 - f_s = ma \quad \Rightarrow \quad 2 - f_s = 2 \times 0.3$$

$$\Rightarrow \quad f_s = 1.4\,\text{N}$$

And $(f_s - f_k)R = I\alpha \quad \Rightarrow \quad f_s - f_k = m(R\alpha)$

$$\Rightarrow \quad 1.4 - f_k = 2 \times 0.3 \quad \Rightarrow \quad f_k = 0.8\,\text{N}$$

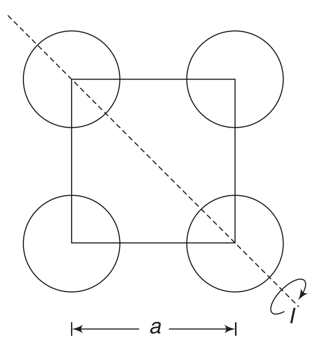
$$\therefore \quad \mu = \frac{f_k}{N_1} = \frac{0.8}{2} = 0.4$$

$$\therefore \quad P = 4$$

18. $$I = 2\left(\frac{2}{5}mr^2\right) + 2\left[\frac{2}{5}mr^2 + m\left(\frac{a}{\sqrt{2}}\right)^2\right]$$

$$= 9 \times 10^{-4} \text{ kgm}^2$$

$\therefore \; N = 9$

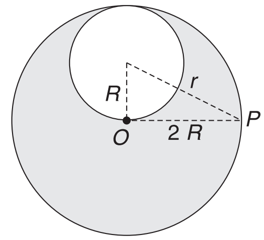


19. Mass of bigger disc = $4M$

Mass of removed part = M

$$I_0 = \frac{1}{2}4M(2R)^2 - \left[\frac{1}{2}MR^2 + MR^2\right]$$

$$= \frac{13}{2}MR^2$$



$$r = \sqrt{(2R)^2 + R^2}$$

$$I_p = \left[\frac{1}{2}4M(2R)^2 + (4M)(2R)^2\right] - \left[\frac{MR^2}{2} + M(4R^2 + R^2)\right]$$

$$= \frac{37}{2}MR^2$$

$$\frac{I_p}{I_0} = \frac{37/2}{13/2} \simeq 3$$

20. Energy is conserved in pure rolling

$$\frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2 = Mgh$$

$$\Rightarrow \quad \frac{1}{2}M(R\omega)^2 + \frac{1}{2}I\omega^2 = Mgh$$

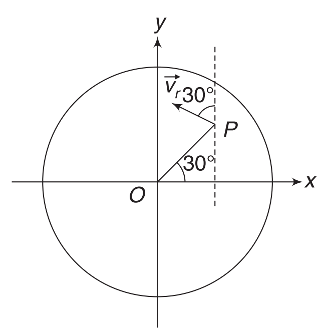
$$\Rightarrow \quad \frac{1}{2}[MR^2 + I]\omega^2 = Mgh$$

$\therefore$ ω is larger for smaller I. Cylinder Q has lower I.

21. The centre (O) must have a velocity towards the right, equal to

$$v = 3R \cdot \omega$$

$$\therefore \quad \vec{v}_0 = (3R\omega)\hat{i}$$



Velocity of P wrt O is as shown.

$$\vec{v}_r = -\frac{\omega}{2}R\sin 30°\,\hat{i} + \frac{\omega}{2}R\cos 30°\,\hat{j}$$

$$= -\frac{\omega R}{4}\hat{i} + \frac{\sqrt{3}}{4}\omega R\,\hat{j}$$

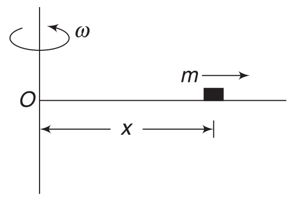
$$\Rightarrow \quad \vec{v}_p - \vec{v}_0 = -\frac{\omega R}{4}\hat{i} + \frac{\sqrt{3}}{4}\omega R\,\hat{j}$$

$$\Rightarrow \quad \vec{v}_p = \left(3R\omega - \frac{\omega R}{4}\right)\hat{i} + \frac{\sqrt{3}}{4}\,\omega R\,\hat{j}$$

$$= \left(\frac{11}{4}R\omega\right)\hat{i} + \left(\frac{\sqrt{3}}{4}R\omega\right)\hat{j}$$

22. At time t,

$$x = vt$$



Moment of inertia of the system about rotation axis (through O) is $I = I_0 + m(vt)^2$

Where $I_0 = MI$ of the rod.

$\therefore$ Angular momentum at time t is

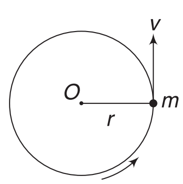
$$L = I\omega = [I_0 + mv^2t^2]\,\omega$$

L is changing as I is increasing. An external torque is needed to change L. [If there is no torque, the rod will slow down as the MI increases.]

$$\therefore \quad \tau = \frac{dL}{dt} = mv^2\omega(2t)$$

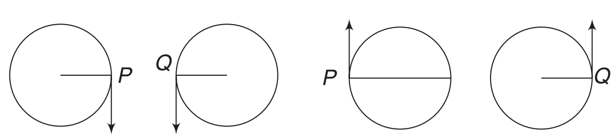
$\therefore$ Torque increases linearly with time. For $t > T$, no torque is needed.

23. Go through solution to Q.10 in AIEEE/JEE Main questions. Obviously, $\vec{L}_P$ Keeps changing directions. $\vec{L}_0$ is along the z-axis always and has fixed magnitude of mvr.



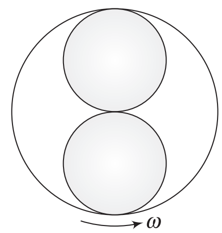
L_0 is directed perpendicular to this plane at all times.

24. One can easily observe that relative velocity will become zero twice in every rotation. Thus, (a) is the obvious choice.



25. Imagine yourself at the origin of a co-ordinate system. You always see the same face of the disc (just like we always see only one face of the moon). As the COM of the disc completes one rotation, the disc is back to its original orientation (in both cases). Thus, angular speed about COM is same as ω.

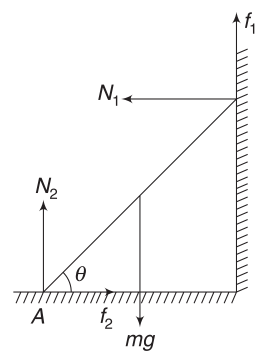
26. Conservation of angular momentum about vertical axis gives



$$\frac{50\,(0.4)^2}{2} \times 10 = \left[\frac{50(0.4)^2}{2} + 2(6.25)(0.2)^2 \times 2\right]\omega$$

$$\Rightarrow \quad \omega = 8\,\text{rad s}^{-1}$$

27. The ladder is about to slip. This means that friction is limiting.



when $\mu_1 = 0$ (in option (A) and (D))

$$N_1 = f_2 \quad \text{and} \quad N_2 = mg$$

Torque about $A = 0$

$$\Rightarrow \quad N_1 l \sin\theta = mg\,\frac{l}{2}\cos\theta$$

$$\Rightarrow \quad N_1 = \frac{mg}{2}\cot\theta$$

$$\Rightarrow \quad N_1 \tan\theta = \frac{mg}{2}$$

when $\mu_2 = 0$

There is no force to balance N_1. Hence, equilibrium is not possible.

when $\mu_1 \neq 0$, $\mu_2 \neq 0$

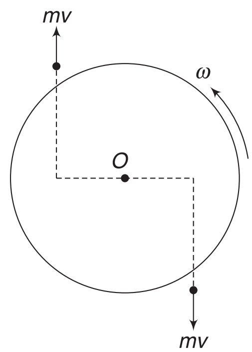
$$N_1 = \mu_2 N_2$$

$$N_2 + \mu_1 N_1 = mg$$

$$\Rightarrow \quad N_2 + \mu_1\mu_2 N_2 = mg \quad \Rightarrow N_2 = \frac{mg}{1 + \mu_1\mu_2}$$

28. Conservation of angular momentum gives

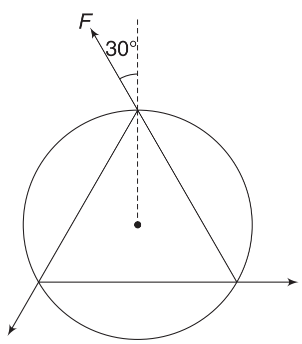
$$2mvr = \frac{MR^2}{2}\,\omega$$



$$\Rightarrow \quad \omega = \frac{4mvr}{MR^2} = \frac{4 \times 5 \times 10^{-2} \times 9 \times \frac{1}{4}}{45 \times 10^{-2} \times \frac{1}{4}}$$

$$= 4 \text{ rad s}^{-1}$$

29. $\tau = 3\,(F \sin 30°)\,R$



Angular impulse

$$= \tau t = 3FRt \sin 30°$$

$$= 3 \times 0.5 \times 0.5 \times 1 \times \frac{1}{2}$$

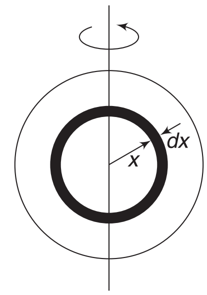
$$= \frac{3}{8} \text{ N ms}$$

$$I\omega = \frac{3}{8}$$

$$\Rightarrow \quad \frac{1}{2} \times 1.5 \times (0.5)^2\,\omega = \frac{3}{8}$$

$$\Rightarrow \quad \omega = 2 \text{ rad s}^{-1}$$

30. Consider a shell of radius x and thickness dx.



Its MI is $dI = \frac{2}{3}\,\rho(4\pi x^2 dx)x^2$

$$\therefore \quad I_A = \int_0^R \frac{2}{3}\,\rho_A\,(4\pi x^4 dx) = \frac{2}{3}\,\frac{k4\pi}{R}\int_0^R x^5 dx$$

$$= \frac{8\pi k}{3R}\,\frac{R^6}{6}$$

Similarly for B; $I_B = \int_0^R \frac{2}{3}\,\rho_B\,(4\pi x^4 dx)$

$$= \frac{8\pi k}{3R^5}\int_0^R x^9 dx = \frac{8\pi k}{3R^5}\,\frac{R^{10}}{10}$$

$$\therefore \quad \frac{I_B}{I_A} = \frac{6}{10}$$

31. Angular momentum is conserved. Initial $MI = MR^2$ (point masses at centre)
MI when one mass is at $\frac{3R}{5}$ from O and the other is at x is

$$I = MR^2 + \frac{M}{8}\left(\frac{3R}{5}\right)^2 + \frac{M}{8}x^2$$

$$\left[MR^2 + \frac{M}{8}\left(\frac{3R}{5}\right)^2 + \frac{M}{8}\,x^2\right]\cdot\frac{8\omega}{9} = MR^2\cdot\omega$$

$$\Rightarrow \quad x = \frac{4}{5}\,R$$

32. $\vec{r} = \alpha t^3\hat{i} + \beta t^2\hat{j} = \frac{10}{3}\,t^3\hat{i} + 5t^2\hat{j}$

$$\vec{v} = \frac{d\vec{r}}{dt} = 10t^2\hat{i} + 10t\,\hat{j}$$

At $t = 1.0$ s, $\vec{v} = (10\hat{i} + 10\hat{j})\,\text{ms}^{-1}$

$$\vec{L} = m(\vec{r} \times \vec{v}) = 0.1\left[\frac{10}{3}\hat{i} + 5\hat{j}\right] \times [10\hat{i} + 10\hat{j}]$$

$$= \frac{10}{3}\,\hat{k} - 5\hat{k} = -\frac{5}{3}\hat{k} \text{ N ms}$$

Force $\vec{F} = m\,\frac{d\vec{v}}{dt} = m(20t\hat{i} + 10\hat{j})$

At $\quad t = 1$; $\vec{F} = 0.1\,(20\,\hat{i} + 10\,\hat{j}) = (2\hat{i} + \hat{j})N.$

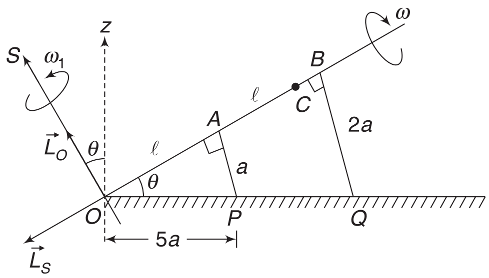
$$\tau = \vec{r} \times \vec{F} = \left(\frac{10}{3}\hat{i} + 5\hat{j}\right) \times (2\hat{i} + \hat{j})$$

$$= \frac{10}{3}\hat{k} - 10\hat{k} = -\frac{20}{3}\hat{k} \text{ Nm}$$

33. Speed of centre (A) of smaller disc is $v_A = \omega a$

[Rolling without slipping]

Note that the entire rod (and centres of the discs) is rotating about axis OS that is perpendicular to the rod. Let angular speed of the assembly about OS be ω_1. Speed of point A can also be written as



A and *B* are centres of the two discs. *C* is COM

$$v_A = \omega_1(OA) = \omega_1 l$$

$$\therefore \quad \omega_1 l = \omega a \Rightarrow \omega_1 = \omega\frac{a}{l}$$

Angular velocity $\vec{\omega}_1$ is directed along OS. Component of this velocity in z direction is angular velocity about z-axis

$$\therefore \ \omega_Z = \omega_1 \cos\theta = \omega\frac{a}{l}\cos\theta = \frac{\omega a}{l}\frac{OA}{OP}$$

$$= \frac{\omega a}{l}\frac{l}{5a} = \frac{\omega}{5} \quad \left[\because \ OP = \sqrt{24^2 a + a^2} = 5a\right]$$

[(B) is correct]

COM of the assembly is at C such that

$$AC = \frac{0 + 4m.l}{5m} = \frac{4l}{5}$$

Angular momentum about COM (or the angular momentum associated with spin) is

$$L_s = I_{CM}\cdot\omega \text{ (Direction is along } \overrightarrow{BO} \text{ as shown)}$$

$$= \left[\frac{ma^2}{2} + \frac{4m(2a)^2}{2}\right]\omega = \frac{17}{2}ma^2$$

[(*D*) is correct]

Angular momentum associated with motion of COM about O (which we have usually called $L_{orbital}$ in our discussion in the book) is

$$L_0 = (5m)\ (v_{CM})(OC) = 5m\ (\omega_1\cdot OC)(OC)$$

$$= 5m\ \omega\frac{a}{l}\left(\frac{9l}{5}\right)^2 \qquad \left[\because OC = l + \frac{4l}{5} = \frac{9l}{5}\right]$$

$$= \frac{81}{5}m\,\omega a\,l = \frac{81\sqrt{24}}{5}ma^2\omega \quad \text{[(A) is incorrect]}$$

Direction of $\vec{L}_0$ is along OS.

To find L_Z we need to add z components of $\vec{L}_0$ and $\vec{L}_S$.

$$L_Z = L_0\cos\theta - L_s\sin\theta$$

On substituting the value

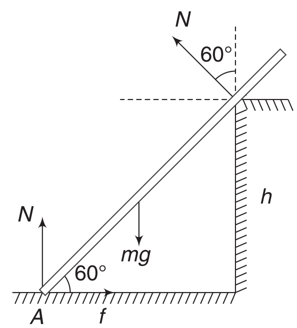
$$L_Z \simeq 80\,ma^2\omega$$

34. $N + N\cos 60° = mg$

$$\Rightarrow \qquad N = \frac{2mg}{3} \qquad \text{...(i)}$$

And $\qquad f = N\sin 60°$

$$= \frac{mg}{\sqrt{3}} = \frac{16\sqrt{3}}{3} \text{ N}$$



Taking torque about A to be zero.

$$mg\ \frac{l}{2}\cos 60° = N\cdot\frac{h}{\sin 60°}$$

$$\Rightarrow \quad \frac{mgl}{4} = \frac{2mg}{3}\frac{2h}{\sqrt{3}} \quad\Rightarrow\quad \frac{h}{l} = \frac{3\sqrt{3}}{16}$$

35. (i) $\vec{F}_{rot} = \vec{F}_{in} + 2m(\vec{v}_{rot} \times \vec{\omega}) + m(\vec{\omega} \times \vec{r}) \times \vec{\omega}$

The third term is directed radially outward (actually it is the centrifugal force).

$$\vec{\omega} \times \vec{r} = (\omega\hat{k}) \times (r\hat{i}) = \omega r\,\hat{j}$$

$$(\vec{\omega} \times \vec{r}) \times \vec{\omega} = (\omega r\,\hat{j}) \times (\omega\hat{k}) = \omega^2 r\,\hat{i}$$

The second term is a force that is perpendicular to the edge of the slot [i.e., along y direction] and gets balanced by the normal force of the slot wall.

F_{in} is zero as mg is balanced by normal force of the floor of the slot.

$\therefore$ In rotating frame,

$$m\ \frac{dv}{dt} = m\omega^2 r \qquad \text{...(a)}$$

$$\Rightarrow \quad v\frac{dv}{dr} = m\omega^2 r \quad \Rightarrow \quad \int_0^v v\,dv = \omega^2 \int_{R/2}^{r} r\,dr$$

$$\Rightarrow \quad \frac{v^2}{2} = \frac{\omega^2}{2}\left(r^2 - \frac{R^2}{4}\right)$$

$$\Rightarrow \quad v = \omega\sqrt{r^2 - \frac{R^2}{4}}$$

$$\Rightarrow \quad \frac{dr}{dt} = \omega\sqrt{r^2 - \frac{R^2}{4}}$$

$$\Rightarrow \quad \int_{R/2}^{r} \frac{dr}{\sqrt{r^2 - \frac{R^2}{4}}} = \omega \int_0^t dt$$

$$\Rightarrow \quad \left[ln\left(r + \sqrt{r^2 - \frac{R^2}{4}}\right)\right]_{\frac{R}{2}}^{r} = \omega t$$

[Refer a Calculus text for integration]

$$\Rightarrow \quad ln\left[\frac{r + \sqrt{r^2 - \frac{R^2}{4}}}{\frac{R}{2}}\right] = \omega t$$

$$\Rightarrow \quad r + \sqrt{r^2 - \frac{R^2}{4}} = \frac{R}{2}e^{\omega t}$$

$$\Rightarrow \quad r^2 - \frac{R^2}{4} = \left(\frac{R}{2}e^{\omega t} - r\right)^2$$

$$\Rightarrow \quad r^2 - \frac{R^2}{4} = \frac{R^2}{4}e^{2\omega t} + r^2 - Rre^{\omega t}$$

$$\Rightarrow \quad rRe^{\omega t} = \frac{R^2}{4}(1 + e^{2\omega t})$$

$$\Rightarrow \quad r = \frac{R}{4}(e^{\omega t} + e^{-\omega t}) \qquad \text{...(b)}$$

Alternatively, one can decide the answer by carefully looking at the options. Options (a) and (c) are incorrect as they tell that r decreases with time. The unbalanced centrifugal force causes the block to move radially outward.

In option (C), $r = \frac{R}{4}(e^{\omega t} + e^{-\omega t})$

$$\frac{d^2r}{dt^2} = \frac{R}{4}\omega^2(e^{\omega t} + e^{-\omega t})$$

at $\quad t = 0, \dfrac{d^2r}{dt^2} = \dfrac{R}{4}\omega^2(1 + 1) = \dfrac{R\omega^2}{2}$

This is correct from equation (a)

(ii) Horizontal normal reaction (in y direction) is equal to the term

$$2m\,\vec{v}_{\text{rot}} \times \vec{\omega} = 2m\frac{d\vec{r}}{dt} \times \omega$$

$$= 2m\frac{R}{4}\omega(e^{\omega t} + e^{-\omega t})\hat{i} \times (\omega\hat{k})$$

$$= -\frac{mR}{2}\omega^2(e^{\omega t} + e^{-\omega t})\hat{j}$$

The normal force by the slot wall balances this

$\therefore \quad N_y = \frac{1}{2}mR\omega^2(e^{\omega t} + e^{-\omega t})$

The normal force by the floor balances the weight

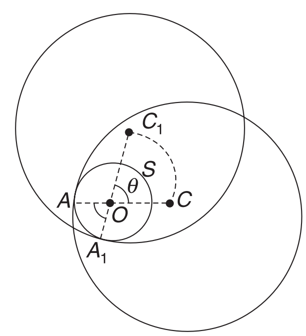
$\therefore \quad N_Z = mg$

$\therefore \quad N = \frac{1}{2}mR\omega^2(e^{\omega t} + e^{-\omega t})\hat{j} + (mg)\hat{k}$

36. The question is not clear.

The official answer key states the answer as (c) or (cd). I am not attempting any solution as I see lots of ambiguity.

37. Once again, the question is ambiguous. In official answer key, first question was awarded bonus marks and answer to the second question was given as (A).



However, let us discuss the problem as it appears interesting.

(i) In the figure shown, the axis of cone (traced by finger) passes through O. Contact point moves on circle S. As the contact point moves from A to A_1, the centre of the ring moves from to C to C_1.

Clearly, $\angle AOA_1 = \angle COC_1$

This implies that the centre of mass (C) of the ring rotates about O with an angular speed ω_0. (equal to the speed of contact point).

In fact, the whole ring is in pure rotation about axis through O.

$\therefore \quad KE = \frac{1}{2}I_0 \times \omega_0^2 = \frac{1}{2}[MR^2 + M(R - r)^2]\omega_0^2$

[$\because$ Distance, $OC = R - r$]

This is the correct value of KE.

We may also write KE as

$$KE = \frac{1}{2} MV_{CM}^2 + \frac{1}{2} I_{CM}\,\omega_0^2$$

$$= \frac{1}{2} M[(R-r)\,\omega_0]^2 + \frac{1}{2}[MR^2]\,\omega_0^2$$

$$= \frac{1}{2} M[(R-r)^2 + R^2]\,\omega_0^2$$

(ii) As discussed above, the COM of the ring is rotating in a circle of radius $(R - r)$ with angular speed ω_0.

Forces on it are:

Normal force by finger (N), Friction by finger (f) and weight (Mg)

For vertical equilibrium:

$$f\cos\theta = N\sin\theta + Mg.$$

$$\Rightarrow \quad f\cos\theta - N\sin\theta = Mg \qquad \text{...(a)}$$

Centripetal force is

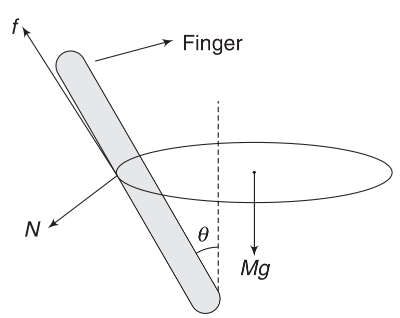
$$f\sin\theta + N\cos\theta = M\omega_0^2\,(R-r) \qquad \text{...(b)}$$

Dividing (b) by (a) gives

$$\frac{\omega_0^2 (R-r)}{g} = \frac{f\sin\theta + N\cos\theta}{f\cos\theta - N\sin\theta}$$

In limiting case $f = \mu N$

$$\therefore \quad \omega_0 = \sqrt{\frac{g}{(R-r)}\left(\frac{\mu\sin\theta + \cos\theta}{\mu\cos\theta - N\sin\,\theta}\right)}$$



We can have answer to this question only if θ is given. If we assume that θ is small (which is not stated in the question) then

$\sin\theta \to 0$ and $\cos\theta \to 1$

$$\therefore \qquad \omega_0 = \sqrt{\frac{g}{(R-r)\mu}}$$

38. Potential energy $V = \dfrac{kr^2}{2}$

Force $\qquad F = -\dfrac{dV}{dr} = -kr$

Negative sign indicates that the force is attractive. This force provides the centripetal force.

$$\frac{mv^2}{R} = kR \Rightarrow v = \sqrt{\frac{k}{m}}\cdot R$$

Angular momentum $L = mvR = \sqrt{mk}\cdot R^2$

39. Acceleration $\vec{a} = \dfrac{\vec{F}}{m} = t\hat{i} + \hat{j}$

$$\frac{d\vec{v}}{dt} = t\hat{i} + \hat{j} \quad \Rightarrow \int_o^v dv = \int_o^t (t\hat{i} + \hat{j})dt$$

or, $$\vec{v} = \frac{t^2}{2}\hat{i} + t\hat{j}$$

$$\Rightarrow \quad \frac{d\vec{r}}{dt} = \frac{t^2}{2}\hat{i} + t\hat{j} \Rightarrow \int_o^{\vec{r}} d\vec{r} = \int_o^t \left(\frac{t^2}{2}\hat{i} + t\hat{j}\right)dt$$

$$\Rightarrow \quad \vec{r} = \frac{t^3}{6}\hat{i} + \frac{t^2}{2}\hat{j}$$

$$\therefore \quad \vec{\tau} = \vec{r}\times\vec{F} = \left(\frac{t^3}{6}\hat{i} + \frac{t^2}{2}\hat{j}\right)\times(t\hat{i} + \hat{j})$$

$$= -\left(\frac{t^3}{3}\hat{k}\right)\text{ N-m}$$

At $t = 1s$

$$\vec{v} = \frac{1}{2}(\hat{i} + 2\hat{j})\text{ m/s}$$

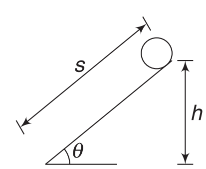
$$\vec{\tau} = -\frac{1}{3}\hat{k} \quad \Rightarrow |\vec{\tau}| = \frac{1}{3}\text{ N-m}$$

40. Length of the incline is

$$s = h/\sin\theta$$

Acceleation $a = \dfrac{g\sin\theta}{1 + \dfrac{I}{mR^2}}$

$$a_{ring} = \frac{g\sin\theta}{2}$$



$$a_{disc} = \frac{2g\,\sin\theta}{3}$$

$$s = \frac{1}{2}at^2 \qquad \Rightarrow \quad t = \sqrt{\frac{2s}{a}}$$

$$t_{ring} = \sqrt{\frac{4s}{g\,\sin\theta}}\ ; \quad t_{disc} = \sqrt{\frac{3s}{g\,\sin\,\theta}}$$

$$\therefore \sqrt{\frac{4s}{10\,\sin\,60°}} - \sqrt{\frac{3s}{10\,\sin\,60°}} = \frac{2-\sqrt{3}}{\sqrt{10}}$$

or $$h = \frac{3}{4} = 0.75\text{ m}$$

41. P represents a path:

$$x = \alpha t \quad \text{and} \quad y = \beta t$$

$$\therefore \quad \frac{y}{x} = \frac{\beta}{\alpha} \quad \Rightarrow y = \left(\frac{\beta}{\alpha}\right)x$$

This is a straight line.

And $\dfrac{dx}{dt} = \alpha \quad \Rightarrow a_x = \dfrac{d^2x}{dt^2} = 0$

$$\frac{dy}{dt} = \beta \quad \Rightarrow a_y = \frac{d^2y}{dt^2} = 0$$

Thus the particle is moving on a straight line with constant velocity.

$\therefore$ P, L, K, U and E are constants.

Similarly, one can show that Q represents an elliptical path. For such a path L and E are constants (think of motion of a planet).

R is a circular path. L, K, U and E are constant

Path S is parabolic

$$x = \alpha t; \quad y = \frac{\beta}{2}t^2$$

$$v_x = \alpha; \quad v_y = \beta t$$

$$\therefore \quad v = \sqrt{\alpha^2 + \beta^2 t^2}$$

$\therefore$ P and K are not constant.

$$a_x = o; \quad a_y = \beta$$

$$\therefore \quad \vec{F} = m\beta\hat{j}$$

$$dV = -\vec{F}\cdot\vec{dr} = -(mb\hat{j})\cdot\left(\alpha\, dt\, \hat{\imath} + \frac{P}{2}(2+dt)\hat{j}\right)$$

$$= -m\beta^2 t\, dt$$

$$\therefore \quad U = -\frac{m\beta^2 t^2}{2} + U_o$$

[U_o = a constant of integration]

Total energy $E = K + U$

$$= \frac{1}{2}m(\alpha^2 + \beta^2 t^2) - \frac{m\beta^2 t^2}{2} + U_o$$

or, $\quad E = \dfrac{1}{2}m\alpha^2 + U_o$ = a constant.

Fluid Mechanism

AIEEE/JEE Main Quesitons

1. In liquid 1, the ball sinks. It implies $\rho_3 > \rho_1$. The ball floats in liquid 2 with only a part submerged.

$$\therefore \quad \rho_2 > \rho_3$$

2. This can be easily understood from pervious question.

3. Bernoulli's equation:

$$\frac{1}{2}\rho v^2 + P_0 + 0 = \frac{1}{2}P\,(0.4)^2 + P_0 + \rho g(0.2\,m)$$

$$\Rightarrow \quad v^2 = 4.16$$

$$\Rightarrow \quad v = 2.04\,\text{ms}^{-1}$$

Continutiy equation:

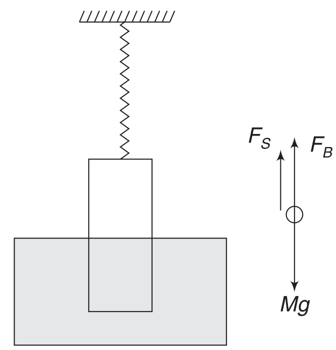
$$\pi\frac{d^2}{4}\cdot v = \pi\frac{(8\times10^{-3})^2}{4}\times 0.4$$

$$\Rightarrow \quad d^2 = 12.8\times10^{-6} \quad \Rightarrow \quad d \simeq 3.6\times10^{-3}\,\text{m}$$

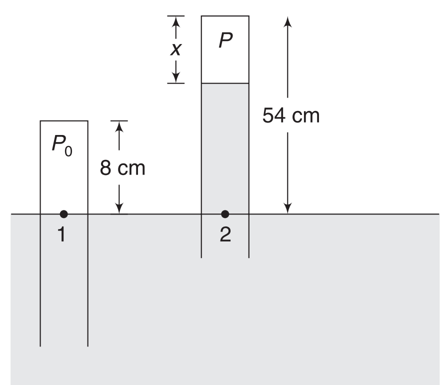
4. $$F_S + F_B = Mg$$

$$kx_0 + \frac{AL}{2}\sigma g = Mg$$

$$\Rightarrow \quad x_0 = \frac{Mg}{k}\left[1 - \frac{L\sigma A}{2M}\right]$$



5. This problem makes use of Boyle's law, which we will study in heat and thermodynamics section.



Air in the tube is sealed at atmospheric pressure (P_0). Its volume is $V_0 = (A\cdot 8)$ where A = cross-sectional area. When the tube is moved up, the air expands and pressure drops to P. Therefore, Hg rises in the tube.

Let final length of air column be x (in cm).

Final volume of air, $V = A\cdot x$

Using Boyle's law, $P_1V_1 = P_2V_2$

$$P_0\,(A \cdot 8\ ) = P(A \cdot x)$$

$$\Rightarrow \qquad P = \frac{76 \times 8}{x}$$

where $P_0 = 76$ cm of Hg.

Now pressure at point 2 must be atmospheric pressure.

$\therefore\ P + (54 - x) = 76$

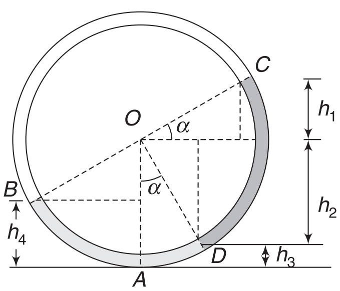
[pressure is being written in unit of height of Hg column]

$$\Rightarrow \quad \frac{76 \times 8}{x} + 54 - x = 76$$

Solving this equation gives $x = 16\,\text{cm}$

6. $\angle BOD = \angle DOC = 90°$

$$h_1 = R\sin\alpha$$
$$h_2 = R\cos\alpha$$
$$h_3 = R(1 - \cos\alpha)$$
$$h_4 = R(1 - \sin\alpha)$$



We will write pressure at A from two sides and equate them.

$$d_1 g h_4 = d_2 g\ (h_1 + h_2) + d_1 g h_3$$

$$\Rightarrow d_1(1 - \sin\alpha) = d_2(\sin\alpha + \cos\alpha) + d_1(1 - \cos\alpha)$$

$$\Rightarrow d_1[\cos\alpha - \sin\alpha] = d_2[\sin\alpha + \cos\alpha]$$

$$\Rightarrow \qquad \frac{d_1}{d_2} = \frac{1 + \tan\alpha}{1 - \tan\alpha}$$

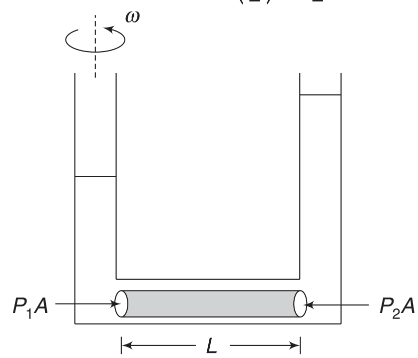
Alternatively, we can check the options under extreme conditions. For example, if $d_2 \to 0$, then $\alpha \to 45°$, $\dfrac{d_1}{d_2} \to \infty$ and the expression given in (c) also tends to infinity when $\alpha \to 45°$.

IIT JEE (Advanced) Questions

1. Consider a cylindrical volume of liquid as shown. Mass of liquid = ρAL. COM of this liquid cylinder is at a distance $\dfrac{L}{2}$ from the rotation axis.

$\therefore$ Centripetal force on considered cylindrical volume of liquid is

$$F_C = (\rho AL)\,\omega^2\left(\frac{L}{2}\right) = \frac{1}{2}\,\rho AL^2\omega^2$$



$$\therefore \qquad P_2A - P_1A = \frac{1}{2}\,\rho AL^2\omega^2$$

$$\Rightarrow \qquad P_2 - P_1 = \frac{1}{2}\,\rho L^2\omega^2$$

$$\Rightarrow \qquad \rho gH = \frac{1}{2}\,\rho L^2\omega^2 \quad \Rightarrow \quad H = \frac{\omega^2 L^2}{2g}$$

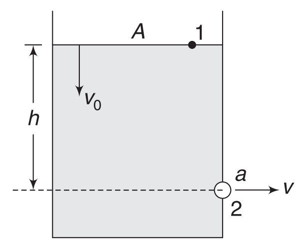
2. $h = 3.000 - 0.525 = 2.475\,\text{m}$

Speed of efflux = v

Speed of the water at the surface in the tank = v_0

$$Av_0 = av$$

$$\Rightarrow \qquad v_0 = \frac{av}{A} \qquad \text{...(i)}$$



Bernoullis's equation between 1 and 2.

$$P_0 + \frac{1}{2}\,\rho v^2 + 0 = P_0 + \frac{1}{2}\,\rho v_0^2 + \rho gh$$

$$\Rightarrow \qquad v^2\left[1 - \frac{a^2}{A^2}\right] = 2gh$$

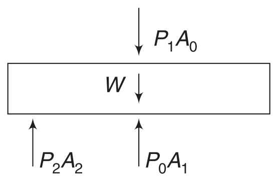
$$\Rightarrow \qquad v = \sqrt{\frac{2gh}{1 - \dfrac{a^2}{A^2}}} = \sqrt{\frac{2 \times 10 \times 2.475}{1 - (0.1)^2}} = 50$$

3. (i) When height of water is h_1, net force on the block is zero. Area of cross-section of the block is

$$A_0 = \pi(2r)^2 = 4\pi r^2$$

Area of lower face that is exposed to atmosphere is

$$A_1 = \pi r^2$$



Area of lower face that is in contact with water is

$$A_2 = A_0 - A_1 = 3\pi r^2$$

$\therefore \quad P_2A_2 + P_0A_1 = P_1A_0 + W$

$$[P_0 + \rho g h_1 + \rho g h][3\pi r^2] + P_0\ \pi r^2$$

$$= [P_0 + \rho g h_1]\ [4\pi r^2] + 4\pi r^2 \cdot h\ \frac{\rho}{3} g$$

$\Rightarrow \quad h_1 = \dfrac{5h}{3}$

(ii) When height of water is h_2, cylinder is again in equilibrium.

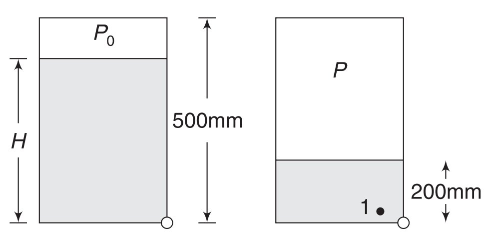
$\therefore \quad P_0 \cdot 4\pi r^2 + \dfrac{\rho}{3} \cdot 4\pi r^2 hg$

$$= (P_0 + \rho g h_2)(3\pi r^2) + P_0 \cdot \pi r^2$$

$\Rightarrow \quad h_2 = \dfrac{4h}{9}$

(iii) for $h < h_2$, the cylinder will not move up.

5. Initially, air above the water is at atmospheric pressure (P_0). As water flows out, the air expands and pressure (P) drops. When pressure in the tank at bottom (point 1) becomes equal to outside atmospheric pressure the flow of water stops.



$$P + \rho g(200\,\text{mm}) = P_0$$

$\Rightarrow \quad P = 1.0 \times 10^5 - 10^3 \times 10 \times 0.2$

$$= 98 \times 10^3\ \text{Nm}^{-2}$$

Using Boyle's law: $PV = P_0V_0$

$$98 \times 10^3 \times A(500 - 200) = 10^5 A\ (500 - H)$$

$\Rightarrow \quad H = 206\,\text{mm}$

$\therefore$ Required answer is $206 - 200 = 6\,\text{mm}$

6. In situation (S), the sphere is denser than liquid. The sphere moves down (its PE decrease) and an equal volume of liquid moves up (its PE increases).

Because sphere is denser, the PE of system ($x + y$) decrease. Since liquid is non viscous, there is no loss in mechanical energy and it remains constant.

In option (t), density of sphere is larger than liquid. Force by liquid (Buoyancy + viscous force) on sphere is Mg upward.

Mechanical energy is dissipated due to viscosity.

7. Let V_w = Volume occupied by water in the shell

V_a = Volume occupied by air in the shell

V_m = Volume of the material of the shell

For equilibrium: $F_B = W$

$$\frac{(V_w + V_a + V_m)}{2} \rho_w g = V_m \rho_m g + V_w \rho_w g$$

[Neglecting weight of air inside the shell]

$\Rightarrow \quad V_w + V_a + V_m = 2V_m \rho_C + 2V_w \qquad \left[\rho_C = \dfrac{\rho_m}{\rho_w}\right]$

$\Rightarrow \quad V_w = V_m(1 - 2\rho_C) + V_a$

If $\quad \rho_C > \dfrac{1}{2}$ then $V_w < V_a$

If $\quad \rho_C < \dfrac{1}{2}$ then $V_w > V_a$

8. (i) $A_1 v_1 = A_2 v_2$

$\Rightarrow \quad \pi(20\ \text{mm})^2 (5\,\text{mm}\,\text{s}^{-1}) = \pi(1\,\text{mm})^2 v_2$

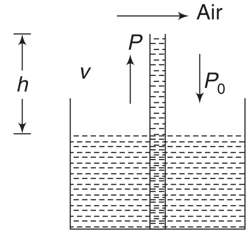
$\Rightarrow \quad v_2 = 2000\,\text{mms}^{-1} = 2\,\text{ms}^{-1}$

(ii) Due to high speed of air in the nozzle, the pressure drops and liquid is forced up by the atmospheric pressure. Let P = pressure inside the nozzles.

$$P + \frac{1}{2} \rho_a v_2^2 = P_0 + \frac{1}{2} \rho_a v_1^2$$

[Pressure inside the tube $\simeq P_0$ and speed $v_1 = 5\ \text{mms}^{-1} \simeq 0$]

$\therefore \quad P = P_0 - \dfrac{1}{2} \rho_a v_2^2$



Neglecting h (see figure), we can write for liquid as

$$P_0 - P = \frac{1}{2}\rho_l v^2$$

$$\Rightarrow \quad \frac{1}{2}\rho_a v_2^2 = \frac{1}{2}\rho_l v^2$$

$$\Rightarrow \quad v = \sqrt{\frac{\rho_a}{\rho_l}}\cdot v_2$$

$\therefore$ Volume flow rate of liquid $\propto \sqrt{\frac{\rho_a}{\rho_l}}$

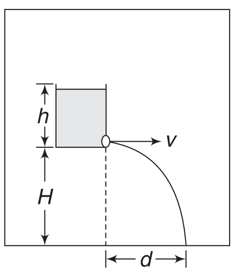
9. Speed of efflux

$$v = \sqrt{2gh}$$

$$\therefore \quad d = v\sqrt{\frac{2H}{g}} = \sqrt{2gh}\sqrt{\frac{2H}{g}}$$

$$d = 2\sqrt{hH}$$

d does not depend on g. Only when the lift falls freely, there will be no efflux.



Surface Tension and Viscosity

AIEEE/JEE Main Questions

1. $g_{\text{eff}} = 0$

In such situation, water can rise up to infinite height. Sine tube length is 20 cm only , water rises to the top of the tube.

2. $v = \frac{2}{9}\frac{r^2}{\eta}(d - \rho)g$

$$\therefore \quad \frac{v_s}{v_g} = \frac{d_s - \rho}{d_g - \rho}$$

$$\Rightarrow \quad v_s = 0.2\left[\frac{10.5 - 1.5.}{19.5 - 1.5}\right] = 0.2 \times \frac{1}{2} = 0.1\,\text{ms}^{-1}$$

3. At terminal speed, acceleration = 0

$$\Rightarrow \quad F = 0$$

$$\Rightarrow \quad F_{\text{viscous}} + F_{\text{Buoyancy}} = \omega$$

$$\Rightarrow \quad kv^2 + V\rho_2 g = V\rho_1 g$$

$$\Rightarrow \quad v = \sqrt{\frac{Vg(\rho_1 - \rho_2)}{k}}$$

4. Surface tension of soap water is less. Therefore, capillary rise is less.

5. Work done is nearly equal to increase in surface energy. [When the size increases, the walls of the bubble are pushed outward against the atmospheric pressure. But this is being done by pushing in more air. The expansion of air itself is negligible]

$$\therefore \; W = [8\pi R^2 - 8\pi r^2]S = 8\pi[(5 \times 10^{-2})^2 - (3 \times 10^{-2})^2] \times 0.03$$

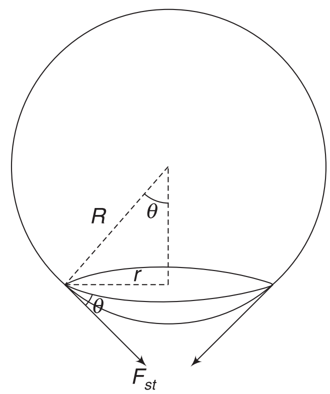
$$= 384\pi \times 10^{-6}\text{ J} = 0.384\pi\text{mJ} \simeq 0.4\pi\text{mJ}.$$

6. The surface tension force on the table acts along a circular line of length $2\pi r$, as shown in figure.

Resultant is vertical having magnitude.

$$F_{\text{st}} = T\cdot 2\pi r\sin\theta = T.2\pi r\cdot\frac{r}{R} = 2\pi T\frac{r^2}{R}$$

The bubble detaches when buoyancy becomes just larger than this surface tension force. For writing buoyancy, we can consider the volume of displaced water as $\frac{4}{3}\pi R^3$ since $r \ll R$



$$\therefore \quad \frac{4}{3}\pi R^3\rho_w g = 2\pi T\frac{r^2}{R}$$

$$\Rightarrow \quad r = R^2\sqrt{\frac{2\rho_w g}{3T}}$$

None of the given answers is correct.

IIT JEE (Advanced) Questions

1. Force due to pressure is

$$F_1 = Pav\cdot A = \left(\frac{P_0 + P_0 + \rho gh}{2}\right)2Rh.$$

$$= 2P_0 Rh + R\rho gh^2$$

Force due to surface tension is $F_2 = 2RT$

$\therefore$ Resultant force is $F = |F_1 - F_2|$

$$= |2P_0 Rh + R\rho gh^2 - 2RT|$$

2. This question makes use of the ideal gas equation

$$PV = nRT$$

At a given temperature, $n \propto P \cdot V$

Pressure of air inside bubble A is

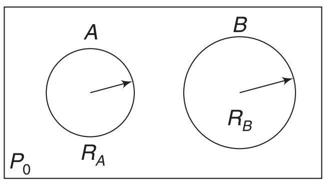
$$P_A = P_0 + \frac{4S}{R_A} = 8 + \frac{4 \times 0.04}{0.02}$$
$$= 16 \text{ Nm}^{-2}$$

Pressure inside bubble B is

$$P_B = P_0 + \frac{4S}{R_B} = 12\,\text{Nm}^{-2}$$

$$\frac{n_A}{n_B} = \frac{P_A V_A}{P_B V_B} = \frac{P_A}{P_B} \cdot \left(\frac{R_A}{R_B}\right)^3 = \frac{16}{12} \times \frac{1}{8} = \frac{1}{6}$$

$$\therefore \quad \frac{n_B}{n_A} = 6$$



3. The calculation of surface tension force on the drop (due to dropper) is similar to Q.6 in AIEEE/JEE Main Problems.

(i) $F_{st} = 2\pi r T \cdot \sin\theta = 2\pi r T \dfrac{r}{R}$

$$= \frac{2\pi r^2 T}{R}$$

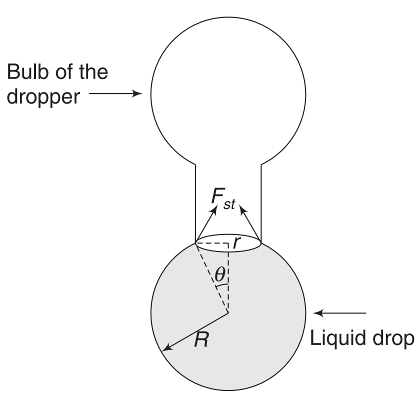
(ii) $\dfrac{2\pi r^2 T}{R} = mg = \dfrac{4}{3}\pi R^3 \rho g$

$$\Rightarrow \quad R^4 = \frac{3}{2}\frac{r^2 T}{\rho \cdot g}$$

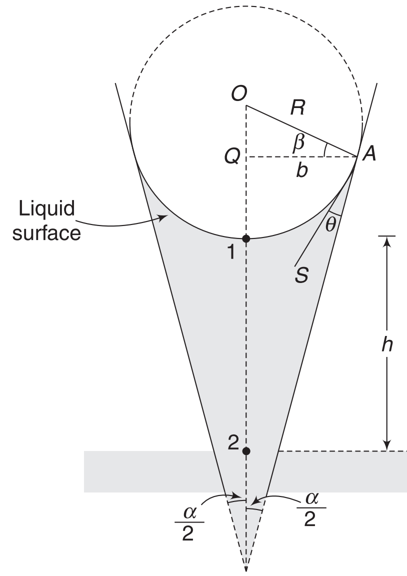
$$= \frac{3}{2} \times \frac{(5 \times 10^{-4})^2 \times 0.11}{10^3 \times 10}$$

$$R \simeq 1.4 \times 10^{-3}\,\text{m}$$

(iii) Surface energy, $U = T \cdot 4\pi R^2 = 2.7 \times 10^{-6}\,\text{J}$



4. The problem is based on finding the geometrical relation between the radius (b) of the tube and radius (R) of curvature of meniscus surface.



$$\angle QAS = 90° - \left(\theta + \frac{\alpha}{2}\right)$$

$$\angle QAS + \beta = 90°$$

$$\therefore \quad \beta = \theta + \frac{\alpha}{2}$$

$$\therefore \quad \frac{b}{R} = \cos\left(\theta + \frac{\alpha}{2}\right)$$

$$\Rightarrow \quad R = \frac{b}{\cos\left(\theta + \frac{\alpha}{2}\right)}$$

$$P_1 = P_0 - \frac{2S}{R}$$

$$P_2 = P_1 + Pgh \quad \Rightarrow \quad P_0 = P_0 - \frac{2S}{R} + \rho g h$$

$$\Rightarrow \quad h = \frac{2S}{R\rho g} = \frac{2S\cos\left(\theta + \frac{\alpha}{2}\right)}{b\rho g}$$

5. Obivously, $\rho_2 > \sigma^2$ and $\rho_1 < \sigma_1$.

For equilibrium of the system of two spheres

$$F_{BP} + F_{BQ} = W_P + W_Q$$

$$\sigma_1 Vg + \sigma_2 Vg = \rho_1 Vg + \rho_2 Vg$$

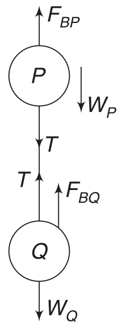
$$\Rightarrow \quad \sigma_1 - \rho_2 = \rho_1 - \sigma_2 \qquad \text{...(i)}$$

When P is alone in L_2, it will move up (since density of L_2 is even larger than L_1). It will attain a terminal velocity in upward direction.

$$v_P = \frac{2r^2}{9\eta_2}(\sigma_2 - \rho_1)$$

For Q in L_1

$$v_Q = \frac{2r^2}{9\eta_1}(\rho_2 - \sigma_1) \text{ downward.}$$



Since $\vec{v}_P$ and $\vec{v}_Q$ are oppositely directed, their dot product is negative.

And $$\left|\frac{v_P}{v_Q}\right| = \frac{\eta_1}{\eta_2}\left|\frac{(\sigma_2 - \rho_1)}{(\rho_2 - \sigma_1)}\right| = \frac{\eta_1}{\eta_2} \quad \text{[Using (i)]}$$

6. $v \propto \dfrac{r^2(d - P)}{\eta}$

$$\therefore \quad \frac{v_P}{v_Q} = \left(\frac{r_P}{r_Q}\right)^2 \frac{\eta_2}{\eta_1}\cdot\left(\frac{d - \rho_1}{d - \rho_2}\right)$$

$$= \left(\frac{2}{1}\right)^2 \times \frac{2}{3} \times \left(\frac{8 - 0.8}{8 - 1.6}\right)$$

$$= 3$$

7. Conservation of volume gives

$$k\cdot\frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3 \quad \Rightarrow \quad kr^3 = R^3 \qquad \text{...(i)}$$

$$U_i = S\cdot 4\pi R^2$$

$$U_f = k(S\cdot 4\pi r^2)$$

Given $k(S.4\pi r^2) - S\cdot 4\pi R^2 = 10^{-3}$

$$4\pi S[kr^2 - R^2] = 10^{-3}$$

$$\Rightarrow \qquad 4\pi\,\frac{0.1}{4\pi}\left[k^{\frac{1}{3}} - 1\right]R^2 = 10^{-3}$$

$$\left[\text{from (i) } r^2 = \frac{R^2}{k^{2/3}}\right]$$

$$\Rightarrow \qquad 10^{\alpha/3} - 1 = 100 \quad \Rightarrow \quad 10^{\alpha/3} = 101$$

$$\Rightarrow \quad \alpha \simeq 6$$

8. Capillary rise $h = \dfrac{2\sigma\cos\theta}{r\rho g}$

For a given material of the tube θ is constant. Therefore, h decreases with increase in r.

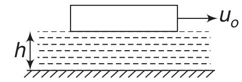
h is dependent on σ.

In a lift that is accelerating up, the effective acceleration is greater than g. Hence, h decreases.

h is not proportional to θ; it is proportional to $\cos\theta$.

9. Velocity gradient $= \dfrac{u_o}{h}$

Viscous force is $F_V = \eta A\,\dfrac{u_o}{h}$



This force is directly proportional to A and inversely proportional to h.

Shear stress $= \dfrac{F_V}{A} = \eta\dfrac{u_o}{h}$

This is proportional to η

Gravitation

AIEEE/JEE Main Questons

1. $PE = -\dfrac{GMm}{R}$

$\therefore$ Work needed to free the particle $= \dfrac{GMm}{R}$

$$= 6.67 \times 10^{-11} \times \frac{100 \times (10 \times 10^{-3})}{0.1}$$

$$= 6.67 \times 10^{-10}\text{ J}$$

2. $v = \sqrt{\dfrac{2GM_p}{R_p}} = \sqrt{\dfrac{2G\,10\,Me}{Re/10}} = 10 \times 11 = 110\text{ kms}^{-1}$

3. Acceleration due to gravity at height h is

$$\frac{g}{\left(1 + \frac{h}{R}\right)^2} = \frac{g}{9}$$

$$\Rightarrow \qquad 1 + \frac{h}{R} = 3 \quad \Rightarrow \quad h = 2R$$

4. Energy needed is $= \dfrac{GMm}{R} = gmR$

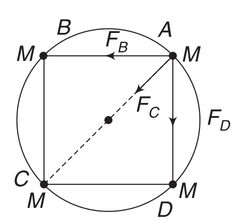
$$= 10 \times (1000) \times (6400 \times 10^3) = 6.4 \times 10^{10}\text{ J}$$

5. $E = E_f - E_i = -\dfrac{GMm}{2(3R)} - \left(-\dfrac{GMn}{R}\right) = \dfrac{5}{6}\,\dfrac{GMm}{R}$

6. Force on mass at A due to masses at B, C and D are

$$F_C = \frac{GM^2}{(2R)^2}$$

$$F_B = F_D = \frac{GM^2}{(AB)^2} = \frac{G\cdot M^2}{(\sqrt{2}\,R)^2} = \frac{GM^2}{2R^2}$$



Resultant force on mass at A is along AC.

$$F = \sqrt{F_B^2 + F_D^2} + F_C = \frac{GM^2}{R^2}\left(\frac{1}{\sqrt{2}} + \frac{1}{4}\right)$$

$$= \frac{GM^2}{4R^2}(2\sqrt{2} + 1)$$

This force provides centripetal force.

$$\frac{Mv^2}{R} = \frac{GM^2}{4R^2}(2\sqrt{2} + 1)$$

$$\Rightarrow \quad v = \frac{1}{2}\sqrt{\frac{GM}{R}(2\sqrt{2} + 1)}$$

7. Removed part has mass $= \dfrac{M}{8}$

$$V_{\text{centre of cavity}} = V_{\text{due to } M} - V_{\text{due to } \frac{M}{8}}$$

$$= -\frac{GM}{R^3}\left[\frac{3}{2}R^2 - \frac{1}{2}\cdot\left(\frac{R}{2}\right)^2\right] - \left[-\frac{3}{2}\,\frac{GM/8}{R/2}\right]$$

$$= -\frac{GM}{R}$$

8. For near surface satellite, orbital speed is

$$v_0 = \sqrt{gR}$$

Escape speed is $v = \sqrt{2gR}$

$$\therefore \quad \Delta v = (\sqrt{2} - 1)\sqrt{gR}$$

9. Variation inside the spherical earth is linear.

IIT JEE (Advanced) Questoins

1. When circular path is 1.

$$\frac{mv^2}{x} = \frac{Gm\left(\frac{4}{3}\pi x^3 \cdot \rho_0\right)}{x^2}$$

[Force is only due to mass inside sphere of radius x]

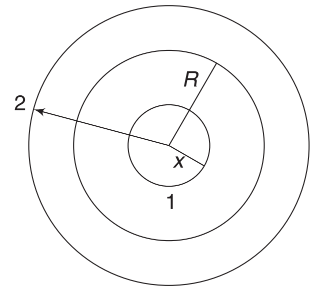
$$\Rightarrow \quad v = \sqrt{\frac{4\pi G\rho_0}{3}}\cdot x$$

$$\therefore \quad v \propto x$$

When path is 2 ($x > R$)

$$\frac{mv^2}{x} = \frac{Gm\left(\frac{4}{3}\pi R^3 \cdot \rho_0\right)}{x^2}$$

$$\Rightarrow \quad v = \sqrt{\frac{4\pi G\rho_0 R^3}{3}}\cdot\frac{1}{\sqrt{x}} \quad \Rightarrow \quad v \propto \frac{1}{\sqrt{x}}$$



3. Consider a ring of radius x and with dx.

Mass of ring element is

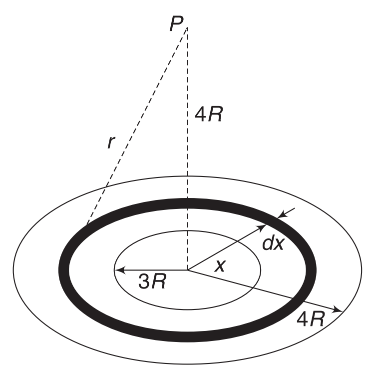
$$dm = \frac{M}{\pi(16R^2 - 9R^2)}\cdot 2\pi x dx$$

$$= \frac{2}{7}\frac{M}{R^2}x\,dx$$

Entire mass dm is at a distance r from point P, where

$$r = \sqrt{16R^2 + x^2}$$

$\therefore$ Potential at P due to dm is



$$dV = -\frac{Gdm}{r} = -\frac{2}{7}\frac{GM}{R^2}\frac{xdx}{\sqrt{16R^2 + x^2}}$$

Potential due to given disc is

$$V = -\frac{2M}{7R^2}\int_{3R}^{4R}\frac{xdx}{\sqrt{16R^2 + x^2}}$$

For integration take $16R^2 + x^2 = t$

$$\Rightarrow \quad 2x\,dx = dt$$

$$\Rightarrow \quad xdx = \frac{1}{2}dt$$

And $t = 25R^2$ when $x = 3R$

$t = 32R^2$ when $x = 4R$

$$\therefore \quad V = -\frac{2GM}{7R^2}\,\frac{1}{2}\int_{25R^2}^{32R^2}\frac{dt}{\sqrt{t}}$$

$$= -\frac{2GM}{7R^2}\left[\sqrt{32R^2} - \sqrt{25R^2}\right]$$

$$= -\frac{2GM}{7R}[4\sqrt{2} - 5]$$

PE of unit mass at P is V.

PE of unit mass at ∞ is zero.

$$\therefore \quad W = 0 - V = \frac{2GM}{7R}(4\sqrt{2} - 5)$$

4. $$\frac{GM_P}{R_P^2} = \frac{\sqrt{6}}{11}g = \frac{\sqrt{6}}{11}\frac{GM_e}{R_e^2}$$

$$\Rightarrow \quad \frac{\frac{4}{3}\pi R_P^3\rho_p}{R_p^2} = \frac{\sqrt{6}}{11}\frac{\frac{4}{3}\pi R_e^3\rho_e}{R_e^2}$$

$$\Rightarrow \quad R_p\rho_p = \frac{\sqrt{6}}{11}R_e\rho_e \quad \Rightarrow \quad R_p\frac{2}{3} = \frac{\sqrt{6}}{11}R_e$$

$$\Rightarrow \quad \frac{R_p}{R_e} = \frac{3\sqrt{6}}{22} \qquad \text{...(i)}$$

Escape speed, $v = \sqrt{\dfrac{2GM}{R}} = \sqrt{\dfrac{2G\frac{4}{3}\pi R^3\cdot\rho}{R}}$

$$\therefore \quad v \propto R\sqrt{\rho}$$

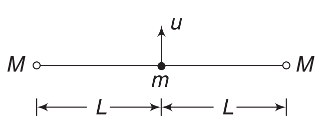
$$\therefore \quad \frac{v_p}{v_e} = \frac{R_p\sqrt{\rho_p}}{R_e\sqrt{\rho_e}} = \frac{3\sqrt{6}}{22}\cdot\sqrt{\frac{2}{3}} = \frac{3}{11}$$

$$\Rightarrow \quad v_p = \frac{3}{11}\times 11 = 3\,\text{km s}^{-1}$$

5. Escape speed $= \sqrt{2}v$

$$\therefore \quad k = \frac{1}{2}m(\sqrt{2}\,v)^2 = mu^2$$

6. Energy of mass m is conserved as gravitational force is conservative.

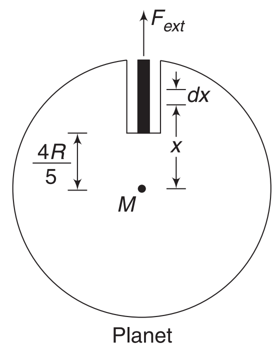


For m to just escape, $\dfrac{1}{2}mu^2 - \dfrac{GMm}{L} - \dfrac{GMm}{L} = 0$

$$\Rightarrow \quad u = 2\sqrt{\frac{GM}{L}}$$

7. Acceleration due to gravity at a distance x from the centre is

$$g = \frac{GM}{R^3}\cdot x$$



Force on a small element of length dx of the wire is

$$dF = (dm)g = (\lambda dx)\frac{GM}{R^3}x$$

Total gravitational force on wire

$$F = \frac{\lambda GM}{R^3}\int_{\frac{4R}{5}}^{R} x\,dx = \frac{\lambda GM}{R^3}\frac{1}{2}\left[R^2 - \frac{16R^2}{25}\right]$$

$$= \frac{9}{50}\frac{\lambda GM}{R}$$

External agent must apply equal force on the wire in radially outward direction to hold it in place

$$\therefore \quad F_{\text{ext}} = \frac{9}{50}\frac{\lambda GM}{R}$$

Volume of planet is $\dfrac{1}{1000}$ times of volume of Earth.

$$\therefore \quad \text{Mass of planet } M = \frac{M_e}{1000}$$

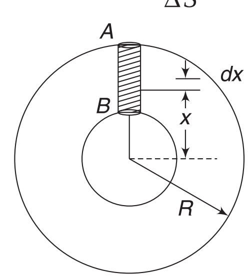
$$\therefore \quad F_{\text{ext}} = \frac{9}{50}\frac{\lambda M_e}{1000\times\frac{R_e}{10}} = \frac{9\lambda}{5000}\frac{GM_e}{R_e}$$

$$= \frac{9\times 10^{-3}}{5000}\times(g_eR_e) = \frac{9\times 10^{-3}}{5000}\times 10\times 6\times 10^6$$

$$= 108 \text{ N}$$

8. Consider a cylinder of fluid extending from surface (A) to a point (B) at a distance r from the centre. Cross section of the cylinder is ΔS. Pressure at B is

$$P = \frac{\text{Weight of cylinder } AB}{\Delta S}$$



To Find weight of cylinder, consider an elemental length dx as shown.

$$dW = \rho(\Delta S\, dx)g$$

Acceleration due to gravity at distance x from the centre is

$$g = \frac{GM}{R^3}\cdot x = \frac{G\,\frac{4}{3}\,\pi R^3\cdot\rho}{R^3}\,x = \frac{4}{3}\,\pi G\rho\cdot x$$

$$\therefore \quad dW = \frac{4}{3}\pi\, G\rho^2\Delta S\, x dx$$

$$\therefore \quad W = \frac{4}{3}\pi\, G\rho^2\Delta S\int_r^R x dx = \frac{2}{3}\pi\, G\rho^2\Delta S(R^2 - r^2)$$

$$\therefore \quad P = \frac{W}{\Delta S} = \frac{2}{3}\,\pi G\rho^2(R^2 - r^2)$$

Using the above expression you can check which of the given options are correct.

9. When bullet is at a distance r from the centre of the planet, force on it is

$$F = \frac{GMm}{r^2}$$

Force becomes one fourth when $r = 2R$.

(R = radius of the planet).

When the bullet is fired with speed v, it goes upto a distance $2R$ from the centre of the planet.

$$\therefore \quad \frac{1}{2}\,mv^2 - \frac{GMm}{R} = -\frac{GMm}{2R}$$

$$\Rightarrow \quad v = \sqrt{\frac{GM}{R}} \quad \Rightarrow \quad v = \frac{v_e}{\sqrt{2}}$$

$$\Rightarrow \quad v_e = \sqrt{2}\,v$$

10. Let acceleration of both point masses be a towards the centre of sphere [Masses are connected with rod. Therefore, they will move together]

For the middle mass

$$ma = \frac{GMm}{(3l)^2} - \frac{Gmm}{l^2} \quad \text{[Tension = 0]} \quad \text{...(i)}$$

For right most mass

$$ma = \frac{GMm}{(4l)^2} + \frac{Gmm}{l^2} \quad \text{...(ii)}$$

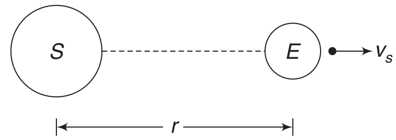
From (i) and (ii)

$$\frac{GMm}{9l^2} - \frac{Gm\cdot m}{l^2} = \frac{GMm}{16l^2} + \frac{Gmm}{l^2}$$

$$\Rightarrow \quad m = \frac{7M}{288}$$

11. $M_s = 3\times 10^5\, M_e$

$r = 2.5\times 10^4\, R_e$



For escaping, the total energy of projectile in the gravitational field of sun + earth shall be zero.

$$\frac{1}{2}\,mv_s^2 - \frac{GM_s m}{r} - \frac{GM_e m}{R_e} = 0$$

$$\Rightarrow \quad \frac{1}{2}v_s^2 = \frac{G\times 3\times 10^5 M_e}{2.5\times 10^4 R_e} + \frac{GM_e}{R_e}$$

$$\Rightarrow \quad v_s = \sqrt{\frac{2\,GM_e}{R_e}}\cdot\sqrt{13} = \sqrt{13}\;v_e \simeq 42\,\text{kms}^{-1}$$

12. Orbital speed is $v = \sqrt{\dfrac{GM}{R}}$

$$\therefore \quad \frac{v_1}{v_2} = \sqrt{\frac{R_2}{R_1}} = 2$$

This ratio is correctly matched in option (b). Just by looking at this one can say that the correct answer is (b)

Elasticity

AIEEE/JEE Main Questions

1. $u = \dfrac{1}{2}\times \text{stress}\times\text{strain} = \dfrac{1}{2}S\left(\dfrac{S}{Y}\right) = \dfrac{S^2}{2Y}$

2. Tension in the wire is still W.

$\therefore$ Extension = l

3. Wire 1: cross section = A, length = $3L$

Wire 2: cross section = $3A$, length = L

For wire 1: $\dfrac{\text{Stress}}{Y} = \text{Strain}$

$$\Rightarrow \quad \frac{F}{AY} = \frac{\Delta x}{3L} \quad\Rightarrow\quad F = \frac{AY\Delta x}{3L} \qquad \text{...(i)}$$

For wire 2: $\dfrac{F'}{3A\cdot Y} = \dfrac{\Delta x}{L}$

$$\Rightarrow \quad F' = \frac{3AY\Delta x}{L} = 9F \qquad \text{[from (i)]}$$

4. Volume (or mass) increases by a factor = 9^3

Cross sectional area of leg increases by a factor = 9^2

$\therefore$ Stress increases by a factor $= \frac{9^3}{9^2} = 9$

5. $$V = \frac{4}{3}\pi r^3$$

$$\Rightarrow \quad \Delta V = \frac{4}{3}\pi(3r^2 \Delta r)$$

$$\therefore \quad \frac{\Delta V}{V} = \frac{3\Delta r}{r}$$

$$K = \frac{P}{\Delta V/V} \quad \Rightarrow \frac{\Delta r}{r} = \frac{mg}{3Ka} \quad \left[\because \quad P = \frac{mg}{a}\right]$$

IIT JEE (Advanced) Questions

1. Volume strain $\frac{\Delta V}{V} = 0.1$

   Volume stress $\Delta P = 0.155 \times 10^5$ Pa.

$$\therefore \quad B = \frac{0.155 \times 10^5}{0.1} = 1.55 \times 10^5 \text{ Pa}$$

2. Tension is same in both wires. Let is be F.

   Let cross-section of the two wires be $4A$ and A.

   For thick wire:

$$\text{Strain} = \frac{\text{Stress}}{Y}$$

$$\Rightarrow \quad \frac{\Delta L_1}{2L} = \frac{F}{4A \cdot Y} \quad \Rightarrow \quad \Delta L_1 = \frac{FL}{2AY}$$

   For thin wire:

$$\frac{\Delta L_2}{L} = \frac{F}{AY} \quad \Rightarrow \quad \Delta L_2 = \frac{FL}{AY}$$

$$\therefore \quad \frac{\Delta L_2}{\Delta L_1} = \frac{2}{1}$$

3. For P fracture happens at higher stress.

   $\therefore$ P has more tensile strength.

   In the graphs, the end points are fracture points.

   P is more ductile as it fractures after a lot of strain. It can be stretched into a wire.